UNIFORM INTERIOR $C^{1,\alpha}$ ESTIMATES FOR INFINITY-HARMONIC FUNCTIONS: BLUEPRINT OF THE LEAN FORMALIZATION

JUAN J. MANFREDI

ABSTRACT. This document collects the natural-language statements and detailed proofs that accompany the Lean 4 formalization of the main results of OpenAI, *Uniform Interior $C^{1,\alpha}$ Estimates for Infinity-Harmonic Functions*, OpenAI Math Release preprint, October 4, 2026. The mathematics is that of the preprint; this document only reorganizes it into the units of the formalization. Each subsection corresponds to the Lean declaration of the same name in the directory `InfHarmonic/`; cross-references point to the results that a proof uses. Section 3 lists the formalized results of the paper: Theorem 1.1, Corollary 1.2 and Theorem 3.1. Within each section, items appear in dependency order.

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1. Definitions

1.1. `Preamble`.

Definition 1.1 (Second derivative in a direction)**.** For a real function ϕ on a finite-dimensional real inner-product space E that is twice continuously differentiable near z, and for $\xi \in E$, we write $D^2\phi(z)[\xi]$ for the value of the second Fréchet derivative of ϕ at z on the pair (ξ, ξ). In Lean this is `iteratedFDeriv` $\mathbb{R}$ 2 ϕ z $(\lambda_, \xi)$, and the hypothesis "ϕ is C^2 near z" is `ContDiffAt` $\mathbb{R}$ 2 ϕ z.

Definition 1.2 (Gradient)**.** For a real function ϕ on a finite-dimensional real inner-product space E, $\nabla\phi(z) \in E$ denotes the Riesz representative of the Fréchet derivative of ϕ at z; equivalently $\langle \nabla\phi(z), h\rangle = D\phi(z)h$ for all h. We write $\nabla g(z)$ for this gradient and, for a vector-valued statement, `HasGradientAt` u g z for the assertion that u is differentiable at z with gradient g.

Definition 1.3 (Local extrema on a set)**.** For $S \subseteq E$ and $z \in S$, we say that ψ has a local maximum on S at z if $\psi(x) \le \psi(z)$ for all x in some neighbourhood of z intersected with S, and a local minimum on S at z if the reverse inequality holds. When $S = E$ these are the unrestricted local extrema.

Definition 1.4 (Suprema and infima of sets of reals)**.** For $A \subseteq \mathbb{R}$, $\sup A$ and $\inf A$ denote the least upper bound and greatest lower bound when these exist. We use the convention that $\sup A = 0$ when A is empty or unbounded above and $\inf A = 0$ when A is empty or unbounded below; consequently every statement about such a supremum or infimum in this development carries the matching boundedness hypothesis explicitly.

Definition 1.5 (Extended reals and upper semicontinuity)**.** $\overline{\mathbb{R}} = [-\infty, +\infty]$ denotes the extended real line, a complete lattice; a real number is identified with its image in $\overline{\mathbb{R}}$, and $\bot = -\infty$. A function g with values in $\overline{\mathbb{R}}$ is upper semicontinuous if $\{g < c\}$ is open for every c.

Definition 1.6 (Locally uniform convergence)**.** For functions g_i, g on E and $S \subseteq E$, we say $g_i \to g$ uniformly on S if $\sup_{x\in S} |g_i(x) - g(x)| \to 0$, and locally uniformly if this holds on every compact S.

Definition 1.7 (Distance to a set)**.** For $S \subseteq E$ and $z \in E$, $\operatorname{dist}(z, S) = \inf_{w\in S} |z - w|$. The frontier (topological boundary) of S is written ∂S.

Definition 1.8 (Real powers, logarithm, square root)**.** For $s > 0$ and $\gamma \in \mathbb{R}$, s^γ denotes the real power; log is the natural logarithm, and $\sqrt{\cdot}$ the nonnegative square root, extended by the value 0 on negative arguments.

Definition 1.9 (The L^2 product and space-time)**.** For finite-dimensional real inner-product spaces F and $\mathbb{R}$, the space $F \times \mathbb{R}$ carries the Euclidean (Hilbert) product inner product, written $\ell^2(F \times \mathbb{R})$; this is the ambient space-time $\mathbb{R}^m \times \mathbb{R}$ of Sections 3–7. We write (h, ℓ) for the element of $\ell^2(F \times \mathbb{R})$ with spatial part h and time part ℓ, and $z = (z_x, z_t)$ for the two components of a space-time point. Likewise ℓ^2-products of finitely many copies of one space E carry the Euclidean product inner product.

Definition 1.10 (Lebesgue measure, null sets and "almost every")**.** On a finite-dimensional real inner-product space E we use the Borel σ-algebra and an additive Haar measure μ, i.e. a nonzero regular translation-invariant Borel measure that is finite on compact sets; on $\mathbb{R}^n$ this is Lebesgue measure up to a positive factor. Any two additive Haar measures on E are positive scalar multiples of one another (`Measure.isAddLeftInvariant_eq_smul_of_regular`), so the statements "$\mu(A) = 0$" ("A is null"), "$\mu(A) > 0$" and "μ-almost every" (abbreviated "a.e.") do not depend on the choice of μ. A set of full measure (null complement) is dense, because a null set has empty interior.

Proposition 1.11 (Compactness, Weierstrass and uniform continuity)**.** *In a finite-dimensional real normed space, a closed bounded set is compact. A continuous real function on a nonempty compact set attains a maximum and a minimum there, and is uniformly continuous there: for every $\eta > 0$ there is $\rho > 0$ such that any two points of the set at distance at most ρ have values differing by at most η.*

Proposition 1.12 (Convex functions are continuous and locally Lipschitz)**.** *Let E be a finite-dimensional real normed space, let $U \subseteq E$ be open and let $g : E \to \mathbb{R}$ be convex on U and real-valued. Then g is continuous on U (`ConvexOn.continuousOn`) and locally Lipschitz on U: every point of U has a neighbourhood on which g is Lipschitz (`ConvexOn.locallyLipschitzOn`).*

Proposition 1.13 (Affine minorant of a convex function)**.** *Let E be a finite-dimensional real normed space and let $g : E \to \mathbb{R}$ be convex and continuous on E. Then g has an affine minorant: there are $p_0 \in E$ and $c_0 \in \mathbb{R}$ with $g(x) \geq \langle p_0, x\rangle + c_0$ for all $x \in E$ (`ConvexOn.exists_affine_le_of_lt`).*

Proposition 1.14 (Rademacher's theorem)**.** *Let E, F be finite-dimensional real normed spaces, let μ be an additive Haar measure on E and let $f : E \to F$ be Lipschitz. Then f is differentiable at μ-almost every point of E (`LipschitzWith.ae_differentiableAt`). If instead f is merely locally Lipschitz on an open set U, the same conclusion holds at almost every point of U: U is a countable union of balls on each of which f is Lipschitz (`LipschitzOnWith.ae_differentiableWithinAt`), and within-differentiability on an open set is differentiability.*

Proposition 1.15 (Measure of a Lipschitz image)**.** *Let E be a finite-dimensional real inner-product space with additive Haar measure μ, put $n = \dim E$, let $A \subseteq E$, let $K \geq 0$ and let $f : E \to E$ be K-Lipschitz on A, i.e. $|f(a) - f(a')| \leq K|a - a'|$ for all $a, a' \in A$. Then*

$$\mu\big(f(A)\big) \leq K^n\, \mu(A).$$

In particular, taking A null, a map that is Lipschitz on a null set sends it to a null set; and a globally Lipschitz $f : E \to E$ sends null sets to null sets.

Both the source and the target of f are the same space E here, and that is essential: no such bound holds for a Lipschitz map between spaces of different dimensions. For instance the 1-Lipschitz coordinate projection $\mathbb{R}^2 \to \mathbb{R}$, $(a_1, a_2) \mapsto a_1$, maps the null line $\{a_2 = 0\}$ onto all of $\mathbb{R}$.

Justification: the n-dimensional Hausdorff measure $\mathcal{H}^n$ is an additive Haar measure on E (`isAddHaarMeasure_hausdorffMeasure`), hence a positive scalar multiple of μ by the uniqueness recorded above, and $\mathcal{H}^n(f(A)) \leq K^n \mathcal{H}^n(A)$ for a map that is K-Lipschitz on A (`LipschitzOnWith.hausdorffMe`). Rescaling by the same factor on both sides returns the statement for μ.

Proposition 1.16 (The area-formula inequality)**.** *Let E be a finite-dimensional real inner-product space with additive Haar measure μ, let $A \subseteq E$ be measurable, let $f : E \to E$ and let $f' : E \to (E \to_L E)$ satisfy: f has the derivative $f'(y)$ at y within A, for every $y \in A$. Then*

$$\mu\big(f(A)\big) \leq \int_A \big|\det f'(y)\big|\, d\mu(y)$$

(`MeasureTheory.addHaar_image_le_lintegral_abs_det_fderiv`).

Proposition 1.17 (Measurability of the differentiability set and of the derivative)**.** *Let E, F be finite-dimensional real normed spaces and let $f : E \to F$. The set $\{y \in E : f \text{ is differentiable at } y\}$ is Borel (`measurableSet_of_differentiableAt`) and the map $y \mapsto Df(y)$ is Borel measurable (`measurable_fderiv`). The determinant $\det : (E \to_L E) \to \mathbb{R}$ is continuous, so for a continuous f the set of points at which f is differentiable with $\det Df = 0$ is Borel.*

Proposition 1.18 (Integrating a continuous gradient along a segment)**.** *Let E be a finite-dimensional real inner-product space and let $g : E \to \mathbb{R}$ be differentiable with continuous gradient ∇g. Then for all $y, h \in E$*

$$g(y + h) - g(y) = \int_0^1 \langle \nabla g(y + th), h\rangle\, dt,$$

the integrand being a continuous function of t (`intervalIntegral.integral_eq_sub_of_hasDerivAt` applied to $t \mapsto g(y + th)$).

Proposition 1.19 (Haar measure of balls)**.** *Let E be a finite-dimensional real inner-product space with additive Haar measure μ and put $n = \dim E$. Then $0 < \mu(B(0, 1)) < \infty$ (`measure_ball_pos`, `measure_closedBall_lt_top`), and for every $x \in E$ and every $r \geq 0$*

$$\mu\big(\overline{B}(x, r)\big) = r^n\, \mu(B(0, 1))$$

(`Measure.addHaar_closedBall`). In particular $\mu(\overline{B}(x,r)) > 0$ for $r > 0$.

Proposition 1.20 (Bolzano–Weierstrass)**.** *In a finite-dimensional real normed space, every bounded sequence has a convergent subsequence.*

Proposition 1.21 (Determinant and invertibility)**.** *Let E be a finite-dimensional real normed space and let $P : E \to E$ be a continuous linear map, with determinant $\det P$ (the determinant of the matrix of P in any basis; `ContinuousLinearMap.det`). The map $P \mapsto \det P$ is continuous. Moreover $\det P \neq 0$ if and only if P is bijective (`LinearMap.det_eq_zero_iff_ker_ne_bot` together with the rank–nullity theorem), and in that case the inverse P^{-1} is again a continuous linear map, so that there is a constant $c > 0$, namely $c = 1/\|P^{-1}\|$ when $P^{-1} \neq 0$, with $|Ph| \geq c|h|$ for all $h \in E$.*

Proposition 1.22 (Second-order Taylor expansion with Peano remainder)**.** *Let V be a finite-dimensional real inner-product space, let $U \subseteq V$ be open, let $\Psi : V \to \mathbb{R}$ be twice continuously differentiable on U and let $\widehat{Z} \in U$. Write $\nabla\Psi(\widehat{Z})$ for the gradient and $D^2\Psi(\widehat{Z})$ for the second derivative, a bilinear form on V. Then $D^2\Psi(\widehat{Z})$ is symmetric (`ContDiffAt.isSymmSndFDerivAt`, `second_derivative_symmetric`), and*

$$\Psi(\widehat{Z}+W) = \Psi(\widehat{Z}) + \langle \nabla\Psi(\widehat{Z}), W\rangle + \tfrac{1}{2}D^2\Psi(\widehat{Z})(W,W) + R(W), \qquad \frac{|R(W)|}{|W|^2} \longrightarrow 0 \quad (W \to 0).$$

Justification of the expansion: for small $|W|$ the segment from $\widehat{Z}$ to $\widehat{Z}+W$ lies in U, and the one-variable second-order Taylor theorem (`taylor_isLittleO`; or the Lagrange form of the remainder) applied to $t \mapsto \Psi(\widehat{Z}+tW)$ on $[0,1]$ gives $\Psi(\widehat{Z}+W) = \Psi(\widehat{Z}) + \langle\nabla\Psi(\widehat{Z}),W\rangle + \frac{1}{2}D^2\Psi(\widehat{Z}+\theta W)(W,W)$ for some $\theta \in (0,1)$; the difference from the displayed expansion is at most $\frac{1}{2}\|D^2\Psi(\widehat{Z}+\theta W) - D^2\Psi(\widehat{Z})\|\,|W|^2$, which is $o(|W|^2)$ because $D^2\Psi$ is continuous at $\widehat{Z}$.

Definition 1.23 (Probability spaces, conditional expectation and filtrations)**.** A *probability space* is a measurable space $(\Omega, \mathcal{M})$ together with a measure μ with $\mu(\Omega) = 1$ (`IsProbabilityMeasure`). We write $\mathbb{E}[X] = \int_\Omega X\,d\mu$ for the integral of an integrable function, in the Bochner sense for functions with values in a finite-dimensional real inner-product space F; `Integrable` is strong measurability together with finiteness of $\mathbb{E}\|X\|$. For a sub-σ-algebra $\mathcal{N} \subseteq \mathcal{M}$, the *conditional expectation* $\mu[X \mid \mathcal{N}]$ (`condExp`, written $\mathbb{E}[X \mid \mathcal{N}]$ below) is the $\mathcal{N}$-measurable integrable function characterized a.e. by $\int_A X\,d\mu = \int_A \mu[X \mid \mathcal{N}]\,d\mu$ for every $A \in \mathcal{N}$; it is set to 0 when X is not integrable. A *filtration* indexed by $\mathbb{N}$ (`Filtration`) is a nondecreasing family $\mathcal{G}_0 \subseteq \mathcal{G}_1 \subseteq \cdots$ of sub-σ-algebras of $\mathcal{M}$, and a family (X_n) is *adapted* to it (`Adapted`, `StronglyAdapted`) if each X_n is $\mathcal{G}_n$-measurable.

Definition 1.24 (Martingales and submartingales)**.** An adapted integrable family (X_n) is a *martingale* for $(\mathcal{G}_n)$ and μ (`Martingale`) if $\mathbb{E}[X_j \mid \mathcal{G}_i] = X_i$ a.e. whenever $i \leq j$, and a *submartingale* (`Submartingale`) if, for real-valued X_n, $X_i \leq \mathbb{E}[X_j \mid \mathcal{G}_i]$ a.e. whenever $i \leq j$. For $\mathbb{N}$-indexed families it suffices to verify the inequality for consecutive indices (`submartingale_nat`).

Definition 1.25 (Stopping times, stopped values and hitting times)**.** A map $\tau : \Omega \to \mathbb{N} \cup \{\infty\}$ is a *stopping time* for $(\mathcal{G}_n)$ (`IsStoppingTime`) if $\{\tau \leq n\} \in \mathcal{G}_n$ for every n. The *stopped value* of a family (X_n) at τ is $X_\tau(\omega) = X_{\tau(\omega)}(\omega)$ (`stoppedValue`). The *hitting time* of a set S in the index window $[n_1, n_2]$ is the least n in that window with $X_n(\omega) \in S$, and n_2 if there is none (`hittingBtwn`); it is a stopping time when (X_n) is adapted and S is measurable (`Adapted.isStoppingTime_hittingBtwn`).

Definition 1.26 (Weak convergence, tightness and infinite products)**.** The space of probability measures on a metrizable space $\mathcal{X}$ carries the topology of weak convergence (`ProbabilityMeasure`), for which $\mu_j \to \mu$ means $\int \varphi\,d\mu_j \to \int \varphi\,d\mu$ for every bounded continuous φ. A set S of measures is *tight* (`IsTightMeasureSet`) if for every $\varepsilon > 0$ there is a compact $K \subseteq \mathcal{X}$ with $\nu(\mathcal{X} \setminus K) \leq \varepsilon$ for every $\nu \in S$ (`isTightMeasureSet_iff_exists_isCompact_measure_compl_le`; the definition itself is the equivalent statement that $\sup_{\nu\in S}\nu$ tends to 0 along the small sets of the cocompact filter). Given measures ν_i on measurable spaces X_i, i ranging over an arbitrary index set, there

is a product measure $\bigotimes_i \nu_i$ on $\prod_i X_i$ (`Measure.infinitePi`), a probability measure when every ν_i is; the fair-sign sequence is the case $X_i = \{-1, 1\}$ with each ν_i uniform.

Proposition 1.27 (Conditional Jensen for the norm and its powers)**.** *Let F be a finite-dimensional real inner-product space, let $\mathcal{N} \subseteq \mathcal{M}$ be a sub-σ-algebra and let $X : \Omega \to F$. Then*

$$\big\|\mathbb{E}[X \mid \mathcal{N}]\big\| \le \mathbb{E}\big[\|X\| \mid \mathcal{N}\big] \quad a.e.$$

(`norm_condExp_le`), and for every $p \ge 1$ such that $\|X\|^p$ is integrable,

$$\big\|\mathbb{E}[X \mid \mathcal{N}]\big\|^p \le \mathbb{E}\big[\|X\|^p \mid \mathcal{N}\big] \quad a.e.$$

(`Integrable.norm_condExp_rpow_le`).

Proposition 1.28 (Doob's weak-type maximal inequality)**.** *Let μ be a finite measure, let $(\mathcal{G}_n)$ be a filtration and let (f_n) be a nonnegative submartingale. Then for every $\varepsilon \ge 0$ and every n,*

$$\varepsilon\,\mu\Big\{\omega : \varepsilon \le \max_{k \le n} f_k(\omega)\Big\} \le \int_{\{\varepsilon \le \max_{k\le n} f_k\}} f_n \, d\mu$$

(`maximal_ineq`). The right side is at most $\mathbb{E}[f_n]$ when $f_n \ge 0$.

Proposition 1.29 (Optional stopping and optional sampling)**.** *Let μ be a finite measure and (f_n) a submartingale for $(\mathcal{G}_n)$. If $\tau \le \pi$ are stopping times bounded by a constant N, then $\mathbb{E}[f_\tau] \le \mathbb{E}[f_\pi]$ (`Submartingale.expected_stoppedValue_mono`). Moreover, for every $\varepsilon \ge 0$ and every n,*

$$\varepsilon\,\mu\Big\{\omega : \varepsilon \le \max_{k \le n} f_k(\omega)\Big\} \le \int_{\{\varepsilon \le \max_{k\le n} f_k\}} f_\sigma \, d\mu,$$

where σ is the hitting time of $[\varepsilon, \infty)$ in the window $[0, n]$ (`smul_le_stoppedValue_hittingBtwn`).

Proposition 1.30 (The layer-cake identity)**.** *Let $(\Omega, \mathcal{M}, \mu)$ be a measure space and let $f : \Omega \to \mathbb{R}$ be almost everywhere nonnegative and almost everywhere measurable. Then*

$$\int_\Omega f \, d\mu = \int_0^\infty \mu\{f > \lambda\} \, d\lambda,$$

as an identity of integrals with values in $[0, \infty]$ (`lintegral_eq_lintegral_meas_lt`). In particular f is integrable if and only if the right side is finite, and then the two sides agree as real numbers.

Proposition 1.31 (Portmanteau)**.** *Let $\mathcal{X}$ be a metrizable space and let $\mu_j \to \mu$ weakly in the space of probability measures on $\mathcal{X}$. Then*

$$\limsup_j \mu_j(C) \le \mu(C) \quad \text{for every closed } C, \qquad \mu(O) \le \liminf_j \mu_j(O) \quad \text{for every open } O$$

(`ProbabilityMeasure.limsup_measure_closed_le_of_tendsto` and `ProbabilityMeasure.le_liminf_measur`

Proposition 1.32 (Prokhorov's theorem)**.** *Let $\mathcal{X}$ be a complete separable metric space and let S be a set of probability measures on $\mathcal{X}$ whose underlying set of measures is tight. Then the closure of S in the topology of weak convergence is compact (`isCompact_closure_of_isTightMeasureSet`). In particular every sequence in S has a weakly convergent subsequence.*

Proposition 1.33 (Spectral theorem in finite dimensions)**.** *Let F be a finite-dimensional real inner-product space of dimension n and let $T : F \to F$ be linear and symmetric, i.e. $\langle Ta, b\rangle = \langle a, Tb\rangle$ for all a, b. Then F has an orthonormal basis $(e_1, \dots, e_n)$ of eigenvectors of T, with real eigenvalues $\lambda_1, \dots, \lambda_n$, so that $Te_i = \lambda_i e_i$ (`LinearMap.IsSymmetric.eigenvectorBasis` and `LinearMap.IsSymmetric.apply_eigenvectorBasis`).*

Proposition 1.34 (Neumann series)**.** *Let R be a normed ring in which geometric series are summable, for instance the ring of continuous linear endomorphisms of a finite-dimensional normed space. If $t \in R$ has $\|t\| < 1$, then $1 - t$ is invertible, with inverse $\sum_{k\ge 0} t^k$ (`Units.oneSub` and `NormedRing.inverse_one_sub`).*

Proposition 1.35 (Pull-out property of the conditional expectation)**.** *Let E, F be finite-dimensional real inner-product spaces, let $B : F \times E \to \mathbb{R}$ be a continuous bilinear map, let $\mathcal{N} \subseteq \mathcal{M}$ be a sub-σ-algebra, and let $X : \Omega \to F$ and $Y : \Omega \to E$ be such that X and $\omega \mapsto B(X(\omega), Y(\omega))$ are integrable and Y is $\mathcal{N}$-measurable. Then*

$$\mathbb{E}\big[B(X,Y) \mid \mathcal{N}\big] = B\big(\mathbb{E}[X \mid \mathcal{N}],\, Y\big) \quad \text{a.e.}$$

(`condExp_bilin_of_aestronglyMeasurable_right`). The case $B(a,b) = \langle a, b\rangle$ with $E = F$ is the one used below.

1.2. `InfLap`.

Definition 1.36 (The infinity Laplacian)**.** Let E be a finite-dimensional real inner-product space, let $\phi : E \to \mathbb{R}$ and let $z \in E$. The *unnormalized infinity Laplacian* of ϕ at z is

$$\Delta_\infty \phi(z) = D^2\phi(z)[\nabla\phi(z)],$$

the value of the second derivative of ϕ at z on the pair $(\nabla\phi(z), \nabla\phi(z))$. For $E = \mathbb{R}^d$ and smooth ϕ this is $\sum_{i,j=1}^d \phi_i\phi_j\phi_{ij} = \langle D^2\phi\,\nabla\phi, \nabla\phi\rangle$.

1.3. `IsInfSubsolutionOn`.

Definition 1.37 (Infinity subsolution)**.** Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ and let $u : E \to \mathbb{R}$. We say u is an *infinity subsolution on* Ω if Ω is open, u is continuous on Ω, and for every $z \in \Omega$ and every $\phi : E \to \mathbb{R}$ that is C^2 near z such that $x \mapsto u(x) - \phi(x)$ has a local maximum on Ω at z, one has

$$\Delta_\infty \phi(z) \geq 0,$$

with Δ_∞ as in `InfLap`. Such a ϕ is a *local upper test for u at z*. All tests are admitted, including those with $\nabla\phi(z) = 0$.

1.4. `IsInfSupersolutionOn`.

Definition 1.38 (Infinity supersolution)**.** Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ and let $u : E \to \mathbb{R}$. We say u is an *infinity supersolution on* Ω if Ω is open, u is continuous on Ω, and for every $z \in \Omega$ and every $\phi : E \to \mathbb{R}$ that is C^2 near z such that $x \mapsto u(x) - \phi(x)$ has a local minimum on Ω at z, one has

$$\Delta_\infty \phi(z) \leq 0,$$

with Δ_∞ as in `InfLap`. Such a ϕ is a *local lower test for u at z*. All tests are admitted, including those with $\nabla\phi(z) = 0$.

1.5. `IsInfHarmonicOn`.

Definition 1.39 (Infinity-harmonic function)**.** Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ and let $u : E \to \mathbb{R}$. We say u is *infinity-harmonic on* Ω if u is both an infinity subsolution on Ω in the sense of `IsInfSubsolutionOn` and an infinity supersolution on Ω in the sense of `IsInfSupersolutionOn`. Explicitly: Ω is open, u is continuous on Ω, every local upper C^2 test ϕ for u at $z \in \Omega$ satisfies $\Delta_\infty\phi(z) \geq 0$, and every local lower test satisfies $\Delta_\infty\phi(z) \leq 0$.

1.6. `ConeSlopeUpper`.

Definition 1.40 (Upper cone slope)**.** Let E be a finite-dimensional real inner-product space, let $w : E \to \mathbb{R}$, let $z \in E$ and let $r \in \mathbb{R}$. The *upper cone slope of w at z of radius r* is

$$L_r^+(w; z) = \sup_{|h| \leq r} \frac{w(z+h) - w(z)}{r},$$

the supremum being over the closed ball $\{h \in E : |h| \leq r\}$. For $r > 0$ and w continuous on a neighbourhood of $\overline{B}_r(z)$ this supremum is a maximum and equals $r^{-1}\max_{|h|\leq r}\{w(z+h) - w(z)\}$.

1.7. `ConeSlopeLower`.

Definition 1.41 (Lower cone slope)**.** Let E be a finite-dimensional real inner-product space, let $w : E \to \mathbb{R}$, let $z \in E$ and let $r \in \mathbb{R}$. The *lower cone slope of w at z of radius r* is

$$L_r^-(w; z) = \sup_{|h| \le r} \frac{-(w(z+h) - w(z))}{r},$$

the supremum being over the closed ball $\{h \in E : |h| \le r\}$. For $r > 0$ and w continuous on a neighbourhood of $\overline{B}_r(z)$ this supremum is a maximum and equals $r^{-1} \max_{|h| \le r}\{-(w(z+h) - w(z))\}$.

1.8. `OscOn`.

Definition 1.42 (Oscillation)**.** Let E be a finite-dimensional real inner-product space, let $u : E \to \mathbb{R}$ and let $S \subseteq E$. The *oscillation of u on S* is

$$\operatorname{osc}_S u = \sup_S u - \inf_S u = \sup u(S) - \inf u(S).$$

Here $\sup u(S)$ and $\inf u(S)$ use the convention of the preamble, so that a statement controlling $\operatorname{osc}_S u$ is meaningful only in the presence of the matching boundedness hypotheses on $u(S)$, which are always stated explicitly.

1.9. `NormalizedInfFamily`.

Definition 1.43 (The normalized family $\mathcal{U}$)**.** Fix an integer $d \ge 1$ and put $n = d+1$, identifying $\mathbb{R}^n$ with the space-time $\ell^2(\mathbb{R}^d \times \mathbb{R})$ of the preamble and writing $e_b = (0, 1)$ for the added unit direction. A function U on $\mathbb{R}^n$ belongs to the *normalized family $\mathcal{U}$* if there is a function $u : \mathbb{R}^d \to \mathbb{R}$ such that

(1) u is continuous and infinity-harmonic on B_1^d in the sense of `IsInfHarmonicOn`;
(2) $u(0) = 0$, the set $u(B_1^d)$ is bounded above and below, and $\operatorname{osc}_{B_1^d} u \le 1$, the oscillation being that of `OscOn`;
(3) $U(y, b) = u(y) + b$ for every $(y, b) \in \mathbb{R}^d \times \mathbb{R}$.

1.10. `FitCylinder`.

Definition 1.44 (Fitting cylinder)**.** Let E be a finite-dimensional real inner-product space, let $z \in E$, let $r, \lambda \in \mathbb{R}$ and let $P \in E$. Writing $N_P = P/|P|$ for the normalized axis direction, the *closed cylinder of axial radius r and transverse radius $\sqrt{\lambda}\, r$ at z with axis P* is

$$Z_\lambda(z, r; P) = \big\{z + h : \ |h \cdot N_P| \le r, \quad |h - (h \cdot N_P)N_P| \le \sqrt{\lambda}\, r\big\},$$

equivalently the set of $Z \in E$ whose increment $h = Z - z$ has axial component of absolute value at most r and transverse component of length at most $\sqrt{\lambda}\, r$.

1.11. `IsCylinderFit`.

Definition 1.45 (Cylinder fit and good parameters)**.** Let E be a finite-dimensional real inner-product space, let $u : E \to \mathbb{R}$, let $z \in E$, let $r, \lambda \in \mathbb{R}$ and let $P \in E$. We say P *realizes the cylinder fit of parameter λ for u at (z, r)* if $P \ne 0$ and

$$|u(Z) - u(z) - P \cdot (Z - z)| \le \lambda r \qquad \big(Z \in Z_\lambda(z, r; P)\big),$$

with $Z_\lambda(z, r; P)$ the cylinder of `FitCylinder`. We call λ *good for u at (z, r)* if some P realizes the cylinder fit of parameter λ for u at (z, r), and *bad* otherwise. Goodness need not be monotone in λ.

1.12. `IsBadScaleData`.

Definition 1.46 (Selected bad-scale data)**.** Fix an integer $d \ge 1$, put $n = d+1$, write $G = B^n_{3/4}$ and $D(z) = \operatorname{dist}(z, \partial G)$, and set $\Lambda = 32$. Let $\gamma_i, s_i, r_i, \lambda_i$ be real sequences, let U_i be real functions on $\mathbb{R}^n$ and let $z_i \in \mathbb{R}^n$. We say that $(\gamma_i, s_i, r_i, \lambda_i, U_i, z_i)_{i \ge 0}$ is *selected bad-scale data in dimension d* if:

(1) $\gamma_i > 0$ and $\gamma_i \to 0$; $s_i > 0$ and $s_i \to 0$; $\lambda_i > 0$ and $\lambda_i \to 0$;
(2) each U_i belongs to the normalized family $\mathcal{U}$ of `NormalizedInfFamily`;
(3) $z_i \in G$ and $0 < r_i \le s_i D(z_i)$;
(4) λ_i is not good for U_i at (z_i, r_i), in the sense of `IsCylinderFit`;
(5) for every fixed $M \ge 2$ there is i_0 such that for all $i \ge i_0$, every center w with $|w - z_i| \le M r_i$ and every radius $0 < \rho \le M r_i$ are admissible, meaning $w \in G$ and $\rho \le D(w)/8$, and $\Lambda\lambda_i$ is good for U_i at (w, ρ).

Monotonicity of the sequences s_i and λ_i, which the selection of Lemma 3.4 also arranges, is not recorded here because it is not used below.

1.13. `BetaFlatness`.

Definition 1.47 (The bad-parameter supremum $\beta_u(z,r)$)**.** Let E be a finite-dimensional real inner-product space, let $u : E \to \mathbb{R}$, let $z \in E$ and let $r \in \mathbb{R}$. Put

$$\beta_u(z,r) = \sup\{\lambda \in (0,1) : \lambda \text{ is not good for } u \text{ at } (z,r)\},$$

with goodness as in `IsCylinderFit`, and with the value 0 when the set is empty. Thus $0 \le \beta_u(z,r) \le 1$, and every parameter in $(\beta_u(z,r), 1)$ is good for u at (z,r).

1.14. `FlatnessModulus`.

Definition 1.48 (The family flatness modulus b)**.** Fix an integer $d \ge 1$, put $n = d + 1$ and identify $\mathbb{R}^n$ with the space-time $\ell^2(\mathbb{R}^d \times \mathbb{R})$. Set

$$G = B^n_{3/4}, \qquad D(z) = \operatorname{dist}(z, \partial G).$$

For $s \in \mathbb{R}$ define

$$b_d(s) = \sup\big\{\beta_U(z,r) : \ U \in \mathcal{U},\ z \in G,\ 0 < r \le sD(z)\big\},$$

with $\mathcal{U}$ the normalized family of `NormalizedInfFamily` and β_U the bad-parameter supremum of `BetaFlatness`, and with the value 0 when the set is empty. For $0 < s \le 1/8$ one has $0 \le b_d \le 1$, and b_d is nondecreasing.

1.15. `Superjet`.

Definition 1.49 (Second-order superjet)**.** Let E be a finite-dimensional real inner-product space, let $g : E \to \mathbb{R}$ and let $z \in E$. The *second-order superjet* $J^{2,+}g(z)$ is the set of pairs (p, X), with $p \in E$ and X a symmetric continuous bilinear form on E, such that

$$g(z+h) \le g(z) + p \cdot h + \tfrac{1}{2} X[h] + o(|h|^2) \qquad (h \to 0),$$

where $X[h] = X(h,h)$. Explicitly: for every $\varepsilon > 0$ there is $\delta > 0$ with

$$g(z+h) \le g(z) + \langle p, h\rangle + \tfrac{1}{2} X(h,h) + \varepsilon |h|^2 \qquad (|h| < \delta).$$

1.16. `ClosedSuperjet`.

Definition 1.50 (Closed superjet)**.** Let E be a finite-dimensional real inner-product space, let $g : E \to \mathbb{R}$ and let $z \in E$. The *closed superjet* $\overline{J}^{2,+}g(z)$ is the set of pairs (p, X), with $p \in E$ and X a continuous bilinear form on E, for which there are sequences $z_k \in E$, $p_k \in E$ and symmetric continuous bilinear forms X_k with

$$(p_k, X_k) \in J^{2,+}g(z_k) \quad \text{for all } k, \qquad z_k \to z, \quad g(z_k) \to g(z), \quad p_k \to p, \quad X_k \to X,$$

where $J^{2,+}$ is the superjet of `Superjet`. Thus closed superjets are limits of superjets at points approaching z, with the function values also converging; the symmetry of X is automatic from the symmetry of the X_k.

1.17. `SupConvolution`.

Definition 1.51 (Sup-convolution)**.** Let E be a finite-dimensional real inner-product space, let $\varepsilon \in \mathbb{R}$ and let $\phi : E \to \mathbb{R}$. The *sup-convolution of ϕ with parameter ε* is the function $\phi^\varepsilon : E \to \mathbb{R}$ given by

$$\phi^\varepsilon(z) = \sup_{w \in E} \left(\phi(w) - \frac{|z-w|^2}{2\varepsilon} \right),$$

the supremum of the set $\{\phi(w) - |z-w|^2/(2\varepsilon) : w \in E\}$.

This set is always nonempty, as the choice $w = z$ shows, but it need not be bounded above, and the preamble's convention assigns the value 0 to the supremum of a set that is not bounded above. The definition therefore carries no implicit finiteness claim: every statement below that uses ϕ^ε as a genuine supremum supplies the required boundedness hypothesis explicitly, and the boundedness itself is established, under a linear growth bound on ϕ and for $\varepsilon > 0$, in the node `SupConvolutionSemiconvex`, which imports this definition.

1.18. `SemiconvexOn`.

Definition 1.52 (Semiconvexity)**.** Let E be a finite-dimensional real inner-product space, let $\lambda \in \mathbb{R}$, let $\Omega \subseteq E$ and let $\phi : E \to \mathbb{R}$. We say that ϕ is *λ-semiconvex on Ω* if the function

$$z \longmapsto \phi(z) + \frac{\lambda |z|^2}{2}$$

is convex on Ω. Here, as throughout, "g is convex on Ω" means that Ω is a convex subset of E and that

$$g(t_1 z_1 + t_2 z_2) \le t_1 g(z_1) + t_2 g(z_2)$$

for all $z_1, z_2 \in \Omega$ and all $t_1, t_2 \ge 0$ with $t_1 + t_2 = 1$.

Thus 0-semiconvexity is convexity, and for $\lambda > 0$ a λ-semiconvex function is one that is convex after the addition of the quadratic $\lambda|z|^2/2$; informally its second derivative, where it exists, is bounded below by $-\lambda$ times the identity.

1.19. `IsMoreauProx`.

Definition 1.53 (Proximal map)**.** Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ and let $u : E \to E$. We say that u is a *proximal map for ψ* if for every $y \in E$ the point $u(y)$ minimizes $x \mapsto \psi(x) + |x-y|^2/2$ over all of E, that is,

$$\psi(u(y)) + \frac{|u(y)-y|^2}{2} \le \psi(x) + \frac{|x-y|^2}{2} \qquad (y, x \in E).$$

The associated *Moreau envelope* of ψ is then the function $y \mapsto \psi(u(y)) + |u(y)-y|^2/2$, whose value is the minimum itself.

The definition imposes no convexity and asserts no existence: convexity of ψ is what makes such a u exist and makes $u(y)$ unique, and those are separate statements.

1.20. `IsBaryConfig`.

Definition 1.54 (Finite horizontal probability configuration)**.** Let F be a finite-dimensional real inner-product space, let n be a nonnegative integer, let $\theta_1, \dots, \theta_n \in \mathbb{R}$, let $y_1, \dots, y_n \in F$ and let $x \in F$. We say (θ, y) is a *finite horizontal probability configuration with mean x* if

$$\theta_i > 0 \ (1 \le i \le n), \qquad \sum_{i=1}^n \theta_i = 1, \qquad \sum_{i=1}^n \theta_i y_i = x.$$

The single-point configuration $n = 1$, $\theta_1 = 1$, $y_1 = x$ is always admissible.

1.21. `BaryRadius`.

Definition 1.55 (Root mean square radius of a configuration)**.** Let F be a finite-dimensional real inner-product space, let n be a nonnegative integer, let $\theta_1, \dots, \theta_n \in \mathbb{R}$, let $y_1, \dots, y_n \in F$ and let $x \in F$. The *root mean square radius about* x is

$$r = \left(\sum_{i=1}^{n} \theta_i |y_i - x|^2 \right)^{1/2}.$$

When $\theta_i > 0$, $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i = x$, this is the radius of that configuration, and $\sum_i \theta_i |y_i|^2 - |x|^2 = r^2$.

1.22. `BaryEnv`.

Definition 1.56 (Barycentric envelope)**.** Let F be a finite-dimensional real inner-product space, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$, let $s \in \mathbb{R}$ and let $z = (x, t)$ be a space-time point. The *barycentric envelope* is

$$\mathcal{C}_s f(x,t) = \sup \left\{ \sum_i \theta_i f(y_i, t) - \frac{\sum_i \theta_i |y_i - x|^2}{2s} \right\},$$

the supremum being over all finite horizontal probability configurations (θ, y) with mean x in the sense of `IsBaryConfig`, and $\left(\sum_i \theta_i |y_i - x|^2\right)^{1/2}$ being the radius of `BaryRadius`. The operation averages the single horizontal slice at the fixed time t, and s controls the variance penalty. The single-point configuration $y_1 = x$ is admissible, so the supremum is at least $f(x, t)$. By the convention of the preamble it is meaningful only together with the hypotheses guaranteeing boundedness above, which are always stated explicitly.

1.23. `SubquadraticGrowth`.

Definition 1.57 (Subquadratic spatial growth on compact time intervals)**.** Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ and let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$. We say f has *subquadratic spatial growth on compact time intervals of* I if for every compact $J \subseteq I$

$$\lim_{M \to \infty} \sup_{\substack{|x| \geq M \\ t \in J}} \frac{|f(x,t)|}{|x|^2} = 0.$$

Explicitly: for every compact $J \subseteq I$ and every $\varepsilon > 0$ there is $M > 0$ such that $|f(x, t)| \leq \varepsilon |x|^2$ whenever $|x| \geq M$ and $t \in J$.

1.24. `HasJensenBound`.

Definition 1.58 (Horizontal Jensen bound)**.** Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let $\kappa \in \mathbb{R}$. We say f *satisfies the horizontal Jensen bound with constant* κ *on* I if

$$\sum_i \theta_i f(y_i, t) - f(x, t) \leq \kappa r$$

for every $t \in I$ and every finite horizontal probability configuration (θ, y) with mean x in the sense of `IsBaryConfig`, where r is its root mean square radius of `BaryRadius`.

1.25. `UnifSubquadraticGrowth`.

Definition 1.59 (Uniform subquadratic spatial growth for a sequence)**.** Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ and let $f_1, f_2, \dots$ be real functions on the space-time $\ell^2(F \times \mathbb{R})$. We say the sequence has *uniform subquadratic spatial growth on compact time intervals of* I if for every compact $J \subseteq I$

$$\lim_{M \to \infty} \sup_j \sup_{\substack{|y| \geq M \\ t \in J}} \frac{|f_j(y,t)|}{|y|^2} = 0.$$

Explicitly: for every compact $J \subseteq I$ and every $\varepsilon > 0$ there is $M > 0$ such that $|f_j(y,t)| \le \varepsilon |y|^2$ for every index j and whenever $|y| \ge M$ and $t \in J$.

1.26. `LimitOp`.

Definition 1.60 (The limiting operator)**.** Let F be a finite-dimensional real inner-product space and let $\phi : F \times \mathbb{R} \to \mathbb{R}$ be defined on the space-time $\ell^2(F \times \mathbb{R})$ of the preamble. For a space-time point z set

$$F[\phi](z) = D^2\phi(z)\big[(\nabla_x \phi(z), 1)\big],$$

the value of the second derivative of ϕ at z on the pair (ξ, ξ) with $\xi = (\nabla_x \phi(z), 1)$, where $\nabla_x \phi(z) \in F$ is the spatial part of the full space-time gradient $\nabla \phi(z)$. For $F = \mathbb{R}^m$ and smooth ϕ this is $D^2\phi[(\nabla_x \phi, 1)]$ in the notation $A[\xi] = \xi^{\mathsf{T}} A \xi$.

1.27. `IsLimitSubOn`.

Definition 1.61 (Subsolution of the limiting equation)**.** Let F be a finite-dimensional real inner-product space, let Ω be a subset of the space-time $\ell^2(F \times \mathbb{R})$ and let v be a real function on space-time. We say v is a *viscosity subsolution of* $F[v] = 0$ *on* Ω if Ω is open, v is continuous on Ω, and for every $z \in \Omega$ and every ϕ that is C^2 near z such that $w \mapsto v(w) - \phi(w)$ has a local maximum on Ω at z one has

$$F[\phi](z) \ge 0,$$

with F as in `LimitOp`. Tests involve all variables (x, t) together, and all tests are admitted.

1.28. `IsFit`.

Definition 1.62 (Fit at a center of a given radius)**.** Let F be a finite-dimensional real inner-product space, let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$, let $C_0, R \in \mathbb{R}$, let $z = (x, t)$ be a space-time point, let $p \in F$ and let $a \in \mathbb{R}$. We say that (p, a) is a *fit for* v *at* z *of radius* R *with constant* C_0 if

$$\left| v(x + h, t + \ell) - v(x, t) - p \cdot h - \left(a - \frac{|p|^2}{2} \right) \ell \right| \le C_0 R \qquad \text{whenever } |\ell| \le R \text{ and } |h - p\ell| \le R.$$

The set of increments appearing here is the sheared box $Q_R(z; p) = \{(x + h, t + \ell) : |\ell| \le R,\ |h - p\ell| \le R\}$, and $q = a - |p|^2/2$ is the ordinary time coefficient of the fitted affine function. The parameters are (p, a) rather than (p, q) because a sheared change of frame then acts by subtracting the frame parameters.

1.29. `IsLimitSuperOn`.

Definition 1.63 (Supersolution of the limiting equation)**.** Let F be a finite-dimensional real inner-product space, let Ω be a subset of the space-time $\ell^2(F \times \mathbb{R})$ and let v be a real function on space-time. We say v is a *viscosity supersolution of* $F[v] = 0$ *on* Ω if Ω is open, v is continuous on Ω, and for every $z \in \Omega$ and every ϕ that is C^2 near z such that $w \mapsto v(w) - \phi(w)$ has a local minimum on Ω at z one has

$$F[\phi](z) \le 0,$$

with F as in `LimitOp`. Tests involve all variables (x, t) together, and all tests are admitted.

1.30. `IsLimitSolutionOn`.

Definition 1.64 (Solution of the limiting equation)**.** Let F, Ω and v be as in `IsLimitSubOn`. We say v is a *viscosity solution of*

$$F[v] := D^2 v[(\nabla_x v, 1)] = 0$$

on Ω if v is both a viscosity subsolution of $F[v] = 0$ on Ω in the sense of `IsLimitSubOn` and a viscosity supersolution of $F[v] = 0$ on Ω in the sense of `IsLimitSuperOn`.

1.31. `IsFitClass`.

Definition 1.65 (The class $\mathcal{A}_{m,C_0}$)**.** Let F be a finite-dimensional real inner-product space and let $C_0 \in \mathbb{R}$. A real function v on the space-time $\ell^2(F \times \mathbb{R})$ belongs to the class $\mathcal{A}_{m,C_0}$, with $m = \dim F$, if

(1) v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there, in the sense of `IsLimitSolutionOn` with Ω the whole space; and
(2) for every space-time center z and every $R > 0$ there are $p \in F$ and $a \in \mathbb{R}$ such that (p, a) is a fit for v at z of radius R with constant C_0 in the sense of `IsFit`.

The fit parameters are allowed to depend on both the center and the radius.

1.32. `FrameChange`.

Definition 1.66 (Sheared change of frame)**.** Let F be a finite-dimensional real inner-product space, let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$, let $z_0 = (x_0, t_0)$ be a space-time point, let $R \in \mathbb{R}$, let $p_0 \in F$ and let $a_0 \in \mathbb{R}$. The *change of frame of v based at z_0 with scale R and parameters (p_0, a_0)* is the function

$$\bar{v}(y, s) = \frac{v\big(x_0 + R(y + p_0 s),\, t_0 + Rs\big) - v(x_0, t_0)}{R} - p_0 \cdot y - \left(a_0 + \frac{|p_0|^2}{2}\right) s.$$

It sends z_0 to the space-time origin, where its value is 0.

1.33. `IsCenteredAt`.

Definition 1.67 (Centered at a point)**.** Let F be a finite-dimensional real inner-product space, let $C_0 \in \mathbb{R}$, let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let z_c be a space-time point. We say that v is *centered at z_c with constant C_0* if

$$v(z_c) = 0$$

and $(0, 0)$ is a fit for v at z_c of radius 1 with constant C_0, in the sense of `IsFit`; explicitly

$$|v(x_c + h, t_c + \ell)| \le C_0 \qquad (|\ell| \le 1,\ |h| \le 1),$$

where $z_c = (x_c, t_c)$.

1.34. `QuadEnvP`.

Definition 1.68 (Forward quadratic envelope)**.** Let F be a finite-dimensional real inner-product space, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$, let $\ell \in \mathbb{R}$ and let $z = (x, t)$ be a space-time point. The *forward quadratic envelope* is

$$\mathcal{P}_\ell f(x, t) = \sup_{y \in F} \left\{ f(y, t + \ell) - \frac{|y - x|^2}{2\ell} \right\} \qquad (\ell \neq 0), \qquad \mathcal{P}_0 f = f.$$

By the convention of the preamble the supremum for $\ell \neq 0$ is meaningful only together with the hypotheses guaranteeing that the set of competitor values is bounded above, which are always stated explicitly.

1.35. `QuadEnvM`.

Definition 1.69 (Backward quadratic envelope)**.** Let F be a finite-dimensional real inner-product space, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$, let $\ell \in \mathbb{R}$ and let $z = (x, t)$ be a space-time point. The *backward quadratic envelope* is

$$\mathcal{M}_\ell f(x, t) = \inf_{y \in F} \left\{ f(y, t - \ell) + \frac{|y - x|^2}{2\ell} \right\} \qquad (\ell \neq 0), \qquad \mathcal{M}_0 f = f.$$

By the convention of the preamble the infimum for $\ell \neq 0$ is meaningful only together with the hypotheses guaranteeing that the set of competitor values is bounded below, which are always stated explicitly.

1.36. `ContactEnv`.

Definition 1.70 (Contact envelope)**.** Let F be a finite-dimensional real inner-product space, let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let $\kappa \in \mathbb{R}$. The *contact envelope* is

$$W(x,t) = \mathcal{C}_t v(x,t) - \frac{\kappa^2 t}{2},$$

where the barycentric parameter is taken equal to the time coordinate and $\mathcal{C}$ is the envelope of `BaryEnv`. We write $W = \mathcal{C}_t v - \kappa^2 t/2$; the construction is used for $t > 0$.

1.37. `JGap`.

Definition 1.71 (The propagation quantity J_ϵ)**.** Let F be a finite-dimensional real inner-product space, let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let $\kappa, \epsilon \in \mathbb{R}$. With $W = \mathcal{C}_t v - \kappa^2 t/2$ the contact envelope of `ContactEnv`, put

$$f = W, \qquad f^\epsilon = \mathcal{P}_\epsilon f, \qquad g_\epsilon = \mathcal{M}_\epsilon v, \qquad J_\epsilon = \frac{f^\epsilon - g_\epsilon}{2\epsilon},$$

with $\mathcal{P}$ and $\mathcal{M}$ the quadratic envelopes of `QuadEnvP` and `QuadEnvM`. Thus

$$J_\epsilon(z) = \frac{\mathcal{P}_\epsilon W(z) - \mathcal{M}_\epsilon v(z)}{2\epsilon}.$$

1.38. `EnvSlopeP`.

Definition 1.72 (Forward envelope slope)**.** Let F, f, ℓ and z be as in `QuadEnvP`. The *forward envelope slope* is

$$\mathcal{S}^+_\ell f(z) = \frac{\mathcal{P}_\ell f(z) - f(z)}{\ell},$$

with $\mathcal{P}_\ell$ the forward quadratic envelope of `QuadEnvP`.

1.39. `EnvSlopeM`.

Definition 1.73 (Backward envelope slope)**.** Let F, f, ℓ and z be as in `QuadEnvM`. The *backward envelope slope* is

$$\mathcal{S}^-_\ell f(z) = \frac{f(z) - \mathcal{M}_\ell f(z)}{\ell},$$

with $\mathcal{M}_\ell$ the backward quadratic envelope of `QuadEnvM`.

1.40. `TimeFlip`.

Definition 1.74 (Time reversal)**.** Let F be a finite-dimensional real inner-product space and let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$. Its *time reversal* is

$$\widetilde{f}(x,t) = -f(x,-t).$$

We write $\widetilde{f}$ for this function and $\widetilde{z} = (x,-t)$ for the reflection of a space-time point $z = (x,t)$ in the spatial hyperplane, so that $\widetilde{f}(z) = -f(\widetilde{z})$.

1.41. `EnvHamiltonian`.

Definition 1.75 (Envelope Hamiltonian)**.** Let F be a finite-dimensional real inner-product space, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let z be a space-time point. The *envelope Hamiltonian* is the extended real number

$$H_f(z) = \inf_{\ell > 0} \ \mathcal{S}^+_\ell f(z) \ \in [-\infty, +\infty),$$

the infimum in $\overline{\mathbb{R}}$ of the forward envelope slopes of `EnvSlopeP` over all positive lengths. The value is never $+\infty$ because each slope is a real number. When $\ell \mapsto \mathcal{S}^+_\ell f(z)$ is nondecreasing, this infimum is exactly the limit $\lim_{\ell \downarrow 0} \mathcal{S}^+_\ell f(z)$.

1.42. `IsStepPair`.

Definition 1.76 (Optimizing step pair)**.** Let F be a finite-dimensional real inner-product space, let v be a real function on the space-time $\ell^2(F\times\mathbb{R})$, let $\kappa,\epsilon\in\mathbb{R}$ and let $z=(x,t)$, z^+, z^- be space-time points. Write $W=\mathcal{C}_t v-\kappa^2 t/2$ for the contact envelope of `ContactEnv` and $f=W$. We say (z^+,z^-) is an *optimizing step pair of size* ϵ *at* z if there are $Y^+,Y^-\in F$ such that

$$\mathcal{P}_{2\epsilon}f(z)=f(Y^+,t+2\epsilon)-\frac{|Y^+-x|^2}{2\cdot 2\epsilon},\qquad \mathcal{M}_{2\epsilon}v(z)=v(Y^-,t-2\epsilon)+\frac{|Y^--x|^2}{2\cdot 2\epsilon},$$
$$z^+=\left(\frac{x+Y^+}{2},\,t+\epsilon\right),\qquad z^-=\left(\frac{x+Y^-}{2},\,t-\epsilon\right),$$

with $\mathcal{P}$, $\mathcal{M}$ as in `QuadEnvP` and `QuadEnvM`. Thus Y^+ attains the forward envelope of f at z at length 2ϵ, Y^- attains the backward envelope of v at z at length 2ϵ, and $z^\pm$ are the intermediate points at times $t\pm\epsilon$ on the straight paths from x to those optimizers. The associated step velocities are

$$Q_+=\frac{x^+-x}{\epsilon},\qquad Q_-=\frac{x-x^-}{\epsilon},$$

where $z^\pm=(x^\pm,t\pm\epsilon)$; both are oriented forward in time.

1.43. `IsContactWalk`.

Definition 1.77 (The stopped contact walk)**.** Let F be a finite-dimensional real inner-product space, let v be a real function on the space-time $\ell^2(F\times\mathbb{R})$, let $\kappa,H\in\mathbb{R}$, let T be a nonnegative integer, let $R\in\mathbb{R}$, let N be a nonnegative integer and let $\epsilon\in\mathbb{R}$. Write W for the contact envelope of `ContactEnv` formed from v and κ, $\mathcal{P}_\epsilon W$ for the forward envelope of `QuadEnvP`, J_ϵ for the quantity of `JGap`, $z_*=(0,1)$ and $\delta_N=2^{-N}$. Let $(\Omega,\mathcal{M},\mu)$ be a measure space with a filtration $(\mathcal{G}_n)_{n\ge0}$ in $\mathcal{M}$, let $\xi_n:\Omega\to\mathbb{R}$, let $z_n,z_n^+,z_n^-:\Omega\to\ell^2(F\times\mathbb{R})$, let $Q_0(n):\Omega\to F$ and let $\tau:\Omega\to\mathbb{N}$.

We say that these data form a *stopped contact walk* for $(\kappa,H,T,R,v,N,\epsilon)$ if all of the following hold.

(1) $\mu(\Omega)=1$.
(2) $\epsilon=2^{-k}$ for some nonnegative integer k, and $0<\epsilon\le\delta_N/4$.
(3) $\xi_n(\omega)\in\{1,-1\}$ for every n and ω, each ξ_n is $\mathcal{G}_{n+1}$-measurable, and each ξ_n is fair given the past: $\mathbb{E}[\xi_n\mid\mathcal{G}_n]=0$ almost everywhere.
(4) Each of z_n, z_n^+, z_n^- and $Q_0(n)$ is $\mathcal{G}_n$-measurable.
(5) $z_0=z_*$ identically.
(6) For every n and every ω with $n<\tau(\omega)$, the pair $(z_n^+(\omega),z_n^-(\omega))$ is an optimizing step pair of size ϵ at $z_n(\omega)$ for v and κ, in the sense of `IsStepPair`.
(7) The walk follows the sign and freezes after stopping:

$$z_{n+1}(\omega)=\begin{cases}z_n^+(\omega) & n<\tau(\omega),\ \xi_n(\omega)=1,\\ z_n^-(\omega) & n<\tau(\omega),\ \xi_n(\omega)\neq1,\\ z_n(\omega) & n\ge\tau(\omega).\end{cases}$$

(8) For every n and ω, writing $z_n(\omega)=(x_n,t_n)$, the function $x'\mapsto\mathcal{P}_\epsilon W(x',t_n)$ has gradient $Q_0(n)(\omega)$ at x_n.
(9) τ is a stopping time for $(\mathcal{G}_n)$ and the function $\omega\mapsto\tau(\omega)$ is μ-integrable.
(10) Before stopping the walk is strictly inside the box: if $n<\tau(\omega)$ then $\delta_N<t_n<T$ and $|x_n|<R$.
(11) At the stop one of the three cutoffs is met: $t_{\tau(\omega)}=\delta_N$ or $t_{\tau(\omega)}=T$ or $|x_{\tau(\omega)}|\ge R$.
(12) Every position up to and including the stop satisfies $|x_n|\le R+1$ for $n\le\tau(\omega)$.
(13) The starting value of the propagation quantity is almost extremal: $H-1\le J_\epsilon(z_*)$.

1.44. `IsMarkData`.

Definition 1.78 (The marks of a stopped contact walk)**.** Let F be a finite-dimensional real inner-product space, let $C_0\in\mathbb{R}$, let v be a real function on the space-time $\ell^2(F\times\mathbb{R})$, let

N be a nonnegative integer, let $(\Omega, \mathcal{M})$ be a measurable space with a filtration $(\mathcal{G}_n)_{n\geq 0}$, let $z_n : \Omega \to \ell^2(F \times \mathbb{R})$, let $\tau : \Omega \to \mathbb{N}$, and let $\sigma_j : \Omega \to \mathbb{N}$, $p_j : \Omega \to F$ and $a_j : \Omega \to \mathbb{R}$ be given for every nonnegative integer j. Write $z_n = (x_n, t_n)$ and $\delta_j = 2^{-j}$.

We say that (σ_j, p_j, a_j) are *marks with fits* for $(C_0, v, N, \mathcal{G}, z, \tau)$ if for every j with $1 \leq j \leq N$ all of the following hold.

(1) σ_j is a stopping time for $(\mathcal{G}_n)$.
(2) $\sigma_j \leq \tau$ pointwise.
(3) σ_j is not preceded by a stop or by a visit to the level δ_j: if $n < \sigma_j(\omega)$ then $n < \tau(\omega)$ and $t_n(\omega) \neq \delta_j$.
(4) At σ_j either the walk stops or the level δ_j is reached: $\sigma_j(\omega) = \tau(\omega)$ or $t_{\sigma_j(\omega)}(\omega) = \delta_j$.
(5) For every n the functions $\mathbf{1}_{\{\sigma_j \leq n\}} p_j$ and $\mathbf{1}_{\{\sigma_j \leq n\}} a_j$ are $\mathcal{G}_n$-measurable.
(6) On the event $\{t_{\sigma_j} = \delta_j\}$ the pair (p_j, a_j) is a fit for v at z_{σ_j} of radius δ_j with constant C_0, in the sense of `IsFit`.
(7) Off that event $p_j = 0$ and $a_j = 0$.

Clauses 3 and 4 say exactly that σ_j is the smaller of τ and the first index at which the time coordinate equals δ_j, so a mark is the first point on its time level, or the terminal point if the walk stops before that level. Clause 5 is the measurability of the choices with respect to the stopped σ-algebra at the marking time.

1.45. `IsSpaceTimeAffine`.

Definition 1.79 (Affine function on space-time)**.** Let F be a finite-dimensional real inner-product space and let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$. We say v is *affine* if there are $q \in F$ and $b, c \in \mathbb{R}$ with

$$v(x,t) = q \cdot x + bt + c \qquad \text{for all } (x,t).$$

1.46. `HolderQuotSet`.

Definition 1.80 (Hölder difference quotients)**.** Let E be a finite-dimensional real inner-product space, let V be a normed space, let $g : E \to V$, let $\alpha \in \mathbb{R}$ and let $S \subseteq E$. The *set of Hölder quotients of g on S at exponent α* is

$$Q_\alpha(g; S) = \left\{ \frac{|g(x) - g(y)|}{|x - y|^\alpha} \;:\; x, y \in S,\ x \neq y \right\} \subseteq \mathbb{R}.$$

1.47. `HolderSeminormOn`.

Definition 1.81 (Hölder seminorm)**.** Let E, V, g, α and S be as in `HolderQuotSet`. The *Hölder seminorm of g on S at exponent α* is

$$[g]_{C^{0,\alpha}(S)} = \sup Q_\alpha(g; S) = \sup_{\substack{x,y \in S \\ x \neq y}} \frac{|g(x) - g(y)|}{|x - y|^\alpha},$$

the supremum of the set of Hölder quotients of `HolderQuotSet`. By the convention of the preamble this supremum is meaningful only together with the hypothesis that $Q_\alpha(g; S)$ is bounded above, which is always stated explicitly.

1.48. `RescaledOp`.

Definition 1.82 (The rescaled operator)**.** Let F be a finite-dimensional real inner-product space, let $\lambda \in \mathbb{R}$ and let ϕ be a real function on the space-time $\ell^2(F \times \mathbb{R})$. For a space-time point z set

$$F_\lambda[\phi](z) = D^2\phi(z)\big[(\nabla_x \phi(z),\, 1 + \lambda \phi_t(z))\big],$$

the value of the second derivative of ϕ at z on the pair (ξ, ξ) with $\xi = (\nabla_x \phi(z), 1 + \lambda \phi_t(z))$, where $\nabla_x \phi(z) \in F$ and $\phi_t(z) \in \mathbb{R}$ are the spatial and time parts of the space-time gradient $\nabla \phi(z)$. For $\lambda = 0$ this is the limiting operator $F[\phi]$ of `LimitOp`.

1.49. `IsRescaledSolutionOn`.

Definition 1.83 (Solution of the rescaled equation)**.** Let F be a finite-dimensional real inner-product space, let $\lambda \in \mathbb{R}$, let Ω be a subset of the space-time $\ell^2(F \times \mathbb{R})$ and let v be a real function on space-time. We say v is a *viscosity solution of*

$$D^2 v\big[(\nabla_x v, 1 + \lambda v_t)\big] = 0$$

on Ω if Ω is open, v is continuous on Ω, and for every $z \in \Omega$ and every ϕ that is C^2 near z:

- if $w \mapsto v(w) - \phi(w)$ has a local maximum on Ω at z, then $F_\lambda[\phi](z) \geq 0$;
- if $w \mapsto v(w) - \phi(w)$ has a local minimum on Ω at z, then $F_\lambda[\phi](z) \leq 0$;

with F_λ the rescaled operator of `RescaledOp`. Tests involve all variables together, and all tests are admitted.

2. Lemmas

2.1. `ConeComparison`.

Lemma 2.1 (Cone comparison and radial monotonicity)**.** *Let E be a finite-dimensional real inner-product space, let $D \subseteq E$, let $w : E \to \mathbb{R}$ be infinity-harmonic on D in the sense of `IsInfHarmonicOn`, let $z \in E$ and let $r > 0$ with $\overline{B}_r(z) \subseteq D$. Then, with the cone slopes of `ConeSlopeUpper` and `ConeSlopeLower`,*

$$w(Z) - w(z) \leq L_r^+(w; z)\,|Z - z| \qquad (Z \in \overline{B}_r(z)), \tag{1}$$

$$-\big(w(Z) - w(z)\big) \leq L_r^-(w; z)\,|Z - z| \qquad (Z \in \overline{B}_r(z)), \tag{2}$$

and for every $r' \geq r$ with $\overline{B}_{r'}(z) \subseteq D$,

$$L_r^+(w; z) \leq L_{r'}^+(w; z), \qquad L_r^-(w; z) \leq L_{r'}^-(w; z). \tag{3}$$

Proof. Throughout, $\overline{B}_\rho(z) = \{Z : |Z - z| \leq \rho\}$.

Step 0: the slopes are finite and nonnegative. Since w is infinity-harmonic on D, it is continuous on D by `IsInfHarmonicOn`; as $\overline{B}_r(z) \subseteq D$ is compact, the set $\{r^{-1}(w(z + h) - w(z)) : |h| \leq r\}$ is a nonempty bounded set of reals, so $L_r^+(w; z)$ is its maximum. Taking $h = 0$ shows $L_r^+(w; z) \geq 0$. The same applies to $L_r^-(w; z)$.

Step 1: $-w$ is infinity-harmonic on D. Let ϕ be C^2 near a point $Z_0 \in D$ and suppose $Z \mapsto (-w)(Z) - \phi(Z)$ has a local maximum on D at Z_0. Then $Z \mapsto w(Z) - (-\phi)(Z)$ has a local minimum on D at Z_0, and $-\phi$ is C^2 near Z_0, so the supersolution inequality of `IsInfHarmonicOn` gives $\Delta_\infty(-\phi)(Z_0) \leq 0$. By the definition of Δ_∞ in `InfLap`, $\nabla(-\phi) = -\nabla\phi$ and $D^2(-\phi) = -D^2\phi$, so

$$\Delta_\infty(-\phi)(Z_0) = -D^2\phi(Z_0)[-\nabla\phi(Z_0)] = -D^2\phi(Z_0)[\nabla\phi(Z_0)] = -\Delta_\infty\phi(Z_0),$$

using that $A[-\xi] = A[\xi]$ for a bilinear form A. Hence $\Delta_\infty\phi(Z_0) \geq 0$, which is the subsolution inequality for $-w$. The symmetric computation with local minima gives the supersolution inequality for $-w$, and $-w$ is continuous on the open set D. Thus $-w$ is infinity-harmonic on D. Note also that, directly from `ConeSlopeUpper` and `ConeSlopeLower`,

$$L_\rho^+(-w; z) = L_\rho^-(w; z), \qquad L_\rho^-(-w; z) = L_\rho^+(w; z) \tag{4}$$

for every ρ, because the defining sets of quotients are exchanged.

Step 2: the perturbed cone barrier. We prove (1). Fix any $L > L_r^+(w; z)$ and choose $\gamma > 0$ so small that

$$\gamma r < L - L_r^+(w; z), \qquad 2\gamma r < L.$$

Define, for $Z \in E$,

$$\Theta(Z) = w(z) + L|Z - z| - \gamma|Z - z|^2.$$

On the sphere $|Z - z| = r$ we have, using Step 0,

$$w(Z) \leq w(z) + rL_r^+(w; z) < w(z) + r\big(L - \gamma r\big) = \Theta(Z),$$

the strict inequality coming from $\gamma r < L - L_r^+(w; z)$. Thus $w - \Theta < 0$ on $\{|Z - z| = r\}$, while $(w - \Theta)(z) = 0$.

Suppose, for contradiction, that $w(Z) > \Theta(Z)$ for some $Z \in \overline{B}_r(z)$. The function $w - \Theta$ is continuous on the compact set $\overline{B}_r(z)$, so it attains a maximum there, and that maximum is strictly positive. By the previous paragraph the maximum is not attained on the sphere, and it is not attained at z since the value there is 0. Let Z_0 be a maximum point; then $|Z_0 - z| =: \rho \in (0, r)$ and Z_0 is interior to $\overline{B}_r(z)$.

The function Θ is C^2 on $E \setminus \{z\}$: indeed $Z \mapsto |Z - z|^2$ is a polynomial, and $Z \mapsto |Z - z|$ is smooth away from z. Since $w - \Theta$ has a local maximum on D at Z_0 (it has a maximum over a neighbourhood of Z_0 contained in $\overline{B}_r(z) \subseteq D$), Θ is a local upper test for w at Z_0, and the subsolution inequality of `IsInfHarmonicOn` gives $\Delta_\infty \Theta(Z_0) \geq 0$.

We compute $\Delta_\infty \Theta(Z_0)$. Write $e = (Z_0 - z)/\rho$, a unit vector. For $g(\rho) = L\rho - \gamma\rho^2$ and $\Theta(Z) = w(z) + g(|Z - z|)$ one has

$$\nabla\Theta(Z_0) = g'(\rho)e = (L - 2\gamma\rho)e.$$

Since $2\gamma\rho < 2\gamma r < L$, the number $L - 2\gamma\rho$ is strictly positive, so $\nabla\Theta(Z_0)$ is a positive multiple of the radial direction e. The second derivative of Θ in the radial direction is

$$D^2\Theta(Z_0)[e] = g''(\rho) = -2\gamma.$$

By bilinearity, $D^2\Theta(Z_0)[\nabla\Theta(Z_0)] = (L - 2\gamma\rho)^2 D^2\Theta(Z_0)[e]$, so

$$\Delta_\infty \Theta(Z_0) = -2\gamma\,(L - 2\gamma\rho)^2 < 0,$$

contradicting $\Delta_\infty \Theta(Z_0) \geq 0$. Hence $w \leq \Theta$ on $\overline{B}_r(z)$, i.e.

$$w(Z) - w(z) \leq L|Z - z| - \gamma|Z - z|^2 \leq L|Z - z| \qquad (Z \in \overline{B}_r(z)).$$

Letting $\gamma \downarrow 0$ is unnecessary here, since the final bound no longer involves γ; letting $L \downarrow L_r^+(w; z)$ gives (1).

Step 3: the lower bound. Apply Step 2 to $-w$, which is infinity-harmonic on D by Step 1, with the same z and r. This gives $(-w)(Z) - (-w)(z) \leq L_r^+(-w; z)|Z - z|$ on $\overline{B}_r(z)$, which by (4) is exactly (2).

Step 4: monotonicity in the radius. Let $r' \geq r$ with $\overline{B}_{r'}(z) \subseteq D$; then $r' > 0$. Applying Step 2 at radius r' gives $w(z + h) - w(z) \leq L_{r'}^+(w; z)|h|$ for all $|h| \leq r'$, in particular for all $|h| \leq r$. Since $L_{r'}^+(w; z) \geq 0$ by Step 0 and $|h| \leq r$,

$$\frac{w(z + h) - w(z)}{r} \leq \frac{L_{r'}^+(w; z)|h|}{r} \leq L_{r'}^+(w; z) \qquad (|h| \leq r).$$

Taking the supremum over $|h| \leq r$ gives $L_r^+(w; z) \leq L_{r'}^+(w; z)$. The same argument applied to $-w$, together with (4), gives $L_r^-(w; z) \leq L_{r'}^-(w; z)$. This is (3). □

2.2. `ConeSlopeContinuousInCenter`.

Lemma 2.2 (Continuity of the cone slopes in the center)**.** *Let E be a finite-dimensional real inner-product space, let $D \subseteq E$ and let $w : E \to \mathbb{R}$ be infinity-harmonic on D in the sense of `IsInfHarmonicOn`. Let $r > 0$ and let $z_0 \in E$ with $\overline{B}_r(z_0) \subseteq D$. Then the functions*

$$z \longmapsto L_r^+(w; z), \qquad z \longmapsto L_r^-(w; z)$$

of `ConeSlopeUpper` and `ConeSlopeLower` are continuous at z_0.

Proof. By `IsInfHarmonicOn` the set D is open and w is continuous on D.

Step 1: a margin. The set $\overline{B}_r(z_0)$ is compact and contained in the open set D, so there is $\eta > 0$ with $\overline{B}_{r+2\eta}(z_0) \subseteq D$. Indeed, otherwise there would be points $\zeta_k \notin D$ with $|\zeta_k - z_0| \leq r + 2/k$; a subsequence converges to some ζ with $|\zeta - z_0| \leq r$, so $\zeta \in \overline{B}_r(z_0) \subseteq D$, while $\zeta \notin D$ because the complement of D is closed, a contradiction. Write $K = \overline{B}_{r+2\eta}(z_0)$, a compact subset of D.

Step 2: uniform continuity and the slope comparison. Since w is continuous on D and $K \subseteq D$ is compact, w is uniformly continuous on K. Let $\varepsilon > 0$. Choose $\delta \in (0, \eta]$ such that

$$|w(a) - w(b)| \leq \frac{\varepsilon r}{2} \qquad \big(a, b \in K,\ |a - b| \leq \delta\big).$$

Let $z \in E$ with $|z - z_0| \le \delta$. For $|h| \le r$ both $z + h$ and $z_0 + h$ lie in K, since $|z + h - z_0| \le r + \delta \le r + \eta$, and $|(z + h) - (z_0 + h)| = |z - z_0| \le \delta$; likewise $z, z_0 \in K$ and $|z - z_0| \le \delta$. Hence

$$\big|\big(w(z+h) - w(z)\big) - \big(w(z_0+h) - w(z_0)\big)\big| \le |w(z+h) - w(z_0+h)| + |w(z) - w(z_0)| \le \varepsilon r.$$

Dividing by $r > 0$,

$$\frac{w(z+h) - w(z)}{r} \le \frac{w(z_0+h) - w(z_0)}{r} + \varepsilon \qquad (|h| \le r),$$

and symmetrically with the roles of z, z_0 exchanged.

Step 3: both suprema are finite. For $|z - z_0| \le \delta$ the set $\{r^{-1}(w(z+h) - w(z)) : |h| \le r\}$ is the image of the compact set $\{|h| \le r\}$ under a continuous function, hence nonempty and bounded; the same holds at z_0. Therefore $L_r^{\pm}(w; z)$ and $L_r^{\pm}(w; z_0)$ are genuine suprema of nonempty bounded sets.

Step 4: conclusion. Taking the supremum over $|h| \le r$ in the two displays of Step 2 gives

$$L_r^+(w; z) \le L_r^+(w; z_0) + \varepsilon, \qquad L_r^+(w; z_0) \le L_r^+(w; z) + \varepsilon,$$

that is $|L_r^+(w; z) - L_r^+(w; z_0)| \le \varepsilon$ whenever $|z - z_0| \le \delta$. Since $\varepsilon > 0$ was arbitrary, $z \mapsto L_r^+(w; z)$ is continuous at z_0. The identical argument with $-(w(z+h) - w(z))$ in place of $w(z+h) - w(z)$, which is the defining family for L_r^- in `ConeSlopeLower`, gives continuity of $z \mapsto L_r^-(w; z)$ at z_0. □

2.3. `ConeSlopeLimitEq`.

Lemma 2.3 (Equality of the small-radius slope limits)**.** *Let E be a finite-dimensional real inner-product space, let $D \subseteq E$ and let $w : E \to \mathbb{R}$ be infinity-harmonic on D in the sense of `IsInfHarmonicOn`. Let $z \in D$. Then there is $L_* \in \mathbb{R}$ with $L_* \ge 0$ such that*

$$\lim_{r \downarrow 0} L_r^+(w; z) = \lim_{r \downarrow 0} L_r^-(w; z) = L_*,$$

with the cone slopes of `ConeSlopeUpper` and `ConeSlopeLower`.

Proof. By `IsInfHarmonicOn`, D is open, so there is $r_0 > 0$ with $\overline{B}_{2r_0}(z) \subseteq D$.

Step 1: both limits exist. Fix $0 < r \le r' \le r_0$. Then $\overline{B}_{r'}(z) \subseteq D$ and $\overline{B}_r(z) \subseteq D$, so `ConeComparison`, applied at the center z with radius r and comparison radius r', gives

$$L_r^+(w; z) \le L_{r'}^+(w; z), \qquad L_r^-(w; z) \le L_{r'}^-(w; z).$$

Thus $r \mapsto L_r^{\pm}(w; z)$ is nondecreasing on $(0, r_0]$. Each value is nonnegative, because the competitor $h = 0$ in `ConeSlopeUpper` and `ConeSlopeLower` contributes the quotient 0, and each value is finite because w is continuous on the compact set $\overline{B}_{r_0}(z) \subseteq D$. A bounded monotone function has a limit at the left endpoint, so

$$L_*^{\pm} := \lim_{r \downarrow 0} L_r^{\pm}(w; z) = \inf_{0 < r \le r_0} L_r^{\pm}(w; z)$$

exist in $\mathbb{R}$ and satisfy $L_*^{\pm} \ge 0$.

Step 2: $L_*^+ \le L_*^-$. Fix R with $0 < R \le r_0$. Let $0 < r \le R/2$. Since w is continuous on the compact set $\overline{B}_r(z)$, the supremum defining $L_r^+(w; z)$ is attained: there is h_r with $|h_r| \le r$ and

$$L_r^+(w; z) = \frac{w(z + h_r) - w(z)}{r}.$$

The vector $-h_r$ satisfies $|-h_r| \le r$, so it is an admissible competitor in the supremum of `ConeSlopeLower` at the center $z + h_r$ and radius r, and contributes the quotient

$$\frac{-\big(w\big((z + h_r) + (-h_r)\big) - w(z + h_r)\big)}{r} = \frac{w(z + h_r) - w(z)}{r} = L_r^+(w; z).$$

Hence

$$L_r^+(w; z) \le L_r^-(w; z + h_r). \tag{5}$$

Also $|z + h_r - z| \le r \le R/2$, so $\overline{B}_R(z + h_r) \subseteq \overline{B}_{2r_0}(z) \subseteq D$ and $\overline{B}_r(z + h_r) \subseteq D$. Applying `ConeComparison` at the center $z+h_r$ with radius r and comparison radius $R \ge r$ gives $L_r^-(w; z + h_r) \le L_R^-(w; z + h_r)$. Combined with (5),

$$L_r^+(w; z) \le L_R^-(w; z + h_r) \qquad (0 < r \le R/2). \tag{6}$$

Now let $r \downarrow 0$. The left side tends to L_*^+ by Step 1. On the right, $|h_r| \le r \to 0$, so $z+h_r \to z$; since $\overline{B}_R(z) \subseteq D$, the function $\zeta \mapsto L_R^-(w; \zeta)$ is continuous at z by `ConeSlopeContinuousInCenter`, so $L_R^-(w; z + h_r) \to L_R^-(w; z)$. Passing to the limit in (6) gives

$$L_*^+ \le L_R^-(w; z) \qquad (0 < R \le r_0).$$

Letting $R \downarrow 0$ and using Step 1 gives $L_*^+ \le L_*^-$.

Step 3: $L_*^- \le L_*^+$. The function $-w$ is infinity-harmonic on D. Indeed, $-w$ is continuous on the open set D; and if ϕ is C^2 near $Z_0 \in D$ and $(-w) - \phi$ has a local maximum on D at Z_0, then $w - (-\phi)$ has a local minimum on D at Z_0, so the supersolution inequality of `IsInfHarmonicOn` gives $\Delta_\infty(-\phi)(Z_0) \le 0$; by `InfLap` and $A[-\xi] = A[\xi]$ for a bilinear form A we have $\Delta_\infty(-\phi) = -\Delta_\infty\phi$, so $\Delta_\infty\phi(Z_0) \ge 0$, and the symmetric computation with local minima gives the other inequality. Directly from `ConeSlopeUpper` and `ConeSlopeLower`,

$$L_\rho^+(-w; z) = L_\rho^-(w; z), \qquad L_\rho^-(-w; z) = L_\rho^+(w; z)$$

for every ρ, because the two defining sets of quotients are exchanged. Applying Step 2 to $-w$ therefore gives $L_*^- \le L_*^+$.

Conclusion. Steps 2 and 3 give $L_*^+ = L_*^-$; call the common value L_*, which is nonnegative by Step 1. □

2.4. `InfHarmonicLocLipschitz`.

Lemma 2.4 (Local Lipschitz bound from one slope radius)**.** *Let E be a finite-dimensional real inner-product space, let $D \subseteq E$ and let $w : E \to \mathbb{R}$ be infinity-harmonic on D in the sense of `IsInfHarmonicOn`. Let $z_0 \in E$, let $\rho \ge 0$, let $\sigma > 0$ and let $\Lambda \in \mathbb{R}$. Assume*

$$\overline{B}_{\rho+\sigma}(z_0) \subseteq D, \qquad |w(a) - w(b)| \le \Lambda\sigma \quad \big(a, b \in \overline{B}_{\rho+\sigma}(z_0)\big).$$

Then

$$|w(a) - w(b)| \le \Lambda|a - b| \qquad \big(a, b \in \overline{B}_\rho(z_0)\big).$$

Proof. Taking $a = b$ in the hypothesis gives $0 \le \Lambda\sigma$, and $\sigma > 0$, so $\Lambda \ge 0$.

Step 1: a Lipschitz estimate for nearby pairs. Let $a \in \overline{B}_\rho(z_0)$. For $|h| \le \sigma$ we have $|a+h-z_0| \le \rho + \sigma$, so $\overline{B}_\sigma(a) \subseteq \overline{B}_{\rho+\sigma}(z_0) \subseteq D$. Hence `ConeComparison` applies at the center a with radius $\sigma > 0$, and gives

$$w(Z) - w(a) \le L_\sigma^+(w; a)|Z - a|, \qquad -(w(Z) - w(a)) \le L_\sigma^-(w; a)|Z - a| \tag{7}$$

for all $Z \in \overline{B}_\sigma(a)$. Next we bound the two slopes. By the definition in `ConeSlopeUpper`, $L_\sigma^+(w; a)$ is the supremum of $\sigma^{-1}(w(a + h) - w(a))$ over $|h| \le \sigma$. For each such h, both $a + h$ and a lie in $\overline{B}_{\rho+\sigma}(z_0)$, so the oscillation hypothesis gives $w(a + h) - w(a) \le \Lambda\sigma$ and therefore $\sigma^{-1}(w(a + h) - w(a)) \le \Lambda$. Taking the supremum, $L_\sigma^+(w; a) \le \Lambda$. The same argument with the definition in `ConeSlopeLower` gives $L_\sigma^-(w; a) \le \Lambda$. Combining with (7),

$$|w(Z) - w(a)| \le \Lambda|Z - a| \qquad \big(a \in \overline{B}_\rho(z_0),\ Z \in \overline{B}_\sigma(a)\big). \tag{8}$$

Step 2: subdivision of a segment. Let $a, b \in \overline{B}_\rho(z_0)$ and put $d = |a - b|$. If $d = 0$ the claim is trivial. Otherwise choose an integer $N \ge 1$ with $d/N \le \sigma$, for instance any $N \ge d/\sigma$, and set

$$a_k = a + \frac{k}{N}(b - a) \qquad (k = 0, 1, \dots, N),$$

so $a_0 = a$ and $a_N = b$. The closed ball $\overline{B}_\rho(z_0)$ is convex, so every a_k lies in it. Moreover $|a_{k+1} - a_k| = d/N \le \sigma$, so $a_{k+1} \in \overline{B}_\sigma(a_k)$ and (8) applies at the center a_k:

$$|w(a_{k+1}) - w(a_k)| \le \Lambda\frac{d}{N} \qquad (0 \le k < N).$$

Summing over k and using the triangle inequality,

$$|w(b)-w(a)| \le \sum_{k=0}^{N-1} |w(a_{k+1})-w(a_k)| \le N\cdot\Lambda\frac{d}{N} = \Lambda d = \Lambda|a-b|,$$

as claimed. □

2.5. `InfHarmonicRescale`.

Lemma 2.5 (Affine rescaling of an infinity-harmonic function)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ and let $u : E \to \mathbb{R}$ be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`. Let $x_0 \in E$ and $A > 0$, and put $T(y) = x_0 + Ay$. Then $u \circ T$ is infinity-harmonic on $T^{-1}(\Omega)$.*

Proof. Write $\widetilde{u} = u \circ T$ and $\widetilde{\Omega} = T^{-1}(\Omega)$. The map T is an affine homeomorphism of E with inverse $T^{-1}(x) = A^{-1}(x - x_0)$, so $\widetilde{\Omega}$ is open because Ω is open by `IsInfHarmonicOn`, and $\widetilde{u}$ is continuous on $\widetilde{\Omega}$ because u is continuous on Ω.

Upper tests. Let $y_0 \in \widetilde{\Omega}$ and let ψ be C^2 near y_0 with $y \mapsto \widetilde{u}(y) - \psi(y)$ having a local maximum on $\widetilde{\Omega}$ at y_0. Put $x_0' = T(y_0) \in \Omega$ and

$$\phi = \psi \circ T^{-1},$$

which is C^2 near x_0' since T^{-1} is affine. For x near x_0' with $x \in \Omega$ we have $T^{-1}(x) \in \widetilde{\Omega}$ near y_0, and

$$u(x) - \phi(x) = \widetilde{u}(T^{-1}x) - \psi(T^{-1}x) \le \widetilde{u}(y_0) - \psi(y_0) = u(x_0') - \phi(x_0'),$$

so $x \mapsto u(x) - \phi(x)$ has a local maximum on Ω at x_0'. The subsolution inequality of `IsInfHarmonicOn` therefore gives $\Delta_\infty\phi(x_0') \ge 0$.

We relate $\Delta_\infty\psi(y_0)$ to $\Delta_\infty\phi(x_0')$. Since $\psi = \phi \circ T$ and T is affine with linear part $A\,\mathrm{Id}$, the chain rule gives $D\psi(y_0)v = A\,D\phi(x_0')v$ for every v, hence, as the inner product identifies derivatives with gradients,

$$\nabla\psi(y_0) = A\,\nabla\phi(x_0'),$$

and, the second derivative of an affine map being zero,

$$D^2\psi(y_0)[v,v] = A^2 D^2\phi(x_0')[v,v] \qquad (v \in E).$$

Consequently, by the definition of Δ_∞ in `InfLap` and bilinearity,

$$\Delta_\infty\psi(y_0) = D^2\psi(y_0)\big[A\nabla\phi(x_0')\big] = A^2\cdot A^2\, D^2\phi(x_0')\big[\nabla\phi(x_0')\big] = A^4\,\Delta_\infty\phi(x_0').$$

Since $A > 0$, we get $\Delta_\infty\psi(y_0) \ge 0$.

Lower tests. The same computation applies with local maxima replaced by local minima and the inequalities reversed: $\phi = \psi \circ T^{-1}$ is a local lower test for u at x_0', the supersolution inequality of `IsInfHarmonicOn` gives $\Delta_\infty\phi(x_0') \le 0$, and $\Delta_\infty\psi(y_0) = A^4\Delta_\infty\phi(x_0') \le 0$.

Thus $\widetilde{u}$ satisfies both viscosity inequalities on $\widetilde{\Omega}$ and is infinity-harmonic there. □

2.6. `InfHarmonicNormalize`.

Lemma 2.6 (Normalization of an infinity-harmonic function)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ and let $u : E \to \mathbb{R}$ be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`. Let $c \in \mathbb{R}$ and $M > 0$. Then $(u - c)/M$ is infinity-harmonic on Ω.*

Proof. Write $\widetilde{u} = (u - c)/M$. The set Ω is open and u is continuous on Ω by `IsInfHarmonicOn`, hence so is $\widetilde{u}$.

Upper tests. Let $z \in \Omega$ and let ψ be C^2 near z with $y \mapsto \widetilde{u}(y) - \psi(y)$ having a local maximum on Ω at z. Put $\phi = M\psi + c$, which is C^2 near z. Then

$$u(y) - \phi(y) = M\big(\widetilde{u}(y) - \psi(y)\big)$$

and $M > 0$, so $y \mapsto u(y) - \phi(y)$ has a local maximum on Ω at z. The subsolution inequality of `IsInfHarmonicOn` gives $\Delta_\infty\phi(z) \ge 0$.

Since $\phi = M\psi + c$ with M and c constants, $\nabla\phi = M\nabla\psi$ and $D^2\phi = M\,D^2\psi$. Hence, by the definition of Δ_∞ in `InfLap` and bilinearity,

$$\Delta_\infty\phi(z) = D^2\phi(z)\big[M\nabla\psi(z)\big] = M\cdot M^2\,D^2\psi(z)\big[\nabla\psi(z)\big] = M^3\,\Delta_\infty\psi(z).$$

As $M > 0$ this gives $\Delta_\infty\psi(z) \geq 0$.

Lower tests. The same computation with local minima and reversed inequalities gives $\Delta_\infty\phi(z) \leq 0$ from the supersolution inequality of `IsInfHarmonicOn`, hence $\Delta_\infty\psi(z) = M^{-3}\Delta_\infty\phi(z) \leq 0$.

Thus $\widetilde{u}$ is infinity-harmonic on Ω. □

2.7. `InfHarmonicStability`.

Lemma 2.7 (Stability of infinity-harmonicity under locally uniform limits)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ be open, let $u_1, u_2, \ldots : E \to \mathbb{R}$ each be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`, and let $u_0 : E \to \mathbb{R}$ be such that $u_j \to u_0$ uniformly on every compact subset of Ω. Then u_0 is infinity-harmonic on Ω.*

Proof. Step 0: continuity of the limit. Let $x \in \Omega$. Since Ω is open there is $\varrho > 0$ with $\overline{B}_\varrho(x) \subseteq \Omega$, and this set is compact, so $u_j \to u_0$ uniformly on it. Each u_j is continuous on Ω by `IsInfHarmonicOn`, hence continuous on $\overline{B}_\varrho(x)$, and a uniform limit of continuous functions on a set is continuous on that set. Thus u_0 is continuous at each point of Ω, i.e. continuous on Ω.

Step 1: strictifying an upper test. Let $z_0 \in \Omega$ and let ϕ be C^2 near z_0 with $x \mapsto u_0(x) - \phi(x)$ having a local maximum on Ω at z_0. Set

$$\phi^\sharp(x) = \phi(x) + |x - z_0|^4.$$

The function $x \mapsto |x-z_0|^4 = \big(|x-z_0|^2\big)^2$ is a polynomial in the coordinates, hence smooth, with

$$\nabla\big(|\cdot - z_0|^4\big)(z_0) = 0, \qquad D^2\big(|\cdot - z_0|^4\big)(z_0) = 0, \tag{9}$$

because $4|x - z_0|^2(x - z_0)$ and its derivative vanish at $x = z_0$. Thus $\phi^\sharp$ is C^2 near z_0 with $\nabla\phi^\sharp(z_0) = \nabla\phi(z_0)$ and $D^2\phi^\sharp(z_0) = D^2\phi(z_0)$, so by `InfLap`

$$\Delta_\infty\phi^\sharp(z_0) = \Delta_\infty\phi(z_0). \tag{10}$$

Choose $\varrho > 0$ so small that $\overline{B}_\varrho(z_0) \subseteq \Omega$, that ϕ is C^2 on a neighbourhood of $\overline{B}_\varrho(z_0)$, and that $u_0 - \phi$ attains its maximum over $\overline{B}_\varrho(z_0)$ at z_0 (possible because the maximum is local on Ω). Then for $x \in \overline{B}_\varrho(z_0)$ with $x \neq z_0$,

$$u_0(x) - \phi^\sharp(x) = \big(u_0(x) - \phi(x)\big) - |x - z_0|^4 < u_0(z_0) - \phi(z_0) = u_0(z_0) - \phi^\sharp(z_0),$$

so z_0 is the *unique* maximum point of $u_0 - \phi^\sharp$ on $\overline{B}_\varrho(z_0)$.

Step 2: nearby maxima of the approximations. The set $\overline{B}_\varrho(z_0)$ is compact and contained in Ω. Each $u_j - \phi^\sharp$ is continuous on it, so it attains a maximum at some $z_j \in \overline{B}_\varrho(z_0)$. We claim $z_j \to z_0$. By compactness it suffices to show every convergent subsequence of (z_j) has limit z_0. Let $z_{j_k} \to \bar{z} \in \overline{B}_\varrho(z_0)$. Uniform convergence of u_j to u_0 on $\overline{B}_\varrho(z_0)$ and continuity of $u_0 - \phi^\sharp$ give

$$(u_{j_k} - \phi^\sharp)(z_{j_k}) \longrightarrow (u_0 - \phi^\sharp)(\bar{z}).$$

On the other hand, for every $x \in \overline{B}_\varrho(z_0)$, $(u_{j_k} - \phi^\sharp)(z_{j_k}) \geq (u_{j_k} - \phi^\sharp)(x) \to (u_0 - \phi^\sharp)(x)$, so $(u_0 - \phi^\sharp)(\bar{z}) \geq (u_0 - \phi^\sharp)(x)$ for all such x. Thus $\bar{z}$ is a maximum point of $u_0 - \phi^\sharp$ on $\overline{B}_\varrho(z_0)$, and by the uniqueness in Step 1 $\bar{z} = z_0$. Hence $z_j \to z_0$, and in particular $|z_j - z_0| < \varrho$ for all large j, so z_j is an interior point of $\overline{B}_\varrho(z_0)$ and therefore $u_j - \phi^\sharp$ has a local maximum on Ω at z_j.

Step 3: passing the inequality. For such j, $\phi^\sharp$ is C^2 near z_j and $u_j - \phi^\sharp$ has a local maximum on Ω at z_j, so the subsolution inequality of `IsInfHarmonicOn` for u_j gives

$$\Delta_\infty\phi^\sharp(z_j) \geq 0.$$

The function $\phi^\sharp$ is C^2 on a neighbourhood of $\overline{B}_\varrho(z_0)$, so $\nabla\phi^\sharp$ and $D^2\phi^\sharp$ are continuous there; by `InfLap` the map $x \mapsto \Delta_\infty\phi^\sharp(x)$ is therefore continuous near z_0. Letting $j \to \infty$ and using $z_j \to z_0$ gives $\Delta_\infty\phi^\sharp(z_0) \geq 0$, and (10) yields $\Delta_\infty\phi(z_0) \geq 0$.

Step 4: lower tests. If instead $u_0 - \phi$ has a local minimum on Ω at z_0, repeat Steps 1–3 with $\phi^\sharp(x) = \phi(x) - |x - z_0|^4$, with minima in place of maxima, and with the supersolution inequality

of `IsInfHarmonicOn`. The identity (9) again gives $\Delta_\infty \phi^\sharp(z_0) = \Delta_\infty \phi(z_0)$, and the conclusion is $\Delta_\infty \phi(z_0) \le 0$.

Together with Step 0 and the openness of Ω, this shows that u_0 is infinity-harmonic on Ω. □

2.8. `AffineBlowups`.

Theorem 2.8 (Affine blow-ups at a fixed center)**.** *Let E be a finite-dimensional real inner-product space, let $D \subseteq E$, let $w : E \to \mathbb{R}$ be infinity-harmonic on D in the sense of `IsInfHarmonicOn`, and let $z \in D$. For a sequence of radii $\rho_i > 0$ with $\rho_i \to 0$ put*

$$V_i(Y) = \frac{w(z + \rho_i Y) - w(z)}{\rho_i} \qquad (Y \in E).$$

Then:

1. *every $V : E \to \mathbb{R}$ which is the limit of such a family V_i, uniformly on every compact subset of E, is a continuous linear function on E; and*
2. *for every sequence of radii $\rho_i > 0$ with $\rho_i \to 0$ there are a strictly increasing sequence of indices $i \mapsto \iota(i)$ and a function $V : E \to \mathbb{R}$ such that $V_{\iota(i)} \to V$ uniformly on every compact subset of E.*

Proof. By `IsInfHarmonicOn` the set D is open and w is continuous on D, so there is $\sigma > 0$ with $\overline{B}_{2\sigma}(z) \subseteq D$.

Step 1: w is Lipschitz on $\overline{B}_\sigma(z)$. The set $\overline{B}_{2\sigma}(z)$ is compact and contained in D, so $\operatorname{osc} = \max_{\overline{B}_{2\sigma}(z)} w - \min_{\overline{B}_{2\sigma}(z)} w$ is finite; put $\Lambda = \operatorname{osc}/\sigma \ge 0$. Then $|w(a) - w(b)| \le \operatorname{osc} = \Lambda\sigma$ for $a, b \in \overline{B}_{2\sigma}(z)$, so `InfHarmonicLocLipschitz`, applied with $z_0 = z$, $\rho = \sigma$, the same σ and this Λ — its hypotheses $\overline{B}_{\rho+\sigma}(z) = \overline{B}_{2\sigma}(z) \subseteq D$ and the oscillation bound hold — gives

$$|w(a) - w(b)| \le \Lambda |a - b| \qquad \big(a, b \in \overline{B}_\sigma(z)\big). \tag{11}$$

Step 2: uniform bounds for the blow-ups. Let $\rho_i > 0$ with $\rho_i \to 0$ and put $V_i(Y) = \big(w(z + \rho_i Y) - w(z)\big)/\rho_i$. Fix $R > 0$. For $|Y| \le R$ and all i with $\rho_i R \le \sigma$ we have $z + \rho_i Y \in \overline{B}_\sigma(z)$, so by (11) with $a = z + \rho_i Y$, $b = z$,

$$|V_i(Y)| \le \Lambda |Y| \le \Lambda R, \qquad |V_i(Y) - V_i(Y')| \le \Lambda |Y - Y'| \qquad (|Y|, |Y'| \le R), \tag{12}$$

the second bound by (11) with $a = z + \rho_i Y$, $b = z + \rho_i Y'$. Since $\rho_i \to 0$, all but finitely many i satisfy $\rho_i R \le \sigma$. Note also $V_i(0) = 0$ for every i.

Step 3: proof of (2). The selection of a locally uniformly convergent subsequence needs a compactness statement which is not among the background items of the preamble, so we prove the instance we use; the argument is the standard Arzelà–Ascoli one, and the only external inputs are the preamble items *Compactness, Weierstrass and uniform continuity* (a closed bounded subset of a finite-dimensional space is compact) and *Bolzano–Weierstrass*.

Claim A. Let $\mathcal{K} \subseteq E$ be closed and bounded, let $A \ge 0$ and $\Lambda \ge 0$, let $S \subseteq \mathbb{N}$ be infinite and let $g_i : E \to \mathbb{R}$, $i \in S$, satisfy

$$|g_i(a)| \le A, \qquad |g_i(a) - g_i(b)| \le \Lambda |a - b| \qquad (a, b \in \mathcal{K},\ i \in S).$$

Then there are an infinite set $S' \subseteq S$ and a function g on $\mathcal{K}$ such that $\sup_{\mathcal{K}} |g_i - g| \to 0$ as $i \to \infty$ along S'.

Proof of Claim A. If $\mathcal{K} = \emptyset$ take $S' = S$ and g the empty function; so assume $\mathcal{K} \ne \emptyset$. Being closed and bounded in the finite-dimensional space E, the set $\mathcal{K}$ is compact. For each integer $k \ge 1$ the balls $B_{1/k}(a)$, $a \in \mathcal{K}$, form an open cover of $\mathcal{K}$, so finitely many of them cover it; let $N_k \subseteq \mathcal{K}$ be a finite nonempty set with $\mathcal{K} \subseteq \bigcup_{b \in N_k} B_{1/k}(b)$, and put $Q = \bigcup_{k \ge 1} N_k$. Then $Q \subseteq \mathcal{K}$ is countable and nonempty, and for every $\varsigma > 0$ the balls $B_\varsigma(b)$, $b \in Q$, cover $\mathcal{K}$: choose k with $1/k \le \varsigma$ and use N_k. Enumerate $Q = \{a_1, a_2, \dots\}$ (if Q is finite the argument below is the same and shorter).

For each k the family $\big(g_i(a_k)\big)_{i \in S}$ is bounded by A, so by Bolzano–Weierstrass we may choose nested infinite sets $S \supseteq J_1 \supseteq J_2 \supseteq \cdots$ such that for every k the values $g_i(a_k)$ converge as $i \to \infty$ along J_k: given J_{k-1} (with $J_0 = S$), pick a convergent subsequence of $\big(g_i(a_k)\big)_{i \in J_{k-1}}$ and let J_k be its set of indices. Choose $i_1 < i_2 < \cdots$ with $i_l \in J_l$, which is possible because each J_l is

infinite, and put $S' = \{i_l : l \geq 1\}$, an infinite subset of S. Since $\{i_l : l \geq k\} \subseteq J_k$ for every k, the sequence $\big(g_{i_l}(a_k)\big)_l$ converges for every k.

Fix $\varepsilon > 0$ and put $\eta = \varepsilon/\big(3(\Lambda+1)\big) > 0$. Cover $\mathcal{K}$ by finitely many balls $B_\eta(b_1), \dots, B_\eta(b_N)$ with $b_\nu \in Q$, as in the first paragraph, and choose l_0 with $|g_{i_l}(b_\nu) - g_{i_{l'}}(b_\nu)| \leq \varepsilon/3$ for all $\nu \leq N$ and all $l, l' \geq l_0$; this is possible because each of the finitely many sequences $\big(g_{i_l}(b_\nu)\big)_l$ converges, hence is Cauchy. Given $a \in \mathcal{K}$, pick ν with $|a - b_\nu| < \eta$; then for $l, l' \geq l_0$,

$$|g_{i_l}(a) - g_{i_{l'}}(a)| \leq |g_{i_l}(a) - g_{i_l}(b_\nu)| + |g_{i_l}(b_\nu) - g_{i_{l'}}(b_\nu)| + |g_{i_{l'}}(b_\nu) - g_{i_{l'}}(a)| \leq 2\Lambda\eta + \frac{\varepsilon}{3} \leq \varepsilon,$$

using $2\Lambda\eta \leq 2\varepsilon/3$. So $\big(g_{i_l}\big)_l$ is uniformly Cauchy on $\mathcal{K}$; for each $a \in \mathcal{K}$ the real sequence $\big(g_{i_l}(a)\big)_l$ is therefore Cauchy and has a limit $g(a)$, and letting $l' \to \infty$ in the display gives $\sup_{\mathcal{K}} |g_{i_l} - g| \leq \varepsilon$ for every $l \geq l_0$. As $\varepsilon > 0$ was arbitrary, $\sup_{\mathcal{K}} |g_i - g| \to 0$ along S'. This proves Claim A.

We now prove (2). Let $\rho_i > 0$ with $\rho_i \to 0$. For each integer $R \geq 1$ there is, since $\rho_i \to 0$, an index m_R with $\rho_i R \leq \sigma$ for all $i \geq m_R$; for such i the bounds (12) give $|V_i| \leq \Lambda R$ and $|V_i(Y) - V_i(Y')| \leq \Lambda |Y - Y'|$ on $\overline{B}_R(0)$, a closed bounded set. Define infinite index sets $S_1 \supseteq S_2 \supseteq \cdots$ recursively: S_R is the set S' produced by Claim A applied with $\mathcal{K} = \overline{B}_R(0)$, $A = \Lambda R$, this Λ, and the index set $\{i \geq m_1\}$ when $R = 1$ and $\{i \in S_{R-1} : i \geq m_R\}$ when $R \geq 2$; these index sets are infinite and their elements i all satisfy $\rho_i R \leq \sigma$, so the hypotheses of Claim A hold. Claim A also supplies functions $V^{(R)}$ on $\overline{B}_R(0)$ with $\sup_{\overline{B}_R(0)} |V_i - V^{(R)}| \to 0$ as $i \to \infty$ along S_R.

Choose $\iota(1) < \iota(2) < \cdots$ with $\iota(j) \in S_j$ for every j, which is possible because each S_j is infinite; thus $\iota : \mathbb{N} \to \mathbb{N}$ is strictly increasing. Fix an integer $R \geq 1$. For $j \geq R$ we have $\iota(j) \in S_j \subseteq S_R$, and $\iota(j) \to \infty$, so

$$\sup_{\overline{B}_R(0)} \big|V_{\iota(j)} - V^{(R)}\big| \longrightarrow 0 \qquad (j \to \infty). \tag{13}$$

In particular, for $R \leq R'$ the functions $V^{(R)}$ and $V^{(R')}$ agree on $\overline{B}_R(0)$, both being the pointwise limit of $V_{\iota(j)}$ there. Hence a single function V on E is well defined by $V(Y) = V^{(R)}(Y)$ for any integer $R \geq \max\{1, |Y|\}$, and (13) says that $V_{\iota(j)} \to V$ uniformly on $\overline{B}_R(0)$ for every integer $R \geq 1$. Since every compact $K \subseteq E$ is bounded, hence contained in some $\overline{B}_R(0)$ with R an integer, $R \geq 1$, the convergence is uniform on every compact subset of E. This is (2).

Step 4: V is infinity-harmonic on E and Λ-Lipschitz. From now on let $\rho_i > 0$, $\rho_i \to 0$, and let V be a function on E such that $V_i \to V$ uniformly on every compact subset of E, as in the hypothesis of (1).

By `InfHarmonicRescale`, applied with $x_0 = z$ and $A = \rho_i > 0$, the function $Y \mapsto w(z + \rho_i Y)$ is infinity-harmonic on $D_i = \{Y : z + \rho_i Y \in D\}$; and by `InfHarmonicNormalize`, applied with $c = w(z)$ and $M = \rho_i > 0$ on the open set D_i, the function V_i is infinity-harmonic on D_i. Moreover $\overline{B}_\sigma(z) \subseteq D$ gives $B_{\sigma/\rho_i}(0) \subseteq D_i$.

Fix $R > 0$ and choose i_0 with $\sigma/\rho_i > R$ for $i \geq i_0$; then each V_{i_0+j}, $j \geq 0$, is infinity-harmonic on the open ball $B_R(0)$. Indeed $B_R(0) \subseteq D_i$ is open, and the restriction of the infinity-harmonicity of `IsInfHarmonicOn` to an open subset is legitimate: continuity restricts, and if $\Omega'' \subseteq \Omega'$ is open, $\zeta \in \Omega''$ and $V_i - \phi$ has a local maximum on Ω'' at ζ, then it has one on Ω' at ζ, because a neighbourhood N of ζ with $(V_i - \phi) \leq (V_i - \phi)(\zeta)$ on $N \cap \Omega''$ may be replaced by $N \cap \Omega''$, itself a neighbourhood of ζ, on which $N \cap \Omega'' \cap \Omega' = N \cap \Omega''$; the same applies to minima.

The shifted sequence $j \mapsto V_{i_0+j}$ still converges to V uniformly on every compact subset of $B_R(0)$, being a tail of the original one. Hence `InfHarmonicStability`, applied with $\Omega = B_R(0)$, shows that V is infinity-harmonic on $B_R(0)$. As $R > 0$ was arbitrary and a local extremum of $V - \phi$ on E at a point ζ is in particular one on $B_R(0)$ for any $R > |\zeta|$, together with the continuity of V on each $B_R(0)$, we conclude that V is infinity-harmonic on E.

Finally, letting $i \to \infty$ in (12) gives

$$|V(Y) - V(Y')| \leq \Lambda |Y - Y'| \quad (Y, Y' \in E), \qquad V(0) = 0. \tag{14}$$

Step 5: the slopes of V at the origin. By ConeSlopeLimitEq, applied to w on D at $z \in D$, there is $L_* \geq 0$ with

$$\lim_{s\downarrow 0} L_s^+(w;z) = \lim_{s\downarrow 0} L_s^-(w;z) = L_*. \tag{15}$$

Fix $R > 0$. By ConeSlopeUpper, for $\rho_i R \leq \sigma$,

$$\sup_{|Y|\leq R} \frac{V_i(Y) - V_i(0)}{R} = \frac{1}{R\rho_i} \sup_{|h|\leq \rho_i R} \big(w(z+h) - w(z)\big) = L^+_{\rho_i R}(w;z),$$

where we substituted $h = \rho_i Y$, which is a bijection of $\{|Y| \leq R\}$ onto $\{|h| \leq \rho_i R\}$, and used $V_i(0) = 0$. The left side is $L_R^+(V_i;0)$ by ConeSlopeUpper. Since $V_i \to V$ uniformly on $\overline{B}_R(0)$ and all these suprema are finite by (12), we get $L_R^+(V_i;0) \to L_R^+(V;0)$; and $\rho_i R \to 0$, so by (15) the right side tends to L_*. Hence, and by the identical computation with $-(w(z+h) - w(z))$ and ConeSlopeLower,

$$L_R^+(V;0) = L_R^-(V;0) = L_* \qquad (R > 0). \tag{16}$$

In particular, for $Y \neq 0$ the competitor Y at radius $R = |Y|$ in ConeSlopeUpper and ConeSlopeLower gives $V(Y)/|Y| \leq L_*$ and $-V(Y)/|Y| \leq L_*$, so

$$|V(Y)| \leq L_*|Y| \qquad (Y \in E), \tag{17}$$

the case $Y = 0$ being $V(0) = 0$.

Step 6: V is globally L_*-Lipschitz. Let $q \in E$ and $T > 0$. By ConeSlopeUpper and (17),

$$L_T^+(V;q) = \frac{1}{T} \sup_{|H|\leq T} \big(V(q+H) - V(q)\big) \leq \frac{L_*(T + |q|) + |V(q)|}{T},$$

and the same bound holds for $L_T^-(V;q)$ by ConeSlopeLower. Fix $T_0 > 0$. Since V is infinity-harmonic on E by Step 4 and $\overline{B}_T(q) \subseteq E$ for every T, the radial monotonicity assertion of ConeComparison, applied at the center q with radius T_0 and comparison radius $T \geq T_0$, gives $L^{\pm}_{T_0}(V;q) \leq L^{\pm}_T(V;q)$. Letting $T \to \infty$ in the displayed bound yields

$$L^+_{T_0}(V;q) \leq L_*, \qquad L^-_{T_0}(V;q) \leq L_* \qquad (q \in E,\ T_0 > 0). \tag{18}$$

Now let $Q, q \in E$ with $Q \neq q$ and put $T_0 = |Q - q| > 0$. The two cone comparison inequalities of ConeComparison at the center q with radius T_0 give $V(Q) - V(q) \leq L^+_{T_0}(V;q)|Q-q|$ and $-(V(Q) - V(q)) \leq L^-_{T_0}(V;q)|Q-q|$, so by (18)

$$|V(Q) - V(q)| \leq L_*|Q - q| \qquad (Q, q \in E). \tag{19}$$

Step 7: the case $L_* = 0$. If $L_* = 0$ then (17) gives $V \equiv 0$, which is the zero continuous linear functional, and (1) holds.

Step 8: the case $L_* > 0$: a distinguished direction. Assume $L_* > 0$. For $R > 0$ the supremum in $L_R^+(V;0)$ is over the compact ball $\overline{B}_R(0)$ of the continuous function $Y \mapsto V(Y)/R$, so it is attained: by (16) there is x_R with $|x_R| \leq R$ and $V(x_R) = L_* R$. Likewise there is y_S with $|y_S| \leq S$ and $-V(y_S) = L_* S$.

For any $R, S > 0$, using (19),

$$L_*(R+S) = V(x_R) - V(y_S) \leq L_*|x_R - y_S| \leq L_*\big(|x_R| + |y_S|\big) \leq L_*(R+S). \tag{20}$$

Since $L_* > 0$, every inequality in (20) is an equality; hence $|x_R| = R$, $|y_S| = S$ and $|x_R - y_S| = |x_R| + |y_S|$.

Equality in $|a - b| \leq |a| + |b|$ for nonzero a, b forces a and $-b$ to be positive multiples of a common unit vector. Indeed, squaring $|a-b|^2 = (|a| + |b|)^2$ gives $-2\langle a, b\rangle = 2|a||b|$, i.e. $\langle a, -b\rangle = |a||-b|$, which is equality in Cauchy–Schwarz and therefore means $-b = (|b|/|a|)\, a$.

Fix $S = 1$ and let $e = -y_1$, a unit vector since $|y_1| = 1$. For each $R > 0$, applying the previous paragraph with $a = x_R$ and $b = y_1$ gives $e = -y_1 = (1/R)x_R$, i.e. $x_R = Re$, so

$$V(Re) = L_* R \qquad (R > 0). \tag{21}$$

Now fix $R = 1$; the same paragraph with $a = x_1 = e$ and $b = y_S$ gives $-y_S = S\,e$, i.e. $y_S = -Se$, so

$$V(-Se) = -L_*S \qquad (S > 0). \tag{22}$$

Together with $V(0) = 0$, (21) and (22) give

$$V(\tau e) = L_*\tau \qquad (\tau \in \mathbb{R}). \tag{23}$$

Step 9: V is linear. Let $Y \in E$. By (19) and (23), for every $\tau \in \mathbb{R}$,

$$L_*\tau - L_*|Y - \tau e| \le V(Y) \le L_*\tau + L_*|Y - \tau e|.$$

For $\tau > 0$ we compute $|Y - \tau e|^2 = \tau^2 - 2\tau\langle e, Y\rangle + |Y|^2$, so

$$|Y - \tau e| = \tau\sqrt{1 - \frac{2\langle e, Y\rangle}{\tau} + \frac{|Y|^2}{\tau^2}} = \tau - \langle e, Y\rangle + O\big(\tau^{-1}\big) \qquad (\tau \to +\infty),$$

by the expansion $\sqrt{1+u} = 1 + u/2 + O(u^2)$ applied with $u = -2\langle e, Y\rangle/\tau + |Y|^2/\tau^2 \to 0$. Hence $L_*\tau - L_*|Y - \tau e| \to L_*\langle e, Y\rangle$ as $\tau \to +\infty$, and the lower bound gives $V(Y) \ge L_*\langle e, Y\rangle$. Taking instead $\tau = -\varsigma$ with $\varsigma \to +\infty$, the same expansion with e replaced by $-e$ gives $|Y + \varsigma e| = \varsigma + \langle e, Y\rangle + O(\varsigma^{-1})$, so

$$L_*\tau + L_*|Y - \tau e| = -L_*\varsigma + L_*|Y + \varsigma e| \longrightarrow L_*\langle e, Y\rangle,$$

and the upper bound gives $V(Y) \le L_*\langle e, Y\rangle$. Therefore

$$V(Y) = L_*\langle e, Y\rangle \qquad (Y \in E),$$

so V is the continuous linear functional $L = L_*\langle e, \cdot\rangle$ on E. This proves (1). □

2.9. `AnisotropicBoxOffsets`.

Lemma 2.9 (Anisotropic box offsets relative to a nearly aligned direction)**.** *Let F be a real inner-product space, let $\ell^2(F \times \mathbb{R})$ be the associated space-time with added unit direction $e_b = (0, 1)$, and let Φ be a linear isometry of $\ell^2(F \times \mathbb{R})$ onto itself. Let $r > 0$, $\theta \ge 0$, $C \ge 0$, $K \ge 1$ and $0 \le s \le 1$ be reals with $s\sqrt{32} = \theta$, let $n \in \ell^2(F \times \mathbb{R})$ satisfy $|n| = 1$, and let R be a real with*

$$16K < R, \qquad K\big(32^{-1/2} + C\big) < R/4. \tag{24}$$

Assume the direction bound

$$\big|\Phi(e_b) - n\big| \le C\theta. \tag{25}$$

Then for every $x \in F$ and $t \in \mathbb{R}$ with $|x| \le K$ and $|t| \le K$, the vector $h = r\,\Phi(s\,x,\, t)$ satisfies

$$|\langle n, h\rangle| \le \frac{Rr}{4}, \qquad \big|h - \langle n, h\rangle\, n\big| \le \frac{\theta\, Rr}{4}.$$

Proof. Throughout, $h = r\,\Phi(s\,x,\, t)$, and $R > 0$ because $16K < R$ and $K \ge 1$.

Step 1: the length of h. Since Φ is a linear isometry and $r > 0$,

$$|h| = r\,\big|(s\,x,\, t)\big|.$$

By the definition of the ℓ^2 product norm, $|(s\,x, t)|^2 = |s\,x|^2 + t^2 = (s|x|)^2 + |t|^2$, and since $s|x| \ge 0$ and $|t| \ge 0$ we have $(s|x| + |t|)^2 = (s|x|)^2 + |t|^2 + 2s|x|\,|t| \ge |(s\,x, t)|^2$; both sides being nonnegative, $|(s\,x, t)| \le s|x| + |t|$. As $0 \le s \le 1$ and $|x| \ge 0$ we get $s|x| \le |x|$, so $s|x| + |t| \le |x| + |t| \le 2K$. Hence

$$|h| \le 2Kr. \tag{26}$$

Step 2: the axial bound. By Cauchy–Schwarz and $|n| = 1$, $|\langle n, h\rangle| \le |n|\,|h| = |h|$. By (26) and $16K < R$ we have $2Kr \le Rr/8$, because multiplying $16K < R$ by $r > 0$ gives $16Kr < Rr$. Since $Rr > 0$, $Rr/8 \le Rr/4$, and therefore $|\langle n, h\rangle| \le Rr/4$, which is the first assertion.

Step 3: $\langle n, h\rangle n$ is the closest multiple of n to h. Let $u \in \mathbb{R}$ and put $\sigma = \langle n, h\rangle$. The vectors $h - \sigma n$ and $(\sigma - u)n$ are orthogonal: using $|n| = 1$,

$$\langle h - \sigma n,\ (\sigma - u)n\rangle = (\sigma - u)\big(\langle h, n\rangle - \sigma|n|^2\big) = (\sigma - u)(\sigma - \sigma) = 0,$$

where $\langle h, n\rangle = \langle n, h\rangle = \sigma$ by symmetry of the real inner product. Since $h - u\,n = (h - \sigma n) + (\sigma - u)n$, the Pythagorean identity gives

$$|h - u\,n|^2 = |h - \sigma n|^2 + |(\sigma - u)n|^2 \geq |h - \sigma n|^2,$$

and as both norms are nonnegative,

$$\big|h - \langle n, h\rangle\, n\big| \leq |h - u\,n| \qquad \text{for every } u \in \mathbb{R}. \tag{27}$$

Step 4: the frame splitting of h. In $F \times \mathbb{R}$ one has the identity $(s\,x,\, t) = s\,(x, 0) + t\,(0, 1) = s\,(x, 0) + t\,e_b$, since both sides have first component $s\,x$ and second component t. Applying the linear map Φ and multiplying by r,

$$h = (rs)\,\Phi(x, 0) + (rt)\,\Phi(e_b). \tag{28}$$

Subtracting $(rt)\,n$ from both sides of (28),

$$h - (rt)\,n = (rs)\,\Phi(x, 0) + (rt)\big(\Phi(e_b) - n\big). \tag{29}$$

Step 5: the transverse bound. Since Φ is a linear isometry and $|(x, 0)| = |x|$ by the definition of the ℓ^2 product norm, $\big|(rs)\Phi(x, 0)\big| = rs\,|x|$, using $r > 0$ and $s \geq 0$; and $\big|(rt)(\Phi(e_b) - n)\big| = r|t|\,\big|\Phi(e_b) - n\big|$, using $r > 0$. So the triangle inequality applied to (29) gives

$$\big|h - (rt)\,n\big| \leq rs\,|x| + r|t|\,\big|\Phi(e_b) - n\big|.$$

From $s\sqrt{32} = \theta$ and $\sqrt{32} > 0$ we get $s = 32^{-1/2}\theta$, so $rs \geq 0$ and, by $|x| \leq K$,

$$rs\,|x| = r\,32^{-1/2}\theta\,|x| \leq r\,32^{-1/2}\theta\,K.$$

By (25), $C\theta \geq 0$ and $|t| \leq K$,

$$r|t|\,\big|\Phi(e_b) - n\big| \leq r|t|\,C\theta \leq rK\,C\theta.$$

Adding the two displays,

$$\big|h - (rt)\,n\big| \leq rK\big(32^{-1/2} + C\big)\theta. \tag{30}$$

Finally, multiplying $K(32^{-1/2} + C) < R/4$ from (24) by $r\theta \geq 0$ gives

$$rK\big(32^{-1/2} + C\big)\theta \leq \frac{\theta\,Rr}{4},$$

and combining this with (30) and with (27) taken at $u = rt$ yields

$$\big|h - \langle n, h\rangle\, n\big| \leq \big|h - (rt)\,n\big| \leq rK\big(32^{-1/2} + C\big)\theta \leq \frac{\theta\,Rr}{4},$$

which is the second assertion. □

2.10. `AsymptEquicontLocalUniformLimit.`

Lemma 2.10 (Locally uniform extraction under an asymptotic modulus of continuity)**.** *Let H be a finite-dimensional real inner-product space and let g_j, $j \geq 0$, be real functions on H. Suppose:*

(i) *for every $\rho > 0$ there are a real number A and an index j_0 such that $|g_j(w)| \leq A$ for every $j \geq j_0$ and every $w \in H$ with $|w| \leq \rho$;*

(ii) *for every $\rho > 0$ and every $\varepsilon > 0$ there are $\eta > 0$ and an index j_0 such that $|g_j(w) - g_j(w')| \leq \varepsilon$ whenever $j \geq j_0$ and $w, w' \in H$ satisfy $|w| \leq \rho$, $|w'| \leq \rho$ and $|w - w'| \leq \eta$.*

Then there are a strictly increasing sequence of indices $l \mapsto \psi(l)$ and a real function w on H such that $g_{\psi(l)} \to w$ uniformly on every compact subset of H.

Proof. Throughout, $\overline{B}_\rho = \{u \in H : |u| \leq \rho\}$ and $B_\sigma(b) = \{u \in H : |u - b| < \sigma\}$. Note that neither hypothesis is assumed for small j, and that the index j_0 in each is allowed to depend on ρ; this is why the extraction below is performed once, against a countable set dense in all of H, rather than ball by ball.

Step 1: a countable dense subset of H. Let $k \geq 1$ be an integer. The set $\overline{B}_k$ is closed and bounded in a finite-dimensional normed space, hence compact. The balls $B_{1/k}(a)$ with $a \in \overline{B}_k$

cover $\overline{B}_k$, so finitely many of them do: there is a finite set $N_k \subseteq \overline{B}_k$ with $\overline{B}_k \subseteq \bigcup_{b\in N_k} B_{1/k}(b)$. Put $Q = \bigcup_{k\geq 1} N_k$, a countable subset of H.

Q is dense in H: given $x \in H$ and $\sigma > 0$, choose an integer $k \geq \max\{|x|, 1/\sigma, 1\}$. Then $x \in \overline{B}_k$, so some $b \in N_k$ satisfies $|x - b| < 1/k \leq \sigma$. Enumerate $Q = \{a_1, a_2, \dots\}$; if Q is finite the argument below is the same and shorter.

Step 2: at each point of Q the whole sequence of values is bounded. Fix k. Apply (i) with $\rho = |a_k| + 1 > 0$: there are A and j_0 with $|g_j(a_k)| \leq A$ for all $j \geq j_0$, since $|a_k| \leq |a_k| + 1$. The finitely many real numbers $|g_j(a_k)|$ with $j < j_0$ admit a common bound B (take $B = 0$ when $j_0 = 0$, and otherwise the maximum of that finite list). Hence, with $A_k = \max\{A, B\}$,

$$|g_j(a_k)| \leq A_k \qquad \text{for } \textit{every } j \geq 0.$$

Step 3: one subsequence converging at every point of Q. By Step 2 the sequence $(g_j(a_k))_{j\geq 0}$ is bounded for each fixed k, so by Bolzano–Weierstrass it has a convergent subsequence. Applying this successively we may choose nested infinite index sets $\mathbb{N} \supseteq J_1 \supseteq J_2 \supseteq \cdots$ such that $g_j(a_k)$ converges as $j \to \infty$ along J_k, for every k: given J_{k-1} (with $J_0 = \mathbb{N}$), the sequence $(g_j(a_k))_{j\in J_{k-1}}$ is bounded, so it has a convergent subsequence, whose index set we take as J_k. Now choose indices $\psi(1) < \psi(2) < \cdots$ with $\psi(l) \in J_l$, which is possible because each J_l is infinite. For each fixed k we have $\{\psi(l) : l \geq k\} \subseteq J_k$ and $\psi(l) \to \infty$, so $(g_{\psi(l)}(a_k))_{l\geq 0}$ is a subsequence of a convergent sequence and therefore converges; call its limit c_k.

Step 4: $(g_{\psi(l)})_l$ is uniformly Cauchy on every $\overline{B}_\rho$. Fix $\rho > 0$ and $\varepsilon > 0$. Apply (ii) with radius $\rho + 1$ and $\varepsilon/4$: there are $\eta > 0$ and an index j_1 with

$$|g_j(u) - g_j(u')| \leq \varepsilon/4 \quad \text{whenever } j \geq j_1,\ |u| \leq \rho + 1,\ |u'| \leq \rho + 1,\ |u - u'| \leq \eta. \tag{31}$$

Put $\delta = \min\{\eta, 1\} > 0$. Since Q is dense, the balls $B_\delta(b)$ with $b \in Q$ cover H and in particular $\overline{B}_\rho$; as $\overline{B}_\rho$ is compact, finitely many suffice: $\overline{B}_\rho \subseteq \bigcup_{\nu=1}^{N} B_\delta(b_\nu)$ with $b_\nu \in Q$, say $b_\nu = a_{k(\nu)}$.

By Step 3, $g_{\psi(l)}(b_\nu) \to c_{k(\nu)}$ for each of the finitely many $\nu \leq N$, and $\psi(l) \geq l \to \infty$ because ψ is strictly increasing. So there is l_0 with

$$\psi(l) \geq j_1 \quad \text{and} \quad |g_{\psi(l)}(b_\nu) - c_{k(\nu)}| < \varepsilon/4 \qquad (l \geq l_0,\ \nu \leq N). \tag{32}$$

Let $x \in \overline{B}_\rho$ and $l, l' \geq l_0$. Choose ν with $|x - b_\nu| < \delta$. Then $|x| \leq \rho \leq \rho + 1$, and $|b_\nu| \leq |x| + |x - b_\nu| \leq \rho + \delta \leq \rho + 1$, and $|x - b_\nu| \leq \delta \leq \eta$; so (31) applies at $j = \psi(l)$ and at $j = \psi(l')$, both of which are $\geq j_1$ by (32), and gives $|g_{\psi(l)}(x) - g_{\psi(l)}(b_\nu)| \leq \varepsilon/4$ and $|g_{\psi(l')}(x) - g_{\psi(l')}(b_\nu)| \leq \varepsilon/4$. From the exact identity

$$g_{\psi(l)}(x) - g_{\psi(l')}(x) = \big(g_{\psi(l)}(x) - g_{\psi(l)}(b_\nu)\big) + \big(g_{\psi(l)}(b_\nu) - c_{k(\nu)}\big) - \big(g_{\psi(l')}(b_\nu) - c_{k(\nu)}\big) - \big(g_{\psi(l')}(x) - g_{\psi(l')}(b_\nu)\big)$$

and (32) we conclude $|g_{\psi(l)}(x) - g_{\psi(l')}(x)| < \varepsilon$. As $x \in \overline{B}_\rho$ was arbitrary, $(g_{\psi(l)})_l$ is uniformly Cauchy on $\overline{B}_\rho$.

Step 5: the limit, and uniform convergence on compact sets. Let $z \in H$ and apply Step 4 with $\rho = |z| + 1 > 0$, noting $z \in \overline{B}_\rho$: the real sequence $(g_{\psi(l)}(z))_l$ is Cauchy, hence convergent by completeness of $\mathbb{R}$. Define $w(z)$ to be its limit; this defines w on all of H.

Fix $\rho > 0$ and $\varepsilon > 0$, and let l_0 be as in Step 4 for this ρ and for $\varepsilon/2$ in place of ε. For $l \geq l_0$ and $x \in \overline{B}_\rho$ we have $|g_{\psi(l)}(x) - g_{\psi(l')}(x)| < \varepsilon/2$ for every $l' \geq l_0$; letting $l' \to \infty$ and using $g_{\psi(l')}(x) \to w(x)$ gives $|g_{\psi(l)}(x) - w(x)| \leq \varepsilon/2 < \varepsilon$. Hence $\sup_{\overline{B}_\rho} |g_{\psi(l)} - w| \leq \varepsilon/2$ for $l \geq l_0$, i.e. $g_{\psi(l)} \to w$ uniformly on $\overline{B}_\rho$.

Finally let $C \subseteq H$ be compact. Then C is bounded, so $C \subseteq \overline{B}_\rho$ for some $\rho > 0$, and uniform convergence on $\overline{B}_\rho$ implies uniform convergence on C. Thus ψ and w have all the asserted properties. □

2.11. `AsymptEquicontOriginNormalizedLimit`.

Lemma 2.11 (Locally uniform extraction, normalized at a starting index and at the origin)**.** *Let H be a finite-dimensional real inner-product space, let g_j, $j \geq 0$, be real functions on H, and let i_1 be an index. Suppose:*

(i) *for every $\rho > 0$ there are a real number A and an index j_0 such that $|g_j(w)| \leq A$ for every $j \geq j_0$ and every $w \in H$ with $|w| \leq \rho$;*

(ii) *for every* $\rho > 0$ *and every* $\varepsilon > 0$ *there are* $\eta > 0$ *and an index* j_0 *such that* $|g_j(w) - g_j(w')| \le \varepsilon$ *whenever* $j \ge j_0$ *and* $w, w' \in H$ *satisfy* $|w| \le \rho$, $|w'| \le \rho$ *and* $|w - w'| \le \eta$;
(iii) $g_i(0) = 0$ *for every* $i \ge i_1$.

Then there are a strictly increasing sequence of indices $l \mapsto \psi(l)$ *and a real function* w *on* H *such that* $\psi(l) \ge i_1$ *for every* l, *the function* w *is continuous on* H, $g_{\psi(l)} \to w$ *uniformly on every compact subset of* H, *and* $w(0) = 0$.

Proof. Apply `AsymptEquicontLocalUniformLimit` to the family g with hypotheses (i) and (ii), which are verbatim that node's two hypotheses. It supplies a strictly increasing $l \mapsto \psi_0(l)$ and a real function w on H with

$$g_{\psi_0(l)} \to w \quad \text{uniformly on every compact subset of } H. \tag{33}$$

Define $\psi(l) = \psi_0(l + i_1)$.

Step 1: ψ is strictly increasing and starts past i_1. If $l < l'$ then $l + i_1 < l' + i_1$, so $\psi(l) = \psi_0(l + i_1) < \psi_0(l' + i_1) = \psi(l')$ because ψ_0 is strictly increasing. A strictly increasing map of the indices satisfies $\psi_0(k) \ge k$ for every k (induction on k: $\psi_0(0) \ge 0$, and $\psi_0(k+1) > \psi_0(k) \ge k$ gives $\psi_0(k+1) \ge k + 1$). Hence $\psi(l) = \psi_0(l + i_1) \ge l + i_1 \ge i_1$.

Step 2: the shifted subsequence still converges, and converges pointwise. Let $C \subseteq H$ be compact. Uniform convergence on C along ψ_0, which is (33), means: for every $\varepsilon > 0$ there is l_0 with $\sup_{u \in C} |g_{\psi_0(l)}(u) - w(u)| \le \varepsilon$ for all $l \ge l_0$. For $l \ge l_0$ we have $l + i_1 \ge l_0$, so the same bound holds along ψ; thus $g_{\psi(l)} \to w$ uniformly on C. Taking $C = \{u\}$, which is compact, gives for every $u \in H$

$$g_{\psi(l)}(u) \to w(u) \qquad (l \to \infty). \tag{34}$$

Step 3: w is continuous. Fix $x \in H$ and $\varepsilon > 0$, and put $\rho = |x| + 2 > 0$. Apply (ii) at this ρ and at $\varepsilon/3 > 0$: there are $\eta > 0$ and j_0 with

$$|g_j(u) - g_j(u')| \le \varepsilon/3 \qquad \text{whenever } j \ge j_0,\ |u| \le \rho,\ |u'| \le \rho,\ |u - u'| \le \eta. \tag{35}$$

Let $\delta = \min\{\eta, 1\} > 0$ and let $y \in H$ satisfy $|y - x| < \delta$. Then $|x| \le \rho$, and $|y| \le |x| + |y - x| \le |x| + 1 \le \rho$, so both x and y are admissible in (35); also $|x - y| < \delta \le \eta$. By Step 1, $\psi(l) \ge l + i_1$, so $\psi(l) \ge j_0$ for every $l \ge j_0$. Therefore (35) applied at $j = \psi(l)$ gives

$$|g_{\psi(l)}(x) - g_{\psi(l)}(y)| \le \varepsilon/3 \qquad \text{for all } l \ge j_0.$$

By (34) at x and at y, the left side converges to $|w(x) - w(y)|$ as $l \to \infty$; a limit of a sequence bounded by $\varepsilon/3$ is bounded by $\varepsilon/3$, so

$$|w(x) - w(y)| \le \varepsilon/3 < \varepsilon.$$

Thus for every x and every $\varepsilon > 0$ there is $\delta > 0$ with $|w(y) - w(x)| < \varepsilon$ whenever $|y - x| < \delta$, which is continuity of w at x. As x was arbitrary, w is continuous on H.

Step 4: $w(0) = 0$. By Step 1, $\psi(l) \ge i_1$ for every l, so hypothesis (iii) gives $g_{\psi(l)}(0) = 0$ for every l. By (34) at $u = 0$, the constant sequence 0 converges to $w(0)$; limits in $\mathbb{R}$ are unique and the constant sequence 0 converges to 0, so $w(0) = 0$.

Steps 1 through 4 give the four asserted properties of ψ and w. □

2.12. `ClosedSuperjetClosed`.

Lemma 2.12 (The closed superjet is a closed set)**.** *Let E be a finite-dimensional real inner-product space, let $f : E \to \mathbb{R}$ and let $z \in E$. Then the closed superjet $\overline{J}^{2,+} f(z)$ of `ClosedSuperjet` is a closed subset of $E \times \mathcal{B}(E)$, where $\mathcal{B}(E)$ denotes the space of continuous bilinear forms on E with the operator norm.*

Proof. Since $E \times \mathcal{B}(E)$ is a metric space, it suffices to show that $\overline{J}^{2,+} f(z)$ contains the limit of every convergent sequence of its elements.

So let $(p^{(m)}, X^{(m)}) \in \overline{J}^{2,+} f(z)$ for $m \ge 1$ with $p^{(m)} \to p$ in E and $X^{(m)} \to X$ in $\mathcal{B}(E)$; we must show $(p, X) \in \overline{J}^{2,+} f(z)$.

A diagonal choice. Fix $m \geq 1$. By `ClosedSuperjet` applied to $(p^{(m)}, X^{(m)}) \in \overline{J}^{2,+} f(z)$ there are sequences $(z_k^{(m)})_k$, $(p_k^{(m)})_k$, $(X_k^{(m)})_k$ with

$$(p_k^{(m)}, X_k^{(m)}) \in J^{2,+} f\big(z_k^{(m)}\big) \quad (k \geq 0),$$

and $z_k^{(m)} \to z$, $f(z_k^{(m)}) \to f(z)$, $p_k^{(m)} \to p^{(m)}$ and $X_k^{(m)} \to X^{(m)}$ as $k \to \infty$, where $J^{2,+}$ is the superjet of `Superjet`. Because these four sequences converge to the displayed limits, and $1/m > 0$, we may choose an index $k(m)$ so large that all four of

$$\big|z_{k(m)}^{(m)} - z\big| < \tfrac{1}{m}, \quad \big|f(z_{k(m)}^{(m)}) - f(z)\big| < \tfrac{1}{m}, \quad \big|p_{k(m)}^{(m)} - p^{(m)}\big| < \tfrac{1}{m}, \quad \big\|X_{k(m)}^{(m)} - X^{(m)}\big\| < \tfrac{1}{m}$$

hold simultaneously: each of the four conditions holds for all sufficiently large k, so any $k(m)$ beyond the largest of the four thresholds works.

Put $\zeta_m = z_{k(m)}^{(m)}$, $\pi_m = p_{k(m)}^{(m)}$ and $\Xi_m = X_{k(m)}^{(m)}$.

The diagonal sequences have the required properties. First, $(\pi_m, \Xi_m) \in J^{2,+} f(\zeta_m)$ for every m, by the displayed membership with $k = k(m)$.

Second, $|\zeta_m - z| < 1/m \to 0$, so $\zeta_m \to z$.

Third, $|f(\zeta_m) - f(z)| < 1/m \to 0$, so $f(\zeta_m) \to f(z)$.

Fourth, $|\pi_m - p| \leq |\pi_m - p^{(m)}| + |p^{(m)} - p| < \frac{1}{m} + |p^{(m)} - p|$, and both terms tend to 0 (the second because $p^{(m)} \to p$), so $\pi_m \to p$.

Fifth, and in the same way, $\|\Xi_m - X\| \leq \|\Xi_m - X^{(m)}\| + \|X^{(m)} - X\| < \frac{1}{m} + \|X^{(m)} - X\| \to 0$, so $\Xi_m \to X$.

These five statements are precisely the requirements in the definition of `ClosedSuperjet` for the pair (p, X) relative to f at z, with defining sequences $(\zeta_m, \pi_m, \Xi_m)_m$. Hence $(p, X) \in \overline{J}^{2,+} f(z)$, and $\overline{J}^{2,+} f(z)$ is closed. □

2.13. `BilinearFormBolzanoWeierstrass.`

Lemma 2.13 (Bolzano–Weierstrass for tuples of continuous bilinear forms)**.** *Let E be a finite-dimensional real inner-product space, let N be a natural number, let $\Lambda \in \mathbb{R}$, and let $\big(X^{(k)}\big)_{k\geq 0}$ be a sequence of N-tuples $X^{(k)} = \big(X_j^{(k)}\big)_{j=1}^N$ of continuous bilinear forms on E such that*

$$\big\|X_j^{(k)}\big\| \leq \Lambda \qquad (k \geq 0,\ 1 \leq j \leq N).$$

Then there are an N-tuple $X = (X_j)_{j=1}^N$ of continuous bilinear forms on E and a strictly increasing map $\varphi : \mathbb{N} \to \mathbb{N}$ such that $X_j^{(\varphi(k))} \to X_j$ as $k \to \infty$ for every j.

Proof. Write $\mathcal{B}(E)$ for the real vector space of continuous bilinear forms on E, normed by the operator norm $\|X\| = \sup\{|X(a,b)| : |a| \leq 1, |b| \leq 1\}$, and $\mathcal{B}(E)^N$ for the space of N-tuples, normed by $\|(X_j)_j\|_\infty = \max_{1\leq j\leq N} \|X_j\|$ (the maximum of the empty family being 0 when $N = 0$).

The ambient space is finite-dimensional. Fix a basis $e_1, \ldots, e_d$ of E, where $d = \dim E < \infty$. A bilinear form on E is determined by its d^2 values $X(e_i, e_l)$, and every choice of these values defines a bilinear form, which is automatically continuous because E is finite-dimensional; the resulting linear bijection $\mathcal{B}(E) \to \mathbb{R}^{d^2}$ shows $\dim \mathcal{B}(E) = d^2 < \infty$. Hence $\dim \mathcal{B}(E)^N = Nd^2 < \infty$, so $\mathcal{B}(E)^N$ is a finite-dimensional real normed space.

The sequence is bounded. By hypothesis $\big\|X_j^{(k)}\big\| \leq \Lambda$ for all k and all j, so $\big\|X^{(k)}\big\|_\infty = \max_j \big\|X_j^{(k)}\big\| \leq \max(\Lambda, 0)$ for every k. Thus $\big(X^{(k)}\big)_k$ is a bounded sequence in $\mathcal{B}(E)^N$.

Extraction. By Bolzano–Weierstrass in a finite-dimensional real normed space (preamble), the bounded sequence $\big(X^{(k)}\big)_k$ has a convergent subsequence: there are a strictly increasing $\varphi : \mathbb{N} \to \mathbb{N}$ and $X \in \mathcal{B}(E)^N$ with $X^{(\varphi(k))} \to X$ in $\mathcal{B}(E)^N$.

Coordinatewise convergence. Fix j. For every k, $\big\|X_j^{(\varphi(k))} - X_j\big\| \leq \big\|X^{(\varphi(k))} - X\big\|_\infty$ by the definition of the maximum norm, and the right-hand side tends to 0; hence $X_j^{(\varphi(k))} \to X_j$. This holds for every j, which is the assertion. □

2.14. `ClosedSuperjetLocal`.

Lemma 2.14 (The closed superjet at a point is local)**.** *Let E be a finite-dimensional real inner-product space, let $f_1, f_2 : E \to \mathbb{R}$, let $b > 0$, and suppose that $f_1(w) = f_2(w)$ for every $w \in E$ with $|w| < b$. Then*

$$\overline{J}^{2,+} f_1(0) = \overline{J}^{2,+} f_2(0),$$

where $\overline{J}^{2,+}$ is the closed superjet of `ClosedSuperjet`.

Proof. By symmetry of the hypothesis in f_1 and f_2 it suffices to prove the inclusion $\overline{J}^{2,+} f_1(0) \subseteq \overline{J}^{2,+} f_2(0)$; exchanging the names of f_1 and f_2 then gives the reverse inclusion and hence the equality.

So let $(p, X) \in \overline{J}^{2,+} f_1(0)$. By `ClosedSuperjet` there are sequences $(z_k)_{k\geq 0}$ in E, $(p_k)_{k\geq 0}$ in E and $(X_k)_{k\geq 0}$ of continuous bilinear forms on E with

$$(p_k, X_k) \in J^{2,+} f_1(z_k) \quad (k \geq 0), \qquad z_k \to 0, \quad f_1(z_k) \to f_1(0), \quad p_k \to p, \quad X_k \to X, \tag{36}$$

where $J^{2,+}$ is the superjet of `Superjet`.

Since $z_k \to 0$ and $b/4 > 0$, there is k_0 with

$$|z_k| < b/4 \qquad (k \geq k_0). \tag{37}$$

Each jet is a superjet of f_2 as well, for $k \geq k_0$. Fix $k \geq k_0$. The pair (p_k, X_k) lies in $J^{2,+} f_1(z_k)$, so by `Superjet` the form X_k is symmetric and for every $\varepsilon' > 0$ there is $\delta > 0$ with

$$f_1(z_k + h) \leq f_1(z_k) + \langle p_k, h\rangle + \tfrac{1}{2} X_k(h, h) + \varepsilon' |h|^2 \qquad (|h| < \delta). \tag{38}$$

Replacing δ by $\delta^* = \min(\delta, b/4) > 0$ only shrinks the range of h, so (38) continues to hold for $|h| < \delta^*$. For such h we have $|h| < b/4$ and hence, by (37),

$$|z_k + h| \leq |z_k| + |h| < b/4 + b/4 = b/2 < b,$$

and also $|z_k| < b/4 < b$. Therefore the hypothesis applies at both points and gives $f_1(z_k) = f_2(z_k)$ and $f_1(z_k + h) = f_2(z_k + h)$. Substituting these two equalities into (38) yields

$$f_2(z_k + h) \leq f_2(z_k) + \langle p_k, h\rangle + \tfrac{1}{2} X_k(h, h) + \varepsilon' |h|^2 \qquad (|h| < \delta^*).$$

As $\varepsilon' > 0$ was arbitrary and X_k is symmetric, the defining condition of `Superjet` holds, i.e.

$$(p_k, X_k) \in J^{2,+} f_2(z_k) \qquad (k \geq k_0). \tag{39}$$

The function values converge to the right limit. For $k \geq k_0$ we have $|z_k| < b/4 < b$, so $f_2(z_k) = f_1(z_k)$; also $|0| = 0 < b$, so $f_2(0) = f_1(0)$. Since $f_1(z_k) \to f_1(0)$ by (36), the shifted sequence satisfies $f_2(z_{k_0+m}) = f_1(z_{k_0+m}) \to f_1(0) = f_2(0)$ as $m \to \infty$.

Conclusion. Define the re-indexed sequences $z'_m = z_{k_0+m}$, $p'_m = p_{k_0+m}$ and $X'_m = X_{k_0+m}$ for $m \geq 0$. Dropping finitely many initial terms changes none of the limits in (36), so $z'_m \to 0$, $p'_m \to p$ and $X'_m \to X$; by (39) we have $(p'_m, X'_m) \in J^{2,+} f_2(z'_m)$ for every m; and by the previous paragraph $f_2(z'_m) \to f_2(0)$. Counted individually these are the five requirements in the definition of `ClosedSuperjet` — the jet membership at every index, and the four convergences $z'_m \to 0$, $f_2(z'_m) \to f_2(0)$, $p'_m \to p$ and $X'_m \to X$ — for the pair (p, X) relative to f_2 at 0, so $(p, X) \in \overline{J}^{2,+} f_2(0)$. □

2.15. `ClosedSuperjetQuartic`.

Lemma 2.15 (Subtracting a quartic does not change the closed superjet at the origin)**.** *Let E be a finite-dimensional real inner-product space and let $f : E \to \mathbb{R}$. Write $s(w) = |w|^4$. Then*

$$\overline{J}^{2,+}(f - s)(0) = \overline{J}^{2,+} f(0),$$

where $\overline{J}^{2,+}$ is the closed superjet of `ClosedSuperjet`.

Proof. Step 1: the derivatives of s and their vanishing at 0. The function $s(w) = |w|^4 = \langle w, w\rangle^2$ is a polynomial in w and therefore infinitely differentiable on E, with

$$\nabla s(w) = 4|w|^2 w, \qquad D^2 s(w)(a,b) = 4|w|^2\langle a,b\rangle + 8\langle w,a\rangle\langle w,b\rangle.$$

In particular $D^2 s(w)$ is a symmetric continuous bilinear form for every w,

$$|\nabla s(w)| = 4|w|^3 \xrightarrow[w\to 0]{} 0, \qquad \|D^2 s(w)\| \le 4|w|^2 + 8|w|^2 = 12|w|^2 \xrightarrow[w\to 0]{} 0,$$

the bound on $\|D^2 s(w)\|$ coming from the Cauchy–Schwarz inequality applied to each of the two terms. Finally s is continuous and $s(0) = 0$.

Step 2: the superjet shift identity. We claim that for every $w_0 \in E$, every $q \in E$ and every symmetric continuous bilinear form X on E,

$$(q, X) \in J^{2,+}(f - s)(w_0) \iff \big(q + \nabla s(w_0),\ X + D^2 s(w_0)\big) \in J^{2,+} f(w_0), \tag{40}$$

where $J^{2,+}$ is the superjet of `Superjet`. Note first that X is symmetric if and only if $X + D^2 s(w_0)$ is, because $D^2 s(w_0)$ is symmetric; so only the inequalities need comparing.

By the second-order Taylor expansion of the smooth function s at w_0 (preamble), for every $\varepsilon'' > 0$ there is $\delta'' > 0$ with

$$\big|s(w_0 + h) - s(w_0) - \langle \nabla s(w_0), h\rangle - \tfrac{1}{2} D^2 s(w_0)(h,h)\big| \le \varepsilon''|h|^2 \qquad (|h| < \delta''). \tag{41}$$

Assume the left-hand side of (40) and let $\varepsilon' > 0$. Pick $\delta_1 > 0$ from `Superjet` for $f - s$ at w_0 with parameter $\varepsilon'/2$, and pick $\delta'' > 0$ from (41) with $\varepsilon'' = \varepsilon'/2$. For $|h| < \min(\delta_1, \delta'')$ we may add

$$f(w_0 + h) - s(w_0 + h) \le f(w_0) - s(w_0) + \langle q, h\rangle + \tfrac{1}{2} X(h,h) + \tfrac{\varepsilon'}{2}|h|^2$$

to the inequality $s(w_0 + h) \le s(w_0) + \langle \nabla s(w_0), h\rangle + \frac{1}{2} D^2 s(w_0)(h,h) + \frac{\varepsilon'}{2}|h|^2$ obtained from (41); the terms $\mp s(w_0 + h)$ and $\mp s(w_0)$ cancel and we obtain

$$f(w_0 + h) \le f(w_0) + \langle q + \nabla s(w_0), h\rangle + \tfrac{1}{2}\big(X + D^2 s(w_0)\big)(h,h) + \varepsilon'|h|^2.$$

As $\varepsilon' > 0$ was arbitrary this is the condition of `Superjet` for f at w_0 with the shifted jet. The converse implication is the same computation with the roles of f and $f - s$ exchanged: subtract, instead of add, the Taylor inequality (41) in the form $s(w_0) + \langle \nabla s(w_0), h\rangle + \frac{1}{2} D^2 s(w_0)(h,h) - \frac{\varepsilon'}{2}|h|^2 \le s(w_0 + h)$. This proves (40).

Step 3: passing to closed superjets. Let $(p, X) \in \overline{J}^{2,+}(f - s)(0)$ with defining sequences (z_k, p_k, X_k) as in `ClosedSuperjet`: $(p_k, X_k) \in J^{2,+}(f - s)(z_k)$, $z_k \to 0$, $(f - s)(z_k) \to (f - s)(0) = f(0)$, $p_k \to p$ and $X_k \to X$. Put $p'_k = p_k + \nabla s(z_k)$ and $X'_k = X_k + D^2 s(z_k)$. By (40) with $w_0 = z_k$ we have $(p'_k, X'_k) \in J^{2,+} f(z_k)$. Since $z_k \to 0$, Step 1 gives $\nabla s(z_k) \to 0$ and $D^2 s(z_k) \to 0$, so $p'_k \to p$ and $X'_k \to X$. Also $s(z_k) \to s(0) = 0$ by the continuity of s, so $f(z_k) = (f - s)(z_k) + s(z_k) \to f(0) + 0 = f(0)$. Thus (z_k, p'_k, X'_k) are defining sequences for (p, X) relative to f at 0, i.e. $(p, X) \in \overline{J}^{2,+} f(0)$.

Conversely let $(p, X) \in \overline{J}^{2,+} f(0)$ with defining sequences (z_k, π_k, Ξ_k). Put $p_k = \pi_k - \nabla s(z_k)$ and $X_k = \Xi_k - D^2 s(z_k)$; then $p_k + \nabla s(z_k) = \pi_k$ and $X_k + D^2 s(z_k) = \Xi_k$, so (40) gives $(p_k, X_k) \in J^{2,+}(f - s)(z_k)$. As before $\nabla s(z_k) \to 0$ and $D^2 s(z_k) \to 0$, so $p_k \to p$ and $X_k \to X$; and $(f - s)(z_k) = f(z_k) - s(z_k) \to f(0) - 0 = (f - s)(0)$. Hence $(p, X) \in \overline{J}^{2,+}(f - s)(0)$. The two inclusions give the asserted equality. □

2.16. `ClosedSuperjetAffineShift`.

Lemma 2.16 (Translating the base point and subtracting an affine function)**.** *Let E be a finite-dimensional real inner-product space, let $g : E \to \mathbb{R}$ and let $z, p \in E$. Define the normalized function*

$$\widehat{g}(w) = g(z + w) - g(z) - \langle p, w\rangle \qquad (w \in E),$$

so that $\widehat{g}(0) = 0$. Then for every $q \in E$ and every continuous bilinear form X on E,

$$(q, X) \in \overline{J}^{2,+}\widehat{g}(0) \iff (q + p, X) \in \overline{J}^{2,+} g(z),$$

where $\overline{J}^{2,+}$ is the closed superjet of `ClosedSuperjet`.

Proof. Step 1: the corresponding statement for superjets, at an arbitrary point. We first show that for every $w_0 \in E$, every $q' \in E$ and every continuous bilinear form X on E,

$$(q', X) \in J^{2,+}\widehat{g}(w_0) \iff (q' + p, X) \in J^{2,+}g(z + w_0), \tag{42}$$

where $J^{2,+}$ is the superjet of `Superjet`. Both memberships require the same condition on X, namely symmetry, so only the inequalities have to be compared. Fix $\varepsilon' > 0$ and $\delta > 0$ and let $h \in E$ with $|h| < \delta$. Substituting the definition of $\widehat{g}$ at the two points $w_0 + h$ and w_0, the inequality of `Superjet` for $\widehat{g}$ at w_0,

$$\widehat{g}(w_0 + h) \le \widehat{g}(w_0) + \langle q', h\rangle + \tfrac{1}{2}X(h,h) + \varepsilon'|h|^2,$$

reads

$$g(z + w_0 + h) - g(z) - \langle p, w_0 + h\rangle \le g(z + w_0) - g(z) - \langle p, w_0\rangle + \langle q', h\rangle + \tfrac{1}{2}X(h,h) + \varepsilon'|h|^2.$$

The terms $-g(z)$ cancel, and $-\langle p, w_0 + h\rangle + \langle p, w_0\rangle = -\langle p, h\rangle$ by the additivity of the inner product in its second argument, so after moving $-\langle p, h\rangle$ to the right-hand side the inequality becomes

$$g\big((z + w_0) + h\big) \le g(z + w_0) + \langle q' + p, h\rangle + \tfrac{1}{2}X(h,h) + \varepsilon'|h|^2.$$

Every step was an equivalence valid for each individual h, so for each pair (ε', δ) the first inequality holds for all $|h| < \delta$ if and only if the last one does. Since the condition of `Superjet` asserts, for every $\varepsilon' > 0$, the existence of some $\delta > 0$ for which the inequality holds on $|h| < \delta$, and the two inequalities have the same (ε', δ) ranges of validity, the two memberships in (42) are equivalent.

Step 2: passing to closed superjets, from left to right. Let $(q, X) \in \overline{J}^{2,+}\widehat{g}(0)$ and pick defining sequences (w_k, q_k, X_k) as in `ClosedSuperjet`: $(q_k, X_k) \in J^{2,+}\widehat{g}(w_k)$, $w_k \to 0$, $\widehat{g}(w_k) \to \widehat{g}(0) = 0$, $q_k \to q$ and $X_k \to X$. Put $z_k = z + w_k$ and $q'_k = q_k + p$. Then:

(i) $(q'_k, X_k) \in J^{2,+}g(z_k)$ for every k, by (42) with $w_0 = w_k$ and $q' = q_k$.
(ii) $z_k = z + w_k \to z + 0 = z$, by the continuity of translation.
(iii) $g(z_k) = \widehat{g}(w_k) + g(z) + \langle p, w_k\rangle$ by the definition of $\widehat{g}$; here $\widehat{g}(w_k) \to 0$ and $\langle p, w_k\rangle \to 0$ by the continuity of the inner product in its second argument, so $g(z_k) \to g(z)$.
(iv) $q'_k = q_k + p \to q + p$ and $X_k \to X$.

By `ClosedSuperjet` these four facts give $(q + p, X) \in \overline{J}^{2,+}g(z)$.

Step 3: passing to closed superjets, from right to left. Conversely let $(q + p, X) \in \overline{J}^{2,+}g(z)$ with defining sequences (z_k, π_k, X_k): $(\pi_k, X_k) \in J^{2,+}g(z_k)$, $z_k \to z$, $g(z_k) \to g(z)$, $\pi_k \to q + p$ and $X_k \to X$. Put $w_k = z_k - z$ and $q_k = \pi_k - p$. Then $z_k = z + w_k$, so by (42) with $w_0 = w_k$ and $q' = q_k$ (note $q_k + p = \pi_k$) we get $(q_k, X_k) \in J^{2,+}\widehat{g}(w_k)$. Moreover $w_k = z_k - z \to 0$; and $\widehat{g}(w_k) = g(z_k) - g(z) - \langle p, w_k\rangle \to g(z) - g(z) - 0 = 0 = \widehat{g}(0)$, again by the continuity of the inner product; and $q_k = \pi_k - p \to q$ and $X_k \to X$. By `ClosedSuperjet`, $(q, X) \in \overline{J}^{2,+}\widehat{g}(0)$. □

2.17. `ContDiffAtSecondOrderPeano`.

Lemma 2.17 (Second-order Taylor expansion with Peano remainder, from a local hypothesis)**.** *Let E be a finite-dimensional real inner-product space, let $\Psi : E \to \mathbb{R}$ and let $Z \in E$. Assume only that Ψ is twice continuously differentiable NEAR Z, i.e. that Ψ is C^2 at Z: some neighbourhood of Z carries a C^2 structure for Ψ, and nothing whatsoever is assumed about Ψ outside such a neighbourhood. Write $\nabla\Psi(Z) \in E$ for the gradient of Ψ at Z and $D^2\Psi(Z) = D(D\Psi)(Z)$ for the second Fréchet derivative at Z, a continuous bilinear form on E. Then for every $\delta > 0$ there is $\rho > 0$ such that*

$$\Big|\Psi(Z + W) - \Psi(Z) - \langle \nabla\Psi(Z), W\rangle - \tfrac{1}{2}D^2\Psi(Z)(W,W)\Big| \le \delta|W|^2 \qquad (|W| < \rho).$$

The estimate is two-sided, i.e. it bounds the absolute value of the remainder and not merely one of its signs.

The hypothesis is local at the single point Z — no gradient map on E and no global continuity is assumed — and the quadratic term is the second Fréchet derivative read as a bilinear form rather than an inner product against the derivative of a gradient map.

Proof. How this differs from `GradientDiffSecondOrder`*, since the two conclusions look alike.* That lemma assumes a gradient map w defined at EVERY point of E, with $w(z)$ a gradient of F at z for every z and with w continuous on all of E, and its quadratic term is $\langle Qh, h\rangle$ for Q the Fréchet derivative of that gradient MAP. The present statement assumes only C^2-ness at the single point Z, which supplies no gradient at all away from one neighbourhood of Z, so `GradientDiffSecondOrder` cannot be applied to a function that is merely C^2 near a point: its hypotheses are not available. Conversely the present statement says nothing about a function whose gradient exists globally but which is nowhere C^2, so it does not imply `GradientDiffSecondOrder` either. The two are genuinely different statements about different hypothesis classes, and the proof below uses the same tool — the fundamental theorem of calculus along a segment — restricted to a ball, which is the whole content of the localization.

Throughout, $D\Psi(y) \in E^*$ denotes the Fréchet derivative of Ψ at y (a continuous linear functional), so that $\langle \nabla\Psi(Z), V\rangle = D\Psi(Z)[V]$ for every $V \in E$ by the definition of the gradient as the Riesz representative of $D\Psi(Z)$.

Step 0: the elementary integral. Since $\tau \mapsto \tau^2/2$ has derivative τ at every τ and $t \mapsto t$ is continuous, hence integrable on $[0,1]$, the fundamental theorem of calculus (preamble) gives

$$\int_0^1 t\,dt = \frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2}. \tag{43}$$

Step 1: a ball on which Ψ is C^2. Being C^2 at a point is a property that propagates to nearby points: if Ψ is C^2 at Z then Ψ is C^2 at every point of some neighbourhood of Z (preamble; this is the standard fact that the C^n property at a point, for finite n, holds on a whole neighbourhood). Fix therefore $r > 0$ with

$$\Psi \text{ is } C^2 \text{ at every } y \text{ with } |y - Z| < r. \tag{44}$$

Two consequences of (44) will be used.

(1a) For every y with $|y-Z| < r$ the function Ψ is differentiable at y, with derivative $D\Psi(y)$; indeed C^2 at y implies C^1 at y, hence differentiable at y.

(1b) For every y with $|y-Z| < r$ the map $u \mapsto D\Psi(u)$ is continuous at y; indeed Ψ being C^2 at y means precisely that $u \mapsto D\Psi(u)$ is C^1 at y, hence continuous at y.

Moreover, Ψ being C^2 at Z makes $u \mapsto D\Psi(u)$ be C^1 at Z, hence differentiable at Z; its derivative there is by definition $D^2\Psi(Z)$, so

$$u \mapsto D\Psi(u) \text{ is differentiable at } Z \text{ with derivative } D^2\Psi(Z). \tag{45}$$

Step 2: the little-o of the first derivative. Let $\delta > 0$. By (45) and the definition of the Fréchet derivative, the difference quotient of $u \mapsto D\Psi(u)$ at Z is $o(|V|)$; in particular, applying the definition with the parameter δ, there is $\rho_1 > 0$ with

$$\big\|D\Psi(Z+V) - D\Psi(Z) - D^2\Psi(Z)(V,\cdot)\big\| \le \delta|V| \qquad (|V| < \rho_1), \tag{46}$$

the norm being the dual norm on E^*.

Put $\rho = \min(r, \rho_1) > 0$ and fix $W \in E$ with $|W| < \rho$.

Step 3: the segment stays in the ball, so the fundamental theorem applies. For $t \in [0,1]$ we have $|(Z + tW) - Z| = |tW| = t|W| \le |W| < r$, so by (44) every point of the segment from Z to $Z + W$ lies in the ball where (1a) and (1b) hold. Consequently:

- the function $\tau \mapsto \Psi(Z + \tau W)$ is differentiable at every $t \in [0,1]$, with derivative $D\Psi(Z + tW)[W]$, by (1a) and the chain rule applied to the affine curve $\tau \mapsto Z + \tau W$ (whose derivative is the constant W);
- the function $t \mapsto D\Psi(Z + tW)[W]$ is continuous on $[0,1]$, being the composition of the continuous curve $t \mapsto Z + tW$ with the map $u \mapsto D\Psi(u)$, continuous on the ball by (1b), followed by the evaluation $L \mapsto L[W]$, which is a continuous linear functional on E^*; hence it is integrable on $[0,1]$.

The fundamental theorem of calculus (preamble) therefore gives

$$\Psi(Z+W) - \Psi(Z) = \int_0^1 D\Psi(Z + tW)[W]\,dt. \tag{47}$$

Step 4: splitting the integrand. For $t \in [0,1]$ set

$$R_t = D\Psi(Z+tW) - D\Psi(Z) - D^2\Psi(Z)(tW,\cdot) \in E^*.$$

Since $D^2\Psi(Z)$ is bilinear, $D^2\Psi(Z)(tW,W) = t\,D^2\Psi(Z)(W,W)$, so by the definition of R_t,

$$D\Psi(Z+tW)[W] = D\Psi(Z)[W] + t\,D^2\Psi(Z)(W,W) + R_t[W] \qquad (t\in[0,1]). \tag{48}$$

The three summands on the right are each continuous in t on $[0,1]$: the first is constant, the second is a constant multiple of t, and the third is the difference of the left-hand side (continuous by Step 3) and the first two; so each is integrable on $[0,1]$ and the integral of the sum is the sum of the integrals. Integrating (48) over $[0,1]$ and using (47), $\int_0^1 dt = 1$ and (43),

$$\Psi(Z+W) - \Psi(Z) = D\Psi(Z)[W] + \tfrac12 D^2\Psi(Z)(W,W) + \int_0^1 R_t[W]\,dt,$$

that is, using $D\Psi(Z)[W] = \langle \nabla\Psi(Z), W\rangle$,

$$\Psi(Z+W) - \Psi(Z) - \langle \nabla\Psi(Z), W\rangle - \tfrac12 D^2\Psi(Z)(W,W) = \int_0^1 R_t[W]\,dt. \tag{49}$$

Step 5: bounding the remainder. Let $t\in[0,1]$. Then $|tW| = t|W| \le |W| < \rho \le \rho_1$, so (46) applies with $V = tW$ and gives $\|R_t\| \le \delta|tW| = \delta t|W|$. Since $R_t \in E^*$,

$$\big|R_t[W]\big| \le \|R_t\|\,|W| \le \delta t|W|^2 \le \delta|W|^2,$$

the last step because $t\le 1$ and $\delta|W|^2 \ge 0$. Therefore, by (49), the triangle inequality for integrals and the monotonicity of the integral over $[0,1]$,

$$\Big|\Psi(Z+W)-\Psi(Z)-\langle\nabla\Psi(Z),W\rangle-\tfrac12 D^2\Psi(Z)(W,W)\Big| = \Big|\int_0^1 R_t[W]\,dt\Big| \le \int_0^1 \big|R_t[W]\big|\,dt \le \int_0^1 \delta|W|^2\,dt = \delta|W|^2,$$

where the integrand $t\mapsto |R_t[W]|$ is integrable because $t\mapsto R_t[W]$ is (Step 4) and the absolute value of an integrable function is integrable. As $\delta>0$ was arbitrary and $\rho = \min(r,\rho_1)$ depends only on δ (and on Ψ, Z), this is the assertion. □

2.18. `SumRuleQuadraticTest.`

Lemma 2.18 (Reduction of the theorem of sums to a quadratic test with a strict global maximum)**.** *Let E be a finite-dimensional real inner-product space, let $N\ge 1$ and write $V = \ell^2\big(\bigoplus_{j=1}^N E\big)$ for the ℓ^2-product of N copies of E, whose elements are written $Y = (y_j)_{j=1}^N$ and whose norm satisfies $|Y|^2 = \sum_j |y_j|^2$. For each j let $O_j \subseteq E$ be open and let $g_j : E\to\mathbb{R}$ be continuous on O_j. Let $\Psi : V\to\mathbb{R}$, let $\widehat{Z}\in V$ with $\widehat{z}_j \in O_j$ for every j, assume that Ψ is C^2 near $\widehat{Z}$, and assume that $Y\mapsto \sum_j g_j(y_j) - \Psi(Y)$ has a local maximum at $\widehat{Z}$. Write $A = D^2\Psi(\widehat{Z})$ and $\widehat{p}_j = D_j\Psi(\widehat{Z})$ for the j-th component of $\nabla\Psi(\widehat{Z})$, and let $\tau>0$. Then there are $a>0$, $C\ge 0$ and functions $G_j : E\to\mathbb{R}$ $(1\le j\le N)$ such that*

(G1) *each G_j is continuous on E and $G_j(w)\le C$ for every $w\in E$;*
(G2) *$G_j(w) = g_j(\widehat{z}_j+w) - g_j(\widehat{z}_j) - \langle \widehat{p}_j, w\rangle - |w|^4$ whenever $|w|\le a$;*
(G3) *for every $W\in V$ with $W\neq 0$,*

$$\sum_j G_j(w_j) - \tfrac12\big(A[W] + \tau|W|^2\big) \le -\sum_j |w_j|^4,$$

where $A[W] = A(W,W)$.

Proof. Write $\widehat{P} = \nabla\Psi(\widehat{Z})$, so that $\widehat{P}$ has components $\widehat{p}_j$ and

$$\langle \widehat{P}, W\rangle = \sum_j \langle \widehat{p}_j, w_j\rangle \qquad (W\in V), \tag{50}$$

by the definition of the ℓ^2-product inner product (preamble). Put

$$\widehat{g}_j(w) = g_j(\widehat{z}_j+w) - g_j(\widehat{z}_j) - \langle \widehat{p}_j, w\rangle,$$

so $\widehat{g}_j(0) = 0$. Let $\bar{A} : V \to V$ be the self-adjoint operator with $\langle \bar{A}\xi, \zeta\rangle = A(\xi, \zeta)$ (it exists and is unique because A is a symmetric continuous bilinear form on the finite-dimensional inner-product space V, and self-adjointness is the symmetry of A), and set $B_\tau = \bar{A} + \tau\,\mathrm{id}$, again self-adjoint, so that

$$\langle B_\tau W, W\rangle = A[W] + \tau|W|^2 \qquad (W \in V). \tag{51}$$

Step 1: a quadratic majorant near 0. By the second-order Taylor expansion of Ψ at $\widehat{Z}$ (preamble), $\Psi(\widehat{Z} + W) = \Psi(\widehat{Z}) + \langle \widehat{P}, W\rangle + \frac{1}{2}A(W, W) + R(W)$ with $|R(W)|/|W|^2 \to 0$ as $W \to 0$. The local maximum at $\widehat{Z}$ gives $\rho' > 0$ with

$$\sum_j g_j(\widehat{z}_j + w_j) - \Psi(\widehat{Z} + W) \le \sum_j g_j(\widehat{z}_j) - \Psi(\widehat{Z}) \qquad (|W| \le \rho').$$

Using (50), for $|W| \le \rho'$,

$$\sum_j \widehat{g}_j(w_j) = \sum_j g_j(\widehat{z}_j + w_j) - \sum_j g_j(\widehat{z}_j) - \langle \widehat{P}, W\rangle \le \Psi(\widehat{Z}+W) - \Psi(\widehat{Z}) - \langle \widehat{P}, W\rangle = \tfrac{1}{2}A(W, W) + R(W).$$

Since $|R(W)|/|W|^2 \to 0$, choose $\rho_\tau \in (0, \rho']$ with $|R(W)| \le (\tau/2)|W|^2$ for $|W| \le \rho_\tau$. Because $\frac{1}{2}A(W, W) + (\tau/2)|W|^2 = \frac{1}{2}\langle B_\tau W, W\rangle$ by (51), we obtain

$$\sum_j \widehat{g}_j(w_j) \le \tfrac{1}{2}\langle B_\tau W, W\rangle \qquad (|W| \le \rho_\tau), \tag{52}$$

with equality at $W = 0$, both sides vanishing because $\widehat{g}_j(0) = 0$.

Step 2: localization. Each O_j is open and $\widehat{z}_j \in O_j$, so for each j there is $r_j > 0$ with $\overline{B}(\widehat{z}_j, r_j) \subseteq O_j$; as there are finitely many j and $N \ge 1$, the number $\rho'' = \min_j r_j > 0$ satisfies $\overline{B}(\widehat{z}_j, \rho'') \subseteq O_j$ for every j. Put

$$a = \frac{\min(\rho_\tau, \rho'')}{1 + \sqrt{N}} > 0.$$

Then $a \le \rho''$, so $\widehat{g}_j$ is defined and continuous on $\overline{B}(0, a)$ (being built from g_j, which is continuous on $O_j \supseteq \overline{B}(\widehat{z}_j, \rho'')$, and from an affine function); and if $|w_j| \le a$ for every j then $|W| \le a\sqrt{N} \le \rho_\tau$, so (52) applies to W. Let $M_j = \max_{\overline{B}(0,a)} |\widehat{g}_j|$, which is finite by the continuity of $\widehat{g}_j$ on the compact set $\overline{B}(0, a)$ (preamble) and satisfies $M_j \ge |\widehat{g}_j(0)| = 0$; put $C = \max_j M_j \ge 0$.

Let $\pi : E \to \overline{B}(0, a)$ be the radial retraction, $\pi(w) = w$ for $|w| \le a$ and $\pi(w) = aw/|w|$ for $|w| > a$; it is continuous, and $|\pi(w)| \le a$ for every w. Write $(s)_+ = \max(s, 0)$, put $K = \|B_\tau\|N(a + 1) + 1$ and define

$$G_j(w) = \widehat{g}_j(\pi(w)) - K(|w| - a)_+ - K(|w| - a)_+^2 - |w|^4.$$

(G1). Each G_j is continuous on E, being a composition and combination of the continuous maps π, $\widehat{g}_j|_{\overline{B}(0,a)}$, $w \mapsto (|w| - a)_+$ and $w \mapsto |w|^4$. The three subtracted terms are nonnegative, so $G_j(w) \le \widehat{g}_j(\pi(w)) \le M_j \le C$ for every w, using $|\pi(w)| \le a$.

(G2). If $|w| \le a$ then $\pi(w) = w$ and $(|w| - a)_+ = 0$, so $G_j(w) = \widehat{g}_j(w) - |w|^4$, which is the asserted identity by the definition of $\widehat{g}_j$.

(G3). Put $\Phi(W) = \sum_j G_j(w_j) - \frac{1}{2}\langle B_\tau W, W\rangle$; by (51) the assertion (G3) is exactly $\Phi(W) \le -\sum_j |w_j|^4$ for $W \ne 0$. Let $W \ne 0$, put $m = \max_j(|w_j| - a)_+ \ge 0$ and $\Pi W = (\pi(w_j))_j$.

If $m = 0$ then $|w_j| \le a$ for every j, so $\pi(w_j) = w_j$ and all penalty terms vanish; since $|W| \le a\sqrt{N} \le \rho_\tau$, (52) gives $\sum_j \widehat{g}_j(w_j) \le \frac{1}{2}\langle B_\tau W, W\rangle$ and hence $\Phi(W) \le -\sum_j |w_j|^4$.

If $m > 0$, then $|\pi(w_j)| \le a$ for every j gives $|\Pi W| \le a\sqrt{N} \le \rho_\tau$, so (52) applied to ΠW gives $\sum_j \widehat{g}_j(\pi(w_j)) \le \frac{1}{2}\langle B_\tau \Pi W, \Pi W\rangle$. Moreover $\sum_j (|w_j| - a)_+ \ge m$ and $\sum_j (|w_j| - a)_+^2 \ge m^2$, since every summand is nonnegative and one of them equals m, resp. m^2. Hence

$$\Phi(W) \le \tfrac{1}{2}\big[\langle B_\tau \Pi W, \Pi W\rangle - \langle B_\tau W, W\rangle\big] - Km - Km^2 - \sum_j |w_j|^4.$$

Since B_τ is self-adjoint,

$$\langle B_\tau W, W\rangle - \langle B_\tau \Pi W, \Pi W\rangle = \langle B_\tau(W - \Pi W), W + \Pi W\rangle.$$

Here $|W-\Pi W|^2=\sum_j |w_j-\pi(w_j)|^2=\sum_j\big((|w_j|-a)_+\big)^2\le Nm^2$, because $|w_j-\pi(w_j)|=(|w_j|-a)_+$ by the definition of π (it is 0 for $|w_j|\le a$ and $|w_j|-a$ otherwise); so $|W-\Pi W|\le\sqrt{N}m$, and $|W+\Pi W|\le|W|+|\Pi W|$. From $|w_j|\le a+m$ for every j we get $|W|\le(a+m)\sqrt{N}$, and $|\Pi W|\le a\sqrt{N}$, so $|W+\Pi W|\le 2a\sqrt{N}+\sqrt{N}m$. By the Cauchy–Schwarz inequality and the definition of the operator norm,

$$\tfrac12\big|\langle B_\tau\Pi W,\Pi W\rangle-\langle B_\tau W,W\rangle\big|\le\tfrac12\|B_\tau\|\sqrt{N}m\big(2a\sqrt{N}+\sqrt{N}m\big)=\|B_\tau\|Nam+\tfrac12\|B_\tau\|Nm^2,$$

and therefore

$$\Phi(W)\le m\big(\|B_\tau\|Na-K\big)+m^2\big(\tfrac12\|B_\tau\|N-K\big)-\sum_j|w_j|^4\le-\sum_j|w_j|^4,$$

because $K=\|B_\tau\|N(a+1)+1\ge\|B_\tau\|Na$ and $K\ge\|B_\tau\|N\ge\frac12\|B_\tau\|N$, while $m>0$ and $m^2>0$. This proves (G3) in both cases. □

2.19. `SumRuleResolvent.`

Lemma 2.19 (The resolvent of a self-adjoint operator, and its comparison with the operator)**.** *Let E be a finite-dimensional real inner-product space, let $B:E\to E$ be a self-adjoint continuous linear operator, i.e. $\langle B\xi,\zeta\rangle=\langle\xi,B\zeta\rangle$ for all $\xi,\zeta\in E$, and let $\varepsilon>0$ satisfy $\varepsilon\|B\|\le\frac12$. Then there is a continuous linear operator $A_\varepsilon:E\to E$ such that*

(R1) *A_ε is self-adjoint: $\langle A_\varepsilon\xi,\zeta\rangle=\langle\xi,A_\varepsilon\zeta\rangle$ for all $\xi,\zeta\in E$;*
(R2) *$A_\varepsilon\xi=B\big(\xi+\varepsilon A_\varepsilon\xi\big)$ for every $\xi\in E$;*
(R3) *$\big(\xi+\varepsilon A_\varepsilon\xi\big)-\varepsilon B\big(\xi+\varepsilon A_\varepsilon\xi\big)=\xi$ for every $\xi\in E$;*
(R4) *$\varepsilon^{-1}\langle\xi,\xi\rangle+\langle A_\varepsilon\xi,\xi\rangle>0$ for every $\xi\neq0$;*
(R5) *$\|A_\varepsilon\|\le2\|B\|$;*
(R6) *for every $\eta\ge2\varepsilon$ and every $\zeta\in E$,*

$$\langle A_\varepsilon\zeta,\zeta\rangle\le\langle B\zeta,\zeta\rangle+\eta\,|B\zeta|^2.$$

Proof. Step 1: $\mathrm{id}-\varepsilon B$ is invertible. From $\varepsilon>0$ and $\varepsilon\|B\|\le\frac12$ we get $\|\varepsilon B\|=\varepsilon\|B\|\le\frac12<1$, so the Neumann series $T=\sum_{k\ge0}(\varepsilon B)^k$ converges in the operator norm and satisfies $T(\mathrm{id}-\varepsilon B)=(\mathrm{id}-\varepsilon B)T=\mathrm{id}$; thus $\mathrm{id}-\varepsilon B$ is invertible with $(\mathrm{id}-\varepsilon B)^{-1}=T$. Each partial sum of the series is a polynomial in B, hence self-adjoint (a sum of powers of a self-adjoint operator is self-adjoint) and commuting with B; since $\xi\mapsto\langle S\xi,\zeta\rangle-\langle\xi,S\zeta\rangle$ and $S\mapsto SB-BS$ are continuous in S, the limit T is again self-adjoint and commutes with B. Also $\|T\|\le\sum_{k\ge0}\|\varepsilon B\|^k\le\sum_{k\ge0}2^{-k}=2$, and more precisely $\|T\|\le(1-\varepsilon\|B\|)^{-1}\le2$.

Define $A_\varepsilon=BT=B(\mathrm{id}-\varepsilon B)^{-1}$, a continuous linear operator on E.

(R1). B and T are self-adjoint and commute, so for all ξ,ζ,

$$\langle BT\xi,\zeta\rangle=\langle T\xi,B\zeta\rangle=\langle\xi,TB\zeta\rangle=\langle\xi,BT\zeta\rangle.$$

The resolvent identity. Using $(\mathrm{id}-\varepsilon B)T=\mathrm{id}$ and the distributivity of composition over addition,

$$\mathrm{id}+\varepsilon A_\varepsilon=\mathrm{id}+\varepsilon BT=(\mathrm{id}-\varepsilon B)T+\varepsilon BT=\big[(\mathrm{id}-\varepsilon B)+\varepsilon B\big]T=T,\tag{53}$$

where the second equality replaced id by $(\mathrm{id}-\varepsilon B)T$.

(R2). By (53), $T\xi=\xi+\varepsilon A_\varepsilon\xi$ for every ξ, so $A_\varepsilon\xi=B(T\xi)=B\big(\xi+\varepsilon A_\varepsilon\xi\big)$.

(R3). Again by (53) and $(\mathrm{id}-\varepsilon B)T=\mathrm{id}$,

$$\big(\xi+\varepsilon A_\varepsilon\xi\big)-\varepsilon B\big(\xi+\varepsilon A_\varepsilon\xi\big)=(\mathrm{id}-\varepsilon B)(T\xi)=\xi.$$

(R4). For $\xi\neq0$, the Cauchy–Schwarz inequality and the definition of the operator norm give $\langle B\xi,\xi\rangle\le|B\xi|\,|\xi|\le\|B\||\xi|^2$, hence

$$\langle(\mathrm{id}-\varepsilon B)\xi,\xi\rangle=|\xi|^2-\varepsilon\langle B\xi,\xi\rangle\ge(1-\varepsilon\|B\|)|\xi|^2\ge\tfrac12|\xi|^2>0.$$

So $\mathrm{id}-\varepsilon B$ is self-adjoint and positive definite. Its inverse T is then positive definite as well: for $x\neq0$ we have $Tx\neq0$ (because T is injective, being invertible), and

$$\langle Tx,x\rangle=\big\langle Tx,(\mathrm{id}-\varepsilon B)Tx\big\rangle=\big\langle(\mathrm{id}-\varepsilon B)(Tx),Tx\big\rangle>0,$$

using $x = (\mathrm{id} - \varepsilon B)Tx$ and the self-adjointness of $\mathrm{id} - \varepsilon B$. Consequently, by (53),

$$\varepsilon^{-1}\langle \xi, \xi\rangle + \langle A_\varepsilon \xi, \xi\rangle = \varepsilon^{-1}\big\langle (\mathrm{id} + \varepsilon A_\varepsilon)\xi, \xi\big\rangle = \varepsilon^{-1}\langle T\xi, \xi\rangle > 0 \qquad (\xi \neq 0).$$

(R5). $\|A_\varepsilon\| = \|BT\| \le \|B\|\,\|T\| \le 2\|B\|$ by Step 1.

(R6). Since B is self-adjoint on the finite-dimensional space E, the spectral theorem supplies an orthonormal basis $(v_i)_{i=1}^n$ of E with $Bv_i = b_i v_i$ and $b_i \in \mathbb{R}$; moreover $|b_i| = |Bv_i| \le \|B\|\,|v_i| = \|B\|$. For each i, $(\mathrm{id} - \varepsilon B)v_i = (1 - \varepsilon b_i)v_i$ with $1 - \varepsilon b_i \ge 1 - \varepsilon\|B\| \ge \frac{1}{2} > 0$, so $Tv_i = (1 - \varepsilon b_i)^{-1} v_i$ and therefore $A_\varepsilon v_i = a_i v_i$ with $a_i = b_i/(1 - \varepsilon b_i)$. Now

$$a_i - b_i = \frac{b_i - b_i(1 - \varepsilon b_i)}{1 - \varepsilon b_i} = \frac{\varepsilon b_i^2}{1 - \varepsilon b_i} \le 2\varepsilon b_i^2 \le \eta b_i^2,$$

using $0 < 1 - \varepsilon b_i$, $1 - \varepsilon b_i \ge \frac{1}{2}$ (so $(1 - \varepsilon b_i)^{-1} \le 2$), $\varepsilon b_i^2 \ge 0$ and $2\varepsilon \le \eta$. Writing $\zeta = \sum_i c_i v_i$ with $c_i \in \mathbb{R}$ and using orthonormality,

$$\langle A_\varepsilon \zeta, \zeta\rangle = \sum_i a_i c_i^2, \qquad \langle B\zeta, \zeta\rangle = \sum_i b_i c_i^2, \qquad |B\zeta|^2 = \sum_i b_i^2 c_i^2.$$

Multiplying the termwise inequality $a_i - b_i \le \eta b_i^2$ by $c_i^2 \ge 0$ and summing over i gives $\langle A_\varepsilon\zeta, \zeta\rangle - \langle B\zeta, \zeta\rangle \le \eta |B\zeta|^2$, which is (R6). □

2.20. `SupConvolutionSeparable`.

Lemma 2.20 (The sup-convolution of a separated sum separates)**.** *Let E be a finite-dimensional real inner-product space, let N be a natural number and write $V = \ell^2\big(\bigoplus_{j=1}^N E\big)$ for the ℓ^2-product of N copies of E, whose elements are written $Z = (z_j)_{j=1}^N$ and whose norm satisfies $|Z|^2 = \sum_j |z_j|^2$. Let $\varepsilon > 0$, let $C \in \mathbb{R}$, and for each j let $G_j : E \to \mathbb{R}$ satisfy $G_j(w) \le C$ for every $w \in E$. Let $\mathcal{G}^\varepsilon$ and G_j^ε denote the sup-convolutions of `SupConvolution` with parameter ε, formed in V for $\mathcal{G}(W) = \sum_j G_j(w_j)$ and in E for each G_j. Then*

$$\mathcal{G}^\varepsilon(Z) = \sum_{j=1}^N G_j^\varepsilon(z_j) \qquad (Z \in V).$$

This is an identity between the two sup-convolutions and asserts nothing about finiteness, domination or semiconvexity of either side; the uniform bound C enters only to make each defining set bounded above, so that each supremum is a genuine least upper bound rather than the value the convention assigns to an unbounded set.

Proof. By the defining formula of `SupConvolution`,

$$\mathcal{G}^\varepsilon(Z) = \sup_{W \in V} f(W), \qquad G_j^\varepsilon(z_j) = \sup_{w \in E} f_j(w),$$

where

$$f(W) = \sum_j G_j(w_j) - \frac{|Z - W|^2}{2\varepsilon}, \qquad f_j(w) = G_j(w) - \frac{|z_j - w|^2}{2\varepsilon}.$$

Step 1: the objective separates. Since the ℓ^2-product norm satisfies $|Y|^2 = \sum_j |y_j|^2$ (preamble) and $(Z - W)_j = z_j - w_j$, we have $|Z - W|^2 = \sum_j |z_j - w_j|^2$, so dividing by 2ε and splitting the finite sum,

$$f(W) = \sum_j G_j(w_j) - \sum_j \frac{|z_j - w_j|^2}{2\varepsilon} = \sum_j f_j(w_j) \qquad (W \in V). \tag{54}$$

Step 2: every supremum here is genuine. Since $\varepsilon > 0$, each subtracted term $|z_j - w|^2/(2\varepsilon)$ is nonnegative, so $f_j(w) \le G_j(w) \le C$ for every w; thus $\{f_j(w) : w \in E\}$ is bounded above by C, and it is nonempty (take $w = z_j$). Likewise, by (54) and the same bound applied to each summand, $f(W) = \sum_j f_j(w_j) \le \sum_j C = NC$ for every W, so $\{f(W) : W \in V\}$ is bounded above by NC and is nonempty (take $W = 0$). Hence all $N + 1$ suprema appearing in the statement are least upper bounds of nonempty sets bounded above, and the junk-value convention of the preamble is not in play for any of them.

Step 3: $\mathcal{G}^\varepsilon(Z) \le \sum_j G_j^\varepsilon(z_j)$. Let $W \in V$. By (54) and, for each j, the fact that $f_j(w_j)$ is at most the supremum of f_j (legitimate by Step 2),

$$f(W) = \sum_j f_j(w_j) \le \sum_j G_j^\varepsilon(z_j).$$

As $W \in V$ was arbitrary and $\{f(W) : W \in V\}$ is nonempty, its least upper bound satisfies $\mathcal{G}^\varepsilon(Z) \le \sum_j G_j^\varepsilon(z_j)$.

Step 4: $\sum_j G_j^\varepsilon(z_j) \le \mathcal{G}^\varepsilon(Z)$. It suffices to show that for every $\theta > 0$,

$$\sum_j G_j^\varepsilon(z_j) \le \mathcal{G}^\varepsilon(Z) + \theta, \tag{55}$$

since a real number that is at most $b + \theta$ for every $\theta > 0$ is at most b.

Suppose first $N = 0$. Then $V = \{0\}$ is the zero space, so $Z = 0$ and $f(0) = 0 - |0-0|^2/(2\varepsilon) = 0$, whence $\mathcal{G}^\varepsilon(Z) = \sup\{0\} = 0$; and the left-hand side of (55) is an empty sum, also 0. So (55) reads $0 \le 0 + \theta$, which holds.

Now suppose $N \ge 1$ and let $\theta > 0$, so that $\theta/N > 0$. Fix j. By Step 2 the number $G_j^\varepsilon(z_j)$ is the least upper bound of the nonempty set $\{f_j(w) : w \in E\}$, so $G_j^\varepsilon(z_j) - \theta/N$, being strictly smaller, is not an upper bound: there is $w_j \in E$ with

$$G_j^\varepsilon(z_j) - \frac{\theta}{N} < f_j(w_j). \tag{56}$$

Let $W = (w_j)_{j=1}^N \in V$. Summing the N strict inequalities (56) over j — legitimate because the index set is nonempty and finite — and using $\sum_j \theta/N = \theta$,

$$\sum_j G_j^\varepsilon(z_j) - \theta = \sum_j \Big(G_j^\varepsilon(z_j) - \frac{\theta}{N} \Big) < \sum_j f_j(w_j) = f(W),$$

the last equality by (54). Since $f(W) \le \mathcal{G}^\varepsilon(Z)$ by Step 2 (the supremum is an upper bound of its own set), we get $\sum_j G_j^\varepsilon(z_j) - \theta < \mathcal{G}^\varepsilon(Z)$, which is (55).

Steps 3 and 4 give the two inequalities, hence the asserted equality. □

2.21. `SupConvolutionSemiconvex`.

Lemma 2.21 (The sup-convolution is finite, dominant and semiconvex)**.** *Let E be a finite-dimensional real inner-product space, let $\varepsilon > 0$, let $C \in \mathbb{R}$ and let $\phi : E \to \mathbb{R}$ satisfy the linear growth bound*

$$\phi(w) \le C + |w| \qquad (w \in E).$$

Let ϕ^ε be the sup-convolution of `SupConvolution`. Then:

1. *for every $z \in E$ the set $\{\phi(w) - |z-w|^2/(2\varepsilon) : w \in E\}$ is bounded above, so that $\phi^\varepsilon(z)$ is a genuine finite supremum;*
2. *$\phi(z) \le \phi^\varepsilon(z)$ for every $z \in E$;*
3. *ϕ^ε is $(1/\varepsilon)$-semiconvex on E in the sense of `SemiconvexOn`.*

Proof. Throughout, fix $\varepsilon > 0$, and for $z \in E$ write

$$S_z = \Big\{ \phi(w) - \frac{|z-w|^2}{2\varepsilon} : w \in E \Big\},$$

so that $\phi^\varepsilon(z) = \sup S_z$ by `SupConvolution`. Each S_z is nonempty, since the choice $w = z$ contributes the element $\phi(z)$.

We shall use repeatedly the elementary identity

$$s - \frac{s^2}{2\varepsilon} = \frac{\varepsilon}{2} - \frac{(s-\varepsilon)^2}{2\varepsilon} \le \frac{\varepsilon}{2} \qquad (s \in \mathbb{R}), \tag{57}$$

which is the expansion of the square and holds for every real s.

(1) Boundedness above. Fix $z \in E$ and let $w \in E$. Put $s = |z - w| \geq 0$. The triangle inequality gives $|w| \leq |z| + |z - w| = |z| + s$, so the growth bound yields

$$\phi(w) - \frac{|z-w|^2}{2\varepsilon} \leq C + |z| + s - \frac{s^2}{2\varepsilon} \leq C + |z| + \frac{\varepsilon}{2},$$

by (57). Thus $C + |z| + \varepsilon/2$ is an upper bound for S_z, which proves (1). In particular $\phi^\varepsilon(z) = \sup S_z$ is a finite real number, it is an upper bound for S_z, and it is the least such.

(2) Domination. Fix $z \in E$. As noted, $\phi(z) \in S_z$ (take $w = z$, so that $|z - w| = 0$). Since $\phi^\varepsilon(z) = \sup S_z$ is an upper bound for S_z by (1), we get $\phi(z) \leq \phi^\varepsilon(z)$.

(3) Semiconvexity. Set $\lambda = 1/\varepsilon > 0$ and

$$G(z) = \phi^\varepsilon(z) + \frac{\lambda|z|^2}{2} = \phi^\varepsilon(z) + \frac{|z|^2}{2\varepsilon}.$$

By `SemiconvexOn` we must show that G is convex on E; the set E is of course convex.

First an algebraic identity. For all $z, w \in E$,

$$-\frac{|z-w|^2}{2\varepsilon} + \frac{|z|^2}{2\varepsilon} = \frac{-|z|^2 + 2\langle z, w\rangle - |w|^2 + |z|^2}{2\varepsilon} = \left\langle \frac{w}{\varepsilon}, z \right\rangle - \frac{|w|^2}{2\varepsilon}.$$

Hence, setting

$$a_w(z) = \left\langle \frac{w}{\varepsilon}, z \right\rangle + \left(\phi(w) - \frac{|w|^2}{2\varepsilon}\right),$$

an affine function of z for each fixed w, we have

$$a_w(z) = \phi(w) - \frac{|z-w|^2}{2\varepsilon} + \frac{|z|^2}{2\varepsilon} \qquad (z, w \in E). \tag{58}$$

Consequently the set $\{a_w(z) : w \in E\}$ is obtained from S_z by adding the constant $|z|^2/(2\varepsilon)$. A nonempty set of reals that is bounded above stays bounded above after the addition of a constant, and its supremum is shifted by that constant; so by (1),

$$G(z) = \sup_{w \in E} a_w(z) \qquad (z \in E), \tag{59}$$

and this supremum is finite and is an upper bound for every $a_w(z)$.

Now let $z_1, z_2 \in E$ and let $t_1, t_2 \geq 0$ with $t_1 + t_2 = 1$; put $z = t_1 z_1 + t_2 z_2$. Fix $w \in E$. Since a_w is affine,

$$a_w(z) = t_1 a_w(z_1) + t_2 a_w(z_2).$$

By (59) applied at z_1 and at z_2 we have $a_w(z_1) \leq G(z_1)$ and $a_w(z_2) \leq G(z_2)$, and $t_1, t_2 \geq 0$, so

$$a_w(z) \leq t_1 G(z_1) + t_2 G(z_2).$$

The right-hand side does not depend on w, so it is an upper bound for the set $\{a_w(z) : w \in E\}$, whose supremum is $G(z)$ by (59). Therefore

$$G(t_1 z_1 + t_2 z_2) = G(z) \leq t_1 G(z_1) + t_2 G(z_2).$$

As z_1, z_2, t_1, t_2 were arbitrary, G is convex on E, which is (3). □

2.22. `SumRuleDualityTest`.

Lemma 2.22 (The sup-convolution duality for a separated quadratic test)**.** *Let E be a finite-dimensional real inner-product space, let N be a natural number and write $V = \ell^2\big(\bigoplus_{j=1}^N E\big)$ for the ℓ^2-product of N copies of E, so that $|W|^2 = \sum_j |w_j|^2$ and $\langle Z, W\rangle = \sum_j \langle z_j, w_j\rangle$. Let $C \geq 0$ and let $G_j : E \to \mathbb{R}$ $(1 \leq j \leq N)$ be continuous with $G_j(w) \leq C$ for every w and $G_j(0) = 0$. Let $B, A_\varepsilon : V \to V$ be self-adjoint continuous linear operators, let $\varepsilon > 0$, and assume*

$$A_\varepsilon \xi = B\big(\xi + \varepsilon A_\varepsilon \xi\big) \qquad (\xi \in V), \tag{60}$$

$$\big(\xi + \varepsilon A_\varepsilon \xi\big) - \varepsilon B\big(\xi + \varepsilon A_\varepsilon \xi\big) = \xi \qquad (\xi \in V), \tag{61}$$

$$\varepsilon^{-1}\langle \xi, \xi\rangle + \langle A_\varepsilon \xi, \xi\rangle > 0 \qquad (\xi \in V,\ \xi \neq 0). \tag{62}$$

Assume finally the strict quadratic test inequality

$$\sum_j G_j(w_j) - \tfrac{1}{2}\langle BW, W\rangle \le -\sum_j |w_j|^4 \qquad (W \in V,\ W \neq 0). \tag{63}$$

Write $\mathcal{G}(W) = \sum_j G_j(w_j)$ *and let* $\mathcal{G}^\varepsilon$ *and* G_j^ε *denote the sup-convolutions of* `SupConvolution` *with parameter* ε*, formed in* V *and in* E *respectively. Then*

$$\mathcal{G}^\varepsilon(Z) = \sum_j G_j^\varepsilon(z_j) \qquad (Z \in V), \tag{64}$$

and the function $\Upsilon(Z) = \mathcal{G}^\varepsilon(Z) - \frac{1}{2}\langle A_\varepsilon Z, Z\rangle$ *satisfies*

$$\Upsilon(0) = 0, \qquad \Upsilon(Z) < 0 \quad (Z \neq 0),$$

is λ*-semiconvex on* V *in the sense of* `SemiconvexOn` *with* $\lambda = \varepsilon^{-1} + \|A_\varepsilon\|$*, and is continuous on* V*.*

Proof. Throughout, $\mathcal{G}$ is continuous on V (a finite sum of continuous functions of the independent coordinates) and $\mathcal{G} \le NC$, hence $\mathcal{G}(W) \le NC + |W|$; likewise $G_j(w) \le C + |w|$. By part (1) of `SupConvolutionSemiconvex`, applied in V to $\mathcal{G}$ with growth constant NC and in E to each G_j with growth constant C, every defining set $\{\mathcal{G}(W) - |Z-W|^2/(2\varepsilon) : W \in V\}$, resp. $\{G_j(w) - |z-w|^2/(2\varepsilon) : w \in E\}$, is bounded above; so $\mathcal{G}^\varepsilon$ and the G_j^ε are finite (and are genuine suprema, not the junk value of an unbounded set).

Step 1: separability (64). First dispose of the degenerate case $N = 0$. Then V is the zero space, the index set is empty, $\mathcal{G}(W) = 0$ for the single element $W = 0$ of V, and by the defining formula of `SupConvolution` $\mathcal{G}^\varepsilon(Z) = \sup_{W\in V}\big(\mathcal{G}(W) - |Z-W|^2/(2\varepsilon)\big) = 0 - 0 = 0$ for the single element $Z = 0$ of V, while $\sum_j G_j^\varepsilon(z_j)$ is an empty sum, also 0; so (64) holds. (The remaining clauses of the lemma are then immediate: A_ε is the zero operator on the zero space, so $\Upsilon \equiv 0$ on $V = \{0\}$, which gives $\Upsilon(0) = 0$, makes the clause "$\Upsilon(Z) < 0$ for $Z \neq 0$" vacuous, makes Υ continuous, and makes $\Upsilon + \frac{\lambda}{2}|\cdot|^2 \equiv 0$ convex on V, i.e. Υ λ-semiconvex in the sense of `SemiconvexOn`.) So assume $N \ge 1$ for the rest of Step 1 and for Steps 2–4.

Since $|Z-W|^2 = \sum_j |z_j - w_j|^2$, the defining formula of `SupConvolution` gives

$$\mathcal{G}^\varepsilon(Z) = \sup_{W\in V}\sum_j\Big(G_j(w_j) - \frac{|z_j-w_j|^2}{2\varepsilon}\Big) = \sum_j \sup_{w_j\in E}\Big(G_j(w_j) - \frac{|z_j-w_j|^2}{2\varepsilon}\Big) = \sum_j G_j^\varepsilon(z_j).$$

The middle equality holds because the expression is a sum of $N \ge 1$ functions of the independent variables $w_1, \dots, w_N$, each bounded above by the previous paragraph: the supremum of the sum is at most the sum of the suprema since each summand is at most its own supremum, and it is at least the sum of the suprema because for every $\theta > 0$ one may choose, for each j, a w_j with $G_j(w_j) - |z_j - w_j|^2/(2\varepsilon) > G_j^\varepsilon(z_j) - \theta/N$ (legitimate since $N \ge 1$, so $\theta/N > 0$, and $G_j^\varepsilon(z_j)$ is a genuine supremum) and add the N inequalities, obtaining a value $> \sum_j G_j^\varepsilon(z_j) - \theta$; as $\theta > 0$ was arbitrary, the supremum of the sum is at least $\sum_j G_j^\varepsilon(z_j)$.

Step 2: the duality identity. For $(Z, W) \in V \times V$ put

$$F(Z,W) = \mathcal{G}(W) - \frac{|Z-W|^2}{2\varepsilon} - \tfrac{1}{2}\langle A_\varepsilon Z, Z\rangle,$$

so that $\Upsilon(Z) = \sup_W F(Z,W)$ by the defining formula of `SupConvolution` applied to $\mathcal{G}$ in V.

(2a) The two operators and the two-sided identity. Put

$$S\xi = \xi + \varepsilon A_\varepsilon \xi, \qquad T\xi = \xi - \varepsilon B\xi \qquad (\xi \in V),$$

two continuous linear operators on V. Both are self-adjoint: for all $x, y \in V$,

$$\langle Sx, y\rangle = \langle x, y\rangle + \varepsilon\langle A_\varepsilon x, y\rangle = \langle x, y\rangle + \varepsilon\langle x, A_\varepsilon y\rangle = \langle x, Sy\rangle,$$

by the self-adjointness of A_ε assumed in the statement, and likewise $\langle Tx, y\rangle = \langle x, Ty\rangle$ by the self-adjointness of B. Hypothesis (61) says exactly

$$T(S\xi) = \xi \qquad (\xi \in V), \tag{65}$$

i.e. $TS = \mathrm{id}$. We claim the identity also holds in the other order. Indeed, for all $x, y \in V$,

$$\langle S(Tx), y\rangle = \langle Tx, Sy\rangle = \langle x, T(Sy)\rangle = \langle x, y\rangle,$$

the first equality by the self-adjointness of S, the second by the self-adjointness of T, and the third by (65) applied to $\xi = y$. Since $\langle S(Tx) - x, y\rangle = 0$ for every $y \in V$, taking $y = S(Tx) - x$ gives $|S(Tx) - x|^2 = 0$, so

$$S(Tx) = x \qquad (x \in V), \tag{66}$$

i.e. $ST = \mathrm{id}$. No appeal to finite-dimensionality is used.

A consequence we shall need twice: applying (60) at the vector $\xi = Tx$ gives $A_\varepsilon(Tx) = B\big(Tx + \varepsilon A_\varepsilon(Tx)\big) = B\big(S(Tx)\big) = Bx$ by (66), that is,

$$A_\varepsilon(Tx) = Bx \qquad (x \in V). \tag{67}$$

(2b) The exact expansion of $F(\cdot, W)$ about $Z_W = TW$. Fix $W \in V$ and put $Z_W = TW$, so that $SZ_W = W$ by (66). For any $h \in V$, expanding the two Z-dependent terms of F exactly,

$$\frac{|Z_W + h - W|^2}{2\varepsilon} = \frac{|Z_W - W|^2}{2\varepsilon} + \frac{\langle Z_W - W, h\rangle}{\varepsilon} + \frac{|h|^2}{2\varepsilon},$$

$$\tfrac{1}{2}\big\langle A_\varepsilon(Z_W + h), Z_W + h\big\rangle = \tfrac{1}{2}\langle A_\varepsilon Z_W, Z_W\rangle + \langle A_\varepsilon Z_W, h\rangle + \tfrac{1}{2}\langle A_\varepsilon h, h\rangle,$$

where the second display uses the SELF-ADJOINTNESS OF A_ε to combine the two cross terms $\frac{1}{2}\langle A_\varepsilon Z_W, h\rangle + \frac{1}{2}\langle A_\varepsilon h, Z_W\rangle$ into $\langle A_\varepsilon Z_W, h\rangle$. Subtracting both displays from $\mathcal{G}(W)$ gives

$$F(Z_W + h, W) = F(Z_W, W) - \Big\langle \frac{Z_W - W}{\varepsilon} + A_\varepsilon Z_W,\, h\Big\rangle - \tfrac{1}{2}\big\langle(\varepsilon^{-1}\mathrm{id} + A_\varepsilon)h, h\big\rangle. \tag{68}$$

The linear term vanishes identically, because

$$\frac{Z_W - W}{\varepsilon} + A_\varepsilon Z_W = \frac{Z_W + \varepsilon A_\varepsilon Z_W - W}{\varepsilon} = \frac{SZ_W - W}{\varepsilon} = 0.$$

Hence $F(Z_W + h, W) = F(Z_W, W) - \frac{1}{2}\langle(\varepsilon^{-1}\mathrm{id} + A_\varepsilon)h, h\rangle$ for every h, and by (62) the subtracted quantity is > 0 for $h \neq 0$. Therefore the supremum of $F(\cdot, W)$ is ATTAINED at Z_W and Z_W is its UNIQUE maximizer:

$$F(Z, W) < F(Z_W, W) \quad (Z \neq Z_W), \qquad \sup_Z F(Z, W) = F(Z_W, W). \tag{69}$$

(2c) The value at the maximizer. From $Z_W = TW = W - \varepsilon BW$ we get $Z_W - W = -\varepsilon BW$, so $|Z_W - W|^2/(2\varepsilon) = \frac{\varepsilon}{2}|BW|^2$; and by (67) with $x = W$ we have $A_\varepsilon Z_W = A_\varepsilon(TW) = BW$, so

$$\langle A_\varepsilon Z_W, Z_W\rangle = \langle BW, TW\rangle = \langle BW, W - \varepsilon BW\rangle = \langle BW, W\rangle - \varepsilon|BW|^2.$$

Therefore

$$\sup_Z F(Z, W) = F(Z_W, W) = \mathcal{G}(W) - \frac{\varepsilon}{2}|BW|^2 - \frac{1}{2}\Big(\langle BW, W\rangle - \varepsilon|BW|^2\Big) = \mathcal{G}(W) - \tfrac{1}{2}\langle BW, W\rangle =: \Phi(W). \tag{70}$$

Step 3: Υ has a strict global maximum at 0 with value 0. By (63) we have $\Phi(W) \le -\sum_j |w_j|^4 < 0$ for $W \neq 0$, while $\Phi(0) = \sum_j G_j(0) - 0 = 0$; so $\sup_W \Phi = 0$, attained exactly at $W = 0$. By (70), $\sup_{Z,W} F = \sup_W \Phi = 0$. Hence $\Upsilon \le 0$ everywhere, and $\Upsilon(0) = \sup_W F(0, W) \ge F(0, 0) = \mathcal{G}(0) - 0 - 0 = 0$, so $\Upsilon(0) = 0$.

Next, for each fixed Z the supremum $\Upsilon(Z) = \sup_W F(Z, W)$ is attained: $F(Z, \cdot)$ is continuous, and $F(Z, W) \le NC - |Z - W|^2/(2\varepsilon) - \frac{1}{2}\langle A_\varepsilon Z, Z\rangle$, whose right-hand side tends to $-\infty$ as $|W - Z| \to \infty$. Choose therefore $R > 0$ so large that this bound is $< F(Z, Z)$ whenever $|W - Z| > R$; then $\sup_W F(Z, W) = \max_{\overline{B}(Z,R)} F(Z, \cdot)$, because every W outside $\overline{B}(Z, R)$ gives a value strictly below the value already attained at $W = Z$, which lies inside the ball. The set $\overline{B}(Z, R)$ is closed, bounded and nonempty, hence compact, so the maximum is attained (preamble).

Now suppose $\Upsilon(Z)=0$ for some Z. By the previous paragraph there is W with $F(Z,W)=0$. Then $\Phi(W)=\sup_{Z'}F(Z',W)\geq F(Z,W)=0$, so $W=0$ by the first paragraph of this step. Thus $F(Z,0)=0$. Now $Z_0=T\cdot 0=0$, so the expansion (68) at $W=0$ with $h=Z$ reads

$$F(Z,0)=F(0,0)-\tfrac{1}{2}\big\langle(\varepsilon^{-1}\mathrm{id}+A_\varepsilon)Z,Z\big\rangle=-\tfrac{1}{2}\big\langle(\varepsilon^{-1}\mathrm{id}+A_\varepsilon)Z,Z\big\rangle,$$

using $F(0,0)=\mathcal{G}(0)-0-0=0$. Since $F(Z,0)=0$ this forces $\langle(\varepsilon^{-1}\mathrm{id}+A_\varepsilon)Z,Z\rangle=0$, and (62) then gives $Z=0$. This proves

$$\Upsilon(0)=0,\qquad \Upsilon(Z)<0\quad (Z\neq 0).$$

Step 4: Υ is λ-semiconvex and continuous. By part (3) of `SupConvolutionSemiconvex`, applied in V to $\mathcal{G}$ with growth constant NC, the function $\mathcal{G}^\varepsilon$ is $(1/\varepsilon)$-semiconvex on V in the sense of `SemiconvexOn`, i.e. $\mathcal{G}^\varepsilon+|\cdot|^2/(2\varepsilon)$ is convex on V. Also

$$Z\mapsto -\tfrac{1}{2}\langle A_\varepsilon Z,Z\rangle+\tfrac{1}{2}\|A_\varepsilon\|\,|Z|^2=\tfrac{1}{2}\big\langle(\|A_\varepsilon\|\mathrm{id}-A_\varepsilon)Z,Z\big\rangle$$

is convex: for a self-adjoint S with $\langle S\xi,\xi\rangle\geq 0$ for all ξ, and for $t_1,t_2\geq 0$ with $t_1+t_2=1$, expanding and using self-adjointness gives

$$\langle S(t_1\xi_1+t_2\xi_2),t_1\xi_1+t_2\xi_2\rangle-t_1\langle S\xi_1,\xi_1\rangle-t_2\langle S\xi_2,\xi_2\rangle=-t_1t_2\langle S(\xi_1-\xi_2),\xi_1-\xi_2\rangle\leq 0,$$

and here $S=\|A_\varepsilon\|\mathrm{id}-A_\varepsilon$ is self-adjoint with $\langle S\xi,\xi\rangle=\|A_\varepsilon\||\xi|^2-\langle A_\varepsilon\xi,\xi\rangle\geq 0$ by Cauchy–Schwarz and the definition of the operator norm. Adding the two convex functions $\mathcal{G}^\varepsilon+|\cdot|^2/(2\varepsilon)$ and $-\frac{1}{2}\langle A_\varepsilon\cdot,\cdot\rangle+\frac{1}{2}\|A_\varepsilon\||\cdot|^2$ shows that $\Upsilon+\frac{\lambda}{2}|\cdot|^2$ is convex on V with $\lambda=\varepsilon^{-1}+\|A_\varepsilon\|>0$; that is, Υ is λ-semiconvex on V.

Finally $\Upsilon+\frac{\lambda}{2}|\cdot|^2$ is a finite convex function on the open set V, hence continuous (preamble); subtracting the continuous function $\frac{\lambda}{2}|\cdot|^2$ shows that Υ is continuous on V. □

2.23. `SupConvolutionMagic`.

Lemma 2.23 (The magic property of the sup-convolution)**.** *Let E be a finite-dimensional real inner-product space, let $\varepsilon>0$, let $C\in\mathbb{R}$ and let $\phi:E\to\mathbb{R}$ be continuous with the linear growth bound*

$$\phi(w)\leq C+|w|\qquad (w\in E).$$

Let ϕ^ε be the sup-convolution of `SupConvolution`, let $z\in E$, let $p\in E$, let X be a continuous bilinear form on E and assume $(p,X)\in J^{2,+}\phi^\varepsilon(z)$ in the sense of `Superjet`. Then the supremum defining $\phi^\varepsilon(z)$ is attained exactly at the point $z+\varepsilon p$:

$$\phi^\varepsilon(z)=\phi(z+\varepsilon p)-\frac{\varepsilon|p|^2}{2},$$

and the jet transfers to ϕ at that point:

$$(p,X)\in J^{2,+}\phi(z+\varepsilon p).$$

Proof. For $y\in E$ write $S_y=\{\phi(w)-|y-w|^2/(2\varepsilon)\,:\,w\in E\}$, so that $\phi^\varepsilon(y)=\sup S_y$ by `SupConvolution`. Part (1) of `SupConvolutionSemiconvex`, applied with this C and this $\varepsilon>0$, says that every S_y is bounded above; each S_y is nonempty because $w=y$ contributes $\phi(y)$. Hence for every y the number $\phi^\varepsilon(y)$ is finite and

$$\phi(w)-\frac{|y-w|^2}{2\varepsilon}\leq\phi^\varepsilon(y)\qquad (y,w\in E).\tag{71}$$

Step 1: the supremum at z is attained. Define $\Theta:E\to\mathbb{R}$ by $\Theta(w)=\phi(w)-|z-w|^2/(2\varepsilon)$, which is continuous, so that $S_z=\Theta(E)$ and $\phi^\varepsilon(z)=\sup_E\Theta$. Note $\Theta(z)=\phi(z)$.

For $w\in E$ put $s=|z-w|$. By the growth bound and $|w|\leq|z|+s$,

$$\Theta(w)\leq C+|z|+s-\frac{s^2}{2\varepsilon}.\tag{72}$$

Since $\varepsilon>0$, the right-hand side tends to $-\infty$ as $s\to+\infty$; choose therefore $S>0$ so large that

$$C+|z|+s-\frac{s^2}{2\varepsilon}<\phi(z)\qquad (s>S).$$

By (72), $\Theta(w) < \phi(z) = \Theta(z)$ whenever $|z - w| > S$. Consequently

$$\sup_{E} \Theta = \sup_{\overline{B}(z,S)} \Theta,$$

because every w outside $\overline{B}(z, S)$ has $\Theta(w) < \Theta(z)$ and $z \in \overline{B}(z, S)$. The set $\overline{B}(z, S)$ is closed and bounded in the finite-dimensional space E, hence compact, and it is nonempty; Θ is continuous, so Θ attains a maximum on $\overline{B}(z, S)$ at some point $\hat{w}$. Then

$$\phi^\varepsilon(z) = \Theta(\hat{w}) = \phi(\hat{w}) - \frac{|z - \hat{w}|^2}{2\varepsilon}. \tag{73}$$

Step 2: a one-sided comparison of increments. Let $h \in E$ be arbitrary. Applying (71) at the point $y = z + h$ with the test point $w = \hat{w} + h$, and noting $(z + h) - (\hat{w} + h) = z - \hat{w}$, we get

$$\phi(\hat{w} + h) - \frac{|z - \hat{w}|^2}{2\varepsilon} \le \phi^\varepsilon(z + h).$$

Subtracting (73) from this gives

$$\phi(\hat{w} + h) - \phi(\hat{w}) \le \phi^\varepsilon(z + h) - \phi^\varepsilon(z) \qquad (h \in E). \tag{74}$$

Step 3: $\hat{w} = z + \varepsilon p$. Apply (71) at the point $y = z + h$ with the test point $w = \hat{w}$:

$$\phi(\hat{w}) - \frac{|z + h - \hat{w}|^2}{2\varepsilon} \le \phi^\varepsilon(z + h).$$

Expanding $|z + h - \hat{w}|^2 = |z - \hat{w}|^2 + 2\langle z - \hat{w}, h\rangle + |h|^2$ and subtracting (73) yields

$$\phi^\varepsilon(z + h) - \phi^\varepsilon(z) \ge -\Big\langle \frac{z - \hat{w}}{\varepsilon}, h \Big\rangle - \frac{|h|^2}{2\varepsilon} \qquad (h \in E). \tag{75}$$

On the other hand, $(p, X) \in J^{2,+}\phi^\varepsilon(z)$; applying the defining property of `Superjet` with the value 1 for the parameter there, we obtain $\delta > 0$ with

$$\phi^\varepsilon(z + h) - \phi^\varepsilon(z) \le \langle p, h\rangle + \tfrac{1}{2}X(h, h) + |h|^2 \qquad (|h| < \delta). \tag{76}$$

Put $u = p + (z - \hat{w})/\varepsilon$. Let $v \in E$ with $|v| = 1$ and let $t \in (0, \delta)$. Taking $h = tv$, so $|h| = t < \delta$, and combining (75) with (76),

$$-t\Big\langle \frac{z - \hat{w}}{\varepsilon}, v \Big\rangle - \frac{t^2}{2\varepsilon} \le t\langle p, v\rangle + \frac{t^2}{2}X(v, v) + t^2.$$

Dividing by $t > 0$ gives

$$-\Big\langle \frac{z - \hat{w}}{\varepsilon}, v \Big\rangle \le \langle p, v\rangle + t\Big(\frac{X(v, v)}{2} + 1 + \frac{1}{2\varepsilon}\Big),$$

and letting $t \downarrow 0$ (the bracket does not depend on t) gives $0 \le \langle u, v\rangle$. This holds for every unit vector v; applying it to $-v$ as well gives $\langle u, v\rangle = 0$ for every unit vector v. If $u \neq 0$ then $v = u/|u|$ is a unit vector with $\langle u, v\rangle = |u| > 0$, a contradiction. Hence $u = 0$, that is,

$$\hat{w} = z + \varepsilon p.$$

Substituting this into (73) and using $|z - \hat{w}| = |\varepsilon p| = \varepsilon|p|$ (as $\varepsilon > 0$), so that $|z - \hat{w}|^2/(2\varepsilon) = \varepsilon^2|p|^2/(2\varepsilon) = \varepsilon|p|^2/2$, we obtain the first assertion:

$$\phi^\varepsilon(z) = \phi(z + \varepsilon p) - \frac{\varepsilon|p|^2}{2}.$$

Step 4: the jet transfers. We verify the two clauses of `Superjet` for the pair (p, X) at the point $\hat{w} = z + \varepsilon p$ with the function ϕ.

Symmetry of X holds because it is already part of the hypothesis $(p, X) \in J^{2,+}\phi^\varepsilon(z)$.

For the expansion, let $\varepsilon' > 0$. By $(p, X) \in J^{2,+}\phi^\varepsilon(z)$ and `Superjet` applied with the parameter ε' there is $\delta' > 0$ with

$$\phi^\varepsilon(z + h) \le \phi^\varepsilon(z) + \langle p, h\rangle + \tfrac{1}{2}X(h, h) + \varepsilon'|h|^2 \qquad (|h| < \delta').$$

Combining this with (74) gives, for every h with $|h| < \delta'$,

$$\phi(\hat{w} + h) \le \phi(\hat{w}) + \big(\phi^\varepsilon(z + h) - \phi^\varepsilon(z)\big) \le \phi(\hat{w}) + \langle p, h\rangle + \tfrac{1}{2}X(h, h) + \varepsilon'|h|^2.$$

Since $\varepsilon' > 0$ was arbitrary, the defining condition of `Superjet` holds for (p, X) at $\hat{w}$ with the function ϕ, that is, $(p, X) \in J^{2,+}\phi(z + \varepsilon p)$. □

2.24. `ConvexSubgradientExists.`

Lemma 2.24 (Existence of subgradients of a convex function)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ be open, let $\psi : E \to \mathbb{R}$ be convex on Ω (so in particular Ω is convex) and let $z \in \Omega$. Then ψ has a subgradient at z: there is $q \in E$ with*

$$\psi(z) + \langle q, x - z\rangle \le \psi(x) \qquad (x \in \Omega).$$

Proof. Step 1: a ball on which ψ is Lipschitz. Since Ω is open and ψ is convex and real-valued on Ω, ψ is continuous on Ω and locally Lipschitz on Ω (preamble). Let $U \subseteq \Omega$ be a neighbourhood of z and $L \ge 0$ a constant with $|\psi(a) - \psi(b)| \le L|a - b|$ for $a, b \in U$. Choose $\rho > 0$ so small that $\overline{B}(z, 2\rho) \subseteq U$; this is possible because U is a neighbourhood of z. Then ψ is L-Lipschitz on $\overline{B}(z, 2\rho) \subseteq \Omega$.

Step 2: differentiability points accumulating at z. The ball $B(z, 2\rho)$ is open and ψ is Lipschitz on it, so by Rademacher's theorem (preamble) ψ is differentiable at almost every point of $B(z, 2\rho)$; write D for the set of those points of $B(z, 2\rho)$ at which ψ is differentiable, so that $B(z, 2\rho) \setminus D$ is null. Fix an additive Haar measure μ on E. For each integer $j \ge 1$ the ball $B(z, \rho/j)$ has $\mu(B(z, \rho/j)) > 0$ (preamble), so it cannot be contained in the null set $B(z, 2\rho) \setminus D$; hence there is $w_j \in D$ with $|w_j - z| < \rho/j$. In particular $w_j \to z$ and $w_j \in B(z, \rho)$ for every j.

Step 3: at a differentiability point the gradient is a subgradient. Let $w \in D$ and let $x \in \Omega$. For $t \in (0, 1]$ the point $w + t(x - w)$ lies in Ω, since Ω is convex and $w, x \in \Omega$, and convexity of ψ on Ω with the weights $1 - t$ and t gives

$$\psi\big(w + t(x - w)\big) \le (1 - t)\psi(w) + t\psi(x),$$

that is,

$$\frac{\psi(w + t(x - w)) - \psi(w)}{t} \le \psi(x) - \psi(w).$$

As $t \downarrow 0$ the left-hand side tends to $\langle \nabla\psi(w), x - w\rangle$, because ψ is differentiable at w with gradient $\nabla\psi(w)$ and $\psi(w + th) = \psi(w) + t\langle \nabla\psi(w), h\rangle + o(t)$ for fixed $h = x - w$. Therefore

$$\psi(w) + \langle \nabla\psi(w), x - w\rangle \le \psi(x) \qquad (w \in D,\ x \in \Omega). \tag{77}$$

Step 4: the gradients at the w_j are bounded by L. Fix j and let $e \in E$ with $|e| = 1$. For $0 < t \le \rho$ the point $w_j + te$ satisfies $|w_j + te - z| \le |w_j - z| + t < \rho + \rho = 2\rho$, so $w_j + te \in B(z, 2\rho) \subseteq \Omega$. Applying (77) with $x = w_j + te$, and then the Lipschitz bound of Step 1,

$$t\langle \nabla\psi(w_j), e\rangle \le \psi(w_j + te) - \psi(w_j) \le Lt.$$

Dividing by $t > 0$ gives $\langle \nabla\psi(w_j), e\rangle \le L$ for every unit vector e. Taking $e = \nabla\psi(w_j)/|\nabla\psi(w_j)|$ when $\nabla\psi(w_j) \ne 0$ yields $|\nabla\psi(w_j)| \le L$; and the bound is trivial when $\nabla\psi(w_j) = 0$. So $|\nabla\psi(w_j)| \le L$ for every j.

Step 5: passing to the limit. The sequence $(\nabla\psi(w_j))_j$ is bounded, so by Bolzano–Weierstrass (preamble) it has a subsequence $(\nabla\psi(w_{j_i}))_i$ converging to some $q \in E$. Let $x \in \Omega$. By (77),

$$\psi(w_{j_i}) + \langle \nabla\psi(w_{j_i}), x - w_{j_i}\rangle \le \psi(x) \qquad (i \ge 1).$$

As $i \to \infty$ we have $w_{j_i} \to z$, hence $\psi(w_{j_i}) \to \psi(z)$ by the continuity of ψ on Ω (Step 1), and $\langle \nabla\psi(w_{j_i}), x - w_{j_i}\rangle \to \langle q, x - z\rangle$ by the joint continuity of the inner product. Therefore $\psi(z) + \langle q, x - z\rangle \le \psi(x)$. Since $x \in \Omega$ was arbitrary, q is the required subgradient. □

2.25. `MaxMultiplierPinsSubgradient.`

Lemma 2.25 (A constrained maximum pins the subgradient of the convexification)**.** *Let E be a finite-dimensional real inner-product space, let $\lambda \in \mathbb{R}$, let $\phi : E \to \mathbb{R}$, let $A, \Omega \subseteq E$, let $z, p, s \in E$ and let $\eta > 0$. Write*

$$\psi(w) = \phi(w) + \frac{\lambda|w|^2}{2} \qquad (w \in E).$$

Assume

(1) $B(z,\eta) \subseteq A \cap \Omega$;
(2) $\phi(w) + \langle p, w\rangle \le \phi(z) + \langle p, z\rangle$ *for every* $w \in A$;
(3) $\psi(z) + \langle s, w - z\rangle \le \psi(w)$ *for every* $w \in \Omega$.

Then

$$s = \lambda z - p.$$

Proof. Set $d = s - (\lambda z - p)$. The proof consists of one algebraic estimate, valid on the ball $B(z,\eta)$, and one choice of test point.

Step 1: the estimate $\langle d, w - z\rangle \le \lambda|w - z|^2/2$ *on* $B(z,\eta)$. Let $w \in B(z,\eta)$. By hypothesis (1) we have $w \in A$ and $w \in \Omega$. Hypothesis (2) applied to this w gives

$$\phi(w) - \phi(z) \le \langle p, z\rangle - \langle p, w\rangle = -\langle p, w - z\rangle,$$

the last equality because $\langle p, w\rangle - \langle p, z\rangle = \langle p, w - z\rangle$ by additivity of the inner product in its second argument. Hypothesis (3) applied to the same w gives

$$\langle s, w - z\rangle \le \psi(w) - \psi(z) = \big(\phi(w) - \phi(z)\big) + \frac{\lambda|w|^2}{2} - \frac{\lambda|z|^2}{2}.$$

Expanding $|w|^2 = |(w - z) + z|^2 = |w - z|^2 + 2\langle w - z, z\rangle + |z|^2$ and using the symmetry of the real inner product,

$$\frac{\lambda|w|^2}{2} - \frac{\lambda|z|^2}{2} = \lambda\langle z, w - z\rangle + \frac{\lambda|w - z|^2}{2}. \tag{78}$$

Combining the three displays,

$$\langle s, w - z\rangle \le -\langle p, w - z\rangle + \lambda\langle z, w - z\rangle + \frac{\lambda|w - z|^2}{2},$$

and since $\langle d, w - z\rangle = \langle s, w - z\rangle + \langle p, w - z\rangle - \lambda\langle z, w - z\rangle$ by linearity of the inner product in its first argument, this is exactly

$$\langle d, w - z\rangle \le \frac{\lambda|w - z|^2}{2} \qquad \big(w \in B(z,\eta)\big). \tag{79}$$

Step 2: the test point. Suppose, for contradiction, that $d \neq 0$, so that $|d| > 0$. Put

$$t = \min\Big\{\frac{\eta}{2|d|},\ \frac{1}{|\lambda| + 1}\Big\} > 0,$$

both entries being strictly positive, and set $w = z + td$. Then $|w - z| = t|d| \le \eta/2 < \eta$, so $w \in B(z,\eta)$ and (79) applies to it. Since $\langle d, td\rangle = t|d|^2$ and $|w - z|^2 = t^2|d|^2$, it reads

$$t|d|^2 \le \frac{\lambda t^2|d|^2}{2} = \frac{\lambda t}{2} \cdot t|d|^2.$$

The number $t|d|^2$ is strictly positive, so dividing by it gives $1 \le \lambda t/2$. On the other hand $\lambda \le |\lambda|$ and $t > 0$ give $\lambda t/2 \le |\lambda| t/2$, while $t \le 1/(|\lambda| + 1)$ and $|\lambda| \ge 0$ give

$$\frac{|\lambda| t}{2} \le \frac{|\lambda|}{2(|\lambda| + 1)} < 1,$$

because $|\lambda| < 2(|\lambda| + 1)$. Hence $1 \le \lambda t/2 < 1$, a contradiction.

Therefore $d = 0$, i.e. $s = \lambda z - p$. □

2.26. `SemiconvexMultiplierLipschitz`.

Lemma 2.26 (The multiplier is 2λ-Lipschitz in the maximum point)**.** *Let E be a finite-dimensional real inner-product space, let $\lambda > 0$, let $\Omega \subseteq E$ be open, let $\phi : E \to \mathbb{R}$ be λ-semiconvex on Ω in the sense of* `SemiconvexOn`*, let $A \subseteq E$, let $z_1, z_2, p_1, p_2 \in E$ and let $\eta > 0$. Assume*

(1) $B(z_1,\eta) \subseteq A \cap \Omega$ *and* $B(z_2,\eta) \subseteq A \cap \Omega$;
(2) $|z_1 - z_2| < \eta$;
(3) *for* $i = 1, 2$, $\phi(w) + \langle p_i, w\rangle \le \phi(z_i) + \langle p_i, z_i\rangle$ *for every* $w \in A$.

Then

$$|p_1 - p_2| \le 2\lambda |z_1 - z_2|.$$

Proof. Write

$$\psi(w) = \phi(w) + \frac{\lambda |w|^2}{2} \qquad (w \in E),$$

so that, by the definition of λ-semiconvexity in `SemiconvexOn`, ψ is convex on Ω and Ω is convex. Put $D = |z_1 - z_2|$.

Throughout we use the identity

$$\frac{\lambda |u|^2}{2} - \frac{\lambda |v|^2}{2} = \lambda \langle v, u - v\rangle + \frac{\lambda |u - v|^2}{2} \qquad (u, v \in E), \tag{80}$$

which follows by expanding $|u|^2 = |(u - v) + v|^2 = |u - v|^2 + 2\langle u - v, v\rangle + |v|^2$ and using the symmetry of the real inner product.

Step 1: a subgradient of ψ at z_2, pinned to $\lambda z_2 - p_2$. We have $z_2 \in B(z_2, \eta) \subseteq A \cap \Omega$ by (1), so $z_2 \in \Omega$. Since Ω is open and ψ is convex on Ω, `ConvexSubgradientExists` applied to Ω, ψ and z_2 produces $s \in E$ with

$$\psi(z_2) + \langle s, w - z_2\rangle \le \psi(w) \qquad (w \in \Omega).$$

The data λ, ϕ, A, Ω, z_2, p_2, s, η satisfy every hypothesis of `MaxMultiplierPinsSubgradient`: $\eta > 0$, $B(z_2, \eta) \subseteq A \cap \Omega$ by (1), the maximality $\phi(w) + \langle p_2, w\rangle \le \phi(z_2) + \langle p_2, z_2\rangle$ on A by (3), and the subgradient inequality just displayed. That lemma therefore gives $s = \lambda z_2 - p_2$, so

$$\psi(z_2) + \langle \lambda z_2 - p_2, w - z_2\rangle \le \psi(w) \qquad (w \in \Omega). \tag{81}$$

Step 2: the degenerate case $D = 0$. If $D = 0$ then $z_1 = z_2$, so (81) with z_1 in place of z_2 reads $\psi(z_1) + \langle s, w - z_1\rangle \le \psi(w)$ for $w \in \Omega$, with the same s. Applying `MaxMultiplierPinsSubgradient` a second time, now with the data z_1, p_1, s, η — admissible because $B(z_1, \eta) \subseteq A \cap \Omega$ by (1) and the maximality of $\phi + \langle p_1, \cdot\rangle$ at z_1 on A holds by (3) — gives $s = \lambda z_1 - p_1$. Comparing with $s = \lambda z_2 - p_2 = \lambda z_1 - p_2$ yields $p_1 = p_2$, hence $|p_1 - p_2| = 0 = 2\lambda D$. So we may assume $D > 0$ from now on.

Step 3: the defect σ and its two-sided bound. Since $|z_1 - z_2| < \eta$ we have $z_1 \in B(z_2, \eta)$, so (1) gives $z_1 \in A$ and $z_1 \in \Omega$. Put

$$\sigma = \psi(z_1) - \psi(z_2) - \langle \lambda z_2 - p_2, z_1 - z_2\rangle.$$

Taking $w = z_1 \in \Omega$ in (81) gives $\sigma \ge 0$. For the upper bound, (3) with $i = 2$ and $w = z_1 \in A$ gives $\phi(z_1) - \phi(z_2) \le \langle p_2, z_2\rangle - \langle p_2, z_1\rangle = -\langle p_2, z_1 - z_2\rangle$. Expanding σ and using $\langle \lambda z_2 - p_2, z_1 - z_2\rangle = \lambda \langle z_2, z_1 - z_2\rangle - \langle p_2, z_1 - z_2\rangle$ together with (80) for $u = z_1$, $v = z_2$,

$$\sigma = \big(\phi(z_1) - \phi(z_2)\big) + \frac{\lambda D^2}{2} + \langle p_2, z_1 - z_2\rangle \le \frac{\lambda D^2}{2}.$$

Hence

$$0 \le \sigma \le \frac{\lambda D^2}{2}. \tag{82}$$

Step 4: the key estimate on $B(z_1, \eta)$. Put $\Delta = (p_1 - p_2) - \lambda(z_1 - z_2)$ and let $w \in B(z_1, \eta)$; by (1), $w \in A$ and $w \in \Omega$. From (3) with $i = 1$,

$$\phi(w) - \phi(z_1) \le -\langle p_1, w - z_1\rangle. \tag{83}$$

From (81) at this w, and subtracting $\lambda |w|^2/2$ from both sides to pass from ψ to ϕ,

$$\phi(w) - \phi(z_1) \ge \big(\psi(z_2) - \psi(z_1)\big) + \langle \lambda z_2 - p_2, w - z_2\rangle - \Big(\frac{\lambda |w|^2}{2} - \frac{\lambda |z_1|^2}{2}\Big).$$

Now $\psi(z_2) - \psi(z_1) = -\sigma - \langle \lambda z_2 - p_2, z_1 - z_2\rangle$ by the definition of σ, and $\langle \lambda z_2 - p_2, w - z_2\rangle - \langle \lambda z_2 - p_2, z_1 - z_2\rangle = \langle \lambda z_2 - p_2, w - z_1\rangle$ because $(w - z_2) - (z_1 - z_2) = w - z_1$; while (80) with $u = w$, $v = z_1$ evaluates the last bracket as $\lambda \langle z_1, w - z_1\rangle + \lambda |w - z_1|^2/2$. Therefore

$$\phi(w) - \phi(z_1) \ge -\sigma + \langle \lambda z_2 - p_2, w - z_1\rangle - \lambda \langle z_1, w - z_1\rangle - \frac{\lambda |w - z_1|^2}{2}.$$

Comparing with (83) and rearranging,

$$\langle \lambda z_2 - p_2, w - z_1\rangle - \lambda\langle z_1, w - z_1\rangle + \langle p_1, w - z_1\rangle \le \sigma + \frac{\lambda|w - z_1|^2}{2}.$$

By linearity of the inner product in its first argument the left-hand side is $\langle \Delta, w - z_1\rangle$, since $(\lambda z_2 - p_2) - \lambda z_1 + p_1 = (p_1 - p_2) - \lambda(z_1 - z_2) = \Delta$. Thus

$$\langle \Delta, w - z_1\rangle \le \sigma + \frac{\lambda|w - z_1|^2}{2} \qquad \big(w \in B(z_1, \eta)\big). \tag{84}$$

Step 5: the bound $|\Delta| \le \lambda D$. If $\Delta = 0$ this is clear, both sides being ≥ 0 and the left-hand side 0. Otherwise $|\Delta| > 0$; take

$$w = z_1 + \frac{D}{|\Delta|}\,\Delta,$$

so that $w - z_1 = (D/|\Delta|)\Delta$ and $|w - z_1| = (D/|\Delta|)|\Delta| = D < \eta$ by (2); hence $w \in B(z_1, \eta)$ and (84) applies. Since $\langle \Delta, (D/|\Delta|)\Delta\rangle = (D/|\Delta|)|\Delta|^2 = D|\Delta|$, it reads

$$D|\Delta| \le \sigma + \frac{\lambda D^2}{2} \le \frac{\lambda D^2}{2} + \frac{\lambda D^2}{2} = \lambda D^2,$$

using (82) for the middle inequality. Dividing by $D > 0$ gives

$$|\Delta| \le \lambda D. \tag{85}$$

(The choice $t = D$ for the step length is what makes the usual optimization in t unnecessary: at $t = D$ the two terms σ/t and $\lambda t/2$ are each at most $\lambda D/2$.)

Step 6: conclusion. Since $p_1 - p_2 = \Delta + \lambda(z_1 - z_2)$, the triangle inequality, $|\lambda(z_1 - z_2)| = \lambda D$ (as $\lambda > 0$) and (85) give

$$|p_1 - p_2| \le |\Delta| + \lambda D \le \lambda D + \lambda D = 2\lambda D = 2\lambda|z_1 - z_2|,$$

which is the assertion. □

2.27. `StrictMaxPerturbLocalization`.

Lemma 2.27 (Localization of the maximizers of a small linear perturbation)**.** *Let E be a finite-dimensional real inner-product space, let $\phi : E \to \mathbb{R}$, let $z_0 \in E$ and let $0 < \rho \le r_1$. Assume that ϕ is continuous on $\overline{B}(z_0, r_1)$ and that z_0 is a strict maximum point of ϕ there:*

$$\phi(z) < \phi(z_0) \qquad \big(z \in \overline{B}(z_0, r_1),\ z \ne z_0\big).$$

Then there exists $\delta_0 > 0$ with the following property. Whenever $p \in E$ satisfies $|p| \le \delta_0$ and $z \in \overline{B}(z_0, r_1)$ satisfies

$$\phi(w) + \langle p, w\rangle \le \phi(z) + \langle p, z\rangle \qquad \big(w \in \overline{B}(z_0, r_1)\big),$$

one has $|z - z_0| < \rho$.

Proof. Note first that $r_1 \ge \rho > 0$, so $r_1 > 0$. Put

$$S = \overline{B}(z_0, r_1) \setminus B(z_0, \rho) = \big\{z \in E :\ \rho \le |z - z_0| \le r_1\big\}.$$

The second description is the first one unwound: $z \in \overline{B}(z_0, r_1)$ says $|z - z_0| \le r_1$, and $z \notin B(z_0, \rho)$ says $\neg(|z - z_0| < \rho)$, i.e. $\rho \le |z - z_0|$.

Case 1: $S = \emptyset$. Take $\delta_0 = 1 > 0$. Let p and z be as in the statement; we must show $|z - z_0| < \rho$. If not, then $\rho \le |z - z_0|$, and also $|z - z_0| \le r_1$ because $z \in \overline{B}(z_0, r_1)$; by the description of S this puts $z \in S$, contradicting $S = \emptyset$. (This is the only case that can occur when $E = \{0\}$, and the hypothesis of the lemma is then vacuous; no nondegeneracy assumption on E is needed anywhere.)

Case 2: $S \ne \emptyset$. The set $\overline{B}(z_0, r_1)$ is closed and bounded in a finite-dimensional space, hence compact (preamble), and $B(z_0, \rho)$ is open; so $S = \overline{B}(z_0, r_1) \setminus B(z_0, \rho)$ is compact, being the intersection of a compact set with a closed set. Also $S \subseteq \overline{B}(z_0, r_1)$, on which ϕ is continuous, so ϕ is continuous on S. A continuous real function on a nonempty compact set attains its maximum (preamble): there is $z_m \in S$ with

$$\phi(z) \le m := \phi(z_m) \qquad (z \in S).$$

We claim $m < \phi(z_0)$. Indeed $z_m \in S \subseteq \overline{B}(z_0, r_1)$, and $z_m \neq z_0$ because $|z_m - z_0| \geq \rho > 0$ while $|z_0 - z_0| = 0$; so the strict maximum hypothesis gives $\phi(z_m) < \phi(z_0)$, i.e. $m < \phi(z_0)$.

Put

$$\delta_0 = \frac{\phi(z_0) - m}{2r_1},$$

which is strictly positive because $\phi(z_0) - m > 0$ and $r_1 > 0$.

Now let $|p| \leq \delta_0$ and let $z \in \overline{B}(z_0, r_1)$ satisfy the maximality hypothesis, and suppose for contradiction that $|z - z_0| \geq \rho$. Since also $|z - z_0| \leq r_1$, we get $z \in S$ and therefore

$$\phi(z) \leq m. \tag{86}$$

The point z_0 lies in $\overline{B}(z_0, r_1)$ because $r_1 \geq 0$, so the maximality hypothesis at $w = z_0$ gives $\phi(z_0) + \langle p, z_0 \rangle \leq \phi(z) + \langle p, z \rangle$, that is,

$$\phi(z_0) - \phi(z) \leq \langle p, z \rangle - \langle p, z_0 \rangle = \langle p, z - z_0 \rangle.$$

By the Cauchy–Schwarz inequality and the hypotheses $|p| \leq \delta_0$ and $|z - z_0| \leq r_1$,

$$\langle p, z - z_0 \rangle \leq |p|\,|z - z_0| \leq \delta_0 r_1 = \frac{\phi(z_0) - m}{2}.$$

Combining this with (86),

$$\phi(z_0) - m \leq \phi(z_0) - \phi(z) \leq \frac{\phi(z_0) - m}{2},$$

whence $\phi(z_0) - m \leq 0$, contradicting $m < \phi(z_0)$. Therefore $|z - z_0| < \rho$, as required. □

2.28. `LipschitzImageHaarBound`.

Lemma 2.28 (Measure of a Lipschitz image)**.** *Let E be a finite-dimensional real inner-product space equipped with an additive Haar measure μ, and put $n = \dim E$. Let $A \subseteq E$, let $K \geq 0$ and let $f : E \to E$ satisfy*

$$|f(a) - f(a')| \leq K|a - a'| \qquad (a, a' \in A).$$

Then

$$\mu\big(f(A)\big) \leq K^n\, \mu(A).$$

Proof. This is the statement recorded in the preamble as the proposition on the measure of a Lipschitz image; we carry out its justification in full, since the assertion is a reduction and not a single imported fact. Write $\mathcal{H}^n$ for the n-dimensional Hausdorff measure on E, and recall that all measures of sets below are taken in $[0, \infty]$, where multiplication is commutative, associative, and monotone in each argument.

Step 1: μ is a scalar multiple of $\mathcal{H}^n$. In the critical dimension $n = \dim E$ the Hausdorff measure $\mathcal{H}^n$ is itself an additive Haar measure on E (`isAddHaarMeasure_hausdorffMeasure`). By the uniqueness of additive Haar measure recorded in the preamble's definition of Haar measure — any two additive Haar measures on E are positive scalar multiples of one another — there is a finite constant $c \geq 0$ with

$$\mu(s) = c\,\mathcal{H}^n(s) \qquad (s \subseteq E). \tag{87}$$

The identity (87) is an identity of measures, so it holds for the two sets $f(A)$ and A, which are the only ones we use; no measurability of A is required.

Step 2: the bound for $\mathcal{H}^n$. For every $d \geq 0$, a map that is K-Lipschitz on A satisfies $\mathcal{H}^d(f(A)) \leq K^d \mathcal{H}^d(A)$ (`LipschitzOnWith.hausdorffMeasure_image_le`); this is exactly the hypothesis $|f(a) - f(a')| \leq K|a - a'|$ for $a, a' \in A$. The dimension $n = \dim E$ is a nonnegative integer, so $d = n$ is admissible and

$$\mathcal{H}^n\big(f(A)\big) \leq K^n\, \mathcal{H}^n(A). \tag{88}$$

Step 3: rescaling by the same factor on both sides. Apply (87) to $s = f(A)$, then multiply (88) by the constant $c \geq 0$, which preserves the inequality, then rearrange the product and apply (87) to $s = A$:

$$\mu\big(f(A)\big) = c\,\mathcal{H}^n\big(f(A)\big) \leq c\,\Big(K^n\, \mathcal{H}^n(A)\Big) = K^n\big(c\,\mathcal{H}^n(A)\big) = K^n\, \mu(A).$$

This is the assertion. Note that the source and the target of f are the same space E, which is what makes the exponent n correct; the preamble records the counterexample that shows no such bound holds between spaces of different dimensions. □

2.29. `JensenLemma`.

Lemma 2.29 (Jensen's lemma)**.** *Let E be a finite-dimensional real inner-product space with an additive Haar measure μ, let $\lambda > 0$ and $r > 0$, let $z_0 \in E$ and let $\phi : E \to \mathbb{R}$ be continuous on $\overline{B}(z_0, r)$ and λ-semiconvex on $B(z_0, r)$ in the sense of `SemiconvexOn`. Assume that z_0 is a strict maximum point of ϕ on the closed ball:*

$$\phi(z) < \phi(z_0) \qquad \big(z \in \overline{B}(z_0, r),\ z \neq z_0\big).$$

Then for every $\delta > 0$

$$\mu\Big(\big\{z \in B(z_0, r)\ :\ \exists\, p \in E,\ |p| \le \delta,\ w \mapsto \phi(w) + \langle p, w\rangle \text{ has a local maximum on } B(z_0, r) \text{ at } z\big\}\Big) > 0.$$

Proof. Write $B = B(z_0, r)$ and fix the three radii

$$r_1 = \frac{r}{2}, \qquad \rho = \frac{r}{8}, \qquad \eta = \frac{3r}{8},$$

so that $0 < \rho \le r_1 < r$, $\eta > 0$ and $\rho + \eta = r_1$. Write

$$\psi(w) = \phi(w) + \frac{\lambda |w|^2}{2} \qquad (w \in E),$$

which is convex on the open convex set B by the definition of λ-semiconvexity in `SemiconvexOn`. Denote by K_δ the set appearing in the statement. Finally put $n = \dim E$.

No nondegeneracy assumption on E is made or needed: the only place where the old argument used a nonzero vector was the construction of the threshold in Step 1, and that construction has been moved into `StrictMaxPerturbLocalization`, which covers the degenerate case internally. The case $n = 0$ is discussed again at the end of Step 5.

Step 1: the localization threshold. Since $r_1 < r$ we have $\overline{B}(z_0, r_1) \subseteq \overline{B}(z_0, r)$, so ϕ is continuous on $\overline{B}(z_0, r_1)$ and

$$\phi(z) < \phi(z_0) \qquad \big(z \in \overline{B}(z_0, r_1),\ z \neq z_0\big),$$

both being restrictions of the corresponding hypotheses of the lemma. Together with $0 < \rho \le r_1$ these are exactly the hypotheses of `StrictMaxPerturbLocalization` for ϕ, z_0, r_1, ρ. That lemma therefore supplies $\delta_0 > 0$ such that

$$\begin{aligned} &\text{if } |p| \le \delta_0,\ z \in \overline{B}(z_0, r_1) \text{ and } \phi(w) + \langle p, w\rangle \le \phi(z) + \langle p, z\rangle \text{ for all } w \in \overline{B}(z_0, r_1), \\ &\qquad\qquad \text{then } |z - z_0| < \rho. \end{aligned} \tag{89}$$

Put

$$d = \min\{\delta, \delta_0\} > 0,$$

so that $d \le \delta$ and $d \le \delta_0$.

Step 2: the set K and its elementary properties. Define

$$K = \Big\{z \in \overline{B}(z_0, r_1)\ :\ \exists\, p \in E,\ |p| \le d,\ \phi(w) + \langle p, w\rangle \le \phi(z) + \langle p, z\rangle \text{ for all } w \in \overline{B}(z_0, r_1)\Big\}.$$

Call such a p a *multiplier for z*.

(2a) $K \subseteq B(z_0, \rho)$. If $z \in K$ with multiplier p, then $|p| \le d \le \delta_0$, so (89) applies and gives $|z - z_0| < \rho$.

(2b) Room around a localized point. If $|z - z_0| < \rho$ then

$$B(z, \eta) \subseteq \overline{B}(z_0, r_1) \cap B.$$

Indeed, for $w \in B(z, \eta)$ the triangle inequality gives

$$|w - z_0| \le |w - z| + |z - z_0| < \eta + \rho = \frac{3r}{8} + \frac{r}{8} = \frac{r}{2} = r_1,$$

so $|w - z_0| \le r_1$, i.e. $w \in \overline{B}(z_0, r_1)$, and also $|w - z_0| < r_1 < r$, i.e. $w \in B$.

(2c) $K \subseteq K_\delta$. Let $z \in K$ with multiplier p. By (2a) $|z - z_0| < \rho$, so (2b) applies to z; in particular $z \in B(z,\eta) \subseteq B$. The set $B(z,\eta)$ is open and contains z, hence is a neighbourhood of z, and $B(z,\eta) \subseteq \overline{B}(z_0, r_1)$ by (2b); therefore

$$\phi(w) + \langle p, w\rangle \le \phi(z) + \langle p, z\rangle \qquad \big(w \in B(z,\eta)\big),$$

and a fortiori for $w \in B(z,\eta) \cap B$. Thus $w \mapsto \phi(w) + \langle p, w\rangle$ has a local maximum on B at z. Since also $|p| \le d \le \delta$, the point z lies in K_δ.

(2d) Every admissible multiplier occurs. Let $p \in E$ with $|p| \le d$. The set $\overline{B}(z_0, r_1)$ is closed and bounded, hence compact (preamble), and it is nonempty since z_0 belongs to it ($r_1 > 0$). The function $w \mapsto \phi(w) + \langle p, w\rangle$ is continuous on it, being the sum of ϕ — continuous on $\overline{B}(z_0, r_1) \subseteq \overline{B}(z_0, r)$ — and of the continuous linear functional $w \mapsto \langle p, w\rangle$. By Weierstrass' theorem (preamble) it attains a maximum at some $z \in \overline{B}(z_0, r_1)$, i.e.

$$\phi(w) + \langle p, w\rangle \le \phi(z) + \langle p, z\rangle \qquad \big(w \in \overline{B}(z_0, r_1)\big).$$

Then $z \in K$ and p is a multiplier for z.

Step 3: the multiplier is unique, so H is well defined. Let $z \in K$ and let p, p' both be multipliers for z. By (2a) and (2b), $B(z,\eta) \subseteq \overline{B}(z_0, r_1) \cap B$; in particular $z \in B$. Since B is open and ψ is convex on B, `ConvexSubgradientExists` applied to B, ψ and z produces $s \in E$ with

$$\psi(z) + \langle s, w - z\rangle \le \psi(w) \qquad (w \in B). \tag{90}$$

Apply `MaxMultiplierPinsSubgradient` with the data

$$\lambda, \quad \phi, \quad A = \overline{B}(z_0, r_1), \quad \Omega = B, \quad z, \quad p, \quad s, \quad \eta.$$

Its three hypotheses hold: $\eta > 0$ and $B(z,\eta) \subseteq \overline{B}(z_0, r_1) \cap B$ as just noted; the maximality of $\phi + \langle p, \cdot\rangle$ at z over A, because p is a multiplier for z; and (90), which is the subgradient inequality for ψ on $\Omega = B$. The conclusion is $s = \lambda z - p$. The same lemma with p' in place of p — the subgradient s being unchanged, since it was produced from z alone — gives $s = \lambda z - p'$. Hence $\lambda z - p = \lambda z - p'$ and so $p = p'$.

Therefore each $z \in K$ has exactly one multiplier, which we call $H(z)$; by definition $|H(z)| \le d$ and

$$\phi(w) + \langle H(z), w\rangle \le \phi(z) + \langle H(z), z\rangle \qquad \big(z \in K,\ w \in \overline{B}(z_0, r_1)\big). \tag{91}$$

Extend H to all of E by setting $H(z) = 0$ for $z \notin K$; this does not affect $H(K)$ or the Lipschitz estimate of Step 4, both of which involve only points of K.

Step 4: H is 2λ-Lipschitz on K. Let $z_1, z_2 \in K$. We verify the hypotheses of `SemiconvexMultiplierLipschi` for the data

$$\lambda, \quad \Omega = B, \quad \phi, \quad A = \overline{B}(z_0, r_1), \quad z_1,\ z_2, \quad p_1 = H(z_1),\ p_2 = H(z_2), \quad \eta.$$

First, $\lambda > 0$ by hypothesis, B is open, and ϕ is λ-semiconvex on B; and $\eta > 0$. Next, by (2a) each z_i satisfies $|z_i - z_0| < \rho$, so (2b) gives $B(z_i, \eta) \subseteq \overline{B}(z_0, r_1) \cap B$ for $i = 1, 2$, which is hypothesis (1) of that lemma. For hypothesis (2),

$$|z_1 - z_2| \le |z_1 - z_0| + |z_0 - z_2| < \rho + \rho = \frac{r}{4} < \frac{3r}{8} = \eta.$$

Hypothesis (3) is (91) for z_1 and for z_2. The conclusion is

$$|H(z_1) - H(z_2)| \le 2\lambda |z_1 - z_2| \qquad (z_1, z_2 \in K). \tag{92}$$

Step 5: the measure estimate. By (92) the map $H : E \to E$ satisfies $|H(a) - H(a')| \le 2\lambda |a - a'|$ for all $a, a' \in K$, and $2\lambda > 0$. So `LipschitzImageHaarBound`, applied with $A = K$, $f = H$ and the constant 2λ, gives

$$\mu\big(H(K)\big) \le (2\lambda)^n \, \mu(K). \tag{93}$$

(No measurability of K is needed: that lemma has none, and below we use only the monotonicity of μ, which holds for arbitrary sets.)

Next, $\overline{B}(0, d) \subseteq H(K)$. Indeed, let $|p| \le d$. By (2d) there is $z \in K$ for which p is a multiplier, and by Step 3 the multiplier of z is unique and equals $H(z)$; hence $p = H(z) \in H(K)$.

By the preamble's computation of the Haar measure of balls,

$$\mu\big(\overline{B}(0,d)\big) = d^n\,\mu\big(B(0,1)\big) \qquad \text{with} \qquad 0 < \mu\big(B(0,1)\big) < \infty,$$

and $d > 0$, so $\mu(\overline{B}(0,d))$ is a strictly positive (finite) number. Combining this with the monotonicity of μ along $\overline{B}(0,d) \subseteq H(K)$ and with (93),

$$0 < \mu\big(\overline{B}(0,d)\big) \le \mu\big(H(K)\big) \le (2\lambda)^n\,\mu(K).$$

The factor $(2\lambda)^n$ is a finite real number, so if $\mu(K)$ were 0 the right-hand side would be 0, contradicting the strict positivity of the left-hand side. Hence

$$\mu(K) > 0.$$

This argument is insensitive to the dimension: when $n = 0$ both d^n and $(2\lambda)^n$ equal 1, the chain reads $0 < \mu(B(0,1)) \le \mu(H(K)) \le \mu(K)$, and the conclusion $\mu(K) > 0$ is unchanged.

Finally $K \subseteq K_\delta$ by (2c), so the monotonicity of μ gives

$$\mu(K_\delta) \ge \mu(K) > 0,$$

which is the assertion of the lemma. □

2.30. `MoreauProxExists`.

Lemma 2.30 (Existence of the proximal map)**.** *Let E be a finite-dimensional real inner-product space and let $\psi : E \to \mathbb{R}$ be convex on E. Then there is $u : E \to E$ which is a proximal map for ψ in the sense of `IsMoreauProx`.*

Proof. Since E is open and ψ is convex and real-valued on it, ψ is continuous on E (preamble), and ψ has an affine minorant: there are $p_0 \in E$ and $c_0 \in \mathbb{R}$ with

$$\psi(x) \ge \langle p_0, x\rangle + c_0 \ge -|p_0|\,|x| + c_0 \qquad (x \in E) \tag{94}$$

(preamble; the second step is the Cauchy–Schwarz inequality).

Fix $y \in E$ and set $\Theta_y(x) = \psi(x) + |x-y|^2/2$, a continuous function of x. We first show that Θ_y attains its infimum over E.

By (94) and $|x-y| \ge |x| - |y|$,

$$\Theta_y(x) \ge -|p_0|\,|x| + c_0 + \frac{(|x| - |y|)^2}{2} \qquad (|x| \ge |y|).$$

The right-hand side is a quadratic polynomial in $|x|$ with positive leading coefficient $1/2$, so it tends to $+\infty$ as $|x| \to \infty$. Choose therefore $R \ge |y|$ so large that

$$-|p_0|\,s + c_0 + \frac{(s - |y|)^2}{2} > \psi(y) \qquad (s > R).$$

Then $\Theta_y(x) > \psi(y) = \Theta_y(y)$ whenever $|x| > R$. Consequently

$$\inf_E \Theta_y = \inf_{\overline{B}(0,R)} \Theta_y,$$

because every x with $|x| > R$ has $\Theta_y(x) > \Theta_y(y)$ while $y \in \overline{B}(0,R)$, so such x cannot lower the infimum below the value already achieved at y inside the ball. The set $\overline{B}(0,R)$ is closed and bounded in the finite-dimensional space E, hence compact, and it is nonempty; Θ_y is continuous, so Θ_y attains a minimum on $\overline{B}(0,R)$ at some point, which we call x_y (preamble). By the displayed identity of infima,

$$\Theta_y(x_y) \le \Theta_y(x) \qquad (x \in E). \tag{95}$$

Choosing one such minimizer x_y for each $y \in E$ defines a function $u : E \to E$, $u(y) = x_y$. By (95), for all $y, x \in E$,

$$\psi(u(y)) + \frac{|u(y) - y|^2}{2} = \Theta_y(u(y)) \le \Theta_y(x) = \psi(x) + \frac{|x-y|^2}{2},$$

which is exactly the defining property of `IsMoreauProx`. □

2.31. `MoreauProxSubgradient.`

Lemma 2.31 (Minty's characterization of the proximal point)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E and let $y, x_0 \in E$. Then the following two statements are equivalent:*

(1) *x_0 minimizes $x \mapsto \psi(x) + |x-y|^2/2$ over E, that is*

$$\psi(x_0) + \frac{|x_0 - y|^2}{2} \le \psi(x) + \frac{|x-y|^2}{2} \qquad (x \in E);$$

(2) *$y - x_0$ is a subgradient of ψ at x_0, that is*

$$\psi(x_0) + \langle y - x_0, x - x_0\rangle \le \psi(x) \qquad (x \in E).$$

Proof. Throughout we use the identity, valid for all x,

$$\frac{|x-y|^2}{2} = \frac{|(x-x_0)+(x_0-y)|^2}{2} = \frac{|x-x_0|^2}{2} + \langle x - x_0, x_0 - y\rangle + \frac{|x_0-y|^2}{2}, \tag{96}$$

which is the expansion of the square of a sum in an inner-product space.

(2) implies (1). Assume (2) and let $x \in E$. Adding $|x-y|^2/2$ to both sides of (2) and using (96),

$$\psi(x) + \frac{|x-y|^2}{2} \ge \psi(x_0) + \langle y - x_0, x - x_0\rangle + \frac{|x-x_0|^2}{2} + \langle x - x_0, x_0 - y\rangle + \frac{|x_0-y|^2}{2}.$$

The two inner products cancel, since $\langle x - x_0, x_0 - y\rangle = -\langle y - x_0, x - x_0\rangle$. Hence

$$\psi(x) + \frac{|x-y|^2}{2} \ge \psi(x_0) + \frac{|x_0-y|^2}{2} + \frac{|x-x_0|^2}{2} \ge \psi(x_0) + \frac{|x_0-y|^2}{2},$$

which is (1).

(1) implies (2). Assume (1), let $x \in E$ and let $t \in (0,1]$. Put $x_t = x_0 + t(x - x_0)$. Applying (1) at the point x_t,

$$\psi(x_0) + \frac{|x_0-y|^2}{2} \le \psi(x_t) + \frac{|x_t-y|^2}{2}. \tag{97}$$

Convexity of ψ on E with the weights $1-t$ and t gives

$$\psi(x_t) \le (1-t)\psi(x_0) + t\psi(x).$$

Also $x_t - y = (x_0 - y) + t(x - x_0)$, so

$$\frac{|x_t-y|^2}{2} = \frac{|x_0-y|^2}{2} + t\langle x_0 - y, x - x_0\rangle + \frac{t^2|x-x_0|^2}{2}.$$

Substituting both into (97) and cancelling $|x_0-y|^2/2$ from the two sides,

$$\psi(x_0) \le (1-t)\psi(x_0) + t\psi(x) + t\langle x_0 - y, x - x_0\rangle + \frac{t^2|x-x_0|^2}{2},$$

that is, after subtracting $(1-t)\psi(x_0)$,

$$t\,\psi(x_0) \le t\,\psi(x) + t\langle x_0 - y, x - x_0\rangle + \frac{t^2|x-x_0|^2}{2}.$$

Dividing by $t > 0$,

$$\psi(x_0) \le \psi(x) + \langle x_0 - y, x - x_0\rangle + \frac{t\,|x-x_0|^2}{2}.$$

The left-hand side and the first two terms on the right do not depend on t, so letting $t \downarrow 0$ gives $\psi(x_0) \le \psi(x) + \langle x_0 - y, x - x_0\rangle$, i.e. $\psi(x_0) + \langle y - x_0, x - x_0\rangle \le \psi(x)$. As x was arbitrary this is (2). □

2.32. `MoreauProxMonotone`.

Lemma 2.32 (Firm nonexpansiveness of the proximal map)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, and let $y_1, y_2, x_1, x_2 \in E$ be such that x_i minimizes $x \mapsto \psi(x) + |x - y_i|^2/2$ over E for $i = 1, 2$, that is*

$$\psi(x_i) + \frac{|x_i - y_i|^2}{2} \le \psi(x) + \frac{|x - y_i|^2}{2} \qquad (x \in E,\ i = 1, 2).$$

Then

$$|x_1 - x_2|^2 + \big|(y_1 - x_1) - (y_2 - x_2)\big|^2 \le |y_1 - y_2|^2.$$

In particular $|x_1 - x_2| \le |y_1 - y_2|$ and $|(y_1 - x_1) - (y_2 - x_2)| \le |y_1 - y_2|$, and the minimizer for a given y is unique: taking $y_1 = y_2$ forces $x_1 = x_2$.

Proof. Put $p_i = y_i - x_i$ for $i = 1, 2$.

Step 1: p_i is a subgradient of ψ at x_i. The hypothesis on x_i is exactly clause (1) of `MoreauProxSubgradient` with $y = y_i$ and $x_0 = x_i$, so clause (2) of that statement gives

$$\psi(x_i) + \langle p_i, x - x_i\rangle \le \psi(x) \qquad (x \in E,\ i = 1, 2). \tag{98}$$

Step 2: monotonicity. Apply (98) with $i = 1$ and $x = x_2$, and with $i = 2$ and $x = x_1$:

$$\psi(x_1) + \langle p_1, x_2 - x_1\rangle \le \psi(x_2), \qquad \psi(x_2) + \langle p_2, x_1 - x_2\rangle \le \psi(x_1).$$

Adding these two inequalities, the terms $\psi(x_1)$ and $\psi(x_2)$ cancel from both sides and we obtain

$$\langle p_1, x_2 - x_1\rangle + \langle p_2, x_1 - x_2\rangle \le 0,$$

that is $\langle p_1 - p_2, x_2 - x_1\rangle \le 0$, or equivalently

$$\langle x_1 - x_2, p_1 - p_2\rangle \ge 0. \tag{99}$$

Step 3: the identity. From $p_i = y_i - x_i$ we get $y_i = x_i + p_i$ and hence $y_1 - y_2 = (x_1 - x_2) + (p_1 - p_2)$. Expanding the square of this sum,

$$|y_1 - y_2|^2 = |x_1 - x_2|^2 + 2\langle x_1 - x_2, p_1 - p_2\rangle + |p_1 - p_2|^2.$$

By (99) the middle term is nonnegative, so

$$|y_1 - y_2|^2 \ge |x_1 - x_2|^2 + |p_1 - p_2|^2 = |x_1 - x_2|^2 + \big|(y_1 - x_1) - (y_2 - x_2)\big|^2,$$

which is the assertion. The two consequences follow by discarding one of the two nonnegative summands on the right, and the uniqueness statement follows by taking $y_1 = y_2$, which makes the right-hand side 0 and forces $|x_1 - x_2| = 0$. □

2.33. `MoreauEnvelopeGradient`.

Lemma 2.33 (The Moreau envelope is differentiable with gradient $y - u(y)$)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx`, and write*

$$\psi_1(y) = \psi(u(y)) + \frac{|u(y) - y|^2}{2}, \qquad v(y) = y - u(y).$$

Then for all $y, y' \in E$

$$\big|\psi_1(y') - \psi_1(y) - \langle v(y), y' - y\rangle\big| \le \tfrac{3}{2}|y' - y|^2.$$

Consequently ψ_1 is differentiable at every y with gradient $\nabla\psi_1(y) = v(y)$.

Proof. Fix $y, y' \in E$ and write $k = y' - y$.

Step 1: the upper bound. Apply the defining inequality of `IsMoreauProx` at the base point y' with the test point $x = u(y)$:

$$\psi_1(y') = \psi(u(y')) + \frac{|u(y') - y'|^2}{2} \le \psi(u(y)) + \frac{|u(y) - y'|^2}{2}.$$

Since $u(y) - y' = (u(y) - y) - k$, expanding the square gives

$$\frac{|u(y) - y'|^2}{2} = \frac{|u(y) - y|^2}{2} - \langle u(y) - y, k\rangle + \frac{|k|^2}{2} = \frac{|u(y) - y|^2}{2} + \langle v(y), k\rangle + \frac{|k|^2}{2},$$

using $v(y) = y - u(y) = -(u(y) - y)$. Adding $\psi(u(y))$ and recalling the definition of $\psi_1(y)$,

$$\psi_1(y') - \psi_1(y) - \langle v(y), k\rangle \le \frac{|k|^2}{2}. \tag{100}$$

Step 2: the same bound with the roles of y and y' exchanged. By the identical computation, with y and y' interchanged (so that k is replaced by $-k$),

$$\psi_1(y) - \psi_1(y') - \langle v(y'), -k\rangle \le \frac{|k|^2}{2},$$

that is

$$\psi_1(y') - \psi_1(y) \ge \langle v(y'), k\rangle - \frac{|k|^2}{2}. \tag{101}$$

Step 3: the lower bound. Subtracting $\langle v(y), k\rangle$ from (101),

$$\psi_1(y') - \psi_1(y) - \langle v(y), k\rangle \ge \langle v(y') - v(y), k\rangle - \frac{|k|^2}{2} \ge -\big|v(y') - v(y)\big|\,|k| - \frac{|k|^2}{2},$$

the last step by the Cauchy–Schwarz inequality. We now bound $|v(y') - v(y)|$. The point $u(y)$ minimizes $x \mapsto \psi(x) + |x - y|^2/2$ and the point $u(y')$ minimizes $x \mapsto \psi(x) + |x - y'|^2/2$, both by `IsMoreauProx`. Hence `MoreauProxMonotone`, applied with $y_1 = y'$, $y_2 = y$, $x_1 = u(y')$, $x_2 = u(y)$, gives

$$|u(y') - u(y)|^2 + \big|(y' - u(y')) - (y - u(y))\big|^2 \le |y' - y|^2,$$

and discarding the first summand, $|v(y') - v(y)| \le |k|$. Therefore

$$\psi_1(y') - \psi_1(y) - \langle v(y), k\rangle \ge -|k|^2 - \frac{|k|^2}{2} = -\tfrac{3}{2}|k|^2. \tag{102}$$

Conclusion. Combining (100) and (102), and using $\frac{1}{2} \le \frac{3}{2}$,

$$\big|\psi_1(y') - \psi_1(y) - \langle v(y), y' - y\rangle\big| \le \tfrac{3}{2}|y' - y|^2,$$

which is the stated estimate. For the final assertion, fix y; the estimate gives

$$\frac{\big|\psi_1(y + k) - \psi_1(y) - \langle v(y), k\rangle\big|}{|k|} \le \tfrac{3}{2}|k| \qquad (k \neq 0),$$

and the right-hand side tends to 0 as $k \to 0$. By the definition of the derivative this says precisely that ψ_1 is differentiable at y with derivative $k \mapsto \langle v(y), k\rangle$, i.e. with gradient $v(y)$. □

2.34. `ProxHitsGradientPoints`.

Lemma 2.34 (The proximal map hits every differentiability point)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, and let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx`. If ψ is differentiable at $x \in E$ with gradient q, then*

$$u(x + q) = x.$$

Proof. The gradient is a subgradient. Let $x' \in E$ and $t \in (0, 1]$. Since ψ is convex on E and $x + t(x' - x) = (1 - t)x + tx'$,

$$\psi\big(x + t(x' - x)\big) \le (1 - t)\psi(x) + t\psi(x'),$$

that is

$$\frac{\psi\big(x + t(x' - x)\big) - \psi(x)}{t} \le \psi(x') - \psi(x).$$

As $t \downarrow 0$ the left-hand side tends to $\langle q, x' - x\rangle$, because q is the gradient of ψ at x and therefore $\psi(x + k) - \psi(x) - \langle q, k\rangle = o(|k|)$ as $k \to 0$, applied with $k = t(x' - x)$. Passing to the limit in the displayed inequality gives

$$\psi(x) + \langle q, x' - x\rangle \le \psi(x') \qquad (x' \in E). \tag{103}$$

Minty's characterization. Put $y = x + q$, so that $y - x = q$. Then (103) states precisely that $\psi(x) + \langle y - x, x' - x\rangle \le \psi(x')$ for all $x' \in E$, which is the right-hand clause of

`MoreauProxSubgradient` with this base point y and with $x_0 = x$. By that equivalence the left-hand clause holds:

$$\psi(x) + \frac{|x-y|^2}{2} \le \psi(x') + \frac{|x'-y|^2}{2} \qquad (x' \in E), \tag{104}$$

i.e. x minimizes $x' \mapsto \psi(x') + |x'-y|^2/2$ over E.

Uniqueness of the minimizer. By `IsMoreauProx` the point $u(y)$ also minimizes $x' \mapsto \psi(x') + |x'-y|^2/2$ over E. Apply `MoreauProxMonotone` with $y_1 = y_2 = y$, $x_1 = x$ and $x_2 = u(y)$: its two hypotheses are (104) and the minimality of $u(y)$, and its conclusion reads

$$|x-u(y)|^2 + \big|(y-x)-(y-u(y))\big|^2 \le |y-y|^2 = 0.$$

Both terms on the left are nonnegative, so $|x-u(y)|^2 = 0$, that is $u(y) = x$. Since $y = x+q$, this is the assertion. □

2.35. `MoreauProxImageFullMeasure`.

Lemma 2.35 (Almost every point lies above an invertible derivative)**.** *Let E be a finite-dimensional real inner-product space equipped with its Borel σ-algebra and an additive Haar measure μ, let $\psi : E \to \mathbb{R}$ be convex on E, and let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx`. Put*

$$G = \big\{y \in E : u \text{ is differentiable at } y \text{ with derivative } P(y),\ \det P(y) \neq 0\big\}.$$

Then $\mu\big(E \setminus u(G)\big) = 0$, and the range of u is dense in E.

Proof. u is 1-Lipschitz. For any $z_1, z_2 \in E$ the points $u(z_1)$ and $u(z_2)$ minimize $x \mapsto \psi(x) + |x-z_1|^2/2$ and $x \mapsto \psi(x) + |x-z_2|^2/2$ respectively, by `IsMoreauProx`, so `MoreauProxMonotone` applied with these data gives

$$|u(z_1)-u(z_2)|^2 + \big|(z_1-u(z_1))-(z_2-u(z_2))\big|^2 \le |z_1-z_2|^2. \tag{105}$$

Dropping the second (nonnegative) term gives $|u(z_1)-u(z_2)| \le |z_1-z_2|$, so u is 1-Lipschitz on E.

The non-differentiability set and its image are null. Let $N = \{y \in E : u \text{ is not differentiable at } y\}$. Since u is 1-Lipschitz on E, Rademacher's theorem (preamble) gives $\mu(N) = 0$. The preamble's bound for the measure of a Lipschitz image, applied with source and target both equal to E and with the 1-Lipschitz map u on the null set N, then gives

$$\mu\big(u(N)\big) = 0. \tag{106}$$

The singular part and its image are null. Put $\Delta = E \setminus N$ and let $P(y)$ denote the derivative of u at $y \in \Delta$, and set

$$S = \{y \in \Delta : \det P(y) = 0\}, \qquad G = \{y \in \Delta : \det P(y) \neq 0\}.$$

By the preamble, Δ is a Borel set and $y \mapsto P(y)$ is Borel measurable on it; the determinant is a continuous function of a linear map (preamble), so S, being the intersection of Δ with the preimage of the closed set $\{0\}$ under a Borel map, is Borel. For $y \in S$ the map u is differentiable at y with derivative $P(y)$, hence a fortiori has derivative $P(y)$ at y within S. The area-formula inequality of the preamble, applied with the measurable set S, the map u and the derivative field P, therefore gives

$$\mu\big(u(S)\big) \le \int_S \big|\det P(y)\big|\, d\mu(y) = 0, \tag{107}$$

since the integrand vanishes identically on S by the definition of S.

Almost every point is a differentiability point of ψ, hence in the range of u. Let $D = \{x \in E : \psi \text{ is differentiable at } x\}$. The function ψ is convex and real-valued on the open set E, hence locally Lipschitz there (preamble), so Rademacher's theorem (preamble) gives

$$\mu(E \setminus D) = 0. \tag{108}$$

If $x \in D$ with gradient $q = \nabla\psi(x)$, then `ProxHitsGradientPoints` gives $u(x+q) = x$; in particular x lies in the range of u. Hence $D \subseteq u(E)$.

Conclusion. Since $E = N \cup S \cup G$ we have $u(E) = u(N) \cup u(S) \cup u(G)$, so every point of D outside $u(G)$ lies in $u(N) \cup u(S)$, and therefore

$$E \setminus u(G) \subseteq (E \setminus D) \cup \big(D \setminus u(G)\big) \subseteq (E \setminus D) \cup u(N) \cup u(S).$$

By (108), (106) and (107) each of the three sets on the right is μ-null, so their union is μ-null and

$$\mu\big(E \setminus u(G)\big) = 0.$$

Finally, $u(E)$ contains D, so $E \setminus u(E) \subseteq E \setminus D$ is μ-null by (108). Now $E \setminus u(E)$ has empty interior: if it had an interior point z, it would contain a closed ball $\overline{B}(z, r)$ with $r > 0$, and $\mu(\overline{B}(z, r)) > 0$ by the preamble's computation of the Haar measure of balls, which together with the monotonicity of μ contradicts $\mu(E \setminus u(E)) = 0$. A set whose complement has empty interior is dense, so $u(E)$ is dense in E. □

2.36. `GradientDiffSecondOrder.`

Lemma 2.36 (Second-order expansion from a differentiable gradient)**.** *Let E be a finite-dimensional real inner-product space, let $F : E \to \mathbb{R}$ and let $w : E \to E$. Assume that for every $z \in E$ the vector $w(z)$ is the gradient of F at z, that w is continuous on E, and that w is differentiable at a point $y \in E$ with derivative the continuous linear map $Q : E \to E$. Then for every $\delta > 0$ there is $\rho > 0$ such that*

$$\Big|F(y + h) - F(y) - \langle w(y), h\rangle - \tfrac{1}{2}\langle Qh, h\rangle\Big| \le \delta|h|^2 \qquad (|h| < \rho).$$

Proof. Write $p = w(y)$ and

$$r(k) = w(y + k) - w(y) - Qk \qquad (k \in E),$$

so that $r(0) = 0$ and, by the assumed differentiability of w at y with derivative Q, $|r(k)|/|k| \to 0$ as $k \to 0$.

Since w is the gradient of F at every point and w is continuous, F is C^1 on E, so integrating the gradient along the segment from y to $y + h$ (preamble) gives, for every $h \in E$,

$$F(y + h) - F(y) = \int_0^1 \langle w(y + th), h\rangle \, dt.$$

The integrand is continuous in t because w is continuous and the inner product is, so the integral exists. By the definition of r applied with $k = th$,

$$\langle w(y + th), h\rangle = \langle p, h\rangle + \langle Q(th), h\rangle + \langle r(th), h\rangle = \langle p, h\rangle + t\langle Qh, h\rangle + \langle r(th), h\rangle,$$

using the linearity of Q and of the inner product in its first argument. Since $\int_0^1 dt = 1$ and $\int_0^1 t\, dt = \frac{1}{2}$, integrating termwise over $t \in [0, 1]$ gives

$$F(y + h) - F(y) = \langle p, h\rangle + \tfrac{1}{2}\langle Qh, h\rangle + \int_0^1 \langle r(th), h\rangle \, dt. \tag{109}$$

By the Cauchy–Schwarz inequality the last integrand is bounded in absolute value by $|r(th)|\,|h|$, so

$$\Big|\int_0^1 \langle r(th), h\rangle \, dt\Big| \le |h| \sup_{0 \le t \le 1} |r(th)|. \tag{110}$$

Now let $\delta > 0$. Because $|r(k)|/|k| \to 0$ as $k \to 0$ and $r(0) = 0$, there is $\rho > 0$ such that $|r(k)| \le \delta|k|$ whenever $|k| < \rho$; the case $k = 0$ is included since $r(0) = 0$. Let $|h| < \rho$ and $t \in [0, 1]$. Then $|th| = t|h| \le |h| < \rho$, so $|r(th)| \le \delta|th| = \delta t|h| \le \delta|h|$. Hence $\sup_{0 \le t \le 1} |r(th)| \le \delta|h|$, and (110) bounds the error term in (109) by $\delta|h|^2$. Rearranging (109) gives

$$\Big|F(y + h) - F(y) - \langle p, h\rangle - \tfrac{1}{2}\langle Qh, h\rangle\Big| \le \delta|h|^2 \qquad (|h| < \rho),$$

which is the assertion since $p = w(y)$. □

2.37. `MoreauCurveExpansion`.

Lemma 2.37 (Second-order expansion along the proximal curve)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, and let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx`. Let $y \in E$ be a point at which u is differentiable with derivative the continuous linear map $P : E \to E$, and write $Q = \mathrm{id} - P$. Then for every $\delta > 0$ there is $\rho > 0$ such that, with $x_h = u(y + h)$,*

$$\Big|\psi(x_h) - \psi(u(y)) - \langle y - u(y), x_h - u(y)\rangle - \tfrac{1}{2}\langle Qh, Ph\rangle\Big| \le \delta|h|^2 \qquad (|h| < \rho).$$

Proof. Write $x = u(y)$, $v(z) = z - u(z)$, $p = v(y) = y - x$, and

$$\psi_1(z) = \psi(u(z)) + \frac{|u(z) - z|^2}{2} = \psi(u(z)) + \frac{|v(z)|^2}{2}.$$

The hypotheses of the gradient lemma hold for ψ_1 and v. By `MoreauEnvelopeGradient`, for all $z, z' \in E$

$$\big|\psi_1(z') - \psi_1(z) - \langle z - u(z), z' - z\rangle\big| \le \tfrac{3}{2}|z' - z|^2,$$

and the right-hand side is $o(|z' - z|)$ as $z' \to z$; hence $v(z) = z - u(z)$ is the gradient of ψ_1 at z, for every $z \in E$. Next, for any $z_1, z_2 \in E$ the points $u(z_1)$ and $u(z_2)$ minimize $x' \mapsto \psi(x') + |x' - z_1|^2/2$ and $x' \mapsto \psi(x') + |x' - z_2|^2/2$ respectively, by `IsMoreauProx`, so `MoreauProxMonotone` applied with these data gives

$$|u(z_1) - u(z_2)|^2 + |v(z_1) - v(z_2)|^2 \le |z_1 - z_2|^2. \tag{111}$$

In particular $|v(z_1) - v(z_2)| \le |z_1 - z_2|$, so v is 1-Lipschitz and hence continuous on E. Finally, $v = \mathrm{id} - u$ and u is differentiable at y with derivative P, so v is differentiable at y with derivative $Q = \mathrm{id} - P$, a continuous linear map.

The expansion of ψ_1. The three facts just established are exactly the hypotheses of `GradientDiffSecondOrde` with $F = \psi_1$, $w = v$ and this Q. That lemma therefore provides, for every $\delta' > 0$, a $\rho' > 0$ with

$$\big|\psi_1(y + h) - \psi_1(y) - \langle p, h\rangle - \tfrac{1}{2}\langle Qh, h\rangle\big| \le \delta'|h|^2 \qquad (|h| < \rho'). \tag{112}$$

The exact identity for $v(y + h)$. Put $\varrho(h) = x_h - x - Ph$, so that $|\varrho(h)|/|h| \to 0$ as $h \to 0$ by the differentiability of u at y, and $\varrho(0) = 0$. Since $Ph + Qh = h$ and $y - x = p$,

$$v(y + h) = (y + h) - x_h = (y - x) + h - (x_h - x) = p + h - Ph - \varrho(h) = p + Qh - \varrho(h). \tag{113}$$

Assembling. By `IsMoreauProx` the proximal point of the base point $y + h$ is $x_h = u(y + h)$, so by the definition of ψ_1, $\psi(x_h) = \psi_1(y + h) - \frac{1}{2}|v(y + h)|^2$; likewise $\psi_1(y) = \psi(x) + \frac{1}{2}|p|^2$. Expanding the square using (113) and the bilinearity of the inner product,

$$\tfrac{1}{2}|v(y + h)|^2 = \tfrac{1}{2}|p|^2 + \langle p, Qh - \varrho(h)\rangle + \tfrac{1}{2}|Qh - \varrho(h)|^2,$$

$$\tfrac{1}{2}|Qh - \varrho(h)|^2 = \tfrac{1}{2}|Qh|^2 - \langle Qh, \varrho(h)\rangle + \tfrac{1}{2}|\varrho(h)|^2.$$

Writing $e_1(h)$ for the error in (112), that is $e_1(h) = \psi_1(y + h) - \psi_1(y) - \langle p, h\rangle - \frac{1}{2}\langle Qh, h\rangle$, and $e_2(h) = \langle Qh, \varrho(h)\rangle - \frac{1}{2}|\varrho(h)|^2$, we get

$$\begin{aligned}
\psi(x_h) &= \Big[\psi(x) + \tfrac{1}{2}|p|^2 + \langle p, h\rangle + \tfrac{1}{2}\langle Qh, h\rangle + e_1(h)\Big] \\
&\quad - \Big[\tfrac{1}{2}|p|^2 + \langle p, Qh\rangle - \langle p, \varrho(h)\rangle + \tfrac{1}{2}|Qh|^2 - \langle Qh, \varrho(h)\rangle + \tfrac{1}{2}|\varrho(h)|^2\Big] \\
&= \psi(x) + \langle p, h - Qh + \varrho(h)\rangle + \tfrac{1}{2}\langle Qh, h - Qh\rangle + e_1(h) + e_2(h) \\
&= \psi(x) + \langle p, x_h - x\rangle + \tfrac{1}{2}\langle Qh, Ph\rangle + e_1(h) + e_2(h),
\end{aligned}$$

where the last line uses $h - Qh = Ph$, hence $h - Qh + \varrho(h) = Ph + \varrho(h) = x_h - x$ and $\langle Qh, h - Qh\rangle = \langle Qh, Ph\rangle$. Therefore

$$\psi(x_h) - \psi(x) - \langle p, x_h - x\rangle - \tfrac{1}{2}\langle Qh, Ph\rangle = e_1(h) + e_2(h). \tag{114}$$

Choice of ρ. Let $\delta > 0$. Choose $\delta' > 0$ so small that $\|Q\|\delta' + \frac{1}{2}\delta'^2 \le \delta/2$; this is possible because the left-hand side tends to 0 as $\delta' \downarrow 0$. By the differentiability of u at y there is $\rho'' > 0$

with $|\varrho(h)| \le \delta'|h|$ for $|h| < \rho''$. For such h, the Cauchy–Schwarz inequality and $|Qh| \le \|Q\|\,|h|$ give

$$|e_2(h)| \le |Qh|\,|\varrho(h)| + \tfrac{1}{2}|\varrho(h)|^2 \le \|Q\|\,|h| \cdot \delta'|h| + \tfrac{1}{2}\delta'^2|h|^2 \le \frac{\delta}{2}|h|^2.$$

Apply (112) with $\delta/2$ in place of δ' to obtain $\rho' > 0$ with $|e_1(h)| \le (\delta/2)|h|^2$ for $|h| < \rho'$. Put $\rho = \min(\rho', \rho'') > 0$. For $|h| < \rho$ both bounds apply, so by (114) the left-hand side of the assertion is at most $|e_1(h)| + |e_2(h)| \le \delta|h|^2$, as required. □

2.38. `ProxSubdifferentialSingleton`.

Lemma 2.38 (The subdifferential is a singleton above an invertible derivative)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, and let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx`. Let $y \in E$ be a point at which u is differentiable with derivative the continuous linear map P, and suppose $\det P \neq 0$. If $s \in E$ satisfies*

$$\psi(u(y)) + \langle s, x - u(y)\rangle \le \psi(x) \qquad (x \in E),$$

then $s = y - u(y)$.

Proof. Write $x = u(y)$ and $p = y - x$, and put $d = s - p$.

p itself satisfies the hypothesis. By `IsMoreauProx` the point $x = u(y)$ minimizes $x' \mapsto \psi(x') + |x' - y|^2/2$ over E. That is the left-hand clause of `MoreauProxSubgradient` with base point y and $x_0 = x$, so the right-hand clause holds:

$$\psi(x) + \langle y - x, x' - x\rangle \le \psi(x') \qquad (x' \in E), \tag{115}$$

i.e. $p = y - x$ satisfies the same inequality as s.

Every point of the segment from p to s satisfies it too. Let $t \in [0,1]$. Multiplying the hypothesis on s by $t \ge 0$, multiplying (115) by $1 - t \ge 0$, and adding, we get for every $x' \in E$

$$\psi(x) + \langle (1-t)p + ts,\, x' - x\rangle \le \psi(x'),$$

because the two left-hand sides combine by linearity of the inner product in its first argument and the two copies of $\psi(x)$ and of $\psi(x')$ combine with coefficients summing to 1. Since $(1-t)p + ts = p + td$, this reads

$$\psi(x) + \langle p + td,\, x' - x\rangle \le \psi(x') \qquad (x' \in E,\ t \in [0,1]). \tag{116}$$

The proximal map is constant along that segment. Fix $t \in [0,1]$ and put $y_t = x + p + td = y + td$, so that $y_t - x = p + td$. Then (116) is exactly the right-hand clause of `MoreauProxSubgradient` with base point y_t and $x_0 = x$, so the left-hand clause holds: x minimizes $x' \mapsto \psi(x') + |x' - y_t|^2/2$ over E. By `IsMoreauProx` the point $u(y_t)$ minimizes the same function. Apply `MoreauProxMonotone` with $y_1 = y_2 = y_t$, $x_1 = x$ and $x_2 = u(y_t)$: its two hypotheses are the two minimality statements just obtained, and its conclusion gives

$$|x - u(y_t)|^2 + \big|(y_t - x) - (y_t - u(y_t))\big|^2 \le |y_t - y_t|^2 = 0,$$

whence $u(y_t) = x$. Thus

$$u(y + td) - u(y) = x - x = 0 \qquad (t \in [0,1]). \tag{117}$$

Conclusion. Suppose $d \neq 0$. Dividing (117) by $t > 0$ and letting $t \downarrow 0$, the differentiability of u at y with derivative P gives

$$Pd = \lim_{t \downarrow 0} \frac{u(y + td) - u(y)}{t} = 0.$$

Since $d \neq 0$, the linear map P is then not injective, hence not bijective, so $\det P = 0$ (preamble: for a linear endomorphism of a finite-dimensional space, $\det P \neq 0$ if and only if P is bijective). This contradicts the hypothesis $\det P \neq 0$. Therefore $d = 0$, that is $s = p = y - u(y)$. □

2.39. `ProxPreimageLocalization`.

Lemma 2.39 (Uniform localization of proximal preimages)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, and let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx`. Let $y \in E$ be a point at which u is differentiable with derivative the continuous linear map P, and suppose $\det P \neq 0$. Then for every $\sigma > 0$ there is $\rho > 0$ such that*

$$y' \in E, \quad |u(y') - u(y)| < \rho \quad \Longrightarrow \quad |y' - y| < \sigma.$$

Proof. Write $x = u(y)$ and $p = y - x$. Suppose the assertion fails for some $\sigma > 0$. Then for every integer $j \geq 0$ the choice $\rho = 1/(j+1)$ fails, so there is $y'_j \in E$ with

$$|u(y'_j) - x| < \frac{1}{j+1}, \qquad |y'_j - y| \geq \sigma. \tag{118}$$

Put $w_j = u(y'_j)$, so $w_j \to x$, and $s_j = y'_j - w_j$.

Each s_j is a subgradient at w_j. By `IsMoreauProx` the point $w_j = u(y'_j)$ minimizes $x' \mapsto \psi(x') + |x' - y'_j|^2/2$ over E. That is the left-hand clause of `MoreauProxSubgradient` with base point y'_j and $x_0 = w_j$, so the right-hand clause holds, and since $y'_j - w_j = s_j$ it reads

$$\psi(w_j) + \langle s_j, x' - w_j \rangle \leq \psi(x') \qquad (x' \in E). \tag{119}$$

The s_j are bounded. The function ψ is convex and real-valued on the open set E, hence locally Lipschitz there (preamble): there are $a > 0$ and $L \geq 0$ with $|\psi(b) - \psi(b')| \leq L|b - b'|$ for all b, b' in the closed ball $\overline{B}(x, 2a)$. Since $w_j \to x$ there is J with $|w_j - x| \leq a$ for all $j \geq J$. Fix such a j, let $e \in E$ with $|e| = 1$ and let $0 < t \leq a$. Then $|w_j + te - x| \leq |w_j - x| + t \leq 2a$, so $w_j + te \in \overline{B}(x, 2a)$, and (119) with $x' = w_j + te$ gives

$$t\langle s_j, e \rangle = \langle s_j, (w_j + te) - w_j \rangle \leq \psi(w_j + te) - \psi(w_j) \leq L\,|te| = Lt.$$

Dividing by $t > 0$ gives $\langle s_j, e \rangle \leq L$ for every unit vector e. Taking $e = s_j/|s_j|$ when $s_j \neq 0$ yields $|s_j| \leq L$; and $|s_j| \leq L$ trivially when $s_j = 0$. So $|s_j| \leq L$ for all $j \geq J$.

Passing to a limit. Consequently $y'_j = w_j + s_j$ is a bounded sequence for $j \geq J$, since $w_j \to x$ is bounded. By Bolzano–Weierstrass (preamble) there is a subsequence (y'_{j_i}) converging to some $\bar{y} \in E$; passing to the limit in the second part of (118) gives $|\bar{y} - y| \geq \sigma$, so in particular

$$\bar{y} \neq y. \tag{120}$$

Along this subsequence $w_{j_i} \to x$, hence $s_{j_i} = y'_{j_i} - w_{j_i} \to \bar{y} - x =: \bar{s}$. Now fix $x' \in E$ and pass to the limit in (119) along the subsequence: ψ is continuous on E, being locally Lipschitz there (preamble), so $\psi(w_{j_i}) \to \psi(x)$, and the inner product is continuous in both arguments, so $\langle s_{j_i}, x' - w_{j_i} \rangle \to \langle \bar{s}, x' - x \rangle$. Therefore

$$\psi(x) + \langle \bar{s}, x' - x \rangle \leq \psi(x') \qquad (x' \in E).$$

Conclusion. This is precisely the hypothesis of `ProxSubdifferentialSingleton` on the vector $\bar{s}$, at the same point y with the same P and $\det P \neq 0$. That lemma gives $\bar{s} = y - u(y) = p$, and hence $\bar{y} = x + \bar{s} = x + p = y$, contradicting (120). So the assertion cannot fail, which proves the lemma. □

2.40. `ProxQuadraticFormComparison`.

Lemma 2.40 (Comparison of the two quadratic forms)**.** *Let E be a finite-dimensional real inner-product space and let $P, P^{-1} : E \to E$ be continuous linear maps with $P^{-1}(Pz) = z$ for all $z \in E$. Write $Q = \mathrm{id} - P$ and $A_0 = QP^{-1}$. Let $c > 0$ satisfy $c|z| \leq |Pz|$ for all $z \in E$, let δ_1 satisfy $0 < \delta_1 \leq c/2$, and let $h, k \in E$ satisfy $|k - Ph| \leq \delta_1 |h|$. Then*

$$|h| \leq \frac{2}{c}|k| \qquad \text{and} \qquad \big|\langle Qh, Ph \rangle - \langle A_0 k, k \rangle\big| \leq C_1 \delta_1 |k|^2,$$

where

$$C_1 = \frac{4\,\|Q\|\,\|P^{-1}\|\,\|P\|}{c^2} + \frac{2\,\|A_0\|}{c}.$$

Proof. The bound on $|h|$. By the triangle inequality, the lower bound on P and the hypothesis on k,

$$|k| \geq |Ph| - |k - Ph| \geq c|h| - \delta_1|h| = (c - \delta_1)|h| \geq \frac{c}{2}|h|,$$

the last step because $\delta_1 \leq c/2$. Since $c > 0$ this gives

$$|h| \leq \frac{2}{c}|k|. \tag{121}$$

Two consequences. Combining (121) with the hypothesis $|k - Ph| \leq \delta_1|h|$,

$$|k - Ph| \leq \delta_1|h| \leq \frac{2\delta_1}{c}|k|. \tag{122}$$

Applying P^{-1} to $Ph - k$ and using $P^{-1}(Ph) = h$ together with the operator norm bound,

$$\big|h - P^{-1}k\big| = \big|P^{-1}(Ph) - P^{-1}k\big| = \big|P^{-1}(Ph - k)\big| \leq \|P^{-1}\|\,|Ph - k| \leq \|P^{-1}\|\frac{2\delta_1}{c}|k|, \tag{123}$$

using (122) and $|Ph - k| = |k - Ph|$ at the last step.

The algebraic identity. Inserting and removing the term $\langle QP^{-1}k, Ph\rangle$, and using $A_0 = QP^{-1}$ and the bilinearity of the inner product,

$$\begin{aligned}\langle Qh, Ph\rangle - \langle A_0k, k\rangle &= \big[\langle Qh, Ph\rangle - \langle QP^{-1}k, Ph\rangle\big] + \big[\langle QP^{-1}k, Ph\rangle - \langle QP^{-1}k, k\rangle\big] \\ &= \big\langle Q\big(h - P^{-1}k\big), Ph\big\rangle + \big\langle QP^{-1}k,\, Ph - k\big\rangle,\end{aligned}$$

where the first bracket uses the linearity of Q and the second the linearity of the inner product in its second argument.

Estimating the two terms. From (121) and the operator norm bound, $|Ph| \leq \|P\|\,|h| \leq (2\|P\|/c)|k|$. Hence by the Cauchy–Schwarz inequality, the operator norm bound for Q and (123),

$$\big|\big\langle Q\big(h - P^{-1}k\big), Ph\big\rangle\big| \leq \|Q\|\,\big|h - P^{-1}k\big| \cdot |Ph| \leq \|Q\| \cdot \|P^{-1}\|\frac{2\delta_1}{c}|k| \cdot \frac{2\|P\|}{c}|k| = \frac{4\,\|Q\|\,\|P^{-1}\|\,\|P\|}{c^2}\,\delta_1|k|^2.$$

Likewise, by Cauchy–Schwarz, $|A_0k| \leq \|A_0\|\,|k|$ and (122),

$$\big|\big\langle QP^{-1}k,\, Ph - k\big\rangle\big| = \big|\langle A_0k,\, Ph - k\rangle\big| \leq \|A_0\|\,|k| \cdot \frac{2\delta_1}{c}|k| = \frac{2\,\|A_0\|}{c}\,\delta_1|k|^2.$$

Adding the two estimates and using the triangle inequality on the identity above gives

$$\big|\langle Qh, Ph\rangle - \langle A_0k, k\rangle\big| \leq \Big(\frac{4\,\|Q\|\,\|P^{-1}\|\,\|P\|}{c^2} + \frac{2\,\|A_0\|}{c}\Big)\delta_1|k|^2 = C_1\delta_1|k|^2,$$

which together with (121) is the assertion. □

2.41. `ProxEnvelopeExpansionAt.`

Lemma 2.41 (Second-order expansion of a convex function above an invertible derivative)**.** *Let E be a finite-dimensional real inner-product space, let $\psi : E \to \mathbb{R}$ be convex on E, and let $u : E \to E$ be a proximal map for ψ in the sense of `IsMoreauProx` whose range is dense in E. Let $y \in E$ be a point at which u is differentiable with derivative the continuous linear map P, let $P^{-1} : E \to E$ be a continuous linear map with $P^{-1}(Pz) = z$ and $P(P^{-1}z) = z$ for all $z \in E$, and suppose* $\det P \neq 0$. *Write $Q = \mathrm{id} - P$ and $A_0 = QP^{-1}$. Then for every $\delta > 0$ there is $\rho > 0$ such that*

$$\Big|\psi\big(u(y) + k\big) - \psi\big(u(y)\big) - \langle y - u(y), k\rangle - \tfrac{1}{2}\langle A_0k, k\rangle\Big| \leq \delta|k|^2 \qquad (|k| < \rho).$$

Proof. Write $x = u(y)$, $p = y - x$, $x_h = u(y + h)$ and

$$g(k) = \psi(x + k) - \psi(x) - \langle p, k\rangle \qquad (k \in E),$$

so that the assertion is that for every $\delta > 0$ there is $\rho > 0$ with $|g(k) - \frac{1}{2}\langle A_0k, k\rangle| \leq \delta|k|^2$ for $|k| < \rho$. Note that g is continuous on E, because ψ is convex and real-valued on the open set E and hence locally Lipschitz, so continuous (preamble), and the inner product is continuous.

A lower bound for P. Since $P^{-1}(Pz) = z$ for every z, the operator norm bound gives $|z| = |P^{-1}(Pz)| \leq \|P^{-1}\|\,|Pz|$. The map P^{-1} is not the zero map when $E \neq \{0\}$, and in any

case $\|P^{-1}\| \geq 0$; if $\|P^{-1}\| = 0$ then $P^{-1} = 0$, so $z = P^{-1}(Pz) = 0$ for every z, i.e. $E = \{0\}$, and then the assertion is trivial because every vector is 0 and both sides vanish. So we may assume $\|P^{-1}\| > 0$ and put $c = 1/\|P^{-1}\| > 0$; the displayed inequality rearranges to

$$c|z| \leq |Pz| \qquad (z \in E). \tag{124}$$

Choice of the parameters. Let $\delta > 0$. Put

$$C_1 = \frac{4\,\|Q\|\,\|P^{-1}\|\,\|P\|}{c^2} + \frac{2\,\|A_0\|}{c},$$

choose δ_1 with $0 < \delta_1 \leq \min(1, c/2)$ and $\frac{1}{2}C_1\delta_1 \leq \delta/2$, which is possible since $C_1 \geq 0$ is a fixed finite number, and choose $\delta_2 > 0$ with $4\delta_2/c^2 \leq \delta/2$. By `MoreauCurveExpansion`, applied at this y with this P and with δ_2 in place of δ, there is $\rho_1 > 0$ with

$$\left|\psi(x_h) - \psi(x) - \langle p, x_h - x\rangle - \tfrac{1}{2}\langle Qh, Ph\rangle\right| \leq \delta_2|h|^2 \qquad (|h| < \rho_1). \tag{125}$$

By the differentiability of u at y with derivative P there is $\rho_2 > 0$ with

$$|x_h - x - Ph| \leq \delta_1|h| \qquad (|h| < \rho_2). \tag{126}$$

By `ProxPreimageLocalization`, applied at this y with this P and $\det P \neq 0$ and with $\sigma = \min(\rho_1, \rho_2) > 0$, there is $\rho_3 > 0$ such that

$$y' \in E,\ |u(y') - x| < \rho_3 \implies |y' - y| < \sigma. \tag{127}$$

The estimate at points of the range. Let $k \in E$ with $|k| < \rho_3$ and suppose first that $x + k$ lies in the range of u, say $u(y') = x + k$. Put $h = y' - y$. Since $|u(y') - x| = |k| < \rho_3$, (127) gives $|h| < \sigma \leq \min(\rho_1, \rho_2)$, and $x_h = u(y + h) = u(y') = x + k$, that is

$$x_h - x = k. \tag{128}$$

By (126) and (128) we have $|k - Ph| \leq \delta_1|h|$. The hypotheses of `ProxQuadraticFormComparison` are therefore met for this P, P^{-1}, c, δ_1, h and k — namely $P^{-1}(Pz) = z$, the lower bound (124), $0 < \delta_1 \leq c/2$, and $|k - Ph| \leq \delta_1|h|$ — and that lemma gives

$$|h| \leq \frac{2}{c}|k| \qquad \text{and} \qquad \left|\langle Qh, Ph\rangle - \langle A_0k, k\rangle\right| \leq C_1\delta_1|k|^2. \tag{129}$$

On the other hand (125) together with (128) and the first part of (129) gives

$$\left|g(k) - \tfrac{1}{2}\langle Qh, Ph\rangle\right| = \left|\psi(x_h) - \psi(x) - \langle p, x_h - x\rangle - \tfrac{1}{2}\langle Qh, Ph\rangle\right| \leq \delta_2|h|^2 \leq \frac{4\delta_2}{c^2}|k|^2, \tag{130}$$

where the first equality uses (128) and the definition of g. Adding half of the second part of (129) to (130), by the triangle inequality,

$$\left|g(k) - \tfrac{1}{2}\langle A_0k, k\rangle\right| \leq \frac{4\delta_2}{c^2}|k|^2 + \tfrac{1}{2}C_1\delta_1|k|^2 \leq \frac{\delta}{2}|k|^2 + \frac{\delta}{2}|k|^2 = \delta|k|^2, \tag{131}$$

by the choices of δ_1 and δ_2. So the asserted inequality holds for every k with $|k| < \rho_3$ such that $x + k$ lies in the range of u.

Removing the restriction, by density. Put $T = \{k \in E : |k| < \rho_3,\ x + k \text{ lies in the range of } u\}$. We claim T is dense in the open ball $B(0, \rho_3)$. Let $k_0 \in B(0, \rho_3)$ and let $\varepsilon > 0$; shrinking ε we may assume $B(k_0, \varepsilon) \subseteq B(0, \rho_3)$, which is possible because $B(0, \rho_3)$ is open. The set $x + B(k_0, \varepsilon)$ is open and nonempty, so by the assumed density of the range of u it contains a point $u(y')$ of that range. Writing $u(y') = x + k$ we get $k \in B(k_0, \varepsilon) \subseteq B(0, \rho_3)$ and $x + k$ in the range of u, so $k \in T$ and $|k - k_0| < \varepsilon$. This proves the claim.

Both $k \mapsto \left|g(k) - \frac{1}{2}\langle A_0k, k\rangle\right|$ and $k \mapsto \delta|k|^2$ are continuous on E: the first because g is continuous, A_0 is continuous linear, the inner product is continuous and the absolute value is continuous; the second because the norm is. By (131) the first is bounded by the second at every point of T. Let $k_0 \in B(0, \rho_3)$. By the density just proved there is a sequence $k_i \in T$ with $k_i \to k_0$, and passing to the limit in $\left|g(k_i) - \frac{1}{2}\langle A_0k_i, k_i\rangle\right| \leq \delta|k_i|^2$, using the continuity of both sides and the fact that weak inequalities are preserved in the limit, gives $\left|g(k_0) - \frac{1}{2}\langle A_0k_0, k_0\rangle\right| \leq \delta|k_0|^2$.

Thus the asserted inequality holds for every k with $|k| < \rho_3$, and $\rho = \rho_3$ proves the lemma. □

2.42. `ConvexSecondOrderAe`.

Theorem 2.42 (Alexandrov's theorem)**.** *Let E be a finite-dimensional real inner-product space with an additive Haar measure μ, and let $\psi : E \to \mathbb{R}$ be convex on E. Then for μ-almost every $x \in E$ there are a vector $q \in E$ and a symmetric continuous bilinear form A on E with $A(h,h) \geq 0$ for all $h \in E$, such that for every $\delta > 0$ there is $\rho > 0$ with*

$$\left|\psi(x+h) - \psi(x) - \langle q, h\rangle - \tfrac{1}{2}A(h,h)\right| \leq \delta|h|^2 \qquad (|h| < \rho).$$

Proof. Notation and standing facts. By `MoreauProxExists` fix $u : E \to E$ which is a proximal map for ψ in the sense of `IsMoreauProx`, and set

$$v(y) = y - u(y), \qquad \psi_1(y) = \psi(u(y)) + \frac{|u(y)-y|^2}{2} = \psi(u(y)) + \frac{|v(y)|^2}{2}.$$

For any $y_1, y_2 \in E$ the points $u(y_1)$ and $u(y_2)$ minimize $x \mapsto \psi(x) + |x - y_1|^2/2$ and $x \mapsto \psi(x) + |x - y_2|^2/2$ respectively, by `IsMoreauProx`, so `MoreauProxMonotone` gives

$$|u(y_1) - u(y_2)|^2 + |v(y_1) - v(y_2)|^2 \leq |y_1 - y_2|^2. \tag{132}$$

In particular u and v are 1-Lipschitz on E, and for each y the minimizer of $x \mapsto \psi(x)+|x-y|^2/2$ is unique and equals $u(y)$ (take $y_1 = y_2$ in `MoreauProxMonotone`). By `MoreauEnvelopeGradient`, ψ_1 is differentiable at every y with $\nabla\psi_1(y) = v(y)$; since v is 1-Lipschitz, $\nabla\psi_1$ is continuous, so ψ_1 is C^1. Finally, the hypothesis of `IsMoreauProx` at the base point y is clause (1) of `MoreauProxSubgradient` with $x_0 = u(y)$, so clause (2) of that statement gives

$$\psi(u(y)) + \langle v(y), x - u(y)\rangle \leq \psi(x) \qquad (x, y \in E). \tag{133}$$

Step 1: a second-order expansion of ψ_1 where u is differentiable. Let y be a point at which u is differentiable, let $P = Du(y)$, and put $Q = \mathrm{id} - P$, $x = u(y)$, $p = v(y)$. Since $v = \mathrm{id} - u$, the map v is differentiable at y with $Dv(y) = Q$; write

$$r(k) = v(y+k) - v(y) - Qk, \qquad \text{so} \qquad \frac{|r(k)|}{|k|} \to 0 \ \ (k \to 0), \quad r(0) = 0. \tag{134}$$

We claim: for every $\delta > 0$ there is $\rho > 0$ with

$$\left|\psi_1(y+h) - \psi_1(y) - \langle p, h\rangle - \tfrac{1}{2}\langle Qh, h\rangle\right| \leq \delta|h|^2 \qquad (|h| < \rho). \tag{135}$$

Indeed ψ_1 is C^1 with $\nabla\psi_1 = v$, so integrating the gradient along the segment from y to $y + h$ (preamble) gives

$$\psi_1(y+h) - \psi_1(y) = \int_0^1 \langle v(y+th), h\rangle\, dt.$$

By (134) with $k = th$, $\langle v(y+th), h\rangle = \langle p, h\rangle + t\langle Qh, h\rangle + \langle r(th), h\rangle$, and $\int_0^1 t\, dt = \frac{1}{2}$, so

$$\psi_1(y+h) - \psi_1(y) = \langle p, h\rangle + \tfrac{1}{2}\langle Qh, h\rangle + \int_0^1 \langle r(th), h\rangle\, dt,$$

and the last integral has absolute value at most $|h| \sup_{0\leq t\leq 1} |r(th)|$. Given $\delta > 0$ choose $\rho > 0$ with $|r(k)| \leq \delta|k|$ for $|k| < \rho$; then for $|h| < \rho$ and $t \in [0,1]$ we have $|th| \leq |h| < \rho$, hence $|r(th)| \leq \delta|th| \leq \delta|h|$, and the integral is bounded by $\delta|h|^2$. This proves (135).

Step 2: the curve identity. Keep y, P, Q, x, p as in Step 1 and set, for $h \in E$,

$$x_h = u(y+h), \qquad \varrho(h) = x_h - x - Ph,$$

so that $|\varrho(h)|/|h| \to 0$ as $h \to 0$ by the differentiability of u at y. Because $Ph + Qh = h$ and $y - x = p$, we have the *exact* identity

$$v(y+h) = (y+h) - x_h = (y-x) + h - (x_h - x) = p + h - Ph - \varrho(h) = p + Qh - \varrho(h). \tag{136}$$

We claim: for every $\delta > 0$ there is $\rho_1 > 0$ with

$$\left|\psi(x_h) - \psi(x) - \langle p, x_h - x\rangle - \tfrac{1}{2}\langle Qh, Ph\rangle\right| \leq \delta|h|^2 \qquad (|h| < \rho_1). \tag{137}$$

Indeed, by the definition of ψ_1 at the base point $y+h$ (whose proximal point is x_h) we have $\psi(x_h) = \psi_1(y+h) - \frac{1}{2}|v(y+h)|^2$, and by (136)

$$\tfrac{1}{2}|v(y+h)|^2 = \tfrac{1}{2}|p|^2 + \langle p, Qh - \varrho(h)\rangle + \tfrac{1}{2}|Qh-\varrho(h)|^2,$$

$$\tfrac{1}{2}|Qh-\varrho(h)|^2 = \tfrac{1}{2}|Qh|^2 - \langle Qh, \varrho(h)\rangle + \tfrac{1}{2}|\varrho(h)|^2.$$

Also $\psi_1(y) = \psi(x) + \frac{1}{2}|p|^2$. Subtracting and using (135),

$$\begin{aligned}\psi(x_h) &= \Big[\psi(x) + \tfrac{1}{2}|p|^2 + \langle p, h\rangle + \tfrac{1}{2}\langle Qh, h\rangle + e_1(h)\Big] \\ &\quad - \Big[\tfrac{1}{2}|p|^2 + \langle p, Qh\rangle - \langle p, \varrho(h)\rangle + \tfrac{1}{2}|Qh|^2 - \langle Qh, \varrho(h)\rangle + \tfrac{1}{2}|\varrho(h)|^2\Big] \\ &= \psi(x) + \langle p, h - Qh + \varrho(h)\rangle + \tfrac{1}{2}\langle Qh, h - Qh\rangle + e_1(h) + e_2(h) \\ &= \psi(x) + \langle p, x_h - x\rangle + \tfrac{1}{2}\langle Qh, Ph\rangle + e_1(h) + e_2(h),\end{aligned}$$

where $e_1(h)$ is the error in (135), $e_2(h) = \langle Qh, \varrho(h)\rangle - \frac{1}{2}|\varrho(h)|^2$, and we used $h - Qh = Ph$, $h - Qh + \varrho(h) = Ph + \varrho(h) = x_h - x$ and $\langle Qh, h - Qh\rangle = \langle Qh, Ph\rangle$. Now fix $\delta > 0$. Choose $\delta' > 0$ so small that $\|Q\|\delta' + \frac{1}{2}\delta'^2 \le \delta/2$, then $\rho' > 0$ with $|\varrho(h)| \le \delta'|h|$ for $|h| < \rho'$; this gives $|e_2(h)| \le \|Q\|\,|h|\,\delta'|h| + \frac{1}{2}\delta'^2|h|^2 \le (\delta/2)|h|^2$. Choosing also $\rho'' > 0$ so that (135) holds with $\delta/2$ in place of δ, and putting $\rho_1 = \min(\rho', \rho'')$, we obtain (137).

Step 3: almost every x is $u(y)$ for a y at which $Du(y)$ is invertible. Let $N = \{y \in E : u$ is not differentiable at $y\}$. Since u is 1-Lipschitz on E, Rademacher's theorem (preamble) gives $\mu(N) = 0$, and the preamble's bound for the measure of a Lipschitz image — applied with source and target both equal to E, and with the 1-Lipschitz map u — gives

$$\mu\big(u(N)\big) = 0. \tag{138}$$

Put $\Delta = E \setminus N$ and $P(y) = Du(y)$ for $y \in \Delta$, and set

$$S = \{y \in \Delta : \det P(y) = 0\}, \qquad G = \{y \in \Delta : \det P(y) \neq 0\}.$$

By the preamble, Δ is a Borel set and $y \mapsto Du(y)$ is Borel measurable; the determinant is continuous (preamble), so S, being the intersection of Δ with the preimage of the closed set $\{0\}$ under a Borel map, is Borel. For $y \in S$ the map u is differentiable at y, hence a fortiori has the derivative $P(y)$ at y within S. The area-formula inequality (preamble), applied with the measurable set S, the map u and the derivative field P, therefore gives

$$\mu\big(u(S)\big) \le \int_S \big|\det P(y)\big|\, d\mu(y) = 0, \tag{139}$$

since the integrand vanishes identically on S.

Let $D = \{x \in E : \psi$ is differentiable at $x\}$. The function ψ is convex and real-valued on the open set E, hence locally Lipschitz there (preamble), so Rademacher's theorem (preamble) gives

$$\mu(E \setminus D) = 0. \tag{140}$$

We claim $D \subseteq u(E)$. Let $x \in D$ and $q = \nabla\psi(x)$. For $x' \in E$ and $t \in (0,1]$, convexity of ψ gives $\psi(x + t(x'-x)) \le (1-t)\psi(x) + t\psi(x')$, i.e.

$$\frac{\psi(x + t(x'-x)) - \psi(x)}{t} \le \psi(x') - \psi(x);$$

letting $t \downarrow 0$ and using differentiability at x, the left-hand side tends to $\langle q, x'-x\rangle$, so

$$\psi(x) + \langle q, x'-x\rangle \le \psi(x') \qquad (x' \in E). \tag{141}$$

Put $y = x+q$, so that $y - x = q$. Then (141) is clause (2) of `MoreauProxSubgradient` with this y and $x_0 = x$, so clause (1) holds: x minimizes $x' \mapsto \psi(x') + |x'-y|^2/2$. By `IsMoreauProx` the point $u(y)$ minimizes the same function, and the minimizer is unique by `MoreauProxMonotone` with $y_1 = y_2 = y$; hence $u(y) = x$ and $x \in u(E)$.

Since $E = N \cup S \cup G$ we have $u(E) = u(N) \cup u(S) \cup u(G)$, so

$$E \setminus u(G) \subseteq (E \setminus D) \cup \big(D \setminus u(G)\big) \subseteq (E \setminus D) \cup u(N) \cup u(S),$$

and by (138), (139), (140) the right-hand side is null. Therefore

$$\mu\big(E \setminus u(G)\big) = 0, \tag{142}$$

and it suffices to establish the asserted expansion at every $x \in u(G)$. We record also, for later use, that $u(E)$ has full measure (it contains D), so $E \setminus u(E)$ is null; a null set has empty interior (preamble), hence

$$u(E) \text{ is dense in } E. \tag{143}$$

Step 4: the subdifferential is a singleton above G, with a uniform localization. Fix $y \in G$ and put $x = u(y)$, $p = v(y)$, $P = Du(y)$; note $y = x + p$. Let

$$\partial\psi(x) = \big\{s \in E : \psi(x) + \langle s, x' - x\rangle \le \psi(x') \text{ for all } x' \in E\big\}.$$

By (133), $p \in \partial\psi(x)$. The set $\partial\psi(x)$ is an intersection, over $x' \in E$, of the closed half-spaces $\{s : \langle s, x' - x\rangle \le \psi(x') - \psi(x)\}$, hence is closed and convex.

(4a) $\partial\psi(x) = \{p\}$. Let $s \in \partial\psi(x)$ and put $d = s - p$. For $t \in [0,1]$ the vector $p + td$ lies in $\partial\psi(x)$ by convexity of $\partial\psi(x)$, so with $y_t = x + p + td = y + td$ the defining property of $\partial\psi(x)$ reads $\psi(x) + \langle y_t - x, x' - x\rangle \le \psi(x')$ for all x', which is clause (2) of `MoreauProxSubgradient` with the base point y_t and $x_0 = x$. Hence x minimizes $x' \mapsto \psi(x') + |x' - y_t|^2/2$, and by the uniqueness of the minimizer, $u(y_t) = x = u(y)$. Thus $u(y + td) - u(y) = 0$ for every $t \in [0,1]$. Dividing by t and letting $t \downarrow 0$, differentiability of u at y gives $Pd = 0$. If $d \neq 0$ then P is not injective, so $\det P = 0$ (preamble: $\det P \neq 0$ if and only if P is bijective), contradicting $y \in G$. Hence $d = 0$ and $s = p$.

(4b) Uniform localization. We claim: for every $\sigma > 0$ there is $\rho_\sigma > 0$ such that

$$w \in E,\ |w - x| < \rho_\sigma,\ y' \in E,\ u(y') = w \quad\Longrightarrow\quad |y' - y| < \sigma.$$

Suppose not. Then there are $\sigma > 0$, points $w_j \to x$ and points y'_j with $u(y'_j) = w_j$ and $|y'_j - y| \ge \sigma$ for all j. Put $s_j = y'_j - w_j = v(y'_j)$; by (133) at the base point y'_j,

$$\psi(w_j) + \langle s_j, x' - w_j\rangle \le \psi(x') \qquad (x' \in E). \tag{144}$$

The s_j are bounded. Indeed ψ is locally Lipschitz on E (preamble), so there are $a > 0$ and $L \ge 0$ with $|\psi(b) - \psi(b')| \le L|b - b'|$ for $b, b' \in \overline{B}(x, 2a)$; for all large j we have $|w_j - x| \le a$, and then for a unit vector e and $0 < t \le a$ the point $w_j + te$ lies in $\overline{B}(x, 2a)$, so (144) with $x' = w_j + te$ gives $t\langle s_j, e\rangle \le \psi(w_j + te) - \psi(w_j) \le Lt$, whence $\langle s_j, e\rangle \le L$ for every unit e and therefore $|s_j| \le L$. Consequently $y'_j = w_j + s_j$ is a bounded sequence, and by Bolzano–Weierstrass (preamble) a subsequence y'_{j_i} converges to some $\bar y$ with $|\bar y - y| \ge \sigma$. Then $s_{j_i} = y'_{j_i} - w_{j_i} \to \bar y - x =: \bar s$. Passing to the limit in (144) along this subsequence, using the continuity of ψ (preamble) and of the inner product, gives $\psi(x) + \langle \bar s, x' - x\rangle \le \psi(x')$ for all x', i.e. $\bar s \in \partial\psi(x) = \{p\}$ by (4a). So $\bar s = p$ and $\bar y = x + p = y$, contradicting $|\bar y - y| \ge \sigma$.

Step 5: the expansion at $x = u(y)$ for $y \in G$. Fix $y \in G$ and keep $x = u(y)$, $p = v(y)$, $P = Du(y)$, $Q = \mathrm{id} - P$. Since $\det P \neq 0$, P is bijective with continuous linear inverse P^{-1}, and there is $c > 0$ with

$$|Ph| \ge c|h| \qquad (h \in E) \tag{145}$$

(preamble). Put $A_0 = QP^{-1}$, a continuous linear map $E \to E$, and

$$A(a,b) = \tfrac{1}{2}\big(\langle A_0 a, b\rangle + \langle A_0 b, a\rangle\big),$$

a symmetric continuous bilinear form on E with $A(k,k) = \langle A_0 k, k\rangle$. Finally set $g(k) = \psi(x + k) - \psi(x) - \langle p, k\rangle$; by (133) at the base point y we have $p \in \partial\psi(x)$, so

$$g(k) \ge 0 \ \ (k \in E), \qquad g(0) = 0, \tag{146}$$

and g is continuous because ψ is.

We claim: for every $\delta > 0$ there is $\rho > 0$ with

$$\big|g(k) - \tfrac{1}{2}\langle A_0 k, k\rangle\big| \le \delta|k|^2 \qquad (|k| < \rho). \tag{147}$$

Let $\delta > 0$. Put

$$C_1 = \frac{4\|Q\|\,\|P^{-1}\|\,\|P\|}{c^2} + \frac{2\|A_0\|}{c},$$

choose δ_1 with $0 < \delta_1 \le \min(1, c/2)$ and $\frac{1}{2}C_1\delta_1 \le \delta/2$, and choose $\delta_2 > 0$ with $4\delta_2/c^2 \le \delta/2$. By Step 2 there is $\rho_1 > 0$ such that (137) holds with δ_2 in place of δ; by the differentiability of u at y there is $\rho_2 > 0$ with

$$|\varrho(h)| = |x_h - x - Ph| \le \delta_1|h| \qquad (|h| < \rho_2); \tag{148}$$

and by Step 4(4b) with $\sigma = \min(\rho_1, \rho_2)$ there is $\rho_3 > 0$ such that every w with $|w - x| < \rho_3$ and every y' with $u(y') = w$ satisfy $|y' - y| < \sigma$.

Let $k \in E$ with $|k| < \rho_3$ and suppose first that $x + k \in u(E)$, say $u(y') = x + k$. Put $h = y' - y$. By the choice of ρ_3 we have $|h| < \sigma \le \min(\rho_1, \rho_2)$, and $u(y + h) = u(y') = x + k$, that is

$$x_h - x = k. \tag{149}$$

By (148) and (149), $|k - Ph| \le \delta_1|h|$. Combining with (145) and $\delta_1 \le c/2$,

$$|k| \ge |Ph| - \delta_1|h| \ge (c - \delta_1)|h| \ge \frac{c}{2}|h|, \qquad \text{so} \qquad |h| \le \frac{2}{c}|k|.$$

Hence

$$|k - Ph| \le \frac{2\delta_1}{c}|k|, \qquad |h - P^{-1}k| = \left|P^{-1}(Ph - k)\right| \le \|P^{-1}\|\frac{2\delta_1}{c}|k|. \tag{150}$$

Now

$$\langle Qh, Ph\rangle - \langle A_0k, k\rangle = \left\langle Q(h - P^{-1}k), Ph\right\rangle + \left\langle QP^{-1}k, Ph - k\right\rangle,$$

as one checks by inserting and removing the term $\langle QP^{-1}k, Ph\rangle$ and using $A_0 = QP^{-1}$. Since $|Ph| \le \|P\|\,|h| \le (2\|P\|/c)|k|$, the Cauchy–Schwarz inequality and (150) give

$$\left|\langle Qh, Ph\rangle - \langle A_0k, k\rangle\right| \le \|Q\|\,\|P^{-1}\|\frac{2\delta_1}{c}|k| \cdot \frac{2\|P\|}{c}|k| + \|A_0\|\,|k| \cdot \frac{2\delta_1}{c}|k| = C_1\delta_1|k|^2. \tag{151}$$

On the other hand (137) (with δ_2) together with (149) and $|h| \le (2/c)|k|$ gives

$$\left|g(k) - \tfrac{1}{2}\langle Qh, Ph\rangle\right| = \left|\psi(x_h) - \psi(x) - \langle p, x_h - x\rangle - \tfrac{1}{2}\langle Qh, Ph\rangle\right| \le \delta_2|h|^2 \le \frac{4\delta_2}{c^2}|k|^2. \tag{152}$$

Adding half of (151) to (152) and using the choices of δ_1, δ_2,

$$\left|g(k) - \tfrac{1}{2}\langle A_0k, k\rangle\right| \le \frac{4\delta_2}{c^2}|k|^2 + \tfrac{1}{2}C_1\delta_1|k|^2 \le \delta|k|^2.$$

So (147) holds for every k with $|k| < \rho_3$ such that $x + k \in u(E)$.

Finally we remove that restriction. Put $T = \{k \in E : |k| < \rho_3,\ x + k \in u(E)\}$. By (143) the set $u(E)$ is dense in E, so T is dense in $B(0, \rho_3)$: given $k_0 \in B(0, \rho_3)$ and $\varepsilon > 0$ with $B(k_0, \varepsilon) \subseteq B(0, \rho_3)$, the open set $x + B(k_0, \varepsilon)$ meets $u(E)$, which produces a point of T within ε of k_0. Both $k \mapsto |g(k) - \frac{1}{2}\langle A_0k, k\rangle|$ and $k \mapsto \delta|k|^2$ are continuous on E, and the inequality (147) holds on the dense subset T of the ball $B(0, \rho_3)$; hence, by passing to limits along sequences in T, it holds at every point of $B(0, \rho_3)$. This proves (147) with $\rho = \rho_3$.

Step 6: nonnegativity of A, and conclusion. Let $x \in u(G)$ and let p and A be as in Step 5 for some $y \in G$ with $u(y) = x$. The form A is symmetric and continuous by construction, and Step 5 is exactly the asserted expansion with $q = p$, since $A(h, h) = \langle A_0h, h\rangle$.

For nonnegativity, let $k_0 \in E$ with $|k_0| = 1$ and let $\delta > 0$. Take ρ from (147) and let $t \in (0, \rho)$. With $k = tk_0$, so $|k| = t < \rho$, (146) and (147) give

$$0 \le g(tk_0) \le \tfrac{1}{2}\langle A_0(tk_0), tk_0\rangle + \delta t^2 = t^2\Big(\tfrac{1}{2}A(k_0, k_0) + \delta\Big),$$

so $\frac{1}{2}A(k_0, k_0) + \delta \ge 0$. As $\delta > 0$ was arbitrary, $A(k_0, k_0) \ge 0$. For general $h \ne 0$, bilinearity gives $A(h, h) = |h|^2A(h/|h|, h/|h|) \ge 0$, and $A(0, 0) = 0$.

By (142) the set of x not of the form $u(y)$ with $y \in G$ is null, so the expansion holds for μ-almost every $x \in E$. □

2.43. `ConvexLocalExtension`.

Lemma 2.43 (Local extension of a convex function to a globally convex one)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ be open, let $\psi : E \to \mathbb{R}$ be convex on Ω (so in particular Ω is convex), let $x_0 \in E$ and let $r > 0$ satisfy $\overline{B}(x_0, 2r) \subseteq \Omega$. Then there is $\psi' : E \to \mathbb{R}$ which is convex on all of E and satisfies*

$$\psi'(x) = \psi(x) \qquad \big(x \in B(x_0, r)\big).$$

Proof. Step 1: a bound for ψ and for its subgradients on the inner ball. The set $\overline{B}(x_0, 2r)$ is closed and bounded in the finite-dimensional space E, hence compact, and it is contained in Ω, on which ψ is continuous (preamble). So

$$M := \max_{\overline{B}(x_0, 2r)} |\psi|$$

is a finite nonnegative real number (preamble).

For each $w \in B(x_0, r)$ we have $w \in \Omega$, so by `ConvexSubgradientExists` there is $q_w \in E$ with

$$\psi(w) + \langle q_w, x - w\rangle \le \psi(x) \qquad (x \in \Omega). \tag{153}$$

Fix one such q_w for each $w \in B(x_0, r)$. These vectors are bounded: let $e \in E$ with $|e| = 1$ and put $x = w + re$. Then $|x - x_0| \le |w - x_0| + r < r + r = 2r$, so $x \in \overline{B}(x_0, 2r) \subseteq \Omega$, and (153) gives

$$r\langle q_w, e\rangle \le \psi(w + re) - \psi(w) \le M - (-M) = 2M.$$

Dividing by $r > 0$ gives $\langle q_w, e\rangle \le 2M/r$ for every unit vector e, hence

$$|q_w| \le \frac{2M}{r} \qquad \big(w \in B(x_0, r)\big) \tag{154}$$

(take $e = q_w/|q_w|$ when $q_w \neq 0$; the bound is trivial otherwise).

Step 2: the extension. For $w \in B(x_0, r)$ let $a_w : E \to \mathbb{R}$ be the affine function $a_w(x) = \psi(w) + \langle q_w, x - w\rangle$, and define

$$\psi'(x) = \sup\big\{a_w(x) : w \in B(x_0, r)\big\} \qquad (x \in E).$$

The index set is nonempty, since $r > 0$ gives $x_0 \in B(x_0, r)$. The supremum is finite for every x: by $|\psi(w)| \le M$ and (154),

$$a_w(x) \le M + \frac{2M}{r}|x - w| \le M + \frac{2M}{r}\big(|x - x_0| + r\big) \qquad \big(w \in B(x_0, r)\big),$$

a bound independent of w. Hence $\psi'(x)$ is a real number, it is an upper bound for every $a_w(x)$, and it is the least such.

Step 3: ψ' is convex on E. The set E is convex. Let $x_1, x_2 \in E$ and $t_1, t_2 \ge 0$ with $t_1 + t_2 = 1$, and let $w \in B(x_0, r)$. Since a_w is affine,

$$a_w(t_1x_1 + t_2x_2) = t_1a_w(x_1) + t_2a_w(x_2) \le t_1\psi'(x_1) + t_2\psi'(x_2),$$

using $a_w(x_i) \le \psi'(x_i)$ and $t_i \ge 0$. The right-hand side does not depend on w, so it is an upper bound for $\{a_w(t_1x_1 + t_2x_2) : w \in B(x_0, r)\}$, whose least upper bound is $\psi'(t_1x_1 + t_2x_2)$. Therefore $\psi'(t_1x_1 + t_2x_2) \le t_1\psi'(x_1) + t_2\psi'(x_2)$, i.e. ψ' is convex on E.

Step 4: $\psi' = \psi$ on $B(x_0, r)$. Let $x \in B(x_0, r)$; then $x \in \Omega$. For every $w \in B(x_0, r)$, (153) applied at this x gives $a_w(x) \le \psi(x)$, so $\psi(x)$ is an upper bound for the set defining $\psi'(x)$ and therefore $\psi'(x) \le \psi(x)$. Conversely the choice $w = x$ is admissible and gives $a_x(x) = \psi(x)$, an element of that set, so $\psi'(x) \ge \psi(x)$. Hence $\psi'(x) = \psi(x)$. □

2.44. `SemiconvexSecondOrderAe`.

Theorem 2.44 (Alexandrov's theorem for semiconvex functions)**.** *Let E be a finite-dimensional real inner-product space with an additive Haar measure μ, let $\lambda \ge 0$, let $\Omega \subseteq E$ be open and let $\phi : E \to \mathbb{R}$ be λ-semiconvex on Ω in the sense of `SemiconvexOn` (so in particular Ω is convex). Then for μ-almost every $x \in E$ the following holds: if $x \in \Omega$, there are a vector $q \in E$ and a*

symmetric continuous bilinear form A on E with $A(h,h) \ge -\lambda|h|^2$ for all $h \in E$, such that for every $\delta > 0$ there is $\rho > 0$ with

$$\Big|\phi(x+h) - \phi(x) - \langle q, h\rangle - \tfrac{1}{2}A(h,h)\Big| \le \delta|h|^2 \qquad (|h| < \rho).$$

Proof. Set

$$\psi(z) = \phi(z) + \frac{\lambda|z|^2}{2} \qquad (z \in E),$$

so that ψ is convex on Ω by the definition of semiconvexity in `SemiconvexOn`, and Ω is convex.

Step 1: a countable cover of Ω by balls well inside it. The space E is a finite-dimensional real normed space, hence separable: fixing a basis $e_1, \dots, e_n$ of E, the countable set of combinations $\sum_i c_i e_i$ with all c_i rational is dense in E. Let $(x_i)_{i\in\mathbb{N}}$ enumerate such a countable dense subset of E, and let

$$\mathcal{F} = \big\{\, (i,s) \;:\; i \in \mathbb{N},\ s \in \mathbb{Q},\ s > 0,\ \overline{B}(x_i, 2s) \subseteq \Omega \,\big\},$$

a countable set. We claim that the balls $B(x_i, s)$ with $(i,s) \in \mathcal{F}$ cover Ω. Let $x \in \Omega$. Since Ω is open there is $r > 0$ with $\overline{B}(x, 5r) \subseteq \Omega$. By density there is i with $|x_i - x| < r$, and there is a rational s with $r < s < 2r$. Then $|x - x_i| < r < s$, so $x \in B(x_i, s)$; and for $z \in \overline{B}(x_i, 2s)$ we have $|z - x| \le |z - x_i| + |x_i - x| \le 2s + r < 4r + r = 5r$, so $\overline{B}(x_i, 2s) \subseteq \overline{B}(x, 5r) \subseteq \Omega$ and $(i,s) \in \mathcal{F}$. This proves the claim.

Step 2: a single null set. Let $(i,s) \in \mathcal{F}$. Applying `ConvexLocalExtension` to the open set Ω, the function ψ (convex on Ω), the centre x_i and the radius $s > 0$ — whose hypothesis $\overline{B}(x_i, 2s) \subseteq \Omega$ holds by the definition of $\mathcal{F}$ — we obtain $\psi_{i,s} : E \to \mathbb{R}$, convex on all of E, with

$$\psi_{i,s} = \psi \quad \text{on } B(x_i, s). \tag{155}$$

By `ConvexSecondOrderAe` applied to $\psi_{i,s}$ there is a μ-null set $Z_{i,s} \subseteq E$ such that every $x \notin Z_{i,s}$ admits a vector $q \in E$ and a symmetric continuous bilinear form A with $A(h,h) \ge 0$ for all h and

$$\Big|\psi_{i,s}(x+h) - \psi_{i,s}(x) - \langle q, h\rangle - \tfrac{1}{2}A(h,h)\Big| \le \delta|h|^2 \qquad (|h| < \rho) \tag{156}$$

for every $\delta > 0$ and a suitable $\rho = \rho(\delta) > 0$. Put $Z = \bigcup_{(i,s)\in\mathcal{F}} Z_{i,s}$, a countable union of null sets, hence $\mu(Z) = 0$.

Step 3: the expansion at every $x \in \Omega \setminus Z$. Let $x \in \Omega \setminus Z$. By Step 1 there is $(i,s) \in \mathcal{F}$ with $x \in B(x_i, s)$; since $B(x_i, s)$ is open there is $\rho_0 > 0$ with $B(x, \rho_0) \subseteq B(x_i, s)$. Since $x \notin Z_{i,s}$, Step 2 provides q and A with $A(h,h) \ge 0$ and (156).

Let $\delta > 0$ and let $\rho = \min(\rho(\delta), \rho_0) > 0$. For $|h| < \rho$ both x and $x + h$ lie in $B(x_i, s)$, so (155) turns (156) into

$$\big|e(h)\big| \le \delta|h|^2, \qquad e(h) := \psi(x+h) - \psi(x) - \langle q, h\rangle - \tfrac{1}{2}A(h,h) \qquad (|h| < \rho). \tag{157}$$

Now translate this into a statement about $\phi = \psi - \lambda|\cdot|^2/2$. Expanding the square,

$$\frac{\lambda|x+h|^2}{2} = \frac{\lambda|x|^2}{2} + \lambda\langle x, h\rangle + \frac{\lambda|h|^2}{2},$$

so

$$\phi(x+h) - \phi(x) = \big(\psi(x+h) - \psi(x)\big) - \lambda\langle x, h\rangle - \frac{\lambda|h|^2}{2}.$$

Substituting $\psi(x+h) - \psi(x) = \langle q, h\rangle + \tfrac{1}{2}A(h,h) + e(h)$ from (157),

$$\phi(x+h) - \phi(x) = \langle q - \lambda x, h\rangle + \tfrac{1}{2}\big(A(h,h) - \lambda|h|^2\big) + e(h).$$

Define

$$q' = q - \lambda x, \qquad A'(a,b) = A(a,b) - \lambda\langle a, b\rangle.$$

Then A' is again a symmetric continuous bilinear form on E (a difference of two such), $A'(h,h) = A(h,h) - \lambda|h|^2 \ge -\lambda|h|^2$ because $A(h,h) \ge 0$, and the display above reads

$$\phi(x+h) - \phi(x) - \langle q', h\rangle - \tfrac{1}{2}A'(h,h) = e(h),$$

so that $|\phi(x+h) - \phi(x) - \langle q', h\rangle - \tfrac{1}{2}A'(h,h)| \le \delta|h|^2$ for $|h| < \rho$ by (157). Since $\delta > 0$ was arbitrary and the pair (q', A') does not depend on δ, the point x admits the asserted expansion.

As $\mu(Z)=0$, the conclusion holds for μ-almost every x: for $x\notin Z$, either $x\notin\Omega$ and there is nothing to prove, or $x\in\Omega\setminus Z$ and the expansion has just been established. □

2.45. `SymmetricFormNormFromDiagonal`.

Lemma 2.45 (A symmetric bilinear form is controlled by its diagonal)**.** *Let E be a finite-dimensional real inner-product space, let S be a continuous bilinear form on E that is symmetric, $S(a,b)=S(b,a)$ for all $a,b\in E$, and let $C\geq 0$ satisfy*

$$\big|S(a,a)\big|\leq C|a|^2\qquad(a\in E).$$

Then $\|S\|\leq C$, where $\|S\|$ is the operator norm, i.e. the least constant K with $|S(a,b)|\leq K|a|\,|b|$ for all $a,b\in E$.

Proof. It suffices to prove

$$\big|S(a,b)\big|\leq C|a|\,|b|\qquad(a,b\in E),\tag{158}$$

since $C\geq 0$ and the operator norm of a continuous bilinear form is the least such constant.

Step 1: the polarization bound. Since S is bilinear and symmetric,

$$S(u+v,u+v)-S(u-v,u-v)=\big(S(u,u)+2S(u,v)+S(v,v)\big)-\big(S(u,u)-2S(u,v)+S(v,v)\big)=4S(u,v),$$

the cross terms being combined by $S(v,u)=S(u,v)$. Hence

$$S(u,v)=\tfrac14\big(S(u+v,u+v)-S(u-v,u-v)\big),$$

and the diagonal hypothesis applied at $u+v$ and at $u-v$ gives

$$\big|S(u,v)\big|\leq\tfrac14\Big(C|u+v|^2+C|u-v|^2\Big)=\tfrac14\,C\cdot 2\big(|u|^2+|v|^2\big)=\frac{C\big(|u|^2+|v|^2\big)}{2},$$

using the parallelogram law $|u+v|^2+|u-v|^2=2(|u|^2+|v|^2)$, which follows by adding the two expansions $|u\pm v|^2=|u|^2\pm 2\langle u,v\rangle+|v|^2$. So

$$\big|S(u,v)\big|\leq\frac{C\big(|u|^2+|v|^2\big)}{2}\qquad(u,v\in E).\tag{159}$$

Step 2: the degenerate cases. If $a=0$ then $S(a,b)=S(0,b)=0$ by linearity of S in its first argument, so $|S(a,b)|=0\leq C|a|\,|b|$, the right-hand side being ≥ 0 because $C\geq 0$. Likewise if $b=0$. So assume $a\neq 0$ and $b\neq 0$, hence $|a|>0$ and $|b|>0$.

Step 3: scaling the two arguments against each other. Put

$$t=\sqrt{\frac{|b|}{|a|}}>0,\qquad\text{so}\qquad t^2=\frac{|b|}{|a|},\qquad t^{-2}=\frac{|a|}{|b|}.$$

By bilinearity, $S(ta,t^{-1}b)=t\,t^{-1}S(a,b)=S(a,b)$. Applying (159) to $u=ta$ and $v=t^{-1}b$, and using $|ta|=t|a|$ and $|t^{-1}b|=t^{-1}|b|$ (both scalars being positive),

$$\big|S(a,b)\big|=\big|S(ta,t^{-1}b)\big|\leq\frac{C\big(t^2|a|^2+t^{-2}|b|^2\big)}{2}=\frac{C\big(\frac{|b|}{|a|}|a|^2+\frac{|a|}{|b|}|b|^2\big)}{2}=\frac{C\big(|a|\,|b|+|a|\,|b|\big)}{2}=C|a|\,|b|,$$

which is (158). (The scaling is exactly what replaces the arithmetic mean $(|u|^2+|v|^2)/2$ of Step 1 by the geometric mean $|a|\,|b|$; without it (159) alone gives only the weaker bound $|S(a,b)|\leq C$ on the unit ball.) □

2.46. `SumRuleJetFromJensen`.

Lemma 2.46 (From Jensen's lemma to a superjet of each summand)**.** *Let E be a finite-dimensional real inner-product space equipped with its Borel σ-algebra, let N be a natural number and write $V=\ell^2\big(\bigoplus_{j=1}^{N}E\big)$ for the ℓ^2-product of N copies of E. Let $C\geq 0$ and let $G_j:E\to\mathbb{R}$ $(1\leq j\leq N)$ be continuous with $G_j(w)\leq C$ for every w. Let $\varepsilon>0$, let $A_\varepsilon:V\to V$ be a*

self-adjoint continuous linear operator, write $\mathcal{G}(W) = \sum_j G_j(w_j)$*, let* $\mathcal{G}^\varepsilon$ *and* G_j^ε *be the sup-convolutions of* `SupConvolution` *with parameter* ε *formed in* V *and in* E*, and let* $\Upsilon : V \to \mathbb{R}$ *satisfy*

$$\Upsilon(Z) = \mathcal{G}^\varepsilon(Z) - \tfrac{1}{2}\langle A_\varepsilon Z, Z\rangle \qquad (Z \in V), \qquad \mathcal{G}^\varepsilon(Z) = \sum_j G_j^\varepsilon(z_j) \qquad (Z \in V).$$

Assume $\Upsilon(0) = 0$*,* $\Upsilon(Z) < 0$ *for* $Z \neq 0$*, that* Υ *is* λ*-semiconvex on* V *in the sense of* `SemiconvexOn` *with* $\lambda = \varepsilon^{-1} + \|A_\varepsilon\|$*, and that* Υ *is continuous on* V*. Let* $R > 0$ *and* $\delta' > 0$*. Then there are* $\widehat{Z}' \in V$*,* $\Pi \in V$ *and symmetric continuous bilinear forms* X_j *on* E $(1 \le j \le N)$ *with*

(J1) $|\widehat{Z}'| < R$ *and* $|\Pi| \le \delta' + \|A_\varepsilon\| R$*;*
(J2) $\|X_j\| \le \varepsilon^{-1} + 2\|A_\varepsilon\|$ *for every* j*;*
(J3) $\big(\Pi_j, X_j\big) \in J^{2,+} G_j\big(\widehat{z}_j' + \varepsilon\Pi_j\big)$ *for every* j*, where* $J^{2,+}$ *is the superjet of* `Superjet`*;*
(J4) $\sum_j X_j[\zeta_j] \le \langle A_\varepsilon \zeta, \zeta\rangle$ *for every* $\zeta \in V$*, where* $X[h] = X(h,h)$*.*

Proof. For $a \in E$ and $1 \le j \le N$ let $\iota_j a \in V$ be the vector with j-th component a and all other components 0; then $|\iota_j a| = |a|$ and $\langle \xi, \iota_j a\rangle = \langle \xi_j, a\rangle$ for every $\xi \in V$, by the definition of the ℓ^2-product inner product (preamble). Fix an additive Haar measure μ_V on V.

Step 1: a good point. The ball $B(0,R)$ is open and convex, and Υ is λ-semiconvex on V, hence on the convex subset $B(0,R)$: the function $\Upsilon + \frac{\lambda}{2}|\cdot|^2$ is convex on V and therefore convex on $B(0,R)$. Also Υ is continuous, hence continuous on $\overline{B}(0,R)$, and 0 is a strict maximum point of Υ on $\overline{B}(0,R)$: for $Z \in \overline{B}(0,R)$ with $Z \neq 0$ we have $\Upsilon(Z) < 0 = \Upsilon(0)$. Finally $\lambda > 0$ because $\varepsilon > 0$ and $\|A_\varepsilon\| \ge 0$. By `JensenLemma`, applied in V with this λ, the radius R, the centre 0 and the parameter δ', the set

$$\mathcal{K} = \big\{ Z \in B(0,R) : \exists\, Q \in V,\ |Q| \le \delta',\ W \mapsto \Upsilon(W) + \langle Q, W\rangle \text{ has a local maximum on } B(0,R) \text{ at } Z \big\}$$

satisfies $\mu_V(\mathcal{K}) > 0$.

By `SemiconvexSecondOrderAe`, applied in V with this $\lambda \ge 0$ and the open set $B(0,R)$, there is a μ_V-null set $\mathcal{N}$ such that every $Z \in B(0,R) \setminus \mathcal{N}$ admits $q_\Upsilon \in V$ and a symmetric continuous bilinear form A_Υ on V with $A_\Upsilon(\zeta,\zeta) \ge -\lambda|\zeta|^2$ for all ζ and, for every $\delta > 0$, some $\rho(\delta) > 0$ with

$$\big|\Upsilon(Z+\zeta) - \Upsilon(Z) - \langle q_\Upsilon, \zeta\rangle - \tfrac{1}{2}A_\Upsilon(\zeta,\zeta)\big| \le \delta|\zeta|^2 \qquad (|\zeta| < \rho(\delta)). \tag{160}$$

The set $\mathcal{K} \setminus \mathcal{N}$ is nonempty: were it empty we would have $\mathcal{K} \subseteq \mathcal{N}$, hence $\mu_V(\mathcal{K}) \le \mu_V(\mathcal{N}) = 0$ by the monotonicity of μ_V, contradicting $\mu_V(\mathcal{K}) > 0$. So pick $\widehat{Z}' \in \mathcal{K} \setminus \mathcal{N}$, a multiplier Q with $|Q| \le \delta'$ witnessing $\widehat{Z}' \in \mathcal{K}$, and expansion data (q_Υ, A_Υ) at $\widehat{Z}'$. In particular $|\widehat{Z}'| < R$.

Step 2: first- and second-order conditions. As $B(0,R)$ is open and $\widehat{Z}' \in B(0,R)$, and the local maximum of $W \mapsto \Upsilon(W) + \langle Q, W\rangle$ on $B(0,R)$ at $\widehat{Z}'$ holds on some neighbourhood of $\widehat{Z}'$ inside $B(0,R)$, there is $\rho_1 > 0$ with $B(\widehat{Z}', \rho_1) \subseteq B(0,R)$ and

$$\Upsilon(\widehat{Z}' + \zeta) - \Upsilon(\widehat{Z}') \le -\langle Q, \zeta\rangle \qquad (|\zeta| < \rho_1). \tag{161}$$

Let $\delta > 0$ and $|\zeta| < \min(\rho_1, \rho(\delta))$. Adding (160) (in the form $\langle q_\Upsilon, \zeta\rangle + \frac{1}{2}A_\Upsilon(\zeta,\zeta) - \delta|\zeta|^2 \le \Upsilon(\widehat{Z}'+\zeta) - \Upsilon(\widehat{Z}')$) to (161) gives

$$\langle q_\Upsilon + Q, \zeta\rangle \le -\tfrac{1}{2}A_\Upsilon(\zeta,\zeta) + \delta|\zeta|^2 \qquad \big(|\zeta| < \min(\rho_1, \rho(\delta))\big). \tag{162}$$

Fix a unit vector $v \in V$ and put $\zeta = tv$ with $t > 0$ small, so that $|\zeta| = t < \min(\rho_1, \rho(\delta))$ once t is small enough. Dividing (162) by t gives $\langle q_\Upsilon + Q, v\rangle \le t\big(-\frac{1}{2}A_\Upsilon(v,v) + \delta\big)$, and letting $t \downarrow 0$ gives $\langle q_\Upsilon + Q, v\rangle \le 0$ for every unit v; applying this to v and to $-v$ gives $\langle q_\Upsilon + Q, v\rangle = 0$ for every unit v. Writing $u = q_\Upsilon + Q$, this says $\langle u, v\rangle = 0$ for every unit v; if $u \neq 0$ then $v = u/|u|$ is a unit vector with $\langle u, v\rangle = |u| > 0$, a contradiction, so $u = 0$. (If $V = \{0\}$ there are no unit vectors at all and the display above is vacuous, but then $u = 0$ holds outright, V having only the zero vector; the normalisation $v = u/|u|$ is legitimate precisely because it is performed only when $u \neq 0$.) Hence

$$q_\Upsilon = -Q. \tag{163}$$

Substituting (163) into (162) leaves $0 \le -\frac12 A_\Upsilon(\zeta,\zeta) + \delta|\zeta|^2$, i.e. $\frac12 A_\Upsilon(\zeta,\zeta) \le \delta|\zeta|^2$ for $|\zeta| < \min(\rho_1, \rho(\delta))$. Now fix $\zeta \in V$ and $\delta > 0$. If $\zeta = 0$ then $A_\Upsilon(\zeta,\zeta) = 0$ by bilinearity and there is nothing to prove. If $\zeta \neq 0$, put $v = \zeta/|\zeta|$, a unit vector, and apply the displayed inequality at tv for $0 < t < \min(\rho_1, \rho(\delta))$: since $A_\Upsilon(tv, tv) = t^2 A_\Upsilon(v,v)$, dividing by $t^2 > 0$ gives $\frac12 A_\Upsilon(v,v) \le \delta$, and $\delta > 0$ was arbitrary, so $A_\Upsilon(v,v) \le 0$; multiplying by $|\zeta|^2 \ge 0$ and using $A_\Upsilon(\zeta,\zeta) = |\zeta|^2 A_\Upsilon(v,v)$ gives

$$A_\Upsilon(\zeta,\zeta) \le 0 \qquad (\zeta \in V). \tag{164}$$

(The normalisation $v = \zeta/|\zeta|$ is performed only when $\zeta \neq 0$, which is what makes the argument correct when $V = \{0\}$: there every ζ is 0 and only the first case occurs.)

Step 3: the expansion of $\mathcal{G}^\varepsilon$ and the forms X_j. Since $\mathcal{G}^\varepsilon(Z) = \Upsilon(Z) + \frac12\langle A_\varepsilon Z, Z\rangle$ and, exactly, using the self-adjointness of A_ε,

$$\tfrac12\big\langle A_\varepsilon(\widehat{Z}' + \zeta), \widehat{Z}' + \zeta\big\rangle = \tfrac12\langle A_\varepsilon \widehat{Z}', \widehat{Z}'\rangle + \langle A_\varepsilon \widehat{Z}', \zeta\rangle + \tfrac12\langle A_\varepsilon\zeta, \zeta\rangle,$$

the estimate (160) becomes, with

$$\Pi := q_\Upsilon + A_\varepsilon \widehat{Z}' = -Q + A_\varepsilon \widehat{Z}', \qquad A' := A_\Upsilon + \langle A_\varepsilon \cdot, \cdot\rangle,$$

a symmetric continuous bilinear form on V,

$$\big|\mathcal{G}^\varepsilon(\widehat{Z}' + \zeta) - \mathcal{G}^\varepsilon(\widehat{Z}') - \langle \Pi, \zeta\rangle - \tfrac12 A'(\zeta,\zeta)\big| \le \delta|\zeta|^2 \qquad (|\zeta| < \rho(\delta)). \tag{165}$$

Here $|\Pi| \le |Q| + |A_\varepsilon \widehat{Z}'| \le \delta' + \|A_\varepsilon\|\,|\widehat{Z}'| \le \delta' + \|A_\varepsilon\| R$, which together with $|\widehat{Z}'| < R$ proves (J1).

Define $X_j(a,b) = A'(\iota_j a, \iota_j b)$ for $a, b \in E$; each X_j is a continuous bilinear form on E, and it is symmetric because A' is. Taking $\zeta = \iota_j h$ in (165), and using the separability hypothesis $\mathcal{G}^\varepsilon(Z) = \sum_i G_i^\varepsilon(z_i)$ — which shows that changing only the j-th component of the argument changes $\mathcal{G}^\varepsilon$ only through the single summand G_j^ε, all other summands being unchanged — together with $\langle \Pi, \iota_j h\rangle = \langle \Pi_j, h\rangle$ and $|\iota_j h| = |h|$, we get

$$\big|G_j^\varepsilon(\widehat{z}_j' + h) - G_j^\varepsilon(\widehat{z}_j') - \langle \Pi_j, h\rangle - \tfrac12 X_j(h,h)\big| \le \delta|h|^2 \qquad (|h| < \rho(\delta)). \tag{166}$$

In particular, given $\varepsilon' > 0$, apply (166) with $\delta = \varepsilon'$ and drop the absolute value on the favourable side: for $|h| < \rho(\varepsilon')$,

$$G_j^\varepsilon(\widehat{z}_j' + h) \le G_j^\varepsilon(\widehat{z}_j') + \langle \Pi_j, h\rangle + \tfrac12 X_j(h,h) + \varepsilon'|h|^2.$$

Since $\varepsilon' > 0$ was arbitrary and X_j is symmetric, the defining condition of `Superjet` holds, so

$$\big(\Pi_j, X_j\big) \in J^{2,+} G_j^\varepsilon(\widehat{z}_j') \qquad (1 \le j \le N). \tag{167}$$

Step 4: the matrix inequality and the norm bound. Let $\zeta \in V$ and $\delta > 0$ with $|\zeta| < \rho(\delta)$. Then $|\zeta_j| \le |\zeta| < \rho(\delta)$ for every j, so summing (166) over j with $h = \zeta_j$ and using the separability hypothesis again, together with $\sum_j \langle \Pi_j, \zeta_j\rangle = \langle \Pi, \zeta\rangle$ and $\sum_j |\zeta_j|^2 = |\zeta|^2$ (both instances of the definition of the ℓ^2-product inner product, preamble), gives

$$\Big|\mathcal{G}^\varepsilon(\widehat{Z}' + \zeta) - \mathcal{G}^\varepsilon(\widehat{Z}') - \langle \Pi, \zeta\rangle - \tfrac12 \sum_j X_j[\zeta_j]\Big| \le \sum_j \delta|\zeta_j|^2 = \delta|\zeta|^2.$$

Comparing with (165) and using the triangle inequality gives $\big|\frac12 A'(\zeta,\zeta) - \frac12\sum_j X_j[\zeta_j]\big| \le 2\delta|\zeta|^2$ for $|\zeta| < \rho(\delta)$. Both $\zeta \mapsto A'(\zeta,\zeta)$ and $\zeta \mapsto \sum_j X_j[\zeta_j]$ are homogeneous of degree two, the second because $(t\zeta)_j = t\zeta_j$ for every j. Fix $\zeta \in V$ and $\delta > 0$. If $\zeta = 0$ then both forms vanish and the two sides of (168) below agree. If $\zeta \neq 0$, put $v = \zeta/|\zeta|$, a unit vector, and apply the previous display at tv with $t \in (0, \rho(\delta))$; dividing by t^2 shows $\big|\frac12 A'(v,v) - \frac12\sum_j X_j[v_j]\big| \le 2\delta$, and $\delta > 0$ was arbitrary, so $\frac12 A'(v,v) = \frac12\sum_j X_j[v_j]$. Multiplying by $|\zeta|^2$ and using the degree-two homogeneity of both forms,

$$\sum_j X_j[\zeta_j] = A'(\zeta,\zeta) = A_\Upsilon(\zeta,\zeta) + \langle A_\varepsilon\zeta, \zeta\rangle \qquad (\zeta \in V). \tag{168}$$

By (164) this gives $\sum_j X_j[\zeta_j] \le \langle A_\varepsilon\zeta,\zeta\rangle$, which is (J4); and together with $A_\Upsilon(\zeta,\zeta) \ge -\lambda|\zeta|^2$ and the two-sided Cauchy–Schwarz estimate

$$\big|\langle A_\varepsilon\zeta,\zeta\rangle\big| \le |A_\varepsilon\zeta|\,|\zeta| \le \|A_\varepsilon\|\,|\zeta|^2, \qquad \text{i.e.} \qquad -\|A_\varepsilon\|\,|\zeta|^2 \le \langle A_\varepsilon\zeta,\zeta\rangle \le \|A_\varepsilon\|\,|\zeta|^2$$

— the lower bound being needed for the left-hand half of what follows — it gives

$$-\big(\lambda+\|A_\varepsilon\|\big)|\zeta|^2 \le \sum_j X_j[\zeta_j] \le \|A_\varepsilon\|\,|\zeta|^2 \qquad (\zeta\in V). \tag{169}$$

Taking $\zeta=\iota_j a$ in (169), so that $\zeta_j=a$ and $\zeta_i=0$ for $i\ne j$ and $|\zeta|=|a|$, yields the bound on the diagonal

$$\big|X_j[a]\big| \le \big(\lambda+\|A_\varepsilon\|\big)|a|^2 \qquad (a\in E).$$

Each X_j is a symmetric continuous bilinear form on E and $\lambda+\|A_\varepsilon\|\ge 0$, since $\lambda>0$; so `SymmetricFormNormFromDiagonal`, applied to $S=X_j$ with the constant $C=\lambda+\|A_\varepsilon\|$, converts this diagonal bound into the same bound on the operator norm:

$$\|X_j\| \le \lambda+\|A_\varepsilon\| = \varepsilon^{-1}+2\|A_\varepsilon\|,$$

which is (J2).

Step 5: transfer to G_j. The function G_j is continuous on E and satisfies $G_j(w)\le C\le C+|w|$, and $\varepsilon>0$; so `SupConvolutionMagic` applies to the jet (167) at the point $\widehat z_j'$ with the vector Π_j and the form X_j, and yields

$$\big(\Pi_j, X_j\big) \in J^{2,+}G_j\big(\widehat z_j' + \varepsilon\Pi_j\big),$$

which is (J3). □

2.47. `SumRuleFixedTauJets`.

Lemma 2.47 (The theorem of sums for the regularized form, with a regularization-free bound)**.** *Let E be a finite-dimensional real inner-product space equipped with its Borel σ-algebra, let $N\ge 1$ and write $V=\ell^2\big(\bigoplus_{j=1}^N E\big)$ for the ℓ^2-product of N copies of E, whose elements are written $Y=(y_j)_{j=1}^N$ and whose norm satisfies $|Y|^2=\sum_j|y_j|^2$. For each j let $O_j\subseteq E$ be open and let $g_j:E\to\mathbb{R}$ be continuous on O_j. Let $\Psi:V\to\mathbb{R}$, let $\widehat Z\in V$ with $\widehat z_j\in O_j$ for every j, assume that Ψ is C^2 near $\widehat Z$ and that $Y\mapsto\sum_j g_j(y_j)-\Psi(Y)$ has a local maximum at $\widehat Z$. Set $A=D^2\Psi(\widehat Z)$ and $\widehat p_j=D_j\Psi(\widehat Z)$, let $\bar A:V\to V$ be the self-adjoint operator with $\langle\bar A\xi,\zeta\rangle=A(\xi,\zeta)$, and let $\eta>0$. Then there is a constant $\Lambda\ge 0$ such that for every $\tau\in(0,1]$ there are symmetric continuous bilinear forms X_j^τ on E $(1\le j\le N)$ with*

(T1) *$\|X_j^\tau\|\le\Lambda$ for every j, with Λ independent of τ;*

(T2) *$\big(\widehat p_j, X_j^\tau\big)\in\overline{J}^{2,+}g_j(\widehat z_j)$ for every j, where $\overline{J}^{2,+}$ is the closed superjet of `ClosedSuperjet`;*

(T3) *writing $B_\tau=\bar A+\tau\,\mathrm{id}$,*

$$\sum_j X_j^\tau[\zeta_j] \le \langle B_\tau\zeta,\zeta\rangle+\eta\,|B_\tau\zeta|^2 = A[\zeta]+\tau|\zeta|^2+\eta\,|B_\tau\zeta|^2 \qquad (\zeta\in V),$$

where $X[h]=X(h,h)$, $A[\zeta]=A(\zeta,\zeta)$ and $|B_\tau\zeta|$ is the norm of the linear functional $\xi\mapsto A(\zeta,\xi)+\tau\langle\zeta,\xi\rangle$.

Proof. For $a\in E$ and $1\le j\le N$ let $\iota_j a\in V$ be the vector with j-th component a and all other components 0, so $|\iota_j a|=|a|$. Put

$$\widehat g_j(w)=g_j(\widehat z_j+w)-g_j(\widehat z_j)-\langle\widehat p_j,w\rangle \qquad (w\in E),$$

a function defined on all of E with $\widehat g_j(0)=0$.

Step 0: the constant Λ, chosen before τ. For $\tau\in(0,1]$ the operator $B_\tau=\bar A+\tau\,\mathrm{id}$ is self-adjoint with $\|B_\tau\|\le\|\bar A\|+\tau\le\|\bar A\|+1$. Put

$$\varepsilon(\tau)=\min\Big(\frac{\eta}{2},\ \frac{1}{2\|B_\tau\|+2}\Big)>0, \qquad \Lambda=\max\Big(\frac{2}{\eta},\ 2\|\bar A\|+4\Big)+4\big(\|\bar A\|+1\big).$$

Then $\Lambda \geq 0$, and Λ depends only on η and $\|\bar{A}\|$, not on τ. We record two consequences of the choice of $\varepsilon(\tau)$, writing $\varepsilon = \varepsilon(\tau)$ from now on:

$$\varepsilon\|B_\tau\| \leq \frac{\|B_\tau\|}{2\|B_\tau\| + 2} \leq \tfrac{1}{2}, \qquad 2\varepsilon \leq \eta. \tag{170}$$

Moreover $\varepsilon^{-1} = \max\big(2/\eta,\ 2\|B_\tau\| + 2\big) \leq \max\big(2/\eta,\ 2\|\bar{A}\| + 4\big)$, so that

$$\varepsilon^{-1} + 2 \cdot 2\|B_\tau\| \leq \max\Big(\frac{2}{\eta},\ 2\|\bar{A}\| + 4\Big) + 4\big(\|\bar{A}\| + 1\big) = \Lambda. \tag{171}$$

Now fix $\tau \in (0, 1]$.

Step 1: the quadratic test. By `SumRuleQuadraticTest`, applied with this τ, there are $a > 0$, $C \geq 0$ and continuous functions $G_j : E \to \mathbb{R}$ with $G_j(w) \leq C$ for all w, with $G_j(w) = \widehat{g}_j(w) - |w|^4$ for $|w| \leq a$, and with

$$\sum_j G_j(w_j) - \tfrac{1}{2}\big(A[W] + \tau|W|^2\big) \leq -\sum_j |w_j|^4 \qquad (W \neq 0). \tag{172}$$

Since $\langle B_\tau W, W\rangle = \langle \bar{A}W, W\rangle + \tau|W|^2 = A[W] + \tau|W|^2$, the inequality (172) is exactly $\sum_j G_j(w_j) - \frac{1}{2}\langle B_\tau W, W\rangle \leq -\sum_j |w_j|^4$ for $W \neq 0$. Taking $w = 0$ in the identity $G_j(w) = \widehat{g}_j(w) - |w|^4$ (legitimate since $|0| = 0 \leq a$) gives $G_j(0) = \widehat{g}_j(0) - 0 = 0$.

Step 2: the resolvent. By (170) we have $\varepsilon > 0$ and $\varepsilon\|B_\tau\| \leq \frac{1}{2}$, so `SumRuleResolvent`, applied in V to the self-adjoint operator B_τ with this ε, supplies a self-adjoint continuous linear operator $A_\varepsilon : V \to V$ with the properties (R1)–(R6) listed there; in particular

$$\|A_\varepsilon\| \leq 2\|B_\tau\|, \qquad \langle A_\varepsilon\zeta, \zeta\rangle \leq \langle B_\tau\zeta, \zeta\rangle + \eta|B_\tau\zeta|^2 \quad (\zeta \in V), \tag{173}$$

the second inequality being (R6) with the present η, which is admissible because $2\varepsilon \leq \eta$ by (170).

Step 3: the duality test. The data assembled so far — the continuous G_j with $G_j \leq C$, $C \geq 0$ and $G_j(0) = 0$; the self-adjoint B_τ and A_ε; the parameter $\varepsilon > 0$; the identities (R2), (R3) and the positivity (R4) from Step 2; and the strict inequality (172) in the form of Step 1 — are exactly the hypotheses of `SumRuleDualityTest`. That lemma gives the separability identity $\mathcal{G}^\varepsilon(Z) = \sum_j G_j^\varepsilon(z_j)$ for $\mathcal{G}(W) = \sum_j G_j(w_j)$, and shows that $\Upsilon(Z) = \mathcal{G}^\varepsilon(Z) - \frac{1}{2}\langle A_\varepsilon Z, Z\rangle$ satisfies $\Upsilon(0) = 0$, $\Upsilon(Z) < 0$ for $Z \neq 0$, is λ-semiconvex on V with $\lambda = \varepsilon^{-1} + \|A_\varepsilon\|$ in the sense of `SemiconvexOn`, and is continuous.

Step 4: jets at shrinking scales. Fix $k \geq 1$ and apply `SumRuleJetFromJensen` with the above $G_j, C, \varepsilon, A_\varepsilon, \Upsilon$ and with $R = \delta' = 1/k$. It produces $\widehat{Z}'^{(k)} \in V$, $\Pi^{(k)} \in V$ and symmetric continuous bilinear forms $X_j^{(k)}$ on E with

$$\big|\widehat{Z}'^{(k)}\big| < \tfrac{1}{k}, \qquad \big|\Pi^{(k)}\big| \leq \tfrac{1}{k} + \|A_\varepsilon\|\tfrac{1}{k} = \frac{1 + \|A_\varepsilon\|}{k}, \tag{174}$$

$$\big\|X_j^{(k)}\big\| \leq \varepsilon^{-1} + 2\|A_\varepsilon\|, \tag{175}$$

$$\big(\Pi_j^{(k)}, X_j^{(k)}\big) \in J^{2,+}G_j\big(\widehat{z}_j'^{(k)} + \varepsilon\Pi_j^{(k)}\big), \tag{176}$$

$$\sum_j X_j^{(k)}[\zeta_j] \leq \langle A_\varepsilon\zeta, \zeta\rangle \qquad (\zeta \in V). \tag{177}$$

By (175), (173) and (171),

$$\big\|X_j^{(k)}\big\| \leq \varepsilon^{-1} + 2\|A_\varepsilon\| \leq \varepsilon^{-1} + 4\|B_\tau\| \leq \Lambda \qquad (k \geq 1,\ 1 \leq j \leq N), \tag{178}$$

a bound independent of both k and τ.

Step 5: the limit $k \to \infty$. By (178) and `BilinearFormBolzanoWeierstrass` there are continuous bilinear forms X_j^τ on E and a strictly increasing $\varphi : \mathbb{N} \to \mathbb{N}$ with $X_j^{(\varphi(k))} \to X_j^\tau$ for every j.

Each X_j^τ is symmetric: evaluation at a fixed pair of vectors is a continuous linear functional on the space of continuous bilinear forms, so for fixed $a, b \in E$ the real sequences $X_j^{(\varphi(k))}(a, b)$ and

$X_j^{(\varphi(k))}(b,a)$ converge to $X_j^\tau(a,b)$ and $X_j^\tau(b,a)$ respectively; these two sequences are equal term by term because each $X_j^{(\varphi(k))}$ is symmetric, so their limits agree. Since the norm is continuous, (178) likewise passes to the limit and gives (T1): $\|X_j^\tau\| \le \Lambda$.

Fix $\zeta \in V$. Evaluation at (ζ_j, ζ_j) is continuous, so $\sum_j X_j^{(\varphi(k))}[\zeta_j] \to \sum_j X_j^\tau[\zeta_j]$; passing to the limit in (177) along the subsequence gives

$$\sum_j X_j^\tau[\zeta_j] \le \langle A_\varepsilon \zeta, \zeta\rangle \qquad (\zeta \in V). \tag{179}$$

Next, put $z_k = \widehat{z}_j^{\prime(\varphi(k))} + \varepsilon \Pi_j^{(\varphi(k))}$, $p_k = \Pi_j^{(\varphi(k))}$ and $\Xi_k = X_j^{(\varphi(k))}$ for a fixed j. From (174) and $\varphi(k) \ge k$ we get $\left|\widehat{z}_j^{\prime(\varphi(k))}\right| \le \left|\widehat{Z}^{\prime(\varphi(k))}\right| < 1/\varphi(k) \le 1/k$ and $|p_k| \le \left|\Pi^{(\varphi(k))}\right| \le (1 + \|A_\varepsilon\|)/k$, so $p_k \to 0$ and $z_k \to 0$. By (176), $(p_k, \Xi_k) \in J^{2,+}G_j(z_k)$; by continuity of G_j, $G_j(z_k) \to G_j(0)$; and $\Xi_k \to X_j^\tau$. These are exactly the four requirements of `ClosedSuperjet`, so

$$\big(0, X_j^\tau\big) \in \overline{J}^{2,+}G_j(0) \qquad (1 \le j \le N). \tag{180}$$

Step 6: back to g_j. By Step 1, $G_j(w) = \widehat{g}_j(w) - |w|^4$ whenever $|w| \le a$, so in particular the two functions G_j and $\widehat{g}_j - |\cdot|^4$ agree on the ball $\{|w| < a\}$ with $a > 0$. By `ClosedSuperjetLocal`, $\overline{J}^{2,+}G_j(0) = \overline{J}^{2,+}\big(\widehat{g}_j - |\cdot|^4\big)(0)$, and by `ClosedSuperjetQuartic` the latter equals $\overline{J}^{2,+}\widehat{g}_j(0)$. Hence (180) gives $\big(0, X_j^\tau\big) \in \overline{J}^{2,+}\widehat{g}_j(0)$. Applying `ClosedSuperjetAffineShift` with $z = \widehat{z}_j$, $p = \widehat{p}_j$, $q = 0$ and $X = X_j^\tau$ — so that the normalized function there is exactly $\widehat{g}_j$ — converts this into

$$\big(\widehat{p}_j, X_j^\tau\big) \in \overline{J}^{2,+}g_j(\widehat{z}_j),$$

which is (T2).

Step 7: the matrix inequality for B_τ. Combining (179) with the second inequality of (173) gives, for every $\zeta \in V$,

$$\sum_j X_j^\tau[\zeta_j] \le \langle A_\varepsilon\zeta,\zeta\rangle \le \langle B_\tau\zeta,\zeta\rangle + \eta|B_\tau\zeta|^2 = A[\zeta] + \tau|\zeta|^2 + \eta|B_\tau\zeta|^2,$$

which is (T3). As $\tau \in (0,1]$ was arbitrary and Λ was fixed in Step 0 before τ, the proof is complete. □

2.48. `SumRuleJets`.

Lemma 2.48 (Theorem of sums)**.** *Let E be a finite-dimensional real inner-product space, let N be a natural number and write $V = \ell^2\big(\bigoplus_{j=1}^N E\big)$ for the ℓ^2-product of N copies of E, whose elements are written $Y = (y_j)_{j=1}^N$ and whose norm satisfies $|Y|^2 = \sum_j |y_j|^2$. For each j let $O_j \subseteq E$ be open and let $g_j : E \to \mathbb{R}$ be continuous on O_j. Let $\Psi : V \to \mathbb{R}$, let $\widehat{Z} \in V$ with $\widehat{z}_j \in O_j$ for every j, and assume that Ψ is C^2 near $\widehat{Z}$ and that*

$$Y \longmapsto \sum_{j=1}^N g_j(y_j) - \Psi(Y)$$

has a local maximum at $\widehat{Z}$. Set $A = D^2\Psi(\widehat{Z})$, a symmetric bilinear form on V, let $\bar{A} : V \to V$ be the self-adjoint operator with $\langle \bar{A}\xi, \zeta\rangle = A(\xi,\zeta)$, so that $|\bar{A}\xi|$ is the norm of the linear functional $\zeta \mapsto A(\xi,\zeta)$ and $|\bar{A}\xi|^2 = \langle \bar{A}^2\xi, \xi\rangle$ is the value at ξ of the quadratic form of $\bar{A}^2$; the matrix inequality below is thus the paper's $\operatorname{diag}(X_1,\dots,X_N) \le A + \eta A^2$ read as an inequality between quadratic forms. Write $D_j\Psi(\widehat{Z})$ for the j-th component of $\nabla\Psi(\widehat{Z})$. Then for every $\eta > 0$ there are symmetric continuous bilinear forms X_j on E such that

$$\big(D_j\Psi(\widehat{Z}), X_j\big) \in \overline{J}^{2,+}g_j(\widehat{z}_j) \quad (1 \le j \le N), \qquad \sum_{j=1}^N X_j[\xi_j] \le A[\xi] + \eta\,|\bar{A}\xi|^2 \quad (\xi \in V),$$

where $\overline{J}^{2,+}$ is the closed superjet of `ClosedSuperjet`, $X[h] = X(h,h)$ and $A[\xi] = A(\xi,\xi)$.

Proof. If $N = 0$ then $V = \{0\}$, the family of X_j is empty and both assertions are vacuous, so assume $N \geq 1$. Fix $\eta > 0$. Write $\widehat{p}_j = D_j\Psi(\widehat{Z})$, so that $\widehat{P} := \nabla\Psi(\widehat{Z})$ has components $\widehat{p}_j$ and

$$\langle \widehat{P}, W\rangle = \sum_j \langle \widehat{p}_j, w_j\rangle \qquad (W \in V), \tag{181}$$

by the definition of the ℓ^2-product inner product (preamble). For $a \in E$ and $1 \leq j \leq N$ let $\iota_j a \in V$ be the vector with j-th component a and all other components 0; then $|\iota_j a| = |a|$ and $\langle \xi, \iota_j a\rangle = \langle \xi_j, a\rangle$.

Part I: four facts about superjets.

(A) Superjets and closed superjets are local. If $f_1, f_2 : E \to \mathbb{R}$ agree on a ball $B(0, b)$ with $b > 0$, then $\overline{J}^{2,+} f_1(0) = \overline{J}^{2,+} f_2(0)$. Indeed, let $(p, X) \in \overline{J}^{2,+} f_1(0)$ with defining sequences (z_k, p_k, X_k) as in `ClosedSuperjet`. Choose k_0 with $|z_k| < b/4$ for $k \geq k_0$. For $k \geq k_0$ and any $\varepsilon' > 0$, the superjet condition of `Superjet` for f_1 at z_k supplies some $\delta > 0$, and replacing δ by $\min(\delta, b/4)$ — which only weakens the condition — we may assume $|h| < \delta$ implies $|z_k + h| < b/2 < b$, so that f_1 and f_2 agree at z_k and at $z_k + h$; hence the same inequality holds for f_2, giving $(p_k, X_k) \in J^{2,+} f_2(z_k)$. Also $f_2(z_k) = f_1(z_k) \to f_1(0) = f_2(0)$. Re-indexing the sequences by $k \geq k_0$ (which changes none of the limits) exhibits $(p, X) \in \overline{J}^{2,+} f_2(0)$. The reverse inclusion is symmetric.

(B) Subtracting $|\cdot|^4$ does not change the closed superjet at 0. Let $s(w) = |w|^4$. Then s is infinitely differentiable with $\nabla s(w) = 4|w|^2 w$ and $D^2 s(w)(a, b) = 4|w|^2\langle a, b\rangle + 8\langle w, a\rangle\langle w, b\rangle$, so $\nabla s(w) \to 0$ and $\|D^2 s(w)\| \to 0$ as $w \to 0$, and $D^2 s(w)$ is symmetric. For any $f : E \to \mathbb{R}$, any $z \in E$ and any pair (p, X) with X symmetric,

$$(p, X) \in J^{2,+}(f - s)(z) \iff \big(p + \nabla s(z),\, X + D^2 s(z)\big) \in J^{2,+} f(z). \tag{182}$$

Indeed, by the second-order Taylor expansion of the smooth s at z (preamble), for every $\varepsilon'' > 0$ there is $\delta'' > 0$ with $|s(z + h) - s(z) - \langle \nabla s(z), h\rangle - \frac{1}{2} D^2 s(z)(h, h)| \leq \varepsilon''|h|^2$ for $|h| < \delta''$; adding this to the defining inequality of `Superjet` for $f - s$ at z with parameter ε' turns it into the inequality for f at z with the shifted jet and parameter $\varepsilon' + \varepsilon''$, and conversely. Since the defining condition quantifies over all positive parameters, the two memberships are equivalent. Passing to closed superjets, the defining sequences (z_k, p_k, X_k) for $f - s$ at 0 correspond to the sequences $(z_k, p_k + \nabla s(z_k), X_k + D^2 s(z_k))$ for f at 0; because $\nabla s(z_k) \to 0$ and $D^2 s(z_k) \to 0$ the limits (p, X) are unchanged, and $(f - s)(z_k) \to (f - s)(0)$ is equivalent to $f(z_k) \to f(0)$ since s is continuous. Hence $\overline{J}^{2,+}(f - s)(0) = \overline{J}^{2,+} f(0)$.

(C) Translating and subtracting an affine function. Fix j and set $\widehat{g}_j(w) = g_j(\widehat{z}_j + w) - g_j(\widehat{z}_j) - \langle \widehat{p}_j, w\rangle$, so $\widehat{g}_j(0) = 0$. For every $w_0 \in E$ and every pair (q, X),

$$(q, X) \in J^{2,+}\widehat{g}_j(w_0) \iff (q + \widehat{p}_j, X) \in J^{2,+} g_j(\widehat{z}_j + w_0),$$

because substituting the definition of $\widehat{g}_j$ into the inequality of `Superjet` at w_0 and cancelling the affine terms $-\langle \widehat{p}_j, w_0 + h\rangle$ and $-\langle \widehat{p}_j, w_0\rangle$ yields exactly the inequality for g_j at $\widehat{z}_j + w_0$ with the gradient increased by $\widehat{p}_j$ and the same δ. Translating the defining sequences accordingly — $w_k \to 0$ iff $\widehat{z}_j + w_k \to \widehat{z}_j$, and $\widehat{g}_j(w_k) \to 0$ iff $g_j(\widehat{z}_j + w_k) \to g_j(\widehat{z}_j)$ — gives

$$(q, X) \in \overline{J}^{2,+}\widehat{g}_j(0) \iff (q + \widehat{p}_j, X) \in \overline{J}^{2,+} g_j(\widehat{z}_j). \tag{183}$$

(D) The closed superjet is a closed set. Let $f : E \to \mathbb{R}$, $z \in E$, and let $(p^{(m)}, X^{(m)}) \in \overline{J}^{2,+} f(z)$ with $p^{(m)} \to p$ and $X^{(m)} \to X$. For each m pick defining sequences $(z_k^{(m)}, p_k^{(m)}, X_k^{(m)})_k$ as in `ClosedSuperjet`, and then $k(m)$ so large that

$$\big|z_{k(m)}^{(m)} - z\big| < \tfrac{1}{m}, \quad \big|f(z_{k(m)}^{(m)}) - f(z)\big| < \tfrac{1}{m}, \quad \big|p_{k(m)}^{(m)} - p^{(m)}\big| < \tfrac{1}{m}, \quad \big\|X_{k(m)}^{(m)} - X^{(m)}\big\| < \tfrac{1}{m}.$$

Setting $\zeta_m = z_{k(m)}^{(m)}$, $\pi_m = p_{k(m)}^{(m)}$, $\Xi_m = X_{k(m)}^{(m)}$ we get $(\pi_m, \Xi_m) \in J^{2,+} f(\zeta_m)$, $\zeta_m \to z$, $f(\zeta_m) \to f(z)$, $\pi_m \to p$ and $\Xi_m \to X$; so $(p, X) \in \overline{J}^{2,+} f(z)$.

Part II: reduction to a quadratic test with a strict global maximum.

Fix $\tau \in (0, 1]$ and let $B_\tau : V \to V$ be the self-adjoint operator $B_\tau = \bar{A} + \tau\,\mathrm{id}$.

(II.1) A quadratic majorant. By the second-order Taylor expansion of Ψ at $\widehat{Z}$ (preamble), $\Psi(\widehat{Z}+W) = \Psi(\widehat{Z}) + \langle \widehat{P}, W\rangle + \frac{1}{2}A(W,W) + R(W)$ with $|R(W)|/|W|^2 \to 0$. The local maximum at $\widehat{Z}$ gives $\rho' > 0$ with $\sum_j g_j(\widehat{z}_j + w_j) - \Psi(\widehat{Z}+W) \le \sum_j g_j(\widehat{z}_j) - \Psi(\widehat{Z})$ for $|W| \le \rho'$. Using (181), for $|W| \le \rho'$,

$$\sum_j \widehat{g}_j(w_j) = \sum_j g_j(\widehat{z}_j+w_j) - \sum_j g_j(\widehat{z}_j) - \langle \widehat{P}, W\rangle \le \Psi(\widehat{Z}+W) - \Psi(\widehat{Z}) - \langle \widehat{P}, W\rangle = \tfrac{1}{2}A(W,W) + R(W).$$

Choose $\rho_\tau \in (0, \rho']$ with $|R(W)| \le (\tau/2)|W|^2$ for $|W| \le \rho_\tau$. Since $\frac{1}{2}A(W,W) + (\tau/2)|W|^2 = \frac{1}{2}\langle B_\tau W, W\rangle$, we obtain

$$\sum_j \widehat{g}_j(w_j) \le \tfrac{1}{2}\langle B_\tau W, W\rangle \qquad (|W| \le \rho_\tau), \tag{184}$$

with equality at $W = 0$ (both sides vanish, as $\widehat{g}_j(0) = 0$).

(II.2) Localization. Each O_j is open and $\widehat{z}_j \in O_j$, so there is $\rho'' > 0$ with $\overline{B}(\widehat{z}_j, \rho'') \subseteq O_j$ for every one of the finitely many j. Put

$$a = \frac{\min(\rho_\tau, \rho'')}{1 + \sqrt{N}} > 0.$$

Then $a \le \rho''$, so $\widehat{g}_j$ is defined and continuous on $\overline{B}(0,a)$; and if $|w_j| \le a$ for all j then $|W| \le a\sqrt{N} \le \rho_\tau$, so (184) applies. Let $M_j = \max_{\overline{B}(0,a)} |\widehat{g}_j|$, finite by continuity and compactness (preamble) and nonnegative because $\widehat{g}_j(0) = 0$; put $C = \max_j M_j \ge 0$.

Let $\pi : E \to \overline{B}(0,a)$ be the radial retraction, $\pi(w) = w$ for $|w| \le a$ and $\pi(w) = aw/|w|$ for $|w| > a$; it is continuous. Write $(s)_+ = \max(s, 0)$ and set $K = \|B_\tau\| N(a+1) + 1$ and

$$G_j(w) = \widehat{g}_j(\pi(w)) - K(|w| - a)_+ - K(|w| - a)_+^2 - |w|^4.$$

Then:

- (G1) G_j is continuous on E and $G_j \le M_j \le C \le C + |w|$ for all w, since the three subtracted terms are nonnegative.
- (G2) $G_j(w) = \widehat{g}_j(w) - |w|^4$ for $|w| \le a$.
- (G3) Writing $\Phi(W) = \sum_j G_j(w_j) - \frac{1}{2}\langle B_\tau W, W\rangle$, we have $\Phi(0) = 0$ and $\Phi(W) \le -\sum_j |w_j|^4 < 0$ for $W \ne 0$.

Only (G3) needs an argument. $\Phi(0) = \sum_j \widehat{g}_j(0) = 0$. Let $W \ne 0$, put $m = \max_j(|w_j| - a)_+ \ge 0$ and $\Pi W = (\pi(w_j))_j$. If $m = 0$ then $|w_j| \le a$ for all j, so $\pi(w_j) = w_j$ and the penalty terms vanish; by (184) (applicable since $|W| \le a\sqrt{N} \le \rho_\tau$), $\Phi(W) \le -\sum_j |w_j|^4$. If $m > 0$, then $|\pi(w_j)| \le a$ for all j gives $|\Pi W| \le a\sqrt{N} \le \rho_\tau$, so (184) applied to ΠW gives $\sum_j \widehat{g}_j(\pi(w_j)) \le \frac{1}{2}\langle B_\tau \Pi W, \Pi W\rangle$; moreover $\sum_j (|w_j| - a)_+ \ge m$ and $\sum_j (|w_j| - a)_+^2 \ge m^2$, whence

$$\Phi(W) \le \tfrac{1}{2}\big[\langle B_\tau \Pi W, \Pi W\rangle - \langle B_\tau W, W\rangle\big] - Km - Km^2 - \sum_j |w_j|^4.$$

Since B_τ is self-adjoint, $\langle B_\tau W, W\rangle - \langle B_\tau \Pi W, \Pi W\rangle = \langle B_\tau(W - \Pi W), W + \Pi W\rangle$. Here $|W - \Pi W|^2 = \sum_j \big((|w_j| - a)_+\big)^2 \le Nm^2$, so $|W - \Pi W| \le \sqrt{N}m$, and $|W + \Pi W| \le |W| + |\Pi W| \le 2a\sqrt{N} + \sqrt{N}m$. Therefore

$$\tfrac{1}{2}\big|\langle B_\tau \Pi W, \Pi W\rangle - \langle B_\tau W, W\rangle\big| \le \tfrac{1}{2}\|B_\tau\|\sqrt{N}m\big(2a\sqrt{N} + \sqrt{N}m\big) = \|B_\tau\| Nam + \tfrac{1}{2}\|B_\tau\| Nm^2,$$

and so

$$\Phi(W) \le m\big(\|B_\tau\| Na - K\big) + m^2\big(\tfrac{1}{2}\|B_\tau\| N - K\big) - \sum_j |w_j|^4 \le -\sum_j |w_j|^4,$$

because $K \ge \|B_\tau\| Na$ and $K \ge \|B_\tau\| N \ge \frac{1}{2}\|B_\tau\| N$ by the choice of K. Finally $\sum_j |w_j|^4 > 0$ since $W \ne 0$, which proves (G3).

Part III: the sup-convolution duality.

Put

$$\varepsilon = \min\Big(\frac{\eta}{2},\ \frac{1}{2\|B_\tau\|+2}\Big) > 0,$$

so that $\varepsilon\|B_\tau\| \le \|B_\tau\|/(2\|B_\tau\|+2) \le \frac12$ and $2\varepsilon \le \eta$; the second property is used only in (V.2). Then $\mathrm{id} - \varepsilon B_\tau$ is invertible, with inverse given by the convergent Neumann series $\sum_{k\ge0}(\varepsilon B_\tau)^k$, which is self-adjoint and commutes with B_τ. Put

$$A_\varepsilon = B_\tau(\mathrm{id} - \varepsilon B_\tau)^{-1},$$

a self-adjoint operator on V, and note the identity

$$\mathrm{id} + \varepsilon A_\varepsilon = \big[(\mathrm{id} - \varepsilon B_\tau) + \varepsilon B_\tau\big](\mathrm{id} - \varepsilon B_\tau)^{-1} = (\mathrm{id} - \varepsilon B_\tau)^{-1}. \tag{185}$$

Since $\langle(\mathrm{id} - \varepsilon B_\tau)\xi,\xi\rangle \ge (1 - \varepsilon\|B_\tau\|)|\xi|^2 \ge \frac12|\xi|^2$, the operator $\mathrm{id} - \varepsilon B_\tau$ is positive definite, hence so is its inverse $\mathrm{id} + \varepsilon A_\varepsilon$, and therefore so is $\varepsilon^{-1}\mathrm{id} + A_\varepsilon = \varepsilon^{-1}(\mathrm{id} + \varepsilon A_\varepsilon)$:

$$\big\langle(\varepsilon^{-1}\mathrm{id} + A_\varepsilon)\xi,\xi\big\rangle > 0 \qquad (\xi \ne 0). \tag{186}$$

Set $\mathcal{G}(W) = \sum_j G_j(w_j)$, a continuous function on V with $\mathcal{G} \le NC$, so that $\mathcal{G}(W) \le NC + |W|$, and let $\mathcal{G}^\varepsilon$ be its sup-convolution on V (SupConvolution). By part (1) of SupConvolutionSemiconvex, applied in V with the growth constant NC, each defining set is bounded above, so $\mathcal{G}^\varepsilon$ is finite; likewise each G_j^ε is finite by (G1) and part (1) of SupConvolutionSemiconvex in E.

(III.1) Separability. Since $|Z - W|^2 = \sum_j |z_j - w_j|^2$,

$$\mathcal{G}^\varepsilon(Z) = \sup_{W\in V}\sum_j\Big(G_j(w_j) - \frac{|z_j - w_j|^2}{2\varepsilon}\Big) = \sum_j \sup_{w_j\in E}\Big(G_j(w_j) - \frac{|z_j - w_j|^2}{2\varepsilon}\Big) = \sum_j G_j^\varepsilon(z_j),$$

the middle step because a supremum of a sum of functions of independent variables, each bounded above, is the sum of the suprema.

(III.2) The duality identity. For $(Z,W) \in V\times V$ put $F(Z,W) = \mathcal{G}(W) - |Z-W|^2/(2\varepsilon) - \frac12\langle A_\varepsilon Z, Z\rangle$, and $\Upsilon(Z) = \mathcal{G}^\varepsilon(Z) - \frac12\langle A_\varepsilon Z, Z\rangle$, so that $\Upsilon(Z) = \sup_W F(Z,W)$.

Fix W and consider $Z \mapsto F(Z,W)$. Its quadratic part is $-\frac12\langle(\varepsilon^{-1}\mathrm{id} + A_\varepsilon)Z, Z\rangle$, which by (186) is negative definite; so $F(\cdot,W)$ is a strictly concave quadratic and has a unique maximum at its critical point, determined by $-(Z-W)/\varepsilon - A_\varepsilon Z = 0$, i.e. $(\mathrm{id} + \varepsilon A_\varepsilon)Z = W$, i.e. $Z_W = (\mathrm{id} - \varepsilon B_\tau)W$ by (185). Writing $T = \mathrm{id} - \varepsilon B_\tau$ we have $Z_W - W = -\varepsilon B_\tau W$, so $|Z_W - W|^2/(2\varepsilon) = \frac{\varepsilon}{2}|B_\tau W|^2$, and

$$\langle A_\varepsilon Z_W, Z_W\rangle = \langle B_\tau T^{-1}TW, TW\rangle = \langle B_\tau W, TW\rangle = \langle B_\tau W, W\rangle - \varepsilon|B_\tau W|^2.$$

Hence

$$\sup_Z F(Z,W) = \mathcal{G}(W) - \frac{\varepsilon}{2}|B_\tau W|^2 - \frac12\Big(\langle B_\tau W, W\rangle - \varepsilon|B_\tau W|^2\Big) = \mathcal{G}(W) - \tfrac12\langle B_\tau W, W\rangle = \Phi(W). \tag{187}$$

(III.3) Υ has a strict global maximum at 0. By (187) and (G3), $\sup_{Z,W} F = \sup_W \Phi = 0$, and $F(Z,W) = 0$ forces $\Phi(W) \ge F(Z,W) = 0$, hence $W = 0$ by (G3), and then $Z = Z_0 = T\cdot 0 = 0$ by the uniqueness of the maximizer of $F(\cdot,0)$. So the joint maximum 0 is attained exactly at $(0,0)$. Consequently $\Upsilon \le 0$ everywhere, and $\Upsilon(0) = \sup_W F(0,W) \ge F(0,0) = 0$, so $\Upsilon(0) = 0$. Finally, for each Z the supremum $\Upsilon(Z) = \sup_W F(Z,W)$ is attained: $F(Z,\cdot)$ is continuous, and $F(Z,W) \le NC - |Z-W|^2/(2\varepsilon) - \frac12\langle A_\varepsilon Z, Z\rangle$, whose right-hand side tends to $-\infty$ as $|W| \to \infty$. Choose therefore $S > 0$ so large that this bound is $< F(Z,Z)$ whenever $|W - Z| > S$; then $\sup_W F(Z,W) = \max_{\overline{B}(Z,S)} F(Z,\cdot)$, since every W outside $\overline{B}(Z,S)$ gives a value below the one already attained at $W = Z$ inside the ball. The ball $\overline{B}(Z,S)$ is closed and bounded, hence compact and nonempty, so the maximum is attained (preamble). Hence $\Upsilon(Z) = 0$ implies $F(Z,W) = 0$ for some W, which forces $Z = 0$. Therefore

$$\Upsilon(0) = 0, \qquad \Upsilon(Z) < 0 \ \ (Z \ne 0). \tag{188}$$

(III.4) Υ is semiconvex and continuous. By part (3) of SupConvolutionSemiconvex, applied in V to $\mathcal{G}$ with growth constant NC, the function $\mathcal{G}^\varepsilon + |\cdot|^2/(2\varepsilon)$ is convex on V. Also $Z \mapsto$

$-\frac{1}{2}\langle A_\varepsilon Z, Z\rangle + \frac{1}{2}\|A_\varepsilon\|\,|Z|^2 = \frac{1}{2}\langle(\|A_\varepsilon\|\mathrm{id} - A_\varepsilon)Z, Z\rangle$ is convex: for a self-adjoint S with $\langle S\xi,\xi\rangle \geq 0$ and $t_1, t_2 \geq 0$, $t_1 + t_2 = 1$, one has

$$\langle S(t_1\xi_1 + t_2\xi_2), t_1\xi_1 + t_2\xi_2\rangle - t_1\langle S\xi_1,\xi_1\rangle - t_2\langle S\xi_2,\xi_2\rangle = -t_1t_2\langle S(\xi_1-\xi_2),\xi_1-\xi_2\rangle \leq 0,$$

and here $S = \|A_\varepsilon\|\mathrm{id} - A_\varepsilon$ satisfies $\langle S\xi,\xi\rangle \geq 0$. Adding the two convex functions shows that $\Upsilon + \frac{\lambda}{2}|\cdot|^2$ is convex on V with

$$\lambda := \varepsilon^{-1} + \|A_\varepsilon\| > 0,$$

that is, Υ is λ-semiconvex on V in the sense of `SemiconvexOn`; the same then holds on any convex subset of V. Moreover $\Upsilon + \frac{\lambda}{2}|\cdot|^2$ is a finite convex function on the open set V, hence continuous (preamble), so Υ is continuous.

Part IV: Jensen's lemma and the second-order expansion.

Let $R > 0$ and $\delta' > 0$. By (188) the point 0 is a strict maximum point of Υ on $\overline{B}(0,R)$, and Υ is continuous there and λ-semiconvex on $B(0,R)$ by (III.4). Fix an additive Haar measure μ_V on V. By `JensenLemma` the set

$$\mathcal{K} = \big\{Z \in B(0,R) : \exists\, Q,\ |Q| \leq \delta',\ \Upsilon + \langle Q,\cdot\rangle \text{ has a local maximum on } B(0,R) \text{ at } Z\big\}$$

satisfies $\mu_V(\mathcal{K}) > 0$. By `SemiconvexSecondOrderAe`, applied in V with this λ and the open convex set $B(0,R)$, there is a μ_V-null set $\mathcal{N}$ such that every $Z \in B(0,R)\setminus\mathcal{N}$ admits $q_\Upsilon \in V$ and a symmetric continuous bilinear form A_Υ on V with $A_\Upsilon(\zeta,\zeta) \geq -\lambda|\zeta|^2$ and, for every $\delta > 0$, some $\rho(\delta) > 0$ with

$$\big|\Upsilon(Z+\zeta) - \Upsilon(Z) - \langle q_\Upsilon,\zeta\rangle - \tfrac{1}{2}A_\Upsilon(\zeta,\zeta)\big| \leq \delta|\zeta|^2 \qquad (|\zeta| < \rho(\delta)). \tag{189}$$

Since $\mu_V(\mathcal{K}) > 0 = \mu_V(\mathcal{N})$, we may pick $\widehat{Z}' \in \mathcal{K}\setminus\mathcal{N}$; fix a multiplier Q with $|Q| \leq \delta'$ and expansion data (q_Υ, A_Υ) at $\widehat{Z}'$.

(IV.1) First- and second-order conditions. As $B(0,R)$ is open and $\widehat{Z}' \in B(0,R)$, there is $\rho_1 > 0$ with $B(\widehat{Z}',\rho_1) \subseteq B(0,R)$ and

$$\Upsilon(\widehat{Z}'+\zeta) - \Upsilon(\widehat{Z}') \leq -\langle Q,\zeta\rangle \qquad (|\zeta| < \rho_1). \tag{190}$$

Let $\delta > 0$ and $|\zeta| < \min(\rho_1, \rho(\delta))$. Combining (189) and (190),

$$\langle q_\Upsilon + Q,\zeta\rangle \leq -\tfrac{1}{2}A_\Upsilon(\zeta,\zeta) + \delta|\zeta|^2.$$

Taking $\zeta = tv$ with $|v| = 1$ and $t \downarrow 0$, and dividing by t, gives $\langle q_\Upsilon + Q, v\rangle \leq 0$ for every unit v, hence

$$q_\Upsilon = -Q. \tag{191}$$

Substituting (191) back, $\frac{1}{2}A_\Upsilon(\zeta,\zeta) \leq \delta|\zeta|^2$ for $|\zeta|$ small; taking $\zeta = tv$, dividing by t^2 and letting $\delta \downarrow 0$ gives

$$A_\Upsilon(\zeta,\zeta) \leq 0 \qquad (\zeta \in V). \tag{192}$$

(IV.2) The expansion of $\mathcal{G}^\varepsilon$ and the forms X_j. Since $\mathcal{G}^\varepsilon(Z) = \Upsilon(Z) + \frac{1}{2}\langle A_\varepsilon Z, Z\rangle$ and, exactly, $\frac{1}{2}\langle A_\varepsilon(\widehat{Z}'+\zeta),\widehat{Z}'+\zeta\rangle = \frac{1}{2}\langle A_\varepsilon\widehat{Z}',\widehat{Z}'\rangle + \langle A_\varepsilon\widehat{Z}',\zeta\rangle + \frac{1}{2}\langle A_\varepsilon\zeta,\zeta\rangle$ (using self-adjointness), (189) becomes: with

$$\Pi := q_\Upsilon + A_\varepsilon\widehat{Z}' = -Q + A_\varepsilon\widehat{Z}', \qquad A' := A_\Upsilon + \langle A_\varepsilon\cdot,\cdot\rangle,$$

a symmetric continuous bilinear form on V,

$$\big|\mathcal{G}^\varepsilon(\widehat{Z}'+\zeta) - \mathcal{G}^\varepsilon(\widehat{Z}') - \langle\Pi,\zeta\rangle - \tfrac{1}{2}A'(\zeta,\zeta)\big| \leq \delta|\zeta|^2 \qquad (|\zeta| < \rho(\delta)). \tag{193}$$

Define, for each j, the continuous bilinear form $X_j(a,b) = A'(\iota_j a, \iota_j b)$ on E; it is symmetric because A' is. Taking $\zeta = \iota_j h$ in (193) and using (III.1) — which shows that changing only the j-th component of Z changes $\mathcal{G}^\varepsilon$ only through G_j^ε — together with $\langle\Pi,\iota_j h\rangle = \langle\Pi_j, h\rangle$ and $|\iota_j h| = |h|$,

$$\big|G_j^\varepsilon(\widehat{z}_j'+h) - G_j^\varepsilon(\widehat{z}_j') - \langle\Pi_j,h\rangle - \tfrac{1}{2}X_j(h,h)\big| \leq \delta|h|^2 \qquad (|h| < \rho(\delta)). \tag{194}$$

In particular, taking $\delta = \varepsilon'$ for an arbitrary $\varepsilon' > 0$, the defining inequality of `Superjet` holds, so

$$(\Pi_j, X_j) \in J^{2,+}G_j^\varepsilon(\widehat{z}_j') \qquad (1 \leq j \leq N). \tag{195}$$

(IV.3) The matrix inequality. Summing (194) over j with $h=\zeta_j$: if $|\zeta|<\rho(\delta)$ then $|\zeta_j|\le|\zeta|<\rho(\delta)$ for every j, so

$$\Big|\mathcal{G}^\varepsilon(\widehat{Z}'+\zeta)-\mathcal{G}^\varepsilon(\widehat{Z}')-\langle\Pi,\zeta\rangle-\tfrac12\sum_j X_j[\zeta_j]\Big|\le\sum_j\delta|\zeta_j|^2=\delta|\zeta|^2,$$

where we used (III.1) again, together with $\sum_j\langle\Pi_j,\zeta_j\rangle=\langle\Pi,\zeta\rangle$ and $\sum_j|\zeta_j|^2=|\zeta|^2$, both instances of the definition of the ℓ^2-product inner product (preamble). Comparing with (193) gives $\big|\frac12A'(\zeta,\zeta)-\frac12\sum_j X_j[\zeta_j]\big|\le2\delta|\zeta|^2$ for $|\zeta|<\rho(\delta)$. Both $\zeta\mapsto A'(\zeta,\zeta)$ and $\zeta\mapsto\sum_j X_j[\zeta_j]$ are homogeneous of degree two, so replacing ζ by tv with $|v|=1$ and dividing by t^2 shows $|\frac12A'(v,v)-\frac12\sum_j X_j[v_j]|\le2\delta$ for every unit v and every $\delta>0$. Hence

$$\sum_j X_j[\zeta_j]=A'(\zeta,\zeta)=A_\Upsilon(\zeta,\zeta)+\langle A_\varepsilon\zeta,\zeta\rangle\qquad(\zeta\in V).\tag{196}$$

By (192) and $A_\Upsilon(\zeta,\zeta)\ge-\lambda|\zeta|^2$,

$$-\big(\lambda+\|A_\varepsilon\|\big)|\zeta|^2\le\sum_j X_j[\zeta_j]\le\langle A_\varepsilon\zeta,\zeta\rangle\le\|A_\varepsilon\|\,|\zeta|^2.\tag{197}$$

Taking $\zeta=\iota_j a$ in (197) gives $|X_j[a]|\le(\lambda+\|A_\varepsilon\|)|a|^2$, and by the polarization identity $X_j(a,b)=\frac14\big(X_j[a+b]-X_j[a-b]\big)$ together with $|a+b|^2+|a-b|^2=2(|a|^2+|b|^2)$ we get $\|X_j\|\le\lambda+\|A_\varepsilon\|$.

(IV.4) Transfer to G_j. By (G1) the function G_j is continuous with $G_j(w)\le C+|w|$, and $\varepsilon>0$; so `SupConvolutionMagic` applies to the jet (195) and yields

$$(\Pi_j,X_j)\in J^{2,+}G_j\big(\widehat{z}'_j+\varepsilon\Pi_j\big).\tag{198}$$

Part V: the limits.

(V.1) Letting $R,\delta'\downarrow0$. Fix $R_0>0$ and apply Part IV with $R=\min(R_0,1/k)$ and $\delta'=1/k$, for $k=1,2,\dots$; this produces $\widehat{Z}'^{(k)}$ with $|\widehat{Z}'^{(k)}|<1/k$, a multiplier $Q^{(k)}$ with $|Q^{(k)}|\le1/k$, the vector $\Pi^{(k)}=-Q^{(k)}+A_\varepsilon\widehat{Z}'^{(k)}$ and symmetric forms $X_j^{(k)}$ satisfying (197) and (198). Then $|\Pi^{(k)}|\le(1+\|A_\varepsilon\|)/k\to0$, so the base points $\widehat{z}_j'^{(k)}+\varepsilon\Pi_j^{(k)}\to0$, and by (IV.3) the forms $X_j^{(k)}$ are bounded by $\lambda+\|A_\varepsilon\|$, a bound independent of k. The space of N-tuples of continuous bilinear forms on the finite-dimensional space E is itself finite-dimensional, so by Bolzano–Weierstrass (preamble) there is a subsequence along which $X_j^{(k)}\to X_j^\infty$ for every j. Each X_j^∞ is symmetric, and passing to the limit in (197),

$$\sum_j X_j^\infty[\zeta_j]\le\langle A_\varepsilon\zeta,\zeta\rangle,\qquad\sum_j X_j^\infty[\zeta_j]\ge-\big(\lambda+\|A_\varepsilon\|\big)|\zeta|^2.\tag{199}$$

Moreover the sequences $z_k=\widehat{z}_j'^{(k)}+\varepsilon\Pi_j^{(k)}\to0$, $p_k=\Pi_j^{(k)}\to0$, $X_k=X_j^{(k)}\to X_j^\infty$ satisfy $(p_k,X_k)\in J^{2,+}G_j(z_k)$ by (198) and $G_j(z_k)\to G_j(0)$ by the continuity of G_j; this is exactly the defining condition of `ClosedSuperjet`, so

$$(0,X_j^\infty)\in\overline{J}^{2,+}G_j(0)\qquad(1\le j\le N).\tag{200}$$

By (G2) the functions G_j and $\widehat{g}_j-|\cdot|^4$ agree on $B(0,a)$, so Part I(A) and Part I(B) give $\overline{J}^{2,+}G_j(0)=\overline{J}^{2,+}\widehat{g}_j(0)$, and then (183) turns (200) into

$$(\widehat{p}_j,X_j^\infty)\in\overline{J}^{2,+}g_j(\widehat{z}_j)\qquad(1\le j\le N).\tag{201}$$

(V.2) Comparing A_ε with $B_\tau+\eta B_\tau^2$. Recall from Part III that $\varepsilon\|B_\tau\|\le\frac12$ and $2\varepsilon\le\eta$. Since B_τ is self-adjoint on the finite-dimensional space V, there is an orthonormal basis (v_i) of V with $B_\tau v_i=b_iv_i$, $b_i\in\mathbb{R}$, $|b_i|\le\|B_\tau\|$. Then $(\mathrm{id}-\varepsilon B_\tau)v_i=(1-\varepsilon b_i)v_i$ with $1-\varepsilon b_i\ge1-\varepsilon\|B_\tau\|\ge\frac12>0$, so $A_\varepsilon v_i=a_iv_i$ with $a_i=b_i/(1-\varepsilon b_i)$ and

$$a_i-b_i=\frac{\varepsilon b_i^2}{1-\varepsilon b_i}\le2\varepsilon b_i^2\le\eta b_i^2.$$

Writing $\zeta = \sum_i c_i v_i$ we have $\langle A_\varepsilon\zeta,\zeta\rangle = \sum_i a_i c_i^2$, $\langle B_\tau\zeta,\zeta\rangle = \sum_i b_i c_i^2$ and $|B_\tau\zeta|^2 = \sum_i b_i^2 c_i^2$, so the termwise inequality gives

$$\langle A_\varepsilon\zeta,\zeta\rangle \le \langle B_\tau\zeta,\zeta\rangle + \eta|B_\tau\zeta|^2 \qquad (\zeta\in V). \tag{202}$$

Combining (199) and (202), and writing $X_j^\tau = X_j^\infty$ to record the dependence on τ,

$$\sum_j X_j^\tau[\zeta_j] \le \langle B_\tau\zeta,\zeta\rangle + \eta|B_\tau\zeta|^2 \qquad (\zeta\in V). \tag{203}$$

(V.3) Letting $\tau\downarrow 0$. For $\tau\in(0,1]$ we have $\|B_\tau\| \le \|\bar A\|+1$, hence $\|A_\varepsilon\| \le \|B_\tau\|/(1-\varepsilon\|B_\tau\|) \le 2(\|\bar A\|+1)$ and $\varepsilon^{-1} = \max\big(2/\eta,\ 2\|B_\tau\|+2\big) \le \max\big(2/\eta,\ 2\|\bar A\|+4\big)$, so the bound $\lambda + \|A_\varepsilon\| = \varepsilon^{-1} + 2\|A_\varepsilon\|$ is bounded by a constant Λ depending only on η and $\|\bar A\|$ and not on $\tau\in(0,1]$. By (197) and (IV.3), therefore, $\|X_j^\tau\|\le\Lambda$ for all $\tau\in(0,1]$ and all j.

Apply the above with $\tau = 1/k$, $k=1,2,\dots$. By Bolzano–Weierstrass (preamble) there is a subsequence along which $X_j^{1/k}\to X_j$ for every j; each X_j is a symmetric continuous bilinear form on E. Along this subsequence:

- $B_{1/k} = \bar A + (1/k)\mathrm{id} \to \bar A$ in operator norm, so the right-hand side of (203) converges to $\langle\bar A\zeta,\zeta\rangle + \eta|\bar A\zeta|^2 = A[\zeta] + \eta|\bar A\zeta|^2$ for each fixed ζ; passing to the limit in (203),
$$\sum_j X_j[\zeta_j] \le A[\zeta] + \eta|\bar A\zeta|^2 \qquad (\zeta\in V).$$
- By (201), $(\widehat p_j, X_j^{1/k}) \in \overline{J}^{2,+}g_j(\widehat z_j)$ for every k, and $X_j^{1/k}\to X_j$; so Part I(D) gives $(\widehat p_j, X_j)\in\overline{J}^{2,+}g_j(\widehat z_j)$.

These are exactly the two assertions of the lemma, with $\widehat p_j = D_j\Psi(\widehat Z)$. □

2.49. `InfSubsolutionClosedSuperjet`.

Lemma 2.49 (The subsolution inequality holds on closed superjets)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega\subseteq E$ and let $u:E\to\mathbb{R}$ be an infinity subsolution on Ω in the sense of `IsInfSubsolutionOn`. Let $z\in\Omega$, let $p\in E$, let X be a continuous bilinear form on E, and assume $(p,X)\in\overline{J}^{2,+}u(z)$, the closed superjet of `ClosedSuperjet`. Then*

$$X[p] = X(p,p) \ge 0.$$

Proof. Step 1: the inequality at a plain superjet. We first prove the following claim. If $z'\in\Omega$, if $q\in E$ and if Y is a continuous bilinear form on E with $(q,Y)\in J^{2,+}u(z')$ in the sense of `Superjet`, then $Y(q,q)\ge 0$.

Fix $\varepsilon>0$ and define $\phi_\varepsilon : E\to\mathbb{R}$ by

$$\phi_\varepsilon(x) = u(z') + \langle q, x-z'\rangle + \tfrac12 Y(x-z',x-z') + \varepsilon|x-z'|^2.$$

This function is a sum of a constant, a continuous linear form in $x-z'$, the diagonal restriction of the continuous bilinear form Y, and ε times the diagonal restriction of the inner product. Each summand is a continuous polynomial of degree at most two on E and hence is C^∞; in particular ϕ_ε is C^2 near z'.

We record its derivatives at z'. Writing $g(x) = \frac12 Y(x-z',x-z')$ we have, for $h,k\in E$,

$$Dg(x)h = \tfrac12\big(Y(h,x-z')+Y(x-z',h)\big) = Y(x-z',h), \qquad D^2g(x)(h,k) = \tfrac12\big(Y(k,h)+Y(h,k)\big) = Y(h,k),$$

where both middle steps use the symmetry $Y(h,k)=Y(k,h)$, which is part of the definition of a superjet in `Superjet`. Likewise the function $x\mapsto\varepsilon|x-z'|^2$ has derivative $h\mapsto 2\varepsilon\langle x-z',h\rangle$ and second derivative $(h,k)\mapsto 2\varepsilon\langle h,k\rangle$. Evaluating at $x=z'$, where $x-z'=0$, gives

$$\nabla\phi_\varepsilon(z') = q, \qquad D^2\phi_\varepsilon(z')(h,k) = Y(h,k) + 2\varepsilon\langle h,k\rangle.$$

Consequently, by the definition of the infinity Laplacian in `InfLap`,

$$\Delta_\infty\phi_\varepsilon(z') = D^2\phi_\varepsilon(z')\big[\nabla\phi_\varepsilon(z')\big] = Y(q,q) + 2\varepsilon|q|^2. \tag{204}$$

Next, ϕ_ε is an upper test for u at z'. Indeed, apply the defining property of $(q,Y)\in J^{2,+}u(z')$ from `Superjet` with the positive number ε: there is $\delta>0$ such that

$$u(z'+h)\le u(z')+\langle q,h\rangle+\tfrac12 Y(h,h)+\varepsilon|h|^2=\phi_\varepsilon(z'+h)\qquad(|h|<\delta).$$

Since $\phi_\varepsilon(z')=u(z')$, this says $u(x)-\phi_\varepsilon(x)\le 0=u(z')-\phi_\varepsilon(z')$ for every x in the ball $B(z',\delta)$. Hence $x\mapsto u(x)-\phi_\varepsilon(x)$ has a local maximum on Ω at z': the required inequality holds on the neighbourhood $B(z',\delta)$ of z' intersected with Ω.

So $z'\in\Omega$, ϕ_ε is C^2 near z', and $u-\phi_\varepsilon$ has a local maximum on Ω at z'. The defining property of `IsInfSubsolutionOn` therefore gives $\Delta_\infty\phi_\varepsilon(z')\ge 0$, that is, by (204),

$$0\le Y(q,q)+2\varepsilon|q|^2.$$

The vector q and the form Y do not depend on ε, so letting $\varepsilon\downarrow 0$ yields $Y(q,q)\ge 0$. This proves the claim.

Step 2: passing to the closure. By `ClosedSuperjet` there are sequences $z_k\in E$, $p_k\in E$ and symmetric continuous bilinear forms X_k on E with

$$(p_k,X_k)\in J^{2,+}u(z_k)\quad\text{for all } k,\qquad z_k\to z,\quad p_k\to p,\quad X_k\to X$$

(the defining sequences also satisfy $u(z_k)\to u(z)$, which we do not need).

By the first clause of `IsInfSubsolutionOn` the set Ω is open, and $z\in\Omega$, so there is $\delta_0>0$ with $B(z,\delta_0)\subseteq\Omega$. Since $z_k\to z$ there is K with $|z_k-z|<\delta_0$, hence $z_k\in\Omega$, for every $k\ge K$. Step 1, applied at the point z_k to the superjet $(p_k,X_k)\in J^{2,+}u(z_k)$, gives

$$0\le X_k(p_k,p_k)\qquad(k\ge K).\tag{205}$$

It remains to pass to the limit in (205). The sequences (p_k) and (X_k) converge, so $M:=\sup_k|p_k|<\infty$ and $|p|\le M$. Decompose

$$X_k(p_k,p_k)-X(p,p)=(X_k-X)(p_k,p_k)+\big(X(p_k,p_k)-X(p,p)\big).$$

For the first term, $|(X_k-X)(p_k,p_k)|\le\|X_k-X\|\,|p_k|^2\le M^2\|X_k-X\|$, which tends to 0 because $X_k\to X$ in the operator norm. For the second, bilinearity gives

$$X(p_k,p_k)-X(p,p)=X(p_k-p,p_k)+X(p,p_k-p),$$

whence $|X(p_k,p_k)-X(p,p)|\le\|X\|\,|p_k-p|\,(|p_k|+|p|)\le 2M\|X\|\,|p_k-p|$, which tends to 0 because $p_k\to p$. Therefore $X_k(p_k,p_k)\to X(p,p)$, and (205) gives $X(p,p)\ge 0$ in the limit. □

2.50. `InfHarmonicNegScale`.

Lemma 2.50 (Negative scaling of infinity-harmonic functions)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega\subseteq E$, let $u:E\to\mathbb{R}$ be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`, and let $t<0$. Then the function $x\mapsto tu(x)$ is again infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`.*

Proof. By `IsInfHarmonicOn` the hypothesis says that u is both an infinity subsolution on Ω in the sense of `IsInfSubsolutionOn` and an infinity supersolution on Ω in the sense of `IsInfSupersolutionOn`, and the conclusion to be proved is the conjunction of the same two properties for tu. Unwinding `IsInfSubsolutionOn` and `IsInfSupersolutionOn`, each of those two properties consists of three clauses: that Ω be open, that the function be continuous on Ω, and a one-sided inequality at every C^2 test. The first clause is part of the hypothesis and is unchanged. For the second, u is continuous on Ω by the hypothesis, hence so is the multiple tu of it by a constant. There remain the two test inequalities.

The scaling identity. Let $s\in\mathbb{R}$, let $z\in E$ and let φ be a C^2 function near z. Then

$$\Delta_\infty(s\varphi)(z)=s^3\,\Delta_\infty\varphi(z).\tag{206}$$

Indeed $\nabla(s\varphi)(z)=s\nabla\varphi(z)$ and $D^2(s\varphi)(z)=sD^2\varphi(z)$, these being the first and second derivatives of a constant multiple; so, by the definition of Δ_∞ in `InfLap` and the bilinearity of the second derivative,

$$\Delta_\infty(s\varphi)(z)=D^2(s\varphi)(z)\big[\nabla(s\varphi)(z)\big]=s\,D^2\varphi(z)\big(s\nabla\varphi(z),s\nabla\varphi(z)\big)=s^3D^2\varphi(z)\big[\nabla\varphi(z)\big]=s^3\Delta_\infty\varphi(z).$$

tu is a subsolution. Let $z \in \Omega$ and let φ be C^2 at z with $tu - \varphi$ having a local maximum on Ω at z, i.e. $tu(x) - \varphi(x) \le tu(z) - \varphi(z)$ for all x in Ω near z. Multiplying this inequality by $t^{-1} < 0$ reverses it:

$$u(z) - t^{-1}\varphi(z) \le u(x) - t^{-1}\varphi(x)$$

for all such x, since $t^{-1}(tu(w) - \varphi(w)) = u(w) - t^{-1}\varphi(w)$ for every w. Thus $t^{-1}\varphi$, which is again C^2 at z, is a lower C^2 test for u at z: $u - t^{-1}\varphi$ has a local minimum on Ω at z. As u is an infinity supersolution on Ω, `IsInfSupersolutionOn` gives $\Delta_\infty(t^{-1}\varphi)(z) \le 0$. By (206) with $s = t^{-1}$ this reads $t^{-3}\Delta_\infty\varphi(z) \le 0$. Since $t < 0$ we have $t^{-1} < 0$ and hence $t^{-3} = t^{-1}(t^{-1})^2 < 0$; dividing by t^{-3} reverses the inequality and yields $\Delta_\infty\varphi(z) \ge 0$. As z and φ were arbitrary, tu is an infinity subsolution on Ω in the sense of `IsInfSubsolutionOn`.

tu is a supersolution. Symmetrically, let $z \in \Omega$ and let φ be C^2 at z with $tu - \varphi$ having a local minimum on Ω at z. Multiplying $tu(z) - \varphi(z) \le tu(x) - \varphi(x)$ by $t^{-1} < 0$ gives $u(x) - t^{-1}\varphi(x) \le u(z) - t^{-1}\varphi(z)$ for x in Ω near z, so $t^{-1}\varphi$ is an upper C^2 test for u at z. As u is an infinity subsolution on Ω, `IsInfSubsolutionOn` gives $0 \le \Delta_\infty(t^{-1}\varphi)(z) = t^{-3}\Delta_\infty\varphi(z)$ by (206), and dividing by $t^{-3} < 0$ yields $\Delta_\infty\varphi(z) \le 0$. Hence tu is an infinity supersolution on Ω in the sense of `IsInfSupersolutionOn`.

The two displayed properties together are exactly infinity-harmonicity of tu on Ω in the sense of `IsInfHarmonicOn`. □

2.51. `SmoothPlateau`.

Lemma 2.51 (A C^2 radial plateau between two radii)**.** *Let E be a finite-dimensional real inner-product space and let $m, M, \rho_1, \rho_2 \in \mathbb{R}$ with $0 < \rho_1 < \rho_2$ and $m \le M$. Then there is a function $B : E \to \mathbb{R}$ of class C^2 on E such that*

$$m \le B(c) \le M \quad (c \in E), \qquad B(c) = m \ \text{ whenever } |c| \le \rho_1, \qquad B(c) = M \ \text{ whenever } |c| \ge \rho_2.$$

Proof. Define $P : \mathbb{R} \to \mathbb{R}$ by

$$P(\tau) = \begin{cases} 0, & \tau \le 0, \\ 6\tau^5 - 15\tau^4 + 10\tau^3, & 0 \le \tau \le 1, \\ 1, & \tau \ge 1. \end{cases}$$

P is well defined, C^2, and takes values in $[0,1]$. The three branches agree where they overlap: at $\tau = 0$ the middle formula gives $6\cdot 0 - 15\cdot 0 + 10\cdot 0 = 0$, which is the value of the first branch, and at $\tau = 1$ it gives $6 - 15 + 10 = 1$, the value of the third. On $(0,1)$ the middle formula is a polynomial, hence C^∞ there, with

$$P'(\tau) = 30\tau^4 - 60\tau^3 + 30\tau^2 = 30\tau^2(\tau-1)^2, \qquad P''(\tau) = 120\tau^3 - 180\tau^2 + 60\tau = 60\tau(2\tau-1)(\tau-1).$$

Both formulas vanish at $\tau = 0$ and at $\tau = 1$: $P'(0) = 0$, $P''(0) = 0$, $P'(1) = 30\cdot 1\cdot 0 = 0$, $P''(1) = 60\cdot 1\cdot 1\cdot 0 = 0$. The constant branches have $P' = P'' = 0$ identically. Hence the one-sided limits of P, P' and P'' at the two junction points $\tau = 0$ and $\tau = 1$ agree from both sides, so P is twice differentiable on all of $\mathbb{R}$ with continuous second derivative, i.e. P is C^2 on $\mathbb{R}$. Moreover $P'(\tau) = 30\tau^2(\tau-1)^2 \ge 0$ on $[0,1]$, so P is nondecreasing there, whence $0 = P(0) \le P(\tau) \le P(1) = 1$ for $\tau \in [0,1]$; together with the two constant branches this gives $0 \le P \le 1$ on $\mathbb{R}$.

The plateau. Since $0 < \rho_1 < \rho_2$ we have $\rho_2^2 - \rho_1^2 > 0$, so we may define

$$B(c) = m + (M - m)\, P\!\left(\frac{|c|^2 - \rho_1^2}{\rho_2^2 - \rho_1^2}\right) \qquad (c \in E).$$

B is C^2. The map $c \mapsto |c|^2 = \langle c, c\rangle$ is a continuous quadratic form on E, hence C^∞; composing with the affine map $s \mapsto (s - \rho_1^2)/(\rho_2^2 - \rho_1^2)$ of $\mathbb{R}$ keeps it C^∞. Thus $c \mapsto (|c|^2 - \rho_1^2)/(\rho_2^2 - \rho_1^2)$ is C^∞ from E to $\mathbb{R}$, and composing it with the C^2 function P gives a C^2 function; finally multiplying by the constant $M - m$ and adding the constant m preserves C^2.

The three displayed properties. Let $c \in E$ and write $\tau(c) = (|c|^2 - \rho_1^2)/(\rho_2^2 - \rho_1^2)$.

- Since $0 \le P \le 1$ and $M - m \ge 0$, we get $m = m + (M-m)\cdot 0 \le B(c) \le m + (M-m)\cdot 1 = M$.

- If $|c| \le \rho_1$ then $|c|^2 \le \rho_1^2$, so $\tau(c) \le 0$ and hence $P(\tau(c)) = 0$ by the first branch; therefore $B(c) = m$.
- If $|c| \ge \rho_2$ then $|c|^2 \ge \rho_2^2$, so $|c|^2 - \rho_1^2 \ge \rho_2^2 - \rho_1^2 > 0$ and hence $\tau(c) \ge 1$, so $P(\tau(c)) = 1$ by the third branch; therefore $B(c) = m + (M - m) = M$.

This is the required function. □

2.52. `ConcaveRadialProfile`.

Lemma 2.52 (The concave radial profile $f(r) = r - \frac{1}{8}r^{3/2}$)**.** *Put*

$$f(r) = r - \tfrac{1}{8}r\sqrt{r} \qquad (r \ge 0),$$

so that $f(r) = r - \frac{1}{8}r^{3/2}$*, put* $f_1(r) = 1 - \frac{3}{16}\sqrt{r}$ *and put* $\nu = 1 - \frac{1}{8}\sqrt{3/2}$*. Then*

(1) $\nu > 0$*;*
(2) f *is of class* C^2 *at every* $r > 0$*;*
(3) *for every* $r > 0$*,* f *is differentiable at* r *with* $f'(r) = f_1(r)$*;*
(4) *for every* $r > 0$*,* f_1 *is differentiable at* r *with* $f_1'(r) = -\dfrac{3}{32\sqrt{r}}$ *(so* $f''(r) = -3/(32\sqrt{r})$*);*
(5) *for every* r *with* $0 < r \le \frac{3}{2}$*,*

$$\nu r \le f(r) \le r, \qquad 0 < f_1(r) \le 1.$$

Proof. Throughout, $\sqrt{\cdot}$ denotes the nonnegative square root, and for $r > 0$ we have $\sqrt{r} > 0$ and $(\sqrt{r})^2 = r$.

A numerical bound used twice. Since $3/2 < 4$ and the square root is strictly increasing on $[0, \infty)$, $\sqrt{3/2} < \sqrt{4} = 2$. In particular

$$0 \le \sqrt{3/2} < 2. \tag{207}$$

(1). By (207), $\frac{1}{8}\sqrt{3/2} < \frac{2}{8} = \frac{1}{4} < 1$, so $\nu = 1 - \frac{1}{8}\sqrt{3/2} > 0$.

(2). Fix $r > 0$. The square root is C^∞ on $(0, \infty)$, hence C^2 at r; the identity $s \mapsto s$ is C^∞; a product of two functions that are C^2 at a point is C^2 there, and so are a constant multiple of such a function and a difference of two of them. Hence $f(s) = s - \frac{1}{8}s\sqrt{s}$ is C^2 at r.

(3). Fix $r > 0$. The square root is differentiable at r with derivative $1/(2\sqrt{r})$. Hence, by the product rule applied to $s \mapsto s$ and $s \mapsto \sqrt{s}$, the function $s \mapsto s\sqrt{s}$ is differentiable at r with derivative

$$1 \cdot \sqrt{r} + r \cdot \frac{1}{2\sqrt{r}} = \sqrt{r} + \frac{(\sqrt{r})^2}{2\sqrt{r}} = \sqrt{r} + \frac{\sqrt{r}}{2} = \frac{3}{2}\sqrt{r},$$

where $r = (\sqrt{r})^2$ was used in the middle step. Dividing by the constant 8 and subtracting from the identity $s \mapsto s$, whose derivative is 1, shows that f is differentiable at r with

$$f'(r) = 1 - \frac{1}{8} \cdot \frac{3}{2}\sqrt{r} = 1 - \frac{3}{16}\sqrt{r} = f_1(r).$$

(4). Fix $r > 0$. Again the square root is differentiable at r with derivative $1/(2\sqrt{r})$; multiplying by the constant 3, dividing by the constant 16 and subtracting from the constant 1 shows that f_1 is differentiable at r with

$$f_1'(r) = -\frac{3}{16} \cdot \frac{1}{2\sqrt{r}} = -\frac{3}{32\sqrt{r}}.$$

(5). Fix r with $0 < r \le \frac{3}{2}$. Monotonicity of the square root gives $0 < \sqrt{r} \le \sqrt{3/2}$.

First, $f(r) = r\big(1 - \frac{1}{8}\sqrt{r}\big)$. Since $\sqrt{r} \le \sqrt{3/2}$ we have $1 - \frac{1}{8}\sqrt{r} \ge 1 - \frac{1}{8}\sqrt{3/2} = \nu$, and $r > 0$, so $f(r) \ge \nu r$. Since $\sqrt{r} > 0$ we have $1 - \frac{1}{8}\sqrt{r} \le 1$, so $f(r) \le r$.

Second, by (207), $\sqrt{r} \le \sqrt{3/2} < 2$, hence $\frac{3}{16}\sqrt{r} < \frac{6}{16} < 1$ and so $f_1(r) = 1 - \frac{3}{16}\sqrt{r} > 0$; and $\sqrt{r} > 0$ gives $f_1(r) \le 1$. □

2.53. `RadialQuotientSecondVariation`.

Lemma 2.53 (Second variation of the quadratic penalty)**.** *Let E be a real inner-product space, let $h, k \in E$ and let $r, s \in \mathbb{R}$ with $r > 0$. Then there is a function $G : \mathbb{R} \to \mathbb{R}$ such that*

(1) *for every τ in some neighbourhood of 0 the function*

$$t \longmapsto \frac{|h + tk|^2}{(r + ts)^2}$$

is differentiable at τ with derivative $G(\tau)$, and

(2) *G is differentiable at 0 with*

$$G'(0) = \frac{2|k|^2}{r^2} - \frac{8s\langle h, k\rangle}{r^3} + \frac{6s^2|h|^2}{r^4}.$$

In other words, along the motion in which the numerator vector moves from h with velocity k and the denominator moves affinely from r with velocity s, the quotient $|h|^2/r^2$ has a second derivative at the base point, and that second derivative is the displayed expression.

Proof. Write $a = \langle h, k\rangle$, $b = |k|^2$ and $p = |h|^2$, and set $D(t) = r + ts$.

The numerator is a quadratic polynomial in t. Expanding the square of the norm,

$$|h + tk|^2 = |h|^2 + 2\langle h, tk\rangle + |tk|^2 = p + 2ta + t^2 b \qquad (t \in \mathbb{R}), \tag{208}$$

using the bilinearity and symmetry of the real inner product and $|tk|^2 = t^2|k|^2$. Call this polynomial $P(t) = p + 2ta + t^2 b$; it is differentiable everywhere, with

$$P'(t) = 2a + 2tb, \qquad P''(t) = 2b. \tag{209}$$

Also D is affine, so $D' \equiv s$, and for each $m \geq 1$ the chain rule gives that $t \mapsto D(t)^m$ is differentiable with derivative $mD(t)^{m-1}s$.

Definition of G. Put

$$G(t) = \frac{P'(t)}{D(t)^2} - \frac{2sP(t)}{D(t)^3} = \frac{2a + 2tb}{(r + ts)^2} - \frac{2s\,(p + 2ta + t^2 b)}{(r + ts)^3}, \tag{210}$$

defined for every t with $D(t) \neq 0$, and extended arbitrarily (say by 0) elsewhere.

Proof of (1). The map $t \mapsto D(t)$ is continuous and $D(0) = r > 0 \neq 0$, so $D(\tau) \neq 0$ for every τ in some neighbourhood $\mathcal{U}$ of 0; shrinking $\mathcal{U}$ we may assume G is given by (210) on $\mathcal{U}$. Fix $\tau \in \mathcal{U}$. By (208) the function in (1) equals $t \mapsto P(t)/D(t)^2$, and $D(\tau)^2 \neq 0$, so the quotient rule applies and gives differentiability at τ with derivative

$$\frac{P'(\tau)D(\tau)^2 - P(\tau)\cdot 2D(\tau)s}{\big(D(\tau)^2\big)^2} = \frac{P'(\tau)}{D(\tau)^2} - \frac{2sP(\tau)}{D(\tau)^3} = G(\tau),$$

the middle equality by cancelling one factor $D(\tau) \neq 0$ in the second term and two in the first. This is (1).

Proof of (2). We differentiate the two summands of (210) at $t = 0$, where $D(0) = r \neq 0$ so both quotient rules apply.

For the first summand, the numerator $P'(t) = 2a + 2tb$ has derivative $2b$ at 0 by (209), and the denominator $D(t)^2$ has value r^2 and derivative $2rs$ at 0; hence

$$\frac{d}{dt}\frac{P'(t)}{D(t)^2}\bigg|_{t=0} = \frac{2b\cdot r^2 - 2a\cdot 2rs}{r^4} = \frac{2b}{r^2} - \frac{4as}{r^3}.$$

For the second summand, the numerator $2sP(t)$ has value $2sp$ and derivative $2sP'(0) = 4as$ at 0, and the denominator $D(t)^3$ has value r^3 and derivative $3r^2 s$ at 0; hence

$$\frac{d}{dt}\frac{2sP(t)}{D(t)^3}\bigg|_{t=0} = \frac{4as\cdot r^3 - 2sp\cdot 3r^2 s}{r^6} = \frac{4as}{r^3} - \frac{6ps^2}{r^4}.$$

Subtracting, G is differentiable at 0 with

$$G'(0) = \Big(\frac{2b}{r^2} - \frac{4as}{r^3}\Big) - \Big(\frac{4as}{r^3} - \frac{6ps^2}{r^4}\Big) = \frac{2b}{r^2} - \frac{8as}{r^3} + \frac{6ps^2}{r^4},$$

which is the displayed expression once a, b, p are written out. This is (2). □

2.54. `RadialTestFirstDerivative`.

Lemma 2.54 (The radial test is C^2 off the diagonal, with its first derivative)**.** *Let V be a real normed space and E a real inner-product space. Let $A : V \to \mathbb{R}$ be a continuous linear functional, let $L_c, L_d, L_h : V \to E$ be continuous linear maps, and let $B : E \to \mathbb{R}$ and $f : \mathbb{R} \to \mathbb{R}$ be functions. Define*

$$\Theta(U) = A(U) + B(L_cU)\, f\big(|L_dU|\big) + \frac{|L_hU|^2}{|L_dU|^2} \qquad (U \in V).$$

Let $W \in V$ with $L_dW \neq 0$, and write

$$r = |L_dW| > 0, \qquad c = L_cW, \qquad h = L_hW, \qquad e = r^{-1}L_dW.$$

Assume that B is C^2 at c, that f is C^2 at r, and that f is differentiable at r with $f'(r) = a_0$. Then

(1) *Θ is C^2 at W, and*
(2) *for every $\zeta \in V$,*

$$D\Theta(W)[\zeta] = A(\zeta) + f(r)\, DB(c)[L_c\zeta] + \Big(B(c)\, a_0 - \frac{2|h|^2}{r^3}\Big)\langle e, L_d\zeta\rangle + \frac{2\langle h, L_h\zeta\rangle}{r^2}.$$

Proof. Since $L_dW \neq 0$ we have $r = |L_dW| > 0$, so $r \neq 0$ and $r^2 \neq 0$; these nonvanishing facts are used at every division below.

Step 1: the squared norms and their derivatives. For any continuous linear $L : V \to E$ the map $U \mapsto LU$ is C^∞, and $x \mapsto |x|^2 = \langle x, x\rangle$ is a continuous quadratic form on E, hence C^∞, with derivative $D(|\cdot|^2)(x)[v] = 2\langle x, v\rangle$ by bilinearity and symmetry of the real inner product. Composing with L (chain rule for a composition with a continuous linear map) shows that $U \mapsto |LU|^2$ is C^∞ with

$$D\big(|L\cdot|^2\big)(W)[\zeta] = 2\langle LW, L\zeta\rangle. \tag{211}$$

Applying this to $L = L_d$ and to $L = L_h$ gives both squared-norm blocks.

Step 2: the radius is C^2 at W, with derivative $\langle e, L_d\cdot\rangle$. The square root is C^∞ on $(0, \infty)$, and $|L_dW|^2 = r^2 > 0$, so the composition $U \mapsto \sqrt{|L_dU|^2}$ is C^2 at W by Step 1 and the chain rule. Since $|x| \geq 0$, we have $\sqrt{|L_dU|^2} = |L_dU|$ for every U; the two functions therefore coincide on V, so $U \mapsto |L_dU|$ is C^2 at W and has the same derivative. The derivative of $\sqrt{\cdot}$ at $s = r^2$ is $1/(2\sqrt{r^2}) = 1/(2r)$, so by (211) with $L = L_d$,

$$D\big(|L_d\cdot|\big)(W)[\zeta] = \frac{1}{2r}\cdot 2\langle L_dW, L_d\zeta\rangle = \frac{\langle L_dW, L_d\zeta\rangle}{r} = \langle e, L_d\zeta\rangle, \tag{212}$$

the last equality because $e = r^{-1}L_dW$ and the inner product is linear in its first argument over $\mathbb{R}$. Note also, for later use, that (212) gives

$$\langle L_dW, L_d\zeta\rangle = r\,\langle e, L_d\zeta\rangle. \tag{213}$$

Step 3: the three summands are C^2 at W. A is continuous linear, hence C^∞. The map $U \mapsto B(L_cU)$ is C^2 at W: it is the composition of the C^∞ map $U \mapsto L_cU$ with B, which is C^2 at $L_cW = c$. The map $U \mapsto f(|L_dU|)$ is C^2 at W: it is the composition of the map of Step 2, which is C^2 at W with value r, with f, which is C^2 at r. A product of two functions that are C^2 at a point is C^2 there, so $U \mapsto B(L_cU)f(|L_dU|)$ is C^2 at W. Finally $U \mapsto |L_hU|^2$ and $U \mapsto |L_dU|^2$ are C^∞ by Step 1 and the second has the nonzero value r^2 at W, so their quotient is C^2 at W. A sum of functions that are C^2 at a point is C^2 there, which proves (1).

Step 4: the derivative of the localizing product. By the chain rule for a composition with a continuous linear map, $U \mapsto B(L_cU)$ has derivative $\zeta \mapsto DB(c)[L_c\zeta]$ at W. By Step 2 and the one-dimensional chain rule, $U \mapsto f(|L_dU|)$ has derivative $\zeta \mapsto a_0\langle e, L_d\zeta\rangle$ at W, where $a_0 = f'(r)$. The product rule therefore gives

$$D\big(B(L_c\cdot)f(|L_d\cdot|)\big)(W)[\zeta] = B(c)\, a_0\,\langle e, L_d\zeta\rangle + f(r)\, DB(c)[L_c\zeta]. \tag{214}$$

Step 5: the derivative of the penalty. Write $N(U) = |L_h U|^2$ and $D(U) = |L_d U|^2$, so $N(W) = |h|^2$, $D(W) = r^2 \neq 0$, and by (211)

$$DN(W)[\zeta] = 2\langle h, L_h\zeta\rangle, \qquad DD(W)[\zeta] = 2\langle L_d W, L_d\zeta\rangle.$$

The quotient rule gives

$$D\Big(\frac{N}{D}\Big)(W)[\zeta] = \frac{DN(W)[\zeta]\,D(W) - N(W)\,DD(W)[\zeta]}{D(W)^2} = \frac{2\langle h, L_h\zeta\rangle\, r^2 - |h|^2 \cdot 2\langle L_d W, L_d\zeta\rangle}{r^4},$$

and substituting (213) and simplifying,

$$D\Big(\frac{|L_h\cdot|^2}{|L_d\cdot|^2}\Big)(W)[\zeta] = \frac{2\langle h, L_h\zeta\rangle}{r^2} - \frac{2|h|^2}{r^3}\langle e, L_d\zeta\rangle. \tag{215}$$

Step 6: conclusion. The derivative of the continuous linear functional A at any point is A itself. Adding this to (214) and (215), and collecting the two terms proportional to $\langle e, L_d\zeta\rangle$, gives for every $\zeta \in V$

$$D\Theta(W)[\zeta] = A(\zeta) + f(r)\,DB(c)[L_c\zeta] + \Big(B(c)\,a_0 - \frac{2|h|^2}{r^3}\Big)\langle e, L_d\zeta\rangle + \frac{2\langle h, L_h\zeta\rangle}{r^2},$$

which is (2). □

2.55. `ProfileProductSecondVariation`.

Lemma 2.55 (Second variation of the localizing product)**.** *Let E be a real inner-product space, let $B : E \to \mathbb{R}$ be of class C^2 on E, let $f, f_1 : \mathbb{R} \to \mathbb{R}$, let $c, u \in E$ and let $r, s, \gamma_2 \in \mathbb{R}$. Assume that*

$$f \text{ is differentiable at } \rho \text{ with } f'(\rho) = f_1(\rho) \quad \text{for every } \rho \text{ in some neighbourhood of } r,$$

and that f_1 is differentiable at r with $f_1'(r) = \gamma_2$. Then there is a function $G : \mathbb{R} \to \mathbb{R}$ such that

(1) *for every τ in some neighbourhood of 0 the function*

$$t \longmapsto B(c + tu)\, f(r + ts)$$

is differentiable at τ with derivative $G(\tau)$, and

(2) *G is differentiable at 0 with*

$$G'(0) = D^2B(c)[u,u]\, f(r) + 2\, DB(c)[u]\, f_1(r)\, s + B(c)\,\gamma_2\, s^2.$$

In other words, along the motion in which the center moves from c with velocity u and the radius moves affinely from r with velocity s, the product $B(c)f(r)$ has a second derivative at the base point, and that second derivative is the displayed Leibniz sum.

Proof. Write $\beta(\tau) = B(c + \tau u)$ and $\gamma(\tau) = f(r + \tau s)$, so that the function in (1) is $\beta\gamma$.

Step 1: the two straight motions. The map $t \mapsto c + tu$ is affine, hence differentiable at every τ with derivative u; likewise $t \mapsto r + ts$ is differentiable at every τ with derivative s. Both maps send 0 to c and to r respectively.

Step 2: β is differentiable everywhere, with $\beta'(\tau) = DB(c + \tau u)[u]$. Since B is C^2 on E it is in particular differentiable at every point, so B is differentiable at $c + \tau u$ with derivative $DB(c + \tau u)$. Composing with the motion of Step 1, the chain rule gives

$$\beta'(\tau) = DB(c + \tau u)[u] \qquad (\tau \in \mathbb{R}); \tag{216}$$

in particular $\beta(0) = B(c)$ and $\beta'(0) = DB(c)[u]$.

Step 3: β' is differentiable at 0, with $\beta''(0) = D^2B(c)[u,u]$. As B is C^2, the map $x \mapsto DB(x)$ is C^1 on E, hence differentiable at c, with derivative $D^2B(c)$, a continuous linear map from E to the space of continuous linear functionals on E. Composing with the motion $t \mapsto c + tu$, which has derivative u at 0 and sends 0 to c, gives that $\tau \mapsto DB(c + \tau u)$ is differentiable at 0 with derivative $D^2B(c)[u]$. Evaluation at u is a continuous linear functional on the space of continuous linear functionals, so composing once more and using (216),

$$\beta''(0) = D^2B(c)[u,u]. \tag{217}$$

Step 4: γ near 0, and γ' at 0. The map $\tau \mapsto r + \tau s$ is continuous and sends 0 to r, so it maps some neighbourhood of 0 into the neighbourhood of r on which f is differentiable with derivative f_1. For τ in that neighbourhood of 0, f is differentiable at $r + \tau s$ with derivative $f_1(r + \tau s)$, and the chain rule with Step 1 gives that γ is differentiable at τ with

$$\gamma'(\tau) = f_1(r + \tau s)\, s. \tag{218}$$

Taking $\tau = 0$: $\gamma(0) = f(r)$ and $\gamma'(0) = f_1(r)s$. Moreover f_1 is differentiable at r with derivative γ_2, so composing with the motion once more,

$$\tau \mapsto f_1(r + \tau s)s \text{ is differentiable at } 0 \text{ with derivative } \gamma_2 s \cdot s = \gamma_2 s^2. \tag{219}$$

Step 5: definition of G and proof of (1). Put

$$G(\tau) = DB(c + \tau u)[u]\, f(r + \tau s) + B(c + \tau u)\, f_1(r + \tau s)\, s = \beta'(\tau)\gamma(\tau) + \beta(\tau)\gamma'(\tau)$$

for τ in the neighbourhood of 0 supplied by Step 4, extended arbitrarily (say by 0) elsewhere. For such τ, β is differentiable at τ by Step 2 and γ is differentiable at τ by Step 4, so the product rule gives that $\beta\gamma$ is differentiable at τ with derivative $\beta'(\tau)\gamma(\tau) + \beta(\tau)\gamma'(\tau) = G(\tau)$. This is (1).

Step 6: proof of (2). On that neighbourhood, $G = G_1 + G_2$ with $G_1(\tau) = \beta'(\tau)\gamma(\tau)$ and $G_2(\tau) = \beta(\tau)\gamma'(\tau)$. By Step 3 the factor β' is differentiable at 0 with derivative $\beta''(0)$, and by Step 4 the factor γ is differentiable at 0 with derivative $\gamma'(0)$; the product rule gives that G_1 is differentiable at 0 with

$$G_1'(0) = \beta''(0)\gamma(0) + \beta'(0)\gamma'(0).$$

By Step 2 the factor β is differentiable at 0 with derivative $\beta'(0)$, and by (219) the factor γ' is differentiable at 0 with derivative $\gamma_2 s^2$; the product rule gives that G_2 is differentiable at 0 with

$$G_2'(0) = \beta'(0)\gamma'(0) + \beta(0)\, \gamma_2 s^2.$$

Adding, G is differentiable at 0 with

$$G'(0) = \beta''(0)\gamma(0) + 2\beta'(0)\gamma'(0) + \beta(0)\, \gamma_2 s^2 = D^2B(c)[u,u]\, f(r) + 2\, DB(c)[u]\, f_1(r)\, s + B(c)\, \gamma_2\, s^2,$$

where the values $\beta(0) = B(c)$, $\beta'(0) = DB(c)[u]$, $\beta''(0) = D^2B(c)[u,u]$, $\gamma(0) = f(r)$ and $\gamma'(0) = f_1(r)s$ were substituted from Steps 2, 3 and 4. This is (2). □

2.56. `SecondVariationAlongLine`.

Lemma 2.56 (First and second variation along a line, read off any local model)**.** *Let P be a real normed space, let $g : P \to \mathbb{R}$, and let $W, \zeta \in P$ be such that g is twice continuously differentiable at W. Suppose given functions $r, r' : \mathbb{R} \to \mathbb{R}$ and a number $c \in \mathbb{R}$ such that*

1. *$g(W + h\zeta) = r(h)$ for all h in a neighbourhood of 0;*
2. *for every h in a neighbourhood of 0, r is differentiable at h with derivative $r'(h)$;*
3. *r' is differentiable at 0 with derivative c.*

Then

$$Dg(W)[\zeta] = r'(0), \qquad D^2g(W)[\zeta,\zeta] = c,$$

the second being the second iterated Fréchet derivative of g at W evaluated on the constant pair (ζ, ζ).

This is the bridge from the second iterated Fréchet derivative, with which the limiting operator of `LimitOp` is stated, to an ordinary one-dimensional second derivative, which is what an explicit test function can be differentiated by hand. It is strictly more general than `SecondVariationAddAffineOnLine`, which discards only a summand whose restriction to the line is affine: here the restriction may be any function with a second derivative at the base point, and it is identified only up to agreement on a neighbourhood of 0, so the algebraic rearrangement of the restriction may use inequalities that hold only near the base point.

Proof. Write $\lambda(h) = W + h\zeta$. Since λ is affine with linear part $h \mapsto h\zeta$, it is differentiable everywhere with $\lambda'(h) = \zeta$.

Step 1: the restriction of g to the line. Because g is twice continuously differentiable at W, it is twice continuously differentiable at every point of some neighbourhood U of W; in particular

g is differentiable on U. Since λ is continuous and $\lambda(0) = W$, the set $\lambda^{-1}(U)$ is a neighbourhood of 0, so for every h in a neighbourhood of 0 the function g is differentiable at $\lambda(h)$, and by the chain rule $h \mapsto g(\lambda(h))$ is differentiable at h with derivative $Dg(W + h\zeta)[\zeta]$.

Step 2: r' is that derivative near 0. Let h be in a neighbourhood of 0 small enough that the agreement in (1) holds on a whole neighbourhood of h — possible because (1) holds on a neighbourhood of 0, and a neighbourhood of 0 contains an open set around 0, each point of which has a neighbourhood inside it. On that neighbourhood r coincides with $h \mapsto g(\lambda(h))$, so by Step 1 r is differentiable at h with derivative $Dg(W + h\zeta)[\zeta]$. By (2) and the uniqueness of the derivative,

$$r'(h) = Dg(W + h\zeta)[\zeta] \qquad \text{for all } h \text{ in a neighbourhood of } 0. \tag{220}$$

Taking $h = 0$ gives $r'(0) = Dg(W)[\zeta]$, the first assertion.

Step 3: one more derivative. Because g is twice continuously differentiable at W, the map $Dg : U \to P^*$ is differentiable at W, with derivative $D^2g(W)$. By the chain rule along λ, the map $h \mapsto Dg(W + h\zeta)$ is differentiable at 0 with derivative $D^2g(W)[\zeta] \in P^*$. Evaluation at ζ, that is $\varphi \mapsto \varphi[\zeta]$, is a continuous linear functional on P^*, so composing with it, the map $h \mapsto Dg(W + h\zeta)[\zeta]$ is differentiable at 0 with derivative $D^2g(W)[\zeta, \zeta]$.

By (220) this map agrees with r' on a neighbourhood of 0, hence r' is differentiable at 0 with the same derivative. Comparing with (3) and using the uniqueness of the derivative, $c = D^2g(W)[\zeta, \zeta]$.

Finally, the second iterated Fréchet derivative of g at W, evaluated on a pair of vectors, is $D^2g(W)$ applied to that pair; on the constant pair (ζ, ζ) it is therefore $D^2g(W)[\zeta, \zeta] = c$. □

2.57. `SecondVariationAddAffineOnLine`.

Lemma 2.57 (Adding a term affine along one motion leaves the second variation alone)**.** *Let P be a real normed space, let $W, \zeta \in P$, and let $g, \psi : P \to \mathbb{R}$ both be C^2 near W. Suppose there are $a, b \in \mathbb{R}$ such that*

$$\psi(W + h\zeta) = a + bh$$

for all real h in a neighbourhood of 0. Then

$$D^2(g + \psi)(W)\big[\zeta, \zeta\big] = D^2g(W)\big[\zeta, \zeta\big].$$

Proof. Write $\gamma(h) = W + h\zeta$ for the straight motion through W with velocity ζ. Since γ is affine in h with linear part $h \mapsto h\zeta$, it is differentiable with $\gamma'(h) = \zeta$ for every h. Also γ is continuous and $\gamma(0) = W$, so for any function u that is C^2 near W — hence differentiable at every point of some neighbourhood of W — the point $\gamma(h)$ lies in that neighbourhood for all h near 0, and therefore u is differentiable at $\gamma(h)$ for all h near 0. We use this for $u = g$, for $u = \psi$ and for $u = g + \psi$.

Put

$$A(h) = D(g + \psi)\big(\gamma(h)\big)[\zeta], \qquad B(h) = Dg\big(\gamma(h)\big)[\zeta].$$

Step 1: $A = B + b$ near 0. Fix h near 0, small enough that g and ψ are both differentiable at $\gamma(h)$ and that the affine identity for ψ holds on a whole neighbourhood of h (the hypothesis holds on a neighbourhood of 0, so it holds on a neighbourhood of each point of a possibly smaller neighbourhood of 0). Consider the real function $s \mapsto g(\gamma(s)) + \psi(\gamma(s))$.

On the one hand, $g + \psi$ is differentiable at $\gamma(h)$, so the chain rule for the composition with γ gives that this function is differentiable at h with derivative $D(g + \psi)(\gamma(h))[\gamma'(h)] = A(h)$.

On the other hand, the chain rule applied to g alone gives that $s \mapsto g(\gamma(s))$ is differentiable at h with derivative $B(h)$, while $s \mapsto \psi(\gamma(s))$ agrees on a neighbourhood of h with the affine function $s \mapsto a + bs$, whose derivative at h is b; two functions agreeing on a neighbourhood of h have the same derivative there, so $s \mapsto \psi(\gamma(s))$ is differentiable at h with derivative b. Summing, the function $s \mapsto g(\gamma(s)) + \psi(\gamma(s))$ is differentiable at h with derivative $B(h) + b$.

A real function has at most one derivative at a point, so $A(h) = B(h) + b$.

Step 2: the derivative of A and of B at 0. Let u be C^2 near W. Then Du is differentiable at W, with derivative $D^2u(W)$, as a map into the space of continuous linear functionals on P. Composing with γ on the right, the map $h \mapsto Du(\gamma(h))$ is differentiable at 0 with derivative $D^2u(W)[\zeta]$,

again a continuous linear functional on P. Composing with the evaluation at the fixed vector ζ — which is bilinear and bounded in the pair (functional, vector), hence differentiable, and is here applied with the second argument constant — the real function $h \mapsto Du(\gamma(h))[\zeta]$ is differentiable at 0 with derivative $D^2u(W)[\zeta,\zeta]$.

Applying this with $u = g + \psi$ gives that A is differentiable at 0 with derivative $D^2(g + \psi)(W)[\zeta,\zeta]$, and with $u = g$ that B is differentiable at 0 with derivative $D^2g(W)[\zeta,\zeta]$.

Step 3: conclusion. By Step 2 the function $h \mapsto B(h) + b$ is differentiable at 0 with derivative $D^2g(W)[\zeta,\zeta]$. By Step 1 it agrees with A on a neighbourhood of 0, so A is also differentiable at 0 with that same derivative. Comparing with the value found for A in Step 2 and using uniqueness of the derivative,

$$D^2(g+\psi)(W)\big[\zeta,\zeta\big] = D^2g(W)\big[\zeta,\zeta\big],$$

as asserted. (Taking $g = 0$, for which $D^2g(W) = 0$, this says in particular that a function affine along the motion $h \mapsto W + h\zeta$ has vanishing second variation in the direction ζ; that is the form in which the statement is used when a summand is to be discarded outright rather than stripped off a remainder.) □

2.58. `ThreePointTestSecondVariation`.

Lemma 2.58 (The second variation of the three-point test is very negative)**.** *Let E be a finite-dimensional real inner-product space and write $\ell^2(E^3)$ for the product $E \times E \times E$ with the Euclidean product norm. Let $L_c, L_d, L_h : \ell^2(E^3) \to E$ and $\alpha : \ell^2(E^3) \to \mathbb{R}$ be continuous linear maps, let $\Psi : \ell^2(E^3) \to \mathbb{R}$, let $Z, \sigma \in \ell^2(E^3)$, let $B : E \to \mathbb{R}$, let $f, f_1 : \mathbb{R} \to \mathbb{R}$, let $c, h, e, p_1, p_2 \in E$ and let $r, a, \delta, C, B_{\mathrm{out}}, K_1, K_2, K_3, M \in \mathbb{R}$. Assume:*

(1) $\Psi(Y) = \alpha(Y) + B(L_cY)\, f(\|L_dY\|) + \|L_hY\|^2/\|L_dY\|^2$ *for every* $Y \in \ell^2(E^3)$;
(2) Ψ *is twice continuously differentiable at* Z;
(3) $L_cZ = c$, $L_dZ = r\,e$ *and* $L_hZ = h$;
(4) $L_c\sigma = \frac{1}{2}(p_1 + p_2)$;
(5) $L_d\sigma = (2a)\,e$;
(6) $L_h\sigma = -\frac{1}{2}f(r)\,\nabla B(c)$;
(7) $\|e\| = 1$;
(8) $0 < r$ *and* $r \le 3/2$;
(9) $0 \le f(r)$ *and* $f(r) \le r$;
(10) f *has derivative* $f_1(\rho)$ *at* ρ *for every* ρ *in some neighbourhood of* r;
(11) f_1 *has derivative* $-3/(32\sqrt{r})$ *at* r;
(12) $0 \le f_1(r)$ *and* $f_1(r) \le 1$;
(13) B *is twice continuously differentiable on* E *and* $\delta/2 \le B(c)$;
(14) $|DB(c)| \le K_1$ *and* $|D^2B(c)| \le K_2$ *as operator norms,* $|\nabla B(c)| \le K_1$, *and* $0 \le K_1$, $0 \le K_2$;
(15) $\|\frac{1}{2}(p_1 + p_2)\| \le K_3$;
(16) $0 < \delta$, $0 < a$, $\delta/4 \le a$ *and* $a \le B_{\mathrm{out}}$;
(17) $0 < C$ *and* $\|h\| \le 2Cr^2$;
(18) $M = \dfrac{K_1^2}{2} + 16B_{\mathrm{out}}CK_1 + 96B_{\mathrm{out}}^2C^2 + \dfrac{3}{2}K_2K_3^2 + 4B_{\mathrm{out}}K_1K_3$.

Then

$$D^2\Psi(Z)[\sigma,\sigma] \le M - \frac{3\delta^3}{256\sqrt{r}}. \tag{221}$$

Proof. Write $\dot c = \frac{1}{2}(p_1 + p_2)$ and $\dot h = -\frac{1}{2}f(r)\nabla B(c)$, so that $L_c\sigma = \dot c$ and $L_h\sigma = \dot h$ by 4 and 6. These are the paper's exact velocities of the joint motion $\tau \mapsto Z + \tau\sigma$, together with $\dot r = 2a$ from 5 and $\ddot r = 0$, the motion being a straight line.

The test along the motion. By linearity of L_d and 3, 5,

$$L_d(Z + \tau\sigma) = L_dZ + \tau L_d\sigma = (r + 2a\tau)\,e \qquad (\tau \in \mathbb{R}),$$

so $\|L_d(Z+\tau\sigma)\| = |r+2a\tau|\,\|e\| = r+2a\tau$ whenever $r+2a\tau \ge 0$, using 7. Likewise $L_c(Z+\tau\sigma) = c + \tau\dot c$, $L_h(Z + \tau\sigma) = h + \tau\dot h$ and $\alpha(Z + \tau\sigma) = \alpha(Z) + \tau\,\alpha(\sigma)$. Since $r > 0$ by 8, the function

$\tau \mapsto r + 2a\tau$ is continuous and positive at $\tau = 0$, hence positive for all τ in a neighbourhood of 0. Setting

$$\rho(\tau) = \big(\alpha(Z) + \tau\,\alpha(\sigma)\big) + B(c + \tau\dot c)\, f(r + 2a\tau) + \frac{\|h + \tau\dot h\|^2}{(r + 2a\tau)^2},$$

hypothesis 1 and the four displayed identities therefore give $\Psi(Z + \tau\sigma) = \rho(\tau)$ for every τ in a neighbourhood of 0.

The second derivative of ρ at 0. Apply `ProfileProductSecondVariation` with the plateau B, the profiles f, f_1, the base point c, the velocity $\dot c$, the radius r, the speed $2a$ and the curvature $-3/(32\sqrt{r})$: its hypotheses are exactly 13, 10 and 11, and it produces a function G_B which is the derivative of $\tau \mapsto B(c + \tau\dot c) f(r + 2a\tau)$ near 0 and whose derivative at 0 is

$$T_1{+}T_2{+}T_3, \qquad T_1 = D^2B(c)[\dot c, \dot c]\, f(r), \quad T_2 = 2\, DB(c)[\dot c]\, f_1(r)(2a), \quad T_3 = B(c)\Big(-\frac{3}{32\sqrt{r}}\Big)(2a)^2.$$

Apply `RadialQuotientSecondVariation` with h, $\dot h$, r, $2a$ and $r > 0$: it produces a function G_Q which is the derivative of $\tau \mapsto \|h + \tau\dot h\|^2/(r + 2a\tau)^2$ near 0 and whose derivative at 0 is

$$T_4 + T_5 + T_6, \qquad T_4 = \frac{2\|\dot h\|^2}{r^2}, \quad T_5 = -\frac{8(2a)\langle h, \dot h\rangle}{r^3}, \quad T_6 = \frac{6(2a)^2\|h\|^2}{r^4}.$$

The affine summand $\tau \mapsto \alpha(Z) + \tau\alpha(\sigma)$ has derivative the constant $\alpha(\sigma)$ everywhere, hence second derivative 0 at 0. So $\rho' = \alpha(\sigma) + G_B + G_Q$ near 0 and $\rho''(0) = \sum_{k=1}^{6} T_k$. By `SecondVariationAlongLine`, applied to Ψ at Z in the direction σ with 2 and the three facts just established,

$$D^2\Psi(Z)[\sigma, \sigma] = T_1 + T_2 + T_3 + T_4 + T_5 + T_6. \tag{222}$$

Bounds on the two velocities. First $\|\dot c\| \le K_3$ by 15. Next, by 9, 8 and 14,

$$\|\dot h\| = \tfrac{1}{2} f(r)\, \|\nabla B(c)\| \le \tfrac{1}{2}\, r\, K_1, \tag{223}$$

where $\frac{1}{2} f(r) \ge 0$ lets us take it out of the norm and then $f(r) \le r$ and $\|\nabla B(c)\| \le K_1$ are multiplied, both factors being nonnegative. Also, from 14 and $\|\dot c\| \le K_3$, two applications of the operator-norm inequality give

$$\big|D^2B(c)[\dot c, \dot c]\big| \le |D^2B(c)|\, \|\dot c\|^2 \le K_2K_3^2, \qquad \big|DB(c)[\dot c]\big| \le |DB(c)|\, \|\dot c\| \le K_1K_3.$$

The two localizing terms. Since $0 \le f(r) \le r \le 3/2$ by 9 and 8,

$$T_1 \le \big|D^2B(c)[\dot c, \dot c]\big|\, f(r) \le K_2K_3^2 \cdot \tfrac{3}{2}.$$

Since $0 < a$ gives $0 < 2a$, and $0 \le f_1(r) \le 1$ with $a \le B_{\text{out}}$ by 12 and 16, we get $0 \le f_1(r)(2a) \le 2a \le 2B_{\text{out}}$, whence

$$T_2 = 2\, DB(c)[\dot c] \cdot f_1(r)(2a) \le 2\, K_1K_3 \cdot 2B_{\text{out}} = 4B_{\text{out}}K_1K_3.$$

The dominant term. Here $T_3 = -\dfrac{3}{8\sqrt{r}}\, B(c)a^2$, which is the paper's $-3B(c)a^2/(8\sqrt{r})$. By 16 we have $a \ge \delta/4 > 0$, so $a^2 \ge \delta^2/16$, and $B(c) \ge \delta/2$ by 13, so $B(c)a^2 \ge \delta^3/32$. Multiplying by the nonnegative number $\frac{3}{8}(\sqrt{r})^{-1}$ gives

$$T_3 = -\frac{3}{8}\frac{B(c)a^2}{\sqrt{r}} \le -\frac{3}{8} \cdot \frac{\delta^3}{32\sqrt{r}} = -\frac{3\delta^3}{256\sqrt{r}}.$$

The three penalty terms. By (223), $\|\dot h\|^2 \le r^2K_1^2/4$, so $T_4 = 2\|\dot h\|^2/r^2 \le K_1^2/2$. For T_5, Cauchy–Schwarz and 17 with (223) give

$$-\langle h, \dot h\rangle \le |\langle h, \dot h\rangle| \le \|h\|\, \|\dot h\| \le 2Cr^2 \cdot \tfrac{1}{2} rK_1 = CK_1r^3,$$

and the right-hand side is nonnegative, so, using $0 < a \le B_{\text{out}}$,

$$T_5 = \frac{16a\big(-\langle h, \dot h\rangle\big)}{r^3} \le \frac{16a\, CK_1r^3}{r^3} = 16aCK_1 \le 16B_{\text{out}}CK_1.$$

For T_6, $(2a)^2 \le 4B_{\text{out}}^2$ and $\|h\|^2 \le 4C^2r^4$ by 16 and 17, both bounds being between nonnegative quantities, so

$$T_6 = \frac{(2a)^2\|h\|^2 \cdot 6}{r^4} \le \frac{6 \cdot 4B_{\text{out}}^2 \cdot 4C^2r^4}{r^4} = 96B_{\text{out}}^2C^2.$$

Conclusion. Adding the six bounds and comparing with the definition 18 of M,

$$\sum_{k=1}^{6} T_k \le \frac{K_1^2}{2} + 16B_{\text{out}}CK_1 + 96B_{\text{out}}^2C^2 + \frac{3}{2}K_2K_3^2 + 4B_{\text{out}}K_1K_3 - \frac{3\delta^3}{256\sqrt{r}} = M - \frac{3\delta^3}{256\sqrt{r}},$$

which with (222) is (221). □

2.59. `ConstrainedThreePointMaximizer.`

Lemma 2.59 (A constrained maximizer of the penalized three-point functional, strictly inside the constraint)**.** *Let E be a finite-dimensional real inner-product space and write $\overline{B}_\rho$ for the closed ball of radius ρ centred at the origin of E. Let $\Phi : E \times E \times E \to \mathbb{R}$, let $x_k, y_k, z_k \in E$ and let $\beta, C' \in \mathbb{R}$. Write*

$$K = \left(\overline{B}_{3/4}\right)^3 \cap \left\{(x', y', z') : \frac{\beta}{2C'} \le |x' - y'|\right\}$$

for the constrained set. Assume:

1. *$0 < \beta$ and $0 < C'$;*
2. *$x_k, y_k, z_k \in \overline{B}_{3/4}$;*
3. *$\dfrac{\beta}{2C'} \le |x_k - y_k|$;*
4. *$\beta \le \Phi(x_k, y_k, z_k)$;*
5. *the function $(x', y', z') \mapsto \Phi(x', y', z')$ is continuous on K;*
6. *$\Phi(x', y', z') \le |x' - y'|\, C'$ for all $x', y', z' \in \overline{B}_{3/4}$ with $x' \ne y'$.*

Then there are $x, y, z \in \overline{B}_{3/4}$ with $x \ne y$ such that

$$\beta \le \Phi(x, y, z), \qquad \Phi(x', y', z') \le \Phi(x, y, z) \quad \text{for every } (x', y', z') \in K, \tag{224}$$

and such that the separation at the maximizer satisfies the lower bound

$$\frac{\beta}{C'} \le |x - y|, \tag{225}$$

which is twice the radius $\beta/(2C')$ defining K.

Proof. Throughout, 1 gives $\beta/(2C') > 0$, being a quotient of positive numbers.

K is compact and nonempty. Since E is finite-dimensional, each closed ball $\overline{B}_{3/4}$ is compact, hence so is the product $(\overline{B}_{3/4})^3$ inside $E \times E \times E$. The map $(x', y', z') \mapsto |x' - y'|$ is continuous, so $\{(x', y', z') : \beta/(2C') \le |x' - y'|\}$ is closed, being the set where a continuous function dominates a constant. Thus K is the intersection of a compact set with a closed set and is compact. It is nonempty because $(x_k, y_k, z_k) \in K$ by 2 and 3.

The maximum is attained. By 5 the functional is continuous on the compact nonempty set K, so it attains a maximum there: there is $(x, y, z) \in K$ with

$$\Phi(x', y', z') \le \Phi(x, y, z) \qquad \text{for every } (x', y', z') \in K. \tag{226}$$

Membership $(x, y, z) \in K$ unpacks to $x, y, z \in \overline{B}_{3/4}$ together with

$$\frac{\beta}{2C'} \le |x - y|. \tag{227}$$

This is the second half of (224).

The maximum is at least β. Applying (226) to the admissible triple $(x_k, y_k, z_k) \in K$ and then 4,

$$\beta \le \Phi(x_k, y_k, z_k) \le \Phi(x, y, z),$$

which is the first half of (224).

The two endpoints are distinct. If $x = y$ then $|x - y| = 0$, and (227) would give $\beta/(2C') \le 0$, contradicting $\beta/(2C') > 0$. Hence $x \ne y$.

The maximizer is strictly inside the constraint. The triple (x, y, z) has all three entries in $\overline{B}_{3/4}$ and satisfies $x \neq y$, so 6 applies to it and gives $\Phi(x, y, z) \leq |x - y|\, C'$. Combining with $\beta \leq \Phi(x, y, z)$ yields $\beta \leq |x-y|\, C'$, and dividing by $C' > 0$ gives (225). Since $\beta/C' = 2 \cdot \beta/(2C')$ is strictly larger than the radius $\beta/(2C')$ that defines K — the latter being positive — the inequality (225) is strictly stronger than the constraint (227), so the maximizer does not sit on the boundary imposed by the constraint. □

2.60. `ThreePointMaximizerGeometry`.

Lemma 2.60 (The geometry at a maximizing triple of the penalized three-point functional)**.** *Let E be a finite-dimensional real inner-product space, write $\overline{B}_\rho$ for the closed ball of radius ρ centred at the origin of E, and write $\ell^2(E^3)$ for $E \times E \times E$ with the Euclidean product norm. Let $\Phi : E \times E \times E \to \mathbb{R}$, let $\Psi : \ell^2(E^3) \to \mathbb{R}$, let $g_0, g_1, g_2 : E \to \mathbb{R}$, let $w, B : E \to \mathbb{R}$, let $f : \mathbb{R} \to \mathbb{R}$, let $x, y, z, c, h \in E$, let $Z \in \ell^2(E^3)$ and let $r, \delta, \nu, C, C', B_{\mathrm{out}}, \beta, r_0 \in \mathbb{R}$. Assume:*

(1) *for all $x', y', z' \in E$,*

$$\Phi(x', y', z') = w(x') + w(y') - 2w(z') - B\Big(\frac{x' + y'}{2}\Big) f\big(|x' - y'|\big) - \frac{\big|z' - \frac{x'+y'}{2}\big|^2}{|x' - y'|^2};$$

(2) $c = \dfrac{x + y}{2}$, $h = z - c$ *and* $r = |x - y|$;
(3) *the three coordinates of Z are x, y and z;*
(4) $\displaystyle\sum_j g_j\big(Y_j\big) - \Psi(Y) = \Phi(Y_0, Y_1, Y_2)$ *for every $Y \in \ell^2(E^3)$, where Y_0, Y_1, Y_2 are the coordinates of Y;*
(5) $x, y, z, c \in \overline{B}_{3/4}$;
(6) $0 < r$ *and* $\nu r \leq f(r)$;
(7) $|w(t)| \leq \dfrac{\delta \nu r_0}{8}$ *for every $t \in \overline{B}_{3/4}$;*
(8) $|w(s) - w(t)| \leq C\,|s - t|$ *for all $s, t \in \overline{B}_{3/4}$;*
(9) $\dfrac{\delta}{2} \leq B(c)$, *and* $B(c) = B_{\mathrm{out}}$ *if* $\dfrac{1}{2} \leq |c|$;
(10) $\Phi(x, y, z) \leq Cr + C^2 r^2 - B(c) f(r)$;
(11) $\beta \leq \Phi(x, y, z)$;
(12) $\Phi(x', y', z') \leq \Phi(x, y, z)$ *for all $x', y', z' \in \overline{B}_{3/4}$ with* $\dfrac{\beta}{2C'} \leq |x' - y'|$;
(13) $\dfrac{\beta}{C'} \leq |x - y|$;
(14) $0 < \beta$, $0 < C'$, $0 < C$, $0 < \delta$, $0 < \nu$, $C' = C + \frac{3}{2}C^2$, $B_{\mathrm{out}} = \dfrac{C'}{\nu} + \dfrac{\delta}{2} + 1$ *and* $\dfrac{\delta}{2} \leq B_{\mathrm{out}}$;
(15) $r_0 \leq \dfrac{1}{4}$, $r_0 \leq \dfrac{1}{16C}$, $r_0 \leq \dfrac{16}{9}$ *and* $r_0 \leq \dfrac{\delta}{64C^2}$.

Then

$$r < r_0, \qquad |h| \leq 2Cr^2, \qquad \frac{3}{16}\sqrt{r} \leq \frac{1}{4}, \qquad 8C^2 r \leq \frac{\delta}{8}, \tag{228}$$

and Z is a local maximum of $Y \mapsto \sum_j g_j(Y_j) - \Psi(Y)$ on $\ell^2(E^3)$.

Proof. By 2 and 1 evaluated at (x, y, z),

$$\Phi(x, y, z) = w(x) + w(y) - 2w(z) - B(c) f(r) - \frac{|h|^2}{r^2}, \tag{229}$$

and by 11 with 14 the maximum is positive: $0 < \beta \leq \Phi(x, y, z)$. Note also that $C^2 > 0$, being the square of the positive C.

The radius is small. Apply 7 at x, y and z, which lie in $\overline{B}_{3/4}$ by 5; this bounds each of $w(x), w(y), w(z)$ between $\pm\delta\nu r_0/8$, so $w(x) + w(y) - 2w(z) \leq 4 \cdot \delta\nu r_0/8 = \delta\nu r_0/2$. Next $0 \leq \nu r$

by 6 and 14, so $0 \le f(r)$ by 6, and therefore

$$\frac{\delta}{2}(\nu r) \le \frac{\delta}{2} f(r) \le B(c) f(r)$$

using 6 for the first inequality (multiplying by $\delta/2 \ge 0$) and 9 for the second (multiplying by $f(r) \ge 0$). Finally $|h|^2/r^2 \ge 0$. Substituting these three facts into (229) and using $\Phi(x,y,z) > 0$,

$$0 < \frac{\delta \nu r_0}{2} - \frac{\delta \nu r}{2},$$

and cancelling the positive factor $\delta\nu/2$ gives $r < r_0$. This is the paper's $r \le 8\varepsilon_i/(\delta\nu)$ with $\varepsilon_i = \delta\nu r_0/8$.

The deviation of the third point. First $\beta/(2C') \le |x-y|$: indeed $\beta/(2C') = \frac{1}{2} \cdot \beta/C'$ and $\beta/C' \ge 0$, so $\beta/(2C') \le \beta/C' \le |x-y|$ by 13. Hence the competitor (x,y,c) is admissible in 12 — its three entries lie in $\overline{B}_{3/4}$ by 5 — and $\Phi(x,y,c) \le \Phi(x,y,z)$. Evaluating 1 at (x,y,c) and using $c - \frac{x+y}{2} = 0$ from 2 gives $\Phi(x,y,c) = w(x) + w(y) - 2w(c) - B(c)f(r)$, so comparing with (229) and cancelling the common terms,

$$\frac{|h|^2}{r^2} \le 2\big(w(c) - w(z)\big).$$

By 8 at $s = c$, $t = z$, both in $\overline{B}_{3/4}$, and $|c - z| = |h|$ from 2, the right-hand side is at most $2C|h|$. Multiplying by $r^2 > 0$ gives $|h|^2 \le 2Cr^2|h|$. If $|h| = 0$ then $|h| \le 2Cr^2$ because $2Cr^2 \ge 0$; otherwise divide by $|h| > 0$. Either way $|h| \le 2Cr^2$.

The two smallness packages. From $r < r_0$ and 15: $r < 1/4$, so $r < 1/2$; and $r < 1/(16C)$, so $r(16C) < 1$ and hence $2Cr^2 < 1/4$ (multiply $r \cdot 16C < 1$ by $r/8 > 0$ and use $r < 1/4$). Likewise $r \le 16/9$ gives $\sqrt{r} \le \sqrt{16/9} = 4/3$, so $\frac{3}{16}\sqrt{r} \le \frac{3}{16} \cdot \frac{4}{3} = \frac{1}{4}$; and $r \le \delta/(64C^2)$ with $64C^2 > 0$ gives $64C^2 r \le \delta$, i.e. $8C^2 r \le \delta/8$. The last two are the asserted bounds of (228); the first two, $r < 1/2$ and $2Cr^2 < 1/4$, are used below and are not among the conclusions.

The midpoint is interior to the localization region. Suppose $\frac{1}{2} \le |c|$. Then $B(c) = B_{\text{out}}$ by 9, so 10 reads $\Phi(x,y,z) \le Cr + C^2r^2 - B_{\text{out}} f(r)$. Now $B_{\text{out}} > 0$, since $0 < \delta/2 \le B_{\text{out}}$ by 14, so $\nu r \le f(r)$ gives $B_{\text{out}}(\nu r) \le B_{\text{out}} f(r)$; and by the definitions of C' and B_{out} in 14,

$$B_{\text{out}}\nu = C' + \Big(\frac{\delta}{2} + 1\Big)\nu, \qquad \text{so} \qquad B_{\text{out}}(\nu r) = \Big(C + \frac{3}{2}C^2 + \Big(\frac{\delta}{2} + 1\Big)\nu\Big) r.$$

Since $r < r_0 \le 1/4 \le 3/2$ we have $C^2 r(\frac{3}{2} - r) \ge 0$, i.e. $Cr + C^2r^2 \le Cr + \frac{3}{2}C^2 r$, and $\big(\frac{\delta}{2}+1\big)\nu r > 0$. Chaining these,

$$0 < \Phi(x,y,z) \le Cr + C^2r^2 - B_{\text{out}} f(r) \le Cr + \frac{3}{2}C^2 r - \Big(C + \frac{3}{2}C^2 + \Big(\frac{\delta}{2}+1\Big)\nu\Big) r = -\Big(\frac{\delta}{2}+1\Big)\nu r < 0,$$

a contradiction. Hence $|c| < 1/2$. This is where the choice of B_{out} in 14 is used, and it is the paper's $|c| < 1/2$ at a positive maximum; the Lean statement does not list it as a conclusion because nothing after this step reads it, only the interiority it produces next.

The three points are interior. By 2, $x - c = \frac{1}{2}(x-y)$ and $y - c = -\frac{1}{2}(x-y)$, so $|x-c| = |y-c| = \frac{1}{2}r$, while $|z-c| = |h|$. For any $t \in E$ with $|t-c| < 1/4$ we get $|t| \le |c| + |t-c| < \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ by the triangle inequality and $|c| < 1/2$. Since $\frac{1}{2}r < \frac{1}{4}$ (from $r < 1/2$) and $|h| \le 2Cr^2 < 1/4$, this applies to $t = x, y, z$ and gives $|x|, |y|, |z| < 3/4$. Moreover $\beta/(2C') < \beta/C' \le |x-y| = r$, strictly, because $\beta > 0$ and $C' > 0$.

The constrained maximum is local. The three coordinate maps $Y \mapsto Y_j$ on $\ell^2(E^3)$ are continuous, hence so are $Y \mapsto |Y_j|$ and $Y \mapsto |Y_0 - Y_1|$. By 3 these take the values $|x|, |y|, |z| < 3/4$ and $|x-y| > \beta/(2C')$ at $Y = Z$, so each of the four strict inequalities persists on a neighbourhood of Z; intersecting the four neighbourhoods, there is a neighbourhood N of Z on which $|Y_0|, |Y_1|, |Y_2| < 3/4$ and $\beta/(2C') < |Y_0 - Y_1|$. For $Y \in N$ the triple (Y_0, Y_1, Y_2) is therefore admissible in 12, so $\Phi(Y_0, Y_1, Y_2) \le \Phi(x,y,z)$, and by 4 together with 3 this says exactly

$$\sum_j g_j(Y_j) - \Psi(Y) \le \sum_j g_j(Z_j) - \Psi(Z).$$

Thus Z is a local maximum of $Y \mapsto \sum_j g_j(Y_j) - \Psi(Y)$, which is the last assertion. □

2.61. `ThreePointJetsAndMatrixInequality`.

Lemma 2.61 (The jet bounds and the matrix inequality at a maximizing triple)**.** *Let E be a finite-dimensional real inner-product space and write $\ell^2(E^3)$ for $E \times E \times E$ with the Euclidean product norm. Let $\Psi : \ell^2(E^3) \to \mathbb{R}$, let $Z, \sigma \in \ell^2(E^3)$, let $g_0, g_1, g_2 : E \to \mathbb{R}$, let $p_0, p_1, p_2, q, e, c, h \in E$, let $B : E \to \mathbb{R}$, let $f, f_1 : \mathbb{R} \to \mathbb{R}$ and let $a, r, \delta, C, K_1, K_3, B_{\mathrm{out}} \in \mathbb{R}$. Assume:*

(1) $a = B(c)\, f_1(r) - \dfrac{2|h|^2}{r^3}$;
(2) $p_1 = q + \frac{1}{2} f(r)\, \nabla B(c) - \dfrac{h}{r^2} + a\, e$ *and* $p_2 = q + \frac{1}{2} f(r)\, \nabla B(c) - \dfrac{h}{r^2} - a\, e$;
(3) *the three coordinates of σ are p_1, p_2 and p_0;*
(4) *the three coordinate blocks of $\nabla\Psi(Z)$ are p_1, p_2 and $-2p_0$;*
(5) *each g_j is an infinity subsolution on the unit ball B_1 of E (in particular continuous there);*
(6) *every coordinate of Z lies in B_1;*
(7) *Ψ is twice continuously differentiable at Z;*
(8) *Z is a local maximum of $Y \mapsto \sum_j g_j(Y_j) - \Psi(Y)$ on $\ell^2(E^3)$;*
(9) $0 < r$ *and* $r \le 3/2$;
(10) $f_1(r) = 1 - \dfrac{3\sqrt{r}}{16}$ *and* $f_1(r) \le 1$;
(11) $\dfrac{3}{16}\sqrt{r} \le \dfrac{1}{4}$ *and* $8C^2 r \le \dfrac{\delta}{8}$;
(12) $|h| \le 2Cr^2$;
(13) $\dfrac{\delta}{2} \le B(c) \le B_{\mathrm{out}}$ *and* $0 < \delta$;
(14) $|\nabla B(c)| \le K_1$, $0 \le f(r) \le r$, $|q| \le C$ *and* $K_3 = C + \frac{3}{4}K_1 + 2C$.

Then

$$\frac{\delta}{4} \le a, \qquad a \le B_{\mathrm{out}}, \qquad \left|\frac{p_1 + p_2}{2}\right| \le K_3, \tag{230}$$

and the second variation of Ψ at Z in the direction σ is nonnegative:

$$0 \le D^2\Psi(Z)[\sigma, \sigma]. \tag{231}$$

Proof. *The lower bound $a \ge \delta/4$.* By 10 and the first half of 11, $f_1(r) = 1 - \frac{3}{16}\sqrt{r} \ge 1 - \frac{1}{4} = \frac{3}{4}$. Next, $|h|^2 \le (2Cr^2)^2$ by 12, both sides being nonnegative, and $r^3 > 0$ by 9, so

$$\frac{2|h|^2}{r^3} \le \frac{2(2Cr^2)^2}{r^3} = 8C^2 r \le \frac{\delta}{8}$$

by the second half of 11. Also $\frac{\delta}{2} \cdot \frac{3}{4} \le \frac{\delta}{2} f_1(r) \le B(c) f_1(r)$: the first inequality multiplies $f_1(r) \ge 3/4$ by $\delta/2 \ge 0$, the second multiplies $B(c) \ge \delta/2$ from 13 by $f_1(r) \ge 3/4 \ge 0$. Substituting both into 1,

$$a = B(c) f_1(r) - \frac{2|h|^2}{r^3} \ge \frac{3\delta}{8} - \frac{\delta}{8} = \frac{\delta}{4}.$$

The upper bound $a \le B_{\mathrm{out}}$. From 13, $0 < \delta$ gives $0 \le \delta/2 \le B(c)$, and $f_1(r) \le 1$ by 10, so $B(c) f_1(r) \le B(c) \le B_{\mathrm{out}}$. Since $2|h|^2/r^3 \ge 0$, 1 gives $a \le B(c) f_1(r) \le B_{\mathrm{out}}$.

The mean endpoint velocity. Subtracting the two identities of 2, the radial parts $\pm a\, e$ cancel in the mean:

$$\frac{p_1 + p_2}{2} = q + \tfrac{1}{2} f(r)\, \nabla B(c) - \frac{h}{r^2},$$

so no bound on e is needed. Now $\left|\frac{1}{2} f(r) \nabla B(c)\right| = \frac{1}{2} f(r)\, |\nabla B(c)| \le \frac{1}{2} \cdot \frac{3}{2} \cdot K_1 = \frac{3}{4} K_1$, using $0 \le f(r) \le r \le 3/2$ from 14 and 9 and $|\nabla B(c)| \le K_1$; and $\left|h/r^2\right| = r^{-2}|h| \le r^{-2} \cdot 2Cr^2 = 2C$ by 12 and $r^2 > 0$. Two applications of the triangle inequality and $|q| \le C$ give

$$\left|\frac{p_1 + p_2}{2}\right| \le |q| + \left|\tfrac{1}{2} f(r) \nabla B(c)\right| + \left|\frac{h}{r^2}\right| \le C + \tfrac{3}{4} K_1 + 2C = K_3$$

by the definition of K_3 in 14.

The matrix inequality, for each regularization parameter. Fix $\eta > 0$. Apply `SumRuleJets` with $N = 3$, the three open sets all equal to B_1, the functions g_j — continuous on B_1 by 5 — the test Ψ, the point Z, the memberships 6, the regularity 7, the local maximality 8, and this η. It produces continuous symmetric bilinear forms X_0, X_1, X_2 on E such that, for each j, the pair $\big((\nabla\Psi(Z))_j, X_j\big)$ lies in the closed superjet of g_j at Z_j, and such that

$$\sum_j X_j[\xi_j,\xi_j] \le D^2\Psi(Z)[\xi,\xi] + \eta\,\big|D^2\Psi(Z)\xi\big|^2 \qquad (\xi \in \ell^2(E^3)). \tag{232}$$

By 4 the three jet gradients are p_1, p_2 and $-2p_0$. Hence `InfSubsolutionClosedSuperjet`, applied to the infinity subsolution g_j on B_1 at the point $Z_j \in B_1$, gives

$$0 \le X_0[p_1,p_1], \qquad 0 \le X_1[p_2,p_2], \qquad 0 \le X_2[-2p_0,-2p_0].$$

The last one is $4X_2[p_0,p_0]$ by bilinearity, so $0 \le X_2[p_0,p_0]$ as well. Evaluating (232) at $\xi = \sigma$ and using 3 to read $\sum_j X_j[\sigma_j,\sigma_j] = X_0[p_1,p_1] + X_1[p_2,p_2] + X_2[p_0,p_0] \ge 0$, we obtain

$$0 \le D^2\Psi(Z)[\sigma,\sigma] + \eta\,\big|D^2\Psi(Z)\sigma\big|^2 \qquad \text{for every } \eta > 0. \tag{233}$$

Letting η decrease to zero. The point Z, the Hessian $D^2\Psi(Z)$ and the direction σ in (233) do not depend on η — only the forms X_j supplied by the sum rule do, and they have already been eliminated — so the inequality holds for a fixed pair of real numbers $A = D^2\Psi(Z)[\sigma,\sigma]$ and $M = |D^2\Psi(Z)\sigma|^2 \ge 0$ and every $\eta > 0$. Suppose $A < 0$. Choosing $\eta = \dfrac{-A}{2(M+1)} > 0$, which is legitimate because $M + 1 > 0$, gives

$$\eta M = \frac{-A\,M}{2(M+1)} \le \frac{-A}{2},$$

since $M \le M + 1$ and $-A > 0$, and therefore $A + \eta M \le A - A/2 = A/2 < 0$, contradicting (233) at that η. Hence $A \ge 0$, which is (231). □

2.62. `ThreePointFlatness`.

Theorem 2.62 (Three-point flatness)**.** *Let E be a finite-dimensional real inner-product space, write $B_\rho = B(0,\rho)$ for the open ball of radius ρ about the origin, and let $L \in \mathbb{R}$. Let $U_i : E \to \mathbb{R}$, $i \in \mathbb{N}$, be infinity-harmonic on B_1 in the sense of `IsInfHarmonicOn` and satisfy the common Lipschitz bound*

$$|U_i(x) - U_i(y)| \le L|x-y| \qquad (x,y \in B_1,\ i \in \mathbb{N}).$$

Let $q_i \in E$ and $b_i \in \mathbb{R}$, and write $\ell_i(x) = \langle q_i, x\rangle + b_i$ for the corresponding affine functions. Assume that $U_i - \ell_i \to 0$ uniformly on B_1, that is,

$$\text{for every } \varepsilon > 0 \text{ there is } i_0 \text{ with } |U_i(x) - \ell_i(x)| \le \varepsilon \quad (i \ge i_0,\ x \in B_1).$$

Then the midpoint defect of U_i, divided by the endpoint separation, tends to zero uniformly on $B_{1/4}$: for every $\varepsilon > 0$ there is i_0 such that

$$\Big|U_i(x) + U_i(y) - 2U_i\big(\tfrac12(x+y)\big)\Big| \le \varepsilon|x-y| \qquad \big(i \ge i_0,\ x,y \in B_{1/4},\ x \ne y\big).$$

Proof. Step 0: the degenerate case and the normalizations. If every element of E is 0 then there are no points x, y with $x \ne y$ and the conclusion holds vacuously with $i_0 = 0$. So assume $E \ne \{0\}$. Then B_1 contains two distinct points $x \ne y$, and the Lipschitz hypothesis applied to them gives $0 \le |U_i(x) - U_i(y)| \le L|x-y|$ with $|x-y| > 0$, so

$$L \ge 0.$$

Write $w_i = U_i - \ell_i$. For $\varepsilon > 0$ let $\mathcal{E}(\varepsilon)$ denote an index i_0 supplied by the flatness hypothesis for that ε, so that

$$|w_i(x)| \le \varepsilon \qquad \big(i \ge \mathcal{E}(\varepsilon),\ x \in B_1\big).$$

The slopes q_i are bounded. Let $i \geq \mathcal{E}(1)$ and let $e \in E$ with $|e| = 1$. The points $\pm e/2$ lie in B_1, and $\ell_i(e/2) - \ell_i(-e/2) = \langle q_i, e\rangle$, so

$$|\langle q_i, e\rangle| \leq \big|U_i(e/2) - U_i(-e/2)\big| + |w_i(e/2)| + |w_i(-e/2)| \leq L|e| + 2 = L + 2.$$

Taking $e = q_i/|q_i|$ when $q_i \neq 0$ gives $|q_i| \leq L + 2$ (the bound being trivial otherwise). Consequently, for $i \geq \mathcal{E}(1)$ and $x, y \in B_1$,

$$|w_i(x) - w_i(y)| \leq |U_i(x) - U_i(y)| + |\langle q_i, x - y\rangle| \leq \big(L + (L+2)\big)|x - y|,$$

so with

$$C := 2L + 2 \geq 2 > 0$$

every w_i with $i \geq \mathcal{E}(1)$ is C-Lipschitz on B_1, and $|q_i| \leq L + 2 \leq C$.

Negation is admissible. We record, for use below, the following scaling property. Let $\Omega \subseteq E$ be open, let $u : E \to \mathbb{R}$ and let $t < 0$. First, for any C^2 function ϕ near a point z and any $s \in \mathbb{R}$,

$$\Delta_\infty(s\phi)(z) = s^3\, \Delta_\infty\phi(z), \tag{234}$$

because $\nabla(s\phi)(z) = s\nabla\phi(z)$ and $D^2(s\phi)(z) = sD^2\phi(z)$, so that $D^2(s\phi)(z)[\nabla(s\phi)(z)] = s\, D^2\phi(z)\big(s\nabla\phi(z), s\nabla\phi(z$ $s^3 D^2\phi(z)[\nabla\phi(z)]$, using the definition of Δ_∞ in `InfLap` and the bilinearity of the second derivative. Now suppose u is an infinity supersolution on Ω in the sense of `IsInfSupersolutionOn`, and let ϕ be C^2 near $z \in \Omega$ with $tu - \phi$ having a local maximum on Ω at z. Dividing the inequality $tu(x) - \phi(x) \leq tu(z) - \phi(z)$ by $t < 0$ reverses it and gives $u(x) - \phi(x)/t \geq u(z) - \phi(z)/t$ for x near z in Ω; that is, ϕ/t is a C^2 lower test for u at z, so $\Delta_\infty(\phi/t)(z) \leq 0$ by `IsInfSupersolutionOn`. By (234) with $s = 1/t$ this reads $t^{-3}\Delta_\infty\phi(z) \leq 0$, and $t^{-3} < 0$, so $\Delta_\infty\phi(z) \geq 0$. As Ω is open and tu is continuous on Ω, this says that tu is an infinity subsolution on Ω in the sense of `IsInfSubsolutionOn`. The symmetric argument, starting from u an infinity subsolution and a lower test for tu, shows that tu is then an infinity supersolution. In particular, if u is infinity-harmonic on Ω then so is $-u$, and if u is infinity-harmonic on Ω then $-2u$ is an infinity subsolution on Ω.

Step 1: the contradiction hypothesis. Suppose the conclusion fails. Then there is $\delta > 0$ such that for every i_0 there are $i \geq i_0$ and $x, y \in B_{1/4}$ with $x \neq y$ and

$$\Big|U_i(x) + U_i(y) - 2U_i\big(\tfrac{1}{2}(x+y)\big)\Big| > \delta|x - y|.$$

Since ℓ_i is affine and $\frac{1}{2}(x+y)$ is the midpoint, $\ell_i(x) + \ell_i(y) - 2\ell_i(\frac{1}{2}(x+y)) = 0$, so the displayed quantity is unchanged when U_i is replaced by w_i. Choosing $i_0 = 1, 2, \ldots$ in turn and discarding repetitions produces a strictly increasing sequence of indices $i_1 < i_2 < \cdots$ and points $x_k, y_k \in B_{1/4}$, $x_k \neq y_k$, with

$$\big|w_{i_k}(x_k) + w_{i_k}(y_k) - 2w_{i_k}(c_k)\big| > \delta|x_k - y_k|, \qquad c_k = \tfrac{1}{2}(x_k + y_k). \tag{235}$$

Passing to a further subsequence we may assume that the sign of $w_{i_k}(x_k) + w_{i_k}(y_k) - 2w_{i_k}(c_k)$ is the same for every k. If that sign is negative, replace the whole family (U_i, q_i, b_i) by $(-U_i, -q_i, -b_i)$: by the scaling property of Step 0 each $-U_i$ is again infinity-harmonic on B_1, the Lipschitz bound with the same L is unchanged, the flatness hypothesis is unchanged, w_i is replaced by $-w_i$, and the quantity in (235) changes sign. We may therefore assume

$$w_{i_k}(x_k) + w_{i_k}(y_k) - 2w_{i_k}(c_k) > \delta|x_k - y_k| \qquad (k \geq 1). \tag{236}$$

Note also that, by the C-Lipschitz bound and $|x_k - c_k| = |y_k - c_k| = \frac{1}{2}|x_k - y_k|$, the left side of (236) is at most

$$\big(w_{i_k}(x_k) - w_{i_k}(c_k)\big) + \big(w_{i_k}(y_k) - w_{i_k}(c_k)\big) \leq C \cdot \tfrac{1}{2}|x_k - y_k| + C \cdot \tfrac{1}{2}|x_k - y_k| = C|x_k - y_k|$$

whenever $i_k \geq \mathcal{E}(1)$. Choosing one such k — possible because $i_k \to \infty$ — and dividing by $|x_k - y_k| > 0$ gives

$$\delta \leq C. \tag{237}$$

Step 2: the two auxiliary profiles. For $0 < r \leq 3/2$ set

$$f(r) = r - \tfrac{1}{8}r^{3/2}, \qquad f'(r) = 1 - \tfrac{3}{16}\sqrt{r}, \qquad f''(r) = -\frac{3}{32\sqrt{r}},$$

so f is C^2 on $(0,\infty)$. Since $\sqrt{r} \le \sqrt{3/2}$ on $(0,3/2]$,

$$\nu r \le f(r) \le r, \qquad \nu := 1 - \tfrac{1}{8}\sqrt{3/2} > 0, \qquad 0 < f'(r) \le 1. \tag{238}$$

Next we build the localizing factor. Let

$$P(\tau) = \begin{cases} 0, & \tau \le 0, \\ 6\tau^5 - 15\tau^4 + 10\tau^3, & 0 \le \tau \le 1, \\ 1, & \tau \ge 1, \end{cases}$$

a C^2 function on $\mathbb{R}$: on $[0,1]$ one computes $P(0) = 0$, $P(1) = 1$, $P'(\tau) = 30\tau^2(\tau-1)^2$ and $P''(\tau) = 60\tau(2\tau-1)(\tau-1)$, so P, P', P'' all match the constant branches in value at $\tau = 0$ and $\tau = 1$; moreover $P' \ge 0$, so $0 \le P \le 1$. Fix

$$B_{\text{out}} := \frac{C + \frac{3}{2}C^2}{\nu} + \frac{\delta}{2} + 1$$

and define $B : E \to \mathbb{R}$ by

$$B(c) = \frac{\delta}{2} + \Big(B_{\text{out}} - \frac{\delta}{2}\Big) P\Big(\frac{16|c|^2 - 1}{3}\Big).$$

Since $c \mapsto |c|^2$ is a continuous quadratic form, hence C^∞, and P is C^2, the function B is C^2 on E. Moreover $B_{\text{out}} - \delta/2 > 0$ and $0 \le P \le 1$ give

$$\frac{\delta}{2} \le B \le B_{\text{out}}, \qquad B \equiv \frac{\delta}{2} \text{ on } \overline{B}(0,1/4), \qquad B \equiv B_{\text{out}} \text{ where } |c| \ge \tfrac{1}{2}, \tag{239}$$

because $|c| \le 1/4$ makes the argument of P at most 0 and $|c| \ge 1/2$ makes it at least 1. Finally put

$$K_1 = \max_{\overline{B}(0,3/4)} |DB|, \qquad K_2 = \max_{\overline{B}(0,3/4)} \|D^2 B\|,$$

finite because DB and D^2B are continuous on the compact set $\overline{B}(0,3/4)$ (preamble).

Step 3: the maximization. Fix k, write $i = i_k$ and assume $i \ge \mathcal{E}(1)$. For a triple $(x,y,z) \in \big(\overline{B}(0,3/4)\big)^3$ with $x \ne y$ put

$$c = \tfrac{1}{2}(x+y), \qquad h = z - c, \qquad r = |x - y| \in (0, 3/2],$$

and

$$\Phi_i(x,y,z) = w_i(x) + w_i(y) - 2w_i(z) - B(c) f(r) - \frac{|h|^2}{r^2}.$$

Note $c \in \overline{B}(0,3/4)$ by convexity, and all three points lie in B_1.

(3a) An upper bound. Writing $w_i(x) + w_i(y) - 2w_i(z) = \big(w_i(x) - w_i(c)\big) + \big(w_i(y) - w_i(c)\big) + 2\big(w_i(c) - w_i(z)\big)$ and using the C-Lipschitz bound together with $|x - c| = |y - c| = r/2$ and $|c - z| = |h|$,

$$w_i(x) + w_i(y) - 2w_i(z) \le Cr + 2C|h|.$$

Since $2C|h| - |h|^2/r^2 \le C^2 r^2$ (the quadratic $s \mapsto 2Cs - s^2/r^2$ is maximal at $s = Cr^2$ with value C^2r^2), we get

$$\Phi_i(x,y,z) \le Cr + C^2r^2 - B(c)f(r) \le Cr + C^2r^2 =: rC', \qquad C' := C + \tfrac{3}{2}C^2, \tag{240}$$

using $B \ge 0$, $f \ge 0$ and $r \le 3/2$ in the last step.

(3b) The supremum is positive. Take $x = x_k$, $y = y_k$, $z = c_k$, so that $h = 0$ and $r = r_k := |x_k - y_k| > 0$. Then $c = c_k \in \overline{B}(0,1/4)$, since $x_k, y_k \in B_{1/4}$, so $B(c_k) = \delta/2$ by (239), and by (236) and $f(r) \le r$,

$$\Phi_i(x_k, y_k, c_k) > \delta r_k - \frac{\delta}{2} f(r_k) \ge \delta r_k - \frac{\delta}{2} r_k = \frac{\delta r_k}{2} =: \beta_k > 0.$$

Hence

$$M_i := \sup\big\{\Phi_i(x,y,z) : x,y,z \in \overline{B}(0,3/4),\ x \ne y\big\}$$

satisfies $M_i \ge \beta_k > 0$, and $M_i \le \frac{3}{2}C' < \infty$ by (240).

(3c) The supremum is attained, at a triple with r bounded below. Put

$$\mathcal{K} = \Big\{(x,y,z) \in \big(\overline{B}(0,3/4)\big)^3 : |x-y| \geq \frac{\beta_k}{2C'}\Big\},$$

a closed and bounded, hence compact, subset of E^3 (preamble). Any admissible triple with $r < \beta_k/(2C')$ has $\Phi_i \leq rC' < \beta_k/2 < \beta_k \leq M_i$ by (240), so such triples do not affect the supremum: M_i equals the supremum of Φ_i over $\mathcal{K}$, which is nonempty because $\beta_k/(2C') = \delta r_k/(4C') \leq r_k$ by (237) and $C' \geq C \geq \delta$, so $(x_k, y_k, c_k) \in \mathcal{K}$. On $\mathcal{K}$ we have $|x-y| > 0$, so Φ_i is continuous there, and a continuous function on a nonempty compact set attains its maximum (preamble). Let (x,y,z) be a maximizing triple, so

$$\Phi_i(x,y,z) = M_i \geq \beta_k > 0. \tag{241}$$

By (240) and (241), $rC' \geq \beta_k$, i.e. $r \geq \beta_k/C' > \beta_k/(2C')$: the maximizing triple is strictly inside the constraint defining $\mathcal{K}$.

Step 4: the geometry at the maximizer. Keep the maximizing triple (x,y,z) of Step 3 and its c, h, r.

(4a) r is small when i is large. Let $\varepsilon > 0$ and suppose $i \geq \mathcal{E}(\varepsilon)$, so $|w_i| \leq \varepsilon$ on B_1. From (241), $B \geq \delta/2$ and $f(r) \geq \nu r$,

$$0 < \Phi_i(x,y,z) \leq 4\varepsilon - B(c)f(r) \leq 4\varepsilon - \frac{\delta\nu r}{2},$$

whence

$$r < \frac{8\varepsilon}{\delta\nu}. \tag{242}$$

(4b) $|h| \leq 2Cr^2$. The triple (x,y,c) is admissible and lies in $\mathcal{K}$ (it has the same x, y), so $\Phi_i(x,y,z) \geq \Phi_i(x,y,c)$. The terms $w_i(x) + w_i(y)$ and $B(c)f(r)$ are identical in the two, and the penalty vanishes at $z = c$, so

$$-2w_i(z) - \frac{|h|^2}{r^2} \geq -2w_i(c), \qquad \text{i.e.} \qquad \frac{|h|^2}{r^2} \leq 2\big(w_i(c) - w_i(z)\big) \leq 2C|h|.$$

Hence $|h|^2 \leq 2C|h|r^2$, and therefore

$$|h| \leq 2Cr^2 \tag{243}$$

(trivially if $h = 0$, and after dividing by $|h| > 0$ otherwise).

(4c) $|c| < 1/2$. If $|c| \geq 1/2$ then $B(c) = B_{\text{out}}$ by (239), so (240), $f(r) \geq \nu r$ and $r \leq 3/2$ give

$$\Phi_i(x,y,z) \leq Cr + C^2r^2 - B_{\text{out}}\nu r \leq r\Big(C + \tfrac{3}{2}C^2 - B_{\text{out}}\nu\Big) < 0$$

by the choice of B_{out}, contradicting (241). So $|c| < 1/2$.

(4d) The maximizer is interior. Assume in addition

$$r < \tfrac{1}{2} \qquad \text{and} \qquad 2Cr^2 < \tfrac{1}{4}. \tag{244}$$

Then $|x| \leq |c| + \frac{1}{2}r < \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ and likewise $|y| < 3/4$, while $|z| \leq |c| + |h| < \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ by (243). So (x,y,z) lies in the open set $\big(B_{3/4}\big)^3$. Together with $r > \beta_k/(2C')$ and the continuity of $(x',y',z') \mapsto |x'-y'|$, there is a neighbourhood $\mathcal{N}$ of (x,y,z) in E^3 contained in $\big(B_{3/4}\big)^3 \cap \{|x'-y'| > \beta_k/(2C')\} \subseteq \mathcal{K}$. Hence

$$\Phi_i \leq \Phi_i(x,y,z) \quad \text{on } \mathcal{N}: \qquad (x,y,z) \text{ is a local maximum of } \Phi_i \text{ on } E^3. \tag{245}$$

Step 5: the sum rule. Set $e = (x-y)/r$, a unit vector, $q = q_i$, and

$$\Psi_0(x',y',z') = \ell_i(x') + \ell_i(y') - 2\ell_i(z') + B(c')f(r') + \frac{|h'|^2}{(r')^2},$$

where $c' = \frac{1}{2}(x'+y')$, $h' = z' - c'$, $r' = |x'-y'|$. This formula defines Ψ_0 on the open set $\mathcal{O} = \{(x',y',z') \in E^3 : x' \neq y'\}$ only, since $f(r')$ and $|h'|^2/(r')^2$ are undefined at $r' = 0$. On $\mathcal{O}$ the maps $(x',y') \mapsto c'$ and $(x',y',z') \mapsto h'$ are linear, r' is C^∞ (being the norm away from 0), f is C^2 on $(0,\infty)$ and B is C^2; so Ψ_0 is C^2 on $\mathcal{O}$, and $(x,y,z) \in \mathcal{O}$ because $r > \beta_k/(2C') > 0$.

The total test. SumRuleJets takes a test defined on all of $V = \ell^2(E \oplus E \oplus E)$, so Ψ_0 must be extended; we name one extension rather than leave it implicit. Put $\rho = \beta_k/(4C') > 0$, so that $r > \beta_k/(2C') = 2\rho$ and, by (4d),

$$\mathcal{N} \subseteq \big\{(x', y', z') : |x' - y'| > 2\rho\big\} \subseteq \mathcal{O}. \tag{246}$$

Define $\Psi : V \to \mathbb{R}$ by truncation outside that slab:

$$\Psi(x', y', z') = \begin{cases} \Psi_0(x', y', z') & |x' - y'| > 2\rho, \\ 0 & |x' - y'| \le 2\rho. \end{cases}$$

This is defined on all of V — the first branch is legitimate because $|x' - y'| > 2\rho > 0$ forces $x' \neq y'$, so $(x', y', z') \in \mathcal{O}$ — and

$$\Psi = \Psi_0 \quad \text{on the open set} \quad \big\{(x', y', z') : |x' - y'| > 2\rho\big\} \supseteq \mathcal{N}, \tag{247}$$

which contains (x, y, z) because $r > 2\rho$. Hence Ψ is C^2 near (x, y, z), and $D\Psi(x, y, z) = D\Psi_0(x, y, z)$ and $D^2\Psi(x, y, z) = D^2\Psi_0(x, y, z)$, two functions agreeing on a neighbourhood of a point having the same derivatives there. So every derivative computation below may be carried out on the formula for Ψ_0, and we write Ψ for both from here on.

Two remarks on why this suffices, and why nothing smoother is wanted. First, SumRuleJets imposes no regularity on its test away from the point: its two hypotheses involving Ψ are that Ψ be C^2 *near* $\widehat{Z}$ and that $\sum_j g_j - \Psi$ have a *local* maximum at $\widehat{Z}$, both local conditions, and its conclusion involves Ψ only through $D_j\Psi(\widehat{Z})$ and $A = D^2\Psi(\widehat{Z})$. The truncation above is therefore enough even though it is discontinuous across $\{|x' - y'| = 2\rho\}$, and in particular no smooth cutoff and no existence statement for one is needed. Second, for the same reason nothing depends on the choice: any map $V \to \mathbb{R}$ agreeing with Ψ_0 on some neighbourhood of (x, y, z) would serve equally, and this one is named only so that no extension has to be invented later.

With

$$g_1 = g_2 = U_i, \qquad g_3 = -2U_i,$$

each continuous on the open set B_1 (by IsInfHarmonicOn, which includes continuity on B_1), we have, on $\mathcal{N}$,

$$g_1(x') + g_2(y') + g_3(z') - \Psi(x', y', z') = \Phi_i(x', y', z'),$$

since $U_i - \ell_i = w_i$ and, by (247), $\Psi = \Psi_0$ there; the right-hand side is defined on $\mathcal{N}$ because $\mathcal{N} \subseteq \big(B_{3/4}\big)^3$ and $x' \neq y'$ on $\mathcal{N}$ by (246). By (245) the right-hand side is $\le \Phi_i(x, y, z)$ on $\mathcal{N}$, so $\sum_j g_j - \Psi$ has a local maximum at (x, y, z) on V, whose components lie in $B_{3/4} \subseteq B_1$.

Differentiating Ψ at (x, y, z) gives, with $\partial c'/\partial x' = \partial c'/\partial y' = \frac{1}{2}\mathrm{id}$, $\partial h'/\partial z' = \mathrm{id}$, $\partial h'/\partial x' = \partial h'/\partial y' = -\frac{1}{2}\mathrm{id}$, $\partial r'/\partial x' = e$ and $\partial r'/\partial y' = -e$,

$$\begin{gathered} p_1 := D_x\Psi = q + \tfrac{1}{2}f(r)DB(c) - \frac{h}{r^2} + ae, \qquad p_2 := D_y\Psi = q + \tfrac{1}{2}f(r)DB(c) - \frac{h}{r^2} - ae, \\ D_z\Psi = -2p_0 \text{ with } p_0 := q - \frac{h}{r^2}, \qquad a := B(c)f'(r) - \frac{2|h|^2}{r^3}. \end{gathered} \tag{248}$$

Indeed $D_z\Psi = -2q + 2h/r^2 = -2p_0$; and in the x' variable the three pieces contribute q, then $\frac{1}{2}f(r)DB(c) + B(c)f'(r)e$, then $-h/r^2 + |h|^2 \cdot (-2r^{-3})e$, whose sum is the stated p_1; the y' computation is identical with e replaced by $-e$, giving p_2.

(5a) Bounds on the data. By (238), (239), (243), $f(r) \le r \le 3/2$ and $|q| \le C$,

$$0 < a \le B_{\mathrm{out}}, \qquad \Big|\tfrac{1}{2}(p_1 + p_2)\Big| = \Big|q + \tfrac{1}{2}f(r)DB(c) - \frac{h}{r^2}\Big| \le C + \tfrac{3}{4}K_1 + 2C =: K_3, \tag{249}$$

where $a \le B(c)f'(r) \le B_{\mathrm{out}}$ because $2|h|^2/r^3 \ge 0$ and $f' \le 1$; the positivity of a is part of (5b).

(5b) $a \ge \delta/4$ for small r. Assume in addition

$$\tfrac{3}{16}\sqrt{r} \le \tfrac{1}{4} \qquad \text{and} \qquad 8C^2 r \le \frac{\delta}{8}. \tag{250}$$

Then $f'(r) \geq 3/4$, and $2|h|^2/r^3 \leq 2(2Cr^2)^2/r^3 = 8C^2 r \leq \delta/8$ by (243), so

$$a \geq \frac{\delta}{2}\cdot\frac{3}{4} - \frac{\delta}{8} = \frac{3\delta}{8} - \frac{\delta}{8} = \frac{\delta}{4} > 0.$$

(5c) The matrix inequality. Let $\eta > 0$. Apply `SumRuleJets` with $N = 3$, the open sets $O_1 = O_2 = O_3 = B_1$, the functions g_1, g_2, g_3, the test Ψ, the point (x, y, z) and this η: all its hypotheses have just been verified. It supplies symmetric continuous bilinear forms X_1, X_2, X_3 on E with, $\overline{J}^{2,+}$ denoting throughout the closed superjet of `ClosedSuperjet`, as in the statements of both `SumRuleJets` and `InfSubsolutionClosedSuperjet`,

$$(p_1, X_1) \in \overline{J}^{2,+}U_i(x), \quad (p_2, X_2) \in \overline{J}^{2,+}U_i(y), \quad (-2p_0, X_3) \in \overline{J}^{2,+}(-2U_i)(z), \tag{251}$$

and, writing $A = D^2\Psi(x, y, z)$ and $\bar{A}$ for the associated self-adjoint operator,

$$X_1[\xi_1] + X_2[\xi_2] + X_3[\xi_3] \leq A[\xi] + \eta|\bar{A}\xi|^2 \qquad \big(\xi = (\xi_1, \xi_2, \xi_3)\big). \tag{252}$$

By Step 0 the function U_i is an infinity subsolution on B_1 and $-2U_i$ is an infinity subsolution on B_1. Therefore `InfSubsolutionClosedSuperjet`, applied three times — to U_i at $x \in B_1$ with (p_1, X_1), to U_i at $y \in B_1$ with (p_2, X_2), and to $-2U_i$ at $z \in B_1$ with $(-2p_0, X_3)$ — gives

$$X_1[p_1] \geq 0, \qquad X_2[p_2] \geq 0, \qquad X_3[-2p_0] \geq 0.$$

The last reads $4X_3[p_0] \geq 0$ by bilinearity, so $X_3[p_0] \geq 0$. Taking $\xi = \sigma := (p_1, p_2, p_0)$ in (252),

$$0 \leq X_1[p_1] + X_2[p_2] + X_3[p_0] \leq A[\sigma] + \eta|\bar{A}\sigma|^2.$$

Neither A nor σ depends on η, so letting $\eta \downarrow 0$ yields

$$A[\sigma] \geq 0. \tag{253}$$

Step 6: the second variation is very negative. Because Ψ is C^2 near (x, y, z),

$$A[\sigma] = \frac{d^2}{d\tau^2}\Psi\big(x + \tau p_1,\ y + \tau p_2,\ z + \tau p_0\big)\Big|_{\tau=0}.$$

Write $c(\tau), h(\tau), r(\tau)$ for the quantities induced by that line. Since $p_1 - p_2 = 2ae$ by (248),

$$(x + \tau p_1) - (y + \tau p_2) = re + 2a\tau e = (r + 2a\tau)e,$$

so $r(\tau) = r + 2a\tau$ for $|\tau|$ small (here $a > 0$ and $r > 0$), giving $\dot{r} = 2a$ and $\ddot{r} = 0$. Also $c(\tau) = c + \tau\dot{c}$ with $\dot{c} = \frac{1}{2}(p_1 + p_2)$, and $h(\tau) = h + \tau\dot{h}$ with

$$\dot{h} = p_0 - \dot{c} = \Big(q - \frac{h}{r^2}\Big) - \Big(q + \tfrac{1}{2}f(r)DB(c) - \frac{h}{r^2}\Big) = -\tfrac{1}{2}f(r)DB(c), \qquad |\dot{h}| \leq \tfrac{1}{2}rK_1,$$

using $f(r) \leq r$ and $c \in \overline{B}(0, 3/4)$. The affine part $\ell_i(x + \tau p_1) + \ell_i(y + \tau p_2) - 2\ell_i(z + \tau p_0)$ is affine in τ and contributes 0.

(6a) The penalty term. With $u(\tau) = |h(\tau)|^2$ and $v(\tau) = r(\tau)^{-2}$ we have $u(0) = |h|^2$, $\dot{u}(0) = 2\langle h, \dot{h}\rangle$, $\ddot{u}(0) = 2|\dot{h}|^2$, $v(0) = r^{-2}$, $\dot{v}(0) = -2\dot{r}r^{-3}$, $\ddot{v}(0) = 6\dot{r}^2 r^{-4}$, so

$$\frac{d^2}{d\tau^2}\frac{|h(\tau)|^2}{r(\tau)^2}\Big|_{\tau=0} = \frac{2|\dot{h}|^2}{r^2} - \frac{8\dot{r}\langle h, \dot{h}\rangle}{r^3} + \frac{6\dot{r}^2|h|^2}{r^4}.$$

By $|\dot{h}| \leq rK_1/2$, $|h| \leq 2Cr^2$, $\dot{r} = 2a \leq 2B_{\mathrm{out}}$, the three terms are bounded in absolute value by $K_1^2/2$, $16B_{\mathrm{out}}CK_1$ and $96B_{\mathrm{out}}^2C^2$ respectively.

(6b) The localizing term. Since $c(\tau)$ and $r(\tau)$ are affine in τ,

$$\frac{d^2}{d\tau^2}\big(B(c(\tau))f(r(\tau))\big)\Big|_{\tau=0} = f(r)D^2B(c)[\dot{c}] + 2f'(r)\dot{r}\,\langle DB(c), \dot{c}\rangle + B(c)f''(r)\dot{r}^{\,2}.$$

The first two terms are bounded in absolute value by $\frac{3}{2}K_2K_3^2$ and $4B_{\mathrm{out}}K_1K_3$ by (238) and (249). The third is

$$B(c)\cdot\Big(-\frac{3}{32\sqrt{r}}\Big)\cdot 4a^2 = -\frac{3B(c)a^2}{8\sqrt{r}} \leq -\frac{3}{8\sqrt{r}}\cdot\frac{\delta}{2}\cdot\frac{\delta^2}{16} = -\frac{3\delta^3}{256\sqrt{r}},$$

using $B(c) \geq \delta/2$ and $a \geq \delta/4$ from (5b).

Collecting, with

$$M := \frac{K_1^2}{2} + 16B_{\text{out}}CK_1 + 96B_{\text{out}}^2C^2 + \tfrac{3}{2}K_2K_3^2 + 4B_{\text{out}}K_1K_3,$$

a constant depending only on C, δ, K_1, K_2, K_3 and B_{out} — and so on nothing that varies with k — we obtain

$$A[\sigma] \le M - \frac{3\delta^3}{256\sqrt{r}}. \tag{254}$$

Step 7: the contradiction. Let

$$r_0 = \min\Big\{\frac{1}{4},\ \frac{1}{16C},\ \frac{16}{9},\ \frac{\delta}{64C^2},\ \Big(\frac{3\delta^3}{256(M+1)}\Big)^2\Big\} > 0,$$

a positive number independent of k. If $r < r_0$ then: $r < 1/2$ and $2Cr^2 < 2C \cdot \frac{1}{16C} \cdot r < \frac{1}{4}$, so (244) holds; $\frac{3}{16}\sqrt{r} \le \frac{3}{16} \cdot \frac{4}{3} = \frac{1}{4}$ and $8C^2r \le 8C^2\delta/(64C^2) = \delta/8$, so (250) holds; and $\sqrt{r} < 3\delta^3/(256(M+1))$, so $3\delta^3/(256\sqrt{r}) > M+1$ and (254) gives $A[\sigma] < -1 < 0$, contradicting (253).

So it remains to produce one index k whose maximizing triple has $r < r_0$. Put $\varepsilon_0 = \delta\nu r_0/8 > 0$ and choose k with $i_k \ge \max\big(\mathcal{E}(1), \mathcal{E}(\varepsilon_0)\big)$, which is possible because $i_k \to \infty$. For this k all the hypotheses used in Steps 3–6 hold, and (242) with $\varepsilon = \varepsilon_0$ gives $r < 8\varepsilon_0/(\delta\nu) = r_0$. This is the required contradiction, so the conclusion cannot fail. □

2.63. `LinearLift`.

Lemma 2.63 (Linear lift)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega \subseteq E$ and let $u : E \to \mathbb{R}$ be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`. Then the function*

$$U(y, b) = u(y) + b$$

on the space-time $\ell^2(E \times \mathbb{R})$ is infinity-harmonic on $\Omega \times \mathbb{R} = \{(y, b) : y \in \Omega\}$, again in the sense of `IsInfHarmonicOn`.

Proof. Write $\widetilde{\Omega} = \Omega \times \mathbb{R}$ for the preimage of Ω under the spatial projection. Since the projection is continuous and Ω is open, $\widetilde{\Omega}$ is open; and U is continuous on $\widetilde{\Omega}$ because u is continuous on Ω and $(y, b) \mapsto b$ is continuous.

Upper tests. Let $(y_0, b_0) \in \widetilde{\Omega}$ and let ϕ be C^2 near (y_0, b_0) with $(y, b) \mapsto U(y, b) - \phi(y, b)$ having a local maximum on $\widetilde{\Omega}$ at (y_0, b_0). We must show $\Delta_\infty\phi(y_0, b_0) \ge 0$, where Δ_∞ is as in `InfLap`.

First, restrict to the vertical line through (y_0, b_0): since $y_0 \in \Omega$, all points (y_0, b) with b near b_0 lie in $\widetilde{\Omega}$, so the function

$$b \longmapsto U(y_0, b) - \phi(y_0, b) = u(y_0) + b - \phi(y_0, b)$$

has a local maximum at b_0. It is differentiable there, so its derivative vanishes:

$$\phi_b(y_0, b_0) = 1. \tag{255}$$

Put $p = \nabla_y\phi(y_0, b_0) \in E$; thus $\nabla\phi(y_0, b_0) = (p, 1)$ by (255).

Case $p \neq 0$. Choose the affine function $\ell(y) = b_0 + \langle p, y - y_0\rangle/|p|^2$ on E; then $\ell(y_0) = b_0$ and $\nabla\ell \cdot p = 1$. Define

$$\psi(y) = \phi(y, \ell(y)) - \ell(y).$$

Then ψ is C^2 near y_0, being the composition of the C^2 function ϕ with the affine map $y \mapsto (y, \ell(y))$, minus an affine function. Moreover ψ is a local upper test for u at y_0: for y near y_0 the point $(y, \ell(y))$ lies in $\widetilde{\Omega}$ and is near (y_0, b_0), so the local maximum property gives

$$\psi(y) - u(y) = \phi(y, \ell(y)) - U(y, \ell(y)) \ge \phi(y_0, b_0) - U(y_0, b_0) = \psi(y_0) - u(y_0),$$

that is, $y \mapsto u(y) - \psi(y)$ has a local maximum on Ω at y_0.

We compute the derivatives of ψ at y_0. By the chain rule for the affine map $y \mapsto (y, \ell(y))$, whose derivative is $v \mapsto (v, \nabla\ell \cdot v)$,

$$\nabla\psi(y_0) = \nabla_y\phi(y_0, b_0) + \phi_b(y_0, b_0)\,\nabla\ell - \nabla\ell = p,$$

using (255). Since the map is affine its second derivative vanishes, and the affine term $-\ell$ has vanishing Hessian, so for every $v \in E$

$$D^2\psi(y_0)[v] = \phi_{yy}(y_0, b_0)[v] + 2(\nabla\ell \cdot v)\,\phi_{yb}(y_0, b_0) \cdot v + (\nabla\ell \cdot v)^2\phi_{bb}(y_0, b_0).$$

Taking $v = p$ and using $\nabla\ell \cdot p = 1$,

$$D^2\psi(y_0)[p] = \phi_{yy}[p] + 2\phi_{yb} \cdot p + \phi_{bb} = D^2\phi(y_0, b_0)[(p, 1)] = \Delta_\infty\phi(y_0, b_0),$$

the last equality because $\nabla\phi(y_0, b_0) = (p, 1)$. As $\nabla\psi(y_0) = p$, the left-hand side is $\Delta_\infty\psi(y_0)$. The upper viscosity inequality for u at y_0, which is available since u is an infinity subsolution on Ω by `IsInfHarmonicOn`, gives $\Delta_\infty\psi(y_0) \geq 0$, i.e. $\Delta_\infty\phi(y_0, b_0) \geq 0$.

Case $p = 0$. Then $\nabla\phi(y_0, b_0) = (0, 1)$, so

$$\Delta_\infty\phi(y_0, b_0) = D^2\phi(y_0, b_0)[(0, 1)] = \phi_{bb}(y_0, b_0).$$

The vertical restriction $b \mapsto u(y_0) + b - \phi(y_0, b)$ has a local maximum at b_0 and is twice differentiable there, so its second derivative is nonpositive: $-\phi_{bb}(y_0, b_0) \leq 0$. Hence $\Delta_\infty\phi(y_0, b_0) \geq 0$.

Lower tests. The same two cases apply with all inequalities reversed: if $U - \phi$ has a local minimum on $\widetilde{\Omega}$ at (y_0, b_0), the vertical restriction again gives $\phi_b(y_0, b_0) = 1$; for $p \neq 0$ the function ψ above is a local lower test for u at y_0 and the supersolution inequality for u, available by `IsInfHarmonicOn`, gives $\Delta_\infty\phi(y_0, b_0) = \Delta_\infty\psi(y_0) \leq 0$; for $p = 0$ the vertical minimum gives $-\phi_{bb}(y_0, b_0) \geq 0$, i.e. $\Delta_\infty\phi(y_0, b_0) = \phi_{bb}(y_0, b_0) \leq 0$.

Thus U satisfies both viscosity inequalities on $\widetilde{\Omega}$ and is infinity-harmonic there. □

2.64. `NormalizedFamilyLipschitz`.

Lemma 2.64 (Uniform bounds for the normalized family). *Fix an integer $d \geq 1$ and put $n = d + 1$. There is a constant $L \geq 0$, depending only on d, such that every $U \in \mathcal{U}$ in the normalized family of `NormalizedInfFamily` satisfies*

$$|U| \leq L \quad \text{on } \overline{B}^n_{7/8}, \qquad |U(a) - U(b)| \leq L|a - b| \quad \big(a, b \in \overline{B}^n_{7/8}\big).$$

Proof. We show that $L = 64$ works.

Step 1: $|u| \leq 1$ on B^d_1. Let $U \in \mathcal{U}$ and let u be as in `NormalizedInfFamily`: u is continuous and infinity-harmonic on B^d_1, $u(0) = 0$, the set $u(B^d_1)$ is bounded above and below, and $\operatorname{osc}_{B^d_1} u \leq 1$. For $y \in B^d_1$ we have $u(y) \leq \sup u(B^d_1)$ and $0 = u(0) \geq \inf u(B^d_1)$, so by the definition of the oscillation in `OscOn`,

$$u(y) = u(y) - u(0) \leq \sup u(B^d_1) - \inf u(B^d_1) = \operatorname{osc}_{B^d_1} u \leq 1.$$

Symmetrically $-u(y) = u(0) - u(y) \leq \operatorname{osc}_{B^d_1} u \leq 1$. Hence $|u| \leq 1$ on B^d_1.

Step 2: the pointwise bound for U. For a space-time point $z = (y, b)$ we have $|y| \leq |z|$ and $|b| \leq |z|$, because $|z|^2 = |y|^2 + b^2$. If $|z| \leq 15/16$ then $|y| \leq 15/16 < 1$, so by Step 1 and $U(y, b) = u(y) + b$,

$$|U(z)| \leq |u(y)| + |b| \leq 1 + \tfrac{15}{16} \leq 2 \qquad \big(z \in \overline{B}^n_{15/16}\big). \tag{256}$$

In particular $|U| \leq 2 \leq 64 = L$ on $\overline{B}^n_{7/8}$.

Step 3: U is infinity-harmonic on the lifted domain. Since u is infinity-harmonic on B^d_1, `LinearLift` shows that $(y, b) \mapsto u(y) + b$ is infinity-harmonic on $B^d_1 \times \mathbb{R}$, the preimage of B^d_1 under the spatial projection. By `NormalizedInfFamily` this function is U, so U is infinity-harmonic on $D := B^d_1 \times \mathbb{R}$.

Step 4: the Lipschitz bound. Apply `InfHarmonicLocLipschitz` with D as in Step 3, $w = U$, $z_0 = 0$, $\rho = 7/8$, $\sigma = 1/16$ and $\Lambda = 64$. Its hypotheses hold: $\rho \geq 0$ and $\sigma > 0$; the set

$\overline{B}^n_{\rho+\sigma} = \overline{B}^n_{15/16}$ is contained in D, since $|z| \le 15/16$ forces $|y| \le 15/16 < 1$; and for $a, b \in \overline{B}^n_{15/16}$ the bound (256) gives

$$|U(a) - U(b)| \le |U(a)| + |U(b)| \le 4 = 64 \cdot \tfrac{1}{16} = \Lambda\sigma.$$

The conclusion of `InfHarmonicLocLipschitz` is

$$|U(a) - U(b)| \le 64\,|a - b| \qquad \big(a, b \in \overline{B}^n_{7/8}\big),$$

as required. The constant 64 does not depend on U or on d. □

2.65. `InfHarmonicMono`.

Lemma 2.65 (Infinity-harmonicity restricts to an open subset)**.** *Let E be a finite-dimensional real inner-product space, let $\Omega' \subseteq \Omega \subseteq E$ with Ω' open, and let $u : E \to \mathbb{R}$ be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`. Then u is infinity-harmonic on Ω' in the sense of `IsInfHarmonicOn`.*

Proof. By `IsInfHarmonicOn` the hypothesis says that u is both an infinity subsolution on Ω in the sense of `IsInfSubsolutionOn` and an infinity supersolution on Ω in the sense of `IsInfSupersolutionOn`, and the conclusion to be proved is the conjunction of the same two properties for Ω'. Unwinding `IsInfSubsolutionOn` and `IsInfSupersolutionOn`, each of those two properties consists of three clauses: that the set be open, that u be continuous on it, and a one-sided inequality at every C^2 test.

The first two clauses. That Ω' is open is a hypothesis. That u is continuous on Ω' follows from its continuity on Ω, which is part of the hypothesis, together with $\Omega' \subseteq \Omega$: the restriction of a function continuous on a set to a subset of that set is continuous on the subset.

The key observation about neighbourhoods. Let $z \in \Omega'$ and let $f : E \to \mathbb{R}$. Since Ω' is open and $z \in \Omega'$, the set Ω' is a neighbourhood of z; hence the neighbourhood filter of z relative to Ω' coincides with the full neighbourhood filter of z. Consequently, if an inequality holds for all x in Ω' near z, it holds for all x near z without restriction, and therefore in particular for all x in Ω near z, the neighbourhood filter of z relative to Ω being coarser than the full one. In other words:

$$\begin{aligned} &\text{if } f(x) \le f(z) \text{ for all } x \in \Omega' \text{ near } z, \text{ then } f(x) \le f(z) \text{ for all } x \in \Omega \text{ near } z;\\ &\text{if } f(z) \le f(x) \text{ for all } x \in \Omega' \text{ near } z, \text{ then } f(z) \le f(x) \text{ for all } x \in \Omega \text{ near } z. \end{aligned} \tag{257}$$

The subsolution inequality. Let $z \in \Omega'$ and let φ be C^2 at z such that $u - \varphi$ has a local maximum on Ω' at z, i.e. $u(x) - \varphi(x) \le u(z) - \varphi(z)$ for all $x \in \Omega'$ near z. By the first half of (257) applied to $f = u - \varphi$, the same inequality holds for all $x \in \Omega$ near z, i.e. $u - \varphi$ has a local maximum on Ω at z. Also $z \in \Omega$ because $\Omega' \subseteq \Omega$. Since u is an infinity subsolution on Ω, `IsInfSubsolutionOn` gives $\Delta_\infty\varphi(z) \ge 0$. As z and φ were arbitrary, u is an infinity subsolution on Ω' in the sense of `IsInfSubsolutionOn`.

The supersolution inequality. Symmetrically, let $z \in \Omega'$ and let φ be C^2 at z such that $u - \varphi$ has a local minimum on Ω' at z. By the second half of (257) applied to $f = u - \varphi$, the function $u - \varphi$ has a local minimum on Ω at z, and $z \in \Omega$. Since u is an infinity supersolution on Ω, `IsInfSupersolutionOn` gives $\Delta_\infty\varphi(z) \le 0$. Hence u is an infinity supersolution on Ω' in the sense of `IsInfSupersolutionOn`.

The two displayed properties together are exactly infinity-harmonicity of u on Ω' in the sense of `IsInfHarmonicOn`. □

2.66. `NormalizedFamilyInfHarmonic`.

Lemma 2.66 (Members of the normalized family are infinity-harmonic on the unit ball)**.** *Fix an integer $d \ge 0$, write $\mathbb{R}^d$ for the Euclidean space of dimension d and identify space-time with $\ell^2(\mathbb{R}^d \times \mathbb{R})$, and let B_1 denote the open unit ball about the origin of space-time. Let $U : \ell^2(\mathbb{R}^d \times \mathbb{R}) \to \mathbb{R}$ belong to the normalized lifted family in the sense of `NormalizedInfFamily`. Then U is infinity-harmonic on B_1 in the sense of `IsInfHarmonicOn`.*

Proof. By `NormalizedInfFamily` there is a function $u : \mathbb{R}^d \to \mathbb{R}$, among whose listed properties are that u is infinity-harmonic on the open unit ball B_1^d of $\mathbb{R}^d$ in the sense of `IsInfHarmonicOn` and that

$$U(z) = u(z_x) + z_t \qquad \big(z = (z_x, z_t) \in \ell^2(\mathbb{R}^d \times \mathbb{R})\big). \tag{258}$$

(The remaining properties listed there — $u(0) = 0$, the boundedness of the image of B_1^d above and below, and the bound on the oscillation — are not used here.)

The lift is infinity-harmonic on the slab. Applying `LinearLift` to the open set B_1^d and the function u, the function $z \mapsto u(z_x) + z_t$ is infinity-harmonic, in the sense of `IsInfHarmonicOn`, on the preimage of B_1^d under the projection $z \mapsto z_x$, that is, on the slab

$$S = \big\{z \in \ell^2(\mathbb{R}^d \times \mathbb{R}) : |z_x| < 1\big\}.$$

By (258) that function is exactly U, so U is infinity-harmonic on S.

The unit ball of space-time lies in the slab. The ℓ^2 norm of space-time satisfies $|z|^2 = |z_x|^2 + |z_t|^2$, so $|z_x|^2 \le |z|^2$ and hence $|z_x| \le |z|$, both being nonnegative. Therefore $|z| < 1$ implies $|z_x| < 1$, i.e.

$$B_1 \subseteq S.$$

Restriction. The ball B_1 is open, it is contained in S, and U is infinity-harmonic on S; so `InfHarmonicMono` applies and gives that U is infinity-harmonic on B_1 in the sense of `IsInfHarmonicOn`. □

2.67. `EquiLipschitzUniformLimit`.

Lemma 2.67 (Uniform limit of an equi-bounded, equi-Lipschitz sequence on a ball)**.** *Let $\mathcal{H}$ be a finite-dimensional real inner-product space, let $a \in \mathcal{H}$, let $\rho > 0$, and put $K = \overline{B}(a, \rho)$ for the closed ball of radius ρ about a. Let $A, \Lambda \in \mathbb{R}$ and let $v_i : \mathcal{H} \to \mathbb{R}$, $i \in \mathbb{N}$, satisfy*

$$|v_i(w)| \le A \qquad (w \in K,\ i \in \mathbb{N})$$

and

$$|v_i(w) - v_i(w')| \le \Lambda |w - w'| \qquad (w, w' \in K,\ i \in \mathbb{N}),$$

the two constants being independent of i. Then there are a strictly increasing $\iota : \mathbb{N} \to \mathbb{N}$ and a function $V : \mathcal{H} \to \mathbb{R}$ such that $v_{\iota(k)} \to V$ uniformly on K as $k \to \infty$. (Nothing is claimed about V off K.)

Proof. Since $\rho > 0$ we have $a \in K$, so K is nonempty; applying the bound at $w = a$ gives $0 \le |v_0(a)| \le A$, so $A \ge 0$.

The Lipschitz constant may be taken nonnegative. Put $M = \max(\Lambda, 0)$, so $M \ge 0$ and $\Lambda \le M$. Since $|w - w'| \ge 0$, the hypothesis gives

$$|v_i(w) - v_i(w')| \le \Lambda |w - w'| \le M |w - w'| \qquad (w, w' \in K,\ i \in \mathbb{N}). \tag{259}$$

A countable dense subset of K. A finite-dimensional real normed space is a separable metric space, and a subspace of a separable metric space is separable; K is nonempty, so it carries a sequence $(Q_k)_{k \in \mathbb{N}}$ of points of K whose range is dense in K. Density means precisely:

$$\text{for every } x \in K \text{ and every } \delta > 0 \text{ there is } k \text{ with } |x - Q_k| < \delta. \tag{260}$$

A subsequence converging at every Q_k. For every i and k we have $Q_k \in K$, hence $|v_i(Q_k)| \le A$, so the real sequence $\big(v_i(Q_k)\big)_i$ lies in $[-A, A]$ for each fixed k. Applying Bolzano–Weierstrass (preamble) to $k = 1$ gives a subsequence along which the values at Q_1 converge; applying it again inside that subsequence for $k = 2$, and continuing, produces nested subsequences, the k-th of which converges at $Q_1, \dots, Q_k$. Taking the diagonal — the j-th term of the j-th nested subsequence — yields a single strictly increasing $\varphi : \mathbb{N} \to \mathbb{N}$ and reals c_k with

$$v_{\varphi(j)}(Q_k) \longrightarrow c_k \qquad (j \to \infty) \tag{261}$$

for every k: for each fixed k the diagonal sequence is, from the index k onwards, a subsequence of the k-th nested subsequence, which converges at Q_k.

The subsequence is uniformly Cauchy on K. Let $\varepsilon > 0$ and put

$$\delta = \frac{\varepsilon}{4(M+1)} > 0,$$

so that

$$M\delta = \frac{\varepsilon}{4} \cdot \frac{M}{M+1} \le \frac{\varepsilon}{4}, \tag{262}$$

using $M \ge 0$. By (260) the balls $B(Q_k, \delta)$, $k \in \mathbb{N}$, cover K; the set K is closed and bounded in a finite-dimensional space, hence compact (preamble), so finitely many of them cover K, say those indexed by a finite set $t \subseteq \mathbb{N}$. By (261) there is N such that

$$\big|v_{\varphi(j)}(Q_k) - c_k\big| < \frac{\varepsilon}{4} \qquad (j \ge N,\ k \in t), \tag{263}$$

because each of the finitely many convergences in (261) eventually puts its term within $\varepsilon/4$ of its limit, and the maximum of the finitely many thresholds works for all of them.

Now let $m, n \ge N$ and $x \in K$. Choose $k \in t$ with $x \in B(Q_k, \delta)$, so $|x - Q_k| < \delta$. Writing

$$v_{\varphi(m)}(x) - v_{\varphi(n)}(x) = \big(v_{\varphi(m)}(x) - v_{\varphi(m)}(Q_k)\big) + \big(v_{\varphi(m)}(Q_k) - c_k\big) - \big(v_{\varphi(n)}(Q_k) - c_k\big) - \big(v_{\varphi(n)}(x) - v_{\varphi(n)}(Q_k)\big),$$

and bounding the first and fourth brackets by $M\delta$ using (259) with $w = x$, $w' = Q_k$ (both in K), and the second and third strictly by $\varepsilon/4$ using (263), we get

$$\big|v_{\varphi(m)}(x) - v_{\varphi(n)}(x)\big| < M\delta + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + M\delta \le \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \varepsilon,$$

by (262). As $\varepsilon > 0$ was arbitrary, the sequence $\big(v_{\varphi(j)}\big)_j$ is uniformly Cauchy on K.

The limit. Fix $z \in K$. Taking $x = z$ in the previous paragraph shows that $\big(v_{\varphi(j)}(z)\big)_j$ is a Cauchy sequence of reals, hence convergent; call its limit $V(z)$. For $z \notin K$ set $V(z) = 0$. A sequence that is uniformly Cauchy on a set and converges pointwise on that set converges uniformly on it to the pointwise limit: given $\varepsilon > 0$, choose N as above for $\varepsilon/2$; for $j \ge N$ and $x \in K$, letting $n \to \infty$ in $|v_{\varphi(j)}(x) - v_{\varphi(n)}(x)| < \varepsilon/2$ gives $|v_{\varphi(j)}(x) - V(x)| \le \varepsilon/2 < \varepsilon$. Hence $v_{\varphi(k)} \to V$ uniformly on K, and $\iota = \varphi$ is strictly increasing, which is the assertion. □

2.68. `MidpointLinearOnBall.`

Lemma 2.68 (The exact midpoint identity on a ball forces linearity)**.** *Let $\mathcal{H}$ be a finite-dimensional real inner-product space, let $R > 0$ and write $\overline{B}_R$ for the closed ball of radius R about the origin of $\mathcal{H}$. Let $W : \mathcal{H} \to \mathbb{R}$ be continuous on $\overline{B}_R$, satisfy $W(0) = 0$, and satisfy the exact midpoint identity*

$$W(A) + W(B) = 2W\big(\tfrac{1}{2}(A+B)\big) \qquad \big(A, B \in \overline{B}_R\big).$$

Then there is $P \in \mathcal{H}$ with

$$W(A) = \langle P, A\rangle \qquad \big(A \in \overline{B}_R\big).$$

Proof. Put $c = R/2 > 0$, so $c \le R$.

Step 1: halving. Taking $B = 0$ in the midpoint identity and using $W(0) = 0$,

$$W(A) = 2W\big(\tfrac{1}{2}A\big) \qquad \big(A \in \overline{B}_R\big). \tag{264}$$

Step 2: additivity on the half-ball. Let $A, B \in \overline{B}_c$. Then $|A + B| \le |A| + |B| \le 2c = R$, so $A + B \in \overline{B}_R$, and $A, B \in \overline{B}_R$ because $c \le R$. The midpoint identity gives $W(A) + W(B) = 2W(\frac{1}{2}(A+B))$, while (264) applied to $A + B \in \overline{B}_R$ gives $W(A+B) = 2W(\frac{1}{2}(A+B))$. Hence

$$W(A+B) = W(A) + W(B) \qquad \big(A, B \in \overline{B}_c\big). \tag{265}$$

Step 3: halving iterated. For every integer $n \ge 0$ and every $A \in \overline{B}_R$,

$$W\big(2^{-n}A\big) = 2^{-n}W(A). \tag{266}$$

This is an induction on n. For $n = 0$ there is nothing to prove. Assume it for n and let $A \in \overline{B}_R$. Then $|\frac{1}{2}A| = \frac{1}{2}|A| \le R$, so $\frac{1}{2}A \in \overline{B}_R$ and the inductive hypothesis applies to it: $W(2^{-n}(\frac{1}{2}A)) = 2^{-n}W(\frac{1}{2}A)$. Since $2^{-(n+1)}A = 2^{-n}(\frac{1}{2}A)$, and $W(\frac{1}{2}A) = \frac{1}{2}W(A)$ by (264), we get $W(2^{-(n+1)}A) = 2^{-n} \cdot \frac{1}{2}W(A) = 2^{-(n+1)}W(A)$.

Step 4: admissible exponents. For $t > 0$ and $A \in \mathcal{H}$ put $n_t(A) = \lceil |A|/t \rceil$. Then $|A|/t \le n_t(A) \le 2^{n_t(A)}$, the second inequality because $m \le 2^m$ for every integer $m \ge 0$; multiplying by $t > 0$,

$$|A| \le 2^{n_t(A)} t. \tag{267}$$

Moreover if $|A| \le 2^n t$ and $n \le m$ then $|A| \le 2^m t$, because $2^n \le 2^m$ and $t \ge 0$. Finally, if $|A| \le 2^n t$ then $|2^{-n}A| = 2^{-n}|A| \le t$.

Step 5: the value $2^n W(2^{-n}A)$ does not depend on n. First, for integers $n, k \ge 0$ and A with $|A| \le 2^n c$, put $A' = 2^{-n}A$; by Step 4, $|A'| \le c \le R$, so $A' \in \overline{B}_R$ and (266) applies to it:

$$2^{n+k} W\big(2^{-(n+k)}A\big) = 2^{n+k} W\big(2^{-k}A'\big) = 2^{n+k} \cdot 2^{-k} W(A') = 2^n W\big(2^{-n}A\big),$$

using $2^{-(n+k)}A = 2^{-k}(2^{-n}A)$. Given now $n, m \ge 0$ with $|A| \le 2^n c$ and $|A| \le 2^m c$, one of $n \le m$ and $m \le n$ holds; writing the larger as the smaller plus k and applying the previous display gives

$$2^n W\big(2^{-n}A\big) = 2^m W\big(2^{-m}A\big). \tag{268}$$

Step 6: the extension. Define $\Lambda : \mathcal{H} \to \mathbb{R}$ by

$$\Lambda(A) = 2^{n_c(A)} W\big(2^{-n_c(A)}A\big),$$

which makes sense by (267). By (268),

$$\Lambda(A) = 2^n W\big(2^{-n}A\big) \qquad \text{whenever } |A| \le 2^n c. \tag{269}$$

Taking $n = 0$ in (269),

$$\Lambda(A) = W(A) \qquad \big(A \in \overline{B}_c\big), \tag{270}$$

and in particular $\Lambda(0) = W(0) = 0$.

Λ *is additive.* Let $A, B \in \mathcal{H}$. By Step 4 applied with $t = c/2$ there are exponents admissible for A and for B; taking n to be the larger of the two, monotonicity gives $|A| \le 2^n(c/2)$ and $|B| \le 2^n(c/2)$. Then $|A| \le 2^n c$, $|B| \le 2^n c$ and $|A + B| \le |A| + |B| \le 2^n(c/2) + 2^n(c/2) = 2^n c$, so (269) applies to all three with the same n. Also $|2^{-n}A| \le c/2 \le c$ and $|2^{-n}B| \le c/2 \le c$ by Step 4, so (265) applies to the pair $2^{-n}A, 2^{-n}B$, whose sum is $2^{-n}(A + B)$. Therefore

$$\Lambda(A + B) = 2^n W\big(2^{-n}(A + B)\big) = 2^n \big(W(2^{-n}A) + W(2^{-n}B)\big) = \Lambda(A) + \Lambda(B).$$

Λ *is continuous.* Fix $A \in \mathcal{H}$. By (267) there is n with $|A| \le 2^n c$, and then $|A| < 2^{n+1} c$ because $2^n c > 0$; rename $n + 1$ as n, so $|A| < 2^n c$. The set $S = \{x : |x| < 2^n c\}$ is open, being the preimage of an open half-line under the continuous norm, and it contains A. On S we have $\Lambda(x) = 2^n W(2^{-n}x)$ by (269); the map $x \mapsto 2^{-n}x$ is continuous and sends S into $\overline{B}_c \subseteq \overline{B}_R$ by Step 4, and W is continuous on $\overline{B}_R$, so $x \mapsto 2^n W(2^{-n}x)$ is continuous on S. Hence Λ is continuous on the neighbourhood S of A, so continuous at A. As A was arbitrary, Λ is continuous on $\mathcal{H}$.

Step 7: Λ *is linear, and the representing vector.* An additive map between real normed spaces that is continuous is $\mathbb{R}$-linear: additivity gives $\Lambda(qx) = q\Lambda(x)$ for every rational q, and continuity extends this to every real scalar by density of $\mathbb{Q}$ in $\mathbb{R}$. So Λ is a continuous linear functional on $\mathcal{H}$. By the Riesz representation of a continuous linear functional on a finite-dimensional inner-product space there is $P \in \mathcal{H}$ with

$$\Lambda(x) = \langle P, x \rangle \qquad (x \in \mathcal{H}). \tag{271}$$

Step 8: conclusion on the full ball. Let $A \in \overline{B}_R$. Then $|\tfrac{1}{2}A| = \tfrac{1}{2}|A| \le R/2 = c$, so $\tfrac{1}{2}A \in \overline{B}_c$ and (270) gives $W(\tfrac{1}{2}A) = \Lambda(\tfrac{1}{2}A)$. Using (264), then (271) together with the linearity of Λ and of the inner product in its second argument,

$$W(A) = 2W\big(\tfrac{1}{2}A\big) = 2\Lambda\big(\tfrac{1}{2}A\big) = 2 \cdot \tfrac{1}{2}\Lambda(A) = \Lambda(A) = \langle P, A \rangle.$$

This is the assertion. □

2.69. `SlowScalesAndBlowupLimit`.

Lemma 2.69 (The slow scales, the blow-up slope at z_*, and the small-scale limit)**.** *Let $d \in \mathbb{N}$ and work in $X = \ell^2(\mathbb{R}^d \times \mathbb{R})$. Let $e_b \in X$ with $e_b = (0,1)$, let $L \in \mathbb{R}$ with $0 \le L$ and assume*

$$|U(z)| \le L \quad \text{and} \quad |U(a) - U(b)| \le L|a-b| \qquad \big(a,b,z \in \overline{B}_{7/8}\big) \tag{272}$$

for every $U : X \to \mathbb{R}$ which is a member of a normalized inf-family in the sense of `NormalizedInfFamily` for this d. Let $U_i : X \to \mathbb{R}$, $z_i \in X$ and $r_i \in \mathbb{R}$ be given for $i \in \mathbb{N}$, and assume:

(1) *each U_i is a member of a normalized inf-family in the sense of `NormalizedInfFamily`;*
(2) *$z_i \in B_{3/4}$ for every i;*
(3) *$0 < r_i$ for every i, and $r_i \to 0$.*

Let $\iota : \mathbb{N} \to \mathbb{N}$ be strictly increasing, let $z_* \in X$ and let $U_\circ : X \to \mathbb{R}$, and assume:

(4) *$|z_*| \le 3/4$ and $z_{\iota(k)} \to z_*$;*
(5) *$U_{\iota(k)} \to U_\circ$ uniformly on $\overline{B}_{7/8}$;*
(6) *$U_\circ$ is infinity-harmonic on $B_{7/8}$;*
(7) *$z_* + RX \in \overline{B}_{7/8}$ whenever $0 < R \le 1/16$ and $|X| \le 1$.*

Then there are a strictly increasing $j : \mathbb{N} \to \mathbb{N}$, radii $R_k \in \mathbb{R}$, a function $W : X \to \mathbb{R}$ and a vector $P_0 \in X$ such that

(8) *$0 < R_k$ and $R_k \le 1/16$ for every k;*
(9) $\dfrac{|z_{\iota(j(k))} - z_*| + r_{\iota(j(k))}}{R_k} \longrightarrow 0$;
(10) *for every $\epsilon' > 0$ there is k_0 such that for all $k \ge k_0$ and all $X \in B_1$,*
$$\left| \frac{U_{\iota(j(k))}(z_* + R_k X) - U_{\iota(j(k))}(z_*)}{R_k} - \langle P_0, X \rangle \right| \le \epsilon';$$
(11) *the rescalings $H \mapsto \dfrac{U_{\iota(j(k))}\big(z_{\iota(j(k))} + r_{\iota(j(k))}H\big) - U_{\iota(j(k))}\big(z_{\iota(j(k))}\big)}{r_{\iota(j(k))}}$ converge to W uniformly on $\overline{B}_2$;*
(12) *W is continuous on $\overline{B}_2$;*
(13) *$W(0) = 0$ and $W(e_b) = 1$.*

Proof. The reference scales. Put $R_n^0 = \dfrac{1}{16} \cdot \dfrac{1}{n+1}$. Then $0 < R_n^0$ and $R_n^0 \le 1/16$, because $1/(n+1) \le 1$; and $R_n^0 \to 0$, being $1/16$ times the null sequence $1/(n+1)$.

Step 4: a simultaneous extraction. Three sequences tend to zero along ι. First $r_{\iota(k)} \to 0$, since $r_i \to 0$ by 3 and ι is strictly increasing, hence tends to infinity. Second $|z_{\iota(k)} - z_*| \to 0$ by 4 and continuity of the norm. Third $\sup_{\overline{B}_{7/8}} |U_{\iota(k)} - U_\circ| \to 0$ by 5. Consequently, for each fixed n the three conditions

$$\sup_{w \in \overline{B}_{7/8}} \big|U_{\iota(k)}(w) - U_\circ(w)\big| \le \frac{R_n^0}{n+1}, \qquad |z_{\iota(k)} - z_*| + r_{\iota(k)} \le \frac{R_n^0}{n+1}, \qquad r_{\iota(k)} \le \frac{1}{16} \tag{273}$$

hold for all sufficiently large k: the three thresholds $R_n^0/(n+1) > 0$, $R_n^0/(n+1) > 0$ and $1/16 > 0$ are positive, so each condition holds eventually, and a finite intersection of eventual conditions is eventual. A sequence of eventually-true conditions admits a simultaneous extraction: there is a strictly increasing $j_0 : \mathbb{N} \to \mathbb{N}$ such that (273) holds at $k = j_0(n)$ for every n. Fix such a j_0.

Step 5: the blow-up slope of $U_\circ$ at z_*. Since $|z_*| \le 3/4 < 7/8$ we have $z_* \in B_{7/8}$, so 6 lets us apply `AffineBlowups` to $U_\circ$ on $D = B_{7/8}$ at z_*. Its second assertion, applied to the positive null sequence (R_n^0), gives a strictly increasing ι_5 and a function V such that

$$X \mapsto \frac{U_\circ\big(z_* + R^0_{\iota_5(i)} X\big) - U_\circ(z_*)}{R^0_{\iota_5(i)}} \longrightarrow V$$

uniformly on every compact subset of X. Its first assertion, applied to the same subsequence of scales and the same V, gives a continuous linear functional Λ with $V = \Lambda$ everywhere. Since X

is a finite-dimensional real inner-product space, the Riesz representation supplies $P_0 \in X$ with $\langle P_0, \cdot\rangle = \Lambda$. Fix this P_0.

Step 7: the small-scale rescalings. For $n \in \mathbb{N}$ put

$$W_n(H) = \frac{U_{\iota(j_0(n))}\big(z_{\iota(j_0(n))} + r_{\iota(j_0(n))}H\big) - U_{\iota(j_0(n))}\big(z_{\iota(j_0(n))}\big)}{r_{\iota(j_0(n))}}.$$

Write $z = z_{\iota(j_0(n))}$ and $r = r_{\iota(j_0(n))}$, so $0 < r \le 1/16$ by 3 and the third condition of (273), and $|z| < 3/4$ by 2. For $|H| \le 2$,

$$|z + rH| \le |z| + r|H| \le \tfrac{3}{4} + 2r \le \tfrac{3}{4} + \tfrac{1}{8} = \tfrac{7}{8},$$

so both $z + rH$ and z lie in $\overline{B}_{7/8}$. Hence (272), available for $U_{\iota(j_0(n))}$ by 1, gives for $H, H' \in \overline{B}_2$

$$|W_n(H)| = \frac{|U(z+rH) - U(z)|}{r} \le \frac{L\,r|H|}{r} \le 2L, \qquad |W_n(H) - W_n(H')| = \frac{|U(z+rH) - U(z+rH')|}{r} \le \frac{L\,r|H}{r}$$

where the two displayed norms are computed from $\|rH\| = r|H|$ and $\|r(H - H')\| = r|H - H'|$, r being positive. So the sequence $\big(W_{\iota_5(k)}\big)_k$ is uniformly bounded by $2L$ and uniformly L-Lipschitz on $\overline{B}_2$, and `EquiLipschitzUniformLimit` with centre 0, radius 2, bound $2L$ and Lipschitz constant L gives a strictly increasing ι_7 and a function W with

$$W_{\iota_5(\iota_7(k))} \longrightarrow W \quad \text{uniformly on } \overline{B}_2. \tag{274}$$

Assembly. Set $j = j_0 \circ \iota_5 \circ \iota_7$ and $R_k = R^0_{\iota_5(\iota_7(k))}$, and keep W and P_0. Then j is strictly increasing, being a composition of strictly increasing maps, and 8 holds because $0 < R^0_n \le 1/16$ for every n. Assertion 11 is (274) written out, since $W_{\iota_5(\iota_7(k))}$ is exactly the rescaling at the index $j(k) = j_0(\iota_5(\iota_7(k)))$.

9. Writing n for an arbitrary index, the second condition of (273) at $k = j_0(n)$ gives

$$0 \le \frac{|z_{\iota(j_0(n))} - z_*| + r_{\iota(j_0(n))}}{R^0_n} \le \frac{R^0_n/(n+1)}{R^0_n} = \frac{1}{n+1},$$

so that quotient tends to zero by comparison with $1/(n+1)$; composing with the strictly increasing $\iota_5 \circ \iota_7$ preserves the limit, which is 9.

10. Let $\epsilon' > 0$. The uniform convergence of Step 5 on the compact set $\overline{B}_1$ gives i_0 with

$$\left|\frac{U_\circ\big(z_* + R^0_{\iota_5(i)}X\big) - U_\circ(z_*)}{R^0_{\iota_5(i)}} - \Lambda X\right| \le \frac{\epsilon'}{2} \qquad \big(i \ge i_0,\ |X| \le 1\big), \tag{275}$$

and choosing a natural number $k_1 > 4/\epsilon'$ gives $2/(n+1) \le \epsilon'/2$ for every $n \ge k_1$. Put $k_0 = \max(i_0, k_1)$ and let $k \ge k_0$ and $X \in B_1$, so $|X| \le 1$. Write $n = \iota_5(\iota_7(k))$; since ι_5 and ι_7 are strictly increasing we have $k \le \iota_7(k) \le n$, hence $n \ge k_1$ and $\iota_7(k) \ge i_0$. By 7 with $R = R^0_n \le 1/16$ the point $z_* + R^0_n X$ lies in $\overline{B}_{7/8}$, and so does z_* because $|z_*| \le 3/4$. The first condition of (273) at $k = j_0(n)$ therefore applies at both points and gives

$$\left|\frac{U_{\iota(j_0(n))}(z_* + R^0_n X) - U_{\iota(j_0(n))}(z_*)}{R^0_n} - \frac{U_\circ(z_* + R^0_n X) - U_\circ(z_*)}{R^0_n}\right| \le \frac{2\,R^0_n/(n+1)}{R^0_n} = \frac{2}{n+1} \le \frac{\epsilon'}{2},$$

the numerator being bounded by the sum of the two pointwise gaps. Combining this with (275) at $i = \iota_7(k)$, where $R^0_{\iota_5(\iota_7(k))} = R^0_n = R_k$, and with $\Lambda X = \langle P_0, X\rangle$ from Step 5, the triangle inequality gives

$$\left|\frac{U_{\iota(j(k))}(z_* + R_k X) - U_{\iota(j(k))}(z_*)}{R_k} - \langle P_0, X\rangle\right| \le \frac{\epsilon'}{2} + \frac{\epsilon'}{2} = \epsilon',$$

which is 10.

12. Each $W_{\iota_5(\iota_7(k))}$ is L-Lipschitz on $\overline{B}_2$ by the estimate of Step 7, and a uniform limit of L-Lipschitz functions on a set is L-Lipschitz on that set; an L-Lipschitz function is continuous there.

13. Both normalizations are read off pointwise from (274) and the uniqueness of limits. At $H = 0$ we have $W_n(0) = \big(U(z + r \cdot 0) - U(z)\big)/r = 0$ for every n, so the constant sequence 0

converges to $W(0)$ and $W(0)=0$. At $H=e_b$, note $|e_b|=1\le 2$, so $e_b\in\overline{B}_2$ and the evaluation is legitimate; and by `NormalizedInfFamily` and 1 there is $u:\mathbb{R}^d\to\mathbb{R}$ with $U_{\iota(j_0(n))}(z)=u(z_x)+z_t$ for every z, where $z=(z_x,z_t)$. Since $e_b=(0,1)$, adding $r\,e_b$ leaves the space coordinate unchanged and increases the time coordinate by r, so

$$W_n(e_b)=\frac{\big(u(z_x)+z_t+r\big)-\big(u(z_x)+z_t\big)}{r}=\frac{r}{r}=1$$

for every n, using $r>0$. Hence the constant sequence 1 converges to $W(e_b)$ and $W(e_b)=1$. □

2.70. `BlowupLimitVanishingMidpointDefect.`

Lemma 2.70 (The small-scale limit has vanishing midpoint defect)**.** *Let $d\in\mathbb{N}$ and work in $X=\ell^2(\mathbb{R}^d\times\mathbb{R})$, writing $\overline{B}_\rho$ and B_ρ for the closed and open balls of radius ρ centred at the origin. Let $U_i:X\to\mathbb{R}$, $z_i\in X$ and $r_i\in\mathbb{R}$ be given for $i\in\mathbb{N}$, let $R_k\in\mathbb{R}$ for $k\in\mathbb{N}$, let $z_*\in X$, let $W:X\to\mathbb{R}$ and let $\kappa:\mathbb{N}\to\mathbb{N}$. Assume:*

(1) *$0<r_i$ for every i and $0<R_k$ for every k;*

(2) $\dfrac{|z_{\kappa(k)}-z_*|+r_{\kappa(k)}}{R_k}\longrightarrow 0$ *as $k\to\infty$;*

(3) *the rescalings*

$$W_k(H)=\frac{U_{\kappa(k)}\big(z_{\kappa(k)}+r_{\kappa(k)}H\big)-U_{\kappa(k)}\big(z_{\kappa(k)}\big)}{r_{\kappa(k)}}$$

converge to W uniformly on $\overline{B}_2$;

(4) *for every $\epsilon'>0$ there is k_0 such that for all $k\ge k_0$ and all $X',Y'\in B_{1/4}$ with $X'\neq Y'$,*

$$\Big|\widehat{U}_k(X')+\widehat{U}_k(Y')-2\widehat{U}_k\big(\tfrac12(X'+Y')\big)\Big|\le\epsilon'\,|X'-Y'|,\qquad \widehat{U}_k(V)=\frac{U_{\kappa(k)}(z_*+R_kV)-U_{\kappa(k)}(z_*)}{R_k}.$$

Then

$$W(H)+W(K)=2\,W\Big(\frac{H+K}{2}\Big)\qquad\big(H,K\in\overline{B}_2\big).\tag{276}$$

Proof. Fix $H,K\in\overline{B}_2$, so $|H|\le 2$ and $|K|\le 2$. The midpoint also lies in $\overline{B}_2$: $|\frac12(H+K)|=\frac12|H+K|\le\frac12(|H|+|K|)\le 2$ by the triangle inequality.

The degenerate case. If $H=K$ then $\frac12(H+H)=H$ and (276) reads $W(H)+W(H)=2W(H)$, which holds. So assume $H\neq K$, whence $|H-K|>0$.

The quantitative claim. We show

$$\Big|W(H)+W(K)-2W\Big(\frac{H+K}{2}\Big)\Big|\le\epsilon'\qquad\text{for every }\epsilon'>0.\tag{277}$$

Fix $\epsilon'>0$ and put $\delta=\dfrac{\epsilon'}{2|H-K|}$, which is positive and satisfies $\delta\,|H-K|=\epsilon'/2$. Apply 4 at this δ and let k_0 be the resulting index.

The substituted points. For $k\in\mathbb{N}$ set

$$X_k=\frac{z_{\kappa(k)}-z_*+r_{\kappa(k)}H}{R_k},\qquad Y_k=\frac{z_{\kappa(k)}-z_*+r_{\kappa(k)}K}{R_k},$$

legitimate because $R_k\neq 0$ by 1. Clearing the denominator in each case gives the three identities

$$z_*+R_kX_k=z_{\kappa(k)}+r_{\kappa(k)}H,\qquad z_*+R_kY_k=z_{\kappa(k)}+r_{\kappa(k)}K,\tag{278}$$

$$z_*+R_k\cdot\tfrac12(X_k+Y_k)=z_{\kappa(k)}+r_{\kappa(k)}\cdot\tfrac12(H+K),$$

the last because $\frac12(X_k+Y_k)=\big(z_{\kappa(k)}-z_*+r_{\kappa(k)}\frac12(H+K)\big)/R_k$; and

$$X_k-Y_k=\frac{r_{\kappa(k)}}{R_k}\,(H-K).\tag{279}$$

So the images of X_k, Y_k and their midpoint under $V\mapsto z_*+R_kV$ are exactly the three points at which W_k is evaluated in (276).

The substituted points are in $B_{1/4}$ for large k. Since $R_k > 0$ and $r_{\kappa(k)} > 0$,

$$|X_k| = \frac{\left|z_{\kappa(k)} - z_* + r_{\kappa(k)}H\right|}{R_k} \le \frac{|z_{\kappa(k)} - z_*| + r_{\kappa(k)}|H|}{R_k} \le \frac{|z_{\kappa(k)} - z_*| + 2r_{\kappa(k)}}{R_k} \le 2\cdot\frac{|z_{\kappa(k)} - z_*| + r_{\kappa(k)}}{R_k},$$

using $|H| \le 2$ and $|z_{\kappa(k)} - z_*| \ge 0$; the same bound holds for Y_k with K in place of H.

One index at which everything holds. By 2 the quantity $(|z_{\kappa(k)} - z_*| + r_{\kappa(k)})/R_k$ is eventually below $1/8$, and by 3 the uniform distance $\sup_{\overline{B}_2} |W - W_k|$ is eventually below $\epsilon'/8$; together with $k \ge k_0$ these are three eventual conditions, so some k satisfies all three:

$$k \ge k_0, \qquad \frac{|z_{\kappa(k)} - z_*| + r_{\kappa(k)}}{R_k} < \frac{1}{8}, \qquad |W(w) - W_k(w)| < \frac{\epsilon'}{8} \quad \big(w \in \overline{B}_2\big). \tag{280}$$

Fix such a k. The second condition and the previous paragraph give $|X_k| \le 2 \cdot \frac{1}{8} = \frac{1}{4}$, in fact $|X_k| < 1/4$, and likewise $|Y_k| < 1/4$, so $X_k, Y_k \in B_{1/4}$. Also $X_k \ne Y_k$: otherwise (279) would give $\frac{r_{\kappa(k)}}{R_k}(H - K) = 0$, and since $r_{\kappa(k)}/R_k > 0$ this forces $H = K$, excluded above. Finally, by (279),

$$|X_k - Y_k| = \frac{r_{\kappa(k)}}{R_k}\,|H - K|. \tag{281}$$

Transferring the three-point estimate. Hypothesis 4 at this k, applied to X_k, Y_k, together with (278) and (281), gives

$$\left|\frac{U_{\kappa(k)}(z_{\kappa(k)} + r_{\kappa(k)}H) - U_{\kappa(k)}(z_*)}{R_k} + \frac{U_{\kappa(k)}(z_{\kappa(k)} + r_{\kappa(k)}K) - U_{\kappa(k)}(z_*)}{R_k} - 2\,\frac{U_{\kappa(k)}\big(z_{\kappa(k)} + r_{\kappa(k)}\frac{1}{2}(H + K)\big) - U_{\kappa(k)}(}{R_k}\right.$$

The three coefficients $1, 1, -2$ sum to zero, so the constants $U_{\kappa(k)}(z_*)$ cancel and the left-hand expression equals

$$\frac{r_{\kappa(k)}}{R_k}\Big(W_k(H) + W_k(K) - 2W_k\big(\tfrac{1}{2}(H + K)\big)\Big),$$

as one sees by multiplying out: each term carries the factor $1/R_k = (r_{\kappa(k)}/R_k)\cdot(1/r_{\kappa(k)})$ and W_k is the same difference divided by $r_{\kappa(k)}$. Dividing the resulting inequality by the positive number $r_{\kappa(k)}/R_k$,

$$\Big|W_k(H) + W_k(K) - 2W_k\big(\tfrac{1}{2}(H + K)\big)\Big| \le \delta\,|H - K| = \frac{\epsilon'}{2}. \tag{282}$$

Passing to the limit. The three points H, K and $\frac{1}{2}(H+K)$ all lie in $\overline{B}_2$, so the third condition of (280) bounds $|W - W_k|$ by $\epsilon'/8$ at each of them. Writing $D = W(H) + W(K) - 2W(\frac{1}{2}(H+K))$ and $D_k = W_k(H) + W_k(K) - 2W_k(\frac{1}{2}(H + K))$, the difference $D - D_k$ is a combination of those three errors with coefficients $1, 1, -2$, so $|D - D_k| \le (1 + 1 + 2) \cdot \epsilon'/8 = \epsilon'/2$. With (282) this gives $|D| \le |D_k| + |D - D_k| \le \epsilon'/2 + \epsilon'/2 = \epsilon'$, which is (277).

Conclusion. If $|D| > 0$ then (277) at $\epsilon' = |D|/2 > 0$ gives $|D| \le |D|/2$, hence $|D| \le 0$, a contradiction. So $|D| \le 0$, hence $D = 0$, which is (276). □

2.71. `FamilyFlatness`.

Lemma 2.71 (Uniform qualitative flatness)**.** *Fix an integer $d \ge 1$, put $n = d + 1$ and identify $\mathbb{R}^n$ with the space-time $\ell^2(\mathbb{R}^d \times \mathbb{R})$. Let b_d be the family flatness modulus of* `FlatnessModulus`*. Then $b_d(s) \to 0$ as $s \downarrow 0$: for every $\varepsilon > 0$ there is s_0 with $0 < s_0 \le 1/8$ and*

$$b_d(s) \le \varepsilon \qquad (0 < s \le s_0).$$

Proof. Write B_ρ for the open and $\overline{B}_\rho$ for the closed ball of radius ρ about the origin of the space-time $\ell^2(\mathbb{R}^d \times \mathbb{R})$, put $G = B_{3/4}$ and $D(z) = \operatorname{dist}(z, \partial G)$, and write $e_b = (0, 1)$, a unit vector. For U in the normalized family $\mathcal{U}$ of `NormalizedInfFamily`, $z \in G$ and $r > 0$ let $\beta_U(z, r)$ be as in `BetaFlatness`, so that by `FlatnessModulus`

$$b_d(s) = \sup\big\{\beta_U(z, r) : U \in \mathcal{U},\ z \in G,\ 0 < r \le sD(z)\big\}. \tag{283}$$

By `BetaFlatness` each $\beta_U(z, r)$ is the supremum of a subset of $(0, 1)$, hence $0 \le \beta_U(z, r) \le 1$; so the set in (283) is bounded above by 1, and therefore $0 \le b_d(s) \le 1$. What this paragraph

establishes is boundedness above, and nothing more: we do *not* assert here that either defining set is nonempty. Nonemptiness is needed only for the approximation property of a supremum, it is used only in Step 2, and it is derived there at the point of use, where the strict positivity of the suprema concerned supplies it. Until then we use only the upper-bound property, which boundedness alone licenses: $\beta_U(z,r) \le b_d(s)$ whenever (U,z,r) is admissible for s as in (283). We also use, as part of the statement of `BetaFlatness` rather than as a consequence of it, that every $\lambda \in (\beta_U(z,r),1)$ is good for U at (z,r), i.e. admits a cylinder fit in the sense of `IsCylinderFit`.

Let $L \ge 0$ be the common constant of `NormalizedFamilyLipschitz`, so that every $U \in \mathcal{U}$ satisfies

$$|U| \le L \quad \text{and} \quad |U(a)-U(b)| \le L|a-b| \qquad \big(a,b \in \overline{B}_{7/8}\big). \tag{284}$$

Apart from the two suprema of (283) and of `BetaFlatness` just discussed, every supremum written below is of the form $\sup_{\overline{B}_\rho} |\cdot|$ over a closed ball of positive radius, hence over a nonempty set, and in each case the function inside the absolute value is bounded on that ball by (284) — together with the Cauchy–Schwarz bound $|\langle Q, X\rangle| \le |Q|\,|X|$ in the instances where a fixed linear functional also appears. Each of those suprema is therefore the supremum of a nonempty set that is bounded above, so it is a genuine supremum, and we use this below without further comment.

Step 0: every member of $\mathcal{U}$ is infinity-harmonic on B_1. Let $U \in \mathcal{U}$. By `NormalizedInfFamily` there is u, infinity-harmonic on the unit ball of $\mathbb{R}^d$, with $U(z) = u(z_x) + z_t$ for every space-time $z = (z_x, z_t)$. By `LinearLift` the function $z \mapsto u(z_x) + z_t$ is infinity-harmonic on the preimage of that ball under $z \mapsto z_x$, i.e. on $\{z : |z_x| < 1\}$. Since $|z_x| \le |z|$, the ball B_1 is contained in that set, so

$$U \text{ is infinity-harmonic on } B_1, \text{ hence on } B_{7/8}. \tag{285}$$

Here we used that infinity-harmonicity passes to an open subset: if $\Omega' \subseteq \Omega$ are open and v is infinity-harmonic on Ω, then for $z \in \Omega'$ and a C^2 function ϕ near z such that $v - \phi$ has a local maximum on Ω' at z, say $v - \phi \le (v-\phi)(z)$ on $\mathcal{N} \cap \Omega'$ for a neighbourhood $\mathcal{N}$ of z, the set $\mathcal{N}' = \mathcal{N} \cap \Omega'$ is again a neighbourhood of z because Ω' is open, and $\mathcal{N}' \cap \Omega = \mathcal{N} \cap \Omega'$; so $v - \phi$ has a local maximum on Ω at z and `IsInfSubsolutionOn` gives $\Delta_\infty \phi(z) \ge 0$. The same argument with minima and `IsInfSupersolutionOn` gives the other inequality, so v is infinity-harmonic on Ω' in the sense of `IsInfHarmonicOn`.

Step 1: a compactness claim. We shall use the following twice.

Claim C. *Let a be a point of a finite-dimensional real inner-product space $\mathcal{H}$, let $\rho > 0$, put $K = \overline{B}(a,\rho)$, and let $v_i : \mathcal{H} \to \mathbb{R}$ satisfy $|v_i| \le A$ on K and $|v_i(w) - v_i(w')| \le \Lambda |w - w'|$ for $w, w' \in K$, with constants $A, \Lambda \ge 0$ independent of i. Then there are a strictly increasing $\iota : \mathbb{N} \to \mathbb{N}$ and $v : \mathcal{H} \to \mathbb{R}$ with $v_{\iota(k)} \to v$ uniformly on K.*

Proof of Claim C. If $\mathcal{H} = \{0\}$ then $K = \{a\}$ is a single point, the bounded sequence $(v_i(a))$ has a convergent subsequence by Bolzano–Weierstrass (preamble), and the constant function with that limit works. So assume $\mathcal{H} \ne \{0\}$. Fixing an orthonormal basis of $\mathcal{H}$, the set of points with rational coordinates is countable and dense, so its intersection with the open ball $B(a,\rho)$ is a countable set $\{w_1, w_2, \dots\}$ dense in $B(a,\rho)$ and hence in its closure K. For each m the real sequence $(v_i(w_m))_i$ lies in $[-A, A]$; applying Bolzano–Weierstrass (preamble) and extracting nested subsequences, and then taking the diagonal, we obtain a strictly increasing ι such that $(v_{\iota(k)}(w_m))_k$ converges for every m. We show $(v_{\iota(k)})_k$ is uniformly Cauchy on K. Let $\varepsilon > 0$ and put $\sigma = \varepsilon/(3(\Lambda+1)) > 0$. The set K is closed and bounded, hence compact (preamble), so the cover of K by the balls $B(w_m, \sigma)$ — which is a cover, by density — has a finite subcover, indexed by $m_1, \dots, m_p$. Choose k_0 with $|v_{\iota(k)}(w_{m_l}) - v_{\iota(k')}(w_{m_l})| < \varepsilon/3$ for all $k, k' \ge k_0$ and all $l \le p$. Given $w \in K$ pick l with $|w - w_{m_l}| < \sigma$; then for $k, k' \ge k_0$

$$|v_{\iota(k)}(w) - v_{\iota(k')}(w)| \le \Lambda\sigma + \frac{\varepsilon}{3} + \Lambda\sigma \le \varepsilon,$$

using the equi-Lipschitz bound twice and $\Lambda\sigma \le \varepsilon/3$. Hence $(v_{\iota(k)})$ is uniformly Cauchy on K and therefore converges uniformly on K to a function v defined on K, which we extend to $\mathcal{H}$ by the value 0 off K. □

Step 2: the contradiction hypothesis. Suppose the conclusion fails. Then there is $\varepsilon_* > 0$ such that for every $s_0 \in (0, 1/8]$ there is $s \in (0, s_0]$ with $b_d(s) > \varepsilon_*$. Put $\delta = \varepsilon_* > 0$ and apply this with $s_0 = \min(1/8, 1/i)$ for $i = 1, 2, \dots$: we obtain $s_i \in (0, 1/8]$ with

$$s_i \le \frac{1}{i} \quad \text{and} \quad b_d(s_i) > \delta. \tag{286}$$

In particular $s_i \to 0$.

Both of the extractions we now perform use the approximation property of a supremum, which is available only for a nonempty set; in both cases positivity supplies nonemptiness. First, `FlatnessModulus` assigns $b_d(s_i)$ the value 0 when the set in (283) is empty, whereas $b_d(s_i) > \delta > 0$ by (286); hence that set is nonempty, and being bounded above by 1 it has a genuine supremum exceeding δ. The approximation property therefore applies and produces $U_i \in \mathcal{U}$, $z_i \in G$ and r_i with $0 < r_i \le s_i D(z_i)$ and

$$\beta_{U_i}(z_i, r_i) > \delta. \tag{287}$$

Second, the same argument runs one level down. By `BetaFlatness` the value $\beta_{U_i}(z_i, r_i)$ is 0 when the set $\{\lambda \in (0,1) : \lambda \text{ is not good for } U_i \text{ at } (z_i, r_i)\}$ defining it is empty, whereas $\beta_{U_i}(z_i, r_i) > \delta > 0$ by (287); hence that set is nonempty too, and it is bounded above by 1. The approximation property applies to it as well and yields a parameter

$$\lambda_i \in (\delta, 1) \quad \text{that is } \textit{not} \text{ good for } U_i \text{ at } (z_i, r_i). \tag{288}$$

Since $D(z_i) \le 3/4$ we get $r_i \le \frac{3}{4} s_i \to 0$.

Step 3: passing to a convergent subsequence. The points z_i lie in the compact set $\overline{B}_{3/4}$, so after passing to a subsequence $z_i \to z_*$ with $|z_*| \le 3/4$. By (284) the functions U_i are bounded by L and L-Lipschitz on $K = \overline{B}_{7/8}$, so Claim C (with $a = 0$, $\rho = 7/8$, $A = \Lambda = L$) gives, after a further subsequence, a function $U_\circ$ with

$$\eta_i := \sup_{\overline{B}_{7/8}} |U_i - U_\circ| \longrightarrow 0. \tag{289}$$

All the hypotheses recorded so far are preserved by passing to subsequences, and we keep the same names. By (285) each U_i is infinity-harmonic on the open set $B_{7/8}$, and (289) gives uniform convergence on every compact subset of $B_{7/8}$; so `InfHarmonicStability` applies and

$$U_\circ \text{ is infinity-harmonic on } B_{7/8}. \tag{290}$$

Note $z_* \in B_{7/8}$, since $|z_*| \le 3/4 < 7/8$.

Step 4: the slow scales. Put $a_i = r_i + |z_i - z_*| + \eta_i$, so $a_i \to 0$ by Steps 2 and 3. Passing to a further subsequence we may assume $a_i \le 4^{-i}$, and we set

$$R_i = 2^{-i-4},$$

so that $R_i \downarrow 0$, $R_i \le 1/16$, and

$$\frac{a_i}{R_i} \le \frac{4^{-i}}{2^{-i-4}} = 16 \cdot 2^{-i} \longrightarrow 0. \tag{291}$$

Since $|z_*| \le 3/4$ and $R_i \le 1/16$ we have

$$|z_*| + R_i \le \tfrac{3}{4} + \tfrac{1}{16} < \tfrac{7}{8}, \tag{292}$$

so $z_* + R_i X \in B_{7/8}$ whenever $|X| \le 1$.

Step 5: the blow-ups at z_* are asymptotically linear. By (290) and $z_* \in B_{7/8}$ we may apply `AffineBlowups` to the single function $U_\circ$ on $D = B_{7/8}$ at the point z_*. Its second assertion, applied to the positive null sequence (R_i), gives a strictly increasing ι and a function V such that the blow-ups

$$\check{U}_k(X) = \frac{U_\circ(z_* + R_{\iota(k)} X) - U_\circ(z_*)}{R_{\iota(k)}}$$

converge to V uniformly on every compact set; its first assertion, applied to the same data, gives a continuous linear functional Λ_0 with $V = \Lambda_0$. By the Riesz representation of a continuous linear functional on a finite-dimensional inner-product space there is P_0 with $\Lambda_0(X) = \langle P_0, X\rangle$. Replace the whole family by the ι-subsequence and keep all names; (291) and all earlier properties persist. Thus

$$\sup_{|X|\le 1}\Big|\frac{U_\circ(z_* + R_iX) - U_\circ(z_*)}{R_i} - \langle P_0, X\rangle\Big| \longrightarrow 0. \tag{293}$$

Define

$$\widehat{U}_i(X) = \frac{U_i(z_* + R_iX) - U_i(z_*)}{R_i}.$$

For $|X| \le 1$ both $z_* + R_iX$ and z_* lie in $\overline{B}_{7/8}$ by (292), so by (289)

$$\big|\widehat{U}_i(X) - \check{U}_i(X)\big| \le \frac{2\eta_i}{R_i} \le \frac{2a_i}{R_i} \longrightarrow 0$$

by (291), where $\check{U}_i$ is as above with R_i in place of $R_{\iota(k)}$. Combining with (293),

$$\sup_{|X|\le 1}\big|\widehat{U}_i(X) - \langle P_0, X\rangle\big| \longrightarrow 0. \tag{294}$$

Step 6: the three-point estimate at the scales R_i. Each $\widehat{U}_i$ is infinity-harmonic on B_1. Indeed U_i is infinity-harmonic on $B_{7/8}$ by (285); `InfHarmonicRescale`, with $x_0 = z_*$ and $A = R_i > 0$, makes $X \mapsto U_i(z_* + R_iX)$ infinity-harmonic on the preimage of $B_{7/8}$ under $X \mapsto z_* + R_iX$, which contains B_1 by (292); and `InfHarmonicNormalize`, with $c = U_i(z_*)$ and $M = R_i > 0$, makes the normalized difference $\widehat{U}_i$ infinity-harmonic on the same set. Also, for $X, Y \in B_1$ the points $z_* + R_iX$ and $z_* + R_iY$ lie in $\overline{B}_{7/8}$, so (284) gives

$$|\widehat{U}_i(X) - \widehat{U}_i(Y)| = \frac{|U_i(z_* + R_iX) - U_i(z_* + R_iY)|}{R_i} \le L|X - Y|.$$

Finally (294) is exactly the hypothesis that $\widehat{U}_i - \ell$ tends to 0 uniformly on B_1 for the affine functions $\ell(X) = \langle P_0, X\rangle + 0$, the same one for every i. So `ThreePointFlatness`, applied with this L, the functions $\widehat{U}_i$, the slopes P_0 and the constants 0, gives: for every $\varepsilon > 0$ there is i_0 with

$$\big|\widehat{U}_i(X) + \widehat{U}_i(Y) - 2\widehat{U}_i\big(\tfrac{1}{2}(X+Y)\big)\big| \le \varepsilon|X - Y| \qquad \big(i \ge i_0,\ X, Y \in B_{1/4},\ X \ne Y\big). \tag{295}$$

Step 7: the small-scale blow-ups. Since $r_i \to 0$ we may assume $r_i \le 1/16$ for every i, so that for $|H| \le 2$

$$|z_i + r_iH| \le |z_i| + 2r_i < \tfrac{3}{4} + \tfrac{1}{8} = \tfrac{7}{8},$$

i.e. $z_i + r_iH \in \overline{B}_{7/8}$. Put

$$W_i(H) = \frac{U_i(z_i + r_iH) - U_i(z_i)}{r_i}.$$

By (284), $|W_i(H)| \le L|H| \le 2L$ and $|W_i(H) - W_i(K)| \le L|H - K|$ for $H, K \in \overline{B}_2$. Claim C (with $a = 0$, $\rho = 2$, $A = 2L$, $\Lambda = L$) gives, after a further subsequence, a function W with

$$\sup_{|H|\le 2}|W_i - W| \longrightarrow 0. \tag{296}$$

In particular $W(0) = \lim W_i(0) = 0$, and, because $U_i(z) = u_i(z_x) + z_t$ for the u_i of `NormalizedInfFamily` and $e_b = (0, 1)$ changes only the time coordinate,

$$W_i(e_b) = \frac{U_i(z_i + r_ie_b) - U_i(z_i)}{r_i} = \frac{\big(u_i((z_i)_x) + (z_i)_t + r_i\big) - \big(u_i((z_i)_x) + (z_i)_t\big)}{r_i} = 1, \tag{297}$$

so $W(e_b) = 1$.

Step 8: W has vanishing midpoint defect. Let $H, K \in \overline{B}_2$ with $H \ne K$ and put, for each i,

$$X_i = \frac{z_i - z_* + r_iH}{R_i}, \qquad Y_i = \frac{z_i - z_* + r_iK}{R_i},$$

so that $z_*+R_iX_i = z_i+r_iH$ and $z_*+R_iY_i = z_i+r_iK$, and $\frac12(X_i+Y_i) = (z_i-z_*+r_i\frac12(H+K))/R_i$, whose image under $X \mapsto z_* + R_iX$ is $z_i + r_i\frac12(H+K)$. Moreover

$$|X_i| \le \frac{|z_i - z_*| + 2r_i}{R_i} \le \frac{3a_i}{R_i} \longrightarrow 0$$

by (291), and likewise for Y_i; so $X_i, Y_i \in B_{1/4}$ for all large i. Also $X_i - Y_i = (r_i/R_i)(H-K) \neq 0$, so $X_i \neq Y_i$.

Now the two midpoint defects are proportional. Since $\widehat{U}_i(X) = (U_i(z_* + R_iX) - U_i(z_*))/R_i$ and the three coefficients $1, 1, -2$ sum to zero, the constants $U_i(z_*)$ cancel and

$$\widehat{U}_i(X_i) + \widehat{U}_i(Y_i) - 2\widehat{U}_i\big(\tfrac12(X_i+Y_i)\big) = \frac{U_i(z_i+r_iH) + U_i(z_i+r_iK) - 2U_i\big(z_i + r_i\frac12(H+K)\big)}{R_i};$$

the same cancellation of $U_i(z_i)$ in the definition of W_i gives

$$W_i(H) + W_i(K) - 2W_i\big(\tfrac12(H+K)\big) = \frac{U_i(z_i+r_iH) + U_i(z_i+r_iK) - 2U_i\big(z_i + r_i\frac12(H+K)\big)}{r_i}.$$

Hence the left sides differ by the factor R_i/r_i. Let $\varepsilon > 0$ and let i_0 be as in (295). For $i \ge i_0$ large enough that $X_i, Y_i \in B_{1/4}$,

$$\big|W_i(H)+W_i(K)-2W_i\big(\tfrac12(H+K)\big)\big| = \frac{R_i}{r_i}\big|\widehat{U}_i(X_i)+\widehat{U}_i(Y_i)-2\widehat{U}_i\big(\tfrac12(X_i+Y_i)\big)\big| \le \frac{R_i}{r_i}\varepsilon|X_i-Y_i| = \varepsilon|H-K|,$$

using $|X_i - Y_i| = (r_i/R_i)|H-K|$. Letting $i \to \infty$ and using (296) at the three points H, K, $\frac12(H+K)$ of $\overline{B}_2$,

$$\big|W(H) + W(K) - 2W\big(\tfrac12(H+K)\big)\big| \le \varepsilon|H-K|.$$

As $\varepsilon > 0$ was arbitrary, and the case $H = K$ is trivial,

$$W(H) + W(K) = 2W\big(\tfrac12(H+K)\big) \qquad \big(H, K \in \overline{B}_2\big). \tag{298}$$

Step 9: W is linear. W is L-Lipschitz on $\overline{B}_2$, being a uniform limit of L-Lipschitz functions, and $W(0) = 0$. Taking $K = 0$ in (298) and using $W(0) = 0$,

$$W(H) = 2W(H/2) \qquad \big(H \in \overline{B}_2\big). \tag{299}$$

For $H, K \in \overline{B}_1$ we have $H + K \in \overline{B}_2$ and $\frac12(H+K) \in \overline{B}_1$, so (298) and (299) give

$$W(H) + W(K) = 2W\big(\tfrac12(H+K)\big) = W(H+K) \qquad \big(H, K \in \overline{B}_1\big). \tag{300}$$

In particular $W(-H) = -W(H)$ for $H \in \overline{B}_1$ (take $K = -H$ and use $W(0) = 0$). Next,

$$W(qH) = qW(H) \qquad \big(q \in [0,1] \text{ a dyadic rational},\ H \in \overline{B}_1\big). \tag{301}$$

We prove (301) by subdividing, not by doubling, and subdivision is the only direction the identities support: (300) is stated for $H, K \in \overline{B}_1$, so taking $K = H$ in it gives $W(2H) = 2W(H)$ *once*, but applying it a second time would require $2H \in \overline{B}_1$, which fails as soon as $|H| > 1/2$. Fix $H \in \overline{B}_1$ and write $q = p2^{-m}$ with integers $m \ge 0$ and $0 \le p \le 2^m$. Halving first: $H \in \overline{B}_1 \subseteq \overline{B}_2$ and $\overline{B}_2$ is closed under $X \mapsto X/2$, so (299) may be applied m times in succession, each time to a point of $\overline{B}_2$, and it gives

$$W\big(2^{-m}H\big) = 2^{-m}W(H).$$

Adding second: form the partial sums $k2^{-m}H$ for $k = 0, 1, \dots, p$. For $0 \le k < p$ we have $\big|k2^{-m}H\big| = k2^{-m}|H| \le q|H| \le |H| \le 1$ and $\big|2^{-m}H\big| \le |H| \le 1$, so the partial sum and the increment both lie in $\overline{B}_1$ and (300) applies at each of the p steps. Hence

$$W\big(p2^{-m}H\big) = pW\big(2^{-m}H\big) = p2^{-m}W(H) = qW(H),$$

which is (301). (This is exactly the bookkeeping used for the basis expansion in the next paragraph, performed one step earlier.) Since the dyadic rationals are dense in $[0,1]$, the continuity of W extends (301) to

$$W(tH) = tW(H) \qquad \big(t \in [-1,1],\ H \in \overline{B}_1\big), \tag{302}$$

the negative values following from oddness.

Let $e_1,\dots,e_n$ be an orthonormal basis of the space-time, and set $P=\sum_j W(e_j)e_j$. Let $H\in\overline{B}_1$ and write $H=\sum_j a_j e_j$ with $a_j=\langle H,e_j\rangle$. Every partial sum $\sum_{j\le k}a_je_j$ has norm at most $|H|\le 1$, and each a_ke_k has norm $|a_k|\le 1$, so (300) applies repeatedly and gives $W(H)=\sum_j W(a_je_j)$; and $W(a_je_j)=a_jW(e_j)$ by (302). Therefore

$$W(H)=\sum_j a_jW(e_j)=\langle P,H\rangle \qquad \big(H\in\overline{B}_1\big),$$

and then (299) extends this to $\overline{B}_2$: $W(H)=2W(H/2)=2\langle P,H/2\rangle=\langle P,H\rangle$. By (297), $\langle P,e_b\rangle=W(e_b)=1$, so in particular

$$W(H)=\langle P,H\rangle\ \big(H\in\overline{B}_2\big),\qquad P\neq 0. \tag{303}$$

Step 10: the contradiction. By (296) and (303) there is i with

$$\sup_{|H|\le 2}\big|U_i(z_i+r_iH)-U_i(z_i)-r_i\langle P,H\rangle\big|=r_i\sup_{|H|\le 2}\big|W_i(H)-\langle P,H\rangle\big|<\delta r_i, \tag{304}$$

and we fix such an i, large enough for all the finitely many "for all large i" requirements above. We claim P realizes the cylinder fit of parameter λ_i for U_i at (z_i,r_i) in the sense of `IsCylinderFit`. First $P\neq 0$ by (303). Next let Z belong to the cylinder $Z_{\lambda_i}(z_i,r_i;P)$ of `FitCylinder` and write $Z=z_i+h$, $N=P/|P|$. By that definition $|\langle N,h\rangle|\le r_i$ and $|h-\langle N,h\rangle N|\le\sqrt{\lambda_i}\,r_i$, and these two components of h are orthogonal, so

$$|h|^2=|\langle N,h\rangle|^2+|h-\langle N,h\rangle N|^2\le r_i^2+\lambda_i r_i^2\le 2r_i^2,$$

using $\lambda_i<1$; hence $|h|\le\sqrt{2}\,r_i\le 2r_i$. Writing $H=h/r_i$, so $|H|\le 2$, (304) gives

$$\big|U_i(Z)-U_i(z_i)-\langle P,Z-z_i\rangle\big|=\big|U_i(z_i+r_iH)-U_i(z_i)-r_i\langle P,H\rangle\big|<\delta r_i\le\lambda_i r_i,$$

the last step by $\delta<\lambda_i$ from (288). So λ_i is good for U_i at (z_i,r_i), contradicting (288). Therefore the conclusion cannot fail. □

2.72. `BadScales`.

Lemma 2.72 (Selection of bad scales)**.** *Fix an integer $d\ge 1$, put $n=d+1$, write $G=B_{3/4}$ for the open ball of radius $3/4$ about the origin of the space-time $\ell^2(\mathbb{R}^d\times\mathbb{R})$ and $D(z)=\operatorname{dist}(z,\partial G)$, and let b_d be the family flatness modulus of `FlatnessModulus`. Suppose that the uniform power bound on b_d fails for every positive exponent: for every $B>0$, every $\gamma\in(0,1]$ and every $s_0\in(0,1/8]$ there is $s\in(0,s_0]$ with*

$$b_d(s)>Bs^{\gamma}.$$

Then there are real sequences $\gamma_i,s_i,r_i,\lambda_i$, functions U_i on the space-time and centers z_i such that $(\gamma_i,s_i,r_i,\lambda_i,U_i,z_i)_{i\ge 0}$ is selected bad-scale data in dimension d in the sense of `IsBadScaleData`.

Proof. Throughout, $\mathcal{U}$ is the normalized family of `NormalizedInfFamily`, $\beta_U(z,r)$ is as in `BetaFlatness`, and by `FlatnessModulus`

$$b_d(s)=\sup\big\{\beta_U(z,r):U\in\mathcal{U},\ z\in G,\ 0<r\le sD(z)\big\}. \tag{305}$$

By `BetaFlatness` each $\beta_U(z,r)$ is the supremum of a subset of $(0,1)$, so $0\le\beta_U(z,r)\le 1$; hence the set in (305) is bounded above by 1 and $0\le b_d(s)\le 1$. What this establishes is boundedness above, and nothing more: we do *not* claim here that the set in (305) is nonempty. Nonemptiness is needed only where the *approximation property* of a supremum is invoked, which happens exactly three times: for the scalar supremum M in Step 1, and in Step 2 for (305) itself and for $\beta_{U_i}(z_i,r_i)$. It is established at each of those three points of use, in Step 2 from the strict positivity (311). (The scalar supremum T of Step 1 is used only as an upper bound, together with $T\ge 0$, and its defining set is visibly nonempty because $1/N\le 1/8$.) We use three consequences repeatedly: $\beta_U(z,r)\le b_d(s)$ whenever (U,z,r) is admissible for s as in (305), which needs only the boundedness just proved; every λ with $\beta_U(z,r)<\lambda<1$ is good for U at (z,r), i.e. admits a cylinder fit in the sense of `IsCylinderFit`, since otherwise λ would lie in the set whose supremum is $\beta_U(z,r)$; and D is 1-Lipschitz, being a distance to a fixed set.

We also record that for $z \in G$

$$D(z) = \tfrac{3}{4} - |z| \in (0, \tfrac{3}{4}], \tag{306}$$

because ∂G is the sphere of radius $3/4$, which is nonempty since the space-time has dimension $d+1 \geq 2$. Indeed the point $\frac{3}{4}|z|^{-1}z$, or any point of the sphere when $z = 0$, lies on the sphere at distance exactly $\frac{3}{4} - |z|$ from z; and no point w of the sphere is nearer, because $|z - w| \geq |w| - |z| = \frac{3}{4} - |z|$.

Step 1: the running maximum. Put $\gamma_i = 1/(i+1)$ for $i \geq 0$, so $0 < \gamma_i \leq 1$ and $\gamma_i \to 0$. Fix i and let

$$g(s) = \frac{b_d(s)}{s^{\gamma_i}} \qquad (0 < s \leq 1/8),$$

a nonnegative function. Let N be an integer with $N \geq \max(8, i+1)$, so that $1/N \leq 1/8$.

The quantity $T = \sup\{g(s) : 1/N \leq s \leq 1/8\}$ is finite: on that interval $b_d(s) \leq 1$ and $s^{-\gamma_i} \leq N^{\gamma_i}$, so $g \leq N^{\gamma_i}$. Also $T \geq 0$. Apply the hypothesis with $B = 2T + 2 > 0$, the exponent γ_i and $s_0 = 1/N$: there is $\sigma \in (0, 1/N]$ with $b_d(\sigma) > (2T+2)\sigma^{\gamma_i}$, that is

$$g(\sigma) > 2T + 2.$$

In particular $g(\sigma) > T$, so $\sigma \notin [1/N, 1/8]$ and therefore

$$0 < \sigma < \tfrac{1}{N}. \tag{307}$$

Let $M = \sup\{g(s) : \sigma \leq s \leq 1/8\}$, again finite because $b_d \leq 1$ and $s^{-\gamma_i} \leq \sigma^{-\gamma_i}$ there. Its defining set is nonempty, and the witness is σ itself: by (307) and $N \geq 8$ we have $\sigma < 1/N \leq 1/8$, so $\sigma \in [\sigma, 1/8]$ and $g(\sigma)$ belongs to the set. Hence $M \geq g(\sigma) > 2$, and M is the supremum of a nonempty set bounded above. Since $M/2 < M$, the approximation property of the supremum is available and gives $s_i \in [\sigma, 1/8]$ with

$$g(s_i) > \tfrac{1}{2}M. \tag{308}$$

Two consequences. First, every $s \in [s_i, 1/8]$ lies in $[\sigma, 1/8]$, so $g(s) \leq M < 2g(s_i)$, i.e.

$$\frac{b_d(s)}{s^{\gamma_i}} \leq 2\frac{b_d(s_i)}{s_i^{\gamma_i}} \qquad (s_i \leq s \leq 1/8). \tag{309}$$

Second, $g(s_i) > \frac{1}{2}M \geq \frac{1}{2}g(\sigma) > T$, while every $s \in [1/N, 1/8]$ has $g(s) \leq T$; so $s_i \notin [1/N, 1/8]$, and as $s_i \in [\sigma, 1/8]$ with $\sigma < 1/N$ by (307), we conclude

$$0 < s_i < \tfrac{1}{N} \leq \frac{1}{i+1}, \qquad s_i \leq \tfrac{1}{8}. \tag{310}$$

Hence $s_i \to 0$. Also $g(s_i) > \frac{1}{2}M > 1 > 0$, so

$$b_d(s_i) > 0. \tag{311}$$

Step 2: the bad triple and the bad parameter. Fix i. The approximation property of a supremum is available only for a nonempty set, and here strict positivity supplies that. By `FlatnessModulus` the value $b_d(s_i)$ is 0 when the set in (305) is empty, whereas $b_d(s_i) > 0$ by (311); hence that set is nonempty, and being bounded above by 1 as recorded above it has a genuine supremum. Since $b_d(s_i)/2 < b_d(s_i)$, again by (311), the approximation property at $s = s_i$ gives $U_i \in \mathcal{U}$, $z_i \in G$ and r_i with

$$0 < r_i \leq s_i D(z_i), \qquad \beta_{U_i}(z_i, r_i) > \tfrac{1}{2}b_d(s_i) > 0. \tag{312}$$

The same argument runs one level down. By `BetaFlatness` the quantity $\beta_{U_i}(z_i, r_i)$ is the supremum of the set of parameters in $(0,1)$ that are not good for U_i at (z_i, r_i), *with the value* 0 *when that set is empty*; since $\beta_{U_i}(z_i, r_i) > 0$ by (312), that set is nonempty, and it is contained in $(0,1)$ and so bounded above by 1. The approximation property is therefore available and gives a parameter

$$\lambda_i \in (0,1) \text{ not good for } U_i \text{ at } (z_i, r_i), \qquad \lambda_i > \tfrac{1}{2}\beta_{U_i}(z_i, r_i). \tag{313}$$

Combining (312) and (313), and using $\lambda_i \le \beta_{U_i}(z_i, r_i) \le b_d(s_i)$ — the first because λ_i belongs to the set whose supremum is $\beta_{U_i}(z_i, r_i)$, the second because (U_i, z_i, r_i) is admissible for s_i — we obtain

$$\tfrac{1}{4} b_d(s_i) < \lambda_i \le b_d(s_i). \tag{314}$$

Step 3: $\lambda_i \to 0$. By (310), $s_i \to 0$ with $s_i > 0$. Let $\varepsilon > 0$ and take s_0 as in `FamilyFlatness` for this ε; for all large i we have $0 < s_i \le s_0$ and therefore $b_d(s_i) \le \varepsilon$. Hence $b_d(s_i) \to 0$, and $\lambda_i \le b_d(s_i)$ from (314) gives

$$\lambda_i \longrightarrow 0. \tag{315}$$

Step 4: the uniformity with $\Lambda = 32$. Fix $M_* \ge 2$. Since $s_i \to 0$, $\gamma_i \to 0$ and $\lambda_i \to 0$, and since $(2M_*)^{\gamma_i} = \exp(\gamma_i \log(2M_*)) \to 1$, there is i_0 such that for every $i \ge i_0$

$$M_* s_i < \tfrac{1}{2}, \qquad 2M_* s_i \le \tfrac{1}{8}, \qquad (2M_*)^{\gamma_i} < 4, \qquad 32\lambda_i < 1. \tag{316}$$

Let $i \ge i_0$, let w satisfy $|w - z_i| \le M_* r_i$ and let $0 < \rho \le M_* r_i$.

(4a) $w \in G$. By (312) and (316),

$$M_* r_i \le M_* s_i D(z_i) < \tfrac{1}{2} D(z_i), \tag{317}$$

so $|w - z_i| < \frac{1}{2} D(z_i)$. By (306), $|w| \le |z_i| + |w - z_i| < |z_i| + (\frac{3}{4} - |z_i|) = \frac{3}{4}$, i.e. $w \in G$.

(4b) $\rho \le \frac{1}{8} D(w)$. Since D is 1-Lipschitz, $D(w) \ge D(z_i) - |w - z_i| \ge D(z_i) - M_* r_i \ge \frac{1}{2} D(z_i)$ by (317), so $D(z_i) \le 2D(w)$ and, using (312) and (316),

$$\rho \le M_* r_i \le M_* s_i D(z_i) \le 2M_* s_i D(w) \le \tfrac{1}{8} D(w). \tag{318}$$

Thus the center w and radius ρ are admissible.

(4c) $32\lambda_i$ *is good for* U_i *at* (w, ρ). The triple (U_i, w, ρ) is admissible for the scale $2M_* s_i$, since $U_i \in \mathcal{U}$, $w \in G$ by (4a), $\rho > 0$, and $\rho \le 2M_* s_i D(w)$ by (318). Therefore

$$\beta_{U_i}(w, \rho) \le b_d(2M_* s_i).$$

Now $2M_* s_i \ge s_i$ because $2M_* \ge 4 \ge 1$, and $2M_* s_i \le 1/8$ by (316); so (309) applies at $s = 2M_* s_i$ and gives

$$b_d(2M_* s_i) \le 2 b_d(s_i) \frac{(2M_* s_i)^{\gamma_i}}{s_i^{\gamma_i}} = 2 b_d(s_i) (2M_*)^{\gamma_i}.$$

By (314), $b_d(s_i) < 4\lambda_i$, and by (316), $(2M_*)^{\gamma_i} < 4$; hence

$$\beta_{U_i}(w, \rho) \le 2 b_d(s_i) (2M_*)^{\gamma_i} < 2 \cdot 4\lambda_i \cdot 4 = 32\lambda_i.$$

Finally $32\lambda_i < 1$ by (316), so $32\lambda_i \in (\beta_{U_i}(w, \rho), 1)$ and therefore $32\lambda_i$ is good for U_i at (w, ρ): there is P realizing the cylinder fit of parameter $32\lambda_i$ for U_i at (w, ρ) in the sense of `IsCylinderFit`.

Conclusion. We check every clause of `IsBadScaleData` for the data $(\gamma_i, s_i, r_i, \lambda_i, U_i, z_i)$ just constructed, with $\Lambda = 32$. We have $\gamma_i = 1/(i+1) > 0$ with $\gamma_i \to 0$; $s_i > 0$ with $s_i \to 0$ by (310); $\lambda_i > 0$ by (313) with $\lambda_i \to 0$ by (315); $U_i \in \mathcal{U}$ and $z_i \in G$ by Step 2; $r_i > 0$ and $r_i \le s_i D(z_i)$ by (312); λ_i is not good for U_i at (z_i, r_i), i.e. no P realizes that cylinder fit, by (313); and the final clause is Step 4, whose threshold i_0 depends on M_* only. □

2.73. `BaryAffineMajorantIntegral`.

Lemma 2.73 (Integrating an affine majorant against a probability measure)**.** *Let F be a finite-dimensional real inner-product space equipped with its Borel σ-algebra, let f be a real function on space-time, let $s > 0$, let $t \in \mathbb{R}$, let $x \in F$, let $\Lambda \in \mathbb{R}$ and let $A_0 \in F$. Assume the affine majorant*

$$f(y, t) - \frac{|y|^2}{2s} \le \Lambda + A_0 \cdot (y - x) \qquad (y \in F). \tag{319}$$

Let μ be a Borel probability measure on F such that $y \mapsto |y - x|^2$, $y \mapsto f(y, t)$ and $y \mapsto y - x$ are all μ-integrable and $\int (y - x)\, d\mu(y) = 0$. Then

$$\int f(y, t)\, d\mu(y) - \frac{\int |y - x|^2\, d\mu(y)}{2s} \le \Lambda + \frac{|x|^2}{2s}.$$

Proof. Throughout, integrals are Bochner integrals with respect to μ, which is a probability measure, so $\mu(F) = 1$ and constants integrate to themselves.

Integrability of the pieces. Expanding the square in the inner product of F,

$$|y|^2 = |(y-x)+x|^2 = |y-x|^2 + 2\,(y-x)\cdot x + |x|^2 \qquad (y \in F). \tag{320}$$

On the right, $y \mapsto |y-x|^2$ is μ-integrable by hypothesis; $y \mapsto (y-x)\cdot x$ is μ-integrable because $y \mapsto y - x$ is and the map $v \mapsto v \cdot x$ is a continuous linear functional on F, hence composing it with an integrable function leaves an integrable function; and the constant $|x|^2$ is integrable because μ is finite. So $y \mapsto |y|^2$ is μ-integrable, and therefore so are

$$g(y) = f(y,t) - \frac{|y|^2}{2s} \qquad \text{and} \qquad y \longmapsto \Lambda + A_0 \cdot (y-x),$$

the first because $y \mapsto f(y,t)$ is integrable by hypothesis and $y \mapsto |y|^2/(2s)$ is a constant multiple of an integrable function, the second because the constant Λ is integrable and $y \mapsto A_0 \cdot (y-x)$ is the composition of the integrable $y \mapsto y - x$ with the continuous linear functional $v \mapsto A_0 \cdot v$.

The linear term integrates to zero. By linearity of the Bochner integral, a continuous linear functional commutes with it, so

$$\int A_0 \cdot (y-x)\, d\mu(y) = A_0 \cdot \int (y-x)\, d\mu(y) = A_0 \cdot 0 = 0, \tag{321}$$

using the hypothesis $\int (y-x)\, d\mu = 0$. The same argument applied to the functional $v \mapsto v \cdot x$ gives

$$\int (y-x)\cdot x\, d\mu(y) = \Big(\int (y-x)\, d\mu(y)\Big) \cdot x = 0. \tag{322}$$

Integrating the majorant. Both sides of (319) are μ-integrable, as just shown, and the inequality holds pointwise, so monotonicity of the integral gives

$$\int g\, d\mu \le \int \big(\Lambda + A_0 \cdot (y-x)\big) d\mu(y) = \Lambda\, \mu(F) + \int A_0 \cdot (y-x)\, d\mu(y) = \Lambda,$$

by additivity, the value of the integral of a constant against a probability measure, and (321).

The second-moment identity. Integrating (320), using additivity of the integral over the three integrable summands, (322) and $\mu(F) = 1$,

$$\int |y|^2\, d\mu(y) = \int |y-x|^2\, d\mu(y) + 2 \cdot 0 + |x|^2.$$

Hence, dividing by $2s > 0$ and using additivity once more,

$$\int g\, d\mu = \int f(y,t)\, d\mu(y) - \frac{\int |y|^2\, d\mu(y)}{2s} = \int f(y,t)\, d\mu(y) - \frac{\int |y-x|^2\, d\mu(y)}{2s} - \frac{|x|^2}{2s}.$$

Conclusion. Comparing the last display with $\int g\, d\mu \le \Lambda$,

$$\int f(y,t)\, d\mu(y) - \frac{\int |y-x|^2\, d\mu(y)}{2s} - \frac{|x|^2}{2s} \le \Lambda,$$

and adding $|x|^2/(2s)$ to both sides gives the assertion. □

2.74. `BaryConfigOfConvexHull`.

Lemma 2.74 (Finite convex combinations and compactness of the convex hull)**.** *Let F be a finite-dimensional real inner-product space and let $E \subseteq F$. Write* conv E *for the convex hull of E, that is, the smallest convex subset of F containing E. Then:*

1. *every $x \in \operatorname{conv} E$ is the mean of a finite horizontal probability configuration supported in E: there are $n \ge 1$, weights $\theta_i > 0$ with $\sum_{i=1}^n \theta_i = 1$ and points $y_1, \dots, y_n \in E$ with $\sum_{i=1}^n \theta_i y_i = x$, so that (θ, y) is a configuration with mean x in the sense of `IsBaryConfig` and $y_i \in E$ for every i;*
2. *if E is compact then* conv E *is compact.*

Proof. Let

$$W = \Big\{ \sum_{i=1}^{n} \theta_i y_i : n \geq 1,\ y_i \in E,\ \theta_i > 0,\ \sum_{i=1}^{n} \theta_i = 1 \Big\}$$

be the set of means of finite horizontal probability configurations supported in E, so that (1) asserts exactly $\operatorname{conv} E \subseteq W$.

$E \subseteq W$. For $y \in E$ take $n = 1$, $\theta_1 = 1 > 0$ and $y_1 = y$; then $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i = y$.

W is convex. Let $u = \sum_{i=1}^{n} \theta_i y_i$ and $u' = \sum_{j=1}^{n'} \theta'_j y'_j$ be two such combinations and let $\tau \in [0,1]$. If $\tau = 0$ then $\tau u + (1-\tau)u' = u' \in W$, and if $\tau = 1$ then it is $u \in W$; so assume $0 < \tau < 1$. Then

$$\tau u + (1-\tau)u' = \sum_{i=1}^{n} (\tau\theta_i) y_i + \sum_{j=1}^{n'} \big((1-\tau)\theta'_j\big) y'_j$$

is a combination of the $n + n' \geq 1$ points $y_1, \dots, y_n, y'_1, \dots, y'_{n'}$ of E with weights $\tau\theta_i > 0$ and $(1-\tau)\theta'_j > 0$, which sum to $\tau \sum_i \theta_i + (1-\tau) \sum_j \theta'_j = \tau + (1-\tau) = 1$. Hence $\tau u + (1-\tau)u' \in W$, and W is convex.

Proof of (1). W is a convex set containing E, and $\operatorname{conv} E$ is the smallest such set; therefore $\operatorname{conv} E \subseteq W$, which is (1).

Every member of $\operatorname{conv} E$ *is a combination of at most* $m+1$ *points, where* $m = \dim F$. Let $u \in \operatorname{conv} E$. By (1) it has a representation $u = \sum_{i=1}^{n} \theta_i y_i$ as above; choose one with n least, and suppose $n > m+1$. The $n-1$ vectors $y_i - y_n$, $1 \leq i \leq n-1$, lie in the m-dimensional space F and $n - 1 > m$, so they are linearly dependent: there are reals $\lambda_1, \dots, \lambda_{n-1}$, not all zero, with $\sum_{i<n} \lambda_i (y_i - y_n) = 0$. Put $\lambda_n = -\sum_{i<n} \lambda_i$, so that

$$\sum_{i=1}^{n} \lambda_i y_i = 0, \qquad \sum_{i=1}^{n} \lambda_i = 0,$$

with not all λ_i zero. Were every $\lambda_i \leq 0$, the second identity would force them all to vanish; so $\lambda_i > 0$ for some i. Put $\tau = \min\{\theta_i/\lambda_i : \lambda_i > 0\}$, a minimum over a nonempty finite set, which is positive because every $\theta_i > 0$; let i_0 attain it and put $\theta'_i = \theta_i - \tau\lambda_i$. Then $\theta'_i \geq 0$ for every i: for $\lambda_i > 0$ because $\tau \leq \theta_i/\lambda_i$, and for $\lambda_i \leq 0$ because $\theta_i > 0$ and $-\tau\lambda_i \geq 0$. Moreover $\sum_i \theta'_i = \sum_i \theta_i - \tau \sum_i \lambda_i = 1$ and $\sum_i \theta'_i y_i = \sum_i \theta_i y_i - \tau \sum_i \lambda_i y_i = u$, while $\theta'_{i_0} = 0$. Discarding the indices with $\theta'_i = 0$ leaves a representation of u with strictly positive weights summing to 1 and with at least one term, since the weights sum to 1; it has fewer than n terms, contradicting minimality. Hence $n \leq m+1$.

Consequently, fixing any $y_* \in E$ — which exists whenever $\operatorname{conv} E \neq \emptyset$, since by (1) a member of $\operatorname{conv} E$ is a combination of points of E and so $E \neq \emptyset$ — and padding a representation with $n \leq m+1$ terms by the terms $\theta_i = 0$, $y_i = y_*$ for $n < i \leq m+1$,

(323)

$$\operatorname{conv} E = \Big\{ \sum_{i=1}^{m+1} \theta_i y_i : \theta \in \Delta,\ y_1, \dots, y_{m+1} \in E \Big\}, \qquad \Delta = \Big\{ \theta \in \mathbb{R}^{m+1} : \theta_i \geq 0,\ \sum_i \theta_i = 1 \Big\}.$$

The padding gives the inclusion from left to right. For the other inclusion, a sum $\sum_{i=1}^{m+1} \theta_i y_i$ with $\theta \in \Delta$ and $y_i \in E$ becomes, after discarding the indices with $\theta_i = 0$, a combination with strictly positive weights summing to 1 and with at least one term, hence lies in W; and $W \subseteq \operatorname{conv} E$ because $\operatorname{conv} E$ is convex and contains E, so it contains every finite convex combination of points of E, by induction on the number of terms: one term is a point of E, and a combination with $n+1$ terms and weights summing to 1 equals $(1-\theta_{n+1})u + \theta_{n+1} y_{n+1}$ with u a combination of n terms when $\theta_{n+1} < 1$, and equals $y_{n+1} \in E$ when $\theta_{n+1} = 1$.

Proof of (2). Assume E compact. If $E = \emptyset$ then $\operatorname{conv} E = \emptyset$, which is compact; so assume $E \neq \emptyset$. $\operatorname{conv} E$ is bounded: E is bounded, say $|y| \leq M_E$ for $y \in E$, and by (323) any member satisfies $|\sum_i \theta_i y_i| \leq \sum_i \theta_i |y_i| \leq M_E \sum_i \theta_i = M_E$. It is closed: let $u_k \in \operatorname{conv} E$ with $u_k \to u$ in F, and write $u_k = \sum_{i=1}^{m+1} \theta_i^k y_i^k$ as in (323). The sequence (θ^k) lies in the bounded set Δ and each (y_i^k) in the bounded set E, so applying Bolzano–Weierstrass (preamble) finitely many times in

succession there is one subsequence along which $\theta^k \to \theta$ and $y_i^k \to y_i$ for every i; then $\theta \in \Delta$ and $y_i \in E$ because Δ and E are closed, and $u = \lim_k \sum_i \theta_i^k y_i^k = \sum_i \theta_i y_i$ by continuity of the finitely many products and sums, so $u \in \operatorname{conv} E$ by (323). Being closed and bounded in the finite-dimensional space F, the set $\operatorname{conv} E$ is compact (preamble). This is (2). □

2.75. `BaryEnvWeakDuality`.

Lemma 2.75 (Barycentric weak duality and the configuration-radius identity)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on space-time, let $s > 0$ and let $z = (x, t)$ be a space-time point. Write*

$$g(y) = f(y,t) - \frac{|y|^2}{2s}, \qquad \mathcal{G}(A;x) = \big\{ g(y) - A \cdot (y - x) : y \in F \big\} \quad (A \in F),$$

and assume that $\mathcal{G}(A;x)$ is bounded above for every $A \in F$. Then, with $\mathcal{C}$ the barycentric envelope of `BaryEnv`, r the radius of `BaryRadius` and "configuration of mean x" meaning a finite horizontal probability configuration with mean x in the sense of `IsBaryConfig`:

(1) *for every configuration (θ, y) of mean x,*

$$\sum_i \theta_i f(y_i,t) - \frac{r(\theta,y)^2}{2s} - \frac{|x|^2}{2s} = \sum_i \theta_i \Big(f(y_i,t) - \frac{|y_i|^2}{2s} \Big);$$

(2) *for every configuration (θ, y) of mean x,*

$$\sum_i \theta_i f(y_i,t) - \frac{r(\theta,y)^2}{2s} \le \mathcal{C}_s f(x,t);$$

(3) *for every $A \in F$,*

$$\mathcal{C}_s f(x,t) - \frac{|x|^2}{2s} \le \sup \mathcal{G}(A;x).$$

Proof. Step 0: the configuration-radius identity, i.e. (1). Let (θ, y) be a configuration of mean x, so by `IsBaryConfig` $\theta_i > 0$ for every i, $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i = x$. Expanding the square in the inner product of F,

$$|y_i - x|^2 = |y_i|^2 - 2\, y_i \cdot x + |x|^2.$$

Multiplying by θ_i and summing over the finitely many indices, and using the two identities $\sum_i \theta_i y_i = x$ and $\sum_i \theta_i = 1$,

(324)
$$\sum_i \theta_i |y_i - x|^2 = \sum_i \theta_i |y_i|^2 - 2\Big(\sum_i \theta_i y_i\Big) \cdot x + \Big(\sum_i \theta_i\Big) |x|^2 = \sum_i \theta_i |y_i|^2 - 2\, x \cdot x + |x|^2 = \sum_i \theta_i |y_i|^2 - |x|^2.$$

The left-hand side of (324) is nonnegative, being a sum of products of the positive numbers θ_i with the nonnegative numbers $|y_i - x|^2$. By `BaryRadius` the radius is $r(\theta, y) = \sqrt{\sum_i \theta_i |y_i - x|^2}$, and the square root of a nonnegative real squares back to it, so

$$r(\theta,y)^2 = \sum_i \theta_i |y_i - x|^2 = \sum_i \theta_i |y_i|^2 - |x|^2,$$

that is $r(\theta, y)^2 + |x|^2 = \sum_i \theta_i |y_i|^2$. Dividing by $2s > 0$ and subtracting from $\sum_i \theta_i f(y_i, t)$,

$$\sum_i \theta_i f(y_i,t) - \frac{r(\theta,y)^2}{2s} - \frac{|x|^2}{2s} = \sum_i \theta_i f(y_i,t) - \frac{\sum_i \theta_i |y_i|^2}{2s} = \sum_i \theta_i \Big(f(y_i,t) - \frac{|y_i|^2}{2s} \Big),$$

the last step by distributing the finite sum. This is (1), and the right-hand side is $\sum_i \theta_i g(y_i)$.

Step 1: weak duality on elements. Fix $A \in F$. The set $\mathcal{G}(A;x)$ is nonempty: taking $y = x$ gives the element $g(x) - A \cdot (x - x) = g(x)$. By hypothesis it is bounded above. Both facts are needed before the supremum may be used at all, because by the convention of the preamble a supremum takes the value 0 when its set is empty or unbounded above; with them, $\Gamma(A;x) := \sup \mathcal{G}(A;x)$ is a genuine finite least upper bound, and in particular

(325)
$$g(y) - A \cdot (y - x) \le \Gamma(A;x) \qquad (y \in F).$$

Let (θ, y) be a configuration of mean x. By (325) at $y = y_i$, rearranged, $g(y_i) \le \Gamma(A;x) + A \cdot (y_i - x)$ for every i. Multiplying by $\theta_i > 0$ and summing,

$$\sum_i \theta_i g(y_i) \le \Big(\sum_i \theta_i\Big)\Gamma(A;x) + A \cdot \Big(\sum_i \theta_i (y_i - x)\Big),$$

where the inner product was taken out of the finite sum by linearity in its second argument. Now $\sum_i \theta_i = 1$ and $\sum_i \theta_i (y_i - x) = \sum_i \theta_i y_i - \big(\sum_i \theta_i\big)x = x - x = 0$, so

$$\sum_i \theta_i g(y_i) \le \Gamma(A;x) \qquad \big((\theta, y) \text{ of mean } x,\ A \in F\big). \tag{326}$$

The barycentric supremum is a genuine least upper bound, i.e. (2). By `BaryEnv` the number $\mathcal{C}_s f(x,t)$ is the supremum of the set

$$\mathcal{B}(x) = \Big\{\sum_i \theta_i f(y_i,t) - \frac{r(\theta,y)^2}{2s} : (\theta, y) \text{ a configuration of mean } x\Big\}.$$

This set is nonempty: $n = 1$, $\theta_1 = 1$, $y_1 = x$ is a configuration of mean x in the sense of `IsBaryConfig`, since $\theta_1 = 1 > 0$, $\sum_i \theta_i = 1$ and $1 \cdot x = x$; its radius is $r = \sqrt{1 \cdot |x - x|^2} = 0$, so it contributes the element $f(x,t)$. It is bounded above: by (1), each element of $\mathcal{B}(x)$ equals $\sum_i \theta_i g(y_i) + |x|^2/(2s)$ for the corresponding configuration, and by (326) with $A = 0$ this is at most the real number $\Gamma(0;x) + |x|^2/(2s)$. Hence $\sup \mathcal{B}(x) = \mathcal{C}_s f(x,t)$ is a genuine finite least upper bound, not the convention value, and being an upper bound of $\mathcal{B}(x)$ it dominates each of its elements. That is exactly (2).

Weak duality on the supremum, i.e. (3). Fix $A \in F$ and let (θ, y) be any configuration of mean x. By (1) and (326),

$$\sum_i \theta_i f(y_i,t) - \frac{r(\theta,y)^2}{2s} = \sum_i \theta_i g(y_i) + \frac{|x|^2}{2s} \le \Gamma(A;x) + \frac{|x|^2}{2s}.$$

So the real number $\Gamma(A;x) + |x|^2/(2s)$ is an upper bound for $\mathcal{B}(x)$. Since $\mathcal{C}_s f(x,t) = \sup \mathcal{B}(x)$ is the *least* upper bound of that nonempty bounded-above set, as just established, $\mathcal{C}_s f(x,t) \le \Gamma(A;x) + |x|^2/(2s)$, i.e. $\mathcal{C}_s f(x,t) - |x|^2/(2s) \le \Gamma(A;x)$. This is (3). □

2.76. `BaryEnvSlopeLocalization`.

Lemma 2.76 (Uniform localization of slopes and maximizing positions)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$, let $\Omega = \{(y,\sigma) : \sigma \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Let $S \subseteq F \times \mathbb{R} \times \mathbb{R}$ be a nonempty compact set of triples (x,t,s) with $s > 0$ and $t \in I$. For such a triple write*

$$g(y) = f(y,t) - \frac{|y|^2}{2s}, \qquad \mathcal{G}(A;x,t,s) = \big\{g(y) - A \cdot (y - x) : y \in F\big\} \quad (A \in F).$$

Then there are $\Lambda_0 \ge 0$ and $M > 0$, depending only on S, such that for every $(x,t,s) \in S$:

(i) *$\mathcal{G}(A;x,t,s)$ is bounded above for every $A \in F$;*
(ii) *every $A \in F$ with $\sup \mathcal{G}(A;x,t,s) \le \sup \mathcal{G}(0;x,t,s) + 1$ satisfies $|A| \le \Lambda_0$;*
(iii) *for every $A \in F$ with $|A| \le \Lambda_0 + 1$,*

$$\sup \mathcal{G}(A;x,t,s) = \sup_{|y| \le M} \big\{g(y) - A \cdot (y - x)\big\},$$

that value is attained at some y_0 with $|y_0| \le M$, and every $y \in F$ with $g(y) - A \cdot (y - x) = \sup \mathcal{G}(A;x,t,s)$ satisfies $|y| \le M$.

Proof. The data attached to S. The three coordinate maps $(x,t,s) \mapsto |x|$, $(x,t,s) \mapsto t$ and $(x,t,s) \mapsto s$ are continuous on $F \times \mathbb{R} \times \mathbb{R}$, so on the nonempty compact set S the first and third attain a maximum and the third also attains a minimum (preamble). Put

$$R_0 = \max_S |x| \ \ (\ge 0), \qquad s_{\max} = \max_S s, \qquad s_{\min} = \min_S s.$$

Every triple of S has $s > 0$, so $s_{\min} > 0$ and hence $s_{\max} \geq s_{\min} > 0$. Let J be the image of S under the continuous projection $(x, t, s) \mapsto t$; then J is compact, being a continuous image of a compact set, and $J \subseteq I$ because every triple of S has $t \in I$.

The far bound. Apply `SubquadraticGrowth` with the compact set $J \subseteq I$ and $\varepsilon = 1/(4s_{\max}) > 0$: there is $M_0 > 0$ with $|f(y,\sigma)| \leq |y|^2/(4s_{\max})$ whenever $|y| \geq M_0$ and $\sigma \in J$. Let $(x, t, s) \in S$ and $|y| \geq M_0$. Then $t \in J$ and $0 < s \leq s_{\max}$, so $1/(4s_{\max}) \leq 1/(4s)$ and

$$g(y) = f(y,t) - \frac{|y|^2}{2s} \leq \frac{|y|^2}{4s_{\max}} - \frac{|y|^2}{2s} \leq \frac{|y|^2}{4s} - \frac{|y|^2}{2s} = -\frac{|y|^2}{4s} \leq -\frac{|y|^2}{4s_{\max}}, \tag{327}$$

the last step because $0 < s \leq s_{\max}$ and $|y|^2 \geq 0$.

The near bound. The set $\mathcal{K} = \{(y,\sigma) : |y| \leq M_0 + R_0 + 1,\ \sigma \in J\}$ is compact: the closed ball $\{|y| \leq M_0 + R_0 + 1\}$ is a closed bounded subset of the finite-dimensional space F and hence compact, J is compact, and the identification of $F \times \mathbb{R}$ with the L^2 product space of the preamble is a homeomorphism. It is nonempty, because $J \neq \emptyset$ and $|0| = 0 \leq M_0 + R_0 + 1$, and it is contained in Ω because $J \subseteq I$. As f is continuous on Ω, the continuous function $|f|$ attains a finite maximum $N_0 \geq 0$ on $\mathcal{K}$. Put $N_1 = N_0 + (M_0 + R_0 + 1)^2/(2s_{\min})$, a finite nonnegative number. For $(x, t, s) \in S$ and $|y| \leq M_0 + R_0 + 1$ we have $t \in J$ and $s \geq s_{\min} > 0$, so

$$|g(y)| \leq |f(y,t)| + \frac{|y|^2}{2s} \leq N_0 + \frac{(M_0 + R_0 + 1)^2}{2s_{\min}} = N_1. \tag{328}$$

Proof of (i). Let $(x, t, s) \in S$ and $A \in F$. We bound $g(y) - A \cdot (y - x)$ for every y, splitting on $|y|$ and using $|x| \leq R_0$ and the Cauchy–Schwarz inequality $|A \cdot v| \leq |A|\,|v|$. If $|y| \leq M_0 + R_0 + 1$ then $|y - x| \leq |y| + |x| \leq M_0 + 2R_0 + 1$, so by (328)

$$g(y) - A \cdot (y - x) \leq N_1 + |A|\,(M_0 + 2R_0 + 1).$$

If $|y| > M_0 + R_0 + 1$ then in particular $|y| \geq M_0$, so (327) applies, and $-A \cdot (y - x) \leq |A|\,|y - x| \leq |A|(|y| + R_0)$; writing $\varrho = |y|$,

$$g(y) - A \cdot (y - x) \leq -\frac{\varrho^2}{4s_{\max}} + |A|\varrho + |A|R_0 \leq s_{\max}|A|^2 + |A|R_0,$$

the last step by completing the square: for every real ϱ,

$$-\frac{\varrho^2}{4s_{\max}} + |A|\varrho = -\frac{\big(\varrho - 2s_{\max}|A|\big)^2}{4s_{\max}} + s_{\max}|A|^2 \leq s_{\max}|A|^2,$$

since $s_{\max} > 0$ and squares are nonnegative. Hence every element of $\mathcal{G}(A; x, t, s)$ is at most the real number

$$\max\big\{N_1 + |A|(M_0 + 2R_0 + 1),\ s_{\max}|A|^2 + |A|R_0\big\}, \tag{329}$$

so $\mathcal{G}(A; x, t, s)$ is bounded above, which is (i). It is also nonempty, containing $g(x)$ at $y = x$; so for every A the number $\Gamma(A; x, t, s) := \sup \mathcal{G}(A; x, t, s)$ is a genuine finite least upper bound, and in particular

$$g(y) - A \cdot (y - x) \leq \Gamma(A; x, t, s) \qquad (y \in F,\ A \in F), \tag{330}$$

which is both the bound on the elements and, read from right to left, the licence to bound Γ from below by inserting any competitor y. The bound (329) is allowed to grow with $|A|$: only finiteness for each fixed A is asserted in (i), and the uniformity over $|A| \leq \Lambda_0 + 1$ needed later is the separate statement (iii).

Proof of (ii), with $\Lambda_0 = 2N_1 + 1$. Let $(x, t, s) \in S$. Taking $A = 0$ in (330) and inserting the competitor $y = 0$, which satisfies $|0| = 0 \leq M_0 + R_0 + 1$, gives by (328) that $\Gamma(0; x, t, s) \geq g(0) \geq -N_1$. Now let $A \neq 0$ and put $y_A = x - A/|A|$, so $|y_A| \leq |x| + 1 \leq R_0 + 1 \leq M_0 + R_0 + 1$ and $-A \cdot (y_A - x) = A \cdot (A/|A|) = |A|$. Hence by (330) and (328),

$$\Gamma(A; x, t, s) \geq g(y_A) - A \cdot (y_A - x) \geq -N_1 + |A|.$$

If moreover $\Gamma(A; x, t, s) \leq \Gamma(0; x, t, s) + 1$, then combining this with the upper bound $\Gamma(0; x, t, s) \leq N_1$ — which holds because by (328) and (327) every $g(y)$ is at most N_1 when $|y| \leq M_0 + R_0 + 1$

and at most $0 \le N_1$ when $|y| > M_0 + R_0 + 1$, so N_1 is an upper bound for $\mathcal{G}(0;x,t,s)$ and the supremum is the least one — we get

$$|A| - N_1 \le \Gamma(A;x,t,s) \le \Gamma(0;x,t,s) + 1 \le N_1 + 1,$$

so $|A| \le 2N_1 + 1 = \Lambda_0$. For $A = 0$ the conclusion $|A| \le \Lambda_0$ is immediate since $\Lambda_0 = 2N_1 + 1 \ge 1 > 0$. This is (ii), and $\Lambda_0 = 2N_1 + 1 \ge 0$ because $N_1 \ge 0$.

Proof of (iii). Let $(x,t,s) \in S$ and $|A| \le \Lambda_0 + 1$. For $|y| \ge M_0$, (327) and Cauchy–Schwarz give

$$g(y) - A \cdot (y - x) \le -\frac{|y|^2}{4s_{\max}} + (\Lambda_0 + 1)\big(|y| + R_0\big), \tag{331}$$

whose right-hand side depends on y only through $|y|$ and tends to $-\infty$ as $|y| \to \infty$, with a rate depending only on $s_{\max}$, Λ_0 and R_0, hence only on S. Concretely, choose

$$M = \max\Big\{M_0 + R_0 + 1,\ 8s_{\max}(\Lambda_0 + 1),\ \sqrt{8s_{\max}\big(N_1 + 1 + (\Lambda_0 + 1)R_0\big)}\Big\},$$

which is $\ge M_0 + R_0 + 1 > 0$ and depends only on S. For $|y| \ge M$ we have $|y| \ge M_0$, and $(\Lambda_0 + 1)|y| \le |y|^2/(8s_{\max})$ because $|y| \ge 8s_{\max}(\Lambda_0 + 1)$, and $N_1 + 1 + (\Lambda_0 + 1)R_0 \le |y|^2/(8s_{\max})$ because $|y|^2 \ge 8s_{\max}(N_1 + 1 + (\Lambda_0 + 1)R_0)$; so (331) gives

$$g(y) - A \cdot (y - x) \le -\frac{|y|^2}{4s_{\max}} + \frac{|y|^2}{8s_{\max}} + (\Lambda_0 + 1)R_0 \le -\frac{|y|^2}{8s_{\max}} + (\Lambda_0 + 1)R_0 \le -(N_1 + 1).$$

On the other hand the competitor $y = x$ has $|x| \le R_0 \le M_0 + R_0 + 1$, so by (328)

$$g(x) - A \cdot (x - x) = g(x) \ge -N_1 > -(N_1 + 1).$$

Therefore

$$g(y) - A \cdot (y - x) < g(x) - A \cdot (x - x) \qquad (|y| \ge M). \tag{332}$$

Write $B_M = \{y \in F : |y| \le M\}$, a nonempty compact set containing x. The function $y \mapsto g(y) - A \cdot (y - x)$ is continuous on B_M: for $|y| \le M$ the point (y,t) lies in Ω, where f is continuous, while $y \mapsto |y|^2/(2s)$ and $y \mapsto A \cdot (y - x)$ are continuous on F because $s > 0$. So it attains a maximum on B_M at some $y_0 \in B_M$ (preamble), and

$$\sup_{|y| \le M} \big\{g(y) - A \cdot (y - x)\big\} = g(y_0) - A \cdot (y_0 - x)$$

is a genuine least upper bound, the set of values over B_M being nonempty and bounded above by $g(y_0) - A \cdot (y_0 - x)$. Since $x \in B_M$, maximality gives $g(x) - A \cdot (x - x) \le g(y_0) - A \cdot (y_0 - x)$, and therefore by (332) every y with $|y| \ge M$ satisfies $g(y) - A \cdot (y - x) < g(y_0) - A \cdot (y_0 - x)$. Combining the two cases $|y| \le M$ and $|y| \ge M$, the value $g(y_0) - A \cdot (y_0 - x)$ is an upper bound for $\mathcal{G}(A;x,t,s)$, and it belongs to that set; so it is the greatest element and

$$\Gamma(A;x,t,s) = g(y_0) - A \cdot (y_0 - x) = \sup_{|y| \le M} \big\{g(y) - A \cdot (y - x)\big\},$$

which is the displayed identity of (iii) together with its attainment at $y_0 \in B_M$. Finally, if $y \in F$ satisfies $g(y) - A \cdot (y - x) = \Gamma(A;x,t,s)$ then $|y| < M$ is forced: otherwise $|y| \ge M$ and the strict inequality of the previous sentence would give $\Gamma(A;x,t,s) < g(y_0) - A \cdot (y_0 - x) = \Gamma(A;x,t,s)$. In particular $|y| \le M$, completing (iii). □

2.77. `BaryGammaMinAttained`.

Lemma 2.77 (Continuity of the slope functional and attainment of its infimum)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$, let $\Omega = \{(y,\sigma) : \sigma \in I\}$, and let f be a real function on space-time continuous on Ω. Let $S \subseteq F \times \mathbb{R} \times \mathbb{R}$ be a nonempty compact set of triples (x,t,s) with $s > 0$ and $t \in I$, and for such a triple write*

$$g(y) = f(y,t) - \frac{|y|^2}{2s}, \qquad \mathcal{G}(A;x,t,s) = \big\{g(y) - A \cdot (y - x) : y \in F\big\}, \qquad \Gamma(A;x,t,s) = \sup \mathcal{G}(A;x,t,s).$$

Let $\Lambda_0 \ge 0$ and $M > 0$ and assume that for every $(x,t,s) \in S$:

(i) *$\mathcal{G}(A;x,t,s)$ is bounded above for every $A \in F$;*

(ii) *every* $A \in F$ *with* $\Gamma(A;x,t,s) \le \Gamma(0;x,t,s)+1$ *satisfies* $|A| \le \Lambda_0$*;*
(iii) $\Gamma(A;x,t,s) = \sup_{|y|\le M}\{g(y) - A\cdot(y-x)\}$ *for every* $A \in F$ *with* $|A| \le \Lambda_0+1$.

Then $(A,x,t,s) \mapsto \Gamma(A;x,t,s)$ *is continuous on* $\{|A| \le \Lambda_0+1\}\times S$*, and there is a function* $G : F\times\mathbb{R}\times\mathbb{R}\to\mathbb{R}$*, continuous on* S*, such that for every* $(x,t,s)\in S$

$$G(x,t,s) \le \Gamma(A;x,t,s) \quad (A\in F), \qquad \text{and} \qquad G(x,t,s) = \Gamma(A_0;x,t,s) \text{ for some } A_0 \text{ with } |A_0| \le \Lambda_0.$$

Proof. A parametric-maximum fact. We first record that the maximum of a continuous function over a fixed nonempty compact set is a continuous function of the remaining parameters. Let P be a metric space, let Q be a nonempty compact metric space and let H be real-valued and continuous on $P\times Q$. For each $\pi\in P$ the function $H(\pi,\cdot)$ is continuous on the nonempty compact Q, so $\max_{q\in Q} H(\pi,q)$ exists (preamble). Let $\pi_k\to\pi$ in P. Testing against a fixed $q\in Q$ gives $\max_{q'} H(\pi_k,q') \ge H(\pi_k,q)$, and $H(\pi_k,q)\to H(\pi,q)$ by continuity of H, so $\liminf_k \max_{q'} H(\pi_k,q') \ge H(\pi,q)$ for every $q\in Q$ and therefore $\liminf_k \max_{q'} H(\pi_k,q') \ge \max_q H(\pi,q)$. Conversely choose for each k a maximizer $q_k\in Q$ of $H(\pi_k,\cdot)$ and let a subsequence be given; by compactness of Q it has a further subsequence along which $q_k\to\bar q\in Q$ (Bolzano–Weierstrass, preamble). Along it $(\pi_k,q_k)\to(\pi,\bar q)$, so by continuity of H,

$$\max_{q'} H(\pi_k,q') = H(\pi_k,q_k) \longrightarrow H(\pi,\bar q) \le \max_q H(\pi,q).$$

Write $a_k = \max_{q'} H(\pi_k,q')$ and $L = \max_q H(\pi,q)$. Every subsequence of (a_k) thus has a further subsequence converging to a real number at most L; by the first bound every subsequential limit of (a_k) is also at least L, so every subsequence of (a_k) has a further subsequence converging to L exactly. In particular no subsequence of (a_k) diverges, so (a_k) is bounded; and a bounded real sequence all of whose subsequences have a further subsequence converging to L converges to L, since otherwise some $\varepsilon>0$ would admit infinitely many k with $|a_k-L|\ge\varepsilon$, and that subsequence would have no further subsequence converging to L. Hence $a_k\to L$. As P is a metric space, sequential continuity is continuity, so $\pi\mapsto\max_{q\in Q} H(\pi,q)$ is continuous on P. Applying this to $-H$ and using $\min_q H(\pi,q) = -\max_q(-H)(\pi,q)$, the map $\pi\mapsto\min_{q\in Q} H(\pi,q)$ is continuous on P as well.

Continuity of Γ. Put $P = \{A\in F : |A|\le\Lambda_0+1\}\times S$ and $Q = \{y\in F : |y|\le M\}$, each with the metric induced from the ambient finite-dimensional space, and

$$H\big((A,x,t,s),y\big) = f(y,t) - \frac{|y|^2}{2s} - A\cdot(y-x).$$

Q is nonempty and compact, being a closed bounded subset of the finite-dimensional space F (preamble). H is continuous on $P\times Q$: the map $((A,x,t,s),y)\mapsto(y,t)$ is continuous and sends $P\times Q$ into Ω, because every triple of S has $t\in I$, and f is continuous on Ω; the map $((A,x,t,s),y)\mapsto|y|^2/(2s)$ is continuous on $P\times Q$ because $s>0$ on S, which is a compact set of triples with positive third coordinate, so s does not vanish there; and $((A,x,t,s),y)\mapsto A\cdot(y-x)$ is continuous, the inner product being continuous in both arguments. By hypothesis (iii), $\Gamma(A;x,t,s) = \max_{y\in Q} H((A,x,t,s),y)$ for every $(A,x,t,s)\in P$ — the supremum over Q in (iii) being a maximum because $H((A,x,t,s),\cdot)$ is continuous on the nonempty compact Q. So by the recorded fact $(A,x,t,s)\mapsto\Gamma(A;x,t,s)$ is continuous on P, which is the first assertion.

The infimum over all slopes is a minimum over a ball. Let $(x,t,s)\in S$. By hypothesis (i) the number $\Gamma(A;x,t,s)$ is a real number for every $A\in F$, so the family $\{\Gamma(A;x,t,s) : A\in F\}$ is a nonempty set of reals, containing $\Gamma(0;x,t,s)$. Put $Q_0 = \{A\in F : |A|\le\Lambda_0\}$, a nonempty compact set containing 0. By the previous paragraph $A\mapsto\Gamma(A;x,t,s)$ is continuous on Q_0, which is contained in $\{|A|\le\Lambda_0+1\}$, so it attains a minimum on Q_0 at some A_0 with $|A_0|\le\Lambda_0$ (preamble). We claim

$$\Gamma(A_0;x,t,s) \le \Gamma(A;x,t,s) \qquad (A\in F). \tag{333}$$

Indeed, if $|A|\le\Lambda_0$ this is the minimality of A_0 on Q_0. If $|A|>\Lambda_0$ then by hypothesis (ii), read in contrapositive, $\Gamma(A;x,t,s) > \Gamma(0;x,t,s)+1 > \Gamma(0;x,t,s) \ge \Gamma(A_0;x,t,s)$, the last inequality being the minimality of A_0 on Q_0 applied at $0\in Q_0$. This is (333).

The continuous minimal value. Apply the recorded parametric fact a second time, with $P = S$, $Q = Q_0$ and $H((x,t,s),A) = \Gamma(A;x,t,s)$, in the minimum form: Q_0 is nonempty and compact, and H is continuous on $S\times Q_0$ because $Q_0\times S \subseteq \{|A| \le \Lambda_0+1\}\times S$ and Γ is continuous there by the first assertion, the exchange of the two factors being a homeomorphism. Hence

$$G(x,t,s) := \min_{|A|\le\Lambda_0} \Gamma(A;x,t,s)$$

defines a function continuous on S; extend it to all of $F\times\mathbb{R}\times\mathbb{R}$ by giving it the value 0 off S, which does not affect continuity on S. For $(x,t,s)\in S$ the minimum is attained at the A_0 produced above, so $G(x,t,s) = \Gamma(A_0;x,t,s)$ with $|A_0| \le \Lambda_0$, and by (333) $G(x,t,s) \le \Gamma(A;x,t,s)$ for every $A \in F$. These are the two displayed properties, and the proof is complete. □

2.78. `BaryEnvHullOfMinSlope`.

Lemma 2.78 (The center lies in the convex hull of the maximizing positions)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on space-time, let $s > 0$, let $t \in \mathbb{R}$ and let $x \in F$. Write*

$$g(y) = f(y,t) - \frac{|y|^2}{2s}, \qquad \mathcal{G}(A) = \big\{g(y) - A\cdot(y-x) : y \in F\big\}, \qquad \Gamma(A) = \sup \mathcal{G}(A) \quad (A \in F),$$

and assume that $\mathcal{G}(A)$ is bounded above for every $A \in F$. Let $\Lambda_0 \ge 0$, let $M > 0$ and let $A_0 \in F$ with $|A_0| \le \Lambda_0$. Assume

(a) *$\Gamma(A_0) \le \Gamma(A)$ for every $A \in F$;*
(b) *for every A with $|A| \le \Lambda_0 + 1$ there is y_0 with $|y_0| \le M$ and $g(y_0) - A\cdot(y_0 - x) = \Gamma(A)$;*
(c) *for every A with $|A| \le \Lambda_0 + 1$, every $y \in F$ with $g(y) - A\cdot(y-x) = \Gamma(A)$ satisfies $|y| \le M$;*
(d) *$A \mapsto \Gamma(A)$ is continuous on $\{|A| \le \Lambda_0 + 1\}$;*
(e) *$y \mapsto f(y,t)$ is continuous on $\{|y| \le M\}$.*

Then x belongs to the convex hull of the set of maximizing positions at A_0,

$$E = \big\{y \in F : |y| \le M,\ g(y) - A_0\cdot(y-x) = \Gamma(A_0)\big\}.$$

Proof. E is nonempty and compact. By (b) applied at $A = A_0$, which is legitimate because $|A_0| \le \Lambda_0 \le \Lambda_0 + 1$, there is $y_0 \in E$, so $E \ne \emptyset$. The function $y \mapsto g(y) - A_0\cdot(y-x)$ is continuous on the compact ball $B_M = \{|y| \le M\}$: indeed $y \mapsto f(y,t)$ is continuous there by (e), while $y \mapsto |y|^2/(2s)$ is continuous because $s > 0$, and $y \mapsto A_0\cdot(y-x)$ is continuous because the inner product is. Hence E, the subset of B_M on which that continuous function equals the constant $\Gamma(A_0)$, is closed in B_M; and B_M is a closed bounded subset of the finite-dimensional space F and therefore compact (preamble). A closed subset of a compact set is compact, so E is compact.

Strict separation from a compact convex set. We record the following. Let $K \subseteq F$ be nonempty, compact and convex and let $w \in F \setminus K$. Then there are a unit vector $e \in F$ and $\delta > 0$ with $e\cdot(y-w) \ge \delta$ for every $y \in K$.

To see this, note that $y \mapsto |y-w|$ is continuous on the nonempty compact set K, so it attains its minimum there at some $p \in K$ (preamble). Put $d = |p-w|$; then $d > 0$, because $d = 0$ would give $w = p \in K$. Let $y \in K$ and $0 < \tau \le 1$. By convexity $(1-\tau)p + \tau y \in K$, so minimality of p gives

$$d^2 \le \big|(1-\tau)p + \tau y - w\big|^2 = \big|p - w + \tau(y-p)\big|^2 = d^2 + 2\tau\,(p-w)\cdot(y-p) + \tau^2|y-p|^2,$$

that is $0 \le 2(p-w)\cdot(y-p) + \tau|y-p|^2$ for every $\tau \in (0,1]$; letting $\tau \downarrow 0$ gives $(p-w)\cdot(y-p) \ge 0$. Hence

$$(p-w)\cdot(y-w) = (p-w)\cdot(y-p) + (p-w)\cdot(p-w) \ge |p-w|^2 = d^2.$$

Taking $e = (p-w)/d$, a unit vector, and $\delta = d > 0$, and dividing the last display by $d > 0$, gives $e\cdot(y-w) \ge \delta$ for every $y \in K$, as recorded.

The separation argument. Suppose, for contradiction, that $x \notin \operatorname{conv} E$. By `BaryConfigOfConvexHull` the set $\operatorname{conv} E$ is compact, E being compact; it is convex, being a convex hull; and it is nonempty,

containing the nonempty set E. So the recorded fact applies with $K = \operatorname{conv} E$ and $w = x$ and gives a unit vector $e \in F$ and $\delta > 0$ with

$$e \cdot (y - x) \geq \delta \qquad (y \in \operatorname{conv} E), \tag{334}$$

in particular for every $y \in E$.

For $0 < h \leq 1$ we have $|A_0 + he| \leq |A_0| + h|e| \leq \Lambda_0 + 1$, so by (b) there is a maximizer y_h with $|y_h| \leq M$ and

$$\Gamma(A_0 + he) = g(y_h) - (A_0 + he) \cdot (y_h - x). \tag{335}$$

Claim: $e \cdot (y_h - x) \geq \delta/2$ for all sufficiently small $h > 0$. If not, there are h_k with $0 < h_k \leq 1$, $h_k \downarrow 0$ and $e \cdot (y_{h_k} - x) < \delta/2$ for every k. The points y_{h_k} lie in the compact ball B_M, so after passing to a subsequence $y_{h_k} \to \bar{y}$ with $|\bar{y}| \leq M$ (Bolzano–Weierstrass, preamble). Along that subsequence, g is continuous on B_M (shown above) and $A_0 + h_k e \to A_0$, so by (335)

$$\Gamma(A_0 + h_k e) = g(y_{h_k}) - (A_0 + h_k e) \cdot (y_{h_k} - x) \longrightarrow g(\bar{y}) - A_0 \cdot (\bar{y} - x),$$

the convergence of the inner-product term following from the joint continuity of the inner product. On the other hand $|A_0 + h_k e| \leq \Lambda_0 + 1$ for every k and $A_0 + h_k e \to A_0$ with $|A_0| \leq \Lambda_0 + 1$, so by (d) $\Gamma(A_0 + h_k e) \to \Gamma(A_0)$. Limits of a convergent real sequence being unique,

$$g(\bar{y}) - A_0 \cdot (\bar{y} - x) = \Gamma(A_0),$$

so $\bar{y} \in E$ — it satisfies the defining equation and $|\bar{y}| \leq M$ — and therefore $e \cdot (\bar{y} - x) \geq \delta$ by (334). But $e \cdot (y_{h_k} - x) < \delta/2$ for every k and $e \cdot (y_{h_k} - x) \to e \cdot (\bar{y} - x)$ by continuity of the inner product, so $e \cdot (\bar{y} - x) \leq \delta/2 < \delta$. This contradiction proves the claim.

Conclusion. Fix h with $0 < h \leq 1$ small enough that $e \cdot (y_h - x) \geq \delta/2$. The number $g(y_h) - A_0 \cdot (y_h - x)$ is an element of $\mathcal{G}(A_0)$, namely its element at $y = y_h$. By the hypothesis of the statement $\mathcal{G}(A_0)$ is bounded above, and it is nonempty, containing $g(x)$ at $y = x$; hence $\Gamma(A_0) = \sup \mathcal{G}(A_0)$ is a genuine finite least upper bound of $\mathcal{G}(A_0)$, rather than the convention value of the preamble, and so it dominates every element of that set:

$$g(y_h) - A_0 \cdot (y_h - x) \leq \Gamma(A_0). \tag{336}$$

This is the one place the boundedness hypothesis is used, and it cannot be replaced by the attainment supplied by (b) and (c): under the preamble's convention a competitor set that is unbounded above has supremum 0, and 0 may itself be a value of $y \mapsto g(y) - A_0 \cdot (y - x)$, so attainment alone does not yield the least-upper-bound property. Therefore, by (335) and bilinearity of the inner product,

$$\Gamma(A_0 + he) = g(y_h) - A_0 \cdot (y_h - x) - h\, e \cdot (y_h - x) \leq \Gamma(A_0) - \frac{h\delta}{2} < \Gamma(A_0),$$

the first inequality by (336) together with $e \cdot (y_h - x) \geq \delta/2$ and $h > 0$, and the second because $h > 0$ and $\delta > 0$. This contradicts (a) applied at $A = A_0 + he$. Therefore $x \in \operatorname{conv} E$. □

2.79. `BaryGammaAffineInCenter`.

Lemma 2.79 (The slope functional is affine in the center)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on space-time, let $s > 0$ and let $t \in \mathbb{R}$. Write $g(y) = f(y,t) - |y|^2/(2s)$, and assume that for every $A \in F$ the set $\{g(y) - A \cdot y : y \in F\}$ is bounded above. Then there is a function $c : F \to \mathbb{R}$ such that*

$$\sup_{y \in F} \big\{ g(y) - A \cdot (y - x) \big\} = c(A) + A \cdot x \qquad (A, x \in F).$$

Proof. Fix $A \in F$ and put

$$\mathcal{C}(A) = \big\{ g(y) - A \cdot y : y \in F \big\}, \qquad \mathcal{G}(A; x) = \big\{ g(y) - A \cdot (y - x) : y \in F \big\} \quad (x \in F).$$

The set $\mathcal{C}(A)$ is nonempty, containing $g(0) - A \cdot 0 = g(0)$ at $y = 0$, and it is bounded above by hypothesis. Hence, by the convention of the preamble being satisfied in the only case it needs to be, $c(A) := \sup \mathcal{C}(A)$ is a genuine finite least upper bound, and

$$g(y) - A \cdot y \leq c(A) \quad (y \in F), \qquad \text{and} \qquad c(A) \leq b \ \text{ whenever } g(y) - A \cdot y \leq b \text{ for all } y \in F. \tag{337}$$

Define $c : F \to \mathbb{R}$ this way for every $A \in F$.

Now fix $x \in F$ as well. By bilinearity of the inner product,

$$g(y) - A \cdot (y - x) = \big(g(y) - A \cdot y\big) + A \cdot x \qquad (y \in F),$$

so $\mathcal{G}(A; x)$ is the translate of $\mathcal{C}(A)$ by the constant $A \cdot x$: every element of $\mathcal{G}(A; x)$ is an element of $\mathcal{C}(A)$ plus $A \cdot x$, and conversely. We check that $c(A) + A \cdot x$ is the least upper bound of $\mathcal{G}(A; x)$.

It is an upper bound: for $y \in F$, the first part of (337) gives $g(y) - A \cdot (y - x) = \big(g(y) - A \cdot y\big) + A \cdot x \le c(A) + A \cdot x$.

It is the least one: let b be any upper bound of $\mathcal{G}(A; x)$, so $\big(g(y) - A \cdot y\big) + A \cdot x \le b$, i.e. $g(y) - A \cdot y \le b - A \cdot x$, for every $y \in F$. By the second part of (337), $c(A) \le b - A \cdot x$, i.e. $c(A) + A \cdot x \le b$.

Moreover $\mathcal{G}(A; x)$ is nonempty, containing $g(x)$ at $y = x$, and bounded above by the real number $c(A) + A \cdot x$ just exhibited; so its supremum is a genuine least upper bound rather than the convention value 0, and being the least upper bound it equals $c(A) + A \cdot x$. As A and x were arbitrary, this is the displayed identity. □

2.80. `BaryEnvDuality`.

Lemma 2.80 (Barycentric envelope: attainment, duality, concavity, measures)**.** *Let F be a finite-dimensional real inner-product space, equipped with its Borel σ-algebra, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(x, t) : t \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Let $s > 0$ and let $z = (x, t)$ with $t \in I$. Then, with $\mathcal{C}$ the barycentric envelope of `BaryEnv`:*

(1) *the supremum defining $\mathcal{C}_s f(x, t)$ is attained by a finite horizontal probability configuration with mean x, in the sense of `IsBaryConfig` and `BaryRadius`;*

(2) *the min-max duality*

$$\mathcal{C}_s f(x, t) - \frac{|x|^2}{2s} = \min_{A \in F} \sup_{y \in F} \left\{ f(y, t) - \frac{|y|^2}{2s} - A \cdot (y - x) \right\} \tag{338}$$

holds, with the minimum attained at some $A_0 \in F$: that is, the left side equals the supremum for $A = A_0$ and is at most the supremum for every $A \in F$;

(3) *$(s, z) \mapsto \mathcal{C}_s f(z)$ is continuous at every (s, z) with $s > 0$ and $t \in I$;*

(4) *for every $s > 0$ and every $t \in I$ the function*

$$x \longmapsto \mathcal{C}_s f(x, t) - \frac{|x|^2}{2s} \tag{339}$$

is concave on F;

(5) *allowing any Borel probability measure μ on F with mean x and finite second moment, and with $f(\cdot, t)$ integrable, does not increase the value:*

$$\int f(y, t)\, d\mu(y) - \frac{\int |y - x|^2\, d\mu(y)}{2s} \le \mathcal{C}_s f(x, t).$$

Proof. Throughout fix $s > 0$ and $t \in I$, and abbreviate

$$g(y) = f(y, t) - \frac{|y|^2}{2s} \qquad (y \in F), \qquad \Gamma(A; x) = \sup_{y \in F} \big\{ g(y) - A \cdot (y - x) \big\} \quad (A, x \in F).$$

We write $\Omega = \{(y, \sigma) : \sigma \in I\}$, an open set on which f is continuous.

Step 0: the left side of the duality is a supremum of g-averages. Let (θ, y) be a finite horizontal probability configuration with mean x in the sense of `IsBaryConfig`. Expanding and using $\sum_i \theta_i y_i = x$ and $\sum_i \theta_i = 1$,

$$\sum_i \theta_i |y_i - x|^2 + |x|^2 = \sum_i \theta_i |y_i|^2 - 2x \cdot x + |x|^2 + |x|^2 = \sum_i \theta_i |y_i|^2.$$

Hence, with $r(\theta, y)$ the radius of `BaryRadius`,

$$\sum_i \theta_i f(y_i, t) - \frac{r(\theta,y)^2}{2s} - \frac{|x|^2}{2s} = \sum_i \theta_i \Big(f(y_i,t) - \frac{|y_i|^2}{2s} \Big) = \sum_i \theta_i g(y_i).$$

So the set of competitor values whose supremum defines $\mathcal{C}_s f(x,t)$ in `BaryEnv` is, term by term, the translate by the constant $|x|^2/(2s)$ of

$$\mathcal{A}(x) = \Big\{ \sum_i \theta_i g(y_i) : (\theta, y) \text{ a configuration of mean } x \Big\},$$

and $\mathcal{A}(x)$ is nonempty, the single-point configuration $y_1 = x$ of `IsBaryConfig` contributing the element $g(x)$. We do not yet pass to suprema: `BaryEnv` says of its own supremum that it is meaningful only together with the hypothesis of boundedness above, which Step 1 supplies.

Step 1: weak duality. Both $\Lambda(x)$ and $\Gamma(A;x)$ are suprema, and by the convention of the preamble a supremum takes the value 0 when its set is empty or unbounded above; so each of them has to be shown to be a genuine least upper bound before it is used as one. For Γ this is (344) in Step 2 below, which does it for *every* $A \in F$; the present step records what follows for Λ, and it is the only thing before Step 2 that the rest of the proof uses.

For any $A \in F$ and any configuration of mean x, the inequality $g(y_i) - A \cdot (y_i - x) \le \Gamma(A;x)$ is the least-upper-bound property of $\Gamma(A;x)$, available by (344); multiplying $g(y_i) \le \Gamma(A;x) + A \cdot (y_i - x)$ by θ_i and summing, and using $\sum_i \theta_i (y_i - x) = 0$,

$$\sum_i \theta_i g(y_i) \le \Gamma(A;x) \qquad \big((\theta, y) \text{ of mean } x,\ A \in F\big). \tag{340}$$

This bounds the *elements* of $\mathcal{A}(x)$; it is not yet a bound on a supremum. Taking $A = 0$ in (340) shows $\mathcal{A}(x)$ is bounded above by the real number $\Gamma(0;x)$, and Step 0 shows it is nonempty; so $\sup \mathcal{A}(x)$ is a genuine least upper bound and is finite, and the same holds for its translate by $|x|^2/(2s)$, whose supremum is therefore $\mathcal{C}_s f(x,t) = \sup \mathcal{A}(x) + |x|^2/(2s)$. Writing

$$\Lambda(x) := \mathcal{C}_s f(x,t) - \frac{|x|^2}{2s} = \sup \mathcal{A}(x),$$

and taking the supremum over configurations in (340),

$$\Lambda(x) \le \Gamma(A;x) \quad (A \in F), \qquad \text{hence} \qquad \Lambda(x) \le \inf_{A \in F} \Gamma(A;x). \tag{341}$$

Every later step uses (341), and (341) rests on (344); Step 2 proves the latter from (342) and (343) without using anything from this step, so the order is not circular.

Step 2: uniform localization of slopes and positions. Fix a nonempty compact set S of triples (x,t,s) with $s > 0$ and $t \in I$ — the applications below take S to be a singleton, in Steps 4 and 6, or the box of Step 5 — and let $R_0 = \max\{|x|\}$, $s_{\max} = \max\{s\}$, $s_{\min} = \min\{s\} > 0$ over S, these three maxima and minimum existing because S is nonempty and compact and the three coordinate functions are continuous (preamble). Let J be the image of S under the continuous projection $(x,t,s) \mapsto t$, so J is compact, and $J \subseteq I$ because every triple of S has $t \in I$. We claim there are $\Lambda_0 < \infty$ and $M < \infty$, depending only on S, such that for all $(x,t,s) \in S$:

(a) $\Gamma(0;x)$ is finite, and every A with $\Gamma(A;x) \le \Gamma(0;x) + 1$ satisfies $|A| \le \Lambda_0$;
(b) for every A with $|A| \le \Lambda_0 + 1$ the supremum defining $\Gamma(A;x)$ is attained, and every maximizer y satisfies $|y| \le M$.

Apply `SubquadraticGrowth` with the compact set $J \subseteq I$ and $\varepsilon = 1/(4s_{\max})$: there is $M_0 > 0$ with $|f(y,\sigma)| \le |y|^2/(4s_{\max})$ for $|y| \ge M_0$, $\sigma \in J$. For such y and $(x,t,s) \in S$, since $s \le s_{\max}$,

$$g(y) \le \frac{|y|^2}{4s} - \frac{|y|^2}{2s} = -\frac{|y|^2}{4s} \le -\frac{|y|^2}{4s_{\max}}. \tag{342}$$

The set $\{(y,\sigma) : |y| \le M_0 + R_0 + 1,\ \sigma \in J\}$ is a compact subset of Ω, so $N_0 := \max |f|$ over it is finite, and therefore

$$|g(y)| \le N_0 + \frac{(M_0 + R_0 + 1)^2}{2s_{\min}} =: N_1 \qquad (|y| \le M_0 + R_0 + 1,\ (x,t,s) \in S). \tag{343}$$

Before using Γ as a supremum at all, we record that it is one, for every slope and not only for the slopes localized in (a) and (b). Let $A \in F$ and $(x,t,s) \in S$, and consider the competitor set $\mathcal{G}(A;x) = \{g(y) - A\cdot(y-x) : y \in F\}$, whose supremum is $\Gamma(A;x)$. It is nonempty, containing $g(x)$ at $y = x$. It is bounded above, by the two bounds just proved: for $|y| \le M_0 + R_0 + 1$ we have $|y-x| \le |y| + |x| \le M_0 + 2R_0 + 1$, so by (343)

$$g(y) - A\cdot(y-x) \le N_1 + |A|\,(M_0 + 2R_0 + 1),$$

while for $|y| > M_0 + R_0 + 1$, which gives $|y| \ge M_0$, (342) and $-A\cdot(y-x) \le |A|(|y| + R_0)$ give

$$g(y) - A\cdot(y-x) \le -\frac{|y|^2}{4s_{\max}} + |A|\,|y| + |A|R_0 \le s_{\max}|A|^2 + |A|R_0,$$

the last step by completing the square: $-r^2/(4s_{\max}) + |A|r = -\big(r - 2s_{\max}|A|\big)^2/(4s_{\max}) + s_{\max}|A|^2 \le s_{\max}|A|^2$ for every real r. Hence

(344)
$$\mathcal{G}(A;x) \neq \emptyset \quad\text{and}\quad \sup \mathcal{G}(A;x) \le \max\big\{N_1 + |A|(M_0+2R_0+1),\ s_{\max}|A|^2 + |A|R_0\big\} < \infty \qquad (A \in F,\ (x,t,s) \in S)$$

so for every $A \in F$ the number $\Gamma(A;x)$ is a genuine finite least upper bound of $\mathcal{G}(A;x)$. In particular

$$g(y) - A\cdot(y-x) \le \Gamma(A;x) \qquad (y \in F,\ A \in F),$$

which is both the bound used on the elements of $\mathcal{G}(A;x)$ and, read from right to left, the licence to bound $\Gamma(A;x)$ from below by inserting any competitor y. Note that the bound in (344) is allowed to grow with $|A|$: only finiteness for each fixed A is claimed here, and that is all that is needed below. The uniformity over $|A| \le \Lambda_0 + 1$ that (b) supplies is a separate and stronger statement, consumed by (345) and the continuity argument of Step 3, and nothing here replaces it.

Four later steps use (344) for arbitrary A: Step 1's (340) and (341); the competitor inequality of the next paragraph; Step 3's reduction of the infimum over all of F to a minimum over a ball; and Step 6's identity $\Gamma(A;x) = c(A) + A\cdot x$.

By (342), (343) and (344), $\Gamma(0;x) = \sup_y g(y) \le \max\{N_1, 0\}$ and $\Gamma(0;x) \ge g(0) \ge -N_1$; so $\Gamma(0;x)$ is finite, with $|\Gamma(0;x)| \le N_1$. For $A \neq 0$ the competitor $y = x - A/|A|$, which has $|y| \le R_0 + 1 \le M_0 + R_0 + 1$, gives, again by (344),

$$\Gamma(A;x) \ge g\Big(x - \frac{A}{|A|}\Big) + |A| \ge |A| - N_1,$$

so $\Gamma(A;x) \le \Gamma(0;x) + 1 \le N_1 + 1$ forces $|A| \le 2N_1 + 1$. Thus (a) holds with $\Lambda_0 = 2N_1 + 1$.

For (b), let $|A| \le \Lambda_0 + 1$. For $|y| \ge M_0$ we get from (342)

$$g(y) - A\cdot(y-x) \le -\frac{|y|^2}{4s_{\max}} + (\Lambda_0 + 1)\big(|y| + R_0\big),$$

and the right-hand side tends to $-\infty$ as $|y| \to \infty$, uniformly over S and over such A. Hence there is $M \ge M_0 + R_0 + 1$, depending only on S, with

$$g(y) - A\cdot(y-x) < -N_1 - 1 \le g(x) - A\cdot(x-x) \qquad (|y| \ge M),$$

the last inequality by (343). Consequently the supremum over all y equals the supremum over the compact ball $\{|y| \le M\}$. On that ball the function $y \mapsto g(y) - A\cdot(y-x)$ is continuous, because $(y,t) \in \Omega$ and f is continuous on Ω; so the supremum is attained there, and by the displayed strict inequality no maximizer has $|y| \ge M$. This is (b).

Step 3: Γ is continuous and the infimum is attained. By Step 2(a), $\inf_A \Gamma(A;x) = \min_{|A| \le \Lambda_0} \Gamma(A;x)$ for $(x,t,s) \in S$: any A outside that ball has $\Gamma(A;x) > \Gamma(0;x) + 1$, while the ball contains $A = 0$. Both sides of that reduction are infima of real numbers over nonempty sets of slopes, since $\Gamma(A;x)$ is a real number for every $A \in F$ by (344). By Step 2(b),

$$\Gamma(A;x) = \max_{|y| \le M}\big\{f(y,t) - \tfrac{|y|^2}{2s} - A\cdot(y-x)\big\} \qquad \big(|A| \le \Lambda_0 + 1,\ (x,t,s) \in S\big), \tag{345}$$

the range being $|A| \le \Lambda_0 + 1$ because that is what Step 2(b) supplies and what Step 4 will need; the function inside the maximum is continuous in (A, x, t, s, y) on the compact set $\{|A| \le \Lambda_0 + 1\} \times S \times \{|y| \le M\}$.

The maximum of a continuous function over a fixed compact set is a continuous function of the remaining parameters. Indeed, let H be continuous on $P \times Q$ with P a metric space and Q nonempty and compact, and let $\pi_k \to \pi$ in P. Testing against each fixed $q \in Q$ gives $\max_{q'} H(\pi_k, q') \ge H(\pi_k, q) \to H(\pi, q)$, so $\liminf_k \max_{q'} H(\pi_k, q') \ge \max_q H(\pi, q)$. Conversely, choose maximizers q_k, which exist because Q is nonempty and compact and $H(\pi_k, \cdot)$ is continuous, and pass to a subsequence along which $q_k \to \bar{q} \in Q$, using the compactness of Q again; then $(\pi_k, q_k) \to (\pi, \bar{q})$, so by the continuity of H, $\max_{q'} H(\pi_k, q') = H(\pi_k, q_k) \to H(\pi, \bar{q}) \le \max_q H(\pi, q)$ along that subsequence. Every subsequence of the real sequence $\max_{q'} H(\pi_k, q')$ thus admits a further subsequence converging to a real number at most $\max_q H(\pi, q)$, and hence $\limsup_k \max_{q'} H(\pi_k, q') \le \max_q H(\pi, q)$. So $\pi \mapsto \max_q H(\pi, q)$ is continuous.

Applied with $P = \{|A| \le \Lambda_0 + 1\} \times S$ and $Q = \{|y| \le M\}$, this and (345) show that Γ is continuous on $\{|A| \le \Lambda_0 + 1\} \times S$. In particular, for fixed $(x, t, s) \in S$ the map $A \mapsto \Gamma(A; x)$ is continuous on the compact ball $\{|A| \le \Lambda_0\}$, so its minimum there is attained at some A_0, which by the first sentence of this step equals $\inf_{A \in F} \Gamma(A; x)$. Applying the recorded fact once more, now with $H = -\Gamma$, $P = S$ and $Q = \{|A| \le \Lambda_0\}$ — legitimate because a minimum is the negative of the maximum of the negated function, and the recorded fact is stated for maxima — gives

$$(x, t, s) \longmapsto \inf_{A \in F} \Gamma(A; x) \quad \text{is continuous on } S. \tag{346}$$

Step 4: strong duality and finite attainment, i.e. (1) and (2). Fix (x, t, s) with $s > 0$ and $t \in I$, and apply Steps 2–3 with $S = \{(x, t, s)\}$. Let A_0 be a minimizing slope and let

$$E = \Big\{ y : |y| \le M,\ g(y) - A_0 \cdot (y - x) = \Gamma(A_0; x) \Big\},$$

which is nonempty by Step 2(b) and compact, being the level set of a continuous function on a compact ball.

The argument that follows uses two facts about convexity in the finite-dimensional space F: strict separation of a point from a compact convex set, and the representation of a convex hull by finite combinations. Neither is a background item of the preamble, so both are proved here, from the preamble items *Compactness, Weierstrass and uniform continuity* and *Bolzano–Weierstrass* together with elementary linear algebra.

Claim A: strict separation from a compact convex set. Let $K \subseteq F$ be nonempty, compact and convex and let $w \in F \setminus K$. Then there are a unit vector $e \in F$ and $\delta > 0$ with $e \cdot (y - w) \ge \delta$ for every $y \in K$.

Proof of Claim A. The function $y \mapsto |y - w|$ is continuous on the nonempty compact set K, so it attains its minimum there at some $p \in K$ (preamble). Put $d = |p - w|$, so $d > 0$ because $w \notin K$. Let $y \in K$ and $0 < \tau \le 1$. By convexity $(1 - \tau)p + \tau y \in K$, so minimality of p gives

$$d^2 \le \big|p - w + \tau(y - p)\big|^2 = d^2 + 2\tau\,(p - w) \cdot (y - p) + \tau^2 |y - p|^2,$$

that is $0 \le 2(p - w) \cdot (y - p) + \tau |y - p|^2$ for every $\tau \in (0, 1]$; letting $\tau \downarrow 0$ gives $(p - w) \cdot (y - p) \ge 0$. Hence

$$(p - w) \cdot (y - w) = (p - w) \cdot (y - p) + |p - w|^2 \ge d^2.$$

Taking $e = (p - w)/d$, a unit vector, and $\delta = d > 0$, and dividing by d, gives $e \cdot (y - w) \ge \delta$ for every $y \in K$. This proves Claim A.

Claim B: finite convex combinations. With E the nonempty compact set just defined, put $m = \dim F$ and let $\operatorname{conv} E$ denote the set of all finite convex combinations $\sum_{i=1}^{n} \theta_i y_i$ with $n \ge 1$, $y_i \in E$, $\theta_i > 0$ and $\sum_i \theta_i = 1$. Then $E \subseteq \operatorname{conv} E$, the set $\operatorname{conv} E$ is convex and compact, and each of its members is, by the definition just given, the mean of a finite horizontal probability configuration supported in E in the sense of `IsBaryConfig`.

Proof of Claim B. Taking $n = 1$ and $\theta_1 = 1$ gives $E \subseteq \operatorname{conv} E$. For convexity, let $u = \sum_i \theta_i y_i$ and $u' = \sum_j \theta'_j y'_j$ be two such combinations and $\tau \in [0, 1]$; then $\tau u + (1 - \tau) u'$ is the combination with weights $\tau\theta_i$ at y_i and $(1 - \tau)\theta'_j$ at y'_j, which are nonnegative and sum to 1, and discarding

the zero weights leaves at least one term, since the weights sum to 1, and a combination of the required form.

Next, every member of $\operatorname{conv} E$ is a combination of at most $m+1$ points. Fix such a member u, choose among its representations one, $u=\sum_{i=1}^{n}\theta_i y_i$, with n least, and suppose $n>m+1$. The $n-1$ vectors y_i-y_n, $1\le i\le n-1$, lie in the m-dimensional space F and $n-1>m$, so they are linearly dependent: there are reals $\lambda_1,\dots,\lambda_{n-1}$, not all zero, with $\sum_{i<n}\lambda_i(y_i-y_n)=0$. Put $\lambda_n=-\sum_{i<n}\lambda_i$, so that

$$\sum_{i=1}^{n}\lambda_i y_i=0,\qquad \sum_{i=1}^{n}\lambda_i=0,$$

with not all λ_i zero. Were every $\lambda_i\le 0$ the second identity would force them all to vanish, so $\lambda_i>0$ for some i. Put $\tau=\min\{\theta_i/\lambda_i:\lambda_i>0\}$, which is positive because every $\theta_i>0$, and let i_0 attain it; put $\theta_i'=\theta_i-\tau\lambda_i$. Then $\theta_i'\ge 0$ for every i — for $\lambda_i>0$ by the choice of τ, and for $\lambda_i\le 0$ because $\theta_i>0$ and $-\tau\lambda_i\ge 0$ — while $\sum_i\theta_i'=1$ and $\sum_i\theta_i' y_i=u$ by the two identities, and $\theta_{i_0}'=0$. Discarding the indices with $\theta_i'=0$ represents u with fewer than n terms, contradicting minimality. So $n\le m+1$.

Consequently, fixing any $y_*\in E$ and padding a representation with $n\le m+1$ terms by the terms $\theta_i=0$, $y_i=y_*$ for $n<i\le m+1$,

(347)

$$\operatorname{conv} E=\Big\{\sum_{i=1}^{m+1}\theta_i y_i:\theta\in\Delta,\ y_1,\dots,y_{m+1}\in E\Big\},\qquad \Delta=\Big\{\theta\in\mathbb{R}^{m+1}:\theta_i\ge 0,\ \sum_i\theta_i=1\Big\};$$

the padding gives one inclusion, and for the other, discarding the zero weights of such a sum leaves a combination of the required form, with at least one term because the weights sum to 1.

Finally $\operatorname{conv} E$ is bounded, since E is compact and hence bounded by some M_E and $|\sum_i\theta_i y_i|\le\sum_i\theta_i|y_i|\le M_E$; and it is closed. For the latter, let $u_k\in\operatorname{conv} E$ with $u_k\to u$, and write $u_k=\sum_{i=1}^{m+1}\theta_i^k y_i^k$ as in (347). The sequence (θ^k) lies in the bounded set Δ and each (y_i^k) in the bounded set E, so by Bolzano–Weierstrass applied finitely many times in succession there is one subsequence along which $\theta^k\to\theta$ and $y_i^k\to y_i$ for every i; then $\theta\in\Delta$ and $y_i\in E$ because Δ and E are closed, and $u=\lim_k\sum_i\theta_i^k y_i^k=\sum_i\theta_i y_i$, so $u\in\operatorname{conv} E$ by (347). Being closed and bounded in the finite-dimensional space F, the set $\operatorname{conv} E$ is compact (preamble). This proves Claim B.

We claim $x\in\operatorname{conv} E$, with $\operatorname{conv} E$ as in Claim B. Suppose not. By Claim B that set is convex and compact, and it is nonempty because it contains the nonempty set E; so Claim A, applied with $K=\operatorname{conv} E$ and $w=x$, gives a unit vector $e\in F$ and $\delta>0$ with

$$e\cdot(y-x)\ge\delta\qquad(y\in\operatorname{conv} E),$$

in particular for $y\in E$. For $0<h\le 1$ let y_h be a maximizer for $\Gamma(A_0+he;x)$, which exists with $|y_h|\le M$ by Step 2(b), since $|A_0+he|\le\Lambda_0+1$. We claim $e\cdot(y_h-x)\ge\delta/2$ for all small $h>0$. If not, there are $h_k\downarrow 0$ with $e\cdot(y_{h_k}-x)<\delta/2$; passing to a subsequence, $y_{h_k}\to\bar y$ with $|\bar y|\le M$, and then

$$\Gamma(A_0+h_ke;x)=g(y_{h_k})-(A_0+h_ke)\cdot(y_{h_k}-x)\longrightarrow g(\bar y)-A_0\cdot(\bar y-x),$$

using continuity of g on $\{|y|\le M\}$, while $\Gamma(A_0+h_ke;x)\to\Gamma(A_0;x)$ by the continuity of Γ in A on $\{|A|\le\Lambda_0+1\}$ from Step 3, a range that contains every A_0+he with $0<h\le 1$ because $|A_0|\le\Lambda_0$ and $|e|=1$. Hence $g(\bar y)-A_0\cdot(\bar y-x)=\Gamma(A_0;x)$, i.e. $\bar y\in E$, so $e\cdot(\bar y-x)\ge\delta$, contradicting $e\cdot(y_{h_k}-x)<\delta/2$ in the limit.

So for small $h>0$,

$$\Gamma(A_0+he;x)=g(y_h)-A_0\cdot(y_h-x)-h\,e\cdot(y_h-x)\le\Gamma(A_0;x)-\frac{h\delta}{2}<\Gamma(A_0;x),$$

contradicting the minimality of A_0. Therefore $x\in\operatorname{conv} E$.

Since $x\in\operatorname{conv} E$, the definition of that set in Claim B gives $n\ge 1$, points $y_1,\dots,y_n\in E$ and weights $\theta_i>0$ with $\sum_i\theta_i=1$ and $\sum_i\theta_i y_i=x$. No separate convex-hull theorem is invoked: membership in $\operatorname{conv} E$ is by definition the existence of such a combination, and Claim B was

needed only for the compactness and convexity that Claim A consumed. So (θ, y) is a finite horizontal probability configuration with mean x in the sense of `IsBaryConfig`, and since each $y_i \in E$,

$$\sum_i \theta_i g(y_i) = \sum_i \theta_i \big(\Gamma(A_0; x) + A_0 \cdot (y_i - x)\big) = \Gamma(A_0; x).$$

Combining with Step 1 and (341),

$$\Gamma(A_0; x) = \sum_i \theta_i g(y_i) \le \Lambda(x) \le \inf_A \Gamma(A; x) = \Gamma(A_0; x),$$

so all are equal. This gives (2), with the minimizing slope A_0, and shows that the configuration (θ, y) attains the supremum in `BaryEnv`, which, by the set identity of Step 0 and the finiteness recorded in Step 1, is (1).

Step 5: continuity, i.e. (3). Let $s_0 > 0$ and $z_0 = (x_0, t_0)$ with $t_0 \in I$. Choose $\rho > 0$ so small that $\rho < s_0$ and $[t_0 - \rho, t_0 + \rho] \subseteq I$, and let $S = \{(x, t, s) : |x - x_0| \le \rho,\ |t - t_0| \le \rho,\ |s - s_0| \le \rho\}$, a compact set of admissible triples. By Steps 3 and 4, on S we have $\mathcal{C}_s f(x, t) = \inf_A \Gamma(A; x) + |x|^2/(2s)$, and the first term is continuous on S by (346) while the second is continuous there since $s > 0$. Hence the restriction of $(s, z) \mapsto \mathcal{C}_s f(z)$ to S is continuous at (s_0, z_0). And S is a neighbourhood of that point, containing every (x, t, s) whose three coordinate offsets from (x_0, t_0, s_0) are at most ρ, so (s_0, z_0) is an interior point of S and continuity of the restriction there is continuity of $(s, z) \mapsto \mathcal{C}_s f(z)$ at (s_0, z_0), which is (3).

Step 6: concavity, i.e. (4). Fix $s > 0$ and $t \in I$, and let $A \in F$. Apply Step 2 with the singleton $S = \{(0, t, s)\}$, which is compact and consists of one admissible triple: the case $x = 0$ of (344) says that $\{g(y) - A \cdot y : y \in F\}$ is nonempty and bounded above, so

$$c(A) := \sup_y \{g(y) - A \cdot y\}$$

is a real number. For every $x \in F$ the set $\{g(y) - A \cdot (y - x) : y \in F\}$ is the translate of that set by the constant $A \cdot x$, so it is nonempty and bounded above as well and its supremum is $c(A) + A \cdot x$; that is,

$$\Gamma(A; x) = c(A) + A \cdot x,$$

a real-valued affine function of x for each fixed A. By Steps 2–4, $\Lambda(x) = \inf_A \Gamma(A; x)$ is finite for every x. An infimum of a family of real-valued affine functions of x is concave wherever it is finite: for $x_0, x_1 \in F$, $\tau \in [0, 1]$ and any A,

$$c(A) + A \cdot \big((1-\tau)x_0 + \tau x_1\big) = (1-\tau)\big(c(A) + A \cdot x_0\big) + \tau\big(c(A) + A \cdot x_1\big) \ge (1-\tau)\Lambda(x_0) + \tau\Lambda(x_1),$$

and taking the infimum over A gives $\Lambda((1-\tau)x_0 + \tau x_1) \ge (1-\tau)\Lambda(x_0) + \tau\Lambda(x_1)$. Since $\Lambda(x) = \mathcal{C}_s f(x, t) - |x|^2/(2s)$ by Step 1, this is (4).

Step 7: measures, i.e. (5). Fix $s > 0$ and $z = (x, t)$ with $t \in I$, and let A_0 be as in Step 4, so that

$$g(y) \le \Gamma(A_0; x) + A_0 \cdot (y - x) = \Lambda(x) + A_0 \cdot (y - x) \qquad (y \in F). \tag{348}$$

Let μ be a Borel probability measure on F with $y \mapsto |y - x|^2$ and $y \mapsto f(y, t)$ both μ-integrable, with $y \mapsto y - x$ μ-integrable, and with $\int (y - x)\, d\mu = 0$. Then $y \mapsto |y|^2 = |y - x|^2 + 2(y - x) \cdot x + |x|^2$ is μ-integrable, so g is μ-integrable and $\int A_0 \cdot (y - x)\, d\mu = A_0 \cdot \int (y - x)\, d\mu = 0$ by linearity of the Bochner integral. Integrating (348),

$$\int g\, d\mu \le \Lambda(x).$$

On the other hand, integrating the identity $|y|^2 = |y - x|^2 + 2(y - x) \cdot x + |x|^2$ and using $\int (y - x)\, d\mu = 0$ gives $\int |y|^2 d\mu = \int |y - x|^2 d\mu + |x|^2$, whence

$$\int g\, d\mu = \int f(y, t)\, d\mu - \frac{\int |y|^2\, d\mu}{2s} = \int f(y, t)\, d\mu - \frac{\int |y - x|^2\, d\mu}{2s} - \frac{|x|^2}{2s}.$$

Comparing the last two displays and using $\Lambda(x) = \mathcal{C}_s f(x,t) - |x|^2/(2s)$ gives

$$\int f(y,t)\,d\mu - \frac{\int |y-x|^2\,d\mu}{2s} \le \mathcal{C}_s f(x,t),$$

which is (5). □

2.81. `BaryEnvSandwich`.

Lemma 2.81 (The barycentric sandwich)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let $\kappa \ge 0$. Assume that f is continuous on $\Omega = \{(x,t) : t \in I\}$, that f satisfies the horizontal Jensen bound with constant κ on I in the sense of `HasJensenBound`, and that f has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Then, for every $s > 0$ and every $z = (x,t)$ with $t \in I$,*

$$0 \le \mathcal{C}_s f(x,t) - f(x,t) \le \frac{\kappa^2 s}{2}, \tag{349}$$

with $\mathcal{C}$ the barycentric envelope of `BaryEnv`.

Proof. Fix $s > 0$ and $z = (x,t)$ with $t \in I$. Write

$$\mathcal{V} = \left\{ \sum_i \theta_i f(y_i,t) - \frac{r(\theta,y)^2}{2s} \right\}$$

for the set of competitor values in `BaryEnv`, where $r(\theta,y) = \left(\sum_i \theta_i |y_i - x|^2\right)^{1/2}$ is the radius of `BaryRadius` and the configurations (θ, y) have mean x in the sense of `IsBaryConfig`, so that $\mathcal{C}_s f(z) = \sup \mathcal{V}$.

Step 1: $\mathcal{V}$ is nonempty and bounded above. The single-point configuration $n = 1$, $\theta_1 = 1$, $y_1 = x$ has mean x and radius 0, and contributes the value $f(x,t)$ to $\mathcal{V}$; so $\mathcal{V} \neq \emptyset$ and

$$f(x,t) \le \sup \mathcal{V}. \tag{350}$$

For boundedness above, apply `SubquadraticGrowth` with the compact set $J = \{t\} \subseteq I$ and $\varepsilon = 1/(16s) > 0$: there is $M_0 > 0$ with $|f(y,t)| \le |y|^2/(16s)$ whenever $|y| \ge M_0$. Put $M = \max\{M_0,\ 2|x|\}$. The set $\{(y,t) : |y| \le M\}$ is compact and contained in Ω, because $t \in I$, and f is continuous on Ω; hence

$$N = \sup_{|y| \le M} |f(y,t)|$$

is finite.

Let (θ, y) be a configuration with mean x. Split the indices according to whether $|y_i| \le M$ or $|y_i| > M$. For $|y_i| > M$ we have $|y_i| \ge 2|x|$, hence $|y_i - x| \ge |y_i| - |x| \ge |y_i|/2$, so

$$f(y_i,t) \le \frac{|y_i|^2}{16s} \le \frac{4|y_i - x|^2}{16s} = \frac{|y_i - x|^2}{4s}.$$

For $|y_i| \le M$ we have $f(y_i,t) \le N$. Therefore

$$\sum_i \theta_i f(y_i,t) \le N + \sum_i \theta_i \frac{|y_i - x|^2}{4s} = N + \frac{r(\theta,y)^2}{4s},$$

using $\sum_i \theta_i = 1$ and bounding each term by the larger of the two estimates. Consequently every element of $\mathcal{V}$ is at most

$$N + \frac{r(\theta,y)^2}{4s} - \frac{r(\theta,y)^2}{2s} \le N,$$

so $\mathcal{V}$ is bounded above by N and $\sup \mathcal{V}$ is a real number.

Step 2: the upper bound. Let (θ, y) be any configuration with mean x, and let $r = r(\theta,y) \ge 0$. The Jensen bound of `HasJensenBound` at the time $t \in I$ gives

$$\sum_i \theta_i f(y_i,t) - f(x,t) \le \kappa r.$$

Therefore the corresponding element of $\mathcal{V}$ satisfies

$$\sum_i \theta_i f(y_i,t) - \frac{r^2}{2s} - f(x,t) \le \kappa r - \frac{r^2}{2s}.$$

For every real r the algebraic identity

$$\kappa r - \frac{r^2}{2s} = \frac{\kappa^2 s}{2} - \frac{(r-\kappa s)^2}{2s}$$

holds, as one checks by expanding the square and using $s \neq 0$. Since $(r-\kappa s)^2 \ge 0$ and $s > 0$, this gives $\kappa r - r^2/(2s) \le \kappa^2 s/2$, with equality exactly when $r = \kappa s$. Hence every element of $\mathcal{V}$ is at most $f(x,t) + \kappa^2 s/2$, and so is its supremum:

$$\mathcal{C}_s f(x,t) \le f(x,t) + \frac{\kappa^2 s}{2}.$$

Together with (350) this is (349). □

2.82. `BaryEnvStability`.

Lemma 2.82 (Stability of the barycentric envelope)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open and let $\Omega = \{(x,t) : t \in I\}$. Let f and $f_1, f_2, \dots$ be real functions on space-time, continuous on Ω, with $f_j \to f$ uniformly on every compact subset of Ω, with f of subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`, and with the sequence having the common growth control of `UnifSubquadraticGrowth` on I. Let $s > 0$. Then $\mathcal{C}_s f_j \to \mathcal{C}_s f$ uniformly on every compact set of centers $z = (x,t)$ with $t \in I$, where $\mathcal{C}$ is the barycentric envelope of `BaryEnv`.*

Proof. Write $\Omega = \{(y,\sigma) : \sigma \in I\}$, an open set on which f and every f_j are continuous, and for a real function g on space-time and a time $t \in I$ put

$$g_t(y) = g(y,t) - \frac{|y|^2}{2s}, \qquad \Gamma_g(A;x,t) = \sup_{y \in F}\bigl\{g_t(y) - A\cdot(y-x)\bigr\} \qquad (A,x \in F).$$

By `BaryEnvDuality`, part (2), applied to f and to each f_j — all of which satisfy its hypotheses, namely continuity on Ω and the growth condition of `SubquadraticGrowth` on I, the latter for f_j because `UnifSubquadraticGrowth` for the sequence implies it for each member — we have, for every $t \in I$, every $x \in F$ and $g \in \{f\} \cup \{f_j\}$,

$$\mathcal{C}_s g(x,t) - \frac{|x|^2}{2s} = \inf_{A \in F} \Gamma_g(A;x,t), \tag{351}$$

the infimum being attained.

Step 1: uniform localization of slopes and positions. Let $\mathcal{K}$ be a compact set of centers $z = (x,t)$ with $t \in I$, put $R_0 = \max\{|x| : (x,t) \in \mathcal{K}\} < \infty$ and let J be the set of time coordinates of $\mathcal{K}$, a compact subset of I.

Apply `UnifSubquadraticGrowth` with J and $\varepsilon = 1/(4s)$: there is $M_1 > 0$ with $|f_j(y,t)| \le |y|^2/(4s)$ for all j, all $|y| \ge M_1$ and all $t \in J$; apply `SubquadraticGrowth` with the same data to get $M_2 > 0$ with the same bound for f. Put $M_3 = \max\{M_1, M_2, R_0 + 1, 1\}$. Then, for $g \in \{f\} \cup \{f_j\}$, $t \in J$ and $|y| \ge M_3$,

$$g_t(y) \le \frac{|y|^2}{4s} - \frac{|y|^2}{2s} = -\frac{|y|^2}{4s} \le 0. \tag{352}$$

The set $\mathcal{C}_3 = \{(y,t) : |y| \le M_3,\ t \in J\}$ is a compact subset of Ω, and the quantity

$$N_0 = \sup_{g \in \{f\}\cup\{f_j\}} \max_{\mathcal{C}_3} |g|$$

is finite. Indeed f is continuous on Ω, hence bounded on the compact $\mathcal{C}_3$; $f_j \to f$ uniformly on $\mathcal{C}_3$, so there is j_0 with $\sup_{\mathcal{C}_3}|f_j - f| \le 1$ for $j \ge j_0$; and each of the finitely many remaining f_j is continuous, hence bounded on $\mathcal{C}_3$. Consequently, for $g \in \{f\} \cup \{f_j\}$, $t \in J$ and $|y| \le M_3$,

$$|g_t(y)| \le N_0 + \frac{M_3^2}{2s} =: N_1. \tag{353}$$

By (352) and (353), for $z=(x,t)\in\mathcal{K}$,

$$\Gamma_g(0;x,t)=\sup_y g_t(y)\le \max\{N_1,0\}=N_1,$$

(enlarging N_1 so that $N_1\ge 0$), and for $A\neq 0$ the competitor $y=x-A/|A|$, which satisfies $|y|\le R_0+1\le M_3$, gives by (353)

$$\Gamma_g(A;x,t)\ge g_t\Big(x-\frac{A}{|A|}\Big)+|A|\ge |A|-N_1.$$

Hence every A with $\Gamma_g(A;x,t)\le\Gamma_g(0;x,t)+1$ satisfies $|A|\le 2N_1+1=:\Lambda_0$, and therefore

$$\inf_{A\in F}\Gamma_g(A;x,t)=\inf_{|A|\le\Lambda_0}\Gamma_g(A;x,t)\qquad (z\in\mathcal{K},\ g\in\{f\}\cup\{f_j\}),\tag{354}$$

because $A=0$ lies in that ball while any A outside it has $\Gamma_g(A;x,t)>\Gamma_g(0;x,t)+1$.

For $|A|\le\Lambda_0$, $z\in\mathcal{K}$ and $|y|\ge M_3$, (352) gives

$$g_t(y)-A\cdot(y-x)\le -\frac{|y|^2}{4s}+\Lambda_0\big(|y|+R_0\big),$$

whose right side tends to $-\infty$ as $|y|\to\infty$ and does not depend on g or z. Hence there is $M\ge M_3$, depending only on Λ_0, R_0, s and N_1, with

$$g_t(y)-A\cdot(y-x)<-N_1\le g_t(x)-A\cdot(x-x)\qquad (|y|\ge M),$$

the last inequality by (353) and $|x|\le R_0\le M_3$. Therefore, for all $g\in\{f\}\cup\{f_j\}$, all $z\in\mathcal{K}$ and all $|A|\le\Lambda_0$,

$$\Gamma_g(A;x,t)=\max_{|y|\le M}\big\{g_t(y)-A\cdot(y-x)\big\},\tag{355}$$

the maximum existing because $y\mapsto g_t(y)-A\cdot(y-x)$ is continuous on the compact ball $\{|y|\le M\}$, where $(y,t)\in\Omega$.

Step 2: conclusion. Let $\delta_j=\sup_{\mathcal{C}}|f_j-f|$, where $\mathcal{C}=\{(y,t):|y|\le M,\ t\in J\}$ is a compact subset of Ω; by hypothesis $\delta_j\to 0$. For $z=(x,t)\in\mathcal{K}$ and $|A|\le\Lambda_0$, the two functions of y inside the maxima in (355) for f_j and for f differ by at most δ_j on $\{|y|\le M\}$, the quadratic and linear terms being identical. Using $|\sup_Q G-\sup_Q H|\le\sup_Q|G-H|$ for bounded functions on a nonempty set Q,

$$\big|\Gamma_{f_j}(A;x,t)-\Gamma_f(A;x,t)\big|\le\delta_j\qquad (|A|\le\Lambda_0,\ z\in\mathcal{K}).$$

Taking infima over $|A|\le\Lambda_0$, and using the same inequality for infima, $\big|\inf_Q G-\inf_Q H\big|\le\sup_Q|G-H|$, we get from (354)

$$\Big|\inf_A\Gamma_{f_j}(A;x,t)-\inf_A\Gamma_f(A;x,t)\Big|\le\delta_j\qquad (z\in\mathcal{K}).$$

By (351) the terms $|x|^2/(2s)$ cancel, so

$$\big|\mathcal{C}_sf_j(x,t)-\mathcal{C}_sf(x,t)\big|\le\delta_j\qquad (z\in\mathcal{K}),$$

and $\delta_j\to 0$. Hence $\mathcal{C}_sf_j\to\mathcal{C}_sf$ uniformly on $\mathcal{K}$. □

2.83. `BaryTestDirections`.

Lemma 2.83 (The common-time directions of the barycentric test function)**.** *Let F be a finite-dimensional real inner-product space and let $n\in\mathbb{N}$. Let $\theta:\{1,\dots,n\}\to\mathbb{R}$ satisfy $\theta_i>0$ for every i and $\sum_i\theta_i=1$, let $y^0=(y_i^0)_i$ be points of F with $\sum_i\theta_iy_i^0=x_0$, let $t_0>0$ and $c\in\mathbb{R}$. Let φ be a real function on the space-time $\ell^2(F\times\mathbb{R})$ which is C^2 at (x_0,t_0). On the product space-time $\ell^2\big(\ell^2(F^n)\times\mathbb{R}\big)$, write a point as $W=(Y,t)$ with $Y=(y_i)_i$, and let Φ be a real function there satisfying, for every such W,*

$$\Phi(Y,t)=\varphi\Big(\sum_j\theta_jy_j,\ t\Big)+ct+\frac{\sum_j\theta_j\big|y_j-\sum_k\theta_ky_k\big|^2}{2t}.$$

Then, writing $Y^0 = (y_i^0)_i$ *and* $\nabla_{y_i}\Phi(Y^0, t_0)$ *for the* i*-th block of the spatial part of* $\nabla\Phi(Y^0, t_0)$*, we have for every* i

$$\theta_i^{-1}\,\nabla_{y_i}\Phi\big(Y^0, t_0\big) = \nabla_x\varphi(x_0, t_0) + \frac{y_i^0 - x_0}{t_0},$$

where $\nabla_x\varphi(x_0, t_0)$ *denotes the spatial part of* $\nabla\varphi(x_0, t_0)$*.*

Proof. Write $P = \ell^2(F^n)$ for the spatial factor of the product space-time, and introduce the continuous linear maps

$$\pi_1(W) = Y, \qquad \pi_2(W) = t, \qquad X(Y) = \sum_j \theta_j y_j, \qquad L(W) = \big(X(Y),\, t\big),$$

so that L maps the product space-time to $\ell^2(F \times \mathbb{R})$, together with, for each index j, the continuous linear map $M_j(W) = y_j - X(Y)$ from the product space-time to F. Each of these is a composition, difference or finite linear combination of coordinate projections, hence continuous and linear. Set

$$N(W) = \sum_j \theta_j \,\big|M_j(W)\big|^2,$$

so that the hypothesis on Φ reads

$$\Phi(W) = \Big(\varphi\big(L(W)\big) + c\,\pi_2(W)\Big) + N(W)\cdot\big(2\pi_2(W)\big)^{-1}. \tag{356}$$

Write $W_0 = (Y^0, t_0)$. Since $\sum_j \theta_j y_j^0 = x_0$ we have

$$L(W_0) = (x_0, t_0), \qquad \pi_2(W_0) = t_0, \qquad M_j(W_0) = y_j^0 - x_0. \tag{357}$$

Also, using $\sum_j \theta_j = 1$ and $\sum_j \theta_j y_j^0 = x_0$,

$$\sum_j \theta_j\big(y_j^0 - x_0\big) = x_0 - x_0 = 0. \tag{358}$$

Step 1: the derivative of Φ *at* W_0*.* Because φ is C^2 at $(x_0, t_0) = L(W_0)$ it is differentiable there, so by the chain rule for a composition with a continuous linear map, $W \mapsto \varphi(L(W))$ is differentiable at W_0 with derivative $D\varphi(L(W_0))\circ L$. The map $W \mapsto c\,\pi_2(W)$ is continuous linear, hence differentiable with derivative $c\,\pi_2$. For each j the map $W \mapsto |M_j(W)|^2$ is differentiable with derivative $\xi \mapsto 2\langle M_j(W_0), M_j\xi\rangle$, since $u \mapsto |u|^2$ has derivative $v \mapsto 2\langle u, v\rangle$ and M_j is continuous linear; multiplying by the constant θ_j and summing over the finitely many j,

$$DN(W_0)[\xi] = \sum_j \theta_j \cdot 2\big\langle M_j(W_0),\, M_j\xi\big\rangle. \tag{359}$$

Finally $t \mapsto (2t)^{-1}$ is differentiable at t_0, because $2t_0 \neq 0$; composing with the continuous linear π_2, the map $W \mapsto (2\pi_2(W))^{-1}$ is differentiable at W_0 with derivative of the form $\mu\,\pi_2$ for some real μ. The product rule applied to (356) therefore gives that Φ is differentiable at W_0 with

$$D\Phi(W_0)[\xi] = D\varphi\big(L(W_0)\big)\big[L\xi\big] + c\,\pi_2\xi + N(W_0)\,\mu\,\pi_2\xi + \frac{DN(W_0)[\xi]}{2t_0}. \tag{360}$$

Step 2: the test direction. Fix an index i and a vector $a \in F$, let $e \in F^n$ be the tuple with $e_i = a$ and $e_j = 0$ for $j \neq i$, and put $\xi = (e, 0)$ in the product space-time. Then $\pi_2\xi = 0$, and since $\sum_j \theta_j e_j = \theta_i a$,

$$L\xi = \big(\theta_i a,\, 0\big), \qquad M_j\xi = e_j - \theta_i a.$$

Step 3: evaluating (360) *on* ξ*.* The two terms carrying the factor $\pi_2\xi$ vanish. For the first term, by definition of the gradient and (357),

$$D\varphi\big(L(W_0)\big)\big[L\xi\big] = \big\langle\nabla\varphi(x_0, t_0),\ (\theta_i a, 0)\big\rangle = \big\langle\nabla_x\varphi(x_0, t_0),\ \theta_i a\big\rangle = \theta_i\big\langle\nabla_x\varphi(x_0, t_0),\ a\big\rangle,$$

the middle equality because the inner product of the ℓ^2 product splits as the sum of the inner products of the two components and the second component of ξ is 0. For the last term, by (359), (357) and Step 2,

$$DN(W_0)[\xi] = 2\sum_j \theta_j\langle y_j^0 - x_0,\ e_j - \theta_i a\rangle = 2\sum_j \theta_j\langle y_j^0 - x_0,\ e_j\rangle - 2\theta_i \sum_j \theta_j\langle y_j^0 - x_0,\ a\rangle.$$

In the first sum only the index $j = i$ contributes, because $e_j = 0$ otherwise, and it contributes $\theta_i\langle y_i^0 - x_0, a\rangle$. The second sum equals $\big\langle \sum_j \theta_j(y_j^0 - x_0),\, a\big\rangle$, which vanishes by (358). Hence $DN(W_0)[\xi] = 2\theta_i\langle y_i^0 - x_0, a\rangle$ and

$$D\Phi(W_0)[\xi] = \theta_i\big\langle \nabla_x\varphi(x_0,t_0), a\big\rangle + \frac{\theta_i\big\langle y_i^0 - x_0, a\big\rangle}{t_0} = \Big\langle \theta_i\Big(\nabla_x\varphi(x_0,t_0) + \frac{y_i^0 - x_0}{t_0}\Big),\ a\Big\rangle.$$

Step 4: reading off the block. By definition of the gradient, $D\Phi(W_0)[\xi] = \langle \nabla\Phi(W_0), \xi\rangle$. The inner product of the product space-time splits as the sum over the two components, and the time component of ξ is 0; the inner product of $P = \ell^2(F^n)$ is the sum over blocks, and only the i-th block of e is nonzero. Hence

$$\big\langle \nabla\Phi(W_0), \xi\big\rangle = \big\langle \nabla_{y_i}\Phi(W_0),\ a\big\rangle.$$

Comparing with Step 3,

$$\big\langle \nabla_{y_i}\Phi(W_0),\ a\big\rangle = \Big\langle \theta_i\Big(\nabla_x\varphi(x_0,t_0) + \frac{y_i^0 - x_0}{t_0}\Big),\ a\Big\rangle$$

for every $a \in F$. As the inner product of F separates points,

$$\nabla_{y_i}\Phi(W_0) = \theta_i\Big(\nabla_x\varphi(x_0,t_0) + \frac{y_i^0 - x_0}{t_0}\Big),$$

and dividing by $\theta_i > 0$ gives the assertion. □

2.84. `SecondFDerivCompLinear.`

Lemma 2.84 (Second derivative of a composition with a linear map)**.** *Let P and Q be real normed spaces, let $L : P \to Q$ be a continuous linear map, let $f : Q \to \mathbb{R}$, let $W \in P$, and assume that f is C^2 near $L(W)$. Then for all $\xi, \eta \in P$,*

$$D^2(f\circ L)(W)[\xi,\eta] = D^2 f\big(L(W)\big)\big[L\xi,\ L\eta\big].$$

Proof. Write CL for the map $g \mapsto g \circ L$ sending a continuous linear functional on Q to one on P. It is linear in g, and it is continuous because $\|g\circ L\| \le \|g\|\,\|L\|$; so CL is itself a continuous linear map from the space of continuous linear functionals on Q to that on P.

Step 1: the first derivative of $f \circ L$ near W. Since f is C^2 near $L(W)$, it is in particular C^1, hence differentiable, at every point of some neighbourhood U of $L(W)$. As L is continuous, the preimage of U is a neighbourhood of W; so for every U' in a neighbourhood of W the function f is differentiable at $L(U')$. For such U' the chain rule for the composition of the differentiable f at $L(U')$ with the continuous linear L — whose derivative at any point is L itself — gives

$$D(f\circ L)(U') = Df\big(L(U')\big)\circ L = CL\Big(Df\big(L(U')\big)\Big).$$

Thus the two functions $U' \mapsto D(f\circ L)(U')$ and $U' \mapsto CL\big(Df(L(U'))\big)$ agree on a neighbourhood of W, and therefore have the same derivative at W:

$$D^2(f\circ L)(W) = D\Big(U' \mapsto CL\big(Df(L(U'))\big)\Big)(W). \tag{361}$$

Step 2: differentiating once more. Because f is C^2 near $L(W)$, the map Df is differentiable at $L(W)$, with derivative $D^2 f(L(W))$; that is,

$$y \longmapsto Df(y) \quad \text{has derivative} \quad D^2 f\big(L(W)\big) \quad \text{at } L(W).$$

Composing with the continuous linear L on the right, the map $U' \mapsto Df(L(U'))$ has derivative $D^2f(L(W)) \circ L$ at W. Composing with the continuous linear CL on the left, the map $U' \mapsto CL\big(Df(L(U'))\big)$ has derivative $CL \circ \big(D^2f(L(W)) \circ L\big)$ at W. With (361) this gives

$$D^2(f \circ L)(W) = CL \circ \Big(D^2f\big(L(W)\big) \circ L\Big).$$

Step 3: evaluating. Apply the displayed identity to $\xi \in P$ and then to $\eta \in P$. The inner composition sends ξ to $D^2f(L(W))(L\xi)$, a continuous linear functional on Q; applying CL to it gives $D^2f(L(W))(L\xi) \circ L$, a continuous linear functional on P; and evaluating that at η gives $D^2f(L(W))(L\xi)(L\eta)$, i.e.

$$D^2(f \circ L)(W)[\xi, \eta] = D^2f\big(L(W)\big)\big[L\xi,\ L\eta\big],$$

which is the assertion. □

2.85. `BaryVarianceAffineAlongMotion`.

Lemma 2.85 (The variance penalty is affine along the common-time motion)**.** *Let F be a finite-dimensional real inner-product space, let $n \in \mathbb{N}$, let $\theta : \{1, \dots, n\} \to \mathbb{R}$ satisfy $\sum_i \theta_i = 1$, let $y^0 = (y_i^0)_i$ be points of F with weighted mean $\sum_i \theta_i y_i^0 = x_0$, let $t_0 > 0$ and let $p \in F$. Put*

$$q_i = p + \frac{y_i^0 - x_0}{t_0}, \qquad y_i(h) = y_i^0 + hq_i,$$

and write $R^2 = \sum_i \theta_i |y_i^0 - x_0|^2$. Then for every real h with $t_0 + h > 0$,

$$\frac{\sum_i \theta_i \big|y_i(h) - \sum_j \theta_j y_j(h)\big|^2}{2(t_0 + h)} = \frac{R^2}{2t_0} + \frac{R^2}{2t_0^2}\, h,$$

an affine function of h.

Proof. Since $t_0 > 0$ and $t_0 + h > 0$, both t_0 and $t_0 + h$ are nonzero, so all the divisions below are legitimate.

Step 1: the weighted mean of the offsets vanishes. Using $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i^0 = x_0$,

$$\sum_i \theta_i \big(y_i^0 - x_0\big) = \sum_i \theta_i y_i^0 - \Big(\sum_i \theta_i\Big) x_0 = x_0 - x_0 = 0.$$

Step 2: the weighted mean of the velocities is p. By the definition of q_i, linearity of the weighted sum, $\sum_i \theta_i = 1$, and Step 1,

$$\sum_i \theta_i q_i = \Big(\sum_i \theta_i\Big) p + \frac{1}{t_0} \sum_i \theta_i \big(y_i^0 - x_0\big) = p + \frac{1}{t_0} \cdot 0 = p.$$

Step 3: the running barycentre. Again by linearity, using $\sum_i \theta_i y_i^0 = x_0$ and Step 2,

$$\sum_j \theta_j y_j(h) = \sum_j \theta_j y_j^0 + h \sum_j \theta_j q_j = x_0 + hp.$$

Step 4: every offset dilates by the same factor. Subtracting Step 3 from the definition of $y_i(h)$ and using $q_i - p = (y_i^0 - x_0)/t_0$,

$$y_i(h) - \sum_j \theta_j y_j(h) = \big(y_i^0 - x_0\big) + h\big(q_i - p\big) = \big(y_i^0 - x_0\big) + \frac{h}{t_0}\big(y_i^0 - x_0\big) = \Big(1 + \frac{h}{t_0}\Big)\big(y_i^0 - x_0\big).$$

The factor $1 + h/t_0$ is the same for every index i: this is the point of the calculation.

Step 5: the variance scales by the square of that factor. Taking norms in Step 4, for each i

$$\big|y_i(h) - \textstyle\sum_j \theta_j y_j(h)\big|^2 = \Big(1 + \frac{h}{t_0}\Big)^2 \big|y_i^0 - x_0\big|^2,$$

since $|\lambda v|^2 = \lambda^2 |v|^2$ for a real scalar λ. Multiplying by θ_i and summing, and pulling the common factor out of the sum,

$$\sum_i \theta_i \big|y_i(h) - \textstyle\sum_j \theta_j y_j(h)\big|^2 = \Big(1 + \frac{h}{t_0}\Big)^2 R^2.$$

Step 6: the collapse. Since $1 + h/t_0 = (t_0 + h)/t_0$,

$$\frac{\big(1 + h/t_0\big)^2 R^2}{2(t_0 + h)} = \frac{R^2(t_0 + h)^2/t_0^2}{2(t_0 + h)} = \frac{R^2(t_0 + h)}{2t_0^2} = \frac{R^2}{2t_0} + \frac{R^2}{2t_0^2}\, h,$$

where the cancellation of one factor $t_0 + h$ uses $t_0 + h \neq 0$. Combining with Step 5 gives the assertion. □

2.86. `BaryTestSecondVariation`.

Lemma 2.86 (The lifted barycentric test, and its second variation)**.** *Let F be a finite-dimensional real inner-product space and let $n \in \mathbb{N}$. Let $\theta : \{1, \dots, n\} \to \mathbb{R}$ satisfy $\sum_i \theta_i = 1$, let $y^0 = (y_i^0)_i$ be points of F with $\sum_i \theta_i y_i^0 = x_0$, let $t_0 > 0$ and $c \in \mathbb{R}$, and let φ be a real function on the space-time $\ell^2(F \times \mathbb{R})$ which is C^2 at (x_0, t_0). On the product space-time $\ell^2\big(\ell^2(F^n) \times \mathbb{R}\big)$, write a point as $W = (Y, t)$ with $Y = (y_i)_i$, and let Φ be a real function there satisfying, for every such W,*

$$\Phi(Y, t) = \varphi\Big(\sum_j \theta_j y_j,\ t\Big) + ct + \frac{\sum_j \theta_j \big|y_j - \sum_k \theta_k y_k\big|^2}{2t}.$$

Put $Y^0 = (y_i^0)_i$, $p = \nabla_x \varphi(x_0, t_0)$ for the spatial part of $\nabla\varphi(x_0, t_0)$, and

$$\zeta = \Big(\big(p + \tfrac{y_i^0 - x_0}{t_0}\big)_{i=1}^n,\ 1\Big).$$

Then Φ is C^2 at (Y^0, t_0), and

$$D^2\Phi\big(Y^0, t_0\big)\big[\zeta, \zeta\big] = F[\varphi](x_0, t_0),$$

with F the limiting operator of `LimitOp`.

Proof. Write $P = \ell^2(F^n)$ and introduce the continuous linear maps

$$\pi_1(W) = Y, \qquad \pi_2(W) = t, \qquad X(Y) = \sum_j \theta_j y_j, \qquad L(W) = \big(X(Y),\, t\big), \qquad M_j(W) = y_j - X(Y),$$

each built from coordinate projections and finite linear combinations. Write $W_0 = (Y^0, t_0)$ and $q_i = p + (y_i^0 - x_0)/t_0$, so that $\zeta = \big((q_i)_i, 1\big)$. Since $\sum_j \theta_j y_j^0 = x_0$ we have $L(W_0) = (x_0, t_0)$ and $\pi_2(W_0) = t_0$. Set

$$g(W) = \varphi\big(L(W)\big) + c\,\pi_2(W), \qquad \psi(W) = \frac{\sum_j \theta_j |M_j(W)|^2}{2\pi_2(W)},$$

so that the hypothesis on Φ says exactly $\Phi = g + \psi$.

Step 1: regularity. The map L is continuous linear and $L(W_0) = (x_0, t_0)$, where φ is C^2; hence $\varphi \circ L$ is C^2 at W_0. The map $W \mapsto c\,\pi_2(W)$ is continuous linear, hence C^∞. So g is C^2 at W_0. For ψ: each M_j is continuous linear and $u \mapsto |u|^2$ is C^∞, so the numerator is C^∞; the denominator $W \mapsto 2\pi_2(W)$ is continuous linear, hence C^∞, and it takes the value $2t_0 \neq 0$ at W_0; so the quotient is C^2 at W_0. Therefore $\Phi = g + \psi$ is C^2 at W_0, which is the first assertion.

Step 2: the two summands other than $\varphi \circ L$ are affine along the motion. Consider the straight motion $h \mapsto W_0 + h\zeta$. In coordinates it is $\big(Y^0 + hq,\ t_0 + h\big)$ with $q = (q_i)_i$, because addition and scalar multiplication on the ℓ^2 product are performed componentwise; in particular the i-th block of its spatial part is $y_i^0 + hq_i$ and $\pi_2(W_0 + h\zeta) = t_0 + h$.

For the time summand, $c\,\pi_2(W_0 + h\zeta) = c(t_0 + h) = ct_0 + ch$, which is affine in h for every h.

For ψ, note that $M_j(W_0 + h\zeta)$ is the j-th block of the spatial part minus the weighted mean of all the blocks, that is $\big(y_j^0 + hq_j\big) - \sum_k \theta_k \big(y_k^0 + hq_k\big)$. Hence `BaryVarianceAffineAlongMotion` applies with these θ, y^0, x_0, t_0 and with this p: its hypotheses $\sum_i \theta_i = 1$, $\sum_i \theta_i y_i^0 = x_0$ and $t_0 > 0$ hold, and the points it moves are exactly the ones just displayed. It gives, for every h with $t_0 + h > 0$ — a set of h containing a neighbourhood of 0, since $t_0 > 0$ —

$$\psi\big(W_0 + h\zeta\big) = \frac{R^2}{2t_0} + \frac{R^2}{2t_0^2}\, h, \qquad R^2 = \sum_i \theta_i \big|y_i^0 - x_0\big|^2,$$

which is affine in h.

Step 3: stripping them. Apply `SecondVariationAddAffineOnLine` on the product space-time at the point W_0 in the direction ζ, first with the pair (g, ψ). Both are C^2 at W_0 by Step 1, and ψ is affine along the motion by Step 2, so

$$D^2\Phi(W_0)[\zeta,\zeta] = D^2\big(g+\psi\big)(W_0)[\zeta,\zeta] = D^2 g(W_0)[\zeta,\zeta].$$

Apply it again with the pair $\big(\varphi\circ L,\ c\,\pi_2\big)$, whose sum is g: both are C^2 at W_0 by Step 1, and $c\,\pi_2$ is affine along the motion by Step 2, so

$$D^2 g(W_0)[\zeta,\zeta] = D^2\big(\varphi\circ L\big)(W_0)[\zeta,\zeta].$$

Step 4: the remaining summand is the operator. Since L is continuous linear and φ is C^2 at $L(W_0) = (x_0, t_0)$, `SecondFDerivCompLinear` gives

$$D^2\big(\varphi\circ L\big)(W_0)[\zeta,\zeta] = D^2\varphi(x_0,t_0)[L\zeta,\ L\zeta].$$

The time component of $L\zeta$ is the time component of ζ, namely 1. Its spatial component is $X\big((q_i)_i\big) = \sum_i \theta_i q_i$, and since $\sum_i \theta_i = 1$ and $\sum_i \theta_i\big(y_i^0 - x_0\big) = \sum_i \theta_i y_i^0 - x_0 = 0$,

$$\sum_i \theta_i q_i = \Big(\sum_i \theta_i\Big)p + \frac{1}{t_0}\sum_i \theta_i\big(y_i^0 - x_0\big) = p.$$

So $L\zeta = (p, 1)$ and

$$D^2\big(\varphi\circ L\big)(W_0)[\zeta,\zeta] = D^2\varphi(x_0,t_0)[(p,1)].$$

As $p = \nabla_x\varphi(x_0,t_0)$ is the spatial part of $\nabla\varphi(x_0,t_0)$, the right-hand side is exactly $F[\varphi](x_0,t_0)$ in the sense of `LimitOp`. Chaining with Step 3 gives the second assertion. □

2.87. `LimitSubClosedSuperjet`.

Lemma 2.87 (The limiting-equation subsolution inequality holds on closed superjets)**.** *Let F be a finite-dimensional real inner-product space, let Ω be a subset of the space-time $\ell^2(F\times\mathbb{R})$, and let v be a viscosity subsolution of $F[v] = 0$ on Ω in the sense of `IsLimitSubOn`. Let $z \in \Omega$, let $p \in \ell^2(F\times\mathbb{R})$, let X be a continuous bilinear form on $\ell^2(F\times\mathbb{R})$, and assume $(p, X) \in \overline{J}^{2,+}v(z)$, the closed superjet of `ClosedSuperjet`. Write $p_x \in F$ for the spatial part of p and $\zeta = (p_x, 1) \in \ell^2(F\times\mathbb{R})$. Then*

$$X[\zeta] = X(\zeta,\zeta) \geq 0.$$

Proof. Step 1: the inequality at a plain superjet. We first prove the following claim. If $z' \in \Omega$, if $q \in \ell^2(F\times\mathbb{R})$ and if Y is a continuous bilinear form on $\ell^2(F\times\mathbb{R})$ with $(q, Y) \in J^{2,+}v(z')$ in the sense of `Superjet`, then $Y(\xi,\xi) \geq 0$, where $\xi = (q_x, 1)$ and q_x is the spatial part of q.

Fix $\varepsilon > 0$ and define ϕ_ε on $\ell^2(F\times\mathbb{R})$ by

$$\phi_\varepsilon(w) = v(z') + \langle q, w - z'\rangle + \tfrac{1}{2}Y(w-z', w-z') + \varepsilon|w-z'|^2.$$

This is a sum of a constant, a continuous linear form in $w - z'$, the diagonal restriction of the continuous bilinear form Y, and ε times the diagonal restriction of the inner product; each summand is a continuous polynomial of degree at most two and hence C^∞. In particular ϕ_ε is C^2 near z'.

Its derivatives at z' are computed exactly as for a quadratic: writing $g(w) = \frac{1}{2}Y(w-z', w-z')$, for all h, k

$$Dg(w)h = \tfrac{1}{2}\big(Y(h, w-z') + Y(w-z', h)\big) = Y(w-z', h), \qquad D^2 g(w)(h,k) = \tfrac{1}{2}\big(Y(k,h) + Y(h,k)\big) = Y(h,k),$$

both middle steps using the symmetry $Y(h,k) = Y(k,h)$ required of a superjet in `Superjet`; and $w \mapsto \varepsilon|w-z'|^2$ has second derivative $(h,k) \mapsto 2\varepsilon\langle h,k\rangle$. Evaluating at $w = z'$,

$$\nabla\phi_\varepsilon(z') = q, \qquad D^2\phi_\varepsilon(z')(h,k) = Y(h,k) + 2\varepsilon\langle h,k\rangle.$$

The spatial part of $\nabla\phi_\varepsilon(z')$ is therefore q_x, so the direction used by the operator of `LimitOp` at z' is exactly $\xi = (q_x, 1)$, and

$$F[\phi_\varepsilon](z') = D^2\phi_\varepsilon(z')[\xi] = Y(\xi,\xi) + 2\varepsilon|\xi|^2. \tag{362}$$

Next, ϕ_ε is an upper test for v at z'. Apply the defining property of $(q,Y)\in J^{2,+}v(z')$ from `Superjet` with the positive number ε: there is $\delta>0$ with

$$v(z'+h)\le v(z')+\langle q,h\rangle+\tfrac12 Y(h,h)+\varepsilon|h|^2=\phi_\varepsilon(z'+h)\qquad(|h|<\delta).$$

Since $\phi_\varepsilon(z')=v(z')$, this says $v(w)-\phi_\varepsilon(w)\le 0=v(z')-\phi_\varepsilon(z')$ for all $w\in B(z',\delta)$, so $w\mapsto v(w)-\phi_\varepsilon(w)$ has a local maximum on Ω at z'.

So $z'\in\Omega$, ϕ_ε is C^2 near z', and $v-\phi_\varepsilon$ has a local maximum on Ω at z'. The defining property of `IsLimitSubOn` gives $F[\phi_\varepsilon](z')\ge 0$, that is, by (362),

$$0\le Y(\xi,\xi)+2\varepsilon|\xi|^2.$$

Neither ξ nor Y depends on ε, so letting $\varepsilon\downarrow 0$ gives $Y(\xi,\xi)\ge 0$, proving the claim.

Step 2: passing to the closure. By `ClosedSuperjet` there are sequences z_k, p_k in $\ell^2(F\times\mathbb{R})$ and symmetric continuous bilinear forms X_k with

$$(p_k,X_k)\in J^{2,+}v(z_k)\quad\text{for all } k,\qquad z_k\to z,\quad p_k\to p,\quad X_k\to X$$

(the defining sequences also satisfy $v(z_k)\to v(z)$, which we do not need).

By the first clause of `IsLimitSubOn` the set Ω is open, and $z\in\Omega$, so there is $\delta_0>0$ with $B(z,\delta_0)\subseteq\Omega$, and since $z_k\to z$ there is K with $z_k\in\Omega$ for all $k\ge K$. Write $\zeta_k=((p_k)_x,1)$. Step 1, applied at z_k to the superjet $(p_k,X_k)\in J^{2,+}v(z_k)$, gives

$$0\le X_k(\zeta_k,\zeta_k)\qquad(k\ge K).\tag{363}$$

We pass to the limit. The map $w\mapsto w_x$ taking a space-time point to its spatial part is linear with norm one, so $(p_k)_x\to p_x$; the map $a\mapsto(a,1)$ is continuous from F to $\ell^2(F\times\mathbb{R})$, since $|(a,1)-(b,1)|=|a-b|$. Hence $\zeta_k\to\zeta$. In particular $M:=\sup_k|\zeta_k|<\infty$ and $|\zeta|\le M$. Now

$$X_k(\zeta_k,\zeta_k)-X(\zeta,\zeta)=(X_k-X)(\zeta_k,\zeta_k)+\big(X(\zeta_k,\zeta_k)-X(\zeta,\zeta)\big).$$

The first term has absolute value at most $\|X_k-X\|\,|\zeta_k|^2\le M^2\|X_k-X\|\to 0$, because $X_k\to X$ in the operator norm. For the second, bilinearity gives

$$X(\zeta_k,\zeta_k)-X(\zeta,\zeta)=X(\zeta_k-\zeta,\zeta_k)+X(\zeta,\zeta_k-\zeta),$$

of absolute value at most $2M\|X\|\,|\zeta_k-\zeta|\to 0$. Therefore $X_k(\zeta_k,\zeta_k)\to X(\zeta,\zeta)$, and (363) gives $X(\zeta,\zeta)\ge 0$ in the limit. □

2.88. `ClosedSuperjetScale`.

Lemma 2.88 (Rescaling a closed superjet)**.** *Let E be a finite-dimensional real inner-product space, let $\theta>0$, let $g:E\to\mathbb{R}$, let $z,p\in E$ and let X be a continuous bilinear form on E. Then*

$$(p,X)\in\overline{J}^{2,+}(\theta g)(z)\quad\Longleftrightarrow\quad\big(\theta^{-1}p,\ \theta^{-1}X\big)\in\overline{J}^{2,+}g(z),$$

where θg denotes the function $w\mapsto\theta\,g(w)$ and $\overline{J}^{2,+}$ is the closed superjet of `ClosedSuperjet`.

Proof. It suffices to prove the single implication

for every $c>0$, every $h:E\to\mathbb{R}$ and every (p',X') : $\qquad(p',X')\in\overline{J}^{2,+}h(z)\implies(cp',cX')\in\overline{J}^{2,+}(ch)(z),$ (364)

because the two directions of the lemma are both instances of it:

Forward. Suppose $(p,X)\in\overline{J}^{2,+}(\theta g)(z)$. Apply (364) with $c=\theta^{-1}>0$ and $h=\theta g$: it gives $(\theta^{-1}p,\theta^{-1}X)\in\overline{J}^{2,+}\big(\theta^{-1}(\theta g)\big)(z)$, and $\theta^{-1}(\theta g)=g$ as functions, since $\theta^{-1}\theta=1$.

Backward. Suppose $(\theta^{-1}p,\theta^{-1}X)\in\overline{J}^{2,+}g(z)$. Apply (364) with $c=\theta$ and $h=g$: it gives $\big(\theta(\theta^{-1}p),\theta(\theta^{-1}X)\big)\in\overline{J}^{2,+}(\theta g)(z)$, and $\theta(\theta^{-1}p)=p$, $\theta(\theta^{-1}X)=X$.

Proof of (364). We first record the corresponding statement for the superjet of `Superjet`: for every $c>0$, every h, every z' and every (p',X'),

$$(p',X')\in J^{2,+}h(z')\implies(cp',cX')\in J^{2,+}(ch)(z').\tag{365}$$

Indeed, cX' is symmetric because $(cX')(a,b) = c\,X'(a,b) = c\,X'(b,a) = (cX')(b,a)$. And given $\varepsilon > 0$, apply the defining condition of `Superjet` for (p', X') at z' with the parameter $\varepsilon/c > 0$: it produces $\delta > 0$ with

$$h(z'+u) \le h(z') + \langle p', u\rangle + \tfrac{1}{2}X'(u,u) + \frac{\varepsilon}{c}|u|^2 \qquad (|u| < \delta).$$

Multiplying through by $c > 0$, which preserves the inequality, and using $c\langle p', u\rangle = \langle cp', u\rangle$ and $c\,X'(u,u) = (cX')(u,u)$ together with $c \cdot (\varepsilon/c) = \varepsilon$, gives

$$(ch)(z'+u) \le (ch)(z') + \langle cp', u\rangle + \tfrac{1}{2}(cX')(u,u) + \varepsilon|u|^2 \qquad (|u| < \delta),$$

which is the defining condition for (cp', cX') at z' with the parameter ε. As $\varepsilon > 0$ was arbitrary, (365) holds.

Now let $(p', X') \in \overline{J}^{2,+}h(z)$. By `ClosedSuperjet` there are sequences (z_k), (p_k), (X_k) with $(p_k, X_k) \in J^{2,+}h(z_k)$ for every k, and with $z_k \to z$, $h(z_k) \to h(z)$, $p_k \to p'$ and $X_k \to X'$. Consider the sequences (z_k), (cp_k), (cX_k). Then:

- $(cp_k, cX_k) \in J^{2,+}(ch)(z_k)$ for every k, by (365) applied at z_k;
- $z_k \to z$, unchanged;
- $(ch)(z_k) = c\,h(z_k) \to c\,h(z) = (ch)(z)$, multiplication by the constant c being continuous on $\mathbb{R}$;
- $cp_k \to cp'$ and $cX_k \to cX'$, multiplication by the constant c being continuous on E and on the space of continuous bilinear forms.

These are exactly the four requirements of `ClosedSuperjet` for the pair (cp', cX') relative to ch at z, so $(cp', cX') \in \overline{J}^{2,+}(ch)(z)$. This proves (364) and hence the lemma. □

2.89. `SumRuleCommonTimeDirection`.

Lemma 2.89 (The sum rule in the common-time direction)**.** *Let F be a finite-dimensional real inner-product space, let N be a natural number, let $\Omega \subseteq \ell^2(F \times \mathbb{R})$ and let v be a viscosity subsolution of $F[v] = 0$ on Ω in the sense of `IsLimitSubOn` (so in particular Ω is open and v is continuous on Ω). Let $\theta_i > 0$ for $1 \le i \le N$, write $P = \ell^2\big(\bigoplus_{i=1}^N \ell^2(F \times \mathbb{R})\big)$ for the ℓ^2-product of N copies of space-time, with elements $W = (w_i)_{i=1}^N$, and let $\Psi : P \to \mathbb{R}$. Let $Z = (z_i)_{i=1}^N \in P$ with $z_i \in \Omega$ for every i, assume Ψ is C^2 near Z, and assume that*

$$W \longmapsto \sum_{i=1}^N \theta_i\, v(w_i) - \Psi(W)$$

has a local maximum at Z. Put

$$q_i = \theta_i^{-1}\big(\nabla\Psi(Z)_i\big)_x \in F, \qquad \zeta = \big((q_i, 1)\big)_{i=1}^N \in P,$$

where $\nabla\Psi(Z)_i \in \ell^2(F \times \mathbb{R})$ is the i-th block of $\nabla\Psi(Z)$ and $(\cdot)_x$ is its spatial part. Then

$$D^2\Psi(Z)[\zeta] \ge 0.$$

Proof. Write $A = D^2\Psi(Z)$, regarded as a continuous bilinear form on P, and note that neither A nor ζ depends on the parameter η introduced below: both are determined by Ψ and Z alone.

Step 1: the sum rule at a fixed η. Fix $\eta > 0$. The data

$$N, \qquad O_i = \Omega, \qquad g_i = \theta_i v, \qquad \Psi, \qquad Z, \qquad \eta$$

satisfy every hypothesis of `SumRuleJets`: each $O_i = \Omega$ is open and each $g_i = \theta_i v$ is continuous on it, both by `IsLimitSubOn` (continuity of $\theta_i v$ being continuity of v multiplied by the constant θ_i); $z_i \in \Omega$ for every i by hypothesis; Ψ is C^2 near Z; the function $W \mapsto \sum_i g_i(w_i) - \Psi(W)$ has a local maximum at Z by hypothesis, since $g_i(w_i) = \theta_i v(w_i)$; and $\eta > 0$. That lemma therefore supplies symmetric continuous bilinear forms X_i on $\ell^2(F \times \mathbb{R})$ with

$$\big(\nabla\Psi(Z)_i,\ X_i\big) \in \overline{J}^{2,+}\big(\theta_i v\big)(z_i) \qquad (1 \le i \le N) \tag{366}$$

and

$$\sum_{i=1}^{N} X_i[\xi_i] \le A[\xi] + \eta\,|A\xi|^2 \qquad \big(\xi = (\xi_i)_i \in P\big), \tag{367}$$

where $|A\xi|$ is the norm of the linear functional $A(\xi,\cdot)$.

Step 2: dividing each jet by θ_i. Fix i. Since $\theta_i > 0$, `ClosedSuperjetScale` applied with $\theta = \theta_i$, $g = v$, $z = z_i$, $p = \nabla\Psi(Z)_i$ and $X = X_i$ turns (366) into

$$\big(\theta_i^{-1}\nabla\Psi(Z)_i,\ \theta_i^{-1}X_i\big) \in \overline{J}^{2,+}v(z_i). \tag{368}$$

The spatial part of $\theta_i^{-1}\nabla\Psi(Z)_i$ is $\theta_i^{-1}\big(\nabla\Psi(Z)_i\big)_x = q_i$, scalar multiplication on $\ell^2(F\times\mathbb{R})$ being componentwise.

Step 3: the viscosity inequality on each rescaled jet. Since $z_i \in \Omega$ and (368) is a closed superjet of v at z_i, `LimitSubClosedSuperjet` applies to it and gives

$$0 \le \big(\theta_i^{-1}X_i\big)\big[(q_i,1)\big],$$

the direction being formed from the spatial part q_i of the jet's gradient and the time component 1, exactly as that lemma prescribes. Now $\big(\theta_i^{-1}X_i\big)[(q_i,1)] = \theta_i^{-1}X_i[(q_i,1)]$, so multiplying by $\theta_i > 0$ gives

$$X_i\big[(q_i,1)\big] \ge 0 \qquad (1 \le i \le N). \tag{369}$$

Step 4: evaluating the matrix inequality on ζ. The i-th block of ζ is $(q_i,1)$, so taking $\xi = \zeta$ in (367) and summing (369) over i gives

$$0 \le \sum_{i=1}^{N} X_i\big[(q_i,1)\big] \le A[\zeta] + \eta\,|A\zeta|^2. \tag{370}$$

Step 5: letting $\eta \downarrow 0$. Step 1 was carried out for an arbitrary $\eta > 0$, and, as recorded at the start of the proof, neither A nor ζ depends on η. So (370) says

$$0 \le A[\zeta] + \eta\,|A\zeta|^2 \qquad \text{for every } \eta > 0.$$

Suppose, for contradiction, that $A[\zeta] < 0$, and put

$$\eta_\star = \frac{-A[\zeta]}{2\big(|A\zeta|^2 + 1\big)} > 0,$$

which is positive because the numerator is positive and the denominator is at least 2. Then

$$\eta_\star|A\zeta|^2 \le \eta_\star\big(|A\zeta|^2 + 1\big) = \frac{-A[\zeta]}{2},$$

so

$$A[\zeta] + \eta_\star|A\zeta|^2 \le A[\zeta] - \frac{A[\zeta]}{2} = \frac{A[\zeta]}{2} < 0,$$

contradicting the displayed inequality at $\eta = \eta_\star$. Hence $A[\zeta] \ge 0$, which is the assertion. □

2.90. `PenalizedMaxConvergence`.

Lemma 2.90 (Penalization converges to the strict constrained maximum)**.** *Let Z be a metric space, let $K \subseteq Z$ be compact, and let $\Theta, \chi : Z \to \mathbb{R}$ be continuous on K with $\chi \ge 0$ on K. Let $k_0 \in K$ satisfy $\chi(k_0) = 0$, and assume that k_0 is a* strict *maximum point of Θ among the points of K at which χ vanishes:*

$$\Theta(k) < \Theta(k_0) \qquad \big(k \in K,\ \chi(k) = 0,\ k \ne k_0\big).$$

Let $(\lambda_m)_{m\ge0}$ be positive reals with $\lambda_m \to \infty$, and let $(k_m)_{m\ge0}$ be points of K with

$$\Theta(k) - \lambda_m\chi(k) \le \Theta(k_m) - \lambda_m\chi(k_m) \qquad (k \in K,\ m \ge 0),$$

that is, k_m maximizes $\Theta - \lambda_m\chi$ over K. Then $k_m \to k_0$.

Proof. Step 1: two consequences of maximality, taking $k = k_0$. The function Θ is continuous on the nonempty compact set K, so it attains a maximum there (preamble), say at $k_\star \in K$; thus $\Theta(k) \le \Theta(k_\star)$ for every $k \in K$. Taking $k = k_0$ in the maximality hypothesis and using $\chi(k_0) = 0$ gives

$$\Theta(k_0) \le \Theta(k_m) - \lambda_m \chi(k_m) \qquad (m \ge 0). \tag{371}$$

From (371) we read off two facts. First, since $\lambda_m > 0$ and $\chi(k_m) \ge 0$ (as $k_m \in K$), the subtracted term is nonnegative, so

$$\Theta(k_0) \le \Theta(k_m) \qquad (m \ge 0). \tag{372}$$

Second, since $\Theta(k_m) \le \Theta(k_\star)$,

$$\lambda_m \chi(k_m) \le \Theta(k_\star) - \Theta(k_0) =: C_0 \qquad (m \ge 0), \tag{373}$$

and $C_0 \ge 0$ because $k_0 \in K$ gives $\Theta(k_0) \le \Theta(k_\star)$.

Step 2: the penalty vanishes. Dividing (373) by $\lambda_m > 0$,

$$0 \le \chi(k_m) \le \frac{C_0}{\lambda_m} \qquad (m \ge 0),$$

the left inequality because $k_m \in K$. Since $\lambda_m \to \infty$ and C_0 is a fixed real number, $C_0/\lambda_m \to 0$; by the squeeze principle,

$$\chi(k_m) \longrightarrow 0 \qquad (m \to \infty). \tag{374}$$

Step 3: every subsequence has a sub-subsequence converging to k_0. It suffices to prove this: a sequence in a metric space converges to k_0 as soon as every subsequence admits a further subsequence converging to k_0.

So let $(n_j)_j$ be indices with $n_j \to \infty$. The points k_{n_j} all lie in the compact set K, so by sequential compactness of a compact metric space (preamble) there are a strictly increasing $(m_j)_j$ and a point $\hat{k} \in K$ with $k_{n_{m_j}} \to \hat{k}$. We claim $\hat{k} = k_0$.

Since every $k_{n_{m_j}}$ lies in K and $k_{n_{m_j}} \to \hat{k}$ with $\hat{k} \in K$, the convergence also holds within K; as χ and Θ are continuous on K, this gives

$$\chi\big(k_{n_{m_j}}\big) \longrightarrow \chi(\hat{k}), \qquad \Theta\big(k_{n_{m_j}}\big) \longrightarrow \Theta(\hat{k}).$$

On the other hand $(n_{m_j})_j \to \infty$, because $n_j \to \infty$ and $(m_j)_j$ is strictly increasing, so (374) applies along this subsequence and gives $\chi(k_{n_{m_j}}) \to 0$. Limits in a metric space are unique, hence

$$\chi(\hat{k}) = 0.$$

Likewise (372) holds for every index, so passing to the limit in $\Theta(k_0) \le \Theta(k_{n_{m_j}})$ gives

$$\Theta(k_0) \le \Theta(\hat{k}).$$

Now suppose $\hat{k} \ne k_0$. Then $\hat{k} \in K$ with $\chi(\hat{k}) = 0$ and $\hat{k} \ne k_0$, so the strict-maximum hypothesis gives $\Theta(\hat{k}) < \Theta(k_0)$, contradicting the previous display. Therefore $\hat{k} = k_0$, and $k_{n_{m_j}} \to k_0$.

As $(n_j)_j$ was an arbitrary sequence of indices tending to infinity, the criterion quoted at the start of this step applies and gives $k_m \to k_0$. □

2.91. `QuarticFlatAtCentre`.

Lemma 2.91 (The quartic penalty is flat to second order at its centre)**.** *Let E be a real inner-product space and let $Z_0 \in E$. Then the function*

$$\kappa(Z) = |Z - Z_0|^4 \qquad (Z \in E)$$

is C^2 on E, and

$$D\kappa(Z_0) = 0, \qquad D^2\kappa(Z_0) = 0.$$

Proof. Write $n(Z) = |Z - Z_0|^2$, so that $\kappa = n^2$, since $|Z - Z_0|^4 = \big(|Z - Z_0|^2\big)^2$.

Step 1: κ is C^2. The map $Z \mapsto Z - Z_0$ is the sum of the identity and a constant, hence C^∞; the squared norm $W \mapsto |W|^2$ is C^∞ on an inner-product space (preamble, it is the quadratic form $W \mapsto \langle W, W\rangle$); so n is C^∞ as a composition, and $\kappa = n^2$ is C^∞ as a product, in particular C^2.

Step 2: the first derivative, at every point. The derivative of $Z \mapsto Z - Z_0$ at any Z is the identity map. The derivative of $W \mapsto |W|^2$ at W is $h \mapsto 2\langle W, h\rangle$. By the chain rule,

$$Dn(Z)h = 2\langle Z - Z_0, h\rangle \qquad (h \in E).$$

By the chain rule for the square, $D\kappa(Z) = 2\, n(Z)\, Dn(Z)$, that is,

$$D\kappa(Z)h = 4\,|Z - Z_0|^2\,\langle Z - Z_0, h\rangle \qquad (Z, h \in E). \tag{375}$$

Taking $Z = Z_0$ in (375), the factor $|Z_0 - Z_0|^2 = 0$ gives $D\kappa(Z_0)h = 0$ for every h, i.e. $D\kappa(Z_0) = 0$.

Step 3: the second derivative at Z_0. By (375) the function $Z \mapsto D\kappa(Z)$ is the product

$$Z \longmapsto g(Z) \cdot \Lambda(Z), \qquad g(Z) = 4\,|Z - Z_0|^2 \in \mathbb{R}, \qquad \Lambda(Z) = \langle Z - Z_0, \cdot\,\rangle,$$

of a scalar function g and a function Λ with values in the space of continuous linear functionals on E. Both factors are differentiable: g by Step 2 applied to the squared norm, and Λ because it is the composition of the continuous linear map $W \mapsto \langle W, \cdot\,\rangle$ with the map $Z \mapsto Z - Z_0$, so that $D\Lambda(Z)h = \langle h, \cdot\,\rangle$.

By the product rule for a scalar function times a vector-valued function,

$$D^2\kappa(Z_0)h = \big(Dg(Z_0)h\big)\,\Lambda(Z_0) + g(Z_0)\,\big(D\Lambda(Z_0)h\big) \qquad (h \in E).$$

Both terms vanish: $\Lambda(Z_0) = \langle Z_0 - Z_0, \cdot\,\rangle = \langle 0, \cdot\,\rangle = 0$ kills the first, and $g(Z_0) = 4|Z_0 - Z_0|^2 = 0$ kills the second. Hence $D^2\kappa(Z_0)h = 0$ for every h, i.e. $D^2\kappa(Z_0) = 0$. □

2.92. `PenalizedLiftGradientBlocks`.

Lemma 2.92 (Spatial blocks of the gradient of a penalised lift)**.** *Let F be a finite-dimensional real inner product space and let $N \in \mathbb{N}$. Write $P = \big(F \times \mathbb{R}\big)^N$ and $Q = F^N \times \mathbb{R}$, each carried with its ℓ^2 product inner product, and write a point of P as $W = (y_i, s_i)_{i<N}$. Let*

$$L : P \to Q, \qquad L(W) = \Big((y_i)_{i<N},\ \tfrac{1}{N}\sum_{i<N} s_i\Big),$$

$$M : P \to \mathbb{R}^N, \qquad M(W) = \Big(s_i - \tfrac{1}{N}\sum_{j<N} s_j\Big)_{i<N},$$

both of which are continuous linear. Let $\lambda \in \mathbb{R}$, let $W \in P$, let $g : Q \to \mathbb{R}$ be C^2 near $L(W)$, and set

$$\Psi(U) = g\big(L(U)\big) + \lambda\,\frac{\|M(U)\|^2}{2}.$$

Then for every $i < N$, writing $\nabla\Psi(W) = (\,p_j, \sigma_j\,)_{j<N} \in P$ and $\nabla g(L(W)) = \big((q_j)_{j<N}, \tau\big) \in Q$,

$$p_i = q_i.$$

Proof. Fix $i < N$ and $a \in F$. Since the inner product on F separates points, it suffices to show $\langle a, p_i\rangle = \langle a, q_i\rangle$.

Step 1: the test direction. Let $\xi \in P$ be the point whose i-th block is $(a, 0)$ and whose every other block is 0. Its time coordinates all vanish, so $\sum_{j<N} 0 = 0$ and therefore

$$M\xi = \big(0 - \tfrac{1}{N}\cdot 0\big)_{j<N} = 0, \qquad L\xi = \big((\delta_{ji}a)_{j<N},\ \tfrac{1}{N}\cdot 0\big) = \big(\iota_i a,\ 0\big),$$

where $\iota_i a \in F^N$ denotes the tuple with a in slot i and 0 elsewhere.

Step 2: the penalty does not see ξ. The penalty $U \mapsto \lambda\|M(U)\|^2/2$ is the composition of the differentiable function $u \mapsto \lambda\|u\|^2/2$ with the continuous linear M, so by the chain rule its derivative at W is a linear functional factoring through M. Since $M\xi = 0$ and a linear map

sends 0 to 0, that derivative annihilates ξ. By the sum rule, and because g is differentiable at $L(W)$ so that $D(g \circ L)(W) = Dg(L(W)) \circ L$,

$$D\Psi(W)[\xi] = Dg\big(L(W)\big)\big[L\xi\big]. \tag{376}$$

Step 3: reading both sides as inner products. By definition of the gradient, $D\Psi(W)[\xi] = \langle \nabla\Psi(W), \xi\rangle_P$. As the inner product of P is the sum over blocks and all blocks of ξ except the i-th vanish,

$$\langle \nabla\Psi(W), \xi\rangle_P = \big\langle (p_i, \sigma_i),\ (a, 0)\big\rangle_{F\times\mathbb{R}} = \langle p_i, a\rangle.$$

Likewise $Dg(L(W))[L\xi] = \langle \nabla g(L(W)), L\xi\rangle_Q$, and since $L\xi = (\iota_i a, 0)$ the time component contributes nothing while the spatial component is again a sum over blocks of which only the i-th survives:

$$\big\langle \nabla g(L(W)), L\xi\big\rangle_Q = \big\langle (q_j)_{j<N},\ \iota_i a\big\rangle_{F^N} + \langle \tau, 0\rangle = \langle q_i, a\rangle.$$

Step 4: conclusion. Combining Step 3 with (376) gives $\langle p_i, a\rangle = \langle q_i, a\rangle$. As $a \in F$ was arbitrary and the inner product is symmetric, $p_i = q_i$, which is the assertion. □

2.93. `PenalizedLiftSecondDeriv`.

Lemma 2.93 (Second derivative of a penalised lift in a direction the penalty misses)**.** *Let P and Q be real normed spaces, let R be a real inner product space, let $L : P \to Q$ and $M : P \to R$ be continuous linear maps, let $\lambda \in \mathbb{R}$, let $W \in P$, and let $g : Q \to \mathbb{R}$ be C^2 near $L(W)$. Define*

$$\Psi(U) = g\big(L(U)\big) + \lambda\,\frac{\|M(U)\|^2}{2}.$$

Then for every $\zeta \in P$ with $M\zeta = 0$,

$$D^2\Psi(W)[\zeta,\zeta] = D^2 g\big(L(W)\big)\big[L\zeta,\ L\zeta\big].$$

Proof. Write $q(u) = \|u\|^2/2$ for $u \in R$, so that the penalty is $\lambda\,(q \circ M)$. The function q is C^2 on all of R, because $u \mapsto \|u\|^2$ is a smooth function on an inner product space; hence $q \circ M$ is C^2 on all of P, being the composition of a C^2 function with a continuous linear map. Likewise $g \circ L$ is C^2 near W, since g is C^2 near $L(W)$ and L is continuous linear.

Step 1: the first derivative splits on a neighbourhood of W. For every U in a neighbourhood of W both $g \circ L$ and $q \circ M$ are differentiable at U; by the sum rule and the rule for a constant multiple,

$$D\Psi(U) = D(g \circ L)(U) + \lambda\, D(q \circ M)(U).$$

Thus the two maps $U \mapsto D\Psi(U)$ and $U \mapsto D(g \circ L)(U) + \lambda\, D(q \circ M)(U)$, both with values in the space of continuous linear functionals on P, agree on a neighbourhood of W.

Step 2: differentiating once more. Because $g \circ L$ and $q \circ M$ are C^2 near W, their first derivatives are differentiable at W, with derivatives $D^2(g \circ L)(W)$ and $D^2(q \circ M)(W)$. Two maps that agree on a neighbourhood of W have the same derivative at W, so by Step 1

$$D^2\Psi(W) = D^2(g \circ L)(W) + \lambda\, D^2(q \circ M)(W).$$

Evaluating at the pair (ζ, ζ),

$$D^2\Psi(W)[\zeta,\zeta] = D^2(g \circ L)(W)[\zeta,\zeta] + \lambda\, D^2(q \circ M)(W)[\zeta,\zeta]. \tag{377}$$

Step 3: the penalty term vanishes. By the formula for the second derivative of a composition with a continuous linear map on the right, applied to q and M,

$$D^2(q \circ M)(W)[\zeta,\zeta] = D^2 q\big(M(W)\big)\big[M\zeta,\ M\zeta\big].$$

By hypothesis $M\zeta = 0$, and $D^2 q(M(W))$ is a bilinear map, hence vanishes as soon as one of its arguments is 0. Therefore $D^2(q \circ M)(W)[\zeta,\zeta] = 0$.

Step 4: conclusion. Applying the same composition formula to g and L gives $D^2(g \circ L)(W)[\zeta,\zeta] = D^2 g(L(W))[L\zeta, L\zeta]$. Substituting this and Step 3 into (377) leaves

$$D^2\Psi(W)[\zeta,\zeta] = D^2 g\big(L(W)\big)\big[L\zeta,\ L\zeta\big],$$

as asserted. □

2.94. `SecondDerivAddFlat.`

Lemma 2.94 (A second-order flat perturbation changes nothing at the centre)**.** *Let Q be a real normed space, let $x \in Q$, let $f : Q \to \mathbb{R}$ be C^2 near x, and let $g : Q \to \mathbb{R}$ be C^2 on all of Q with*

$$Dg(x) = 0 \qquad \text{and} \qquad D^2g(x) = 0.$$

Then

$$D(f+g)(x) = Df(x) \qquad \text{and} \qquad D^2(f+g)(x) = D^2f(x).$$

Proof. Since g is C^2 on all of Q it is differentiable at every point y, with derivative $Dg(y)$.

Step 1: the first derivative. As f is C^2 near x it is in particular differentiable at x. By the sum rule the function $f+g$ is differentiable at x with

$$D(f+g)(x) = Df(x) + Dg(x).$$

The hypothesis $Dg(x) = 0$ gives $D(f+g)(x) = Df(x)$.

Step 2: the first derivative splits near x. For every y in a neighbourhood of x the function f is still C^2, hence differentiable, so the same sum rule applies at y:

$$D(f+g)(y) = Df(y) + Dg(y).$$

Thus the two maps $y \mapsto D(f+g)(y)$ and $y \mapsto Df(y) + Dg(y)$, with values in the space of continuous linear functionals on Q, agree on a neighbourhood of x.

Step 3: the second derivative. Because f is C^2 near x and g is C^2, the maps Df and Dg are each differentiable at x, with derivatives $D^2f(x)$ and $D^2g(x)$. Hence $y \mapsto Df(y) + Dg(y)$ is differentiable at x with derivative $D^2f(x) + D^2g(x)$; and since by Step 2 it agrees with $y \mapsto D(f+g)(y)$ on a neighbourhood of x, the latter has the same derivative at x:

$$D^2(f+g)(x) = D^2f(x) + D^2g(x).$$

The hypothesis $D^2g(x) = 0$ gives $D^2(f+g)(x) = D^2f(x)$, which completes the proof. □

2.95. `BilinearEvalContinuousAt.`

Lemma 2.95 (Joint continuity of evaluating a varying form on a varying direction)**.** *Let Z be a topological space, let E be a real normed space, let $B : Z \to \mathcal{L}\big(E, \mathcal{L}(E,\mathbb{R})\big)$ assign to each $w \in Z$ a continuous bilinear form on E, and let $u : Z \to E$. If B and u are both continuous at $z \in Z$, then so is*

$$w \longmapsto B(w)\big[u(w),\, u(w)\big].$$

Proof. The evaluation map

$$\mathcal{L}(E,G) \times E \longrightarrow G, \qquad (C,x) \longmapsto C(x)$$

is bilinear and bounded, since $\|C(x)\| \le \|C\|\,\|x\|$; hence it is continuous, for every real normed space G.

Step 1. Apply this with $G = \mathcal{L}(E,\mathbb{R})$. The map $w \mapsto (B(w), u(w))$ is continuous at z, because a map into a product is continuous at a point exactly when both of its components are, and both B and u are continuous at z by hypothesis. Composing with the evaluation map, which is continuous at the point $(B(z), u(z))$, shows that

$$w \longmapsto B(w)\big[u(w),\, \cdot\,\big] \in \mathcal{L}(E,\mathbb{R})$$

is continuous at z.

Step 2. Apply the same evaluation map with $G = \mathbb{R}$. By Step 1 the map $w \mapsto \big(B(w)[u(w),\cdot\,],\, u(w)\big)$ is continuous at z, again because each component is. Composing with the evaluation map, continuous at the point $\big(B(z)[u(z),\cdot\,], u(z)\big)$, gives the continuity at z of

$$w \longmapsto B(w)\big[u(w),\, u(w)\big],$$

which is the assertion. □

2.96. `CommonTimeSubsolution`.

Lemma 2.96 (Common-time subsolutions)**.** *Let F be a finite-dimensional real inner-product space, let $N \geq 1$, let $I \subseteq \mathbb{R}$, and let v be a viscosity subsolution of $F[v] = 0$ on the space-time slab $\{z : z_t \in I\} \subseteq \ell^2(F \times \mathbb{R})$ in the sense of `IsLimitSubOn` (so in particular that slab is open, equivalently I is open, and v is continuous on it). Fix weights $\theta_i > 0$ with $\sum_{i=1}^N \theta_i = 1$, write $G = \ell^2\big(\bigoplus_{i=1}^N F\big)$ for the ℓ^2-product of N copies of F with elements $Y = (y_i)_{i=1}^N$, and define*

$$V(Y,t) = \sum_{i=1}^N \theta_i\, v(y_i,t) \qquad \big((Y,t) \in \ell^2(G \times \mathbb{R})\big).$$

Let $\Phi : \ell^2(G \times \mathbb{R}) \to \mathbb{R}$, let $Y_0 \in G$ and $t_0 \in I$, assume Φ is C^2 near (Y_0,t_0), and assume Φ touches V from above at (Y_0,t_0), i.e. that $V - \Phi$ has a local maximum on $\{(Y,t) : t \in I\}$ at (Y_0,t_0). Then

$$D^2\Phi(Y_0,t_0)\Big[\Big(\big(\theta_i^{-1}\nabla_{y_i}\Phi(Y_0,t_0)\big)_{i=1}^N,\ 1\Big)\Big] \geq 0,$$

where $\nabla_{y_i}\Phi(Y_0,t_0) \in F$ is the i-th spatial block of $\nabla\Phi(Y_0,t_0)$.

Proof. Throughout, $Z_0 = (Y_0,t_0)$, and for $z \in \ell^2(F \times \mathbb{R})$ we write $z_x \in F$ and $z_t \in \mathbb{R}$ for its spatial and time parts. Put

$$\Omega = \{z \in \ell^2(F \times \mathbb{R}) : z_t \in I\}, \qquad \mathcal{D} = \{(Y,t) \in \ell^2(G \times \mathbb{R}) : t \in I\},$$

both open, the second because it is the preimage of the open set I under the continuous time projection. Write $P = \ell^2\big(\bigoplus_{i=1}^N \ell^2(F \times \mathbb{R})\big)$ for the ℓ^2-product of N copies of space-time, with elements $W = \big((y_i,t_i)\big)_{i=1}^N$; this is the product space the sum rule works in. A point of P may equally be written as a pair $(Y,T) \in \ell^2(G \times \mathbb{R}^N)$ with $Y = (y_i)_i$ and $T = (t_i)_i$; the map exchanging the two groupings is linear and preserves the norm, since both squared norms equal $\sum_i\big(|y_i|^2 + t_i^2\big)$, and we use the two descriptions interchangeably.

Step 1: making the maximum strict. Set

$$\Phi^\sharp(Y,t) = \Phi(Y,t) + |Y - Y_0|^4 + |t - t_0|^4.$$

The added function $\kappa(Y,t) = |Y - Y_0|^4 + |t - t_0|^4$ is C^∞, vanishes at Z_0 together with its first and second derivatives there — indeed $|W|^4 = (\langle W, W\rangle)^2$ has gradient $4|W|^2W$ and second derivative $(a,b) \mapsto 4|W|^2\langle a,b\rangle + 8\langle W,a\rangle\langle W,b\rangle$, both vanishing at $W = 0$, and the same for $|t - t_0|^4$ — so

$$\Phi^\sharp(Z_0) = \Phi(Z_0), \qquad \nabla\Phi^\sharp(Z_0) = \nabla\Phi(Z_0), \qquad D^2\Phi^\sharp(Z_0) = D^2\Phi(Z_0), \tag{378}$$

and $\Phi^\sharp$ is C^2 wherever Φ is. Moreover $\kappa \geq 0$ with $\kappa(Z) > 0$ for $Z \neq Z_0$, so $V - \Phi^\sharp = (V - \Phi) - \kappa$ has a *strict* local maximum on $\mathcal{D}$ at Z_0: there is $\rho_0 > 0$ such that $\overline{B}(Z_0,\rho_0) \subseteq \mathcal{D}$, Φ is C^2 on a neighbourhood of $\overline{B}(Z_0,\rho_0)$, $V - \Phi \leq (V - \Phi)(Z_0)$ on $\overline{B}(Z_0,\rho_0)$, and consequently

$$\big(V - \Phi^\sharp\big)(Z) < \big(V - \Phi^\sharp\big)(Z_0) \qquad \big(Z \in \overline{B}(Z_0,\rho_0),\ Z \neq Z_0\big). \tag{379}$$

Fix once and for all a radius $\rho \in (0, \rho_0/4)$ so small that

$$\overline{K} := \Big\{\big((y_i,t_i)\big)_{i=1}^N : |y_i - y_i^0| \leq \rho \text{ and } |t_i - t_0| \leq \rho \text{ for all } i\Big\} \tag{380}$$

satisfies $(y_i,t_i) \in \Omega$ for every i and every point of $\overline{K}$, and such that every diagonal point $\big((y_i,t)\big)_i$ of $\overline{K}$ has $(Y,t) \in \overline{B}(Z_0,\rho_0)$. Both are possible: $t_0 \in I$ with I open gives $[t_0 - \rho, t_0 + \rho] \subseteq I$ for small ρ, which yields the first; and for a diagonal point $|(Y,t) - Z_0|^2 = \sum_i |y_i - y_i^0|^2 + |t - t_0|^2 \leq (N+1)\rho^2$, so $\rho \leq \rho_0/\sqrt{N+1}$ yields the second. The set $\overline{K}$ is closed and bounded, hence compact (preamble), and it is nonempty because it contains the diagonal point $D_0 = \big((y_i^0,t_0)\big)_i$.

Step 2: separating the time variables. For $T = (t_1,\dots,t_N) \in \mathbb{R}^N$ put

$$\bar t = \frac{1}{N}\sum_{i=1}^N t_i, \qquad \chi(T) = \frac{1}{2}\sum_{i=1}^N (t_i - \bar t)^2 \ \geq 0,$$

and, for $\lambda > 0$,

$$\Psi_\lambda(Y,T) = \Phi^\sharp(Y,\bar t) + \lambda\chi(T), \qquad \Theta\big((y_i,t_i)_i\big) = \sum_i \theta_i\, v(y_i,t_i) - \Phi^\sharp(Y,\bar t),$$

so that $\Theta - \lambda\chi$ is the function $W \mapsto \sum_i \theta_i v(y_i,t_i) - \Psi_\lambda(Y,T)$ on P.

(2a) Θ and χ are continuous on $\overline{K}$. Each $v(y_i,t_i)$ is continuous on $\overline{K}$ because $(y_i,t_i) \in \Omega$ there by Step 1 and v is continuous on Ω by `IsLimitSubOn`; the map $(Y,T) \mapsto (Y,\bar t)$ is linear, hence continuous, and it carries $\overline{K}$ into $\overline{B}(Z_0,\rho_0)$ — indeed $|\bar t - t_0| \le \max_i |t_i - t_0| \le \rho$, so $(Y,\bar t)$ is a diagonal point of $\overline{K}$ in the sense of Step 1 — where $\Phi^\sharp$ is C^2 and in particular continuous. So Θ is continuous on $\overline{K}$, and χ is a polynomial in T, hence continuous, and nonnegative by its definition as a sum of squares.

(2b) D_0 is a strict maximum of Θ on the zero set of χ. First χ vanishes at D_0, all of whose time coordinates equal t_0. Now let $W = \big((y_i,t_i)\big)_i \in \overline{K}$ with $\chi(W) = 0$ and $W \ne D_0$. From $\chi(W) = \frac12 \sum_i (t_i - \bar t)^2 = 0$ we get $t_i = \bar t$ for every i; write t for this common value and $Y = (y_i)_i$, so that W is the diagonal point $\big((y_i,t)\big)_i$ and

$$\Theta(W) = \sum_i \theta_i v(y_i,t) - \Phi^\sharp(Y,t) = \big(V - \Phi^\sharp\big)(Y,t),$$

by the definition of V in the statement. By Step 1 the point (Y,t) lies in $\overline{B}(Z_0,\rho_0)$. Moreover $(Y,t) \ne Z_0$: a diagonal point of $\overline{K}$ is determined by the pair (Y,t), so $(Y,t) = Z_0 = (Y_0,t_0)$ would force $W = \big((y_i^0,t_0)\big)_i = D_0$, which is excluded. Hence (379) applies and gives

$$\Theta(W) = \big(V - \Phi^\sharp\big)(Y,t) < \big(V - \Phi^\sharp\big)(Z_0) = \Theta(D_0),$$

the last equality because D_0 is the diagonal point with $Y = Y_0$, $t = t_0$.

(2c) The penalized maximizers and their limit. Put $\lambda_m = m+1$ for $m \ge 0$; these are positive and tend to ∞. For each m the function $\Theta - \lambda_m\chi$ is continuous on the nonempty compact set $\overline{K}$, so it attains a maximum there (preamble), at a point $W^{(m)} = \big((y_i^{(m)},t_i^{(m)})\big)_i \in \overline{K}$; write Y_m, T_m, $\bar t_m$ for the associated data. Thus

$$\Theta(W) - \lambda_m\chi(W) \le \Theta\big(W^{(m)}\big) - \lambda_m\chi\big(W^{(m)}\big) \qquad \big(W \in \overline{K}\big).$$

The data $\overline{K}$ (compact), Θ and χ (continuous on $\overline{K}$ by (2a), with $\chi \ge 0$ there), $D_0 \in \overline{K}$ with $\chi(D_0) = 0$ and the strict-maximum property (2b), the weights $\lambda_m > 0$ with $\lambda_m \to \infty$, and the maximizers $W^{(m)}$ are exactly the hypotheses of `PenalizedMaxConvergence`, applied in the metric space P. That lemma gives

$$W^{(m)} \longrightarrow D_0 \qquad (m \to \infty), \tag{381}$$

and hence also $(Y_m,\bar t_m) \to Z_0$, because $(Y,T) \mapsto (Y,\bar t)$ is linear and continuous and sends D_0 to Z_0. In particular there is m_0 such that for $m \ge m_0$ the maximizer lies in the interior of $\overline{K}$, i.e. $|y_i^{(m)} - y_i^0| < \rho$ and $|t_i^{(m)} - t_0| < \rho$ for every i.

Step 3: the sum rule at a fixed large λ. Fix $m \ge m_0$ and write $\lambda = \lambda_m$. Since $W^{(m)}$ is interior to $\overline{K}$ and $\Theta - \lambda\chi$ is the restriction to $\overline{K}$ of the globally defined function $W \mapsto \sum_i \theta_i v(y_i,t_i) - \Psi_\lambda(Y,T)$ on P, the point $W^{(m)}$ is a local maximum of that global function on all of P.

The test Ψ_λ is C^2 near $W^{(m)}$: it is the sum of $\lambda\chi$, a quadratic polynomial in T, and the composition of $\Phi^\sharp$ — which is C^2 near $(Y_m,\bar t_m)$ by Step 1 and (381) — with the linear map $L(Y,T) = (Y,\bar t)$.

Put

$$q_i = \theta_i^{-1}\big(\nabla\Psi_\lambda(W^{(m)})_i\big)_x, \qquad \zeta_\lambda = \big((q_i,1)\big)_{i=1}^N \in P, \qquad A_\lambda = D^2\Psi_\lambda\big(W^{(m)}\big),$$

the last written in the separate variables $\big((y_i,t_i)\big)_i$. The data N, Ω, v, the weights $\theta_i > 0$, the test Ψ_λ and the base point $W^{(m)}$ — whose i-th component $(y_i^{(m)},t_i^{(m)})$ lies in Ω by Step 1 — satisfy every hypothesis of `SumRuleCommonTimeDirection`: v is a subsolution on Ω by hypothesis, Ψ_λ

is C^2 near $W^{(m)}$ as just shown, and $W \mapsto \sum_i \theta_i v(y_i,t_i) - \Psi_\lambda(W)$ has a local maximum at $W^{(m)}$ as just shown. That lemma therefore gives

$$A_\lambda[\zeta_\lambda] \geq 0. \tag{382}$$

Finally we identify q_i. The i-th block of $\nabla\Psi_\lambda(W^{(m)})$ has spatial part $\nabla_{y_i}\Psi_\lambda(Y_m,T_m)$, and $\nabla_{y_i}\Psi_\lambda(Y_m,T_m) = \nabla_{y_i}\Phi^\sharp(Y_m,\bar t_m)$ because $\lambda\chi$ does not involve Y and the linear map L is the identity in the Y variables; so

$$q_i = \theta_i^{-1}\nabla_{y_i}\Phi^\sharp(Y_m,\bar t_m). \tag{383}$$

Step 4: removing the penalty. With $L(Y,T) = (Y,\bar t)$, a linear map from $\ell^2(G\times\mathbb{R}^N)$ to $\ell^2(G\times\mathbb{R})$, we have $\Psi_\lambda = \Phi^\sharp\circ L + \lambda\chi$, so by the chain rule for a composition with a linear map and the bilinearity of second derivatives,

$$A_\lambda[\xi] = D^2\Phi^\sharp(Y_m,\bar t_m)\big[L\xi\big] + \lambda D^2\chi\big[\xi_T\big] \qquad \big(\xi = (\xi_Y,\xi_T)\big),$$

where $\xi_T \in \mathbb{R}^N$ collects the time components of ξ and χ is a quadratic form in T alone.

The time components of $\zeta_\lambda = ((q_i,1))_i$ form the vector $(1,1,\dots,1)\in\mathbb{R}^N$. For this vector $\bar t = \frac{1}{N}\sum_i 1 = 1$ and each $t_i - \bar t = 0$, so χ vanishes at it; being a quadratic form, $\chi(T) = \frac12 D^2\chi[T]$, hence $D^2\chi[(1,\dots,1)] = 2\chi(1,\dots,1) = 0$. Also $L\zeta_\lambda = \big((q_i)_i,\ \frac{1}{N}\sum_i 1\big) = \big((q_i)_i,1\big)$. Therefore (382) reads

$$D^2\Phi^\sharp(Y_m,\bar t_m)\Big[\big((q_i)_{i=1}^N,\ 1\big)\Big] \geq 0. \tag{384}$$

Step 5: letting $\lambda\to\infty$. By (381), $(Y_m,\bar t_m)\to Z_0$ as $m\to\infty$. Since $\Phi^\sharp$ is C^2 on a neighbourhood of Z_0, the maps $Z\mapsto\nabla\Phi^\sharp(Z)$ and $Z\mapsto D^2\Phi^\sharp(Z)$ are continuous there. Hence, by (383),

$$q_i = \theta_i^{-1}\nabla_{y_i}\Phi^\sharp(Y_m,\bar t_m) \longrightarrow \theta_i^{-1}\nabla_{y_i}\Phi^\sharp(Z_0) = \theta_i^{-1}\nabla_{y_i}\Phi(Z_0) =: q_i^0$$

for each i, using (378) and $\theta_i>0$, and $D^2\Phi^\sharp(Y_m,\bar t_m)\to D^2\Phi^\sharp(Z_0) = D^2\Phi(Z_0)$ in the norm of bilinear forms, again by (378). The map $\big(A,\xi\big)\mapsto A[\xi]$ is continuous in the pair, because

$$\big|A[\xi] - A_0[\xi_0]\big| \leq \|A-A_0\|\,|\xi|^2 + \|A_0\|\,|\xi-\xi_0|\big(|\xi|+|\xi_0|\big),$$

so passing to the limit in (384) along $m\to\infty$, with $\xi_m = \big((q_i)_i,1\big)\to\big((q_i^0)_i,1\big)$, gives

$$D^2\Phi(Z_0)\Big[\Big(\big(\theta_i^{-1}\nabla_{y_i}\Phi(Z_0)\big)_{i=1}^N,\ 1\Big)\Big] \geq 0,$$

which is the assertion. (No bound on the time components of $\nabla\Psi_\lambda(W^{(m)})$ was needed: those components never entered the direction ζ_λ, whose time entries are the constant 1 prescribed by `SumRuleCommonTimeDirection`.) □

2.97. `BarycentricSubsolution`.

Corollary 2.97 (A barycentric subsolution)**.** *Let F be a finite-dimensional real inner-product space, write $z=(z_x,z_t)$ for a point of the space-time $\ell^2(F\times\mathbb{R})$ and put $\Omega = \{z : z_t>0\}$. Let f be a real function on the space-time which is a viscosity subsolution of $F[f]=0$ on Ω in the sense of `IsLimitSubOn` — so in particular Ω is open and f is continuous on Ω — and which has subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`. Let $c\in\mathbb{R}$. Then*

$$z \longmapsto \mathcal{C}_{z_t} f(z) - c\,z_t,$$

with $\mathcal{C}$ the barycentric envelope of `BaryEnv`, is a viscosity subsolution of $F[\cdot]=0$ on Ω in the sense of `IsLimitSubOn`.

Proof. Write $V(z) = \mathcal{C}_{z_t}f(z) - c\,z_t$. We verify the three clauses of `IsLimitSubOn` for V on Ω. That Ω is open is the first clause of `IsLimitSubOn` for f, which is a hypothesis. Throughout, the hypotheses of `BaryEnvDuality` hold for f with the open time set $I=(0,\infty)$: the slab Ω is exactly the preimage of I under $z\mapsto z_t$, f is continuous there, and the growth hypothesis is the one assumed.

Step 1: the defining supremum is finite, and dominates every configuration. Claim. *Let $t > 0$ and $x \in F$. Then the set of numbers*

$$\sum_i \theta_i f(y_i, t) - \frac{\sum_i \theta_i |y_i - x|^2}{2t},$$

taken over all finite horizontal probability configurations (θ, y) with mean x in the sense of `IsBaryConfig`*, is nonempty and bounded above. Consequently $\mathcal{C}_t f(x,t)$ is a genuine supremum and every such number is at most $\mathcal{C}_t f(x,t)$.*

Proof of the Claim. Nonemptiness: the one-point configuration $n = 1$, $\theta_1 = 1$, $y_1 = x$ satisfies `IsBaryConfig` and contributes $f(x,t)$. For the upper bound, put $h(y) = f(y,t) - |y-x|^2/(2t)$. Since $\theta_i > 0$ and $\sum_i \theta_i = 1$, and since $\sum_i \theta_i |y_i - x|^2/(2t) = \sum_i \theta_i \big(|y_i - x|^2/(2t)\big)$, the displayed number equals $\sum_i \theta_i h(y_i)$, a convex combination of values of h, and is therefore at most $\sup_F h$. It remains to bound $\sup_F h$.

Apply `SubquadraticGrowth` with the compact subinterval $J = \{t\} \subseteq (0,\infty)$ and $\varepsilon = 1/(16t) > 0$: there is $M > 0$ with $|f(y,t)| \le |y|^2/(16t)$ whenever $|y| \ge M$. Put $M' = \max(M, 2|x|)$. For $|y| \ge M'$ we have $|y - x| \ge |y| - |x| \ge |y|/2$, so

$$h(y) \le \frac{|y|^2}{16t} - \frac{(|y|/2)^2}{2t} = \frac{|y|^2}{16t} - \frac{|y|^2}{8t} \le 0.$$

For $|y| \le M'$ the point (y,t) lies in Ω, where f is continuous, and $\overline{B}(0, M')$ is compact (preamble), so h is bounded above on that ball by some C_1. Hence $\sup_F h \le \max(0, C_1) < \infty$.

Thus the set in `BaryEnv` defining $\mathcal{C}_t f(x,t)$ is nonempty and bounded above, so its supremum is a real number which is an upper bound for all its elements. □

Step 2: continuity of V on Ω. Let $z_0 \in \Omega$ and put $s = (z_0)_t > 0$. The third clause of `BaryEnvDuality` says that $(s, z) \mapsto \mathcal{C}_s f(z)$ is continuous at (s, z_0) whenever $s > 0$ and $(z_0)_t \in I$, which holds here. The map $z \mapsto (z_t, z)$ is continuous and sends z_0 to (s, z_0), so the composite $z \mapsto \mathcal{C}_{z_t} f(z)$ is continuous at z_0; subtracting the continuous function $z \mapsto c\, z_t$ shows V is continuous at z_0. As $z_0 \in \Omega$ was arbitrary and Ω is open, V is continuous on Ω.

Step 3: the viscosity inequality. Let $z_0 = (x_0, t_0) \in \Omega$, so $t_0 > 0$, let φ be C^2 near z_0, and suppose $w \mapsto V(w) - \varphi(w)$ has a local maximum on Ω at z_0. We must show $F[\varphi](z_0) \ge 0$, with F as in `LimitOp`.

(3a) An attaining configuration. The first clause of `BaryEnvDuality`, applied with $s = t_0 > 0$ and $z = z_0$ (whose time t_0 lies in I), gives n, weights $\theta : \{1, \dots, n\} \to \mathbb{R}$ and points $y^0 = (y_i^0)$ with (θ, y^0) a finite horizontal probability configuration of mean x_0 in the sense of `IsBaryConfig` and

$$\mathcal{C}_{t_0} f(z_0) = \sum_i \theta_i f(y_i^0, t_0) - \frac{R^2}{2t_0}, \qquad R^2 := \sum_i \theta_i |y_i^0 - x_0|^2, \tag{385}$$

where we used that the radius of `BaryRadius` is the square root of the nonnegative number R^2, so that its square is R^2. By `IsBaryConfig` we have $\theta_i > 0$, $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i^0 = x_0$; in particular the index set is nonempty, since an empty sum cannot equal 1, so $n \ge 1$. These weights are now held fixed.

(3b) The test function on the product. For $Y = (y_i)$ in the ℓ^2-product P of n copies of F put

$$X(Y) = \sum_i \theta_i y_i, \qquad Q(Y) = \sum_i \theta_i |y_i - X(Y)|^2, \qquad \Phi(Y,t) = \varphi(X(Y), t) + ct + \frac{Q(Y)}{2t}.$$

This is exactly the lifted test of `BaryTestSecondVariation`, whose hypotheses hold here: $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i^0 = x_0$ by (3a), $t_0 > 0$, φ is C^2 at $(x_0, t_0) = z_0$, and Φ is given by the displayed formula at every point of the product space-time. Its first conclusion is that Φ is C^2 at (Y^0, t_0), where $Y^0 = (y_i^0)_i$.

(3c) The local maximum on the product. Let $t > 0$ and let Y be arbitrary. The pair (θ, Y) is a finite horizontal probability configuration with mean $X(Y)$, since $\theta_i > 0$, $\sum_i \theta_i = 1$ and

$\sum_i \theta_i y_i = X(Y)$ by definition. So by Step 1, applied at the point $(X(Y), t)$,

$$\sum_i \theta_i f(y_i, t) - \frac{Q(Y)}{2t} \le \mathcal{C}_t f(X(Y), t),$$

and therefore

$$\sum_i \theta_i f(y_i, t) - \Phi(Y, t) \le \mathcal{C}_t f(X(Y), t) - ct - \varphi(X(Y), t) = V\big(X(Y), t\big) - \varphi\big(X(Y), t\big). \tag{386}$$

By hypothesis there is a neighbourhood $\mathcal{O}$ of z_0 with $V - \varphi \le (V - \varphi)(z_0)$ on $\mathcal{O} \cap \Omega$. The map $(Y, t) \mapsto (X(Y), t)$ is linear and continuous, being built from coordinate projections and a finite linear combination, and it sends (Y^0, t_0) to z_0 because $\sum_i \theta_i y_i^0 = x_0$; so there is a neighbourhood $\mathcal{N}$ of (Y^0, t_0) whose image under that map lies in $\mathcal{O}$, and if moreover $t > 0$ then $(X(Y), t) \in \Omega$. Hence by (386)

$$\sum_i \theta_i f(y_i, t) - \Phi(Y, t) \le (V - \varphi)(z_0) \qquad \big((Y, t) \in \mathcal{N},\ t > 0\big).$$

At (Y^0, t_0) there is equality: $X(Y^0) = x_0$ and $Q(Y^0) = R^2$, so by (385)

$$\sum_i \theta_i f(y_i^0, t_0) - \Phi(Y^0, t_0) = \Big(\sum_i \theta_i f(y_i^0, t_0) - \frac{R^2}{2t_0}\Big) - \varphi(z_0) - ct_0 = \mathcal{C}_{t_0} f(z_0) - ct_0 - \varphi(z_0) = (V - \varphi)(z_0).$$

Therefore $(Y, t) \mapsto \sum_i \theta_i f(y_i, t) - \Phi(Y, t)$ has a local maximum on $\{(Y, t) : t \in I\}$ at (Y^0, t_0).

(3d) Applying the common-time rule. All the hypotheses of `CommonTimeSubsolution` now hold, with this $n \ge 1$, the open time set $I = (0, \infty)$, the subsolution f, the weights θ, the test Φ, the point (Y^0, t_0) with $t_0 \in I$, the C^2 property of (3b) and the local maximum of (3c). It gives

$$D^2\Phi(Y^0, t_0)\big[\zeta, \zeta\big] \ge 0, \qquad \zeta := \big((q_i)_i,\, 1\big), \quad q_i := \theta_i^{-1} \nabla_{y_i} \Phi(Y^0, t_0). \tag{387}$$

(3e) The directions. Write $p = \nabla_x \varphi(z_0)$ for the spatial part of $\nabla \varphi(z_0)$. The hypotheses of `BaryTestDirections` are the ones already checked in (3b), together with the positivity $\theta_i > 0$ from (3a). That lemma therefore gives

$$q_i = p + \frac{y_i^0 - x_0}{t_0} \qquad (1 \le i \le n), \tag{388}$$

so that the direction ζ of (387) is exactly the direction $\big(\big(p + (y_i^0 - x_0)/t_0\big)_i, 1\big)$ named in `BaryTestSecondVariation`.

(3f) The second variation is the operator. The second conclusion of `BaryTestSecondVariation`, whose hypotheses were checked in (3b), reads

$$D^2\Phi\big(Y^0, t_0\big)\Big[\big(\big(p + \tfrac{y_i^0 - x_0}{t_0}\big)_i, 1\big),\ \big(\big(p + \tfrac{y_i^0 - x_0}{t_0}\big)_i, 1\big)\Big] = F[\varphi](z_0).$$

By (388) the left-hand side is $D^2\Phi(Y^0, t_0)[\zeta, \zeta]$, so (387) gives $F[\varphi](z_0) \ge 0$.

As $z_0 \in \Omega$ and the test φ were arbitrary, the third clause of `IsLimitSubOn` holds for V on Ω; with Steps 1 and 2 this completes the verification. The argument was carried out for an arbitrary $c \in \mathbb{R}$, and in particular covers $c = 0$. □

2.98. `FitComparisonSameCenter`.

Lemma 2.98 (Comparison of fits at one center across scales)**.** *Let F be a finite-dimensional real inner-product space and let $1 \le C_0 < \infty$. There is a constant $C > 0$, depending only on C_0, with the following property. Let v be a real function on the space-time $\ell^2(F \times \mathbb{R})$ and let z be a space-time point such that v admits a fit at z of every positive radius with constant C_0, in the sense of `IsFit`. Then, for all $R, R' > 0$ and all fits (p, a) for v at z of radius R and (p', a') for v at z of radius R', both with constant C_0,*

$$|p - p'| + |a - a'| \le C\Big(1 + \big|\log(R'/R)\big|\Big). \tag{389}$$

Proof. Write

$$\ell_{p,a}(h,\ell) = p\cdot h + \Big(a - \frac{|p|^2}{2}\Big)\ell$$

for the affine increment attached to a fit, so that `IsFit` reads $\big|v(z+(h,\ell)) - v(z) - \ell_{p,a}(h,\ell)\big| \le C_0 R$ whenever $|\ell| \le R$ and $|h - p\ell| \le R$.

Step 1: comparable radii. We first show that there is $C_1 > 0$, depending only on C_0, such that

$$|p - p'| + |a - a'| \le C_1 \tag{390}$$

whenever (p,a) and (p',a') are fits for v at z of radii R and R' with $0 < R \le R' \le 2R$.

Consider the increments $(h,0)$ with $|h| \le R$. The box conditions for the first fit read $|0| \le R$ and $|h - p\cdot 0| = |h| \le R$, both satisfied; those for the second read $|0| \le R'$ and $|h| \le R'$, satisfied because $R \le R'$. So both estimates of `IsFit` apply at $(h,0)$, and subtracting them, using $\ell_{p,a}(h,0) = p\cdot h$, $\ell_{p',a'}(h,0) = p'\cdot h$ and $R' \le 2R$,

$$|(p-p')\cdot h| \le C_0 R + C_0 R' \le 3C_0 R \qquad (|h| \le R).$$

If $p \ne p'$, taking $h = R(p-p')/|p-p'|$, for which $|h| = R$ and $(p-p')\cdot h = R|p-p'|$, and dividing by $R > 0$ gives

$$|p - p'| \le 3C_0 \le 4C_0, \tag{391}$$

and this is trivial when $p = p'$.

Put $c = (2 + 8C_0)^{-1}$, so $0 < c \le \frac12$ and $c \le 1/(8C_0)$, and consider the increment $(h,\ell) = (cRp,\, cR)$. It lies in the first box: $|\ell| = cR \le R$ and $|h - p\ell| = 0 \le R$. It lies in the second box too: $|\ell| = cR \le R \le R'$ and, by (391), $|h - p'\ell| = cR|p - p'| \le 4C_0 cR \le \frac12 R \le R'$. At this increment,

$$\big(\ell_{p,a} - \ell_{p',a'}\big)(cRp, cR) = cR\Big[(p-p')\cdot p + a - a' - \frac{|p|^2 - |p'|^2}{2}\Big] = cR\Big[a - a' + \frac{|p-p'|^2}{2}\Big],$$

because $(p-p')\cdot p - \frac12(|p|^2 - |p'|^2) = |p|^2 - p\cdot p' - \frac12|p|^2 + \frac12|p'|^2 = \frac12|p-p'|^2$. Subtracting the two estimates of `IsFit` at this increment, whose errors sum to at most $C_0R + C_0R' \le 3C_0R$, gives $cR\big|a - a' + \frac12|p-p'|^2\big| \le 3C_0R$, hence by (391)

$$|a - a'| \le \frac{3C_0}{c} + \frac{|p-p'|^2}{2} \le 3C_0(2 + 8C_0) + 8C_0^2. \tag{392}$$

Adding (391) and (392) gives (390) with $C_1 = 4C_0 + 3C_0(2+8C_0) + 8C_0^2 > 0$.

Step 2: a dyadic chain between two arbitrary radii. Let now $R, R' > 0$ be arbitrary and let (p,a), (p',a') be fits at z of these radii. Put

$$s_0 = \min\{R, R'\}, \qquad s_1 = \max\{R, R'\}, \qquad \varrho = \frac{s_1}{s_0} \ge 1,$$

and let $k \ge 0$ be the least integer with $2^k \ge \varrho$. Define

$$\rho_j = s_0\, 2^j \quad (0 \le j \le k-1), \qquad \rho_k = s_1,$$

so $\rho_0 = s_0$ and $\rho_k = s_1$ (when $k = 0$ we have $\varrho = 1$, hence $s_0 = s_1$ and the chain is the single radius s_0). All ρ_j are positive, and consecutive radii satisfy $\rho_j \le \rho_{j+1} \le 2\rho_j$: for $j + 1 \le k - 1$ the ratio is exactly 2; and for $j = k - 1$ with $k \ge 1$,

$$\frac{\rho_k}{\rho_{k-1}} = \frac{s_1}{s_0\, 2^{k-1}} = \frac{\varrho}{2^{k-1}},$$

which is ≥ 1 because $2^{k-1} < \varrho$ by minimality of k, and ≤ 2 because $2^k \ge \varrho$.

Suppose $k \ge 1$. For $1 \le j \le k-1$ use the hypothesis to choose a fit (p_j, a_j) for v at z of radius $\rho_j > 0$ with constant C_0, and at the two ends put

$$(p_0,a_0) = \begin{cases} (p,a) & \text{if } s_0 = R,\\ (p',a') & \text{otherwise,}\end{cases} \qquad (p_k,a_k) = \begin{cases} (p',a') & \text{if } s_0 = R,\\ (p,a) & \text{otherwise,}\end{cases}$$

which are fits at the radii $\rho_0 = s_0$ and $\rho_k = s_1$ respectively. Applying (390) to each of the k consecutive pairs (ρ_j, ρ_{j+1}) and summing with the triangle inequality,

$$|p - p'| + |a - a'| = |p_0 - p_k| + |a_0 - a_k| \le \sum_{j=0}^{k-1} \big(|p_{j+1} - p_j| + |a_{j+1} - a_j|\big) \le k\, C_1.$$

If $k = 0$ then $R = R'$, and (390) applied once with $R = R'$ gives $|p - p'| + |a - a'| \le C_1$. In all cases

$$|p - p'| + |a - a'| \le (k + 1)\, C_1. \tag{393}$$

Step 3: bounding k. We claim $k < 1 + \big|\log(R'/R)\big|/\log 2$. If $k = 0$ this is immediate since the absolute value is nonnegative. If $k \ge 1$, minimality of k gives $2^{k-1} < \varrho$, i.e. $(k-1)\log 2 < \log \varrho$; and

$$\log \varrho = \log \frac{\max\{R, R'\}}{\min\{R, R'\}} = \big|\log(R'/R)\big|,$$

since $\log(R'/R)$ and $\log(R/R')$ are negatives of one another. Dividing by $\log 2 > 0$ gives the claim.

Combining with (393) and using $2 + u/\log 2 \le (2 + 1/\log 2)(1 + u)$ for $u = |\log(R'/R)| \ge 0$,

$$|p - p'| + |a - a'| \le C_1\Big(2 + \frac{|\log(R'/R)|}{\log 2}\Big) \le C_1\Big(2 + \frac{1}{\log 2}\Big)\Big(1 + \big|\log(R'/R)\big|\Big).$$

Thus (389) holds with $C = C_1(2 + 1/\log 2) > 0$, which depends only on C_0. □

2.99. `FrameInvariance`.

Lemma 2.99 (Changes of frame)**.** *Let F be a finite-dimensional real inner-product space, let $0 \le C_0 < \infty$, let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`, let $z_0 = (x_0, t_0)$ be a space-time point, let $R > 0$ and let $(p_0, a_0) \in F \times \mathbb{R}$. Let $\bar v$ be the change of frame of `FrameChange`,*

$$\bar v(y, s) = \frac{v\big(x_0 + R(y + p_0 s),\, t_0 + Rs\big) - v(x_0, t_0)}{R} - p_0 \cdot y - \Big(a_0 + \frac{|p_0|^2}{2}\Big)\, s.$$

Then $\bar v \in \mathcal{A}_{m,C_0}$. Moreover, writing $T(y, s) = (x_0 + R(y + p_0 s), t_0 + Rs)$ for the corresponding old point, if (p, a) is a fit for v at $T(\zeta)$ of radius $R\rho$ with constant C_0, in the sense of `IsFit`, then $(p - p_0, a - a_0)$ is a fit for $\bar v$ at ζ of radius ρ with constant C_0.

Proof. Step 1: the equation. Write $\zeta = (y, s)$ for a space-time point in the new frame, let

$$A(y, s) = -p_0 \cdot y - \Big(a_0 + \frac{|p_0|^2}{2}\Big) s$$

be the affine part of the formula of `FrameChange`, and let B be the linear map $B(h, \ell) = (h + p_0\ell, \ell)$ on $F \times \mathbb{R}$. Then $T(\zeta) = z_0 + R\, B\zeta$, so the linear part of the affine map T is $R\, B$, and the formula of `FrameChange` reads

$$\bar v = \frac{v \circ T - v(z_0)}{R} + A. \tag{394}$$

(1a) T is an affine homeomorphism, and $\bar v$ is continuous. The map B is linear and invertible with $B^{-1}(h, \ell) = (h - p_0\ell, \ell)$, because $B(h - p_0\ell, \ell) = (h, \ell)$ and $B^{-1}(h + p_0\ell, \ell) = (h, \ell)$; both are continuous, being built from vector addition and scalar multiplication. Since $R > 0$, the linear part $R\, B$ of T is a linear homeomorphism of space-time onto itself, so T is an affine homeomorphism of space-time onto itself with

$$T^{-1}(Z) = B^{-1}\big(R^{-1}(Z - z_0)\big).$$

In particular T and T^{-1} are continuous, T maps open sets to open sets, and T carries every neighbourhood of a point ζ onto a neighbourhood of $T(\zeta)$.

By clause (1) of `IsFitClass` the function v is continuous on all of space-time. In (394) the composite $v \circ T$ is therefore continuous, $v(z_0)$ is a constant, $R > 0$, and A is linear and hence continuous; so $\bar v$ is continuous on all of space-time. The whole space is open, which is the

openness requirement that `IsLimitSubOn` and `IsLimitSuperOn` place on Ω when Ω is the whole space.

(1b) The adjoint of B. The space-time inner product is $\langle (h,\ell),(g,k)\rangle = \langle h,g\rangle + \ell k$, so for all (h,ℓ) and (g,k)

$$\big\langle B(h,\ell),(g,k)\big\rangle = \langle h + p_0\ell, g\rangle + \ell k = \langle h,g\rangle + \ell\big(\langle p_0,g\rangle + k\big) = \big\langle (h,\ell),\big(g,\ \langle p_0,g\rangle + k\big)\big\rangle,$$

whence

$$B^{\mathsf T}(g,k) = \big(g,\ \langle p_0,g\rangle + k\big), \tag{395}$$

which is the identity in the spatial slot. For a symmetric bilinear form M on space-time write $B^{\mathsf T}MB$ for the pulled-back form $(\xi,\eta)\mapsto M[B\xi,B\eta]$; then, taking $\eta=\xi$,

$$\big(B^{\mathsf T}MB\big)[\xi] = M[B\xi] \qquad (\xi\in F\times\mathbb{R}). \tag{396}$$

(1c) The frame change of a C^2 test, and the identity for F. Let Z_0 be a space-time point, put $\zeta_0 = T^{-1}(Z_0)$, and let ψ be C^2 near Z_0. Define the frame change of ψ by

$$\phi = \frac{\psi\circ T - v(z_0)}{R} + A, \tag{397}$$

built from ψ by the formula (394), with the additive constant kept as $v(z_0)$. If ψ is C^2 on the open set $N\ni Z_0$, then $T^{-1}(N)$ is open and contains ζ_0 by (1a), and $\psi\circ T$ is C^2 there because T is affine; since A is affine and $R>0$, the function ϕ is C^2 near ζ_0.

Write $\nabla\psi(Z_0) = (\nabla_x\psi,\psi_t)$ for the space-time gradient of ψ at Z_0, split into spatial and time parts as in `LimitOp`, and note that A is linear with $\nabla A = \big(-p_0,\ -(a_0+|p_0|^2/2)\big)$. Because T is affine with linear part $R\,B$, and the second derivative of an affine map vanishes, the chain rule applied to (397) gives, for all $\xi,\eta\in F\times\mathbb{R}$,

$$D\phi(\zeta_0)\xi = \frac{D\psi(Z_0)(R\,B\xi)}{R} + \langle\nabla A,\xi\rangle = D\psi(Z_0)(B\xi) + \langle\nabla A,\xi\rangle, \qquad D^2\phi(\zeta_0)[\xi,\eta] = \frac{D^2\psi(Z_0)[R\,B\xi,\ R\,B\eta]}{R}.$$

The inner product identifies derivatives with gradients, so the first identity reads $\langle\nabla\phi(\zeta_0),\xi\rangle = \langle\nabla\psi(Z_0),B\xi\rangle + \langle\nabla A,\xi\rangle = \langle B^{\mathsf T}\nabla\psi(Z_0) + \nabla A,\xi\rangle$ for every ξ, so by (395)

$$\nabla\phi(\zeta_0) = B^{\mathsf T}\nabla\psi(Z_0) + \nabla A = \Big(\nabla_x\psi - p_0,\ \langle p_0,\nabla_x\psi\rangle + \psi_t - a_0 - \frac{|p_0|^2}{2}\Big),$$

whose spatial part is

$$\nabla_y\phi(\zeta_0) = \nabla_x\psi(Z_0) - p_0. \tag{398}$$

The second identity, since $R^{-1}R^2 = R$, reads in the notation of (396)

$$D^2\phi(\zeta_0) = R\,B^{\mathsf T}\,D^2\psi(Z_0)\,B. \tag{399}$$

These are the two identities that the change of frame induces, now asserted for a C^2 test pair rather than for v and $\bar v$, which by hypothesis are only continuous, so that neither D^2v nor $D^2\bar v$ need exist. Adding a constant to ψ changes neither $\nabla\psi(Z_0)$ nor $D^2\psi(Z_0)$, hence changes neither (398) nor (399): the particular constant $-v(z_0)/R$ in (397) is immaterial to what follows.

By (398) and the definition of B,

$$B\big(\nabla_y\phi(\zeta_0),1\big) = \big(\nabla_y\phi(\zeta_0) + p_0,\ 1\big) = \big(\nabla_x\psi(Z_0),1\big), \tag{400}$$

so by the definition of F in `LimitOp`, then (399), (396) and (400),

$$F[\phi](\zeta_0) = D^2\phi(\zeta_0)\big[(\nabla_y\phi(\zeta_0),1)\big] = R\,D^2\psi(Z_0)\big[B(\nabla_y\phi(\zeta_0),1)\big] = R\,D^2\psi(Z_0)\big[(\nabla_x\psi(Z_0),1)\big] = R\,F[\psi](Z_0). \tag{401}$$

(1d) The test transfer, both signs. Let ζ_0 be any space-time point and let ϕ be C^2 near ζ_0 such that $\zeta\mapsto\bar v(\zeta)-\phi(\zeta)$ has a local maximum at ζ_0. Put $Z_0 = T(\zeta_0)$ and

$$\psi = R\,(\phi - A)\circ T^{-1} + v(z_0). \tag{402}$$

If ϕ is C^2 on the open set $M\ni\zeta_0$, then $T(M)$ is open and contains Z_0 by (1a), and ψ is C^2 there because T^{-1} and A are affine; so ψ is C^2 near Z_0, and $T^{-1}(Z_0)=\zeta_0$.

First, ϕ is the frame change of this ψ in the sense of (397). Indeed, evaluating (402) at $Z = T\zeta$, so that $T^{-1}Z = \zeta$, gives $\psi(T\zeta) = R\big(\phi(\zeta) - A(\zeta)\big) + v(z_0)$, whence

$$\frac{\psi(T\zeta) - v(z_0)}{R} + A(\zeta) = \big(\phi(\zeta) - A(\zeta)\big) + A(\zeta) = \phi(\zeta) \qquad (\zeta \in F \times \mathbb{R}).$$

So (1c) applies to the pair (ψ, ϕ) at Z_0 and ζ_0, and (401) holds for them.

Second, by (394) evaluated at $\zeta = T^{-1}Z$ we have $v(Z) = R\big(\bar v(T^{-1}Z) - A(T^{-1}Z)\big) + v(z_0)$, while (402) says $\psi(Z) = R\big(\phi(T^{-1}Z) - A(T^{-1}Z)\big) + v(z_0)$; subtracting, the two copies of $R\,A(T^{-1}Z)$ and of $v(z_0)$ cancel and

$$v(Z) - \psi(Z) = R\Big(\bar v\big(T^{-1}Z\big) - \phi\big(T^{-1}Z\big)\Big) \tag{403}$$

for every space-time point Z.

Now transfer the local maximum. Let N_0 be a neighbourhood of ζ_0 with $\bar v - \phi \le (\bar v - \phi)(\zeta_0)$ on N_0. By (1a) the set $T(N_0)$ is a neighbourhood of Z_0, and for $Z \in T(N_0)$ we have $T^{-1}Z \in N_0$, so by (403) and $R > 0$

$$v(Z) - \psi(Z) = R\,\big(\bar v - \phi\big)\big(T^{-1}Z\big) \le R\,\big(\bar v - \phi\big)(\zeta_0) = v(Z_0) - \psi(Z_0),$$

the last equality by (403) at $Z = Z_0$. Hence $Z \mapsto v(Z) - \psi(Z)$ has a local maximum at Z_0. By clause (1) of `IsFitClass` and `IsLimitSolutionOn`, v is a viscosity subsolution of $F[v] = 0$ on the whole space in the sense of `IsLimitSubOn`; its subsolution inequality, applied at the point Z_0 with the test ψ, which is C^2 near Z_0 and for which $v - \psi$ has a local maximum there, gives $F[\psi](Z_0) \ge 0$. Since $R > 0$, (401) yields

$$F[\phi](\zeta_0) = R\,F[\psi](Z_0) \ge 0.$$

If instead $\bar v - \phi$ has a local *minimum* at ζ_0, the same construction applies verbatim with the two inequalities reversed: $v - \psi$ has a local minimum at Z_0 by (403) and $R > 0$; by clause (1) of `IsFitClass` and `IsLimitSolutionOn`, v is a viscosity supersolution of $F[v] = 0$ on the whole space in the sense of `IsLimitSuperOn`, so its supersolution inequality at Z_0 with the test ψ gives $F[\psi](Z_0) \le 0$; and (401) yields $F[\phi](\zeta_0) = R\,F[\psi](Z_0) \le 0$.

(1e) $\bar v$ solves the equation. By (1a) the whole space is open and $\bar v$ is continuous on it. By (1d), for every space-time point ζ_0 and every ϕ that is C^2 near ζ_0: if $\bar v - \phi$ has a local maximum at ζ_0 then $F[\phi](\zeta_0) \ge 0$, and if $\bar v - \phi$ has a local minimum at ζ_0 then $F[\phi](\zeta_0) \le 0$. With Ω the whole space, a local maximum of $\bar v - \phi$ on Ω at ζ_0, as `IsLimitSubOn` requires, is an ordinary local maximum at ζ_0, and likewise for local minima in `IsLimitSuperOn`. So the three conditions of each of those two definitions hold, and $\bar v$ is a viscosity subsolution and a viscosity supersolution of $F[\bar v] = 0$ on all of space-time, that is, by `IsLimitSolutionOn`, a viscosity solution of $F[\bar v] = 0$ there. Together with the continuity of $\bar v$ this is clause (1) of `IsFitClass` for $\bar v$.

Step 2: the fits. Let (p, a) be a fit for v at $Z = T(\zeta)$ of radius $R\rho$. Let $h \in F$ and $\ell \in \mathbb{R}$ with

$$|\ell| \le \rho, \qquad |h - (p - p_0)\ell| \le \rho. \tag{404}$$

The increment (h, ℓ) in the new variables corresponds to the old increment $R\,B(h,\ell) = \big(R(h + p_0\ell), R\ell\big)$, and

$$T(\zeta + (h,\ell)) = Z + \big(R(h + p_0\ell), R\ell\big).$$

The old box conditions for the fit (p, a) at radius $R\rho$ are $|R\ell| \le R\rho$ and $|R(h + p_0\ell) - pR\ell| \le R\rho$, i.e. $|\ell| \le \rho$ and $|h - (p - p_0)\ell| \le \rho$; these are exactly (404). So `IsFit` for v gives

$$\Big| v\big(T(\zeta + (h,\ell))\big) - v(Z) - p \cdot R(h + p_0\ell) - \Big(a - \frac{|p|^2}{2}\Big) R\ell \Big| \le C_0 R\rho. \tag{405}$$

By the formula of `FrameChange`,

$$\bar v(\zeta + (h,\ell)) - \bar v(\zeta) = \frac{v\big(T(\zeta + (h,\ell))\big) - v(Z)}{R} - p_0 \cdot h - \Big(a_0 + \frac{|p_0|^2}{2}\Big)\ell.$$

Subtracting the new affine increment $(p - p_0) \cdot h + \big(a - a_0 - |p - p_0|^2/2\big)\ell$ and using the identity

$$\Big(a_0 + \frac{|p_0|^2}{2}\Big) + \Big(a - a_0 - \frac{|p - p_0|^2}{2}\Big) = a - \frac{|p|^2}{2} + p \cdot p_0,$$

which holds because $|p-p_0|^2 = |p|^2 - 2p\cdot p_0 + |p_0|^2$, we obtain

$$\bar v(\zeta+(h,\ell))-\bar v(\zeta)-(p-p_0)\cdot h-\Big(a-a_0-\frac{|p-p_0|^2}{2}\Big)\ell = \frac{1}{R}\Big[\,v\big(T(\zeta+(h,\ell))\big)-v(Z)-p\cdot R(h+p_0\ell)-\Big(a-\frac{|p|^2}{2}\Big)R\ell\,\Big].$$

Its absolute value is therefore at most $C_0R\rho/R = C_0\rho$ by (405). By `IsFit` this says that $(p-p_0, a-a_0)$ is a fit for $\bar v$ at ζ of radius ρ with constant C_0.

Step 3: $\bar v$ has fits at every center and radius. Let ζ be any space-time point and $\rho>0$. Since $v\in\mathcal{A}_{m,C_0}$, `IsFitClass` provides a fit (p,a) for v at $T(\zeta)$ of radius $R\rho>0$. By Step 2, $(p-p_0,a-a_0)$ is a fit for $\bar v$ at ζ of radius ρ. Together with Step 1 this shows $\bar v\in\mathcal{A}_{m,C_0}$. □

2.100. `CenteredGrowth`.

Lemma 2.100 (Growth of a centered member)**.** *Let F be a finite-dimensional real inner-product space and let $1\le C_0<\infty$. Write $L(K)=1+\log K$ for $K\ge 2$ and $\mathcal{B}_K=\{(x,t):|x|\le K,\ |t|\le K\}$. There is $C>0$, depending only on C_0 and $\dim F$, such that for every $v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` which is centered, in the sense of `IsCenteredAt` with constant C_0, at $(0,0)$ or at $(0,1)$,*

$$|v(x,t)|\le C\,K\,L(K)^2 \qquad \big((x,t)\in\mathcal{B}_K,\ K\ge 2\big). \tag{406}$$

Proof. Write $L(K)=1+\log K$, so $L(K)\ge 1+\log 2>1$ for $K\ge 2$. Let $C_1^0>0$ be the constant of `FitComparisonSameCenter` for this C_0 and put $C_1=\max\{C_1^0,1\}$. The conclusion of `FitComparisonSameCenter` is an upper bound, so it continues to hold with $C_1\ge C_1^0$ in place of C_1^0; we therefore work throughout with a constant $C_1\ge 1$ which still depends only on C_0. We fix C_1 now, before any other constant is chosen, because the choice of A in Step 1 below depends on it. We first record the consequence we shall use: if $w\in\mathcal{A}_{m,C_0}$ is centered at the origin with constant C_0, then for every $r>0$ and every fit (p,a) for w at the origin of radius r with constant C_0,

$$|p|+|a|\le C_1\big(1+|\log r|\big). \tag{407}$$

Indeed, by `IsFitClass` the function w admits a fit at the origin of every positive radius with constant C_0, and by `IsCenteredAt` the pair $(0,0)$ is such a fit of radius 1. Applying `FitComparisonSameCenter` at the center $z=0$ with the two radii r and $R'=1$ and the fits (p,a) and $(0,0)$ gives $|p-0|+|a-0|\le C_1\big(1+|\log(1/r)|\big)=C_1\big(1+|\log r|\big)$, since $\log(1/r)=-\log r$.

Step 1: choice of the constant A. Because $(\log\sigma)/\sigma\to 0$ as $\sigma\to\infty$, also $(3+\log A)/A\to 0$ as $A\to\infty$. Since $C_1\ge 1$, the number $1/(8C_1)$ is positive, so there is $A_0>0$ with

$$\frac{3+\log A}{A}\le\frac{1}{8C_1} \qquad (A\ge A_0).$$

Fix once and for all

$$A=\max\{4,\ A_0\},$$

so that $A\ge 4$ and, applying the displayed bound at this $A\ge A_0$,

$$A\ge 8\,C_1(3+\log A). \tag{408}$$

The choice depends only on C_1, hence only on C_0 and $\dim F$, and it is made once, for all v and all K. Put

$$C_2=C_1(3+\log A),$$

so that (408) reads $A\ge 8C_2$, and $C_2\ge C_1\ge 1$ because $3+\log A\ge 3+\log 4>1$.

Step 2: the case of centering at the origin. Let $w\in\mathcal{A}_{m,C_0}$ be centered at $(0,0)$ with constant C_0 in the sense of `IsCenteredAt`, so $w(0,0)=0$, and let $K\ge 2$. Put

$$R=A\,K\,L(K)\ \ (\ge A\cdot 2\cdot 1\ge 8>1).$$

By `IsFitClass` there is a fit (p_R,a_R) for w at the origin of radius R with constant C_0. Since $R>1$ we have $|\log R|=\log R$, and

$$\log R=\log A+\log K+\log L(K)\le \log A+L(K)+L(K)=\log A+2L(K),$$

using $\log K \le 1 + \log K = L(K)$ and $\log L(K) \le L(K)$ (valid because $\log \sigma \le \sigma$ for $\sigma > 0$). Hence by (407) and $L(K) \ge 1$,

$$|p_R| + |a_R| \le C_1\big(1 + \log A + 2L(K)\big) \le C_1(3 + \log A)L(K) = C_2 L(K). \tag{409}$$

The middle inequality holds because

$$(3 + \log A)L(K) = 2L(K) + L(K) + (\log A)L(K) \ge 2L(K) + 1 + \log A,$$

where the second summand is bounded below using $L(K) \ge 1$ and the third using $(\log A)L(K) \ge \log A$, which needs $\log A \ge 0$ and holds since $A \ge 4$.

Now let $z = (x, t)$ with $|x| \le K$ and $|t| \le K$. We check that the increment (x, t) from the origin lies in the sheared box of the fit (p_R, a_R) of radius R, i.e. that $|t| \le R$ and $|x - p_R t| \le R$. First,

$$|t| \le K \le \frac{A\,K\,L(K)}{4} = \frac{R}{4} \le R,$$

because $AL(K) \ge 4$ by $A \ge 4$ and $L(K) \ge 1$. Second, by (409) and $|t| \le K$,

$$|x - p_R t| \le |x| + |p_R||t| \le K + C_2 L(K) K \le 2C_2 K L(K),$$

using $1 \le C_2 L(K)$; and $2C_2 K L(K) \le \frac{A}{4} K L(K) = \frac{R}{4} \le R$ because $A \ge 8C_2$. So both box conditions hold, and `IsFit` applied with $h = x$, $\ell = t$ at the center $(0, 0)$ gives, using $w(0, 0) = 0$,

$$\Big| w(x, t) - p_R \cdot x - \Big(a_R - \frac{|p_R|^2}{2}\Big) t \Big| \le C_0 R.$$

Therefore, by Cauchy–Schwarz, $|x| \le K$, $|t| \le K$ and (409),

$$|w(x, t)| \le C_0 R + |p_R|\,K + \Big(|a_R| + \frac{|p_R|^2}{2}\Big) K \le C_0 A K L(K) + C_2 L(K) K + C_2 L(K) K + \frac{C_2^2 L(K)^2}{2} K.$$

Since $L(K) \ge 1$ we have $L(K) \le L(K)^2$, so with

$$C_3 = C_0 A + 2C_2 + \tfrac{1}{2} C_2^2$$

we obtain

$$|w(x, t)| \le C_3\, K\, L(K)^2 \qquad (|x| \le K,\ |t| \le K,\ K \ge 2). \tag{410}$$

Step 3: the case of centering at $(0, 1)$. Let $v \in \mathcal{A}_{m, C_0}$ be centered at $(0, 1)$ with constant C_0. Apply `FrameInvariance` with $z_0 = (0, 1)$, $R = 1$, $p_0 = 0$ and $a_0 = 0$. By `FrameChange`, and using $v(0, 1) = 0$ from `IsCenteredAt`, the resulting function is

$$\bar v(y, s) = \frac{v\big(0 + 1 \cdot (y + 0 \cdot s),\ 1 + 1 \cdot s\big) - v(0, 1)}{1} - 0 \cdot y - \Big(0 + \frac{|0|^2}{2}\Big) s = v(y, 1 + s),$$

and `FrameInvariance` gives $\bar v \in \mathcal{A}_{m, C_0}$.

We check that $\bar v$ is centered at the origin with constant C_0. First $\bar v(0, 0) = v(0, 1) = 0$. Second, by `IsCenteredAt` the pair $(0, 0)$ is a fit for v at $(0, 1)$ of radius 1 with constant C_0, so by `IsFit`, for all h with $|h| \le 1$ and all ℓ with $|\ell| \le 1$,

$$\big| v(0 + h, 1 + \ell) - v(0, 1) - 0 \cdot h - (0 - 0)\ell \big| \le C_0 \cdot 1,$$

that is $|\bar v(h, \ell)| = |v(h, 1 + \ell)| \le C_0$. Since $\bar v(0, 0) = 0$, this is exactly the statement of `IsFit` saying that $(0, 0)$ is a fit for $\bar v$ at the origin of radius 1 with constant C_0. By `IsCenteredAt`, $\bar v$ is centered at the origin with constant C_0.

Let now $|x| \le K$ and $|t| \le K$ with $K \ge 2$, and put $s = t - 1$, so $v(x, t) = \bar v(x, s)$. Then $|x| \le K \le 2K$ and $|s| \le |t| + 1 \le K + 1 \le 2K$, and $2K \ge 2$, so (410) applied to $\bar v$ at the box size $2K$ gives

$$|v(x, t)| = |\bar v(x, s)| \le C_3\,(2K)\,L(2K)^2.$$

Finally $L(2K) = 1 + \log 2 + \log K \le (1 + \log 2)(1 + \log K) = (1 + \log 2)L(K)$, because $1 + \log 2 + \log K \le (1 + \log 2) + (1 + \log 2)\log K$ and $\log K \ge 0$. Hence

$$|v(x, t)| \le 2(1 + \log 2)^2\, C_3\, K\, L(K)^2.$$

Conclusion. Taking $C = \max\{C_3,\ 2(1 + \log 2)^2 C_3\} + 1 = 2(1 + \log 2)^2 C_3 + 1 > 0$, which depends only on C_0 and $\dim F$, both cases give (406). □

2.101. `FitParamBound`.

Lemma 2.101 (Bound for fit parameters on a box)**.** *Let F be a finite-dimensional real inner-product space and let $1 \le C_0 < \infty$. Write $L(K) = 1 + \log K$ for $K \ge 2$ and $\mathcal{B}_K = \{(x,t) : |x| \le K,\ |t| \le K\}$. There is $C > 0$, depending only on C_0 and $\dim F$, such that for every $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` which is centered at $(0,0)$ or at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, every $K \ge 2$, every $z \in \mathcal{B}_K$, every $r > 0$ and every fit (p,a) for v at z of radius r with constant C_0 in the sense of `IsFit`,*

$$|p| + |a| \le C\big(L(K) + |\log r|\big). \tag{411}$$

Proof. Write $L(K) = 1 + \log K \ge 1$ for $K \ge 2$, and write $z_c = (0,\theta)$ with $\theta \in \{0,1\}$ for the centering point, so that by `IsCenteredAt` $v(z_c) = 0$ and $(0,0)$ is a fit for v at z_c of radius 1 with constant C_0. By `IsFitClass`, v admits a fit at every center of every positive radius with constant C_0; in particular the hypothesis of `FitComparisonSameCenter` holds at every center. Let $C_1 > 0$ be the constant of `FitComparisonSameCenter` for this C_0. All constants below depend only on C_0 and $\dim F$.

Step 1: a fit of a large radius at the centering point. Let $A \ge 8$ be a constant, depending only on C_0 and $\dim F$, to be fixed in Step 2, and set

$$R = A\,K\,L(K) \quad \ge 8 \cdot 2 \cdot 1 = 16 > 1.$$

By `IsFitClass` choose a fit (p_R, a_R) for v at z_c of radius R. Applying `FitComparisonSameCenter` at the center z_c with the radii R and 1 and the fits (p_R, a_R) and $(0,0)$,

$$|p_R| + |a_R| \le C_1\big(1 + |\log(1/R)|\big) = C_1\big(1 + \log R\big),$$

using $R > 1$. Now $\log R = \log A + \log K + \log L(K) \le \log A + 2L(K)$, since $\log K \le 1 + \log K = L(K)$ and $\log L(K) \le L(K)$ (because $\log \sigma \le \sigma$ for $\sigma > 0$). Hence, using $L(K) \ge 1$,

$$|p_R| + |a_R| \le C_4\,L(K), \qquad C_4 := C_1\big(1 + \log A + 2\big) + 1 \ \ (\ge 1). \tag{412}$$

Step 2: the box of (p_R, a_R) contains $\mathcal{B}_K$ with margin. Let $z = (x,t)$ with $|x| \le K$ and $|t| \le K$. The increment from z_c to z is $(x,\ t-\theta)$, and $|t - \theta| \le K + 1 \le 2K$. We claim

$$|t - \theta| \le \frac{R}{4}, \qquad \big|x - p_R(t-\theta)\big| \le \frac{R}{4}. \tag{413}$$

For the first, $|t-\theta| \le 2K$ while $R/4 = AKL(K)/4 \ge AK/4 \ge 2K$ because $A \ge 8$. For the second, by (412) and $C_4 L(K) \ge 1$,

$$\big|x - p_R(t-\theta)\big| \le K + 2K\,|p_R| \le K + 2K\,C_4 L(K) \le 3C_4\,K\,L(K),$$

and this is at most $R/4 = AKL(K)/4$ as soon as $A \ge 12C_4$. Since $C_4 = C_1(3 + \log A) + 1$ depends on A only through $\log A$, the requirement

$$A \ge \max\big\{8,\ 12\big(C_1(3 + \log A) + 1\big)\big\}$$

has the form $A \ge \alpha + \beta \log A$ with $\alpha, \beta > 0$ depending only on C_0 and $\dim F$, and is therefore satisfiable because $\log A / A \to 0$. Fix such an A once and for all. Then (412) and (413) hold for every $K \ge 2$ and every $z \in \mathcal{B}_K$, with $C_4 \ge 1$ depending only on C_0 and $\dim F$.

Step 3: fits of radius R at a center in $\mathcal{B}_K$. Let $z = (x,t) \in \mathcal{B}_K$ and let (p_z, a_z) be any fit for v at z of radius R with constant C_0. Write $\ell_{p,a}(h,\ell) = p \cdot h + (a - |p|^2/2)\ell$, so that `IsFit` at a center ζ and radius ρ reads $\big|v(\zeta + (h,\ell)) - v(\zeta) - \ell_{p,a}(h,\ell)\big| \le C_0\rho$ on the corresponding box.

(a) The slopes. Let $h \in F$ with $|h| \le R/2$. The increment $(h, 0)$ from z lies in the box of (p_z, a_z) of radius R, since $|0| \le R$ and $|h - p_z \cdot 0| = |h| \le R$; hence

$$\big|v(x+h,t) - v(x,t) - p_z \cdot h\big| \le C_0 R. \tag{414}$$

The increments $(x, t-\theta)$ and $(x+h, t-\theta)$ from z_c both lie in the box of (p_R, a_R) of radius R: by (413) we have $|t-\theta| \le R/4 \le R$ and $|x - p_R(t-\theta)| \le R/4 \le R$, and

$$\big|x + h - p_R(t-\theta)\big| \le \frac{R}{4} + \frac{R}{2} \le R.$$

Subtracting the two corresponding estimates of `IsFit`, and noting that ℓ_{p_R,a_R} is linear so that its values at the two increments differ by $\ell_{p_R,a_R}(h,0) = p_R \cdot h$, we obtain

$$\big|v(x+h,t) - v(x,t) - p_R \cdot h\big| \le 2C_0 R \qquad (|h| \le R/2).$$

Subtracting this from (414) gives $|(p_z - p_R)\cdot h| \le 3C_0 R$ for all $|h| \le R/2$; taking $h = (R/2)(p_z - p_R)/|p_z - p_R|$ when $p_z \neq p_R$, and dividing by $R/2$,

$$|p_z - p_R| \le 6C_0. \tag{415}$$

(b) The time coefficients. Put $c' = \big(4(1+6C_0)\big)^{-1}$, so $0 < c' \le 1/4$ and $6C_0 c' \le 1/4$, and consider the increment $(p_z c'R,\, c'R)$ from z. It lies in the box of (p_z, a_z) of radius R: $|c'R| \le R$ and $|p_z c'R - p_z c'R| = 0 \le R$. The same space-time point, written as an increment from z_c, is $\big(x + p_z c'R,\ t - \theta + c'R\big)$, and it lies in the box of (p_R, a_R) of radius R: by (413), $|t - \theta + c'R| \le R/4 + R/4 \le R$, and by (413) and (415)

$$\big|x + p_z c'R - p_R(t - \theta + c'R)\big| \le \big|x - p_R(t-\theta)\big| + c'R\,|p_z - p_R| \le \frac{R}{4} + \frac{R}{4} \le R.$$

Finally $(x, t-\theta)$ lies in that box by (413).

Apply `IsFit` for (p_z, a_z) at the increment $(p_z c'R, c'R)$ from z, and `IsFit` for (p_R, a_R) at the two increments $(x + p_z c'R,\, t - \theta + c'R)$ and $(x,\, t-\theta)$ from z_c. The three estimates involve v at the two space-time points $z + (p_z c'R, c'R)$ and z only, so the values of v cancel on forming the combination (first estimate) $-$ (second) $+$ (third), and the total error is at most $3C_0R$. Since ℓ_{p_R,a_R} is linear in the increment, the difference of its values at the second and third increments is $\ell_{p_R,a_R}(p_z c'R, c'R)$. Hence

$$\big|(\ell_{p_z,a_z} - \ell_{p_R,a_R})(p_z c'R,\, c'R)\big| \le 3C_0 R.$$

Expanding, and using $(p_z - p_R)\cdot p_z - \tfrac12\big(|p_z|^2 - |p_R|^2\big) = \tfrac12 |p_z - p_R|^2$,

$$(\ell_{p_z,a_z} - \ell_{p_R,a_R})(p_z c'R, c'R) = c'R\Big[a_z - a_R + \tfrac12 |p_z - p_R|^2\Big],$$

so dividing by $c'R > 0$ and using (415),

$$|a_z - a_R| \le \frac{3C_0}{c'} + 18C_0^2. \tag{416}$$

Combining (412), (415) and (416), and using $L(K) \ge 1$, there is $C_5 > 0$, depending only on C_0 and $\dim F$, with

$$|p_z| + |a_z| \le C_5\, L(K) \tag{417}$$

for every $z \in \mathcal{B}_K$ and every fit at z of radius exactly R.

Step 4: arbitrary radii at a center in $\mathcal{B}_K$. Let $z \in \mathcal{B}_K$, let $r > 0$ and let (p, a) be a fit for v at z of radius r with constant C_0. By `IsFitClass` choose a fit (p_z, a_z) for v at z of radius R. Applying `FitComparisonSameCenter` at the center z with the two radii r and R and these two fits,

$$|p - p_z| + |a - a_z| \le C_1\Big(1 + \big|\log(R/r)\big|\Big).$$

Hence, with (417),

$$|p| + |a| \le C_5 L(K) + C_1\Big(1 + \big|\log(R/r)\big|\Big).$$

Finally $\big|\log(R/r)\big| \le \log R + |\log r| \le \log A + 2L(K) + |\log r|$ by Step 1 and $R > 1$, so with $L(K) \ge 1$ there is $C > 0$, depending only on C_0 and $\dim F$, with

$$|p| + |a| \le C\big(L(K) + |\log r|\big),$$

which is (411). □

2.102. `SpatialFitModulus`.

Lemma 2.102 (Spatial fit modulus)**.** *Let F be a finite-dimensional real inner-product space and let $1 \le C_0 < \infty$. Write $L(K) = 1 + \log K$ for $K \ge 2$ and $\mathcal{B}_K = \{(x,t) : |x| \le K,\ |t| \le K\}$. There is $C > 0$, depending only on C_0 and $\dim F$, such that for every $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` which is centered at $(0,0)$ or at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, every $K \ge 2$, every $(x,t) \in \mathcal{B}_K$ and every $r > 0$,*

$$|v(x+h,t) - v(x,t)| \le Cr\big(L(K) + |\log r|\big) \qquad (|h| \le r). \tag{418}$$

Proof. Let $C_1 > 0$ be the constant of `FitParamBound` for this C_0, and let v, K, $z = (x,t) \in \mathcal{B}_K$ and $r > 0$ be as in the statement.

Since $v \in \mathcal{A}_{m,C_0}$ and $r > 0$, `IsFitClass` provides a fit (p,a) for v at z of radius r with constant C_0. By `FitParamBound` applied at this center and radius,

$$|p| + |a| \le C_1\big(L(K) + |\log r|\big). \tag{419}$$

Let $|h| \le r$ and consider the increment (h,ℓ) with $\ell = 0$. Its box conditions in `IsFit` read $|\ell| = 0 \le r$ and $|h - \ell p| = |h| \le r$, both of which hold. Hence

$$\big|v(x+h,t) - v(x,t) - p \cdot h\big| \le C_0 r,$$

the time term dropping out because $\ell = 0$. Therefore, by Cauchy–Schwarz and $|h| \le r$,

$$|v(x+h,t) - v(x,t)| \le |p|\,|h| + C_0 r \le \big(|p| + C_0\big) r.$$

Now $L(K) = 1 + \log K \ge 1 + \log 2 \ge 1$ for $K \ge 2$, so $C_0 \le C_0\big(L(K) + |\log r|\big)$, and with (419)

$$|v(x+h,t) - v(x,t)| \le (C_1 + C_0)\, r\,\big(L(K) + |\log r|\big).$$

Thus (418) holds with $C = C_1 + C_0$, which depends only on C_0 and $\dim F$. □

2.103. `QuarticBumpJetAndRegularity`.

Lemma 2.103 (A quartic bump preserves the two-jet and leaves a C^2 neighbourhood)**.** *Let F be a finite-dimensional real inner-product space, let Ω be an open subset of the space-time $\ell^2(F \times \mathbb{R})$, let $z \in \Omega$, let $s \in \mathbb{R}$ and let ϕ be a real function on $\ell^2(F \times \mathbb{R})$ that is C^2 at z, i.e. twice continuously differentiable on some neighbourhood of z. Put*

$$\phi^\sharp(\zeta) = \phi(\zeta) + s\,\big(|\zeta - z|^2\big)^2 \qquad \big(\zeta \in \ell^2(F \times \mathbb{R})\big).$$

Then both of the following hold.

(1) *The limiting operator of `LimitOp` takes the same value at z on $\phi^\sharp$ as on ϕ:*

$$F[\phi^\sharp](z) = F[\phi](z). \tag{420}$$

(2) *There is a set $V \subseteq \ell^2(F \times \mathbb{R})$ which is open, contains z and is contained in Ω, and on which $\phi^\sharp$ is C^2.*

Proof. Write $b(\zeta) = s\big(|\zeta - z|^2\big)^2$ for the bump, so that $\phi^\sharp = \phi + b$.

Step 1: the bump is C^2 on all of $\ell^2(F \times \mathbb{R})$. The translation $g(\zeta) = \zeta - z$ is the sum of the identity and a constant, hence smooth, and the squared norm $w \mapsto |w|^2$ of an inner-product space is smooth. Therefore $\zeta \mapsto |\zeta - z|^2 = |g(\zeta)|^2$ is C^2 as a composition of C^2 maps, its square is C^2, and the constant multiple $b = s \cdot \big(|g|^2\big)^2$ is C^2.

Step 2: the first derivative of the bump, and $Db(z) = 0$. Since g has derivative the identity map at every point, the chain rule gives $D\big(|g|^2\big)(\zeta) = 2\langle \zeta - z, \cdot \rangle$ for every ζ. Applying the chain rule once more to the outer squaring and multiplying by the constant s,

$$Db(\zeta) = s \cdot 2\big(|\zeta - z|^2\big) \cdot D\big(|g|^2\big)(\zeta) = \big(4s\,|\zeta - z|^2\big)\,\langle \zeta - z, \cdot \rangle \qquad \big(\zeta \in \ell^2(F \times \mathbb{R})\big). \tag{421}$$

At $\zeta = z$ the factor $|z - z|^2$ vanishes, so

$$Db(z) = 0, \qquad \text{and hence} \qquad \nabla b(z) = 0, \tag{422}$$

the gradient being the Riesz representative of the derivative.

Step 3: the second derivative of the bump vanishes at z. By (421) the map $\zeta \mapsto Db(\zeta)$ is the pointwise product of the scalar function $c(\zeta) = 4s|\zeta - z|^2$ and the linear-functional-valued map

$L(\zeta) = \langle \zeta - z, \cdot \rangle$. Both are differentiable: c by Step 1 and L because it is the composition of the bounded linear map $w \mapsto \langle w, \cdot \rangle$ with g, whose derivative at any point is $u \mapsto \langle u, \cdot \rangle$. The product rule for a scalar function times a map into a normed space therefore gives, for every $u \in \ell^2(F \times \mathbb{R})$,

$$D^2 b(z)[u] = \big(Dc(z)[u]\big)\, L(z) + c(z)\, \big(DL(z)[u]\big).$$

Now $L(z) = \langle z - z, \cdot \rangle = 0$ and $c(z) = 4s|z - z|^2 = 0$, so both summands vanish and

$$D^2 b(z) = 0. \tag{423}$$

Step 4: proof of (420). By hypothesis ϕ is C^2 on a neighbourhood of z, and b is C^2 everywhere by Step 1, so $\phi^\sharp = \phi + b$ is C^2 on a neighbourhood of z; in particular ϕ, b and $\phi^\sharp$ are differentiable at z and twice differentiable at z. Derivatives are additive, so by (422) and (423)

$$\nabla \phi^\sharp(z) = \nabla \phi(z) + \nabla b(z) = \nabla \phi(z), \qquad D^2 \phi^\sharp(z) = D^2 \phi(z) + D^2 b(z) = D^2 \phi(z).$$

By `LimitOp`, $F[\psi](z) = D^2\psi(z)\big[(\nabla_x \psi(z), 1)\big]$ is determined by the two quantities $\nabla\psi(z)$ and $D^2\psi(z)$ alone. Since $\phi^\sharp$ and ϕ agree in both, $F[\phi^\sharp](z) = F[\phi](z)$, which is (420).

Step 5: the neighbourhood V. Because ϕ is C^2 at z, there is a neighbourhood u of z with ϕ of class C^2 on u; shrinking, choose an open V_0 with $z \in V_0 \subseteq u$. Put $V = V_0 \cap \Omega$. Then V is open, being the intersection of the two open sets V_0 and Ω; $z \in V$ because $z \in V_0$ and $z \in \Omega$ by hypothesis; and $V \subseteq \Omega$. Finally ϕ is C^2 on V, since $V \subseteq V_0 \subseteq u$ and the class C^2 passes to subsets, and b is C^2 on V by Step 1; hence their sum $\phi^\sharp$ is C^2 on V. This is assertion 2, and the proof is complete. □

2.104. `MaximizerSequenceTendsToStrictMax`.

Lemma 2.104 (Maximizers of a uniformly convergent sequence concentrate at a strict maximum point)**.** *Let F be a finite-dimensional real inner-product space, let $z \in \ell^2(F \times \mathbb{R})$, let $\varrho > 0$, let $s \in \mathbb{R}$ with $|s| = 1$, let $v_1, v_2, \dots$ be real functions on $\ell^2(F \times \mathbb{R})$ and let v_0 and ψ be real functions on $\ell^2(F \times \mathbb{R})$. Write $B = \overline{B}_\varrho(z)$ for the closed ball of radius ϱ about z, and put*

$$g(y) = s\big(v_0(y) - \psi(y)\big), \qquad g_j(y) = s\big(v_j(y) - \psi(y)\big) \qquad (j \in \mathbb{N}).$$

Assume:

1. *v_0 is continuous on B and ψ is continuous on B;*
2. *$v_j \to v_0$ uniformly on B;*
3. *g has a strict maximum at z relative to B, in the sense that $g(w) < g(z)$ for every $w \in B$ with $w \neq z$;*
4. *for every $j \in \mathbb{N}$ there is some $w \in B$ at which g_j attains its maximum over B.*

Then there is a sequence $(\zeta_j)_{j \in \mathbb{N}}$ in $\ell^2(F \times \mathbb{R})$ such that $\zeta_j \in B$ for every j, such that g_j attains its maximum over B at ζ_j for every j, and such that $\zeta_j \to z$.

Proof. The ball B is compact, being a closed ball in a finite-dimensional normed space, and $z \in B$ because $\operatorname{dist}(z, z) = 0 \le \varrho$.

Choice of the sequence. By hypothesis 4, for every j the set of points of B at which g_j attains its maximum over B is nonempty; choose one such point and call it ζ_j. This gives a sequence with $\zeta_j \in B$ and with g_j attaining its maximum over B at ζ_j for every j, which are the first two assertions. It remains to prove $\zeta_j \to z$.

Reduction. By the metric characterization of convergence it suffices to show that for every $\varepsilon > 0$ one has $\operatorname{dist}(\zeta_j, z) < \varepsilon$ for all sufficiently large j. Fix $\varepsilon > 0$ and set $r = \min\{\varepsilon, \varrho\}$, so that $r > 0$ (both $\varepsilon > 0$ and $\varrho > 0$) and $r \le \varepsilon$. We distinguish two cases.

Case 1: every $w \in B$ satisfies $\operatorname{dist}(w, z) < r$. Then for every j we have $\zeta_j \in B$, hence $\operatorname{dist}(\zeta_j, z) < r \le \varepsilon$, and the required statement holds for all j.

Case 2: some $w_0 \in B$ satisfies $\operatorname{dist}(w_0, z) \ge r$. Put

$$A = \{w \in B : \ r \le \operatorname{dist}(w, z)\}.$$

Then $A \neq \emptyset$, since $w_0 \in A$. Moreover A is compact: the set $\{w : \ r \le \operatorname{dist}(w, z)\}$ is closed, because $w \mapsto \operatorname{dist}(w, z)$ is continuous and the inequality $r \le \cdot$ defines a closed condition, and A is the intersection of the compact set B with that closed set.

By hypothesis 1 the function $v_0 - \psi$ is continuous on B, hence so is its constant multiple g; since $A \subseteq B$, g is continuous on A. A continuous real function on a nonempty compact set attains its maximum, so there is $m \in A$ with

$$g(w) \le g(m) \qquad (w \in A). \tag{424}$$

Because $m \in A$ we have $\operatorname{dist}(m, z) \ge r > 0$, so $m \ne z$; as also $m \in B$, hypothesis 3 gives $g(m) < g(z)$. Write

$$M = g(z), \qquad \mu = g(m), \qquad \delta = \frac{M - \mu}{2} > 0.$$

By hypothesis 2 and the definition of uniform convergence on B, there is J such that for every $j \ge J$,

$$|v_j(w) - v_0(w)| < \delta \qquad (w \in B). \tag{425}$$

Fix $j \ge J$; we claim $\operatorname{dist}(\zeta_j, z) < r$, which gives $\operatorname{dist}(\zeta_j, z) < \varepsilon$ since $r \le \varepsilon$ and finishes the proof.

Suppose instead that $\operatorname{dist}(\zeta_j, z) \ge r$. Since $\zeta_j \in B$, this means $\zeta_j \in A$. For any $w \in B$, using $|s| = 1$ and (425),

$$|g_j(w) - g(w)| = \big|s\big(v_j(w) - \psi(w)\big) - s\big(v_0(w) - \psi(w)\big)\big| = |s|\,|v_j(w) - v_0(w)| = |v_j(w) - v_0(w)| < \delta. \tag{426}$$

Applying (426) at $w = z \in B$ gives $g_j(z) > g(z) - \delta = M - \delta$, and applying it at $w = \zeta_j \in B$ gives $g_j(\zeta_j) < g(\zeta_j) + \delta$. Since $\zeta_j \in A$, (424) gives $g(\zeta_j) \le \mu$, whence $g_j(\zeta_j) < \mu + \delta$. Finally ζ_j maximizes g_j over B and $z \in B$, so $g_j(z) \le g_j(\zeta_j)$. Combining the three displayed inequalities,

$$M - \delta < g_j(z) \le g_j(\zeta_j) < \mu + \delta,$$

so $M - \mu < 2\delta = M - \mu$, a contradiction. Therefore $\operatorname{dist}(\zeta_j, z) < r \le \varepsilon$ for every $j \ge J$, as required. □

2.105. `LimitStability`.

Lemma 2.105 (Stability of the limiting equation under locally uniform limits)**.** *Let F be a finite-dimensional real inner-product space, let Ω be an open subset of the space-time $\ell^2(F \times \mathbb{R})$, let $v_1, v_2, \dots$ each be a viscosity solution of $F[v_j] = 0$ on Ω in the sense of `IsLimitSolutionOn`, and let v_0 be a real function on space-time such that $v_j \to v_0$ uniformly on every compact subset of Ω. Then v_0 is a viscosity solution of $F[v_0] = 0$ on Ω.*

Proof. Step 0: continuity of the limit. Let $\zeta \in \Omega$. Since Ω is open there is $\varrho > 0$ with $\overline{B}_\varrho(\zeta) \subseteq \Omega$, a compact set, so $v_j \to v_0$ uniformly on it. By `IsLimitSolutionOn` each v_j is continuous on Ω, hence on $\overline{B}_\varrho(\zeta)$, and a uniform limit of continuous functions on a set is continuous on that set. Thus v_0 is continuous at every point of Ω, i.e. continuous on Ω.

Step 1: strictifying an upper test. Let $\zeta_0 \in \Omega$ and let ϕ be C^2 near ζ_0 with $\zeta \mapsto v_0(\zeta) - \phi(\zeta)$ having a local maximum on Ω at ζ_0. Set

$$\phi^\sharp(\zeta) = \phi(\zeta) + |\zeta - \zeta_0|^4.$$

The function $\zeta \mapsto |\zeta - \zeta_0|^4 = \big(|\zeta - \zeta_0|^2\big)^2$ is a polynomial in the coordinates, hence smooth, and

$$\nabla\big(|\cdot - \zeta_0|^4\big)(\zeta_0) = 0, \qquad D^2\big(|\cdot - \zeta_0|^4\big)(\zeta_0) = 0, \tag{427}$$

since its gradient is $4|\zeta - \zeta_0|^2(\zeta - \zeta_0)$, which vanishes at ζ_0 together with its derivative. Hence $\phi^\sharp$ is C^2 near ζ_0 with $\nabla\phi^\sharp(\zeta_0) = \nabla\phi(\zeta_0)$ and $D^2\phi^\sharp(\zeta_0) = D^2\phi(\zeta_0)$, so by `LimitOp`

$$F[\phi^\sharp](\zeta_0) = F[\phi](\zeta_0). \tag{428}$$

Choose $\varrho > 0$ so small that $\overline{B}_\varrho(\zeta_0) \subseteq \Omega$, that ϕ is C^2 on a neighbourhood of $\overline{B}_\varrho(\zeta_0)$, and that $v_0 - \phi$ attains its maximum over $\overline{B}_\varrho(\zeta_0)$ at ζ_0; the last is possible because the maximum is local on Ω. Then for $\zeta \in \overline{B}_\varrho(\zeta_0)$ with $\zeta \ne \zeta_0$,

$$v_0(\zeta) - \phi^\sharp(\zeta) = \big(v_0(\zeta) - \phi(\zeta)\big) - |\zeta - \zeta_0|^4 < v_0(\zeta_0) - \phi(\zeta_0) = v_0(\zeta_0) - \phi^\sharp(\zeta_0),$$

so ζ_0 is the *unique* maximum point of $v_0 - \phi^\sharp$ on $\overline{B}_\varrho(\zeta_0)$.

Step 2: nearby maxima of the approximations. The set $\overline{B}_\varrho(\zeta_0)$ is compact and contained in Ω, and each $v_j - \phi^\sharp$ is continuous on it, so it attains a maximum at some $\zeta_j \in \overline{B}_\varrho(\zeta_0)$. We claim $\zeta_j \to \zeta_0$. By compactness it suffices to show that every convergent subsequence has limit ζ_0. Let $\zeta_{j_k} \to \bar\zeta \in \overline{B}_\varrho(\zeta_0)$. Uniform convergence of v_j to v_0 on $\overline{B}_\varrho(\zeta_0)$ and continuity of $v_0 - \phi^\sharp$ there give

$$(v_{j_k} - \phi^\sharp)(\zeta_{j_k}) \longrightarrow (v_0 - \phi^\sharp)(\bar\zeta).$$

On the other hand, for every $\zeta \in \overline{B}_\varrho(\zeta_0)$, $(v_{j_k} - \phi^\sharp)(\zeta_{j_k}) \geq (v_{j_k} - \phi^\sharp)(\zeta) \to (v_0 - \phi^\sharp)(\zeta)$, so $(v_0 - \phi^\sharp)(\bar\zeta) \geq (v_0 - \phi^\sharp)(\zeta)$ for all such ζ. Thus $\bar\zeta$ is a maximum point of $v_0 - \phi^\sharp$ on $\overline{B}_\varrho(\zeta_0)$, and by the uniqueness in Step 1, $\bar\zeta = \zeta_0$. Hence $\zeta_j \to \zeta_0$; in particular $|\zeta_j - \zeta_0| < \varrho$ for all large j, so ζ_j is an interior point of $\overline{B}_\varrho(\zeta_0)$ and therefore $v_j - \phi^\sharp$ has a local maximum on Ω at ζ_j.

Step 3: passing the inequality. For such j, $\phi^\sharp$ is C^2 near ζ_j and $v_j - \phi^\sharp$ has a local maximum on Ω at ζ_j, so the subsolution inequality of `IsLimitSubOn`, available because v_j is a solution by `IsLimitSolutionOn`, gives

$$F[\phi^\sharp](\zeta_j) \geq 0.$$

Since $\phi^\sharp$ is C^2 on a neighbourhood of $\overline{B}_\varrho(\zeta_0)$, its gradient and second derivative are continuous there; by `LimitOp` the map $\zeta \mapsto F[\phi^\sharp](\zeta)$, which is the second derivative of $\phi^\sharp$ at ζ evaluated on the pair (ξ_ζ, ξ_ζ) with $\xi_\zeta = (\nabla_x \phi^\sharp(\zeta), 1)$, is therefore continuous near ζ_0. Letting $j \to \infty$ and using $\zeta_j \to \zeta_0$ gives $F[\phi^\sharp](\zeta_0) \geq 0$, and (428) yields $F[\phi](\zeta_0) \geq 0$.

Step 4: lower tests. If instead $v_0 - \phi$ has a local minimum on Ω at ζ_0, repeat Steps 1–3 with $\phi^\sharp(\zeta) = \phi(\zeta) - |\zeta - \zeta_0|^4$, with minima in place of maxima, and with the supersolution inequality of `IsLimitSuperOn`. Identity (427) again gives $F[\phi^\sharp](\zeta_0) = F[\phi](\zeta_0)$, and the conclusion is $F[\phi](\zeta_0) \leq 0$.

Together with Step 0 and the openness of Ω, this verifies both clauses of `IsLimitSolutionOn` for v_0. □

2.106. `TimeFitModulus`.

Lemma 2.106 (Time fit modulus)**.** *Let F be a finite-dimensional real inner-product space and let $1 \leq C_0 < \infty$. Write $L(K) = 1 + \log K$ for $K \geq 2$ and $\mathcal{B}_K = \{(x,t) : |x| \leq K,\ |t| \leq K\}$. There is $C > 0$, depending only on C_0 and $\dim F$, such that for every $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` which is centered at $(0,0)$ or at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, every $K \geq 2$, every $(x,t) \in \mathcal{B}_K$ and every τ with $0 < |\tau| \leq 1/2$,*

$$|v(x,t+\tau) - v(x,t)| \leq C|\tau|\big(L(K) + |\log|\tau||\big)^2. \tag{429}$$

Proof. Write $L(K) = 1 + \log K \geq 1$ for $K \geq 2$. Let $C_1 > 0$ be the constant of `FitParamBound` for this C_0, so that for every $v \in \mathcal{A}_{m,C_0}$ centered at $(0,0)$ or $(0,1)$ with constant C_0, every $K \geq 2$, every $z \in \mathcal{B}_K$, every $\varrho > 0$ and every fit (p,a) for v at z of radius ϱ with constant C_0,

$$|p| + |a| \leq C_1\big(L(K) + |\log\varrho|\big). \tag{430}$$

Step 1: choice of the auxiliary constant A'. Since $(3 + \log A')/A' \to 0$ as $A' \to \infty$, we may fix once and for all a real number $A' \geq 1$ with

$$A' \geq C_1\big(3 + \log A'\big), \tag{431}$$

and this choice depends only on C_1, hence only on C_0 and $\dim F$. It is made before K and τ vary.

Step 2: the fit radius. Let v, $z = (x,t) \in \mathcal{B}_K$ and τ with $0 < |\tau| \leq 1/2$ be as in the statement. Put

$$B = L(K) + \big|\log|\tau|\big|, \qquad \varrho = A'\,|\tau|\,B.$$

Since $L(K) \geq 1$ we have $B \geq 1$, and $\varrho > 0$ because $A' \geq 1$, $|\tau| > 0$ and $B \geq 1$. By `IsFitClass` choose a fit (p,a) for v at z of radius ϱ with constant C_0.

We estimate $|\log\varrho|$. We have $\log\varrho = \log A' + \log|\tau| + \log B$, so

$$|\log\varrho| \leq \log A' + \big|\log|\tau|\big| + \log B \leq \log A' + B + B = \log A' + 2B,$$

using $\big|\log|\tau|\big| \le B$ and $\log B \le B$ (valid since $\log\sigma \le \sigma$ for $\sigma > 0$). Hence by (430), and using $L(K) \le B$ and $B \ge 1$,

$$|p| + |a| \le C_1\big(L(K) + \log A' + 2B\big) \le C_1\big(3 + \log A'\big)B \le A'B, \tag{432}$$

the last step by (431).

Step 3: the vertical increment lies in the fitted box. Consider the increment $(h,\ell) = (0,\tau)$ from z. Its box conditions in `IsFit` at radius ϱ are $|\tau| \le \varrho$ and $|0 - \tau p| \le \varrho$. The first holds because $A'B \ge 1$ gives $\varrho = A'|\tau|B \ge |\tau|$. The second holds because, by (432),

$$|\tau p| = |\tau|\,|p| \le |\tau|\,A'B = \varrho.$$

Step 4: conclusion. By `IsFit` at the increment $(0,\tau)$,

$$\Big|v(x,t+\tau) - v(x,t) - \Big(a - \frac{|p|^2}{2}\Big)\tau\Big| \le C_0\varrho,$$

the spatial term vanishing because $h = 0$. Hence

$$|v(x,t+\tau) - v(x,t)| \le \Big(|a| + \frac{|p|^2}{2}\Big)|\tau| + C_0\varrho.$$

By (432) we have $|a| \le A'B$ and $|p|^2 \le (A'B)^2$, so, using $B \ge 1$ and hence $B \le B^2$,

$$\Big(|a| + \frac{|p|^2}{2}\Big)|\tau| \le \Big(A'B + \frac{(A')^2B^2}{2}\Big)|\tau| \le \Big(A' + \frac{(A')^2}{2}\Big)|\tau|\,B^2,$$

while $C_0\varrho = C_0A'|\tau|B \le C_0A'|\tau|B^2$. Adding,

$$|v(x,t+\tau) - v(x,t)| \le C\,|\tau|\,B^2 = C\,|\tau|\,\big(L(K) + |\log|\tau||\big)^2$$

with $C = A' + \frac{1}{2}(A')^2 + C_0A' > 0$, which depends only on C_0 and $\dim F$. This is (429). □

2.107. `ClassCompactness`.

Lemma 2.107 (Compactness of the centered class)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and let z_c be either $(0,0)$ or $(0,1)$. Let $v_1, v_2, \ldots \in \mathcal{A}_{m,C_0}$, in the sense of `IsFitClass`, each centered at z_c with constant C_0 in the sense of `IsCenteredAt`. Then there are a strictly increasing sequence of indices $j \mapsto \iota(j)$ and a function w on space-time such that $v_{\iota(j)} \to w$ uniformly on every compact set, $w \in \mathcal{A}_{m,C_0}$, and w is centered at z_c with constant C_0.*

Proof. Write $L(K) = 1 + \log K \ge 1$ for $K \ge 2$ and $\mathcal{B}_K = \{(x,t) : |x| \le K,\ |t| \le K\}$. Let C_g, C_s, C_τ and C_p be the constants of `CenteredGrowth`, `SpatialFitModulus`, `TimeFitModulus` and `FitParamBound` respectively, for this C_0. Each v_j belongs to $\mathcal{A}_{m,C_0}$ and is centered at z_c with constant C_0, so all four lemmas apply to every v_j with the same constants.

Step 1: uniform local bounds. By `CenteredGrowth`, for every $K \ge 2$,

$$|v_j(z)| \le C_gKL(K)^2 \qquad (z \in \mathcal{B}_K,\ j \in \mathbb{N}). \tag{433}$$

Step 2: uniform equicontinuity on each box. Fix $K \ge 2$ and put $K' = K + 1 \ge 2$. Let $z = (x,t)$ and $z' = (x',t')$ lie in $\mathcal{B}_K$ with $|x - x'| \le \varsigma$ and $|t - t'| \le \varsigma$ for some $0 < \varsigma \le 1/2$. The intermediate point (x,t') lies in $\mathcal{B}_{K'}$, as do z and z'. Writing $h = x' - x$ and $\tau = t' - t$,

$$|v_j(z') - v_j(z)| \le |v_j(x',t') - v_j(x,t')| + |v_j(x,t') - v_j(x,t)|.$$

If $h \ne 0$, `SpatialFitModulus`, applied at the center $(x,t') \in \mathcal{B}_{K'}$ with the radius $r = |h| \le \varsigma$, bounds the first term by $C_s\varsigma\big(L(K') + |\log\varsigma|\big)$ when $|h| = \varsigma$; in general, applying it with $r = \varsigma$ and the increment h, which satisfies $|h| \le \varsigma = r$, gives

$$|v_j(x+h,t') - v_j(x,t')| \le C_s\,\varsigma\big(L(K') + |\log\varsigma|\big),$$

and this also covers $h = 0$. If $\tau \ne 0$, `TimeFitModulus`, applied at the center $(x,t) \in \mathcal{B}_{K'}$ with the time increment τ, for which $0 < |\tau| \le \varsigma \le 1/2$, bounds the second term by $C_\tau|\tau|\big(L(K') + |\log|\tau||\big)^2$. The function $\sigma \mapsto \sigma\big(L(K') + |\log\sigma|\big)^2$ is bounded on $(0,\varsigma]$ by its supremum there,

and tends to 0 as $\varsigma \to 0$, because $\sigma(\log\sigma)^2 \to 0$; the same holds for $\sigma \mapsto \sigma(L(K') + |\log\sigma|)$. Hence, setting

$$\omega_K(\varsigma) = C_s\,\varsigma\big(L(K') + |\log\varsigma|\big) + C_\tau \sup_{0<\sigma\le\varsigma} \sigma\big(L(K') + |\log\sigma|\big)^2,$$

we obtain

$$|v_j(z') - v_j(z)| \le \omega_K(\varsigma) \qquad \big(z, z' \in \mathcal{B}_K,\ |z - z'| \le \varsigma \le \tfrac{1}{2},\ j \in \mathbb{N}\big), \tag{434}$$

with $\omega_K(\varsigma) \to 0$ as $\varsigma \downarrow 0$ and ω_K independent of j. (We used $|x-x'| \le |z-z'|$ and $|t-t'| \le |z-z'|$.)

Step 3: a compactness claim and a diagonal subsequence. The extraction of a locally uniformly convergent subsequence needs a compactness statement which is not among the background items of the preamble, so we prove the instance we use; the argument is the standard Arzelà–Ascoli one, and the only external inputs are the preamble items *Compactness, Weierstrass and uniform continuity* (a closed bounded subset of a finite-dimensional space is compact) and *Bolzano–Weierstrass*.

Claim A. Let $\mathcal{K}$ be a closed bounded subset of space-time, let $A \ge 0$, let $S \subseteq \mathbb{N}$ be infinite, let g_j, $j \in S$, be real functions on space-time with $|g_j| \le A$ on $\mathcal{K}$, and let $\omega : (0, \frac{1}{2}] \to [0, \infty)$ be a function with $\omega(\varsigma) \to 0$ as $\varsigma \downarrow 0$ and

$$|g_j(a) - g_j(b)| \le \omega(\varsigma) \qquad \big(a, b \in \mathcal{K},\ |a - b| \le \varsigma \le \tfrac{1}{2},\ j \in S\big).$$

Then there are an infinite set $S' \subseteq S$ and a function g on $\mathcal{K}$ such that $\sup_{\mathcal{K}} |g_j - g| \to 0$ as $j \to \infty$ along S'.

Proof of Claim A. If $\mathcal{K} = \emptyset$ take $S' = S$ and g the empty function; so assume $\mathcal{K} \ne \emptyset$. Being closed and bounded in the finite-dimensional space-time, $\mathcal{K}$ is compact. For each integer $k \ge 1$ the balls $B_{1/k}(a)$, $a \in \mathcal{K}$, form an open cover of $\mathcal{K}$, so finitely many of them cover it; let $N_k \subseteq \mathcal{K}$ be a finite nonempty set with $\mathcal{K} \subseteq \bigcup_{b\in N_k} B_{1/k}(b)$, and put $Q = \bigcup_{k\ge1} N_k$. Then $Q \subseteq \mathcal{K}$ is countable and nonempty, and for every $\varsigma > 0$ the balls $B_\varsigma(b)$, $b \in Q$, cover $\mathcal{K}$: choose k with $1/k \le \varsigma$ and use N_k. Enumerate $Q = \{a_1, a_2, \dots\}$ (if Q is finite the argument below is the same and shorter).

For each k the family $\big(g_j(a_k)\big)_{j\in S}$ is bounded by A, so by Bolzano–Weierstrass we may choose nested infinite sets $S \supseteq J_1 \supseteq J_2 \supseteq \cdots$ such that for every k the values $g_j(a_k)$ converge as $j \to \infty$ along J_k: given J_{k-1} (with $J_0 = S$), pick a convergent subsequence of $\big(g_j(a_k)\big)_{j\in J_{k-1}}$ and let J_k be its set of indices. Choose $j_1 < j_2 < \cdots$ with $j_l \in J_l$, which is possible because each J_l is infinite, and put $S' = \{j_l : l \ge 1\}$, an infinite subset of S. Since $\{j_l : l \ge k\} \subseteq J_k$ for every k, the sequence $\big(g_{j_l}(a_k)\big)_l$ converges for every k.

Fix $\varepsilon > 0$ and choose $\eta \in (0, \frac{1}{2}]$ with $\omega(\eta) \le \varepsilon/3$, which is possible because $\omega(\varsigma) \to 0$ as $\varsigma \downarrow 0$. Cover $\mathcal{K}$ by finitely many balls $B_\eta(b_1), \dots, B_\eta(b_N)$ with $b_\nu \in Q$, as in the first paragraph, and choose l_0 with $|g_{j_l}(b_\nu) - g_{j_{l'}}(b_\nu)| \le \varepsilon/3$ for all $\nu \le N$ and all $l, l' \ge l_0$; this is possible because each of the finitely many sequences $\big(g_{j_l}(b_\nu)\big)_l$ converges, hence is Cauchy. Given $a \in \mathcal{K}$, pick ν with $|a - b_\nu| < \eta$; then for $l, l' \ge l_0$,

$$|g_{j_l}(a) - g_{j_{l'}}(a)| \le |g_{j_l}(a) - g_{j_l}(b_\nu)| + |g_{j_l}(b_\nu) - g_{j_{l'}}(b_\nu)| + |g_{j_{l'}}(b_\nu) - g_{j_{l'}}(a)| \le 2\omega(\eta) + \frac{\varepsilon}{3} \le \varepsilon.$$

So $\big(g_{j_l}\big)_l$ is uniformly Cauchy on $\mathcal{K}$; for each $a \in \mathcal{K}$ the real sequence $\big(g_{j_l}(a)\big)_l$ is therefore Cauchy and has a limit $g(a)$, and letting $l' \to \infty$ in the display gives $\sup_{\mathcal{K}} |g_{j_l} - g| \le \varepsilon$ for every $l \ge l_0$. As $\varepsilon > 0$ was arbitrary, $\sup_{\mathcal{K}} |g_j - g| \to 0$ along S'. This proves Claim A.

Each box $\mathcal{B}_K$ is closed and bounded, and for an integer $K \ge 2$ the hypotheses of Claim A hold on $\mathcal{B}_K$ for every infinite index set, with $A = C_g K L(K)^2$ by (433) and $\omega = \omega_K$ by (434), the latter being valid for all $j \in \mathbb{N}$ and all $0 < \varsigma \le \frac{1}{2}$ with $\omega_K(\varsigma) \to 0$ as $\varsigma \downarrow 0$ by Step 2. Define infinite index sets $S_2 \supseteq S_3 \supseteq \cdots$ recursively: S_K is the set S' produced by Claim A applied on $\mathcal{B}_K$ with the index set $\mathbb{N}$ when $K = 2$ and with the index set S_{K-1} when $K \ge 3$. Claim A also supplies functions $w^{(K)}$ on $\mathcal{B}_K$ with $\sup_{\mathcal{B}_K} |v_j - w^{(K)}| \to 0$ as $j \to \infty$ along S_K.

Choose $\iota(1) < \iota(2) < \cdots$ with $\iota(j) \in S_{j+1}$ for every j, which is possible because each S_{j+1} is infinite; thus $\iota : \mathbb{N} \to \mathbb{N}$ is strictly increasing. Fix an integer $K \ge 2$. For $j \ge K - 1$ we have

$\iota(j) \in S_{j+1} \subseteq S_K$, and $\iota(j) \to \infty$, so

$$\sup_{\mathcal{B}_K} \big| v_{\iota(j)} - w^{(K)} \big| \longrightarrow 0 \qquad (j \to \infty). \tag{435}$$

In particular, for $2 \le K \le K'$ the functions $w^{(K)}$ and $w^{(K')}$ agree on $\mathcal{B}_K$, both being the pointwise limit of $v_{\iota(j)}$ there. Hence a single function w on space-time is well defined by $w(z) = w^{(K)}(z)$ for any integer $K \ge 2$ with $z \in \mathcal{B}_K$, and (435) says that $v_{\iota(j)} \to w$ uniformly on $\mathcal{B}_K$ for every integer $K \ge 2$. Every compact subset of space-time is bounded, hence contained in some $\mathcal{B}_K$ with $K \ge 2$ an integer, so the convergence is uniform on every compact set. This is the first assertion.

Step 4: w solves the equation. By `IsFitClass` each $v_{\iota(j)}$ is a viscosity solution of $F[\cdot] = 0$ on all of space-time, in the sense of `IsLimitSolutionOn`. The whole space is open, and $v_{\iota(j)} \to w$ uniformly on every compact subset of it by Step 3, so `LimitStability` gives that w is a viscosity solution of $F[w] = 0$ on all of space-time; in particular w is continuous.

Step 5: w has fits at every center and radius. Fix a space-time point $z = (x, t)$ and $r > 0$, and pick an integer $K \ge \max\{2, |x|, |t|\}$, so $z \in \mathcal{B}_K$. By `IsFitClass` choose, for each j, a fit (p_j, a_j) for $v_{\iota(j)}$ at z of radius r with constant C_0. By `FitParamBound`,

$$|p_j| + |a_j| \le C_p\big(L(K) + |\log r|\big) \qquad (j \in \mathbb{N}),$$

a bound independent of j. Since $F \times \mathbb{R}$ is finite-dimensional, a subsequence $j \mapsto j_k$ has $(p_{j_k}, a_{j_k}) \to (p, a)$ for some $p \in F$, $a \in \mathbb{R}$.

Let $h \in F$ and $\ell \in \mathbb{R}$ with $|\ell| \le r$ and $|h - \ell p| \le r$. Put

$$h_k = \ell\, p_{j_k} + (h - \ell p),$$

so $h_k \to h$ and $|h_k - \ell p_{j_k}| = |h - \ell p| \le r$; together with $|\ell| \le r$ this means (h_k, ℓ) satisfies the box conditions of `IsFit` for the fit (p_{j_k}, a_{j_k}) at z of radius r. Hence

$$\Big| v_{\iota(j_k)}\big(z + (h_k, \ell)\big) - v_{\iota(j_k)}(z) - \langle p_{j_k}, h_k \rangle - \Big(a_{j_k} - \frac{|p_{j_k}|^2}{2}\Big)\ell \Big| \le C_0 r. \tag{436}$$

We pass to the limit. The points $\zeta_k = z + (h_k, \ell)$ converge to $\zeta = z + (h, \ell)$, and all of them, together with ζ, lie in a fixed compact set; since $v_{\iota(j)} \to w$ uniformly there and w is continuous by Step 4,

$$\big| v_{\iota(j_k)}(\zeta_k) - w(\zeta) \big| \le \big| v_{\iota(j_k)}(\zeta_k) - w(\zeta_k) \big| + \big| w(\zeta_k) - w(\zeta) \big| \longrightarrow 0,$$

so $v_{\iota(j_k)}(\zeta_k) \to w(\zeta)$; likewise $v_{\iota(j_k)}(z) \to w(z)$. Also $\langle p_{j_k}, h_k \rangle \to \langle p, h \rangle$ and $a_{j_k} - |p_{j_k}|^2/2 \to a - |p|^2/2$. Letting $k \to \infty$ in (436),

$$\Big| w\big(z + (h, \ell)\big) - w(z) - \langle p, h \rangle - \Big(a - \frac{|p|^2}{2}\Big)\ell \Big| \le C_0 r.$$

As (h, ℓ) was an arbitrary increment satisfying the box conditions for (p, a) at radius r, this says by `IsFit` that (p, a) is a fit for w at z of radius r with constant C_0. The subsequence used depends on z and r, which is irrelevant: the function w is fixed and only the existence of a fit at each (z, r) is asserted. Together with Step 4 and `IsFitClass`, $w \in \mathcal{A}_{m, C_0}$.

Step 6: w is centered at z_c. By `IsCenteredAt`, $v_{\iota(j)}(z_c) = 0$ for every j, and $v_{\iota(j)} \to w$ pointwise, so $w(z_c) = 0$. Again by `IsCenteredAt` the pair $(0, 0)$ is a fit for each $v_{\iota(j)}$ at z_c of radius 1 with constant C_0, i.e. by `IsFit`

$$\big| v_{\iota(j)}\big(z_c + (h, \ell)\big) - v_{\iota(j)}(z_c) \big| \le C_0 \qquad (|\ell| \le 1,\ |h| \le 1),$$

the affine terms vanishing because $p = 0$ and $a = 0$. Letting $j \to \infty$ pointwise gives $\big| w(z_c + (h, \ell)) - w(z_c) \big| \le C_0$ on the same set of increments, which by `IsFit` says that $(0, 0)$ is a fit for w at z_c of radius 1 with constant C_0. Hence w is centered at z_c with constant C_0 in the sense of `IsCenteredAt`. □

2.108. `ContactAtLimitingMarks`.

Lemma 2.108 (Contact at every mark of the limiting law)**.** *Let F be a finite-dimensional real inner-product space — the ambient convention of this development, used here only to carry the space-time $X = \ell^2(F \times \mathbb{R})$ — let $v : X \to \mathbb{R}$ and $\kappa, H, U_B \in \mathbb{R}$, let W be the contact envelope of `ContactEnv` formed from v and κ, and let J_ϵ be the propagation quantity of `JGap` formed from the same data. Write t_z for the time coordinate of $z \in X$ and $\delta_j = 2^{-j}$. Assume:*

(1) *v is continuous on X, W is continuous on $\{t_z > 0\}$, and $W \le v$ there;*
(2) *for every compact $K \subseteq X$ with $t_z > 0$ on K and every $\delta > 0$ there is $\epsilon_0 > 0$ such that*
$$\big|2\epsilon J_\epsilon(z) - \big(W(z) - v(z)\big)\big| \le \delta \qquad (0 < \epsilon < \epsilon_0,\ z \in K);$$
(3) *$\mathcal{B} \subseteq X$ is compact and $J_\epsilon(z) \le U_B$ whenever $z \in \mathcal{B}$, $t_z > 0$ and $0 < \epsilon \le 1$;*
(4) *$\mathcal{S}$ is a topological space with a σ-algebra for which the open sets are measurable and closed sets admit outer approximations, $m_j : \mathcal{S} \to X$ is continuous for every $j \in \mathbb{N}$, and λ, $(\lambda_k)_{k\in\mathbb{N}}$ are Borel probability measures on $\mathcal{S}$ with $\lambda_{n_h} \to \lambda$ weakly along an index map $n : \mathbb{N} \to \mathbb{N}$ satisfying $h \le n_h$;*
(5) *$A_j \subseteq \mathcal{S}$ is closed with $\lambda(A_j) = 1$ for every $j \ge 1$, and $\delta_j \le t_{m_j(q)}$ for every $q \in A_j$;*
(6) *for each k, (Ω_k, μ_k) is a probability space, $\Psi_k : \Omega_k \to \mathcal{S}$ is measurable and pushes μ_k to λ_k, i.e. $\lambda_k(A) = \mu_k(\Psi_k^{-1}A)$ for measurable $A \subseteq \mathcal{S}$; there are maps $Z_k : \mathbb{N} \times \Omega_k \to X$ with $t_{Z_k(n,\omega)} > 0$ always, indices $\varsigma_j^k : \Omega_k \to \mathbb{N}$, and lengths ϵ_k with $0 < \epsilon_k \le \delta_{k+1}/4$;*
(7) *for $1 \le j \le k+1$ and every $\omega \in \Omega_k$,*
$$m_j(\Psi_k\omega) = Z_k\big(\varsigma_j^k(\omega), \omega\big) \in \mathcal{B};$$
(8) *for $1 \le j \le k+1$ the function $\omega \mapsto J_{\epsilon_k}\big(Z_k(\varsigma_j^k(\omega), \omega)\big)$ is μ_k-integrable with*
$$H - 1 \le \int J_{\epsilon_k}\big(Z_k(\varsigma_j^k(\omega), \omega)\big)\, d\mu_k(\omega).$$

Then for λ-almost every $q \in \mathcal{S}$,
$$W\big(m_j(q)\big) = v\big(m_j(q)\big) \qquad \text{for every } j \ge 1.$$

Proof. Fix $j \ge 1$; since the conclusion is a countable conjunction it suffices to prove it for this j almost surely. By 5 the event A_j has full measure and $t_{m_j(q)} \ge \delta_j > 0$ on it, so by 1 the defect $v(m_j(q)) - W(m_j(q))$ is nonnegative there. It therefore suffices to show that it is almost surely not positive, and for that, by the Archimedean property — every positive real exceeds $1/(n+1)$ for some $n \in \mathbb{N}$ — it suffices to show

$$\lambda\Big\{q :\ \frac{1}{n+1} < v\big(m_j(q)\big) - W\big(m_j(q)\big)\Big\} = 0 \qquad (n \in \mathbb{N}), \tag{437}$$

since the set where the defect is positive is the union over n of these sets.

The open set. Fix n and suppose, for contradiction, that the set in (437) has positive measure. Put $\alpha = \dfrac{1}{2(n+1)} > 0$, so that $2\alpha = 1/(n+1)$. Let $f(q) = v(m_j(q)) - W(m_j(q))$, let
$$V = \big\{q :\ \delta_j/2 < t_{m_j(q)}\big\}, \qquad O = V \cap f^{-1}\big((\alpha, \infty)\big).$$
The map $q \mapsto t_{m_j(q)}$ is continuous by 4, so V is open; every $q \in V$ has $t_{m_j(q)} > \delta_j/2 > 0$, so f is continuous on V by 1, and therefore O is open.

O has positive measure. Indeed, intersecting the set of (437) with the conull set A_j does not change its measure, and every q in that intersection lies in O: its mark time is at least $\delta_j > \delta_j/2$ by 5, and $f(q) > 2\alpha > \alpha$. So $\lambda(O) > 0$.

The compact set and the gap bound. Put
$$K = \big(\mathcal{B} \cap \{z :\ \delta_j/2 \le t_z\}\big) \cap \{z :\ \alpha \le v(z) - W(z)\}.$$
The first bracket is closed, being the intersection of the compact — hence closed — set $\mathcal{B}$ with a closed half space of the continuous coordinate $z \mapsto t_z$; every one of its points has $t_z \ge \delta_j/2 > 0$, so $v - W$ is continuous on it by 1, and therefore K, the preimage of the closed half line $[\alpha, \infty)$

under that restriction, is closed as well. Being a closed subset of the compact $\mathcal{B}$, K is compact, and $t_z > 0$ on K.

Apply 2 to K with $\delta = \alpha/2 > 0$: there is $\epsilon_0 > 0$ such that for $0 < \epsilon < \epsilon_0$ and $z \in K$,

$$2\epsilon J_\epsilon(z) \le \big(W(z) - v(z)\big) + \frac{\alpha}{2} \le -\alpha + \frac{\alpha}{2} = -\frac{\alpha}{2},$$

using $\alpha \le v(z) - W(z)$ on K. Doubling, $4\epsilon J_\epsilon(z) \le -\alpha$, and dividing by $4\epsilon > 0$ — if $(4\epsilon)x \le -a$ with $\epsilon > 0$ and $a > 0$ then $x \le -a/(4\epsilon)$, by multiplying with the positive number $(4\epsilon)^{-1}$ —

$$J_\epsilon(z) \le -\frac{\alpha}{4\epsilon} \qquad (0 < \epsilon < \epsilon_0,\ z \in K). \tag{438}$$

One index. By the open-set half of the portmanteau theorem, available by 4, $\lambda(O) \le \liminf_h \lambda_{n_h}(O)$. Put $a = \lambda(O)/2$. Then $0 < a < \lambda(O)$ and $a < \infty$, so $a < \lambda_{n_h}(O)$ for all large h. Let $D = \max\{|U_B| - H + 1,\, 1\} \ge 1 > 0$. Since $(1/2)^m \to 0$ there are m_0, m_1 with

$$\Big(\frac{1}{2}\Big)^{m_0} < \frac{\alpha a}{4D}, \qquad \Big(\frac{1}{2}\Big)^{m_1} < \epsilon_0,$$

the first right-hand side being positive because $\alpha > 0$, $a > 0$ and $D > 0$. Choose h with $j \le h$, $m_0 \le h$, $m_1 \le h$ and $a < \lambda_{n_h}(O)$, and write $k = n_h$. Then $h \le k$ by 4, so $1 \le j \le k+1$, and by 6

$$\epsilon_k \le \Big(\frac{1}{2}\Big)^{k+1}/4 \le \Big(\frac{1}{2}\Big)^{m_0} < \frac{\alpha a}{4D}, \qquad \epsilon_k \le \Big(\frac{1}{2}\Big)^{k+1}/4 \le \Big(\frac{1}{2}\Big)^{m_1} < \epsilon_0, \tag{439}$$

both middle inequalities because $m_i \le h \le k < k+1$ and $(1/2)^{\cdot}$ is nonincreasing with $(1/2)^{k+1} > 0$. Put $p = \lambda_k(O)$, so that $p = \mu_k(\Psi_k^{-1}O)$ by 6; then $a < p$ and $p \le 1$, since μ_k is a probability measure.

The pointwise bound on Ω_k. Let $\omega \in \Omega_k$ and write $Z = Z_k(\varsigma_j^k(\omega), \omega)$, which by 7 equals $m_j(\Psi_k\omega)$ and lies in $\mathcal{B}$. Put $c = U_B + \alpha/(4\epsilon_k)$.

If $\Psi_k\omega \in O$ then $Z \in K$: it lies in $\mathcal{B}$; its time satisfies $t_Z = t_{m_j(\Psi_k\omega)} > \delta_j/2$, hence $\ge \delta_j/2$; and $v(Z) - W(Z) = f(\Psi_k\omega) > \alpha$, hence $\ge \alpha$. So (438) applies at ϵ_k, which is admissible by (439) and $\epsilon_k > 0$, and gives $J_{\epsilon_k}(Z) \le -\alpha/(4\epsilon_k) = U_B - c$.

If $\Psi_k\omega \notin O$ then 3 applies: $Z \in \mathcal{B}$, $t_Z > 0$ by 6, and $0 < \epsilon_k \le 1$ because $\epsilon_k \le (1/2)^{k+1}/4 \le 1$. So $J_{\epsilon_k}(Z) \le U_B$. In both cases

$$J_{\epsilon_k}(Z) \le U_B - c\,\mathbf{1}_{\Psi_k^{-1}O}(\omega). \tag{440}$$

Integration. The left-hand side of (440) is integrable by 8; the right-hand side is integrable as well, being the constant U_B minus c times the indicator of the measurable set $\Psi_k^{-1}O$ on a probability space. Integrating (440) and using $\int \mathbf{1}_{\Psi_k^{-1}O}\, d\mu_k = p$ together with 8,

$$H - 1 \le \int J_{\epsilon_k}(Z)\, d\mu_k \le U_B - cp = U_B(1-p) - \frac{\alpha}{4\epsilon_k}\, p,$$

the last step by the definition of c.

Finally $U_B(1-p) \le |U_B|$, because $0 \le 1-p$ and $U_B \le |U_B|$ and $|U_B| \ge 0$ — if $0 \le 1-p \le 1$ then $U_B(1-p) \le |U_B|(1-p) \le |U_B|$ — and $\dfrac{\alpha a}{4\epsilon_k} \le \dfrac{\alpha}{4\epsilon_k}\, p$ because $a \le p$ and $\alpha/(4\epsilon_k) > 0$. Hence

$$H - 1 \le |U_B| - \frac{\alpha a}{4\epsilon_k}, \qquad \text{i.e.} \qquad \frac{\alpha a}{4\epsilon_k} \le |U_B| - H + 1 \le D,$$

so $\alpha a \le 4D\epsilon_k$ after multiplying by $4\epsilon_k > 0$. But (439) gives $4D\epsilon_k < \alpha a$, since $4D > 0$. This contradiction proves (437).

Conclusion. By (437) and the countable union, almost every q has $v(m_j(q)) - W(m_j(q)) \le 0$; intersecting with the conull A_j, almost every q also has $t_{m_j(q)} \ge \delta_j > 0$ and hence $W(m_j(q)) \le v(m_j(q))$ by 1. The two inequalities give $W(m_j(q)) = v(m_j(q))$ almost surely, and intersecting the countably many full-measure events over $j \ge 1$ completes the proof. □

2.109. `UniformSubquadratic`.

Lemma 2.109 (Uniform subquadratic growth of the centered class)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$, and let z_c be either $(0,0)$ or $(0,1)$. Then, for every compact $J \subseteq \mathbb{R}$,*

$$\lim_{M\to\infty} \sup_{\substack{v\in\mathcal{A}_{m,C_0} \text{ centered at } z_c \\ |x|\ge M,\ t\in J}} \frac{|v(x,t)|}{|x|^2} = 0, \tag{441}$$

where $\mathcal{A}_{m,C_0}$ is the class of `IsFitClass` and "centered at z_c" is meant with constant C_0 in the sense of `IsCenteredAt`. Explicitly: for every compact $J \subseteq \mathbb{R}$ and every $\varepsilon > 0$ there is $M > 0$ such that $|v(x,t)| \le \varepsilon|x|^2$ for every such v and whenever $|x| \ge M$ and $t \in J$.

Proof. Let $C_1 > 0$ be the constant of `CenteredGrowth` for this C_0. Let $J \subseteq \mathbb{R}$ be compact and let $\varepsilon > 0$.

Since J is compact it is bounded, so there is $T \ge 2$ with $|t| \le T$ for all $t \in J$. For $M \ge T$ and a point $z = (x,t)$ with $|x| \ge M$ and $t \in J$, set $K = |x|$. Then $K \ge M \ge T \ge 2$, $|x| \le K$ and $|t| \le T \le K$, so $z \in \mathcal{B}_K$ and `CenteredGrowth` applies to any $v \in \mathcal{A}_{m,C_0}$ centered at z_c — its hypothesis is that v be centered at $(0,0)$ or at $(0,1)$, and z_c is one of those two points by assumption, so that disjunction holds in either case — giving

$$|v(x,t)| \le C_1 K\big(1+\log K\big)^2.$$

Dividing by $|x|^2 = K^2$,

$$\frac{|v(x,t)|}{|x|^2} \le C_1 \frac{(1+\log K)^2}{K}.$$

The function $K \mapsto C_1(1+\log K)^2/K$ tends to 0 as $K \to \infty$. Indeed, for $K \ge 1$ we have $\log K \ge 0$ and $(\log K - 1)^2 \ge 0$, so $2\log K \le 1 + (\log K)^2$ and therefore

$$(1+\log K)^2 = 1+2\log K+(\log K)^2 \le 2+2(\log K)^2, \qquad \text{whence} \qquad C_1\frac{(1+\log K)^2}{K} \le 2C_1\Big(\frac{1}{K}+\frac{(\log K)^2}{K}\Big),$$

and both $1/K$ and $(\log K)^2/K$ tend to 0 as $K \to \infty$. Hence there is $M_0 \ge 2$ with $C_1(1+\log K)^2/K \le \varepsilon$ for all $K \ge M_0$. Taking $M = \max\{T, M_0\} > 0$ gives $|v(x,t)| \le \varepsilon|x|^2$ for every centered member v of the class and all $|x| \ge M$, $t \in J$. The constant M depends only on C_0, $\dim F$, J and ε, not on v, which is the asserted uniformity. This is the explicit form of the statement.

The displayed limit. It remains to read the display (441) off that form. This step is not formality: the display is a statement about a supremum, and by the convention of the preamble a supremum carries the value 0 when its set is empty or unbounded above, so the boundedness has to be exhibited. Write

$$S(M) = \Big\{ \frac{|v(x,t)|}{|x|^2} \ :\ v \in \mathcal{A}_{m,C_0} \text{ centered at } z_c \text{ with constant } C_0,\ |x| \ge M,\ t \in J \Big\},$$

the set over which the supremum in (441) is taken, so that the display asserts $\sup S(M) \to 0$ as $M \to \infty$.

Fix $\varepsilon > 0$ and write $M_\varepsilon = \max\{T, M_0\}$ for the radius produced above for this ε, which the previous paragraph called M; in the present paragraph the letter M is free again and ranges over the argument of the supremum. Let $M \ge M_\varepsilon$. Every element of $S(M)$ is of the form $|v(x,t)|/|x|^2$ with v centered, $t \in J$ and $|x| \ge M \ge M_\varepsilon$, so the bound of the previous paragraph applies at that very point and gives $|v(x,t)| \le \varepsilon|x|^2$; since $|x| \ge M > 0$ we may divide, and $|v(x,t)|/|x|^2 \le \varepsilon$. Also $|v(x,t)|/|x|^2 \ge 0$. Hence

$$S(M) \subseteq [0,\varepsilon] \qquad (M \ge M_\varepsilon).$$

In particular $S(M)$ is bounded above, so $\sup S(M)$ is a genuine least upper bound rather than the convention value whenever $S(M) \ne \emptyset$, and then $0 \le \sup S(M) \le \varepsilon$; and if $S(M) = \emptyset$ then $\sup S(M) = 0$ by the convention, so $0 \le \sup S(M) \le \varepsilon$ in that case too. Thus $0 \le \sup S(M) \le \varepsilon$ for every $M \ge M_\varepsilon$. As $\varepsilon > 0$ was arbitrary, $\sup S(M) \to 0$ as $M \to \infty$, which is (441). □

2.110. `ContactEnvPositiveTimeRegularity`.

Lemma 2.110 (Continuity and subquadratic growth of the contact envelope on positive times)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and $0 \le \kappa < \infty$, and let v belong to the class $\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`, be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let $W = \mathcal{C}_t v - \kappa^2 t/2$ be the contact envelope of `ContactEnv`. Then W is continuous on $\{z : z_t > 0\}$, W has subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`, and for every $\sigma > 0$ the function*

$$y \longmapsto W(y,\sigma) - \frac{|y|^2}{2\sigma}$$

is concave on F.

Proof. Throughout, $z = (z_x, z_t)$ denotes a point of the space-time $\ell^2(F \times \mathbb{R})$.

Step 1: v is continuous and subquadratic on all of space-time. By `IsFitClass`, v is a viscosity solution of $F[v] = 0$ on all of space-time in the sense of `IsLimitSolutionOn`, whose first component is the subsolution property of `IsLimitSubOn`; the second clause of that property is precisely that v is continuous on all of space-time. For the growth, let $J \subseteq \mathbb{R}$ be compact and let $\varepsilon > 0$. Apply `UniformSubquadratic` with the constant C_0 and the centering point $(0,1)$ — which is one of the two admissible centering points in that statement — at this J and this ε: it yields $M > 0$ such that every member of the class centered at $(0,1)$ with constant C_0, in particular our v, satisfies $|v(z)| \le \varepsilon |z_x|^2$ whenever $|z_x| \ge M$ and $z_t \in J$. As J and ε were arbitrary and every J is contained in $\mathbb{R}$, this is `SubquadraticGrowth` for v with the time set $\mathbb{R}$.

Step 2: the sandwich at the running time. The hypotheses of `BaryEnvSandwich` hold for v with the time set $I = \mathbb{R}$, the constant $\kappa \ge 0$, the continuity of Step 1 and the Jensen bound assumed. It gives, for every parameter $\sigma > 0$ and every z,

$$v(z) \le \mathcal{C}_\sigma v(z) \le v(z) + \frac{\kappa^2 \sigma}{2}.$$

Specialising the parameter to $\sigma = z_t$ at any z with $z_t > 0$, and subtracting $\kappa^2 z_t/2$, the definition of W in `ContactEnv` gives

$$v(z) - \frac{\kappa^2 z_t}{2} \le W(z) \le v(z) \qquad (z_t > 0). \tag{442}$$

Step 3: continuity on positive times. The hypotheses of `BaryEnvDuality` hold for v with the open time set $I = \mathbb{R}$, by Step 1. Its third part says that the map $(\sigma, \zeta) \mapsto \mathcal{C}_\sigma v(\zeta)$ is continuous at (σ, ζ) whenever $\sigma > 0$ and $\zeta_t \in \mathbb{R}$, the latter being automatic. Fix z_0 with $(z_0)_t > 0$. The map $\zeta \mapsto (\zeta_t, \zeta)$ is continuous, being the pair of the continuous time coordinate and the identity, and it sends z_0 to $((z_0)_t, z_0)$ with $(z_0)_t > 0$; so the composite $\zeta \mapsto \mathcal{C}_{\zeta_t} v(\zeta)$ is continuous at z_0. The map $\zeta \mapsto \kappa^2 \zeta_t/2$ is continuous, so the difference, which is W by `ContactEnv`, is continuous at z_0. As z_0 was an arbitrary point with positive time, W is continuous on $\{z : z_t > 0\}$.

Step 4: subquadratic growth on compact subintervals of $(0,\infty)$. Let $J \subseteq (0,\infty)$ be compact and let $\varepsilon > 0$. If $J = \emptyset$ the conclusion holds with $M' = 1$, there being no admissible z. So assume $J \ne \emptyset$. Then J is a nonempty compact subset of $\mathbb{R}$, so it has a greatest element $\tau_{\max} \in J$, and $\tau_{\max} > 0$ because $J \subseteq (0,\infty)$.

Apply Step 1 with this J and with $\varepsilon/2 > 0$ to get $M > 0$ such that

$$|v(z)| \le \frac{\varepsilon}{2} |z_x|^2 \qquad (|z_x| \ge M,\ z_t \in J). \tag{443}$$

Put $A = \kappa^2 \tau_{\max}/\varepsilon \ge 0$ and $M' = \max\bigl(M,\ \sqrt{A} + 1\bigr)$, so that $M' \ge M$ and $M' > 0$. Since $M' \ge \sqrt{A} + 1 \ge 0$ we get $M'^2 \ge (\sqrt{A} + 1)^2 = A + 2\sqrt{A} + 1 \ge A$, hence

$$\frac{\kappa^2 \tau_{\max}}{2} = \frac{\varepsilon}{2} A \le \frac{\varepsilon}{2} M'^2. \tag{444}$$

Now let z satisfy $|z_x| \geq M'$ and $z_t \in J$. Then $|z_x| \geq M$, so (443) applies; and $z_t > 0$ because $J \subseteq (0,\infty)$, so (442) applies. From (442),

$$W(z) \leq v(z) \leq |v(z)|, \qquad -W(z) \leq -v(z) + \frac{\kappa^2 z_t}{2} \leq |v(z)| + \frac{\kappa^2 \tau_{\max}}{2},$$

the last step using $z_t \leq \tau_{\max}$ and $\kappa^2 \geq 0$. Hence

$$|W(z)| \leq |v(z)| + \frac{\kappa^2 \tau_{\max}}{2} \leq \frac{\varepsilon}{2}|z_x|^2 + \frac{\varepsilon}{2}M'^2 \leq \frac{\varepsilon}{2}|z_x|^2 + \frac{\varepsilon}{2}|z_x|^2 = \varepsilon |z_x|^2,$$

using (443), then (444), then $M' \leq |z_x|$ together with $\varepsilon > 0$. As J and ε were arbitrary, this is `SubquadraticGrowth` for W with the time set $(0,\infty)$.

Step 5: concavity after the quadratic correction. Let $\sigma > 0$. The fourth part of `BaryEnvDuality`, whose hypotheses were checked in Step 3, says that for every parameter $\sigma > 0$ and every time $\tau \in I = \mathbb{R}$ the function

$$y \longmapsto \mathcal{C}_\sigma v(y,\tau) - \frac{|y|^2}{2\sigma}$$

is concave on F. Apply it with this parameter σ and the time $\tau = \sigma$, which lies in $\mathbb{R}$.

By `ContactEnv`, $W(y,\sigma) = \mathcal{C}_\sigma v(y,\sigma) - \kappa^2\sigma/2$, so for every $y \in F$

$$W(y,\sigma) - \frac{|y|^2}{2\sigma} = \Big(\mathcal{C}_\sigma v(y,\sigma) - \frac{|y|^2}{2\sigma}\Big) - \frac{\kappa^2\sigma}{2}.$$

The two functions therefore differ by the constant $\kappa^2\sigma/2$, which does not depend on y. Subtracting a constant preserves concavity: if $y_0, y_1 \in F$ and $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$, then writing G for the concave function above and $c = \kappa^2\sigma/2$,

$$\alpha\big(G(y_0) - c\big) + \beta\big(G(y_1) - c\big) = \alpha G(y_0) + \beta G(y_1) - c \leq G(\alpha y_0 + \beta y_1) - c,$$

where the first equality uses $\alpha c + \beta c = (\alpha+\beta)c = c$ and the inequality is the concavity of G. Hence $y \mapsto W(y,\sigma) - |y|^2/(2\sigma)$ is concave on F, which is the third assertion. □

2.111. `ContactEnvelopeProperties`.

Lemma 2.111 (The six properties of the contact envelope)**.** *Let F be a finite-dimensional real inner-product space — the ambient convention of this development — and work in the space-time $X = \ell^2(F \times \mathbb{R})$, writing a point as $z = (x,t)$ with $x \in F$ the space coordinate and $t \in \mathbb{R}$ the time coordinate, and writing $P = \{z \in X : \ t > 0\}$ for the positive-time half space. Let $C_0 \in \mathbb{R}$, let $\kappa > 0$ and let $v : X \to \mathbb{R}$. Assume:*

(1) *$v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`, with $m = \dim F$;*
(2) *v satisfies the horizontal Jensen bound with constant κ on $I = \mathbb{R}$ in the sense of `HasJensenBound`;*
(3) *v has subquadratic spatial growth on compact time intervals of $I = \mathbb{R}$ in the sense of `SubquadraticGrowth`.*

Let $W = \mathcal{C}_t v - \kappa^2 t/2$ be the contact envelope of `ContactEnv` formed from v and κ, so $W(z) = \mathcal{C}_t v(z) - \kappa^2 t/2$ at $z = (x,t)$, the barycentric radius being taken equal to the time. Then

(a) *for every $z = (x,t)$ with $t > 0$,*

$$v(z) - \frac{\kappa^2 t}{2} \leq W(z) \leq v(z);$$

(b) *W is continuous on P, and for every $t > 0$ the function $x \mapsto W(x,t) - |x|^2/(2t)$ is concave on all of F;*
(c) *W has subquadratic spatial growth on compact time intervals of $I = (0,\infty)$;*
(d) *W is a viscosity subsolution of the limiting equation on P in the sense of `IsLimitSubOn`;*
(e) *at every $z = (x,t)$ with $t > 0$ and $W(z) = v(z)$,*

$$\frac{\mathcal{C}_t v(z) - v(z)}{t} = \frac{\kappa^2}{2}.$$

Proof. The six conclusions are proved and stated in this order; the construction that supplies 1–3 in the application plays no role here and is not re-narrated.

Preliminaries. The time coordinate $z \mapsto t$ is continuous on X, being the second projection composed with the linear homeomorphism $\ell^2(F \times \mathbb{R}) \cong F \times \mathbb{R}$; hence P, the preimage of $(0,\infty)$ under it, is open. From 1 we use only that v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there; the two further hypotheses 2 and 3 are stated at $I = \mathbb{R}$, so they hold a fortiori at $I = (0,\infty)$, as does the continuity of v on the preimage of $(0,\infty)$. These restricted forms are what the cited nodes consume below.

(a) The sandwich. `BaryEnvSandwich`, applied with $I = (0,\infty)$, $f = v$ and the constant $\kappa \geq 0$, and with the continuity, Jensen and growth hypotheses just restricted, gives

$$v(z) \leq \mathcal{C}_s v(z) \leq v(z) + \frac{\kappa^2 s}{2} \tag{445}$$

for every $s > 0$ and every $z = (x,t)$ with $t > 0$. Take $s = t$, which is legitimate since $t > 0$, and subtract $\kappa^2 t/2$ throughout: by `ContactEnv` the middle term becomes $W(z)$ and (a) follows.

(b) Continuity and semiconcavity. Apply `BaryEnvDuality` with the *open* interval $I = (0,\infty)$ and $f = v$, whose hypotheses are the restricted continuity and growth of the preliminaries. One of its clauses speaks about probability measures on F and therefore requires a Borel measurable structure on F; the present statement carries none and needs none, so F is given its Borel σ-algebra for the duration of this step, and nothing below depends on that choice.

The continuity clause gives that $(s,z) \mapsto \mathcal{C}_s v(z)$ is continuous at every pair (s, z_0) with $s > 0$ and the time of z_0 positive. Fix $z_0 \in P$ and compose that clause at (t_0, z_0), where $t_0 > 0$ is the time of z_0, with the continuous map $z \mapsto (t, z)$: the composite $z \mapsto \mathcal{C}_t v(z)$ is continuous at z_0. The function $z \mapsto \kappa^2 t/2$ is continuous as well, so their difference, which is W by `ContactEnv`, is continuous at z_0; as $z_0 \in P$ was arbitrary, W is continuous on P.

The concavity clause, applied with the barycentric radius s and the time both equal to a fixed $t > 0$, gives that $x \mapsto \mathcal{C}_t v(x,t) - |x|^2/(2t)$ is concave on F. Adding the constant $-\kappa^2 t/2$ preserves concavity, and the result is $x \mapsto W(x,t) - |x|^2/(2t)$, because the time coordinate of (x,t) is t. This is the second half of (b).

(c) Growth. Let $J \subseteq (0,\infty)$ be compact and let $\varepsilon > 0$. Being compact, J is bounded above, say by T. Put

$$c_J = \max\Big\{0,\ \frac{\kappa^2 T}{2}\Big\} \geq 0.$$

By 3 restricted to $I = (0,\infty)$, applied to J and to $\varepsilon/2 > 0$, there is $M_1 > 0$ with

$$|v(x,t)| \leq \frac{\varepsilon}{2}|x|^2 \qquad (|x| \geq M_1,\ t \in J). \tag{446}$$

Set $M = \max\{M_1, \sqrt{2c_J/\varepsilon}\}$, which is positive because $M_1 > 0$. Let $z = (x,t)$ with $|x| \geq M$ and $t \in J$; then $t > 0$, since $J \subseteq (0,\infty)$. From $|x| \geq M_1$ we get (446). From $\sqrt{2c_J/\varepsilon} \leq |x|$ and $2c_J/\varepsilon \geq 0$ we get, squaring, $2c_J/\varepsilon \leq |x|^2$, that is

$$c_J \leq \frac{\varepsilon}{2}|x|^2. \tag{447}$$

Also $t \leq T$ and $\kappa^2 \geq 0$ give $\kappa^2 t/2 \leq \kappa^2 T/2 \leq c_J$. Now (a) bounds $W(z)$ on both sides: on one side $W(z) \leq v(z) \leq (\varepsilon/2)|x|^2 \leq \varepsilon|x|^2$, and on the other

$$W(z) \ \geq\ v(z) - \frac{\kappa^2 t}{2} \ \geq\ -\frac{\varepsilon}{2}|x|^2 - c_J \ \geq\ -\varepsilon|x|^2,$$

using (446) and (447). Hence $|W(x,t)| \leq \varepsilon|x|^2$ whenever $|x| \geq M$ and $t \in J$, which is (c).

(d) The subsolution property. First, v is a viscosity subsolution of the limiting equation on P in the sense of `IsLimitSubOn`: the set P is open and v is continuous there by the preliminaries, and if ϕ is C^2 near a point $z \in P$ and $v - \phi$ has a local maximum at z relative to P, then, P being open and containing z, the neighbourhood filter within P at z is the full neighbourhood filter; so $v - \phi$ has a local maximum at z relative to all of X, and $F[\phi](z) \geq 0$ follows from the subsolution half of the equation satisfied by v on the whole space.

Now `BarycentricSubsolution`, applied with $f = v$, with this subsolution property on P, with the restricted growth on $I = (0, \infty)$ and with $c = \kappa^2/2$, shows that $z \mapsto \mathcal{C}_t v(z) - (\kappa^2/2)\, t$ is a viscosity subsolution of the limiting equation on P. That function is W pointwise, by `ContactEnv` and the identity $(\kappa^2/2)t = \kappa^2 t/2$, so (d) follows.

(e) The gap at a contact. Let $z = (x, t)$ with $t > 0$ and $W(z) = v(z)$. By `ContactEnv` this reads $\mathcal{C}_t v(z) - \kappa^2 t/2 = v(z)$, hence $\mathcal{C}_t v(z) - v(z) = \kappa^2 t/2$, and dividing by $t > 0$ gives

$$\frac{\mathcal{C}_t v(z) - v(z)}{t} = \frac{\kappa^2}{2},$$

which is (e).

Collecting (a)–(e) gives the six conclusions in the stated order. □

2.112. `QuadEnvFarCompetitors`.

Lemma 2.112 (Localization of the quadratic envelopes: far competitors lose)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$, let $\Omega = \{(y, s) : s \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Let $R_0 \geq 0$, let $J \subseteq I$ be nonempty and compact, and let $\ell_0 > 0$. Then there are real numbers $M \geq \max\{R_0, 1\}$ and N with*

(i) *$|f(y, s)| \leq N$ whenever $|y| \leq M$ and $s \in J$;*

(ii) *for all $x \in F$ with $|x| \leq R_0$, all $s \in J$, all ℓ with $0 < \ell \leq \ell_0$ and all $y \in F$ with $|y| > M$,*

$$f(y, s) - \frac{|y - x|^2}{2\ell} < f(x, s), \qquad f(x, s) < f(y, s) + \frac{|y - x|^2}{2\ell}.$$

Proof. The growth radius. Apply `SubquadraticGrowth` with the compact set $J \subseteq I$ and $\varepsilon = 1/(8\ell_0) > 0$: there is $M_0 > 0$ with

$$|f(y, s)| \leq \frac{|y|^2}{8\ell_0} \qquad (|y| \geq M_0,\ s \in J). \tag{448}$$

Set $M_1 = \max\{M_0, 4R_0, 1\}$, so $M_1 \geq 1 > 0$.

The far-competitor gap. Let $|y| > M_1$, $|x| \leq R_0$ and $0 < \ell \leq \ell_0$. From $4R_0 \leq M_1 < |y|$ we get $R_0 < \frac{1}{4}|y|$, so by the triangle inequality

$$|y - x| \geq |y| - |x| \geq |y| - R_0 > |y| - \tfrac{1}{4}|y| = \tfrac{3}{4}|y|.$$

All three quantities $|y - x|$, $\frac{3}{4}|y|$ are nonnegative, so squaring the strict inequality $|y - x| > \frac{3}{4}|y|$ preserves it: $|y - x|^2 > \frac{9}{16}|y|^2$. Dividing by $2\ell > 0$,

$$\frac{|y - x|^2}{2\ell} > \frac{9|y|^2}{32\ell}. \tag{449}$$

Also $|y| > M_1 \geq M_0$ and $0 < \ell \leq \ell_0$, so (448) and $1/(8\ell_0) \leq 1/(8\ell)$ give

$$-\frac{|y|^2}{8\ell} \leq -\frac{|y|^2}{8\ell_0} \leq f(y, s) \leq \frac{|y|^2}{8\ell_0} \leq \frac{|y|^2}{8\ell} \qquad (s \in J). \tag{450}$$

The two constants. The set $\mathcal{K}_1 = \{(y, s) : |y| \leq M_1,\ s \in J\}$ is compact: the closed ball $\{|y| \leq M_1\}$ is a closed bounded subset of the finite-dimensional space F and hence compact, J is compact, and the identification of $F \times \mathbb{R}$ with the L^2 product space of the preamble is a homeomorphism, so $\mathcal{K}_1$ is the image of a product of two compact sets under a homeomorphism. It is nonempty, because $J \neq \emptyset$ and $|0| = 0 \leq M_1$, and it is contained in Ω, because $J \subseteq I$. Since f is continuous on Ω, the continuous function $|f|$ attains a finite maximum on $\mathcal{K}_1$; call it N_1. Being a value of $|f|$, $N_1 \geq 0$, and

$$|f(x', s)| \leq N_1 \qquad (|x'| \leq M_1,\ s \in J). \tag{451}$$

Put

$$M = \max\Big\{M_1,\ \sqrt{\tfrac{32}{5} N_1 \ell_0} + 1\Big\},$$

which satisfies $M \geq M_1 \geq \max\{R_0, 1\}$, and let N be the finite maximum of $|f|$ on $\mathcal{K}_2 = \{(y,s) : |y| \leq M,\ s \in J\}$, which is again a nonempty compact subset of Ω by the same argument with M in place of M_1. Then (i) holds by construction.

Proof of (ii). Let $|y| > M$, $|x| \leq R_0$, $s \in J$ and $0 < \ell \leq \ell_0$. Since $M \geq M_1$ we have $|y| > M_1$, so (449) and (450) are available. Moreover $|y| > M \geq \sqrt{\frac{32}{5}N_1\ell_0}+1 > \sqrt{\frac{32}{5}N_1\ell_0} \geq 0$, so squaring gives $|y|^2 > \frac{32}{5}N_1\ell_0 \geq \frac{32}{5}N_1\ell$, the last step because $N_1 \geq 0$ and $\ell \leq \ell_0$. Dividing by $\frac{32}{5}\ell > 0$,

$$\frac{5|y|^2}{32\ell} > N_1. \tag{452}$$

Finally $|x| \leq R_0 \leq M_1$ and $s \in J$, so (451) gives $-N_1 \leq f(x,s) \leq N_1$.

For the first inequality of (ii), combine the upper bound of (450) with (449):

$$f(y,s) - \frac{|y-x|^2}{2\ell} < \frac{|y|^2}{8\ell} - \frac{9|y|^2}{32\ell} = \frac{4|y|^2 - 9|y|^2}{32\ell} = -\frac{5|y|^2}{32\ell} < -N_1 \leq f(x,s),$$

the middle strict inequality being (452) multiplied by -1.

For the second inequality of (ii), combine the lower bound of (450) with (449):

$$f(y,s) + \frac{|y-x|^2}{2\ell} > -\frac{|y|^2}{8\ell} + \frac{9|y|^2}{32\ell} = \frac{-4|y|^2 + 9|y|^2}{32\ell} = \frac{5|y|^2}{32\ell} > N_1 \geq f(x,s),$$

using (452) again. This is (ii), with the same M and N for both inequalities. □

2.113. `QuadEnvBallMax`.

Lemma 2.113 (The forward envelope as a maximum over a fixed ball)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on space-time, let $\ell > 0$, let $z = (x,t)$ be a space-time point and let $M \in \mathbb{R}$ satisfy $|x| \leq M$. Write $B_M = \{y \in F : |y| \leq M\}$ and assume*

(a) *$y \mapsto f(y, t+\ell)$ is continuous on B_M;*

(b) *$f(y,t+\ell) - \dfrac{|y-x|^2}{2\ell} < f(x,t+\ell)$ for every $y \in F$ with $|y| > M$.*

Then the forward quadratic envelope of `QuadEnvP` satisfies

$$\mathcal{P}_\ell f(z) = \sup\Big\{ f(y,t+\ell) - \frac{|y-x|^2}{2\ell} \ :\ y \in B_M \Big\},$$

and this value is attained in B_M: there is $y^* \in F$ with $|y^*| \leq M$ and

$$\mathcal{P}_\ell f(z) = f(y^*, t+\ell) - \frac{|y^*-x|^2}{2\ell}.$$

Proof. Write

$$G(y) = f(y,t+\ell) - \frac{|y-x|^2}{2\ell} \qquad (y \in F),$$

so that, by `QuadEnvP` and $\ell \neq 0$, the envelope $\mathcal{P}_\ell f(z)$ is the supremum of the competitor set $\mathcal{V} = \{G(y) : y \in F\}$, while the supremum in the assertion is that of $\mathcal{V}_M = \{G(y) : y \in B_M\}$.

A maximizer on the ball. B_M is a closed bounded subset of the finite-dimensional space F, hence compact, and it is nonempty because $|x| \leq M$ gives $x \in B_M$. The function $y \mapsto |y-x|^2/(2\ell)$ is continuous on F, since $\ell > 0$ makes the denominator a nonzero constant; together with (a) this makes G continuous on B_M. A continuous real function on a nonempty compact set attains its maximum, so there is $y^* \in B_M$ with

$$G(y) \leq G(y^*) \qquad (y \in B_M). \tag{453}$$

In particular $\mathcal{V}_M$ is nonempty and bounded above by $G(y^*)$, and $G(y^*) \in \mathcal{V}_M$, so $G(y^*)$ is the greatest element of $\mathcal{V}_M$ and $\sup \mathcal{V}_M = G(y^*)$. Here the supremum is a genuine least upper bound and not the value 0 that the convention of the preamble assigns to an empty or unbounded-above set, precisely because nonemptiness and the upper bound $G(y^*)$ have just been exhibited.

The ball is not restrictive. Since $x \in B_M$ and $|x - x| = 0$, (453) at $y = x$ gives

$$f(x,t+\ell) = G(x) \leq G(y^*). \tag{454}$$

Let $y \in F$ be arbitrary. If $|y| \le M$ then $G(y) \le G(y^*)$ by (453). If $|y| > M$ then (b) and (454) give

$$G(y) < f(x, t+\ell) \le G(y^*).$$

Hence $G(y) \le G(y^*)$ for every $y \in F$, i.e. $\mathcal{V}$ is bounded above by $G(y^*)$; and $\mathcal{V}$ is nonempty, containing $G(x)$. Since also $G(y^*) \in \mathcal{V}$, the number $G(y^*)$ is the greatest element of $\mathcal{V}$, so $\sup \mathcal{V} = G(y^*)$; again the boundedness and nonemptiness needed for the preamble's convention not to intervene have been exhibited.

Conclusion. By `QuadEnvP`, $\mathcal{P}_\ell f(z) = \sup \mathcal{V} = G(y^*) = \sup \mathcal{V}_M$, which is the displayed identity, and the witness $y^* \in B_M$, i.e. $|y^*| \le M$, satisfies $\mathcal{P}_\ell f(z) = G(y^*) = f(y^*, t+\ell) - |y^* - x|^2/(2\ell)$, which is the attainment statement. □

2.114. `QuadEnvBallMin`.

Lemma 2.114 (The backward envelope as a minimum over a fixed ball)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on space-time, let $\ell > 0$, let $z = (x, t)$ be a space-time point and let $M \in \mathbb{R}$ satisfy $|x| \le M$. Write $B_M = \{y \in F : |y| \le M\}$ and assume*

(a) *$y \mapsto f(y, t-\ell)$ is continuous on B_M;*

(b) *$f(x, t-\ell) < f(y, t-\ell) + \dfrac{|y-x|^2}{2\ell}$ for every $y \in F$ with $|y| > M$.*

Then the backward quadratic envelope of `QuadEnvM` satisfies

$$\mathcal{M}_\ell f(z) = \inf\Big\{ f(y, t-\ell) + \frac{|y-x|^2}{2\ell} \ : \ y \in B_M \Big\},$$

and this value is attained in B_M: there is $y^* \in F$ with $|y^*| \le M$ and

$$\mathcal{M}_\ell f(z) = f(y^*, t-\ell) + \frac{|y^*-x|^2}{2\ell}.$$

Proof. Write

$$G(y) = f(y, t-\ell) + \frac{|y-x|^2}{2\ell} \qquad (y \in F),$$

so that, by `QuadEnvM` and $\ell \neq 0$, the envelope $\mathcal{M}_\ell f(z)$ is the infimum of the competitor set $\mathcal{V} = \{G(y) : y \in F\}$, while the infimum in the assertion is that of $\mathcal{V}_M = \{G(y) : y \in B_M\}$.

A minimizer on the ball. B_M is a closed bounded subset of the finite-dimensional space F, hence compact, and it is nonempty because $|x| \le M$ gives $x \in B_M$. The function $y \mapsto |y-x|^2/(2\ell)$ is continuous on F, since $\ell > 0$ makes the denominator a nonzero constant; together with (a) this makes G continuous on B_M. A continuous real function on a nonempty compact set attains its minimum, so there is $y^* \in B_M$ with

$$G(y^*) \le G(y) \qquad (y \in B_M). \tag{455}$$

In particular $\mathcal{V}_M$ is nonempty and bounded below by $G(y^*)$, and $G(y^*) \in \mathcal{V}_M$, so $G(y^*)$ is the least element of $\mathcal{V}_M$ and $\inf \mathcal{V}_M = G(y^*)$. Here the infimum is a genuine greatest lower bound and not the value 0 that the convention of the preamble assigns to an empty or unbounded-below set, precisely because nonemptiness and the lower bound $G(y^*)$ have just been exhibited.

The ball is not restrictive. Since $x \in B_M$ and $|x - x| = 0$, (455) at $y = x$ gives

$$G(y^*) \le G(x) = f(x, t-\ell). \tag{456}$$

Let $y \in F$ be arbitrary. If $|y| \le M$ then $G(y^*) \le G(y)$ by (455). If $|y| > M$ then (b) and (456) give

$$G(y^*) \le f(x, t-\ell) < G(y).$$

Hence $G(y^*) \le G(y)$ for every $y \in F$, i.e. $\mathcal{V}$ is bounded below by $G(y^*)$; and $\mathcal{V}$ is nonempty, containing $G(x)$. Since also $G(y^*) \in \mathcal{V}$, the number $G(y^*)$ is the least element of $\mathcal{V}$, so $\inf \mathcal{V} = G(y^*)$; again the boundedness and nonemptiness needed for the preamble's convention not to intervene have been exhibited.

Conclusion. By `QuadEnvM`, $\mathcal{M}_\ell f(z) = \inf \mathcal{V} = G(y^*) = \inf \mathcal{V}_M$, which is the displayed identity, and the witness $y^* \in B_M$, i.e. $|y^*| \le M$, satisfies $\mathcal{M}_\ell f(z) = G(y^*) = f(y^*, t-\ell) + |y^* - x|^2/(2\ell)$, which is the attainment statement. □

2.115. `QuadEnvBoxRepresentation`.

Lemma 2.115 (One ball localizes the envelopes on a box of lengths and centers)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(y,s) : s \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Let $\ell_0 > 0$ and let $z_0 = (x_0, t_0)$ be a space-time point. Then:*

(1) *if $t_0 + \ell_0 \in I$, there are $\rho > 0$ and $M > 0$ such that every pair (ℓ, z) with $z = (x,t)$ and*

$$|\ell - \ell_0| \le \rho, \qquad |x - x_0| \le \rho, \qquad |t - t_0| \le \rho$$

satisfies $\ell > 0$, $t + \ell \in I$ and

$$\mathcal{P}_\ell f(z) = \sup\Big\{ f(y, t+\ell) - \frac{|y-x|^2}{2\ell} \ : \ |y| \le M \Big\},$$

with $\mathcal{P}_\ell f$ the forward envelope of `QuadEnvP`; and moreover $(\ell, z) \mapsto \mathcal{P}_\ell f(z)$ is continuous at (ℓ_0, z_0);

(2) *if $t_0 - \ell_0 \in I$, there are $\rho > 0$ and $M > 0$ such that every pair (ℓ, z) with $z = (x,t)$ and $|\ell - \ell_0| \le \rho$, $|x - x_0| \le \rho$, $|t - t_0| \le \rho$ satisfies $\ell > 0$, $t - \ell \in I$ and*

$$\mathcal{M}_\ell f(z) = \inf\Big\{ f(y, t-\ell) + \frac{|y-x|^2}{2\ell} \ : \ |y| \le M \Big\},$$

with $\mathcal{M}_\ell f$ the backward envelope of `QuadEnvM`; and moreover $(\ell, z) \mapsto \mathcal{M}_\ell f(z)$ is continuous at (ℓ_0, z_0).

Proof. An elementary parametric-extremum fact. We first record that the maximum of a continuous function over a fixed compact set depends continuously on the remaining parameters. Let P be a metric space, let Q be a nonempty compact metric space and let H be real-valued and continuous on $P \times Q$. For each $\pi \in P$ the function $H(\pi, \cdot)$ is continuous on the nonempty compact Q, so $\max_{q \in Q} H(\pi, q)$ exists. Let $\pi_k \to \pi$ in P. Testing against each fixed $q \in Q$ gives $\max_{q'} H(\pi_k, q') \ge H(\pi_k, q)$ and $H(\pi_k, q) \to H(\pi, q)$ by continuity of H, hence $\liminf_k \max_{q'} H(\pi_k, q') \ge H(\pi, q)$ for every q, and therefore $\liminf_k \max_{q'} H(\pi_k, q') \ge \max_q H(\pi, q)$. Conversely, choose for each k a maximizer $q_k \in Q$ of $H(\pi_k, \cdot)$, and let a subsequence be given; by compactness of Q it has a further subsequence along which $q_k \to \bar{q} \in Q$. Along that further subsequence $(\pi_k, q_k) \to (\pi, \bar{q})$, so by continuity of H

$$\max_{q'} H(\pi_k, q') = H(\pi_k, q_k) \to H(\pi, \bar{q}) \le \max_q H(\pi, q).$$

Thus every subsequence of the real sequence $a_k = \max_{q'} H(\pi_k, q')$ has a further subsequence converging to a real number that is at most $L := \max_q H(\pi, q)$. By the first bound every subsequential limit of (a_k) is also at least L, so every subsequence of (a_k) has a further subsequence converging to L exactly; in particular no subsequence of (a_k) diverges, so (a_k) is bounded. A bounded real sequence all of whose subsequences have a further subsequence converging to L converges to L: otherwise some $\varepsilon > 0$ admits infinitely many k with $|a_k - L| \ge \varepsilon$, and that subsequence would have no further subsequence converging to L. Hence $\max_{q'} H(\pi_k, q') \to \max_q H(\pi, q)$. As P is a metric space, sequential continuity is continuity, so $\pi \mapsto \max_{q \in Q} H(\pi, q)$ is continuous on P. Applying this to $-H$ and using $\min_q H(\pi, q) = -\max_q (-H)(\pi, q)$, the map $\pi \mapsto \min_{q \in Q} H(\pi, q)$ is continuous on P as well.

Proof of (1). Assume $t_0 + \ell_0 \in I$. Since I is open there is $\sigma > 0$ with $(t_0 + \ell_0 - \sigma,\ t_0 + \ell_0 + \sigma) \subseteq I$. Put

$$\rho = \min\Big\{ \frac{\sigma}{4},\ \frac{\ell_0}{4} \Big\} > 0, \qquad J = [\, t_0 + \ell_0 - 2\rho,\ t_0 + \ell_0 + 2\rho \,].$$

Then $2\rho \le \sigma/2 < \sigma$, so $J \subseteq (t_0 + \ell_0 - \sigma, t_0 + \ell_0 + \sigma) \subseteq I$, and J is a nonempty compact interval. Let

$$\mathcal{P} = \big\{ (\ell, z) : z = (x,t),\ |\ell - \ell_0| \le \rho,\ |x - x_0| \le \rho,\ |t - t_0| \le \rho \big\}.$$

For $(\ell, z) \in \mathcal{P}$ we have $\ell \geq \ell_0 - \rho \geq \frac{3}{4}\ell_0 > 0$, $|x| \leq |x_0| + |x - x_0| \leq |x_0| + \rho$, $\ell \leq \ell_0 + \rho$ and

$$|(t + \ell) - (t_0 + \ell_0)| \leq |t - t_0| + |\ell - \ell_0| \leq 2\rho,$$

so $t + \ell \in J \subseteq I$. This already gives the assertions $\ell > 0$ and $t + \ell \in I$ for all $(\ell, z) \in \mathcal{P}$.

Apply `QuadEnvFarCompetitors` with $R_0 = |x_0| + \rho$ (≥ 0), with this J (nonempty, compact and contained in I) and with the length bound $\ell_0 + \rho > 0$. It produces $M \geq \max\{R_0, 1\}$, so $M > 0$ and $M \geq R_0$, and a constant N, such that in particular

$$f(y, s) - \frac{|y - x'|^2}{2\ell} < f(x', s) \qquad \big(|x'| \leq R_0,\ s \in J,\ 0 < \ell \leq \ell_0 + \rho,\ |y| > M\big). \tag{457}$$

Fix $(\ell, z) \in \mathcal{P}$ with $z = (x, t)$. Then $|x| \leq R_0 \leq M$, and $y \mapsto f(y, t + \ell)$ is continuous on $B_M = \{|y| \leq M\}$: for $|y| \leq M$ the point $(y, t+\ell)$ lies in Ω, because $t+\ell \in I$, and f is continuous on Ω. Moreover (457) applied with $x' = x$ (admissible since $|x| \leq R_0$), $s = t + \ell \in J$ and this $\ell \in (0, \ell_0 + \rho]$ says exactly that $f(y, t+\ell) - |y - x|^2/(2\ell) < f(x, t+\ell)$ whenever $|y| > M$. Hence `QuadEnvBallMax` applies at (ℓ, z) with this M and yields

$$\mathcal{P}_\ell f(z) = \sup\Big\{ f(y, t + \ell) - \frac{|y - x|^2}{2\ell} : |y| \leq M \Big\} = \max_{y \in B_M} \Big(f(y, t + \ell) - \frac{|y - x|^2}{2\ell} \Big), \tag{458}$$

the second equality because `QuadEnvBallMax` also asserts that the value is attained at some point of B_M. This is the displayed identity of (1), for every $(\ell, z) \in \mathcal{P}$, with the ρ and M just constructed.

For the continuity, put

$$H\big((\ell, z), y\big) = f(y, t + \ell) - \frac{|y - x|^2}{2\ell} \qquad \big((\ell, z) \in \mathcal{P},\ y \in B_M\big).$$

The set B_M is a closed bounded subset of the finite-dimensional space F, hence compact, and it is nonempty. The set $\mathcal{P}$, with the metric induced from $\mathbb{R} \times F \times \mathbb{R}$, is a metric space. The map $((\ell, z), y) \mapsto (y, t+\ell)$ is continuous and maps $\mathcal{P} \times B_M$ into Ω, as shown above, and f is continuous on Ω; the map $((\ell, z), y) \mapsto |y - x|^2/(2\ell)$ is continuous on $\mathcal{P} \times B_M$ because $\ell \geq \frac{3}{4}\ell_0 > 0$ there, so the denominator does not vanish. Hence H is continuous on $\mathcal{P} \times B_M$, and by the parametric-extremum fact recorded at the start the map $(\ell, z) \mapsto \max_{y \in B_M} H((\ell, z), y)$ is continuous on $\mathcal{P}$. By (458) that map is $(\ell, z) \mapsto \mathcal{P}_\ell f(z)$ restricted to $\mathcal{P}$.

Finally, the set $\mathcal{U} = \{(\ell, z) : |\ell - \ell_0| < \rho,\ |x - x_0| < \rho,\ |t - t_0| < \rho\}$ is open in $\mathbb{R} \times F \times \mathbb{R}$, contains (ℓ_0, z_0) and is contained in $\mathcal{P}$. The function $(\ell, z) \mapsto \mathcal{P}_\ell f(z)$, defined on all of $\mathbb{R} \times F \times \mathbb{R}$, agrees on the neighbourhood $\mathcal{U}$ of (ℓ_0, z_0) with the restriction to $\mathcal{P}$, which was just shown continuous at (ℓ_0, z_0) relative to $\mathcal{P}$. Since $\mathcal{U}$ is a neighbourhood of (ℓ_0, z_0) contained in $\mathcal{P}$, relative continuity at (ℓ_0, z_0) on $\mathcal{P}$ gives continuity of the unrestricted map at (ℓ_0, z_0). This proves (1).

Proof of (2). Assume $t_0 - \ell_0 \in I$. The argument is the same with the time level $t - \ell$ in place of $t + \ell$, minima in place of maxima, and the second inequality of `QuadEnvFarCompetitors` in place of the first. In detail: choose $\sigma > 0$ with $(t_0 - \ell_0 - \sigma, t_0 - \ell_0 + \sigma) \subseteq I$, put $\rho = \min\{\sigma/4, \ell_0/4\} > 0$ and $J = [\,t_0 - \ell_0 - 2\rho,\ t_0 - \ell_0 + 2\rho\,] \subseteq I$, a nonempty compact interval, and let $\mathcal{P}$ be as above. For $(\ell, z) \in \mathcal{P}$ we again have $\ell \geq \frac{3}{4}\ell_0 > 0$, $\ell \leq \ell_0 + \rho$, $|x| \leq |x_0| + \rho$ and

$$|(t - \ell) - (t_0 - \ell_0)| \leq |t - t_0| + |\ell - \ell_0| \leq 2\rho,$$

so $t - \ell \in J \subseteq I$. Apply `QuadEnvFarCompetitors` with $R_0 = |x_0| + \rho$, this J and the length bound $\ell_0 + \rho$, obtaining $M \geq \max\{R_0, 1\}$ and N with, in particular,

$$f(x', s) < f(y, s) + \frac{|y - x'|^2}{2\ell} \qquad \big(|x'| \leq R_0,\ s \in J,\ 0 < \ell \leq \ell_0 + \rho,\ |y| > M\big).$$

For $(\ell, z) \in \mathcal{P}$ this says, at $x' = x$ and $s = t - \ell \in J$, that $f(x, t - \ell) < f(y, t - \ell) + |y - x|^2/(2\ell)$ whenever $|y| > M$; and $y \mapsto f(y, t - \ell)$ is continuous on B_M because $(y, t - \ell) \in \Omega$ there. So `QuadEnvBallMin` applies and gives

$$\mathcal{M}_\ell f(z) = \inf\Big\{ f(y, t - \ell) + \frac{|y - x|^2}{2\ell} : |y| \leq M \Big\} = \min_{y \in B_M} \Big(f(y, t - \ell) + \frac{|y - x|^2}{2\ell} \Big)$$

for every $(\ell, z) \in \mathcal{P}$, the second equality because `QuadEnvBallMin` asserts attainment in B_M. The function $H((\ell, z), y) = f(y, t-\ell)+|y-x|^2/(2\ell)$ is continuous on $\mathcal{P}\times B_M$ for the same reasons as before, so by the minimum half of the parametric-extremum fact the map $(\ell, z) \mapsto \mathcal{M}_\ell f(z)$ is continuous on $\mathcal{P}$, hence — by the same neighbourhood argument, with $\mathcal{U} \subseteq \mathcal{P}$ an open set containing (ℓ_0, z_0) — continuous at (ℓ_0, z_0). This proves (2). □

2.116. `QuadEnvOptimizerRate`.

Lemma 2.116 (Rate of collapse of the envelope optimizers)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on space-time, let $\ell > 0$, let $z = (x, t)$ be a space-time point and let $M, N \in \mathbb{R}$ satisfy $|x| \le M$ and $N \ge 0$. Assume*

(a) *$|f(y, t + \ell)| \le N$ and $|f(y, t - \ell)| \le N$ for every $y \in F$ with $|y| \le M$;*
(b) *for every $y \in F$ with $|y| > M$,*

$$f(y, t + \ell) - \frac{|y - x|^2}{2\ell} < f(x, t + \ell), \qquad f(x, t - \ell) < f(y, t - \ell) + \frac{|y - x|^2}{2\ell}.$$

Then, with the envelopes of `QuadEnvP` and `QuadEnvM`:

(1) *$f(x, t + \ell) \le \mathcal{P}_\ell f(z)$, and every $y \in F$ with $\mathcal{P}_\ell f(z) = f(y, t + \ell) - |y - x|^2/(2\ell)$ satisfies $|y| \le M$ and $|y - x| \le 2\sqrt{N\ell}$;*
(2) *$\mathcal{M}_\ell f(z) \le f(x, t-\ell)$, and every $y \in F$ with $\mathcal{M}_\ell f(z) = f(y, t-\ell)+|y-x|^2/(2\ell)$ satisfies $|y| \le M$ and $|y - x| \le 2\sqrt{N\ell}$.*

Proof. (1). Write $G(y) = f(y, t + \ell) - |y - x|^2/(2\ell)$ for $y \in F$, so that by `QuadEnvP` and $\ell \neq 0$ the envelope $\mathcal{P}_\ell f(z)$ is the supremum of $\mathcal{V} = \{G(y) : y \in F\}$.

The set $\mathcal{V}$ is bounded above by N. Indeed, if $|y| \le M$ then $|y - x|^2/(2\ell) \ge 0$ and (a) give $G(y) \le f(y, t+\ell) \le |f(y, t+\ell)| \le N$; and if $|y| > M$ then the first inequality of (b) together with (a) applied at x — legitimate because $|x| \le M$ — gives $G(y) < f(x, t + \ell) \le |f(x, t + \ell)| \le N$. The set $\mathcal{V}$ is also nonempty, since $G(x) \in \mathcal{V}$. Hence $\sup \mathcal{V}$ is a genuine least upper bound rather than the value 0 that the convention of the preamble assigns to an empty or unbounded-above set, and being an upper bound of $\mathcal{V}$ it dominates the element $G(x)$:

$$f(x, t + \ell) = G(x) \le \sup \mathcal{V} = \mathcal{P}_\ell f(z), \tag{459}$$

which is the first claim of (1).

Now let $y \in F$ satisfy $\mathcal{P}_\ell f(z) = G(y)$. If $|y| > M$, the first inequality of (b) and (459) would give

$$G(y) < f(x, t + \ell) \le \mathcal{P}_\ell f(z) = G(y),$$

which is impossible; hence $|y| \le M$. Consequently, using the defining equation of y, then (459), then (a) at y and at x (both of norm $\le M$),

$$\frac{|y - x|^2}{2\ell} = f(y, t+\ell)-G(y) = f(y, t+\ell)-\mathcal{P}_\ell f(z) \le f(y, t+\ell)-f(x, t+\ell) \le N+N = 2N. \tag{460}$$

Multiplying by $2\ell > 0$ gives $|y - x|^2 \le 4N\ell$. Since $|y - x| \ge 0$ and $4N\ell \ge 0$ (because $N \ge 0$ and $\ell > 0$), taking nonnegative square roots preserves the inequality: $|y - x| \le \sqrt{4N\ell} = 2\sqrt{N\ell}$. This completes (1).

(2). This is the mirror computation, with the time argument $t - \ell$ in place of $t + \ell$ and an infimum in place of the supremum; it is not the forward statement applied to $-f$. Write $G(y) = f(y, t-\ell)+|y-x|^2/(2\ell)$ for $y \in F$, so that by `QuadEnvM` and $\ell \neq 0$ the envelope $\mathcal{M}_\ell f(z)$ is the infimum of $\mathcal{V} = \{G(y) : y \in F\}$.

The set $\mathcal{V}$ is bounded below by $-N$. Indeed, if $|y| \le M$ then $|y - x|^2/(2\ell) \ge 0$ and (a) give $G(y) \ge f(y, t - \ell) \ge -|f(y, t - \ell)| \ge -N$; and if $|y| > M$ then the second inequality of (b) together with (a) applied at x gives $G(y) > f(x, t - \ell) \ge -|f(x, t - \ell)| \ge -N$. The set $\mathcal{V}$ is nonempty, containing $G(x)$. Hence $\inf \mathcal{V}$ is a genuine greatest lower bound rather than the preamble's convention value, and being a lower bound of $\mathcal{V}$ it is dominated by the element $G(x)$:

$$\mathcal{M}_\ell f(z) = \inf \mathcal{V} \le G(x) = f(x, t - \ell), \tag{461}$$

which is the first claim of (2).

Now let $y \in F$ satisfy $\mathcal{M}_\ell f(z) = G(y)$. If $|y| > M$, the second inequality of (b) and (461) would give

$$G(y) > f(x, t-\ell) \geq \mathcal{M}_\ell f(z) = G(y),$$

which is impossible; hence $|y| \leq M$. Consequently, using the defining equation of y, then (461), then (a) at y and at x,

$$\frac{|y-x|^2}{2\ell} = G(y) - f(y, t-\ell) = \mathcal{M}_\ell f(z) - f(y, t-\ell) \leq f(x, t-\ell) - f(y, t-\ell) \leq N + N = 2N, \tag{462}$$

so $|y-x|^2 \leq 4N\ell$ and, as before, $|y-x| \leq 2\sqrt{N\ell}$. This completes (2). □

2.117. `QuadEnvAttained`.

Lemma 2.117 (Quadratic envelopes: attainment, continuity, collapse)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(x,t) : t \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Then, with the envelopes of `QuadEnvP` and `QuadEnvM`:*

1. *for every $\ell > 0$ and every $z = (x,t)$ with $t \in I$ and $t + \ell \in I$ the supremum defining $\mathcal{P}_\ell f(z)$ is attained at some $y \in F$, and for every $\ell > 0$ and every z with $t \in I$ and $t - \ell \in I$ the infimum defining $\mathcal{M}_\ell f(z)$ is attained;*
2. *the maps $(\ell, z) \mapsto \mathcal{P}_\ell f(z)$ and $(\ell, z) \mapsto \mathcal{M}_\ell f(z)$ are jointly continuous in the length and the center at every (ℓ_0, z_0) with $\ell_0 > 0$, $t_0 \in I$ and $t_0 + \ell_0 \in I$, respectively $t_0 - \ell_0 \in I$, where $z_0 = (x_0, t_0)$;*
3. *the convergence to the function as the length tends to zero is uniform on compact sets of admissible centers: for every compact set K of points $z = (x,t)$ with $t \in I$ and every $\eta > 0$ there is $\delta > 0$ with*

$$|\mathcal{P}_\ell f(z) - f(z)| \leq \eta, \qquad |\mathcal{M}_\ell f(z) - f(z)| \leq \eta \qquad (0 < \ell < \delta,\ z \in K);$$

   in particular $\mathcal{P}_\ell f(z) \to f(z)$ and $\mathcal{M}_\ell f(z) \to f(z)$ as $\ell \downarrow 0$ at each such z;
4. *for every $z = (x,t)$ with $t \in I$ and every $\eta > 0$ there is $\delta > 0$ such that for all $0 < \ell < \delta$ every optimizer y for $\mathcal{P}_\ell f(z)$ and every optimizer y for $\mathcal{M}_\ell f(z)$ satisfies $|y - x| \leq \eta$.*

Proof. Throughout, $z = (x,t)$ denotes a space-time point. We use freely that $\Omega = \{(y,s) : s \in I\}$ is open because I is open, and that f is continuous on Ω.

Step 1: a localization lemma. We first prove the following. Let $R_0 \geq 0$, let $J \subseteq I$ be nonempty and compact, and let $\ell_0 > 0$. Then there are $M \geq \max\{R_0, 1\}$ and $N < \infty$ with

(i) $|f(y,s)| \leq N$ whenever $|y| \leq M$ and $s \in J$;
(ii) for all x with $|x| \leq R_0$, all $s \in J$, all $0 < \ell \leq \ell_0$ and all y with $|y| > M$,

$$f(y,s) - \frac{|y-x|^2}{2\ell} < f(x,s), \qquad f(y,s) + \frac{|y-x|^2}{2\ell} > f(x,s).$$

To see this, apply `SubquadraticGrowth` with the compact set $J \subseteq I$ and $\varepsilon = 1/(8\ell_0) > 0$: there is $M_0 > 0$ with

$$|f(y,s)| \leq \frac{|y|^2}{8\ell_0} \qquad (|y| \geq M_0,\ s \in J). \tag{463}$$

Set $M_1 = \max\{M_0, 4R_0, 1\}$. For $|y| > M_1$ and $|x| \leq R_0$ we have $|y-x| \geq |y| - R_0 \geq |y| - \frac{1}{4}|y| = \frac{3}{4}|y|$, so for $0 < \ell \leq \ell_0$ and $s \in J$, using (463) and $\ell \leq \ell_0$,

$$f(y,s) - \frac{|y-x|^2}{2\ell} \leq \frac{|y|^2}{8\ell} - \frac{9|y|^2}{32\ell} = -\frac{5|y|^2}{32\ell}. \tag{464}$$

The set $\mathcal{K}_1 = \{(y,s) : |y| \leq M_1,\ s \in J\}$ is compact and contained in Ω, so $N_1 = \max_{\mathcal{K}_1} |f|$ is finite. Put

$$M = \max\Big\{M_1,\ \sqrt{\tfrac{32}{5} N_1 \ell_0} + 1\Big\},$$

and let $N = \max_{\mathcal{K}_2} |f|$, where $\mathcal{K}_2 = \{(y,s) : |y| \leq M,\ s \in J\}$ is again a compact subset of Ω; this is (i), and $M \geq \max\{R_0, 1\}$.

For (ii), let $|y| > M$, $|x| \le R_0$, $s \in J$ and $0 < \ell \le \ell_0$. Then $|y| > M_1$, so (464) applies; and $|y|^2 > \frac{32}{5} N_1 \ell_0 \ge \frac{32}{5} N_1 \ell$, so $\frac{5|y|^2}{32\ell} > N_1$. Since $|x| \le R_0 \le M_1$ and $s \in J$ we have $f(x,s) \ge -N_1$, and therefore

$$f(y,s) - \frac{|y-x|^2}{2\ell} < -N_1 \le f(x,s),$$

which is the first inequality.

For the second inequality we apply the first to $-f$, and we must check that it returns the same M, since (ii) asserts both inequalities for one M. The function $-f$ is continuous on Ω, and it has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth` because $|-f| = |f|$ and that definition constrains the function only through its absolute value. Moreover every constant produced above depends on f only through $|f|$, so the construction for $-f$ can be run with the constants already chosen. In detail: (463) is a bound on $|f|$, so the M_0 chosen for f also witnesses the conclusion of `SubquadraticGrowth` for $-f$ with the same J and ε, and we take that M_0; then $M_1 = \max\{M_0, 4R_0, 1\}$ is unchanged, being built from M_0, R_0 and 1; and N_1 and N are unchanged, being the maxima of $|f|$ over $\mathcal{K}_1$ and $\mathcal{K}_2$, whence $M = \max\{M_1, \sqrt{(32/5)N_1\ell_0} + 1\}$ is unchanged as well. In particular (i) holds for $-f$ with that same N. The first inequality applied to $-f$ therefore reads

$$-f(y,s) - \frac{|y-x|^2}{2\ell} < -f(x,s) \qquad \big(|y| > M,\ |x| \le R_0,\ s \in J,\ 0 < \ell \le \ell_0\big),$$

with this same M; multiplying by -1 gives the second inequality of (ii).

Step 2: attainment, i.e. (1). Let $\ell > 0$ and $z = (x,t)$ with $t \in I$ and $t + \ell \in I$. Apply Step 1 with $R_0 = |x|$, $J = \{t+\ell\}$ (nonempty, compact, contained in I) and $\ell_0 = \ell$, obtaining M and N. By Step 1(ii) with $s = t+\ell$, no competitor y with $|y| > M$ gives a value $f(y,t+\ell) - |y-x|^2/(2\ell)$ exceeding the value $f(x,t+\ell)$ at $y = x$. Hence the set of competitor values

$$\mathcal{V} = \Big\{ f(y,t+\ell) - \frac{|y-x|^2}{2\ell} : y \in F \Big\}$$

appearing in `QuadEnvP` satisfies $\sup \mathcal{V} = \max_{|y| \le M} \big(f(y,t+\ell) - |y-x|^2/(2\ell) \big)$, and this maximum exists because the function $y \mapsto f(y,t+\ell) - |y-x|^2/(2\ell)$ is continuous on the compact ball $\{|y| \le M\}$: there $(y, t+\ell) \in \Omega$, where f is continuous. In particular $\mathcal{V}$ is nonempty and bounded above, so by `QuadEnvP` $\mathcal{P}_\ell f(z)$ is this maximum and is attained at some y with $|y| \le M$. This is the first assertion of (1).

For the second, let $t \in I$ and $t - \ell \in I$, and apply the same reasoning with $J = \{t-\ell\}$ and the second inequality of Step 1(ii): no competitor with $|y| > M$ gives a value $f(y,t-\ell) + |y-x|^2/(2\ell)$ below the value at $y = x$, so the infimum in `QuadEnvM` is the minimum over $\{|y| \le M\}$ of a continuous function and is attained there.

Step 3: joint continuity in the length and the center, i.e. (2). We first record the elementary fact that the maximum of a continuous function over a fixed compact set depends continuously on the remaining parameters. Let H be continuous on $P \times Q$ with P a metric space and Q compact, and let $\pi_k \to \pi$ in P. Testing against each fixed $q \in Q$ gives $\max_{q'} H(\pi_k, q') \ge H(\pi_k, q) \to H(\pi, q)$, hence $\liminf_k \max_{q'} H(\pi_k, q') \ge \max_q H(\pi, q)$. Conversely, choose maximizers q_k, which exist because Q is compact and $H(\pi_k, \cdot)$ is continuous, and pass to a subsequence along which $q_k \to \bar{q} \in Q$, using the compactness of Q again; then $(\pi_k, q_k) \to (\pi, \bar{q})$, so by the continuity of H, $\max_{q'} H(\pi_k, q') = H(\pi_k, q_k) \to H(\pi, \bar{q}) \le \max_q H(\pi, q)$ along that subsequence. Every subsequence of the real sequence $\max_{q'} H(\pi_k, q')$ therefore has a further subsequence converging to a real number that is at most $\max_q H(\pi, q)$, and hence $\limsup_k \max_{q'} H(\pi_k, q') \le \max_q H(\pi, q)$. Hence $\pi \mapsto \max_q H(\pi, q)$ is continuous.

Now fix $\ell_0 > 0$ and $z_0 = (x_0, t_0)$ with $t_0 \in I$ and $t_0 + \ell_0 \in I$. Since I is open there is $\rho > 0$ with

$$\rho < \tfrac{1}{2}\ell_0, \qquad J := [t_0 + \ell_0 - 2\rho,\ t_0 + \ell_0 + 2\rho] \subseteq I.$$

Put

$$\mathcal{P} = \big\{ (\ell, z) : |\ell - \ell_0| \le \rho,\ |x - x_0| \le \rho,\ |t - t_0| \le \rho \big\},$$

a compact neighbourhood of (ℓ_0, z_0) in $\mathbb{R} \times F \times \mathbb{R}$. For $(\ell, z) \in \mathcal{P}$ we have $\ell \geq \ell_0 - \rho \geq \frac{1}{2}\ell_0 > 0$, $|x| \leq |x_0| + \rho$, and $t + \ell \in J \subseteq I$. Apply Step 1 with $R_0 = |x_0| + \rho$, this J and $\ell_0^{\max} = \ell_0 + \rho$, obtaining M. By Step 1(ii) and the argument of Step 2, for every $(\ell, z) \in \mathcal{P}$,

$$\mathcal{P}_\ell f(z) = \max_{|y| \leq M} \Big(f(y, t + \ell) - \frac{|y - x|^2}{2\ell} \Big).$$

The function $H\big((\ell, z), y\big) = f(y, t + \ell) - |y - x|^2/(2\ell)$ is continuous on $\mathcal{P} \times \{|y| \leq M\}$: indeed $(y, t+\ell) \in \Omega$ there, where f is continuous, and $\ell \geq \frac{1}{2}\ell_0 > 0$ keeps the quadratic term continuous. By the fact recorded above, $(\ell, z) \mapsto \mathcal{P}_\ell f(z)$ is continuous on $\mathcal{P}$, hence continuous at (ℓ_0, z_0).

The assertion for $\mathcal{M}$ at a point with $t_0 \in I$ and $t_0 - \ell_0 \in I$ follows by the same argument with $J = [t_0 - \ell_0 - 2\rho,\ t_0 - \ell_0 + 2\rho]$, minima in place of maxima, and the second inequality of Step 1(ii).

Step 4: uniform convergence at length zero, i.e. (3), and the optimizer collapse (4). Let $\mathcal{K}$ be a compact set of points $z = (x, t)$ with $t \in I$, and let $\eta > 0$. The set T of time coordinates of $\mathcal{K}$ is a compact subset of the open set I, so there is $L > 0$ with

$$J := \{s : \operatorname{dist}(s, T) \leq L\} \subseteq I,$$

and J is compact. Put $R_0 = \max\{|x| : (x, t) \in \mathcal{K}\}$ and apply Step 1 with R_0, this J and $\ell_0 = L$, obtaining $M \geq R_0$ and N; write $\mathcal{K}_2 = \{(y, s) : |y| \leq M,\ s \in J\}$, a compact subset of Ω on which $|f| \leq N$.

Fix $z = (x, t) \in \mathcal{K}$ and $0 < \ell \leq L$. Then $t, t + \ell, t - \ell \in J$, because $t \in T$ and $\operatorname{dist}(t \pm \ell, T) \leq \ell \leq L$. Note also that $|x| \leq R_0 \leq M$, so $z \in \mathcal{K}_2$.

Let y_ℓ denote *any* $y \in F$ with $\mathcal{P}_\ell f(z) = f(y, t + \ell) - |y - x|^2/(2\ell)$; by Step 2 at least one exists, and every such y satisfies $|y| \leq M$, since by Step 1(ii) with $s = t + \ell$ a competitor with $|y| > M$ has value strictly below $f(x, t + \ell)$, which is at most $\mathcal{P}_\ell f(z)$. The competitor $y = x$ gives

$$f(x, t + \ell) \leq \mathcal{P}_\ell f(z) = f(y_\ell, t + \ell) - \frac{|y_\ell - x|^2}{2\ell} \leq f(y_\ell, t + \ell). \tag{465}$$

Both $(x, t + \ell)$ and $(y_\ell, t + \ell)$ lie in $\mathcal{K}_2$, so

$$\frac{|y_\ell - x|^2}{2\ell} = f(y_\ell, t + \ell) - \mathcal{P}_\ell f(z) \leq f(y_\ell, t + \ell) - f(x, t + \ell) \leq 2N, \tag{466}$$

that is $|y_\ell - x| \leq 2\sqrt{N\ell}$.

The same bound holds for the backward envelope. It is the mirror computation, not the forward one applied to $-f$: the time argument is $t - \ell$ rather than $t + \ell$ and the extremum is an infimum. Let y'_ℓ denote any $y \in F$ with $\mathcal{M}_\ell f(z) = f(y, t - \ell) + |y - x|^2/(2\ell)$; by Step 2 at least one exists, and every such y satisfies $|y| \leq M$, since by the second inequality of Step 1(ii) with $s = t - \ell$ a competitor with $|y| > M$ has value strictly above $f(x, t - \ell)$, which is at least $\mathcal{M}_\ell f(z)$. The competitor $y = x$ gives

$$f(x, t - \ell) \geq \mathcal{M}_\ell f(z) = f(y'_\ell, t - \ell) + \frac{|y'_\ell - x|^2}{2\ell} \geq f(y'_\ell, t - \ell), \tag{467}$$

and both $(x, t - \ell)$ and $(y'_\ell, t - \ell)$ lie in $\mathcal{K}_2$, so

$$\frac{|y'_\ell - x|^2}{2\ell} = \mathcal{M}_\ell f(z) - f(y'_\ell, t - \ell) \leq f(x, t - \ell) - f(y'_\ell, t - \ell) \leq 2N, \tag{468}$$

that is $|y'_\ell - x| \leq 2\sqrt{N\ell}$.

For (4), take $\mathcal{K} = \{z\}$ above, which is a compact set of points with time coordinate in I because $t \in I$, and put $\delta = \min\{L,\ \eta^2/(4N + 1)\} > 0$. Let $0 < \ell < \delta$. Every optimizer y for $\mathcal{P}_\ell f(z)$ is one of the y_ℓ and every optimizer for $\mathcal{M}_\ell f(z)$ is one of the y'_ℓ, both of which were quantified over all optimizers, so in either case

$$|y - x| \leq 2\sqrt{N\ell} \leq 2\sqrt{\frac{N\eta^2}{4N + 1}} = \eta \cdot 2\sqrt{\frac{N}{4N + 1}} \leq \eta,$$

the last step because $N \geq 0$ gives $N/(4N + 1) \leq 1/4$. This is (4).

For (3), the set $\mathcal{K}_2$ is a compact subset of Ω, so f is uniformly continuous there: there is $\delta_1 > 0$ with

$$|f(a) - f(b)| \le \eta \qquad (a, b \in \mathcal{K}_2,\ |a-b| \le \delta_1). \tag{469}$$

Put $\delta = \min\{L,\ \delta_1/2,\ \delta_1^2/(32N+1)\} > 0$. Let $0 < \ell < \delta$ and $z = (x,t) \in \mathcal{K}$. By (466) and the choice of δ we have $|y_\ell - x| \le 2\sqrt{N\ell} \le \delta_1/2$ and $\ell \le \delta_1/2$, so the space-time distances from $(y_\ell, t+\ell)$ and from $(x, t+\ell)$ to (x,t) are at most $\big(|y_\ell - x|^2 + \ell^2\big)^{1/2} \le \delta_1$ and $\ell \le \delta_1$ respectively, and all three points lie in $\mathcal{K}_2$. Hence (465) and (469) give

$$\mathcal{P}_\ell f(z) - f(x,t) \le f(y_\ell, t+\ell) - f(x,t) \le \eta, \qquad \mathcal{P}_\ell f(z) - f(x,t) \ge f(x, t+\ell) - f(x,t) \ge -\eta,$$

so $|\mathcal{P}_\ell f(z) - f(z)| \le \eta$. For the backward envelope, (468) and the same choice of δ give $|y'_\ell - x| \le 2\sqrt{N\ell} \le \delta_1/2$, so the space-time distances from $(y'_\ell, t-\ell)$ and from $(x, t-\ell)$ to (x,t) are again at most δ_1 and these points again lie in $\mathcal{K}_2$; hence (467) and (469) give

$$\mathcal{M}_\ell f(z) - f(x,t) \ge f(y'_\ell, t-\ell) - f(x,t) \ge -\eta, \qquad \mathcal{M}_\ell f(z) - f(x,t) \le f(x, t-\ell) - f(x,t) \le \eta,$$

so $|\mathcal{M}_\ell f(z) - f(z)| \le \eta$. Both bounds hold for every $z \in \mathcal{K}$ and every $0 < \ell < \delta$, which is (3). □

2.118. `TimeReversal`.

Lemma 2.118 (Time reversal)**.** *Let F be a finite-dimensional real inner-product space, let $C_0 \in \mathbb{R}$ and let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$. Let $\widetilde{f}$ be its time reversal of `TimeFlip`, and for a space-time point $z = (x,t)$ write $\widetilde{z} = (x, -t)$. Then:*

1. *for every Ω, if f is a viscosity subsolution of $F[f] = 0$ on Ω in the sense of `IsLimitSubOn`, then $\widetilde{f}$ is a viscosity supersolution of $F[\widetilde{f}] = 0$ on $\widetilde{\Omega} = \{z : \widetilde{z} \in \Omega\}$ in the sense of `IsLimitSuperOn`, and conversely with sub- and supersolutions exchanged;*
2. *for every ℓ and every z,*
$$\mathcal{P}_\ell \widetilde{f}(z) = -\mathcal{M}_\ell f(\widetilde{z}), \qquad \mathcal{M}_\ell \widetilde{f}(z) = -\mathcal{P}_\ell f(\widetilde{z}),$$
with the envelopes of `QuadEnvP` and `QuadEnvM`;
3. *if (p, a) is a fit for f at z of radius R with constant C_0 in the sense of `IsFit`, then $(-p, a)$ is a fit for $\widetilde{f}$ at $\widetilde{z}$ of radius R with the same constant; consequently if $f \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` then $\widetilde{f} \in \mathcal{A}_{m,C_0}$;*
4. *for every κ, the function $\widetilde{f}$ satisfies the horizontal Jensen bound with constant κ on all times, in the sense of `HasJensenBound`, if and only if*
$$f(x,t) - \sum_i \theta_i f(y_i, t) \le \kappa r$$
for every time t and every finite horizontal probability configuration (θ, y) with mean x and radius r, in the senses of `IsBaryConfig` and `BaryRadius`. Thus time reversal changes the sign of a horizontal Jensen defect.

Proof. Let $\Pi(x,t) = (x, -t)$, an involutive linear isometry of space-time with $\Pi^{-1} = \Pi$, so $\widetilde{f} = -f \circ \Pi$ and $\widetilde{\Omega} = \Pi^{-1}(\Omega)$ is open whenever Ω is.

(1). Let ϕ be C^2 near $z \in \widetilde{\Omega}$ with $w \mapsto \widetilde{f}(w) - \phi(w)$ having a local minimum on $\widetilde{\Omega}$ at z. Put $\psi = -\phi \circ \Pi$, which is C^2 near $\Pi z \in \Omega$. For w' near Πz we have $\Pi w' \in \widetilde{\Omega}$ near z and

$$f(w') - \psi(w') = -\big(\widetilde{f}(\Pi w') - \phi(\Pi w')\big) \le -\big(\widetilde{f}(z) - \phi(z)\big) = f(\Pi z) - \psi(\Pi z),$$

so $f - \psi$ has a local maximum on Ω at Πz, and `IsLimitSubOn` gives $F[\psi](\Pi z) \ge 0$.

Now compute $F[\psi]$ at Πz in terms of $F[\phi]$ at z. Since Π is linear with $\Pi^2 = \mathrm{Id}$ and $\psi = -\phi \circ \Pi$, the chain rule gives $\nabla\psi(\Pi z) = -\Pi\,\nabla\phi(z)$ and $D^2\psi(\Pi z)[\xi] = -D^2\phi(z)[\Pi\xi]$ for every ξ. Writing $\nabla\phi(z) = (\nabla_x\phi, \phi_t)$ we get $\nabla\psi(\Pi z) = (-\nabla_x\phi, \phi_t)$, so its spatial part is $-\nabla_x\phi$ and

$$F[\psi](\Pi z) = D^2\psi(\Pi z)\big[(-\nabla_x\phi, 1)\big] = -D^2\phi(z)\big[\Pi(-\nabla_x\phi, 1)\big] = -D^2\phi(z)\big[(-\nabla_x\phi, -1)\big] = -F[\phi](z),$$

the last step because $A[-\xi] = A[\xi]$ for a bilinear form A. Hence $F[\phi](z) \le 0$, which is the supersolution inequality for $\widetilde{f}$ at z. Continuity of $\widetilde{f}$ on $\widetilde{\Omega}$ follows from continuity of f on Ω. The converse implication is the same computation with maxima and minima exchanged.

(2). Fix ℓ and $z=(x,t)$. By `TimeFlip` and `QuadEnvP`,

$$\mathcal{P}_\ell \widetilde{f}(x,t)=\sup_y\left\{-f(y,-t-\ell)-\frac{|y-x|^2}{2\ell}\right\}=-\inf_y\left\{f(y,-t-\ell)+\frac{|y-x|^2}{2\ell}\right\}=-\mathcal{M}_\ell f(x,-t),$$

using $\sup(-A)=-\inf A$ for a set A of reals, and the time argument $(-t)-\ell$ in `QuadEnvM` at the point $\widetilde{z}=(x,-t)$. The second identity is the same computation with the roles of sup and inf exchanged.

(3). Let (p,a) be a fit for f at $z=(x,t)$ of radius R. Let $h\in F$ and $\ell\in\mathbb{R}$ with $|\ell|\le R$ and $|h-\ell(-p)|\le R$. Apply `IsFit` for f with the increment $(h,-\ell)$: the conditions $|-\ell|\le R$ and $|h-(-\ell)p|\le R$ hold, so

$$\left|f(x+h,t-\ell)-f(x,t)-p\cdot h-\Big(a-\frac{|p|^2}{2}\Big)(-\ell)\right|\le C_0R.$$

Multiplying by -1 inside the absolute value and using $\widetilde{f}(x+h,-t+\ell)=-f(x+h,t-\ell)$, $\widetilde{f}(\widetilde{z})=-f(z)$ and $|-p|=|p|$,

$$\left|\widetilde{f}\big(\widetilde{z}+(h,\ell)\big)-\widetilde{f}(\widetilde{z})-(-p)\cdot h-\Big(a-\frac{|-p|^2}{2}\Big)\ell\right|\le C_0R,$$

which is `IsFit` for $\widetilde{f}$ at $\widetilde{z}$ of radius R with parameters $(-p,a)$. If $f\in\mathcal{A}_{m,C_0}$, then by part (1) applied with Ω the whole space (whose reflection is the whole space) $\widetilde{f}$ is a viscosity solution everywhere, and the fit just constructed exists at every center and radius because $z\mapsto\widetilde{z}$ is a bijection; so $\widetilde{f}\in\mathcal{A}_{m,C_0}$ by `IsFitClass`.

(4). Fix κ. By `TimeFlip`, for a configuration (θ,y) with mean x and any time t,

$$\sum_i\theta_i\widetilde{f}(y_i,t)-\widetilde{f}(x,t)=f(x,-t)-\sum_i\theta_i f(y_i,-t).$$

Since $t\mapsto-t$ is a bijection of $\mathbb{R}$ and the radius of `BaryRadius` does not involve t, the condition of `HasJensenBound` for $\widetilde{f}$ with $I=\mathbb{R}$ is equivalent to the displayed reversed-defect inequality for f at every time and configuration. □

2.119. `ViscosityConvexOnIcc`.

Lemma 2.119 (The one-dimensional viscosity criterion for convexity)**.** *Let $T>0$ and let $G:\mathbb{R}\to\mathbb{R}$ be continuous on $[0,T]$. Suppose that for every $\ell_0\in(0,T)$ and all $A,B,C\in\mathbb{R}$, if the function $\tau\mapsto G(\tau)-\big(A\tau^2+B\tau+C\big)$ has a local maximum on $[0,T]$ at ℓ_0, then $A\ge0$. Then G is convex on $[0,T]$.*

Proof. The interval $[0,T]$ is convex, so by the adjacent-slope characterisation of convexity it suffices to show that for all $a,b\in[0,T]$ and all τ with $a<\tau<b$,

$$\frac{G(\tau)-G(a)}{\tau-a}\le\frac{G(b)-G(\tau)}{b-\tau}.$$

Suppose this fails for some such a,τ,b. Then $a<b$, and the three quantities $b-a$, $\tau-a$, $b-\tau$ are all positive.

Step 1: the chord lies strictly below G at τ. Let

$$\lambda(r)=\frac{(b-r)G(a)+(r-a)G(b)}{b-a}$$

be the affine function with $\lambda(a)=G(a)$ and $\lambda(b)=G(b)$. The failed inequality, multiplied by the positive number $(\tau-a)(b-\tau)$, reads

$$\big(G(b)-G(\tau)\big)(\tau-a)<\big(G(\tau)-G(a)\big)(b-\tau),$$

that is, after moving the terms containing $G(\tau)$ to one side,

$$G(b)(\tau-a)+G(a)(b-\tau)<G(\tau)\big((b-\tau)+(\tau-a)\big)=G(\tau)(b-a).$$

Dividing by $b-a>0$ gives $\lambda(\tau)<G(\tau)$.

Step 2: the gap has a positive maximum. Put $h=G-\lambda$. Since $[a,b]\subseteq[0,T]$, by the hypothesis G is continuous on $[a,b]$, and λ is continuous, so h is continuous on the nonempty compact set

$[a,b]$ and attains a maximum there, say at τ_1. Put $m = h(\tau_1)$. Since $h(\tau) = G(\tau) - \lambda(\tau) > 0$ by Step 1 and $h(\tau) \le m$, we get $m > 0$.

Step 3: a strictly concave perturbation whose maximum is still positive. Put $\eta = m/(b-a)^2 > 0$ and

$$K(r) = h(r) + \eta\,(r-a)(r-b) \qquad (r \in [a,b]).$$

For every $r \in [a,b]$ we have $(r-a)(r-b) \ge -(b-a)^2/4$: indeed $(2r-a-b)^2 \ge 0$ expands to $4r^2 - 4r(a+b) + (a+b)^2 \ge 0$, while $-4(r-a)(r-b) = -4r^2 + 4r(a+b) - 4ab$, and adding the two gives $-4(r-a)(r-b) \le (a+b)^2 - 4ab = (b-a)^2$. Multiplying by $\eta > 0$ and using $\eta(b-a)^2/4 = m/4$,

$$K(\tau_1) = m + \eta(\tau_1 - a)(\tau_1 - b) \ge m - \frac{m}{4} = \frac{3m}{4} > \frac{m}{2}.$$

K is continuous on $[a,b]$, so it attains a maximum there, say at τ_2, and $K(\tau_2) \ge K(\tau_1) > m/2 > 0$.

Step 4: the maximum is attained strictly inside. At the endpoints λ agrees with G and the perturbation vanishes, so $K(a) = h(a) + 0 = G(a) - \lambda(a) = 0$ and likewise $K(b) = 0$. Since $K(\tau_2) > 0$, we conclude $\tau_2 \ne a$ and $\tau_2 \ne b$, so $a < \tau_2 < b$. As $0 \le a$ and $b \le T$ this gives $\tau_2 \in (0,T)$.

Step 5: the quadratic upper test. Expanding λ and the perturbation, set

$$A = -\eta, \qquad B = \frac{G(b) - G(a)}{b-a} + \eta(a+b), \qquad C = \frac{bG(a) - aG(b)}{b-a} - \eta ab + K(\tau_2),$$

so that for every $u \in \mathbb{R}$

$$Au^2 + Bu + C = -\eta(u-a)(u-b) + \lambda(u) + K(\tau_2),$$

using $\lambda(u) = \big[(G(b) - G(a))u + bG(a) - aG(b)\big]/(b-a)$. Consequently

$$G(u) - \big(Au^2 + Bu + C\big) = \big(G(u) - \lambda(u)\big) + \eta(u-a)(u-b) - K(\tau_2) = K(u) - K(\tau_2). \tag{470}$$

The open interval (a,b) is a neighbourhood of τ_2, so for every r in $(a,b) \cap [0,T]$ we have $r \in [a,b]$ and hence $K(r) \le K(\tau_2)$ by the choice of τ_2; by (470) this says

$$G(r) - \big(Ar^2 + Br + C\big) \le 0 = G(\tau_2) - \big(A\tau_2^2 + B\tau_2 + C\big).$$

Therefore $\tau \mapsto G(\tau) - \big(A\tau^2 + B\tau + C\big)$ has a local maximum on $[0,T]$ at τ_2, and $\tau_2 \in (0,T)$.

Step 6: the contradiction. The hypothesis applied with these A, B, C and this $\ell_0 = \tau_2$ gives $A \ge 0$. But $A = -\eta < 0$, a contradiction. Hence the adjacent-slope inequality holds for all admissible a, τ, b, and G is convex on $[0,T]$. □

2.120. `EnvelopeCompetitorBounds.`

Lemma 2.120 (The envelopes are genuine suprema and infima)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$, let f be a real function on the space-time $\ell^2(F \times \mathbb{R})$ which is continuous on $\Omega_I = \{(y,\tau) : \tau \in I\}$ and has subquadratic spatial growth on compact time subintervals of I in the sense of `SubquadraticGrowth`. Let $z = (x,t)$ with $t \in I$. Then:*

(1) *for every $\ell > 0$ with $t + \ell \in I$, the forward envelope $\mathcal{P}_\ell f(z)$ of `QuadEnvP` dominates every competitor and is the least such bound:*

$$f(y, t+\ell) - \frac{|y-x|^2}{2\ell} \le \mathcal{P}_\ell f(z) \quad (y \in F), \qquad \Big(\forall y \;\; f(y,t+\ell) - \frac{|y-x|^2}{2\ell} \le b\Big) \Longrightarrow \mathcal{P}_\ell f(z) \le b;$$

(2) *for every $\ell > 0$ with $t - \ell \in I$, the backward envelope $\mathcal{M}_\ell f(z)$ of `QuadEnvM` is dominated by every competitor and is the greatest such bound:*

$$\mathcal{M}_\ell f(z) \le f(y, t-\ell) + \frac{|y-x|^2}{2\ell} \quad (y \in F), \qquad \Big(\forall y \;\; b \le f(y,t-\ell) + \frac{|y-x|^2}{2\ell}\Big) \Longrightarrow b \le \mathcal{M}_\ell f(z);$$

(3) *for every $s > 0$, the barycentric envelope $\mathcal{C}_s f(z)$ of* `BaryEnv` *dominates the value of every finite horizontal probability configuration (θ_i, y_i) with mean x in the sense of* `IsBaryConfig`*, with radius ρ the root mean square radius of* `BaryRadius`*, and is the least such bound:*

$$\sum_i \theta_i f(y_i,t) - \frac{\rho^2}{2s} \le \mathcal{C}_s f(z), \qquad \Big(\forall \text{ configurations } \sum_i \theta_i f(y_i,t) - \frac{\rho^2}{2s} \le b\Big) \Longrightarrow \mathcal{C}_s f(z) \le b.$$

Proof. The three envelopes of `QuadEnvP`, `QuadEnvM` and `BaryEnv` are, for the parameters considered here, the supremum, the infimum and the supremum of the three sets

$$\mathcal{P} = \Big\{ f(y,t+\ell) - \frac{|y-x|^2}{2\ell} : y \in F \Big\},$$
$$\mathcal{M} = \Big\{ f(y,t-\ell) + \frac{|y-x|^2}{2\ell} : y \in F \Big\},$$
$$\mathcal{C} = \Big\{ \sum_i \theta_i f(y_i,t) - \frac{\rho^2}{2s} : (\theta_i, y_i) \text{ a configuration with mean } x \text{ and radius } \rho \Big\},$$

respectively. By the preamble's convention a supremum is the least upper bound and an infimum the greatest lower bound exactly when the set is nonempty and bounded on the relevant side, and otherwise the value is 0 and carries no information. So each of the three assertions is precisely the statement that the corresponding set is nonempty and bounded above, respectively below; once that is known, the two displayed properties are the defining properties of the least upper bound and the greatest lower bound. We prove the three boundedness statements. Note that in each case the time argument of f lies in I, by the hypotheses $t+\ell \in I$, $t-\ell \in I$ and $t \in I$, so f is continuous at the points used.

A remark on why this is not a restatement of attainment. The assertions above quantify over *all* competitors. A statement that the supremum is *attained* at some single y — as in the node `QuadEnvAttained`, part (1), or in the uniqueness clause of the node `DPSpatialRegularity` — names one optimizer and does not, under the preamble's convention, entail that the value $\mathcal{P}_\ell f(z)$ dominates the competitor at any other point: if $\mathcal{P}$ were unbounded above then $\mathcal{P}_\ell f(z)$ would be 0, which may coincide with the value at some particular y while failing the first display. Nor does boundedness entail attainment, since a bounded set need not contain its supremum. The two statements are therefore independent, and only the present one licenses a comparison of the envelope with a chosen competitor.

Part (1). The set $\mathcal{P}$ is nonempty: the choice $y = x$ contributes $f(x, t+\ell)$. For the upper bound, apply `SubquadraticGrowth` with the compact set $J = \{t+\ell\} \subseteq I$ and the accuracy $\varepsilon = \dfrac{1}{4\ell} > 0$: there is $M > 0$ with

$$|f(y,t+\ell)| \le \frac{|y|^2}{4\ell} \qquad (|y| \ge M). \tag{471}$$

The function $f(\cdot, t+\ell)$ is continuous on the closed ball $\overline{B}_M(0)$ of F, because every point $(y, t+\ell)$ with y in that ball lies in Ω_I; that ball is compact, so there is $C_M < \infty$ with

$$|f(y,t+\ell)| \le C_M \qquad (|y| \le M). \tag{472}$$

Enlarging C_M we may assume $C_M \ge 0$. From $|y|^2 \le 2|y-x|^2 + 2|x|^2$ we get

$$|y-x|^2 \ge \frac{|y|^2}{2} - |x|^2 \qquad (y \in F). \tag{473}$$

Now let $y \in F$. If $|y| \ge M$ then by (471) and (473)

$$f(y,t+\ell) - \frac{|y-x|^2}{2\ell} \le \frac{|y|^2}{4\ell} - \frac{1}{2\ell}\Big(\frac{|y|^2}{2} - |x|^2\Big) = \frac{|x|^2}{2\ell},$$

since $\dfrac{|y|^2}{4\ell} - \dfrac{|y|^2}{4\ell} = 0$. If $|y| \le M$ then by (472) and $|y-x|^2 \ge 0$,

$$f(y, t+\ell) - \frac{|y-x|^2}{2\ell} \le C_M.$$

In both cases the value is at most $C_M + \dfrac{|x|^2}{2\ell}$, a bound independent of y. Hence $\mathcal{P}$ is bounded above, which proves part (1).

Part (2). Apply part (1) to the function $-f$ and the time $t-\ell$. The function $-f$ is continuous on Ω_I and satisfies the hypothesis of `SubquadraticGrowth` with the same data, because $|-f| = |f|$. Part (1)'s boundedness argument, run with $t-\ell \in I$ in place of $t+\ell$, shows that

$$\Big\{ -f(y, t-\ell) - \frac{|y-x|^2}{2\ell} : y \in F \Big\}$$

is nonempty and bounded above, by $C'_M + \dfrac{|x|^2}{2\ell}$ where C'_M bounds $|f(\cdot, t-\ell)|$ on $\overline{B}_M(0)$ for the M supplied by `SubquadraticGrowth` at $J = \{t-\ell\}$ and $\varepsilon = 1/(4\ell)$. Multiplying by -1 reverses the inequalities and shows that $\mathcal{M}$ is nonempty and bounded below, by $-C'_M - \dfrac{|x|^2}{2\ell}$. This proves part (2).

Part (3). The set $\mathcal{C}$ is nonempty: by `IsBaryConfig` the one-point configuration $n = 1$, $\theta_1 = 1$, $y_1 = x$ is admissible, since its weight is positive, the weights sum to 1 and its weighted mean is x; by `BaryRadius` its radius is $\rho = 0$, so it contributes $f(x,t)$. For the upper bound, apply `SubquadraticGrowth` with the compact set $J = \{t\} \subseteq I$ and the accuracy $\varepsilon = \dfrac{1}{4s} > 0$ to get $M > 0$ with

$$|f(y,t)| \le \frac{|y|^2}{4s} \qquad (|y| \ge M), \tag{474}$$

and let $C_M \ge 0$ bound the continuous $|f(\cdot,t)|$ on the compact ball $\overline{B}_M(0)$. Let $(\theta_i, y_i)_{i \le n}$ be a configuration with mean x and radius ρ, so that $\theta_i > 0$, $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i = x$ by `IsBaryConfig`, and $\rho^2 = \sum_i \theta_i |y_i - x|^2$ by `BaryRadius`. Expanding the squared norms and using these two identities,

$$\sum_i \theta_i |y_i - x|^2 = \sum_i \theta_i |y_i|^2 - 2\Big\langle \sum_i \theta_i y_i,\, x \Big\rangle + \Big(\sum_i \theta_i\Big)|x|^2 = \sum_i \theta_i |y_i|^2 - |x|^2,$$

that is

$$\sum_i \theta_i |y_i|^2 = \rho^2 + |x|^2. \tag{475}$$

For each index i, (474) in the case $|y_i| \ge M$ and the bound C_M in the case $|y_i| \le M$ give

$$f(y_i, t) \le C_M + \frac{|y_i|^2}{4s},$$

since both terms on the right are nonnegative. Multiplying by $\theta_i > 0$, summing, and using $\sum_i \theta_i = 1$ together with (475),

$$\sum_i \theta_i f(y_i, t) \le C_M + \frac{\rho^2 + |x|^2}{4s}.$$

Therefore the configuration value satisfies

$$\sum_i \theta_i f(y_i, t) - \frac{\rho^2}{2s} \le C_M + \frac{|x|^2}{4s} + \rho^2\Big(\frac{1}{4s} - \frac{1}{2s}\Big) \le C_M + \frac{|x|^2}{4s},$$

because $\dfrac{1}{4s} - \dfrac{1}{2s} = -\dfrac{1}{4s} < 0$ and $\rho^2 \ge 0$. This bound is independent of the configuration, so $\mathcal{C}$ is bounded above, which proves part (3). $\square$

2.121. `QuadEnvPathConvex`.

Lemma 2.121 (Path convexity of the quadratic envelopes)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(x,t) : t \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`, and let $T > 0$.*

(1) *If f is a viscosity subsolution of $F[f] = 0$ on Ω in the sense of `IsLimitSubOn`, then for every affine path $x(\ell) = x_0 + d\ell$ with $x_0, d \in F$ and every t with $t + \ell \in I$ for all $\ell \in [0,T]$, the function*

$$\ell \longmapsto \mathcal{P}_\ell f\big(x(\ell), t\big) + |d|^2 \ell \log \ell \tag{476}$$

is convex on $[0,T]$, with $\ell \log \ell$ given its continuous value 0 at $\ell = 0$ and $\mathcal{P}_0 f = f$ as in `QuadEnvP`. In particular, taking $d = 0$, $\ell \mapsto \mathcal{P}_\ell f(x,t)$ is convex on $[0,T]$.

(2) *If f is a viscosity supersolution of $F[f] = 0$ on Ω in the sense of `IsLimitSuperOn`, then for every $x \in F$ and every t with $t - \ell \in I$ for all $\ell \in [0,T]$, the function $\ell \mapsto \mathcal{M}_\ell f(x,t)$ is concave on $[0,T]$, with $\mathcal{M}_0 f = f$ as in `QuadEnvM`.*

Proof. Part (1). Assume f is a viscosity subsolution of $F[f] = 0$ on Ω, fix $x_0, d \in F$ and t with $t + \ell \in I$ for all $\ell \in [0,T]$, and put

$$x(\ell) = x_0 + \ell d, \qquad g(\ell) = \mathcal{P}_\ell f\big(x(\ell), t\big), \qquad G(\ell) = g(\ell) + |d|^2\, \ell \log \ell \quad (\ell \in [0,T]),$$

with $\ell \log \ell$ given its continuous value 0 at $\ell = 0$. Note that $g(0) = \mathcal{P}_0 f(x_0, t) = f(x_0, t)$ by `QuadEnvP`, so $G(0) = f(x_0, t)$.

Step 1: G is continuous on $[0,T]$. For $\ell \in (0,T]$ we have $t \in I$ and $t + \ell \in I$, so part (2) of `QuadEnvAttained` gives joint continuity of $(\sigma, \zeta) \mapsto \mathcal{P}_\sigma f(\zeta)$ at $(\ell, (x(\ell), t))$; composing with the continuous map $\sigma \mapsto (\sigma, (x(\sigma), t))$ shows that g is continuous at ℓ. Hence g, and therefore G, is continuous on $(0,T]$.

For continuity at 0, let $\eta > 0$. The set $\mathcal{K} = \{(x(\sigma), t) : \sigma \in [0,T]\}$ is compact (the image of $[0,T]$ under a continuous map) and all its time coordinates equal $t \in I$, so part (3) of `QuadEnvAttained` provides $\delta > 0$ with $|\mathcal{P}_\ell f(\zeta) - f(\zeta)| \le \eta$ for all $0 < \ell < \delta$ and all $\zeta \in \mathcal{K}$; in particular $|g(\ell) - f(x(\ell), t)| \le \eta$ for $0 < \ell < \min\{\delta, T\}$. Since f is continuous on the open set Ω and $(x(\ell), t) \to (x_0, t) \in \Omega$ as $\ell \downarrow 0$, we get $f(x(\ell), t) \to f(x_0, t)$, so $\limsup_{\ell \downarrow 0} |g(\ell) - f(x_0, t)| \le \eta$. As $\eta > 0$ was arbitrary, $g(\ell) \to f(x_0, t) = g(0)$. Finally $\ell \log \ell \to 0$, so $G(\ell) \to G(0)$.

Step 2: every quadratic upper test at an interior length has nonnegative leading coefficient. Let $\ell_0 \in (0,T)$ and $A, B, C \in \mathbb{R}$, and suppose that $\tau \mapsto G(\tau) - \big(A\tau^2 + B\tau + C\big)$ has a local maximum on $[0,T]$ at ℓ_0. We show $A \ge 0$. Put

$$K = G(\ell_0) - \big(A\ell_0^2 + B\ell_0 + C\big), \qquad \widetilde{\psi}(\ell) = A\ell^2 + B\ell + C + K - |d|^2\, \ell \log \ell.$$

Since $\ell_0 > 0$, the function $\widetilde{\psi}$ is C^2 on a neighbourhood of ℓ_0 — the logarithm being smooth away from 0 — with

$$\widetilde{\psi}''(\ell) = 2A - \frac{|d|^2}{\ell}. \tag{477}$$

The local-maximum hypothesis says that for $\tau \in [0,T]$ near ℓ_0,

$$G(\tau) - \big(A\tau^2 + B\tau + C\big) \le G(\ell_0) - \big(A\ell_0^2 + B\ell_0 + C\big) = K,$$

that is $G(\tau) \le A\tau^2 + B\tau + C + K$. Subtracting $|d|^2 \tau \log \tau$ from both sides and recalling $G = g + |d|^2 \cdot \log(\cdot)$, this reads $g(\tau) \le \widetilde{\psi}(\tau)$. At $\tau = \ell_0$ the same computation is an equality, because $A\ell_0^2 + B\ell_0 + C + K = G(\ell_0)$ by the definition of K. Shrinking the neighbourhood so that it lies inside $(0,T)$ — possible because $(0,T)$ is open and $\ell_0 \in (0,T)$ — we obtain

$$g(\ell) \le \widetilde{\psi}(\ell) \ \text{ for all } \ell \in (0,T) \text{ near } \ell_0, \qquad g(\ell_0) = \widetilde{\psi}(\ell_0). \tag{478}$$

By part (1) of `QuadEnvAttained`, applied at the center $(x(\ell_0), t)$ and the length $\ell_0 > 0$, whose time arguments t and $t + \ell_0$ lie in I, there is $y_0 \in F$ with

$$g(\ell_0) = \mathcal{P}_{\ell_0} f\big(x(\ell_0), t\big) = f\big(y_0, t + \ell_0\big) - \frac{|y_0 - x(\ell_0)|^2}{2\ell_0}. \tag{479}$$

Put

$$q = \frac{y_0 - x(\ell_0)}{\ell_0}, \qquad \text{so that} \qquad y_0 - x(\ell_0) = \ell_0 q,$$

and define, for space-time points $\zeta = (y, s)$ with $s - t$ near ℓ_0,

$$\widehat{\Psi}(y, s) = \widetilde{\psi}(s - t) + \frac{|y - x(s - t)|^2}{2(s - t)}.$$

Since $\widetilde{\psi}$ is C^2 near ℓ_0, since $\ell_0 > 0$ so that the denominator $2(s - t)$ does not vanish near $\zeta_0 := (y_0, t + \ell_0)$, and since $x(\cdot)$ is affine, $\widehat{\Psi}$ is C^2 near ζ_0.

The local maximum. We claim that $f - \widehat{\Psi}$ has a local maximum on Ω at ζ_0. At ζ_0 its value is, by (479) and $g(\ell_0) = \widetilde{\psi}(\ell_0)$,

$$f(y_0, t + \ell_0) - \widetilde{\psi}(\ell_0) - \frac{|y_0 - x(\ell_0)|^2}{2\ell_0} = g(\ell_0) - \widetilde{\psi}(\ell_0) = 0.$$

Now let $\zeta = (y, s) \in \Omega$ with $\ell := s - t$ so close to ℓ_0 that $\ell \in (0, T)$ and (478) holds at ℓ; note $t + \ell = s \in I$ because $\zeta \in \Omega$, and $t \in I$ by the admissibility hypothesis taken at $\ell = 0$. The number $f(y, t+\ell) - |y - x(\ell)|^2/(2\ell)$ is one element of the competitor set whose supremum defines $\mathcal{P}_\ell f(x(\ell), t)$ in `QuadEnvP`. A supremum dominates its elements only once the set is known to be nonempty and bounded above: $\mathcal{P}_\ell f$ is literally a supremum, and the supremum of a set that is empty or unbounded above is a junk value, so the domination has to be licensed rather than assumed. The licence is part (1) of `EnvelopeCompetitorBounds`, whose first clause is exactly the statement that $\mathcal{P}_\ell f(z)$ dominates every competitor value; we apply it at this length $\ell > 0$ and this centre $z = (x(\ell), t)$, its hypotheses being the continuity and subquadratic growth of f already assumed, together with $t \in I$ and $t + \ell \in I$, both just checked. Therefore

$$f(y, s) - \widehat{\Psi}(y, s) = f(y, t + \ell) - \frac{|y - x(\ell)|^2}{2\ell} - \widetilde{\psi}(\ell) \le \mathcal{P}_\ell f\big(x(\ell), t\big) - \widetilde{\psi}(\ell) = g(\ell) - \widetilde{\psi}(\ell) \le 0,$$

the last step by (478). This proves the claim. Consequently, by `IsLimitSubOn`,

$$F[\widehat{\Psi}](\zeta_0) \ge 0. \tag{480}$$

The value of the operator. By `LimitOp`, $F[\widehat{\Psi}](\zeta_0)$ is the second derivative of $\widehat{\Psi}$ at ζ_0 on the pair (ξ, ξ), where $\xi = (p, 1)$ and p is the spatial block of $\nabla\widehat{\Psi}(\zeta_0)$. Both the identification of p and the evaluation of the second derivative are instances of `SecondVariationAlongLine`, which computes the first and second variation of a C^2 function along a straight line from any one-variable function agreeing with the restriction near the base point.

First, $p = q$. Indeed, fix $a \in F$ and restrict $\widehat{\Psi}$ to the purely spatial line $\tau \mapsto \zeta_0 + \tau(a, 0)$. The time coordinate is constant equal to $t + \ell_0$ along it, so, using $y_0 - x(\ell_0) = \ell_0 q$,

$$\widehat{\Psi}\big(\zeta_0 + \tau(a, 0)\big) = \widetilde{\psi}(\ell_0) + \frac{|\ell_0 q + \tau a|^2}{2\ell_0} = \widetilde{\psi}(\ell_0) + \frac{\ell_0^2|q|^2 + 2\tau\ell_0\langle q, a\rangle + \tau^2|a|^2}{2\ell_0},$$

a quadratic polynomial in τ whose derivative at $\tau = 0$ is $\langle q, a\rangle$. Hence $D\widehat{\Psi}(\zeta_0)[(a, 0)] = \langle q, a\rangle$ for every $a \in F$. Since the inner product of space-time splits as $\langle (v, b), (a, c)\rangle = \langle v, a\rangle + bc$, testing $\nabla\widehat{\Psi}(\zeta_0)$ against $(a, 0)$ sees only its spatial block, so that block is characterised by $\langle p, a\rangle = \langle q, a\rangle$ for all a, i.e. $p = q$.

Second, restrict $\widehat{\Psi}$ to the line $\tau \mapsto \zeta_0 + \tau\xi$ with $\xi = (q, 1)$. Its space-time coordinates are $(y_0 + \tau q,\ t + \ell_0 + \tau)$, so the time offset is $\ell_0 + \tau$ and, using $x(\ell) = x_0 + \ell d$ together with $y_0 - x(\ell_0) = \ell_0 q$,

$$y_0 + \tau q - x(\ell_0 + \tau) = \ell_0 q + \tau(q - d) = (\ell_0 + \tau)(q - d) + \ell_0 d.$$

Expanding the square and dividing by $2(\ell_0 + \tau)$, which is nonzero for τ near 0,

$$R(\tau) := \widehat{\Psi}\big(\zeta_0 + \tau\xi\big) = \widetilde{\psi}(\ell_0 + \tau) + \frac{(\ell_0 + \tau)|q - d|^2}{2} + \ell_0\langle q - d, d\rangle + \frac{\ell_0^2|d|^2}{2(\ell_0 + \tau)}.$$

The second and third summands are respectively affine and constant in τ, and

$$\frac{d^2}{d\tau^2}\,\frac{\ell_0^2|d|^2}{2(\ell_0+\tau)} = \frac{\ell_0^2|d|^2}{(\ell_0+\tau)^3}, \qquad \text{which at } \tau = 0 \text{ equals } \frac{|d|^2}{\ell_0}.$$

Therefore, by (477),

$$R''(0) = \widetilde{\psi}''(\ell_0) + \frac{|d|^2}{\ell_0} = \Big(2A - \frac{|d|^2}{\ell_0}\Big) + \frac{|d|^2}{\ell_0} = 2A,$$

and `SecondVariationAlongLine` gives $F[\widehat{\Psi}](\zeta_0) = R''(0) = 2A$. With (480) this is $2A \geq 0$, i.e. $A \geq 0$, as claimed.

Step 3: from the test criterion to convexity. The hypotheses of `ViscosityConvexOnIcc` for G on $[0,T]$ are exactly continuity of G on $[0,T]$, which is Step 1, and the implication just proved in Step 2. That lemma therefore applies with this $T > 0$ and this G, and gives that G is convex on $[0,T]$, which is (476). Taking $d = 0$ makes the correction term vanish and $x(\ell) \equiv x_0$, so $\ell \mapsto \mathcal{P}_\ell f(x_0,t)$ is convex on $[0,T]$.

Part (2). Assume now that f is a viscosity supersolution of $F[f] = 0$ on Ω, and let $x \in F$ and t be such that $t - \ell \in I$ for all $\ell \in [0,T]$. Let $\widetilde{f}$ be the time reversal of `TimeFlip` and put $-I = \{-\sigma : \sigma \in I\}$, an open set, with $\widetilde{\Omega} = \{(y,\sigma) : \sigma \in -I\}$.

We check that the hypotheses of Part (1) hold for $\widetilde{f}$ on $-I$.

- By `TimeReversal`, part (1) applied with Ω there taken to be our Ω, the function $\widetilde{f}$ is a viscosity subsolution of $F[\widetilde{f}] = 0$ on the preimage of Ω under the reflection $(y,\sigma) \mapsto (y,-\sigma)$, which is exactly $\widetilde{\Omega}$.
- $\widetilde{f}$ is continuous on $\widetilde{\Omega}$: by `TimeFlip`, $\widetilde{f}(\zeta) = -f(\widetilde{\zeta})$, and $\zeta \mapsto \widetilde{\zeta}$ is a homeomorphism carrying $\widetilde{\Omega}$ onto Ω, where f is continuous.
- $\widetilde{f}$ has subquadratic spatial growth on compact time intervals of $-I$ in the sense of `SubquadraticGrowth`: let $J \subseteq -I$ be compact and $\varepsilon > 0$. Then $-J \subseteq I$ is compact, so `SubquadraticGrowth` for f provides $M > 0$ with $|f(y,\sigma)| \leq \varepsilon|y|^2$ whenever $|y| \geq M$ and $\sigma \in -J$. For $|y| \geq M$ and $\sigma \in J$ we have $-\sigma \in -J$, hence $|\widetilde{f}(y,\sigma)| = |f(y,-\sigma)| \leq \varepsilon|y|^2$.

Apply Part (1) to $\widetilde{f}$ on $-I$ with $x_0 = x$, $d = 0$, base time $-t$ and the same T; the admissibility hypothesis requires $-t + \ell \in -I$ for all $\ell \in [0,T]$, which is precisely $t - \ell \in I$ for all $\ell \in [0,T]$. It gives that $\ell \mapsto \mathcal{P}_\ell \widetilde{f}(x,-t)$ is convex on $[0,T]$.

By `TimeReversal`, part (2), applied at the space-time point $(x,-t)$, whose reflection is (x,t),

$$\mathcal{P}_\ell \widetilde{f}(x,-t) = -\mathcal{M}_\ell f(x,t) \qquad (\ell \in \mathbb{R}).$$

So $\ell \mapsto -\mathcal{M}_\ell f(x,t)$ is convex on $[0,T]$, i.e. $\ell \mapsto \mathcal{M}_\ell f(x,t)$ is concave on $[0,T]$, with $\mathcal{M}_0 f = f$ as in `QuadEnvM`. This is Part (2). □

2.122. `EnvelopeSlopeOrder`.

Lemma 2.122 (Envelope slopes of a subsolution)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(x,t) : t \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`, and assume in addition that f is a viscosity subsolution of $F[f] = 0$ on Ω in the sense of `IsLimitSubOn`. Then, with the slopes of `EnvSlopeP` and `EnvSlopeM`:*

(1) *for every $z = (x,t)$ and all $0 < \ell \leq \ell'$ such that $t + \sigma \in I$ for all $\sigma \in [0,\ell']$,*

$$\mathcal{S}^+_\ell f(z) \leq \mathcal{S}^+_{\ell'} f(z);$$

(2) *for every $z = (x,t)$, every $h > 0$ with $t + \sigma \in I$ for all $\sigma \in [-h,h]$, and every $\varepsilon > 0$, there is $\delta > 0$ with*

$$\mathcal{S}^-_\ell f(z) \leq \mathcal{S}^+_h f(z) + \varepsilon \qquad (0 < \ell < \delta).$$

Equivalently, in the limit form: for every $z=(x,t)$ and every $h_0>0$ with $t+\sigma\in I$ for all $\sigma\in[-h_0,h_0]$, item (1) makes $h\mapsto \mathcal{S}_h^+ f(z)$ nondecreasing on $(0,h_0]$, so that $\lim_{h\downarrow 0}\mathcal{S}_h^+ f(z)$ exists in $[-\infty,\infty)$, and item (2) gives

$$\limsup_{\ell\downarrow 0}\mathcal{S}_\ell^- f(z)\le \lim_{h\downarrow 0}\mathcal{S}_h^+ f(z), \tag{481}$$

the right side being allowed the value $-\infty$.

Proof. (1). Fix $z=(x,t)$ and $0<\ell\le\ell'$ with $t+\sigma\in I$ for all $\sigma\in[0,\ell']$. By QuadEnvPathConvex with $T=\ell'$, $d=0$ and $x_0=x$, the function $g(\sigma)=\mathcal{P}_\sigma f(x,t)$ is convex on $[0,\ell']$, and $g(0)=f(z)$ because $\mathcal{P}_0 f=f$ in QuadEnvP. By EnvSlopeP,

$$\mathcal{S}_\sigma^+ f(z)=\frac{g(\sigma)-g(0)}{\sigma}\qquad(\sigma>0),$$

which is the slope of the chord of g from 0 to σ. For a convex function the chord slope from a fixed left endpoint is nondecreasing in the right endpoint, so $\mathcal{S}_\ell^+ f(z)\le\mathcal{S}_{\ell'}^+ f(z)$.

(2). Fix $z=(x,t)$, $h>0$ with $t+\sigma\in I$ for all $\sigma\in[-h,h]$, and $\varepsilon>0$.

For $0<\ell<h$, the time arguments t and $t-\ell$ lie in I, so by part (1) of QuadEnvAttained the infimum defining $\mathcal{M}_\ell f(z)$ is attained at some $y_\ell\in F$; put $\widetilde{z}_\ell=(y_\ell,t-\ell)$. Attainment and the formula of QuadEnvM give $\mathcal{M}_\ell f(z)=f(\widetilde{z}_\ell)+|y_\ell-x|^2/(2\ell)$, so

$$f(z)-\mathcal{M}_\ell f(z)=f(z)-f(\widetilde{z}_\ell)-\frac{|y_\ell-x|^2}{2\ell}.$$

On the other hand x, at time $(t-\ell)+\ell=t$, is an admissible competitor in the supremum defining $\mathcal{P}_\ell f(\widetilde{z}_\ell)$ of QuadEnvP. That a competitor's value is at most the envelope is not automatic: the envelope is a supremum, and the supremum of a set that is empty or unbounded above is a junk value, so attainment at one competitor does not by itself license domination at the others. The licence is part (1) of EnvelopeCompetitorBounds, whose first clause states exactly that $\mathcal{P}_\ell f$ dominates every competitor value; we apply it at the center $\widetilde{z}_\ell$ and the length $\ell>0$, its hypotheses being the continuity and subquadratic growth of f already assumed together with the memberships $t-\ell\in I$ and $(t-\ell)+\ell=t\in I$, both of which hold. This gives

$$\mathcal{P}_\ell f(\widetilde{z}_\ell)\ge f(z)-\frac{|x-y_\ell|^2}{2\ell}.$$

Subtracting $f(\widetilde{z}_\ell)$ from both sides and dividing by $\ell>0$, the two displays above give, by EnvSlopeM and EnvSlopeP,

$$\mathcal{S}_\ell^- f(z)\le\mathcal{S}_\ell^+ f(\widetilde{z}_\ell). \tag{482}$$

By part (4) of QuadEnvAttained, applied at z with the tolerance η, every optimizer y_ℓ for $\mathcal{M}_\ell f(z)$ satisfies $|y_\ell-x|\le\eta$ once ℓ is small; since also $t-\ell\to t$, this gives $\widetilde{z}_\ell\to z$ as $\ell\downarrow 0$.

Fix ℓ with $0<\ell<h$, and apply part (1) at the center $\widetilde{z}_\ell=(y_\ell,t-\ell)$ with lengths $\ell\le h$. Its hypothesis requires $(t-\ell)+\sigma\in I$ for all $\sigma\in[0,h]$, which holds because $(t-\ell)+\sigma=t+(\sigma-\ell)$ and $\sigma-\ell\in[-h,h]$. Therefore

$$\mathcal{S}_\ell^+ f(\widetilde{z}_\ell)\le\mathcal{S}_h^+ f(\widetilde{z}_\ell).$$

Combining with (482),

$$\mathcal{S}_\ell^- f(z)\le\mathcal{S}_h^+ f(\widetilde{z}_\ell)=\frac{\mathcal{P}_h f(\widetilde{z}_\ell)-f(\widetilde{z}_\ell)}{h}\qquad(0<\ell<h).$$

Now $\zeta\mapsto\mathcal{P}_h f(\zeta)$ is continuous at z: part (2) of QuadEnvAttained, whose hypotheses $h>0$, $t\in I$ and $t+h\in I$ hold, gives joint continuity of $(\ell,\zeta)\mapsto\mathcal{P}_\ell f(\zeta)$ at (h,z), and restricting to the slice $\ell=h$ gives continuity in the center. Also f is continuous at z because $z\in\Omega$ and Ω is open. Hence $\zeta\mapsto\mathcal{S}_h^+ f(\zeta)$ is continuous at z, so there is $\delta_1>0$ such that $|\zeta-z|<\delta_1$ implies $\mathcal{S}_h^+ f(\zeta)\le\mathcal{S}_h^+ f(z)+\varepsilon$. Choose $\delta\in(0,h)$ so small that $|\widetilde{z}_\ell-z|<\delta_1$ for all $0<\ell<\delta$, which is possible by the previous paragraph. Then

$$\mathcal{S}_\ell^- f(z)\le\mathcal{S}_h^+ f(z)+\varepsilon\qquad(0<\ell<\delta),$$

as required.

Finally we derive (481) from (1) and (2). Fix $z = (x, t)$ and $h_0 > 0$ with $t + \sigma \in I$ for all $\sigma \in [-h_0, h_0]$.

For $0 < h \le h' \le h_0$ the hypothesis of (1) at the center z with the lengths $h \le h'$ asks that $t+\sigma \in I$ for all $\sigma \in [0, h']$, which holds because $[0, h'] \subseteq [-h_0, h_0]$; so (1) gives $\mathcal{S}^+_h f(z) \le \mathcal{S}^+_{h'} f(z)$, i.e. $h \mapsto \mathcal{S}^+_h f(z)$ is a nondecreasing real-valued function on $(0, h_0]$. A nondecreasing real-valued function on $(0, h_0]$ converges as $h \downarrow 0$, in $\overline{\mathbb{R}}$, to

$$L = \inf_{0<h\le h_0} \mathcal{S}^+_h f(z) \in [-\infty, \infty),$$

the value $-\infty$ occurring exactly when it is unbounded below, and the value $+\infty$ being impossible because $L \le \mathcal{S}^+_{h_0} f(z) < +\infty$. Hence $\lim_{h\downarrow 0} \mathcal{S}^+_h f(z)$ exists in $[-\infty, \infty)$ and equals L.

Now fix $h \in (0, h_0]$ and $\varepsilon > 0$. The hypothesis of (2) at z with this h asks that $t + \sigma \in I$ for all $\sigma \in [-h, h]$, which holds because $[-h, h] \subseteq [-h_0, h_0]$; so (2) supplies $\delta > 0$ with $\mathcal{S}^-_\ell f(z) \le \mathcal{S}^+_h f(z) + \varepsilon$ for all $0 < \ell < \delta$. Therefore $\limsup_{\ell\downarrow 0} \mathcal{S}^-_\ell f(z) \le \mathcal{S}^+_h f(z) + \varepsilon$. Letting $\varepsilon \downarrow 0$ gives $\limsup_{\ell\downarrow 0} \mathcal{S}^-_\ell f(z) \le \mathcal{S}^+_h f(z)$ for every $h \in (0, h_0]$, and taking the infimum over $h \in (0, h_0]$ gives

$$\limsup_{\ell\downarrow 0} \mathcal{S}^-_\ell f(z) \le L = \lim_{h\downarrow 0} \mathcal{S}^+_h f(z),$$

which is (481). □

2.123. `EnvHamiltonianSolution`.

Lemma 2.123 (The envelope Hamiltonian of a solution)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(x, t) : t \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`, and assume in addition that f is a viscosity solution of $F[f] = 0$ on Ω in the sense of `IsLimitSolutionOn`. Then, with the slopes of `EnvSlopeP` and `EnvSlopeM` and the Hamiltonian H_f of `EnvHamiltonian`:*

(1) *for every $z = (x, t)$ and all $0 < \ell \le \ell'$ with $t + \sigma \in I$ for all $\sigma \in [-\ell', \ell']$, both slopes are nondecreasing:*

$$\mathcal{S}^+_\ell f(z) \le \mathcal{S}^+_{\ell'} f(z), \qquad \mathcal{S}^-_\ell f(z) \le \mathcal{S}^-_{\ell'} f(z);$$

(2) *for every $z = (x, t)$ with $t + \sigma \in I$ for every $\sigma \in \mathbb{R}$,*

$$\inf_{\ell>0} \mathcal{S}^-_\ell f(z) = \inf_{\ell>0} \mathcal{S}^+_\ell f(z) = H_f(z), \tag{483}$$

so that by (1) the two monotone small-length limits exist and agree: $\lim_{\ell\downarrow 0} \mathcal{S}^+_\ell f(z) = \lim_{\ell\downarrow 0} \mathcal{S}^-_\ell f(z) = H_f(z)$;

(3) *$H_f(z) \in [-\infty, \infty)$ for every z, that is, H_f never takes the value $+\infty$;*

(4) *if moreover every length is admissible throughout the domain, in the sense that $t+\sigma \in I$ for every $t \in I$ and every $\sigma \in \mathbb{R}$ — the domain-level form of the hypothesis carried pointwise in (2) — then H_f is upper semicontinuous on Ω: the set $\{\zeta \in \Omega : H_f(\zeta) < c\}$ is open in Ω for every $c \in \overline{\mathbb{R}}$.*

Proof. By `IsLimitSolutionOn`, f is both a viscosity subsolution and a viscosity supersolution of $F[f] = 0$ on Ω.

(1). The bound for $\mathcal{S}^+$ is part (1) of `EnvelopeSlopeOrder`, whose hypotheses hold with $[0, \ell'] \subseteq [-\ell', \ell']$. For $\mathcal{S}^-$, let $\widetilde{f}$ be the time reversal of `TimeFlip`. By `TimeReversal`, $\widetilde{f}$ is a viscosity subsolution of $F[\widetilde{f}] = 0$ on the reflected domain $\widetilde{\Omega} = \{(x, t) : -t \in I\}$, it is continuous there, and it satisfies the growth condition of `SubquadraticGrowth` on $-I = \{-t : t \in I\}$, because that condition is invariant under reflecting the time variable and changing the sign of the function. The set $-I$ is open, being the image of the open set I under the homeomorphism $s \mapsto -s$ of $\mathbb{R}$, and $\widetilde{\Omega} = \{(x, t) : t \in -I\}$, so part (1) of `EnvelopeSlopeOrder` is applicable to $\widetilde{f}$ with $-I$ in place of I. Also by `TimeReversal`, $\mathcal{P}_\ell \widetilde{f}(\widetilde{z}) = -\mathcal{M}_\ell f(z)$ and $\widetilde{f}(\widetilde{z}) = -f(z)$, so by `EnvSlopeP` and `EnvSlopeM`

$$\mathcal{S}^+_\ell \widetilde{f}(\widetilde{z}) = \frac{\mathcal{P}_\ell \widetilde{f}(\widetilde{z}) - \widetilde{f}(\widetilde{z})}{\ell} = \frac{f(z) - \mathcal{M}_\ell f(z)}{\ell} = \mathcal{S}^-_\ell f(z).$$

Applying part (1) of EnvelopeSlopeOrder to $\widetilde{f}$ at $\widetilde{z}=(x,-t)$ with the lengths $\ell\le\ell'$, whose hypothesis $-t+\sigma\in -I$ for $\sigma\in[0,\ell']$ follows from $t-\sigma\in I$, i.e. from the assumption with $-\sigma\in[-\ell',0]$, gives $\mathcal{S}^-_\ell f(z)\le\mathcal{S}^-_{\ell'}f(z)$.

(2). Fix z with $t+\sigma\in I$ for every σ, so all the length hypotheses above are satisfied for every $\ell,\ell'>0$, and write $S^\pm(\ell)=\mathcal{S}^\pm_\ell f(z)$, nondecreasing in ℓ by (1). By EnvHamiltonian, $H_f(z)=\inf_{\ell>0}S^+(\ell)$, and by monotonicity this equals $\lim_{\ell\downarrow0}S^+(\ell)$ in $\overline{\mathbb{R}}$; similarly $\inf_{\ell>0}S^-(\ell)=\lim_{\ell\downarrow0}S^-(\ell)$.

Part (2) of EnvelopeSlopeOrder, applied to f as a subsolution, gives for every $h>0$ and every $\varepsilon>0$ a $\delta>0$ with $S^-(\ell)\le S^+(h)+\varepsilon$ for $0<\ell<\delta$. Hence $\inf_{\ell>0}S^-(\ell)\le S^+(h)+\varepsilon$ for all $h>0,\varepsilon>0$, so $\inf_{\ell>0}S^-\le\inf_{\ell>0}S^+=H_f(z)$.

Applying the same part (2) of EnvelopeSlopeOrder to $\widetilde{f}$, which is a subsolution by TimeReversal, and translating back with the identities of (1) together with $\mathcal{S}^-_\ell\widetilde{f}(\widetilde{z})=\mathcal{S}^+_\ell f(z)$ (which follows from $\mathcal{M}_\ell\widetilde{f}(\widetilde{z})=-\mathcal{P}_\ell f(z)$ in TimeReversal), gives $\inf_{\ell>0}S^+\le\inf_{\ell>0}S^-$. The two infima are therefore equal, which is (483).

(3). Let z be arbitrary. By QuadEnvP and EnvSlopeP, $\mathcal{S}^+_1f(z)=\mathcal{P}_1f(z)-f(z)$ is a real number: the supremum defining $\mathcal{P}_1f(z)$ is a real number whether or not the competitor set is bounded above, by the preamble convention assigning the value 0 to the supremum of a set of reals that is empty or unbounded above (the time arguments t and $t+1$ of z need not lie in I here, so no attainment statement is invoked). By EnvHamiltonian, $H_f(z)$ is the infimum in $\overline{\mathbb{R}}$ of the real numbers $\mathcal{S}^+_\ell f(z)$ over the nonempty set $\ell>0$. The element $\mathcal{S}^+_1f(z)$ of that family is a real upper bound for the infimum, so $H_f(z)\le\mathcal{S}^+_1f(z)<+\infty$ and $H_f(z)\ne+\infty$.

(4). Assume in addition that $t+\sigma\in I$ for every $t\in I$ and every $\sigma\in\mathbb{R}$. Fix once and for all $c\in\overline{\mathbb{R}}$ and $z_0=(x_0,t_0)\in\Omega$ with $H_f(z_0)<c$; it suffices to produce an open set V with $z_0\in V$, $V\subseteq\Omega$ and $H_f(\zeta)<c$ for every $\zeta\in V$, since then every point of $\{\zeta\in\Omega:H_f(\zeta)<c\}$ is interior to it relative to Ω, which is the assertion.

Step 1: monotonicity at every point of Ω, for every pair of lengths. Let $\zeta=(\xi,\tau)$ with $\tau\in I$ and let $0<h\le h'$. The hypothesis of part (1) of EnvelopeSlopeOrder at the center ζ with the lengths $h\le h'$ asks that $\tau+\sigma\in I$ for all $\sigma\in[0,h']$, and this is the added hypothesis applied at $\tau\in I$ with the values $\sigma\in[0,h']$. Its remaining hypotheses hold: I is open, f is continuous on Ω and satisfies SubquadraticGrowth on I by assumption, and f is a viscosity subsolution of $F[f]=0$ on Ω as recorded at the start of this proof. Hence

$$\mathcal{S}^+_hf(\zeta)\le\mathcal{S}^+_{h'}f(\zeta)\qquad(\tau\in I,\ 0<h\le h'). \tag{484}$$

Step 2: the local length range and neighbourhood. We have $t_0\in I$ because $z_0\in\Omega$. Since I is open there is $h_0>0$ with $(t_0-h_0,t_0+h_0)\subseteq I$. Put

$$N=\{\zeta=(\xi,\tau):\ |\tau-t_0|<h_0\},$$

which is an open subset of space-time, because $\zeta\mapsto\tau$ is continuous, contains z_0, and is contained in Ω, because $|\tau-t_0|<h_0$ forces $\tau\in(t_0-h_0,t_0+h_0)\subseteq I$.

Step 3: each fixed forward slope is continuous on N. Fix $h>0$ and let $\zeta_0=(\xi_0,\tau_0)\in N$. Then $\tau_0\in I$ by Step 2, and $\tau_0+h\in I$ by the added hypothesis. Part (2) of QuadEnvAttained, whose hypotheses $h>0$, $\tau_0\in I$ and $\tau_0+h\in I$ therefore all hold, gives joint continuity of $(\ell,\zeta)\mapsto\mathcal{P}_\ell f(\zeta)$ at (h,ζ_0); composing with the continuous map $\zeta\mapsto(h,\zeta)$ gives continuity of $\zeta\mapsto\mathcal{P}_hf(\zeta)$ at ζ_0. Also f is continuous at ζ_0, since $\zeta_0\in\Omega$, f is continuous on Ω and Ω is open. Since $h>0$ is a fixed constant, $\mathcal{S}^+_hf=(\mathcal{P}_hf-f)/h$ by EnvSlopeP is continuous at ζ_0. As $\zeta_0\in N$ was arbitrary, $\zeta\mapsto\mathcal{S}^+_hf(\zeta)$ is continuous on N for every $h>0$.

Step 4: the Hamiltonian is the infimum over the restricted range. We claim

$$H_f(\zeta)=\inf_{0<h\le h_0}\mathcal{S}^+_hf(\zeta)\qquad(\zeta\in N). \tag{485}$$

Fix $\zeta=(\xi,\tau)\in N$, so $\tau\in I$, and write $A=\{\mathcal{S}^+_hf(\zeta):h>0\}$ and $B=\{\mathcal{S}^+_hf(\zeta):0<h\le h_0\}$, so that $H_f(\zeta)=\inf A$ in $\overline{\mathbb{R}}$ by EnvHamiltonian and the right side of (485) is $\inf B$. Since $B\subseteq A$ we have $\inf A\le\inf B$. Conversely, let $h>0$. If $h\le h_0$ then $\mathcal{S}^+_hf(\zeta)\in B$, so $\inf B\le\mathcal{S}^+_hf(\zeta)$. If $h>h_0$ then (484) at ζ with the lengths $h_0\le h$ gives $\mathcal{S}^+_{h_0}f(\zeta)\le\mathcal{S}^+_hf(\zeta)$, and $\mathcal{S}^+_{h_0}f(\zeta)\in B$,

so again $\inf B \le \mathcal{S}^+_h f(\zeta)$. Thus $\inf B$ is a lower bound for A, whence $\inf B \le \inf A$. This proves (485).

Step 5: the required neighbourhood. Apply (485) at $\zeta = z_0$, which is legitimate because $z_0 \in N$: the infimum in $\overline{\mathbb{R}}$ of the nonempty set $\{\mathcal{S}^+_h f(z_0) : 0 < h \le h_0\}$ of real numbers equals $H_f(z_0)$ and is therefore $< c$. Hence c is not a lower bound for that set, and there is $h_* \in (0, h_0]$ with $\mathcal{S}^+_{h_*} f(z_0) < c$. By Step 3 the real-valued function $\zeta \mapsto \mathcal{S}^+_{h_*} f(\zeta)$ is continuous on the open set N, so

$$V = \{\zeta \in N : \mathcal{S}^+_{h_*} f(\zeta) < c\}$$

is open, and $z_0 \in V$. For every $\zeta \in V$ we have $\zeta \in N$, so (485) applies at ζ and, h_* being a member of the index set $(0, h_0]$, gives $H_f(\zeta) \le \mathcal{S}^+_{h_*} f(\zeta) < c$. Also $V \subseteq N \subseteq \Omega$ by Step 2. So V has the three properties required at the start of (4), and H_f is upper semicontinuous on Ω. □

2.124. `ContactFiniteSlope`.

Lemma 2.124 (A finite slope at the extremal contact)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$, let $\kappa > 0$ and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $z_* = (0,1)$ with constant C_0 in the sense of `IsCenteredAt`, satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`, and satisfy*

$$\mathcal{C}_1 v(0,1) - v(0,1) = \frac{\kappa^2}{2}$$

with the barycentric envelope of `BaryEnv`. Then the envelope Hamiltonian of `EnvHamiltonian` is finite at z_*:

$$H_v(z_*) > -\infty.$$

Proof. Write $z_* = (0,1)$; by `IsCenteredAt` we have $v(z_*) = 0$.

Step 0: standing facts. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there; by `UniformSubquadratic`, applied with the centering point $(0,1)$, it has subquadratic spatial growth on compact time intervals of $I = \mathbb{R}$ in the sense of `SubquadraticGrowth`. Hence `QuadEnvAttained`, `QuadEnvPathConvex`, `EnvelopeCompetitorBounds`, `EnvHamiltonianSolution` and `BaryEnvDuality` all apply to v with $I = \mathbb{R}$.

We record the licence under which every supremum and infimum below is treated as a genuine least upper or greatest lower bound, since a supremum of a set that is empty or unbounded above carries a convention rather than that meaning. For $\ell > 0$ and any centre $\zeta = (\xi, \tau)$, part (1) of `EnvelopeCompetitorBounds` applied to v with $I = \mathbb{R}$ states both that

$$v(w, \tau + \ell) - \frac{|w - \xi|^2}{2\ell} \le \mathcal{P}_\ell v(\zeta) \qquad (w \in F)$$

and that $\mathcal{P}_\ell v(\zeta)$ is at most every upper bound for those values; part (2) states the dual pair, that

$$\mathcal{M}_\ell v(\zeta) \le v(w, \tau - \ell) + \frac{|w - \xi|^2}{2\ell} \qquad (w \in F)$$

and that $\mathcal{M}_\ell v(\zeta)$ is at least every lower bound for those values. Its admissibility requirements $\tau \in I$ and $\tau \pm \ell \in I$ hold for every ζ and every $\ell > 0$ because $I = \mathbb{R}$. Each use below of "a competitor bounds the envelope" or "the envelope is the least upper bound" is an instance of these four clauses.

Step 1: the attaining configuration is extremal. By `BaryEnvDuality`, part (1), applied with $s = 1 > 0$ and the center z_*, whose time 1 lies in $\mathbb{R}$, there is a finite horizontal probability configuration (θ, y) with mean 0, in the sense of `IsBaryConfig`, attaining the envelope:

$$\mathcal{C}_1 v(0,1) = \sum_i \theta_i v(y_i, 1) - \frac{r^2}{2}, \qquad r = \Big(\sum_i \theta_i |y_i|^2\Big)^{1/2},$$

with r the radius of `BaryRadius` about the mean 0. Put $D = \sum_i \theta_i v(y_i,1) - v(z_*) = \sum_i \theta_i v(y_i, 1)$. The contact hypothesis reads $\mathcal{C}_1 v(0,1) - v(z_*) = \kappa^2/2$, i.e.

$$D - \frac{r^2}{2} = \frac{\kappa^2}{2}. \tag{486}$$

By HasJensenBound at the time 1 we have $D \le \kappa r$, so by the algebraic identity $\kappa r - r^2/2 = \kappa^2/2 - (r-\kappa)^2/2$,

$$\frac{\kappa^2}{2} = D - \frac{r^2}{2} \le \kappa r - \frac{r^2}{2} = \frac{\kappa^2}{2} - \frac{(r-\kappa)^2}{2}.$$

Hence $(r-\kappa)^2 \le 0$, so

$$r = \kappa, \qquad D = \kappa r = \kappa^2, \tag{487}$$

the second equality from (486) and $r = \kappa$.

Step 2: all y_i are forward optimizers from one point. By BaryEnvDuality, part (2), at $(s,z) = (1, z_*)$ there is $A \in F$ with

$$\mathcal{C}_1 v(0,1) - \frac{|0|^2}{2} = \sup_{y \in F}\Big\{ v(y,1) - \frac{|y|^2}{2} - \langle A, y - 0\rangle \Big\} =: \Gamma.$$

Put $g(y) = v(y,1) - |y|^2/2$ and $b = -A$. Before using Γ as an upper bound we check that the family it is the supremum of is nonempty and bounded above. Nonemptiness is witnessed by $y = 0$. For the bound, the algebraic identity

$$v(y,1) - \frac{|y|^2}{2} - \langle A, y - 0\rangle = \Big(v(y,1) - \frac{|y-b|^2}{2} \Big) + \frac{|A|^2}{2} \qquad (y \in F), \tag{488}$$

which holds because $|y-b|^2 = |y+A|^2 = |y|^2 + 2\langle A, y\rangle + |A|^2$, exhibits each member of that family as a member of the forward competitor family for $\mathcal{P}_1 v(b,0)$ shifted by the constant $|A|^2/2$. By the first clause of part (1) of EnvelopeCompetitorBounds, recorded in Step 0, at the centre $(b,0)$ and the length 1, every member of that competitor family is at most $\mathcal{P}_1 v(b,0)$; hence every member of the Γ-family is at most $\mathcal{P}_1 v(b,0) + |A|^2/2$. So the family is nonempty and bounded above, Γ is its least upper bound, and in particular

$$g(y) - \langle A, y\rangle \le \Gamma \qquad (y \in F). \tag{489}$$

Since $\sum_i \theta_i |y_i|^2 - |0|^2 = r^2$, the attainment in Step 1 gives $\sum_i \theta_i g(y_i) = \mathcal{C}_1 v(0,1) = \Gamma$, while (489) gives $g(y_i) - \langle A, y_i\rangle \le \Gamma$ for every i, and $\sum_i \theta_i \langle A, y_i\rangle = \langle A, 0\rangle = 0$. Therefore

$$\sum_i \theta_i \Big(g(y_i) - \langle A, y_i\rangle \Big) = \Gamma,$$

a weighted average with strictly positive weights of numbers each at most Γ; hence each equals Γ:

$$g(y_i) - \langle A, y_i\rangle = \Gamma \qquad \text{for every } i.$$

Substituting (488) into (489) and into the displayed equalities, and cancelling the constant $|A|^2/2$,

$$v(y,1) - \frac{|y-b|^2}{2} \le \Gamma - \frac{|A|^2}{2} \quad (y \in F), \qquad v(y_i,1) - \frac{|y_i - b|^2}{2} = \Gamma - \frac{|A|^2}{2} \quad \text{for every } i.$$

The first display says that $\Gamma - |A|^2/2$ is an upper bound for the forward competitor family for $\mathcal{P}_1 v(b,0)$ at the centre $(b,0)$ and the length 1, whose time arguments are 0 and $0+1=1$, so by the second clause of part (1) of EnvelopeCompetitorBounds we get $\mathcal{P}_1 v(b,0) \le \Gamma - |A|^2/2$; the second display exhibits a member of that family equal to $\Gamma - |A|^2/2$, which by the first clause of the same part is at most $\mathcal{P}_1 v(b,0)$. The two inequalities together give

$$h(1) := \mathcal{P}_1 v(b,0) = \Gamma - \frac{|A|^2}{2}, \qquad v(y_i,1) = h(1) + \frac{|y_i - b|^2}{2} \quad \text{for every } i. \tag{490}$$

Step 3: h is finite and convex on $[0,1]$. Put $h(\sigma) = \mathcal{P}_\sigma v(b,0)$ for $0 \le \sigma \le 1$. For $\sigma > 0$ part (1) of QuadEnvAttained shows the defining supremum is attained, so $h(\sigma) \in \mathbb{R}$; and $h(0) = v(b,0)$ by QuadEnvP. By QuadEnvPathConvex, part (1), applied with $d = 0$, $x_0 = b$, base time 0 and $T = 1$ — the admissibility hypothesis being vacuous since $I = \mathbb{R}$ — the function h is convex on $[0,1]$.

Step 4: a lower bound for the backward slopes at the y_i. Fix i and let $0 < \ell < 1$. Consider the competitor $y_{i,\ell}^- = b + (1-\ell)(y_i - b)$ at the time $1 - \ell$ for $\mathcal{M}_\ell v(y_i, 1)$. Since $y_{i,\ell}^- - y_i = -\ell(y_i - b)$,

the first clause of part (2) of `EnvelopeCompetitorBounds`, at the centre $(y_i, 1)$ and the length ℓ, gives

$$\mathcal{M}_\ell v(y_i,1) \le v\big(y_{i,\ell}^-, 1-\ell\big) + \frac{\ell\,|y_i-b|^2}{2}. \tag{491}$$

On the other hand $y_{i,\ell}^-$ is a competitor at the time $0+(1-\ell) = 1-\ell$ for $\mathcal{P}_{1-\ell}v(b,0)$, and $y_{i,\ell}^- - b = (1-\ell)(y_i-b)$, so by the first clause of part (1) of `EnvelopeCompetitorBounds`, at the centre $(b,0)$ and the length $1-\ell>0$,

$$v\big(y_{i,\ell}^-, 1-\ell\big) - \frac{(1-\ell)|y_i-b|^2}{2} \le \mathcal{P}_{1-\ell}v(b,0) = h(1-\ell),$$

i.e. $v(y_{i,\ell}^-, 1-\ell) \le h(1-\ell) + (1-\ell)|y_i-b|^2/2$. Substituting into (491), the two quadratic costs add to $|y_i-b|^2/2$:

$$\mathcal{M}_\ell v(y_i,1) \le h(1-\ell) + \frac{|y_i-b|^2}{2}.$$

Combining with the equality in (490) and `EnvSlopeM`,

$$\mathcal{S}_\ell^- v(y_i,1) = \frac{v(y_i,1)-\mathcal{M}_\ell v(y_i,1)}{\ell} \ge \frac{h(1)-h(1-\ell)}{\ell} \qquad (0<\ell<1). \tag{492}$$

For a convex function on $[0,1]$ the chord slope from σ to 1, namely $\big(h(1)-h(\sigma)\big)/(1-\sigma)$, is nondecreasing in σ; taking $\sigma = 1-\ell \ge 0$ and comparing with $\sigma = 0$ gives $\big(h(1)-h(1-\ell)\big)/\ell \ge h(1)-h(0)$. Hence, writing $\beta := h(1)-h(0) \in \mathbb{R}$,

$$\mathcal{S}_\ell^- v(y_i,1) \ge \beta \qquad (0<\ell<1). \tag{493}$$

For $\ell \ge 1$, part (1) of `EnvHamiltonianSolution` applied at the center $(y_i,1)$ with the lengths $\frac12 \le \ell$ — its admissibility hypothesis is vacuous since $I=\mathbb{R}$ — gives $\mathcal{S}_\ell^- v(y_i,1) \ge \mathcal{S}_{1/2}^- v(y_i,1) \ge \beta$. So $\mathcal{S}_\ell^- v(y_i,1) \ge \beta$ for every $\ell>0$, and therefore, by part (2) of `EnvHamiltonianSolution`,

$$H_v(y_i,1) = \inf_{\ell>0} \mathcal{S}_\ell^- v(y_i,1) \ge \beta \qquad \text{for every } i. \tag{494}$$

In particular, since the forward slopes satisfy $\mathcal{S}_\ell^+ v(y_i,1) \ge \inf_{\ell'>0} \mathcal{S}_{\ell'}^+ v(y_i,1) = H_v(y_i,1)$ by `EnvHamiltonian` and part (2) of `EnvHamiltonianSolution`,

$$\mathcal{S}_\ell^+ v(y_i,1) \ge \beta \qquad (\ell>0,\ \text{every } i). \tag{495}$$

Step 5: transferring the bound to the barycenter. Fix $\ell>0$. By part (1) of `QuadEnvAttained`, for each i choose a maximizer $\widehat{y}_i$ for $\mathcal{P}_\ell v(y_i,1)$, so

$$\mathcal{P}_\ell v(y_i,1) = v\big(\widehat{y}_i, 1+\ell\big) - \frac{|\widehat{y}_i-y_i|^2}{2\ell}. \tag{496}$$

Put $\widehat{x} = \sum_i \theta_i \widehat{y}_i$, $\widehat{\xi}_i = \widehat{y}_i - \widehat{x}$ and $r' = \big(\sum_i \theta_i |\widehat{\xi}_i|^2\big)^{1/2}$, and recall that the mean of (θ, y) is 0, so $\sum_i \theta_i y_i = 0$ and $\sum_i \theta_i |y_i|^2 = r^2$.

Since $\widehat{y}_i - y_i = (\widehat{\xi}_i - y_i) + \widehat{x}$ and $\sum_i \theta_i(\widehat{\xi}_i - y_i) = 0-0 = 0$, expanding gives the exact orthogonal decomposition

$$\sum_i \theta_i |\widehat{y}_i - y_i|^2 = |\widehat{x}|^2 + \sum_i \theta_i |\widehat{\xi}_i - y_i|^2.$$

The reverse triangle inequality in the weighted Euclidean space, i.e. in $L^2(\theta)$, gives $\big(\sum_i \theta_i |\widehat{\xi}_i - y_i|^2\big)^{1/2} \ge |r'-r|$, so

$$\sum_i \theta_i |\widehat{y}_i - y_i|^2 \ge |\widehat{x}|^2 + (r'-r)^2. \tag{497}$$

The configuration $(\theta, \widehat{y})$ has mean $\widehat{x}$ and radius r', so `HasJensenBound` at the time $1+\ell$ gives $\sum_i \theta_i v(\widehat{y}_i, 1+\ell) \le v(\widehat{x}, 1+\ell) + \kappa r'$. Also $\widehat{x}$ is a competitor at the time $1+\ell$ for $\mathcal{P}_\ell v(0,1)$, so

by the first clause of part (1) of `EnvelopeCompetitorBounds` at the centre z_* and the length ℓ, $v(\widehat{x},1+\ell)\le \mathcal{P}_\ell v(z_*)+|\widehat{x}|^2/(2\ell)$. Averaging (496) with the weights θ_i and using (497),

$$\sum_i \theta_i \mathcal{P}_\ell v(y_i,1) \le v(\widehat{x},1+\ell)+\kappa r' - \frac{|\widehat{x}|^2+(r'-r)^2}{2\ell}$$
$$\le \mathcal{P}_\ell v(z_*)+\kappa r' - \frac{(r'-r)^2}{2\ell}.$$

By (487) we have $\kappa r=\kappa^2$, so $\kappa r'=\kappa^2+\kappa(r'-r)$, and for every real u $\kappa u-u^2/(2\ell)\le \kappa^2\ell/2$, as the identity $\kappa u-u^2/(2\ell)=\kappa^2\ell/2-(u-\kappa\ell)^2/(2\ell)$ shows. With $u=r'-r$,

$$\sum_i \theta_i \mathcal{P}_\ell v(y_i,1) \le \mathcal{P}_\ell v(z_*)+\kappa^2+\frac{\kappa^2\ell}{2}. \tag{498}$$

Now subtract the values. By `EnvSlopeP` and (487),

$$\ell\sum_i \theta_i \mathcal{S}_\ell^+ v(y_i,1)=\sum_i \theta_i \mathcal{P}_\ell v(y_i,1)-\sum_i \theta_i v(y_i,1)=\sum_i \theta_i \mathcal{P}_\ell v(y_i,1)-\kappa^2,$$

while $\ell\,\mathcal{S}_\ell^+ v(z_*)=\mathcal{P}_\ell v(z_*)-v(z_*)=\mathcal{P}_\ell v(z_*)$. Therefore (498) reads

$$\ell\sum_i \theta_i \mathcal{S}_\ell^+ v(y_i,1)\le \ell\,\mathcal{S}_\ell^+ v(z_*)+\frac{\kappa^2\ell}{2},$$

and dividing by $\ell>0$,

$$\mathcal{S}_\ell^+ v(z_*)\ge \sum_i \theta_i \mathcal{S}_\ell^+ v(y_i,1)-\frac{\kappa^2}{2}. \tag{499}$$

Step 6: conclusion. By (495) and $\sum_i\theta_i=1$, the right side of (499) is at least $\beta-\kappa^2/2$. Hence

$$\mathcal{S}_\ell^+ v(z_*)\ge \beta-\frac{\kappa^2}{2}\qquad(\ell>0),$$

and by `EnvHamiltonian`

$$H_v(z_*)=\inf_{\ell>0}\mathcal{S}_\ell^+ v(z_*)\ge \beta-\frac{\kappa^2}{2}>-\infty.$$

In particular $H_v(z_*)\neq-\infty$, which is the assertion. □

2.125. `ContactEnvelopeSlopeLimit`.

Lemma 2.125 (The forward slopes of the contact envelope converge to the envelope Hamiltonian)**.** *Let F be a finite-dimensional real inner-product space — the ambient convention of this development — and work in the space-time $X=\ell^2(F\times\mathbb{R})$, writing a point as $z=(x,t)$ and $P=\{z\in X:\ t>0\}$ for the positive-time half space. Put $z_*=(0,1)$. Let $C_0\ge 1$, let $\kappa>0$ and let $v:X\to\mathbb{R}$. Assume:*

(1) *$v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`, with $m=\dim F$;*
(2) *v is centered at z_* with constant C_0 in the sense of `IsCenteredAt`;*
(3) *v satisfies the horizontal Jensen bound with constant κ on $I=\mathbb{R}$ in the sense of `HasJensenBound`;*
(4) *v has subquadratic spatial growth on compact time intervals of $I=\mathbb{R}$ in the sense of `SubquadraticGrowth`;*
(5) *$W(z_*)=0$, where $W=\mathcal{C}_t v-\kappa^2 t/2$ is the contact envelope of `ContactEnv` formed from v and κ;*
(6) *W is continuous on P;*
(7) *W has subquadratic spatial growth on compact time intervals of $I=(0,\infty)$;*
(8) *W is a viscosity subsolution of the limiting equation on P in the sense of `IsLimitSubOn`;*
(9) *$v(z)-\kappa^2 t/2\le W(z)\le v(z)$ for every $z=(x,t)$ with $t>0$.*

Then there is a real number H such that the envelope Hamiltonian of ***EnvHamiltonian*** *satisfies*

$$H_v(z_*) = H \qquad \text{and} \qquad \lim_{\ell \downarrow 0} \mathcal{S}_\ell^+ W(z_*) = H,$$

the limit being over $\ell \to 0$ within $(0,\infty)$ and $\mathcal{S}_\ell^+$ the forward envelope slope of ***EnvSlopeP****.*

Proof. Hypotheses 5–9 are the four properties of the contact envelope supplied by `ContactEnvelopeProperties` that this argument reads, together with the contact; the construction of such a v in the application is not re-narrated. Both conclusions are returned together because H is produced in the course of the proof.

Write $t(z)$ for the time coordinate and note $t(z_*) = 1$ and, more generally, $t\big((y,c)\big) = c$: these are the only facts about the coordinates used below. By 2 and `IsCenteredAt`,

$$v(z_*) = 0, \tag{500}$$

and by 1 and `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there. Also, by `ContactEnv` at $t = 1$, hypothesis 5 reads $\mathcal{C}_1 v(z_*) - \kappa^2/2 = 0$, so with (500)

$$\mathcal{C}_1 v(z_*) - v(z_*) = \frac{\kappa^2}{2}. \tag{501}$$

Step A: $H_v(z_*)$ is a real number. `EnvHamiltonianSolution` applies with $I = \mathbb{R}$ and $f = v$, its hypotheses being the continuity just recorded, 4 and the solution property, the open set being all of space-time. Its third clause gives $H_v(z_*) \neq +\infty$, and its second clause, whose hypothesis $1 + \sigma \in \mathbb{R}$ is vacuous here, gives

$$\inf_{\ell > 0} \mathcal{S}_\ell^- v(z_*) = H_v(z_*), \tag{502}$$

with $\mathcal{S}_\ell^-$ the backward envelope slope of `EnvSlopeM`. By 1, 2, 3, $\kappa > 0$ and (501), the hypotheses of `ContactFiniteSlope` hold, so $H_v(z_*) \neq -\infty$. A value of $\overline{\mathbb{R}}$ that is neither $+\infty$ nor $-\infty$ is a real number; let H be its real value, so that

$$H_v(z_*) = H. \tag{503}$$

That is the first conclusion. `ContactFiniteSlope` carries the paper's whole lower-bound argument — the $r = \kappa$ squeeze, the duality minimizing slope, the positive-weight attainment and the completed square — so none of it is redone here.

Step B: the two one-sided slope comparisons. The function v satisfies the hypotheses of `EnvelopeCompetitorBounds` with $I = \mathbb{R}$, and W satisfies them with $I = (0,\infty)$ by 6 and 7. The relevant times are $1 + \ell > 0$ for $\ell > 0$ and $1 - \ell > 0$ for $0 < \ell < 1$, so all four envelopes below are at admissible parameters.

Let $\ell > 0$. The competitor set defining $\mathcal{P}_\ell$ at z_* in `QuadEnvP` involves only values at the time $1 + \ell > 0$, where $W \le v$ by 9; so for each $y \in F$,

$$W(y, 1+\ell) - \frac{|y|^2}{2\ell} \;\le\; v(y, 1+\ell) - \frac{|y|^2}{2\ell} \;\le\; \mathcal{P}_\ell v(z_*),$$

the second inequality being the domination half of clause 1 of `EnvelopeCompetitorBounds` for v. The extremality half of that clause for W then gives $\mathcal{P}_\ell W(z_*) \le \mathcal{P}_\ell v(z_*)$. Dividing by $\ell > 0$ after subtracting the values, which are both 0 by 5 and (500), `EnvSlopeP` turns this into

$$\mathcal{S}_\ell^+ W(z_*) \le \mathcal{S}_\ell^+ v(z_*) \qquad (\ell > 0). \tag{504}$$

Let $0 < \ell < 1$. The competitor set defining $\mathcal{M}_\ell$ at z_* in `QuadEnvM` involves only values at the time $1 - \ell > 0$, where again $W \le v$; so for each $y \in F$,

$$\mathcal{M}_\ell W(z_*) \;\le\; W(y, 1-\ell) + \frac{|y|^2}{2\ell} \;\le\; v(y, 1-\ell) + \frac{|y|^2}{2\ell},$$

the first inequality being the domination half of clause 2 of `EnvelopeCompetitorBounds` for W. The extremality half of that clause for v then gives $\mathcal{M}_\ell W(z_*) \le \mathcal{M}_\ell v(z_*)$, and `EnvSlopeM` with the two vanishing values turns this into

$$\mathcal{S}_\ell^- v(z_*) \le \mathcal{S}_\ell^- W(z_*) \qquad (0 < \ell < 1). \tag{505}$$

Step C: H against the slopes of v. From (502) and (503), H is the infimum of the backward slopes, so

$$H \le \mathcal{S}_\ell^- v(z_*) \qquad (\ell > 0). \tag{506}$$

In the other direction, $H_v(z_*)$ is by `EnvHamiltonian` the infimum of the *forward* slopes of v at z_*, so for every $\varepsilon > 0$ there is $\ell > 0$ with

$$\mathcal{S}_\ell^+ v(z_*) < H + \varepsilon. \tag{507}$$

Indeed, were $H + \varepsilon \le \mathcal{S}_\ell^+ v(z_*)$ for every $\ell > 0$, then $H + \varepsilon$ would be a lower bound for that family and hence $H + \varepsilon \le H_v(z_*) = H$ by (503), contradicting $\varepsilon > 0$.

Step D: H is a lower bound for the forward slopes of W. By 6, 7 and 8 the hypotheses of `EnvelopeSlopeOrder` hold for W on the open interval $I = (0, \infty)$. Its first clause, whose hypothesis $1 + \sigma > 0$ for $\sigma \in [0, \ell']$ is immediate, gives that

$$\mathcal{S}_\ell^+ W(z_*) \le \mathcal{S}_{\ell'}^+ W(z_*) \qquad (0 < \ell \le \ell'). \tag{508}$$

Fix h with $0 < h < 1$ and suppose, for contradiction, that $\mathcal{S}_h^+ W(z_*) < H$. Apply the second clause of `EnvelopeSlopeOrder` at z_* with this h and with $\varepsilon = (H - \mathcal{S}_h^+ W(z_*))/2 > 0$; its hypothesis $1 + \sigma > 0$ for $|\sigma| \le h$ holds because $h < 1$. It supplies $\delta > 0$ such that $\mathcal{S}_\ell^- W(z_*) \le \mathcal{S}_h^+ W(z_*) + \varepsilon$ whenever $0 < \ell < \delta$. Choose $\ell = \min\{\delta/2, 1/2\}$, so $0 < \ell < \delta$ and $\ell < 1$. Then (506), (505) and that bound give

$$H \;\le\; \mathcal{S}_\ell^- v(z_*) \;\le\; \mathcal{S}_\ell^- W(z_*) \;\le\; \mathcal{S}_h^+ W(z_*) + \frac{H - \mathcal{S}_h^+ W(z_*)}{2} = \frac{H + \mathcal{S}_h^+ W(z_*)}{2} \;<\; H,$$

the last inequality because $\mathcal{S}_h^+ W(z_*) < H$. This contradiction shows

$$H \le \mathcal{S}_h^+ W(z_*) \qquad (0 < h < 1). \tag{509}$$

For $h \ge 1$ the same bound follows from (509) at $1/2$ together with the monotonicity (508) from $1/2$ to h. Hence

$$H \le \mathcal{S}_h^+ W(z_*) \qquad (h > 0). \tag{510}$$

Step E: the limit. We verify the two halves of the order characterization of a limit for the function $\ell \mapsto \mathcal{S}_\ell^+ W(z_*)$ along $\ell \downarrow 0$.

If $a < H$, then by (510) $a < \mathcal{S}_\ell^+ W(z_*)$ for *every* $\ell > 0$, so this holds in particular for all ℓ near 0 with $\ell > 0$.

If $a > H$, apply (507) with $\varepsilon = a - H > 0$: there is $\ell_0 > 0$ with $\mathcal{S}_{\ell_0}^+ v(z_*) < H + (a - H) = a$, and then (504) gives $\mathcal{S}_{\ell_0}^+ W(z_*) < a$. For every ℓ with $0 < \ell < \ell_0$, the monotonicity (508) gives $\mathcal{S}_\ell^+ W(z_*) \le \mathcal{S}_{\ell_0}^+ W(z_*) < a$; and the interval $(0, \ell_0)$ is a neighbourhood of 0 within $(0, \infty)$.

Therefore $\mathcal{S}_\ell^+ W(z_*) \to H$ as $\ell \downarrow 0$, which with (503) is the second conclusion, for the same H. □

2.126. `ContactGapLimit`.

Lemma 2.126 (The contact gap in the limit of small steps)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and $0 \le \kappa < \infty$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Write $W = \mathcal{C}_t v - \kappa^2 t/2$ for the contact envelope of `ContactEnv` and J_ϵ for the propagation quantity of `JGap`. Then:*

(1) *for every $\epsilon > 0$ and every space-time point z,*

$$\begin{aligned} 2\epsilon J_\epsilon(z) = W(z) - v(z) &+ \big(\mathcal{P}_\epsilon W(z) - W(z)\big) \\ &+ \big(v(z) - \mathcal{M}_\epsilon v(z)\big); \end{aligned} \tag{511}$$

(2) *$2\epsilon J_\epsilon \to W - v$ uniformly on every compact subset of $F \times (0, \infty)$: for every such compact K and every $\delta > 0$ there is $\epsilon_0 > 0$ with*

$$\big|2\epsilon J_\epsilon(z) - \big(W(z) - v(z)\big)\big| \le \delta \qquad (0 < \epsilon < \epsilon_0,\ z \in K).$$

Proof. (1). By `JGap`, for $\epsilon > 0$,

$$2\epsilon J_\epsilon(z) = \mathcal{P}_\epsilon W(z) - \mathcal{M}_\epsilon v(z),$$

and the right side of (511) is

$$\big(W(z) - v(z)\big) + \big(\mathcal{P}_\epsilon W(z) - W(z)\big) + \big(v(z) - \mathcal{M}_\epsilon v(z)\big) = \mathcal{P}_\epsilon W(z) - \mathcal{M}_\epsilon v(z),$$

the terms $W(z)$, $v(z)$ cancelling in pairs. This proves the identity, for every $\epsilon > 0$ and every z.

(2). Let $K \subseteq F \times (0,\infty)$ be compact and let $\delta > 0$. By (511) it suffices to make both $|\mathcal{P}_\epsilon W(z) - W(z)|$ and $|v(z) - \mathcal{M}_\epsilon v(z)|$ at most $\delta/2$ uniformly on K for all small $\epsilon > 0$.

For v we apply part (3) of `QuadEnvAttained` with $I = \mathbb{R}$. Its hypotheses hold: v is continuous on all of space-time and is a solution of the limiting equation by `IsFitClass`, and it has subquadratic spatial growth on compact time intervals by `UniformSubquadratic`, applied with the centering point $(0,1)$. Hence there is $\delta_1 > 0$ with $|\mathcal{M}_\epsilon v(z) - v(z)| \le \delta/2$ for all $0 < \epsilon < \delta_1$ and all $z \in K$, the compact set K having all its time coordinates in $\mathbb{R}$.

For W we apply part (3) of `QuadEnvAttained` with $I = (0,\infty)$, and its two hypotheses for W are exactly the first two conclusions of `ContactEnvPositiveTimeRegularity`, which holds under the hypotheses on C_0, κ and v assumed here: that lemma states that W is continuous on $\{(y,\sigma) : \sigma > 0\}$ and that W has subquadratic spatial growth on compact time intervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`. Neither is a verbatim transfer from v, which is why they are cited rather than rederived here. The growth statement is not: the sandwich of `BaryEnvSandwich` gives only $v(x,t) - \kappa^2 t/2 \le W(x,t) \le v(x,t)$, so $|W|$ is bounded by $|v|$ plus an additive constant on a compact time interval, and an additive constant is absorbed into $\varepsilon|y|^2$ only once $|y|$ is large, so the growth radius must be enlarged. The continuity statement is not either: $W(z) = \mathcal{C}_\sigma v(z) - \kappa^2\sigma/2$ with σ the time coordinate of z itself, so it is the continuity of $(s,z) \mapsto \mathcal{C}_s v(z)$ from part (3) of `BaryEnvDuality` composed along $z \mapsto (\sigma, z)$, not a restriction of it to a fixed parameter. Since K is a compact subset of $F \times (0,\infty)$, all its time coordinates lie in $(0,\infty)$, so there is $\delta_2 > 0$ with $|\mathcal{P}_\epsilon W(z) - W(z)| \le \delta/2$ for all $0 < \epsilon < \delta_2$ and all $z \in K$.

Taking $\epsilon_0 = \min\{\delta_1, \delta_2\} > 0$ and using (511) gives the claim. □

2.127. `ContactWalkTimeRange`.

Lemma 2.127 (The walk's time coordinate stays in the box)**.** Let F be a finite-dimensional real inner-product space, let $\kappa, H \in \mathbb{R}$, let v be a real function on space-time, let $T \in \mathbb{N}$, $R \in \mathbb{R}$, $N \in \mathbb{N}$ and $\epsilon \in \mathbb{R}$, and let $(\Omega, \mathcal{M}, \mu)$ be a measure space with a filtration $(\mathcal{G}_n)_{n\in\mathbb{N}}$ carrying functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of `IsContactWalk`. Write $\delta_N = 2^{-N}$ and $z_n = (x_n, t_n)$, and assume $N \ge 1$ and $T \ge 1$. Then

$$\delta_N \le t_n(\omega) \le T \qquad \text{for every } n \in \mathbb{N} \text{ and every } \omega \in \Omega.$$

Proof. Clause numbers refer to `IsContactWalk`.

(G1) The grid. By clause 2 there is an integer $k \ge 0$ with $\epsilon = 2^{-k}$, and $0 < \epsilon \le \delta_N/4 = 2^{-N-2}$; since $i \mapsto 2^{-i}$ is decreasing, $k \ge N + 2$. Consequently $2^{-i} = 2^{k-i}\epsilon$ is an integer multiple of ϵ for every $i \le k$, because $2^{k-i}2^i = 2^k$ and $2^{-i}2^i = 1$. In particular $\delta_N = 2^{k-N}\epsilon$ and $1 = 2^k\epsilon$ are integer multiples of ϵ, using $N \le k$, and so is $T = (T2^k)\epsilon$.

(G2) The time coordinate moves by 0 *or* $\pm\epsilon$. Let n and ω be given. If $n \ge \tau(\omega)$ then clause 7 gives $z_{n+1} = z_n$, so $t_{n+1} = t_n$. If $n < \tau(\omega)$ then clause 6 makes (z_n^+, z_n^-) an optimizing step pair of size ϵ at z_n, so by `IsStepPair` the time coordinates of z_n^+ and z_n^- are $t_n + \epsilon$ and $t_n - \epsilon$; clause 7 says z_{n+1} is one of the two. In all cases $t_{n+1} \in \{t_n,\ t_n + \epsilon,\ t_n - \epsilon\}$.

(G3) Every time coordinate is an integer multiple of ϵ. By induction on n: clause 5 gives $t_0 = 1$, an integer multiple of ϵ by (G1), and the inductive step is (G2), since $m\epsilon \pm \epsilon$ is again an integer multiple of ϵ.

A strict inequality between multiples of ϵ *leaves a gap* ϵ. If $x = m_1\epsilon$ and $c = m_2\epsilon$ with $m_1, m_2 \in \mathbb{Z}$ and $x < c$, then dividing by $\epsilon > 0$ gives $m_1 < m_2$ in $\mathbb{R}$, hence in $\mathbb{Z}$, hence $m_1 + 1 \le m_2$; multiplying by $\epsilon > 0$ gives $x + \epsilon \le c$.

(G4) The conclusion. We induct on n. For $n = 0$ clause 5 gives $t_0 = 1$, and $\delta_N \le 2^{-1} = \frac{1}{2} \le 1 \le T$ by $N \ge 1$ and $T \ge 1$.

For the step assume $\delta_N \le t_n(\omega) \le T$. If $n \ge \tau(\omega)$ then clause 7 gives $t_{n+1}(\omega) = t_n(\omega)$ and the bounds persist. If $n < \tau(\omega)$ then clause 10 gives the strict inequalities $\delta_N < t_n(\omega) < T$. By (G1), (G3) and the gap statement just proved, applied once to $\delta_N < t_n(\omega)$ and once to $t_n(\omega) < T$,

$$\delta_N + \epsilon \le t_n(\omega), \qquad t_n(\omega) + \epsilon \le T.$$

Now examine the three cases of (G2). If $t_{n+1}(\omega) = t_n(\omega)$ the bounds are the inductive hypothesis. If $t_{n+1}(\omega) = t_n(\omega) + \epsilon$ then $t_{n+1}(\omega) \le T$ by the second display, and $t_{n+1}(\omega) \ge t_n(\omega) \ge \delta_N$ because $\epsilon > 0$ by clause 2. If $t_{n+1}(\omega) = t_n(\omega) - \epsilon$ then $t_{n+1}(\omega) \ge \delta_N$ by the first display, and $t_{n+1}(\omega) \le t_n(\omega) \le T$ because $\epsilon > 0$. In every case $\delta_N \le t_{n+1}(\omega) \le T$. □

2.128. `TailEventSelection`.

Lemma 2.128 (Selection from events of uniformly positive measure)**.** Let $(X, \mathcal{M}, \mu)$ be a finite measure space, let $c > 0$ be an element of $[0, \infty]$, let $E_j \in \mathcal{M}$ for $j \in \mathbb{N}$ satisfy

$$\mu(E_j) \ge c \qquad (j \ge 1),$$

and let $G \subseteq X$ satisfy $\mu(X \setminus G) = 0$. Then there is a point $x \in G$ for which the set $\{j \ge 1 : x \in E_j\}$ is infinite.

Proof. No independence of the events E_j is assumed or used.

For $h \in \mathbb{N}$ put

$$U_h = \bigcup_{j \ge h} E_j \in \mathcal{M}.$$

The tail unions are antitone with measure at least c. If $a \le b$ then every $j \ge b$ satisfies $j \ge a$, so $U_b \subseteq U_a$; thus $h \mapsto U_h$ is antitone. For each h the index $\max(h, 1)$ satisfies both $\max(h, 1) \ge h$ and $\max(h, 1) \ge 1$, so $E_{\max(h,1)} \subseteq U_h$ and the hypothesis applies to it, giving

$$c \le \mu\big(E_{\max(h,1)}\big) \le \mu(U_h).$$

The intersection of the tail unions has measure at least c. The U_h are measurable, antitone, and $\mu(U_0) \le \mu(X) < \infty$ because μ is finite. By continuity of a finite measure along a decreasing sequence of measurable sets,

$$\mu\Big(\bigcap_{h \in \mathbb{N}} U_h\Big) = \inf_{h \in \mathbb{N}} \mu(U_h) \ge c,$$

the last inequality because c is a lower bound for every $\mu(U_h)$ by the previous paragraph.

A point of G in the intersection. Since $\mu(X \setminus G) = 0$ we have $\mu(A \cap G) = \mu(A)$ for every measurable A, so

$$\mu\Big(\Big(\bigcap_h U_h\Big) \cap G\Big) = \mu\Big(\bigcap_h U_h\Big) \ge c > 0.$$

A set of nonzero measure is nonempty, so we may choose $x \in \big(\bigcap_h U_h\big) \cap G$. In particular $x \in G$.

The index set is infinite. Let $S = \{j \ge 1 : x \in E_j\}$ and suppose S were bounded above by some $b \in \mathbb{N}$. Since $x \in U_{b+1}$ there is $j \ge b + 1$ with $x \in E_j$; this j satisfies $j \ge b + 1 \ge 1$, hence $j \in S$, hence $j \le b$, contradicting $j \ge b + 1$. So S is not bounded above, and a subset of $\mathbb{N}$ that is not bounded above is infinite. □

2.129. `MonotoneLimitFiniteAE`.

Lemma 2.129 (An increasing sequence with bounded integrals is a.e. bounded)**.** Let $(X, \mathcal{M}, \mu)$ be a measure space, let $M \in \mathbb{R}$, and let $g_i : X \to \mathbb{R}$, $i \in \mathbb{N}$, be measurable functions such that for every $i \in \mathbb{N}$ and every $x \in X$

$$0 \le g_i(x), \qquad g_i(x) \le g_{i+1}(x),$$

such that each g_i is μ-integrable, and such that

$$\int g_i \, d\mu \le M \qquad (i \in \mathbb{N}).$$

Then for μ-almost every $x \in X$ the set $\{g_i(x) : i \in \mathbb{N}\}$ is bounded above.

Proof. Put $f_i(x) = \iota\big(g_i(x)\big)$, where $\iota : \mathbb{R} \to [0,\infty]$ sends s to s for $s \geq 0$ and to 0 for $s < 0$. Each f_i is measurable, and $i \mapsto f_i$ is nondecreasing: for $a \leq b$ an induction on b starting at $b = a$ and using $g_n(x) \leq g_{n+1}(x)$ at each step gives $g_a(x) \leq g_b(x)$, and ι is nondecreasing.

The upper integrals are uniformly bounded. Fix i. Since g_i is integrable and $g_i \geq 0$ everywhere, the upper integral of f_i equals ι of the integral of g_i:

$$\int^* f_i \, d\mu = \iota\Big(\int g_i \, d\mu\Big) \leq \iota(M),$$

the inequality because ι is nondecreasing and $\int g_i \, d\mu \leq M$.

The pointwise supremum has finite upper integral. By the monotone convergence theorem for upper integrals of a nondecreasing sequence of measurable $[0,\infty]$-valued functions,

$$\int^* \Big(\sup_i f_i\Big) d\mu = \sup_i \int^* f_i \, d\mu \leq \iota(M) < \infty.$$

Conclusion. A measurable $[0,\infty]$-valued function with finite upper integral is finite almost everywhere, so for μ-almost every x we have $s(x) = \sup_i f_i(x) < \infty$. Fix such an x and let $r \in [0,\infty)$ be the real number $s(x)$. For every i we have $\iota(g_i(x)) = f_i(x) \leq s(x)$, and since $g_i(x) \geq 0$ this says exactly $g_i(x) \leq r$. Hence r is an upper bound for $\{g_i(x) : i \in \mathbb{N}\}$, which is therefore bounded above. □

2.130. `JUpperBound`.

Lemma 2.130 (Upper bound for the propagation quantity)**.** *Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$ and $0 \leq \kappa < \infty$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let $W = \mathcal{C}_t v - \kappa^2 t/2$ be the contact envelope of `ContactEnv` and J_ϵ the propagation quantity of `JGap`. Then*

$$J_\epsilon(z) \leq \frac{1}{2}\big(\mathcal{S}_\epsilon^+ v(z) + \mathcal{S}_\epsilon^- v(z)\big) \tag{512}$$

for every $\epsilon > 0$ and every $z = (x,t)$ with $t > 0$, the slopes being those of `EnvSlopeP` and `EnvSlopeM`.

Proof. Fix $\epsilon > 0$ and $z = (x,t)$ with $t > 0$.

Step 1: $W \leq v$ at time $t + \epsilon$. By `BaryEnvSandwich`, applied with $I = \mathbb{R}$ at each positive parameter s, we have $\mathcal{C}_s v(y,\sigma) - v(y,\sigma) \leq \kappa^2 s/2$ for every y, σ and $s > 0$; taking $s = \sigma > 0$ and using `ContactEnv` gives $W(y,\sigma) \leq v(y,\sigma)$ for every $y \in F$ and every $\sigma > 0$. The hypotheses of `BaryEnvSandwich` hold: v is continuous on all of space-time and satisfies the horizontal Jensen bound with constant $\kappa \geq 0$ at every time by `IsFitClass` and `HasJensenBound`, and v has subquadratic spatial growth on compact time intervals by `UniformSubquadratic`, applied with the centering point $(0,1)$ and the constant C_0. In particular, since $t > 0$ and $\epsilon > 0$, we have $t + \epsilon > 0$ and $W(y, t+\epsilon) \leq v(y, t+\epsilon)$ for all $y \in F$.

Step 2: monotonicity of the forward envelope. By `QuadEnvP`, for $\epsilon \neq 0$,

$$\mathcal{P}_\epsilon W(z) = \sup_{y \in F}\left\{W(y,t+\epsilon) - \frac{|y-x|^2}{2\epsilon}\right\}, \qquad \mathcal{P}_\epsilon v(z) = \sup_{y \in F}\left\{v(y,t+\epsilon) - \frac{|y-x|^2}{2\epsilon}\right\}.$$

Write

$$A_W = \left\{W(y,t+\epsilon) - \frac{|y-x|^2}{2\epsilon} : y \in F\right\}, \qquad A_v = \left\{v(y,t+\epsilon) - \frac{|y-x|^2}{2\epsilon} : y \in F\right\}$$

for the two competitor sets, so that $\mathcal{P}_\epsilon W(z) = \sup A_W$ and $\mathcal{P}_\epsilon v(z) = \sup A_v$.

The lemma `QuadEnvAttained` applies to v with $I = \mathbb{R}$: the interval $I = \mathbb{R}$ is open, and v is continuous on $\Omega = \{(y,\sigma) : \sigma \in \mathbb{R}\}$ and has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`, which are exactly the two facts verified in Step 1. Its part (1) is used once, at the centre $z = (x,t)$ with the length $\epsilon > 0$, where the admissibility conditions $t \in \mathbb{R}$ and $t - \epsilon \in \mathbb{R}$ are vacuous because $I = \mathbb{R}$ contains every real

number; the *backward* instance gives that the infimum defining $\mathcal{M}_\epsilon v(z)$ is attained, and that is the instance used in Step 3. The corresponding forward statement is not needed: what Step 2 requires of $\mathcal{P}_\epsilon v(z)$ is domination of the competitors, not attainment, and that is supplied below by `EnvelopeCompetitorBounds` instead.

We now check that $\mathcal{P}_\epsilon W(z)$ is a genuine least upper bound, and compare the two suprema. First, taking the competitor $y = x$ shows that A_W contains the real number $W(x, t+\epsilon) - 0 = W(x, t+\epsilon)$, so A_W is nonempty. Second, by Step 1 we have $W(y, t+\epsilon) \le v(y, t+\epsilon)$ for every $y \in F$, since $t+\epsilon > 0$ was established there from $t > 0$ and $\epsilon > 0$; subtracting the common term $|y-x|^2/(2\epsilon)$, each element of A_W is at most the corresponding element of A_v. Third, every element of A_v is at most $\mathcal{P}_\epsilon v(z)$. This is NOT read off the attainment just named: attainment supplies one element of A_v whose value equals $\mathcal{P}_\epsilon v(z)$, and since $\mathcal{P}_\epsilon v(z)$ is a supremum, which the preamble's convention sets to 0 when the set is empty or unbounded above, an equality with one element does not by itself bound the others. The domination is instead part (1) of `EnvelopeCompetitorBounds`, whose first clause states exactly that $\mathcal{P}_\ell v(z)$ dominates every competitor value; we apply it at this centre z and this length $\epsilon > 0$, its hypotheses being the continuity and subquadratic growth of v verified in Step 1 together with $t \in I$ and $t+\epsilon \in I$, both automatic since $I = \mathbb{R}$. In particular A_v is nonempty and bounded above by $\mathcal{P}_\epsilon v(z)$. Fourth, chaining the second and third observations, every element of A_W is at most the real number $\mathcal{P}_\epsilon v(z)$, so A_W is bounded above by $\mathcal{P}_\epsilon v(z)$. Fifth, A_W is therefore nonempty and bounded above, so $\sup A_W$ is the genuine least upper bound of A_W and the convention of the preamble – which sets $\sup A = 0$ when A is empty or unbounded above – is not invoked: that branch is never entered, and $\mathcal{P}_\epsilon W(z) = \sup A_W$ is a real number. Being the least upper bound of a set bounded above by $\mathcal{P}_\epsilon v(z)$, it satisfies $\sup A_W \le \mathcal{P}_\epsilon v(z)$. Hence

$$\mathcal{P}_\epsilon W(z) \le \mathcal{P}_\epsilon v(z).$$

Step 3: conclusion. By `JGap`, $J_\epsilon(z) = \big(\mathcal{P}_\epsilon W(z) - \mathcal{M}_\epsilon v(z)\big)/(2\epsilon)$. By the backward instance of part (1) of `QuadEnvAttained` named in Step 2, the infimum defining $\mathcal{M}_\epsilon v(z)$ is attained, so $\mathcal{M}_\epsilon v(z)$ is a real number and the convention of the preamble is not invoked for it either. Subtracting that fixed real number from both sides of the inequality of Step 2 preserves it, and dividing by 2ϵ preserves it because $\epsilon > 0$. Hence

$$J_\epsilon(z) = \frac{\mathcal{P}_\epsilon W(z) - \mathcal{M}_\epsilon v(z)}{2\epsilon} \le \frac{\mathcal{P}_\epsilon v(z) - \mathcal{M}_\epsilon v(z)}{2\epsilon}.$$

Inserting $v(z)$ and splitting,

$$\frac{\mathcal{P}_\epsilon v(z) - \mathcal{M}_\epsilon v(z)}{2\epsilon} = \frac{1}{2}\left(\frac{\mathcal{P}_\epsilon v(z) - v(z)}{\epsilon} + \frac{v(z) - \mathcal{M}_\epsilon v(z)}{\epsilon}\right) = \frac{1}{2}\big(\mathcal{S}_\epsilon^+ v(z) + \mathcal{S}_\epsilon^- v(z)\big),$$

by `EnvSlopeP` and `EnvSlopeM`. This is (512). □

2.131. `JBoxUpperBound`.

Lemma 2.131 (A uniform upper bound for the propagation quantity on a compact set)**.** Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and $\kappa \ge 0$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $z_* = (0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let J_ϵ be the propagation quantity of `JGap` formed from v and κ, and let $\mathcal{B}$ be a compact subset of space-time. Then there is a constant $U \in \mathbb{R}$ with

$$J_\epsilon(z) \le U \qquad \text{for every } z = (x,t) \in \mathcal{B} \text{ with } t > 0 \text{ and every } 0 < \epsilon \le 1.$$

Proof. Write $\mathcal{P}_\ell$, $\mathcal{M}_\ell$ for the quadratic envelopes and $\mathcal{S}_\ell^+$, $\mathcal{S}_\ell^-$ for the envelope slopes of `EnvSlopeP` and `EnvSlopeM`.

The two standing facts about v. By `IsFitClass` the function v is a viscosity solution of the limiting equation on all of space-time, and in particular it is continuous there. By `UniformSubquadratic`, applied with the centering point $(0,1)$ and the constant C_0, for every compact $J \subseteq \mathbb{R}$ and every $\varepsilon > 0$ there is $M > 0$ such that $|v(x,t)| \le \varepsilon |x|^2$ whenever $|x| \ge M$ and $t \in J$; that is, v has subquadratic spatial growth on compact time subintervals of $\mathbb{R}$ in the sense

of SubquadraticGrowth. Hence QuadEnvAttained and EnvHamiltonianSolution are available for v with $I = \mathbb{R}$, for which every time and every shifted time is admissible.

The averaged slope at length one is continuous. Put

$$\Phi(z) = \tfrac{1}{2}\big(\mathcal{S}_1^+ v(z) + \mathcal{S}_1^- v(z)\big).$$

By EnvSlopeP and EnvSlopeM this equals $\big(\mathcal{P}_1 v(z) - \mathcal{M}_1 v(z)\big)/2$. By parts (3) and (4) of QuadEnvAttained for v with $I = \mathbb{R}$, the maps $(\ell, z) \mapsto \mathcal{P}_\ell v(z)$ and $(\ell, z) \mapsto \mathcal{M}_\ell v(z)$ are continuous at $(1, z_0)$ for every z_0, the required admissibility of t_0 and $t_0 \pm 1$ being automatic for $I = \mathbb{R}$. Composing with the continuous map $z \mapsto (1, z)$ shows that $z \mapsto \mathcal{P}_1 v(z)$ and $z \mapsto \mathcal{M}_1 v(z)$ are continuous at every point, hence continuous; therefore Φ is continuous on all of space-time.

The constant. If $\mathcal{B} = \emptyset$ the assertion holds with $U = 0$, there being no $z \in \mathcal{B}$. Otherwise $\mathcal{B}$ is a nonempty compact set and Φ is continuous on it, so Φ attains a maximum on $\mathcal{B}$: there is $z_m \in \mathcal{B}$ with $\Phi(z) \le \Phi(z_m)$ for every $z \in \mathcal{B}$. Put $U = \Phi(z_m)$.

The bound. Let $z = (x, t) \in \mathcal{B}$ with $t > 0$ and let $0 < \epsilon \le 1$. By JUpperBound,

$$J_\epsilon(z) \le \tfrac{1}{2}\big(\mathcal{S}_\epsilon^+ v(z) + \mathcal{S}_\epsilon^- v(z)\big),$$

its hypotheses being the present $C_0 \ge 1$, $\kappa \ge 0$, $v \in \mathcal{A}_{m,C_0}$, the centering, the Jensen bound, $\epsilon > 0$ and $t > 0$. By part (1) of EnvHamiltonianSolution for v with $I = \mathbb{R}$, applied with $\ell = \epsilon$ and $\ell' = 1$ — legitimate since $0 < \epsilon \le 1$ and every $t + \sigma$ lies in $I = \mathbb{R}$ —

$$\mathcal{S}_\epsilon^+ v(z) \le \mathcal{S}_1^+ v(z), \qquad \mathcal{S}_\epsilon^- v(z) \le \mathcal{S}_1^- v(z).$$

Averaging the two inequalities and combining with the previous display gives $J_\epsilon(z) \le \Phi(z)$, and $\Phi(z) \le \Phi(z_m) = U$ because $z \in \mathcal{B}$. □

2.132. `ContactWalkMarkLevels`.

Lemma 2.132 (The marks sit on their own time level)**.** Let F be a finite-dimensional real inner-product space, let $C_0, \kappa, H \in \mathbb{R}$, let v be a real function on space-time, let $T \in \mathbb{N}$, $R \in \mathbb{R}$, $N \in \mathbb{N}$ and $\epsilon \in \mathbb{R}$, and let $(\Omega, \mathcal{M}, \mu)$ be a measure space with a filtration $(\mathcal{G}_n)_{n\in\mathbb{N}}$ carrying functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of IsContactWalk, together with functions σ_j, p_j, a_j which are mark data for $(C_0, v, N, \mathcal{G}, z, \tau)$ in the sense of IsMarkData. Write $\delta_i = 2^{-i}$ and $z_n = (x_n, t_n)$. Then for every j with $1 \le j \le N$:

(i) $t_{\sigma_j} \ge \delta_j$ everywhere on Ω;
(ii) $t_{\sigma_j} = \delta_j$ at every point where $t_\tau = \delta_N$.

Proof. Clause numbers refer to IsContactWalk unless IsMarkData is named. Fix j with $1 \le j \le N$.

(G1) The grid. By clause 2 there is an integer $k \ge 0$ with $\epsilon = 2^{-k}$, and $0 < \epsilon \le \delta_N/4 = 2^{-N-2}$; since $i \mapsto 2^{-i}$ is decreasing, $k \ge N + 2$. Consequently, for every $i \le k$ the number $\delta_i = 2^{-i}$ is an integer multiple of ϵ, namely $\delta_i = 2^{k-i}\epsilon$, because $2^{k-i}2^i = 2^k$ and $2^{-i}2^i = 1$. In particular $1 = \delta_0 = 2^k\epsilon$ and $\delta_j = 2^{k-j}\epsilon$ are integer multiples of ϵ, using $j \le N \le k$.

(G2) The time coordinate moves by 0 or $\pm\epsilon$. Let n and ω be given. If $n \ge \tau(\omega)$ then clause 7 gives $z_{n+1} = z_n$, so $t_{n+1} = t_n$. If $n < \tau(\omega)$ then clause 6 makes (z_n^+, z_n^-) an optimizing step pair of size ϵ at z_n, so by IsStepPair the time coordinates of z_n^+ and z_n^- are $t_n + \epsilon$ and $t_n - \epsilon$; clause 7 says that z_{n+1} is one of the two, so $t_{n+1} \in \{t_n + \epsilon,\ t_n - \epsilon\}$. In all cases $t_{n+1} \in \{t_n,\ t_n + \epsilon,\ t_n - \epsilon\}$.

(G3) Every time coordinate is an integer multiple of ϵ. By induction on n. For $n = 0$, clause 5 gives $z_0 = (0, 1)$, so $t_0 = 1$, which is an integer multiple of ϵ by (G1). If $t_n = m\epsilon$ with $m \in \mathbb{Z}$ then by (G2) t_{n+1} is one of $m\epsilon$, $(m+1)\epsilon$, $(m-1)\epsilon$.

(G4) Two multiples of ϵ in a half-open interval of length ϵ coincide. Let $x = m_1\epsilon$ and $c = m_2\epsilon$ with $m_1, m_2 \in \mathbb{Z}$ and $c - \epsilon < x \le c$. Dividing by $\epsilon > 0$ gives $m_2 - 1 < m_1 \le m_2$ in $\mathbb{R}$, hence in $\mathbb{Z}$, so $m_1 = m_2$ and $x = c$.

(G5) A discrete intermediate value property. Let $\omega \in \Omega$, let c be an integer multiple of ϵ, let $b \in \mathbb{N}$ and suppose $t_b(\omega) \le c \le t_0(\omega)$. Then $t_n(\omega) = c$ for some $n \le b$. We induct on b. For $b = 0$ we have $t_0(\omega) \le c \le t_0(\omega)$, so $t_0(\omega) = c$. For the step, suppose the claim holds for b and let $t_{b+1}(\omega) \le c \le t_0(\omega)$. If $t_b(\omega) \le c$ the inductive hypothesis gives $n \le b \le b+1$ with $t_n(\omega) = c$. Otherwise $t_{b+1}(\omega) \le c < t_b(\omega)$; of the three possibilities of (G2) only $t_{b+1}(\omega) = t_b(\omega) - \epsilon$ is

compatible with $t_{b+1}(\omega) \le c < t_b(\omega)$, so $t_{b+1}(\omega) = t_b(\omega) - \epsilon > c - \epsilon$. Thus $t_{b+1}(\omega)$ and c are integer multiples of ϵ — by (G3) and the hypothesis on c — lying in $(c-\epsilon, c]$, so $t_{b+1}(\omega) = c$ by (G4), and $n = b+1$ serves.

(i). Note $\delta_j \le \delta_0 = 1 = t_0(\omega)$ for every ω, by clause 5. Suppose $t_{\sigma_j}(\omega) < \delta_j$ for some ω. Since δ_j is an integer multiple of ϵ by (G1) and $t_{\sigma_j(\omega)}(\omega) \le \delta_j \le t_0(\omega)$, (G5) applied with $b = \sigma_j(\omega)$ and $c = \delta_j$ gives an $n \le \sigma_j(\omega)$ with $t_n(\omega) = \delta_j$. As $t_{\sigma_j}(\omega) < \delta_j$ we have $n \ne \sigma_j(\omega)$, so $n < \sigma_j(\omega)$; but then clause 3 of `IsMarkData` gives $t_n(\omega) \ne \delta_j$, a contradiction. Hence $t_{\sigma_j} \ge \delta_j$ everywhere.

(ii). Let ω satisfy $t_\tau(\omega) = \delta_N$. Since $j \le N$ we have $\delta_N \le \delta_j$, so $t_{\tau(\omega)}(\omega) = \delta_N \le \delta_j \le 1 = t_0(\omega)$, and (G5) with $b = \tau(\omega)$ and $c = \delta_j$ gives an $n \le \tau(\omega)$ with $t_n(\omega) = \delta_j$. Let n_0 be the least index with $t_{n_0}(\omega) = \delta_j$; then $n_0 \le n \le \tau(\omega)$. If $n_0 < \sigma_j(\omega)$ then clause 3 of `IsMarkData` applied at n_0 would give $t_{n_0}(\omega) \ne \delta_j$; hence $\sigma_j(\omega) \le n_0$. By clause 4 of `IsMarkData` either $t_{\sigma_j}(\omega) = \delta_j$, which is the claim, or $\sigma_j(\omega) = \tau(\omega)$, in which case $n_0 \le \tau(\omega) = \sigma_j(\omega) \le n_0$ forces $\sigma_j(\omega) = n_0$ and therefore $t_{\sigma_j}(\omega) = \delta_j$ again. □

2.133. `FitSetClosed`.

Lemma 2.133 (Fits of a fixed radius form a closed set)**.** Let F be a finite-dimensional real inner-product space, let $C_0 \in \mathbb{R}$, let v be a continuous real function on space-time and let $r > 0$. Then the set of triples (z, p, a) in $(F \times \mathbb{R}) \times F \times \mathbb{R}$ such that (p, a) is a fit for v at z of radius r with constant C_0, in the sense of `IsFit`, is closed.

Proof. By `IsFit`, (p, a) is a fit for v at z of radius r with constant C_0 if and only if

$$\Big| v\big(z + (h,\ell)\big) - v(z) - \langle p, h\rangle - \big(a - \tfrac{1}{2}|p|^2\big)\ell \Big| \le C_0 r \tag{513}$$

for every increment (h, ℓ) with $|\ell| \le r$ and $|h - \ell p| \le r$. The constraint on the increments involves p, so the domain of the quantified condition varies with the point of the set under consideration. We remove that dependence by reparameterizing the increments: for $\theta \in \mathbb{R}$ and $\zeta \in F$ put

$$\ell(\theta) = r\theta, \qquad h(\theta, \zeta, p) = r\theta\, p + r\zeta.$$

The reparameterization is a bijection onto the admissible increments. Fix (z, p, a). If $|\theta| \le 1$ and $|\zeta| \le 1$ then $|\ell(\theta)| = r|\theta| \le r$ and $|h(\theta,\zeta,p) - \ell(\theta)p| = |r\zeta| = r|\zeta| \le r$, so $(h(\theta,\zeta,p), \ell(\theta))$ is an admissible increment. Conversely, if (h, ℓ) satisfies $|\ell| \le r$ and $|h - \ell p| \le r$, put $\theta = \ell/r$ and $\zeta = r^{-1}(h - \ell p)$, which is legitimate because $r > 0$; then $|\theta| = |\ell|/r \le 1$ and $|\zeta| = |h - \ell p|/r \le 1$, while $\ell(\theta) = r(\ell/r) = \ell$ and $h(\theta,\zeta,p) = \ell p + (h - \ell p) = h$. Hence, for every (z, p, a), the condition of (513) over all admissible increments is equivalent to the same condition over the increments $(h(\theta,\zeta,p), \ell(\theta))$ with $|\theta| \le 1$ and $|\zeta| \le 1$.

Each reparameterized condition is closed. Fix θ with $|\theta| \le 1$ and ζ with $|\zeta| \le 1$, and consider

$$S_{\theta,\zeta} = \Big\{ (z,p,a) : \big| v\big(z + (r\theta p + r\zeta,\ r\theta)\big) - v(z) - \langle p,\ r\theta p + r\zeta\rangle - \big(a - \tfrac{1}{2}|p|^2\big) r\theta \big| \le C_0 r \Big\}.$$

The map $(z,p,a) \mapsto v(z + (r\theta p + r\zeta, r\theta)) - v(z) - \langle p, r\theta p + r\zeta\rangle - (a - \frac{1}{2}|p|^2) r\theta$ is continuous, being built from the continuous v, the continuous vector space operations, the continuous inner product and the continuous norm, with θ, ζ and r fixed; composing with the continuous absolute value and taking the preimage of the closed half-line $(-\infty, C_0 r]$ shows that $S_{\theta,\zeta}$ is closed.

Conclusion. By the previous two paragraphs the set in question is

$$\bigcap_{|\theta| \le 1} \ \bigcap_{|\zeta| \le 1} S_{\theta,\zeta},$$

an intersection of closed sets, hence closed. □

2.134. `QuadShiftCompleteSquare`.

Lemma 2.134 (Completing the square across a two-leg quadratic cost)**.** *Let F be a finite-dimensional real inner-product space and let $t, \ell > 0$. Then for all $y, x \in F$, writing $s = t + \ell$,*

$$\frac{|y|^2}{2s} - \frac{|y - x|^2}{2\ell} - \frac{|x|^2}{2t} = -\frac{t}{2\ell s}\Big| y - \frac{s}{t} x \Big|^2.$$

Proof. Since $t > 0$ and $\ell > 0$ we have $s = t + \ell > 0$, so each of t, ℓ and s is nonzero and every division below is legitimate.

Both sides are quadratic in (y, x), so it suffices to expand each in the three quantities $|y|^2$, $\langle y, x\rangle$ and $|x|^2$ and compare coefficients. In an inner product space $|u-w|^2 = |u|^2 - 2\langle u, w\rangle + |w|^2$, so

$$|y - x|^2 = |y|^2 - 2\langle y, x\rangle + |x|^2,$$

and, applying the same identity with $w = (s/t)x$ and using $\langle y, (s/t)x\rangle = (s/t)\langle y, x\rangle$ together with $|(s/t)x|^2 = (s/t)^2|x|^2$,

$$\Big|y - \frac{s}{t}x\Big|^2 = |y|^2 - 2\,\frac{s}{t}\langle y, x\rangle + \frac{s^2}{t^2}|x|^2.$$

Substituting these two expansions, the left-hand side becomes

$$\Big(\frac{1}{2s} - \frac{1}{2\ell}\Big)|y|^2 + \frac{1}{\ell}\langle y, x\rangle + \Big(-\frac{1}{2\ell} - \frac{1}{2t}\Big)|x|^2,$$

and the right-hand side becomes

$$-\frac{t}{2\ell s}|y|^2 + \frac{t}{2\ell s}\cdot\frac{2s}{t}\langle y, x\rangle - \frac{t}{2\ell s}\cdot\frac{s^2}{t^2}|x|^2.$$

We compare the three coefficients.

For $|y|^2$: on the left $\dfrac{1}{2s} - \dfrac{1}{2\ell} = \dfrac{\ell - s}{2\ell s} = \dfrac{-t}{2\ell s}$, using $s = t + \ell$ so that $\ell - s = -t$; this is the coefficient on the right.

For $\langle y, x\rangle$: on the left it is $\dfrac{1}{\ell}$, and on the right $\dfrac{t}{2\ell s}\cdot\dfrac{2s}{t} = \dfrac{1}{\ell}$.

For $|x|^2$: on the left $-\dfrac{1}{2\ell} - \dfrac{1}{2t} = -\dfrac{t+\ell}{2\ell t} = -\dfrac{s}{2\ell t}$, again by $s = t + \ell$; on the right $-\dfrac{t}{2\ell s}\cdot\dfrac{s^2}{t^2} = -\dfrac{s}{2\ell t}$.

All three coefficients agree, so the two sides are equal for all $y, x \in F$. □

2.135. `PartialSupConcaveOfAttained`.

Lemma 2.135 (An attained partial supremum of a jointly concave function is concave)**.** *Let E and G be real vector spaces, let $\Phi : E \times G \to \mathbb{R}$ be concave on $E \times G$, and let $\Lambda : E \to \mathbb{R}$ satisfy*

$$\Phi(x, y) \le \Lambda(x) \qquad (x \in E,\ y \in G),$$

and suppose that for every $x \in E$ there is $y \in G$ with $\Lambda(x) = \Phi(x, y)$. Then Λ is concave on E.

Proof. The set E is convex, so it remains to verify the concavity inequality. Let $x_0, x_1 \in E$ and let $a, b \ge 0$ with $a + b = 1$.

By the attainment hypothesis applied at x_0 and at x_1, choose $y_0, y_1 \in G$ with

$$\Lambda(x_0) = \Phi(x_0, y_0), \qquad \Lambda(x_1) = \Phi(x_1, y_1).$$

The points (x_0, y_0) and (x_1, y_1) both lie in $E \times G$, on which Φ is concave, so

$$a\,\Phi(x_0, y_0) + b\,\Phi(x_1, y_1) \le \Phi\big(a(x_0, y_0) + b(x_1, y_1)\big).$$

Addition and scalar multiplication on the product $E \times G$ are performed coordinatewise, so

$$a(x_0, y_0) + b(x_1, y_1) = \big(ax_0 + bx_1,\ ay_0 + by_1\big).$$

Finally the pair $\big(ax_0 + bx_1,\ ay_0 + by_1\big)$ is one admissible pair for the upper bound hypothesis at the point $ax_0 + bx_1$, which therefore gives

$$\Phi\big(ax_0 + bx_1,\ ay_0 + by_1\big) \le \Lambda\big(ax_0 + bx_1\big).$$

Chaining the three displays and substituting the two attainment identities,

$$a\,\Lambda(x_0) + b\,\Lambda(x_1) = a\,\Phi(x_0, y_0) + b\,\Phi(x_1, y_1) \le \Phi\big(ax_0 + bx_1,\ ay_0 + by_1\big) \le \Lambda\big(ax_0 + bx_1\big),$$

which is the required inequality. Hence Λ is concave on E. □

2.136. `QuadEnvPConcaveCorrection`.

Lemma 2.136 (The forward envelope is concave after its own quadratic correction)**.** *Let F be a finite-dimensional real inner-product space, let f be a real function on the space-time $\ell^2(F\times\mathbb{R})$, let $t,\ell>0$ and put $s=t+\ell$. Assume*

(1) *the function $y\mapsto f(y,s)-|y|^2/(2s)$ is concave on F;*
(2) *for every $x\in F$ there is $y\in F$ with $\mathcal{P}_\ell f(x,t)=f(y,s)-|y-x|^2/(2\ell)$, where $\mathcal{P}_\ell$ is the forward envelope of `QuadEnvP`;*
(3) *for all $x,y\in F$ one has $f(y,s)-|y-x|^2/(2\ell)\le\mathcal{P}_\ell f(x,t)$.*

Then

$$x\longmapsto \mathcal{P}_\ell f(x,t)-\frac{|x|^2}{2t}$$

is concave on F.

Proof. Put $a=\dfrac{t}{2\ell s}$ and $c=\dfrac{s}{t}$; since $t,\ell>0$ we have $s>0$ and $a>0$. Define

$$\Phi(x,y)=\Big(f(y,s)-\frac{|y|^2}{2s}\Big)-a\,|y-cx|^2\qquad(x,y\in F),$$

and write $\Lambda(x)=\mathcal{P}_\ell f(x,t)-|x|^2/(2t)$.

Step 1: a convex-combination identity for the squared norm. Let $u_0,u_1\in F$ and let $\alpha,\beta\in\mathbb{R}$ with $\alpha+\beta=1$. Expanding with $|u+w|^2=|u|^2+2\langle u,w\rangle+|w|^2$, $|\lambda u|^2=\lambda^2|u|^2$ and $\langle\lambda u,\mu w\rangle=\lambda\mu\langle u,w\rangle$,

$$|\alpha u_0+\beta u_1|^2=\alpha^2|u_0|^2+2\alpha\beta\langle u_0,u_1\rangle+\beta^2|u_1|^2.$$

Substituting $\beta=1-\alpha$ and $\alpha=1-\beta$ in the first and third coefficients, $\alpha^2=\alpha-\alpha\beta$ and $\beta^2=\beta-\alpha\beta$, so

$$|\alpha u_0+\beta u_1|^2=\alpha|u_0|^2+\beta|u_1|^2-\alpha\beta|u_0-u_1|^2,\tag{514}$$

using $|u_0-u_1|^2=|u_0|^2-2\langle u_0,u_1\rangle+|u_1|^2$.

Step 2: Φ is jointly concave on $F\times F$. The set $F\times F$ is convex. Let $p_0=(x_0,y_0)$ and $p_1=(x_1,y_1)$ lie in $F\times F$ and let $\alpha,\beta\ge0$ with $\alpha+\beta=1$.

The first summand of Φ depends only on the second coordinate, and $(\alpha p_0+\beta p_1)$ has second coordinate $\alpha y_0+\beta y_1$ because operations on the product are coordinatewise; so hypothesis (1) applied at y_0 and y_1 gives

$$\alpha\Big(f(y_0,s)-\frac{|y_0|^2}{2s}\Big)+\beta\Big(f(y_1,s)-\frac{|y_1|^2}{2s}\Big)\le f(\alpha y_0+\beta y_1,s)-\frac{|\alpha y_0+\beta y_1|^2}{2s}.$$

For the second summand put $u_i=y_i-cx_i$. Since the coordinates of $\alpha p_0+\beta p_1$ are $\alpha x_0+\beta x_1$ and $\alpha y_0+\beta y_1$, and since $w\mapsto w-c\cdot$ is linear in the pair,

$$(\alpha y_0+\beta y_1)-c(\alpha x_0+\beta x_1)=\alpha u_0+\beta u_1.$$

Hence by (514)

$$-a\big|(\alpha y_0+\beta y_1)-c(\alpha x_0+\beta x_1)\big|^2=-a\alpha|u_0|^2-a\beta|u_1|^2+a\alpha\beta|u_0-u_1|^2\ge\alpha\big(-a|u_0|^2\big)+\beta\big(-a|u_1|^2\big),$$

because $a>0$, $\alpha\ge0$, $\beta\ge0$ and $|u_0-u_1|^2\ge0$ make the extra term $a\alpha\beta|u_0-u_1|^2$ nonnegative.

Adding the two displays gives $\alpha\Phi(p_0)+\beta\Phi(p_1)\le\Phi(\alpha p_0+\beta p_1)$, so Φ is concave on $F\times F$.

Step 3: Φ is the competitor value, shifted by a constant. Fix $x,y\in F$. By `QuadShiftCompleteSquare`, applied with these $t,\ell>0$ and these y,x,

$$\frac{|y|^2}{2s}-\frac{|y-x|^2}{2\ell}-\frac{|x|^2}{2t}=-a\big|y-cx\big|^2,$$

since $a=t/(2\ell s)$ and $c=s/t$. Substituting this for the penalty term in the definition of Φ and cancelling $|y|^2/(2s)$,

$$\Phi(x,y)=\Big(f(y,s)-\frac{|y-x|^2}{2\ell}\Big)-\frac{|x|^2}{2t}.\tag{515}$$

Step 4: Λ is an attained partial supremum of Φ. By (515) and hypothesis (3), for all $x, y \in F$

$$\Phi(x,y) = \Big(f(y,s) - \frac{|y-x|^2}{2\ell}\Big) - \frac{|x|^2}{2t} \le \mathcal{P}_\ell f(x,t) - \frac{|x|^2}{2t} = \Lambda(x).$$

And given $x \in F$, hypothesis (2) supplies $y \in F$ with $\mathcal{P}_\ell f(x,t) = f(y,s) - |y-x|^2/(2\ell)$; subtracting $|x|^2/(2t)$ and using (515) gives $\Lambda(x) = \Phi(x,y)$.

So Φ is concave on $F \times F$ by Step 2, Λ is an upper bound for $\Phi(x,\cdot)$ at every x, and that upper bound is attained at every x. Therefore `PartialSupConcaveOfAttained`, applied with $E = G = F$, this Φ and this Λ, gives that Λ is concave on F, which is the assertion. □

2.137. `DPSpatialRegularity`.

Lemma 2.137 (Spatial regularity of the forward contact envelope)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and $0 \le \kappa < \infty$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let $f = W = \mathcal{C}_t v - \kappa^2 t/2$ be the contact envelope of `ContactEnv`, and write $f^\ell = \mathcal{P}_\ell f$ for the forward envelope of `QuadEnvP`. Then for every $t, \ell > 0$ and every $x \in F$ the optimization defining $f^\ell(x,t)$ has a unique maximizer $y = y(x,t,\ell)$, the function $f^\ell(\cdot,t)$ is differentiable at x with*

$$\nabla_x f^\ell(x,t) = \frac{y-x}{\ell}, \tag{516}$$

and for every $h \in F$ the two supporting inequalities hold:

$$\begin{aligned} f^\ell(x,t) + \nabla_x f^\ell(x,t)\cdot h - \frac{|h|^2}{2\ell} &\le f^\ell(x+h,t) \\ &\le f^\ell(x,t) + \nabla_x f^\ell(x,t)\cdot h + \frac{|h|^2}{2t}. \end{aligned} \tag{517}$$

In particular the spatial Hessian of $f^\ell(\cdot,t)$ is bounded below by $-\ell^{-1}I$ in the semiconvexity sense and above by $t^{-1}I$ in the semiconcavity sense.

Proof. Write $f = W = \mathcal{C}_t v - \kappa^2 t/2$ for the contact envelope of `ContactEnv`, fix $t, l > 0$ and $x \in F$, and put $s = t + l > 0$.

Step 0: regularity of the integrand on positive times. The hypotheses of `ContactEnvPositiveTimeRegularity` are exactly those assumed here: $1 \le C_0$, $0 \le \kappa$, v in the class $\mathcal{A}_{m,C_0}$ of `IsFitClass`, centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, and satisfying the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. That lemma therefore gives three facts about f, which are the only properties of f used below:

(1) f is continuous on $\{z : z_t > 0\}$;
(2) f has subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`;
(3) for every $\sigma > 0$ the function $y \mapsto f(y,\sigma) - |y|^2/(2\sigma)$ is concave on F.

Step 1: $f(\cdot,s) - |\cdot|^2/(2s)$ is concave. Apply part (3) of Step 0 with $\sigma = s = t + l > 0$:

$$y \longmapsto f(y,s) - \frac{|y|^2}{2s} \tag{518}$$

is concave on F.

Step 2: strict concavity of the objective, hence uniqueness. Write, for $y \in F$,

$$\Xi(y) = f(y,s) - \frac{|y-x|^2}{2l} = \Big(f(y,s) - \frac{|y|^2}{2s}\Big) + \Big(\frac{|y|^2}{2s} - \frac{|y-x|^2}{2l}\Big).$$

The first bracket is concave by (518). The second is a quadratic function of y whose quadratic part is $\frac{1}{2}\big(\frac{1}{s} - \frac{1}{l}\big)|y|^2 = -\frac{t}{2ls}|y|^2$, strictly concave because $t, l, s > 0$. Hence Ξ is strictly concave: for $y_0 \ne y_1$ and $\theta \in (0,1)$, $\Xi((1-\theta)y_0 + \theta y_1) > (1-\theta)\Xi(y_0) + \theta\Xi(y_1)$. In particular Ξ has at most one maximizer.

Step 3: existence of the maximizer. Apply `QuadEnvAttained` to f with $I = (0,\infty)$: its hypotheses are parts (1) and (2) of Step 0. Part (1), at the center (x,t) and the length $l > 0$, whose time arguments t and $t + l = s$ both lie in $(0,\infty)$, gives $y \in F$ with

$$\mathcal{P}_l f(x,t) = f(y,s) - \frac{|y-x|^2}{2l} = \Xi(y) = \max_{y' \in F} \Xi(y'). \tag{519}$$

By Step 2 this maximizer is unique, which is the first two assertions of the lemma.

The hypotheses of part (1) just applied constrain only the two time arguments, namely t and $t + l = s$, and both lie in $I = (0,\infty)$ independently of the spatial center. So the identical application of part (1) at the center (x',t), for an arbitrary $x' \in F$ and the same length $l > 0$, gives a point $y' \in F$ at which the supremum defining $\mathcal{P}_l f(x',t)$ in `QuadEnvP` is attained:

$$\mathcal{P}_l f(x',t) = f(y',s) - \frac{|y'-x'|^2}{2l} = \max_{w \in F}\Big\{ f(w,s) - \frac{|w-x'|^2}{2l} \Big\} \qquad (x' \in F). \tag{520}$$

Two consequences are recorded here once and used below without further comment. First, $\mathcal{P}_l f(x',t)$ is a real number equal to a maximum at every center, so it is a genuine least upper bound of the competitor values there and not the junk value fixed by the preamble convention for a supremum over an empty or unbounded-above set. Second, it therefore dominates the value of each single competitor:

$$\mathcal{P}_l f(x',t) \geq f(w,s) - \frac{|w-x'|^2}{2l} \qquad (x', w \in F). \tag{521}$$

Step 4: the lower supporting inequality. Let $h \in F$. Keeping the maximizer y of (519) fixed and using it as a competitor at the center $x + h$, (521) with $x' = x + h$ and $w = y$ gives

$$\mathcal{P}_l f(x+h,t) \geq f(y,s) - \frac{|y-x-h|^2}{2l} = \mathcal{P}_l f(x,t) + \frac{|y-x|^2 - |y-x-h|^2}{2l},$$

the equality by (519). Since $|y-x-h|^2 = |y-x|^2 - 2\langle y-x, h\rangle + |h|^2$, the fraction equals $\langle (y-x)/l, h\rangle - |h|^2/(2l)$, so

$$\mathcal{P}_l f(x+h,t) \geq \mathcal{P}_l f(x,t) + \Big\langle \frac{y-x}{l}, h \Big\rangle - \frac{|h|^2}{2l}. \tag{522}$$

Step 5: the upper supporting inequality. The three hypotheses of `QuadEnvPConcaveCorrection`, for this f and these $t, l > 0$, are available: its hypothesis (1) is (518); its hypothesis (2), attainment of the defining supremum at every center, is (520); and its hypothesis (3), domination of every competitor at every center, is (521). That lemma therefore gives that

$$\Lambda(x') := \mathcal{P}_l f(x',t) - \frac{|x'|^2}{2t}$$

is concave on F. It is real-valued at every $x' \in F$ by (520), which exhibits $\mathcal{P}_l f(x',t)$ as a maximum of real numbers.

Since Λ is real-valued and concave on F, the function $\psi := -\Lambda$ is real-valued and convex on F; and F is an open subset of F containing x. So `ConvexSubgradientExists`, applied with the open set $\Omega = F$, the function $\psi = -\Lambda$ and the point $z = x \in F$, provides $q_0 \in F$ with

$$-\Lambda(x) + \langle q_0, \chi - x\rangle \leq -\Lambda(\chi) \qquad (\chi \in F).$$

Multiplying by -1 and writing $q := -q_0$ gives

$$\Lambda(\chi) \leq \Lambda(x) + \langle q, \chi - x\rangle \qquad (\chi \in F).$$

Taking $\chi = x + h$ and adding $|x+h|^2/(2t) = |x|^2/(2t) + \langle x, h\rangle/t + |h|^2/(2t)$ to both sides,

$$\mathcal{P}_l f(x+h,t) \leq \mathcal{P}_l f(x,t) + \langle q', h\rangle + \frac{|h|^2}{2t}, \qquad q' := q + \frac{x}{t}. \tag{523}$$

Step 6: identification of q' and differentiability. Let $e \in F$ be a unit vector and $\rho > 0$. Applying (522) and (523) with $h = \rho e$,

$$\rho\Big\langle \frac{y-x}{l}, e \Big\rangle - \frac{\rho^2}{2l} \leq \mathcal{P}_l f(x+\rho e,t) - \mathcal{P}_l f(x,t) \leq \rho\langle q', e\rangle + \frac{\rho^2}{2t},$$

so $\langle (y-x)/l - q', e\rangle \le \rho\big(\frac{1}{2t}+\frac{1}{2l}\big)$; letting $\rho \downarrow 0$ gives $\langle (y-x)/l - q', e\rangle \le 0$. As e ranges over all unit vectors, this forces $q' = (y-x)/l$. Hence (522) and (523) read

$$-\frac{|h|^2}{2l} \le \mathcal{P}_l f(x+h,t) - \mathcal{P}_l f(x,t) - \Big\langle \frac{y-x}{l}, h\Big\rangle \le \frac{|h|^2}{2t} \qquad (h \in F),$$

which are the two supporting inequalities (517), and which show that $x' \mapsto \mathcal{P}_l f(x',t)$ is differentiable at x with gradient $(y-x)/l$, since the middle term is $O(|h|^2) = o(|h|)$. This is (516). □

2.138. `OptimizerLocalization`.

Lemma 2.138 (Uniform localization of the envelope optimizers)**.** Let F be a finite-dimensional real inner-product space and let v, W be real functions on space-time $F \times \mathbb{R}$ such that

(1) v is continuous on all of space-time and has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`;
(2) W is continuous on $\{(x,t) : t > 0\}$ and has subquadratic spatial growth on compact time subintervals of $(0,\infty)$, again in the sense of `SubquadraticGrowth`.

Let $T \in \mathbb{R}$, let $R \ge 2$, let $N \ge 1$ be an integer and write $\delta_N = 2^{-N}$. Then there is $A \ge 1$ such that for every ℓ with $0 < \ell \le \min\{1, \delta_N/2\}$, every (x,t) with $|x| \le R$ and $\delta_N \le t \le T$, and every $y \in F$, both implications

$$\mathcal{P}_\ell W(x,t) = W(y, t+\ell) - \frac{|y-x|^2}{2\ell} \implies |y-x| \le 2\sqrt{A\ell},$$

$$\mathcal{M}_\ell v(x,t) = v(y, t-\ell) + \frac{|y-x|^2}{2\ell} \implies |y-x| \le 2\sqrt{A\ell}$$

hold, with $\mathcal{P}$ the forward quadratic envelope of `QuadEnvP` and $\mathcal{M}$ the backward quadratic envelope of `QuadEnvM`.

Proof. Put $J = [\delta_N/2, T+1]$, a compact subset of $\mathbb{R}$, and note that every $s \in J$ satisfies $s \ge \delta_N/2 > 0$, so J is also a compact subset of $(0,\infty)$. For $r \ge 0$ the set $\mathcal{B}_r = \{(y,s) : |y| \le r,\ s \in J\}$ is compact, being the image of the compact product of a closed ball and J under the (continuous) identification of $F \times \mathbb{R}$ with space-time, and every point of $\mathcal{B}_r$ has positive time coordinate, so W is continuous on $\mathcal{B}_r$ by hypothesis (2) and v is continuous on $\mathcal{B}_r$ by hypothesis (1). Consequently $|W|$ and $|v|$ are bounded on each $\mathcal{B}_r$, a continuous function on a compact set being bounded.

The two implications are proved separately, the first with a constant $A^+ \ge 1$ and the second with a constant $A^- \ge 1$; we then set $A = \max\{A^+, A^-\} \ge 1$, which serves both, because $A^\pm \le A$ and $\ell > 0$ give $2\sqrt{A^\pm \ell} \le 2\sqrt{A\ell}$.

Fix ℓ with $0 < \ell \le \min\{1, \delta_N/2\}$ and (x,t) with $|x| \le R$ and $\delta_N \le t \le T$. Then

$$t + \ell \in J, \qquad t - \ell \in J, \tag{524}$$

because $t + \ell \le T + 1$ and $t + \ell \ge t \ge \delta_N \ge \delta_N/2$ for the first, and $t - \ell \le t \le T \le T + 1$ and $t - \ell \ge \delta_N - \delta_N/2 = \delta_N/2$ for the second. We also record the elementary inequality

$$\frac{|y|^2}{2} - |x|^2 \le |y-x|^2 \qquad (x, y \in F), \tag{525}$$

which follows from $|y| \le |y-x| + |x|$ and $(|y-x| - |x|)^2 \ge 0$.

The forward implication. Apply hypothesis (2) with the compact set $J \subseteq (0,\infty)$ and the accuracy $1/8$: there is $M_1^+ > 0$ with

$$|W(y,s)| \le \frac{|y|^2}{8} \qquad (|y| \ge M_1^+,\ s \in J). \tag{526}$$

Let $A_1^+ \ge 1$ bound $|W|$ on $\mathcal{B}_{\max\{M_1^+, R\}}$ and put

$$M_2^+ = \max\big\{M_1^+,\ \sqrt{8A_1^+ + 4R^2},\ R\big\},$$

and let $A^+ \ge 1$ bound $|W|$ on $\mathcal{B}_{M_2^+}$.

Let y satisfy $\mathcal{P}_\ell W(x,t) = W(y,t+\ell) - |y-x|^2/(2\ell)$. By the domination half of part (1) of `EnvelopeCompetitorBounds`, applied to W with $I=(0,\infty)$ at the center (x,t) and the length ℓ — its hypotheses hold: $\ell>0$, and $t\in I$ and $t+\ell\in I$ because $t\geq \delta_N>0$ and $\ell>0$, while the continuity and growth hypotheses are hypothesis (2) — the competitor $y'=x$ gives $W(x,t+\ell)-|x-x|^2/(2\ell)=W(x,t+\ell)\leq \mathcal{P}_\ell W(x,t)$, so

$$\frac{|y-x|^2}{2\ell}\leq W(y,t+\ell)-W(x,t+\ell). \tag{527}$$

We first show $|y|\leq M_2^+$. If $|y|\leq M_1^+$ this is immediate from the definition of M_2^+. Suppose $|y|>M_1^+$. Since $0<\ell\leq 1$ and $|y-x|^2\geq 0$ we have $|y-x|^2/2\leq |y-x|^2/(2\ell)$, so (525), (527), (526) at $(y,t+\ell)$ — legitimate by (524) — and the bound $W(x,t+\ell)\geq -A_1^+$ — legitimate because $|x|\leq R\leq \max\{M_1^+,R\}$ and $t+\ell\in J$ — give

$$\frac{|y|^2}{4}-\frac{|x|^2}{2}\leq \frac{|y-x|^2}{2}\leq \frac{|y-x|^2}{2\ell}\leq W(y,t+\ell)-W(x,t+\ell)\leq \frac{|y|^2}{8}+A_1^+.$$

With $|x|\leq R$ this yields $|y|^2/8\leq A_1^+ + R^2/2$, i.e. $|y|^2\leq 8A_1^+ + 4R^2$, hence $|y|\leq \sqrt{8A_1^+ + 4R^2}\leq M_2^+$.

Now $|y|\leq M_2^+$ and $|x|\leq R\leq M_2^+$ and $t+\ell\in J$, so both $|W(y,t+\ell)|\leq A^+$ and $|W(x,t+\ell)|\leq A^+$, and (527) gives $|y-x|^2/(2\ell)\leq 2A^+$, that is $|y-x|^2\leq 4A^+\ell$. Since $A^+\leq A$ and $\ell>0$ this gives $|y-x|^2\leq 4A\ell$ and therefore $|y-x|=\sqrt{|y-x|^2}\leq \sqrt{4A\ell}=2\sqrt{A\ell}$, as claimed.

The backward implication. This is not a substitution instance of the forward one: it uses part (2) of `EnvelopeCompetitorBounds` rather than part (1), and the continuity and growth of v on all of space-time rather than those of W on $\{t>0\}$. Apply hypothesis (1) with the compact set $J\subseteq\mathbb{R}$ and the accuracy $1/8$: there is $M_1^->0$ with

$$|v(y,s)|\leq \frac{|y|^2}{8}\qquad (|y|\geq M_1^-,\ s\in J). \tag{528}$$

Let $A_1^-\geq 1$ bound $|v|$ on $\mathcal{B}_{\max\{M_1^-,R\}}$, put $M_2^-=\max\{M_1^-,\sqrt{8A_1^-+4R^2},R\}$ and let $A^-\geq 1$ bound $|v|$ on $\mathcal{B}_{M_2^-}$.

Let y satisfy $\mathcal{M}_\ell v(x,t)=v(y,t-\ell)+|y-x|^2/(2\ell)$. By the domination half of part (2) of `EnvelopeCompetitorBounds`, applied to v with $I=\mathbb{R}$ at the center (x,t) and the length ℓ — its hypotheses hold: $\ell>0$, the conditions $t\in I$ and $t-\ell\in I$ are vacuous for $I=\mathbb{R}$, and the continuity and growth hypotheses are hypothesis (1) — the competitor $y'=x$ gives $\mathcal{M}_\ell v(x,t)\leq v(x,t-\ell)+|x-x|^2/(2\ell)=v(x,t-\ell)$, so

$$\frac{|y-x|^2}{2\ell}\leq v(x,t-\ell)-v(y,t-\ell). \tag{529}$$

Exactly as before, if $|y|>M_1^-$ then $0<\ell\leq 1$, (525), (529), (528) at $(y,t-\ell)$ — legitimate by (524) — and $v(x,t-\ell)\leq A_1^-$ give

$$\frac{|y|^2}{4}-\frac{|x|^2}{2}\leq \frac{|y-x|^2}{2}\leq \frac{|y-x|^2}{2\ell}\leq v(x,t-\ell)-v(y,t-\ell)\leq A_1^-+\frac{|y|^2}{8},$$

whence $|y|^2\leq 8A_1^-+4R^2$ and $|y|\leq M_2^-$; and if $|y|\leq M_1^-$ then $|y|\leq M_2^-$ outright. With $|y|\leq M_2^-$, $|x|\leq R\leq M_2^-$ and $t-\ell\in J$ both $|v(y,t-\ell)|\leq A^-$ and $|v(x,t-\ell)|\leq A^-$, so (529) gives $|y-x|^2\leq 4A^-\ell\leq 4A\ell$ and therefore $|y-x|\leq 2\sqrt{A\ell}$.

Setting $A=\max\{A^+,A^-\}\geq 1$, both implications hold with this single A, which depends only on N, R, T and the data v, W. □

2.139. `FairSignEscape`.

Lemma 2.139 (The fair-sign space with a forced escape block)**.** Let $q\in\mathbb{N}$. Then there are a measurable space $(\Omega,\mathcal{M})$, a measure μ on it with $\mu(\Omega)=1$, a filtration $(\mathcal{G}_n)_{n\in\mathbb{N}}$ of sub-σ-algebras of $\mathcal{M}$, a sequence of functions $\xi_n:\Omega\to\mathbb{R}$ and a function $G:\Omega\to\mathbb{N}$ such that

(1) $\xi_n(\omega)=1$ or $\xi_n(\omega)=-1$ for every $n\in\mathbb{N}$ and every $\omega\in\Omega$;

(2) ξ_n is $\mathcal{G}_{n+1}$-measurable for every $n \in \mathbb{N}$;
(3) $\mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ almost everywhere, for every $n \in \mathbb{N}$;
(4) $\xi_{qG(\omega)+i}(\omega) = 1$ for every $\omega \in \Omega$ and every $i < q$;
(5) G is integrable and $\int_\Omega G\, d\mu \le 2^q$.

Proof. Write $p = 2^{-q} \in (0, 1]$.

Step 1: the product space. Let $B = \{\mathrm{t}, \mathrm{f}\}$ be a two-point set with its discrete σ-algebra and let ν be the uniform probability measure on B, so that $\nu\{\mathrm{t}\} = \nu\{\mathrm{f}\} = \frac{1}{2}$ and $\nu(B) = 1$. Let $\Omega_0 = B^{\mathbb{N}}$ carry the product σ-algebra and let $\mu_0 = \bigotimes_{k\in\mathbb{N}} \nu$ be the product measure of the Preamble. Then μ_0 is a probability measure, and for every finite $I \subseteq \mathbb{N}$ and every family of measurable sets $(T_k)_{k\in I}$ in B,

$$\mu_0\big(\{\omega \in \Omega_0 : \omega_k \in T_k \text{ for all } k \in I\}\big) = \prod_{k\in I} \nu(T_k). \tag{530}$$

In particular the measure of a cylinder prescribing $|I|$ coordinates exactly is $2^{-|I|}$, and (530) shows that the coordinate maps $\omega \mapsto \omega_k$ are independent: if I and J are disjoint finite sets of indices, then every event determined by the coordinates in I is independent of every event determined by the coordinates in J, because both sides of (530) factorize over $I \cup J$.

Step 2: the signs on Ω_0. Put $Z_0 \equiv 0$ and, for $k \ge 1$,

$$Z_k(\omega) = \begin{cases} +1, & \omega_k = \mathrm{t}, \\ -1, & \omega_k = \mathrm{f}. \end{cases}$$

Each Z_k is a function of the single coordinate ω_k and B is discrete, so each Z_k is measurable, indeed strongly measurable, being real valued with finite range. Let $\mathcal{F}_n = \sigma(Z_k : k \le n)$ be the natural filtration of the sequence $(Z_k)_{k\in\mathbb{N}}$; since Z_0 is constant, $\mathcal{F}_n$ is generated by the coordinates $\omega_1, \dots, \omega_n$ and in particular $\mathcal{F}_n$ is independent of $\sigma(Z_{n+1})$ by Step 1, $\sigma(Z_{n+1})$ being contained in the σ-algebra generated by the single coordinate ω_{n+1}. Moreover, by (530) with $I = \{k\}$ the law of ω_k under μ_0 is ν, so for $k \ge 1$

$$\mu_0\{Z_k = 1\} = \mu_0\{Z_k = -1\} = \tfrac{1}{2}, \qquad \int_{\Omega_0} Z_k\, d\mu_0 = \tfrac{1}{2} - \tfrac{1}{2} = 0. \tag{531}$$

Step 3: the escape blocks. For $m \in \mathbb{N}$ let

$$A_m = \{\omega \in \Omega_0 : \omega_{qm+i+1} = \mathrm{t} \text{ for all } i < q\}, \qquad E_m = \bigcap_{j<m} A_j^c.$$

Each A_m is a cylinder prescribing the q coordinates with indices in $(qm, q(m+1)]$, so it is measurable and, by (530), $\mu_0(A_m) = 2^{-q} = p$ and $\mu_0(A_m^c) = 1 - p$. The set E_m is determined by the coordinates with indices $\le qm$, because A_j for $j < m$ involves only coordinates with indices in $(qj, q(j+1)] \subseteq [1, qm]$, while A_m is determined by the coordinates with indices $> qm$. Hence E_m and A_m^c are independent by Step 1, and

$$\mu_0(E_{m+1}) = \mu_0(E_m \cap A_m^c) = \mu_0(E_m)\,(1 - p).$$

Since $E_0 = \Omega_0$ has measure 1, induction on m gives

$$\mu_0(E_m) = (1 - p)^m \qquad (m \in \mathbb{N}). \tag{532}$$

Step 4: the escape set has full measure. Let $S = \bigcup_{m\in\mathbb{N}} A_m$, a measurable subset of Ω_0. If $\omega \notin S$ then $\omega \notin A_j$ for every j, so $\omega \in E_m$ for every m; thus $\mu_0(S^c) \le (1 - p)^m$ for every m by (532). As $p > 0$ we have $0 \le 1 - p < 1$, so $(1 - p)^m \to 0$ and therefore $\mu_0(S^c) = 0$, that is $\mu_0(S) = 1$.

Let $\Omega = S$, let $\mathcal{M}$ be the σ-algebra induced on S by the product σ-algebra, that is the collection of the sets $\iota^{-1}(B) = B \cap S$ for measurable $B \subseteq \Omega_0$, where $\iota : S \to \Omega_0$ is the inclusion, and let μ be the measure induced by μ_0 on $(\Omega, \mathcal{M})$, so that $\mu(\iota^{-1}(B)) = \mu_0(\iota(\iota^{-1}(B))) = \mu_0(B \cap S)$ for every measurable B; this is legitimate because S is measurable, so ι is a measurable

embedding and the image of a measurable subset of S is measurable. Since $\mu_0(S^c)=0$ we get the transport identity

$$\mu(\iota^{-1}(B))=\mu_0(B)\qquad (B\subseteq\Omega_0 \text{ measurable}), \tag{533}$$

and in particular $\mu(\Omega)=\mu(\iota^{-1}(\Omega_0))=\mu_0(\Omega_0)=1$.

Step 5: the signs on Ω and their conditional means. Set $\xi_n=Z_{n+1}\circ\iota$ for $n\in\mathbb{N}$ and let $\mathcal{G}_n=\sigma(Z_k\circ\iota : k\le n)$ be the natural filtration on Ω of the restricted sequence; as the inverse image of a generated σ-algebra is generated by the inverse images,

$$\mathcal{G}_n=\{\iota^{-1}(B) : B\in\mathcal{F}_n\}. \tag{534}$$

Claim (1) holds by the definition of Z_k for $k\ge 1$. Claim (2) holds because $\xi_n=Z_{n+1}\circ\iota$ is measurable for $\sigma(Z_{n+1}\circ\iota)$, which is one of the generators of $\mathcal{G}_{n+1}$ since $n+1\le n+1$.

For claim (3) fix n and note first that, by (534), (533) and Step 2, the σ-algebras $\sigma(\xi_n)$ and $\mathcal{G}_n$ are independent under μ: if $A=\iota^{-1}(A')$ with $A'\in\sigma(Z_{n+1})$ and $B=\iota^{-1}(B')$ with $B'\in\mathcal{F}_n$, then

$$\mu(A\cap B)=\mu(\iota^{-1}(A'\cap B'))=\mu_0(A'\cap B')=\mu_0(A')\,\mu_0(B')=\mu(A)\,\mu(B).$$

Also ξ_n is bounded and measurable, hence integrable for the probability measure μ, and by (533) and (531)

$$\mathbb{E}[\xi_n]=\mu\{\xi_n=1\}-\mu\{\xi_n=-1\}=\mu_0\{Z_{n+1}=1\}-\mu_0\{Z_{n+1}=-1\}=0.$$

Now let $B\in\mathcal{G}_n$. Since ξ_n takes only the values ±1,

$$\int_B \xi_n\,d\mu=\mu(B\cap\{\xi_n=1\})-\mu(B\cap\{\xi_n=-1\})=\mu(B)\big(\mu\{\xi_n=1\}-\mu\{\xi_n=-1\}\big)=0=\int_B 0\,d\mu,$$

where the middle equality used the independence just proved. As the zero function is $\mathcal{G}_n$-measurable and integrable, the characterization of conditional expectation in the Preamble gives $\mathbb{E}[\xi_n\mid\mathcal{G}_n]=0$ almost everywhere, which is claim (3).

Step 6: the escape index. For $\omega\in\Omega=S$ the set $\{m:\omega\in A_m\}$ is nonempty; let $G(\omega)$ be its least element. If $i<q$ then $\omega\in A_{G(\omega)}$ gives $\omega_{qG(\omega)+i+1}=\mathrm{t}$, that is $\xi_{qG(\omega)+i}(\omega)=Z_{qG(\omega)+i+1}(\omega)=1$, which is claim (4). By the definition of a least element,

$$\{G=m\}=\iota^{-1}(E_m\cap A_m),\qquad \{G>m\}=\iota^{-1}(E_{m+1})\qquad (m\in\mathbb{N}),$$

so G is measurable, and by (533) and (532)

$$\mu\{G>m\}=\mu_0(E_{m+1})=(1-p)^{m+1}\qquad (m\in\mathbb{N}). \tag{535}$$

Since $G\ge 0$ is measurable and integer valued, the layer-cake identity of the Preamble, applied to G and evaluated by splitting $(0,\infty)=\bigcup_{m\in\mathbb{N}}(m,m+1]$ on each piece of which $\lambda\mapsto\mu\{G>\lambda\}=\mu\{G>m\}$ is constant, gives

$$\int_\Omega G\,d\mu=\int_0^\infty \mu\{G>\lambda\}\,d\lambda=\sum_{m\in\mathbb{N}}\mu\{G>m\}=\sum_{m\in\mathbb{N}}(1-p)^{m+1}=\frac{1-p}{p}\le\frac{1}{p}=2^q,$$

the geometric series being summable because $0\le 1-p<1$. In particular the right-hand side of the layer-cake identity is finite, so G is integrable and $\int_\Omega G\,d\mu\le 2^q$, which is claim (5). □

2.140. `ContactWalkExists`.

Lemma 2.140 (Existence of the stopped contact walks)**.** *Let F be a finite-dimensional real inner-product space, let $1\le C_0<\infty$, let $\kappa>0$ and $H\in\mathbb{R}$, and let $v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $z_*=(0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let W be the contact envelope of `ContactEnv` formed from v and κ, and assume*

1. *W is continuous on $\{(x,t):t>0\}$ and has subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`;*
2. *$W(z_*)=0$ and $v(z_*)=0$;*
3. *the envelope Hamiltonian of `EnvHamiltonian` satisfies $H_v(z_*)=H$, and the forward envelope slopes of `EnvSlopeP` satisfy $\mathcal{S}_\ell^+W(z_*)\to H$ as $\ell\downarrow 0$.*

Write $\delta_j = 2^{-j}$. Then for every integer $T \geq 2$, every real $R \geq 2$ and every integer $N \geq 1$ there are a step size $\epsilon > 0$, a measure space $(\Omega, \mathcal{M}, \mu)$ with a filtration $(\mathcal{G}_n)$, and functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ on it which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of `IsContactWalk` *and are such that, for every n, the range of z_n is at most countable.*

Proof. Throughout, $z_* = (0, 1)$, $f^\ell = \mathcal{P}_\ell W$ is the forward envelope of `QuadEnvP`, $g_\ell = \mathcal{M}_\ell v$ is the backward envelope of `QuadEnvM`, and $J_\ell = (f^\ell - g_\ell)/(2\ell)$ is the quantity of `JGap`. States are written $z = (x, t)$.

Step 0: standing facts about v. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of the limiting equation there. By `UniformSubquadratic`, applied with the centering point $(0, 1)$, v has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`. Hence `QuadEnvAttained` and `EnvHamiltonianSolution` are available for v with $I = \mathbb{R}$ (which is open), and, by hypothesis (1), `QuadEnvAttained` is available for W with $I = (0, \infty)$; also `OptimizerLocalization` is available for the pair (v, W).

Step 1: a lower bound for $J_\epsilon(z_*)$ at small lengths. For $\ell > 0$, hypothesis (2) gives $W(z_*) = v(z_*) = 0$, so by `JGap`, `EnvSlopeP` and `EnvSlopeM`,

(536)
$$J_\ell(z_*) = \frac{\mathcal{P}_\ell W(z_*) - \mathcal{M}_\ell v(z_*)}{2\ell} = \frac{1}{2}\Big(\frac{\mathcal{P}_\ell W(z_*) - W(z_*)}{\ell} + \frac{v(z_*) - \mathcal{M}_\ell v(z_*)}{\ell}\Big) = \frac{\mathcal{S}_\ell^+ W(z_*) + \mathcal{S}_\ell^- v(z_*)}{2}.$$

By the second clause of `EnvHamiltonianSolution` for v with $I = \mathbb{R}$ — whose hypothesis $t+\sigma \in I$ for all σ is vacuous for $I = \mathbb{R}$ —

$$\inf_{\ell > 0} \mathcal{S}_\ell^- v(z_*) = H_v(z_*) = H$$

by hypothesis (3), and therefore

$$H \leq \mathcal{S}_\ell^- v(z_*) \qquad (\ell > 0), \tag{537}$$

an infimum being a lower bound for each member of the family. By hypothesis (3) again, $\mathcal{S}_\ell^+ W(z_*) \to H$ as $\ell \downarrow 0$, and $H - 2 < H$, so there is $\ell_0 > 0$ with

$$H - 2 \leq \mathcal{S}_\ell^+ W(z_*) \qquad (0 < \ell \leq \ell_0). \tag{538}$$

Combining (536), (537) and (538),

$$J_\ell(z_*) \geq \frac{(H-2) + H}{2} = H - 1 \qquad (0 < \ell \leq \ell_0). \tag{539}$$

Step 2: the choice of ϵ. Fix $T \geq 2$, $R \geq 2$ and $N \geq 1$ and write $\delta = \delta_N = 2^{-N}$, so $0 < \delta \leq 1/2$. By `OptimizerLocalization`, applied to v and W with this T, R and N — its hypotheses are Step 0 for v and hypothesis (1) for W — there is $A \geq 1$ such that for every ℓ with $0 < \ell \leq \min\{1, \delta/2\}$, every (x, t) with $|x| \leq R$ and $\delta \leq t \leq T$ and every y,

$$\mathcal{P}_\ell W(x,t) = W(y, t+\ell) - \frac{|y-x|^2}{2\ell} \implies |y - x| \leq 2\sqrt{A\ell}, \tag{540}$$

and the same implication for $\mathcal{M}_\ell v(x,t) = v(y, t-\ell) + |y-x|^2/(2\ell)$. Since $1/2 < 1$, there is an integer k_1 with $(1/2)^{k_1} < \min\{\ell_0, 1/(32A)\}$; put $k = \max\{N+2, k_1\}$ and

$$\epsilon = 2^{-k}. \tag{541}$$

Then $\epsilon > 0$ and, because $(1/2)^j$ is nonincreasing in j,

$$\epsilon \leq 2^{-(N+2)} = \frac{\delta}{4}, \qquad \epsilon \leq \ell_0, \qquad 32A\epsilon \leq 1. \tag{542}$$

In particular $\epsilon \leq 1/2$ and $2\epsilon \leq \min\{1, \delta/2\}$, so (540) is applicable with $\ell = 2\epsilon$, and then $2\sqrt{2A\epsilon} \leq 1/2$ because $2A\epsilon \leq 1/16$. Moreover each of 1, T and δ is an integer multiple of ϵ:

$$1 = 2^k\epsilon, \qquad T = (T2^k)\epsilon, \qquad \delta = 2^{k-N}\epsilon, \tag{543}$$

the second because T is an integer and the third because $k \geq N+2 \geq N$. We use these through the following consequence: if $a = m_1\epsilon$ and $b = m_2\epsilon$ with $m_1, m_2 \in \mathbb{Z}$ and $a < b$, then $m_1 < m_2$, hence $m_1 + 1 \leq m_2$ and

$$a + \epsilon \leq b. \tag{544}$$

Step 3: the probability space and the signs. Put $q = \lceil (T-\delta)/\epsilon \rceil + 1$, a positive integer. By `FairSignEscape` applied with this q there are a measure space $(\Omega, \mathcal{M}, \mu)$ with $\mu(\Omega) = 1$, a filtration $(\mathcal{G}_n)_{n\in\mathbb{N}}$, functions $\xi_n : \Omega \to \mathbb{R}$ and a function $G : \Omega \to \mathbb{N}$ such that each ξ_n takes only the values ± 1, is $\mathcal{G}_{n+1}$-measurable and satisfies $\mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ a.e., such that

$$\xi_{qG(\omega)+i}(\omega) = 1 \qquad (\omega \in \Omega,\ i < q), \tag{545}$$

and such that G is integrable with $\int_\Omega G\, d\mu \leq 2^q$. Write $b_k(\omega) \in \{0,1\}$ for the sign bit of $\xi_k(\omega)$, that is $b_k = 1$ if $\xi_k = 1$ and $b_k = 0$ otherwise; then

$$\{b_k = 1\} = \{\xi_k = 1\} \in \mathcal{G}_{k+1}, \qquad \{b_k = 0\} = \{\xi_k = 1\}^{\mathrm{c}} \in \mathcal{G}_{k+1}, \tag{546}$$

because ξ_k is $\mathcal{G}_{k+1}$-measurable and $\{1\}$ is a measurable subset of $\mathbb{R}$. Consequently, for every n and every sequence of bits (s_k), the cylinder $\{\omega : b_k(\omega) = s_k \text{ for all } k < n\}$ is a finite intersection of sets of $\mathcal{G}_n$, hence lies in $\mathcal{G}_n$, the filtration being nondecreasing.

The walk. Call a state $z = (x,t)$ *live* if $\delta < t < T$ and $|x| < R$. We define two maps on states. If $z = (x,t)$ has $t > 0$ then, by `QuadEnvAttained` part (1) applied to W with $I = (0,\infty)$, the length 2ϵ and the center z — admissible since $t > 0$ and $t + 2\epsilon > 0$ — there is a point $Y^+(z)$ with

$$\mathcal{P}_{2\epsilon}W(z) = W\big(Y^+(z), t + 2\epsilon\big) - \frac{|Y^+(z) - x|^2}{2 \cdot 2\epsilon}, \tag{547}$$

and we fix one such point; by part (2) of the same lemma applied to v with $I = \mathbb{R}$ — where both admissibility conditions are vacuous — there is a point $Y^-(z)$ with

$$\mathcal{M}_{2\epsilon}v(z) = v\big(Y^-(z), t - 2\epsilon\big) + \frac{|Y^-(z) - x|^2}{2 \cdot 2\epsilon}, \tag{548}$$

and we fix one such point as well. Put

$$z^+ = \Big(\tfrac{1}{2}\big(x + Y^+(z)\big),\, t + \epsilon\Big), \qquad z^- = \Big(\tfrac{1}{2}\big(x + Y^-(z)\big),\, t - \epsilon\Big)$$

if z is live, and $z^\pm = z$ otherwise. By `IsStepPair` and (547)–(548), (z^+, z^-) is an optimizing step pair of size ϵ at a live z for v and κ. Also, if $t > 0$ then by `DPSpatialRegularity` applied at the time t, the length $\epsilon > 0$ and the center x — its hypotheses on C_0, κ, v are the standing ones and its two numerical hypotheses are $t > 0$ and $\epsilon > 0$ — the function $\mathcal{P}_\epsilon W(\cdot, t)$ is differentiable at x with some gradient there, and we let $Q_0(z)$ be one such gradient.

Now define $Z_n(\omega)$ by recursion: $Z_0(\omega) = z_*$ and

$$Z_{n+1}(\omega) = \begin{cases} Z_n(\omega)^+, & Z_n(\omega) \text{ live and } b_n(\omega) = 1, \\ Z_n(\omega)^-, & Z_n(\omega) \text{ live and } b_n(\omega) = 0, \\ Z_n(\omega), & Z_n(\omega) \text{ not live}, \end{cases}$$

and put $z_n = Z_n$, $z_n^\pm = Z_n^\pm$ and $Q_0(n) = Q_0(Z_n)$. An immediate induction on n shows that $Z_n(\omega)$ depends only on $b_0(\omega), \dots, b_{n-1}(\omega)$: the recursion step reads only b_n. Hence each of z_n, $z_n^\pm$ and $Q_0(n)$ takes at most 2^n values and is constant on each of the 2^n cylinders of the first n bits; writing such a function as the finite sum of the (constant) values times the indicators of those cylinders, which lie in $\mathcal{G}_n$ by (546), each of them is $\mathcal{G}_n$-measurable, indeed strongly measurable, being a finite sum of indicators times constants. In particular the range of z_n is finite, hence at most countable, for every n, which is the last assertion of the statement.

The numerical invariants. We claim that for every ω and n

$$t_n \in \epsilon\mathbb{Z}, \qquad \delta \leq t_n \leq T, \tag{549}$$

where $z_n = (x_n, t_n)$. For $n = 0$ we have $t_0 = 1$, which lies in $\epsilon\mathbb{Z}$ by (543), and $\delta \leq 1/2 < 1 \leq T$. Assume (549) at n. If z_n is not live then $z_{n+1} = z_n$ and there is nothing to prove. If z_n is live then $t_{n+1} = t_n \pm \epsilon$, so $t_{n+1} \in \epsilon\mathbb{Z}$; moreover $\delta < t_n < T$ with all three of δ, t_n, T in $\epsilon\mathbb{Z}$, so (544)

upgrades these to $\delta+\epsilon \le t_n \le T-\epsilon$, whence $\delta \le t_n-\epsilon \le t_{n+1} \le t_n+\epsilon \le T$. This proves (549); in particular $t_n \ge \delta > 0$ for every n, including at and after the stop, which is what makes the definitions of $Y^{\pm}$ and Q_0 legitimate at every Z_n.

The steps are short. Let $z=(x,t)$ be live. Then $|x| \le R$ and $\delta \le t \le T$ by (549), so (540) applies with $\ell = 2\epsilon$ to the optimizers $Y^{\pm}(z)$ of (547)–(548) and gives $|Y^{\pm}(z)-x| \le 2\sqrt{2A\epsilon} \le 1/2$. The spatial parts of $z^{\pm}$ are the midpoints $\frac{1}{2}(x+Y^{\pm}(z))$, so

$$\left|\tfrac{1}{2}(x+Y^{\pm}(z))-x\right| = \tfrac{1}{2}|Y^{\pm}(z)-x| \le \tfrac{1}{4}. \tag{550}$$

The stopping index. We claim that for every ω there is an n with $Z_n(\omega)$ not live. Suppose, to the contrary, that $Z_n(\omega)$ is live for every $n \le qG(\omega)+q$. By (545) we have $b_{qG(\omega)+i}(\omega)=1$ for every $i<q$, so the recursion gives, by induction on $j \le q$,

$$t_{qG(\omega)+j} = t_{qG(\omega)} + j\epsilon,$$

each step being a $+\epsilon$ step. At $j=q$ this yields $t_{qG(\omega)+q} = t_{qG(\omega)} + q\epsilon \ge \delta + q\epsilon$ by (549). But $q > (T-\delta)/\epsilon$ by the definition of q and $\epsilon > 0$, so $q\epsilon > T-\delta$ and therefore $t_{qG(\omega)+q} > T$, contradicting (549). Hence there is an n with $Z_n(\omega)$ not live, and we let $\tau(\omega)$ be the least such n. If we had $\tau(\omega) > qG(\omega)+q$ then every $n \le qG(\omega)+q$ would satisfy $n < \tau(\omega)$ and hence, by the minimality of $\tau(\omega)$, would carry a live state — which the contradiction just derived excludes. Therefore

$$\tau(\omega) \le qG(\omega)+q. \tag{551}$$

By minimality, $Z_n(\omega)$ is live for $n<\tau(\omega)$; and for $n \ge \tau(\omega)$ an induction using the third line of the recursion gives $Z_n(\omega) = Z_{\tau(\omega)}(\omega)$, so $Z_n(\omega)$ is not live there either. Thus

$$Z_n(\omega) \text{ is live} \iff n < \tau(\omega). \tag{552}$$

τ is a stopping time and is integrable. The set of states that are live is $\{(x,t) : t \in (\delta,T)\} \cap \{(x,t) : |x| < R\}$, so, $t \mapsto t$ and $x \mapsto |x|$ being continuous, the event $\{Z_n \text{ live}\}$ is $\mathcal{G}_n$-measurable, Z_n being $\mathcal{G}_n$-measurable. By (552),

$$\{\tau \le n\} = \bigcup_{m \le n} \{Z_m \text{ not live}\},$$

a finite union of sets of $\mathcal{G}_m \subseteq \mathcal{G}_n$ with $m \le n$, hence in $\mathcal{G}_n$; so τ is a stopping time. Likewise $\{\tau = n\} = \{Z_n \text{ not live}\} \cap \bigcap_{m<n}\{Z_m \text{ live}\}$ is measurable for every n, so τ, having countable range, is measurable; and $0 \le \tau \le qG+q$ pointwise by (551) with $qG+q$ integrable, because G is integrable and μ is finite. Hence τ is integrable.

The positions stay in $\overline{B}_{R+1}(0)$. Let $n \le \tau(\omega)$. If $n < \tau(\omega)$ then $Z_n(\omega)$ is live, so $|x_n| < R \le R+1$. If $n = \tau(\omega) = 0$ then $x_n = 0$ and $|x_n| = 0 \le R+1$. If $n = \tau(\omega) = m+1$ then $Z_m(\omega)$ is live, so $|x_m| < R$, and by (550) and the triangle inequality $|x_{m+1}| \le |x_{m+1}-x_m| + |x_m| \le \frac{1}{4} + R \le R+1$.

The stop is exactly at a cutoff. At $n=\tau(\omega)$ the state is not live, that is, not all of $\delta < t_\tau$, $t_\tau < T$, $|x_\tau| < R$ hold. If $\delta < t_\tau$ and $t_\tau < T$ then $R \le |x_\tau|$. If $t_\tau \ge T$ then $t_\tau = T$ by (549); if $t_\tau \le \delta$ then $t_\tau = \delta$ by (549). So in every case $t_\tau = \delta$ or $t_\tau = T$ or $R \le |x_\tau|$.

Conclusion. The data ϵ, $(\Omega,\mathcal{M},\mu)$, $(\mathcal{G}_n)$, ξ, z, $z^{\pm}$, Q_0, τ satisfy every clause of `IsContactWalk`: $\mu(\Omega)=1$ and the three clauses on ξ come from `FairSignEscape`; $\epsilon = 2^{-k}$ is an inverse power of two with $0 < \epsilon \le \delta_N/4$ by (541) and (542); the four measurability clauses were proved after the definition of the walk; $z_0 = z_*$ by definition; the step-pair clause for $n<\tau$ holds by (547)–(548) and (552); the recursion clause is the definition of Z_{n+1} rewritten with (552) and with $b_n = 1$ replaced by $\xi_n = 1$; the gradient clause is the definition of Q_0, legitimate because $t_n > 0$ by (549); τ is an integrable stopping time as just shown; the live bounds for $n<\tau$ are (552) together with the definition of liveness; the exit alternative and the bound $|x_n| \le R+1$ for $n \le \tau$ are the previous two paragraphs; the countability of the range of each z_n was recorded when the walk was defined; and $H-1 \le J_\epsilon(z_*)$ is (539) at $\ell=\epsilon$, admissible because $0 < \epsilon \le \ell_0$ by (542). □

2.141. `QuadEnvComposition`.

Lemma 2.141 (Composition of quadratic envelopes)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open, let $\Omega = \{(x,t) : t \in I\}$, and let f be a real function on space-time which is continuous on Ω and has subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth`. Then, whenever $\ell, h > 0$ and the time arguments t, $t+h$, $t+\ell+h$ all lie in I,*

$$\mathcal{P}_\ell\big(\mathcal{P}_h f\big)(x,t) = \mathcal{P}_{\ell+h} f(x,t), \tag{553}$$

and likewise, whenever $\ell, h > 0$ and t, $t-h$, $t-\ell-h$ all lie in I,

$$\mathcal{M}_\ell\big(\mathcal{M}_h f\big)(x,t) = \mathcal{M}_{\ell+h} f(x,t),$$

with the envelopes of `QuadEnvP` and `QuadEnvM`. Here $\mathcal{P}_h f$ and $\mathcal{M}_h f$ are regarded as functions of their center and time, the time being shifted by h in the respective direction.

Proof. Throughout, $z = (x,t)$ is a space-time point with $x \in F$ and $t \in \mathbb{R}$, and $\ell, h > 0$ are fixed. For a fixed pair $(x,w) \in F \times F$ we write

$$Q(y) = \frac{|y-x|^2}{2\ell} + \frac{|w-y|^2}{2h} \qquad (y \in F)$$

for the combined cost. By the convention of the preamble a supremum over a set which is empty or unbounded above, and an infimum over a set which is empty or unbounded below, take the junk value 0; accordingly every extremum occurring below is shown to be over a nonempty set carrying the relevant one-sided bound *before* that extremum is used. Steps 1–5 prove (553) and Step 6 proves the backward identity; Step 6 also explains why the backward identity is not the forward one applied to $-f$.

Step 1: the parallel-axis identity. Fix $x, w \in F$ and set

$$a = \frac{1}{2\ell}, \qquad b = \frac{1}{2h}, \qquad y_* = \frac{a\,x + b\,w}{a+b}.$$

Since $a+b = \dfrac{\ell+h}{2\ell h}$, we have $\dfrac{a}{a+b} = \dfrac{h}{\ell+h}$ and $\dfrac{b}{a+b} = \dfrac{\ell}{\ell+h}$, so $y_* = \dfrac{h\,x + \ell\,w}{\ell+h}$ is the convex combination of x and w with weights $h/(\ell+h)$ and $\ell/(\ell+h)$. Now $Q(y) = a|y-x|^2 + b|y-w|^2$, and expanding both squares by bilinearity and symmetry of the inner product,

$$Q(y) = (a+b)|y|^2 - 2\langle y,\, a\,x + b\,w\rangle + a|x|^2 + b|w|^2.$$

Because $a+b > 0$ we may complete the square, obtaining

$$Q(y) = (a+b)\Big|y - \frac{a\,x+b\,w}{a+b}\Big|^2 - \frac{|a\,x+b\,w|^2}{a+b} + a|x|^2 + b|w|^2.$$

For the constant term, $|a\,x+b\,w|^2 = a^2|x|^2 + 2ab\langle x,w\rangle + b^2|w|^2$, so

$$(a+b)\big(a|x|^2 + b|w|^2\big) - |a\,x+b\,w|^2 = ab|x|^2 - 2ab\langle x,w\rangle + ab|w|^2 = ab\,|w-x|^2,$$

the $a^2|x|^2$ and $b^2|w|^2$ terms cancelling, and hence

$$a|x|^2 + b|w|^2 - \frac{|a\,x+b\,w|^2}{a+b} = \frac{ab}{a+b}\,|w-x|^2.$$

Finally $\dfrac{ab}{a+b} = \dfrac{1}{4\ell h}\cdot\dfrac{2\ell h}{\ell+h} = \dfrac{1}{2(\ell+h)}$. Substituting the last two displays and $a+b = \frac{\ell+h}{2\ell h}$ into the completed square gives, for every $y \in F$,

$$\frac{|y-x|^2}{2\ell} + \frac{|w-y|^2}{2h} = \frac{|w-x|^2}{2(\ell+h)} + \frac{\ell+h}{2\ell h}\,|y-y_*|^2. \tag{554}$$

Three consequences are used below, each immediate from (554).

(1a) Since $\frac{\ell+h}{2\ell h} > 0$ and $|y-y_*|^2 \geq 0$, for *every* $y \in F$

$$\frac{|y-x|^2}{2\ell} + \frac{|w-y|^2}{2h} \;\geq\; \frac{|w-x|^2}{2(\ell+h)}.$$

(1b) Taking $y = y_*$ in (554) gives $Q(y_*) = \frac{|w-x|^2}{2(\ell+h)}$. Hence $\{Q(y) : y \in F\}$ is nonempty and, by (1a), bounded below by $\frac{|w-x|^2}{2(\ell+h)}$, which is one of its elements; therefore

$$\inf_{y\in F} Q(y) = \frac{|w-x|^2}{2(\ell+h)},$$

the infimum being attained at y_* and, since $\frac{\ell+h}{2\ell h} > 0$ forces $|y-y_*|^2 = 0$ for any other minimizer, only there.

(1c) Consistently with (1b), $y_* - x = \frac{\ell}{\ell+h}(w-x)$ and $w - y_* = \frac{h}{\ell+h}(w-x)$ by direct computation from $y_* = \frac{hx+\ell w}{\ell+h}$, and substituting these two into Q gives $\frac{1}{2\ell}\frac{\ell^2|w-x|^2}{(\ell+h)^2} + \frac{1}{2h}\frac{h^2|w-x|^2}{(\ell+h)^2} = \frac{|w-x|^2}{2(\ell+h)^2}(\ell + h) = \frac{|w-x|^2}{2(\ell+h)}$.

Step 2: the single-cost family is nonempty and bounded above. Assume from here to the end of Step 5 that t, $t+h$ and $t+\ell+h$ lie in I. Put $s = t+\ell+h$ and define

$$G(w) = f(w,s) - \frac{|w-x|^2}{2(\ell+h)} \qquad (w \in F).$$

We claim there is $B \in \mathbb{R}$ with $G(w) \le B$ for all $w \in F$. The singleton $J = \{s\}$ is compact and $J \subseteq I$ because $s = t+\ell+h \in I$ by hypothesis, so `SubquadraticGrowth` applies to J with $\varepsilon = \frac{1}{8(\ell+h)} > 0$ and yields $M > 0$ such that

$$|f(w,s)| \le \frac{|w|^2}{8(\ell+h)} \qquad (|w| \ge M).$$

Set $R = \max\{M, 2|x|\}$. If $|w| \ge R$ then $|x| \le |w|/2$, so by the triangle inequality $|w-x| \ge |w| - |x| \ge |w|/2 \ge 0$ and hence $|w-x|^2 \ge |w|^2/4$; combining,

$$\frac{|w-x|^2}{2(\ell+h)} \;\ge\; \frac{|w|^2}{8(\ell+h)} \;\ge\; f(w,s),$$

so $G(w) \le 0$. If $|w| \le R$, note that F is finite dimensional and $\overline{B}_R = \{w \in F : |w| \le R\}$ is closed and bounded, hence compact, and nonempty because $R \ge M > 0$ gives $0 \in \overline{B}_R$; the map $w \mapsto (w,s)$ is continuous and takes $\overline{B}_R$ into $\Omega = \{(y,\sigma) : \sigma \in I\}$ because $s \in I$, so $w \mapsto f(w,s)$ is continuous on $\overline{B}_R$ by the continuity hypothesis on f, and $w \mapsto \frac{|w-x|^2}{2(\ell+h)}$ is continuous. Thus G is continuous on the nonempty compact set $\overline{B}_R$ and attains a finite maximum B_0 there. With $B = \max\{B_0, 0\}$ we get $G(w) \le B$ for every $w \in F$. The family is also nonempty, F containing x.

Step 3: every extremum in the iterated envelope is legitimate. Define

$$H(y,w) = f(w,s) - \frac{|w-y|^2}{2h} - \frac{|y-x|^2}{2\ell} \qquad (y,w \in F).$$

Applying (1a) at the pair (x,w), we have for every $y \in F$

$$\frac{|y-x|^2}{2\ell} + \frac{|w-y|^2}{2h} \;\ge\; \frac{|w-x|^2}{2(\ell+h)},$$

so for all $y, w \in F$

$$H(y,w) \;\le\; f(w,s) - \frac{|w-x|^2}{2(\ell+h)} = G(w) \;\le\; B. \tag{555}$$

Consequently:

- for each fixed $y \in F$ the set $\left\{f(w,s) - \frac{|w-y|^2}{2h} : w \in F\right\}$ is nonempty (take $w = y$) and bounded above by $B + \frac{|y-x|^2}{2\ell}$, by (555);
- the set $\{H(y,w) : (y,w) \in F \times F\}$ is nonempty (take $y = w = x$) and bounded above by B;
- for each fixed $w \in F$ the set $\{H(y,w) : y \in F\}$ is nonempty (take $y = x$) and bounded above by B;

- the set $\{f(w,s) - \frac{|w-x|^2}{2(\ell+h)} : w \in F\}$ is nonempty and bounded above by B, by Step 2.

By QuadEnvP with length $h \neq 0$, at the center y and time $t+\ell$, the envelope $\mathcal{P}_h f(y, t+\ell)$ is the supremum of the first of these sets, since the time argument of f is $(t+\ell)+h = s$; the first bullet makes it a genuine supremum, in particular finite. Note that f is evaluated only at time $s = t+\ell+h$, so no information about $t+\ell$ is required here or anywhere below. By QuadEnvP with length $\ell \neq 0$ at the center (x,t),

$$\mathcal{P}_\ell\big(\mathcal{P}_h f\big)(x,t) = \sup_{y\in F}\left\{\mathcal{P}_h f(y,t+\ell) - \frac{|y-x|^2}{2\ell}\right\}, \tag{556}$$

and this outer set is nonempty and bounded above by B: for each y, subtracting the constant $\frac{|y-x|^2}{2\ell}$ from the first bullet's upper bound gives $\mathcal{P}_h f(y,t+\ell) - \frac{|y-x|^2}{2\ell} \leq B$.

Step 4: the interchange, in three separate moves.

(4i) Translating the inner supremum by a w-independent constant. Fix $y \in F$. For a nonempty set $S \subseteq \mathbb{R}$ bounded above and $c \in \mathbb{R}$ one has $\sup\{u - c : u \in S\} = \sup S - c$, since $u \leq \sup S$ for all $u \in S$ gives $\leq$, and $\sup S \leq \sup\{u-c : u \in S\} + c$ follows because $\sup\{u-c : u \in S\} + c$ is an upper bound for S. Applying this with S the first bullet's set and $c = \frac{|y-x|^2}{2\ell}$,

$$\mathcal{P}_h f(y,t+\ell) - \frac{|y-x|^2}{2\ell} = \sup_{w\in F}\left\{f(w,s) - \frac{|w-y|^2}{2h} - \frac{|y-x|^2}{2\ell}\right\} = \sup_{w\in F} H(y,w).$$

(4ii) Iterated supremum equals supremum over the product. All three families in the second and third bullets are nonempty and bounded above, so $S^* = \sup_{(y,w)\in F\times F} H(y,w)$, $\sup_w H(y,w)$ for each y, and $\sup_y H(y,w)$ for each w all exist in $\mathbb{R}$. For each y and each w, $H(y,w) \leq S^*$, so $\sup_w H(y,w) \leq S^*$; the set $\{\sup_w H(y,w) : y \in F\}$ is therefore nonempty and bounded above by S^*, giving $\sup_y \sup_w H(y,w) \leq S^*$. Conversely, for any (y,w), $H(y,w) \leq \sup_{w'} H(y,w') \leq \sup_{y'}\sup_{w'} H(y',w')$, so $\sup_{y'}\sup_{w'} H(y',w')$ is an upper bound for the product family and $S^* \leq \sup_y \sup_w H(y,w)$. Hence $\sup_y \sup_w H(y,w) = S^*$, and the identical argument with the roles of the two coordinates exchanged gives $\sup_w \sup_y H(y,w) = S^*$. Combining with (556) and (4i),

$$\mathcal{P}_\ell\big(\mathcal{P}_h f\big)(x,t) = \sup_{y\in F}\sup_{w\in F} H(y,w) = S^* = \sup_{w\in F}\sup_{y\in F} H(y,w).$$

(4iii) Pulling out the y-independent term and passing to the infimum. Fix $w \in F$. The quantity $f(w,s)$ does not depend on y, and $H(y,w) = f(w,s) - Q(y)$ with Q as in Step 1. By (1b) the set $\{Q(y) : y \in F\}$ is nonempty and bounded below, so for a nonempty $T \subseteq \mathbb{R}$ bounded below we may use $\sup\{-v : v \in T\} = -\inf T$: indeed $-v \leq -\inf T$ for every $v \in T$ gives $\leq$, and any upper bound β for $\{-v : v \in T\}$ makes $-\beta$ a lower bound for T, so $-\beta \leq \inf T$, i.e. $-\inf T \leq \beta$; as $\sup\{-v : v \in T\}$ is the least upper bound, the two halves give equality. Applying this with $T = \{Q(y) : y \in F\}$, which is nonempty and bounded below by (1b),

$$\sup_{y\in F}\big(-Q(y)\big) = -\inf_{y\in F} Q(y) = -\frac{|w-x|^2}{2(\ell+h)},$$

the second equality by (1b). In particular $\{-Q(y) : y \in F\}$ is nonempty and bounded above, so the translation rule of (4i) applies to it with $c = -f(w,s)$; since $H(y,w) = \big(-Q(y)\big) - \big(-f(w,s)\big)$, that rule gives

$$\sup_{y\in F} H(y,w) = \sup_{y\in F}\big(-Q(y)\big) + f(w,s) = f(w,s) - \frac{|w-x|^2}{2(\ell+h)} = G(w).$$

Step 5: conclusion of the forward identity. Combining (4ii) and (4iii),

$$\mathcal{P}_\ell\big(\mathcal{P}_h f\big)(x,t) = \sup_{w\in F} G(w) = \sup_{w\in F}\left\{f(w,t+\ell+h) - \frac{|w-x|^2}{2(\ell+h)}\right\},$$

a supremum over a nonempty set bounded above by Step 2. On the other hand $\ell + h > 0$, so by `QuadEnvP` with length $\ell + h \neq 0$ at the center (x, t),

$$\mathcal{P}_{\ell+h} f(x,t) = \sup_{w \in F} \left\{ f\big(w, t + (\ell + h)\big) - \frac{|w - x|^2}{2(\ell + h)} \right\},$$

and $t + (\ell + h) = t + \ell + h = s$, so the two right-hand sides are suprema of the same set. This is (553). (The argument used only $t + \ell + h \in I$; the hypotheses $t \in I$ and $t + h \in I$ belong to the statement's admissibility convention and are not needed for this identity.)

Step 6: the backward identity, written out. Assume now $\ell, h > 0$ and t, $t - h$, $t - \ell - h \in I$, and put $\sigma = t - \ell - h$. This is *not* the forward case applied to $-f$: unwinding `QuadEnvP` for $-f$ gives $-\mathcal{P}_\ell(-f)(x,t) = \inf_y\{f(y, t + \ell) + \frac{|y-x|^2}{2\ell}\}$, whose time argument is $t + \ell$, whereas `QuadEnvM` shifts time by $-\ell$; reducing one to the other would additionally require reflecting the time variable about a base point, which replaces I by its reflection and so is unavailable under hypotheses stated for the fixed set I. Note also that the cost signs are *not* reversed below: the two cost terms are added, exactly as in Q, and the same infimum (1b) is used, with no appeal to $\sup(-T) = -\inf T$.

By `QuadEnvM` with length $h \neq 0$ at the center y and time $t - \ell$, the time argument of f is $(t - \ell) - h = \sigma$, so

$$\mathcal{M}_h f(y, t - \ell) = \inf_{w \in F} \left\{ f(w, \sigma) + \frac{|w - y|^2}{2h} \right\},$$

and by `QuadEnvM` with length $\ell \neq 0$ at (x, t),

$$\mathcal{M}_\ell\big(\mathcal{M}_h f\big)(x,t) = \inf_{y \in F} \left\{ \mathcal{M}_h f(y, t - \ell) + \frac{|y - x|^2}{2\ell} \right\}.$$

Set $\widetilde{G}(w) = f(w, \sigma) + \frac{|w-x|^2}{2(\ell+h)}$ and $\widetilde{H}(y, w) = f(w, \sigma) + \frac{|w-y|^2}{2h} + \frac{|y-x|^2}{2\ell}$. Because an infimum over a set unbounded below is likewise the junk value 0, this direction needs its own *lower* bound, which is the mirror of Step 2. The singleton $\{\sigma\}$ is compact and contained in I since $\sigma = t - \ell - h \in I$ by hypothesis, so `SubquadraticGrowth` with $\varepsilon = \frac{1}{8(\ell+h)} > 0$ gives $M > 0$ with $|f(w, \sigma)| \leq \frac{|w|^2}{8(\ell+h)}$ for $|w| \geq M$. With $R = \max\{M, 2|x|\}$, if $|w| \geq R$ then as in Step 2 $|w - x|^2 \geq |w|^2/4$, hence

$$\frac{|w - x|^2}{2(\ell + h)} \;\geq\; \frac{|w|^2}{8(\ell + h)} \;\geq\; -f(w, \sigma),$$

so $\widetilde{G}(w) \geq 0$; and on $\overline{B}_R = \{w \in F : |w| \leq R\}$, which is closed, bounded and nonempty exactly as in Step 2, hence compact and nonempty, the function $\widetilde{G}$ is continuous, because $\sigma \in I$ puts (w, σ) in Ω for every such w, so $\widetilde{G}$ attains a finite minimum b_0 there. With $b = \min\{b_0, 0\}$ we get $\widetilde{G}(w) \geq b$ for all $w \in F$. By (1a), $\frac{|y-x|^2}{2\ell} + \frac{|w-y|^2}{2h} \geq \frac{|w-x|^2}{2(\ell+h)}$, so

$$\widetilde{H}(y, w) \;\geq\; \widetilde{G}(w) \;\geq\; b \qquad (y, w \in F).$$

Hence, mirroring Step 3: for each fixed y the set $\{f(w, \sigma) + \frac{|w-y|^2}{2h} : w \in F\}$ is nonempty ($w = y$) and bounded below by $b - \frac{|y-x|^2}{2\ell}$, so $\mathcal{M}_h f(y, t - \ell)$ is a genuine infimum; the families $\{\widetilde{H}(y, w) : (y, w) \in F \times F\}$, $\{\widetilde{H}(y, w) : y \in F\}$ for fixed w, and $\{\widetilde{G}(w) : w \in F\}$ are nonempty and bounded below by b; and the outer set $\{\mathcal{M}_h f(y, t - \ell) + \frac{|y-x|^2}{2\ell} : y \in F\}$ is nonempty and bounded below by b.

The three moves of Step 4 now run with every supremum replaced by an infimum and each inequality reversed, the corresponding elementary facts being $\inf\{u + c : u \in S\} = \inf S + c$ for nonempty S bounded below and $c \in \mathbb{R}$, and the equality of the iterated infimum with the infimum over the product for a nonempty family bounded below (same two-inequality argument as in (4ii), with lower bounds in place of upper bounds). This gives

$$\mathcal{M}_\ell\big(\mathcal{M}_h f\big)(x,t) = \inf_{y \in F} \inf_{w \in F} \widetilde{H}(y, w) = \inf_{w \in F} \inf_{y \in F} \widetilde{H}(y, w),$$

and for fixed w, since $f(w,\sigma)$ does not depend on y and $\widetilde{H}(y,w) = f(w,\sigma) + Q(y)$,

$$\inf_{y\in F} \widetilde{H}(y,w) = f(w,\sigma) + \inf_{y\in F} Q(y) = f(w,\sigma) + \frac{|w-x|^2}{2(\ell+h)} = \widetilde{G}(w),$$

by the translation rule and (1b) – the same infimum as in the forward case, with no sign change. Therefore

$$\mathcal{M}_\ell\big(\mathcal{M}_h f\big)(x,t) = \inf_{w\in F}\left\{ f(w, t-\ell-h) + \frac{|w-x|^2}{2(\ell+h)} \right\},$$

an infimum over a nonempty set bounded below. Finally $\ell + h > 0$, so by `QuadEnvM` with length $\ell + h \neq 0$ at (x,t),

$$\mathcal{M}_{\ell+h} f(x,t) = \inf_{w\in F}\left\{ f\big(w, t-(\ell+h)\big) + \frac{|w-x|^2}{2(\ell+h)} \right\},$$

and $t-(\ell+h) = t-\ell-h = \sigma$, so the two infima are over the same set and $\mathcal{M}_\ell(\mathcal{M}_h f)(x,t) = \mathcal{M}_{\ell+h} f(x,t)$. □

2.142. `DPStepIdentities`.

Lemma 2.142 (Half-step identities for an optimizing step pair)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and $0 \le \kappa < \infty$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Write $f = W$ for the contact envelope of `ContactEnv`, $f^\ell = \mathcal{P}_\ell f$, $g_\epsilon = \mathcal{M}_\epsilon v$ as in `JGap`. Let $\epsilon > 0$ and let $z = (x,t)$ with $t > \epsilon$, and let (z^+, z^-) be an optimizing step pair of size ϵ at z in the sense of `IsStepPair`, with $z^\pm = (x^\pm, t \pm \epsilon)$. Put*

$$Q_+ = \frac{x^+ - x}{\epsilon}, \qquad Q_- = \frac{x - x^-}{\epsilon}. \tag{557}$$

Then

$$\begin{aligned} f^{2\epsilon}(z) &= f^\epsilon(z^+) - \frac{\epsilon}{2}|Q_+|^2, \\ \mathcal{M}_{2\epsilon} v(z) &= g_\epsilon(z^-) + \frac{\epsilon}{2}|Q_-|^2, \end{aligned} \tag{558}$$

and

$$Q_+ = \nabla_x f^{2\epsilon}(z),$$

the gradient being that of `DPSpatialRegularity`. Although z^- is the backward step, Q_- is oriented forward from x^- to x; the spatial increment for that step is $-\epsilon Q_-$.

Proof. Write $f = W = \mathcal{C}_t v - \kappa^2 t/2$ for the contact envelope of `ContactEnv`, $z = (x,t)$, and recall $t > \epsilon > 0$.

Step 0: hypotheses for the envelope lemmas. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there; by `UniformSubquadratic`, applied with the centering point $(0,1)$, it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`. By `BaryEnvSandwich` with $I = \mathbb{R}$, applied at each parameter $\sigma > 0$, and `ContactEnv`,

$$v(y,\tau) - \frac{\kappa^2 \tau}{2} \le f(y,\tau) \le v(y,\tau) \qquad (\tau > 0), \tag{559}$$

so, exactly as for v, the function f has subquadratic spatial growth on compact subintervals of $(0,\infty)$: let $J \subseteq (0,\infty)$ be compact and let $\varepsilon' > 0$. If $J = \emptyset$ the required conclusion holds vacuously with $M' = 1$, since there is no $\tau \in J$; so assume $J \neq \emptyset$, and then $\max J$ exists because J is a nonempty compact set of reals. Apply `SubquadraticGrowth` for v with J and $\varepsilon'/2$ to get $M' > 0$, then enlarge M' so that $\kappa^2 \max J/2 \le (\varepsilon'/2)M'^2$; for $|y| \ge M'$ and $\tau \in J$ the two bounds of (559) then give $|f(y,\tau)| \le \varepsilon'|y|^2$. Also f is continuous on $\{(y,\tau) : \tau > 0\}$, because `BaryEnvDuality`, part (3), applied to v with $I = \mathbb{R}$, makes $(\sigma,\zeta) \mapsto \mathcal{C}_\sigma v(\zeta)$ continuous at every (σ,ζ) with $\sigma > 0$, and `ContactEnv` composes this with the continuous map sending ζ to its time coordinate.

Hence `QuadEnvAttained` applies to f with $I = (0, \infty)$ and to v with $I = \mathbb{R}$: in both cases the hypotheses of that lemma – continuity on $\Omega = \{(y, \tau) : \tau \in I\}$ and subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth` – are exactly the two facts just established. Also `DPSpatialRegularity` applies to v and κ, since its hypotheses on v, C_0, κ and f are the hypotheses of the present lemma.

Step 1: the optimizers and the velocities. By `IsStepPair` there are $Y^+, Y^- \in F$ with

$$\mathcal{P}_{2\epsilon} f(z) = f\big(Y^+, t + 2\epsilon\big) - \frac{|Y^+ - x|^2}{2 \cdot 2\epsilon}, \qquad \mathcal{M}_{2\epsilon} v(z) = v\big(Y^-, t - 2\epsilon\big) + \frac{|Y^- - x|^2}{2 \cdot 2\epsilon}, \tag{560}$$

and $z^\pm = (x^\pm, t \pm \epsilon)$ with $x^+ = \frac{1}{2}(x + Y^+)$ and $x^- = \frac{1}{2}(x + Y^-)$. By (557),

$$Q_+ = \frac{x^+ - x}{\epsilon} = \frac{Y^+ - x}{2\epsilon}, \qquad Q_- = \frac{x - x^-}{\epsilon} = \frac{x - Y^-}{2\epsilon},$$

so

$$Y^+ = x + 2\epsilon Q_+, \quad x^+ = x + \epsilon Q_+, \qquad Y^- = x - 2\epsilon Q_-, \quad x^- = x - \epsilon Q_-. \tag{561}$$

Consequently the quadratic costs split evenly along the straight paths:

$$\frac{|Y^+ - x|^2}{4\epsilon} = \epsilon |Q_+|^2, \quad \frac{|Y^+ - x^+|^2}{2\epsilon} = \frac{|x^+ - x|^2}{2\epsilon} = \frac{\epsilon |Q_+|^2}{2}, \tag{562}$$

and likewise with Q_- in place of Q_+.

We also record once the elementary splitting inequality used in Steps 2 and 3: for all $a, b \in F$,

$$\frac{|a + b|^2}{4\epsilon} \le \frac{|a|^2}{2\epsilon} + \frac{|b|^2}{2\epsilon}, \tag{563}$$

because expanding the inner products gives $2|a|^2 + 2|b|^2 - |a+b|^2 = |a|^2 - 2\langle a, b\rangle + |b|^2 = |a-b|^2 \ge 0$, and dividing by $4\epsilon > 0$ yields (563). The slack is exactly $|a - b|^2/(4\epsilon)$, so (563) is an equality precisely when $a = b$.

Step 2: the forward identity. We first show

$$\mathcal{P}_\epsilon f(z^+) = f\big(Y^+, t + 2\epsilon\big) - \frac{\epsilon |Q_+|^2}{2}. \tag{564}$$

Throughout this step we use `QuadEnvAttained`, part (1), for f with $I = (0, \infty)$, which Step 0 licenses. Two instances are needed, and both have admissible time arguments because $t > \epsilon > 0$:

(2a) at the center $z^+ = (x^+, t + \epsilon)$ and the length ϵ, where $t + \epsilon \in (0, \infty)$ and $(t + \epsilon) + \epsilon = t + 2\epsilon \in (0, \infty)$;

(2b) at the center $z = (x, t)$ and the length 2ϵ, where $t \in (0, \infty)$ and $t + 2\epsilon \in (0, \infty)$.

Each instance supplies attainment: the supremum defining the envelope is equal to the value at some competitor. That is all `QuadEnvAttained` gives, and it is NOT the same as domination. The envelope is a supremum, and the supremum of a set that is empty or unbounded above is a junk value, so an equality with the value at one competitor does not by itself place the other competitors below the envelope – the inference runs the other way, since attainment in the sense of a genuine maximum presupposes boundedness rather than supplying it. The domination used below is therefore taken from part (1) of `EnvelopeCompetitorBounds`, whose first clause states that $\mathcal{P}_\ell f(\zeta)$ dominates every competitor value, applied at the same two instances: at the centre z^+ with length ϵ for (2a), and at the centre z with length 2ϵ for (2b). Its hypotheses are the continuity and subquadratic growth of f on $(0, \infty)$ established in Step 0, together with the time memberships just checked. Only domination is needed; the leastness half of the supremum is not used in this step.

For "$\ge$": by (561) the point Y^+ is a competitor in `QuadEnvP` for $\mathcal{P}_\epsilon f(z^+)$, whose time argument is $(t + \epsilon) + \epsilon = t + 2\epsilon$, and its cost is $|Y^+ - x^+|^2/(2\epsilon) = \epsilon |Q_+|^2/2$ by (562). By instance (2a) this competitor value is at most the envelope:

$$\mathcal{P}_\epsilon f(z^+) \ge f\big(Y^+, t + 2\epsilon\big) - \frac{\epsilon |Q_+|^2}{2}. \tag{565}$$

For "≤": by instance (2a) the supremum defining $\mathcal{P}_\epsilon f(z^+)$ is attained at some $y_1 \in F$, that is,

$$\mathcal{P}_\epsilon f(z^+) = f\big(y_1, t+2\epsilon\big) - \frac{|y_1 - x^+|^2}{2\epsilon}. \tag{566}$$

The same point y_1 is a competitor in `QuadEnvP` for $\mathcal{P}_{2\epsilon} f(z)$, whose time argument is also $t+2\epsilon$ and whose cost is $|y_1 - x|^2/(2\cdot 2\epsilon) = |y_1 - x|^2/(4\epsilon)$; by instance (2b) that competitor value is at most the envelope, so

$$f\big(y_1, t+2\epsilon\big) \le \mathcal{P}_{2\epsilon} f(z) + \frac{|y_1 - x|^2}{4\epsilon}. \tag{567}$$

Apply (563) with $a = y_1 - x^+$ and $b = x^+ - x$, whose sum is $y_1 - x$:

$$\frac{|y_1 - x|^2}{4\epsilon} \le \frac{|y_1 - x^+|^2}{2\epsilon} + \frac{|x^+ - x|^2}{2\epsilon}. \tag{568}$$

Substituting (567) into (566) and then using (568), the two terms $|y_1 - x^+|^2/(2\epsilon)$ cancel:

$$\mathcal{P}_\epsilon f(z^+) \le \mathcal{P}_{2\epsilon} f(z) + \frac{|y_1 - x|^2}{4\epsilon} - \frac{|y_1 - x^+|^2}{2\epsilon} \le \mathcal{P}_{2\epsilon} f(z) + \frac{|x^+ - x|^2}{2\epsilon} = \mathcal{P}_{2\epsilon} f(z) + \frac{\epsilon |Q_+|^2}{2},$$

the last equality by (562). By (560) and (562), $\mathcal{P}_{2\epsilon} f(z) = f(Y^+, t+2\epsilon) - \epsilon|Q_+|^2$, so

$$\mathcal{P}_\epsilon f(z^+) \le f\big(Y^+, t+2\epsilon\big) - \epsilon|Q_+|^2 + \frac{\epsilon|Q_+|^2}{2} = f\big(Y^+, t+2\epsilon\big) - \frac{\epsilon|Q_+|^2}{2}.$$

With (565) this proves (564). Note that no property of the composed function $\mathcal{P}_\epsilon f$ is used anywhere above: every supremum appearing is an envelope of f itself.

Combining (560), (562) and (564),

$$\mathcal{P}_{2\epsilon} f(z) = f\big(Y^+, t+2\epsilon\big) - \epsilon|Q_+|^2 = \Big(\mathcal{P}_\epsilon f(z^+) + \frac{\epsilon|Q_+|^2}{2}\Big) - \epsilon|Q_+|^2 = \mathcal{P}_\epsilon f(z^+) - \frac{\epsilon}{2}|Q_+|^2,$$

which is the first line of (558).

Step 3: the backward identity. Symmetrically, we show

$$\mathcal{M}_\epsilon v(z^-) = v\big(Y^-, t-2\epsilon\big) + \frac{\epsilon|Q_-|^2}{2}. \tag{569}$$

Throughout this step we use `QuadEnvAttained`, part (1), for v with $I = \mathbb{R}$, which Step 0 licenses. Because $I = \mathbb{R}$, every time argument is admissible with no sign condition – which matters here, since $t > \epsilon$ leaves $t - 2\epsilon$ possibly negative. Two instances are needed:

(3a) at the center $z^- = (x^-, t-\epsilon)$ and the length ϵ, where $t-\epsilon \in \mathbb{R}$ and $(t-\epsilon)-\epsilon = t-2\epsilon \in \mathbb{R}$;

(3b) at the center $z = (x,t)$ and the length 2ϵ, where $t \in \mathbb{R}$ and $t - 2\epsilon \in \mathbb{R}$.

Each instance supplies attainment: the infimum defining the envelope is equal to the value at some competitor. As in Step 2 that is not domination, and for the same reason with the inequalities reversed, since an infimum over a set that is empty or unbounded below is equally the junk value. The domination used below is part (2) of `EnvelopeCompetitorBounds`, whose first clause states that $\mathcal{M}_\ell v(\zeta)$ is dominated by every competitor value, applied at the centre z^- with length ϵ for (3a) and at the centre z with length 2ϵ for (3b); its hypotheses are the continuity and subquadratic growth of v on $\mathbb{R}$ established in Step 0. Only domination is needed here too.

For "≤": by (561) the point Y^- is a competitor in `QuadEnvM` for $\mathcal{M}_\epsilon v(z^-)$, whose time argument is $(t-\epsilon)-\epsilon = t-2\epsilon$, with cost $|Y^- - x^-|^2/(2\epsilon) = \epsilon|Q_-|^2/2$ by (562). By instance (3a) this competitor value is at least the envelope:

$$\mathcal{M}_\epsilon v(z^-) \le v\big(Y^-, t-2\epsilon\big) + \frac{\epsilon|Q_-|^2}{2}. \tag{570}$$

For "≥": by instance (3a) the infimum defining $\mathcal{M}_\epsilon v(z^-)$ is attained at some $w_0 \in F$, that is,

$$\mathcal{M}_\epsilon v(z^-) = v\big(w_0, t-2\epsilon\big) + \frac{|w_0 - x^-|^2}{2\epsilon}. \tag{571}$$

The same point w_0 is a competitor in `QuadEnvM` for $\mathcal{M}_{2\epsilon}v(z)$, whose time argument is also $t-2\epsilon$ and whose cost is $|w_0-x|^2/(2\cdot 2\epsilon)=|w_0-x|^2/(4\epsilon)$; by instance (3b) that competitor value is at least the envelope, so

$$v\big(w_0,t-2\epsilon\big)\ge \mathcal{M}_{2\epsilon}v(z)-\frac{|w_0-x|^2}{4\epsilon}. \tag{572}$$

Apply (563) with $a=w_0-x^-$ and $b=x^--x$, whose sum is w_0-x:

$$\frac{|w_0-x|^2}{4\epsilon}\le \frac{|w_0-x^-|^2}{2\epsilon}+\frac{|x^--x|^2}{2\epsilon}. \tag{573}$$

Substituting (572) into (571) and then using (573), the two terms $|w_0-x^-|^2/(2\epsilon)$ cancel:

$$\mathcal{M}_\epsilon v(z^-)\ge \mathcal{M}_{2\epsilon}v(z)-\frac{|w_0-x|^2}{4\epsilon}+\frac{|w_0-x^-|^2}{2\epsilon}\ge \mathcal{M}_{2\epsilon}v(z)-\frac{|x^--x|^2}{2\epsilon}=\mathcal{M}_{2\epsilon}v(z)-\frac{\epsilon|Q_-|^2}{2},$$

the last equality by (562). By (560) and (562), $\mathcal{M}_{2\epsilon}v(z)=v(Y^-,t-2\epsilon)+\epsilon|Q_-|^2$, so

$$\mathcal{M}_\epsilon v(z^-)\ge v\big(Y^-,t-2\epsilon\big)+\epsilon|Q_-|^2-\frac{\epsilon|Q_-|^2}{2}=v\big(Y^-,t-2\epsilon\big)+\frac{\epsilon|Q_-|^2}{2}.$$

With (570) this proves (569); again every infimum used is an envelope of v itself, never of a composed function, and w_0 is not claimed to be unique. Combining with (560) and (562),

$$\mathcal{M}_{2\epsilon}v(z)=v\big(Y^-,t-2\epsilon\big)+\epsilon|Q_-|^2=\mathcal{M}_\epsilon v(z^-)+\frac{\epsilon}{2}|Q_-|^2,$$

which is the second line of (558).

Step 4: the forward velocity is the gradient. Apply `DPSpatialRegularity` to v and κ at the time $t>0$, the length $2\epsilon>0$ and the point x. It supplies a point $y\in F$ which attains $\mathcal{P}_{2\epsilon}f(x,t)$, is the *unique* such point, and satisfies

$$x'\mapsto \mathcal{P}_{2\epsilon}f(x',t) \text{ is differentiable at } x \text{ with gradient } \frac{y-x}{2\epsilon}.$$

By (560) the point Y^+ also attains $\mathcal{P}_{2\epsilon}f(x,t)$, so uniqueness gives $Y^+=y$, and by (561) $(y-x)/(2\epsilon)=Q_+$. Hence $Q_+=\nabla_x f^{2\epsilon}(z)$, which is the last assertion. □

2.143. `DPDrift`.

Lemma 2.143 (The one-step drift)**.** *Let F be a finite-dimensional real inner-product space, let $1\le C_0<\infty$ and $0\le\kappa<\infty$, and let $v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Write $f=W$ for the contact envelope of `ContactEnv`, $f^\ell=\mathcal{P}_\ell f$ for the forward envelope of `QuadEnvP`, $g_\epsilon=\mathcal{M}_\epsilon v$ for the backward envelope of `QuadEnvM`, and J_ϵ for the quantity of `JGap`. Let $\epsilon>0$, let $z=(x,t)$ with $t>\epsilon$, and let (z^+,z^-) be an optimizing step pair of size ϵ at z in the sense of `IsStepPair`, with $z^\pm=(x^\pm,t\pm\epsilon)$. Put*

$$Q_+=\frac{x^+-x}{\epsilon},\qquad Q_-=\frac{x-x^-}{\epsilon},$$

and let $Q_0\in F$ be the reference velocity of `DPSpatialRegularity` at z and length ϵ, i.e. a vector at which $f^\epsilon(\cdot,t)$ has gradient Q_0 at x:

$$Q_0=\nabla_x f^\epsilon(x,t).$$

Define

$$\begin{aligned} D&=f(z)+f^{2\epsilon}(z)-2f^\epsilon(z),\\ e&=f^\epsilon(z^-)+\frac{\epsilon}{2}|Q_-|^2-f(z),\\ \widetilde{D}&=2g_\epsilon(z)-v(z)-\mathcal{M}_{2\epsilon}v(z),\\ \widetilde{e}&=v(z)+\frac{\epsilon}{2}|Q_+|^2-g_\epsilon(z^+). \end{aligned}$$

Then all four quantities are nonnegative, and

$$\begin{aligned}\frac{J_\epsilon(z^+)+J_\epsilon(z^-)}{2}-J_\epsilon(z)&=\frac{D+e+\widetilde{D}+\widetilde{e}}{4\epsilon}\\ &\geq\frac{D+e}{4\epsilon}\\ &\geq c\big(|Q_+-Q_0|^2+|Q_--Q_0|^2\big),\qquad c=\frac{1}{16(1+\log 2)}.\end{aligned}\tag{574}$$

Proof. Throughout, $z=(x,t)$ with $t>\epsilon>0$, $x^\pm$ are the spatial parts of $z^\pm$, and $f=W$ denotes the contact envelope of `ContactEnv`, so that $f(y,\tau)=\mathcal{C}_\tau v(y,\tau)-\kappa^2\tau/2$ for $\tau>0$. Note $t-\epsilon>0$ and $t+\epsilon>0$, so every value of f written below is taken at a positive time, and v is defined at every time.

Step 0: standing facts about v and f. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of $F[v]=0$ there; in particular it is a viscosity supersolution on all of space-time in the sense of `IsLimitSuperOn` and a subsolution in the sense of `IsLimitSubOn`, by `IsLimitSolutionOn`. By `UniformSubquadratic`, applied with the centering point $(0,1)$, v has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`; the same property with $I=(0,\infty)$ follows, since shrinking I only shrinks the family of compact subintervals tested. By `BaryEnvSandwich`, applied with $I=\mathbb{R}$, $f=v$ and the constant κ, we have for every $\sigma>0$ and every (y,τ)

$$0\leq\mathcal{C}_\sigma v(y,\tau)-v(y,\tau)\leq\frac{\kappa^2\sigma}{2},\tag{575}$$

and taking $\sigma=\tau>0$ with `ContactEnv` gives

$$v(y,\tau)-\frac{\kappa^2\tau}{2}\leq f(y,\tau)\leq v(y,\tau)\qquad(\tau>0).\tag{576}$$

Consequently f has subquadratic spatial growth on compact subintervals of $(0,\infty)$: given $J\subseteq(0,\infty)$ compact and $\varepsilon'>0$, put $\tau_{\max}=\max J$, apply `SubquadraticGrowth` for v with J and $\varepsilon'/2$ to obtain $M'>0$, and enlarge M' so that $\kappa^2\tau_{\max}/2\leq(\varepsilon'/2)M'^2$; then (576) gives $|f(y,\tau)|\leq\varepsilon'|y|^2$ for $|y|\geq M'$ and $\tau\in J$. Moreover f is continuous on $\{(y,\tau):\tau>0\}$: by `BaryEnvDuality`, part (3), applied to v with $I=\mathbb{R}$, the map $(\sigma,\zeta)\mapsto\mathcal{C}_\sigma v(\zeta)$ is continuous at every pair with $\sigma>0$; composing with the continuous map $\zeta\mapsto(\tau,\zeta)$, where τ is the time coordinate of ζ, and subtracting the continuous function $\kappa^2\tau/2$ gives the claim. Finally v is a viscosity subsolution on the open set $\{\tau>0\}$: if ϕ is twice continuously differentiable at a point ζ of positive time and $v-\phi$ has a local maximum at ζ relative to $\{\tau>0\}$, then, $\{\tau>0\}$ being open and containing ζ, a neighbourhood witnessing that relative maximum may be intersected with $\{\tau>0\}$ and is then a neighbourhood of ζ in the whole space on which the inequality holds, so $v-\phi$ has a local maximum at ζ in the whole space and $0\leq F[\phi](\zeta)$ follows. Hence `BarycentricSubsolution`, applied to v with this subsolution property, with the subquadratic growth on $(0,\infty)$ and with $c=\kappa^2/2$, shows that

$$f\text{ is a viscosity subsolution of }F[\cdot]=0\text{ on }\{\tau>0\},\tag{577}$$

since $\mathcal{C}_\tau v(\zeta)-(\kappa^2/2)\tau$ is exactly f by `ContactEnv`.

Claim B: the envelopes are genuine suprema and infima. Let $I\subseteq\mathbb{R}$, let g be continuous on $\{(y,\tau):\tau\in I\}$ with subquadratic spatial growth on compact subintervals of I in the sense of `SubquadraticGrowth`, let $\ell>0$ and let $\zeta=(\xi,\tau)$ with $\tau\in I$.

(i) If $\tau+\ell\in I$, the set $\{g(y,\tau+\ell)-|y-\xi|^2/(2\ell):y\in F\}$ of `QuadEnvP` is nonempty and bounded above, so $\mathcal{P}_\ell g(\zeta)$ is finite, is at least every member of that set, and is at most every upper bound for it.

(ii) If $\tau-\ell\in I$, the set $\{g(y,\tau-\ell)+|y-\xi|^2/(2\ell):y\in F\}$ of `QuadEnvM` is nonempty and bounded below, so $\mathcal{M}_\ell g(\zeta)$ is finite, is at most every member, and is at least every lower bound.

Proof. (i) The choice $y=\xi$ shows the set is nonempty. Apply `SubquadraticGrowth` with the compact set $J=\{\tau+\ell\}\subseteq I$ and $\varepsilon'=1/(4\ell)>0$ to get $M>0$ with $|g(y,\tau+\ell)|\leq|y|^2/(4\ell)$

for $|y| \geq M$; and $g(\cdot, \tau + \ell)$ is continuous, hence bounded in absolute value by some C_M on the closed ball of radius M. From $|y|^2 \leq 2|y-\xi|^2 + 2|\xi|^2$ we get $|y-\xi|^2 \geq |y|^2/2 - |\xi|^2$, so for every y

$$g(y, \tau+\ell) - \frac{|y-\xi|^2}{2\ell} \leq C_M + \frac{|y|^2}{4\ell} - \frac{|y|^2}{4\ell} + \frac{|\xi|^2}{2\ell} = C_M + \frac{|\xi|^2}{2\ell},$$

an upper bound independent of y. (ii) Apply (i) to $-g$, which is continuous with the same growth property, at the time $\tau - \ell$, and multiply the resulting bound by -1. This proves Claim B.

By Step 0, Claim B applies to $g = f$ with $I = (0, \infty)$ at every center of positive time whose shifted time is also positive, and to $g = v$ with $I = \mathbb{R}$ at every center and length.

Step 1: $D \geq 0$. Apply `QuadEnvPathConvex`, part (1), to f with $I = (0, \infty)$, $T = 2\epsilon > 0$, the subsolution property (577), the continuity and growth of Step 0, the base point $x_0 = x$, the velocity $d' = 0$ and the time t; its hypothesis $t + \ell \in (0, \infty)$ for $\ell \in [0, 2\epsilon]$ holds since $t > 0$. Since $\|d'\| = 0$, the conclusion is that $\ell \mapsto \mathcal{P}_\ell f(x,t)$ is convex on $[0, 2\epsilon]$. The midpoint inequality at $\ell = 0$ and $\ell = 2\epsilon$, whose midpoint is ϵ, gives

$$\mathcal{P}_\epsilon f(x,t) \leq \tfrac{1}{2}\mathcal{P}_0 f(x,t) + \tfrac{1}{2}\mathcal{P}_{2\epsilon} f(x,t),$$

and $\mathcal{P}_0 f(x,t) = f(z)$ by `QuadEnvP`. Multiplying by 2 and rearranging gives $f(z) + f^{2\epsilon}(z) - 2f^\epsilon(z) \geq 0$, i.e. $D \geq 0$.

Step 2: $\widetilde{D} \geq 0$. Apply `QuadEnvPathConvex`, part (2), to v with $I = \mathbb{R}$, $T = 2\epsilon > 0$, the supersolution property of Step 0, and the continuity and growth of Step 0, at the point (x,t); its hypothesis $t - \ell \in \mathbb{R}$ is vacuous. The conclusion is that $\ell \mapsto \mathcal{M}_\ell v(x,t)$ is concave on $[0, 2\epsilon]$, so the midpoint inequality gives

$$\mathcal{M}_\epsilon v(x,t) \geq \tfrac{1}{2}\mathcal{M}_0 v(x,t) + \tfrac{1}{2}\mathcal{M}_{2\epsilon} v(x,t),$$

and $\mathcal{M}_0 v(x,t) = v(z)$ by `QuadEnvM`. Multiplying by 2 and rearranging gives $2g_\epsilon(z) - v(z) - \mathcal{M}_{2\epsilon} v(z) \geq 0$, i.e. $\widetilde{D} \geq 0$.

Step 3: $e \geq 0$, and a competitor bound used later. The center $z^- = (x^-, t - \epsilon)$ has positive time and $(t-\epsilon) + \epsilon = t > 0$, so Claim B(i) applies to f at z^- with length ϵ. For every $y \in F$ the point y is a member of the set of `QuadEnvP` for $f^\epsilon(z^-)$, whose time argument is t; hence

$$f^\epsilon(z^-) \geq f(y,t) - \frac{|y - x^-|^2}{2\epsilon} \qquad (y \in F). \tag{578}$$

Taking $y = x$ and using $x - x^- = \epsilon Q_-$, so that $|x - x^-|^2/(2\epsilon) = \epsilon|Q_-|^2/2$, gives $f^\epsilon(z^-) \geq f(z) - \epsilon|Q_-|^2/2$, i.e. $e \geq 0$.

Step 4: $\widetilde{e} \geq 0$. The center $z^+ = (x^+, t + \epsilon)$ has $(t+\epsilon) - \epsilon = t$, and Claim B(ii) applies to v at z^+ with length ϵ and $I = \mathbb{R}$. The point $y = x$ is a member of the set of `QuadEnvM` for $g_\epsilon(z^+)$, whose time argument is t, so

$$g_\epsilon(z^+) \leq v(z) + \frac{|x - x^+|^2}{2\epsilon} = v(z) + \frac{\epsilon|Q_+|^2}{2},$$

using $x^+ - x = \epsilon Q_+$. This is exactly $\widetilde{e} \geq 0$.

Step 5: the drift identity. By `DPStepIdentities`, applied with the present data, the two half-step identities hold:

$$f^{2\epsilon}(z) = f^\epsilon(z^+) - \frac{\epsilon}{2}|Q_+|^2, \qquad \mathcal{M}_{2\epsilon} v(z) = g_\epsilon(z^-) + \frac{\epsilon}{2}|Q_-|^2. \tag{579}$$

The first gives $f^\epsilon(z^+) = f^{2\epsilon}(z) + \epsilon|Q_+|^2/2$, and the definition of e gives $f^\epsilon(z^-) = f(z) + e - \epsilon|Q_-|^2/2$. Adding and subtracting $2f^\epsilon(z)$,

$$f^\epsilon(z^+) + f^\epsilon(z^-) - 2f^\epsilon(z) = D + e + \frac{\epsilon}{2}\big(|Q_+|^2 - |Q_-|^2\big). \tag{580}$$

Likewise the second identity in (579) gives $g_\epsilon(z^-) = \mathcal{M}_{2\epsilon} v(z) - \epsilon|Q_-|^2/2$, and the definition of $\widetilde{e}$ gives $g_\epsilon(z^+) = v(z) + \epsilon|Q_+|^2/2 - \widetilde{e}$. Adding and subtracting $2g_\epsilon(z)$,

$$g_\epsilon(z^+) + g_\epsilon(z^-) - 2g_\epsilon(z) = -\widetilde{D} - \widetilde{e} + \frac{\epsilon}{2}\big(|Q_+|^2 - |Q_-|^2\big), \tag{581}$$

because $-\widetilde{D} = v(z) + \mathcal{M}_{2\epsilon}v(z) - 2g_\epsilon(z)$. The half-step cost difference $\epsilon(|Q_+|^2 - |Q_-|^2)/2$ is the same in (580) and (581), so subtracting the second from the first cancels it:

$$\big(f^\epsilon - g_\epsilon\big)(z^+) + \big(f^\epsilon - g_\epsilon\big)(z^-) - 2\big(f^\epsilon - g_\epsilon\big)(z) = D + e + \widetilde{D} + \widetilde{e}.$$

By JGap, $J_\epsilon = (f^\epsilon - g_\epsilon)/(2\epsilon)$, so dividing by 4ϵ turns the left side into $\big(J_\epsilon(z^+) + J_\epsilon(z^-)\big)/2 - J_\epsilon(z)$ and proves the first line of (574). Since $\widetilde{D} \geq 0$ and $\widetilde{e} \geq 0$ by Steps 2 and 4, and $4\epsilon > 0$, the second line follows.

Step 6: the velocity estimate. Let $b, d \in F$ be arbitrary. Apply QuadEnvPathConvex, part (1), to f with $I = (0, \infty)$, $T = 2\epsilon$, the base point $x_0 = x + d$, the velocity $d' = (b - d)/(2\epsilon)$ and the time t; the hypotheses are those of Step 1. Writing

$$\Phi(\ell) = \mathcal{P}_\ell f\big(x + d + \ell d', t\big) + \frac{|b - d|^2}{4\epsilon^2}\,\ell \log \ell \qquad (0 \leq \ell \leq 2\epsilon),$$

where $\|d'\|^2 = |b - d|^2/(4\epsilon^2)$ and $\ell \log \ell$ takes the value 0 at $\ell = 0$, the conclusion is that Φ is convex on $[0, 2\epsilon]$. Now $x + d + 0 \cdot d' = x + d$, $x + d + 2\epsilon d' = x + b$ and $x + d + \epsilon d' = x + (b + d)/2$, and $\mathcal{P}_0 f = f$ by QuadEnvP, so

$$\Phi(0) = f(x + d, t), \qquad \Phi(2\epsilon) = f^{2\epsilon}(x + b, t) + \frac{|b - d|^2}{2\epsilon}\log(2\epsilon),$$

$$\Phi(\epsilon) = f^\epsilon\Big(x + \frac{b + d}{2}, t\Big) + \frac{|b - d|^2}{4\epsilon}\log \epsilon.$$

The midpoint inequality $2\Phi(\epsilon) \leq \Phi(0) + \Phi(2\epsilon)$ and $\log(2\epsilon) - \log \epsilon = \log 2$ therefore give

$$f(x + d, t) + f^{2\epsilon}(x + b, t) - 2f^\epsilon\Big(x + \frac{b + d}{2}, t\Big) \geq -\frac{\log 2}{2\epsilon}|b - d|^2. \tag{582}$$

We bound the left side of (582) from above by three inequalities. First, (578) with $y = x + d$, together with $x + d - x^- = d + \epsilon Q_-$ and

$$\frac{|d + \epsilon Q_-|^2}{2\epsilon} = \frac{|d|^2}{2\epsilon} + Q_- \cdot d + \frac{\epsilon}{2}|Q_-|^2,$$

and with $f^\epsilon(z^-) = f(z) + e - \epsilon|Q_-|^2/2$ from the definition of e, gives

$$f(x + d, t) \leq f(z) + e + Q_- \cdot d + \frac{|d|^2}{2\epsilon}. \tag{583}$$

Second, DPSpatialRegularity, applied at the time $t > 0$, the length $2\epsilon > 0$ and the center x, provides the unique maximizer $y_{2\epsilon}$ with $\nabla_x f^{2\epsilon}(x, t) = (2\epsilon)^{-1}(y_{2\epsilon} - x)$ and the two supporting inequalities at that gradient. By DPStepIdentities the same function has gradient Q_+ at x, and a function has at most one gradient at a point, so $(2\epsilon)^{-1}(y_{2\epsilon} - x) = Q_+$. The upper supporting inequality with $h = b$ is therefore

$$f^{2\epsilon}(x + b, t) \leq f^{2\epsilon}(z) + Q_+ \cdot b + \frac{|b|^2}{2t}. \tag{584}$$

Third, DPSpatialRegularity at the time t, the length $\epsilon > 0$ and the center x provides the unique maximizer y_ϵ with $\nabla_x f^\epsilon(x, t) = \epsilon^{-1}(y_\epsilon - x)$ and the same two inequalities; by hypothesis $f^\epsilon(\cdot, t)$ has gradient Q_0 at x, so again by uniqueness of the gradient $\epsilon^{-1}(y_\epsilon - x) = Q_0$. The lower supporting inequality with $h = (b + d)/2$ is

$$f^\epsilon(z) + Q_0 \cdot \frac{b + d}{2} - \frac{|b + d|^2}{8\epsilon} \leq f^\epsilon\Big(x + \frac{b + d}{2}, t\Big),$$

that is, after multiplying by -2,

$$-2f^\epsilon\Big(x + \frac{b + d}{2}, t\Big) \leq -2f^\epsilon(z) - Q_0 \cdot (b + d) + \frac{|b + d|^2}{4\epsilon}. \tag{585}$$

Adding (583), (584) and (585), and using $D = f(z) + f^{2\epsilon}(z) - 2f^\epsilon(z)$, the left side of (582) is at most

$$D + e + (Q_+ - Q_0) \cdot b + (Q_- - Q_0) \cdot d + \frac{|b|^2}{2t} + \frac{|d|^2}{2\epsilon} + \frac{|b + d|^2}{4\epsilon}.$$

Combining with (582) and rearranging,

$$D+e \ge (Q_0-Q_+)\cdot b+(Q_0-Q_-)\cdot d-\frac{|b|^2}{2t}-\frac{|d|^2}{2\epsilon}-\frac{|b+d|^2}{4\epsilon}-\frac{\log 2}{2\epsilon}|b-d|^2.$$

Since $t>\epsilon$ we have $|b|^2/(2t)\le |b|^2/(2\epsilon)$, and $|b\pm d|^2\le 2(|b|^2+|d|^2)$ gives $|b+d|^2/(4\epsilon)\le (|b|^2+|d|^2)/(2\epsilon)$ and $(\log 2/(2\epsilon))|b-d|^2\le(\log 2/\epsilon)(|b|^2+|d|^2)$. The four subtracted terms therefore total at most

$$\Big(\frac{1}{2\epsilon}+\frac{1}{2\epsilon}+\frac{\log 2}{\epsilon}\Big)\big(|b|^2+|d|^2\big)=\frac{1+\log 2}{\epsilon}\big(|b|^2+|d|^2\big),$$

so that

$$D+e\ge (Q_0-Q_+)\cdot b+(Q_0-Q_-)\cdot d-\frac{1+\log 2}{\epsilon}\big(|b|^2+|d|^2\big) \tag{586}$$

for all $b,d\in F$. Put $A=Q_0-Q_+$, $B=Q_0-Q_-$, $\mu=\epsilon/(2(1+\log 2))>0$, and choose $b=\mu A$, $d=\mu B$ in (586). Then

$$D+e\ge \mu\big(|A|^2+|B|^2\big)-\frac{1+\log 2}{\epsilon}\mu^2\big(|A|^2+|B|^2\big)=\Big(\mu-\frac{\mu}{2}\Big)\big(|A|^2+|B|^2\big)=\frac{\epsilon}{4(1+\log 2)}\big(|A|^2+|B|^2\big),$$

where the middle equality uses $((1+\log 2)/\epsilon)\mu^2=\mu/2$. Dividing by $4\epsilon>0$ and recalling $|A|=|Q_+-Q_0|$, $|B|=|Q_--Q_0|$,

$$\frac{D+e}{4\epsilon}\ge\frac{1}{16(1+\log 2)}\big(|Q_+-Q_0|^2+|Q_--Q_0|^2\big),$$

which is the last line of (574) with $c=1/(16(1+\log 2))$. This completes the proof. □

2.144. `FittedEnvelopeBounds`.

Lemma 2.144 (Envelope bounds for fitted solutions)**.** *Let F be a finite-dimensional real inner-product space and let $1\le C_0<\infty$. Write $L(K)=1+\log K$ for $K\ge 2$ and $\mathcal{B}_K=\{(x,t):|x|\le K,\ |t|\le K\}$. There is $C>0$, depending only on C_0 and $\dim F$, such that for every $v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` which is centered at $(0,0)$ or at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, every $K\ge 2$ and every $z\in\mathcal{B}_K$:*

$$\mathcal{S}^+_\ell v(z)\le CL(K)^4,\qquad \mathcal{S}^-_\ell v(z)\le CL(K)^4\qquad(0<\ell\le K), \tag{587}$$

with the slopes of `EnvSlopeP` and `EnvSlopeM`; and every optimizing displacement $y-x$ for $\mathcal{P}_Kv(z)$ or for $\mathcal{M}_Kv(z)$, in the sense of `QuadEnvP` and `QuadEnvM`, has length at most $CKL(K)^2$.

Proof. Write $L(K)=1+\log K\ge 1$ for $K\ge 2$. Let $C_g>0$ be the constant of `CenteredGrowth` for this C_0, so that

$$|v(\zeta)|\le C_g\,\Theta\,L(\Theta)^2\qquad(\zeta\in\mathcal{B}_\Theta,\ \Theta\ge 2) \tag{588}$$

for every $v\in\mathcal{A}_{m,C_0}$ centered at $(0,0)$ or at $(0,1)$ with constant C_0. Fix such a v, let $K\ge 2$ and let $z=(x,t)\in\mathcal{B}_K$.

Step 0: the envelopes at length K are attained. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of $F[v]=0$ there. By `UniformSubquadratic`, applied with the centering point of v, for every compact $J\subseteq\mathbb{R}$ and every $\varepsilon>0$ there is $M>0$ with $|v(y,\sigma)|\le\varepsilon|y|^2$ whenever $|y|\ge M$ and $\sigma\in J$; that is, v has subquadratic spatial growth on compact time intervals of $I=\mathbb{R}$ in the sense of `SubquadraticGrowth`. Hence `QuadEnvAttained` applies to v with $I=\mathbb{R}$, and its part (1) with the length $K>0$ gives $y^+,y^-\in F$ with

$$\mathcal{P}_Kv(z)=v(y^+,t+K)-\frac{|y^+-x|^2}{2K},\qquad \mathcal{M}_Kv(z)=v(y^-,t-K)+\frac{|y^--x|^2}{2K}. \tag{589}$$

The same applies to any optimizer, so the displacement bound below applies to every optimizing displacement at length K.

Step 1: the optimizing displacements at length K. Let y be any optimizer for $\mathcal{P}_Kv(z)$, and write $|y-x|=BK$ with $B\ge 0$. The point x itself, at time $(t+K)$, is a competitor in the supremum of `QuadEnvP` defining $\mathcal{P}_Kv(z)$, and its cost $|x-x|^2/(2K)$ vanishes. That a competitor's value is at most the envelope is not read off the definition: the envelope is a supremum, and the

supremum of a set that is empty or unbounded above is a junk value, so the domination must be licensed. The licence is part (1) of `EnvelopeCompetitorBounds`, whose first clause states that $\mathcal{P}_\ell v(z)$ dominates every competitor value; we apply it at this centre z and the length $K>0$, its hypotheses being the continuity and subquadratic growth of v established in Step 0 together with $t\in I$ and $t+K\in I$, which are automatic since $I=\mathbb{R}$. This gives $\mathcal{P}_K v(z)\geq v(x,t+K)$, so

$$\frac{|y-x|^2}{2K}\leq v(y,t+K)-v(x,t+K)\leq |v(y,t+K)|+|v(x,t+K)|. \tag{590}$$

If $B\leq 2$ we are done, since $|y-x|\leq 2K\leq 2KL(K)^2$. So assume $B>2$ and put $\Theta=(B+2)K$; then $\Theta\geq 4K\geq 8\geq 2$. We have $|y|\leq |x|+BK\leq (B+1)K$, $|x|\leq K$ and $|t+K|\leq 2K$, so both $(y,t+K)$ and $(x,t+K)$ lie in $\mathcal{B}_\Theta$. By (588) and (590),

$$\frac{B^2K}{2}\leq 2C_g\,\Theta\,L(\Theta)^2=2C_g(B+2)K\big(1+\log((B+2)K)\big)^2\leq 4C_gBK\big(1+\log(2BK)\big)^2,$$

using $B+2<2B$ for $B>2$. Dividing by $BK/2>0$,

$$B\leq 8C_g\big(1+\log 2+\log B+\log K\big)^2.$$

Since $1+\log 2+\log K\leq (1+\log 2)(1+\log K)=(1+\log 2)L(K)$, and $(u+w)^2\leq 2u^2+2w^2$,

$$B\leq a\,L(K)^2+b\,(\log B)^2,\qquad a:=16C_g(1+\log 2)^2,\quad b:=16C_g. \tag{591}$$

Because $(\log B)^2/B\to 0$ as $B\to\infty$, we may fix $B_0\geq 2$, depending only on b, with $b(\log B)^2\leq B/2$ for all $B\geq B_0$. If $B\geq B_0$ then (591) gives $B\leq aL(K)^2+B/2$, hence $B\leq 2aL(K)^2$; and if $B<B_0$ then $B<B_0\leq B_0L(K)^2$ since $L(K)\geq 1$. In all cases

$$|y-x|=BK\leq C_6\,K\,L(K)^2,\qquad C_6:=\max\{2a,\,B_0,\,2\}, \tag{592}$$

which also covers the case $B\leq 2$.

For an optimizer y of $\mathcal{M}_K v(z)$ the same reasoning applies with the mirror clause: part (2) of `EnvelopeCompetitorBounds`, whose first clause states that $\mathcal{M}_\ell v(z)$ is dominated by every competitor value, applied at this centre and this length with the competitor x of vanishing cost, gives $\mathcal{M}_K v(z)\leq v(x,t-K)$, while `QuadEnvM` and attainment give $\mathcal{M}_K v(z)=v(y,t-K)+|y-x|^2/(2K)$; hence

$$\frac{|y-x|^2}{2K}\leq v(x,t-K)-v(y,t-K)\leq |v(x,t-K)|+|v(y,t-K)|,$$

and the argument above applies verbatim with $t-K$ in place of $t+K$, giving the same bound (592). This is the displacement assertion.

Step 2: the slopes at length K. Let $y=y^+$ be as in (589), so $|y-x|\leq C_6KL(K)^2$ by (592). Then $|y|\leq |x|+C_6KL(K)^2\leq (1+C_6)KL(K)^2$ and $|t+K|\leq 2K$, so $(y,t+K)\in\mathcal{B}_{K_1}$ with

$$K_1:=C_7\,K\,L(K)^2,\qquad C_7:=3+C_6,$$

and $K_1\geq 3K\geq 6\geq 2$. Moreover

$$L(K_1)=1+\log C_7+\log K+2\log L(K)\leq (1+\log C_7)+L(K)+2L(K)\leq (3+\log C_7)L(K),$$

using $\log K\leq L(K)$, $\log L(K)\leq L(K)$ and $1\leq L(K)$. Hence by (588)

$$|v(y,t+K)|\leq C_gK_1L(K_1)^2\leq C_gC_7(3+\log C_7)^2\,K\,L(K)^4=:C_8\,K\,L(K)^4, \tag{593}$$

and also $|v(z)|\leq C_gKL(K)^2\leq C_gKL(K)^4$ by (588) with $\Theta=K$.

Dropping the nonpositive cost in (589) gives $\mathcal{P}_K v(z)\leq v(y,t+K)$, so by `EnvSlopeP`, (593) and the bound on $|v(z)|$,

$$\mathcal{S}_K^+ v(z)=\frac{\mathcal{P}_K v(z)-v(z)}{K}\leq\frac{|v(y,t+K)|+|v(z)|}{K}\leq (C_8+C_g)L(K)^4.$$

Similarly $\mathcal{M}_K v(z)\geq v(y^-,t-K)$ by (589), and (592) applied to y^- together with (588) gives $|v(y^-,t-K)|\leq C_8KL(K)^4$, so by `EnvSlopeM`

$$\mathcal{S}_K^- v(z)=\frac{v(z)-\mathcal{M}_K v(z)}{K}\leq\frac{|v(z)|+|v(y^-,t-K)|}{K}\leq (C_8+C_g)L(K)^4.$$

Step 3: all lengths $0 < \ell \le K$. The hypotheses of `EnvHamiltonianSolution` hold for v with $I = \mathbb{R}$: the set $\mathbb{R}$ is open, v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there by `IsFitClass`, and v has the growth property of `SubquadraticGrowth` on $\mathbb{R}$ by Step 0. Its part (1), applied at the center z with the lengths $\ell \le K$ — its hypothesis $t + \sigma \in \mathbb{R}$ for $\sigma \in [-K, K]$ being automatic — gives

$$\mathcal{S}^{+}_{\ell} v(z) \le \mathcal{S}^{+}_{K} v(z), \qquad \mathcal{S}^{-}_{\ell} v(z) \le \mathcal{S}^{-}_{K} v(z) \qquad (0 < \ell \le K).$$

Combining with Step 2, both slopes are at most $(C_8 + C_g)L(K)^4$ for every $0 < \ell \le K$.

Conclusion. Taking $C = \max\{C_8 + C_g,\ C_6\} + 1 > 0$, which depends only on C_0 and $\dim F$, gives (587) and the displacement bound. □

2.145. `OptionalStoppingErrorBudget`.

Lemma 2.145 (Optional stopping and the accumulated error budget)**.** *Let $(\Omega, \mathcal{M}, \mu)$ be a probability space, let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a filtration of sub-σ-algebras of $\mathcal{M}$, let $J, \mathcal{E} : \mathbb{N} \times \Omega \to \mathbb{R}$ be two processes written J_n and $\mathcal{E}_n$, let $\tau : \Omega \to \mathbb{N}$ and let $c, J_0, \Lambda, J^{\mathrm{u}}, J^{\mathrm{l}}$ be real numbers. Assume:*

(1) *$c > 0$ and $\Lambda \ge 0$;*
(2) *J_n and $\mathcal{E}_n$ are $\mathcal{G}_n$-measurable for every n;*
(3) *$\mathcal{E}_n(\omega) \ge 0$ for every n and every ω;*
(4) *τ is a stopping time for $(\mathcal{G}_n)$;*
(5) *$J_0(\omega) = J_0$ for every ω — the starting value is the constant J_0;*
(6) *$-J^{\mathrm{l}} \le J_n(\omega) \le J^{\mathrm{u}}$ for every n and every ω;*
(7) *for almost every ω and every $n < \tau(\omega)$ one has $c\,\mathcal{E}_n(\omega) \le \Lambda$;*
(8) *the process*

$$M_n = J_n - c \sum_{k<n} \mathbf{1}_{\{k<\tau\}} \mathcal{E}_k$$

is a submartingale for $(\mathcal{G}_n)$ and μ.

Then:

(i) *for every stopping time σ for $(\mathcal{G}_n)$ with $\sigma \le \tau$ pointwise, the stopped value $\omega \mapsto J_{\sigma(\omega)}(\omega)$ is μ-integrable and*

$$J_0 \le \int J_{\sigma(\omega)}(\omega)\, d\mu(\omega);$$

(ii) *the accumulated error $\omega \mapsto \sum_{k<\tau(\omega)} \mathcal{E}_k(\omega)$ is μ-integrable;*
(iii) *its mean obeys the budget*

$$\int \sum_{k<\tau(\omega)} \mathcal{E}_k(\omega)\, d\mu(\omega) \le \frac{J^{\mathrm{u}} - J_0}{c}.$$

Proof. Nothing below uses more about the data than 1–8; in particular the budget is stated with the symbolic numerator $J^{\mathrm{u}} - J_0$, and an application supplies its own bound J^{u} and its own starting value J_0.

Step 0: random indices, boundedness and integrability. Since each $\mathcal{G}_n$ is contained in $\mathcal{M}$, hypothesis 2 makes every J_i and every partial sum $\omega \mapsto \sum_{k<i} \mathcal{E}_k(\omega)$ measurable. Next, if $\rho : \Omega \to \mathbb{N}$ is measurable with $\rho \le h$ pointwise and $g_i : \Omega \to \mathbb{R}$ is measurable for each i, then $\omega \mapsto g_{\rho(\omega)}(\omega)$ is measurable, because

$$g_{\rho(\omega)}(\omega) = \sum_{i \le h} \mathbf{1}_{\{\rho = i\}}(\omega)\, g_i(\omega),$$

a finite sum of measurable functions: for a fixed ω exactly one indicator is 1, namely $i = \rho(\omega)$, which lies in the index range because $\rho(\omega) \le h$. We use this with $\rho = \sigma \wedge h$ for a stopping time σ, which is legitimate because a $\mathbb{N}$-valued stopping time is measurable: $\{\sigma = i\} \in \mathcal{G}_i \subseteq \mathcal{M}$ for every i and $\mathbb{N}$ is countable.

Put $\Lambda_1 = \max\{J^{\mathrm{u}}, J^{\mathrm{l}}\}$. By 6,

$$|J_i(\omega)| \le \Lambda_1 \qquad (i \in \mathbb{N},\ \omega \in \Omega), \tag{594}$$

since $J_i \le J^{\mathrm{u}} \le \Lambda_1$ and $-\Lambda_1 \le -J^{\mathrm{l}} \le J_i$. As μ is a probability measure, the constant Λ_1 is integrable, so (594) shows that

$$\omega \mapsto J_{\rho(\omega)}(\omega) \text{ is integrable} \tag{595}$$

for every $\rho : \Omega \to \mathbb{N}$ for which that function is measurable.

Step 1: optional stopping at a truncated time. Let σ be a stopping time with $\sigma \le \tau$ and let $h \in \mathbb{N}$. Then $\sigma \wedge h$ is again a stopping time: $\{\sigma \wedge h \le i\} = \{\sigma \le i\} \cup \{h \le i\}$, and the second set is $\emptyset$ or Ω according as $i < h$ or $h \le i$; and it is bounded by h. The constant time 0 is a stopping time with $0 \le \sigma \wedge h \le h$, so optional stopping for submartingales, applied to (M_n) of 8 at the bounded stopping times 0 and $\sigma \wedge h$, gives

$$\int M_0 \, d\mu \;\le\; \int M_{\sigma(\omega)\wedge h}(\omega) \, d\mu(\omega). \tag{596}$$

We compute both sides. The sum in M_0 is empty, so $M_0 = J_0$, which is the constant J_0 by 5; as $\mu(\Omega) = 1$, the left side of (596) is J_0. For the right side, if $k < \sigma(\omega) \wedge h$ then $k < \sigma(\omega) \le \tau(\omega)$, so every indicator in the sum equals 1 and

$$M_{\sigma(\omega)\wedge h}(\omega) = J_{\sigma(\omega)\wedge h}(\omega) - c \sum_{k<\sigma(\omega)\wedge h} \mathcal{E}_k(\omega).$$

Both terms are integrable: the first by (595) and Step 0, the second because it is measurable by Step 0 and, for almost every ω,

$$0 \le \sum_{k<\sigma(\omega)\wedge h} \mathcal{E}_k(\omega) \le \sum_{k<\sigma(\omega)\wedge h} \frac{\Lambda}{c} \le h \cdot \frac{\Lambda}{c},$$

using 3 for the first inequality and, for the second, hypothesis 7 together with $k < \tau(\omega)$ and $c > 0$; the last step uses $\sigma(\omega) \wedge h \le h$ and $\Lambda/c \ge 0$, which holds by 1. So the integral of the right side splits, and (596) becomes

$$J_0 + c \int \sum_{k<\sigma(\omega)\wedge h} \mathcal{E}_k(\omega) \, d\mu(\omega) \;\le\; \int J_{\sigma(\omega)\wedge h}(\omega) \, d\mu(\omega). \tag{597}$$

Step 2: conclusion (i). Let $\sigma \le \tau$ be a stopping time. For each fixed ω the sequence $h \mapsto J_{\sigma(\omega)\wedge h}(\omega)$ is eventually constant, equal to $J_{\sigma(\omega)}(\omega)$ as soon as $h \ge \sigma(\omega)$; so it converges to $J_{\sigma(\omega)}(\omega)$. Each function $\omega \mapsto J_{\sigma(\omega)\wedge h}(\omega)$ is measurable by Step 0, so the pointwise limit $\omega \mapsto J_{\sigma(\omega)}(\omega)$ is measurable, and it is integrable by (595). By (594) the whole family is dominated by the integrable constant Λ_1, so dominated convergence gives

$$\int J_{\sigma(\omega)\wedge h}(\omega) \, d\mu(\omega) \;\longrightarrow\; \int J_{\sigma(\omega)}(\omega) \, d\mu(\omega) \qquad (h \to \infty).$$

For each h, the error integral in (597) is nonnegative by 3 and $c > 0$, so $J_0 \le \int J_{\sigma(\cdot)\wedge h} \, d\mu$. A limit of numbers that are all at least J_0 is at least J_0, whence $J_0 \le \int J_{\sigma(\cdot)} \, d\mu$. This is (i).

Step 3: the truncated budget. Take $\sigma = \tau$ in (597), which is admissible since $\tau \le \tau$. By 6 and $\mu(\Omega) = 1$,

$$\int J_{\tau(\omega)\wedge h}(\omega) \, d\mu(\omega) \le \int J^{\mathrm{u}} \, d\mu = J^{\mathrm{u}},$$

the integral comparison being legitimate because the left integrand is integrable by (595) and the right one is constant. Combining with (597) gives $c \int \sum_{k<\tau(\omega)\wedge h} \mathcal{E}_k \, d\mu \le J^{\mathrm{u}} - J_0$, hence, dividing by $c > 0$,

$$\int \sum_{k<\tau(\omega)\wedge h} \mathcal{E}_k(\omega) \, d\mu(\omega) \;\le\; \frac{J^{\mathrm{u}} - J_0}{c} \qquad (h \in \mathbb{N}). \tag{598}$$

Since the left side is nonnegative — the integrand is a finite sum of nonnegative terms — the right side is nonnegative as well:

$$0 \le \frac{J^{\mathrm{u}} - J_0}{c}. \tag{599}$$

Step 4: conclusions (ii) and (iii). Write $\iota : \mathbb{R} \to [0,\infty]$ for the map sending s to s when $s \geq 0$ and to 0 otherwise, and $\int^*$ for the upper integral of a measurable $[0,\infty]$-valued function. Put

$$S_h(\omega) = \sum_{k<\tau(\omega)\wedge h} \mathcal{E}_k(\omega), \qquad S(\omega) = \sum_{k<\tau(\omega)} \mathcal{E}_k(\omega),$$

both nonnegative by 3. Each S_h is measurable by Step 0, and so is S, as their pointwise limit: $S_h(\omega) = S(\omega)$ as soon as $h \geq \tau(\omega)$, exactly as in Step 2.

The functions $\iota \circ S_h$ are measurable and nondecreasing in h: for $h \leq h'$ we have $\tau(\omega)\wedge h \leq \tau(\omega)\wedge h'$, so the index set of S_h is contained in that of $S_{h'}$ and the extra terms are nonnegative; and ι is nondecreasing. Their pointwise supremum is $\iota \circ S$, because $\iota(S_h(\omega)) = \iota(S(\omega))$ once $h \geq \tau(\omega)$. Moreover each S_h is nonnegative and integrable by Step 1, so its upper integral is ι of its integral. By the monotone convergence theorem for upper integrals of a nondecreasing sequence of measurable $[0,\infty]$-valued functions,

$$\int^* \iota \circ S \, d\mu = \sup_h \int^* \iota \circ S_h \, d\mu = \sup_h \iota\Big(\int S_h \, d\mu\Big) \leq \iota\Big(\frac{J^{\mathrm{u}} - J_0}{c}\Big) < \infty,$$

the inequality by (598) and the monotonicity of ι.

Thus S is a nonnegative measurable function whose upper integral is finite, so S is integrable, which is (ii). Finally, for a nonnegative integrable function the integral is the real value of that upper integral, and $\iota\big((J^{\mathrm{u}} - J_0)/c\big) = (J^{\mathrm{u}} - J_0)/c$ by (599), so the display gives

$$\int \sum_{k<\tau(\omega)} \mathcal{E}_k(\omega) \, d\mu(\omega) \leq \frac{J^{\mathrm{u}} - J_0}{c},$$

which is (iii). □

2.146. `JGapLowerBoundAndContinuity`.

Lemma 2.146 (A lower bound for J_ϵ on a box, and the continuity of its time retract)**.** *Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$ and $\kappa \in \mathbb{R}$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`. Let W be the contact envelope of `ContactEnv` formed from v and κ, and assume that W is continuous on $\{(x,t) : t > 0\}$ and has subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`. Write J_ϵ for the quantity of `JGap`, formed from this v and κ. Finally let $R_b, T_b, \delta, \epsilon$ be real numbers with*

$$0 \leq R_b, \qquad 0 \leq T_b, \qquad 0 < \delta \leq \tfrac{1}{2}, \qquad 0 < \epsilon \leq \tfrac{\delta}{4}.$$

Then there is a constant $\Lambda_0 \geq 0$ such that

(i) *$J_\epsilon(w) \geq -\Lambda_0$ for every $w = (y,s)$ with $|y| \leq R_b$ and $\delta - \epsilon \leq s \leq T_b + \epsilon$; and*

(ii) *the retracted gap*

$$w = (y,s) \longmapsto J_\epsilon\big(y, \max\{s, \delta/2\}\big)$$

is continuous on all of space-time.

Proof. The box. Put $J = [\delta/2, T_b + 1]$, a compact subset of $(0,\infty)$ — note $\delta/2 > 0$ — and $\mathcal{B} = \{(y,s) : |y| \leq R_b,\ s \in J\}$, a compact set, being the image of the product of a closed ball of F and J under the (continuous) identification of $F \times \mathbb{R}$ with space-time. The ball is compact because F is finite-dimensional.

Let $w = (y,s)$ be as in (i). Then

$$s \ \geq\ \delta - \epsilon \ \geq\ \delta - \frac{\delta}{4} \ =\ \frac{3\delta}{4} \ >\ 0,$$

using $\epsilon \leq \delta/4$ and $\delta > 0$; so the time coordinate of w is positive. Moreover both shifted times lie in J:

$$s + \epsilon \ \geq\ \delta \ \geq\ \frac{\delta}{2}, \qquad s + \epsilon \ \leq\ T_b + 2\epsilon \ \leq\ T_b + \frac{\delta}{2} \ \leq\ T_b + 1,$$

the first line from $\delta - \epsilon \le s$ and $\delta > 0$, the second from $s \le T_b + \epsilon$, $2\epsilon \le \delta/2$ and $\delta \le 1/2$; and

$$s - \epsilon \;\ge\; \delta - 2\epsilon \;\ge\; \frac{\delta}{2}, \qquad s - \epsilon \;\le\; T_b \;\le\; T_b + 1,$$

again from $\epsilon \le \delta/4$ and from $s \le T_b + \epsilon$. So the two evaluation points $(y, s+\epsilon)$ and $(y, s-\epsilon)$ lie in $\mathcal{B}$.

The constant. The function W is continuous on $\mathcal{B}$ by hypothesis, every point of that set having positive time coordinate, and v is continuous on all of space-time by `IsFitClass`; a continuous function on a compact set is bounded, so there are K_W, K_v with $|W| \le K_W$ and $|v| \le K_v$ on $\mathcal{B}$. Both are nonnegative, because $\mathcal{B}$ is not empty: it contains the point $(0, \delta/2)$, as $|0| = 0 \le R_b$ and $\delta/2 \in J$ — the latter because $\delta/2 \le 1/4 \le T_b + 1$, using $\delta \le 1/2$ and $T_b \ge 0$. Put

$$\Lambda_0 = \frac{K_W + K_v}{2\epsilon} \;\ge\; 0,$$

nonnegative since $K_W + K_v \ge 0$ and $\epsilon > 0$.

(i) The lower bound. Let $w = (y, s)$ be as in (i). By the domination half of part (1) of `EnvelopeCompetitorBounds` for W with $I = (0, \infty)$, at the center w and the length ϵ — admissible because $s > 0$ and $s + \epsilon > 0$, its continuity and growth hypotheses being the standing ones for W — the competitor $y' = y$ gives

$$W(y, s+\epsilon) \;\le\; \mathcal{P}_\epsilon W(w),$$

the quadratic penalty $|y' - y|^2/(2\epsilon)$ vanishing at $y' = y$. By the domination half of part (2) of `EnvelopeCompetitorBounds` for v with $I = \mathbb{R}$ — admissible, the two time conditions being vacuous, with the continuity of v from `IsFitClass` and its subquadratic spatial growth on compact time intervals of $\mathbb{R}$ from `UniformSubquadratic` applied with the centering point $(0, 1)$, which is where the fit class and the centering hypothesis are used — the same competitor gives

$$\mathcal{M}_\epsilon v(w) \;\le\; v(y, s-\epsilon).$$

Hence, by `JGap` and $\epsilon > 0$,

$$J_\epsilon(w) = \frac{\mathcal{P}_\epsilon W(w) - \mathcal{M}_\epsilon v(w)}{2\epsilon} \;\ge\; \frac{W(y, s+\epsilon) - v(y, s-\epsilon)}{2\epsilon} \;\ge\; \frac{-K_W - K_v}{2\epsilon} \;=\; -\Lambda_0,$$

the last inequality because both evaluation points lie in $\mathcal{B}$, so $W(y, s+\epsilon) \ge -K_W$ and $v(y, s-\epsilon) \le K_v$. This is (i).

(ii) The retract is continuous. By parts (3) and (4) of `QuadEnvAttained`, applied to W with $I = (0, \infty)$ — an open set, with the standing continuity and growth hypotheses — and to v with $I = \mathbb{R}$ — where the admissibility conditions are vacuous, the continuity and growth of v being as in the previous paragraph — the maps $w \mapsto \mathcal{P}_\epsilon W(w)$ and $w \mapsto \mathcal{M}_\epsilon v(w)$ are continuous at every w whose time coordinate is positive. By `JGap`, J_ϵ is the difference of these two divided by the constant 2ϵ, so J_ϵ is continuous at every such w.

Let $\rho(y, s) = (y, \max\{s, \delta/2\})$. It is continuous, being built from the two coordinate projections of space-time — continuous for the ℓ^2 norm — the continuous map $r \mapsto \max\{r, \delta/2\}$ and the identification of $F \times \mathbb{R}$ with space-time. Every value of ρ has time coordinate $\max\{s, \delta/2\} \ge \delta/2 > 0$. So at an arbitrary point w of space-time, J_ϵ is continuous at $\rho(w)$ by the previous paragraph and ρ is continuous at w, whence $J_\epsilon \circ \rho$ is continuous at w. As w was arbitrary, $J_\epsilon \circ \rho$ is continuous, which is (ii). □

2.147. `ContactWalkBudgets`.

Lemma 2.147 (The two budgets of a stopped contact walk)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$, let $\kappa > 0$ and $H \in \mathbb{R}$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $z_* = (0, 1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let W be the contact envelope of `ContactEnv` formed from v and κ, and assume that W is continuous on $\{(x, t) : t > 0\}$ and has subquadratic spatial growth on compact subintervals of $(0, \infty)$ in the sense of `SubquadraticGrowth`. Write J_ϵ for the quantity of `JGap`, $\delta_j = 2^{-j}$ and $L(K) = 1 + \log K$.*

Then there is a constant $C_* > 0$ such that for every integer $T \geq 2$, every real $R \geq 2$, every integer $N \geq 1$, every ϵ, every measure space $(\Omega, \mathcal{M}, \mu)$ with a filtration $(\mathcal{G}_n)$, and all functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ on it which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of `IsContactWalk`*, the following hold, where*

$$Q_+(n) = \frac{x_n^+ - x_n}{\epsilon}, \qquad Q_-(n) = \frac{x_n - x_n^-}{\epsilon}, \qquad \mathcal{E}_n = |Q_+(n) - Q_0(n)|^2 + |Q_-(n) - Q_0(n)|^2,$$

and $B = R + T + 2$:

(i) *for every stopping time σ for $(\mathcal{G}_n)$ with $\sigma \leq \tau$ pointwise, the function $J_\epsilon(z_\sigma)$ is μ-integrable and $\mathbb{E} J_\epsilon(z_\sigma) \geq J_\epsilon(z_*)$;*

(ii) *the function $\sum_{n<\tau} \mathcal{E}_n$ is μ-integrable and $\mathbb{E} \sum_{n<\tau} \mathcal{E}_n \leq C_* L(B)^4$.*

Proof. Step 0: the constants. Let $C_1 > 0$ be the constant of `FittedEnvelopeBounds` for this C_0 and put

$$c = \frac{1}{16(1 + \log 2)} > 0, \qquad C_* = \frac{C_1 + |H| + 1}{c} > 0,$$

so that c is the constant appearing in `DPDrift`. Both depend only on C_0, $\dim F$ and H, hence neither depends on T, R, N or on the walk. Fix now T, R, N, ϵ and a stopped contact walk as in the statement, and write $\delta = \delta_N$, $B = R + T + 2 \geq 6$ and $L = L(B) \geq 1$. By clauses 2 and 3 of `IsContactWalk` we have $0 < \epsilon \leq \delta/4$, so $\epsilon \leq 1/8$ and $\epsilon \leq B$, and μ is a probability measure by clause 1.

Step 1: the walk stays in a compact box. We claim that for every n and every ω

$$\delta - \epsilon \leq t_n \leq T + \epsilon, \qquad |x_n| \leq R + 1, \tag{600}$$

where $z_n = (x_n, t_n)$. For the time bounds we induct on n. At $n = 0$, clause 6 gives $z_0 = z_*$, so $t_0 = 1$, and $\delta \leq 1/2 < 1 \leq T$ because $N \geq 1$ and $T \geq 2$. Assume the bounds at n. If $n \geq \tau(\omega)$ then clause 7 gives $z_{n+1} = z_n$ and there is nothing to prove. If $n < \tau(\omega)$ then clause 7 gives $z_{n+1} \in \{z_n^+, z_n^-\}$, and by clause 6 together with `IsStepPair` the time coordinates of $z_n^\pm$ are $t_n \pm \epsilon$; clause 10 gives $\delta < t_n < T$, so $t_{n+1} \geq t_n - \epsilon > \delta - \epsilon$ and $t_{n+1} \leq t_n + \epsilon < T + \epsilon$. In particular $t_n \geq \delta - \epsilon > 0$ for every n, since $\epsilon \leq \delta/4$. For the spatial bound: clause 12 gives $|x_n| \leq R + 1$ for $n \leq \tau$, and for $n > \tau$ clause 7 gives $z_n = z_\tau$ by induction, so the same bound holds.

Step 2: two-sided bounds for J_ϵ on the box. Let $w = (y, s)$ satisfy $|y| \leq R + 1$ and $\delta - \epsilon \leq s \leq T + \epsilon$; then $s > 0$ and $|s| \leq T + 1 \leq B$ and $|y| \leq R + 1 \leq B$.

The upper bound. By `JUpperBound`, applied with $\epsilon > 0$ and $s > 0$, $J_\epsilon(w) \leq \frac{1}{2}(\mathcal{S}_\epsilon^+ v(w) + \mathcal{S}_\epsilon^- v(w))$ with the envelope slopes of `EnvSlopeP` and `EnvSlopeM`; and by the first clause of `FittedEnvelopeBounds`, applied to v with the centering point $(0, 1)$, the box size $K = B \geq 2$ and the length $\epsilon \leq B$, both slopes are at most $C_1 L^4$. Hence

$$J_\epsilon(w) \leq C_1 L^4. \tag{601}$$

The lower bound. The standing hypotheses on v, on W and on the scales are exactly those of `JGapLowerBoundAndContinuity`, applied with the spatial radius $R + 1$, the time T, the scale δ and the length ϵ: here $0 \leq R + 1$ and $0 \leq T$ because $R \geq 2$ and $T \geq 2$, while $0 < \delta \leq 1/2$ and $0 < \epsilon \leq \delta/4$ hold by Step 0. Its first conclusion supplies a constant $\Lambda_0 \geq 0$ with

$$J_\epsilon(w) \geq -\Lambda_0 \tag{602}$$

for every $w = (y, s)$ with $|y| \leq R + 1$ and $\delta - \epsilon \leq s \leq T + \epsilon$, which is every w as above. Put $\Lambda_1 = \max\{C_1 L^4, \Lambda_0\}$; by (601), (602) and Step 1, $|J_\epsilon(z_n)| \leq \Lambda_1$ for every n and ω.

Step 3: measurability. Let $\rho(y, s) = (y, \max\{s, \delta/2\})$. By the second conclusion of `JGapLowerBoundAndContin` applied with the same data, the composite $J_\epsilon \circ \rho$ is continuous on all of space-time, and it agrees with J_ϵ at every point whose time coordinate is at least $\delta/2$, because there $\max\{s, \delta/2\} = s$ — in particular, by Step 1 and $\epsilon \leq \delta/4$, at every $z_n(\omega)$. Since each z_n is $\mathcal{G}_n$-measurable by clause 4 of `IsContactWalk`, the function $\omega \mapsto J_\epsilon(z_n(\omega)) = (J_\epsilon \circ \rho)(z_n(\omega))$ is $\mathcal{G}_n$-measurable, being a continuous function of a $\mathcal{G}_n$-measurable one; the same argument applies to $z_n^\pm$ with $J_\epsilon \circ \rho$ in place of J_ϵ. Likewise each $\mathcal{E}_n$ is $\mathcal{G}_n$-measurable, being a continuous function of the $\mathcal{G}_n$-measurable z_n, $z_n^\pm$ and $Q_0(n)$, and $\mathcal{E}_n \geq 0$, being a sum of two squares. Finally, τ is a stopping time by clause

9, so $\{n<\tau\}\in\mathcal{G}_n$ for every n, and τ is measurable because $\{\tau=i\}\in\mathcal{G}_i$ for every i and τ has countable range.

Step 4: almost everywhere the step states lie in the same box. Fix n and let G be the set of ω with $|x_n^+(\omega)|\le R+1$ and $\delta-\epsilon\le t_n^+(\omega)\le T+\epsilon$, a $\mathcal{G}_n$-measurable set by Step 3. Put $S=\{n<\tau\}\setminus G\in\mathcal{G}_n$. On $S\cap\{\xi_n=1\}$ clause 7 gives $z_{n+1}=z_n^+$, so (600) applied at $n+1$ puts ω in G; hence $S\cap\{\xi_n=1\}=\emptyset$. On the other hand ξ_n is bounded and measurable, hence integrable, and clause 3 of `IsContactWalk` gives $\mathbb{E}[\xi_n\mid\mathcal{G}_n]=0$ a.e., so by the characterizing property of the conditional expectation $\int_S\xi_n\,d\mu=0$. But $\xi_n\in\{1,-1\}$ by clause 3, so $\xi_n=-1$ on S and therefore $\int_S\xi_n\,d\mu=-\mu(S)$. Hence $\mu(S)=0$, that is, almost everywhere on $\{n<\tau\}$ the state z_n^+ satisfies the bounds of (600). The same argument with the sign -1 in place of $+1$ — on $S'\cap\{\xi_n=-1\}$ clause 7 gives $z_{n+1}=z_n^-$, and on S' we then have $\xi_n=+1$, so $\int_{S'}\xi_n\,d\mu=\mu(S')$ — gives the same conclusion for z_n^-. Consequently, by Step 2, almost everywhere on $\{n<\tau\}$

$$|J_\epsilon(z_n^+)|\le\Lambda_1,\qquad |J_\epsilon(z_n^-)|\le\Lambda_1. \tag{603}$$

Step 5: the one-step drift. Let $n<\tau(\omega)$. Then clause 10 gives $t_n>\delta\ge 4\epsilon>\epsilon$, clause 6 says that (z_n^+,z_n^-) is an optimizing step pair of size ϵ at z_n in the sense of `IsStepPair`, and clause 8 says that $\mathcal{P}_\epsilon W(\cdot,t_n)$ has gradient $Q_0(n)$ at x_n. So `DPDrift` applies at z_n with this step pair and this reference velocity, with the four gaps $D,e,\tilde D,\tilde e$ given by its defining equations. Its sixth and seventh conclusions read

$$\frac{D+e}{4\epsilon}\le\frac{J_\epsilon(z_n^+)+J_\epsilon(z_n^-)}{2}-J_\epsilon(z_n),\qquad c\,\mathcal{E}_n\le\frac{D+e}{4\epsilon},$$

and together

$$c\,\mathcal{E}_n\le\frac{J_\epsilon(z_n^+)+J_\epsilon(z_n^-)}{2}-J_\epsilon(z_n)\qquad(n<\tau). \tag{604}$$

With (603) and $|J_\epsilon(z_n)|\le\Lambda_1$ this gives the a.e. bound

$$c\,\mathcal{E}_n\le\Lambda:=C_1L^4+\Lambda_0\qquad\text{a.e. on }\{n<\tau\}, \tag{605}$$

and since the index set is countable, almost every ω satisfies (605) for every $n<\tau(\omega)$ simultaneously.

Step 6: the drift submartingale and optional stopping. For $n\ge0$ put

$$M_n=J_\epsilon(z_n)-c\sum_{k<n}\mathbf{1}_{\{k<\tau\}}\mathcal{E}_k.$$

Each M_n is $\mathcal{G}_n$-measurable by Step 3, and integrable: μ is a probability measure, $|J_\epsilon(z_n)|\le\Lambda_1$ by Step 2, and by (605) the sum has at most n terms each at most Λ/c almost everywhere.

We claim (M_n) is a submartingale. Set

$$A_n=\mathbf{1}_{\{n<\tau\}}\Big(\frac{J_\epsilon(z_n^+)+J_\epsilon(z_n^-)}{2}-J_\epsilon(z_n)-c\,\mathcal{E}_n\Big),\qquad D_n=\mathbf{1}_{\{n<\tau\}}\frac{J_\epsilon(z_n^+)-J_\epsilon(z_n^-)}{2},$$

both $\mathcal{G}_n$-measurable by Step 3 (and both read through $J_\epsilon\circ\rho$, which is where their measurability comes from). Then

$$M_{n+1}-M_n=A_n+\xi_nD_n. \tag{606}$$

Indeed, on $\{n\ge\tau\}$ clause 7 gives $z_{n+1}=z_n$ and both sides vanish; on $\{n<\tau\}$ clause 7 gives $z_{n+1}=z_n^+$ where $\xi_n=1$ and $z_{n+1}=z_n^-$ where $\xi_n=-1$, and since $\xi_n\in\{1,-1\}$ by clause 3,

$$J_\epsilon(z_{n+1})=\frac{J_\epsilon(z_n^+)+J_\epsilon(z_n^-)}{2}+\xi_n\frac{J_\epsilon(z_n^+)-J_\epsilon(z_n^-)}{2}$$

in both cases, which is (606) after subtracting $J_\epsilon(z_n)+c\,\mathcal{E}_n$. By (604), $A_n\ge0$ almost everywhere. By (603), D_n is bounded by Λ_1 almost everywhere, so the pull-out property of the conditional expectation for a bounded $\mathcal{G}_n$-measurable factor and clause 3 give $\mathbb{E}[\xi_nD_n\mid\mathcal{G}_n]=D_n\,\mathbb{E}[\xi_n\mid\mathcal{G}_n]=0$ a.e. Since A_n is $\mathcal{G}_n$-measurable and integrable, $\mathbb{E}[A_n\mid\mathcal{G}_n]=A_n$ a.e., and therefore $\mathbb{E}[M_{n+1}-M_n\mid\mathcal{G}_n]=A_n\ge0$ almost everywhere. As each M_n is adapted and integrable, (M_n) is a submartingale.

Let σ be a stopping time with $\sigma \le \tau$ and let $h \in \mathbb{N}$. The constant 0 and $\sigma \wedge h$ are stopping times with $0 \le \sigma \wedge h \le h$, so optional stopping for submartingales gives $\mathbb{E}M_0 \le \mathbb{E}M_{\sigma\wedge h}$. Now $M_0 = J_\epsilon(z_*)$ by clause 6, and for $k < \sigma \wedge h \le \tau$ the indicators in the sum equal 1, so

$$J_\epsilon(z_*) + c\,\mathbb{E}\sum_{k<\sigma\wedge h} \mathcal{E}_k \le \mathbb{E}J_\epsilon(z_{\sigma\wedge h}), \tag{607}$$

both expectations existing: the integrand on the right is bounded by Λ_1, and the sum on the left is nonnegative and bounded by $h\Lambda/c$ almost everywhere, while both are measurable because $\sigma \wedge h \le h$ makes them finite sums of the measurable $\mathbf{1}_{\{\sigma\wedge h = i\}}$ times measurable functions.

Step 7: $h \to \infty$, and the two conclusions. Fix σ as above. For every ω we have $\sigma \wedge h = \sigma$ as soon as $h \ge \sigma(\omega)$, so $J_\epsilon(z_{\sigma\wedge h}) \to J_\epsilon(z_\sigma)$ pointwise and $\sum_{k<\sigma\wedge h}\mathcal{E}_k$ increases to $\sum_{k<\sigma}\mathcal{E}_k$ pointwise. In particular $J_\epsilon(z_\sigma)$ is measurable, being a pointwise limit of measurable functions, and $|J_\epsilon(z_\sigma)| \le \Lambda_1$ by Step 2, so it is integrable: this is the first half of (i). Dominated convergence, legitimate because all these functions are bounded by the constant Λ_1 and μ is a probability measure, gives $\mathbb{E}J_\epsilon(z_{\sigma\wedge h}) \to \mathbb{E}J_\epsilon(z_\sigma)$. Dropping the nonnegative error term in (607) leaves $J_\epsilon(z_*) \le \mathbb{E}J_\epsilon(z_{\sigma\wedge h})$ for every h, and letting $h \to \infty$ gives $J_\epsilon(z_*) \le \mathbb{E}J_\epsilon(z_\sigma)$. This is (i).

For (ii) take $\sigma = \tau$ in (607). By (601) and Step 1, $\mathbb{E}J_\epsilon(z_{\tau\wedge h}) \le C_1L^4$, and by clause 13 of `IsContactWalk`, $J_\epsilon(z_*) \ge H - 1$. Hence for every h

$$c\,\mathbb{E}\sum_{k<\tau\wedge h} \mathcal{E}_k \le C_1L^4 - (H-1) \le \big(C_1 + |H| + 1\big)L^4,$$

using $L \ge 1$ and $-H \le |H|$, that is $\mathbb{E}\sum_{k<\tau\wedge h}\mathcal{E}_k \le C_*L^4$. The functions $\sum_{k<\tau\wedge h}\mathcal{E}_k$ are nonnegative, measurable and nondecreasing in h with pointwise limit $\sum_{k<\tau}\mathcal{E}_k$, so by monotone convergence — applied to the integrals with values in $[0,\infty]$, which is what makes the conclusion carry integrability with it —

$$\int_\Omega \sum_{k<\tau}\mathcal{E}_k\,d\mu = \sup_h \int_\Omega \sum_{k<\tau\wedge h}\mathcal{E}_k\,d\mu \le C_*L^4 < \infty.$$

The function $\sum_{k<\tau}\mathcal{E}_k$ is therefore integrable, with $\mathbb{E}\sum_{k<\tau}\mathcal{E}_k \le C_*L^4$. This is (ii). □

2.148. `ContactWalkMarkData`.

Lemma 2.148 (Mark data for a stopped contact walk)**.** *Let F be a finite-dimensional real inner-product space, let $C_0, \kappa, H \in \mathbb{R}$ and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`. Let $T \in \mathbb{N}$, $R \in \mathbb{R}$, $N \in \mathbb{N}$ and $\epsilon \in \mathbb{R}$, and let $(\Omega, \mathcal{M}, \mu)$ be a measure space with a filtration $(\mathcal{G}_n)_{n\in\mathbb{N}}$ carrying functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of `IsContactWalk`. Assume that for every n the range of z_n is at most countable. Then there are functions σ_j, p_j, a_j which are mark data for $(C_0, v, N, \mathcal{G}, z, \tau)$ in the sense of `IsMarkData`.*

Proof. Write $\delta_j = 2^{-j}$ and $z_n = (x_n, t_n)$. Throughout, clause numbers refer to `IsContactWalk`: clause 1 says $\mu(\Omega) = 1$, clause 4 that each z_n is $\mathcal{G}_n$-measurable, and clause 9 that τ is a stopping time for $(\mathcal{G}_n)$.

Step 0: preliminaries. Since $\mu(\Omega) = 1$ by clause 1, Ω is nonempty. Since each z_n is $\mathcal{G}_n$-measurable by clause 4 and the time coordinate $w \mapsto t(w)$ is continuous, each function $\omega \mapsto t_n(\omega)$ is $\mathcal{G}_n$-measurable, so

$$\{t_m = \delta_j\} \in \mathcal{G}_m \qquad (m, j \in \mathbb{N}). \tag{608}$$

Also, for every state c,

$$\{\omega : z_i(\omega) = c\} \in \mathcal{G}_i, \tag{609}$$

because the set where two $\mathcal{G}_i$-measurable functions agree — here z_i and the constant c — is $\mathcal{G}_i$-measurable. Finally, since τ is a stopping time, $\{\tau \le n\} \in \mathcal{G}_n$ and $\{\tau = k\} \in \mathcal{G}_k$ for every k, the latter because τ has countable range; consequently, for $m \le n$,

$$\{m \le \tau\} = \Big(\bigcup_{k<m}\{\tau = k\}\Big)^{\mathrm{c}} \in \mathcal{G}_n, \tag{610}$$

a finite union of sets of $\mathcal{G}_k \subseteq \mathcal{G}_n$ with $k < m \le n$.

Step 1: the mark index and clauses 1–4 of `IsMarkData`. Fix j. If there is an $n \le \tau(\omega)$ with $t_n(\omega) = \delta_j$, let $\sigma_j(\omega)$ be the least such n; otherwise put $\sigma_j(\omega) = \tau(\omega)$. This is the index $\tau \wedge \inf\{n : t_n = \delta_j\}$ of the paper, read as: a mark is the first point on its time level, or the terminal point if the walk stops before that level.

Then $\sigma_j \le \tau$: in the first case the least such n is itself $\le \tau(\omega)$, and in the second case $\sigma_j = \tau$. This is clause 2.

Let $n < \sigma_j(\omega)$. In the first case $n < \sigma_j(\omega) \le \tau(\omega)$, so $n \le \tau(\omega)$, and minimality of $\sigma_j(\omega)$ rules out $t_n(\omega) = \delta_j$; in the second case $n < \tau(\omega)$, and no $n' \le \tau$ at all has $t_{n'} = \delta_j$, in particular not n. Either way $n < \tau(\omega)$ and $t_n(\omega) \ne \delta_j$, which is clause 3.

In the first case $t_{\sigma_j(\omega)}(\omega) = \delta_j$ by the definition of $\sigma_j(\omega)$, and in the second $\sigma_j(\omega) = \tau(\omega)$; this is clause 4.

For clause 1 we claim

$$\{\sigma_j \le n\} = \{\tau \le n\} \cup \bigcup_{m \le n} \big(\{m \le \tau\} \cap \{t_m = \delta_j\}\big). \tag{611}$$

For the inclusion $\subseteq$: if ω is in the first case, then $m = \sigma_j(\omega) \le n$ satisfies $m \le \tau(\omega)$ and $t_m(\omega) = \delta_j$, so ω lies in the union; if ω is in the second case, then $\tau(\omega) = \sigma_j(\omega) \le n$. For $\supseteq$: if $\tau(\omega) \le n$ then $\sigma_j(\omega) \le \tau(\omega) \le n$ by clause 2; and if $m \le n$ satisfies $m \le \tau(\omega)$ and $t_m(\omega) = \delta_j$, then ω is in the first case and $\sigma_j(\omega) \le m \le n$ by minimality. Every set on the right of (611) lies in $\mathcal{G}_n$, by $\{\tau \le n\} \in \mathcal{G}_n$, by (610) and by (608) together with $\mathcal{G}_m \subseteq \mathcal{G}_n$ for $m \le n$. Hence σ_j is a stopping time, which is clause 1. Since σ_j has countable range, it also follows that

$$\{\sigma_j = i\} \in \mathcal{G}_i \qquad (i \in \mathbb{N}). \tag{612}$$

Step 2: every function of the state at a fixed time is adapted. Fix i. By hypothesis the range of z_i is countable, and it is nonempty because Ω is, so there is a map $e : \mathbb{N} \to F \times \mathbb{R}$ whose range is exactly the range of z_i. For $\omega \in \Omega$ let $\mathrm{cd}(\omega)$ be the least m with $e(m) = z_i(\omega)$; this exists because $z_i(\omega)$ lies in the range of e. Then $e(\mathrm{cd}(\omega)) = z_i(\omega)$ for every ω, and cd is $\mathcal{G}_i$-measurable, because it takes values in the countable set $\mathbb{N}$ and, for every m,

$$\{\mathrm{cd} = m\} = \{z_i = e(m)\} \cap \bigcap_{m' < m} \{z_i = e(m')\}^{\mathrm{c}} \in \mathcal{G}_i$$

by (609). Now let g be any function on states with values in a finite-dimensional real inner-product space. For $k \in \mathbb{N}$ put

$$g_k(\omega) = \sum_{m < k} \mathbf{1}_{\{\mathrm{cd}=m\}}(\omega)\, g(e(m)),$$

a finite sum of indicators of sets of $\mathcal{G}_i$ times constants, hence $\mathcal{G}_i$-measurable. For each ω and each $k > \mathrm{cd}(\omega)$ exactly one summand is nonzero, namely the one with $m = \mathrm{cd}(\omega)$, so $g_k(\omega) = g(e(\mathrm{cd}(\omega))) = g(z_i(\omega))$; thus $g_k \to g \circ z_i$ pointwise, and a pointwise limit of $\mathcal{G}_i$-measurable functions with values in a metrizable space is $\mathcal{G}_i$-measurable. Hence

$$\omega \mapsto g(z_i(\omega)) \text{ is } \mathcal{G}_i\text{-measurable} \qquad (i \in \mathbb{N}). \tag{613}$$

Step 3: the fits and clauses 7–8. Fix j and let w be a state. Since $\delta_j > 0$, the second clause of `IsFitClass` provides a pair $(p(w), a(w))$ with $\mathrm{IsFit}(C_0, v, w, \delta_j, p(w), a(w))$ in the sense of `IsFit`; fix one such pair for each w (and each j). Define

$$G(w) = \begin{cases} (p(w), a(w)), & t(w) = \delta_j, \\ (0,0), & t(w) \ne \delta_j, \end{cases} \qquad (p_j(\omega), a_j(\omega)) = G\big(z_{\sigma_j(\omega)}(\omega)\big).$$

If $t_{\sigma_j(\omega)}(\omega) = \delta_j$ then $(p_j(\omega), a_j(\omega)) = (p(w), a(w))$ at $w = z_{\sigma_j(\omega)}(\omega)$, so $(p_j(\omega), a_j(\omega))$ is a fit for v at that state of radius δ_j, which is clause 7; and if $t_{\sigma_j(\omega)}(\omega) \ne \delta_j$ then $p_j(\omega) = 0$ and $a_j(\omega) = 0$, which is clause 8.

Step 4: clauses 5–6. Fix j and n. We claim

$$\mathbf{1}_{\{\sigma_j \le n\}}\, G\big(z_{\sigma_j}\big) = \sum_{i \le n} \mathbf{1}_{\{\sigma_j = i\}}\, G\big(z_i\big). \tag{614}$$

Indeed, at an ω with $\sigma_j(\omega) \le n$ exactly the summand $i = \sigma_j(\omega)$ survives and equals $G(z_{\sigma_j(\omega)}(\omega))$, while at an ω with $\sigma_j(\omega) > n$ no summand survives and both sides are 0. The sets $\{\sigma_j = i\}$ with $i \le n$ are pairwise disjoint, lie in $\mathcal{G}_i \subseteq \mathcal{G}_n$ by (612), and cover $\{\sigma_j \le n\}$; these are the sets $A_{i,c} = \{\sigma_j = i\} \cap \{z_i = c\}$ of the paper's countable tree of histories, grouped by i. By (613) each $\omega \mapsto G(z_i(\omega))$ is $\mathcal{G}_i$-measurable, hence $\mathcal{G}_n$-measurable for $i \le n$, so the right-hand side of (614) is a finite sum of products of indicators of $\mathcal{G}_n$-measurable sets with $\mathcal{G}_n$-measurable functions, and is therefore $\mathcal{G}_n$-measurable. Taking the two components of (614) — the projections onto F and onto $\mathbb{R}$ are continuous — gives that $\omega \mapsto \mathbf{1}_{\{\sigma_j \le n\}}(\omega) p_j(\omega)$ and $\omega \mapsto \mathbf{1}_{\{\sigma_j \le n\}}(\omega) a_j(\omega)$, that is the functions equal to p_j and a_j on $\{\sigma_j \le n\}$ and to 0 off it, are $\mathcal{G}_n$-measurable. These are clauses 5 and 6.

All eight clauses of `IsMarkData` hold for every j with $1 \le j \le N$ — indeed for every j — so (σ_j, p_j, a_j) is mark data for $(C_0, v, N, \mathcal{G}, z, \tau)$. □

2.149. `StoppedWalkBudgets`.

Lemma 2.149 (The stopped walks and their expectation bounds)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$, let $\kappa > 0$ and $H \in \mathbb{R}$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $z_* = (0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let W be the contact envelope of `ContactEnv` formed from v and κ, and assume the standing properties of the extremal contact pair:*

(1) *W is continuous on $\{(x,t) : t > 0\}$ and has subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`;*
(2) *W is a viscosity subsolution of the limiting equation on $\{(x,t) : t > 0\}$ in the sense of `IsLimitSubOn`;*
(3) *$v(x,t) - \kappa^2 t/2 \le W(x,t) \le v(x,t)$ for $t > 0$, and $W(z_*) = v(z_*) = 0$;*
(4) *$x \mapsto W(x,t) - |x|^2/(2t)$ is concave on F for every $t > 0$;*
(5) *if $t > 0$ and $W(x,t) = v(x,t)$ then $(\mathcal{C}_t v(x,t) - v(x,t))/t = \kappa^2/2$, with $\mathcal{C}$ the barycentric envelope of `BaryEnv`;*
(6) *the envelope Hamiltonian of `EnvHamiltonian` satisfies $H_v(z_*) = H$, and the forward envelope slopes of `EnvSlopeP` satisfy $\mathcal{S}^+_\ell W(z_*) \to H$ as $\ell \downarrow 0$.*

Write J_ϵ for the quantity of `JGap`, $\delta_j = 2^{-j}$ and $L(K) = 1 + \log K$.

Then there is a constant $C_* > 0$ such that for every integer $T \ge 2$, every $R \ge 2$ and every integer $N \ge 1$ there are $\epsilon > 0$, a measure space $(\Omega, \mathcal{M}, \mu)$ with a filtration $(\mathcal{G}_n)$, functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of `IsContactWalk`, such that, writing

$$Q_+(n) = \frac{x_n^+ - x_n}{\epsilon}, \qquad Q_-(n) = \frac{x_n - x_n^-}{\epsilon}, \qquad \mathcal{E}_n = |Q_+(n) - Q_0(n)|^2 + |Q_-(n) - Q_0(n)|^2,$$

the following two bounds hold with $B = R + T + 2$:

(i) *for every stopping time σ for $(\mathcal{G}_n)$ with $\sigma \le \tau$ pointwise, the function $J_\epsilon(z_\sigma)$ is μ-integrable and*

$$\mathbb{E} J_\epsilon(z_\sigma) \ge J_\epsilon(z_*); \tag{615}$$

(ii) *the function $\sum_{n<\tau} \mathcal{E}_n$ is μ-integrable and*

$$\mathbb{E} \sum_{n<\tau} \mathcal{E}_n \le B_0, \qquad B_0 = C_* L(B)^4; \tag{616}$$

(iii) *there are marks with fits (σ_j, p_j, a_j) for $(C_0, v, N, \mathcal{G}, z, \tau)$ in the sense of `IsMarkData`.*

Proof. Hypotheses (1) and (6) and the normalizations in (3) are the only parts of the standing package used below; hypotheses (2), (4), (5) and the sandwich inequalities of (3) are not needed for this assembly.

The constant. By `ContactWalkBudgets`, applied to C_0, κ, H and v — its hypotheses are that v lies in $\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`, is centered at z_* with constant C_0 in the sense of `IsCenteredAt` and satisfies the horizontal Jensen bound with constant κ in the sense

of HasJensenBound, together with hypothesis (1) for the contact envelope W of ContactEnv — there is a constant $C_* > 0$ with the property stated there. That node produces C_* before T, R and N are quantified, so the single C_* serves every box; this is the uniformity the present statement asserts. Fix now an integer $T \geq 2$, a real $R \geq 2$ and an integer $N \geq 1$.

The walk. By ContactWalkExists, applied with these T, R and N — its hypotheses are the same three assumptions on v, hypothesis (1) for W, the normalizations $W(z_*) = 0$ and $v(z_*) = 0$ from hypothesis (3), and hypothesis (6), namely $H_v(z_*) = H$ for the envelope Hamiltonian of EnvHamiltonian and $\mathcal{S}_\ell^+ W(z_*) \to H$ for the forward envelope slopes of EnvSlopeP — there are $\epsilon > 0$, a measure space $(\Omega, \mathcal{M}, \mu)$ with a filtration $(\mathcal{G}_n)$, and functions ξ_n, z_n, z_n^+, z_n^-, $Q_0(n)$ and τ on it which form a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of IsContactWalk, and such that the range of each z_n is at most countable.

Conclusions (i) and (ii). ContactWalkBudgets quantifies the walk universally: its conclusion applies to every ϵ and every stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$, in particular to the one just produced. Its two conclusions at this T, R, N, ϵ and this walk are verbatim conclusions (i) and (ii) of the present statement, with the same constant C_* and the same $B = R + T + 2$.

Conclusion (iii). By ContactWalkMarkData, applied to the same walk — its hypotheses are that v lies in $\mathcal{A}_{m,C_0}$, that the data form a stopped contact walk in the sense of IsContactWalk, and that the range of each z_n is at most countable, the last being the final assertion of ContactWalkExists — there are marks with fits (σ_j, p_j, a_j) for $(C_0, v, N, \mathcal{G}, z, \tau)$ in the sense of IsMarkData. This is conclusion (iii), and the lemma follows with the constant C_* fixed above. □

2.150. DPLogVelocity.

Lemma 2.150 (Logarithmic bound for the reference velocity)**.** *Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$ and $0 \leq \kappa < \infty$, and write $L(K) = 1 + \log K$ for $K \geq 2$. There is $C > 0$, depending only on C_0, $\dim F$ and κ, such that the following holds. Let $v \in \mathcal{A}_{m,C_0}$ in the sense of IsFitClass be centered at $(0,1)$ with constant C_0 in the sense of IsCenteredAt and satisfy the horizontal Jensen bound with constant κ at every time in the sense of HasJensenBound, and let $W = \mathcal{C}_t v - \kappa^2 t/2$ be the contact envelope of ContactEnv. Then, whenever $K \geq 2$, $|x| \leq K$, $|t| \leq K$, $0 < \epsilon \leq 1$ and $t > \epsilon$, the reference velocity of DPSpatialRegularity satisfies*

$$|Q_0| = |\nabla_x \mathcal{P}_\epsilon W(x,t)| \leq C\big(L(K) + |\log t|\big). \tag{617}$$

Proof. Write $L(\Theta) = 1 + \log \Theta$ for $\Theta \geq 2$ and $f = W = \mathcal{C}_t v - \kappa^2 t/2$ for the contact envelope of ContactEnv. Let $C_s > 0$ be the constant of SpatialFitModulus for this C_0. The constant C produced below is explicit in C_s, κ and absolute numbers, hence depends only on C_0, $\dim F$ and κ.

Let v, $K \geq 2$, x, t, ϵ and Q_0 be as in the statement, so $|x| \leq K$, $|t| \leq K$, $0 < \epsilon \leq 1$ and $t > \epsilon > 0$. Put $s = t + \epsilon$, so

$$2\epsilon < s \leq K + 1, \qquad 1 < \frac{s}{t} < 2, \qquad s > 0. \tag{618}$$

The middle statement holds because $s = t + \epsilon$ with $0 < \epsilon < t$.

Step 0: standing facts. By IsFitClass the function v is continuous on all of space-time and is a viscosity solution of $F[v] = 0$ there; by UniformSubquadratic, applied with the centering point $(0,1)$, it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of SubquadraticGrowth. Hence BaryEnvSandwich with $I = \mathbb{R}$ applies at every parameter $\sigma > 0$ and, with ContactEnv, gives

$$v(y,\tau) - \frac{\kappa^2 \tau}{2} \leq f(y,\tau) \leq v(y,\tau) \qquad (\tau > 0). \tag{619}$$

Step 1: an upper supporting slope for $f(\cdot, s)$. By BaryEnvDuality, part (4), applied to v with $I = \mathbb{R}$ at the parameter $s > 0$ and the time s, the function $y \mapsto \mathcal{C}_s v(y,s) - |y|^2/(2s)$ is concave

on F; by `ContactEnv`, $f(y,s) = \mathcal{C}_s v(y,s) - \kappa^2 s/2$, so

$$y \longmapsto f(y,s) - \frac{|y|^2}{2s}$$

is concave and finite on all of F. A finite concave function on a finite-dimensional real inner-product space has a supporting affine function at each point: its hypograph is convex with nonempty interior and the graph point is a boundary point, so the supporting hyperplane theorem provides a nonzero (q_0,β_0) with $\langle q_0,\chi\rangle + \beta_0\beta \le \langle q_0,x\rangle + \beta_0\Lambda(x)$ for all (χ,β) in the hypograph, where Λ denotes the concave function; taking $\chi = x$ and $\beta \to -\infty$ forces $\beta_0 \ge 0$, and $\beta_0 = 0$ is impossible since then $\langle q_0, \chi - x\rangle \le 0$ for all χ with $q_0 \ne 0$. Dividing by $\beta_0 > 0$ gives a supergradient. Adding back the quadratic $|\cdot|^2/(2s)$, there is $p \in F$ with

$$f(x+h,s) \le f(x,s) + p\cdot h + \frac{|h|^2}{2s} \qquad (h \in F). \tag{620}$$

Step 2: a bound for $|p|$. If $p = 0$ then $|p| = 0$ and the bound (623) below is trivial, so assume $p \ne 0$ and insert $h = -sp/|p|$, for which $|h| = s$, into (620):

$$f(x+h,s) - f(x,s) \le -s|p| + \frac{s}{2}. \tag{621}$$

In the other direction, (619) at the two points $(x+h,s)$ and (x,s), both with time $s > 0$, gives

$$f(x+h,s) - f(x,s) \ge v(x+h,s) - v(x,s) - \frac{\kappa^2 s}{2}. \tag{622}$$

We bound $v(x+h,s) - v(x,s)$ from below using `SpatialFitModulus`. Put $K_1 = 2K+1 \ge 5 \ge 2$. Then $|x| \le K \le K_1$ and $|s| \le K+1 \le K_1$ by (618), so $(x,s) \in \mathcal{B}_{K_1}$; and $|h| = s \le K+1 \le K_1$, so `SpatialFitModulus` applied at the center (x,s) with the radius $r = s > 0$ and the increment h, for which $|h| = s = r$, gives

$$|v(x+h,s) - v(x,s)| \le C_s\, s\,\big(L(K_1) + |\log s|\big).$$

By (618) we have $1 < s/t < 2$, so $|\log s| \le |\log t| + \log 2$; and $L(K_1) = 1 + \log(2K+1) \le 1 + \log(4K) = 1 + \log 4 + \log K \le (1+\log 4)L(K)$ because $L(K) \ge 1$ and $\log K \le L(K)$. Hence

$$v(x+h,s) - v(x,s) \ge -C_s\, s\,\big((1+\log 4)L(K) + |\log t| + \log 2\big).$$

Combining with (621) and (622) and dividing by $s > 0$,

$$|p| \le \frac{1}{2} + \frac{\kappa^2}{2} + C_s\big((1+\log 4)L(K) + |\log t| + \log 2\big),$$

so, using $L(K) \ge 1$ to absorb the constants,

$$|p| \le C'\big(L(K) + |\log t|\big) \tag{623}$$

with $C' = \frac{1}{2} + \frac{\kappa^2}{2} + C_s(1+\log 4) + C_s \log 2$, which depends only on C_0, $\dim F$ and κ.

Step 3: the optimizing displacement and the gradient. By `DPSpatialRegularity`, applied to v and κ at the time $t > 0$, the length $\epsilon > 0$ and the point x, there is a unique $y \in F$ attaining $\mathcal{P}_\epsilon f(x,t)$, and $x' \mapsto \mathcal{P}_\epsilon f(x',t)$ is differentiable at x with gradient $(y-x)/\epsilon$. Since HasGradientAt determines the gradient uniquely, the hypothesis on Q_0 gives

$$Q_0 = \frac{y-x}{\epsilon}. \tag{624}$$

Put $h = y - x$. Comparing the optimizer with the competitor $y' = x$ in `QuadEnvP`,

$$f(x+h,s) - \frac{|h|^2}{2\epsilon} = \mathcal{P}_\epsilon f(x,t) \ge f(x,s),$$

the time argument being $t + \epsilon = s$ in both cases, hence

$$\frac{|h|^2}{2\epsilon} \le f(x+h,s) - f(x,s) \le p\cdot h + \frac{|h|^2}{2s}$$

by (620). If $h = 0$ then $Q_0 = 0$ and we are done. Otherwise, since $s > 2\epsilon$ by (618),

$$\frac{|h|^2}{2\epsilon} - \frac{|h|^2}{2s} = \frac{|h|^2(s-\epsilon)}{2\epsilon s} \ge \frac{|h|^2}{2\epsilon}\cdot\frac{1}{2} = \frac{|h|^2}{4\epsilon},$$

because $s-\epsilon > s/2$. Hence $|h|^2/(4\epsilon) \le p\cdot h \le |p|\,|h|$, so $|h|/\epsilon \le 4|p|$, i.e. by (624)

$$|Q_0| = \frac{|h|}{\epsilon} \le 4|p|.$$

Combining with (623) gives

$$|Q_0| \le 4C'\big(L(K) + |\log t|\big),$$

which is (617) with $C = 4C'$, a constant depending only on C_0, $\dim F$ and κ. □

2.151. `WalkPathwiseBounds`.

Lemma 2.151 (Step 3a: pathwise bounds along a stopped contact walk)**.** *Let F be a finite-dimensional real inner-product space and write $\mathcal{H} = \ell^2(F\times\mathbb{R})$. Let $C_0, \kappa \in \mathbb{R}$ with $1 \le C_0$, let $v : \mathcal{H} \to \mathbb{R}$, and assume the standing properties of the contact pair: $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`, v is centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, v has the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`, the contact envelope $W = \mathcal{W}[v,\kappa]$ of `ContactEnv` is continuous on $\{t>0\}$ and has subquadratic spatial growth there in the sense of `SubquadraticGrowth`, and the sandwich*

$$v(z) - \frac{\kappa^2 t}{2} \le W(z) \le v(z) \qquad (z=(x,t),\ t>0)$$

holds. Let $C_2 \in \mathbb{R}$ with $C_2 > 0$ satisfy the logarithmic velocity bound of `DPLogVelocity` at the constants (C_0,κ): for every $u \in \mathcal{A}_{m,C_0}$ centered at $(0,1)$ with the Jensen bound at κ, every $K \ge 2$, every $x \in F$ and $t,\epsilon \in \mathbb{R}$ with $|x| \le K$, $|t| \le K$, $0 < \epsilon \le 1$ and $\epsilon < t$, and every $Q_0 \in F$ which is the gradient at x of $x' \mapsto \mathcal{P}_\epsilon(\mathcal{W}[u,\kappa])(x',t)$,

$$|Q_0| \le C_2\big((1+\log K) + |\log t|\big).$$

Let $R \in \mathbb{R}$ with $10 \le R$, let $\epsilon \in \mathbb{R}$ with $0 < \epsilon \le \frac{1}{8}$, and let $q \in \mathbb{N}$ with $\epsilon\, q = 8$.

Let Ω be a set — no measurable structure is assumed — and let $\xi : \mathbb{N}\times\Omega \to \mathbb{R}$, $z, z^+, z^- : \mathbb{N}\times\Omega \to \mathcal{H}$, $Q_0 : \mathbb{N}\times\Omega \to F$, $\tau : \Omega \to \mathbb{N}$ and $S : \mathbb{N}\times\Omega \to \mathbb{Z}$. Write $z_n = (x_n, t_n)$, $z_n^\pm = (x_n^\pm, \cdot)$. Assume:

1. *the walk recursion: $z_{n+1} = z_n^+$ if $n < \tau$ and $\xi_n = 1$, $z_{n+1} = z_n^-$ if $n < \tau$ and $\xi_n \ne 1$, and $z_{n+1} = z_n$ if $n \ge \tau$;*
2. *for $n < \tau$ the triple (z_n, z_n^+, z_n^-) is a step pair for (v,κ,ϵ) in the sense of `IsStepPair`;*
3. *$Q_0(n)$ is the gradient at x_n of $x' \mapsto \mathcal{P}_\epsilon W(x', t_n)$;*
4. *$|x_n| \le R+1$ whenever $n \le \tau$;*
5. *$t_n = \epsilon\, S_n$ for every n and every ω;*
6. *$4 \le S_n \le q$ for every n and every ω.*

Then there are constants $\Lambda_Q, \Lambda_E \in \mathbb{R}$ with $\Lambda_E \ge 0$ such that, for every $n \in \mathbb{N}$ and every $\omega \in \Omega$,

$$|x_n| \le R+1, \tag{625}$$

$$4\epsilon \le t_n \le 8, \tag{626}$$

$$\big|\log t_n\big| \le \big|\log(4\epsilon)\big| + \log 8, \tag{627}$$

$$|Q_0(n)| \le \Lambda_Q, \tag{628}$$

and, for every n and ω with $n < \tau(\omega)$,

$$\big|\epsilon^{-1}(x_n^+ - x_n) - Q_0(n)\big|^2 + \big|\epsilon^{-1}(x_n - x_n^-) - Q_0(n)\big|^2 \le \Lambda_E. \tag{629}$$

Proof. The walk is frozen after τ. We first show

$$z_n = z_\tau \qquad (\tau \le n), \tag{630}$$

by induction on n. If $n = 0$ then $\tau \le 0$ forces $\tau = 0$ and the claim is trivial. For the step, suppose $\tau \le n+1$. If $\tau = n+1$ the claim is again trivial. Otherwise $\tau \le n$, so the inductive hypothesis gives $z_n = z_\tau$, and $n \ge \tau$ means $\neg(n < \tau)$, so the third case of 1 gives $z_{n+1} = z_n = z_\tau$.

Proof of (625). For $n \le \tau$ this is 4. For $n > \tau$, (630) gives $z_n = z_\tau$, so $x_n = x_\tau$ and 4 at the index τ (which satisfies $\tau \le \tau$) applies.

Proof of (626). By 5, $t_n = \epsilon S_n$, and by 6, $4 \le S_n \le q$. Since $\epsilon > 0$, multiplying through by ϵ gives $4\epsilon \le \epsilon S_n \le \epsilon q$, and $\epsilon q = 8$ by hypothesis. This is (626).

Proof of (627). Fix n, ω and put $t = t_n$. By (626) we have $0 < 4\epsilon \le t \le 8$, the left inequality strict because $\epsilon > 0$. The logarithm is monotone on $(0, \infty)$, so $\log(4\epsilon) \le \log t \le \log 8$. Both bounds are dominated in absolute value by the right side of (627): if $\log t \ge 0$ then $|\log t| = \log t \le \log 8 \le |\log(4\epsilon)| + \log 8$, since $|\log(4\epsilon)| \ge 0$; and if $\log t < 0$ then $|\log t| = -\log t \le -\log(4\epsilon) \le |\log(4\epsilon)| \le |\log(4\epsilon)| + \log 8$, since $\log 8 \ge 0$.

Proof of (628). Put

$$\Lambda_Q = C_2\Big((1 + \log(R + 10)) + \big(|\log(4\epsilon)| + \log 8\big)\Big).$$

Fix n, ω and apply the bound assumed of C_2 to $u = v$ at the box size $K = R + 10$. Its hypotheses hold: $K \ge 2$ because $R \ge 10$; $|x_n| \le R + 1 \le R + 10$ by (625); $|t_n| \le 8 \le R + 10$ by (626) and $R \ge 10$, the absolute value being t_n itself since $t_n \ge 4\epsilon > 0$; $0 < \epsilon \le \frac{1}{8} \le 1$; $\epsilon < t_n$ because $t_n \ge 4\epsilon > \epsilon$; and $Q_0(n)$ is the required gradient by 3. The conclusion is $|Q_0(n)| \le C_2\big((1 + \log(R + 10)) + |\log t_n|\big)$, and (627) together with $C_2 > 0$ bounds the right side by Λ_Q. Note that Λ_Q depends only on C_2, R and ϵ, not on n or ω.

The per-step displacement bound. The two envelope competitor bounds of `EnvelopeCompetitorBounds`, applied to v on $\mathbb{R}$ and to W on $(0, \infty)$ — legitimate because v is continuous with subquadratic growth by the standing properties, and W is continuous on $\{t > 0\}$ with subquadratic growth there by hypothesis — supply, together with the growth bound for v on a compact slab and the sandwich, a constant $B_Y \ge 0$, depending only on the standing data and on ϵ, with the following property: whenever $n < \tau$ and $Y^{\pm}$ are the step displacements of 2, their norms obey $|Y^{\pm}|^2 \le B_Y$. Combining this with 2, which expresses $x_n^+ - x_n$ and $x_n - x_n^-$ in terms of $Y^{\pm}$ at the scale 2ϵ, gives for $n < \tau$

$$\big|\epsilon^{-1}(x_n^+ - x_n) - Q_0(n)\big| \le \frac{\sqrt{B_Y}}{2\epsilon} + \Lambda_Q, \qquad \big|\epsilon^{-1}(x_n - x_n^-) - Q_0(n)\big| \le \frac{\sqrt{B_Y}}{2\epsilon} + \Lambda_Q, \tag{631}$$

the term Λ_Q entering through (628) and the triangle inequality.

Proof of (629). Put

$$\Lambda_E = 2\Big(\frac{\sqrt{B_Y}}{2\epsilon} + \Lambda_Q\Big)^2,$$

which is nonnegative, being twice a square. Fix $n < \tau$ and write a, b for the two norms in (629). Both are nonnegative and both are at most $\sqrt{B_Y}/(2\epsilon) + \Lambda_Q$ by (631); squaring the inequality between nonnegative reals gives $a^2, b^2 \le (\sqrt{B_Y}/(2\epsilon) + \Lambda_Q)^2$, and adding the two yields $a^2 + b^2 \le \Lambda_E$. This is (629), and Λ_E depends only on B_Y, ϵ and Λ_Q, hence not on n or ω. □

2.152. `DoobL2Maximal`.

Lemma 2.152 (Doob's L^2 maximal inequality, vector valued)**.** *Let F be a finite-dimensional real inner-product space, let $(\Omega, \mathcal{M}, \mu)$ be a probability space, let $(\mathcal{G}_n)_{n \ge 0}$ be a filtration in $\mathcal{M}$, and let $M = (M_n)_{n \ge 0}$ be an F-valued martingale for $(\mathcal{G}_n)$ and μ such that $\|M_n\|^2$ is integrable for every n. Fix $N \ge 0$ and put*

$$S = \max_{0 \le k \le N} \|M_k\|.$$

Then S is integrable,

$$\mathbb{E}\, S \le 2\big(\mathbb{E}\|M_N\|^2\big)^{1/2}, \tag{632}$$

and the second moment splits orthogonally over the increments:

$$\mathbb{E}\|M_N\|^2 = \mathbb{E}\|M_0\|^2 + \sum_{k=0}^{N-1} \mathbb{E}\|M_{k+1} - M_k\|^2. \tag{633}$$

Proof. Write $C = \mathbb{E}\|M_N\|^2$, which is a finite nonnegative real number by hypothesis, and $d_k = M_{k+1} - M_k$.

Step 1: $(\|M_n\|^2)_n$ is a nonnegative submartingale. Each $\|M_n\|^2$ is nonnegative, and it is $\mathcal{G}_n$-measurable because M_n is (the family M is adapted, being a martingale) and $a \mapsto \|a\|^2$ is continuous. It is integrable by hypothesis. For the submartingale inequality it is enough to compare consecutive indices. Fix n. Since $\|M_{n+1}\|^2$ is integrable, conditional Jensen for the norm power, with $p = 2$ and the sub-σ-algebra $\mathcal{G}_n$, gives

$$\big\|\mathbb{E}[M_{n+1} \mid \mathcal{G}_n]\big\|^2 \le \mathbb{E}\big[\|M_{n+1}\|^2 \mid \mathcal{G}_n\big] \qquad \text{a.e.},$$

and the martingale property gives $\mathbb{E}[M_{n+1} \mid \mathcal{G}_n] = M_n$ a.e. Hence $\|M_n\|^2 \le \mathbb{E}[\|M_{n+1}\|^2 \mid \mathcal{G}_n]$ a.e., which is the required inequality.

Step 2: the weak-type bound. Let $\lambda > 0$. Since $u \mapsto u^2$ is strictly increasing on $[0, \infty)$ and all the quantities $\|M_k\|$ are nonnegative,

$$\{\lambda \le S\} = \Big\{\lambda^2 \le \max_{0\le k\le N} \|M_k\|^2\Big\}.$$

Apply Doob's weak-type maximal inequality to the nonnegative submartingale $(\|M_k\|^2)$ of Step 1, at the level $\varepsilon = \lambda^2 \ge 0$ and the index N:

$$\lambda^2\, \mu\Big\{\lambda^2 \le \max_{0\le k\le N} \|M_k\|^2\Big\} \le \int_{\{\lambda^2 \le \max_{k\le N} \|M_k\|^2\}} \|M_N\|^2 \, d\mu \le \mathbb{E}\|M_N\|^2 = C,$$

the last inequality because $\|M_N\|^2 \ge 0$, so that enlarging the domain of integration to Ω only increases the integral. Dividing by $\lambda^2 > 0$,

$$\mu\{\lambda \le S\} \le \frac{C}{\lambda^2} \qquad (\lambda > 0). \tag{634}$$

Step 3: integrability of S and the two-parameter bound. S is a maximum of finitely many nonnegative measurable functions, hence nonnegative and measurable. By the layer-cake identity,

$$\int_\Omega S \, d\mu = \int_0^\infty \mu\{S > \lambda\}\, d\lambda,$$

as an identity in $[0, \infty]$. Fix $a > 0$ and split the right-hand integral at a. For $0 < \lambda < a$ we use $\mu\{S > \lambda\} \le \mu(\Omega) = 1$, so that part contributes at most a. For $\lambda > a$ we use $\{S > \lambda\} \subseteq \{\lambda \le S\}$ and (634), so that part contributes at most

$$\int_a^\infty \frac{C}{\lambda^2}\, d\lambda = \frac{C}{a}.$$

Hence

$$\int_0^\infty \mu\{S > \lambda\}\, d\lambda \le a + \frac{C}{a} < \infty \qquad (a > 0). \tag{635}$$

In particular the right-hand side of the layer-cake identity is finite, so S is integrable and $\mathbb{E}\, S \le a + C/a$ for every $a > 0$.

Step 4: the optimal choice of a, giving (632). If $C = 0$ then (635) gives $\mathbb{E}\, S \le a$ for every $a > 0$, hence $\mathbb{E}\, S \le 0 = 2\sqrt{C}$. If $C > 0$, take $a = \sqrt{C} > 0$; then

$$\mathbb{E}\, S \le \sqrt{C} + \frac{C}{\sqrt{C}} = 2\sqrt{C},$$

because $\sqrt{C} \cdot \sqrt{C} = C$. This is (632).

Step 5: orthogonality of the increments, giving (633). Fix k with $0 \le k < N$. Expanding $M_{k+1} = M_k + d_k$ in the inner product,

$$\|M_{k+1}\|^2 = \|M_k\|^2 + 2\langle d_k, M_k\rangle + \|d_k\|^2 \qquad \text{pointwise on } \Omega. \tag{636}$$

All four terms are integrable. Indeed $\|M_k\|^2$ and $\|M_{k+1}\|^2$ are so by hypothesis; from $\|a-b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ we get $\|d_k\|^2 \le 2\|M_{k+1}\|^2 + 2\|M_k\|^2$, and $\|d_k\|^2$ is measurable and nonnegative, so it is integrable; and by Cauchy–Schwarz and $2uw \le u^2 + w^2$,

$$|\langle d_k, M_k\rangle| \le \|d_k\|\, \|M_k\| \le \tfrac{1}{2}\big(\|d_k\|^2 + \|M_k\|^2\big),$$

so $\langle d_k, M_k\rangle$ is integrable as well; d_k itself is integrable since $\|d_k\| \le \frac{1}{2}(1 + \|d_k\|^2)$.

Next we check

$$\mathbb{E}[d_k \mid \mathcal{G}_k] = 0 \qquad \text{a.e.} \tag{637}$$

The constant function 0 is $\mathcal{G}_k$-measurable and integrable, and for every $A \in \mathcal{G}_k$

$$\int_A d_k\,d\mu = \int_A M_{k+1}\,d\mu - \int_A M_k\,d\mu = \int_A \mathbb{E}[M_{k+1} \mid \mathcal{G}_k]\,d\mu - \int_A M_k\,d\mu = \int_A M_k\,d\mu - \int_A M_k\,d\mu = 0 = \int_A 0\,d\mu,$$

where the first equality is additivity of the integral (both M_{k+1} and M_k are integrable), the second is the characterizing property of $\mathbb{E}[M_{k+1} \mid \mathcal{G}_k]$ applied to $A \in \mathcal{G}_k$, and the third is the martingale property $\mathbb{E}[M_{k+1} \mid \mathcal{G}_k] = M_k$ a.e. Since the conditional expectation is characterized a.e. by this family of identities, and d_k is integrable, (637) follows. The inner product of F is a continuous bilinear map $F \times F \to \mathbb{R}$, the function M_k is $\mathcal{G}_k$-measurable, and d_k and $\langle d_k, M_k\rangle$ are integrable; so the pull-out property of the conditional expectation applies and gives, with (637),

$$\mathbb{E}\big[\langle d_k, M_k\rangle \mid \mathcal{G}_k\big] = \big\langle \mathbb{E}[d_k \mid \mathcal{G}_k],\, M_k\big\rangle = 0 \qquad \text{a.e.}$$

Integrating this over Ω and using the characterizing property of the conditional expectation with $A = \Omega \in \mathcal{G}_k$, which gives $\mathbb{E}\big[\mathbb{E}[X \mid \mathcal{G}_k]\big] = \mathbb{E}[X]$ for integrable X, we obtain $\mathbb{E}\langle d_k, M_k\rangle = 0$. Taking expectations in (636) therefore yields

$$\mathbb{E}\|M_{k+1}\|^2 = \mathbb{E}\|M_k\|^2 + \mathbb{E}\|d_k\|^2 \qquad (0 \le k < N).$$

Summing over $k = 0, \dots, N-1$, the terms $\mathbb{E}\|M_k\|^2$ telescope and leave (633). (For $N = 0$ both sides of (633) are $\mathbb{E}\|M_0\|^2$ and the empty sum is zero.) □

2.153. `SqrtSumLogSquareBounds.`

Lemma 2.153 (A telescoping square-root sum bound and a logarithmic square bound)**.** *The following two elementary inequalities hold.*

(1) *For every real $a > 0$ and every integer $M \ge 0$,*

$$\sum_{m=0}^{M-1} \frac{1}{\sqrt{a(m+1)}} \le \frac{2\sqrt{aM}}{a}. \tag{638}$$

(2) *For every real s with $0 < s \le 8$,*

$$|\log s|^2 \le \frac{16}{\sqrt{s}} + (\log 8)^2. \tag{639}$$

Proof. Part 1: the telescoping comparison. Fix $a > 0$ and $M \ge 0$. We first prove the termwise estimate

$$\frac{1}{\sqrt{a(m+1)}} \le \frac{2\big(\sqrt{a(m+1)} - \sqrt{am}\big)}{a} \qquad (m \ge 0 \text{ an integer}). \tag{640}$$

Write $u = \sqrt{a(m+1)}$ and $w = \sqrt{am}$. Since $a > 0$ and $m + 1 \ge 1 > 0$ we have $a(m+1) > 0$, so $u > 0$ and $u^2 = a(m+1)$; since $m \ge 0$ we have $am \ge 0$, so $w \ge 0$ and $w^2 = am$. Consequently

$$u^2 - w^2 = a(m+1) - am = a, \tag{641}$$

and $w \le u$ because $am \le a(m+1)$ and the square root is monotone on $[0, \infty)$. Both $u > 0$ and $a > 0$, so (640) is equivalent, after multiplying both sides by the positive number ua, to

$$a \le 2u(u - w),$$

and by (641) this is in turn equivalent to $u^2 - w^2 \le 2u^2 - 2uw$, i.e. to $0 \le u^2 - 2uw + w^2 = (u-w)^2$, which holds. This proves (640).

Next we evaluate the sum of the right-hand sides of (640). Put $g(k) = \sqrt{ak}$ for integers $k \ge 0$. For each integer $m \ge 0$,

$$\frac{2\big(\sqrt{a(m+1)} - \sqrt{am}\big)}{a} = \frac{2}{a}\big(g(m+1) - g(m)\big),$$

so by the telescoping identity $\sum_{m=0}^{M-1}\big(g(m+1)-g(m)\big)=g(M)-g(0)$ and by $g(0)=\sqrt{0}=0$,

$$\sum_{m=0}^{M-1}\frac{2\big(\sqrt{a(m+1)}-\sqrt{am}\big)}{a}=\frac{2}{a}\big(g(M)-g(0)\big)=\frac{2\sqrt{aM}}{a}. \tag{642}$$

Summing (640) over $m=0,\dots,M-1$ and using monotonicity of finite sums together with (642) gives

$$\sum_{m=0}^{M-1}\frac{1}{\sqrt{a(m+1)}}\le\sum_{m=0}^{M-1}\frac{2\big(\sqrt{a(m+1)}-\sqrt{am}\big)}{a}=\frac{2\sqrt{aM}}{a},$$

which is (638).

Part 2: the logarithmic square bound. Fix s with $0<s\le 8$. Since $8\ge 1$ we have $\log 8\ge 0$. We distinguish two cases.

Case $s\le 1$. Set $u=\sqrt{\sqrt{s}}$. As $s>0$ we have $\sqrt{s}>0$ and hence $u>0$, and $u^2=\sqrt{s}$. From $\log\sqrt{x}=\frac{1}{2}\log x$ for $x\ge 0$, applied twice,

$$\log u=\tfrac{1}{2}\log\sqrt{s}=\tfrac{1}{4}\log s. \tag{643}$$

The inequality $\log x\le x-1$, valid for all $x>0$, applied to $x=u^{-1}>0$ gives $\log(u^{-1})\le u^{-1}-1$, i.e. $-\log u\le\frac{1}{u}-1\le\frac{1}{u}$. Combining this with (643) and multiplying by $4>0$,

$$-\log s\le\frac{4}{u}. \tag{644}$$

Because $0<s\le 1$ we have $\log s\le 0$, so $|\log s|=-\log s$ and $-\log s\ge 0$. Both sides of (644) are therefore nonnegative, and squaring a valid inequality between nonnegative reals preserves it:

$$(-\log s)^2\le\Big(\frac{4}{u}\Big)^2=\frac{16}{u^2}=\frac{16}{\sqrt{s}},$$

the last step by $u^2=\sqrt{s}$. Since $(\log 8)^2\ge 0$, we conclude

$$|\log s|^2=(-\log s)^2\le\frac{16}{\sqrt{s}}\le\frac{16}{\sqrt{s}}+(\log 8)^2,$$

which is (639) in this case.

Case $s>1$. Then $\log s\ge 0$, so $|\log s|=\log s$, and $s\le 8$ with $s>0$ gives $\log s\le\log 8$ by monotonicity of the logarithm on $(0,\infty)$. Both $\log s$ and $\log 8$ are nonnegative with $\log s\le\log 8$, so $(\log s)^2\le(\log 8)^2$. Finally $16/\sqrt{s}\ge 0$ because $s>0$, whence

$$|\log s|^2=(\log s)^2\le(\log 8)^2\le\frac{16}{\sqrt{s}}+(\log 8)^2.$$

This is (639) in the remaining case, and the two cases exhaust $0<s\le 8$. □

2.154. `LatticeLogGreenFunction`.

Lemma 2.154 (A lattice Green function dominating the logarithmic weight)**.** *Let $\epsilon\in\mathbb{R}$, let $q,j_0\in\mathbb{N}$ and let $\Lambda_0\in\mathbb{R}$. Assume*

(1) $\epsilon>0$;
(2) $\epsilon\, j_0=1$;
(3) $\epsilon\, q=8$;
(4) $\Lambda_0=32\sqrt{8}+8(\log 8)^2$.

Then there is a function $G:\mathbb{Z}\to\mathbb{R}$ such that

(5) $0\le G(j)$ *whenever* $0<j\le q$;
(6) *for every integer j with $0<j<q$,*

$$\frac{G(j+1)+G(j-1)}{2}-G(j)\le-\epsilon^2\big|\log(\epsilon j)\big|^2; \tag{645}$$

(7) $G(j_0)\le 2\Lambda_0$.

Proof. Throughout write $u(m) = \epsilon^2 |\log(\epsilon m)|^2$ for $m \in \mathbb{N}$, the logarithm being the one of the formalization, which takes the value 0 at 0; the term $m = 0$ of the sums below is therefore $\epsilon^2 |\log 0|^2 = 0$. Nothing in the statement evaluates the weight at 0 — (645) is asserted only for $j \geq 1$, where $\epsilon j > 0$ — so this convention and the division convention $1/\sqrt{0} = 0$ affect only the bookkeeping of the sums.

q is positive. If $q = 0$ then $\epsilon q = 0$, contradicting 3. So $0 < q$ as a natural number and $0 < q$ as a real number.

Step 1: the weight sum. We claim

$$\sum_{m=0}^{q-1} u(m) \leq \epsilon \Lambda_0. \tag{646}$$

Fix m with $0 \leq m \leq q-1$. If $m = 0$ then $u(0) = 0$ and the right-hand side of the termwise bound below is nonnegative, so the bound holds. If $m \geq 1$ then $\epsilon m > 0$ because $\epsilon > 0$ and $m \geq 1$, and

$$\epsilon m \leq \epsilon(q-1) = \epsilon q - \epsilon = 8 - \epsilon \leq 8$$

using 3 and $\epsilon > 0$. Hence part 2 of `SqrtSumLogSquareBounds` applies at $s = \epsilon m$ and gives $|\log(\epsilon m)|^2 \leq 16/\sqrt{\epsilon m} + (\log 8)^2$; multiplying by $\epsilon^2 \geq 0$,

$$u(m) \leq 16\epsilon^2 \cdot \frac{1}{\sqrt{\epsilon m}} + \epsilon^2 (\log 8)^2 \qquad (0 \leq m \leq q-1). \tag{647}$$

Summing (647) over $0 \leq m \leq q-1$ and splitting the constant term off,

$$\sum_{m=0}^{q-1} u(m) \leq 16\epsilon^2 \sum_{m=0}^{q-1} \frac{1}{\sqrt{\epsilon m}} + q\,\epsilon^2 (\log 8)^2. \tag{648}$$

Write $q = q' + 1$, which is legitimate since $q \geq 1$. The $m = 0$ term of $\sum_{m=0}^{q-1} 1/\sqrt{\epsilon m}$ vanishes, and reindexing the remaining terms by $m \mapsto m+1$ turns the sum into $\sum_{m=0}^{q'-1} 1/\sqrt{\epsilon(m+1)}$, to which part 1 of `SqrtSumLogSquareBounds` applies with $a = \epsilon$ and $M = q'$:

$$\sum_{m=0}^{q-1} \frac{1}{\sqrt{\epsilon m}} \leq \frac{2\sqrt{\epsilon q'}}{\epsilon}. \tag{649}$$

Now $q' \leq q$, so $\epsilon q' \leq \epsilon q = 8$ and hence $\sqrt{\epsilon q'} \leq \sqrt{8}$ by monotonicity of the square root. Therefore, using $\epsilon > 0$ to simplify $16\epsilon^2 \cdot 2\sqrt{\epsilon q'}/\epsilon = 32\epsilon\sqrt{\epsilon q'}$,

$$16\epsilon^2 \sum_{m=0}^{q-1} \frac{1}{\sqrt{\epsilon m}} \leq 32\epsilon\sqrt{\epsilon q'} \leq 32\epsilon\sqrt{8},$$

while by 3

$$q\,\epsilon^2 (\log 8)^2 = (q\epsilon) \cdot \epsilon (\log 8)^2 = 8\epsilon (\log 8)^2.$$

Adding the two and using 4, $32\epsilon\sqrt{8} + 8\epsilon(\log 8)^2 = \epsilon\big(32\sqrt{8} + 8(\log 8)^2\big) = \epsilon\Lambda_0$, which with (648) proves (646).

Step 2: the lattice Green kernel. For $j, m \in \mathbb{Z}$ put

$$K(j,m) = \frac{(j \wedge m)\big(q - (j \vee m)\big)}{q},$$

which is well defined because $q \neq 0$. Two properties are needed.

First, $K(j,m) \geq 0$ whenever $0 \leq j \leq q$ and $0 \leq m \leq q$: then $j \wedge m \geq 0$, since both arguments are, and $j \vee m \leq q$, since both arguments are, so $q - (j \vee m) \geq 0$; the quotient of the product of two nonnegative numbers by the positive q is nonnegative.

Second, for every integer j with $0 < j < q$ and every $m \in \mathbb{Z}$,

$$\frac{K(j+1,m) + K(j-1,m)}{2} - K(j,m) = \begin{cases} -\tfrac{1}{2}, & m = j, \\ 0, & m \neq j. \end{cases} \tag{650}$$

We check the three cases of the trichotomy between m and j; in each the minima and maxima occurring in the three terms are determined, after which the identity is an algebraic computation with $q \neq 0$.

If $m < j$ then $m \le j-1$, so m is the minimum and $j+1$, $j-1$, j are respectively the maxima in the three terms, and the left side of (650) equals

$$\frac{m\big[(q-(j+1))+(q-(j-1))-2(q-j)\big]}{2q} = \frac{m\cdot 0}{2q} = 0.$$

If $m > j$ then $j+1 \le m$, so $j+1$, $j-1$, j are the minima and m is the maximum in all three terms, and the left side equals

$$\frac{\big[(j+1)+(j-1)-2j\big](q-m)}{2q} = 0.$$

If $m = j$ then $K(j+1,j) = j(q-j-1)/q$, $K(j-1,j) = (j-1)(q-j)/q$ and $K(j,j) = j(q-j)/q$, so the left side equals

$$\frac{j(q-j-1)+(j-1)(q-j)-2j(q-j)}{2q} = \frac{-j-(q-j)}{2q} = \frac{-q}{2q} = -\frac{1}{2}.$$

Step 3: the Green function. Define

$$G(j) = \sum_{m=0}^{q-1} 2\,u(m)\,K(j,m) \qquad (j \in \mathbb{Z}),$$

the summation index m being read as an integer inside K.

5. Let $0 < j \le q$. For each m in the range $0 \le m \le q-1$ we have $0 \le m \le q$ as integers, so $K(j,m) \ge 0$ by the first property of Step 2, while $2u(m) \ge 0$ because $u(m) = \epsilon^2|\log(\epsilon m)|^2$ is a product of squares. Each summand is therefore nonnegative, hence so is $G(j)$.

6. Let $0 < j < q$. By linearity of the finite sum,

$$\frac{G(j+1)+G(j-1)}{2} - G(j) = \sum_{m=0}^{q-1} 2\,u(m)\Big(\frac{K(j+1,m)+K(j-1,m)}{2} - K(j,m)\Big),$$

and by (650) the m-th summand is $-u(m)$ when $m = j$ and 0 otherwise. Since $0 < j < q$, the integer j equals exactly one index of the summation range $\{0,\dots,q-1\}$, so the sum collapses to $-u(j)$, that is to $-\epsilon^2|\log(\epsilon j)|^2$. This is (645), with equality.

7. For $0 \le m \le q-1$ we have $j_0 \wedge m \le j_0$ and $0 \le q-(j_0 \vee m) \le q$, the latter because $j_0 \vee m \ge 0$; also $j_0 \wedge m \ge 0$. Hence

$$K(j_0,m) = \frac{(j_0 \wedge m)\big(q-(j_0 \vee m)\big)}{q} \le \frac{j_0 \cdot q}{q} = j_0.$$

Multiplying by $2u(m) \ge 0$ and summing,

$$G(j_0) \le \sum_{m=0}^{q-1} 2\,u(m)\,j_0 = 2j_0 \sum_{m=0}^{q-1} u(m) \le 2j_0 \cdot \epsilon\Lambda_0 = 2\Lambda_0\,(\epsilon j_0) = 2\Lambda_0,$$

where the middle inequality is (646) multiplied by the nonnegative number $2j_0$, and the last step is 2. □

2.155. `IntegralSqrtProductCauchySchwarz.`

Lemma 2.155 (Cauchy–Schwarz for the integral of a product of square roots)**.** *Let Ω be a measurable space, let μ be a measure on Ω, and let $f, g : \Omega \to \mathbb{R}$ satisfy $f(\omega) \ge 0$ and $g(\omega) \ge 0$ for every $\omega \in \Omega$. Assume that f and g are both μ-integrable. Then the function $\omega \mapsto \sqrt{f(\omega)}\,\sqrt{g(\omega)}$ is μ-integrable and*

$$\int \sqrt{f(\omega)}\,\sqrt{g(\omega)}\,d\mu(\omega) \le \sqrt{\int f\,d\mu}\,\sqrt{\int g\,d\mu}. \tag{651}$$

Proof. Step 1: a pointwise bound with a free parameter. We claim that for every real $l > 0$ and every $\omega \in \Omega$,

$$\sqrt{f(\omega)}\,\sqrt{g(\omega)} \le \frac{l\,f(\omega) + g(\omega)/l}{2}. \tag{652}$$

Fix $l > 0$ and ω, and put $a = \sqrt{f(\omega)}$ and $b = \sqrt{g(\omega)}$. Then $a \ge 0$ and $b \ge 0$, and since $f(\omega) \ge 0$ and $g(\omega) \ge 0$ we have $a^2 = f(\omega)$ and $b^2 = g(\omega)$. Expanding the inequality $0 \le (l\,a - b)^2 = l^2a^2 - 2l\,ab + b^2$ gives

$$2l\,ab \le l^2a^2 + b^2 = l^2 f(\omega) + g(\omega).$$

Since $l \ne 0$ we have $\big(g(\omega)/l\big) \cdot l = g(\omega)$, so dividing the last inequality by $2l > 0$ gives $ab \le \big(l\,f(\omega) + g(\omega)/l\big)/2$, which is (652).

Step 2: integrability of the product. Taking $l = 1$ in (652) gives the pointwise domination

$$0 \le \sqrt{f(\omega)}\,\sqrt{g(\omega)} \le \frac{1 \cdot f(\omega) + g(\omega)/1}{2} \qquad (\omega \in \Omega),$$

the left inequality because square roots are nonnegative. The dominating function $\big(1 \cdot f + g/1\big)/2$ is μ-integrable, being a constant multiple of the sum of the constant multiple $1 \cdot f$ of the integrable f and the constant multiple $g/1$ of the integrable g. Moreover $\omega \mapsto \sqrt{f(\omega)}\,\sqrt{g(\omega)}$ is μ-almost-everywhere strongly measurable: f and g are, being integrable; the square root is continuous on $\mathbb{R}$, so composing it with each of them preserves almost-everywhere strong measurability; and the product of two such functions is again one. A μ-almost-everywhere strongly measurable function whose norm is dominated pointwise by an integrable function is integrable, and here the norm $\big|\sqrt{f(\omega)}\sqrt{g(\omega)}\big|$ equals $\sqrt{f(\omega)}\sqrt{g(\omega)}$ itself because that quantity is nonnegative. Hence $\sqrt{f}\,\sqrt{g}$ is μ-integrable, which is the first assertion.

Step 3: integrating the parametric bound. Write

$$A = \int f\,d\mu, \qquad B = \int g\,d\mu.$$

Monotonicity of the integral applied to $f \ge 0$ and to $g \ge 0$ gives $A \ge 0$ and $B \ge 0$. Fix $l > 0$. Both sides of (652) are μ-integrable — the left-hand side by Step 2, the right-hand side because $l\,f$ and g/l are constant multiples of integrable functions — so monotonicity of the integral and then its linearity give

$$\int \sqrt{f}\,\sqrt{g}\,d\mu \le \int \frac{l\,f + g/l}{2}\,d\mu = \frac{l\,A + B/l}{2} \qquad (l > 0). \tag{653}$$

Step 4: optimizing in l. We distinguish three cases according to the signs of A and B.

Case $A = 0$. Here $\sqrt{A}\,\sqrt{B} = 0$, so (651) asserts $\int \sqrt{f}\sqrt{g}\,d\mu \le 0$. It suffices to prove $\int \sqrt{f}\sqrt{g}\,d\mu \le 0 + \delta$ for every $\delta > 0$. Given $\delta > 0$, set $l = (B+1)/\delta$, which is positive because $B \ge 0$ and $\delta > 0$. Then

$$\frac{B/l}{2} = \frac{B\,\delta}{(B+1)\cdot 2},$$

and this is at most δ: the inequality $B\,\delta \le \delta\,(B+1)\cdot 2$ holds because $B \ge 0$ and $\delta > 0$, and $(B+1)\cdot 2 > 0$. With $A = 0$, (653) for this l reads $\int \sqrt{f}\sqrt{g}\,d\mu \le (l \cdot 0 + B/l)/2 = (B/l)/2 \le \delta$. As $\delta > 0$ was arbitrary, $\int \sqrt{f}\sqrt{g}\,d\mu \le 0 = \sqrt{A}\,\sqrt{B}$.

Case $A > 0$ and $B = 0$. Here again $\sqrt{A}\,\sqrt{B} = 0$, and we argue symmetrically. Given $\delta > 0$, set $l = \delta/(A+1) > 0$. Then

$$\frac{l\,A}{2} = \frac{\delta\,A}{(A+1)\cdot 2} \le \delta,$$

because $\delta A \le \delta(A+1)\cdot 2$ (as $A > 0$ and $\delta > 0$) and $(A+1)\cdot 2 > 0$. With $B = 0$, (653) for this l reads $\int \sqrt{f}\sqrt{g}\,d\mu \le (l\,A + 0)/2 \le \delta$, and letting $\delta \downarrow 0$ gives $\int \sqrt{f}\sqrt{g}\,d\mu \le 0 = \sqrt{A}\,\sqrt{B}$.

Case $A > 0$ and $B > 0$. Then $\sqrt{A} > 0$ and $\sqrt{B} > 0$, so $l = \sqrt{B}/\sqrt{A}$ is a legitimate positive choice in (653). Using $\sqrt{A} \cdot \sqrt{A} = A$ and $\sqrt{B} \cdot \sqrt{B} = B$, which hold because $A \ge 0$ and $B \ge 0$,

$$l\,A = \frac{\sqrt{B}}{\sqrt{A}}\,A = \sqrt{A}\,\sqrt{B}, \qquad \frac{B}{l} = \frac{B\,\sqrt{A}}{\sqrt{B}} = \sqrt{A}\,\sqrt{B}.$$

Substituting these two identities into (653) gives

$$\int \sqrt{f}\,\sqrt{g}\,d\mu \le \frac{\sqrt{A}\,\sqrt{B}+\sqrt{A}\,\sqrt{B}}{2} = \sqrt{A}\,\sqrt{B},$$

which is (651). The three cases exhaust the possibilities, since $A \ge 0$ and $B \ge 0$, so the proof is complete. □

2.156. `StoppedPortionTravelBound`.

Lemma 2.156 (The travel inequality for a stopped portion in a sheared frame)**.** *Let F be a finite-dimensional real inner-product space, let $(\Omega, \mathcal{M}, \mu)$ be a probability space with Ω nonempty, let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a filtration of $\mathcal{M}$, let $\epsilon > 0$, let $\xi : \mathbb{N}\times\Omega \to \mathbb{R}$, let $z, z^+, z^- : \mathbb{N}\times\Omega \to \ell^2(F\times\mathbb{R})$, let $Q_0 : \mathbb{N}\times\Omega \to F$, let $\tau : \Omega \to \mathbb{N}$ and let $\Lambda_Q, \Lambda_E \in \mathbb{R}$ with $\Lambda_E \ge 0$. Write $z_n = (x_n, t_n)$, $z_n^+ = (x_n^+, t_n^+)$ and $z_n^- = (x_n^-, t_n^-)$. Assume:*

1. *$\xi_n(\omega) = 1$ or $\xi_n(\omega) = -1$ for every n and every ω;*
2. *ξ_n is $\mathcal{G}_{n+1}$-measurable for every n;*
3. *$\mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ μ-almost everywhere, for every n;*
4. *z_n, z_n^+, z_n^- and $Q_0(n)$ are $\mathcal{G}_n$-measurable for every n;*
5. *the recursion $z_{n+1} = z_n^+$ if $n < \tau$ and $\xi_n = 1$, $z_{n+1} = z_n^-$ if $n < \tau$ and $\xi_n \ne 1$, and $z_{n+1} = z_n$ if $n \ge \tau$;*
6. *$\omega \mapsto \tau(\omega)$ is μ-integrable as a real-valued function;*
7. *the time step $t_{n+1} = t_n + \epsilon\,\xi_n$ when $n < \tau$, and $t_{n+1} = t_n$ when $n \ge \tau$;*
8. *$|Q_0(n)(\omega)| \le \Lambda_Q$ for every n and every ω;*
9. *for every n and every ω with $n < \tau(\omega)$,*

$$\mathcal{E}_n := \left|\epsilon^{-1}(x_n^+ - x_n) - Q_0(n)\right|^2 + \left|\epsilon^{-1}(x_n - x_n^-) - Q_0(n)\right|^2 \le \Lambda_E;$$

10. *every $\zeta : \Omega \to \mathbb{N}$ which is a $(\mathcal{G}_n)$-stopping time (in the sense that $\{\zeta \le n\} \in \mathcal{G}_n$ for all n) is $\mathcal{M}$-measurable;*
11. *if $G_h : \Omega \to \mathbb{R}$ is measurable for every $h \in \mathbb{N}$ and $g : \Omega \to \mathbb{R}$ is such that for every ω there is H with $G_h(\omega) = g(\omega)$ for all $h \ge H$, then g is measurable;*
12. *if $\zeta : \Omega \to \mathbb{N}$ is measurable and $G_m : \Omega \to \mathbb{R}$ is measurable for every m, then $\omega \mapsto G_{\zeta(\omega)}(\omega)$ is measurable.*

Then for all $(\mathcal{G}_n)$-stopping times $\rho, \sigma : \Omega \to \mathbb{N}$ with $\rho \le \sigma \le \tau$ pointwise, every $r > 0$, every $C_p \in \mathbb{R}$ and every $p : \Omega \to F$ with $|p| \le C_p$ such that $\mathbf{1}_{\{\rho\le n\}}p$ is $\mathcal{G}_n$-measurable for every n, the four functions

$$\sup_{\rho\le n\le\sigma} \frac{|x_n - x_\rho - (t_n - t_\rho)p|}{r}, \qquad \sum_{\rho\le n<\sigma} \left(\frac{\epsilon}{r}\right)^2 |Q_0(n) - p|^2, \qquad \left(\frac{\epsilon}{r}\right)^2 (\sigma - \rho), \qquad \sum_{\rho\le n<\sigma} \mathcal{E}_n$$

are all μ-integrable, and

$$\begin{aligned}\mathbb{E}\sup_{\rho\le n\le\sigma} \frac{|x_n - x_\rho - (t_n - t_\rho)p|}{r} \le{}& 2\left(\mathbb{E}\sum_{\rho\le n<\sigma} \left(\frac{\epsilon}{r}\right)^2 |Q_0(n) - p|^2\right)^{1/2} \\ &+ \left(\mathbb{E}\left(\frac{\epsilon}{r}\right)^2 (\sigma - \rho)\right)^{1/2} \left(\mathbb{E}\sum_{\rho\le n<\sigma} \mathcal{E}_n\right)^{1/2}.\end{aligned} \tag{654}$$

Proof. Let $\rho \le \sigma \le \tau$, $r > 0$, C_p and p be as in the statement, and put $\epsilon' = \epsilon/r$, which is positive. For $n \ge 0$ set

$$y_n = \frac{x_n - x_\rho - (t_n - t_\rho)p}{r},$$

so $y_\rho = 0$ and the left side of (654) is $\mathbb{E}\sup_{\rho\le n\le\sigma} |y_n|$. Since Ω is nonempty we may fix a point of it and conclude from $|p| \le C_p$ and $|Q_0(0)| \le \Lambda_Q$ that $C_p \ge 0$ and $\Lambda_Q \ge 0$.

The increments. Write $Q_+(n) = \epsilon^{-1}(x_n^+ - x_n)$ and $Q_-(n) = \epsilon^{-1}(x_n - x_n^-)$, and set $Q_{\xi_n}(n) = Q_+(n)$ when $\xi_n = 1$ and $Q_{\xi_n}(n) = Q_-(n)$ otherwise. For $n < \tau$, hypothesis 5 gives $x_{n+1} = x_n^+$

when $\xi_n = 1$ and $x_{n+1} = x_n^-$ when $\xi_n \neq 1$, so in the first case $x_{n+1} - x_n = x_n^+ - x_n = \epsilon Q_+(n) = \epsilon \xi_n Q_{\xi_n}(n)$ (using $\xi_n = 1$), and in the second case, where $\xi_n = -1$ by 1, $x_{n+1} - x_n = x_n^- - x_n = -\epsilon Q_-(n) = \epsilon \xi_n Q_{\xi_n}(n)$. Thus

$$x_{n+1} - x_n = \epsilon\, \xi_n\, Q_{\xi_n}(n) \qquad (n < \tau), \tag{655}$$

where the division by ϵ is legitimate because $\epsilon > 0$. Combining (655) with the time step 7, for $\rho \le n < \sigma$ (so $n < \tau$, as $\sigma \le \tau$),

$$y_{n+1} - y_n = \frac{(x_{n+1} - x_n) - (t_{n+1} - t_n)p}{r} = \frac{\epsilon \xi_n Q_{\xi_n}(n) - \epsilon \xi_n p}{r} = \epsilon' \xi_n \big(Q_{\xi_n}(n) - p\big) = d_n + e_n,$$

where

$$d_n = \mathbf{1}_{\{\rho \le n < \sigma\}} \epsilon' \xi_n \big(Q_0(n) - p\big), \qquad e_n = \mathbf{1}_{\{\rho \le n < \sigma\}} \epsilon' \xi_n \big(Q_{\xi_n}(n) - Q_0(n)\big),$$

and we extend d_n, e_n by these formulas for all n, so that both vanish outside $\{\rho \le n < \sigma\}$. Put $M_n = \sum_{k<n} d_k$, so $M_0 = 0$ and, for $\rho \le n \le \sigma$, telescoping from $y_\rho = 0$ and using that $d_k = e_k = 0$ for $k < \rho$,

$$y_n = \sum_{k<n} (d_k + e_k), \qquad |y_n| \le |M_n| + \sum_{\rho \le k < \sigma} |e_k|, \tag{656}$$

the second line by the triangle inequality and $|e_k| \ge 0$.

(M_n) is a square-integrable martingale. The indicator $\mathbf{1}_{\{\rho \le k < \sigma\}}$ is $\mathcal{G}_k$-measurable, because ρ and σ are stopping times, so $\{\rho \le k\} \in \mathcal{G}_k$ and $\{\sigma \le k\} \in \mathcal{G}_k$, whence $\{\rho \le k < \sigma\} = \{\rho \le k\} \cap \{\sigma \le k\}^c \in \mathcal{G}_k$. The factor $\mathbf{1}_{\{\rho \le k < \sigma\}}(Q_0(k) - p)$ is therefore $\mathcal{G}_k$-measurable, by that observation, the $\mathcal{G}_k$-measurability of $Q_0(k)$ from 4 and the hypothesis that $\mathbf{1}_{\{\rho \le k\}} p$ is $\mathcal{G}_k$-measurable; and it is bounded in norm by $\epsilon'(\Lambda_Q + C_p)$, by 8 and $|p| \le C_p$. Since also $|\xi_k| = 1$ by 1, each d_k is bounded by $\epsilon'(\Lambda_Q + C_p)$, so M_n is $\mathcal{G}_n$-measurable (a finite sum of $\mathcal{G}_k$-measurable terms with $k < n$, using 2 for the $\mathcal{G}_{k+1}$-measurability of ξ_k) and bounded for each n; as μ is a probability measure, $|M_n|^2$ is integrable. The pull-out property of the conditional expectation, applied to the $\mathcal{G}_k$-measurable bounded factor, together with $\mathbb{E}[\xi_k \mid \mathcal{G}_k] = 0$ from 3, gives $\mathbb{E}[d_k \mid \mathcal{G}_k] = 0$ almost everywhere, hence $\mathbb{E}[M_{k+1} \mid \mathcal{G}_k] = M_k$. So (M_n) is a martingale.

The martingale part. Apply `DoobL2Maximal` to (M_n) at the index h:

$$\mathbb{E} \sup_{0 \le n \le h} |M_n| \le 2\big(\mathbb{E}|M_h|^2\big)^{1/2}, \qquad \mathbb{E}|M_h|^2 = \mathbb{E}|M_0|^2 + \sum_{k<h} \mathbb{E}|d_k|^2 = \sum_{k<h} \mathbb{E}|d_k|^2,$$

the last equality because $M_0 = 0$. Since $|\xi_k| = 1$ we have $|d_k|^2 = \mathbf{1}_{\{\rho \le k < \sigma\}} (\epsilon')^2 |Q_0(k) - p|^2$, so, the sum being finite and the terms nonnegative,

$$\sum_{k<h} \mathbb{E}|d_k|^2 = \mathbb{E} \sum_{\substack{k<h \\ \rho \le k < \sigma}} (\epsilon')^2 |Q_0(k) - p|^2 \le \mathbb{E} \sum_{\rho \le k < \sigma} (\epsilon')^2 |Q_0(k) - p|^2.$$

The quantities $\sup_{0 \le n \le h} |M_n|$ are nonnegative and nondecreasing in h, with pointwise limit $\sup_{n \ge 0} |M_n|$, which is finite because $d_k = 0$ for $k \ge \sigma$ and $\sigma(\omega)$ is a natural number. Monotone convergence therefore gives

$$\mathbb{E} \sup_{n \ge 0} |M_n| \le 2 \left(\mathbb{E} \sum_{\rho \le k < \sigma} (\epsilon')^2 |Q_0(k) - p|^2 \right)^{1/2}. \tag{657}$$

The error part. Pathwise, by the Cauchy–Schwarz inequality on the index set $\{\rho \le k < \sigma\}$, which has $\sigma - \rho$ elements, together with $|\xi_k| = 1$ and $|Q_{\xi_k}(k) - Q_0(k)|^2 \le \mathcal{E}_k$ (each of the two terms of $\mathcal{E}_k$ dominating the relevant one),

$$\sum_{\rho \le k < \sigma} |e_k| = \epsilon' \sum_{\rho \le k < \sigma} |Q_{\xi_k}(k) - Q_0(k)| \le \big((\epsilon')^2 (\sigma - \rho)\big)^{1/2} \left(\sum_{\rho \le k < \sigma} \mathcal{E}_k \right)^{1/2}. \tag{658}$$

Taking expectations and applying `IntegralSqrtProductCauchySchwarz` to the two nonnegative integrable functions $(\epsilon')^2(\sigma-\rho)$ and $\sum_{\rho\le k<\sigma}\mathcal{E}_k$ — whose integrability is established in the next paragraph —

$$\mathbb{E}\sum_{\rho\le k<\sigma}|e_k| \le \Big(\mathbb{E}(\epsilon')^2(\sigma-\rho)\Big)^{1/2}\left(\mathbb{E}\sum_{\rho\le k<\sigma}\mathcal{E}_k\right)^{1/2}. \tag{659}$$

The four integrability assertions. All four functions are nonnegative. They are measurable: ρ and σ are $\mathcal{M}$-measurable by 10; for a fixed index n the integrands are built from the $\mathcal{G}_n$-measurable z_n, $z_n^{\pm}$, $Q_0(n)$ and from p (whose strong measurability follows from the $\mathcal{G}_n$-measurability of $\mathbf{1}_{\{\rho\le n\}}p$ and the fact that this family is eventually equal to p pointwise) by finitely many continuous operations; the random index sets $\{\rho\le n<\sigma\}$ and $\{\rho\le n\le\sigma\}$ are determined by events of the filtration, as above; and the passage from the fixed-index families to the random-range sums and the random-range supremum is measurable by 11 and 12, each such sum or supremum being eventually constant in the truncation index because $\sigma\le\tau$ is finite at each ω. For finiteness of their integrals:

- $(\epsilon')^2(\sigma-\rho)\le(\epsilon')^2\tau$ pointwise, because $\sigma-\rho\le\sigma\le\tau$, and τ is integrable by 6; so this function is integrable.
- $\sum_{\rho\le n<\sigma}\mathcal{E}_n\le\Lambda_E\,\tau$ pointwise: the index set is contained in $\{n<\tau\}$, on which $\mathcal{E}_n\le\Lambda_E$ by 9, and it has at most τ elements; the right side is integrable by 6.
- $\sum_{\rho\le n<\sigma}(\epsilon')^2|Q_0(n)-p|^2\le(\epsilon')^2(\Lambda_Q+C_p)^2\,\tau$ pointwise, by 8, $|p|\le C_p$ and the triangle inequality, and the same count of the index set; the right side is integrable by 6.
- For the supremum, (656) gives $\sup_{\rho\le n\le\sigma}|y_n|\le\sup_{n\ge0}|M_n|+\sum_{\rho\le k<\sigma}|e_k|$ pointwise. The first term is nonnegative and measurable and its integral is finite by (657), whose right side is finite by the previous item; the second is integrable by (658) together with the first two items and the pointwise Cauchy–Schwarz bound. A nonnegative measurable function dominated pointwise by a sum of integrable functions is integrable.

Conclusion. Taking the supremum in (656) over $\rho\le n\le\sigma$, then expectations, and inserting (657) and (659), yields (654). □

2.157. `StoppedLatticeSupersolutionBound`.

Lemma 2.157 (Claim T′: the supersolution bound along a stopped lattice portion)**.** *Let $(\Omega,\mathcal{M},\mu)$ be a probability space, let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a filtration of $\mathcal{M}$, let $\xi:\mathbb{N}\times\Omega\to\mathbb{R}$, let $\Xi:\mathbb{N}\times\Omega\to\mathbb{Z}$, let $S:\mathbb{N}\times\Omega\to\mathbb{Z}$, let $\tau,\sigma,\varsigma:\Omega\to\mathbb{N}$ and let $a,b,q\in\mathbb{N}$. Assume:*

(1) *$\xi_n(\omega)=1$ or $\xi_n(\omega)=-1$ for all n,ω;*
(2) *ξ_n is $\mathcal{G}_{n+1}$-measurable for every n;*
(3) *$\mathbb{E}[\xi_n\mid\mathcal{G}_n]=0$ μ-almost everywhere, for every n;*
(4) *$\Xi_n(\omega)=1$ if $\xi_n(\omega)=1$ and $\Xi_n(\omega)=-1$ otherwise;*
(5) *$S_{n+1}=S_n+\Xi_n$ on $\{n<\tau\}$ and $S_{n+1}=S_n$ on $\{n\ge\tau\}$;*
(6) *$a\le S_n(\omega)\le q$ for all n,ω;*
(7) *σ and ς are $(\mathcal{G}_n)$-stopping times and both are $\mathcal{M}$-measurable as maps $\Omega\to\mathbb{N}$;*
(8) *$\sigma\le\varsigma$ and $\varsigma\le\tau$ pointwise;*
(9) *for every $(\mathcal{G}_n)$-stopping time $\zeta:\Omega\to\mathbb{N}$ and every n, $\{\zeta\le n\}\in\mathcal{G}_n$;*
(10) *if $\zeta,\eta:\Omega\to\mathbb{N}$ are measurable and $G_{n,m}:\Omega\to\mathbb{R}$ is measurable for all n,m, then $\omega\mapsto G_{\zeta(\omega),\eta(\omega)}(\omega)$ is measurable;*
(11) *every $g:\mathbb{Z}\to\mathbb{R}$ is bounded on $\{i\in\mathbb{Z}:a\le i\le q\}$;*
(12) *for every $g:\mathbb{Z}\to\mathbb{R}$ and every $k\le n$, the function $\omega\mapsto g(S_k(\omega))$ is $\mathcal{G}_n$-measurable, and for every g and k it is $\mathcal{M}$-measurable;*
(13) *$a<S_n(\omega)<b$ whenever $\sigma(\omega)\le n<\varsigma(\omega)$.*

Let $h,u:\mathbb{Z}\to\mathbb{R}$ and $A\subseteq\Omega$ satisfy $u(i)\ge0$ for every $i\in\mathbb{Z}$, the discrete supersolution inequality

$$\frac{h(i+1)+h(i-1)}{2}-h(i)\le-u(i)\qquad(a<i<b), \tag{660}$$

$A \in \mathcal{M}$, and $A \cap \{\sigma \le n\} \in \mathcal{G}_n$ for every n. Then for every $n \in \mathbb{N}$,

(661)
$$\int \mathbf{1}_A(\omega)\Big(h\big(S_{(\sigma(\omega)\vee n)\wedge\varsigma(\omega)}(\omega)\big) + \sum_{\sigma(\omega)\le m<n\wedge\varsigma(\omega)} u\big(S_m(\omega)\big)\Big)d\mu(\omega) \le \int \mathbf{1}_A(\omega)\, h\big(S_{\sigma(\omega)}(\omega)\big)d\mu(\omega).$$

Proof. By 11 choose C_h, C_h^+, C_h^- and C_u bounding $|h(i)|$, $|h(i+1)|$, $|h(i-1)|$ and $|u(i)|$ respectively for $a \le i \le q$; by 6 every value $S_m(\omega)$ lies in that range, so all four bounds apply at every $S_m(\omega)$. Write $\iota = \mathbf{1}_A$, which is measurable because $A \in \mathcal{M}$, satisfies $0 \le \iota \le 1$ and in particular $|\iota| \le 1$. For $n \in \mathbb{N}$ set

$$Y_n(\omega) = \iota(\omega)\Big(h\big(S_{(\sigma(\omega)\vee n)\wedge\varsigma(\omega)}(\omega)\big) + \sum_{\sigma(\omega)\le m<n\wedge\varsigma(\omega)} u\big(S_m(\omega)\big)\Big),$$

so that (661) asserts $\int Y_n\, d\mu \le \int \iota\, h(S_\sigma)\, d\mu$.

Each Y_n is integrable. The clipped index $(\sigma \vee n) \wedge \varsigma$ is a measurable $\mathbb{N}$-valued function of ω built from the measurable σ, ς of 7, so $\omega \mapsto h(S_{(\sigma\vee n)\wedge\varsigma(\omega)}(\omega))$ is measurable by 10 applied to the doubly-indexed measurable family $(i, m, \omega) \mapsto h(S_{(i\vee n)\wedge m}(\omega))$, each member of which is measurable by 12. The sum is a finite sum, over the σ, ς-determined range, of the measurable functions $\omega \mapsto u(S_m(\omega))$, hence measurable by the same two hypotheses. So Y_n is measurable; it is bounded by $C_h + n\, C_u$, since the sum has at most n terms and each is at most C_u in absolute value; and a bounded measurable function is integrable because μ is a probability measure.

The value at $n = 0$. Here $(\sigma\vee 0)\wedge\varsigma = \sigma\wedge\varsigma = \sigma$ by 8, and the summation range $[\sigma, 0\wedge\varsigma) = [\sigma, 0)$ is empty. Hence

$$\int Y_0\, d\mu = \int \iota\, h(S_\sigma)\, d\mu. \tag{662}$$

The one-step identity. Fix n. We claim that pointwise

$$Y_{n+1} - Y_n = \iota \cdot D_n + \iota \cdot c_n\, \xi_n, \tag{663}$$

where, writing $P_n = \{\sigma \le n < \varsigma\}$,

$$D_n = \mathbf{1}_{P_n}\Big(\frac{h(S_n+1) + h(S_n-1)}{2} - h(S_n) + u(S_n)\Big), \qquad c_n = \mathbf{1}_{P_n}\,\frac{h(S_n+1) - h(S_n-1)}{2}.$$

We check this in three cases, which exhaust Ω.

On $\{\sigma > n\}$ we have $\sigma \ge n+1$, so $(\sigma\vee n)\wedge\varsigma = \sigma\wedge\varsigma = \sigma$ and $(\sigma\vee(n+1))\wedge\varsigma = \sigma\wedge\varsigma = \sigma$, using $\sigma \le \varsigma$; and both summation ranges are empty because $n\wedge\varsigma \le n < \sigma$ and $(n+1)\wedge\varsigma \le n+1 \le \sigma$. Hence $Y_{n+1} = Y_n$, and P_n does not occur, so both terms on the right of (663) vanish.

On $\{\sigma \le n\} \cap \{\varsigma \le n\}$ we have $(\sigma \vee n) \wedge \varsigma = n \wedge \varsigma = \varsigma$ and likewise $(\sigma \vee (n+1)) \wedge \varsigma = \varsigma$, and both summation ranges equal $[\sigma, \varsigma)$. Hence $Y_{n+1} = Y_n$, and again P_n does not occur.

On P_n we have $(\sigma \vee n) \wedge \varsigma = n$ and $(\sigma \vee (n+1)) \wedge \varsigma = n+1$, and the summation range gains exactly the top index n, contributing the single term $u(S_n)$. Also $n < \varsigma \le \tau$ by 8, so 5 gives $S_{n+1} = S_n + \Xi_n$. By 1 and 4 there are two subcases: if $\xi_n = 1$ then $\Xi_n = 1$ and $S_{n+1} = S_n + 1$, and

$$h(S_n+1) = \frac{h(S_n+1) + h(S_n-1)}{2} + \frac{h(S_n+1) - h(S_n-1)}{2}\cdot 1;$$

if $\xi_n = -1$ then $\Xi_n = -1$ and $S_{n+1} = S_n - 1$, and

$$h(S_n-1) = \frac{h(S_n+1) + h(S_n-1)}{2} + \frac{h(S_n+1) - h(S_n-1)}{2}\cdot (-1).$$

In both subcases $h(S_{n+1}) = \frac{h(S_n+1)+h(S_n-1)}{2} + c_n\xi_n$, whence $Y_{n+1} - Y_n = \iota(h(S_{n+1}) - h(S_n) + u(S_n)) = \iota D_n + \iota c_n \xi_n$. This proves (663).

The martingale term integrates to zero. The factor ιc_n is $\mathcal{G}_n$-measurable: $\iota\mathbf{1}_{\{\sigma\le n\}}$ is, because $A \cap \{\sigma \le n\} \in \mathcal{G}_n$ by hypothesis; $\mathbf{1}_{\{n<\varsigma\}}$ is, because $\{n < \varsigma\}$ is the complement of $\{\varsigma \le n\} \in \mathcal{G}_n$ by 9; and $\omega \mapsto h(S_n(\omega) \pm 1)$ is $\mathcal{G}_n$-measurable by 12 applied to $g = h(\cdot \pm 1)$ with $k = n$. It is bounded in absolute value by $(C_h^+ + C_h^-)/2$, using $|\iota| \le 1$ and the bounds fixed at the start. The variable ξ_n is bounded by 1 in absolute value by 1, hence integrable, and $\mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ almost everywhere by 3. The pull-out property of the conditional expectation for a bounded

$\mathcal{G}_n$-measurable factor therefore gives $\mathbb{E}[\iota c_n \xi_n \mid \mathcal{G}_n] = \iota c_n \, \mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ almost everywhere, and since $\iota c_n \xi_n$ is integrable (bounded and measurable, μ being a probability measure),

$$\int \iota \, c_n \, \xi_n \, d\mu = \int \mathbb{E}[\iota c_n \xi_n \mid \mathcal{G}_n] \, d\mu = 0. \tag{664}$$

The drift term is nonpositive. Pointwise, $\iota D_n \le 0$. Indeed off P_n we have $D_n = 0$; and on P_n hypothesis 13 gives $a < S_n < b$, so (660) at $i = S_n$ reads $\frac{h(S_n+1)+h(S_n-1)}{2} - h(S_n) \le -u(S_n)$, i.e. $\frac{h(S_n+1)+h(S_n-1)}{2} - h(S_n) + u(S_n) \le 0$; multiplying by $\iota \ge 0$ preserves the inequality. The function ιD_n is measurable and bounded, hence integrable, so by monotonicity of the integral

$$\int \iota \, D_n \, d\mu \le 0. \tag{665}$$

The decrease. Both ιD_n and $\iota c_n \xi_n$ are integrable, so by additivity of the integral, (663), (665) and (664),

$$\int Y_{n+1} \, d\mu - \int Y_n \, d\mu = \int \iota D_n \, d\mu + \int \iota c_n \xi_n \, d\mu \le 0,$$

that is, $\int Y_{n+1} \, d\mu \le \int Y_n \, d\mu$ for every n.

Conclusion. We prove (661) by induction on n. For $n = 0$ it is the equality (662). If it holds for n, then $\int Y_{n+1} \, d\mu \le \int Y_n \, d\mu \le \int \iota \, h(S_\sigma) \, d\mu$ by the previous paragraph and the inductive hypothesis. This proves (661) for every n. □

2.158. `StoppedWalkLowerExitProbability`.

Lemma 2.158 (The stopped walk reaches the lower time level with probability at least 3/4)**.** *Let F be a finite-dimensional real inner-product space, let $(\Omega, \mathcal{M}, \mu)$ be a probability space, let $z : \mathbb{N} \times \Omega \to \ell^2(F \times \mathbb{R})$, let $S : \mathbb{N} \times \Omega \to \mathbb{Z}$, let $\tau : \Omega \to \mathbb{N}$, let $\epsilon, m, R \in \mathbb{R}$ and let $q, N \in \mathbb{N}$. Write $z_n = (x_n, t_n)$. Assume:*

1. *$\epsilon > 0$, $\epsilon q = 8$ and $R > 0$;*
2. *$t_n(\omega) = \epsilon S_n(\omega)$ for every n and every ω;*
3. *$x_0(\omega) = 0$ for every ω;*
4. *$0 \le S_n(\omega) \le q$ for every n and every ω;*
5. *$S_n(\omega) = S_{\tau(\omega)}(\omega)$ whenever $\tau(\omega) \le n$;*
6. *$\omega \mapsto S_n(\omega)$ and $\omega \mapsto |x_n(\omega)|$ are $\mathcal{M}$-measurable for every n, and so is τ;*
7. *if $\zeta : \Omega \to \mathbb{N}$ is measurable and $G_k : \Omega \to \mathbb{R}$ is measurable for every k, then $\omega \mapsto G_{\zeta(\omega)}(\omega)$ is measurable;*
8. *$\displaystyle\int \frac{S_n(\omega)}{q} \, d\mu(\omega) \le \frac{1}{8}$ for every $n \in \mathbb{N}$;*
9. *for every ω, at least one of $t_{\tau(\omega)}(\omega) = 2^{-N}$, $t_{\tau(\omega)}(\omega) = 8$ and $R \le |x_{\tau(\omega)}(\omega)|$ holds;*
10. *the function $\omega \mapsto \max_{0 \le n \le \tau(\omega)} |x_n(\omega) - x_0(\omega)|$ is μ-integrable and $\displaystyle\int \max_{0 \le n \le \tau(\omega)} |x_n(\omega) - x_0(\omega)| \, d\mu(\omega) \le m$;*
11. *$8m \le R$.*

Then the set $\{\omega : t_{\tau(\omega)}(\omega) = 2^{-N}\}$ belongs to $\mathcal{M}$ and

$$\mu\big(\{\omega : t_{\tau(\omega)}(\omega) = 2^{-N}\}\big) \ge \frac{3}{4}. \tag{666}$$

Proof. Throughout write $B = \{t_\tau = 2^{-N}\}$, $E = \{t_\tau = 8\}$ and $A = \{R \le |x_\tau|\}$, and write $\Sigma(\omega) = \max_{0 \le n \le \tau(\omega)} |x_n(\omega) - x_0(\omega)|$, the maximum being over the nonempty finite index set $\{0, 1, \dots, \tau(\omega)\}$. Note first that $q > 0$: if $q = 0$ then $\epsilon q = 0 \ne 8$, contradicting 1.

The three terminal sets are measurable. By 6 the map τ is measurable and each $\omega \mapsto S_n(\omega)$ is measurable, so 7 applied to $\zeta = \tau$ and $G_n(\omega) = S_n(\omega)$ makes $\omega \mapsto S_{\tau(\omega)}(\omega)$ measurable. By 2 at $n = \tau(\omega)$ we have $t_{\tau(\omega)}(\omega) = \epsilon S_{\tau(\omega)}(\omega)$, so $\omega \mapsto t_{\tau(\omega)}(\omega)$ is measurable as a constant multiple of a measurable function; hence B and E, being preimages of the single points 2^{-N} and 8, lie in $\mathcal{M}$. This proves the first assertion. Likewise 7 applied to $\zeta = \tau$ and $G_n(\omega) = |x_n(\omega)|$, measurable by 6, makes $\omega \mapsto |x_{\tau(\omega)}(\omega)|$ measurable, so A, the preimage of $[R, \infty)$, lies in $\mathcal{M}$.

Markov's inequality. We use the following form. Let $A' \in \mathcal{M}$, let $f : \Omega \to \mathbb{R}$ be μ-integrable with $f \geq 0$ everywhere, and let $c > 0$ satisfy $f \geq c$ on A'. Then $\mathbf{1}_{A'} c \leq f$ pointwise — on A' this is the hypothesis $f \geq c$ and off A' it is $0 \leq f$ — and $\mathbf{1}_{A'} c$ is integrable because $A' \in \mathcal{M}$ and μ is finite, so monotonicity of the integral gives $\mu(A')\, c = \int \mathbf{1}_{A'} c \, d\mu \leq \int f \, d\mu$.

The mean terminal lattice height. We claim

$$\int \frac{S_{\tau(\omega)}(\omega)}{q} \, d\mu(\omega) \leq \frac{1}{8}. \tag{667}$$

Consider the functions $f_n(\omega) = S_n(\omega)/q$. Each is measurable by 6, and 4 together with $q > 0$ gives $0 \leq f_n \leq 1$, so $|f_n| \leq 1$ with the constant 1 integrable, μ being a probability measure. For each fixed ω we have $f_n(\omega) = S_{\tau(\omega)}(\omega)/q$ for every $n \geq \tau(\omega)$ by 5, so $f_n(\omega) \to S_{\tau(\omega)}(\omega)/q$ as $n \to \infty$, the sequence being eventually constant. By the dominated convergence theorem $\int f_n \, d\mu \to \int S_\tau/q \, d\mu$, and 8 says $\int f_n \, d\mu \leq 1/8$ for every n; passing to the limit gives (667).

The upper time level has probability at most $1/8$. Apply Markov's inequality with $A' = E$, $f = S_\tau/q$ and $c = 1$. The function f is measurable (previous paragraph), nonnegative and bounded by 1 by 4 and $q > 0$, hence integrable. On E we have $t_\tau = 8 = \epsilon q$ by 1, while $t_\tau = \epsilon S_\tau$ by 2; since $\epsilon \neq 0$ we may cancel it and obtain $S_\tau = q$ there, so $f = 1 = c$ on E. Markov's inequality and (667) therefore give

$$\mu(E) = \mu(E) \cdot 1 \leq \int \frac{S_\tau}{q} \, d\mu \leq \frac{1}{8}. \tag{668}$$

The horizontal cutoff has probability at most $1/8$. For every ω the index $\tau(\omega)$ lies in $\{0, \ldots, \tau(\omega)\}$, so $|x_{\tau(\omega)}(\omega) - x_0(\omega)| \leq \Sigma(\omega)$; by 3 the left-hand side is $|x_{\tau(\omega)}(\omega)|$, whence

$$|x_{\tau(\omega)}(\omega)| \leq \Sigma(\omega) \qquad (\omega \in \Omega), \tag{669}$$

and in particular $\Sigma \geq 0$ everywhere. On A we have $R \leq |x_\tau| \leq \Sigma$ by (669). So Markov's inequality applies with $A' = A$, $f = \Sigma$ and $c = R > 0$ — Σ being integrable by 10 — and together with the bound in 10 it gives $\mu(A)\, R \leq \int \Sigma \, d\mu \leq m$. By 11 we have $m \leq R/8$, so $\mu(A)\, R \leq R/8$, and dividing by $R > 0$ yields

$$\mu(A) \leq \frac{1}{8}. \tag{670}$$

The union bound. If $\omega \notin B$ then the first alternative of 9 fails at ω, so one of the other two holds, i.e. $\omega \in E \cup A$. Hence $\Omega \setminus B \subseteq E \cup A$, and monotonicity and subadditivity of μ give $\mu(\Omega \setminus B) \leq \mu(E) + \mu(A) \leq 1/8 + 1/8 = 1/4$ by (668) and (670). Since $B \in \mathcal{M}$ and $\mu(\Omega) = 1$, additivity gives $\mu(B) + \mu(\Omega \setminus B) = 1$, so $\mu(B) = 1 - \mu(\Omega \setminus B) \geq 1 - 1/4 = 3/4$, which is (666). □

2.159. `StoppedWalkTravel`.

Lemma 2.159 (The travel inequality and the lower hitting probability)**.** *Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$, let $\kappa > 0$ and $H \in \mathbb{R}$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of* ***`IsFitClass`*** *be centered at $z_* = (0,1)$ with constant C_0 in the sense of* ***`IsCenteredAt`*** *and satisfy the horizontal Jensen bound with constant κ at every time in the sense of* ***`HasJensenBound`****. Let W be the contact envelope of* ***`ContactEnv`*** *formed from v and κ, and assume the six standing properties of the extremal contact pair: W is continuous on $\{t > 0\}$ with subquadratic spatial growth on compact subintervals of $(0, \infty)$ in the sense of* ***`SubquadraticGrowth`****; W is a viscosity subsolution of the limiting equation there in the sense of* ***`IsLimitSubOn`****; $v - \kappa^2 t/2 \leq W \leq v$ for $t > 0$ and $W(z_*) = v(z_*) = 0$; $x \mapsto W(x,t) - |x|^2/(2t)$ is concave for every $t > 0$; the barycentric gap of* ***`BaryEnv`*** *is $\kappa^2 t/2$ at every contact; and the envelope Hamiltonian of* ***`EnvHamiltonian`*** *satisfies $H_v(z_*) = H$ with $\mathcal{S}^+_\ell W(z_*) \to H$ as $\ell \downarrow 0$, the forward envelope slope being that of* ***`EnvSlopeP`****. Let $C_* > 0$. Write $\delta_j = 2^{-j}$ and $L(K) = 1 + \log K$.*

Then there are an integer $T \geq 8$ and a real $R \geq 2$ with the following property. Let $N \geq 1$, let $\epsilon > 0$ and let $(\Omega, \mathcal{M}, \mu)$, $(\mathcal{G}_n)$, ξ, z, z^+, z^-, Q_0, τ be a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$

in the sense of `IsContactWalk`. *Write* $z_n = (x_n, t_n)$,

$$Q_+(n) = \frac{x_n^+ - x_n}{\epsilon}, \qquad Q_-(n) = \frac{x_n - x_n^-}{\epsilon}, \qquad \mathcal{E}_n = |Q_+(n) - Q_0(n)|^2 + |Q_-(n) - Q_0(n)|^2,$$

and $B = R + T + 2$, *and assume the error budget*

$$\mathbb{E}\sum_{n<\tau} \mathcal{E}_n \le C_* L(B)^4. \tag{671}$$

Then:

(i) *(travel inequality) for all stopping times* $\rho \le \sigma$ *for* $(\mathcal{G}_n)$ *with* $\sigma \le \tau$ *pointwise, every* $r > 0$ *and every* $p : \Omega \to F$ *which is bounded, say* $|p| \le C_p$, *and satisfies that* $\mathbf{1}_{\{\rho \le n\}} p$ *is* $\mathcal{G}_n$*-measurable for every* n, *the four functions*

$$\sup_{\rho \le n \le \sigma} \frac{|x_n - x_\rho - (t_n - t_\rho)p|}{r}, \qquad \sum_{\rho \le n < \sigma} \Big(\frac{\epsilon}{r}\Big)^2 |Q_0(n) - p|^2, \qquad \Big(\frac{\epsilon}{r}\Big)^2 (\sigma - \rho), \qquad \sum_{\rho \le n < \sigma} \mathcal{E}_n$$

are all μ*-integrable, and*

$$\begin{aligned} \mathbb{E} \sup_{\rho \le n \le \sigma} \frac{|x_n - x_\rho - (t_n - t_\rho)p|}{r} &\le 2 \left(\mathbb{E} \sum_{\rho \le n < \sigma} \Big(\frac{\epsilon}{r}\Big)^2 |Q_0(n) - p|^2 \right)^{1/2} \\ &\quad + \left(\mathbb{E}\Big(\frac{\epsilon}{r}\Big)^2 (\sigma - \rho) \right)^{1/2} \left(\mathbb{E} \sum_{\rho \le n < \sigma} \mathcal{E}_n \right)^{1/2}; \end{aligned} \tag{672}$$

(ii) *(lower hitting probability) the event* $\{t_\tau = \delta_N\}$ *belongs to* $\mathcal{M}$ *and the walk reaches the lower time level with probability at least* $3/4$*:*

$$\{t_\tau = \delta_N\} \in \mathcal{M}, \qquad \mathbb{P}\big(t_\tau = \delta_N\big) \ge \frac{3}{4}. \tag{673}$$

Proof. Throughout, W is the contact envelope of `ContactEnv` and $\mathcal{P}_\epsilon W$ the forward envelope of `QuadEnvP`. By `IsFitClass` the function v is continuous on all of space-time, and by `UniformSubquadratic` with the centering point $(0, 1)$ it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`.

Step 0: the constants, and the choice of T *and* R. Let $C_2 > 0$ be the constant of `DPLogVelocity` for this C_0 and κ. Put

$$C_T = 4C_2^2\Big(T + \frac{T}{2}\big(2 + T(\log T)^2\big)\Big), \qquad C_T' = 2\sqrt{C_T} + \sqrt{T\,C_*},$$

for an integer $T \ge 2$; both depend only on T, C_* and the fixed data $C_0, \kappa, \dim F$, and in particular not on R or N. Take $T = 8$. Since $L(R+10)^2/R \to 0$ as $R \to \infty$, we may fix $R \ge 2$ with

$$\frac{C_8'\, L(R+10)^2}{R} \le \frac{1}{8}. \tag{674}$$

Fix these $T = 8$ and R for the rest of the proof, and let $N \ge 1$, ϵ and the walk be as in the statement, so $B = R + T + 2 = R + 10 \ge 6$ and $L(B) \ge 1$. By `IsContactWalk`, clause 2, $\epsilon = 2^{-k}$ with $0 < \epsilon \le \delta_N/4 \le 1/8$, and since $N \ge 1$ forces $k \ge N + 2 \ge 3$ the number ϵ divides 1 and divides the integer T. Put

$$S_n = \frac{t_n}{\epsilon}, \qquad q = \frac{T}{\epsilon} \in \mathbb{N}.$$

Step 1: the time walk. By `IsStepPair` the two steps at a live state $z_n = (x_n, t_n)$ have time coordinates $t_n + \epsilon$ and $t_n - \epsilon$, so by `IsContactWalk`, clause 7, and $\xi_n \in \{1, -1\}$,

$$t_{n+1} = t_n + \epsilon\xi_n \ \text{ on } \{n < \tau\}, \qquad t_{n+1} = t_n \ \text{ on } \{n \ge \tau\}. \tag{675}$$

Since $t_0 = 1$ and ϵ divides 1, induction gives $S_n \in \mathbb{Z}$ for every n, and $S_{n+1} = S_n + \xi_n$ on $\{n < \tau\}$. By clauses 10 and 11, $\delta_N/\epsilon < S_n < q$ for $n < \tau$ and $S_\tau \in [\delta_N/\epsilon, q]$; hence

$$0 < S_{n\wedge\tau} \le q \qquad (n \ge 0). \tag{676}$$

Claim T. Let $h : \mathbb{Z} \to \mathbb{R}$ vanish outside $\{0, \dots, q\}$, be nonnegative there, and let $u : \mathbb{Z} \to [0, \infty)$ satisfy

$$\tfrac{1}{2}\big(h(j+1) + h(j-1)\big) - h(j) = -u(j) \qquad (0 < j < q). \tag{677}$$

Then

$$\mathbb{E} \sum_{n<\tau} u(S_n) \le h(S_0) = h(1/\epsilon).$$

Moreover, if $\ell : \mathbb{Z} \to \mathbb{R}$ is bounded and satisfies $\frac{1}{2}(\ell(j+1) + \ell(j-1)) = \ell(j)$ for $0 < j < q$, then $\mathbb{E}\,\ell(S_\tau) = \ell(1/\epsilon)$.

Proof of Claim T. Put $M_n = h(S_{n\wedge\tau}) + \sum_{m<n\wedge\tau} u(S_m)$. Each M_n is $\mathcal{G}_n$-measurable, because τ is a stopping time and each z_m is $\mathcal{G}_m$-measurable by `IsContactWalk`, clauses 4 and 9; and each M_n is bounded, by $\sup h$ and $n \sup u$ on the finite range $\{0, \dots, q\}$, hence integrable. On $\{n \ge \tau\}$ we have $M_{n+1} = M_n$. On $\{n < \tau\}$ we have $S_{n+1} = S_n \pm 1$ according to $\xi_n = \pm 1$, and $0 < S_n < q$, so writing

$$h(S_{n+1}) = \frac{1+\xi_n}{2} h(S_n{+}1) + \frac{1-\xi_n}{2} h(S_n{-}1) = \frac{h(S_n+1) + h(S_n-1)}{2} + \frac{\xi_n}{2}\big(h(S_n{+}1) - h(S_n{-}1)\big),$$

where the coefficient of ξ_n is bounded and $\mathcal{G}_n$-measurable, the pull-out property of the conditional expectation and $\mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ (`IsContactWalk`, clause 3) give

$$\mathbb{E}[h(S_{n+1}) \mid \mathcal{G}_n] = \frac{h(S_n+1) + h(S_n-1)}{2} = h(S_n) - u(S_n)$$

almost everywhere on $\{n < \tau\}$, by (677). Adding $u(S_n)$ shows $\mathbb{E}[M_{n+1} \mid \mathcal{G}_n] = M_n$ there. Hence (M_n) is a martingale, so $\mathbb{E}M_n = M_0 = h(S_0)$ for every n, and since $h \ge 0$ on the range (676), $\mathbb{E}\sum_{m<n\wedge\tau} u(S_m) \le h(S_0)$. The sums are nonnegative and nondecreasing in n with pointwise limit $\sum_{m<\tau} u(S_m)$, because τ is finite, so monotone convergence gives the first assertion. For the second, the same computation with $u = 0$ shows that $\ell(S_{n\wedge\tau})$ is a martingale, so $\mathbb{E}\,\ell(S_{n\wedge\tau}) = \ell(S_0)$ for every n; as τ is finite we have $\ell(S_{n\wedge\tau}) \to \ell(S_\tau)$ pointwise, and ℓ is bounded while μ is a probability measure, so dominated convergence gives $\mathbb{E}\,\ell(S_\tau) = \ell(S_0)$. This proves Claim T.

Step 2: three consequences of Claim T.

(a) The time budget. Take $h_0(j) = j(q-j)$ on $\{0, \dots, q\}$ and 0 outside, and $u \equiv 1$ on $\{1, \dots, q-1\}$, $u = 0$ elsewhere. Then $h_0 \ge 0$ on $\{0, \dots, q\}$, and for $0 < j < q$

$$\tfrac{1}{2}\big(h_0(j+1) + h_0(j-1)\big) - h_0(j) = \tfrac{1}{2}\big(2j(q-j) - 2\big) - j(q-j) = -1,$$

since $(j+1)(q-j-1) + (j-1)(q-j+1) = 2j(q-j) - 2$. Every $n < \tau$ has $0 < S_n < q$ by Step 1, so $u(S_n) = 1$ there and Claim T gives $\mathbb{E}\tau \le h_0(1/\epsilon) = (1/\epsilon)(q - 1/\epsilon) \le T/\epsilon^2$, that is

$$\mathbb{E}\big(\epsilon^2 \tau\big) \le T. \tag{678}$$

In particular τ is integrable, as also recorded in `IsContactWalk`, clause 9.

(b) Visit counts. Fix $0 < k < q$ and put $G_k(j) = \dfrac{2\min(j,k)\,(q - \max(j,k))}{q}$ on $\{0, \dots, q\}$ and 0 outside. Then $G_k \ge 0$, $G_k(0) = G_k(q) = 0$, G_k is affine in j on $\{0, \dots, k\}$ and on $\{k, \dots, q\}$, so $\frac{1}{2}(G_k(j+1) + G_k(j-1)) - G_k(j) = 0$ for $0 < j < q$ with $j \ne k$, while at $j = k$

$$\tfrac{1}{2}\big(G_k(k-1) + G_k(k+1)\big) - G_k(k) = \frac{(k-1)(q-k) + k(q-k-1)}{q} - \frac{2k(q-k)}{q} = -1.$$

So (677) holds with $u = \mathbf{1}_{\{k\}}$, and Claim T gives

$$\mathbb{E}\,\#\{n < \tau : S_n = k\} \le G_k(1/\epsilon) \le \frac{q}{2}, \tag{679}$$

the last step because $\min(j,k)(q - \max(j,k)) \le k(q-k) \le q^2/4$ for every j.

(c) The upper time level. Take $\ell(j) = j/q$ for $0 \le j \le q$, $\ell(j) = 0$ for $j < 0$ and $\ell(j) = 1$ for $j > q$; it is bounded and affine on $\{0, \dots, q\}$, hence satisfies the mean-value identity for

$0 < j < q$. Claim T gives $\mathbb{E}\,\ell(S_\tau) = \ell(1/\epsilon) = 1/T$. By (676) we have $0 \le \ell(S_\tau) \le 1$ with $\ell(S_\tau) = 1$ exactly when $S_\tau = q$, i.e. when $t_\tau = T$; hence

$$\mathbb{P}(t_\tau = T) \le \mathbb{E}\,\ell(S_\tau) = \frac{1}{T}. \tag{680}$$

Step 3: the velocity budget. Every z_n with $n \le \tau$ satisfies $|x_n| \le R + 1 \le B$ and $\delta_N \le t_n \le T \le B$ by `IsContactWalk`, clauses 10–12, and the walk is frozen afterwards, so this holds for every n. Since $B \ge 2$, $0 < \epsilon \le 1$ and $\epsilon < \delta_N \le t_n$, `DPLogVelocity` applies at each state with $K = B$ and gives, $Q_0(n)$ being the gradient of $\mathcal{P}_\epsilon W(\cdot, t_n)$ at x_n by `IsContactWalk`, clause 8,

$$|Q_0(n)| \le C_2\big(L(B) + |\log t_n|\big) \qquad (n \ge 0). \tag{681}$$

In particular $|Q_0(n)| \le C_2(L(B)+N\log 2+\log T)$ is bounded, uniformly in n, since $\delta_N \le t_n \le T$.

By $|a+b|^2 \le 2|a|^2 + 2|b|^2$ and (681), $\epsilon^2|Q_0(n)|^2 \le 2C_2^2\epsilon^2 L(B)^2 + 2C_2^2\epsilon^2|\log t_n|^2$. Summing over $n < \tau$ and taking expectations, the first part contributes $2C_2^2 L(B)^2\mathbb{E}(\epsilon^2\tau) \le 2C_2^2 L(B)^2 T$ by (678). For the second, $t_n = S_n\epsilon$ with $0 < S_n < q$ for $n < \tau$, so, grouping the summands by the level $k = S_n$ and using (679),

$$\mathbb{E}\sum_{n<\tau}\epsilon^2|\log t_n|^2 = \epsilon^2\sum_{k=1}^{q-1}|\log(k\epsilon)|^2\,\mathbb{E}\,\#\{n<\tau : S_n = k\} \le \frac{\epsilon^2 q}{2}\sum_{k=1}^{q-1}|\log(k\epsilon)|^2 = \frac{T\epsilon}{2}\sum_{k=1}^{q-1}|\log(k\epsilon)|^2.$$

Split the last sum at $k\epsilon \le 1$ and $k\epsilon > 1$. On $(0,1]$ the function $s \mapsto |\log s|^2$ is nonnegative and decreasing, so its right-endpoint grid sum obeys $\epsilon\sum_{k:k\epsilon\le 1}|\log(k\epsilon)|^2 \le \int_0^1 |\log s|^2\,ds = 2$, the value of the integral following from the substitution $s = e^{-v}$, which turns it into $\int_0^\infty v^2 e^{-v}\,dv = 2$. For $1 < k\epsilon \le T$ we have $|\log(k\epsilon)|^2 \le (\log T)^2$ and there are at most $q = T/\epsilon$ such k, so $\epsilon\sum_{k:k\epsilon>1}|\log(k\epsilon)|^2 \le T(\log T)^2$. Hence

$$\mathbb{E}\sum_{n<\tau}\epsilon^2|\log t_n|^2 \le \frac{T}{2}\big(2 + T(\log T)^2\big),$$

and altogether, using $L(B) \ge 1$,

$$\mathbb{E}\sum_{n<\tau}\epsilon^2|Q_0(n)|^2 \le 2C_2^2\Big(TL(B)^2 + \frac{T}{2}\big(2 + T(\log T)^2\big)\Big) \le \frac{C_T}{2}L(B)^2, \tag{682}$$

with C_T as in Step 0.

Step 4: the travel inequality, i.e. (i). Let $\rho \le \sigma \le \tau$, $r > 0$ and p be as in (i), and put $\epsilon' = \epsilon/r$. For $n \ge 0$ set

$$y_n = \frac{x_n - x_\rho - (t_n - t_\rho)p}{r},$$

so $y_\rho = 0$ and the left side of (672) is $\mathbb{E}\sup_{\rho\le n\le\sigma}|y_n|$. For $n < \tau$ we have, by `IsContactWalk`, clause 7, and the definitions of $Q_\pm$,

$$x_{n+1} - x_n = \epsilon\xi_n Q_{\xi_n}(n), \qquad Q_1 = Q_+, \quad Q_{-1} = Q_-,$$

because $x_{n+1} = x_n^+$ when $\xi_n = 1$ and $x_{n+1} = x_n^-$ when $\xi_n = -1$, while $x_n^+ - x_n = \epsilon Q_+(n)$ and $x_n - x_n^- = \epsilon Q_-(n)$. With (675) this gives, for $\rho \le n < \sigma$,

$$y_{n+1} - y_n = \epsilon'\xi_n\big(Q_{\xi_n}(n) - p\big) = d_n + e_n,$$

where

$$d_n = \mathbf{1}_{\{\rho\le n<\sigma\}}\epsilon'\xi_n\big(Q_0(n) - p\big), \qquad e_n = \mathbf{1}_{\{\rho\le n<\sigma\}}\epsilon'\xi_n\big(Q_{\xi_n}(n) - Q_0(n)\big),$$

and we extend d_n, e_n by these formulas for all n, so that both vanish outside $\{\rho \le n < \sigma\}$. Put $M_n = \sum_{k<n} d_k$, so $M_0 = 0$ and, for $\rho \le n \le \sigma$,

$$y_n = \sum_{k<n}(d_k + e_k), \qquad |y_n| \le |M_n| + \sum_{\rho\le k<\sigma}|e_k|. \tag{683}$$

(M_n) is a square-integrable martingale. The indicator $\mathbf{1}_{\{\rho\le k<\sigma\}}$ is $\mathcal{G}_k$-measurable, because ρ and σ are stopping times, so $\{\rho \le k\}$ and $\{\sigma \le k\}$ lie in $\mathcal{G}_k$. The factor $\mathbf{1}_{\{\rho\le k<\sigma\}}(Q_0(k) - p)$ is therefore $\mathcal{G}_k$-measurable, by that observation, the $\mathcal{G}_k$-measurability of $Q_0(k)$ (`IsContactWalk`, clause 4) and the hypothesis that $\mathbf{1}_{\{\rho\le k\}}p$ is $\mathcal{G}_k$-measurable; and it is bounded, by Step 3 and

$|p| \le C_p$. Hence M_n is $\mathcal{G}_n$-measurable and bounded for each n, so $|M_n|^2$ is integrable, and the pull-out property of the conditional expectation together with $\mathbb{E}[\xi_k \mid \mathcal{G}_k] = 0$ gives $\mathbb{E}[d_k \mid \mathcal{G}_k] = 0$ almost everywhere, hence $\mathbb{E}[M_{k+1} \mid \mathcal{G}_k] = M_k$. So (M_n) is a martingale.

Apply `DoobL2Maximal` to (M_n) at the index h: it gives

$$\mathbb{E} \sup_{0 \le n \le h} |M_n| \le 2\big(\mathbb{E}|M_h|^2\big)^{1/2}, \qquad \mathbb{E}|M_h|^2 = \mathbb{E}|M_0|^2 + \sum_{k<h} \mathbb{E}|d_k|^2 = \sum_{k<h} \mathbb{E}|d_k|^2.$$

Since $|\xi_k| = 1$ we have $|d_k|^2 = \mathbf{1}_{\{\rho \le k < \sigma\}}(\epsilon')^2 |Q_0(k) - p|^2$, so, the sum being finite and the terms nonnegative,

$$\sum_{k<h} \mathbb{E}|d_k|^2 = \mathbb{E} \sum_{\substack{k<h \\ \rho \le k < \sigma}} (\epsilon')^2 |Q_0(k) - p|^2 \le \mathbb{E} \sum_{\rho \le k < \sigma} (\epsilon')^2 |Q_0(k) - p|^2.$$

The quantities $\sup_{0 \le n \le h} |M_n|$ are nonnegative and nondecreasing in h, with pointwise limit $\sup_{n \ge 0} |M_n|$, which is finite because $d_k = 0$ for $k \ge \sigma$ and σ is finite. Monotone convergence therefore gives

$$\mathbb{E} \sup_{n \ge 0} |M_n| \le 2 \left(\mathbb{E} \sum_{\rho \le k < \sigma} (\epsilon')^2 |Q_0(k) - p|^2 \right)^{1/2}. \tag{684}$$

The error part. Pathwise, by the Cauchy–Schwarz inequality on the index set $\{\rho \le k < \sigma\}$, which has $\sigma - \rho$ elements, and $|Q_{\xi_k}(k) - Q_0(k)|^2 \le \mathcal{E}_k$,

$$\sum_{\rho \le k < \sigma} |e_k| = \epsilon' \sum_{\rho \le k < \sigma} |Q_{\xi_k}(k) - Q_0(k)| \le \big((\epsilon')^2(\sigma - \rho)\big)^{1/2} \left(\sum_{\rho \le k < \sigma} \mathcal{E}_k \right)^{1/2}. \tag{685}$$

Taking expectations and applying the Cauchy–Schwarz inequality in expectation to the two factors on the right,

$$\mathbb{E} \sum_{\rho \le k < \sigma} |e_k| \le \Big(\mathbb{E}(\epsilon')^2(\sigma - \rho)\Big)^{1/2} \left(\mathbb{E} \sum_{\rho \le k < \sigma} \mathcal{E}_k \right)^{1/2}. \tag{686}$$

The four integrability assertions. All four functions are nonnegative and measurable: each is built from the $\mathcal{G}_n$-measurable z_n, $z_n^{\pm}$, $Q_0(n)$ and p and from the stopping times ρ, σ through finitely many operations on each ω, the index sets $\{\rho \le n < \sigma\}$ and $\{\rho \le n \le \sigma\}$ being determined by events of the filtration. For finiteness of their integrals:

- $(\epsilon/r)^2(\sigma - \rho) \le (\epsilon/r)^2 \tau$ pointwise, and τ is integrable by (678) (equivalently by `IsContactWalk`, clause 9), so this function is integrable.
- $\sum_{\rho \le n < \sigma} \mathcal{E}_n \le \sum_{n < \tau} \mathcal{E}_n$ pointwise, because the index set is a subset of $\{n < \tau\}$ and the summands are nonnegative, and the latter is integrable with integral at most $C_* L(B)^4$ by (671).
- $\sum_{\rho \le n < \sigma} (\epsilon/r)^2 |Q_0(n) - p|^2 \le (\epsilon/r)^2 \big(\Lambda_Q + C_p\big)^2 (\sigma - \rho)$ pointwise, where $\Lambda_Q = C_2(L(B) + N \log 2 + \log T)$ is the uniform bound on $|Q_0(n)|$ from Step 3 and $|p| \le C_p$; the right side is an integrable function by the first item.
- For the supremum, (683) gives $\sup_{\rho \le n \le \sigma} |y_n| \le \sup_{n \ge 0} |M_n| + \sum_{\rho \le k < \sigma} |e_k|$ pointwise. The first term is integrable because (684) bounds its integral by a finite number — the right side there is finite by the previous item — and it is nonnegative and measurable; the second is integrable by (685) and the two previous items together with the pointwise Cauchy–Schwarz bound. A nonnegative measurable function dominated by a sum of integrable functions is integrable.

Taking the supremum in (683) over $\rho \le n \le \sigma$, then expectations, and inserting (684) and (686), yields (672). This proves (i).

Step 5: the lower hitting probability, i.e. (ii). Apply (i) with $\rho = 0$, $\sigma = \tau$, $p = 0$ (so $C_p = 0$, and $\mathbf{1}_{\{0\le n\}}p = 0$ is $\mathcal{G}_n$-measurable) and $r = 1$. Since $x_0 = 0$ and $t_0 - t_0 = 0$, the left side is $\mathbb{E}\sup_{0\le n\le\tau}|x_n|$, and by (682), (678) and (671),

(687)
$$\mathbb{E}\sup_{0\le n\le\tau}|x_n| \le 2\Big(\frac{C_T}{2}L(B)^2\Big)^{1/2} + T^{1/2}\big(C_*L(B)^4\big)^{1/2} \le 2\sqrt{C_T}\,L(B) + \sqrt{TC_*}\,L(B)^2 \le C_T'L(B)^2,$$

using $L(B)\ge 1$ and the definition of C_T' in Step 0.

By `IsContactWalk`, clause 11, at least one of $t_\tau = \delta_N$, $t_\tau = T$ and $|x_\tau|\ge R$ holds, so

$$\mathbb{P}(t_\tau \ne \delta_N) \le \mathbb{P}(t_\tau = T) + \mathbb{P}\big(|x_\tau|\ge R\big).$$

The first term is at most $1/T = 1/8$ by (680). For the second, $|x_\tau|\le\sup_{0\le n\le\tau}|x_n|$, and for a nonnegative random variable S and $R>0$ the pointwise inequality $R\mathbf{1}_{\{S\ge R\}}\le S$ integrates to $\mathbb{P}(S\ge R)\le \mathbb{E}S/R$; with $S=\sup_{0\le n\le\tau}|x_n|$, (687) and $T=8$, $B=R+10$, this gives

$$\mathbb{P}\big(|x_\tau|\ge R\big) \le \frac{C_8'L(R+10)^2}{R} \le \frac{1}{8}$$

by the choice (674) of R. Hence $\mathbb{P}(t_\tau\ne\delta_N)\le 1/4$ and, μ being a probability measure, $\mathbb{P}(t_\tau = \delta_N)\ge 3/4$.

The event is measurable. Indeed

$$\{t_\tau = \delta_N\} = \bigcup_{n\ge 0}\big(\{\tau = n\}\cap\{t_n = \delta_N\}\big),$$

because τ takes values in $\mathbb{N}$, and each term lies in $\mathcal{M}$: $\{\tau=n\}\in\mathcal{G}_n\subseteq\mathcal{M}$ because τ is a stopping time for $(\mathcal{G}_n)$ by clause 9 of `IsContactWalk`, and $\{t_n=\delta_N\}\in\mathcal{G}_n\subseteq\mathcal{M}$ because z_n is $\mathcal{G}_n$-measurable by clause 4 of `IsContactWalk` and $z\mapsto t$ is continuous, hence Borel, on $\ell^2(F\times\mathbb{R})$. A countable union of measurable sets is measurable. The same decomposition is what makes the probabilities written above meaningful, since the events $\{t_\tau=T\}$ and $\{|x_\tau|\ge R\}$ and the random variable $\sup_{0\le n\le\tau}|x_n| = \sup_{n\ge0}|x_{n\wedge\tau}|$ are measurable by the same argument. This is (673) and completes the proof. □

2.160. `FrameEnvelopeCovariance`.

Lemma 2.160 (Covariance of the envelope slopes in a fitted frame)**.** *Let F be a finite-dimensional real inner-product space, let $1\le C_0<\infty$ and $0\le\kappa<\infty$, and let $v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let $r>0$, let $x_0\in F$, and let (p_0,a_0) be a fit for v at (x_0,r) of radius r with constant C_0, in the sense of `IsFit`. Define the contact frame by*

(688)
$$\begin{aligned} s &= \frac{t}{r}, \qquad y = \frac{x - x_0 - p_0(t-r)}{r},\\ x &= x_0 + r\big(y + p_0(s-1)\big), \qquad t = rs,\\ \bar v(y,s) &= \frac{v(x,t)-v(x_0,r)}{r} - p_0\cdot y - \left(a_0 + \frac{|p_0|^2}{2}\right)(s-1), \end{aligned}$$

so that $\bar v$ is the change of frame of `FrameChange` based at (x_0,r) with scale r and parameters (p_0,a_0), composed with the translation of the time coordinate by one unit. Then $\bar v\in\mathcal{A}_{m,C_0}$, it is centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`, it satisfies the horizontal Jensen bound with the same constant κ, and for every $\ell>0$, at corresponding points,

(689)
$$\mathcal{S}^{\pm}_{\ell}\bar v(y,s) = \mathcal{S}^{\pm}_{r\ell}v(x,t) - a_0,$$

with the slopes of `EnvSlopeP` and `EnvSlopeM`.

Proof. Write $T(y,s) = \big(x_0 + r(y + p_0(s-1)),\, rs\big)$ for the frame map, which is an affine bijection of space-time because $r>0$, with inverse $T^{-1}(x,t) = \big((x - x_0 - p_0(t-r))/r,\ t/r\big)$; the formulas (688) say exactly that $\bar v = \bar v(y,s)$ is given in terms of v at the corresponding point $(x,t) = T(y,s)$. Note $T(0,1) = (x_0,r)$.

Step 1: $\bar v$ is the change of frame followed by a time translation. By `FrameChange` with base point $z_0 = (x_0, r)$, scale r and parameters (p_0, a_0), the function

$$\widehat v(y,\sigma) = \frac{v\big(x_0 + r(y + p_0\sigma),\ r + r\sigma\big) - v(x_0,r)}{r} - p_0\cdot y - \Big(a_0 + \frac{|p_0|^2}{2}\Big)\sigma$$

satisfies $\bar v(y,s) = \widehat v(y, s-1)$, since substituting $\sigma = s-1$ gives $r + r\sigma = rs$ and reproduces (688).

Step 2: class, centering and Jensen bound. By `FrameInvariance`, applied to $v \in \mathcal{A}_{m,C_0}$ with $z_0 = (x_0, r)$, $R = r > 0$ and (p_0, a_0), we have $\widehat v \in \mathcal{A}_{m,C_0}$. Its second assertion, applied at the new center $\zeta = (0,0)$ — whose corresponding old point is $T(0,1) = (x_0, r)$ — with $\rho = 1$ and the fit (p_0, a_0) for v at (x_0, r) of radius $r\cdot 1 = r$, shows that $(p_0 - p_0, a_0 - a_0) = (0,0)$ is a fit for $\widehat v$ at $(0,0)$ of radius 1 with constant C_0; and $\widehat v(0,0) = 0$ directly from the displayed formula. So by `IsCenteredAt` the function $\widehat v$ is centered at the origin with constant C_0.

It remains to transport this across the time translation $\Theta(y,s) = (y, s-1)$, for which $\bar v = \widehat v \circ \Theta$ by Step 1.

- *Equation.* Θ is a translation, so for a C^2 function ϕ on space-time, $\phi\circ\Theta^{-1}$ is C^2 and, at corresponding points, has the same gradient and the same second derivative as ϕ; in particular the operator of `LimitOp` satisfies $F[\phi\circ\Theta^{-1}](\Theta\zeta) = F[\phi](\zeta)$. Also $\bar v - \phi = (\widehat v - \phi\circ\Theta^{-1})\circ\Theta$ and Θ is a homeomorphism, so local maxima and minima correspond. Assembling: let ζ_0 be any space-time point and let ϕ be C^2 near ζ_0. If $\bar v - \phi$ has a local maximum at ζ_0, then $\widehat v - \phi\circ\Theta^{-1}$ has a local maximum at $\Theta\zeta_0$, so the subsolution property of $\widehat v$ in `IsLimitSubOn` applied to the test function $\phi\circ\Theta^{-1}$ gives $0 \le F[\phi\circ\Theta^{-1}](\Theta\zeta_0) = F[\phi](\zeta_0)$; and if $\bar v - \phi$ has a local minimum at ζ_0, the supersolution property of $\widehat v$ in `IsLimitSuperOn` applied to the same test function gives $F[\phi](\zeta_0) = F[\phi\circ\Theta^{-1}](\Theta\zeta_0) \le 0$. Moreover $\bar v = \widehat v\circ\Theta$ is continuous because $\widehat v$ is, and the whole space is open, which is the first of the three conditions in each of `IsLimitSubOn` and `IsLimitSuperOn`; with Ω the whole space, a local maximum of $\bar v - \phi$ on Ω at ζ_0, as `IsLimitSubOn` requires, is an ordinary local maximum at ζ_0, and likewise for local minima in `IsLimitSuperOn`. So the three conditions of each of those two definitions hold, and $\bar v$ is both a viscosity subsolution and a viscosity supersolution of $F[\bar v] = 0$ on all of space-time, that is, by `IsLimitSolutionOn`, a viscosity solution of $F[\bar v] = 0$ there.
- *Fits.* For any ζ, $\rho > 0$, p and a, the increments (h,ℓ) appearing in `IsFit` are the same for $\bar v$ at ζ and for $\widehat v$ at $\Theta\zeta$, because Θ is a translation and $\bar v(\zeta + (h,\ell)) = \widehat v(\Theta\zeta + (h,\ell))$. So (p,a) is a fit for $\bar v$ at ζ of radius ρ exactly when it is a fit for $\widehat v$ at $\Theta\zeta$ of radius ρ. Since $\widehat v$ has fits at every center and radius, so does $\bar v$; hence $\bar v \in \mathcal{A}_{m,C_0}$ by `IsFitClass`. Taking $\zeta = (0,1)$, so $\Theta\zeta = (0,0)$, the centered fit $(0,0)$ of radius 1 for $\widehat v$ at the origin is a fit for $\bar v$ at $(0,1)$ of radius 1, and $\bar v(0,1) = \widehat v(0,0) = 0$. By `IsCenteredAt`, $\bar v$ is centered at $(0,1)$ with constant C_0.

For the Jensen bound, fix a time s and a finite horizontal probability configuration (θ, y) with mean y_* in the sense of `IsBaryConfig`. Put $t = rs$ and $x_i = x_0 + r(y_i + p_0(s-1))$, and $x_* = x_0 + r(y_* + p_0(s-1))$. Since $\sum_i \theta_i y_i = y_*$ and $\sum_i \theta_i = 1$, we get $\sum_i \theta_i x_i = x_*$, so (θ, x) is a finite horizontal probability configuration at time t with mean x_*; moreover $x_i - x_* = r(y_i - y_*)$, so by `BaryRadius` its radius is r times the radius $\bar r$ of (θ, y). From (688),

$$\bar v(y_i, s) = \frac{v(x_i,t) - v(x_0,r)}{r} - p_0\cdot y_i - \Big(a_0 + \frac{|p_0|^2}{2}\Big)(s-1),$$

and the terms other than $v(x_i,t)/r$ are affine in y_i; averaging with the weights θ_i and subtracting the same expression at y_*, the affine terms cancel because the configuration has mean y_*, and we get

$$\sum_i \theta_i \bar v(y_i,s) - \bar v(y_*,s) = \frac{1}{r}\Big(\sum_i \theta_i v(x_i,t) - v(x_*,t)\Big). \tag{690}$$

The hypothesis `HasJensenBound` for v at the time t bounds the right side by $\kappa\cdot(r\bar r)/r = \kappa\bar r$. As s and (θ, y) were arbitrary, `HasJensenBound` holds for $\bar v$ with the same constant κ at every time.

Step 3: the envelope value identities. We first record that the envelopes of $\bar v$ are genuine extrema, so that no convention for the supremum or infimum of an unbounded set is ever in play. By Step 2, $\bar v \in \mathcal{A}_{m,C_0}$ and $\bar v$ is centered at $(0,1)$ with constant C_0 in the sense of `IsCenteredAt`; these are exactly the hypotheses of `UniformSubquadratic` with $z_c = (0,1)$, which therefore gives, for every compact $J \subseteq \mathbb{R}$ and every $\varepsilon > 0$, an $M > 0$ with $|\bar v(y',s')| \le \varepsilon |y'|^2$ whenever $|y'| \ge M$ and $s' \in J$. That is precisely subquadratic spatial growth on compact time intervals of $I = \mathbb{R}$ in the sense of `SubquadraticGrowth`, and $\bar v$ is continuous on all of space-time, which is $\Omega_{\mathbb{R}}$, by clause (1) of `IsFitClass`. So `EnvelopeCompetitorBounds` applies to $\bar v$ with $I = \mathbb{R}$; since every real time lies in I, its admissibility requirements $s \in I$, $s + \ell \in I$ and $s - \ell \in I$ hold for every (y,s) and every $\ell > 0$, with no sign condition on s or restriction on ℓ. Clause (1) then says that $\mathcal{P}_\ell \bar v(y,s)$ is the least upper bound of the competitor values

$$\Big\{ \bar v(y', s+\ell) - \frac{|y'-y|^2}{2\ell} \ : \ y' \in F \Big\}, \tag{691}$$

and clause (2) says that $\mathcal{M}_\ell \bar v(y,s)$ is the greatest lower bound of

$$\Big\{ \bar v(y', s-\ell) + \frac{|y'-y|^2}{2\ell} \ : \ y' \in F \Big\}. \tag{692}$$

In particular (691) is nonempty and bounded above, and (692) is nonempty and bounded below.

Put

$$A(y,s) = \frac{v(x_0,r)}{r} + p_0 \cdot y + \Big(a_0 + \frac{|p_0|^2}{2}\Big)(s-1),$$

so that (688) reads

$$\frac{v(x,t)}{r} = \bar v(y,s) + A(y,s) \qquad \big((x,t) = T(y,s)\big). \tag{693}$$

Let $\ell > 0$ and let (y,s) correspond to (x,t). We claim

$$\begin{aligned} \frac{\mathcal{P}_{r\ell} v(x,t)}{r} &= \mathcal{P}_\ell \bar v(y,s) + A(y,s) + a_0 \ell, \\ \frac{\mathcal{M}_{r\ell} v(x,t)}{r} &= \mathcal{M}_\ell \bar v(y,s) + A(y,s) - a_0 \ell. \end{aligned} \tag{694}$$

For the first identity, parameterize the competitors in `QuadEnvP` for $\mathcal{P}_{r\ell} v(x,t)$: a competitor is a point x' at time $t + r\ell$, and we write $x' = x_0 + r(y' + p_0(s+\ell-1))$ for a unique $y' \in F$, i.e. $(y', s+\ell)$ is the point corresponding to $(x', t+r\ell)$ under T; as y' ranges over F so does x'. The old displacement is

$$x' - x = r\big(y' - y + p_0 \ell\big),$$

so the old cost divided by r is

$$\frac{|x'-x|^2}{2r\ell\, r} = \frac{|y'-y+p_0\ell|^2}{2\ell} = \frac{|y'-y|^2}{2\ell} + p_0 \cdot (y'-y) + \frac{|p_0|^2 \ell}{2}.$$

By (693), $v(x', t+r\ell)/r = \bar v(y', s+\ell) + A(y', s+\ell)$, and

$$A(y', s+\ell) - A(y,s) = p_0 \cdot (y'-y) + \Big(a_0 + \frac{|p_0|^2}{2}\Big)\ell.$$

Therefore the old competitor value divided by r is

$$\frac{v(x', t+r\ell)}{r} - \frac{|x'-x|^2}{2r\ell\, r} = \bar v(y', s+\ell) - \frac{|y'-y|^2}{2\ell} + A(y,s) + a_0 \ell.$$

The correspondence $x' \leftrightarrow y'$ is a bijection of F onto F, hence a bijection of the competitor index sets, and the last display says that the competitor value for $\mathcal{P}_{r\ell} v(x,t)$ at x' equals $r\Psi(u)$, where

$$u = \bar v(y', s+\ell) - \frac{|y'-y|^2}{2\ell}, \qquad \Psi(u) = u + A(y,s) + a_0 \ell$$

is the competitor value for $\mathcal{P}_\ell \bar v(y,s)$ at y' shifted by the constant $A(y,s) + a_0\ell$, which is independent of the competitor. So the set of competitor values for $\mathcal{P}_{r\ell} v(x,t)$ is the image of (691) under $u \mapsto r\Psi(u)$, an increasing affine bijection of $\mathbb{R}$ because $r > 0$. An increasing affine bijection of $\mathbb{R}$ carries a nonempty set that is bounded above to a nonempty set that is bounded above, and

carries its least upper bound to the least upper bound of the image; this is how nonemptiness and boundedness transport across the bijection, so the conclusion of `EnvelopeCompetitorBounds` for $\bar v$ is what makes the v-side family genuinely bounded, even though no growth hypothesis is available for v itself. Therefore the competitor family for $\mathcal{P}_{r\ell}v(x,t)$ is nonempty and bounded above with least upper bound $r\big(\mathcal{P}_\ell\bar v(y,s)+A(y,s)+a_0\ell\big)$. The supremum defining $\mathcal{P}_{r\ell}v(x,t)$ in `QuadEnvP` is taken over exactly that family, and a supremum of a nonempty set that is bounded above is its least upper bound, so $\mathcal{P}_{r\ell}v(x,t)=r\big(\mathcal{P}_\ell\bar v(y,s)+A(y,s)+a_0\ell\big)$. Dividing by $r>0$ gives the first identity of (694).

For the second identity, the same parameterization with x' at time $t-r\ell$ and $(y',s-\ell)$ gives $x'-x=r(y'-y-p_0\ell)$, so the old cost divided by r is

$$\frac{|y'-y-p_0\ell|^2}{2\ell}=\frac{|y'-y|^2}{2\ell}-p_0\cdot(y'-y)+\frac{|p_0|^2\ell}{2},$$

while $A(y',s-\ell)-A(y,s)=p_0\cdot(y'-y)-(a_0+|p_0|^2/2)\ell$. Adding the cost, as in `QuadEnvM`, the old competitor value divided by r is

$$\bar v(y',s-\ell)+\frac{|y'-y|^2}{2\ell}+A(y,s)-a_0\ell,$$

and the same reasoning applies in the dual direction. The correspondence $x'\leftrightarrow y'$ is again a bijection of F onto F, and by the last display the set of competitor values for $\mathcal{M}_{r\ell}v(x,t)$ is the image of (692) under the increasing affine bijection $u\mapsto r\big(u+A(y,s)-a_0\ell\big)$ of $\mathbb{R}$. Such a map carries a nonempty set that is bounded below to a nonempty set that is bounded below and carries its greatest lower bound to the greatest lower bound of the image, so nonemptiness and boundedness again transport from (692), which `EnvelopeCompetitorBounds` supplied for $\bar v$; the competitor family for $\mathcal{M}_{r\ell}v(x,t)$ is thus nonempty and bounded below with greatest lower bound $r\big(\mathcal{M}_\ell\bar v(y,s)+A(y,s)-a_0\ell\big)$. The infimum defining $\mathcal{M}_{r\ell}v(x,t)$ in `QuadEnvM` is taken over exactly that family, and an infimum of a nonempty set that is bounded below is its greatest lower bound, so dividing by $r>0$ gives the second identity of (694).

Step 4: the slope covariance. Subtracting (693) from the first line of (694),

$$\frac{\mathcal{P}_{r\ell}v(x,t)-v(x,t)}{r}=\mathcal{P}_\ell\bar v(y,s)-\bar v(y,s)+a_0\ell;$$

dividing by $\ell>0$ and using `EnvSlopeP` on both sides, together with $(r\ell)^{-1}\big(\mathcal{P}_{r\ell}v(x,t)-v(x,t)\big)=\mathcal{S}^+_{r\ell}v(x,t)$,

$$\mathcal{S}^+_{r\ell}v(x,t)=\mathcal{S}^+_\ell\bar v(y,s)+a_0.$$

Subtracting the second line of (694) from (693),

$$\frac{v(x,t)-\mathcal{M}_{r\ell}v(x,t)}{r}=\bar v(y,s)-\mathcal{M}_\ell\bar v(y,s)+a_0\ell,$$

and dividing by ℓ and using `EnvSlopeM` gives $\mathcal{S}^-_{r\ell}v(x,t)=\mathcal{S}^-_\ell\bar v(y,s)+a_0$. Both identities are (689). □

2.161. `FrameGapCovariance`.

Lemma 2.161 (Covariance of the barycentric and contact gaps)**.** *Let F be a finite-dimensional real inner-product space, let $1\le C_0<\infty$ and $0\le\kappa<\infty$, let $v\in\mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`, let $r>0$, let $x_0\in F$, let (p_0,a_0) be a fit for v at (x_0,r) of radius r with constant C_0 in the sense of `IsFit`, and let $\bar v$ be the contact frame*

$$\bar v(y,s)=\frac{v(x,t)-v(x_0,r)}{r}-p_0\cdot y-\left(a_0+\frac{|p_0|^2}{2}\right)(s-1),\qquad x=x_0+r\big(y+p_0(s-1)\big),\quad t=rs,$$

that is, the change of frame of `FrameChange` based at (x_0,r) with scale r and parameters (p_0,a_0), composed with the translation of the time coordinate by one unit. Put

$$\bar W(y,s)=\mathcal{C}_s\bar v(y,s)-\frac{\kappa^2 s}{2}\qquad(s>0),$$

the contact envelope of `ContactEnv` *for* $\bar v$*, while* $W = \mathcal{C}_t v - \kappa^2 t/2$ *is that for* v*. Then, at corresponding points* $(y,s) \leftrightarrow (x,t)$ *of the contact frame, i.e. with* $x = x_0 + r(y + p_0(s-1))$ *and* $t = rs$*,*

$$\begin{aligned} \mathcal{C}_\sigma \bar v(y,s) - \bar v(y,s) &= \frac{\mathcal{C}_{r\sigma} v(x,t) - v(x,t)}{r} \qquad (\sigma > 0), \\ \bar W(y,s) - \bar v(y,s) &= \frac{W(x,t) - v(x,t)}{r} \qquad (s > 0), \end{aligned} \tag{695}$$

with the barycentric envelope of `BaryEnv`*.*

Proof. Put

$$A(y,s) = \frac{v(x_0,r)}{r} + p_0 \cdot y + \left(a_0 + \frac{|p_0|^2}{2}\right)(s-1), \qquad \frac{v(x,t)}{r} = \bar v(y,s) + A(y,s),$$

the second identity being the defining formula of the contact frame $\bar v$ rearranged.

The barycentric gap. Fix $\sigma > 0$ and corresponding points $(y,s) \leftrightarrow (x,t)$, i.e. $x = x_0 + r(y + p_0(s-1))$ and $t = rs$. A finite horizontal probability configuration for $\bar v$ at (y,s), with weights θ_i, positions y_i and mean y in the sense of `IsBaryConfig`, corresponds, at the fixed time $t = rs$, to the configuration with the same weights, positions $x_i = x_0 + r(y_i + p_0(s-1))$ and mean x; this correspondence is a bijection between the configurations at the two points, since $y \mapsto x$ is an invertible affine map at fixed time with linear part $r\,\mathrm{Id}$. Under it the spatial offsets satisfy $x_i - x = r(y_i - y)$, so by `BaryRadius` the squared radius is multiplied by r^2, and the weighted values satisfy

$$\sum_i \theta_i \frac{v(x_i,t)}{r} = \sum_i \theta_i \bar v(y_i,s) + \sum_i \theta_i A(y_i,s) = \sum_i \theta_i \bar v(y_i,s) + A(y,s),$$

because $A(\cdot,s)$ is affine in the spatial variable and the configuration has mean y, so the weighted average of the affine terms is the value at the mean. Hence, dividing the defining expression of `BaryEnv` at parameter $r\sigma$ by r,

$$\sum_i \theta_i \frac{v(x_i,t)}{r} - \frac{r^2\,\bar r^2}{2r\sigma r} = \sum_i \theta_i \bar v(y_i,s) + A(y,s) - \frac{\bar r^2}{2\sigma},$$

where $\bar r$ is the radius of the new configuration. Taking suprema over the corresponding configurations gives

$$\frac{\mathcal{C}_{r\sigma} v(x,t)}{r} = \mathcal{C}_\sigma \bar v(y,s) + A(y,s).$$

Subtracting the displayed formula for $v(x,t)/r$ proves the first identity in (695).

The contact gap. Let $s > 0$, so $t = rs > 0$. Take $\sigma = s$ in the first identity. The old time and the old barycentric parameter in W are both $t = rs$, while the correction $\kappa^2 t/(2r) = \kappa^2 s/2$ matches the correction in $\bar W$. Hence, by `ContactEnv`,

$$\frac{W(x,t)}{r} = \frac{\mathcal{C}_t v(x,t)}{r} - \frac{\kappa^2 t}{2r} = \mathcal{C}_s \bar v(y,s) + A(y,s) - \frac{\kappa^2 s}{2} = \bar W(y,s) + A(y,s),$$

and subtracting the formula for $v(x,t)/r$ gives the second identity in (695). This is why the frame uses $s = t/r$: it preserves the time-zero boundary in the definition of W. □

2.162. `FrameVelocityCovariance`.

Lemma 2.162 (Covariance of the step velocities in a fitted frame)**.** *Let* F *be a finite-dimensional real inner-product space, let* $1 \le C_0 < \infty$ *and* $0 \le \kappa < \infty$*, and let* $v \in \mathcal{A}_{m,C_0}$ *in the sense of* `IsFitClass` *be centered at* $(0,1)$ *with constant* C_0 *in the sense of* `IsCenteredAt` *and satisfy the horizontal Jensen bound with constant* κ *at every time in the sense of* `HasJensenBound`*. Write* $f = W$ *for the contact envelope of* `ContactEnv`*. Let* $r > 0$*, let* $x_0, p_0 \in F$ *and* $a_0 \in \mathbb{R}$*, and let* (p_0, a_0) *be a fit for* v *at* (x_0, r) *of radius* r *in the sense of* `IsFit`*. Let* $\bar v$ *be the contact frame of*

FrameChange based at (x_0, r) with scale r and parameters (p_0, a_0), preceded by the translation by $(0, 1)$, that is

$$\bar v(y,s) = \frac{v(x,t) - v(x_0,r)}{r} - p_0 \cdot y - \Big(a_0 + \frac{|p_0|^2}{2}\Big)(s-1), \qquad x = x_0 + r\big(y + p_0(s-1)\big), \quad t = rs,$$

and write $\bar W$ for the contact envelope of ContactEnv formed from $\bar v$ and the same κ.

Let $\epsilon > 0$, let $z = (x,t)$ with $t > \epsilon$, let (z^+, z^-) be an optimizing step pair of size ϵ at z for v and κ in the sense of IsStepPair, with $z^\pm = (x^\pm, t \pm \epsilon)$, and let $Q_0 \in F$ be the reference velocity of DPSpatialRegularity at z and length ϵ, i.e. a vector at which $\mathcal{P}_\epsilon W(\cdot, t)$ has gradient Q_0 at x. Put $\epsilon' = \epsilon/r$ and let $\zeta = (y,s)$, $\zeta^+ = (y^+, s^+)$, $\zeta^- = (y^-, s^-)$ be the images of z, z^+, z^- under the inverse frame map

$$(x',t') \longmapsto \Big(\frac{x' - x_0 - (t'-r)p_0}{r},\ \frac{t'}{r}\Big).$$

Then (ζ^+, ζ^-) is an optimizing step pair of size ϵ' at ζ for $\bar v$ and κ in the sense of IsStepPair, and the velocities satisfy

$$\begin{gathered} Q'_+ = \frac{y^+ - y}{\epsilon'} = Q_+ - p_0, \qquad Q'_- = \frac{y - y^-}{\epsilon'} = Q_- - p_0, \\ Q'_0 = \nabla_y \mathcal{P}_{\epsilon'} \bar W(y,s) = Q_0 - p_0, \\ Q'_+ - Q'_0 = Q_+ - Q_0, \qquad Q'_- - Q'_0 = Q_- - Q_0, \end{gathered} \tag{696}$$

where $Q_+ = (x^+ - x)/\epsilon$ and $Q_- = (x - x^-)/\epsilon$. These identities hold at corresponding states globally, with the stated positive-time restriction for W and $\bar W$; the states need not lie in the fitting box, and no contact at (x_0, r) is assumed.

Proof. Throughout, $z_0 = (x_0, r)$, $z = (x,t)$ with $t > \epsilon > 0$, $x^\pm$ are the spatial parts of $z^\pm$, and $f = W$ is the contact envelope of ContactEnv, so $W(y',\tau) = \mathcal{C}_\tau v(y',\tau) - \kappa^2\tau/2$ for $\tau > 0$. Put $\epsilon' = \epsilon/r > 0$.

Step 0: standing facts about v and W. By IsFitClass the function v is continuous on all of space-time; by UniformSubquadratic, applied with the centering point $(0,1)$, it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of SubquadraticGrowth. By BaryEnvSandwich with $I = \mathbb{R}$, $f = v$ and the constant κ, for every $\sigma > 0$ and every (y',τ),

$$0 \le \mathcal{C}_\sigma v(y',\tau) - v(y',\tau) \le \frac{\kappa^2\sigma}{2},$$

and taking $\sigma = \tau > 0$ with ContactEnv gives

$$v(y',\tau) - \frac{\kappa^2\tau}{2} \le W(y',\tau) \le v(y',\tau) \qquad (\tau > 0). \tag{697}$$

Hence W has subquadratic spatial growth on compact subintervals of $(0,\infty)$: for $J \subseteq (0,\infty)$ compact and $\varepsilon_1 > 0$, put $\tau_{\max} = \max J$, apply SubquadraticGrowth for v with J and $\varepsilon_1/2$ to get $M' > 0$, and enlarge M' so that $\kappa^2\tau_{\max}/2 \le (\varepsilon_1/2)M'^2$; then (697) gives $|W(y',\tau)| \le \varepsilon_1|y'|^2$ for $|y'| \ge M'$, $\tau \in J$. Moreover W is continuous on $\{(y',\tau) : \tau > 0\}$: by BaryEnvDuality, part (3), applied to v with $I = \mathbb{R}$, the map $(\sigma,\xi) \mapsto \mathcal{C}_\sigma v(\xi)$ is continuous at every pair with $\sigma > 0$; composing with the continuous map $\xi \mapsto (\tau,\xi)$, where τ is the time coordinate of ξ, and subtracting the continuous $\kappa^2\tau/2$, gives the claim.

Claim B: the envelopes are genuine suprema and infima. Let $I \subseteq \mathbb{R}$, let g be continuous on $\{(y',\tau) : \tau \in I\}$ with subquadratic spatial growth on compact subintervals of I in the sense of SubquadraticGrowth, let $\ell > 0$ and let $\xi = (\xi_1, \tau)$ with $\tau \in I$.

(i) If $\tau + \ell \in I$, the set $\{g(y', \tau+\ell) - |y' - \xi_1|^2/(2\ell) : y' \in F\}$ of QuadEnvP is nonempty and bounded above, so $\mathcal{P}_\ell g(\xi)$ is finite, is at least every member of that set, and is at most every upper bound for it.

(ii) If $\tau - \ell \in I$, the set $\{g(y', \tau-\ell) + |y' - \xi_1|^2/(2\ell) : y' \in F\}$ of QuadEnvM is nonempty and bounded below, so $\mathcal{M}_\ell g(\xi)$ is finite, is at most every member, and is at least every lower bound.

Proof. (i) The choice $y' = \xi_1$ shows the set is nonempty. Apply `SubquadraticGrowth` with the compact set $J = \{\tau + \ell\} \subseteq I$ and $\varepsilon_1 = 1/(4\ell) > 0$ to get $M > 0$ with $|g(y', \tau + \ell)| \le |y'|^2/(4\ell)$ for $|y'| \ge M$, and let C_M bound the continuous $|g(\cdot, \tau + \ell)|$ on the closed ball of radius M. From $|y'|^2 \le 2|y' - \xi_1|^2 + 2|\xi_1|^2$ we get $|y' - \xi_1|^2 \ge |y'|^2/2 - |\xi_1|^2$, so for every y'

$$g(y', \tau + \ell) - \frac{|y' - \xi_1|^2}{2\ell} \le C_M + \frac{|y'|^2}{4\ell} - \frac{|y'|^2}{4\ell} + \frac{|\xi_1|^2}{2\ell} = C_M + \frac{|\xi_1|^2}{2\ell}.$$

(ii) Apply (i) to $-g$ at the time $\tau - \ell$ and multiply by -1. This proves Claim B. By Step 0 it applies to $g = W$ with $I = (0, \infty)$ and to $g = v$ with $I = \mathbb{R}$.

Step 1: the frame map and the corresponding states. For (y', s') put

$$\Theta(y', s') = \big(x_0 + r(y' + (s' - 1)p_0),\ rs'\big), \qquad \Psi(x', t') = \Big(\frac{x' - x_0 - (t' - r)p_0}{r},\ \frac{t'}{r}\Big).$$

Both are affine maps of space-time, and they are mutually inverse: with $(y', s') = \Psi(x', t')$ one has $rs' = t'$ and $x_0 + r(y' + (s' - 1)p_0) = x_0 + (x' - x_0 - (t' - r)p_0) + (t' - r)p_0 = x'$, and the computation in the other order is the same read backwards. In particular Θ carries the time s' to rs', so, r being positive, it carries positive times to positive times and conversely. For each fixed s' the spatial part

$$T_{s'}(y') = x_0 + r\big(y' + (s' - 1)p_0\big)$$

is an affine bijection of F with $T_{s'}(y') - T_{s'}(y'') = r(y' - y'')$, and $T_{s'}$ has derivative $r\,\mathrm{id}$ at every point.

By hypothesis $\zeta = \Psi(z)$, $\zeta^+ = \Psi(z^+)$ and $\zeta^- = \Psi(z^-)$, so $\Theta(\zeta) = z$, $\Theta(\zeta^\pm) = z^\pm$ and

$$s = \frac{t}{r} > \frac{\epsilon}{r} = \epsilon' > 0, \qquad s^+ = \frac{t + \epsilon}{r} = s + \epsilon', \qquad s^- = \frac{t - \epsilon}{r} = s - \epsilon' > 0. \tag{698}$$

Writing $y = (x - x_0 - (t - r)p_0)/r$ we have $T_s(y) = x$.

Step 2: $\bar W$ is the same frame change applied to W. By `FrameChange` and the definition of $\bar v$,

$$\bar v(y', s') = \frac{v(\Theta(y', s')) - v(z_0)}{r} - p_0 \cdot y' - \Big(a_0 + \frac{|p_0|^2}{2}\Big)(s' - 1). \tag{699}$$

By `FrameGapCovariance`, second part, applied with this C_0, κ, v, r, x_0, p_0, a_0 and the fit (p_0, a_0) at (x_0, r) of radius r, we have for every (y', s') with $s' > 0$

$$\bar W(y', s') - \bar v(y', s') = \frac{W(\Theta(y', s')) - v(\Theta(y', s'))}{r}.$$

Adding (699) and cancelling $v(\Theta(y', s'))/r$,

$$\bar W(y', s') = \frac{W(\Theta(y', s')) - v(z_0)}{r} - p_0 \cdot y' - \Big(a_0 + \frac{|p_0|^2}{2}\Big)(s' - 1) \qquad (s' > 0), \tag{700}$$

i.e. $\bar W$ is obtained from W by the same frame formula that produces $\bar v$ from v.

Step 3: the competitor correspondence and the envelope laws. Fix (y', s') with $s' > 0$, let $\sigma > 0$ and put $\ell = r\sigma > 0$ and $(x', t') = \Theta(y', s')$, so $t' = rs' > 0$. For $y'' \in F$ put $X'' = T_{s' + \sigma}(y'')$, the spatial part of $\Theta(y'', s' + \sigma)$, whose time is $r(s' + \sigma) = t' + \ell$; by Step 1 the map $y'' \mapsto X''$ is a bijection of F, and

$$X'' - x' = r(y'' - y') + r\sigma p_0, \qquad y'' - y' = \frac{X'' - x'}{r} - \sigma p_0. \tag{701}$$

Using (701),

$$\frac{|y'' - y'|^2}{2\sigma} = \frac{|X'' - x'|^2}{2\sigma r^2} - \frac{(X'' - x') \cdot p_0}{r} + \frac{\sigma |p_0|^2}{2} = \frac{|X'' - x'|^2}{2\ell r} - \frac{(X'' - x') \cdot p_0}{r} + \frac{\sigma |p_0|^2}{2},$$

since $\sigma r^2 = \ell r$, and $p_0 \cdot y'' = p_0 \cdot y' + (X'' - x') \cdot p_0/r - \sigma |p_0|^2$. Substituting both into (700) at $(y'', s' + \sigma)$, the two terms $\pm (X'' - x') \cdot p_0/r$ cancel and

$$-\Big(a_0 + \frac{|p_0|^2}{2}\Big)(s' + \sigma - 1) + \sigma |p_0|^2 - \frac{\sigma |p_0|^2}{2} = -\Big(a_0 + \frac{|p_0|^2}{2}\Big)(s' - 1) - a_0 \sigma,$$

so that, with $K_+ = -v(z_0)/r - p_0 \cdot y' - (a_0 + |p_0|^2/2)(s'-1) - a_0\sigma$,

$$\bar W(y'', s'+\sigma) - \frac{|y''-y'|^2}{2\sigma} = \frac{1}{r}\Big(W(X'', t'+\ell) - \frac{|X''-x'|^2}{2\ell}\Big) + K_+. \tag{702}$$

Thus the set of competitor values in `QuadEnvP` for $\mathcal{P}_\sigma \bar W(y', s')$ is the image of the corresponding set for $\mathcal{P}_\ell W(x', t')$ under the increasing affine bijection $w \mapsto w/r + K_+$ of $\mathbb{R}$. The latter set is nonempty and bounded above by Claim B(i) applied to W with $I = (0, \infty)$, since $t' > 0$ and $t' + \ell > 0$; hence so is the former, and the two suprema correspond:

$$\mathcal{P}_\sigma \bar W(y', s') = \frac{\mathcal{P}_\ell W(x', t')}{r} + K_+ \qquad (\ell = r\sigma). \tag{703}$$

Moreover, by (702) and (703), a point y'' attains the supremum defining $\mathcal{P}_\sigma \bar W(y', s')$ if and only if the corresponding X'' attains the one defining $\mathcal{P}_\ell W(x', t')$.

The same computation with $s' - \sigma$ in place of $s' + \sigma$ and v in place of W gives, for $X'' = T_{s'-\sigma}(y'')$ at time $t' - \ell$,

$$\bar v(y'', s'-\sigma) + \frac{|y''-y'|^2}{2\sigma} = \frac{1}{r}\Big(v(X'', t'-\ell) + \frac{|X''-x'|^2}{2\ell}\Big) + K_-, \tag{704}$$

with $K_- = -v(z_0)/r - p_0 \cdot y' - (a_0 + |p_0|^2/2)(s'-1) + a_0\sigma$. Here (701) is replaced by $X'' - x' = r(y''-y') - r\sigma p_0$, so the signs of the two $(X''-x')\cdot p_0/r$ terms are reversed and again cancel, while $-\sigma|p_0|^2 + (a_0 + |p_0|^2/2)\sigma + \sigma|p_0|^2/2 = a_0\sigma$; note that (699) holds at every time, so no positivity is needed on this side. By Claim B(ii) applied to v with $I = \mathbb{R}$, the set for $\mathcal{M}_\ell v(x', t')$ is nonempty and bounded below, hence so is the one for $\mathcal{M}_\sigma \bar v(y', s')$, the correspondence $w \mapsto w/r + K_-$ is again an increasing affine bijection, and

$$\mathcal{M}_\sigma \bar v(y', s') = \frac{\mathcal{M}_\ell v(x', t')}{r} + K_- \qquad (\ell = r\sigma), \tag{705}$$

with the same equivalence of attaining points.

Step 4: the images are an optimizing step pair of size ϵ'. By `IsStepPair`, applied to the hypothesis that (z^+, z^-) is an optimizing step pair of size ϵ at z for v and κ, there are $Y^+, Y^- \in F$ with

$$\mathcal{P}_{2\epsilon} W(z) = W(Y^+, t+2\epsilon) - \frac{|Y^+ - x|^2}{2\cdot 2\epsilon}, \qquad \mathcal{M}_{2\epsilon} v(z) = v(Y^-, t-2\epsilon) + \frac{|Y^- - x|^2}{2\cdot 2\epsilon},$$
$$x^+ = \tfrac{1}{2}(x + Y^+), \qquad x^- = \tfrac{1}{2}(x + Y^-).$$

Apply Step 3 at $(y', s') = \zeta = (y, s)$, which has $s > 0$ by (698), with $\sigma = 2\epsilon'$, so $\ell = 2\epsilon$ and $(x', t') = \Theta(\zeta) = z$. Put

$$Y'^+ = \frac{Y^+ - x_0 - (t + 2\epsilon - r)p_0}{r}, \qquad Y'^- = \frac{Y^- - x_0 - (t - 2\epsilon - r)p_0}{r},$$

so that $T_{s+2\epsilon'}(Y'^+) = Y^+$ and $T_{s-2\epsilon'}(Y'^-) = Y^-$ by Step 1; equivalently $Y'^\pm$ are the spatial parts of $\Psi(Y^\pm, t \pm 2\epsilon)$, the times $t \pm 2\epsilon$ corresponding to $s \pm 2\epsilon'$. Since Y^+ attains the supremum defining $\mathcal{P}_{2\epsilon} W(z)$, the last sentence of Step 3 shows that Y'^+ attains the supremum defining $\mathcal{P}_{2\epsilon'} \bar W(\zeta)$, that is

$$\mathcal{P}_{2\epsilon'} \bar W(\zeta) = \bar W\big(Y'^+, s + 2\epsilon'\big) - \frac{|Y'^+ - y|^2}{2\cdot 2\epsilon'},$$

and likewise, by (704) and (705) with $\sigma = 2\epsilon'$, that Y'^- attains the infimum defining $\mathcal{M}_{2\epsilon'} \bar v(\zeta)$:

$$\mathcal{M}_{2\epsilon'} \bar v(\zeta) = \bar v\big(Y'^-, s - 2\epsilon'\big) + \frac{|Y'^- - y|^2}{2\cdot 2\epsilon'}.$$

It remains to identify the midpoints. Using $y = (x - x_0 - (t-r)p_0)/r$,

$$\frac{y + Y'^+}{2} = \frac{1}{r}\Big(\frac{x + Y^+}{2} - x_0 - \frac{(t-r) + (t + 2\epsilon - r)}{2} p_0\Big) = \frac{x^+ - x_0 - \big((t+\epsilon) - r\big)p_0}{r},$$

which is the spatial part of $\Psi(z^+) = \zeta^+$, whose time is $(t+\epsilon)/r = s + \epsilon'$ by (698). The same computation with Y^- and -2ϵ gives $(y + Y'^-)/2 = \big(x^- - x_0 - ((t-\epsilon) - r)p_0\big)/r$, the spatial part of

$\Psi(z^-)=\zeta^-$, whose time is $s-\epsilon'$. Hence $\zeta^+=\big(\frac12(y+Y'^+),\,s+\epsilon'\big)$ and $\zeta^-=\big(\frac12(y+Y'^-),\,s-\epsilon'\big)$, and Y'^+, Y'^- witness `IsStepPair` for $\bar v$, κ, ϵ' at ζ with the pair (ζ^+,ζ^-).

Step 5: the velocity identities. By Step 4, $y^\pm$ are the spatial parts of $\Psi(z^\pm)$, so

$$y^+-y=\frac{(x^+-x_0-(t+\epsilon-r)p_0)-(x-x_0-(t-r)p_0)}{r}=\frac{x^+-x-\epsilon p_0}{r},$$

and dividing by $\epsilon'=\epsilon/r$, i.e. multiplying by r/ϵ,

$$Q'_+=\frac{y^+-y}{\epsilon'}=\frac{x^+-x}{\epsilon}-p_0=Q_+-p_0.$$

Likewise $y-y^-=\big(x-x^--\epsilon p_0\big)/r$, so $Q'_-=(y-y^-)/\epsilon'=(x-x^-)/\epsilon-p_0=Q_--p_0$.

For the reference velocity, apply (703) with $\sigma=\epsilon'$, hence $\ell=\epsilon$, at the variable center (y',s) with s fixed: since $\Theta(y',s)=(T_s(y'),t)$ with $t=rs$ independent of y', and since K_+ depends on y' only through the term $-p_0\cdot y'$,

$$\mathcal{P}_{\epsilon'}\bar W(y',s)=\frac{\mathcal{P}_\epsilon W\big(T_s(y'),t\big)}{r}-\frac{v(z_0)}{r}-p_0\cdot y'-\Big(a_0+\frac{|p_0|^2}{2}\Big)(s-1)-a_0\epsilon'$$

for every $y'\in F$. By hypothesis the function $x'\mapsto\mathcal{P}_\epsilon W(x',t)$ has gradient Q_0 at $x=T_s(y)$. The map T_s is affine with derivative $r\,\mathrm{id}$, which is self-adjoint, so by the chain rule the function $y'\mapsto\mathcal{P}_\epsilon W(T_s(y'),t)$ is differentiable at y with gradient rQ_0, and dividing by r gives gradient Q_0. The term $-p_0\cdot y'$ has gradient $-p_0$ everywhere and the remaining terms are constant in y'. Adding the gradients,

$$Q'_0=\nabla_y\mathcal{P}_{\epsilon'}\bar W(y,s)=Q_0-p_0.$$

Step 6: the discrepancies. Subtracting the identity for Q'_0 from those for $Q'_\pm$ gives

$$Q'_+-Q'_0=(Q_+-p_0)-(Q_0-p_0)=Q_+-Q_0,\qquad Q'_--Q'_0=Q_--Q_0,$$

which is the last line of (696). No hypothesis beyond $r>0$, the fit at (x_0,r) of radius r, and $t>\epsilon$ was used: in particular no containment of the states in a fitting box and no contact of W with v at (x_0,r). □

2.163. `ExpectedPortionLengthBound`.

Lemma 2.163 (The expected length of a portion of the lattice walk)**.** *Let $(\Omega,\mathcal{M},\mu)$ be a probability space, let $S_n:\Omega\to\mathbb{Z}$ for $n\in\mathbb{N}$, let $\sigma,\varsigma,\tau:\Omega\to\mathbb{N}$ and let $a,b\in\mathbb{N}$. Assume:*

(1) *σ and ς are measurable;*
(2) *$\varsigma(\omega)\le\tau(\omega)$ for every ω;*
(3) *$\omega\mapsto\tau(\omega)$ is μ-integrable;*
(4) *for every measurable $\zeta:\Omega\to\mathbb{N}$ and every family $G_m:\Omega\to\mathbb{R}$ of measurable functions, $\omega\mapsto G_{\zeta(\omega)}(\omega)$ is measurable, and likewise for two indices: for measurable $\zeta,\eta:\Omega\to\mathbb{N}$ and a doubly indexed measurable family $G_{n,m}$, the function $\omega\mapsto G_{\zeta(\omega),\eta(\omega)}(\omega)$ is measurable;*
(5) *for every $g:\mathbb{Z}\to\mathbb{R}$ and every $k\in\mathbb{N}$ the function $\omega\mapsto g(S_k(\omega))$ is measurable;*
(6) *$a<S_n(\omega)<b$ whenever $\sigma(\omega)\le n<\varsigma(\omega)$;*
(7) *for every pair $h,u:\mathbb{Z}\to\mathbb{R}$ with $u\ge0$ everywhere and*

$$\tfrac12\big(h(i+1)+h(i-1)\big)-h(i)\le-u(i)\qquad(a<i<b),$$

and for every $n\in\mathbb{N}$,

$$\int\Big(h\big(S_{\min(\max(\sigma,n),\varsigma)}\big)+\sum_{m=\sigma}^{\min(n,\varsigma)-1}u(S_m)\Big)\,d\mu\;\le\;\int h\big(S_\sigma\big)\,d\mu,$$

all functions being evaluated at the same ω.

Then the portion length $\varsigma-\sigma$, formed with truncated subtraction in $\mathbb{N}$ and read as a real number, is μ-integrable and

$$\int(\varsigma-\sigma)\,d\mu\;\le\;\frac{b^2}{4}.$$

Proof. Hypothesis 7 is the supersolution comparison along the portion; in the application it is the whole-space instance of Claim T′ of the paper, and the construction that produces S, σ and ς plays no part here. Throughout, $\varsigma-\sigma$ and $\min(n,\varsigma)-\sigma$ denote truncated differences of natural numbers, so they are nonnegative, and $\varsigma-\sigma\le\tau$ pointwise by 2.

Step 1: the two conclusions are integrable. The function $\omega\mapsto\big(\varsigma(\omega)-\sigma(\omega)\big)$ is measurable by the two-index case of 4, applied to $\zeta=\sigma$, $\eta=\varsigma$ and the constant family $G_{i,m}=(m-i)$; and $0\le\varsigma-\sigma\le\tau$ pointwise, by 2. Since τ is integrable by 3, domination gives that $\varsigma-\sigma$ is integrable. The same argument, with $G_{i,m}=(\min(n,m)-i)$, shows that each truncation $\min(n,\varsigma)-\sigma$ is measurable and integrable, using $\min(n,\varsigma)-\sigma\le\varsigma-\sigma\le\tau$.

Step 2: the test pair. Put

$$h(i)=\begin{cases}(b-i)(i-a), & a\le i\le b,\\ 0, & \text{otherwise},\end{cases}\qquad u(i)=\begin{cases}1, & a<i<b,\\ 0, & \text{otherwise}.\end{cases}$$

Then $u\ge0$ everywhere. Also $h\ge0$ everywhere: off $[a,b]$ it vanishes, and for $a\le i\le b$ both factors are nonnegative. And h is bounded by $b^2/4$:

$$h(i)\le\frac{b^2}{4}\qquad(i\in\mathbb{Z}).\tag{706}$$

Indeed, off $[a,b]$ the value is $0\le b^2/4$, while for $a\le i\le b$ we have $a\ge0$ and $b-i\ge0$, so $h(i)=(b-i)(i-a)\le(b-i)i$, and $(b-i)i\le b^2/4$ because $b^2-4(b-i)i=(b-2i)^2\ge0$.

For $a<i<b$ all three points $i-1,i,i+1$ lie in $[a,b]$, so there $h(i)=-i^2+(a+b)i-ab$ and

$$\tfrac12\big(h(i+1)+h(i-1)\big)-h(i)=\tfrac12\big(-(i+1)^2-(i-1)^2+2i^2\big)=-1=-u(i),$$

the linear terms cancelling. So (h,u) satisfies the hypothesis of 7.

Step 3: the comparison at a finite horizon. Fix $n\in\mathbb{N}$. By 6, for every ω and every m with $\sigma(\omega)\le m<\min(n,\varsigma(\omega))$ — hence $m<\varsigma(\omega)$ — we have $a<S_m(\omega)<b$ and therefore $u(S_m(\omega))=1$. Consequently

$$\sum_{m=\sigma(\omega)}^{\min(n,\varsigma(\omega))-1}u(S_m(\omega))=\min(n,\varsigma(\omega))-\sigma(\omega),$$

the number of terms. Since $h\ge0$, the integrand on the left of 7 is therefore at least $\min(n,\varsigma)-\sigma$ pointwise.

The left integrand is integrable: its first summand is measurable by the two-index case of 4 with $G_{i,m}(\omega)=h(S_{\min(\max(i,n),m)}(\omega))$, whose members are measurable by 5, and its second summand is measurable by the same case with $G_{i,m}(\omega)=\sum_{k=i}^{\min(n,m)-1}u(S_k(\omega))$, a finite sum of functions measurable by 5; and the sum of the two is bounded in absolute value by the constant $b^2/4+n$, by (706), $h\ge0$ and $0\le\min(n,\varsigma)-\sigma\le n$. Hence monotonicity of the integral and 7 give

$$\int\big(\min(n,\varsigma)-\sigma\big)\,d\mu\;\le\;\int h\big(S_\sigma\big)\,d\mu.\tag{707}$$

The right-hand side is at most $b^2/4$: the function $\omega\mapsto h(S_{\sigma(\omega)}(\omega))$ is measurable by the one-index case of 4 with $G_m(\omega)=h(S_m(\omega))$, it is integrable because $0\le h(S_\sigma)\le b^2/4$ by (706), and integrating that bound over the probability space Ω gives $\int h(S_\sigma)\,d\mu\le b^2/4$. With (707),

$$\int\big(\min(n,\varsigma)-\sigma\big)\,d\mu\;\le\;\frac{b^2}{4}\qquad(n\in\mathbb{N}).\tag{708}$$

Step 4: the horizon tends to infinity. For each fixed ω the sequence $n\mapsto\min(n,\varsigma(\omega))-\sigma(\omega)$ is nondecreasing, because $n\mapsto\min(n,\varsigma(\omega))$ is, and it is eventually constant equal to $\varsigma(\omega)-\sigma(\omega)$, namely for $n\ge\varsigma(\omega)$; so it converges to $\varsigma(\omega)-\sigma(\omega)$. Each term is integrable by Step 1, the limit $\varsigma-\sigma$ is integrable by Step 1 as well, and the convergence is monotone, so the integrals converge:

$$\int\big(\min(n,\varsigma)-\sigma\big)\,d\mu\;\longrightarrow\;\int(\varsigma-\sigma)\,d\mu\qquad(n\to\infty).$$

By (708) every term of the converging sequence is at most $b^2/4$, hence so is the limit:

$$\int (\varsigma - \sigma)\, d\mu \le \frac{b^2}{4}.$$

Together with the integrability of Step 1 this is the assertion. □

2.164. `MarkCutoffProbabilities`.

Lemma 2.164 (Step 8: the two cutoff probabilities at a mark)**.** *Let F be a finite-dimensional real inner-product space, write $\mathcal{H} = \ell^2(F\times\mathbb{R})$, let $(\Omega, \mathcal{M}, \mu)$ be a probability space and let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a filtration of $\mathcal{M}$. Let $z : \mathbb{N}\times\Omega \to \mathcal{H}$, $S : \mathbb{N}\times\Omega \to \mathbb{Z}$, $\sigma : \mathbb{N}\times\Omega \to \mathbb{N}$, $p : \mathbb{N}\times\Omega \to F$, $j \in \mathbb{N}$, $\varsigma : \Omega \to \mathbb{N}$ and $T, Y : \mathbb{N}\times\Omega \to \mathbb{R}$. Let $a, b, c, k \in \mathbb{N}$ and $\epsilon, r, K, C', B_0, L_K \in \mathbb{R}$. Write $\sigma_j = \sigma(j,\cdot)$, $p_j = p(j,\cdot)$, and $\mathcal{R}_j = \{\omega : t_{\sigma_j(\omega)} = r\}$ for the event that the j-th mark sits at its own level, where $z_n = (x_n, t_n)$. Assume:*

1. *$\epsilon > 0$ and $r > 0$; $c = 2^{k-j}$ and $b = 8\cdot 2^{k-j}$; $\epsilon c = r$ and $\epsilon b = 8r$; $a \le c$; $0 \le a$ as a real; and $b - a > 0$ as a real;*
2. *$2 \le K$ and the choice-of-K inequality $8C'\big(L_K + \sqrt{B_0}\big) \le K$;*
3. *$t_n = \epsilon\, S_n$ for all n, ω, and each $\omega \mapsto t_n(\omega)$ is strongly measurable;*
4. *σ_j is a $(\mathcal{G}_n)$-stopping time, both σ_j and ς are $\mathcal{M}$-measurable, and $\sigma_j \le \varsigma$ pointwise;*
5. *the measurability tools of the ambient argument: for every $(\mathcal{G}_n)$-stopping time ζ and every n, $\{\zeta \le n\} \in \mathcal{G}_n$; evaluating a measurable family of real functions at one or at two measurable random indices is measurable; and $\omega \mapsto g(S_k(\omega))$ is measurable for every $g : \mathbb{Z} \to \mathbb{R}$ and every k;*
6. *T_m is $\mathcal{G}_m$-measurable with $T_m = t_{\sigma_j}$ on $\{\sigma_j \le m\}$, and Y_m is $\mathcal{G}_m$-measurable with*
$$Y_m = r^{-1}\big|x_m - x_{\sigma_j} - (t_m - r)p_j\big| \qquad \text{on } \{\sigma_j \le m\};$$
7. *Markov's inequality in the form: for every $A \in \mathcal{M}$, every μ-integrable $f \ge 0$ and every $\kappa > 0$ with $f \ge \kappa$ on A, one has $\mu(A)\,\kappa \le \int f\, d\mu$;*
8. *the travel bound of Step 7: the running supremum*
$$\Upsilon(\omega) = \sup_{\sigma_j(\omega)\le n\le \varsigma(\omega)} r^{-1}\big|x_n - x_{\sigma_j} - (t_n - t_{\sigma_j})p_j\big|$$
is μ-integrable and $\int \Upsilon\, d\mu \le C'(L_K + \sqrt{B_0})$;
9. *the conclusion of Claim T' for σ_j, ς on the window (a, b), as stated in `StoppedLatticeSupersolutionBo`u for all $h, u : \mathbb{Z} \to \mathbb{R}$ and every $A \in \mathcal{M}$ with $u \ge 0$, with $\frac{1}{2}(h(i+1)+h(i-1))-h(i) \le -u(i)$ for $a < i < b$, and with $A \cap \{\sigma_j \le n\} \in \mathcal{G}_n$ for every n, one has for every n*
$$\int \mathbf{1}_A\Big(h\big(S_{(\sigma_j\vee n)\wedge\varsigma}\big) + \sum_{\sigma_j\le m<n\wedge\varsigma} u(S_m)\Big)d\mu \le \int \mathbf{1}_A\, h\big(S_{\sigma_j}\big)\, d\mu.$$

Then both cutoff probabilities are at most $\frac{1}{8}$:

$$\mu\big(\mathcal{R}_j \cap \{K \le Y_\varsigma\}\big) \le \frac{1}{8} \qquad \text{and} \qquad \mu\big(\mathcal{R}_j \cap \{8r \le t_\varsigma\}\big) \le \frac{1}{8}, \tag{709}$$

both measures being read as real numbers.

Proof. Measurability of the mark event. The map $\omega \mapsto t_{\sigma_j(\omega)}(\omega)$ is measurable, by 5 applied to the measurable index σ_j of 4 and to the family $\omega \mapsto t_n(\omega)$, measurable by 3; so $\mathcal{R}_j$, its preimage of the single point r, lies in $\mathcal{M}$. Moreover $\mathcal{R}_j$ is adapted at σ_j: for each n,

$$\mathcal{R}_j \cap \{\sigma_j \le n\} = \big(T_n^{-1}\{r\}\big) \cap \{\sigma_j \le n\} \in \mathcal{G}_n,$$

because T_n is $\mathcal{G}_n$-measurable and agrees with t_{σ_j} on $\{\sigma_j \le n\}$ by 6, and because $\{\sigma_j \le n\} \in \mathcal{G}_n$ by 5 and 4.

The first cutoff. Put $\mathcal{B}^1 = \mathcal{R}_j \cap \{K \le Y_\varsigma\}$, a measurable set by the previous paragraph together with the measurability of $\omega \mapsto Y_{\varsigma(\omega)}(\omega)$, which is 5 applied to the measurable index ς and the family Y_m of 6. On $\mathcal{B}^1$ we have $K \le \Upsilon$: indeed $\sigma_j \le \varsigma$ by 4, so ς is one of the indices in the supremum, and at $m = \varsigma$ the summand equals Y_ς by 6 — using $t_{\sigma_j} = r$ on $\mathcal{R}_j$, which is exactly

what makes the two displayed expressions agree — so $Y_\varsigma \le \Upsilon$, and $K \le Y_\varsigma$ on $\mathcal{B}^1$. The function Υ is nonnegative, being a supremum of absolute values, and μ-integrable by 8; and $K > 0$ by 2. Markov's inequality 7 therefore gives $\mu(\mathcal{B}^1)\,K \le \int \Upsilon\,d\mu \le C'(L_K + \sqrt{B_0})$ by 8. Dividing by $K > 0$ and using the choice of K from 2, which says $8C'\big((1+\log(K+10)) + \sqrt{B_0}\big) \le K$ and hence a fortiori $8C'(L_K + \sqrt{B_0}) \le K$ at the value L_K the travel bound is stated with,

$$\mu(\mathcal{B}^1) \le \frac{C'(L_K + \sqrt{B_0})}{K} \le \frac{1}{8}.$$

The second cutoff. Let

$$\ell(i) = \begin{cases} 0, & i < a, \\ \dfrac{i-a}{b-a}, & a \le i \le b, \\ 1, & i > b, \end{cases}$$

well defined since $b - a > 0$ by 1. Then $\ell \ge 0$, ℓ is bounded by 1, and ℓ is affine on $\{a, \dots, b\}$, so its second difference vanishes for $a < i < b$; hence $\frac{1}{2}(\ell(i+1) + \ell(i-1)) - \ell(i) = 0 \le -u(i)$ there with $u \equiv 0$. Apply 9 with $h = \ell$, $u \equiv 0$ and $A = \mathcal{R}_j$, whose two hypotheses — $\mathcal{R}_j \in \mathcal{M}$ and $\mathcal{R}_j \cap \{\sigma_j \le n\} \in \mathcal{G}_n$ — were verified above. For each n the empty-sum term drops and we get

$$\int \mathbf{1}_{\mathcal{R}_j}\,\ell\big(S_{(\sigma_j \vee n)\wedge\varsigma}\big)d\mu \le \int \mathbf{1}_{\mathcal{R}_j}\,\ell\big(S_{\sigma_j}\big)\,d\mu. \tag{710}$$

We bound the right side. On $\mathcal{R}_j$ we have $t_{\sigma_j} = r$, so $\epsilon S_{\sigma_j} = r = \epsilon c$ by 3 and 1; cancelling $\epsilon > 0$ gives $S_{\sigma_j} = c$. Hence, on $\mathcal{R}_j$, using $a \le c \le b$ — the first from 1, the second because $c \le 8c = b$ in the reals by $\epsilon b = 8r = 8\epsilon c$ and $\epsilon > 0$ —

$$\ell\big(S_{\sigma_j}\big) = \frac{c-a}{b-a} \le \frac{c}{b} = \frac{1}{8},$$

the inequality because $a \ge 0$ makes $(c-a)/(b-a) \le c/b$ whenever $0 \le a \le c \le b$ and $b > a$, and the last step by $b = 8c$. Since $\mathbf{1}_{\mathcal{R}_j} \le 1$ and the integrand is nonnegative, the right side of (710) is at most $\frac{1}{8}$.

For the left side, take n large enough that $(\sigma_j \vee n) \wedge \varsigma = \varsigma$ — which holds once $n \ge \varsigma$ at the given ω, since then $\sigma_j \vee n \ge n \ge \varsigma$ — and observe that on $\mathcal{R}_j \cap \{8r \le t_\varsigma\}$ we have $\epsilon S_\varsigma = t_\varsigma \ge 8r = \epsilon b$, so $S_\varsigma \ge b$ and $\ell(S_\varsigma) = 1$. As $\ell \ge 0$ everywhere, this gives the pointwise bound $\mathbf{1}_{\mathcal{R}_j \cap \{8r \le t_\varsigma\}} \le \mathbf{1}_{\mathcal{R}_j}\,\ell(S_\varsigma)$. Integrating and using (710) at such n together with the bound on its right side,

$$\mu\big(\mathcal{R}_j \cap \{8r \le t_\varsigma\}\big) \le \int \mathbf{1}_{\mathcal{R}_j}\,\ell\big(S_\varsigma\big)\,d\mu \le \frac{1}{8}.$$

This is the second half of (709), and the proof is complete. □

2.165. `LogGridSumBound`.

Lemma 2.165 (Step 6(b,c): the logarithmic grid sum along a stopped portion)**.** *Let $(\Omega, \mathcal{M}, \mu)$ be a probability space, let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a filtration of $\mathcal{M}$, let $S : \mathbb{N} \times \Omega \to \mathbb{Z}$, let $\sigma, \varsigma, \tau : \Omega \to \mathbb{N}$, let $a, b, q \in \mathbb{N}$, let $\epsilon', \Lambda \in \mathbb{R}$ — we write ϵ' for the quotient ϵ/r throughout — and let $w : \mathbb{Z} \to \mathbb{R}$. Assume:*

1. *$\epsilon' > 0$, $0 < b$, $4 \le a$, $0 \le a$ as a real number, and $b - a > 0$ as a real number;*
2. *$\epsilon'\,b = 8$;*
3. *$\Lambda = 32\sqrt{8} + 8(\log 8)^2$, and $\Lambda \ge 0$;*
4. *the lattice weight is*

$$w(i) = \begin{cases} (\epsilon')^2\big|\log(\epsilon' i)\big|^2, & a < i < b, \\ 0, & \text{otherwise}; \end{cases}$$

5. *σ is a $(\mathcal{G}_n)$-stopping time, and σ and ς are both $\mathcal{M}$-measurable as maps $\Omega \to \mathbb{N}$;*
6. *$\varsigma \le \tau$ pointwise and $\omega \mapsto \tau(\omega)$ is μ-integrable as a real-valued function;*
7. *$a \le S_n(\omega) \le q$ for all n and ω;*
8. *for every $(\mathcal{G}_n)$-stopping time $\zeta : \Omega \to \mathbb{N}$ and every n, $\{\zeta \le n\} \in \mathcal{G}_n$;*

(9) *evaluating a measurable family of real functions at a measurable random index is measurable, in both the one-index and the two-index form: if* $\zeta,\eta:\Omega\to\mathbb{N}$ *are measurable and each* G_m*, resp. each* $G_{n,m}$*, is measurable, then so are* $\omega\mapsto G_{\zeta(\omega)}(\omega)$ *and* $\omega\mapsto G_{\zeta(\omega),\eta(\omega)}(\omega)$*;*

(10) *every* $g:\mathbb{Z}\to\mathbb{R}$ *is bounded on* $\{i\in\mathbb{Z}:a\le i\le q\}$*;*

(11) *for every* $g:\mathbb{Z}\to\mathbb{R}$ *and every* k*, the function* $\omega\mapsto g(S_k(\omega))$ *is* $\mathcal{M}$*-measurable;*

(12) *the conclusion of Claim* T' *holds for* σ,ς *on the window* (a,b)*: for all* $h,u:\mathbb{Z}\to\mathbb{R}$ *and every* $A\in\mathcal{M}$ *with* $u\ge 0$ *everywhere, with* $\frac12(h(i+1)+h(i-1))-h(i)\le -u(i)$ *for* $a<i<b$*, and with* $A\cap\{\sigma\le n\}\in\mathcal{G}_n$ *for every* n*, one has for every* n

$$\int \mathbf{1}_A\Big(h\big(S_{(\sigma\vee n)\wedge\varsigma}\big)+\sum_{\sigma\le m<n\wedge\varsigma}u\big(S_m\big)\Big)d\mu\le\int\mathbf{1}_A\,h\big(S_\sigma\big)\,d\mu.$$

Then the function $\omega\mapsto\sum_{\sigma(\omega)\le m<\varsigma(\omega)}w(S_m(\omega))$ *is* μ*-integrable and*

$$\int\Big(\sum_{\sigma\le m<\varsigma}w\big(S_m\big)\Big)d\mu\le 4\Lambda. \tag{711}$$

Proof. Step 1: the weight is nonnegative and supported in the window. By 4, $w(i)=(\epsilon')^2|\log(\epsilon' i)|^2\ge 0$ when $a<i<b$, being a product of a square and a square, and $w(i)=0$ otherwise; so $w\ge 0$ everywhere and w vanishes off the open window.

Step 2: the total weight over the grid. We claim

$$\sum_{m=0}^{b-1}w(m)\le\epsilon'\,\Lambda. \tag{712}$$

Since $0<b$ by 1 we may write $b=b'+1$. For each $m\in\{0,\dots,b-1\}$ we bound $w(m)$ by

$$w(m)\le\big((\epsilon')^2\cdot 16\big)\frac{1}{\sqrt{\epsilon' m}}+(\epsilon')^2(\log 8)^2.$$

Indeed, if m is outside the window then $w(m)=0$ and the right side is nonnegative. If $a<m<b$ then $s:=\epsilon' m$ satisfies $s>0$, because $\epsilon'>0$ and $m>a\ge 4>0$, and $s<\epsilon' b=8$ by 2, so $0<s\le 8$; the second bound of `SqrtSumLogSquareBounds` gives $|\log s|^2\le 16/\sqrt{s}+(\log 8)^2$, and multiplying by $(\epsilon')^2>0$ is exactly the displayed inequality.

Summing over $m\in\{0,\dots,b-1\}$ and applying the first bound of `SqrtSumLogSquareBounds` with $a=\epsilon'$ and $M=b$ — which states $\sum_{m<b}1/\sqrt{\epsilon'(m+1)}\le 2\sqrt{\epsilon' b}/\epsilon'$ — together with $\epsilon' b=8$ from 2, the first group contributes at most $(\epsilon')^2\cdot 16\cdot 2\sqrt{8}/\epsilon'=32\epsilon'\sqrt{8}$, while the second contributes $b\,(\epsilon')^2(\log 8)^2=\epsilon'(\epsilon' b)(\log 8)^2=8\epsilon'(\log 8)^2$. Adding and using 3,

$$\sum_{m=0}^{b-1}w(m)\le 32\epsilon'\sqrt{8}+8\epsilon'(\log 8)^2=\epsilon'\big(32\sqrt{8}+8(\log 8)^2\big)=\epsilon'\Lambda,$$

which is (712).

Step 3: the Green function of the window. For $i,m\in\mathbb{Z}$ put

$$K(i,m)=\begin{cases}\dfrac{\big(\min(i,m)-a\big)\big(b-\max(i,m)\big)}{b-a}, & a\le i\le b,\\[2mm] 0, & \text{otherwise,}\end{cases}$$

which is legitimate since $b-a>0$ by 1. For $a\le m\le b$ the function $K(\cdot,m)$ is nonnegative on $\{a,\dots,b\}$, because $\min(i,m)\ge a$ and $\max(i,m)\le b$ there; it is bounded there by $(b-a)/4$, since $(\min-a)(b-\max)\le\big((b-a)/2\big)^2$; and, being affine in i on $\{a,\dots,m\}$ and on $\{m,\dots,b\}$ and vanishing at $i=a$ and $i=b$, its second difference in i vanishes for $a<i<b$ with $i\ne m$ and equals -1 at $i=m$. Explicitly, with $Q=b-a$ and $\alpha=m-a$,

$$\tfrac12\big(K(m-1,m)+K(m+1,m)\big)-K(m,m)=\frac{(\alpha-1)(Q-\alpha)+\alpha(Q-\alpha-1)}{2Q}-\frac{\alpha(Q-\alpha)}{Q}=-\frac12,$$

because $(\alpha-1)(Q-\alpha)+\alpha(Q-\alpha-1)=2\alpha(Q-\alpha)-Q$.

Now set $H(i)=\sum_m w(m)\,K(i,m)$, the sum running over the finitely many m with $a\le m\le b$. Then $H\ge 0$ on the window, by Step 1 and the nonnegativity of K; H is bounded by 4Λ, because by Step 2 and the bound $K\le (b-a)/4\le b/4$ we get $H(i)\le \big(\sum_m w(m)\big)\cdot\frac{b}{4}\le \epsilon'\Lambda\cdot\frac{b}{4}=\frac{(\epsilon' b)\Lambda}{4}=2\Lambda\le 4\Lambda$ using 2 and $\Lambda\ge 0$; and taking the second difference of H termwise, which is legitimate for a finite sum, the computation above gives

$$\tfrac{1}{2}\big(H(i+1)+H(i-1)\big)-H(i)=-w(i)\qquad (a<i<b), \tag{713}$$

since only the single term $m=i$ contributes $-\frac{1}{2}w(i)\cdot 2$. In particular H is a discrete supersolution for the defect w on the window, in the sense required by 12.

Step 4: measurability and integrability. By 10 fix C with $|w(i)|\le C$ for $a\le i\le q$; by 7 every value $S_m(\omega)$ lies in that range, so $|w(S_m(\omega))|\le C$ always. For each n the partial sum $\omega\mapsto\sum_{\sigma\le m<n\wedge\varsigma}w(S_m(\omega))$ is measurable: each $\omega\mapsto w(S_m(\omega))$ is measurable by 11, the summation range is determined by the measurable σ and ς of 5, and the passage to the random range is measurable by 9. The full sum $\omega\mapsto\sum_{\sigma\le m<\varsigma}w(S_m(\omega))$ is measurable for the same reason. Pointwise it is bounded by $|C|\,\tau(\omega)$, since the index set $[\sigma,\varsigma)$ is contained in $[0,\tau)$ by 6 and so has at most $\tau(\omega)$ elements, each contributing at most $|C|$. As τ is μ-integrable by 6, a measurable function dominated in absolute value by the integrable $|C|\tau$ is integrable; this is the first assertion.

Step 5: conclusion. Apply 12 with $h=H$, $u=w$ and $A=\Omega$. Its hypotheses hold: $w\ge 0$ by Step 1; the supersolution inequality is (713), an equality and so in particular a $\le$; $\Omega\in\mathcal{M}$; and $\Omega\cap\{\sigma\le n\}=\{\sigma\le n\}\in\mathcal{G}_n$ by 8, σ being a stopping time by 5. Since $\mathbf{1}_\Omega\equiv 1$, the conclusion reads, for every n,

$$\int\Big(H\big(S_{(\sigma\vee n)\wedge\varsigma}\big)+\sum_{\sigma\le m<n\wedge\varsigma}w\big(S_m\big)\Big)d\mu\le\int H\big(S_\sigma\big)\,d\mu\le 4\Lambda,$$

the last step because $H\le 4\Lambda$ pointwise by Step 3 and μ is a probability measure. Dropping the first summand, which is nonnegative because $H\ge 0$ on the window and H vanishes outside it, gives

$$\int\Big(\sum_{\sigma\le m<n\wedge\varsigma}w\big(S_m\big)\Big)d\mu\le 4\Lambda\qquad(n\in\mathbb{N}).$$

The integrands are nonnegative and nondecreasing in n, and for $n\ge\varsigma(\omega)$ the range $[\sigma,n\wedge\varsigma)$ equals $[\sigma,\varsigma)$, so they are eventually constant at each ω with limit $\sum_{\sigma\le m<\varsigma}w(S_m)$. Monotone convergence therefore yields (711). □

2.166. `MarkFrameTravelBound`.

Lemma 2.166 (The frame travel bound on a mark's portion)**.** *Let F be a finite-dimensional real inner-product space, let $(\Omega,\mathcal{M},\mu)$ be a measure space, let $(\mathcal{G}_n)_{n\in\mathbb{N}}$ be a filtration of $\mathcal{M}$, let $z,z^+,z^-:\mathbb{N}\times\Omega\to\ell^2(F\times\mathbb{R})$, let $Q_0:\mathbb{N}\times\Omega\to F$, let $S:\mathbb{N}\times\Omega\to\mathbb{Z}$, let $\sigma:\mathbb{N}\times\Omega\to\mathbb{N}$, let $p:\mathbb{N}\times\Omega\to F$, let $\varsigma,\tau:\Omega\to\mathbb{N}$, let $j,b\in\mathbb{N}$, let $\epsilon,\rho,C_p,B_0,C',C_M,C_2,L_K,\Lambda\in\mathbb{R}$ and let $u:\mathbb{Z}\to\mathbb{R}$. Write $z_n=(x_n,t_n)$ and, for a stopping time ϱ, a slope field q and a scale ρ,*

$$\mathcal{T}[\varrho,\varsigma,q]=\sup_{\varrho\le n\le\varsigma}\frac{\big|x_n-x_\varrho-(t_n-t_\varrho)\,q\big|}{\rho}$$

for the running displacement in the sheared frame of slope q and scale ρ. Assume:

(1) *the travel package of* `StoppedWalkTravel`*: for all $(\mathcal{G}_n)$-stopping times $\varrho\le\varsigma'\le\tau$, every $\rho'>0$, every $C\in\mathbb{R}$ and every $q:\Omega\to F$ with $|q|\le C$ such that $\mathbf{1}_{\{\varrho\le n\}}q$ is $\mathcal{G}_n$-measurable for all n, the four functions*

$$\mathcal{T}[\varrho,\varsigma',q],\qquad\sum_{\varrho\le n<\varsigma'}\Big(\frac{\epsilon}{\rho'}\Big)^2|Q_0(n)-q|^2,\qquad\Big(\frac{\epsilon}{\rho'}\Big)^2(\varsigma'-\varrho),\qquad\sum_{\varrho\le n<\varsigma'}\mathcal{E}_n$$

are μ-integrable, where $\mathcal{E}_n=|\epsilon^{-1}(x_n^+-x_n)-Q_0(n)|^2+|\epsilon^{-1}(x_n-x_n^-)-Q_0(n)|^2$, and

$$\int\mathcal{T}[\varrho,\varsigma',q]\,d\mu\le 2\Big(\int\sum_{\varrho\le n<\varsigma'}\tfrac{\epsilon^2}{\rho'^2}|Q_0(n)-q|^2d\mu\Big)^{1/2}+\Big(\int\tfrac{\epsilon^2}{\rho'^2}(\varsigma'-\varrho)\,d\mu\Big)^{1/2}\Big(\int\sum_{\varrho\le n<\varsigma'}\mathcal{E}_n\,d\mu\Big)^{1/2};$$

(2) $|p_j(\omega)| \le C_p$ *for every* ω, *and* $\mathbf{1}_{\{\sigma_j \le n\}} p_j$ *is* $\mathcal{G}_n$*-measurable for every* n;
(3) σ_j, ς *and* τ *are* $(\mathcal{G}_n)$*-stopping times with* $\sigma_j \le \varsigma \le \tau$ *pointwise;*
(4) *the portion length* $\omega \mapsto \varsigma(\omega) - \sigma_j(\omega)$ *and the portion weight sum* $\omega \mapsto \sum_{\sigma_j \le m < \varsigma} u(S_m)$ *are* μ*-integrable, and* $\displaystyle\int \sum_{\sigma_j \le m < \varsigma} u(S_m)\, d\mu \le 4\Lambda$;
(5) $u(S_n(\omega)) = \big(\frac{\epsilon}{\rho}\big)^2 \big|\log\big(\frac{\epsilon}{\rho} S_n(\omega)\big)\big|^2$ *whenever* $\sigma_j(\omega) \le n < \varsigma(\omega)$;
(6) *if* $t_{\sigma_j(\omega)}(\omega) = \rho$ *and* $\sigma_j(\omega) \le n < \varsigma(\omega)$ *then* $|Q_0(n)(\omega) - p_j(\omega)| \le C_2\big(L_K + \big|\log\big(\frac{\epsilon}{\rho} S_n(\omega)\big)\big|\big)$;
(7) *if* $t_{\sigma_j(\omega)}(\omega) \ne \rho$ *then* $\varsigma(\omega) = \sigma_j(\omega)$;
(8) $\displaystyle\int \big(\varsigma - \sigma_j\big) d\mu \le \frac{b^2}{4}$;
(9) $\displaystyle\int \sum_{n < \tau} \mathcal{E}_n\, d\mu \le B_0$;
(10) $0 < \rho$, $\dfrac{\epsilon}{\rho} b = 8$, $C_M = 2C_2^2(16 + 4\Lambda)$, $0 \le C_M$, $0 < C_2$, $0 \le \Lambda$, $2\sqrt{C_M} \le C'$, $4 \le C'$ *and* $1 \le L_K$.

Then $\mathcal{T}[\sigma_j, \varsigma, p_j]$ *is* μ*-integrable and*

$$\int \mathcal{T}[\sigma_j, \varsigma, p_j]\, d\mu \le C'\big(L_K + \sqrt{B_0}\big). \tag{714}$$

Proof. The two instantiations. Apply 1 at $\varrho = \sigma_j$, $\varsigma' = \varsigma$, $q = p_j$, $C = C_p$ and $\rho' = \rho$; its hypotheses are 3, 2 and $\rho > 0$ from 10. This gives the four integrability facts for the portion — the first of them is the integrability asserted in the conclusion — and the travel inequality

$$\int \mathcal{T}[\sigma_j, \varsigma, p_j]\, d\mu \le 2\sqrt{V} + \sqrt{T}\,\sqrt{E}, \tag{715}$$

where V, T and E are the three integrals on its right at this instantiation. Apply 1 a second time at $\varrho = 0$, $\varsigma' = \tau$, $q = 0$, $C = 0$ and $\rho' = 1$ — the zero map is trivially bounded by 0 and $\mathbf{1}_{\{0 \le n\}} 0$ is measurable — and keep only its fourth integrability fact, the integrability of $\omega \mapsto \sum_{n < \tau} \mathcal{E}_n$.

The velocity term: $V \le C_M L_K^2$. Fix ω. If $t_{\sigma_j(\omega)}(\omega) \ne \rho$ then $\varsigma(\omega) = \sigma_j(\omega)$ by 7, the summation range is empty and the summand bound below holds trivially, both sides being 0 on the left and nonnegative on the right. Otherwise, for $\sigma_j(\omega) \le m < \varsigma(\omega)$, 6 gives $|Q_0(m) - p_j| \le C_2(L_K + |\log(\frac{\epsilon}{\rho} S_m)|)$, so squaring and using $(a+b)^2 \le 2a^2 + 2b^2$,

$$|Q_0(m) - p_j|^2 \le C_2^2\Big(2L_K^2 + 2\big|\log\big(\tfrac{\epsilon}{\rho} S_m\big)\big|^2\Big),$$

and multiplying by $(\epsilon/\rho)^2 \ge 0$ and using 5,

$$\Big(\frac{\epsilon}{\rho}\Big)^2 |Q_0(m) - p_j|^2 \le 2C_2^2 L_K^2 \Big(\frac{\epsilon}{\rho}\Big)^2 + 2C_2^2\, u(S_m).$$

Summing over $\sigma_j(\omega) \le m < \varsigma(\omega)$ and splitting the constant term off,

$$\sum_{\sigma_j \le m < \varsigma} \Big(\frac{\epsilon}{\rho}\Big)^2 |Q_0(m) - p_j|^2 \le 2C_2^2 L_K^2 \Big(\frac{\epsilon}{\rho}\Big)^2 (\varsigma - \sigma_j) + 2C_2^2 \sum_{\sigma_j \le m < \varsigma} u(S_m).$$

Both sides are integrable — the left by the first instantiation, the right by 4 — so integrating preserves the inequality. For the first term, 8 and $(\epsilon/\rho)^2 \ge 0$ give $(\epsilon/\rho)^2 \int (\varsigma - \sigma_j)\, d\mu \le (\epsilon/\rho)^2 b^2/4$, which is 16 because $\big((\epsilon/\rho) b\big)^2 = 64$ by 10; for the second, 4 gives at most $2C_2^2 \cdot 4\Lambda$. Hence, using $1 \le L_K^2$ (from $1 \le L_K$) to absorb the second term into a multiple of L_K^2 and $0 \le 8C_2^2\Lambda$,

$$V \le 2C_2^2 L_K^2 \cdot 16 + 8C_2^2\Lambda \le 2C_2^2(16 + 4\Lambda) L_K^2 = C_M L_K^2$$

by the definition of C_M in 10.

The error term: $E \le B_0$. Pointwise, the portion's index range $[\sigma_j, \varsigma)$ is contained in $[0, \tau)$ by 3, and each summand $\mathcal{E}_n$ is nonnegative, so $\sum_{\sigma_j \le n < \varsigma} \mathcal{E}_n \le \sum_{n < \tau} \mathcal{E}_n$. Both sides are integrable — the left by the first instantiation, the right by the second — so $E \le \int \sum_{n < \tau} \mathcal{E}_n\, d\mu \le B_0$ by 9.

The time term: $T \le 16$. Since $\sigma_j \le \varsigma$ pointwise, the real-valued difference $\varsigma - \sigma_j$ agrees with the cast of the natural subtraction, so $T = (\epsilon/\rho)^2 \int (\varsigma - \sigma_j)\, d\mu \le (\epsilon/\rho)^2 b^2/4 = 16$ exactly as in the velocity term.

Assembly. The square root is monotone, so $\sqrt{V} \le \sqrt{C_M L_K^2} = \sqrt{C_M}\, L_K$ — using $L_K \ge 1 > 0$ — $\sqrt{T} \le \sqrt{16} = 4$ and $\sqrt{E} \le \sqrt{B_0}$. Since $\sqrt{T} \ge 0$ and $\sqrt{E} \ge 0$, the product is bounded by $4\sqrt{B_0}$. Substituting into (715),

$$\int \mathcal{T}[\sigma_j, \varsigma, p_j]\, d\mu \le 2\sqrt{C_M}\, L_K + 4\sqrt{B_0} \le C' L_K + C'\sqrt{B_0} = C'\big(L_K + \sqrt{B_0}\big),$$

using $2\sqrt{C_M} \le C'$ with $L_K \ge 0$ and $4 \le C'$ with $\sqrt{B_0} \ge 0$. This is (714). □

2.167. `MarkLocalization`.

Lemma 2.167 (Marks and localization in fitted frames)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$, let $\kappa > 0$ and $H \in \mathbb{R}$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of* ***`IsFitClass`*** *be centered at $z_* = (0,1)$ with constant C_0 in the sense of* ***`IsCenteredAt`*** *and satisfy the horizontal Jensen bound with constant κ at every time in the sense of* ***`HasJensenBound`****. Let W be the contact envelope of* ***`ContactEnv`*** *formed from v and κ, and assume the six standing properties of the extremal contact pair: W is continuous on $\{t > 0\}$ with subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of* ***`SubquadraticGrowth`****; W is a viscosity subsolution of the limiting equation there in the sense of* ***`IsLimitSubOn`****; $v - \kappa^2 t/2 \le W \le v$ for $t > 0$ and $W(z_*) = v(z_*) = 0$; $x \mapsto W(x,t) - |x|^2/(2t)$ is concave for every $t > 0$; the barycentric gap of* ***`BaryEnv`*** *is $\kappa^2 t/2$ at every contact; and the envelope Hamiltonian of* ***`EnvHamiltonian`*** *satisfies $H_v(z_*) = H$ with $\mathcal{S}^+_\ell W(z_*) \to H$ as $\ell \downarrow 0$, the forward envelope slope being that of* ***`EnvSlopeP`****. Let $C_* > 0$. Write $\delta_j = 2^{-j}$ and $L(\Theta) = 1 + \log\Theta$.*

Then there are an integer $T \ge 8$ and reals $R \ge 2$, $K \ge 2$ and $C_\Gamma > 0$ with the following property. Let $N \ge 1$, let $\epsilon > 0$ and let $(\Omega, \mathcal{M}, \mu)$, $(\mathcal{G}_n)$, ξ, z, z^+, z^-, Q_0, τ be a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ in the sense of ***`IsContactWalk`****. Write $z_n = (x_n, t_n)$, $B = R + T + 2$,*

$$Q_+(n) = \frac{x_n^+ - x_n}{\epsilon}, \qquad Q_-(n) = \frac{x_n - x_n^-}{\epsilon}, \qquad \mathcal{E}_n = |Q_+(n) - Q_0(n)|^2 + |Q_-(n) - Q_0(n)|^2,$$

and assume the error budget

$$\mathbb{E} \sum_{n<\tau} \mathcal{E}_n \le C_* L(B)^4. \tag{716}$$

Let (σ_j, p_j, a_j) be marks with fits for $(C_0, v, N, \mathcal{G}, z, \tau)$ in the sense of ***`IsMarkData`****. Then for every j with $1 \le j \le N$, writing $\Gamma_j = C_\Gamma(L(B) + j)$:*

(i) *$|p_j| + |a_j| \le \Gamma_j$ pointwise, and $\Gamma_j \ge 1$;*
(ii) *the localization event*

$$\mathcal{L}_j = \Big\{ t_\tau = \delta_N \text{ and } \big| x_\tau - x_{\sigma_j} - (t_\tau - \delta_j) p_j \big| \le K \delta_j \Big\}$$

belongs to $\mathcal{M}$ and, uniformly in j and N,

$$\mathcal{L}_j \in \mathcal{M}, \qquad \mathbb{P}\big(t_\tau = \delta_N \quad \text{and} \quad \big| x_\tau - x_{\sigma_j} - (t_\tau - \delta_j) p_j \big| \le K \delta_j\big) \ge \frac{1}{2}. \tag{717}$$

Proof. Throughout, W is the contact envelope of `ContactEnv`, states are written $z_n = (x_n, t_n)$, and $\delta_j = 2^{-j}$.

Step 0: the constants. Let $C_f > 0$ be the constant of `FitParamBound` for this C_0, let $C_2 > 0$ be the constant of `DPLogVelocity` for this C_0 and κ, and put

$$M = 8, \qquad C_\Gamma = \max\{C_f, 1\} > 0.$$

Apply `StoppedWalkTravel` with the present standing data and this C_*: it supplies an integer $T \ge 8$ and a real $R \ge 2$ such that every stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon)$ with

$N \geq 1$ satisfying (716) obeys both of its conclusions, namely the travel inequality in the portion generality and

$$\mathbb{P}\big(t_\tau = \delta_N\big) \geq \frac{3}{4}. \tag{718}$$

Fix these T, R and put $B = R + T + 2 \geq 12$, so $L(B) = 1 + \log B > 1$. The constant $K \geq 2$ is fixed in Step 7; everything before that is proved for every $K \geq 2$ simultaneously, with constants not depending on K, N or j.

Now let $N \geq 1$, let ϵ and the walk be as in the statement, let (σ_j, p_j, a_j) be marks with fits, and fix j with $1 \leq j \leq N$. Put $r = \delta_j$ and

$$\mathcal{R}_j = \{t_{\sigma_j} = r\}, \qquad \epsilon' = \frac{\epsilon}{r}, \qquad y_n = \frac{x_n - x_{\sigma_j} - (t_n - r)p_j}{r}, \qquad s_n = \frac{t_n}{r}.$$

By `IsMarkData`, clause 5, each $\mathbf{1}_{\{\sigma_j \leq n\}} p_j$ is $\mathcal{G}_n$-measurable, so $p_j = \sum_{n \geq 0} \mathbf{1}_{\{\sigma_j = n\}} \mathbf{1}_{\{\sigma_j \leq n\}} p_j$ is $\mathcal{M}$-measurable, and likewise a_j; also z_{σ_j} and z_τ are measurable, because σ_j and τ are stopping times and each z_n is $\mathcal{G}_n$-measurable by `IsContactWalk`, clauses 4 and 9. Hence $\mathcal{R}_j$ and the event in (717) are measurable, and all expectations below are of measurable functions.

Step 1: four grid facts, from `IsContactWalk`. By `IsContactWalk`, clause 2, $\epsilon = 2^{-k}$ for some nonnegative integer k and $0 < \epsilon \leq \delta_N/4 = 2^{-N-2}$; hence $k \geq N + 2$.

(a) Every t_n lies on the grid $\epsilon\mathbb{Z}$, and $\delta_i \in \epsilon\mathbb{Z}$ for $i \leq N$. Indeed $\delta_i = 2^{-i} = 2^{k-i}\epsilon$ with $k - i \geq 2$ a nonnegative integer, and $1 = 2^k\epsilon$. For the states: $t_0 = 1$ by clause 5, and by clause 7 together with `IsStepPair` — whose two steps at a live state have time coordinates $t_n + \epsilon$ and $t_n - \epsilon$ — we have

$$t_{n+1} = t_n + \epsilon\xi_n \ \text{ on } \{n < \tau\}, \qquad t_{n+1} = t_n \ \text{ on } \{n \geq \tau\}, \tag{719}$$

with $\xi_n \in \{1, -1\}$ by clause 3. Induction on n gives $t_n \in \epsilon\mathbb{Z}$. Consequently $S_n := t_n/\epsilon$ is an integer for every n, with $S_{n+1} = S_n + \xi_n$ on $\{n < \tau\}$.

(b) $t_{\sigma_j} \geq \delta_j$, with equality on $\{t_\tau = \delta_N\}$. Suppose first $t_{\sigma_j} < \delta_j$. Since $j \geq 1$ we have $t_0 = 1 > \frac{1}{2} \geq \delta_j$, so there is a least index $n_1 \leq \sigma_j$ with $t_{n_1} \leq \delta_j$, and $n_1 \geq 1$ with $t_{n_1 - 1} > \delta_j$. If $n_1 - 1 \geq \tau$ then $t_{n_1} = t_{n_1 - 1} > \delta_j$ by (719), contradicting $t_{n_1} \leq \delta_j$. So $n_1 - 1 < \tau$ and $t_{n_1} \geq t_{n_1 - 1} - \epsilon > \delta_j - \epsilon$ by (719); as t_{n_1} and δ_j both lie on $\epsilon\mathbb{Z}$ by (a), the two inequalities $\delta_j - \epsilon < t_{n_1} \leq \delta_j$ force $t_{n_1} = \delta_j$. Hence $n_1 < \sigma_j$ would contradict `IsMarkData`, clause 3, which gives $t_n \neq \delta_j$ for $n < \sigma_j$; so $n_1 = \sigma_j$ and $t_{\sigma_j} = \delta_j$, contradicting $t_{\sigma_j} < \delta_j$. Therefore $t_{\sigma_j} \geq \delta_j$.

On $\{t_\tau = \delta_N\}$ we have $t_\tau = \delta_N \leq \delta_j$, since $j \leq N$. By `IsMarkData`, clause 4, either $t_{\sigma_j} = \delta_j$ — which is the assertion — or $\sigma_j = \tau$, in which case $t_{\sigma_j} = t_\tau \leq \delta_j$ and the inequality just proved gives $t_{\sigma_j} = \delta_j$ as well. So

$$\{t_\tau = \delta_N\} \subseteq \mathcal{R}_j. \tag{720}$$

(c) $\epsilon' \leq 1/4$, and ϵ' is an inverse power of two dividing 1 and M. Since $j \leq N$ we have $\delta_N \leq \delta_j = r$, so $\epsilon' = \epsilon/r \leq \epsilon/\delta_N \leq 1/4$; and $\epsilon' = 2^{-k}/2^{-j} = 2^{j-k}$ with $k - j \geq 2$, so $\epsilon' = 2^{j-k}$ divides 1 and divides the integer M.

(d) Before any step of the portion the primed time exceeds $4\epsilon'$. If $n < \tau$ then $t_n > \delta_N$ by `IsContactWalk`, clause 10, so

$$s_n = \frac{t_n}{r} > \frac{\delta_N}{r} \geq \frac{4\epsilon}{r} = 4\epsilon', \tag{721}$$

using $\delta_N \geq 4\epsilon$ from clause 2.

Step 2: the fit-parameter bound, i.e. (i). On $\mathcal{R}_j$, `IsMarkData`, clause 6, gives that (p_j, a_j) is a fit for v at $z_{\sigma_j} = (x_{\sigma_j}, r)$ of radius r with constant C_0, in the sense of `IsFit`. By `IsContactWalk`, clause 12, $|x_{\sigma_j}| \leq R + 1 \leq B$, and $|r| = \delta_j \leq 1 \leq B$, while $B \geq 2$ and $r > 0$. Hence `FitParamBound`, applied to v with the centering point $(0, 1)$ — the second alternative of its hypothesis — the box size $K' = B$ and the radius r, gives

$$|p_j| + |a_j| \leq C_f\big(L(B) + |\log r|\big) = C_f\big(L(B) + j\log 2\big) \leq C_f\big(L(B) + j\big) \leq \Gamma_j$$

on $\mathcal{R}_j$, using $\log 2 < 1$ and $C_f \le C_\Gamma$. Off $\mathcal{R}_j$, clause 7 of `IsMarkData` gives $p_j = 0$ and $a_j = 0$, so the bound holds there too. Finally $L(B) > 1$ and $j \ge 1$ give $\Gamma_j = C_\Gamma(L(B) + j) \ge 1 \cdot 2 > 1$, so $\Gamma_j \ge 1$. This is (i).

Step 3: the fitted frame. Fix $\omega \in \mathcal{R}_j$ and write $p_0 = p_j(\omega)$, $a_0 = a_j(\omega)$, $x_0 = x_{\sigma_j}(\omega)$, so that (p_0, a_0) is a fit for v at (x_0, r) of radius r and $r > 0$. Let

$$\bar v_\omega(y,s) = \frac{v(x,t) - v(x_0,r)}{r} - p_0 \cdot y - \Big(a_0 + \frac{|p_0|^2}{2}\Big)(s-1), \qquad x = x_0 + r\big(y + p_0(s-1)\big), \quad t = rs,$$

be the contact frame of `FrameChange` based at (x_0, r) with scale r and parameters (p_0, a_0), preceded by the translation by $z_* = (0,1)$, and let $\bar W_\omega$ be the contact envelope of `ContactEnv` formed from $\bar v_\omega$ and the same κ. By `FrameEnvelopeCovariance`, whose hypotheses $1 \le C_0$, $0 \le \kappa$, $v \in \mathcal{A}_{m,C_0}$, the Jensen bound, $r > 0$ and the fit are all available, its first three conclusions give

(722)
$$\bar v_\omega \in \mathcal{A}_{m,C_0}, \quad \bar v_\omega \text{ is centered at } (0,1) \text{ with constant } C_0, \quad \bar v_\omega \text{ has the Jensen bound with the same } \kappa.$$

These are exactly the standing hypotheses carried by v, so every statement below that was quoted for a centered member of the class with a Jensen bound applies to $\bar v_\omega$ and $\bar W_\omega$ with the same constants; this is the only place the frame enters the constants, and it enters not at all.

Let n satisfy $\sigma_j(\omega) \le n < \tau(\omega)$. Then $t_n > \delta_N \ge 4\epsilon > \epsilon$ by Step 1(d), the pair (z_n^+, z_n^-) is an optimizing step pair of size ϵ at z_n for v and κ by `IsContactWalk`, clause 6, and $\mathcal{P}_\epsilon W(\cdot, t_n)$ has gradient $Q_0(n)$ at x_n by clause 8. The images of z_n, $z_n^\pm$ under the inverse frame map $(x', t') \mapsto \big((x' - x_0 - (t'-r)p_0)/r,\ t'/r\big)$ are $\zeta_n = (y_n, s_n)$ and $\zeta_n^\pm$. So `FrameVelocityCovariance` applies at this state and gives: the pair (ζ_n^+, ζ_n^-) is an optimizing step pair of size ϵ' at ζ_n for $\bar v_\omega$ and κ; the primed velocities are

$$Q'_\pm(n) = Q_\pm(n) - p_0, \qquad Q'_0(n) = Q_0(n) - p_0 = \nabla_y \mathcal{P}_{\epsilon'} \bar W_\omega(y_n, s_n); \tag{723}$$

and the discrepancies are unchanged,

$$\big|Q'_+(n) - Q'_0(n)\big| = \big|Q_+(n) - Q_0(n)\big|, \qquad \big|Q'_-(n) - Q'_0(n)\big| = \big|Q_-(n) - Q_0(n)\big|. \tag{724}$$

No contact of W with v at the mark is used, and the fit and the frame are determined by $\mathcal{G}_{\sigma_j}$-measurable data, hence fixed once the mark is reached.

Step 4: the portion and its stopping index. Let $K \ge 2$. Define $\tau'_j(\omega)$, for $\omega \in \mathcal{R}_j$, to be the least $n \ge \sigma_j(\omega)$ with

$$n = \tau(\omega) \quad \text{or} \quad s_n(\omega) \ge M \quad \text{or} \quad |y_n(\omega)| \ge K,$$

which exists because $n = \tau(\omega) \ge \sigma_j(\omega)$ qualifies; and put $\tau'_j = \sigma_j$ off $\mathcal{R}_j$. Then

$$\sigma_j \le \tau'_j \le \tau, \tag{725}$$

and for $\sigma_j \le n < \tau'_j$ we have $n < \tau$, $s_n < M$ and $|y_n| < K$. Moreover τ'_j is a stopping time: $\{\tau'_j \le n\}$ is a finite union, over $i \le n$, of intersections of $\{\sigma_j = i\}$ with events determined by $z_0, \dots, z_n$ and by $\mathbf{1}_{\{\sigma_j \le i\}} p_j$, all of which lie in $\mathcal{G}_n$ by `IsContactWalk`, clause 4, and `IsMarkData`, clauses 1 and 5. Off $\mathcal{R}_j$ the portion is empty: $\tau'_j = \sigma_j$, and there $p_j = 0$, so $y_{\sigma_j} = 0$ and every sum over $\sigma_j \le n < \tau'_j$ is empty. On $\mathcal{R}_j$ we have $t_{\sigma_j} = r$, hence $y_{\sigma_j} = 0$ as well. In all cases

$$y_{\sigma_j} = 0, \qquad y_n = \frac{x_n - x_{\sigma_j} - (t_n - t_{\sigma_j})p_j}{r} \quad (\sigma_j \le n \le \tau'_j), \tag{726}$$

the second identity because on $\mathcal{R}_j$ we may replace r by t_{σ_j} and off $\mathcal{R}_j$ we have $p_j = 0$.

Step 5: the frame error budget. By (724), for $\sigma_j \le n < \tau'_j$ the primed squared discrepancy equals $\mathcal{E}_n$. Since $\{n : \sigma_j \le n < \tau'_j\} \subseteq \{n : n < \tau\}$ by (725) and the summands are nonnegative, we get pointwise $\sum_{\sigma_j \le n < \tau'_j} \mathcal{E}_n \le \sum_{n<\tau} \mathcal{E}_n$, and on an empty portion the left side is 0. Taking expectations and using (716),

$$\mathbb{E} \sum_{\sigma_j \le n < \tau'_j} \big(|Q'_+(n) - Q'_0(n)|^2 + |Q'_-(n) - Q'_0(n)|^2\big) = \mathbb{E} \sum_{\sigma_j \le n < \tau'_j} \mathcal{E}_n \le B_0 := C_* L(B)^4. \tag{727}$$

This is an unconditional estimate; it asserts nothing after conditioning on the history at the mark.

Claim T' (the conditional time walk). Put $a = \delta_N/\epsilon$ and $b = M/\epsilon'$, which are integers by Step 1(a) and (c), and note $0 \le a$. Let $h : \mathbb{Z} \to \mathbb{R}$ vanish outside $\{a, \dots, b\}$ and be nonnegative there, and let $u : \mathbb{Z} \to [0, \infty)$ satisfy

$$\tfrac{1}{2}\big(h(i+1) + h(i-1)\big) - h(i) = -u(i) \qquad (a < i < b). \tag{728}$$

Then

$$\mathbb{E}\Big[\sum_{\sigma_j \le n < \tau_j'} u(S_n) \;\Big|\; \mathcal{G}_{\sigma_j} \Big] \le h(S_{\sigma_j}) \qquad \text{a.e.}$$

Moreover, if $\ell : \mathbb{Z} \to \mathbb{R}$ is bounded and satisfies $\frac{1}{2}(\ell(i+1) + \ell(i-1)) = \ell(i)$ for $a < i < b$, then $\mathbb{E}[\ell(S_{\tau_j'}) \mid \mathcal{G}_{\sigma_j}] = \ell(S_{\sigma_j})$ almost everywhere.

Proof of Claim T'. Write $\rho = \sigma_j$ and $\varsigma = \tau_j'$. We first record that

$$a < S_n < b \qquad (\rho \le n < \varsigma). \tag{729}$$

Indeed such an n has $n < \tau$ by Step 4, so $t_n > \delta_N$ by `IsContactWalk`, clause 10, i.e. $S_n > a$; and $s_n < M$ by Step 4, i.e. $S_n = s_n/\epsilon' < M/\epsilon' = b$. (If the portion is empty, (729) is vacuous.)

For $n \ge 0$ put

$$Y_n = h\big(S_{(\rho \vee n) \wedge \varsigma}\big) + \sum_{\rho \le m < n \wedge \varsigma} u(S_m),$$

a bounded function for each n, since h and u are bounded on the finite set $\{a, \dots, b\}$ and vanish or are irrelevant elsewhere, and the sum has at most n terms.

Define

$$\mathcal{G}_\rho = \big\{ A \subseteq \Omega : A \cap \{\rho \le n\} \in \mathcal{G}_n \text{ for every } n \big\},$$

the σ-algebra of events known at the stopping time ρ. It is a σ-algebra: it contains Ω because $\{\rho \le n\} \in \mathcal{G}_n$, ρ being a stopping time; it is closed under complements because $A^{\mathrm{c}} \cap \{\rho \le n\} = \{\rho \le n\} \setminus (A \cap \{\rho \le n\})$; and it is closed under countable unions because intersection distributes over them. It is contained in $\mathcal{M}$, since $A = \bigcup_n (A \cap \{\rho \le n\})$ and each $A \cap \{\rho \le n\} \in \mathcal{G}_n \subseteq \mathcal{M}$. The only property of $\mathcal{G}_\rho$ used below is the defining one.

Let $A \in \mathcal{G}_\rho$. We claim $\int_A Y_{n+1}\, d\mu = \int_A Y_n\, d\mu$ for every n. Split A into three measurable pieces.

On $\{\rho > n\}$: then $\rho \ge n+1$, so $(\rho \vee n) \wedge \varsigma = \rho \wedge \varsigma = \rho$ and $(\rho \vee (n+1)) \wedge \varsigma = \rho$ as well, using $\rho \le \varsigma$; both sums are empty. Hence $Y_{n+1} = Y_n = h(S_\rho)$ there.

On $\{\rho \le n\} \cap \{\varsigma \le n\}$: then $(\rho \vee n) \wedge \varsigma = \varsigma$ and $(\rho \vee (n+1)) \wedge \varsigma = \varsigma$, and both sums equal $\sum_{\rho \le m < \varsigma} u(S_m)$. Hence $Y_{n+1} = Y_n$ there.

On $E_n := A \cap \{\rho \le n < \varsigma\}$: here $(\rho \vee n) \wedge \varsigma = n$ and $(\rho \vee (n+1)) \wedge \varsigma = n+1$, and the sum gains exactly the term $u(S_n)$, so

$$Y_{n+1} - Y_n = h(S_{n+1}) - h(S_n) + u(S_n).$$

The event E_n lies in $\mathcal{G}_n$: $A \cap \{\rho \le n\} \in \mathcal{G}_n$ by the definition of $\mathcal{G}_\rho$, and $\{n < \varsigma\}$ is the complement of $\{\varsigma \le n\} \in \mathcal{G}_n$, ς being a stopping time by Step 4. On E_n we have $n < \tau$, so $S_{n+1} = S_n + \xi_n$ with $\xi_n \in \{1, -1\}$, and $a < S_n < b$ by (729); writing

$$h(S_{n+1}) = \frac{1+\xi_n}{2} h(S_n{+}1) + \frac{1-\xi_n}{2} h(S_n{-}1) = \frac{h(S_n+1) + h(S_n-1)}{2} + \frac{\xi_n}{2}\big(h(S_n{+}1) - h(S_n{-}1)\big),$$

the coefficient of ξ_n is bounded and $\mathcal{G}_n$-measurable, so the pull-out property of the conditional expectation and $\mathbb{E}[\xi_n \mid \mathcal{G}_n] = 0$ (`IsContactWalk`, clause 3) give $\int_{E_n} h(S_{n+1})\, d\mu = \int_{E_n} \frac{1}{2}(h(S_n + 1) + h(S_n - 1))\, d\mu = \int_{E_n} (h(S_n) - u(S_n))\, d\mu$ by (728). Hence $\int_{E_n} (Y_{n+1} - Y_n)\, d\mu = 0$, which proves the claim.

Therefore $\int_A Y_n\, d\mu = \int_A Y_0\, d\mu = \int_A h(S_\rho)\, d\mu$ for every n, since $Y_0 = h(S_{\rho \wedge \varsigma}) = h(S_\rho)$ and the sum in Y_0 is empty. As $h \ge 0$ on the range (729) and h vanishes elsewhere, dropping the h term

gives $\int_A \sum_{\rho\le m<n\wedge\varsigma} u(S_m)\,d\mu \le \int_A h(S_\rho)\,d\mu$. The sums are nonnegative and nondecreasing in n, with pointwise limit $X := \sum_{\rho\le m<\varsigma} u(S_m)$ because ς is finite, so monotone convergence gives

$$\int_A X\,d\mu \le \int_A h(S_\rho)\,d\mu \qquad (A \in \mathcal{G}_\rho). \tag{730}$$

Taking $A = \Omega$ shows X is integrable, since it is nonnegative, measurable and has finite integral. Now both $\mathbb{E}[X \mid \mathcal{G}_\rho]$ and $h(S_\rho)$ are $\mathcal{G}_\rho$-measurable and integrable, and (730) says $\int_A \mathbb{E}[X \mid \mathcal{G}_\rho]\,d\mu \le \int_A h(S_\rho)\,d\mu$ for every $A \in \mathcal{G}_\rho$; choosing $A = \{\mathbb{E}[X \mid \mathcal{G}_\rho] > h(S_\rho)\}$ forces that event to be null, so $\mathbb{E}[X \mid \mathcal{G}_\rho] \le h(S_\rho)$ a.e. For the last assertion, run the same computation with $u = 0$ and $h = \ell$: the three-piece argument used neither $\ell \ge 0$ nor the vanishing of ℓ outside $\{a,\dots,b\}$, only the mean-value identity on (a,b) and (729), so $\int_A \ell(S_{(\rho\vee n)\wedge\varsigma})\,d\mu = \int_A \ell(S_\rho)\,d\mu$ for every n; since ς is finite, $(\rho\vee n)\wedge\varsigma \to \varsigma$ pointwise, and ℓ is bounded while μ is a probability measure by `IsContactWalk`, clause 1, so dominated convergence gives $\int_A \ell(S_\varsigma)\,d\mu = \int_A \ell(S_\rho)\,d\mu$ for every $A \in \mathcal{G}_\rho$, i.e. the stated conditional identity. This proves Claim T′.

Step 6: the conditional time and velocity budgets. We use three instances of Claim T′. Note first that $\mathcal{R}_j \in \mathcal{G}_{\sigma_j}$ in the sense of Claim T′: for every n,

$$\mathcal{R}_j \cap \{\sigma_j \le n\} = \bigcup_{i\le n} \big(\{\sigma_j = i\} \cap \{t_i = \delta_j\}\big),$$

and each $\{\sigma_j = i\}$ lies in $\mathcal{G}_i \subseteq \mathcal{G}_n$ because σ_j is a stopping time by `IsMarkData`, clause 1, while $\{t_i = \delta_j\} \in \mathcal{G}_i \subseteq \mathcal{G}_n$ because z_i is $\mathcal{G}_i$-measurable by `IsContactWalk`, clause 4. We also note that $S_{\sigma_j} = r/\epsilon = 1/\epsilon'$ on $\mathcal{R}_j$, while off $\mathcal{R}_j$ the portion is empty so every sum below vanishes there; since $\mathcal{R}_j$ is $\mathcal{G}_{\sigma_j}$-measurable we may therefore insert the indicator $\mathbf{1}_{\mathcal{R}_j}$ on the right of each conditional bound.

(a) The number of steps. Take $h_1(i) = (b-i)(i-a)$ on $\{a,\dots,b\}$ and 0 outside, and $u = \mathbf{1}_{(a,b)}$. Then $h_1 \ge 0$ on $\{a,\dots,b\}$ and, for $a < i < b$, writing $h_1(i) = -i^2 + (a+b)i - ab$, the second difference of $-i^2$ is -2, so $\frac12(h_1(i+1)+h_1(i-1)) - h_1(i) = -1 = -u(i)$. By (729) the summand $u(S_n)$ equals 1 for every n in the portion, so Claim T′ gives, with $n' = \tau'_j - \sigma_j$ the number of steps of the portion and $S_{\sigma_j} = 1/\epsilon'$ on $\mathcal{R}_j$,

$$\mathbb{E}\big[n' \mid \mathcal{G}_{\sigma_j}\big] \le \mathbf{1}_{\mathcal{R}_j}\, h_1(1/\epsilon') = \mathbf{1}_{\mathcal{R}_j}\Big(\frac{M}{\epsilon'} - \frac{1}{\epsilon'}\Big)\Big(\frac{1}{\epsilon'} - a\Big) \le \mathbf{1}_{\mathcal{R}_j}\frac{M}{(\epsilon')^2},$$

using $a \ge 0$ and $M/\epsilon' - 1/\epsilon' \le M/\epsilon'$. Hence

$$\mathbb{E}\big[(\epsilon')^2 n' \mid \mathcal{G}_{\sigma_j}\big] \le M\,\mathbf{1}_{\mathcal{R}_j} \qquad \text{a.e.} \tag{731}$$

(b) Visit counts. Fix an integer i_0 with $a < i_0 < b$ and take

$$G_{i_0}(i) = \frac{2\min(i-a,\, i_0-a)\big(b - \max(i, i_0)\big)}{b-a}$$

on $\{a,\dots,b\}$ and 0 outside. Then $G_{i_0} \ge 0$, $G_{i_0}(a) = G_{i_0}(b) = 0$, and G_{i_0} is affine in i on $\{a,\dots,i_0\}$ and on $\{i_0,\dots,b\}$, so its second difference vanishes for $a < i < b$, $i \ne i_0$, while at $i = i_0$, writing $q = b-a$, $\alpha = i_0 - a$,

$$\tfrac12\big(G_{i_0}(i_0-1) + G_{i_0}(i_0+1)\big) - G_{i_0}(i_0) = \frac{(\alpha-1)(q-\alpha) + \alpha(q-\alpha-1)}{q} - \frac{2\alpha(q-\alpha)}{q} = -1,$$

because $(\alpha-1)(q-\alpha) + \alpha(q-\alpha-1) = 2\alpha(q-\alpha) - q$. So (728) holds with $u = \mathbf{1}_{\{i_0\}}$, and Claim T′ gives, with $V_{i_0} = \#\{n : \sigma_j \le n < \tau'_j,\ S_n = i_0\}$,

$$\mathbb{E}\big[V_{i_0} \mid \mathcal{G}_{\sigma_j}\big] \le \mathbf{1}_{\mathcal{R}_j} G_{i_0}(1/\epsilon') \le \mathbf{1}_{\mathcal{R}_j}\frac{b-a}{2} \le \mathbf{1}_{\mathcal{R}_j}\frac{M}{2\epsilon'}, \tag{732}$$

the middle step because $\min(i-a,\alpha)(b-\max(i,i_0)) \le \alpha(q-\alpha) \le q^2/4$ for every i, and the last because $b - a \le b = M/\epsilon'$.

(c) The velocity budget. Let $\sigma_j \le n < \tau'_j$ and $\omega \in \mathcal{R}_j$. Then $|y_n| < K \le K + M + 2$ and $0 < s_n < M \le K + M + 2$ by Step 4, $0 < \epsilon' \le 1/4 \le 1$ by Step 1(c), $\epsilon' < s_n$ by (721), and $K + M + 2 \ge 2$. By (722) the function $\bar v_\omega$ is a member of $\mathcal{A}_{m,C_0}$ centered at $(0,1)$ with the

Jensen bound at the same κ, and by (723) the vector $Q_0'(n)$ is the gradient of $\mathcal{P}_{\epsilon'}\bar{W}_\omega(\cdot, s_n)$ at y_n. Hence DPLogVelocity, applied to $\bar{v}_\omega$ with the box size $K+M+2$, gives

$$|Q_0'(n)| \le C_2\big(L(K+M+2) + |\log s_n|\big), \tag{733}$$

with C_2 the constant of DPLogVelocity, which depends only on C_0, $\dim F$ and κ — in particular not on the frame, which is why the bound is uniform in j, N and ω.

From $|\alpha+\beta|^2 \le 2|\alpha|^2+2|\beta|^2$ and (733), $(\epsilon')^2|Q_0'(n)|^2 \le 2C_2^2(\epsilon')^2L(K+M+2)^2+2C_2^2(\epsilon')^2|\log s_n|^2$. Summing over the portion, the first part contributes $2C_2^2L(K+M+2)^2(\epsilon')^2n'$, whose conditional expectation is at most $2C_2^2L(K+M+2)^2M\mathbf{1}_{\mathcal{R}_j}$ by (731). For the second, $s_n = S_n\epsilon'$ and $a < S_n < b$, so grouping the summands by the level $i_0 = S_n$ and using (732),

$$\mathbb{E}\Big[\sum_{\sigma_j\le n<\tau_j'} (\epsilon')^2|\log s_n|^2 \;\Big|\; \mathcal{G}_{\sigma_j}\Big] = (\epsilon')^2\sum_{i_0=a+1}^{b-1} \big|\log(i_0\epsilon')\big|^2\mathbb{E}\big[V_{i_0} \mid \mathcal{G}_{\sigma_j}\big] \le \mathbf{1}_{\mathcal{R}_j}\frac{M\epsilon'}{2}\sum_{i_0=a+1}^{b-1}\big|\log(i_0\epsilon')\big|^2.$$

The levels $s = i_0\epsilon'$ with $a < i_0 < b$ lie in $(0, M)$. Split the sum. For $i_0\epsilon' \le 1$ the function $s \mapsto |\log s|^2$ is nonnegative and decreasing on $(0,1]$, so its right-endpoint grid sum obeys $\epsilon'\sum_{i_0\epsilon'\le 1}|\log(i_0\epsilon')|^2 \le \int_0^1 |\log s|^2\,ds = 2$, the value of the integral following from the substitution $s = e^{-v}$, which turns it into $\int_0^\infty v^2e^{-v}\,dv = 2$. For $1 < i_0\epsilon' < M$ we have $|\log(i_0\epsilon')|^2 \le (\log M)^2$ and there are at most $b = M/\epsilon'$ such i_0, so $\epsilon'\sum_{i_0\epsilon'>1}|\log(i_0\epsilon')|^2 \le M(\log M)^2$. Therefore, using $L(K+M+2) \ge 1$,

$$\mathbb{E}\Big[\sum_{\sigma_j\le n<\tau_j'} (\epsilon')^2|Q_0'(n)|^2 \;\Big|\; \mathcal{G}_{\sigma_j}\Big] \le C_M L(K+M+2)^2\,\mathbf{1}_{\mathcal{R}_j}, \qquad C_M = 2C_2^2\Big(M+\frac{M}{2}\big(2+M(\log M)^2\big)\Big). \tag{734}$$

Integrating (731) and (734) over Ω removes the indicators and gives the unconditional bounds

$$\mathbb{E}\big[(\epsilon')^2n'\big] \le M, \qquad \mathbb{E}\sum_{\sigma_j\le n<\tau_j'} (\epsilon')^2|Q_0'(n)|^2 \le C_M L(K+M+2)^2. \tag{735}$$

The first of these two unconditional bounds is exactly ExpectedPortionLengthBound, applied to the lattice data of Claim T′ — the process S, the indices $\sigma_j \le \tau_j' \le \tau$, the window (a,b) and the whole-space instance of the comparison — with the test pair h_1, u of (a); the conditional form (731) is kept here because the velocity estimate above reads it before the integration.

Step 7: the frame travel bound and the choice of K. Apply the travel inequality of StoppedWalkTravel, available by Step 0, with the stopping times $\rho = \sigma_j$ and $\varsigma = \tau_j'$, which satisfy $\sigma_j \le \tau_j' \le \tau$ by (725), with $p = p_j$, $C_p = \Gamma_j$ — admissible by Step 2 — and with $r = \delta_j > 0$; the required measurability of $\mathbf{1}_{\{\sigma_j\le n\}}p_j$ is IsMarkData, clause 5. By (726) its left-hand side is exactly $\mathbb{E}\sup_{\sigma_j\le n\le\tau_j'}|y_n|$, and by (723) its first right-hand term involves $|Q_0(n)-p_j| = |Q_0'(n)|$ with the factor $(\epsilon/r)^2 = (\epsilon')^2$. So the clause asserts that $\sup_{\sigma_j\le n\le\tau_j'}|y_n|$ is integrable and

$$\mathbb{E}\sup_{\sigma_j\le n\le\tau_j'}|y_n| \le 2\left(\mathbb{E}\sum_{\sigma_j\le n<\tau_j'}(\epsilon')^2|Q_0'(n)|^2\right)^{1/2} + \big(\mathbb{E}(\epsilon')^2(\tau_j'-\sigma_j)\big)^{1/2}\left(\mathbb{E}\sum_{\sigma_j\le n<\tau_j'}\mathcal{E}_n\right)^{1/2}.$$

Inserting (735) and (727),

$$\mathbb{E}\sup_{\sigma_j\le n\le\tau_j'}|y_n| \le 2\sqrt{C_M}\,L(K+M+2) + \sqrt{M}\sqrt{B_0} \le C_M'\big(L(K+M+2)+\sqrt{B_0}\big), \tag{736}$$

with $C_M' = \max\{2\sqrt{C_M}, \sqrt{M}\}$, a constant depending only on M, C_0, κ and $\dim F$ — and not on K, N or j.

Since $B_0 = C_*L(B)^4$ and M are now fixed and $L(K+M+2)/K \to 0$ as $K\to\infty$, we may fix once and for all

$$K \ge 2 \quad\text{with}\quad \frac{C_M'\big(L(K+M+2)+\sqrt{B_0}\big)}{K} \le \frac{1}{8}. \tag{737}$$

This choice uses only M, B_0 and the fixed data, so it is uniform in N and j, as required by the order of the quantifiers in the statement.

Step 8: the two cutoff probabilities. Put

$$\mathcal{B}_j^1 = \mathcal{R}_j \cap \{|y_{\tau_j'}| \geq K\}, \qquad \mathcal{B}_j^2 = \mathcal{R}_j \cap \{s_{\tau_j'} \geq M\}.$$

On $\mathcal{B}_j^1$ we have $|y_{\tau_j'}| \leq \sup_{\sigma_j \leq n \leq \tau_j'} |y_n|$, so the pointwise inequality $K\mathbf{1}_{\mathcal{B}_j^1} \leq \sup_{\sigma_j \leq n \leq \tau_j'} |y_n|$ holds; integrating it — legitimate because that supremum is integrable by Step 7 — and using (736) and (737),

$$\mathbb{P}(\mathcal{B}_j^1) \leq \frac{\mathbb{E}\sup_{\sigma_j \leq n \leq \tau_j'} |y_n|}{K} \leq \frac{C'_M(L(K+M+2)+\sqrt{B_0})}{K} \leq \frac{1}{8}. \tag{738}$$

For the other cutoff take $\ell(i) = (i-a)/(b-a)$ for $a \leq i \leq b$, $\ell(i) = 0$ for $i < a$ and $\ell(i) = 1$ for $i > b$. It is bounded and affine on $\{a, \dots, b\}$, hence satisfies the mean-value identity for $a < i < b$, so the last assertion of Claim T′ applies: $\mathbb{E}[\ell(S_{\tau_j'}) \mid \mathcal{G}_{\sigma_j}] = \ell(S_{\sigma_j})$ a.e. On $\mathcal{R}_j$ we have $S_{\sigma_j} = 1/\epsilon'$ and $b = M/\epsilon'$, so, with $a \geq 0$ and $M \geq 1$,

$$\ell(S_{\sigma_j}) = \frac{1/\epsilon' - a}{M/\epsilon' - a} \leq \frac{1}{M},$$

the inequality because $M(1/\epsilon' - a) \leq M/\epsilon' - a$ reduces to $Ma \geq a$. Also $s_{\tau_j'} \geq M$ means $S_{\tau_j'} \geq b$, where $\ell = 1$, so $\mathbf{1}_{\mathcal{B}_j^2} \leq \mathbf{1}_{\mathcal{R}_j}\ell(S_{\tau_j'})$ pointwise, ℓ being nonnegative. Since $\mathcal{R}_j \in \mathcal{G}_{\sigma_j}$,

$$\mathbb{P}(\mathcal{B}_j^2) \leq \mathbb{E}\big[\mathbf{1}_{\mathcal{R}_j}\ell(S_{\tau_j'})\big] = \mathbb{E}\Big[\mathbf{1}_{\mathcal{R}_j}\mathbb{E}\big[\ell(S_{\tau_j'}) \mid \mathcal{G}_{\sigma_j}\big]\Big] = \mathbb{E}\big[\mathbf{1}_{\mathcal{R}_j}\ell(S_{\sigma_j})\big] \leq \frac{1}{M} = \frac{1}{8}, \tag{739}$$

the middle equality by the characterizing property of the conditional expectation applied to the set $\mathcal{R}_j \in \mathcal{G}_{\sigma_j}$.

Step 9: the combination, i.e. (ii). Let $\mathcal{G} = \{t_\tau = \delta_N\}$, so $\mathbb{P}(\mathcal{G}) \geq 3/4$ by (718) and $\mathcal{G} \subseteq \mathcal{R}_j$ by (720). Let $\omega \in \mathcal{G} \setminus (\mathcal{B}_j^1 \cup \mathcal{B}_j^2)$. Then $\omega \in \mathcal{R}_j$, and since $\omega \notin \mathcal{B}_j^1 \cup \mathcal{B}_j^2$ we have $|y_{\tau_j'}| < K$ and $s_{\tau_j'} < M$. By the definition of τ_j' in Step 4, at least one of the three alternatives holds at $n = \tau_j'$; the second and third have just been excluded, so $\tau_j'(\omega) = \tau(\omega)$. Hence $|y_\tau| < K$, which by (726) and $t_{\sigma_j} = \delta_j$ on $\mathcal{R}_j$ reads

$$\big|x_\tau - x_{\sigma_j} - (t_\tau - \delta_j)p_j\big| < K\delta_j.$$

Together with $t_\tau = \delta_N$ this places ω in the event of (717). Therefore that event contains $\mathcal{G} \setminus (\mathcal{B}_j^1 \cup \mathcal{B}_j^2)$ and, by (718), (738) and (739),

$$\mathbb{P}\Big(t_\tau = \delta_N \text{ and } \big|x_\tau - x_{\sigma_j} - (t_\tau - \delta_j)p_j\big| \leq K\delta_j\Big) \geq \mathbb{P}(\mathcal{G}) - \mathbb{P}(\mathcal{B}_j^1) - \mathbb{P}(\mathcal{B}_j^2) \geq \frac{3}{4} - \frac{1}{8} - \frac{1}{8} = \frac{1}{2}.$$

The event $\mathcal{L}_j$ is measurable, which is the remaining half of (ii). Both τ and σ_j take values in $\mathbb{N}$, so

$$\mathcal{L}_j = \bigcup_{n \geq 0} \bigcup_{m \geq 0} \big(\{\tau = n\} \cap \{\sigma_j = m\} \cap A_{n,m}\big), \qquad A_{n,m} = \{t_n = \delta_N\} \cap \big\{\big|x_n - x_m - (t_n - \delta_j)p_j\big| \leq K\delta_j\big\},$$

a countable union. Here $\{\tau = n\} \in \mathcal{G}_n \subseteq \mathcal{M}$ because τ is a stopping time for $(\mathcal{G}_n)$ by clause 9 of `IsContactWalk`, and $\{\sigma_j = m\} \in \mathcal{G}_m \subseteq \mathcal{M}$ because σ_j is a stopping time by clause 1 of `IsMarkData`. The maps z_n and z_m are $\mathcal{G}_n$- and $\mathcal{G}_m$-measurable by clause 4 of `IsContactWalk`, the coordinate projections of $\ell^2(F \times \mathbb{R})$ are continuous and hence Borel, and on $\{\sigma_j = m\} \cap \{\sigma_j \leq m\} = \{\sigma_j = m\}$ the function p_j agrees with $\mathbf{1}_{\{\sigma_j \leq m\}}p_j$, which is $\mathcal{G}_m$-measurable by clause 5 of `IsMarkData`; so $\{\sigma_j = m\} \cap A_{n,m} \in \mathcal{G}_{\max(n,m)} \subseteq \mathcal{M}$, the norm and the linear operations being continuous. A countable union of measurable sets is measurable, so $\mathcal{L}_j \in \mathcal{M}$; the same decomposition applied to $\mathcal{B}_j^1$, $\mathcal{B}_j^2$ and $\mathcal{G}$ is what makes the probabilities used above meaningful.

This is (717), and with Step 2 it completes the proof. □

2.168. `ContactPropagation`.

Lemma 2.168 (Propagation of contact to time zero)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$, let $\kappa > 0$ and $H \in \mathbb{R}$, and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at $z_* = (0,1)$ with constant C_0 in the sense of `IsCenteredAt` and satisfy the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound`. Let W be the contact envelope of `ContactEnv` formed from v and κ, and assume the six standing properties of the extremal contact pair: W is continuous on $\{t > 0\}$ with subquadratic spatial growth on compact subintervals of $(0,\infty)$ in the sense of `SubquadraticGrowth`; W is a viscosity subsolution of the limiting equation there in the sense of `IsLimitSubOn`; $v - \kappa^2 t/2 \le W \le v$ for $t > 0$ and $W(z_*) = v(z_*) = 0$; $x \mapsto W(x,t) - |x|^2/(2t)$ is concave for every $t > 0$; the barycentric gap of `BaryEnv` is $\kappa^2 t/2$ at every contact; and the envelope Hamiltonian of `EnvHamiltonian` satisfies $H_v(z_*) = H$ with $\mathcal{S}^+_\ell W(z_*) \to H$ as $\ell \downarrow 0$, the forward envelope slope being that of `EnvSlopeP`. Write $\delta_j = 2^{-j}$.*

Then there are a constant $K \ge 0$, a point $x_{\mathrm{end}} \in F$ and an infinite set $\mathcal{J} \subseteq \{1,2,3,\dots\}$ such that

$$H_v(x_{\mathrm{end}}, 0) > -\infty \tag{740}$$

and such that for every $j \in \mathcal{J}$ there are $x_j, p_j \in F$ and $a_j \in \mathbb{R}$ with (p_j, a_j) a fit for v at (x_j, δ_j) of radius δ_j with constant C_0 in the sense of `IsFit`, and with

$$W(x_j,\delta_j) = v(x_j,\delta_j), \qquad \big|x_{\mathrm{end}} - x_j + \delta_j p_j\big| \le K\delta_j. \tag{741}$$

Proof. Throughout, W is the contact envelope of `ContactEnv`, $\mathcal{P}_\ell$, $\mathcal{M}_\ell$ are the quadratic envelopes of `QuadEnvP` and `QuadEnvM`, $\mathcal{S}^\pm_\ell$ are the envelope slopes of `EnvSlopeP` and `EnvSlopeM`, H_v is the envelope Hamiltonian of `EnvHamiltonian`, and J_ϵ is the propagation quantity of `JGap` formed from v and κ. We write $L(\Theta) = 1 + \log\Theta$ and $\delta_j = 2^{-j}$.

Two standing facts about v are used repeatedly. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of the limiting equation there, and by `UniformSubquadratic`, applied with the centering point $(0,1)$ and the constant C_0, it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`. Hence `QuadEnvAttained` and `EnvHamiltonianSolution` are available for v with $I = \mathbb{R}$, for which every time and every shifted time is admissible.

Step 1: the constants, the box, and the walks. Let $C_* > 0$ be the constant of `StoppedWalkBudgets` for the present C_0, κ, H, v. Feed this C_* to `MarkLocalization` for the same data and let $T \ge 8$, $R \ge 2$, $K \ge 2$ and $C_\Gamma > 0$ be the box, radius and fit-parameter constant it provides. Put

$$B = R + T + 2, \qquad \Gamma_j = C_\Gamma\big(L(B) + j\big) \ \ (j \ge 1), \qquad \mathcal{B} = \{(x,t) : |x| \le R+1,\ 0 \le t \le T\}.$$

By conclusion (i) of `MarkLocalization` every $\Gamma_j \ge 1$; also Γ_j is nondecreasing in j. The set $\mathcal{B}$ is a closed bounded subset of space-time, hence compact.

Fix $N \ge 1$. Apply `StoppedWalkBudgets` with this T, this R and this N. It produces a step size $\epsilon_N > 0$, a measure space $(\Omega_N, \mathcal{M}_N, \mu_N)$ with a filtration $(\mathcal{G}^N_n)$, and functions ξ^N, z^N, $z^{N,+}$, $z^{N,-}$, Q^N_0, τ_N forming a stopped contact walk for $(\kappa, H, T, R, v, N, \epsilon_N)$ in the sense of `IsContactWalk`, together with

$$\mathbb{E}\, J_{\epsilon_N}\big(z^N_\sigma\big) \ge J_{\epsilon_N}(z_*) \qquad \text{for every stopping time } \sigma \le \tau_N, \tag{742}$$

the integrand being μ_N-integrable, the error budget $\mathbb{E}\sum_{n<\tau_N} \mathcal{E}^N_n \le C_* L(B)^4$, and marks with fits $(\sigma^N_j, p^N_j, a^N_j)$ for $(C_0, v, N, \mathcal{G}^N, z^N, \tau_N)$ in the sense of `IsMarkData`. The error budget just stated is exactly the error-budget hypothesis of `MarkLocalization` for this walk, with the same C_*, T, R and B, so that lemma applies to it with these marks, giving for every $1 \le j \le N$

$$\big|p^N_j\big| + \big|a^N_j\big| \le \Gamma_j \quad \text{pointwise}, \tag{743}$$

$$\mathcal{L}^N_j = \Big\{ t^N_{\tau_N} = \delta_N \text{ and } \big|x^N_{\tau_N} - x^N_{\sigma^N_j} - \big(t^N_{\tau_N} - \delta_j\big)p^N_j\big| \le K\delta_j \Big\} \in \mathcal{M}_N, \qquad \mu_N\big(\mathcal{L}^N_j\big) \ge \tfrac{1}{2}. \tag{744}$$

Here and below $z_n^N = (x_n^N, t_n^N)$, and a subscripted random index means evaluation at that index, e.g. $z^N_{\tau_N}(\omega) = z^N_{\tau_N(\omega)}(\omega)$. By clause 2 of `IsContactWalk` we have $0 < \epsilon_N \le \delta_N/4$, so

$$\epsilon_N \le 2^{-N-2} \longrightarrow 0 \qquad (N \to \infty). \tag{745}$$

Finally, clause 13 of `IsContactWalk` gives the starting bound

$$J_{\epsilon_N}(z_*) \ge H - 1 \qquad (N \ge 1). \tag{746}$$

Step 2: four facts about the time grid. Fix N and write $\epsilon = \epsilon_N$, $\tau = \tau_N$, dropping the superscript N from the walk.

(G1) By clause 2 of `IsContactWalk`, $\epsilon = 2^{-k}$ for an integer $k \ge 0$ and $\epsilon \le \delta_N/4 = 2^{-N-2}$, so $k \ge N+2$. Hence 1, T and every δ_j with $j \le N$ are integer multiples of ϵ: indeed $1 = 2^k\epsilon$, $T = T2^k\epsilon$ and $\delta_j = 2^{k-j}\epsilon$ with $k - j \ge 2$.

(G2) By clause 5, $t_0 = 1$. For $n < \tau(\omega)$, clause 6 makes (z_n^+, z_n^-) an optimizing step pair of size ϵ at z_n, so by `IsStepPair` the time coordinates of $z_n^\pm$ are $t_n \pm \epsilon$; clause 7 then gives $t_{n+1} = t_n \pm \epsilon$. For $n \ge \tau(\omega)$, clause 7 gives $t_{n+1} = t_n$. In particular $|t_{n+1} - t_n| \in \{0, \epsilon\}$ for every n.

(G3) By (G1), (G2) and induction, every $t_n(\omega)$ is an integer multiple of ϵ.

(G4) $\delta_N \le t_n(\omega) \le T$ for every n and ω. Indeed $t_0 = 1$ and $\delta_N \le \frac{1}{2} < 1 \le T$ since $N \ge 1$ and $T \ge 8$. If $\delta_N \le t_n \le T$ and $n \ge \tau(\omega)$ then $t_{n+1} = t_n$. If $n < \tau(\omega)$ then clause 10 gives $\delta_N < t_n < T$, and since t_n, δ_N and T are integer multiples of ϵ by (G1) and (G3), in fact $\delta_N + \epsilon \le t_n \le T - \epsilon$; by (G2), $\delta_N \le t_{n+1} \le T$.

(G5) (discrete intermediate value) If $a \le b$, if c is an integer multiple of ϵ and if $t_b(\omega) \le c \le t_a(\omega)$, then $t_n(\omega) = c$ for some n with $a \le n \le b$. Indeed the finite sequence $t_a, t_{a+1}, \dots, t_b$ consists of integer multiples of ϵ by (G3) and has consecutive differences 0 or $\pm\epsilon$ by (G2); the first index n in $[a, b]$ with $t_n \le c$ satisfies $t_n \ge t_{n-1} - \epsilon > c - \epsilon$ when $n > a$, hence $t_n = c$ because t_n and c are multiples of ϵ in $(c - \epsilon, c]$, while for $n = a$ we get $t_a \le c \le t_a$.

Two consequences for the marks, both for $1 \le j \le N$. Write $\sigma_j = \sigma_j^N$.

(M1) $t_{\sigma_j} \ge \delta_j$ everywhere. Suppose $t_{\sigma_j}(\omega) < \delta_j$. Since $\delta_j \le \frac{1}{2} < 1 = t_0(\omega)$ and δ_j is a multiple of ϵ by (G1), (G5) with $a = 0$, $b = \sigma_j(\omega)$ and $c = \delta_j$ gives $n \le \sigma_j(\omega)$ with $t_n(\omega) = \delta_j$. As $t_{\sigma_j}(\omega) \ne \delta_j$ we have $n < \sigma_j(\omega)$, contradicting clause 3 of `IsMarkData`.

(M2) On $\{t_\tau = \delta_N\}$ one has $t_{\sigma_j} = \delta_j$. Fix such an ω. Then $t_\tau(\omega) = \delta_N \le \delta_j \le 1 = t_0(\omega)$, so (G5) with $a = 0$, $b = \tau(\omega)$, $c = \delta_j$ gives an $n \le \tau(\omega)$ with $t_n(\omega) = \delta_j$; let n_0 be the least such. Every $m < n_0$ has $t_m(\omega) \ne \delta_j$ and $m < n_0 \le \tau(\omega)$. If $\sigma_j(\omega) > n_0$ then clause 3 of `IsMarkData`, applied at $n = n_0$, would give $t_{n_0}(\omega) \ne \delta_j$; so $\sigma_j(\omega) \le n_0$. By clause 4 of `IsMarkData` either $t_{\sigma_j}(\omega) = \delta_j$, and we are done, or $\sigma_j(\omega) = \tau(\omega)$, in which case $n_0 \le \tau(\omega) = \sigma_j(\omega) \le n_0$, so $\sigma_j(\omega) = n_0$ and again $t_{\sigma_j}(\omega) = \delta_j$.

Step 3: one compact space carrying terminal point, flag and all marks. Put

$$\mathcal{K}_j = \mathcal{B} \times \overline{B}_{\Gamma_j}(0) \times [-\Gamma_j, \Gamma_j] \quad (j \ge 1), \qquad \mathcal{X} = \mathcal{B} \times \{0,1\} \times \prod_{j \ge 1} \mathcal{K}_j.$$

Each factor is a nonempty compact metric space, so $\mathcal{X}$ is a nonempty compact metrizable space, hence a complete separable metric space, and the coordinate projections are continuous. Denote the coordinates of a point of $\mathcal{X}$ by $z_{\mathrm{end}} = (x_{\mathrm{end}}, t_{\mathrm{end}})$, η and $(z_j, p_j, a_j)_{j \ge 1}$, writing $z_j = (x_j, t_j)$.

Fix $N \ge 1$ and define $\Psi_N : \Omega_N \to \mathcal{X}$ by

$$\Psi_N(\omega) = \Big(z^N_{\tau_N}(\omega),\ \mathbf{1}_{\{t^N_{\tau_N} = \delta_N\}}(\omega),\ \big(w_j^N(\omega)\big)_{j \ge 1} \Big), \qquad w_j^N = \begin{cases} (z^N_{\sigma_j^N}, p_j^N, a_j^N) & 1 \le j \le N, \\ (z_*, 0, 0) & j > N. \end{cases}$$

This is well defined. The terminal point lies in $\mathcal{B}$: clause 12 of `IsContactWalk` gives $|x_n^N| \le R+1$ for $n \le \tau_N$, and (G4) gives $0 < \delta_N \le t_n^N \le T$ for every n. The same two facts put every $z^N_{\sigma_j^N}$ in $\mathcal{B}$, because $\sigma_j^N \le \tau_N$ by clause 2 of `IsMarkData`; the dummy entry $z_* = (0, 1)$ lies in $\mathcal{B}$ since $R + 1 \ge 3$ and $1 \le T$; and (743) puts (p_j^N, a_j^N) in $\overline{B}_{\Gamma_j}(0) \times [-\Gamma_j, \Gamma_j]$ for $j \le N$, while $(0, 0)$ lies there for $j > N$ because $\Gamma_j \ge 1 > 0$.

Ψ_N is measurable. For a stopping time σ for $(\mathcal{G}_n^N)$ with values in $\mathbb{N}$ and an adapted sequence (Y_n), the stopped value Y_σ is $\mathcal{M}_N$-measurable, because $\{\sigma = n\} \in \mathcal{G}_n^N \subseteq \mathcal{M}_N$, each Y_n is $\mathcal{G}_n^N$-measurable, and $Y_\sigma = \sum_n \mathbf{1}_{\{\sigma=n\}} Y_n$ is a countable sum of measurable functions with pairwise disjoint supports. Apply this to $\sigma = \tau_N$, a stopping time by clause 9 of `IsContactWalk`, and to $\sigma = \sigma_j^N$, a stopping time by clause 1 of `IsMarkData`, with $Y_n = z_n^N$, which is adapted by clause 4 of `IsContactWalk`; this makes $z_{\tau_N}^N$ and $z_{\sigma_j^N}^N$ measurable. The flag is measurable because $\{t_{\tau_N}^N = \delta_N\}$ is the preimage of a closed set under the measurable map $z_{\tau_N}^N$ followed by the continuous time projection. Finally $p_j^N = \sum_m \mathbf{1}_{\{\sigma_j^N = m\}} \big(\mathbf{1}_{\{\sigma_j^N \le m\}} p_j^N\big)$ is measurable, each $\mathbf{1}_{\{\sigma_j^N \le m\}} p_j^N$ being $\mathcal{G}_m^N$-measurable by clause 5 of `IsMarkData`, and likewise a_j^N by that same clause 5, which asserts the $\mathcal{G}_n$-measurability of $\mathbf{1}_{\{\sigma_j \le n\}} p_j$ and of $\mathbf{1}_{\{\sigma_j \le n\}} a_j$ together. A map into a countable product is measurable as soon as all its coordinates are.

Let λ_N be the law of Ψ_N, that is, the image of μ_N under Ψ_N; it is a Borel probability measure on $\mathcal{X}$, since $\mu_N(\Omega_N) = 1$ by clause 1 of `IsContactWalk`. By construction, for every Borel $A \subseteq \mathcal{X}$,

$$\lambda_N(A) = \mu_N\big(\Psi_N^{-1}(A)\big), \tag{747}$$

and $\int \varphi \, d\lambda_N = \int \varphi \circ \Psi_N \, d\mu_N$ for every bounded Borel φ on $\mathcal{X}$.

Step 4: one limiting law. The single compact set $\mathcal{X}$ witnesses that $S = \{\lambda_N : N \ge 1\}$ is tight. By Prokhorov's theorem, a Preamble item, some subsequence λ_{N_h} converges weakly to a Borel probability measure λ on $\mathcal{X}$; we may and do take $N_h \to \infty$. All limits below are along this subsequence, and $\mathbb{P}$, $\mathbb{E}$ refer to λ. Weak convergence is preserved by pushing forward under a continuous map, so for each fixed j the marginals $\nu_{N_h}^j$ of λ_{N_h} under (z_j) converge weakly to the corresponding marginal ν^j of λ on $\mathcal{B}$.

Step 5: the terminal time vanishes on the flag. The function $\eta\, t_{\mathrm{end}}$ is continuous on $\mathcal{X}$, because $\{0,1\}$ is discrete and t_{end} is a continuous coordinate, and it is bounded and nonnegative there, since $0 \le t_{\mathrm{end}} \le T$ on $\mathcal{B}$. Under λ_N it equals δ_N on $\{\eta = 1\}$ and 0 elsewhere, by the definition of the flag, so $\int \eta\, t_{\mathrm{end}} \, d\lambda_N \le \delta_N \to 0$. Weak convergence applied to this bounded continuous function gives $\int \eta\, t_{\mathrm{end}} \, d\lambda = 0$, and the integrand being nonnegative,

$$\eta\, t_{\mathrm{end}} = 0 \qquad \lambda\text{-almost surely.} \tag{748}$$

Step 6: levels and fits pass to the limit. Fix $j \ge 1$ and consider the three subsets of $\mathcal{X}$

$$F_j^1 = \{t_j \ge \delta_j\}, \qquad F_j^2 = \{\eta = 0\} \cup \{t_j = \delta_j\}, \qquad F_j^3 = \{\eta = 0\} \cup \mathcal{Q}_j,$$

where $\mathcal{Q}_j$ is the set of points of $\mathcal{X}$ whose j-th mark coordinate (z_j, p_j, a_j) satisfies the fitting inequality of `IsFit` for v at z_j with radius δ_j and constant C_0.

F_j^1 and F_j^2 are closed, the time coordinate being continuous and $\{\eta = 0\}$ being closed and open. F_j^3 is closed as well. To see that $\mathcal{Q}_j$ is closed, parametrize the increments admitted by `IsFit` at radius δ_j: the conditions $|\ell| \le \delta_j$ and $|h - \ell p_j| \le \delta_j$ say exactly that

$$\ell = \delta_j \theta, \qquad h = \delta_j\big(\theta p_j + \zeta\big), \qquad |\theta| \le 1,\ |\zeta| \le 1,$$

with $\theta \in \mathbb{R}$ and $\zeta \in F$ ranging over a fixed compact set that does not depend on the point of $\mathcal{K}_j$. For each fixed (θ, ζ) the inequality

$$\Big| v\big(z_j + (h,\ell)\big) - v(z_j) - \langle p_j, h\rangle - \Big(a_j - \frac{|p_j|^2}{2}\Big)\ell \Big| \le C_0 \delta_j$$

is a closed condition on (z_j, p_j, a_j), because v is continuous on all of space-time and h, ℓ depend continuously on p_j; and $\mathcal{Q}_j$ is the intersection of these closed conditions over (θ, ζ), hence closed.

For every $N \ge j$ we have $\lambda_N(F_j^1) = \lambda_N(F_j^2) = \lambda_N(F_j^3) = 1$. Indeed, by (747) it suffices to check the corresponding statements about Ψ_N. The j-th mark coordinate of $\Psi_N(\omega)$ is $(z_{\sigma_j^N}^N, p_j^N, a_j^N)(\omega)$ because $j \le N$. Its time coordinate is $\ge \delta_j$ by (M1), giving F_j^1. If moreover the flag is 1, that is $t_{\tau_N}^N(\omega) = \delta_N$, then (M2) gives $t_{\sigma_j^N}^N(\omega) = \delta_j$, giving F_j^2, and then clause 6 of `IsMarkData` says that $(p_j^N, a_j^N)(\omega)$ is a fit for v at $z_{\sigma_j^N}^N(\omega)$ of radius δ_j with constant C_0, giving F_j^3.

By the closed-set direction of the Portmanteau Theorem, a Preamble item, $\lambda(F) \geq \limsup_h \lambda_{N_h}(F)$ for closed F; applied to the three sets, whose λ_{N_h}-measures are 1 once $N_h \geq j$, this gives $\lambda(F_j^1) = \lambda(F_j^2) = \lambda(F_j^3) = 1$. Intersecting over $j \geq 1$,

$$t_j \geq \delta_j \text{ for all } j, \qquad \text{and on } \{\eta = 1\}: \ t_j = \delta_j \text{ and } (p_j, a_j) \text{ is a fit at } z_j \text{ of radius } \delta_j, \tag{749}$$

λ-almost surely.

Step 7: a uniform upper bound for J_ϵ on the box. For $\ell > 0$ put

$$\Phi_\ell^{\mathrm{sl}}(z) = \tfrac{1}{2}\big(\mathcal{S}_\ell^+ v(z) + \mathcal{S}_\ell^- v(z)\big).$$

By `EnvSlopeP` and `EnvSlopeM` this is $\big(\mathcal{P}_\ell v(z) - \mathcal{M}_\ell v(z)\big)/(2\ell)$, which by part (2) of `QuadEnvAttained` applied to v with $I = \mathbb{R}$ is a continuous real function of z on all of space-time, every time being admissible. Put

$$U_B = \max_{z \in \mathcal{B}} \Phi_1^{\mathrm{sl}}(z) < \infty,$$

the maximum of a continuous function on the nonempty compact set $\mathcal{B}$. By part (1) of `EnvHamiltonianSolution` for v with $I = \mathbb{R}$, both slopes are nondecreasing in the length, so

$$\Phi_\ell^{\mathrm{sl}} \leq \Phi_{\ell'}^{\mathrm{sl}} \qquad (0 < \ell \leq \ell'), \tag{750}$$

and by `JUpperBound`, $J_\epsilon(z) \leq \Phi_\epsilon^{\mathrm{sl}}(z)$ for every $\epsilon > 0$ and every $z = (x, t)$ with $t > 0$. Combining the two, for $0 < \epsilon \leq \ell$ and $t > 0$,

$$J_\epsilon(z) \leq \Phi_\ell^{\mathrm{sl}}(z), \qquad \text{in particular} \qquad J_\epsilon(z) \leq U_B \quad (z \in \mathcal{B},\ t > 0,\ \epsilon \leq 1). \tag{751}$$

Since $\epsilon_N \leq 2^{-N-2} \leq 1$ by (745), the second bound applies to every walk and every state it visits with positive time; by (G4) all of them have $t \geq \delta_N > 0$.

Step 8: the envelope Hamiltonian is finite at the terminal point. Fix $\ell > 0$. For every N with $\epsilon_N \leq \ell$, apply (742) with $\sigma = \tau_N$ and then the first bound of (751) at $z_{\tau_N}^N$, whose time is $\geq \delta_N > 0$ by (G4), and (746):

$$\int \Phi_\ell^{\mathrm{sl}}(z_{\mathrm{end}})\, d\lambda_N = \mathbb{E}_{\mu_N} \Phi_\ell^{\mathrm{sl}}\big(z_{\tau_N}^N\big) \geq \mathbb{E}_{\mu_N} J_{\epsilon_N}\big(z_{\tau_N}^N\big) \geq J_{\epsilon_N}(z_*) \geq H - 1.$$

The function $\Phi_\ell^{\mathrm{sl}}(z_{\mathrm{end}})$ is continuous and bounded on $\mathcal{X}$, being a continuous function of a continuous coordinate on a compact space, so weak convergence along the subsequence gives

$$\int \Phi_\ell^{\mathrm{sl}}(z_{\mathrm{end}})\, d\lambda \geq H - 1 \qquad (\ell > 0), \tag{752}$$

the hypothesis $\epsilon_{N_h} \leq \ell$ being satisfied for all large h by (745).

Take $\ell_i \downarrow 0$ with $\ell_i \leq 1$. By part (2) of `EnvHamiltonianSolution` for v with $I = \mathbb{R}$, each of $\mathcal{S}_{\ell_i}^+ v(z)$ and $\mathcal{S}_{\ell_i}^- v(z)$ decreases to $H_v(z)$ as $i \to \infty$, for every z; hence $\Phi_{\ell_i}^{\mathrm{sl}}(z)$ decreases to $H_v(z) \in [-\infty, \infty)$, the value being $< +\infty$ by part (3) of the same lemma. By (750) and the definition of U_B, the functions

$$g_i = U_B - \Phi_{\ell_i}^{\mathrm{sl}}(z_{\mathrm{end}})$$

are nonnegative Borel functions on $\mathcal{X}$ increasing pointwise to $U_B - H_v(z_{\mathrm{end}}) \in [0, \infty]$. By the monotone convergence theorem and (752),

$$\int \big(U_B - H_v(z_{\mathrm{end}})\big)\, d\lambda = \lim_i \int g_i\, d\lambda \leq U_B - (H - 1) < \infty.$$

An integrand with finite integral is finite almost everywhere, so

$$H_v(z_{\mathrm{end}}) > -\infty \qquad \lambda\text{-almost surely.} \tag{753}$$

Step 9: contact at every limiting mark. This step is `ContactAtLimitingMarks`, applied as follows. The ambient data are v, κ and the envelope W: v is continuous and W is continuous on $\{t > 0\}$ with $W \leq v$ there, by the standing hypotheses. The uniform gap hypothesis is part (2) of `ContactGapLimit` for the present C_0, κ, v, whose own hypotheses are those of the statement. The box is $\mathcal{B}$ with the bound U_B of (751), admissible since $\epsilon \leq 1$ for every walk by (745). The state space is the compact $\mathcal{X}$ of Step 3 with the mark maps $q \mapsto z_j(q)$, each continuous because it is a coordinate projection; the limiting law is λ and the pre-limit laws are the λ_N of Step 3,

with the subsequence $N_h \geq h$ of Step 4 along which $\lambda_{N_h} \to \lambda$ weakly. The full-measure closed events are the $\mathcal{A}_j$ of Step 6, on which $t_j \geq \delta_j$ by construction.

On the walk side, Ψ_N pushes μ_N to λ_N by Step 3; the mark coordinate of Ψ_N is $z^N_{\sigma^N_j}$ for $1 \leq j \leq N+1$, again by Step 3, and that point lies in $\mathcal{B}$ by (M1) because $\sigma^N_j \leq \tau_N$ by clause 2 of `IsMarkData`; its time is positive by (G4); and the averaged lower bound

$$H - 1 \leq \mathbb{E}_{\mu_N} J_{\epsilon_N}\big(z^N_{\sigma^N_j}\big),$$

with the integrability it needs, is (742) at $\sigma = \sigma^N_j$ — legitimate because σ^N_j is a stopping time bounded by τ_N, by clauses 1 and 2 of `IsMarkData` — combined with the starting bound $H - 1 \leq J_{\epsilon_N}(z_*)$ of Step 1. The cited node therefore gives

$$W(z_j) = v(z_j) \quad \text{for every } j \geq 1, \qquad \lambda\text{-almost surely.} \tag{754}$$

Step 10: the localization events and the selection. For $j \geq 1$ put

$$E_j = \Big\{\eta = 1 \text{ and } \big|x_{\mathrm{end}} - x_j - (t_{\mathrm{end}} - \delta_j)p_j\big| \leq K\delta_j\Big\} \subseteq \mathcal{X}.$$

This set is closed: $\{\eta = 1\}$ is closed, and the displayed norm is a continuous function of the coordinates z_{end}, z_j and p_j. For $N \geq j$ we have $\Psi_N^{-1}(E_j) = \mathcal{L}^N_j$, since the flag equals 1 exactly on $\{t^N_{\tau_N} = \delta_N\}$ and the remaining coordinates of Ψ_N are $z^N_{\tau_N}$, $z^N_{\sigma^N_j}$ and p^N_j. Hence $\lambda_N(E_j) = \mu_N(\mathcal{L}^N_j) \geq \frac{1}{2}$ by (747) and (744). The closed-set direction of Portmanteau then gives

$$\lambda(E_j) \geq \limsup_h \lambda_{N_h}(E_j) \geq \frac{1}{2} \qquad (j \geq 1).$$

These events need not be independent, and none is needed. For every $h \geq 1$ the tail union $\bigcup_{j \geq h} E_j$ contains E_h, so it has λ-measure at least $1/2$; these unions decrease in h, and λ is finite, so by continuity of a finite measure along a decreasing sequence

$$\lambda\Big(\limsup_{j\to\infty} E_j\Big) = \lim_{h\to\infty} \lambda\Big(\bigcup_{j \geq h} E_j\Big) \geq \frac{1}{2}. \tag{755}$$

The four full-measure events (748), (749), (753) and (754) intersected with $\limsup_j E_j$ still has λ-measure at least $1/2 > 0$, so it is nonempty. Choose one point $\omega^\star$ of $\mathcal{X}$ in it, and let x_{end}, t_{end}, η, (z_j, p_j, a_j) now denote its coordinates, with $z_j = (x_j, t_j)$. Let

$$\mathcal{J} = \{j \geq 1 : \omega^\star \in E_j\},$$

an infinite set by the definition of $\limsup_j E_j$, all of whose elements are ≥ 1.

Since $\mathcal{J} \neq \emptyset$ and every E_j forces $\eta = 1$, we have $\eta = 1$. Hence (748) gives $t_{\mathrm{end}} = 0$, and (749) gives, for every $j \geq 1$, $t_j = \delta_j$ and that (p_j, a_j) is a fit for v at $z_j = (x_j, \delta_j)$ of radius δ_j with constant C_0 in the sense of `IsFit`. By (754), $W(x_j, \delta_j) = v(x_j, \delta_j)$ for every $j \geq 1$. By (753) and $t_{\mathrm{end}} = 0$,

$$H_v(x_{\mathrm{end}}, 0) = H_v(z_{\mathrm{end}}) > -\infty,$$

which is (740). Finally, for $j \in \mathcal{J}$ the defining inequality of E_j reads, with $t_{\mathrm{end}} = 0$,

$$\big|x_{\mathrm{end}} - x_j - (0 - \delta_j)p_j\big| = \big|x_{\mathrm{end}} - x_j + \delta_j p_j\big| \leq K\delta_j.$$

Together with the previous display this is (741) at every $j \in \mathcal{J}$. The constant $K \geq 2 \geq 0$, the point x_{end}, the infinite set $\mathcal{J}$ and the triples (x_j, p_j, a_j) for $j \in \mathcal{J}$ are therefore as required, and the proof is complete. □

2.169. `CylinderComparisonSameCenter.`

Lemma 2.169 (Comparison of cylinder fits: common center)**.** *Let E be a finite-dimensional real inner-product space and fix $0 < c_* \leq C_* < \infty$. There is $C > 0$, depending only on c_* and C_*, with the following property. Let $u : E \to \mathbb{R}$, let $z \in E$, let $0 < \theta < 1$, let $0 < H \leq H' \leq 2H$, and let $P, P' \in E$ with*

$$c_* \leq |P|, |P'| \leq C_*.$$

Suppose P realizes the cylinder fit of parameter θ^2 for u at (z, H) and P' realizes the cylinder fit of parameter θ^2 for u at (z, H'), both in the sense of `IsCylinderFit`*. Write $n = P/|P|$ and $n' = P'/|P'|$. Then*

$$|P - P'| + |n - n'| \le C\theta, \qquad \big||P| - |P'|\big| \le C\theta^2. \tag{756}$$

Proof. All the cylinders below are those of `FitCylinder`, with transverse radius $\sqrt{\theta^2}\,R = \theta R$ at axial radius R, since $\theta > 0$.

Step 1: both cylinders contain $\overline{B}_{\theta H}(z)$. Let $|h| \le \theta H$. For the first cylinder $Z_{\theta^2}(z, H; P)$ the axial component satisfies $|h \cdot n| \le |h| \le \theta H \le H$, since $\theta < 1$, and the transverse component satisfies $|h - (h \cdot n)n| \le |h| \le \theta H$. For the second cylinder $Z_{\theta^2}(z, H'; P')$ we similarly get $|h \cdot n'| \le \theta H \le H \le H'$ and transverse length at most $\theta H \le \theta H'$. So $z + h$ lies in both.

Step 2: the vector bound. By `IsCylinderFit` the two fits give, for $|h| \le \theta H$,

$$|u(z + h) - u(z) - P \cdot h| \le \theta^2 H, \qquad |u(z + h) - u(z) - P' \cdot h| \le \theta^2 H' \le 2\theta^2 H.$$

Subtracting, $|(P - P') \cdot h| \le 3\theta^2 H$ for all $|h| \le \theta H$. If $P \ne P'$, take $h = \theta H(P - P')/|P - P'|$, which has $|h| = \theta H$; then $(P - P') \cdot h = \theta H|P - P'|$ and hence

$$|P - P'| \le 3\theta. \tag{757}$$

If $P = P'$ this holds trivially.

Step 3: the direction bound. For nonzero vectors a, b one has the elementary inequality

$$\left|\frac{a}{|a|} - \frac{b}{|b|}\right| = \frac{\big|\,|b|a - |a|b\,\big|}{|a||b|} \le \frac{\big|\,|b|(a - b)\,\big| + \big|(|b| - |a|)b\,\big|}{|a||b|} \le \frac{2|a - b|}{|a|},$$

using $\big||b| - |a|\big| \le |a - b|$. With $a = P$, $b = P'$ and $|P| \ge c_*$ this and (757) give

$$|n - n'| \le \frac{2|P - P'|}{c_*} \le \frac{6\theta}{c_*}. \tag{758}$$

Together with (757), the first bound in (756) holds with $C \ge 3 + 6/c_*$.

Step 4: a point on the axis in both cylinders. Put $\gamma = \min\{1/2,\ c_*/12\}$ and consider the point $z + \gamma H n$. In the first cylinder its axial component is $\gamma H \le H$ and its transverse component is 0, so it lies in $Z_{\theta^2}(z, H; P)$. In the second cylinder its axial component has absolute value at most $\gamma H \le H \le H'$, and its transverse component, relative to n', is $\gamma H(n - (n \cdot n')n')$, whose length is at most $\gamma H|n - n'|$, because $n - (n \cdot n')n'$ is the orthogonal projection of $n - n'$ onto the orthogonal complement of n'. By (758),

$$\gamma H|n - n'| \le \gamma H \frac{6\theta}{c_*} \le \frac{c_*}{12} \cdot \frac{6\theta H}{c_*} = \frac{\theta H}{2} \le \theta H',$$

so $z + \gamma H n \in Z_{\theta^2}(z, H'; P')$ as well.

Step 5: the axial component of the slope difference. Applying both fits at $Z = z + \gamma H n$ and subtracting as in Step 2, using $(P - P') \cdot (\gamma H n) = \gamma H\,(P - P') \cdot n$,

$$\gamma H\,\big|(P - P') \cdot n\big| \le \theta^2 H + 2\theta^2 H = 3\theta^2 H,$$

hence

$$\big|(P - P') \cdot n\big| \le \frac{3\theta^2}{\gamma}. \tag{759}$$

Step 6: the norm bound. Since $P \cdot n = |P|$ and $P' \cdot n = |P'|(n' \cdot n)$,

$$|P| - |P'| = (P - P') \cdot n + |P'|(n' \cdot n) - |P'| = (P - P') \cdot n - |P'|\big(1 - n' \cdot n\big).$$

As $|n| = |n'| = 1$ we have $|n - n'|^2 = 2 - 2\,n \cdot n'$, i.e. $1 - n' \cdot n = |n - n'|^2/2$. Therefore, by (759), (758) and $|P'| \le C_*$,

$$\big||P| - |P'|\big| \le \frac{3\theta^2}{\gamma} + C_* \frac{|n - n'|^2}{2} \le \left(\frac{3}{\gamma} + \frac{18C_*}{c_*^2}\right)\theta^2.$$

Taking $C = \max\{3 + 6/c_*,\ 3/\gamma + 18C_*/c_*^2\}$, which depends only on c_* and C_* and is positive, gives both bounds in (756). □

2.170. `CylinderComparisonShifted`.

Lemma 2.170 (Comparison of cylinder fits: displaced centers)**.** *Let E be a finite-dimensional real inner-product space and fix $0 < c_* \le C_* < \infty$. There is $C > 0$, depending only on c_* and C_*, with the following property. Let $u : E \to \mathbb{R}$, let $z, z' \in E$, let $0 < \theta < 1$, let $H > 0$, and let $P, P' \in E$ with*

$$c_* \le |P|, |P'| \le C_*.$$

Write $n = P/|P|$ and $n' = P'/|P'|$, and suppose that z' lies in the first cylinder with axial and transverse offsets at most $H/4$ and $\theta H/4$:

$$|(z'-z)\cdot n| \le \frac{H}{4}, \qquad \big|(z'-z) - \big((z'-z)\cdot n\big)n\big| \le \frac{\theta H}{4}.$$

Suppose P realizes the cylinder fit of parameter θ^2 for u at (z, H) and P' realizes the cylinder fit of parameter θ^2 for u at (z', H), both in the sense of `IsCylinderFit`. Then

$$|P - P'| + |n - n'| \le C\theta, \qquad \big||P| - |P'|\big| \le C\theta^2.$$

Proof. All cylinders below are those of `FitCylinder`, with transverse radius θR at axial radius R, since $\theta > 0$. Write $\mathcal{C} = Z_{\theta^2}(z, H; P)$ and $\mathcal{C}' = Z_{\theta^2}(z', H; P')$.

Step 1: $\overline{B}_{\theta H/2}(z') \subseteq \mathcal{C} \cap \mathcal{C}'$. Let $|h| \le \theta H/2$ and put $Z = z' + h$. For $\mathcal{C}'$ the axial component is $|h \cdot n'| \le |h| \le \theta H/2 \le H$ (using $\theta < 1$) and the transverse component has length at most $|h| \le \theta H/2 \le \theta H$; so $Z \in \mathcal{C}'$. For $\mathcal{C}$, write $w = z' - z$ and decompose $Z - z = w + h$. Its axial component satisfies

$$|(w+h)\cdot n| \le |w\cdot n| + |h| \le \frac{H}{4} + \frac{\theta H}{2} \le H,$$

and, since orthogonal projection onto the orthogonal complement of n is a contraction and is linear, its transverse component has length at most

$$\big|w - (w\cdot n)n\big| + |h| \le \frac{\theta H}{4} + \frac{\theta H}{2} \le \theta H.$$

So $Z \in \mathcal{C}$ as well. In particular $z' \in \mathcal{C}$, by taking $h = 0$.

Step 2: the vector bound. By `IsCylinderFit` the first fit gives $|u(Z) - u(z) - P\cdot(Z - z)| \le \theta^2 H$ for $Z \in \mathcal{C}$. Applying this at $Z = z' + h$ and at $Z = z'$ (both in $\mathcal{C}$ by Step 1) and subtracting gives, for $|h| \le \theta H/2$,

$$|u(z'+h) - u(z') - P\cdot h| \le 2\theta^2 H.$$

The second fit gives directly $|u(z'+h) - u(z') - P'\cdot h| \le \theta^2 H$. Subtracting, $|(P - P')\cdot h| \le 3\theta^2 H$ for all $|h| \le \theta H/2$. If $P \ne P'$, take $h = (\theta H/2)(P - P')/|P - P'|$ to get

$$|P - P'| \le 6\theta, \tag{760}$$

and the bound is trivial if $P = P'$. For nonzero vectors a, b one has the elementary inequality

$$\left|\frac{a}{|a|} - \frac{b}{|b|}\right| = \frac{\big|\,|b|a - |a|b\,\big|}{|a||b|} \le \frac{\big|\,|b|(a-b)\,\big| + \big|(|b| - |a|)b\,\big|}{|a||b|} \le \frac{2|a-b|}{|a|},$$

using $\big||b| - |a|\big| \le |a - b|$. With $a = P$, $b = P'$ and $|P| \ge c_*$ this and (760) give

$$|n - n'| \le \frac{2|P - P'|}{c_*} \le \frac{12\theta}{c_*}. \tag{761}$$

Step 3: a segment in both cylinders. Put $\gamma = \min\{1/4,\ c_*/48\}$ and consider the points $z' + sn'$ for $|s| \le \gamma H$. In $\mathcal{C}'$ the axial component is s, with $|s| \le \gamma H \le H$, and the transverse component is 0; so these points lie in $\mathcal{C}'$. In $\mathcal{C}$, with $w = z' - z$ as above, the increment from z is $w + sn'$, whose axial component satisfies

$$|(w + sn')\cdot n| \le \frac{H}{4} + |s| \le \frac{H}{4} + \frac{H}{4} \le H,$$

and whose transverse component has length at most

$$\big|w - (w\cdot n)n\big| + |s|\,\big|n' - (n'\cdot n)n\big| \le \frac{\theta H}{4} + \gamma H|n - n'|,$$

because $n' - (n' \cdot n)n$ is the orthogonal projection of $n' - n$ onto the orthogonal complement of n, so its length is at most $|n - n'|$. By (761),

$$\gamma H|n - n'| \le \frac{c_*}{48} \cdot \frac{12\theta H}{c_*} = \frac{\theta H}{4},$$

so the transverse component has length at most $\theta H/2 \le \theta H$. Hence $z' + sn' \in \mathcal{C}$ for $|s| \le \gamma H$.

Step 4: the axial component of the slope difference. Repeating the subtraction of Step 2 at the points $Z = z' + sn'$, which lie in both cylinders by Step 3, gives for $|s| \le \gamma H$

$$|s|\,\big|(P - P') \cdot n'\big| = \big|(P - P') \cdot (sn')\big| \le 3\theta^2 H.$$

Taking $s = \gamma H$,

$$\big|(P - P') \cdot n'\big| \le \frac{3\theta^2}{\gamma}. \tag{762}$$

Step 5: the norm bound. Since $P' \cdot n' = |P'|$ and $P \cdot n' = |P|(n \cdot n')$,

$$|P'| - |P| = (P' - P) \cdot n' + |P|(n \cdot n') - |P| = (P' - P) \cdot n' - |P|\big(1 - n \cdot n'\big),$$

and $1 - n \cdot n' = |n - n'|^2/2$ because $|n| = |n'| = 1$. Hence, by (762), (761) and $|P| \le C_*$,

$$\big||P| - |P'|\big| \le \frac{3\theta^2}{\gamma} + C_* \frac{|n - n'|^2}{2} \le \left(\frac{3}{\gamma} + \frac{72C_*}{c_*^2}\right)\theta^2.$$

Taking $C = \max\{6 + 12/c_*,\ 3/\gamma + 72C_*/c_*^2\}$, which depends only on c_* and C_* and is positive, gives both asserted bounds. □

2.171. `CylinderFitDoublingChain`.

Lemma 2.171 (Comparison along a doubling chain)**.** *Let E be a finite-dimensional real inner-product space and let c_1, C_1 be reals with $0 < c_1 \le C_1$. There is a constant $C \ge 0$, depending only on c_1 and C_1, with the following property. Let u be a real function on E, let $z \in E$, let $r > 0$, let θ satisfy $0 < \theta < 1$ and let $R \ge 2$. Let $P \in E$ satisfy $c_1 \le |P| \le C_1$ and realize a cylinder fit of parameter θ^2 for u at (z, r) in the sense of `IsCylinderFit`. Suppose moreover that for every ρ with $0 < \rho \le Rr$ there is $Q \in E$ with $c_1 \le |Q| \le C_1$ realizing a cylinder fit of parameter θ^2 for u at (z, ρ). Then there is $P_R \in E$ with $c_1 \le |P_R| \le C_1$ realizing a cylinder fit of parameter θ^2 for u at (z, Rr) and satisfying*

$$\left|\frac{P_R}{|P_R|} - \frac{P}{|P|}\right| \le C(1 + \log R)\,\theta, \qquad \big||P_R| - |P|\big| \le C(1 + \log R)\,\theta^2.$$

Proof. Apply `CylinderComparisonSameCenter` with $c_* = c_1$ and $C_* = C_1$ and let $C_a > 0$ be the constant it supplies, so that for every real function u on E, every $z \in E$, every $\theta \in (0, 1)$, all radii $0 < H \le H' \le 2H$ and all slopes Q, Q' with norms in $[c_1, C_1]$ realizing cylinder fits of parameter θ^2 for u at (z, H) and (z, H') respectively,

$$|Q - Q'| + \left|\frac{Q}{|Q|} - \frac{Q'}{|Q'|}\right| \le C_a\theta, \qquad \big||Q| - |Q'|\big| \le C_a\theta^2. \tag{763}$$

We show that $C = 2C_a$ works; it is nonnegative because $C_a > 0$. Fix u, z, r, θ, R and P as in the statement.

Step 1: the chain of radii. The set of integers $k \ge 0$ with $R \le 2^{k+1}$ is nonempty: choosing an integer k with $R < k$ and using $k \le 2^k \le 2^{k+1}$ gives such a k. Let n be its least element, so that

$$R \le 2^{n+1}, \qquad 2^n < R. \tag{764}$$

Indeed the first is the defining property of n; for the second, if $n = 0$ then $2^0 = 1 < 2 \le R$, and if $n \ge 1$ then minimality gives $\neg(R \le 2^{(n-1)+1}) = \neg(R \le 2^n)$, i.e. $2^n < R$.

Define $\rho_k = 2^k r$ for $0 \le k \le n$ and $\rho_k = Rr$ for $k > n$. Then:

- $\rho_k > 0$ for every k, since $r > 0$, $2^k > 0$ and $R \ge 2 > 0$;
- $\rho_0 = r$ and $\rho_{n+1} = Rr$;
- $\rho_k \le Rr$ for every k: for $k \le n$ this is $2^k \le 2^n < R$ by (764), and for $k > n$ it is an equality;

- $\rho_k \le \rho_{k+1} \le 2\rho_k$ for every $k \le n$. If $k < n$ then $k+1 \le n$ and the two radii are $2^k r$ and $2^{k+1} r = 2 \cdot 2^k r$. If $k = n$ then the radii are $2^n r$ and Rr, and (764) gives $2^n r < Rr$ and $Rr \le 2^{n+1} r = 2 \cdot 2^n r$.

Step 2: the slopes along the chain. For each k the radius ρ_k satisfies $0 < \rho_k \le Rr$, so the hypothesis supplies a slope Q_k with $c_1 \le |Q_k| \le C_1$ realizing a cylinder fit of parameter θ^2 for u at (z, ρ_k). Put $\widehat{Q}_0 = P$ and $\widehat{Q}_k = Q_k$ for $k \ge 1$. Then for every k the vector $\widehat{Q}_k$ realizes a cylinder fit of parameter θ^2 for u at (z, ρ_k) — for $k = 0$ because $\rho_0 = r$ and P realizes the fit at (z, r) by hypothesis, and for $k \ge 1$ by the choice of Q_k — and $c_1 \le |\widehat{Q}_k| \le C_1$, for $k = 0$ by hypothesis and for $k \ge 1$ by the choice of Q_k.

Step 3: summing the chain. We prove by induction on k that for $0 \le k \le n+1$

$$\Big| \frac{\widehat{Q}_k}{|\widehat{Q}_k|} - \frac{P}{|P|} \Big| \le k\, C_a \theta, \qquad \big| |\widehat{Q}_k| - |P| \big| \le k\, C_a \theta^2. \tag{765}$$

For $k = 0$ both sides vanish, because $\widehat{Q}_0 = P$. Assume (765) for some k with $k+1 \le n+1$, so $k \le n$. By Step 1, $0 < \rho_k \le \rho_{k+1} \le 2\rho_k$, and by Step 2 the slopes $\widehat{Q}_k, \widehat{Q}_{k+1}$ have norms in $[c_1, C_1]$ and realize cylinder fits of parameter θ^2 at (z, ρ_k) and (z, ρ_{k+1}). So (763) applies and gives, discarding the nonnegative term $|\widehat{Q}_k - \widehat{Q}_{k+1}|$ from its left side,

$$\Big| \frac{\widehat{Q}_k}{|\widehat{Q}_k|} - \frac{\widehat{Q}_{k+1}}{|\widehat{Q}_{k+1}|} \Big| \le C_a \theta, \qquad \big| |\widehat{Q}_k| - |\widehat{Q}_{k+1}| \big| \le C_a \theta^2.$$

The triangle inequality for the norm of E, applied to the three vectors $\widehat{Q}_{k+1}/|\widehat{Q}_{k+1}|$, $\widehat{Q}_k/|\widehat{Q}_k|$ and $P/|P|$, together with the inductive hypothesis, gives the first half of (765) for $k+1$; the triangle inequality for the absolute value, applied to $|\widehat{Q}_{k+1}|$, $|\widehat{Q}_k|$ and $|P|$, gives the second half.

Step 4: the logarithmic factor. Taking $k = n+1$ in (765), it remains to show $n+1 \le 2(1 + \log R)$. First, $\log 2 > 1/2$: the strict inequality $x + 1 < e^x$ at $x = -1/2$ gives $1/2 < e^{-1/2}$, hence $\log(1/2) < -1/2$ by monotonicity of log, and $\log(1/2) = -\log 2$. Next, $2^n < R$ by (764) and $2^n > 0$, so monotonicity of log and $\log(2^n) = n \log 2$ give $n \log 2 \le \log R$. Since $\log 2 > 1/2$ and $n \ge 0$ we get $n/2 \le n \log 2 \le \log R$, i.e. $n \le 2 \log R$, and therefore

$$n + 1 \le 2 \log R + 1 \le 2 + 2 \log R = 2(1 + \log R),$$

using $\log R \ge 0$, which holds because $R \ge 2 \ge 1$. Multiplying (765) at $k = n+1$ by this bound — legitimate because $C_a \theta \ge 0$ and $C_a \theta^2 \ge 0$ — and recalling $\rho_{n+1} = Rr$, the vector $P_R = \widehat{Q}_{n+1}$ has all the asserted properties with $C = 2C_a$. □

2.172. `CylinderFitSlopeBounds.`

Lemma 2.172 (Slope bounds for a cylinder fit)**.** *Fix an integer $d \ge 1$, put $n = d + 1$ and write $e_b = (0, 1)$ for the added unit direction. Let $U \in \mathcal{U}$ be in the normalized family of `NormalizedInfFamily`, let $L \in \mathbb{R}$, let z be a space-time point and let $r > 0$ and $0 < \lambda \le 1$. Assume the Lipschitz estimate*

$$|U(z + h) - U(z)| \le L|h| \qquad (|h| \le r),$$

and let P realize the cylinder fit of parameter λ for U at (z, r) in the sense of `IsCylinderFit`. Then

$$|P| \le L + \sqrt{\lambda}, \qquad |P \cdot e_b - 1| \le \sqrt{\lambda}. \tag{766}$$

Proof. Write $N = P/|P|$ for the axis direction; note $P \ne 0$ by `IsCylinderFit`. Put $\rho = \sqrt{\lambda}\, r$; since $0 < \lambda \le 1$ we have $0 < \rho \le r$.

Step 1: the cylinder contains $\overline{B}_\rho(z)$. Let $|h| \le \rho$. Then $|h \cdot N| \le |h| \le \rho \le r$ and $|h - (h \cdot N)N| \le |h| \le \rho = \sqrt{\lambda}\, r$, the first inequality by Cauchy–Schwarz with $|N| = 1$ and the second because orthogonal projection does not increase length. By `FitCylinder`, $z + h \in Z_\lambda(z, r; P)$. Hence by `IsCylinderFit`

$$|U(z + h) - U(z) - P \cdot h| \le \lambda r \qquad (|h| \le \rho). \tag{767}$$

Step 2: the norm bound. Take $h = \rho N$ in (767), which is admissible since $|\rho N| = \rho$, and also $|h| \le r$ so the Lipschitz hypothesis applies. Then $P \cdot h = \rho |P|$, so

$$\rho |P| \le |U(z+h) - U(z)| + \lambda r \le L\rho + \lambda r.$$

Dividing by $\rho = \sqrt{\lambda}\, r > 0$ and using $\lambda r/\rho = \sqrt{\lambda}$ gives $|P| \le L + \sqrt{\lambda}$, the first bound in (766).

Step 3: the exact increment in the direction e_b. Write $z = (y_0, b_0)$. By `NormalizedInfFamily` there is u with $U(y,b) = u(y) + b$ for all (y,b). Taking $h = \rho e_b = (0, \rho)$, which satisfies $|h| = \rho \le r$, we get exactly

$$U(z+h) - U(z) = \big(u(y_0) + b_0 + \rho\big) - \big(u(y_0) + b_0\big) = \rho,$$

so the increment of U in the direction e_b equals the increment length. Substituting into (767) with this h, and using $P \cdot h = \rho\,(P \cdot e_b)$,

$$\rho\,|1 - P \cdot e_b| = |\rho - P \cdot h| = |U(z+h) - U(z) - P \cdot h| \le \lambda r.$$

Dividing by $\rho = \sqrt{\lambda}\, r > 0$ gives $|P \cdot e_b - 1| \le \sqrt{\lambda}$, the second bound in (766). □

2.173. `DyadicFitTaylorEstimate`.

Lemma 2.173 (The dyadic slope tail and its Hölder–Taylor estimate)**.** *Let E be a finite-dimensional real inner-product space and let $A_0 > 0$, $\gamma > 0$ and $r_0 > 0$ be reals. Write*

$$r_j = 2^{-j} r_0, \qquad \rho_j = \sqrt{A_0}\, r_j^{1+\gamma/2} \qquad (j \ge 0), \qquad \alpha = \frac{\gamma}{2+\gamma}.$$

Then there is a real $C > 0$, depending only on A_0, γ and r_0, with the following property. Let $U : E \to \mathbb{R}$, let $z \in E$ and let $(P_j)_{j \ge 0}$ be vectors of E such that

$$\big|U(z+h) - U(z) - \langle P_j, h\rangle\big| \le A_0 r_j^{1+\gamma} \qquad \text{whenever } j \ge 0 \text{ and } |h| \le \rho_j. \tag{768}$$

Then there is $P_\infty \in E$ with

$$|P_j - P_\infty| \le C r_j^{\gamma/2} \qquad (j \ge 0)$$

and

$$\big|U(z+h) - U(z) - \langle P_\infty, h\rangle\big| \le C|h|^{1+\alpha} \qquad \big(|h| \le \rho_0\big).$$

Proof. Throughout put $p = 1 + \gamma/2 > 0$, so that $\rho_j = \sqrt{A_0}\, r_j^p$ and $\alpha = (\gamma/2)/p$, whence

$$p\,\alpha = \gamma/2. \tag{769}$$

Put $t = (1/2)^p$ and $q = (1/2)^{\gamma/2}$; since $0 < 1/2 < 1$ and $p > 0$, $\gamma/2 > 0$, both lie in $(0,1)$. For any real e and any $j \ge 0$,

$$r_j^e = \big(2^{-j} r_0\big)^e = \big((1/2)^j\big)^e r_0^e = \big((1/2)^e\big)^j r_0^e,$$

the first step because both factors are positive and the second because $((1/2)^j)^e = (1/2)^{je} = ((1/2)^e)^j$. In particular

$$\rho_j = B\, t^j \quad \text{with } B = \sqrt{A_0}\, r_0^p > 0, \qquad r_j^{\gamma/2} = q^j a \quad \text{with } a = r_0^{\gamma/2} > 0. \tag{770}$$

Consequently $\rho_j > 0$, $\rho_{j+1} = t\rho_j \le \rho_j$, and $\rho_j \to 0$ since $0 < t < 1$.

Step 1: consecutive slopes. Fix $j \ge 0$ and put $v = P_j - P_{j+1}$. If $v = 0$ the bound below is trivial, so assume $v \ne 0$ and set

$$h = \frac{\rho_{j+1}}{|v|}\, v, \qquad \text{so} \qquad |h| = \rho_{j+1}, \qquad \langle v, h\rangle = \frac{\rho_{j+1}}{|v|}\,|v|^2 = \rho_{j+1}|v|,$$

using $|v| > 0$ and $\langle v, v\rangle = |v|^2$. Since $|h| = \rho_{j+1} \le \rho_j$, hypothesis (768) applies to this h both at the index j and at the index $j+1$. Writing

$$A = U(z+h) - U(z) - \langle P_j, h\rangle, \qquad B' = U(z+h) - U(z) - \langle P_{j+1}, h\rangle,$$

linearity of the inner product in its first argument gives $\langle v, h\rangle = B' - A$, while (768) gives $|A| \le A_0 r_j^{1+\gamma}$ and $|B'| \le A_0 r_{j+1}^{1+\gamma}$. As $r_{j+1} \le r_j$ and $1 + \gamma > 0$ we have $r_{j+1}^{1+\gamma} \le r_j^{1+\gamma}$, so

$$\rho_{j+1}|v| = B' - A \le |B'| + |A| \le 2A_0 r_j^{1+\gamma}.$$

Now $\gamma/2+p = 1+\gamma$ and $r_j > 0$, so $r_j^{1+\gamma} = r_j^{\gamma/2} r_j^p$, and $r_j^p = \rho_j/\sqrt{A_0}$; hence, using $\sqrt{A_0}\sqrt{A_0} = A_0$,

$$A_0 r_j^{1+\gamma} = \sqrt{A_0}\,(q^j a)\rho_j. \tag{771}$$

Since $\rho_{j+1} = t\rho_j$, dividing the previous display by $\rho_{j+1} > 0$ and substituting (771) gives

$$|P_j - P_{j+1}| \le C_1 a\, q^j, \qquad C_1 = \frac{2\sqrt{A_0}}{t}, \tag{772}$$

and (772) holds trivially when $v = 0$ as well.

Step 2: the limit slope and its tail. By (772) the distances $|P_j - P_{j+1}|$ are bounded by $(C_1 a)q^j$ with $0 < q < 1$, so $\sum_j |P_j - P_{j+1}|$ is dominated by a convergent geometric series and (P_j) is a Cauchy sequence. A finite-dimensional real inner-product space is complete, so the sequence converges; call the limit P_∞. The standard geometric tail estimate for such a sequence gives, for every j,

$$|P_j - P_\infty| \le \frac{C_1 a\, q^j}{1-q} = \frac{C_2}{a}\,(q^j a), \qquad C_2 = \frac{C_1 a}{1-q}, \tag{773}$$

which by (770) is the first asserted bound with constant C_2/a.

Step 3: the scale selection. Let $h \in E$ with $|h| \le \rho_0$. If $h = 0$ both sides of the second assertion are 0, since $\langle P_\infty, 0\rangle = 0$ and $0^{1+\alpha} = 0$ for $1+\alpha > 0$; so assume $h \ne 0$, whence $|h| > 0$. Because $\rho_m \to 0$ there is an m with $\rho_m < |h|$; let m be the least such index. Then $m \ne 0$, since $\rho_0 \ge |h|$, so $m = j+1$ for some $j \ge 0$, and minimality of m gives

$$\rho_{j+1} < |h| \le \rho_j. \tag{774}$$

Step 4: the estimate at that scale. Since $|h| \le \rho_j$, (768) applies at the index j. Writing $\langle P_j - P_\infty, h\rangle = \big(U(z+h) - U(z) - \langle P_\infty, h\rangle\big) - \big(U(z+h) - U(z) - \langle P_j, h\rangle\big)$, the triangle inequality, (768) and the Cauchy–Schwarz bound $|\langle P_j - P_\infty, h\rangle| \le |P_j - P_\infty|\,|h|$ give

$$\big|U(z+h) - U(z) - \langle P_\infty, h\rangle\big| \le A_0 r_j^{1+\gamma} + |P_j - P_\infty|\,|h|.$$

For the first term, $\rho_{j+1} = t\rho_j < |h|$ by (774) gives $\rho_j < |h|/t$, so by (771) and $q^j a > 0$,

$$A_0 r_j^{1+\gamma} = \sqrt{A_0}\,(q^j a)\rho_j \le \frac{\sqrt{A_0}}{t}\,(q^j a)|h|.$$

For the second, (773) gives $|P_j - P_\infty|\,|h| \le (C_2/a)(q^j a)|h|$. Hence, with $C_3 = \sqrt{A_0}/t + C_2/a$,

$$\big|U(z+h) - U(z) - \langle P_\infty, h\rangle\big| \le C_3 (q^j a)|h|. \tag{775}$$

Step 5: converting the scale into $|h|^\alpha$. From $\rho_j < |h|/t$ and $\rho_j = \sqrt{A_0}\, r_j^p$ we get $r_j^p < |h|/(t\sqrt{A_0})$. Both sides are positive and $\alpha > 0$, so raising to the power α preserves the inequality:

$$\big(r_j^p\big)^\alpha \le \Big(\frac{|h|}{t\sqrt{A_0}}\Big)^\alpha = \Big(\frac{1}{t\sqrt{A_0}}\Big)^\alpha |h|^\alpha,$$

the last step because $(xy)^\alpha = x^\alpha y^\alpha$ for $x, y \ge 0$. By (769) and (770), $\big(r_j^p\big)^\alpha = r_j^{p\alpha} = r_j^{\gamma/2} = q^j a$, so

$$q^j a \le \Big(\frac{1}{t\sqrt{A_0}}\Big)^\alpha |h|^\alpha.$$

Substituting into (775) and using $|h|^\alpha |h| = |h|^{1+\alpha}$, which holds because $|h| > 0$,

$$\big|U(z+h) - U(z) - \langle P_\infty, h\rangle\big| \le C_4 |h|^{1+\alpha}, \qquad C_4 = C_3\Big(\frac{1}{t\sqrt{A_0}}\Big)^\alpha.$$

Taking $C = \max\{1, C_2/a, C_4\}$, which is positive and depends only on A_0, γ and r_0, both assertions hold: the first because $C \ge C_2/a$ and $r_j^{\gamma/2} \ge 0$, the second because $C \ge C_4$ and $|h|^{1+\alpha} \ge 0$. Neither C_1, C_2, C_3 nor C_4 depends on U, on z or on the sequence (P_j). □

2.174. `NormalizedSlopeCoefficients`.

Lemma 2.174 (Frame coordinates of a nearly aligned slope)**.** *Let F be a finite-dimensional real inner-product space, let $\ell^2(F\times\mathbb{R})$ be the associated space-time with added unit direction $e_b=(0,1)$, and let Φ be a linear isometry of $\ell^2(F\times\mathbb{R})$ onto itself. Let $c>0$ and $\lambda>0$ be reals, let $Q\in\ell^2(F\times\mathbb{R})$ satisfy $|Q|>0$, and let* dir *and* nrm *be reals with*

$$\Big|\frac{Q}{|Q|}-\Phi(e_b)\Big|\le \mathrm{dir},\qquad \big||Q|-c\big|\le \mathrm{nrm}.$$

Then there are $p'\in F$ and $q'\in\mathbb{R}$ such that

$$Q=\Phi\big(c\sqrt{\lambda}\,p',\ c(1+\lambda q')\big)$$

and

$$c\sqrt{\lambda}\,|p'|\le |Q|\,\mathrm{dir},\qquad c\lambda\,|q'|\le \mathrm{nrm}+\tfrac12|Q|\,\mathrm{dir}^2.$$

Proof. Write $e=\Phi(e_b)$ and $p=Q/|Q|$, which is legitimate because $|Q|>0$. Since Φ is a linear isometry and $|e_b|=1$, we have $|e|=1$; also $|p|=1$. The first hypothesis reads $|p-e|\le\mathrm{dir}$, and in particular $0\le|p-e|\le\mathrm{dir}$.

Step 1: the coordinates of Q in the frame. Put $Q^\flat=\Phi^{-1}Q$ and let $(Q_x,Q_t)\in F\times\mathbb{R}$ be its two components, so that $Q^\flat=(Q_x,Q_t)$ and $Q=\Phi(Q_x,Q_t)$. Two identities will be used.

First, $Q_t=\langle Q,e\rangle$. Indeed $Q_t=\langle Q^\flat,e_b\rangle$, because in $\ell^2(F\times\mathbb{R})$ the inner product of (a,s) with $(0,1)$ is $\langle a,0\rangle+s\cdot1=s$; and $\langle Q^\flat,e_b\rangle=\langle\Phi Q^\flat,\Phi e_b\rangle=\langle Q,e\rangle$ because Φ is a linear isometry and $\Phi Q^\flat=Q$.

Second, $|Q_x|^2+Q_t^2=|Q|^2$. Indeed $|Q^\flat|^2=|Q_x|^2+|Q_t|^2$ by the definition of the ℓ^2 product norm, $|Q_t|^2=Q_t^2$ for a real number, and $|Q^\flat|=|\Phi^{-1}Q|=|Q|$ because Φ^{-1} is a linear isometry.

Step 2: the quadratic angular identity. Since $|p|=|e|=1$,

$$|p-e|^2=|p|^2-2\langle p,e\rangle+|e|^2=2-2\langle p,e\rangle,$$

so $\langle p,e\rangle=1-|p-e|^2/2$. As $Q=|Q|p$ and the inner product is linear in its first argument, Step 1 gives

$$Q_t=\langle Q,e\rangle=|Q|\,\langle p,e\rangle=|Q|\Big(1-\frac{|p-e|^2}{2}\Big). \tag{776}$$

Step 3: the transverse estimate, from the direction bound alone. Substituting (776) into $|Q_x|^2=|Q|^2-Q_t^2$ and writing $w=|p-e|^2\ge0$,

$$|Q_x|^2=|Q|^2-|Q|^2\Big(1-\frac{w}{2}\Big)^2=|Q|^2\Big(w-\frac{w^2}{4}\Big)\le|Q|^2w=\big(|Q|\,|p-e|\big)^2,$$

the inequality because $|Q|^2w^2/4\ge0$. Both $|Q_x|$ and $|Q|\,|p-e|$ are nonnegative, and $|Q|\,|p-e|\le|Q|\,\mathrm{dir}$ because $|Q|\ge0$; hence

$$|Q_x|\le|Q|\,|p-e|\le|Q|\,\mathrm{dir}. \tag{777}$$

Note that only the direction bound has been used.

Step 4: the longitudinal estimate, from the norm bound and the angular error. By (776),

$$Q_t-c=\big(|Q|-c\big)-\frac{|Q|\,|p-e|^2}{2}.$$

The subtracted term is nonnegative, and $|p-e|^2\le\mathrm{dir}^2$ because $0\le|p-e|\le\mathrm{dir}$, so $|Q|\,|p-e|^2/2\le|Q|\,\mathrm{dir}^2/2$ using $|Q|\ge0$. Combining these with $\big||Q|-c\big|\le\mathrm{nrm}$ gives both inequalities $-(\mathrm{nrm}+|Q|\mathrm{dir}^2/2)\le Q_t-c\le\mathrm{nrm}+|Q|\mathrm{dir}^2/2$, that is

$$|Q_t-c|\le\mathrm{nrm}+\tfrac12|Q|\,\mathrm{dir}^2. \tag{778}$$

Here BOTH the norm bound and the quadratic angular error are used; the angular error is quadratic in dir, which is why it does not spoil the estimate.

Step 5: the coefficients. Since $c>0$ and $\lambda>0$ we have $c\sqrt{\lambda}>0$ and $c\lambda>0$. Put

$$p'=\frac{Q_x}{c\sqrt{\lambda}},\qquad q'=\frac{Q_t-c}{c\lambda}.$$

Then $c\sqrt{\lambda}\,p' = Q_x$ and $c(1+\lambda q') = c + (Q_t - c) = Q_t$, so

$$\Phi\big(c\sqrt{\lambda}\,p',\ c(1+\lambda q')\big) = \Phi(Q_x, Q_t) = \Phi Q^\flat = Q,$$

which is the asserted identity. Moreover $c\sqrt{\lambda}\,|p'| = |Q_x|$ and $c\lambda\,|q'| = |Q_t - c|$, so (777) and (778) are exactly the two asserted inequalities. □

2.175. `ShearedBoxInFitCylinder.`

Lemma 2.175 (The sheared box lies in the fitting cylinder of the normalized slope)**.** *Let F be a real inner-product space, let $\ell^2(F\times\mathbb{R})$ be the associated space-time, and let Φ be a linear isometry of $\ell^2(F\times\mathbb{R})$ onto itself. Let $z\in\ell^2(F\times\mathbb{R})$, let $c>0$, $\lambda>0$, $S>0$ and $r>0$ be reals, let $p'\in F$, $q'\in\mathbb{R}$, $h\in F$ and $\ell\in\mathbb{R}$, and let C be a real with*

$$|p'| + |q'| \le C. \tag{779}$$

Write $A = 1+\lambda q'$ and assume

$$\tfrac{1}{2} \le A, \qquad \lambda\big(C^2 + C + 1\big) \le \tfrac{1}{100}. \tag{780}$$

If

$$|\ell| \le S/2, \qquad \Big|h - \frac{\ell}{A}\,p'\Big| \le S, \tag{781}$$

then

$$z + r\,\Phi\big(\sqrt{\lambda}\,h,\ \ell\big) \in Z_{32\lambda}\big(z,\ Sr;\ \Phi(c\sqrt{\lambda}\,p',\ cA)\big).$$

Proof. Note first that $C\ge 0$, since $|p'|+|q'|\ge 0$ and (779) holds; hence also $|p'|\le C$, because $|q'|\ge 0$. From (780), $A\ge 1/2>0$, and since $\lambda>0$ and $C^2, C\ge 0$ the second part of (780) gives

$$\lambda C^2 \le \tfrac{1}{100}, \qquad \lambda C \le \tfrac{1}{100}. \tag{782}$$

Write $g = h - (\ell/A)p'$, so $|g|\le S$ by (781). Put

$$e = (\sqrt{\lambda}\,p',\ A) \in \ell^2(F\times\mathbb{R}), \qquad N = |e|, \qquad u = (\sqrt{\lambda}\,h,\ \ell), \qquad \mathrm{res} = (\sqrt{\lambda}\,g,\ 0).$$

Step 1: the length N of the frame vector. By the definition of the ℓ^2 product norm and $|\sqrt{\lambda}\,p'| = \sqrt{\lambda}\,|p'|$,

$$N^2 = \lambda|p'|^2 + A^2. \tag{783}$$

Since $\lambda|p'|^2\ge 0$ we get $N^2\ge A^2$, and both $N\ge 0$ and $A>0$, so $N\ge A>0$; in particular $N\ge 1/2$, hence $N^{-1}\le 2$. For an upper bound, $|p'|\le C\le 2CA$ because $A\ge 1/2$ and $C\ge 0$, so $\lambda|p'|^2\le 4\lambda C^2A^2$, and therefore by (783)

$$N^2 \le A^2\big(1+4\lambda C^2\big) \le A^2\big(1+2\lambda C^2\big)^2,$$

the last step because $(1+2x)^2 = 1+4x+4x^2 \ge 1+4x$ for $x=\lambda C^2\ge 0$. Both N and $A(1+2\lambda C^2)$ are nonnegative, so

$$N \le A\big(1+2\lambda C^2\big). \tag{784}$$

Step 2: the exact shear decomposition. In $F\times\mathbb{R}$,

$$u = \frac{\ell}{A}\,e + \mathrm{res}. \tag{785}$$

Indeed the second components are $(\ell/A)A + 0 = \ell$, using $A\ne 0$; and the first components are

$$\frac{\ell}{A}\big(\sqrt{\lambda}\,p'\big) + \sqrt{\lambda}\Big(h - \frac{\ell}{A}\,p'\Big) = \sqrt{\lambda}\,h,$$

the two multiples of p' cancelling. This is an identity, with no error term.

Step 3: the normalized axis. Since Φ is a linear isometry and $c>0$, the slope $P = \Phi(c\sqrt{\lambda}\,p', cA) = \Phi(c\,e) = c\,\Phi(e)$ has $|P| = cN > 0$, so

$$\frac{P}{|P|} = \frac{c\,\Phi(e)}{cN} = N^{-1}\Phi(e) =: n,$$

and $|n| = N^{-1}|\Phi(e)| = N^{-1}N = 1$. Also $\Phi(e) = N\,n$. The increment of the asserted point relative to z is $D = r\,\Phi(u)$, and applying Φ to (785) and multiplying by r gives

$$D = \Big(\frac{r\ell N}{A}\Big)n + r\,\Phi(\mathrm{res}). \tag{786}$$

Step 4: the axial bound $|D \cdot n| \le Sr$. Taking the inner product of (786) with n and using $|n| = 1$,

$$D \cdot n = \frac{r\ell N}{A} + r\,\big(\Phi(\mathrm{res}) \cdot n\big). \tag{787}$$

For the second term, $\Phi(\mathrm{res})\cdot n = N^{-1}\big(\Phi(\mathrm{res})\cdot\Phi(e)\big) = N^{-1}(\mathrm{res}\cdot e)$ because Φ is a linear isometry, and by the definition of the ℓ^2 inner product

$$\mathrm{res}\cdot e = \big(\sqrt{\lambda}\,g\big)\cdot\big(\sqrt{\lambda}\,p'\big) + 0\cdot A = \lambda\,(g\cdot p'),$$

using $\sqrt{\lambda}\sqrt{\lambda} = \lambda$, which holds since $\lambda \ge 0$. By Cauchy–Schwarz, $|g\cdot p'| \le |p'|\,|g| \le CS$, so with $N^{-1} \le 2$,

$$\big|\Phi(\mathrm{res})\cdot n\big| \le N^{-1}\lambda\,|g\cdot p'| \le 2\lambda CS.$$

For the first term, $|\ell| \le S/2$ by (781) and $A > 0$, so by (784)

$$\Big|\frac{\ell N}{A}\Big| = \frac{|\ell|\,N}{A} \le \frac{S}{2}\cdot\frac{N}{A} \le \frac{S}{2}\big(1 + 2\lambda C^2\big).$$

Hence from (787) and the triangle inequality, using $r > 0$,

$$|D\cdot n| \le r\Big[\frac{S}{2}\big(1+2\lambda C^2\big) + 2\lambda CS\Big] = rS\Big[\frac{1}{2} + \lambda C^2 + 2\lambda C\Big].$$

By (782), $\lambda C^2 + 2\lambda C \le 1/100 + 2/100 = 3/100 \le 1/2$, so the bracket is at most 1 and $|D\cdot n| \le Sr$. This is the strict axial margin: the first term alone already uses only $(1+2\lambda C^2)/2$ of the available length.

Step 5: the transverse bound $|D - (D\cdot n)n| \le \sqrt{32\lambda}\,Sr$. Subtracting $(D\cdot n)n$ from (786) and using (787), the multiple of n coming from the first term cancels exactly, leaving

$$D - (D\cdot n)n = r\Big[\Phi(\mathrm{res}) - \big(\Phi(\mathrm{res})\cdot n\big)n\Big].$$

For any vector x and any unit vector n the vectors $x - (x\cdot n)n$ and $(x\cdot n)n$ are orthogonal, since $\big(x-(x\cdot n)n\big)\cdot\big((x\cdot n)n\big) = (x\cdot n)\big(x\cdot n - (x\cdot n)|n|^2\big) = 0$; their sum is x, so by the Pythagorean identity $|x-(x\cdot n)n|^2 + |(x\cdot n)n|^2 = |x|^2$ and therefore $|x-(x\cdot n)n| \le |x|$. Applying this with $x = \Phi(\mathrm{res})$ and using that Φ is an isometry,

$$\big|D - (D\cdot n)n\big| \le r\,\big|\Phi(\mathrm{res})\big| = r\,|\mathrm{res}| = r\sqrt{\lambda}\,|g| \le r\sqrt{\lambda}\,S,$$

where $|\mathrm{res}| = \sqrt{\lambda}\,|g|$ by the definition of the ℓ^2 product norm applied to $(\sqrt{\lambda}\,g, 0)$. Finally $\sqrt{32\lambda} = \sqrt{32}\sqrt{\lambda} \ge \sqrt{\lambda}$ because $\sqrt{32} \ge 1$ and $\sqrt{\lambda} > 0$, so $r\sqrt{\lambda}\,S \le \sqrt{32\lambda}\,(Sr)$.

Steps 4 and 5 are exactly the two conditions defining membership of the point $z + D = z + r\Phi(\sqrt{\lambda}\,h, \ell)$ in the cylinder $Z_{32\lambda}(z, Sr; P)$ with $P = \Phi(c\sqrt{\lambda}\,p', cA)$. □

2.176. `EnclosingAndNearbyFits`.

Lemma 2.176 (Step 5: the enclosing fit and the nearby fits)**.** *Let $d \in \mathbb{N}$ and write $\mathcal{H} = \ell^2(\mathbb{R}^d \times \mathbb{R})$. Let $r, \lambda : \mathbb{N} \to \mathbb{R}$, let $U_i : \mathcal{H} \to \mathbb{R}$, let $z_i, P_i \in \mathcal{H}$, let Φ_i be a surjective linear isometry of $\mathcal{H}$ onto itself, let $\theta, c : \mathbb{N} \to \mathbb{R}$, let $Z_i : \mathcal{H} \to \mathcal{H}$ (all four families indexed by $i \in \mathbb{N}$), let $\mathcal{B} : \mathbb{R} \to \mathcal{P}(\mathcal{H})$, let* Rad $: \mathbb{R} \to \mathbb{R}$*, let* Mof $: \mathbb{R}\times\mathbb{R} \to \mathbb{R}$*, let* iadm $: \mathbb{R}\times\mathbb{R} \to \mathbb{N}$*, let $L, C_1, C_2 \in \mathbb{R}$ and let $i_2 \in \mathbb{N}$. Assume:*

1. *$r_i > 0$ and $\lambda_i > 0$ for every i, and $L \ge 0$;*
2. *$\theta_i = \sqrt{32\lambda_i}$ for every i, and $\theta_i \le \frac{1}{2}$ for every $i \ge i_2$;*

(3) $C_1 \ge 1$, *and the two comparison principles hold with constant* C_1*: for* $i \ge i_2$*, for concentric cylinders of radii* $0 < H \le H' \le 2H$ *at a common centre, and for slopes* Q, Q' *with* $\frac{1}{2} \le |Q|, |Q'| \le L+1$ *realizing cylinder fits of parameter* $32\lambda_i$ *for* U_i *at those two radii in the sense of* `IsCylinderFit`*,*

$$|Q - Q'| + \big|Q/|Q| - Q'/|Q'|\big| \le C_1\theta_i, \qquad \big||Q| - |Q'|\big| \le C_1\theta_i^2,$$

and the same two bounds hold for fits at a common radius $H > 0$ *about two centres* w, w' *whose offset satisfies the quarter-bounds* $|\langle Q/|Q|, w' - w\rangle| \le H/4$ *and* $\big|(w' - w) - \langle Q/|Q|, w' - w\rangle\, Q/|Q|\big| \le \theta_i H/4$*;*

(4) $C_2 \ge 0$*, and the chain estimate holds: for all* $R, M \in \mathbb{R}$ *with* $2 \le R$*,* $2 \le M$ *and* $R \le M$ *there is* $i_0 \ge i_2$ *such that for every* $i \ge i_0$ *there is* P_R *realizing a cylinder fit of parameter* $32\lambda_i$ *for* U_i *at* (z_i, Rr_i) *with* $\frac{1}{2} \le |P_R| \le L+1$*,*

$$\big|P_R/|P_R| - P_i/c_i\big| \le C_2(1 + \log R)\,\theta_i, \qquad \big||P_R| - c_i\big| \le C_2(1 + \log R)\,\theta_i^2;$$

(5) $\operatorname{Rad} K \ge 2$ *whenever* $K \ge 1$*;*

(6) *for* $K \ge 1$ *and* $S > 0$*,* $\operatorname{Mof}(K, S)$ *dominates each of* $2K$*,* $\operatorname{Rad} K$ *and* S*; and* $\operatorname{iadm}(K, S) \ge i_2$ *for all* K, S*;*

(7) *fits exist at box points up to that radius: for* $K \ge 1$*,* $S > 0$*,* $i \ge \operatorname{iadm}(K, S)$*,* $w \in \mathcal{B}_K$ *and* $0 < \rho \le \operatorname{Mof}(K, S)\, r_i$ *there is* Q *realizing a cylinder fit of parameter* $32\lambda_i$ *for* U_i *at* $(Z_i(w), \rho)$ *with* $\frac{1}{2} \le |Q| \le L+1$*;*

(8) *the coefficients normalize: for* $i \ge i_2$*, every* Q *with* $\frac{1}{2} \le |Q| \le L+1$*, and every* $\delta \ge 0$ *with* $\big|Q/|Q| - P_i/c_i\big| \le \delta\theta_i$ *and* $\big||Q| - c_i\big| \le \delta\theta_i^2$*, there are* $p' \in \mathbb{R}^d$ *and* $q' \in \mathbb{R}$ *with* $Q = \Phi_i\big(c_i\sqrt{\lambda_i}\,p',\, c_i(1 + \lambda_i q')\big)$ *and* $|p'| + |q'| \le 100(1+L)(\delta + \delta^2)$*;*

(9) *the quarter-bounds of Claim 4a: for* $K \ge 1$*,* $i \ge i_2$ *and every* P_R *with* $|P_R| \ge \frac{1}{2}$ *and* $\big|P_R/|P_R| - P_i/c_i\big| \le C_2(1 + \log \operatorname{Rad} K)\theta_i$*, every* $w \in \mathcal{B}_K$ *satisfies*

$$\big|\langle P_R/|P_R|,\, Z_i(w) - z_i\rangle\big| \le \frac{\operatorname{Rad} K \cdot r_i}{4}, \qquad \Big|\big(Z_i(w) - z_i\big) - \langle P_R/|P_R|, Z_i(w) - z_i\rangle\, P_R/|P_R|\Big| \le \frac{\theta_i \operatorname{Rad} K \cdot r_i}{4}.$$

Then there are $C_R, C_N : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ *and* $i_R, i_N : \mathbb{R}\times\mathbb{R} \to \mathbb{N}$ *with* $C_R \ge 0$*,* $C_N \ge 0$*,* $i_R(K, S) \ge \operatorname{iadm}(K, S)$ *and* $i_N(K, S) \ge i_R(K, S)$*, such that for all* $K \ge 1$*,* $S > 0$ *the following hold.*

- (the enclosing fit) *For every* $i \ge i_R(K, S)$ *there are* $P_R \in \mathcal{H}$*,* $p_R \in \mathbb{R}^d$ *and* $q_R \in \mathbb{R}$ *such that* P_R *realizes a cylinder fit of parameter* $32\lambda_i$ *for* U_i *at* $(z_i, \operatorname{Rad} K \cdot r_i)$*, with* $\frac{1}{2} \le |P_R| \le L+1$*,* $P_R = \Phi_i\big(c_i\sqrt{\lambda_i}\,p_R,\, c_i(1 + \lambda_i q_R)\big)$*,* $|p_R| + |q_R| \le C_R(K, S)$*,* $Z_i(w)$ *lies in the cylinder of* `FitCylinder` *at* $(z_i, \operatorname{Rad} K \cdot r_i)$ *with parameter* $32\lambda_i$ *and slope* P_R *for every* $w \in \mathcal{B}_K$*, the two quarter-bounds of 9 hold for every* $w \in \mathcal{B}_K$*, and* $\big|P_R/|P_R| - P_i/c_i\big| \le C_2(1 + \log \operatorname{Rad} K)\theta_i$ *together with* $\big||P_R| - c_i\big| \le C_2(1 + \log \operatorname{Rad} K)\theta_i^2$*.*
- (the nearby fits) *For every* $i \ge i_N(K, S)$ *and every* $w \in \mathcal{B}_K$ *there are* $Q \in \mathcal{H}$*,* $p' \in \mathbb{R}^d$ *and* $q' \in \mathbb{R}$ *such that* Q *realizes a cylinder fit of parameter* $32\lambda_i$ *for* U_i *at* $(Z_i(w), S\, r_i)$*, with* $Q = \Phi_i\big(c_i\sqrt{\lambda_i}\,p',\, c_i(1 + \lambda_i q')\big)$ *and* $|p'| + |q'| \le C_N(K, S)$*.*

Proof. Throughout fix $K \ge 1$ and $S > 0$, and abbreviate $R = \operatorname{Rad} K$, which satisfies $R \ge 2$ by 5.

Part 1: the enclosing fit. Apply 4 with this R and with $M = \operatorname{Mof}(K, S)$; the hypotheses $2 \le R$, $2 \le M$ and $R \le M$ hold by 5 and 6, the middle one because $M \ge 2K \ge 2$. It supplies a threshold $i_0 \ge i_2$ and, for each $i \ge i_0$, a slope P_R realizing a cylinder fit of parameter $32\lambda_i$ at (z_i, Rr_i) with $\frac{1}{2} \le |P_R| \le L+1$ and the two comparison bounds against P_i/c_i at the constant $C_2(1 + \log R)$. Set $i_R(K, S) = \max\{i_0, \operatorname{iadm}(K, S)\}$, so $i_R(K, S) \ge \operatorname{iadm}(K, S)$ as required.

Fix $i \ge i_R(K, S)$. Since $i \ge \operatorname{iadm}(K, S) \ge i_2$ by 6, hypothesis 9 applies to this P_R — its two conditions $|P_R| \ge \frac{1}{2}$ and the direction bound are exactly what the previous paragraph produced — and gives the two quarter-bounds for every $w \in \mathcal{B}_K$. Those bounds in particular place $Z_i(w)$ inside the cylinder of `FitCylinder` at (z_i, Rr_i) with parameter $32\lambda_i$ and slope P_R: that cylinder is defined by an axial bound of Rr_i and a transverse bound of $\theta_i\, Rr_i$ relative to the direction $P_R/|P_R|$, and a quarter of each is a fortiori within the whole. Finally 8, applied at $Q = P_R$ with $\delta = C_2(1 + \log R) \ge 0$ — nonnegative because $C_2 \ge 0$ and $R \ge 2 > 1$, so $\log R \ge 0$ — yields

p_R and q_R with $P_R = \Phi_i\big(c_i\sqrt{\lambda_i}p_R,\, c_i(1+\lambda_i q_R)\big)$ and $|p_R|+|q_R| \le 100(1+L)(\delta+\delta^2)$. Take $C_R(K,S)$ to be the maximum of 0 and that last quantity; it is nonnegative and independent of i, because C_2, L and R are.

Part 2: the nearby fits. Put $m = \min\{S,R\}$, which is positive, and

$$\rho_S = \max\{2, S/m\}, \qquad \rho_R = \max\{2, R/m\}.$$

For $a \in \{S,R\}$ one has $a \le \rho_a m \le 2a$: if $a/m \ge 2$ then $\rho_a m = a$; and if $a/m < 2$ then $\rho_a = 2$, so $\rho_a m = 2m \le 2a$ and $\rho_a m = 2m \ge a$ because $m \ge a/2$. Both clamped ratios are at least 2, and $\rho_a m \le 2\max\{S,R\}$. Set $i_N(K,S) = \max\{i_R(K,S), \mathrm{iadm}(K, 2\max\{S,R\})\}$, so $i_N(K,S) \ge i_R(K,S)$ as required.

Fix $i \ge i_N(K,S)$ and $w \in \mathcal{B}_K$. Every radius $\rho \in (0, \rho_a m r_i]$ satisfies $\rho \le 2\max\{S,R\}r_i \le \mathrm{Mof}(K, 2\max\{S,R\})\, r_i$ by 6, so 7 supplies at $(Z_i(w), \rho)$ a fit of parameter $32\lambda_i$ with slope in $[\frac{1}{2}, L+1]$. In particular let Q_m be such a slope at the base radius mr_i.

The chain estimate 4 is stated at the centre z_i; the doubling step we need at the centre $Z_i(w)$ is furnished by the same cited node, `CylinderFitDoublingChain`, at the slope window $[\frac{1}{2}, L+1]$, whose hypotheses are met here: $mr_i > 0$ by 1, the clamped ratios are ≥ 2, fits exist at every radius up to $\rho_a m r_i$ by the previous paragraph, and $0 < \theta_i \le \frac{1}{2} < 1$ by 2 with $\theta_i^2 = 32\lambda_i$. Running it out of the base mr_i with ratio ρ_S gives a slope at $(Z_i(w), \rho_S m r_i)$ whose direction and length differ from those of Q_m by at most $C_2(1+\log\rho_S)\theta_i$ and $C_2(1+\log\rho_S)\theta_i^2$; running it with ρ_R does the same at $(Z_i(w), \rho_R m r_i)$. Because $Sr_i \le \rho_S m r_i \le 2Sr_i$, the first comparison principle of 3 at the concentric pair $(Sr_i, \rho_S m r_i)$ transfers those bounds to a slope Q at $(Z_i(w), Sr_i)$; likewise it transfers them to a slope Q' at $(Z_i(w), Rr_i)$. Both target radii are at least mr_i, so the two legs are the same argument and no case distinction between $S \le R$ and $S > R$ arises.

The offsets of the centre $Z_i(w)$ from z_i obey the quarter-bounds of Part 1, so the second comparison principle of 3, at the common radius $H = Rr_i$ and the centres z_i and $Z_i(w)$, compares Q' with the enclosing slope P_R. Summing the six legs — Q to the first chain's outer slope ($\kappa = C_1$), that slope to Q_m ($\kappa = C_2(1+\log\rho_S)$), Q' to the second chain's outer slope ($\kappa = C_1$), that slope to Q_m ($\kappa = C_2(1+\log\rho_R)$), Q' to P_R ($\kappa = C_1$), and P_R to P_i ($\kappa = C_2(1+\log R)$) — and using the triangle inequality separately for the normalized directions and for the lengths, we obtain

$$\delta(K,S) = C_2\big(1+\log\rho_S\big) + C_2\big(1+\log\rho_R\big) + 3C_1 + C_2\big(1+\log R\big)$$

with $\big|Q/|Q| - P_i/c_i\big| \le \delta(K,S)\,\theta_i$ and $\big||Q| - c_i\big| \le \delta(K,S)\,\theta_i^2$. This $\delta(K,S)$ is independent of i and of w, because C_1 and C_2 are and because m, ρ_S, ρ_R and R are built from K and S alone; and it is nonnegative, since $C_1 \ge 1$, $C_2 \ge 0$ and each logarithm is of a quantity ≥ 2.

Finally $i \ge i_R(K,S) \ge \mathrm{iadm}(K,S) \ge i_2$ and $\frac{1}{2} \le |Q| \le L+1$, so 8 applies at this Q with $\delta = \delta(K,S)$ and produces p' and q' with $Q = \Phi_i\big(c_i\sqrt{\lambda_i}p',\, c_i(1+\lambda_i q')\big)$ and $|p'|+|q'| \le 100(1+L)\big(\delta(K,S)+\delta(K,S)^2\big)$. Take $C_N(K,S)$ to be the maximum of 0 and that bound; it is nonnegative and independent of i and of w. This is the second assertion, and the proof is complete. □

2.177. `FrameConfigCovariance`.

Lemma 2.177 (Changes of frame act on configurations)**.** Let F be a finite-dimensional real inner-product space, let g be a real function on space-time, let $z_0 \in F \times \mathbb{R}$, let $R > 0$, $p_0 \in F$, $a_0 \in \mathbb{R}$ and $s \in \mathbb{R}$, and let $\bar{g}$ be the sheared change of frame of `FrameChange` based at z_0 with scale R and parameters (p_0, a_0). Write $T(y) = x_0 + R(y + sp_0)$, where x_0 is the space coordinate of z_0, and $t = t_0 + Rs$, where t_0 is its time coordinate. Then:

(i) for every N, every family of weights θ and positions y and every $\bar{y} \in F$, the family $(\theta_i, T(y_i))$ is a configuration with mean $T(\bar{y})$ in the sense of `IsBaryConfig` if and only if (θ_i, y_i) is a configuration with mean $\bar{y}$;

(ii) the radius of `BaryRadius` of the family $(\theta_i, T(y_i))$ about $T(\bar{y})$ equals R times the radius of (θ_i, y_i) about $\bar{y}$;

(iii) T is surjective: every $Y \in F$ is $T(y)$ for some $y \in F$;

(iv) for every configuration (θ_i, y_i) with mean $\bar{y}$,

$$\sum_i \theta_i \bar{g}(y_i, s) - \bar{g}(\bar{y}, s) = \frac{1}{R}\Big(\sum_i \theta_i g(T(y_i), t) - g(T(\bar{y}), t)\Big).$$

Proof. This is about CONFIGURATIONS, and it is neither implied by nor implies the two nodes that transport other frame-covariant data. `FrameInvariance` transports FITS: it preserves the class and carries a fit at the sheared point of radius $R\rho$ to a fit at the new center of radius ρ with the frame parameters subtracted. `FrameGapCovariance` transports the barycentric ENVELOPE gap, and only in the contact frame, where the base point is (x_0, r), the scale is the same r, and a fit of radius r at that point is assumed; the statement above has an arbitrary base point, an arbitrary scale, and no fit hypothesis, and it speaks about a single configuration rather than about the supremum over all of them. Items (i) to (iii) have no counterpart in either node.

Throughout, write x_0 and t_0 for the coordinates of z_0, so that by `FrameChange`

$$\bar{g}(y, s) = \frac{g\big(T(y), t\big) - g(z_0)}{R} - \langle p_0, y\rangle - \Big(a_0 + \frac{|p_0|^2}{2}\Big)s, \qquad T(y) = x_0 + R(y + sp_0), \quad t = t_0 + Rs.$$

A weighted mean is carried to the weighted mean of the images. Let θ be weights with $\sum_i \theta_i = 1$ and let y be positions. Then

$$\sum_i \theta_i T(y_i) = \sum_i \theta_i x_0 + R\sum_i \theta_i y_i + R\Big(\sum_i \theta_i\Big) sp_0 = x_0 + R\Big(\sum_i \theta_i y_i + sp_0\Big) = T\Big(\sum_i \theta_i y_i\Big),$$

the first equality by expanding T and the linearity of the weighted sum, and the second by $\sum_i \theta_i = 1$.

(i). By `IsBaryConfig` a configuration with mean x consists of weights that are positive and sum to 1 together with positions whose weighted mean is x; the positivity and the sum are untouched by T. If (θ_i, y_i) has mean $\bar{y}$, the display above gives $\sum_i \theta_i T(y_i) = T(\bar{y})$, so $(\theta_i, T(y_i))$ has mean $T(\bar{y})$. Conversely, if $\sum_i \theta_i T(y_i) = T(\bar{y})$ then by the display $T\big(\sum_i \theta_i y_i\big) = T(\bar{y})$; cancelling the summand x_0 and then the factor $R \neq 0$ and the summand sp_0 gives $\sum_i \theta_i y_i = \bar{y}$.

(ii). For each i, $T(y_i) - T(\bar{y}) = R\big((y_i + sp_0) - (\bar{y} + sp_0)\big) = R(y_i - \bar{y})$, so $|T(y_i) - T(\bar{y})|^2 = R^2|y_i - \bar{y}|^2$. Hence $\sum_i \theta_i |T(y_i) - T(\bar{y})|^2 = R^2 \sum_i \theta_i |y_i - \bar{y}|^2$, and taking square roots, which is legitimate because $R^2 \geq 0$ and $R \geq 0$, the radius of `BaryRadius` satisfies the asserted identity.

(iii). Given $Y \in F$, put $y = R^{-1}(Y - x_0) - sp_0$, which is defined because $R \neq 0$. Then $y + sp_0 = R^{-1}(Y - x_0)$ and $T(y) = x_0 + R \cdot R^{-1}(Y - x_0) = Y$.

(iv). Let (θ_i, y_i) be a configuration with mean $\bar{y}$, so $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i = \bar{y}$. Taking the weighted mean of the displayed formula for $\bar{g}$ at the positions y_i,

$$\sum_i \theta_i \bar{g}(y_i, s) = \frac{1}{R}\sum_i \theta_i g\big(T(y_i), t\big) - \frac{g(z_0)}{R} - \Big\langle p_0, \sum_i \theta_i y_i\Big\rangle - \Big(a_0 + \frac{|p_0|^2}{2}\Big)s,$$

where the constant terms contributed $\sum_i \theta_i = 1$ times their value and the inner product was collected by linearity in its second argument. Using $\sum_i \theta_i y_i = \bar{y}$ and subtracting the same formula evaluated at $\bar{y}$, the three terms $-g(z_0)/R$, $-\langle p_0, \bar{y}\rangle$ and $-\big(a_0 + |p_0|^2/2\big)s$ cancel, leaving

$$\sum_i \theta_i \bar{g}(y_i, s) - \bar{g}(\bar{y}, s) = \frac{1}{R}\Big(\sum_i \theta_i g\big(T(y_i), t\big) - g\big(T(\bar{y}), t\big)\Big),$$

which is (iv). □

2.178. `NonaffineSliceExists`.

Lemma 2.178 (A nonaffine horizontal slice)**.** *Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$ and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at the origin with constant C_0 in the sense of `IsCenteredAt`. If v is not affine in the sense of `IsSpaceTimeAffine`, then some horizontal slice of v has a nonzero midpoint Jensen defect: there are a time t and points $y_1, y_2 \in F$ with*

$$v\left(\frac{y_1 + y_2}{2}, t\right) \neq \frac{v(y_1, t) + v(y_2, t)}{2}.$$

Proof. We argue by contraposition: assuming that every horizontal slice of v satisfies the midpoint identity,

$$v\Big(\frac{y_1+y_2}{2},t\Big)=\frac{v(y_1,t)+v(y_2,t)}{2}\qquad (t\in\mathbb{R},\ y_1,y_2\in F), \tag{788}$$

we show that v is affine in the sense of `IsSpaceTimeAffine`, which contradicts the hypothesis.

Step 0: standing facts about v. By `IsFitClass` the function v is continuous on all of space-time and is a viscosity solution of $F[v]=0$ there; by `IsLimitSolutionOn`, whose definition of that notion is precisely the conjunction of the two one-sided conditions, v is therefore both a viscosity subsolution and a viscosity supersolution in the senses of `IsLimitSubOn` and `IsLimitSuperOn`. By `UniformSubquadratic`, applied with the centering point $(0,0)$, for every compact $J\subseteq\mathbb{R}$ and every $\varepsilon>0$ there is $M>0$ with $|v(y,\sigma)|\le\varepsilon|y|^2$ for $|y|\ge M$, $\sigma\in J$; that is, v has subquadratic spatial growth on compact time intervals of $I=\mathbb{R}$ in the sense of `SubquadraticGrowth`. Hence `QuadEnvPathConvex` applies to v with $I=\mathbb{R}$ and any $T>0$.

Step 1: each slice is affine. Fix $t\in\mathbb{R}$ and put $g(y)=v(y,t)$, a continuous function on F. Let $h(y)=g(y)-g(0)$. From (788) with $y_2=0$ we get $h(y/2)=h(y)/2$ for every y, and then (788) itself gives

$$\frac{h(y_1+y_2)}{2}=h\Big(\frac{y_1+y_2}{2}\Big)=\frac{h(y_1)+h(y_2)}{2},$$

so h is additive. An additive map is $\mathbb{Q}$-homogeneous: additivity gives $h(ny)=nh(y)$ for $n\in\mathbb{N}$ by induction, $h(0)=0$, $h(-y)=-h(y)$, and $h(y/n)=h(y)/n$ by applying the first identity to y/n. Since h is continuous and $\mathbb{Q}$ is dense in $\mathbb{R}$, h is $\mathbb{R}$-homogeneous, hence $\mathbb{R}$-linear. By the Riesz representation of a continuous linear functional on the finite-dimensional inner-product space F, there is $p(t)\in F$ with

$$v(y,t)=\langle p(t),y\rangle+c(t)\qquad (y\in F),\qquad c(t):=v(0,t). \tag{789}$$

Step 2: p and c are continuous. $c(t)=v(0,t)$ is continuous in t because v is continuous. Fix an orthonormal basis $e_1,\dots,e_m$ of F. By (789), $\langle p(t),e_i\rangle=v(e_i,t)-v(0,t)$, so $p(t)=\sum_i\big(v(e_i,t)-v(0,t)\big)e_i$ is continuous in t.

Step 3: the forward envelope of an affine slice. Let $\ell>0$ and (x,t) be arbitrary. By (789) and `QuadEnvP`,

$$\mathcal{P}_\ell v(x,t)=\sup_{y\in F}\Big\{\langle p(t+\ell),y\rangle+c(t+\ell)-\frac{|y-x|^2}{2\ell}\Big\}.$$

Writing $y=x+\ell p(t+\ell)+w$ and expanding, the expression in braces equals

$$\langle p(t+\ell),x\rangle+c(t+\ell)+\frac{\ell|p(t+\ell)|^2}{2}-\frac{|w|^2}{2\ell},$$

which is maximal exactly at $w=0$. Hence

$$\mathcal{P}_\ell v(x,t)=\langle p(t+\ell),x\rangle+c(t+\ell)+\frac{\ell}{2}\big|p(t+\ell)\big|^2\qquad (\ell>0), \tag{790}$$

and by `QuadEnvP` the same formula holds at $\ell=0$, where it reads $\mathcal{P}_0v(x,t)=v(x,t)=\langle p(t),x\rangle+c(t)$.

Step 4: p is affine in t. Fix t and $T>0$. By `QuadEnvPathConvex`, part (1), applied with $d=0$, $x_0=x$ and this T — the admissibility hypothesis being vacuous since $I=\mathbb{R}$ — the function $\ell\mapsto\mathcal{P}_\ell v(x,t)$ is convex on $[0,T]$. Let $\ell_0,\ell_1\in[0,T]$, $\theta\in[0,1]$ and $\ell_\theta=(1-\theta)\ell_0+\theta\ell_1$. Putting $A(\ell)=p(t+\ell)$ and $B(\ell)=c(t+\ell)+\frac{\ell}{2}|p(t+\ell)|^2$, convexity and (790) give, for every $x\in F$,

$$\big\langle A(\ell_\theta)-(1-\theta)A(\ell_0)-\theta A(\ell_1),\,x\big\rangle\le(1-\theta)B(\ell_0)+\theta B(\ell_1)-B(\ell_\theta).$$

The right side does not depend on x. If the vector $\Delta=A(\ell_\theta)-(1-\theta)A(\ell_0)-\theta A(\ell_1)$ were nonzero, then taking $x=\lambda\Delta$ with $\lambda\to+\infty$ would make the left side tend to $+\infty$, a contradiction. Hence $\Delta=0$:

$$A(\ell_\theta)=(1-\theta)A(\ell_0)+\theta A(\ell_1)\qquad (\ell_0,\ell_1\in[0,T],\ \theta\in[0,1]).$$

Since $T>0$ was arbitrary, this holds for all $\ell_0,\ell_1\ge0$. Taking $\ell_0=0$, $\ell_1=1$ and $\theta=\ell\in[0,1]$ gives $A(\ell)=A(0)+\ell\big(A(1)-A(0)\big)$ for $\ell\in[0,1]$; and for $\ell>1$, taking $\ell_0=0$, $\ell_1=\ell$ and $\theta=1/\ell$

gives $A(1) = (1-1/\ell)A(0) + (1/\ell)A(\ell)$, i.e. again $A(\ell) = A(0) + \ell(A(1) - A(0))$. Hence, as $t \in \mathbb{R}$ was arbitrary,

$$p(t+\ell) = p(t) + \ell\big(p(t+1) - p(t)\big) \qquad (t \in \mathbb{R},\ \ell \geq 0). \tag{791}$$

Note that (791) is licensed only for *nonnegative* offsets ℓ, because the convexity it came from is convexity on $[0,T]$. Passing from it to a formula valid on all of $\mathbb{R}$ is therefore a step to be carried out, not a remark, and we record it as a statement about an arbitrary map, because Step 6 needs it a second time.

Base-point lemma. Let V be a real vector space and let $q : \mathbb{R} \to V$ satisfy

$$q(t+\ell) = q(t) + \ell\big(q(t+1) - q(t)\big) \qquad (t \in \mathbb{R},\ \ell \geq 0). \tag{792}$$

Then $q(\tau) = q(0) + \tau\big(q(1) - q(0)\big)$ for every $\tau \in \mathbb{R}$.

To see this, fix $\tau \in \mathbb{R}$ and choose the single base point $s = \min\{\tau, 0\} - 1$. Then $s \leq \tau - 1$ and $s \leq -1$, so the three offsets

$$\tau - s \geq 1, \qquad 0 - s \geq 1, \qquad 1 - s \geq 2$$

are all nonnegative, and each of the three instantiations below is admissible in (792). Put $m = q(s+1) - q(s)$. Applying (792) at $t = s$ with offsets $\tau - s$, $0 - s$ and $1 - s$ in turn, and simplifying $s + (\tau - s) = \tau$, $s + (0 - s) = 0$ and $s + (1 - s) = 1$,

$$q(\tau) = q(s) + (\tau - s)m, \qquad q(0) = q(s) + (0 - s)m, \qquad q(1) = q(s) + (1 - s)m.$$

Subtracting the second display from the third gives $q(1) - q(0) = \big((1 - s) - (0 - s)\big)m = m$: the slope m at the base point *is* $q(1) - q(0)$, which is what the base point being unknown might otherwise have cost us. Subtracting the second from the first gives $q(\tau) - q(0) = \big((\tau - s) - (0 - s)\big)m = \tau m$. Combining the two, $q(\tau) = q(0) + \tau\big(q(1) - q(0)\big)$, which proves the lemma. No case split on the sign of τ occurs, and the case $\tau \geq 0$ is covered along with the rest: every instantiation stays inside the nonnegative range of offsets that (792) licenses. This is the sense in which the affine interpolation identity holds on every interval, rather than only forward from one fixed base point.

Applying the base-point lemma with $V = F$ and $q = p$, whose hypothesis (792) is exactly (791), and setting $p_0 = p(0)$ and $p_1 = p(1) - p(0)$,

$$p(\tau) = p_0 + \tau p_1 \qquad (\tau \in \mathbb{R}). \tag{793}$$

Step 5: p is constant. Let $C_g > 0$ be the constant of `CenteredGrowth` for this C_0; since $v \in \mathcal{A}_{m,C_0}$ is centered at the origin, for $K \geq 2$, $|v(\zeta)| \leq C_g K L(K)^2$ on $\mathcal{B}_K$, where $L(K) = 1 + \log K$. If $F = \{0\}$ then $p(\tau) \in F = \{0\}$ for every τ, so $p_1 = p(1) - p(0) = 0$ and p is constant outright, which is the assertion of this step; assume therefore $F \neq \{0\}$, so that F contains a unit vector. Fix a unit vector $e \in F$, let $K \geq 2$ and let $|\tau| \leq K$. By (789), $v(Ke, \tau) - v(-Ke, \tau) = 2K\langle p(\tau), e\rangle$, and both points lie in $\mathcal{B}_K$, so

$$\big|\langle p(\tau), e\rangle\big| \leq \frac{|v(Ke,\tau)| + |v(-Ke,\tau)|}{2K} \leq C_g L(K)^2.$$

Taking the supremum over unit vectors e gives $|p(\tau)| \leq C_g L(K)^2$ for $|\tau| \leq K$. In particular $|p(K)| \leq C_g(1 + \log K)^2$ for every $K \geq 2$. By (793), $|p(K)| \geq K|p_1| - |p_0|$; if $p_1 \neq 0$ this grows linearly in K, while $C_g(1 + \log K)^2$ grows slower than linearly, so $K|p_1| - |p_0| \leq C_g(1 + \log K)^2$ fails for large K. Hence $p_1 = 0$ and $p(\tau) \equiv p_0 =: p$.

Step 6: c is convex and concave, hence affine. With p constant, (790) becomes

$$\mathcal{P}_\ell v(x,t) = \langle p, x\rangle + c(t+\ell) + \frac{\ell}{2}|p|^2.$$

Fix t and $T > 0$; convexity of $\ell \mapsto \mathcal{P}_\ell v(x,t)$ on $[0,T]$, from `QuadEnvPathConvex` part (1) as in Step 4, and the fact that adding the affine function $\ell \mapsto \langle p, x\rangle + \frac{\ell}{2}|p|^2$ changes neither convexity nor its failure, show that $\ell \mapsto c(t+\ell)$ is convex on $[0,T]$. Since t and T are arbitrary, c is convex on $\mathbb{R}$.

For the reverse, compute the backward envelope: by (789) and `QuadEnvM`, for $\ell > 0$,

$$\mathcal{M}_\ell v(x,t) = \inf_y \Big\{ \langle p, y\rangle + c(t-\ell) + \frac{|y-x|^2}{2\ell} \Big\} = \langle p, x\rangle + c(t-\ell) - \frac{\ell}{2}|p|^2,$$

the infimum being attained at $y = x - \ell p$, as the substitution $y = x - \ell p + w$ shows. By `QuadEnvPathConvex`, part (2), applied with $I = \mathbb{R}$ and any $T > 0$ — its hypotheses hold because v is a viscosity supersolution with the required continuity and growth by Step 0 — the function $\ell \mapsto \mathcal{M}_\ell v(x,t)$ is concave on $[0,T]$. Removing the affine function $\ell \mapsto \langle p, x\rangle - \frac{\ell}{2}|p|^2$ shows that $\ell \mapsto c(t-\ell)$ is concave on $[0,T]$ for every t and T, i.e. c is concave on $\mathbb{R}$.

A function on $\mathbb{R}$ that is both convex and concave is affine. Convexity and concavity together give equality in every chord inequality, so

$$c\big((1-\theta)\tau_0 + \theta\tau_1\big) = (1-\theta)c(\tau_0) + \theta c(\tau_1) \qquad (\tau_0, \tau_1 \in \mathbb{R},\ \theta \in [0,1]). \tag{794}$$

We check that c satisfies the hypothesis (792) of the base-point lemma of Step 4, with $V = \mathbb{R}$. Let $t \in \mathbb{R}$ and $\ell \geq 0$. If $0 \leq \ell \leq 1$, then $t+\ell = (1-\ell)t + \ell(t+1)$ is a convex combination of t and $t+1$, so (794) with $\tau_0 = t$, $\tau_1 = t+1$ and $\theta = \ell$ gives $c(t+\ell) = (1-\ell)c(t) + \ell c(t+1) = c(t) + \ell\big(c(t+1) - c(t)\big)$. If $\ell > 1$, then $1/\ell \in (0,1)$ and $t+1 = (1-1/\ell)t + (1/\ell)(t+\ell)$, so (794) with $\tau_0 = t$, $\tau_1 = t+\ell$ and $\theta = 1/\ell$ gives $c(t+1) = (1-1/\ell)c(t) + (1/\ell)c(t+\ell)$; multiplying by ℓ and rearranging yields the same identity $c(t+\ell) = c(t) + \ell\big(c(t+1) - c(t)\big)$. So (792) holds for $q = c$, and the base-point lemma gives

$$c(\tau) = c(0) + \tau\big(c(1) - c(0)\big) \qquad (\tau \in \mathbb{R}). \tag{795}$$

Conclusion. Writing $\beta = c(1) - c(0)$ and $\gamma = c(0)$, (789), (795) and Step 5 give

$$v(x,t) = \langle p, x\rangle + \beta t + \gamma \qquad \big((x,t) \text{ arbitrary}\big),$$

so v is affine in the sense of `IsSpaceTimeAffine`. This contradicts the hypothesis that v is not affine, and therefore (788) must fail for some t, y_1, y_2, which is the assertion. □

2.179. `JensenSupremumFinite`.

Lemma 2.179 (The extremal Jensen constant)**.** *Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$ and let $v^{(0)} \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass` be centered at the origin with constant C_0 in the sense of `IsCenteredAt` and not affine in the sense of `IsSpaceTimeAffine`. Then there are a function $v \in \mathcal{A}_{m,C_0}$, centered at the origin and not affine, and a number κ with $0 < \kappa < \infty$ such that v satisfies the horizontal Jensen bound with constant κ on all times, in the sense of `HasJensenBound`, and κ is the least such constant:*

$$\kappa = \sup_{\substack{t,\ \text{finite configurations} \\ r>0}} \frac{\sum_i \theta_i v(y_i,t) - v(x,t)}{r},$$

the supremum over all times and all finite horizontal probability configurations with mean x and radius $r > 0$, in the senses of `IsBaryConfig` and `BaryRadius`.

Proof. Step 1: a configuration with nonzero defect. By `NonaffineSliceExists`, applied to $v^{(0)}$, there are a time t_1 and points $w_1, w_2 \in F$ with

$$v^{(0)}\Big(\frac{w_1+w_2}{2}, t_1\Big) \neq \frac{v^{(0)}(w_1,t_1) + v^{(0)}(w_2,t_1)}{2}.$$

In particular $w_1 \neq w_2$. Put $n = 2$, $\theta_1 = \theta_2 = \frac{1}{2}$, $y_1 = w_1$, $y_2 = w_2$ and $x_1 = \frac{1}{2}(w_1 + w_2)$. Then $\theta_i > 0$, $\sum_i \theta_i = 1$ and $\sum_i \theta_i y_i = x_1$, so (θ, y) is a finite horizontal probability configuration with mean x_1 in the sense of `IsBaryConfig`. Its radius, in the sense of `BaryRadius`, is

$$r_1 = \Big(\tfrac{1}{2}|w_1 - x_1|^2 + \tfrac{1}{2}|w_2 - x_1|^2\Big)^{1/2} = \frac{|w_1 - w_2|}{2} > 0,$$

since $w_1 - x_1 = \frac{1}{2}(w_1 - w_2)$ and $w_2 - x_1 = -\frac{1}{2}(w_1 - w_2)$. The defect

$$D_1 = \sum_i \theta_i v^{(0)}(y_i, t_1) - v^{(0)}(x_1, t_1)$$

is nonzero by the displayed inequality.

Step 2: normalizing the sign of the defect. If $D_1 > 0$, put $v = v^{(0)}$. If $D_1 < 0$, put $v = \widetilde{v^{(0)}}$, the time reversal of `TimeFlip`. We check that in the second case v has all the required properties.

By `TimeReversal`, part (3), $v \in \mathcal{A}_{m,C_0}$, and $(0,0)$, being a fit for $v^{(0)}$ at the origin of radius 1 by `IsCenteredAt`, becomes the fit $(-0,0) = (0,0)$ for v at the reflection of the origin, which is the origin, of radius 1; moreover $v(0,0) = -v^{(0)}(0,0) = 0$ by `TimeFlip`. So v is centered at the origin with constant C_0 in the sense of `IsCenteredAt`. It is not affine: if $v(y,s) = \langle q, y\rangle + \beta s + \gamma$ for all (y,s), then by `TimeFlip` $v^{(0)}(y,\sigma) = -v(y,-\sigma) = -\langle q,y\rangle + \beta\sigma - \gamma$ would be affine in the sense of `IsSpaceTimeAffine`, contrary to hypothesis. Finally, by `TimeFlip`, for the same configuration at the time $-t_1$,

$$\sum_i \theta_i v(y_i, -t_1) - v(x_1, -t_1) = -\Big(\sum_i \theta_i v^{(0)}(y_i, t_1) - v^{(0)}(x_1, t_1)\Big) = -D_1 > 0.$$

Thus in either case $v \in \mathcal{A}_{m,C_0}$ is centered at the origin with constant C_0, is not affine, and there are a time τ_1, a configuration (θ, y) with mean x_1 and radius $r_1 > 0$, and a defect

$$D = \sum_i \theta_i v(y_i, \tau_1) - v(x_1, \tau_1) > 0.$$

Step 3: the set of defect quotients. Let

$$\mathcal{Q} = \left\{ \frac{\sum_i \theta_i' v(y_i', \tau) - v(x', \tau)}{r'} \right\},$$

the set of quotients over all times τ and all finite horizontal probability configurations (θ', y') with mean x', in the sense of `IsBaryConfig`, whose radius r', in the sense of `BaryRadius`, is strictly positive. By Step 2, $D/r_1 \in \mathcal{Q}$, so $\mathcal{Q}$ is nonempty and contains a positive number.

Step 4: $\mathcal{Q}$ is bounded above. Let $C_g > 0$ be the constant of `CenteredGrowth` for this C_0. We first record a consequence: if $w \in \mathcal{A}_{m,C_0}$ is centered at the origin with constant C_0, then

$$|w(y,0)| \le C_1\big(1 + |y|^2\big) \qquad (y \in F), \tag{796}$$

with C_1 depending only on C_0 and $\dim F$. Indeed, if $|y| \le 2$ take $K = 2$ in `CenteredGrowth`: $(y,0) \in \mathcal{B}_2$ and $|w(y,0)| \le 2C_g(1+\log 2)^2$. If $|y| > 2$ take $K = |y| \ge 2$: then $(y,0) \in \mathcal{B}_K$ and $|w(y,0)| \le C_g|y|(1+\log|y|)^2$; since $c_1 := \sup_{K\ge 2}(1+\log K)^2/K$ is finite, because $(\log K)^2/K \to 0$ and the function is continuous on $[2,\infty)$, this is at most $C_g c_1 |y|^2$. So (796) holds with $C_1 = \max\{2C_g(1+\log 2)^2,\ C_g c_1\}$.

Now let τ, (θ', y'), x' and $r' > 0$ be as in Step 3, with defect $D' = \sum_i \theta_i' v(y_i', \tau) - v(x', \tau)$. By `IsFitClass` choose a fit (p, a) for v at (x', τ) of radius $r' > 0$ with constant C_0, and let

$$\bar v = \text{the change of frame of } \texttt{FrameChange} \text{ based at } (x', \tau) \text{ with scale } r' \text{ and parameters } (p, a).$$

By `FrameInvariance`, $\bar v \in \mathcal{A}_{m,C_0}$; and its second assertion, applied at the new center $\zeta = 0$, whose corresponding old point is (x', τ), with $\rho = 1$ and the fit (p, a) at radius $r' \cdot 1 = r'$, shows that $(p - p, a - a) = (0,0)$ is a fit for $\bar v$ at the origin of radius 1. Together with $\bar v(0,0) = 0$, which is immediate from `FrameChange`, this makes $\bar v$ centered at the origin with constant C_0 in the sense of `IsCenteredAt`.

Put $u_i = (y_i' - x')/r'$. By `FrameChange`, at the new time 0 the point u_i corresponds to the old point $x' + r'u_i = y_i'$ at time τ, and

$$\bar v(u_i, 0) = \frac{v(y_i', \tau) - v(x', \tau)}{r'} - \langle p, u_i\rangle.$$

Since $\sum_i \theta_i' u_i = (x' - x')/r' = 0$ and $\sum_i \theta_i' = 1$, averaging with the weights θ_i' gives

$$\sum_i \theta_i'\, \bar v(u_i, 0) = \frac{D'}{r'}, \tag{797}$$

the linear term dropping out. Moreover $\sum_i \theta'_i |u_i|^2 = \sum_i \theta'_i |y'_i - x'|^2/(r')^2 = 1$ by `BaryRadius`. Hence, by (796) applied to $\bar{v}$,

$$\frac{D'}{r'} = \sum_i \theta'_i \, \bar{v}(u_i, 0) \le \sum_i \theta'_i \, C_1\big(1 + |u_i|^2\big) = 2C_1.$$

So every element of $\mathcal{Q}$ is at most $2C_1$, a constant depending only on C_0 and $\dim F$.

Step 5: the extremal constant. By Steps 3 and 4 the set $\mathcal{Q}$ is nonempty and bounded above, so $\kappa := \sup \mathcal{Q}$ is a real number, and $\kappa \ge D/r_1 > 0$.

We check that v satisfies the horizontal Jensen bound with constant κ at every time, in the sense of `HasJensenBound`. Let $\tau \in \mathbb{R}$ and let (θ', y') be a configuration with mean x' and radius r'. If $r' > 0$ the quotient D'/r' lies in $\mathcal{Q}$, hence is at most κ, i.e. $D' \le \kappa r'$. If $r' = 0$, then by `BaryRadius` $\sum_i \theta'_i |y'_i - x'|^2 = 0$ with all $\theta'_i > 0$, so $y'_i = x'$ for every i, and $D' = \sum_i \theta'_i v(x', \tau) - v(x', \tau) = 0 = \kappa r'$. In both cases $D' \le \kappa r'$.

Finally κ is the least such constant: if κ' is any number for which the horizontal Jensen bound of `HasJensenBound` holds for v at every time, then for every configuration with radius $r' > 0$ we get $D'/r' \le \kappa'$, so every element of $\mathcal{Q}$ is at most κ' and hence $\kappa = \sup \mathcal{Q} \le \kappa'$.

This exhibits v and κ with all the asserted properties. □

2.180. `EnvelopeContact`.

Theorem 2.180 (Extremal envelope contact)**.** *Let F be a finite-dimensional real inner-product space, write m for its dimension and let $1 \le C_0 < \infty$. Suppose that $\mathcal{A}_{m,C_0}$, in the sense of `IsFitClass`, contains a member $v^{(0)}$ that is not affine on space-time in the sense of `IsSpaceTimeAffine`. Then there are $\kappa \in (0, \infty)$, a real number H and a function $v \in \mathcal{A}_{m,C_0}$ centered at $z_* = (0, 1)$ with constant C_0 in the sense of `IsCenteredAt`, such that v satisfies the horizontal Jensen bound with constant κ at every time, in the sense of `HasJensenBound` with $I = \mathbb{R}$, and such that the contact envelope*

$$W(x,t) = \mathcal{C}_t v(x,t) - \frac{\kappa^2 t}{2} \qquad (t > 0)$$

of `ContactEnv`, formed from the barycentric envelope of `BaryEnv`, has the following properties.

1. *W is continuous on $\{(x,t) : t > 0\}$ and has subquadratic spatial growth on compact subintervals of $(0, \infty)$, in the sense of `SubquadraticGrowth`.*
2. *W is a viscosity subsolution of the limiting equation on $\{(x,t) : t > 0\}$, in the sense of `IsLimitSubOn`.*
3. *For every $t > 0$,*
$$v(x,t) - \frac{\kappa^2 t}{2} \le W(x,t) \le v(x,t),$$
and $W(z_*) = v(z_*) = 0$.
4. *For every $t > 0$ the function $x \mapsto W(x,t) - |x|^2/(2t)$ is concave on F.*
5. *At every contact with $t > 0$ the barycentric gap is exactly $\kappa^2 t/2$: if $W(x,t) = v(x,t)$ and $t > 0$ then*
$$\frac{\mathcal{C}_t v(x,t) - v(x,t)}{t} = \frac{\kappa^2}{2}.$$
6. *The envelope Hamiltonian of `EnvHamiltonian` is finite at z_*, with value $H_v(z_*) = H$, and the forward envelope slopes of `EnvSlopeP` satisfy*
$$\lim_{\ell \downarrow 0} \mathcal{S}^+_\ell W(z_*) = H.$$

Proof. Throughout, $z_* = (0, 1)$, a *configuration* means a finite horizontal probability configuration in the sense of `IsBaryConfig`, and its *radius* is the root mean square radius of `BaryRadius`. For a function g in $\mathcal{A}_{m,C_0}$ we use freely that g is continuous on all of space-time and is a viscosity solution of the limiting equation $F[g] = D^2 g[(\nabla_x g, 1)] = 0$ there, by `IsFitClass` and `IsLimitSolutionOn`. By `UniformSubquadratic`, applied with $z_c = (0, 0)$ and with $z_c = (0, 1)$, every member of $\mathcal{A}_{m,C_0}$ centered at one of those two points has subquadratic spatial growth on compact time intervals uniformly over the class; in particular each such member satisfies

SubquadraticGrowth with $I = \mathbb{R}$, hence also with $I = (0, \infty)$, since decreasing I only decreases the family of compact subintervals tested.

The envelope bound pairs. Whenever a real function f on space-time is continuous on $\Omega_I = \{(x,t) : t \in I\}$ and has subquadratic spatial growth on compact subintervals of I in the sense of SubquadraticGrowth, we apply EnvelopeCompetitorBounds to f with that I. Its three conjuncts are exactly the three facts used below, and we name them in that order. For $s > 0$ and $z = (x,t)$ with $t \in I$ write

$$V(f,s,z) = \Big\{ \sum_i \theta_i f(y_i,t) - \frac{\rho^2}{2s} : (\theta_i, y_i) \text{ a configuration with mean } x \text{ and radius } \rho \Big\}$$

for the set of barycentric competitor values at z, the configurations being those of IsBaryConfig and the radius that of BaryRadius.

(B1) The third conjunct of EnvelopeCompetitorBounds: every element of $V(f,s,z)$ is at most $\mathcal{C}_s f(z)$, and $\mathcal{C}_s f(z)$ is at most every real number that dominates every element of $V(f,s,z)$.

(B2) Its first conjunct: for $\ell > 0$ with $t \in I$ and $t + \ell \in I$, every $f(y, t+\ell) - |y-x|^2/(2\ell)$ with $y \in F$ is at most $\mathcal{P}_\ell f(z)$, and $\mathcal{P}_\ell f(z)$ is at most every real number dominating all of them.

(B3) Its second conjunct: for $\ell > 0$ with $t \in I$ and $t - \ell \in I$, $\mathcal{M}_\ell f(z)$ is at most every $f(y, t-\ell) + |y-x|^2/(2\ell)$ with $y \in F$, and every real number below all of them is at most $\mathcal{M}_\ell f(z)$.

Two remarks on how these are used. First, the domination half of each pair exhibits the envelope as an upper bound of its competitor set in (B1) and (B2) and as a lower bound in (B3); so each of those sets is bounded on the relevant side by a real number. Nonemptiness is a separate and immediate matter: the one-point configuration $N = 1$, $\theta_1 = 1$, $y_1 = x$ has mean x and radius 0 and is admissible by IsBaryConfig, so $f(z) \in V(f,s,z)$, and the choice $y = x$ is always available in (B2) and (B3). Second, no separate finiteness assertion is needed anywhere below: $\mathcal{C}_s$, $\mathcal{P}_\ell$ and $\mathcal{M}_\ell$ are real-valued by BaryEnv, QuadEnvP and QuadEnvM together with the preamble's convention for the supremum or infimum of an empty or unbounded set, and what that convention withholds – namely that the value is the genuine least upper bound, respectively greatest lower bound, of its competitor set rather than the conventional substitute – is exactly what the two halves of each pair supply. *Changes of frame act on configurations.* Let g be a real function on space-time, let $z_0 = (x_0, t_0)$, $R > 0$, $p_0 \in F$, $a_0 \in \mathbb{R}$, and let $\bar{g}$ be the change of frame of FrameChange based at z_0 with scale R and parameters (p_0, a_0),

$$\bar{g}(y,s) = \frac{g\big(x_0 + R(y + p_0 s),\, t_0 + Rs\big) - g(z_0)}{R} - p_0 \cdot y - \Big(a_0 + \frac{|p_0|^2}{2}\Big)s.$$

Fix $s \in \mathbb{R}$ and put $T_s(y) = x_0 + R(y + p_0 s)$ and $t = t_0 + Rs$, so that $\bar{g}(y,s) = [g(T_s(y), t) - g(z_0)]/R - p_0 \cdot y - (a_0 + |p_0|^2/2)s$. We write (F0) for FrameConfigCovariance applied to these data. Its parts (i), (ii) and (iii) say that T_s carries a family of weights and positions with mean $\bar{y}$ onto one with mean $T_s(\bar{y})$ and back, in the sense of IsBaryConfig, keeping the weights, multiplying the radius of BaryRadius by R, and hitting every point of F; and its part (iv) says that for every configuration (θ_i, y_i) with mean $\bar{y}$,

$$\sum_i \theta_i \bar{g}(y_i, s) - \bar{g}(\bar{y}, s) = \frac{1}{R}\Big(\sum_i \theta_i g(T_s(y_i), t) - g(T_s(\bar{y}), t) \Big).$$

Three consequences will be used.

(F1) If g satisfies the Jensen bound with constant $\kappa \geq 0$ at every time (HasJensenBound with $I = \mathbb{R}$), then so does $\bar{g}$: by (F0)(iv), together with the radius identity (F0)(ii), for a configuration with mean y and radius ρ,

$$\sum_i \theta_i \bar{g}(y_i, s) - \bar{g}(y, s) = \frac{1}{R}\Big(\sum_i \theta_i g(T_s y_i, t) - g(T_s y, t) \Big) \leq \frac{\kappa R \rho}{R} = \kappa \rho.$$

(F2) Suppose that g and $\bar{g}$ are continuous on space-time with subquadratic spatial growth on compact time intervals. Then for every $\sigma > 0$ and every (y, s), with $Z = (T_s(y), t)$,

$$\mathcal{C}_\sigma \bar{g}(y,s) - \bar{g}(y,s) = \frac{\mathcal{C}_{R\sigma} g(Z) - g(Z)}{R}.$$

Indeed, by (F0)(iv), the element of $V(\bar{g}, \sigma, (y,s))$ attached to a configuration with mean y and radius ρ, minus $\bar{g}(y,s)$, equals

$$\frac{1}{R}\Big(\sum_i \theta_i g(T_s y_i, t) - g(Z)\Big) - \frac{\rho^2}{2\sigma} = \frac{1}{R}\Big(\sum_i \theta_i g(T_s y_i, t) - g(Z) - \frac{(R\rho)^2}{2R\sigma}\Big),$$

because $(R\rho)^2/(2R^2\sigma) = \rho^2/(2\sigma)$; and by (F0)(i),(ii),(iii) the map $(\theta_\bullet, y_\bullet) \mapsto (\theta_\bullet, T_s y_\bullet)$ is a bijection of the configurations occurring in $V(\bar{g}, \sigma, (y,s))$ onto those occurring in $V(g, R\sigma, Z)$, carrying radius ρ to $R\rho$. Thus $V(\bar{g}, \sigma, (y,s)) - \bar{g}(y,s)$ is the image of $V(g, R\sigma, Z) - g(Z)$ under multiplication by the positive number $1/R$. Both sets are nonempty, as noted above, and the domination half of (B1) exhibits each of $\mathcal{C}_\sigma \bar{g}(y,s)$ and $\mathcal{C}_{R\sigma} g(Z)$ as an upper bound of its own set while the extremality half makes it the least such bound; a positive dilation of the reals carries the least upper bound of a nonempty set bounded above to the least upper bound of the dilated set, so the two suprema correspond in the same way, which is (F2).

Both applications of (B1) are legitimate: the hypotheses of `EnvelopeCompetitorBounds` hold for g with the ambient I and for $\bar{g}$ with the corresponding one, by the continuity and growth assumptions in force wherever (F2) is invoked.

(F3) If $\bar{g}$ is affine on space-time in the sense of `IsSpaceTimeAffine` then so is g. Indeed the definition of $\bar{g}$ rearranges to

$$g\big(x_0 + R(y + p_0 s), t_0 + Rs\big) = g(z_0) + R\Big[\bar{g}(y,s) + p_0 \cdot y + \Big(a_0 + \frac{|p_0|^2}{2}\Big)s\Big],$$

whose right side is an affine function of (y, s) when $\bar{g}$ is, while $(y,s) \mapsto (x_0 + R(y + p_0 s), t_0 + Rs)$ is an affine bijection of space-time, with inverse $(x,t) \mapsto \big((x - x_0)/R - p_0(t - t_0)/R,\ (t - t_0)/R\big)$. Composing with that inverse exhibits g as an affine function.

Claim T: translation in time. Let $c \in \mathbb{R}$, let g be a real function on space-time and put $g_c(x,t) = g(x, t - c)$. Then:

(i) g_c is continuous iff g is, and g_c is a viscosity solution of the limiting equation on all of space-time iff g is (`IsLimitSolutionOn`);

(ii) (p, a) is a fit for g_c at (x,t) of radius ρ with constant C_0 iff it is a fit for g at $(x, t-c)$ of radius ρ with constant C_0 (`IsFit`);

(iii) g_c satisfies the Jensen bound with constant κ at every time iff g does (`HasJensenBound` with $I = \mathbb{R}$);

(iv) $\mathcal{C}_s g_c(x,t) = \mathcal{C}_s g(x, t-c)$ for every $s > 0$ and (x,t);

(v) consequently $g_c \in \mathcal{A}_{m,C_0}$ iff $g \in \mathcal{A}_{m,C_0}$, and g_c is centered at (x,t) iff g is centered at $(x, t-c)$.

Proof. (ii) holds because `IsFit` involves only the base point and increments (h, ℓ) from it, and the base point of g_c at (x,t) is the base point of g at $(x, t-c)$, with identical increments. (iii) and (iv) hold because `HasJensenBound` and `BaryEnv` involve, for each assertion, only one horizontal slice, and the slice of g_c at time t is the slice of g at time $t - c$; in (iii) the time quantifier ranges over all of $\mathbb{R}$ in both cases. For (i), write $\tau(x,t) = (x, t-c)$, so $g_c = g \circ \tau$ and τ is a bijective isometry of space-time whose inverse is the translation $(x,t) \mapsto (x, t+c)$; continuity is clear. If ϕ is twice continuously differentiable at $\hat{z}$ and $g_c - \phi$ has a local maximum at $\hat{z}$, then $g - \phi \circ \tau^{-1}$ has a local maximum at $\tau(\hat{z})$ and $\phi \circ \tau^{-1}$ is twice continuously differentiable at $\tau(\hat{z})$, with the same gradient and second derivative as ϕ at $\hat{z}$, because τ^{-1} is a translation. Hence $F[\phi \circ \tau^{-1}](\tau\hat{z}) = F[\phi](\hat{z})$ by `LimitOp`, and the subsolution inequality of `IsLimitSubOn` transfers; minima and `IsLimitSuperOn` are identical, and the converse direction follows by replacing c with $-c$. Part (v) combines (i), (ii) with the definition of `IsFitClass` and `IsCenteredAt`. This proves Claim T.

Step 1: a centered nonaffine member and the extremal constant. By `IsFitClass` choose a fit (p_0, a_0) for $v^{(0)}$ at the origin of radius 1, and let w be the change of frame of `FrameChange` of $v^{(0)}$ based at $z_0 = 0$ with scale $R = 1$ and parameters (p_0, a_0). By the first clause of `FrameInvariance`, $w \in \mathcal{A}_{m,C_0}$. Its second clause, with $\zeta = 0$ and $\rho = 1$, has as hypothesis exactly a fit for $v^{(0)}$ at the origin of radius $1 \cdot 1$, namely (p_0, a_0), and yields that $(p_0 - p_0, a_0 - a_0) = (0, 0)$ is a fit for w at the origin of radius 1; and $w(0) = 0$ directly from `FrameChange`. So w is centered at the origin with constant C_0 in the sense of `IsCenteredAt`. Moreover w is not affine: otherwise $v^{(0)}$ would be affine by (F3).

Now `JensenSupremumFinite`, applied to C_0 and w, supplies a function $\check{v} \in \mathcal{A}_{m,C_0}$, centered at the origin and not affine, together with a constant κ such that

$$0 < \kappa < \infty, \qquad \check{v} \text{ satisfies the Jensen bound with constant } \kappa \text{ at every time,} \tag{798}$$

and such that $\kappa \le \kappa'$ for every κ' with which $\check{v}$ satisfies that bound. This κ is the extremal constant of the proposition.

Step 2: near-attaining configurations. For every $\varepsilon > 0$ there are a time t and a configuration (θ_i, y_i) with mean x and radius $\rho > 0$ such that

$$\sum_i \theta_i \check{v}(y_i, t) - \check{v}(x, t) > (\kappa - \varepsilon)\rho.$$

Otherwise $\check{v}$ would satisfy the Jensen bound with the constant $\kappa - \varepsilon$: for configurations of positive radius this is the failure of the displayed inequality, and for a configuration of radius 0 all y_i equal x, because $\sum_i \theta_i |y_i - x|^2 = 0$ with every $\theta_i > 0$, so the left side vanishes and the required inequality reads $0 \le 0$. The minimality in Step 1, applied to the admissible constant $\kappa' = \kappa - \varepsilon$, would then give $\kappa \le \kappa - \varepsilon$, which is false for $\varepsilon > 0$. Choose accordingly, for each integer $j \ge 0$, a time t_j and a configuration $(\theta_{j,i}, y_{j,i})_{i \le N_j}$ with mean x_j and radius $\rho_j > 0$ such that

$$D_j := \sum_i \theta_{j,i} \check{v}(y_{j,i}, t_j) - \check{v}(x_j, t_j) > \Big(\kappa - \frac{1}{j+1}\Big)\rho_j. \tag{799}$$

Step 3: normalizing each configuration. Fix j. By `IsFitClass` choose a fit (p_j, a_j) for $\check{v}$ at (x_j, t_j) of radius $\rho_j > 0$, and let v_j be the change of frame `FrameChange` of $\check{v}$ based at $z_j = (x_j, t_j)$ with scale $R = \rho_j$ and parameters (p_j, a_j). As in Step 1, `FrameInvariance` gives $v_j \in \mathcal{A}_{m,C_0}$ and, with $\zeta = 0$ and $\rho = 1$, that $(0, 0)$ is a fit for v_j at the origin of radius 1, while $v_j(0) = 0$; so v_j is centered at the origin. By (F1) and (798), v_j satisfies the Jensen bound with the same constant κ at every time.

Put $\hat{y}_{j,i} = (y_{j,i} - x_j)/\rho_j$. In (F0) for this frame change, take $s = 0$; then $T_0(y) = x_j + \rho_j y$ and $t = t_j$, so $T_0(\hat{y}_{j,i}) = y_{j,i}$ and $T_0(0) = x_j$. By (F0)(i),(ii), $(\theta_{j,i}, \hat{y}_{j,i})$ is a configuration with mean 0 and radius 1, since its image under T_0 has mean x_j and radius $\rho_j = \rho_j \cdot 1$. By (F0)(iv) and $v_j(0, 0) = 0$,

$$\sum_i \theta_{j,i} v_j(\hat{y}_{j,i}, 0) = \sum_i \theta_{j,i} v_j(\hat{y}_{j,i}, 0) - v_j(0, 0) = \frac{D_j}{\rho_j} > \kappa - \frac{1}{j+1},$$

while the Jensen bound for v_j at time 0 with this configuration gives $\sum_i \theta_{j,i} v_j(\hat{y}_{j,i}, 0) \le \kappa \cdot 1$. Hence

$$\sum_i \theta_{j,i} v_j(\hat{y}_{j,i}, 0) \longrightarrow \kappa \qquad (j \to \infty). \tag{800}$$

Step 4: compactness and the attainment of the sandwich. By `ClassCompactness`, applied with $z_c = (0, 0)$ to the sequence (v_j), there are a strictly increasing ι and a function $\hat{v} \in \mathcal{A}_{m,C_0}$, centered at the origin, with $v_{\iota(j)} \to \hat{v}$ uniformly on every compact set. For each fixed time and each fixed configuration the Jensen inequality for $v_{\iota(j)}$ passes to the limit of the finitely many values involved, so $\hat{v}$ satisfies the Jensen bound with constant κ at every time (`HasJensenBound` with $I = \mathbb{R}$).

Claim 4a. $\sum_i \theta_{\iota(j),i}\hat v(\hat y_{\iota(j),i},0)\to\kappa$. *Proof.* Let $\delta>0$ and put $\varepsilon=\delta/8$. By `UniformSubquadratic` with $z_c=(0,0)$, the compact time set $J=\{0\}$ and this ε, there is $M>0$ such that $|g(y,0)|\le\varepsilon|y|^2$ whenever $|y|\ge M$, for every $g\in\mathcal{A}_{m,C_0}$ centered at the origin; this applies to every v_j and to $\hat v$. Splitting the weighted sum over the indices with $|\hat y_{j,i}|<M$ and those with $|\hat y_{j,i}|\ge M$, and using $\sum_i\theta_{j,i}|\hat y_{j,i}|^2=1$ (the configuration has mean 0 and radius 1),

$$\Big|\sum_i \theta_{j,i}\big(\hat v-v_j\big)(\hat y_{j,i},0)\Big|\le \sup_{|y|\le M}\big|\hat v(y,0)-v_j(y,0)\big|+2\varepsilon\sum_i\theta_{j,i}|\hat y_{j,i}|^2\le \sup_{|y|\le M}\big|\hat v(y,0)-v_j(y,0)\big|+2\varepsilon.$$

The set $\{(y,0):|y|\le M\}$ is compact, so the first term tends to 0 along ι, and for large j the right side is at most $\delta/2+\delta/4<\delta$. Together with (800) this proves Claim 4a.

Put $R=1/\kappa>0$. The configuration $(\theta_{\iota(j),i},\hat y_{\iota(j),i})$ has mean 0 and radius 1, so it occurs in the set $V(\hat v,R,(0,0))$, and the domination half of (B1) applies to it, the hypotheses of `EnvelopeCompetitorBounds` holding for $\hat v$ with $I=\mathbb{R}$ by the opening paragraph. Hence

$$\mathcal{C}_R\hat v(0,0)\ \ge\ \sum_i\theta_{\iota(j),i}\hat v(\hat y_{\iota(j),i},0)-\frac{1}{2R},$$

and letting $j\to\infty$, using Claim 4a, $\hat v(0,0)=0$ and $1/(2R)=\kappa/2$,

$$\mathcal{C}_R\hat v(0,0)-\hat v(0,0)\ \ge\ \kappa-\frac{\kappa}{2}=\frac{\kappa}{2}.$$

In the opposite direction `BaryEnvSandwich`, applied with $I=\mathbb{R}$, $f=\hat v$ and the constant κ, gives $\mathcal{C}_R\hat v(0,0)-\hat v(0,0)\le\kappa^2R/2=\kappa/2$. Therefore

$$\mathcal{C}_{1/\kappa}\hat v(0,0)-\hat v(0,0)=\frac{\kappa}{2}. \tag{801}$$

Step 5: the final frame and the normalized contact. By `IsFitClass` choose a fit (p_*,a_*) for $\hat v$ at the origin of radius $R=1/\kappa$, let g be the change of frame `FrameChange` of $\hat v$ based at $z_0=0$ with scale R and parameters (p_*,a_*), and define

$$v(y,t)=g(y,t-1)=\frac{\hat v\big(R(y+p_*(t-1)),R(t-1)\big)-\hat v(0,0)}{R}-p_*\cdot y-\Big(a_*+\frac{|p_*|^2}{2}\Big)(t-1),$$

the single change of frame and time translation of the proposition. By `FrameInvariance`, $g\in\mathcal{A}_{m,C_0}$, and its second clause with $\zeta=0$, $\rho=1$ — whose hypothesis is the chosen fit for $\hat v$ at the origin of radius $R\cdot1$ — shows that $(0,0)$ is a fit for g at the origin of radius 1, while $g(0)=0$; so g is centered at the origin. By (F1), g satisfies the Jensen bound with constant κ at every time. By Claim T with $c=1$ it follows that

(802)
$v\in\mathcal{A}_{m,C_0}$, v is centered at $z_*=(0,1)$, v satisfies the Jensen bound with constant κ at every time,

and in particular $v(z_*)=0$ by `IsCenteredAt`.

Both g and $\hat v$ are continuous with subquadratic spatial growth on compact time intervals, by the opening paragraph applied to the centered members g and $\hat v$ of the class. So (F2) applies with $\sigma=1$ and $(y,s)=(0,0)$, for which $T_0(0)=0$ and $t=0$, that is $Z=(0,0)$:

$$\mathcal{C}_1g(0,0)-g(0,0)=\frac{\mathcal{C}_R\hat v(0,0)-\hat v(0,0)}{R}=\kappa\cdot\frac{\kappa}{2}=\frac{\kappa^2}{2},$$

using (801) and $1/R=\kappa$. By Claim T(iv) with $c=1$, $\mathcal{C}_1v(0,1)=\mathcal{C}_1g(0,0)$, and $v(0,1)=g(0,0)$; hence

$$\mathcal{C}_1v(0,1)-v(0,1)=\frac{\kappa^2}{2}. \tag{803}$$

Step 6: the contact envelope. Let W be the contact envelope of `ContactEnv` formed from v and κ, so $W(x,t)=\mathcal{C}_tv(x,t)-\kappa^2t/2$ for $t>0$. By (802) and the opening paragraph, v is continuous on space-time, satisfies `SubquadraticGrowth` with $I=\mathbb{R}$ and with $I=(0,\infty)$, and is a viscosity solution of the limiting equation everywhere.

(a) The sandwich and the contact. `BaryEnvSandwich`, applied with $I=(0,\infty)$, $f=v$ and the constant κ, gives for every $s>0$ and every $z=(x,t)$ with $t>0$ that $v(z)\le \mathcal{C}_s v(z)\le v(z)+\kappa^2 s/2$. Taking $s=t$ and subtracting $\kappa^2 t/2$,

$$v(x,t)-\frac{\kappa^2 t}{2}\le W(x,t)\le v(x,t)\qquad (t>0). \tag{804}$$

At z_*, (803) gives $W(z_*)=\mathcal{C}_1 v(0,1)-\kappa^2/2=v(0,1)=0$, so $W(z_*)=v(z_*)=0$.

(b) Continuity and semiconcavity. Apply `BaryEnvDuality` with $I=(0,\infty)$ and $f=v$; its hypotheses are the continuity and growth just recorded. Its continuity clause gives that $(s,z)\mapsto \mathcal{C}_s v(z)$ is continuous at every pair (s,z_0) with $s>0$ and the time of z_0 positive; composing with the continuous map $z\mapsto (t(z),z)$, where $t(z)$ is the time coordinate, shows that $z\mapsto \mathcal{C}_{t(z)}v(z)$ is continuous on $\{t>0\}$, and subtracting the continuous function $\kappa^2 t/2$ gives the continuity of W there. Its concavity clause, applied with $s=t$ for a fixed $t>0$, gives that $x\mapsto \mathcal{C}_t v(x,t)-|x|^2/(2t)$ is concave on F; subtracting the constant $\kappa^2 t/2$, so is $x\mapsto W(x,t)-|x|^2/(2t)$, for every $t>0$.

(c) Growth. Let $J\subseteq(0,\infty)$ be compact and let $\varepsilon>0$. As J is bounded there is c_J with $\kappa^2 t/2\le c_J$ for $t\in J$. By `SubquadraticGrowth` for v with $I=(0,\infty)$ and $\varepsilon/2$ there is $M_1>0$ with $|v(x,t)|\le(\varepsilon/2)|x|^2$ whenever $|x|\ge M_1$ and $t\in J$. By (804), $|W(x,t)|\le |v(x,t)|+\kappa^2 t/2$ for $t>0$, so for $t\in J$ and $|x|\ge \max\{M_1,\sqrt{2c_J/\varepsilon}\}$ we get $|W(x,t)|\le(\varepsilon/2)|x|^2+c_J\le \varepsilon|x|^2$. This is `SubquadraticGrowth` for W with $I=(0,\infty)$.

(d) The subsolution property. First, v is a viscosity subsolution of the limiting equation on the open set $\{t>0\}$ in the sense of `IsLimitSubOn`: it is continuous there, and if ϕ is twice continuously differentiable at a point z with positive time and $v-\phi$ has a local maximum at z relative to $\{t>0\}$, then, because $\{t>0\}$ is open and contains z, a neighbourhood of z witnessing that relative maximum may be intersected with $\{t>0\}$ and is then a neighbourhood of z in the whole space on which the inequality holds; so $v-\phi$ has a local maximum at z relative to the whole space, and $0\le F[\phi](z)$ follows from `IsLimitSubOn` for v on the whole space. Now `BarycentricSubsolution`, applied with $f=v$, with this subsolution property, with `SubquadraticGrowth` on $I=(0,\infty)$, and with $c=\kappa^2/2$, shows that $z\mapsto \mathcal{C}_{t(z)}v(z)-(\kappa^2/2)t(z)$, which is exactly W by `ContactEnv`, is a viscosity subsolution of the limiting equation on $\{t>0\}$.

(e) The gap at a contact. If $t>0$ and $W(x,t)=v(x,t)$ then `ContactEnv` gives $\mathcal{C}_t v(x,t)-v(x,t)=\kappa^2 t/2$, hence

$$\frac{\mathcal{C}_t v(x,t)-v(x,t)}{t}=\frac{\kappa^2}{2}>0.$$

Step 7: the slope at the contact. Since v is continuous, has subquadratic spatial growth with $I=\mathbb{R}$ and solves the limiting equation on all of space-time, `EnvHamiltonianSolution` applies with $I=\mathbb{R}$ and $f=v$. Its third clause gives $H_v(z_*)\ne+\infty$, where H_v is the envelope Hamiltonian of `EnvHamiltonian`; its first clause gives that $\ell\mapsto \mathcal{S}^+_\ell v(z_*)$ and $\ell\mapsto \mathcal{S}^-_\ell v(z_*)$ are nondecreasing on $(0,\infty)$ (`EnvSlopeP`, `EnvSlopeM`); and its second clause, whose hypothesis holds because $I=\mathbb{R}$, gives

$$\inf_{\ell>0}\mathcal{S}^-_\ell v(z_*)=H_v(z_*)=\inf_{\ell>0}\mathcal{S}^+_\ell v(z_*), \tag{805}$$

the second equality being the definition in `EnvHamiltonian`. By (802) and (803), the hypotheses of `ContactFiniteSlope` hold for $C_0,\kappa>0$ and v, so $H_v(z_*)\ne-\infty$. Hence there is a real number H with $H_v(z_*)=H$, and by (805) every $\mathcal{S}^\pm_\ell v(z_*)$ is a real number not smaller than H.

Next we compare the slopes of W and v at z_*, where $W(z_*)=v(z_*)$ by Step 6(a). By Step 6(b),(c) the function W satisfies the hypotheses of `EnvelopeCompetitorBounds` with $I=(0,\infty)$, and v satisfies them with $I=\mathbb{R}$; the relevant times $1\pm\ell$ lie in $(0,\infty)$ for $0<\ell<1$, and $1+\ell\in(0,\infty)$ for every $\ell>0$. So (B2) applies to $\mathcal{P}_\ell W(z_*)$ and $\mathcal{P}_\ell v(z_*)$ for every $\ell>0$, and (B3) to $\mathcal{M}_\ell W(z_*)$ and $\mathcal{M}_\ell v(z_*)$ for $0<\ell<1$: each of the four is the genuine least upper bound, respectively greatest lower bound, of its competitor set. All four are real numbers, and hence so are $\mathcal{S}^+_\ell W(z_*)$ and, for $0<\ell<1$, $\mathcal{S}^-_\ell W(z_*)$, by the second remark of the opening paragraph.

For $\ell>0$, the set defining $\mathcal{P}_\ell$ at z_* in `QuadEnvP` involves only values at time $1+\ell>0$, where $W\le v$ by (804); so each of its elements for W is at most the element for v with the same y,

and the latter set is bounded above by the domination half of (B2) for v; the extremality half of (B2) for W then gives the comparison. Hence $\mathcal{P}_\ell W(z_*) \le \mathcal{P}_\ell v(z_*)$ and, by `EnvSlopeP` and $W(z_*) = v(z_*)$,

$$\mathcal{S}_\ell^+ W(z_*) \le \mathcal{S}_\ell^+ v(z_*) \qquad (\ell > 0). \tag{806}$$

For $0 < \ell < 1$, the set defining $\mathcal{M}_\ell$ at z_* in `QuadEnvM` involves only values at time $1 - \ell > 0$, where again $W \le v$; so $\mathcal{M}_\ell W(z_*)$, a lower bound for the W-set, is a lower bound for the v-set as well, whence $\mathcal{M}_\ell W(z_*) \le \mathcal{M}_\ell v(z_*)$ and, by `EnvSlopeM` and $W(z_*) = v(z_*)$,

$$\mathcal{S}_\ell^- W(z_*) \ge \mathcal{S}_\ell^- v(z_*) \qquad (0 < \ell < 1). \tag{807}$$

Fix $h \in (0,1)$ and $\varepsilon > 0$. The second clause of `EnvelopeSlopeOrder` applied to W at z_* with these h and ε — its hypothesis $1 + \sigma > 0$ for $|\sigma| \le h$ holds as $h < 1$ — supplies $\delta > 0$ with $\mathcal{S}_\ell^- W(z_*) \le \mathcal{S}_h^+ W(z_*) + \varepsilon$ for $0 < \ell < \delta$. Pick one ℓ with $0 < \ell < \min\{\delta, 1\}$. Then, using (805) and (807),

$$H = \inf_{\ell' > 0} \mathcal{S}_{\ell'}^- v(z_*) \le \mathcal{S}_\ell^- v(z_*) \le \mathcal{S}_\ell^- W(z_*) \le \mathcal{S}_h^+ W(z_*) + \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, $H \le \mathcal{S}_h^+ W(z_*)$ for every $h \in (0,1)$; and the first clause of `EnvelopeSlopeOrder` applied to W — whose hypothesis $1 + \sigma > 0$ for $\sigma \in [0, h']$ always holds — shows that $h \mapsto \mathcal{S}_h^+ W(z_*)$ is nondecreasing on $(0, \infty)$, so $H \le \mathcal{S}_h^+ W(z_*)$ for every $h > 0$. Conversely (806) and (805) give $\inf_{h>0} \mathcal{S}_h^+ W(z_*) \le \inf_{h>0} \mathcal{S}_h^+ v(z_*) = H$. Hence

$$\inf_{h > 0} \mathcal{S}_h^+ W(z_*) = H,$$

and since $h \mapsto \mathcal{S}_h^+ W(z_*)$ is a nondecreasing real-valued function on $(0, \infty)$ with infimum H, it converges to H as $h \downarrow 0$:

$$\lim_{\ell \downarrow 0} \mathcal{S}_\ell^+ W(z_*) = H = H_v(z_*).$$

Collecting (798), (802), Step 6(a)–(e) and Step 7 gives all the assertions, with this κ, this H and this v. □

2.181. `FarOptimizerBound`.

Lemma 2.181 (A far position cannot beat the undisplaced competitor)**.** Let F be a finite-dimensional real inner-product space, let $C_V > 0$ and let W be a real function on space-time satisfying the box bound

$$|W(\zeta)| \le C_V\, K\, L(K)^2 \qquad (\zeta \in \mathcal{B}_K,\ K \ge 2),$$

where $L(K) = 1 + \log K$ and $\mathcal{B}_K = \{(y,s) \in F \times \mathbb{R} : |y| \le K,\ |s| \le K\}$. Fix $x \in F$ and $t \in \mathbb{R}$. Then there is a constant $C < \infty$ such that for every length $\ell \ge 1$ and every $y \in F$ with

$$W(x, t + \ell) \le W(y, t + \ell) - \frac{|y - x|^2}{2\ell} \tag{808}$$

one has

$$|y| \le C\, \ell\, (1 + \log \ell)^2.$$

Proof. The regime here is the opposite of the one in the small-length optimizer estimate of the limiting-tools section, the node `OptimizerLocalization`, which is deliberately not a dependency of this statement: that lemma constrains the optimizers of the quadratic envelopes at *small* lengths, below a threshold depending on the level, and gives the sharp rate $\sqrt{\ell}$; the statement above concerns *large* lengths $\ell \ge 1$ and gives only the much weaker rate $\ell(1 + \log \ell)^2$, which is all that the box growth of W supports and all that is needed where this bound is used. Neither statement implies the other, and the hypothesis here is a bare scalar inequality rather than an envelope optimality property, so no envelope machinery enters this proof.

Set $c_1 = |x| + |t| + 3$, so $c_1 \ge 3$ and $\log(2c_1) \ge 0$, and put

$$c_2 = 16 C_V c_1 \big(1 + \log(2c_1)\big)^2 > 0, \qquad c_3 = 16 C_V c_1 > 0, \qquad b_0 = \max\big\{2, (32 c_3)^2\big\}.$$

We prove the conclusion with $C = |x| + 2 + (2c_2 + b_0)$, which is positive.

An elementary absorption of the logarithm. For $u \geq 1$, $\log u = 4\log\sqrt{\sqrt{u}} \leq 4\sqrt{\sqrt{u}}$, by applying $\log\sqrt{a} = \frac{1}{2}\log a$ twice and then $\log s \leq s-1 \leq s$ at $s = \sqrt{\sqrt{u}} > 0$. Since $\log u \geq 0$, squaring gives $(\log u)^2 \leq 16\sqrt{u}$. Hence if $u \geq 1$ and $u \geq (32c_3)^2$ then $\sqrt{u} \geq 32c_3$, so

$$c_3(\log u)^2 \leq 16c_3\sqrt{u} \leq \frac{\sqrt{u}\cdot\sqrt{u}}{2} = \frac{u}{2}. \tag{809}$$

Setting up. Fix $\ell \geq 1$ and $y \in F$ with (808). Then $L(\ell) = 1+\log\ell \geq 1$, so $L(\ell)^2 \geq 1$. Put $\beta = |y-x|/\ell \geq 0$, so $|y-x| = \beta\ell$ and $|y| \leq |x| + \beta\ell$. It suffices to produce a $D \in [0,\, 2+2c_2+b_0]$ with $\beta \leq DL(\ell)^2$: indeed then $\beta\ell \leq DL(\ell)^2\ell \leq (2+2c_2+b_0)L(\ell)^2\ell$, while $|x| \leq |x|\ell L(\ell)^2$ because $\ell \geq 1$ and $L(\ell)^2 \geq 1$, and adding the two gives $|y| \leq C\ell L(\ell)^2$.

The near regime. If $\beta \leq 2$, take $D = 2$: then $\beta \leq 2 \leq 2L(\ell)^2$, and $2 \leq 2+2c_2+b_0$ because $c_2 > 0$ and $b_0 \geq 2$.

The far regime. Assume $\beta > 2$. Then $\beta\ell \geq \beta > 2$, and since $c_1 \geq 3$,

$$c_1 \leq c_1(\beta\ell), \qquad \beta\ell \leq c_1(\beta\ell), \qquad 6 \leq c_1(\beta\ell).$$

Put $\Theta = 2c_1(\beta\ell)$, so $\Theta \geq 12 \geq 2$. Also $|x| \leq c_1$ and $|t| \leq c_1$ by the definition of c_1, hence

$$|y| \leq |x| + \beta\ell \leq c_1(\beta\ell) + c_1(\beta\ell) = \Theta, \qquad |x| \leq c_1 \leq \Theta,$$

and, using $\ell \leq \beta\ell$, $|t+\ell| \leq |t| + \ell \leq c_1(\beta\ell) + c_1(\beta\ell) = \Theta$. So both $(y, t+\ell)$ and $(x, t+\ell)$ lie in $\mathcal{B}_\Theta$, and the box bound gives

$$|W(y,t+\ell)| \leq C_V\Theta L(\Theta)^2, \qquad |W(x,t+\ell)| \leq C_V\Theta L(\Theta)^2.$$

Rearranging (808) and using $|y-x|^2/(2\ell) = \beta(\beta\ell)/2$,

$$\frac{\beta(\beta\ell)}{2} \leq W(y,t+\ell) - W(x,t+\ell) \leq |W(y,t+\ell)| + |W(x,t+\ell)| \leq 2C_V\Theta L(\Theta)^2 = 4C_Vc_1(\beta\ell)L(\Theta)^2.$$

Both sides are products of $(\beta\ell)/2 > 0$ with β and with $8C_Vc_1L(\Theta)^2$ respectively, so cancelling that positive factor gives

$$\beta \leq 8C_Vc_1L(\Theta)^2. \tag{810}$$

Expanding the logarithm of Θ. Since $\Theta = (2c_1)(\beta\ell)$ with $2c_1 > 0$ and $\beta\ell > 0$, $\log\Theta = \log(2c_1) + \log\beta + \log\ell$, and $\log\beta \geq 0$ because $\beta > 2 > 1$. As $\log(2c_1) \geq 0$ and $\log\ell \geq 0$,

$$1 + \log\Theta \leq \big(1+\log(2c_1)\big)\big(1+\log\ell\big) + \log\beta,$$

because the product on the right expands to $1+\log(2c_1)+\log\ell+\log(2c_1)\log\ell$ and the last term is nonnegative. Both sides being nonnegative, squaring and then using $(u+w)^2 \leq 2u^2+2w^2$ gives

$$L(\Theta)^2 \leq 2\big(1+\log(2c_1)\big)^2L(\ell)^2 + 2(\log\beta)^2.$$

Substituting into (810) and multiplying out,

$$\beta \leq c_2L(\ell)^2 + c_3(\log\beta)^2. \tag{811}$$

Absorbing. If $\beta \geq b_0$ then $\beta \geq 2 \geq 1$ and $\beta \geq (32c_3)^2$, so (809) applies at $u = \beta$ and gives $c_3(\log\beta)^2 \leq \beta/2$; with (811) this yields $\beta \leq 2c_2L(\ell)^2$, so $D = 2c_2$ works, and $2c_2 \leq 2+2c_2+b_0$. If $\beta < b_0$ then $\beta < b_0 \leq b_0L(\ell)^2$ since $L(\ell)^2 \geq 1$ and $b_0 \geq 0$, so $D = b_0$ works, and $b_0 \leq 2+2c_2+b_0$. In both cases the reduction of the previous paragraph applies and completes the proof. □

2.182. `FitClassGrowth`.

Lemma 2.182 (Box growth and subquadratic growth for an arbitrary class member)**.** Let F be a finite-dimensional real inner-product space, let $1 \leq C_0 < \infty$ and let $w \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`. Write $L(K) = 1+\log K$ and $\mathcal{B}_K = \{(y,s) \in F\times\mathbb{R} : |y| \leq K,\ |s| \leq K\}$. Then there is a constant $C > 0$ such that

$$\big|w(\zeta) - w(0)\big| \leq C\,K\,L(K)^2 \qquad (\zeta \in \mathcal{B}_K,\ K \geq 2),$$

and w has subquadratic spatial growth on compact time subintervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`.

Proof. No centering is assumed: the point of the statement is that an arbitrary member of the class obeys the centered growth estimate once its value at the origin is subtracted.

The normalizing frame change. By the second clause of `IsFitClass` choose a fit (p_1, a_1) for w at the origin of radius 1 with constant C_0. Let W be the sheared change of frame of `FrameChange` based at $z_0 = 0$ with scale $R = 1$ and parameters (p_1, a_1). Unwinding the definition of `FrameChange` at the point $(y - sp_1, s)$, whose sheared image is (y, s),

$$W(y - sp_1,\, s) = w(y, s) - w(0) - \langle p_1, y - sp_1\rangle - \Big(a_1 + \frac{|p_1|^2}{2}\Big)s \qquad (y \in F,\ s \in \mathbb{R}). \tag{812}$$

By the first assertion of `FrameInvariance`, $W \in \mathcal{A}_{m,C_0}$.

W is centered at the origin. Taking $(y, s) = (0, 0)$ in (812) gives $W(0) = w(0) - w(0) - 0 - 0 = 0$. For the fit, apply the second assertion of `FrameInvariance` with new center $\zeta = 0$ and radius $\rho = 1$: its hypothesis asks for a fit for w at the point $z_0 + R\big((\zeta_y + \zeta_s p_1),\, \zeta_s\big)$, which for $z_0 = 0$, $R = 1$ and $\zeta = 0$ is the origin, and at radius $R\rho = 1$; that is exactly the fit (p_1, a_1) chosen above. The conclusion is that $(p_1 - p_1, a_1 - a_1) = (0, 0)$ is a fit for W at the origin of radius 1. With $W(0) = 0$ this is precisely W being centered at the origin with constant C_0 in the sense of `IsCenteredAt`.

The box bound. Let $C_g > 0$ be the constant of `CenteredGrowth` for C_0, so that $|W(\zeta)| \le C_g K L(K)^2$ for $\zeta \in \mathcal{B}_K$ and $K \ge 2$, W being a centered member of the class. Let $K \ge 2$ and let $\zeta = (y, s) \in \mathcal{B}_K$. Note $L(K) \ge 1$, hence $L(K)^2 \ge 1$.

Put $K' = K(1 + |p_1|)$. Then $K' \ge K \ge 2$, and $|y - sp_1| \le |y| + |s||p_1| \le K + K|p_1| = K'$ while $|s| \le K \le K'$, so $(y - sp_1, s) \in \mathcal{B}_{K'}$ and $|W(y - sp_1, s)| \le C_g K' L(K')^2$. Since $\log K' = \log K + \log(1 + |p_1|)$ and $\log(1 + |p_1|) \ge 0$ and $L(K) \ge 1$,

$$L(K') = L(K) + \log(1 + |p_1|) \le \big(1 + \log(1 + |p_1|)\big) L(K),$$

and $L(K') \ge 0$, so squaring and multiplying by $C_g K' = C_g K(1 + |p_1|) \ge 0$,

$$|W(y - sp_1, s)| \le C_g(1 + |p_1|)\big(1 + \log(1 + |p_1|)\big)^2\, K\, L(K)^2.$$

By Cauchy–Schwarz and $|y - sp_1| \le K(1 + |p_1|)$, $|\langle p_1, y - sp_1\rangle| \le |p_1|(1 + |p_1|)K \le |p_1|(1 + |p_1|)KL(K)^2$, the last step by $L(K)^2 \ge 1$. Finally $\big|\big(a_1 + \tfrac12|p_1|^2\big)s\big| \le \big(|a_1| + \tfrac12|p_1|^2\big)K \le \big(|a_1| + \tfrac12|p_1|^2\big)KL(K)^2$, again by $|s| \le K$ and $L(K)^2 \ge 1$. Rearranging (812) as

$$w(y, s) - w(0) = W(y - sp_1, s) + \langle p_1, y - sp_1\rangle + \Big(a_1 + \frac{|p_1|^2}{2}\Big)s$$

and adding the three bounds through the triangle inequality gives the box bound with

$$C = C_g(1 + |p_1|)\big(1 + \log(1 + |p_1|)\big)^2 + |p_1|(1 + |p_1|) + |a_1| + \tfrac12|p_1|^2,$$

which is positive because its first summand is.

Subquadratic growth. Let $J \subseteq \mathbb{R}$ be compact and let $\varepsilon > 0$. A compact subset of $\mathbb{R}$ is bounded, so there is $K_J \ge 1$ with $|s| \le K_J$ for every $s \in J$. Apply `UniformSubquadratic` with the centering point 0, the compact set J and the tolerance $\varepsilon/8$: it returns $M_1 > 0$ such that every member of the class centered at the origin satisfies $|U(\eta)| \le (\varepsilon/8)|\eta_y|^2$ whenever $|\eta_y| \ge M_1$ and $\eta_s \in J$. Since W is such a member, this applies to W.

Put

$$A = |w(0)| + |p_1|\,K_J|p_1| + \big(|a_1| + \tfrac12|p_1|^2\big)K_J \ (\ge 0), \qquad M = \max\Big\{1,\ M_1 + K_J|p_1|,\ K_J|p_1|,\ \frac{2(A + |p_1|)}{\varepsilon}\Big\} > 0.$$

Let $\zeta = (y, s)$ with $|y| \ge M$ and $s \in J$. Then $|s| \le K_J$, so $|sp_1| \le K_J|p_1|$, and therefore

$$|y - sp_1| \le |y| + K_J|p_1| \le 2|y|, \qquad |y - sp_1| \ge |y| - K_J|p_1| \ge M_1,$$

using $|y| \ge K_J|p_1|$ for the first and $|y| \ge M_1 + K_J|p_1|$ for the second. Hence the uniform estimate applies at $(y - sp_1, s)$, whose time coordinate is $s \in J$, and

$$|W(y - sp_1, s)| \le \frac{\varepsilon}{8}|y - sp_1|^2 \le \frac{\varepsilon}{8} \cdot 4|y|^2 = \frac{\varepsilon}{2}|y|^2.$$

For the remaining terms of (812), Cauchy–Schwarz and $|y - sp_1| \le |y| + K_J|p_1|$ give $|\langle p_1, y - sp_1\rangle| \le |p_1||y| + |p_1|K_J|p_1|$, and $\big|\big(a_1 + \tfrac{1}{2}|p_1|^2\big)s\big| \le \big(|a_1| + \tfrac{1}{2}|p_1|^2\big)K_J$, so

$$|\langle p_1, y - sp_1\rangle| + \Big|\Big(a_1 + \frac{|p_1|^2}{2}\Big)s\Big| + |w(0)| \le A + |p_1|\,|y|.$$

Since $|y| \ge 1$ and $A \ge 0$ we have $A + |p_1||y| \le (A + |p_1|)|y|$, and $|y| \ge 2(A + |p_1|)/\varepsilon$ gives $2(A + |p_1|) \le \varepsilon|y|$, hence $(A + |p_1|)|y| \le \frac{\varepsilon}{2}|y|^2$. Writing $w(y, s) = W(y - sp_1, s) + \langle p_1, y - sp_1\rangle + \big(a_1 + \frac{1}{2}|p_1|^2\big)s + w(0)$ and applying the triangle inequality,

$$|w(y, s)| \le \frac{\varepsilon}{2}|y|^2 + \frac{\varepsilon}{2}|y|^2 = \varepsilon|y|^2.$$

As J and ε were arbitrary, w has subquadratic spatial growth on compact time subintervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`. □

2.183. `FitCylinderToNormalizedBox`.

Lemma 2.183 (A cylinder of an anisotropic slope sits in one fixed normalized box)**.** *Let F be a real inner-product space, let $\ell^2(F \times \mathbb{R})$ be the associated space-time, and let Φ be a linear isometry of $\ell^2(F \times \mathbb{R})$ onto itself. Let $z \in \ell^2(F \times \mathbb{R})$, let $c > 0$, $\lambda > 0$ and $r > 0$ be reals, let $p \in F$ and $q \in \mathbb{R}$, and write*

$$A = 1 + \lambda q, \qquad N = \big(\lambda|p|^2 + A^2\big)^{1/2}.$$

Assume

$$\tfrac{1}{2} \le N. \tag{813}$$

If $x \in F$ and $t \in \mathbb{R}$ are such that

$$z + r\,\Phi\big(\sqrt{\lambda}\,x,\ t\big) \in Z_\lambda\big(z,\ r;\ \Phi(c\sqrt{\lambda}\,p,\ cA)\big), \tag{814}$$

then

$$|x| \le 2|p| + 1, \qquad |t| \le 1 + \sqrt{\lambda}.$$

Proof. Put

$$e = \big(\sqrt{\lambda}\,p,\ A\big), \qquad u = \big(\sqrt{\lambda}\,x,\ t\big),$$

both in $\ell^2(F \times \mathbb{R})$.

Step 1: the length N of the frame vector, and its two consequences. By the definition of the ℓ^2 product norm and $|\sqrt{\lambda}\,p| = \sqrt{\lambda}\,|p|$ together with $\sqrt{\lambda}\sqrt{\lambda} = \lambda$ (legitimate since $\lambda \ge 0$),

$$|e|^2 = \lambda|p|^2 + A^2. \tag{815}$$

Both $|e|$ and N are nonnegative and have the same square by (815) and the definition of N, so $|e| = N$; in particular $N \ge 1/2 > 0$ by (813). Two separate consequences of (815) will be used, and it is worth separating them:

$$|A| \le N, \tag{816}$$

because $\lambda|p|^2 \ge 0$ gives $A^2 \le N^2$, hence $|A|^2 \le N^2$ and $|A| \le N$ as both sides are nonnegative; and the hypothesis (813) itself, which is what will bound $|p|/N$.

Step 2: the slope's unit vector. Since Φ is linear, $\Phi(c\sqrt{\lambda}\,p, cA) = \Phi(c\,e) = c\,\Phi(e)$, and since Φ is an isometry and $c > 0$ this vector has norm $cN > 0$. Hence its normalized direction is

$$n = \frac{c\,\Phi(e)}{cN} = N^{-1}\Phi(e), \qquad |n| = N^{-1}|\Phi(e)| = N^{-1}N = 1.$$

The increment of the point in (814) relative to z is $D = r\,\Phi(u)$.

Step 3: what the two cylinder conditions say. By the definition of the cylinder, (814) is the conjunction of

$$|D \cdot n| \le r \tag{817}$$

and

$$\big|D - (D \cdot n)n\big| \le \sqrt{\lambda}\,r. \tag{818}$$

Since Φ is a linear isometry it preserves inner products, so $D \cdot n = r\,N^{-1}\big(\Phi(u) \cdot \Phi(e)\big) = r\,N^{-1}(u \cdot e)$, and by the definition of the ℓ^2 inner product together with $\sqrt{\lambda}\sqrt{\lambda} = \lambda$,

$$e \cdot u = \lambda\,(p \cdot x) + A\,t.$$

Put

$$\sigma = \frac{e \cdot u}{N^2}.$$

Step 4: the axial condition bounds σ. Dividing (817) by $r > 0$ gives $N^{-1}|e \cdot u| \le 1$, and multiplying by $N \ge 0$ gives $|e \cdot u| \le N$. Therefore

$$|\sigma|\,N = \frac{|e \cdot u|}{N^2}\,N = \frac{|e \cdot u|}{N} \le 1. \tag{819}$$

Step 5: the transverse condition bounds the residual, componentwise. Because $n = N^{-1}\Phi(e)$ and $D \cdot n = r\,N^{-1}(e \cdot u)$,

$$(D \cdot n)\,n = r\,N^{-1}(e \cdot u) \cdot N^{-1}\Phi(e) = r\,\frac{e \cdot u}{N^2}\,\Phi(e) = r\,\Phi(\sigma\,e),$$

so by linearity of Φ

$$D - (D \cdot n)n = r\,\Phi(u) - r\,\Phi(\sigma e) = r\,\Phi(u - \sigma e).$$

As Φ is an isometry and $r > 0$, (818) becomes $|u - \sigma e| \le \sqrt{\lambda}$. Now in $F \times \mathbb{R}$

$$u - \sigma e = \big(\sqrt{\lambda}\,x - \sigma\sqrt{\lambda}\,p,\ t - \sigma A\big) = \big(\sqrt{\lambda}\,(x - \sigma p),\ t - \sigma A\big),$$

so by the definition of the ℓ^2 product norm, and $\sqrt{\lambda}\sqrt{\lambda} = \lambda$ once more,

$$|u - \sigma e|^2 = \lambda\,|x - \sigma p|^2 + (t - \sigma A)^2.$$

Squaring $|u - \sigma e| \le \sqrt{\lambda}$, which is legitimate as both sides are nonnegative, and using $(\sqrt{\lambda})^2 = \lambda$,

$$\lambda\,|x - \sigma p|^2 + (t - \sigma A)^2 \le \lambda. \tag{820}$$

Step 6: the spatial bound, which is where $N \ge 1/2$ is used. Dropping the nonnegative term $(t - \sigma A)^2$ from (820) gives $\lambda|x - \sigma p|^2 \le \lambda$, and dividing by $\lambda > 0$ gives $|x - \sigma p|^2 \le 1$; as $|x - \sigma p| \ge 0$ this yields $|x - \sigma p| \le 1$. Next, $|\sigma| \le 2$: indeed $|\sigma|N \le 1$ by (819) and $N \ge 1/2$ by (813), so $|\sigma|/2 \le |\sigma|N \le 1$. Hence $|\sigma p| = |\sigma|\,|p| \le 2|p|$, and by the triangle inequality applied to $x = (x - \sigma p) + \sigma p$,

$$|x| \le |x - \sigma p| + |\sigma p| \le 1 + 2|p|.$$

Step 7: the time bound, which uses only $N \ge |A|$. Dropping the nonnegative term $\lambda|x - \sigma p|^2$ from (820) gives $(t - \sigma A)^2 \le \lambda = (\sqrt{\lambda})^2$, and as both $|t - \sigma A|$ and $\sqrt{\lambda}$ are nonnegative this yields $|t - \sigma A| \le \sqrt{\lambda}$. For the other term,

$$|\sigma A| = |\sigma|\,|A| \le |\sigma|\,N \le 1$$

by (816) and (819), the first inequality using $|\sigma| \ge 0$. Hence by the triangle inequality applied to $t = (t - \sigma A) + \sigma A$,

$$|t| \le |t - \sigma A| + |\sigma A| \le \sqrt{\lambda} + 1.$$

Steps 6 and 7 are the two asserted bounds. Note that the hypothesis (813) entered only in Step 6: the time bound holds with no lower bound on N at all. □

2.184. `NonaffineLimitEquicontinuousSubsequence`.

Lemma 2.184 (A locally uniformly convergent subsequence of the normalized solutions)**.** *Let $d \in \mathbb{N}$ and work in $X = \ell^2(\mathbb{R}^d \times \mathbb{R})$, writing a point as $w = (w_x, w_t)$ and $\overline{B}_\rho$ for the closed ball of radius ρ centred at the origin. Let $\lambda : \mathbb{N} \to \mathbb{R}$, let $v_i : X \to \mathbb{R}$ for $i \in \mathbb{N}$, let $i_1 \in \mathbb{N}$, let $\mathcal{B} : \mathbb{R} \to \mathcal{P}(X)$, let $A : \mathbb{R} \to \mathbb{R}$, $i_4 : \mathbb{R} \to \mathbb{N}$, $C_3 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and $i_5 : \mathbb{R} \times \mathbb{R} \to \mathbb{N}$. Assume:*

1. *$\mathcal{B}_K = \{w \in X :\ |w_x| \le K$ and $|w_t| \le K\}$ for every $K \in \mathbb{R}$;*
2. *$v_i(0) = 0$ for every $i \ge i_1$;*
3. *for every $K \ge 1$, every $i \ge i_4(K)$ and every $w \in \mathcal{B}_K$, $|v_i(w)| \le A(K)$;*
4. *$0 \le C_3(K, S)$ for all $K, S \in \mathbb{R}$;*

(5) *for every* $K \geq 1$, *every* $S > 0$, *every* $i \geq i_5(K,S)$ *and every* $w \in \mathcal{B}_K$ *there are* $p' \in \mathbb{R}^d$ *and* $q' \in \mathbb{R}$ *with*

$$|p'| + |q'| \leq C_3(K,S), \qquad \tfrac{1}{2} \leq 1 + \lambda_i q',$$

such that for all $h \in \mathbb{R}^d$ *and* $l \in \mathbb{R}$ *with* $|l| \leq S/2$ *and* $\big|h - \frac{l}{1+\lambda_i q'} p'\big| \leq S$,

$$\big|v_i\big(w + (h,l)\big) - v_i(w) - \langle p', h\rangle - q' l\big| \leq 64S. \tag{821}$$

Then there are a strictly increasing $\psi : \mathbb{N} \to \mathbb{N}$ *with* $i_1 \leq \psi(j)$ *for every* j, *and a continuous* $v_\infty : X \to \mathbb{R}$ *with* $v_\infty(0) = 0$, *such that* $v_{\psi(j)} \to v_\infty$ *uniformly on every compact subset of* X.

Proof. The sup box contains the Euclidean ball. If $y \in \overline{B}_\rho$, i.e. $|y| \leq \rho$, then $|y_x| \leq \rho$ and $|y_t| \leq \rho$: indeed $|y|^2 = |y_x|^2 + |y_t|^2$ with both summands nonnegative, so each is at most $|y|^2 \leq \rho^2$, and the square root is monotone on $[0,\infty)$. We use this three times below, and by 1 it says exactly that $\overline{B}_\rho \subseteq \mathcal{B}_\rho$.

We verify the two hypotheses of `AsymptEquicontOriginNormalizedLimit` for the family (v_i) and the index i_1; its third hypothesis is 2, and its conclusion is the assertion of this lemma.

The uniform bound on each ball. Let $\rho > 0$ and put $K = \max(1,\rho)$, so $1 \leq K$ and $\rho \leq K$. Take $A = A(K)$ and the threshold $i_4(K)$. For $j \geq i_4(K)$ and $y \in \overline{B}_\rho$ the paragraph above gives $|y_x| \leq \rho \leq K$ and $|y_t| \leq \rho \leq K$, so $y \in \mathcal{B}_K$ by 1; hence $|v_j(y)| \leq A(K)$ by 3. This is the first hypothesis.

The asymptotic modulus of continuity. Let $\rho > 0$ and $\epsilon > 0$. Put

$$K = \max(1,\rho), \qquad S = \frac{\epsilon}{128}, \qquad C = C_3(K,S),$$

so $1 \leq K$, $\rho \leq K$, $S > 0$, $64S = \epsilon/2$, and $C \geq 0$ by 4; in particular $1 + 2C > 0$ and $4(C+1) > 0$. Only now choose the radius

$$\eta = \min\Big(\frac{S}{2},\ \min\Big(\frac{S}{1+2C},\ \frac{\epsilon}{4(C+1)}\Big)\Big),$$

which is positive, being the minimum of three positive numbers, and satisfies

$$\eta \leq \frac{S}{2}, \qquad \eta(1+2C) \leq S, \qquad \eta \cdot 4(C+1) \leq \epsilon, \tag{822}$$

the last two after clearing the positive denominators. Take the threshold $i_5(K,S)$. The order of these choices is what makes the argument work: the error $64S$ of (821) is pinned to $\epsilon/2$ by the choice of S alone, which fixes C; the remaining coefficient terms are then beaten by shrinking η, whose admissible size may depend on C.

Let $j \geq i_5(K,S)$ and let $y, y' \in \overline{B}_\rho$ with $|y - y'| \leq \eta$. As above $y \in \mathcal{B}_K$, so 5 applies at K, S, j and y and yields p', q' with $|p'| + |q'| \leq C$ and $1 + \lambda_j q' \geq 1/2$. In particular $1 + \lambda_j q' > 0$, and since $|p'| \geq 0$ and $|q'| \geq 0$ the bound splits into $|p'| \leq C$ and $|q'| \leq C$.

Put $h = y'_x - y_x$ and $l = y'_t - y_t$. Then $y + (h,l) = y'$, because $(y' - y)_x = h$ and $(y' - y)_t = l$ and addition is coordinatewise. Also $|y' - y| = |y - y'| \leq \eta$, so the first paragraph applied to $y' - y$ gives

$$|h| \leq \eta, \qquad |l| \leq \eta. \tag{823}$$

The offset is admissible. First $|l| \leq \eta \leq S/2$ by (823) and (822). Second, from $1 + \lambda_j q' \geq 1/2$ we get $\frac{1}{1+\lambda_j q'} \leq 2$, so

$$\Big|\frac{l}{1+\lambda_j q'}\, p'\Big| = \frac{|l|}{1+\lambda_j q'}\, |p'| \leq 2\eta \cdot C,$$

and therefore, by the triangle inequality and (823),

$$\Big|h - \frac{l}{1+\lambda_j q'} p'\Big| \leq |h| + 2\eta C \leq \eta + 2\eta C = \eta(1+2C) \leq S$$

by (822). So (821) applies at this (h,l) and, after substituting $y + (h,l) = y'$, gives

$$\big|v_j(y') - v_j(y) - \langle p', h\rangle - q' l\big| \leq 64S = \frac{\epsilon}{2}. \tag{824}$$

The two coefficient terms. By Cauchy–Schwarz and (823), $|\langle p', h\rangle| \le |p'|\,|h| \le C\eta$; and $|q'l| = |q'|\,|l| \le C\eta$. Their sum is at most $2C\eta$, and $4C\eta \le 4(C+1)\eta \le \epsilon$ by (822) and $\eta > 0$, so $2C\eta \le \epsilon/2$.

Conclusion of the modulus. Combining the last two paragraphs,

$$\big|v_j(y) - v_j(y')\big| \le \big|v_j(y') - v_j(y) - \langle p', h\rangle - q'l\big| + |\langle p', h\rangle| + |q'l| \le \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This is the second hypothesis, with the radius η and the threshold $i_5(K, S)$. Note that the two hypotheses carry their own thresholds — one inside the bound and one inside the ϵ binder of the modulus — so the fact that i_5 depends on both K and S costs nothing.

The extraction. With both hypotheses verified and 2 in hand, `AsymptEquicontOriginNormalizedLimit` gives the strictly increasing ψ with $\psi(j) \ge i_1$, the continuous limit v_∞ with $v_\infty(0) = 0$, and the uniform convergence of $v_{\psi(j)}$ to v_∞ on every compact subset of X. □

2.185. `LimitIsFitAtEveryScale`.

Lemma 2.185 (Step 11: the limit is a fit of class 256 at every centre and scale)**.** *Let $d \in \mathbb{N}$ and write $\mathcal{H} = \ell^2(\mathbb{R}^d \times \mathbb{R})$. Let $\lambda : \mathbb{N} \to \mathbb{R}$, let $v_i : \mathcal{H} \to \mathbb{R}$ for each $i \in \mathbb{N}$, let $\mathcal{B} : \mathbb{R} \to \mathcal{P}(\mathcal{H})$, let $C_3 : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, let $i_5 : \mathbb{R} \times \mathbb{R} \to \mathbb{N}$, let $\psi : \mathbb{N} \to \mathbb{N}$ and let $w : \mathcal{H} \to \mathbb{R}$. Assume:*

1. *$\lambda_i > 0$ for every i, and $\lambda_i \to 0$ as $i \to \infty$;*
2. *the boxes are the coordinate boxes, $\mathcal{B}_K = \{y \in \mathcal{H} : |y_x| \le K \text{ and } |y_t| \le K\}$, where $y = (y_x, y_t)$;*
3. *$C_3(K, S) \ge 0$ for all K, S;*
4. *ψ is strictly monotone;*
5. *$v_{\psi(j)} \to w$ uniformly on every compact subset of $\mathcal{H}$ as $j \to \infty$;*
6. *(the uniform bound on each box) for every $K \ge 1$, every $S > 0$, every $i \ge i_5(K, S)$ and every $y \in \mathcal{B}_K$ there are $p' \in \mathbb{R}^d$ and $q' \in \mathbb{R}$ with*

$$|p'| + |q'| \le C_3(K, S), \qquad \tfrac{1}{2} \le 1 + \lambda_i q',$$

such that for all $h \in \mathbb{R}^d$ and $\ell \in \mathbb{R}$ with $|\ell| \le S/2$ and $\big|h - \big(\ell/(1 + \lambda_i q')\big)p'\big| \le S$,

$$\big|v_i(y + (h, \ell)) - v_i(y) - p' \cdot h - q'\ell\big| \le 64\, S.$$

Then for every centre $\hat{z} \in \mathcal{H}$ and every radius $R' > 0$ there are $p \in \mathbb{R}^d$ and $a \in \mathbb{R}$ such that (p, a) is a fit for w at $\hat{z}$ of radius R' with constant 256, in the sense of `IsFit`.

Proof. Fix a centre $\hat{z} = (\hat{x}, \hat{t}) \in \mathcal{H}$ and a radius $R' > 0$. Put

$$K = \max\{1, |\hat{x}|, |\hat{t}|\}, \qquad S = 4R',$$

so $K \ge 1$ and $S > 0$, and $\hat{z} \in \mathcal{B}_K$ by 2, because $|\hat{x}| \le K$ and $|\hat{t}| \le K$ by the definition of K.

The per-index coefficients. For every j with $\psi(j) \ge i_5(K, S)$, hypothesis 6 applied at the index $i = \psi(j)$ and the point $y = \hat{z}$ supplies $p'_j \in \mathbb{R}^d$ and $q'_j \in \mathbb{R}$ with

$$|p'_j| + |q'_j| \le C_3(K, S), \qquad \tfrac{1}{2} \le A_j := 1 + \lambda_{\psi(j)} q'_j, \tag{825}$$

and with the increment bound of 6. Since ψ is strictly monotone by 4 we have $\psi(j) \ge j$, so all $j \ge i_5(K, S)$ qualify; discard the finitely many smaller indices.

Extraction. By (825) the sequences (p'_j) and (q'_j) are bounded, the first in $\mathbb{R}^d$ and the second in $\mathbb{R}$, so by Bolzano–Weierstrass — available because $\mathbb{R}^d$ is finite-dimensional, hence its closed balls are compact — there are $p \in \mathbb{R}^d$, $q \in \mathbb{R}$ and a strictly monotone $l \mapsto j_l$ with $p'_{j_l} \to p$ and $q'_{j_l} \to q$. Because $\lambda_i \to 0$ by 1 and $|q'_j| \le C_3(K, S)$ uniformly, $\lambda_{\psi(j)} q'_j \to 0$ and therefore

$$A_{j_l} \to 1, \qquad \text{hence} \qquad \frac{p'_{j_l}}{A_{j_l}} \to p, \tag{826}$$

the quotient being legitimate throughout since $A_j \ge \tfrac{1}{2} > 0$.

The admissible increments. Let $h \in \mathbb{R}^d$ and $\ell \in \mathbb{R}$ satisfy $|\ell| \le R'$ and $|h - \ell p| \le R'$. Then $|\ell| \le R' = S/4 \le S/2$, and by the triangle inequality

$$\Big|h - \frac{p'_{j_l}}{A_{j_l}}\,\ell\Big| \le |h - \ell p| + |\ell|\,\Big|p - \frac{p'_{j_l}}{A_{j_l}}\Big| \le R' + R'\Big|p - \frac{p'_{j_l}}{A_{j_l}}\Big|.$$

By (826) the last factor is at most 1 for all large l, and the threshold depends only on the sequence, not on (h,ℓ); for such l the left side is at most $2R' \le S$. Hypothesis 6 therefore applies at the index $\psi(j_l)$, the centre $\hat z \in \mathcal{B}_K$ and this pair (h,ℓ), giving

$$\big|v_{\psi(j_l)}(\hat z + (h,\ell)) - v_{\psi(j_l)}(\hat z) - p'_{j_l}\cdot h - q'_{j_l}\ell\big| \le 64\,S = 256\,R'. \tag{827}$$

Passing to the limit. The set

$$\mathcal{K} = \big\{\hat z + (h,\ell) : |\ell| \le R',\ |h - \ell p| \le R'\big\}$$

is closed and bounded in $\mathcal{H}$, hence compact: boundedness holds because $|\ell| \le R'$ and $|h| \le |h - \ell p| + |\ell||p| \le R'(1 + |p|)$. By 5, $v_{\psi(j_l)} \to w$ uniformly on $\mathcal{K}$, and in particular pointwise at the two points $\hat z + (h,\ell)$ and $\hat z$, both of which lie in $\mathcal{K}$ (the second by taking $h = 0$, $\ell = 0$). Since also $p'_{j_l} \to p$ and $q'_{j_l} \to q$, the quantity inside the absolute value in (827) converges to $w(\hat z + (h,\ell)) - w(\hat z) - p\cdot h - q\ell$, and a non-strict inequality is preserved in the limit:

$$\big|w(\hat z + (h,\ell)) - w(\hat z) - p\cdot h - q\ell\big| \le 256\,R' \tag{828}$$

whenever $|\ell| \le R'$ and $|h - \ell p| \le R'$.

Conclusion. Put $a = q + |p|^2/2$, so that $q = a - |p|^2/2$. Then (828) is exactly the assertion that (p,a) is a fit for w at $\hat z$ of radius R' with constant 256 in the sense of `IsFit`. As $\hat z$ and R' were arbitrary, this proves the lemma. The auxiliary subsequence $l \mapsto j_l$ depends on $\hat z$ and R', which is harmless: `IsFit` is asserted separately for each centre and each radius, and w itself does not depend on the extraction. □

2.186. `NormalizedLimitNotAffine.`

Lemma 2.186 (Step 12: the limit of a normalized blow-up family is not space-time affine)**.** *Let $d \in \mathbb{N}$ and write $\mathcal{H} = \ell^2(\mathbb{R}^d \times \mathbb{R})$ for the space-time of the preamble. Let $r, \lambda : \mathbb{N} \to \mathbb{R}$, let $U_i : \mathcal{H} \to \mathbb{R}$ and $v_i : \mathcal{H} \to \mathbb{R}$ for each $i \in \mathbb{N}$, let $z_i \in \mathcal{H}$, let Φ_i be a surjective linear isometry of $\mathcal{H}$ onto itself, let $Z_i : \mathcal{H} \to \mathcal{H}$, let $c : \mathbb{N} \to \mathbb{R}$, let $L \in \mathbb{R}$ with $L \ge 0$, let $i_1 \le i_2$ be natural numbers, let $\psi : \mathbb{N} \to \mathbb{N}$ and let $v : \mathcal{H} \to \mathbb{R}$. Assume:*

(1) $\lambda_i > 0$ *for every i, and* $\lambda_i \to 0$ *as* $i \to \infty$;
(2) $r_i > 0$ *for every i;*
(3) *for every i there is* no P *realizing a cylinder fit of parameter λ_i for U_i at (z_i, r_i) in the sense of* `IsCylinderFit`;
(4) *the normalization map is $Z_i(w) = z_i + r_i\,\Phi_i\big(\sqrt{\lambda_i}\,x,\ s\big)$ for every i and every $w \in \mathcal{H}$ with coordinates $w = (x,s)$;*
(5) $\frac12 \le c_i \le L + 1$ *for every $i \ge i_2$;*
(6) $32\lambda_i \le 1$ *for every $i \ge i_2$;*
(7) $v_i(0) = 0$ *for every $i \ge i_1$;*
(8) *for every $i \ge i_1$, every $w \in \mathcal{H}$, every $h \in \mathbb{R}^d$ and $l \in \mathbb{R}$, every $p' \in \mathbb{R}^d$ and $q' \in \mathbb{R}$, and every $Q \in \mathcal{H}$ with $Q = \Phi_i\big(c_i\sqrt{\lambda_i}\,p',\ c_i(1 + \lambda_i q')\big)$,*

$$U_i\big(Z_i(w{+}(h,l))\big) - U_i\big(Z_i(w)\big) - Q\cdot\big(Z_i(w{+}(h,l)) - Z_i(w)\big) = r_i c_i \lambda_i\big(v_i(w{+}(h,l)) - v_i(w) - p'\cdot h - q' l\big);$$

(9) ψ *is strictly monotone;*
(10) $v_{\psi(j)} \to v$ *uniformly on every compact subset of $\mathcal{H}$ as $j \to \infty$;*
(11) $v(0) = 0$.

Then v is not space-time affine in the sense of `IsSpaceTimeAffine`.

Proof. Suppose v were space-time affine. By `IsSpaceTimeAffine` there are $p \in \mathbb{R}^d$ and $q, c \in \mathbb{R}$ with $v(x,t) = p\cdot x + qt + c$ for all (x,t). Taking $(x,t) = 0$ and using 11 gives $0 = v(0) = p\cdot 0 + q\cdot 0 + c = c$, so $c = 0$ and

$$v(x,t) = p\cdot x + qt \qquad \big((x,t) \in \mathcal{H}\big). \tag{829}$$

The candidate slopes. Put, for $i \ge i_2$,

$$\widetilde{P}_i = \Phi_i\big(c_i\sqrt{\lambda_i}\,p,\ c_i(1+\lambda_i q)\big), \qquad N_i = \big(\lambda_i|p|^2 + (1+\lambda_i q)^2\big)^{1/2},$$

so that $|\widetilde{P}_i| = c_i N_i$, because Φ_i is an isometry and the coordinate vector $\big(c_i\sqrt{\lambda_i}p,\ c_i(1+\lambda_i q)\big)$ has norm $c_i N_i$ by $(\sqrt{\lambda_i})^2 = \lambda_i$, which uses $\lambda_i > 0$ from 1. Since $\lambda_i \to 0$ by 1 we may fix $i_3 \ge i_2$ with $\lambda_i < \frac{1}{2(|q|+1)}$ for $i \ge i_3$; then $\lambda_i|q| < \frac{1}{2}$, hence $1+\lambda_i q \ge \frac{1}{2}$ and therefore $N_i^2 \ge (1+\lambda_i q)^2 \ge \frac{1}{4}$, i.e. $N_i \ge \frac{1}{2}$. With $c_i \ge \frac{1}{2}$ from 5 this gives $|\widetilde{P}_i| = c_i N_i \ge \frac{1}{4} > 0$, so

$$\widetilde{P}_i \neq 0 \qquad (i \ge i_3). \tag{830}$$

The fixed box. Fix $i \ge i_3$, let Z lie in the cylinder $Z_{\lambda_i}(z_i, r_i; \widetilde{P}_i)$ of `FitCylinder`, and let (x,t) be the normalized coordinates of Z, that is $Z = Z_i(x,t)$. Such (x,t) exists and is unique because Φ_i is a surjective isometry and $r_i\sqrt{\lambda_i} > 0$ by 2 and 1; concretely, (x,t) is recovered from 4 by applying Φ_i^{-1} and dividing by r_i and $\sqrt{\lambda_i}$. Then

$$|x| \le 2|p| + 1, \qquad |t| \le 1 + \sqrt{\lambda_i} \le 2. \tag{831}$$

The first two bounds are `FitCylinderToNormalizedBox`, applied with the isometry Φ_i, the centre z_i, the scales c_i, λ_i and r_i, the coefficients p and q, and the normalized coordinates (x,t) of Z. Its hypotheses hold: $c_i \ge \frac{1}{2} > 0$ by 5; $\lambda_i > 0$ and $r_i > 0$ by 1 and 2; its lower bound $\big(\lambda_i|p|^2 + (1+\lambda_i q)^2\big)^{1/2} \ge \frac{1}{2}$ is exactly $N_i \ge \frac{1}{2}$, established above; and its membership hypothesis asks that $z_i + r_i\Phi_i\big(\sqrt{\lambda_i}\,x,\ t\big)$ lie in $Z_{\lambda_i}\big(z_i,\ r_i;\ \Phi_i(c_i\sqrt{\lambda_i}\,p,\ c_i(1+\lambda_i q))\big)$, which is the assumption on Z once $Z = Z_i(x,t)$ is written out using 4 and $\widetilde{P}_i$ is substituted. The cited node's two bounds rest on two separate uses of N_i: the time bound needs only $N_i \ge |1+\lambda_i q|$, which $N_i^2 = \lambda_i|p|^2 + (1+\lambda_i q)^2 \ge (1+\lambda_i q)^2$ gives at once, whereas the spatial bound is where $N_i \ge \frac{1}{2}$ enters, through $|p|/N_i \le 2|p|$. The last step $1 + \sqrt{\lambda_i} \le 2$ uses $\lambda_i \le 1$, which follows from $32\lambda_i \le 1$ by 6 and $32 \ge 1$. Thus, with $K_0 = 2|p| + 3$, every point of every such cylinder has normalized coordinates in the closed ball of radius K_0 about the origin, independently of i.

The residual is an identity. Apply 8 at the index i (legitimate since $i \ge i_3 \ge i_2 \ge i_1$), at $w = 0$, with $(h,l) = (x,t)$, with $p' = p$ and $q' = q$, and with $Q = \widetilde{P}_i$, which is exactly the required form. Using $Z_i(0 + (x,t)) = Z_i(x,t) = Z$, $Z_i(0) = z_i$ (immediate from 4, since $\sqrt{\lambda_i}\cdot 0 = 0$ and Φ_i is linear) and $v_i(0) = 0$ from 7, this reads

$$U_i(Z) - U_i(z_i) - \widetilde{P}_i \cdot (Z - z_i) = r_i c_i \lambda_i\big(v_i(x,t) - p\cdot x - qt\big) = r_i c_i \lambda_i\big(v_i(x,t) - v(x,t)\big),$$

the last equality by (829). Note that this is an identity, not an estimate. Taking absolute values and using $c_i \le L + 1$ from 5 together with $r_i, \lambda_i > 0$,

$$\big|U_i(Z) - U_i(z_i) - \widetilde{P}_i \cdot (Z - z_i)\big| \le r_i\lambda_i\,(L+1)\,\big|v_i(x,t) - v(x,t)\big|. \tag{832}$$

Passing to the subsequence. The closed ball of radius K_0 about the origin is compact, so by 10 there is j_0 such that for every $j \ge j_0$,

$$\sup\big\{|v_{\psi(j)}(y) - v(y)| : |y| \le K_0\big\} < \frac{1}{L+1},$$

the right side being positive because $L \ge 0$. Since ψ is strictly monotone by 9, $\psi(j) \ge j$ for every j, so we may fix a single index j with $j \ge j_0$ and $\psi(j) \ge i_3$, and put $i = \psi(j)$. For every $Z \in Z_{\lambda_i}(z_i, r_i; \widetilde{P}_i)$ the normalized coordinates (x,t) of Z satisfy $|(x,t)| \le K_0$ by (831), so (832) gives

$$\big|U_i(Z) - U_i(z_i) - \widetilde{P}_i \cdot (Z - z_i)\big| \le r_i\lambda_i\,(L+1)\cdot\frac{1}{L+1} = \lambda_i r_i.$$

Together with $\widetilde{P}_i \neq 0$ from (830), this says precisely that $\widetilde{P}_i$ realizes a cylinder fit of parameter λ_i — not $32\lambda_i$ — for U_i at (z_i, r_i), in the sense of `IsCylinderFit`. That contradicts 3, which denies such a fit at every index. Hence v is not space-time affine. □

2.187. `RescaledEquation`.

Lemma 2.187 (The rescaled equation)**.** *Let F be a finite-dimensional real inner-product space, let Ω be a subset of the space-time $\ell^2(F\times\mathbb{R})$ and let U be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`. Let z_0 be a space-time point, let $r,c,\lambda>0$, let Φ be a linear isometry of $\ell^2(F\times\mathbb{R})$ onto itself, and define T by*

$$T(x,t)=z_0+r\,\Phi\big(\sqrt{\lambda}\,x,\ t\big).$$

Suppose v is a real function on space-time with

$$U\big(T(x,t)\big)=U(z_0)+rc\big(t+\lambda v(x,t)\big)\qquad\text{for all }(x,t). \tag{833}$$

Then v is a viscosity solution of

$$D^2v\big[(\nabla_x v,\ 1+\lambda v_t)\big]=0$$

on $T^{-1}(\Omega)$, in the sense of `IsRescaledSolutionOn`.

Proof. Write $S(x,t)=(\sqrt{\lambda}\,x,\,t)$ for the anisotropic stretch and $L=r\,\Phi\circ S$, so that $T(\zeta)=z_0+L\zeta$ is affine with linear part L. Let $e_t=(0,1)$ denote the unit time direction, so that the time coordinate of a space-time point ζ is $\pi_t(\zeta)=\langle e_t,\zeta\rangle$.

Step 0: T is an affine homeomorphism, $T^{-1}(\Omega)$ is open and v is continuous there. The map S is linear and invertible, with $S^{-1}(x,t)=(\lambda^{-1/2}x,t)$, because $\lambda>0$; Φ is a linear isometric bijection; and $r>0$. Hence L is a linear homeomorphism of space-time and T is an affine homeomorphism with $T^{-1}(Z)=L^{-1}(Z-z_0)$. By `IsInfHarmonicOn` the set Ω is open and U is continuous on Ω, so $T^{-1}(\Omega)$ is open. Solving (833) for v, using $rc\lambda>0$,

$$v(\zeta)=\frac{U(T\zeta)-U(z_0)}{rc\lambda}-\frac{\pi_t(\zeta)}{\lambda},$$

which is continuous on $T^{-1}(\Omega)$ since U is continuous on Ω and T is continuous.

Step 1: adjoint identities for L. The inner product on space-time is $\langle(x,t),(x',t')\rangle=\langle x,x'\rangle+tt'$, so S is self-adjoint: $\langle S(x,t),(x',t')\rangle=\sqrt{\lambda}\langle x,x'\rangle+tt'=\langle(x,t),S(x',t')\rangle$. Also $S^2(x,t)=(\lambda x,t)$. Since Φ is a surjective linear isometry, $\Phi^*\Phi$ is the identity. Therefore

$$L^*L=r^2S^*\Phi^*\Phi S=r^2S^2,\qquad\text{so}\qquad(L^*L)^{-1}(x,t)=\Big(\frac{x}{r^2\lambda},\,\frac{t}{r^2}\Big). \tag{834}$$

Step 2: the transformed test. Let $\zeta_0\in T^{-1}(\Omega)$, let ϕ be C^2 near ζ_0, and suppose that $\zeta\mapsto v(\zeta)-\phi(\zeta)$ has a local maximum on $T^{-1}(\Omega)$ at ζ_0. Put $Z_0=T\zeta_0\in\Omega$ and

$$\Psi(Z)=U(z_0)+rc\Big(\pi_t\big(T^{-1}Z\big)+\lambda\,\phi\big(T^{-1}Z\big)\Big).$$

Since T^{-1} is affine and π_t is linear, Ψ is C^2 near Z_0. By (833) applied at $\zeta=T^{-1}Z$,

$$U(Z)-\Psi(Z)=rc\lambda\Big(v\big(T^{-1}Z\big)-\phi\big(T^{-1}Z\big)\Big)\qquad(Z\in\text{space-time}),$$

and $rc\lambda>0$. As T is a homeomorphism carrying $T^{-1}(\Omega)$ onto Ω and ζ_0 to Z_0, the function $U-\Psi$ therefore has a local maximum on Ω at Z_0. By `IsInfHarmonicOn` the subsolution inequality gives

$$\Delta_\infty\Psi(Z_0)\ge 0. \tag{835}$$

Step 3: computing $\Delta_\infty\Psi(Z_0)$. Write $\nabla\phi(\zeta_0)=(\nabla_x\phi,\phi_t)$ for the space-time gradient of ϕ at ζ_0. Since $\pi_t\circ T^{-1}$ is affine with linear part $\pi_t\circ L^{-1}$, and since the second derivative of an affine map vanishes, the chain rule gives for every ξ

$$D\Psi(Z_0)\xi=rc\Big(\langle e_t,L^{-1}\xi\rangle+\lambda\,\langle\nabla\phi(\zeta_0),L^{-1}\xi\rangle\Big),\qquad D^2\Psi(Z_0)[\xi,\xi]=rc\lambda\,D^2\phi(\zeta_0)\big[L^{-1}\xi,L^{-1}\xi\big].$$

Let $G=\nabla\Psi(Z_0)$, the Riesz representative of $D\Psi(Z_0)$. Substituting $\xi=L\eta$ in the first display,

$$\langle L^*G,\eta\rangle=\langle G,L\eta\rangle=rc\,\big\langle e_t+\lambda\nabla\phi(\zeta_0),\,\eta\big\rangle\qquad(\eta\in\text{space-time}),$$

so $L^*G = rc\big(e_t + \lambda\nabla\phi(\zeta_0)\big)$ and hence, by (834),

$$L^{-1}G = (L^*L)^{-1}L^*G = rc\,(L^*L)^{-1}\big(\lambda\nabla_x\phi,\ 1+\lambda\phi_t\big) = rc\Big(\frac{\lambda\nabla_x\phi}{r^2\lambda},\ \frac{1+\lambda\phi_t}{r^2}\Big) = \frac{c}{r}\big(\nabla_x\phi,\ 1+\lambda\phi_t\big),$$

where we used $e_t + \lambda\nabla\phi(\zeta_0) = (\lambda\nabla_x\phi,\ 1+\lambda\phi_t)$. Therefore, by the definition of Δ_∞ in `InfLap`, bilinearity of $D^2\Psi(Z_0)$, and the definition of F_λ in `RescaledOp`,

$$\Delta_\infty\Psi(Z_0) = D^2\Psi(Z_0)[G,G] = rc\lambda\,D^2\phi(\zeta_0)\big[L^{-1}G, L^{-1}G\big] = \frac{\lambda c^3}{r}\,F_\lambda[\phi](\zeta_0). \tag{836}$$

Step 4: conclusion. The factor $\lambda c^3/r$ is strictly positive, so (835) and (836) give $F_\lambda[\phi](\zeta_0) \geq 0$. If instead $v-\phi$ has a local *minimum* on $T^{-1}(\Omega)$ at ζ_0, the same computation applies verbatim: $U-\Psi$ then has a local minimum on Ω at Z_0, the supersolution inequality of `IsInfHarmonicOn` gives $\Delta_\infty\Psi(Z_0) \leq 0$, and (836) yields $F_\lambda[\phi](\zeta_0) \leq 0$.

Together with Step 0 this verifies every clause of `IsRescaledSolutionOn` for v on $T^{-1}(\Omega)$. □

2.188. `RescaledLimitStability`.

Lemma 2.188 (Stability of the rescaled equation as its parameter degenerates)**.** *Let F be a finite-dimensional real inner-product space, let Ω be an open subset of the space-time $\ell^2(F\times\mathbb{R})$, and let $(\lambda_j)_{j\geq 0}$ be real numbers with $\lambda_j \to 0$. Let $(v_j)_{j\geq 0}$ be real functions on the space-time such that each v_j is a viscosity solution of $D^2v_j[(\nabla_x v_j, 1+\lambda_j\partial_t v_j)] = 0$ on Ω in the sense of `IsRescaledSolutionOn`, and let v_* be a real function on the space-time such that $v_j \to v_*$ uniformly on every compact subset of Ω. Then v_* is a viscosity solution of $F[v_*] = 0$ on Ω in the sense of `IsLimitSolutionOn`.*

Proof. Write $R_\lambda[\phi](\zeta)$ for the rescaled operator $D^2\phi(\zeta)[(\nabla_x\phi(\zeta), 1+\lambda\partial_t\phi(\zeta))]$ of `RescaledOp` and $F[\phi](\zeta)$ for the limiting operator $D^2\phi(\zeta)[(\nabla_x\phi(\zeta),1)]$ of `LimitOp`. Note the two agree at $\lambda = 0$:

$$R_0[\phi](\zeta) = F[\phi](\zeta), \tag{837}$$

because $1+0\cdot\partial_t\phi(\zeta) = 1$ and the two operators are then the second derivative of ϕ at ζ evaluated on the same pair of vectors.

Step 0: continuity of the limit. Let $\zeta\in\Omega$. Since Ω is open there is $\varrho>0$ with $\overline{B}_\varrho(\zeta)\subseteq\Omega$, a compact set, so $v_j\to v_*$ uniformly on it. By `IsRescaledSolutionOn` each v_j is continuous on Ω, hence on $\overline{B}_\varrho(\zeta)$, and a uniform limit of continuous functions on a set is continuous on that set. Thus v_* is continuous at every point of Ω, i.e. continuous on Ω.

Step 1: strictifying an upper test. Let $\zeta_0\in\Omega$ and let ϕ be C^2 near ζ_0 with $\zeta\mapsto v_*(\zeta)-\phi(\zeta)$ having a local maximum on Ω at ζ_0. Set

$$\phi^\sharp(\zeta) = \phi(\zeta) + |\zeta-\zeta_0|^4.$$

The function $\zeta\mapsto|\zeta-\zeta_0|^4 = \big(|\zeta-\zeta_0|^2\big)^2$ is a polynomial in the coordinates, hence smooth, and

$$\nabla\big(|\cdot-\zeta_0|^4\big)(\zeta_0) = 0, \qquad D^2\big(|\cdot-\zeta_0|^4\big)(\zeta_0) = 0, \tag{838}$$

since its gradient is $4|\zeta-\zeta_0|^2(\zeta-\zeta_0)$, which vanishes at ζ_0 together with its derivative. Hence $\phi^\sharp$ is C^2 near ζ_0 with $\nabla\phi^\sharp(\zeta_0) = \nabla\phi(\zeta_0)$ and $D^2\phi^\sharp(\zeta_0) = D^2\phi(\zeta_0)$, so by `LimitOp`

$$F[\phi^\sharp](\zeta_0) = F[\phi](\zeta_0). \tag{839}$$

Choose $\varrho>0$ so small that $\overline{B}_\varrho(\zeta_0)\subseteq\Omega$, that ϕ is C^2 on an open set V with $\overline{B}_\varrho(\zeta_0)\subseteq V\subseteq\Omega$, and that $v_*-\phi$ attains its maximum over $\overline{B}_\varrho(\zeta_0)$ at ζ_0; the last is possible because the maximum is local on Ω and Ω is open. Then for $\zeta\in\overline{B}_\varrho(\zeta_0)$ with $\zeta\neq\zeta_0$,

$$v_*(\zeta)-\phi^\sharp(\zeta) = \big(v_*(\zeta)-\phi(\zeta)\big) - |\zeta-\zeta_0|^4 < v_*(\zeta_0)-\phi(\zeta_0) = v_*(\zeta_0)-\phi^\sharp(\zeta_0),$$

so ζ_0 is the *unique* maximum point of $v_*-\phi^\sharp$ on $\overline{B}_\varrho(\zeta_0)$.

Step 2: nearby maxima of the approximations. The set $\overline{B}_\varrho(\zeta_0)$ is compact and contained in Ω, and each $v_j-\phi^\sharp$ is continuous on it, so it attains a maximum at some $\zeta_j\in\overline{B}_\varrho(\zeta_0)$. We claim $\zeta_j\to\zeta_0$. By compactness it suffices to show that every convergent subsequence has limit ζ_0. Let

$\zeta_{j_k} \to \bar\zeta \in \overline{B}_\varrho(\zeta_0)$. Uniform convergence of v_j to v_* on $\overline{B}_\varrho(\zeta_0)$ and continuity of $v_* - \phi^\sharp$ there give

$$(v_{j_k} - \phi^\sharp)(\zeta_{j_k}) \longrightarrow (v_* - \phi^\sharp)(\bar\zeta).$$

On the other hand, for every $\zeta \in \overline{B}_\varrho(\zeta_0)$, $(v_{j_k} - \phi^\sharp)(\zeta_{j_k}) \geq (v_{j_k} - \phi^\sharp)(\zeta) \to (v_* - \phi^\sharp)(\zeta)$, so $(v_* - \phi^\sharp)(\bar\zeta) \geq (v_* - \phi^\sharp)(\zeta)$ for all such ζ. Thus $\bar\zeta$ is a maximum point of $v_* - \phi^\sharp$ on $\overline{B}_\varrho(\zeta_0)$, and by the uniqueness in Step 1, $\bar\zeta = \zeta_0$. Hence $\zeta_j \to \zeta_0$; in particular $|\zeta_j - \zeta_0| < \varrho$ for all large j, so ζ_j is an interior point of $\overline{B}_\varrho(\zeta_0)$ and therefore $v_j - \phi^\sharp$ has a local maximum on Ω at ζ_j.

Step 3: passing the inequality, in the point and in the parameter. For such j, $\phi^\sharp$ is C^2 near ζ_j and $v_j - \phi^\sharp$ has a local maximum on Ω at ζ_j, so the subsolution clause of `IsRescaledSolutionOn` for v_j gives

$$R_{\lambda_j}[\phi^\sharp](\zeta_j) \geq 0. \tag{840}$$

It remains to pass to the limit in *both* arguments, the point ζ_j and the parameter λ_j. Since $\phi^\sharp$ is C^2 on the open set V, both $\zeta \mapsto \nabla\phi^\sharp(\zeta)$ and $\zeta \mapsto D^2\phi^\sharp(\zeta)$ are continuous on V; write $\nabla_x\phi^\sharp(\zeta)$ and $\partial_t\phi^\sharp(\zeta)$ for the two components of $\nabla\phi^\sharp(\zeta)$, which are therefore continuous on V as well. For large j we have $\zeta_j \in \overline{B}_\varrho(\zeta_0) \subseteq V$, and $\zeta_j \to \zeta_0$, so

$$\nabla_x\phi^\sharp(\zeta_j) \to \nabla_x\phi^\sharp(\zeta_0), \qquad \partial_t\phi^\sharp(\zeta_j) \to \partial_t\phi^\sharp(\zeta_0), \qquad D^2\phi^\sharp(\zeta_j) \to D^2\phi^\sharp(\zeta_0),$$

the last in the space of continuous symmetric bilinear forms. Since $\lambda_j \to 0$ and $\partial_t\phi^\sharp(\zeta_j)$ converges, hence is bounded,

$$1 + \lambda_j\partial_t\phi^\sharp(\zeta_j) \longrightarrow 1,$$

and therefore the vectors $\xi_j = \big(\nabla_x\phi^\sharp(\zeta_j), 1 + \lambda_j\partial_t\phi^\sharp(\zeta_j)\big)$ converge to $\xi_0 = \big(\nabla_x\phi^\sharp(\zeta_0), 1\big)$. Evaluation of a continuous bilinear form on a pair of vectors is jointly continuous in the form and the vectors, so by `RescaledOp` and `LimitOp`

$$R_{\lambda_j}[\phi^\sharp](\zeta_j) = D^2\phi^\sharp(\zeta_j)[\xi_j] \longrightarrow D^2\phi^\sharp(\zeta_0)[\xi_0] = F[\phi^\sharp](\zeta_0).$$

Passing to the limit in (840) gives $F[\phi^\sharp](\zeta_0) \geq 0$, and (839) yields $F[\phi](\zeta_0) \geq 0$, which is the subsolution inequality of `IsLimitSubOn` at ζ_0.

Step 4: lower tests. If instead $v_* - \phi$ has a local minimum on Ω at ζ_0, repeat Steps 1–3 with $\phi^\sharp(\zeta) = \phi(\zeta) - |\zeta - \zeta_0|^4$, with minima in place of maxima, and with the supersolution clause of `IsRescaledSolutionOn`, which gives $R_{\lambda_j}[\phi^\sharp](\zeta_j) \leq 0$ in place of (840). The limit passage of Step 3 is unchanged, identity (838) again gives $F[\phi^\sharp](\zeta_0) = F[\phi](\zeta_0)$, and the conclusion is $F[\phi](\zeta_0) \leq 0$, the supersolution inequality of `IsLimitSuperOn` at ζ_0.

Together with Step 0 and the openness of Ω, this verifies both clauses of `IsLimitSolutionOn` for v_*. □

2.189. `LimitSolutionLocalToGlobal`.

Lemma 2.189 (Being a limit solution is a local property)**.** *Let F be a finite-dimensional real inner-product space, let $\Omega \subseteq \ell^2(F \times \mathbb{R})$ be open, let $(O_k)_{k\in\iota}$ be a family of subsets of $\ell^2(F \times \mathbb{R})$ with $O_k \subseteq \Omega$ for every k and such that every $z \in \Omega$ lies in some O_k, and let $v : \ell^2(F \times \mathbb{R}) \to \mathbb{R}$ be continuous on Ω. If v is a limit solution on O_k for every k, then v is a limit solution on Ω.*

Proof. By the definition of a limit solution, we must produce the two halves: that v is a limit subsolution on Ω and a limit supersolution on Ω. Each half consists of three conjuncts: that the set is open, that v is continuous on it, and a one-sided inequality at every point of the set. The first conjunct is the hypothesis that Ω is open and the second is the hypothesis that v is continuous on Ω; only the third needs an argument, and the two halves are proved by the same argument with the inequality and the extremum reversed.

The subsolution half. Let $z \in \Omega$, let $\varphi : \ell^2(F \times \mathbb{R}) \to \mathbb{R}$ be twice continuously differentiable at z, and suppose that $v - \varphi$ has a local maximum at z relative to Ω. We must show $0 \leq \mathcal{L}\varphi(z)$, where $\mathcal{L}$ is the limit operator.

Choose k with $z \in O_k$, which the covering hypothesis provides. Since $O_k \subseteq \Omega$, the neighbourhood filter of z within O_k is finer than the neighbourhood filter of z within Ω: every set that is a neighbourhood of z relative to Ω is a neighbourhood of z relative to O_k, because a relative

neighbourhood is the intersection of an ambient neighbourhood with the set, and intersecting with the smaller set O_k gives a subset of the corresponding relative neighbourhood in O_k. A local maximum relative to a set is a statement that the inequality $v-\varphi \le (v-\varphi)(z)$ holds on some relative neighbourhood of z; passing to a finer filter preserves such a statement. Hence $v-\varphi$ has a local maximum at z relative to O_k as well.

Now v is a limit subsolution on O_k by hypothesis, and its third conjunct applies at the point $z \in O_k$, the test function φ, which is twice continuously differentiable at z, and this local maximum relative to O_k. It yields $0 \le \mathcal{L}\varphi(z)$, as required. Note that the hypothesis is used only through this third conjunct; the openness and continuity conjuncts of the local statements are not needed, since the global ones are hypotheses here.

The supersolution half. Let $z \in \Omega$, let φ be twice continuously differentiable at z, and suppose $v-\varphi$ has a local minimum at z relative to Ω. Choose k with $z \in O_k$. The same filter comparison, applied to the inequality $v-\varphi \ge (v-\varphi)(z)$, shows that $v-\varphi$ has a local minimum at z relative to O_k. The third conjunct of the statement that v is a limit supersolution on O_k then gives $\mathcal{L}\varphi(z) \le 0$.

Together with the openness of Ω and the continuity of v on Ω, the two halves say exactly that v is a limit solution on Ω. □

2.190. `NonaffineLimitIsLimitSolution`.

Lemma 2.190 (The normalized limit solves the limiting equation on all of space-time)**.** *Let $d \in \mathbb{N}$ and work in $X = \ell^2(\mathbb{R}^d \times \mathbb{R})$, writing a point as $w = (w_x, w_t)$ and $\overline{B}_\rho$, B_ρ for the closed and open balls of radius ρ centred at the origin. Let $r, \lambda, c : \mathbb{N} \to \mathbb{R}$, let $U_i : X \to \mathbb{R}$, let $z_i \in X$, let $\Phi_i : X \to X$ be linear isometric equivalences, let $v_i : X \to \mathbb{R}$, let $i_1, i_2 \in \mathbb{N}$, let $\psi : \mathbb{N} \to \mathbb{N}$ and let $v_\infty : X \to \mathbb{R}$. Write*

$$Z_i(w) = z_i + r_i\,\Phi_i\big(\sqrt{\lambda_i}\,w_x,\ w_t\big)$$

for the normalizing map of (3.6). Assume:

1. *$0 < r_i$ for every i, and $r_i \to 0$;*
2. *$0 < \lambda_i$ for every i, $\lambda_i \to 0$, and $\lambda_i \le 1$ for every $i \ge i_2$;*
3. *$z_i \in B_{3/4}$ for every i;*
4. *each U_i is a member of a normalized inf-family in the sense of `NormalizedInfFamily` for this d;*
5. *$0 < c_i$ for every $i \ge i_1$;*
6. *for every $i \ge i_1$ and every $w \in X$,*

$$U_i\big(Z_i(w)\big) = U_i(z_i) + r_i\,c_i\big(w_t + \lambda_i\,v_i(w)\big); \tag{841}$$

7. *ψ is strictly increasing with $i_1 \le \psi(j)$ for every j;*
8. *v_∞ is continuous, and $v_{\psi(j)} \to v_\infty$ uniformly on every compact subset of X.*

Then v_∞ is a limit solution on all of X.

Proof. Throughout we use the elementary fact that the ℓ^2 norm dominates each coordinate: if $|y| \le \rho$ then $|y_x| \le \rho$ and $|y_t| \le \rho$, because $|y|^2 = |y_x|^2 + |y_t|^2$ and both summands are nonnegative.

Stage 1: the rescaled equation on the normalized domain. Fix $i \ge i_1$. By 4 there are $u_i : \mathbb{R}^d \to \mathbb{R}$, infinity-harmonic on the unit ball of $\mathbb{R}^d$, with $U_i(y) = u_i(y_x) + y_t$ for every $y \in X$. By `LinearLift` applied to u_i on that ball, the function $y \mapsto u_i(y_x) + y_t$ — which is U_i — is infinity-harmonic on the lifted ball

$$\Omega = \{y \in X :\ y_x \in B_1^{\mathbb{R}^d}\},$$

the preimage of the unit ball of $\mathbb{R}^d$ under $y \mapsto y_x$. Now (841) is exactly the increment hypothesis of `RescaledEquation` at the data Ω, U_i, z_i, r_i, c_i, λ_i, Φ_i, v_i, whose positivity requirements are 1, 5 and 2. That node gives

$$v_i \text{ is a rescaled solution with parameter } \lambda_i \text{ on } Z_i^{-1}(\Omega) \qquad (i \ge i_1). \tag{842}$$

We also record that a rescaled solution restricts to any *open* subset: if $\Omega' \subseteq \Omega''$ with Ω' open and g is a rescaled solution with parameter ℓ on Ω'', then it is one on Ω'. Indeed the continuity and openness requirements are immediate, and for the two viscosity inequalities it suffices that a local maximum of $g-\varphi$ relative to Ω' at a point of Ω' is a local maximum relative to Ω'': since Ω' is open, the neighbourhood filter within Ω' at such a point is the full neighbourhood filter, which is finer than the filter within Ω''.

Stage 2: each bounded ball, after a tail shift. Fix $k \in \mathbb{N}$ and put $\rho = k+1 > 0$. We claim v_∞ is a limit solution on B_ρ.

Since $r_i \to 0$ by 1, there is m with $r_i < \dfrac{1}{8\rho}$ for all $i \geq m$. Put $m_2 = \max(\max(m, i_2), i_1)$, so $m \leq m_2$, $i_2 \leq m_2$ and $i_1 \leq m_2$.

Every point of B_ρ is carried into Ω for $i \geq m_2$. Let $i \geq m_2$ and $w \in B_\rho$, so $|w| < \rho$. By 2 we have $\lambda_i \leq 1$, hence $\sqrt{\lambda_i} \leq 1$ and $\sqrt{\lambda_i} \geq 0$. The coordinate bound gives $|w_x| \leq |w|$ and $|w_t| \leq |w|$, so

$$\big|(\sqrt{\lambda_i}\, w_x,\, w_t)\big|^2 = \lambda_i |w_x|^2 + |w_t|^2 \leq |w|^2 + |w|^2 \leq (2|w|)^2,$$

whence $|(\sqrt{\lambda_i} w_x, w_t)| \leq 2|w|$. Since Φ_i is a linear isometry and $r_i > 0$,

$$\big|r_i \Phi_i(\sqrt{\lambda_i} w_x, w_t)\big| = r_i \big|(\sqrt{\lambda_i} w_x, w_t)\big| \leq 2 r_i |w| < 2 r_i \rho < \frac{2\rho}{8\rho} = \frac{1}{4},$$

using $|w| < \rho$ and $r_i < 1/(8\rho)$. By 3, $|z_i| < 3/4$, so $|Z_i(w)| \leq |z_i| + |r_i \Phi_i(\cdots)| < \frac{3}{4} + \frac{1}{4} = 1$, and therefore $|(Z_i(w))_x| \leq |Z_i(w)| < 1$, i.e. $Z_i(w) \in \Omega$. Thus

$$B_\rho \subseteq Z_i^{-1}(\Omega) \qquad (i \geq m_2). \tag{843}$$

The shifted tail. Consider the sequence $j \mapsto \psi(j + m_2)$. Since ψ is strictly increasing, $j + m_2 \leq \psi(j + m_2)$, so $m_2 \leq \psi(j + m_2)$ for every j and $\psi(j + m_2) \to \infty$; consequently $\lambda_{\psi(j+m_2)} \to 0$ by 2. For each j the index $\psi(j + m_2)$ is at least i_1 by 7, so (842) applies, and it is at least m_2, so (843) applies; by the restriction remark of Stage 1, with B_ρ open, $v_{\psi(j+m_2)}$ is a rescaled solution with parameter $\lambda_{\psi(j+m_2)}$ on B_ρ. Finally, for every compact $C \subseteq B_\rho$ the convergence $v_{\psi(j)} \to v_\infty$ uniformly on C of 8 still holds after shifting the index by m_2, because a shifted tail of a convergent net converges to the same limit. All the hypotheses of `RescaledLimitStability` on the open set B_ρ are therefore met, and it gives

$$v_\infty \text{ is a limit solution on } B_{k+1} \qquad (k \in \mathbb{N}). \tag{844}$$

Stage 3: the gluing. The balls B_{k+1}, $k \in \mathbb{N}$, are all contained in X and cover it: given $z \in X$, the Archimedean property gives $k \in \mathbb{N}$ with $|z| < k$, hence $|z| < k+1$ and $z \in B_{k+1}$. The set X is open, v_∞ is continuous hence continuous on X by 8, and (844) is the per-member hypothesis. So `LimitSolutionLocalToGlobal`, applied with $\Omega = X$ and the family $(B_{k+1})_{k\in\mathbb{N}}$, gives that v_∞ is a limit solution on X.

This last step is where the gluing is genuinely needed: `RescaledLimitStability` cannot be invoked at $\Omega = X$ directly, because its hypothesis is universally quantified over the index while the normalized domains $Z_i^{-1}(\Omega)$ only exhaust X as $i \to \infty$. □

2.191. `NonaffineLimit`.

Theorem 2.191 (The normalized nonaffine limit)**.** *Let $d \geq 1$ be an integer, put $n = d+1$ and identify $\mathbb{R}^n$ with the space-time $\ell^2(\mathbb{R}^d \times \mathbb{R})$, whose added unit direction is written $e_b = (0,1)$. Let*

$$(\gamma_i, s_i, r_i, \lambda_i, U_i, z_i)_{i\geq 0}$$

be bad-scale data for d in the sense of `IsBadScaleData`. Let i_1 be an index and suppose that for every $i \geq i_1$ the vector P_i realizes a cylinder fit of parameter $32\lambda_i$ for U_i at (z_i, r_i) in the sense of `IsCylinderFit`. Put $c_i = |P_i|$, let Φ_i be a linear isometry of the space-time onto itself with

$$\Phi_i(e_b) = P_i/c_i \qquad (i \geq i_1),$$

and let v_i be given by the anisotropic normalization

$$U_i\big(z_i + r_i \Phi_i(\sqrt{\lambda_i}\, x, t)\big) = U_i(z_i) + r_i c_i \big(t + \lambda_i v_i(x,t)\big) \tag{845}$$

for every (x,t) and every $i \geq i_1$. Then there are a constant C_0 with $0 \leq C_0 < \infty$, a strictly increasing sequence of indices $j \mapsto \psi(j)$ with $\psi(j) \geq i_1$ for every j, and a continuous function v on the space-time, such that $v_{\psi(j)} \to v$ uniformly on every compact set, $v \in \mathcal{A}_{d,C_0}$ in the sense of `IsFitClass`, and v is not affine on space-time in the sense of `IsSpaceTimeAffine`.

Proof. Write $\Lambda = 32$, $\theta_i = \sqrt{\Lambda\lambda_i}$, $G = B_{3/4}$ for the open ball of radius $3/4$ about the origin of the space-time and $D(w) = \operatorname{dist}(w, \partial G)$. For (x,t) and $i \geq i_1$ put

$$Z_i(x,t) = z_i + r_i \Phi_i(\sqrt{\lambda_i}\, x, t),$$

so that (845) reads $U_i(Z_i(x,t)) = U_i(z_i) + r_i c_i(t + \lambda_i v_i(x,t))$, and write $\mathcal{B}_K = \{(x,t) : |x| \leq K,\ |t| \leq K\}$ for $K \geq 1$.

Three symbols are reused below with different meanings in different steps, and each is redefined where it is used, so that no step relies on another step's meaning: Q denotes an arbitrary vector of the space-time in Steps 5 and 6, while the base slope of the radius chain in Step 5 is written Q_m and no step writes a chain slope as a bare Q; and (p,q) denotes the limit coefficients of Step 11 and the affine coefficients of Step 12. The letter N is no longer among the reused ones: Step 8's normalizing length left this proof together with the shear computation it belonged to, so the only occurrences are N_i in Step 12 and the neighbourhood N' in Step 10, which are distinct symbols. One further clash is deliberate and harmless: the constant C_1 of Step 2 is a comparison constant of this proof, whereas the symbols c_1 and C_1 appearing in the citations of `CylinderFitDoublingChain` in Steps 3 and 5 are that node's OWN two parameters, instantiated there at $(c_1, C_1) = (1/2, L+1)$; they are unrelated.

Step 0: the standing data. By `IsBadScaleData` we have: $\lambda_i > 0$ and $\lambda_i \to 0$; $r_i > 0$; $z_i \in G$; each U_i lies in the normalized family of `NormalizedInfFamily`; no vector realizes a cylinder fit of parameter λ_i for U_i at (z_i, r_i); and, for every $M \geq 2$, there is $i_0(M)$ such that for every $i \geq i_0(M)$, every w with $|w - z_i| \leq M r_i$ and every ρ with $0 < \rho \leq M r_i$,

(846)
$$w \in G, \qquad \rho \leq D(w)/8, \qquad \text{some } P \text{ realizes a cylinder fit of parameter } \Lambda\lambda_i \text{ for } U_i \text{ at } (w,\rho).$$

Call a pair (w,ρ) *admissible* when $w \in G$ and $0 < \rho \leq D(w)/8$. Let $L \geq 0$ be the constant of `NormalizedFamilyLipschitz`, so that every member of the normalized family satisfies $|U(a) - U(b)| \leq L|a-b|$ for $a, b \in \overline{B}_{7/8}$. Since $\lambda_i \to 0$ we may fix $i_2 \geq \max\{i_1, i_0(2)\}$ such that

(847)
$$\Lambda\lambda_i \leq 1, \qquad \theta_i \leq 1/2 \qquad (i \geq i_2);$$

in particular $0 < \theta_i < 1$ for $i \geq i_2$. Taking $M = 2$, $w = z_i$ and $\rho = r_i$ in (846) shows that (z_i, r_i) is admissible for $i \geq i_2$. Finally, putting $(x,t) = (0,0)$ in (845) gives $U_i(z_i) = U_i(z_i) + r_i c_i \lambda_i v_i(0,0)$ with $r_i c_i \lambda_i > 0$, so

(848)
$$v_i(0,0) = 0 \qquad (i \geq i_1).$$

Step 1: two-sided slope bounds. Let $i \geq i_2$, let (w,ρ) be admissible and let P realize a cylinder fit of parameter $\Lambda\lambda_i$ for U_i at (w,ρ). Then

(849)
$$\tfrac{1}{2} \leq |P| \leq L + 1.$$

Indeed $w \in G$ and $\rho \leq D(w)/8 \leq (3/4)/8 = 3/32$, so $|h| \leq \rho$ implies $w + h \in B_{3/4+3/32} \subseteq \overline{B}_{7/8}$ and therefore $|U_i(w+h) - U_i(w)| \leq L|h|$ by `NormalizedFamilyLipschitz`. With $\Lambda\lambda_i \in (0,1]$ by (847), `CylinderFitSlopeBounds` applies and gives

$$|P| \leq L + \sqrt{\Lambda\lambda_i} = L + \theta_i \leq L + 1, \qquad |P \cdot e_b - 1| \leq \theta_i \leq \tfrac{1}{2},$$

so also $|P| \geq |P \cdot e_b| \geq 1 - \theta_i \geq 1/2$. Applying this to P_i at the admissible pair (z_i, r_i) gives

(850)
$$\tfrac{1}{2} \leq c_i \leq L + 1 \qquad (i \geq i_2).$$

Step 2: one comparison constant. Apply `CylinderComparisonSameCenter` and `CylinderComparisonShifte` with $c_* = 1/2$ and $C_* = L + 1$, and let $C_1 \geq 1$ be the larger of the two constants they supply. By (849) and (847), for $i \geq i_2$ and $\theta = \theta_i \in (0,1)$ their hypotheses on the slope norms hold for every cylinder fit of parameter $\Lambda\lambda_i = \theta_i^2$ at an admissible pair. Thus, writing $n = P/|P|$ and $n' = P'/|P'|$:

(a) if P, P' realize fits of parameter $\Lambda\lambda_i$ for U_i at (w, H) and (w, H') with $0 < H \le H' \le 2H$, both pairs admissible, then $|P - P'| + |n - n'| \le C_1\theta_i$ and $\big||P| - |P'|\big| \le C_1\theta_i^2$;
(b) if P, P' realize fits of parameter $\Lambda\lambda_i$ for U_i at (w, H) and (w', H), both pairs admissible, with $|(w' - w)\cdot n| \le H/4$ and $|(w' - w) - ((w' - w)\cdot n)n| \le \theta_i H/4$, then the same two bounds hold.

Step 3: the chain of doublings at z_i. Apply `CylinderFitDoublingChain` with $c_1 = 1/2$ and $C_1 = L + 1$, legitimate because $0 < 1/2 \le L + 1$ by $L \ge 0$, and let $C_2 \ge 0$ be the constant it supplies; it depends only on L. Fix $R \ge 2$, let $M \ge \max\{2, R\}$ and let $i \ge \max\{i_2, i_0(M)\}$.

The hypotheses of that node hold for $u = U_i$, $z = z_i$, $r = r_i$, $\theta = \theta_i$, this R and the base slope $P = P_i$. Indeed $r_i > 0$; $0 < \theta_i < 1$ by (847); $R \ge 2$; $1/2 \le |P_i| = c_i \le L + 1$ by (850); P_i realizes a cylinder fit of parameter $\Lambda\lambda_i = \theta_i^2$ for U_i at (z_i, r_i) by the hypothesis of the present theorem; and for every ρ with $0 < \rho \le Rr_i$ we have $\rho \le Mr_i$ because $R \le M$ and $r_i > 0$, so (846) makes (z_i, ρ) admissible and supplies a slope realizing a cylinder fit of parameter $\Lambda\lambda_i = \theta_i^2$ for U_i at (z_i, ρ), whose norm lies in $[1/2, L + 1]$ by (849).

Write P_R for the slope the conclusion supplies at (z_i, Rr_i), so that P_R realizes a cylinder fit of parameter $\Lambda\lambda_i$ there with $1/2 \le |P_R| \le L + 1$, and put $n_R = P_R/|P_R|$. Using $c_i = |P_i|$, that conclusion is

$$|n_R - P_i/c_i| \le C_2(1 + \log R)\,\theta_i, \qquad \big||P_R| - c_i\big| \le C_2(1 + \log R)\,\theta_i^2. \tag{851}$$

The factor $1 + \log R$ is not a constant, and Steps 4 and 5 use that it grows only logarithmically in R.

Step 4: a box and its enclosing fit. Fix $K \ge 1$ and $S > 0$. We may fix $R \ge 2$ with

$$R > 16K, \qquad K\big(\Lambda^{-1/2} + C_2(1 + \log R)\big) < R/4; \tag{852}$$

here is the derivation, since the second inequality is what Step 4 and Claim 4a both rest on. First, for every $x \ge 1$,

$$\log x \le 2\sqrt{x}, \tag{853}$$

as follows. For $y > 0$ one has $\log y \le y - 1$: the function log is concave on $(0, \infty)$, with $\log 1 = 0$ and derivative 1 at $y = 1$, so its graph lies below its tangent line at 1, which is $y \mapsto y - 1$. Taking $y = \sqrt{x} > 0$ gives $\log\sqrt{x} \le \sqrt{x} - 1$, and $\log\sqrt{x} = \frac{1}{2}\log x$, so $\log x \le 2\sqrt{x} - 2 \le 2\sqrt{x}$. Put

$$B = 4K\big(\Lambda^{-1/2} + C_2\big), \qquad A = 8KC_2,$$

both nonnegative because $K \ge 1$, $C_2 \ge 0$ and $\Lambda^{-1/2} > 0$. For $R \ge 1$, multiplying the second inequality of (852) by 4 and using (853),

$$4K\big(\Lambda^{-1/2} + C_2(1 + \log R)\big) = B + 4KC_2\log R \le B + 8KC_2\sqrt{R} = B + A\sqrt{R},$$

so it suffices to have $R > B + A\sqrt{R}$. That holds as soon as $\sqrt{R} > A + \sqrt{B} + 1$: then $R = \sqrt{R}\cdot\sqrt{R} > (A + \sqrt{B} + 1)\sqrt{R} \ge A\sqrt{R} + \sqrt{B}\cdot\sqrt{R} \ge A\sqrt{R} + B$, the last step because $\sqrt{R} > \sqrt{B}$. Hence any

$$R > \max\big\{2,\ 16K,\ (A + \sqrt{B} + 1)^2\big\}$$

satisfies both parts of (852), and such an R exists. It depends on K alone, since C_2 depends only on L and $\Lambda = 32$ is fixed; that is what Step 7 uses, and then fix $M = 2\max\{2, 2K, R, S\} + 1$, which is past R, past $2K$ and past $2\max\{R, S\}$ – the last because the radius chain of Step 5 overshoots $\max\{R, S\}r_i$ by at most one doubling. Let $i_3 = i_3(K, S) \ge \max\{i_2, i_0(M)\}$; it will be enlarged finitely often below, always by conditions depending only on K and S. Because Φ_i is a linear isometry and $\lambda_i \le 1$,

$$|Z_i(x,t) - z_i| = r_i\big|(\sqrt{\lambda_i}x, t)\big| \le r_i\big(\sqrt{\lambda_i}|x| + |t|\big) \le 2Kr_i \le Mr_i \tag{854}$$

for $(x, t) \in \mathcal{B}_K$. Hence by (854) and (846), for $i \ge i_3$ and $(x, t) \in \mathcal{B}_K$, the point $Z_i(x, t)$ lies in G and every pair $(Z_i(x, t), \rho)$ with $0 < \rho \le Mr_i$ is admissible and carries a fit of parameter $\Lambda\lambda_i$ for U_i.

Claim 4a. For $i \ge i_3$ and $(x, t) \in \mathcal{B}_K$ the point $Z_i(x, t)$ satisfies

$$|(Z_i(x,t) - z_i)\cdot n_R| \le Rr_i/4, \qquad \big|(Z_i(x,t) - z_i) - ((Z_i(x,t) - z_i)\cdot n_R)n_R\big| \le \theta_i Rr_i/4.$$

This is `AnisotropicBoxOffsets` applied with the isometry Φ_i, the base radius r_i, the anisotropy $s = \sqrt{\lambda_i}$, the scale $\theta = \theta_i$, the constant $C = C_2(1 + \log R)$, the box radius K, the outer radius R, the direction $n = n_R$, and (x, t) the point of $\mathcal{B}_K$; its conclusion is about $h = r_i\Phi_i(\sqrt{\lambda_i}x, t)$, which is $Z_i(x, t) - z_i$ by the definition of Z_i. Its hypotheses hold: $r_i > 0$ and $\theta_i \geq 0$; $0 \leq \sqrt{\lambda_i} \leq 1$ because $\lambda_i > 0$ and $\Lambda\lambda_i \leq 1$ with $\Lambda = 32 \geq 1$ by (847); $s\sqrt{\Lambda} = \sqrt{\lambda_i}\sqrt{32} = \sqrt{32\lambda_i} = \theta_i$ since $\Lambda = 32$; $K \geq 1$; $C \geq 0$ because $C_2 \geq 0$ and $\log R \geq 0$ for $R \geq 2$; $|n_R| = 1$ by construction; the direction bound $|\Phi_i(e_b) - n_R| = |P_i/c_i - n_R| \leq C\theta_i$ is (851), using $\Phi_i(e_b) = P_i/c_i$; the two inequalities $16K < R$ and $K(\Lambda^{-1/2} + C) < R/4$ are exactly (852); and $|x| \leq K$, $|t| \leq K$ because $(x, t) \in \mathcal{B}_K$.

Step 5: normalized coefficients of the nearby fits. Let $i \geq i_3$ and $(x, t) \in \mathcal{B}_K$, and put $w = Z_i(x, t)$. We construct a fit at (w, Sr_i) and bound its slope. We pass between the radii Sr_i and Rr_i at the centre w by two applications of `CylinderFitDoublingChain`, both run *out of the common base radius* mr_i with $m = \min\{S, R\}$. Put

$$\rho_S = \max\{2, S/m\}, \qquad \rho_R = \max\{2, R/m\}.$$

Because $0 < m \leq S$ and $0 < m \leq R$, each clamped ratio satisfies $a \leq \rho_a m \leq 2a$ for $a \in \{S, R\}$: if $a/m \geq 2$ then $\rho_a m = a$, and if $a/m < 2$ then $a \leq 2m = \rho_a m \leq 2a$. Both $\rho_S \geq 2$ and $\rho_R \geq 2$, as the cited node requires, and $\rho_a m \leq 2\max\{R, S\} \leq M$, so by (846) every radius in $(0, \rho_a mr_i]$ is admissible at w and carries a fit of parameter $\Lambda\lambda_i$ with slope in $[1/2, L + 1]$ by (849). The node's two remaining hypotheses also hold: $mr_i > 0$, and $0 < \theta_i < 1$ by (847), which is the scale in which it states its conclusion since $\theta_i^2 = \Lambda\lambda_i$. Let Q_m be a slope at (w, mr_i) realizing a cylinder fit of parameter $\Lambda\lambda_i$ there with $1/2 \leq |Q_m| \leq L + 1$, which the previous sentence supplies at the radius $mr_i \in (0, \rho_a mr_i]$; the norm bounds are the cited node's requirement on its base slope, at its parameters $(c_1, C_1) = (1/2, L + 1)$, the same instantiation as in Step 3. The node applied at base mr_i with ratio ρ_S gives a slope at $(w, \rho_S mr_i)$ whose direction and length differ from those of Q_m by at most $C_2(1 + \log\rho_S)\theta_i$ and $C_2(1 + \log\rho_S)\theta_i^2$ respectively, with the same constant C_2 that Step 3 obtained from this node — it depends only on L, and in particular not on the radius, the centre or the ratio. Since $Sr_i \leq \rho_S mr_i \leq 2Sr_i$, Step 2(a) at the pair $(Sr_i, \rho_S mr_i)$ transfers those bounds to a slope P at (w, Sr_i). The same with ρ_R in place of ρ_S transfers them to a slope P' at (w, Rr_i). Both target radii are at least mr_i, so the two applications are the same argument and no case distinction between $S \leq R$ and $S > R$ is needed; the clamp at 2 costs exactly the one extra same-centre comparison per leg accounted for above. The quantities m, ρ_S and ρ_R are built from S and R alone, and R was fixed from K alone in Step 4, so all three depend only on K and S.

Finally the centre offsets of P' relative to the enclosing fit P_R at (z_i, Rr_i) are at most a quarter of Rr_i and of $\theta_i Rr_i$ by Claim 4a, so Step 2(b) applies to the pair (P_R, P') at the common radius $H = Rr_i$.

The chain from P back to P_i therefore has exactly SIX legs, and we sum them rather than leaving the sum to the reader. Each leg is a pair of inequalities, one of the form $|\cdot| \leq \kappa\theta_i$ for the normalized directions and one of the form $|\cdot| \leq \kappa\theta_i^2$ for the lengths, with κ as listed:

(1) P at (w, Sr_i) against the first chain's outer slope at $(w, \rho_S mr_i)$, by Step 2(a): $\kappa = C_1$;
(2) that outer slope against Q_m at (w, mr_i), by `CylinderFitDoublingChain`: $\kappa = C_2(1 + \log\rho_S)$;
(3) P' at (w, Rr_i) against the second chain's outer slope at $(w, \rho_R mr_i)$, by Step 2(a): $\kappa = C_1$;
(4) that outer slope against the same Q_m, by `CylinderFitDoublingChain`: $\kappa = C_2(1 + \log\rho_R)$;
(5) P' against P_R at the common radius Rr_i, by Step 2(b): $\kappa = C_1$;
(6) P_R against P_i, by (851): $\kappa = C_2(1 + \log R)$.

Legs (1) and (2) join P to Q_m and legs (3) and (4) join P' to Q_m, so (1)–(4) join P to P' through the common base; legs (5) and (6) then carry P' on to P_i. Adding the six, and using the triangle inequality for the directions and for the lengths separately, gives a constant $C_{K,S}$ independent of i, of (x, t) and of the choices, namely

$$C_{K,S} = 3C_1 + C_2\big(1 + \log\rho_S\big) + C_2\big(1 + \log\rho_R\big) + C_2\big(1 + \log R\big),$$

independent of i because C_1 and C_2 are and because ρ_S, ρ_R and R depend only on K and S, with

$$\big|P/|P| - P_i/c_i\big| \le C_{K,S}\theta_i, \qquad \big||P| - c_i\big| \le C_{K,S}\theta_i^2. \tag{855}$$

The same bounds hold for the enclosing slope P_R at (z_i, Rr_i) by (851) alone, after enlarging $C_{K,S}$.

Write $e = P_i/c_i = \Phi_i(e_b)$ and, for a vector Q, let (Q_x, Q_t) be its coordinates in the frame Φ_i, that is $\Phi_i^{-1}Q = (Q_x, Q_t)$. The normalized coefficients

$$p_i' = \frac{P_x}{c_i\sqrt{\lambda_i}}, \qquad q_i' = \frac{P_t/c_i - 1}{\lambda_i} \tag{856}$$

of the slope P at (w, Sr_i) are well defined. Indeed Φ_i is a linear isometry of the space-time onto itself, hence a bijection, so $\Phi_i^{-1}P$ exists and is unique and the pair (P_x, P_t) of its two components is determined by P; and the two divisions are legitimate because $c_i \ge 1/2 > 0$ by (850) and $\lambda_i > 0$, so $c_i\sqrt{\lambda_i} > 0$ and $c_i\lambda_i > 0$. Conversely, and this is the form the next step uses,

$$P = \Phi_i\big(c_i\sqrt{\lambda_i}\,p_i',\ c_i(1+\lambda_i q_i')\big),$$

because $c_i\sqrt{\lambda_i}\,p_i' = P_x$ and $c_i(1+\lambda_i q_i') = c_i + (P_t - c_i) = P_t$ by (856), so that the argument of Φ_i is $\Phi_i^{-1}P$. The pair (p_i', q_i') is the UNIQUE one satisfying this identity: if $\Phi_i(c_i\sqrt{\lambda_i}\,a, c_i(1+\lambda_i b)) = P$ then applying Φ_i^{-1} and comparing components gives $c_i\sqrt{\lambda_i}\,a = P_x$ and $c_i(1+\lambda_i b) = P_t$, and dividing by the two positive scalars gives $a = p_i'$ and $b = q_i'$.

Nothing in (856) or in this paragraph is particular to P. For ANY vector Q of the space-time the same two quotients, formed from the frame coordinates (Q_x, Q_t) of Q, define its normalized coefficients; they are well defined for the same two reasons, namely that Φ_i is a bijection and that $c_i\sqrt{\lambda_i} > 0$ and $c_i\lambda_i > 0$; and they are the unique pair (a, b) with $Q = \Phi_i\big(c_i\sqrt{\lambda_i}\,a,\ c_i(1+\lambda_i b)\big)$, by the same comparison of components. Steps 7, 8 and 12 use (856) at the slopes P_R, P and $\widetilde{P_i}$ respectively, and this sentence is what licenses that.

Apply `NormalizedSlopeCoefficients` with the isometry Φ_i, with $c = c_i$, $\lambda = \lambda_i$, with the vector $Q = P$, and with dir $= C_{K,S}\theta_i$ and nrm $= C_{K,S}\theta_i^2$. Its hypotheses hold: $c_i > 0$ and $\lambda_i > 0$; $|P| \ge 1/2 > 0$ by (849); and its two displayed hypotheses $\big|P/|P| - \Phi_i(e_b)\big| \le$ dir and $\big||P| - c_i\big| \le$ nrm are exactly the two halves of (855), because $\Phi_i(e_b) = P_i/c_i = e$. Its conclusion produces coefficients satisfying the identity of the previous display, hence equal to (p_i', q_i'), and bounds them by

$$c_i\sqrt{\lambda_i}\,|p_i'| \le |P|\,C_{K,S}\theta_i, \qquad c_i\lambda_i\,|q_i'| \le C_{K,S}\theta_i^2 + \tfrac{1}{2}|P|\,C_{K,S}^2\theta_i^2.$$

The first bound draws ONLY on the direction half of (855); the second draws on the norm half TOGETHER WITH the quadratic angular error $\frac{1}{2}|P|\,\big|P/|P| - e\big|^2$, which is why the two coefficients are controlled by different estimates. Dividing the two bounds by $c_i\sqrt{\lambda_i}$ and by $c_i\lambda_i$, both positive, and using $c_i \ge 1/2$ and $|P| \le L+1$ from (849), $\theta_i = \sqrt{\Lambda}\sqrt{\lambda_i}$ and $\theta_i^2 = \Lambda\lambda_i$, we obtain

$$|p_i'| + |q_i'| \le C_3, \qquad C_3 = C_3(K,S) = 2\sqrt{\Lambda}(L+1)C_{K,S} + 2\Lambda\Big(C_{K,S} + \frac{(L+1)C_{K,S}^2}{2}\Big), \tag{857}$$

a bound independent of $i \ge i_3$, of $(x,t) \in \mathcal{B}_K$ and of the chosen fits.

Step 6: the exact increment identity. Let $i \ge i_1$, let Q be any vector with coordinates (Q_x, Q_t) in the frame Φ_i and normalized coefficients (p', q') as in (856), so $Q_x = c_i\sqrt{\lambda_i}\,p'$ and $Q_t = c_i(1+\lambda_i q')$. For any (x,t) and (h,ℓ) put $w = Z_i(x,t)$ and $Z = Z_i(x+h, t+\ell)$. Then $Z - w = r_i\Phi_i(\sqrt{\lambda_i}h, \ell)$, so

$$Q\cdot(Z-w) = r_i\big(\sqrt{\lambda_i}\,Q_x\cdot h + \ell Q_t\big) = r_i c_i\big(\lambda_i\,p'\cdot h + \ell + \lambda_i q'\ell\big),$$

while (845) gives $U_i(Z) - U_i(w) = r_i c_i(\ell + \lambda_i(v_i(x+h,t+\ell) - v_i(x,t)))$. Subtracting,

$$U_i(Z) - U_i(w) - Q\cdot(Z-w) = r_i c_i\lambda_i\big(v_i(x+h,t+\ell) - v_i(x,t) - p'\cdot h - q'\ell\big). \tag{858}$$

With $(x,t)=(0,0)$, so that $w=z_i$, and (848), this reads

$$U_i(Z_i(h,\ell))-U_i(z_i)-Q\cdot(Z_i(h,\ell)-z_i)=r_ic_i\lambda_i\big(v_i(h,\ell)-p'\cdot h-q'\ell\big). \tag{859}$$

Step 7: a uniform bound on each box. Let $K\ge 1$ and let R,M,i_3 be as in Step 4 for this K and $S=R$. Let $(p'_{i,R},q'_{i,R})$ be the normalized coefficients of the ENCLOSING slope P_R at (z_i,Rr_i), which (856) defines at any slope, as the paragraph following it records. The sentence after (855) says that the two inequalities of (855) hold for P_R as well, by (851) alone and after enlarging $C_{K,S}$; so `NormalizedSlopeCoefficients`, applied to P_R with those same (dir, nrm) exactly as in Step 5, gives $|p'_{i,R}|+|q'_{i,R}|\le C_3(K,R)$ at $S=R$. By Claim 4a every $Z_i(x,t)$ with $(x,t)\in\mathcal{B}_K$ lies in the cylinder of that fit, so `IsCylinderFit` bounds the left side of (859) by $\Lambda\lambda_i\,Rr_i$. Taking $Q=P_R$ and $(p',q')=(p'_{i,R},q'_{i,R})$ in (859), which is legitimate because those are the normalized coefficients of P_R, and taking $(h,\ell)=(x,t)$ there, that left side equals $r_ic_i\lambda_i\big|v_i(x,t)-p'_{i,R}\cdot x-q'_{i,R}t\big|$ in absolute value. Dividing by $r_ic_i\lambda_i$, which is positive because $r_i>0$, $c_i\ge 1/2$ by (850) and $\lambda_i>0$,

$$\big|v_i(x,t)-p'_{i,R}\cdot x-q'_{i,R}t\big|\le\frac{\Lambda\lambda_i\,Rr_i}{r_ic_i\lambda_i}=\frac{\Lambda R}{c_i}\le 2\Lambda R\qquad\big((x,t)\in\mathcal{B}_K,\ i\ge i_3\big),$$

the final step because $c_i\ge 1/2$ gives $1/c_i\le 2$, using (850). Consequently

$$\sup_{\mathcal{B}_K}|v_i|\le A_K:=2\Lambda R+C_3(K,R)K\qquad(i\ge i_3), \tag{860}$$

a bound depending only on K, since R was chosen from K alone.

Step 8: the increment estimate. Fix $K\ge 1$ and $S>0$ and let R,M,i_3,C_3 be as in Steps 4 and 5. Enlarge i_3 so that

$$\lambda_i\big(C_3^2+C_3+1\big)\le\tfrac{1}{100}\qquad(i\ge i_3), \tag{861}$$

which is possible because $\lambda_i\to 0$ by `IsBadScaleData` while $C_3=C_3(K,S)$ does not depend on i. Let $i\ge i_3$ and $(x,t)\in\mathcal{B}_K$, put $w=Z_i(x,t)$, let P be the fit at (w,Sr_i) of Step 5 with normalized coefficients (p'_i,q'_i), and set $A=1+\lambda_iq'_i$. By (857) we have $|q'_i|\le C_3$, so $|\lambda_iq'_i|\le\lambda_iC_3\le 1/100$ by (861) and hence

$$\tfrac12\le A\le\tfrac32. \tag{862}$$

Claim 8a. If $|\ell|\le S/2$ and $|h-(p'_i/A)\ell|\le S$, then $Z=Z_i(x+h,t+\ell)$ lies in the cylinder of the fit P at (w,Sr_i), in the sense of `FitCylinder`.

This is `ShearedBoxInFitCylinder`, applied with the isometry Φ_i, the centre $z=w$, the scales $c=c_i$, $\lambda=\lambda_i$, this S and $r=r_i$, the coefficients $p'=p'_i$ and $q'=q'_i$, the coefficient bound $C=C_3$, and the increment (h,ℓ). Its hypotheses hold: $c_i\ge 1/2>0$ by (850); $\lambda_i>0$; $S>0$ and $r_i>0$; $|p'_i|+|q'_i|\le C_3$ by (857); $1/2\le 1+\lambda_iq'_i=A$ by (862); $\lambda_i(C_3^2+C_3+1)\le 1/100$ by (861); and the two domain conditions $|\ell|\le S/2$ and $|h-(p'_i/A)\ell|\le S$ are the hypotheses of the claim. Its conclusion places the point $w+r_i\Phi_i\big(\sqrt{\lambda_i}h,\ell\big)$ in the cylinder $Z_{32\lambda_i}\big(w,\ Sr_i;\ \Phi_i(c_i\sqrt{\lambda_i}p'_i,\ c_iA)\big)$. That point is Z, because

$$Z_i(x+h,t+\ell)=z_i+r_i\Phi_i\big(\sqrt{\lambda_i}(x+h),t+\ell\big)=Z_i(x,t)+r_i\Phi_i\big(\sqrt{\lambda_i}h,\ell\big)$$

by the linearity of Φ_i; that slope is P, by the identity displayed after (856); and that parameter is $\Lambda\lambda_i$, because $\Lambda=32$. This proves Claim 8a.

Combining Claim 8a with `IsCylinderFit` and (858) we obtain, for $i\ge i_3$, $(x,t)\in\mathcal{B}_K$ and all (h,ℓ) with $|\ell|\le S/2$ and $|h-(p'_i/A)\ell|\le S$,

$$\big|v_i(x+h,t+\ell)-v_i(x,t)-p'_i\cdot h-q'_i\ell\big|\le\frac{\Lambda\lambda_iSr_i}{r_ic_i\lambda_i}=\frac{\Lambda S}{c_i}\le 2\Lambda S. \tag{863}$$

The coefficient 2Λ of S does not depend on K or S.

Step 9: equicontinuity, and the convergent subsequence. Fix $K\ge 1$ and $\varepsilon>0$. Choose $S>0$ with $2\Lambda S<\varepsilon/3$, let $R,M,i_3=i_3(K,S)$ and $C_3=C_3(K,S)$ be as in Step 8, and put

$$\eta=\min\Big\{\frac{S}{2},\ \frac{S}{2(1+2C_3)},\ \frac{\varepsilon}{6(1+C_3)}\Big\}>0.$$

If $(x,t) \in \mathcal{B}_K$, $|h| \le \eta$ and $|\ell| \le \eta$, then $|\ell| \le S/2$ and, since $A \ge 1/2$ and $|p_i'| \le C_3$, $|h-(p_i'/A)\ell| \le |h|+2C_3|\ell| \le \eta(1+2C_3) \le S$; moreover $|p_i'\cdot h + q_i'\ell| \le C_3(|h|+|\ell|) \le 2C_3\eta \le \varepsilon/3$. Hence (863) gives

$$\big|v_i(x+h,t+\ell) - v_i(x,t)\big| \le \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon \qquad (i \ge i_3(K,S)). \tag{864}$$

The extraction. Let $\rho > 0$ and let $K \ge 1$ be an integer with $K \ge \rho$. If $|(x,t)| \le \rho$ then $|x| \le |(x,t)| \le \rho \le K$ and $|t| \le |(x,t)| \le \rho \le K$, so

$$\{a : |a| \le \rho\} \subseteq \mathcal{B}_K \qquad (K \text{ an integer},\ K \ge \max\{\rho, 1\}). \tag{865}$$

We verify the three hypotheses of `AsymptEquicontOriginNormalizedLimit` for the family $(v_j)_{j\ge 0}$ on the space-time, which is a finite-dimensional real inner-product space, with starting index i_1.

Hypothesis (i). Let $\rho > 0$, take an integer $K \ge \max\{\rho, 1\}$ and put $A = A_K$ as in (860) and $j_0 = i_4(K) := i_3(K, R(K))$. If $j \ge j_0$ and $|w| \le \rho$ then $w \in \mathcal{B}_K$ by (865), so $|v_j(w)| \le A_K$ by (860).

Hypothesis (ii). Let $\rho > 0$ and $\varepsilon > 0$, take an integer $K \ge \max\{\rho, 1\}$, and let S, η and $i_3(K,S)$ be the quantities produced in the FIRST paragraph of this step for this K and this ε (the paragraph establishing (864), not the preceding one), so that (864) holds with them. Put $j_0 = i_3(K,S)$ and let $j \ge j_0$. Let $w = (x,t)$ and $w' = (x',t')$ satisfy $|w| \le \rho$, $|w'| \le \rho$ and $|w - w'| \le \eta$, and put $h = x' - x$, $\ell = t' - t$. Then $|h| \le |w - w'| \le \eta$ and $|\ell| \le |w - w'| \le \eta$, and $(x,t) \in \mathcal{B}_K$ by (865), so (864) applies and gives

$$|v_j(w) - v_j(w')| = \big|v_j(x,t) - v_j(x+h,t+\ell)\big| \le \varepsilon.$$

The two index thresholds are independent of each other: hypothesis (i) of the cited node carries its own, here $i_4(K)$, and hypothesis (ii) carries a separate one inside its ε quantifier, here $i_3(K,S)$ with S chosen from ε; the node nowhere requires a common threshold, so the (K,S)-dependence of i_3 costs nothing.

Hypothesis (iii). $v_i(0,0) = 0$ for every $i \ge i_1$, which is (848).

All three hypotheses hold, so `AsymptEquicontOriginNormalizedLimit` provides a strictly increasing sequence of indices $j \mapsto \psi(j)$ and a real function v on the space-time such that $\psi(j) \ge i_1$ for every j, the function v is CONTINUOUS, $v_{\psi(j)} \to v$ uniformly on every compact subset of the space-time, and

$$v(0,0) = 0. \tag{866}$$

Those are exactly the five properties of the pair (ψ, v) that Steps 10 to 12 use. In particular the continuity of v and the normalization (866) are part of the cited node's conclusion and need no separate argument here; local uniform convergence alone would not give either.

Step 10: the limiting equation. By `NormalizedInfFamily` each U_i has the form $U_i(y,b) = u_i(y) + b$ with u_i infinity-harmonic on the unit ball B_1^d of $\mathbb{R}^d$ in the sense of `IsInfHarmonicOn`; by `LinearLift`, U_i is infinity-harmonic on $\Omega = \{(y,b) : y \in B_1^d\}$. Hence `RescaledEquation`, applied for $i \ge i_2$ with $z_0 = z_i$, $r = r_i$, $c = c_i$ — which is positive because $c_i \ge 1/2$ by (850), and this is the only place the restriction $i \ge i_2$ is needed here — $\lambda = \lambda_i > 0$, the isometry Φ_i and the function v_i — whose hypothesis is precisely (845) — shows that v_i is a viscosity solution of $D^2\phi[(\nabla_x\phi, 1+\lambda_i\phi_t)] = 0$ on the open set $\Omega_i = Z_i^{-1}(\Omega)$, in the sense of `IsRescaledSolutionOn`. By Step 4, $Z_i(\mathcal{B}_K) \subseteq G$ for $i \ge i_4(K)$, and the spatial part of a point of $G = B_{3/4}$ lies in $B_{3/4}^d \subseteq B_1^d$; hence

$$\mathcal{B}_K \subseteq \Omega_i \qquad (i \ge i_4(K)). \tag{867}$$

We show that v is a viscosity solution of $F[v] = D^2v[(\nabla_x v, 1)] = 0$ on the whole space-time, in the sense of `IsLimitSolutionOn`, where $F[\phi](z) = D^2\phi(z)[(\nabla_x\phi(z), 1)]$ is the limiting operator of `LimitOp`; v is continuous by Step 9.

Fix a point $\hat{z}$ of the space-time, let $K_1 \ge 1$ be an integer with $K_1 > |\hat{z}|$, and let $W = W(\hat{z})$ be the interior of $\mathcal{B}_{K_1}$, that is $W = \{(x,t) : |x| < K_1,\ |t| < K_1\}$. This set is open, being the intersection of the preimages of the open sets $\{|x| < K_1\}$ and $\{|t| < K_1\}$ under the two

coordinate projections of the space-time, which are continuous. It contains $\hat{z}$, because the two coordinates of $\hat{z}$ are each at most $|\hat{z}| < K_1$ in norm, and $W \subseteq \mathcal{B}_{K_1}$. Since ψ is strictly increasing we have $\psi(j) \geq j$, so there is j_1 with $\psi(j) \geq i_4(K_1)$ for every $j \geq j_1$; then by (867)

$$W \subseteq \mathcal{B}_{K_1} \subseteq \Omega_{\psi(j)} \qquad (j \geq j_1). \tag{868}$$

Restriction to W. For $j \geq j_1$ the function $v_{\psi(j)}$ is a viscosity solution of the rescaled equation with parameter $\lambda_{\psi(j)}$ on W, in the sense of `IsRescaledSolutionOn`. Indeed W is open; $v_{\psi(j)}$ is continuous on $\Omega_{\psi(j)}$ by `IsRescaledSolutionOn` and hence on W by (868); and if $\zeta \in W$, if ϕ is twice continuously differentiable at ζ and if $v_{\psi(j)} - \phi$ has a local maximum on W at ζ, say $v_{\psi(j)}(a) - \phi(a) \leq v_{\psi(j)}(\zeta) - \phi(\zeta)$ for all $a \in N' \cap W$ with N' a neighbourhood of ζ, then $N' \cap W$ is itself a neighbourhood of ζ because W is open, and $N' \cap W \subseteq \Omega_{\psi(j)}$ by (868); so $v_{\psi(j)} - \phi$ has a local maximum on $\Omega_{\psi(j)}$ at ζ, and the subsolution clause of `IsRescaledSolutionOn` on $\Omega_{\psi(j)}$ gives $D^2\phi(\zeta)\big[\big(\nabla_x\phi(\zeta), 1 + \lambda_{\psi(j)}\partial_t\phi(\zeta)\big)\big] \geq 0$, which is exactly the required inequality on W. Local minima and the supersolution clause transfer in exactly the same way.

The limit on W. Put $g_j = v_{\psi(j_1+j)}$ and $\mu_j = \lambda_{\psi(j_1+j)}$ for $j \geq 0$. By the previous paragraph each g_j is a viscosity solution of the rescaled equation with parameter μ_j on W in the sense of `IsRescaledSolutionOn`. By `IsBadScaleData`, $\lambda_i \to 0$, and $j \mapsto \psi(j_1 + j)$ is strictly increasing, so $\mu_j \to 0$. By Step 9, $v_{\psi(j)} \to v$ uniformly on every compact set, and $(g_j)_j$ is the tail $j \mapsto v_{\psi(j_1+j)}$ of that sequence, so $g_j \to v$ uniformly on every compact set, in particular on every compact subset of W. Hence `RescaledLimitStability`, applied with $\Omega = W$, the parameters μ_j, the solutions g_j and the limit v, shows that v is a viscosity solution of $F[v] = 0$ on W in the sense of `IsLimitSolutionOn`.

Gluing. The previous two paragraphs give, for *every* point $\hat{z}$ of the space-time, an open set $W(\hat{z})$ containing $\hat{z}$ on which v solves $F[v] = 0$ in the sense of `IsLimitSolutionOn`. Apply `LimitSolutionLocalToGlobal` with Ω the whole space-time, with the family $\big(W(\hat{z})\big)_{\hat{z}}$ indexed by the points of the space-time, and with this v. Its hypotheses hold: the space-time is open; each $W(\hat{z})$ is a subset of it; every point $\hat{z}$ of it lies in the member $W(\hat{z})$ of the family, as noted when $W(\hat{z})$ was constructed; v is continuous on it by Step 9; and v solves $F[v] = 0$ on each member by the previous two paragraphs. Its conclusion is that v solves the limiting equation on the whole space-time.

This gluing step cannot be avoided by invoking `RescaledLimitStability` on the whole space-time directly: that node's hypothesis asks for a family (g_j) each member of which solves the rescaled equation on the SAME set Ω, whereas the sets Ω_i on which v_i solves it only exhaust the space-time as i grows, by (867).

Step 11: the fits. Fix a centre $\hat{z} = (\hat{x}, \hat{t})$ and a radius $R' > 0$. Put $K = \max\{|\hat{x}|, |\hat{t}|, 1\}$ and $S = 4R'$, and let R, M, i_3, C_3 be as in Step 8 for this pair (K, S). For each j with $\psi(j) \geq i_3$, Step 5 provides a fit at $(Z_{\psi(j)}(\hat{x}, \hat{t}), Sr_{\psi(j)})$ whose normalized coefficients (p'_j, q'_j) satisfy $|p'_j| + |q'_j| \leq C_3$ by (857). By Bolzano–Weierstrass there is a subsequence $l \mapsto j_l$ along which $p'_{j_l} \to p$ and $q'_{j_l} \to q$ for some $p \in \mathbb{R}^d$, $q \in \mathbb{R}$. Since $\lambda_{\psi(j)} \to 0$ and $|q'_j| \leq C_3$, the quantities $A_j = 1 + \lambda_{\psi(j)} q'_j$ tend to 1, so $p'_{j_l}/A_{j_l} \to p$.

Let (h, ℓ) satisfy $|\ell| \leq R'$ and $|h - \ell p| \leq R'$. Then $|\ell| \leq R' \leq S/2$ and

$$\Big|h - \frac{p'_{j_l}}{A_{j_l}}\ell\Big| \leq |h - \ell p| + |\ell|\,\Big|p - \frac{p'_{j_l}}{A_{j_l}}\Big| \leq R' + R'\Big|p - \frac{p'_{j_l}}{A_{j_l}}\Big| \leq 2R' \leq S$$

for all large l, uniformly in such (h, ℓ). So (863) applies at the centre $(\hat{x}, \hat{t}) \in \mathcal{B}_K$ and gives

$$\big|v_{\psi(j_l)}(\hat{x} + h, \hat{t} + \ell) - v_{\psi(j_l)}(\hat{x}, \hat{t}) - p'_{j_l} \cdot h - q'_{j_l}\ell\big| \leq 2\Lambda S = 8\Lambda R'.$$

The set $\{(\hat{x} + h, \hat{t} + \ell) : |\ell| \leq R',\ |h - \ell p| \leq R'\}$ is closed and bounded, hence compact, so $v_{\psi(j_l)} \to v$ uniformly on it. Letting $l \to \infty$,

$$\big|v(\hat{x} + h, \hat{t} + \ell) - v(\hat{x}, \hat{t}) - p \cdot h - q\ell\big| \leq C_0 R', \qquad C_0 := 8\Lambda = 256,$$

whenever $|\ell| \leq R'$ and $|h - \ell p| \leq R'$. Putting $a = q + |p|^2/2$, so that $q = a - |p|^2/2$, this is exactly the statement that (p, a) is a fit for v at $\hat{z}$ of radius R' with constant C_0, in the sense of `IsFit`. Since $\hat{z}$ and R' were arbitrary and v solves the limiting equation by Step 10, $v \in \mathcal{A}_{d,C_0}$ in the

sense of `IsFitClass`, with $0 \le C_0 = 256 < \infty$. The auxiliary subsequence $l \mapsto j_l$ depends on $\hat{z}$ and R'; the function v does not, and `IsFit` is asserted separately for each centre and radius.

Step 12: v is not affine. Suppose it were. By `IsSpaceTimeAffine` there are $p \in \mathbb{R}^d$ and $q, c \in \mathbb{R}$ with $v(x,t) = p \cdot x + qt + c$ for all (x,t), and (866) forces $c = 0$. Put, for $i \ge i_2$,

$$\widetilde{P}_i = c_i \Phi_i\big(\sqrt{\lambda_i}\, p,\ 1 + \lambda_i q\big), \qquad N_i = \big(\lambda_i |p|^2 + (1+\lambda_i q)^2\big)^{1/2},$$

so that $|\widetilde{P}_i| = c_i N_i$ and the coordinates of $\widetilde{P}_i$ in the frame Φ_i are $(c_i\sqrt{\lambda_i}p,\ c_i(1+\lambda_i q))$; by (856) its normalized coefficients are exactly $p_i' = p$ and $q_i' = q$. Since $\lambda_i \to 0$ we may enlarge i_2 so that $\lambda_i(1 + |q| + |p|^2) \le 1/2$ for $i \ge i_2$; then $1 + \lambda_i q \ge 1/2$, $1/2 \le N_i \le 2$ and $\widetilde{P}_i \neq 0$.

Let Z lie in the cylinder $Z_{\lambda_i}(z_i, r_i; \widetilde{P}_i)$ of `FitCylinder` and let (x,t) be its normalized coordinates, that is $Z = Z_i(x,t)$, which determines (x,t) uniquely because Φ_i is invertible and $r_i\sqrt{\lambda_i} > 0$. Then

$$|x| \le 2|p| + 1, \qquad |t| \le 1 + \sqrt{\lambda_i} \le 2.$$

The first two bounds are `FitCylinderToNormalizedBox`, applied with the isometry Φ_i, the centre $z = z_i$, the scales $c = c_i$, $\lambda = \lambda_i$ and $r = r_i$, the coefficients p and q, and the normalized coordinates (x,t) of Z. Its hypotheses hold: $c_i \ge 1/2 > 0$ by (850); $\lambda_i > 0$ and $r_i > 0$ by `IsBadScaleData`; its lower bound $\big(\lambda_i|p|^2 + (1+\lambda_i q)^2\big)^{1/2} \ge 1/2$ is exactly $N_i \ge 1/2$, established in the previous paragraph; and its membership hypothesis asks that $z_i + r_i\Phi_i\big(\sqrt{\lambda_i}\, x,\ t\big)$ lie in $Z_{\lambda_i}\big(z_i,\ r_i;\ \Phi_i(c_i\sqrt{\lambda_i}\, p,\ c_i(1+\lambda_i q))\big)$, which is the assumption on Z once $Z = Z_i(x,t)$ and $\widetilde{P}_i = \Phi_i\big(c_i\sqrt{\lambda_i}\, p,\ c_i(1+\lambda_i q)\big)$ are written out. The cited node's two bounds rest on two SEPARATE uses of N_i: the time bound needs only $N_i \ge |1+\lambda_i q|$, which $N_i^2 = \lambda_i|p|^2 + (1+\lambda_i q)^2 \ge (1+\lambda_i q)^2$ gives at once, whereas the spatial bound is where $N_i \ge 1/2$ enters, through $|p|/N_i \le 2|p|$; the over-estimate $2|p|$ in place of the sharper $|p|/N_i$ is deliberate and costs nothing here. The last step $1 + \sqrt{\lambda_i} \le 2$ uses $\lambda_i \le 1$, which follows from $\Lambda\lambda_i \le 1$ by (847) and $\Lambda = 32 \ge 1$. Thus every point of every such cylinder has normalized coordinates in the fixed box $\mathcal{B}_{K_0}$ with K_0 any integer $\ge 2|p| + 2$, independently of i.

By (859) applied with $Q = \widetilde{P}_i$, whose normalized coefficients are (p,q),

$$\big|U_i(Z) - U_i(z_i) - \widetilde{P}_i \cdot (Z - z_i)\big| = r_i c_i \lambda_i \big|v_i(x,t) - p\cdot x - qt\big| = r_i c_i \lambda_i \big|v_i(x,t) - v(x,t)\big|.$$

Hence, using (850) and the previous paragraph, for every index j with $\psi(j) \ge i_2$,

$$\sup\Big\{\big|U_{\psi(j)}(Z) - U_{\psi(j)}(z_{\psi(j)}) - \widetilde{P}_{\psi(j)} \cdot (Z - z_{\psi(j)})\big| : Z \in Z_{\lambda_{\psi(j)}}(z_{\psi(j)}, r_{\psi(j)}; \widetilde{P}_{\psi(j)})\Big\}$$

is at most $r_{\psi(j)}\lambda_{\psi(j)}(L+1)\sup_{\mathcal{B}_{K_0}} |v_{\psi(j)} - v|$, and the last supremum tends to 0 by Step 9. For all large j it is therefore at most $\lambda_{\psi(j)} r_{\psi(j)}$, which together with $\widetilde{P}_{\psi(j)} \neq 0$ says precisely that $\widetilde{P}_{\psi(j)}$ realizes a cylinder fit of parameter $\lambda_{\psi(j)}$ for $U_{\psi(j)}$ at $(z_{\psi(j)}, r_{\psi(j)})$ in the sense of `IsCylinderFit`. This contradicts `IsBadScaleData`, by which no vector realizes such a fit at any index. Hence v is not affine, and $C_0 = 256$, ψ and v have all the asserted properties. □

2.192. `QuadEnvStability`.

Lemma 2.192 (Stability of the quadratic envelopes)**.** *Let F be a finite-dimensional real inner-product space, let $I \subseteq \mathbb{R}$ be open and let $\Omega = \{(x,t) : t \in I\}$. Let f and $f_1, f_2, \dots$ be real functions on space-time, continuous on Ω, with $f_j \to f$ uniformly on every compact subset of Ω, with f of subquadratic spatial growth on compact time intervals of I in the sense of `SubquadraticGrowth` and with the sequence having the common growth control of `UnifSubquadraticGrowth` on I:*

$$\lim_{M\to\infty} \sup_j \sup_{\substack{|y|\ge M \\ t\in J}} \frac{|f_j(y,t)|}{|y|^2} = 0 \qquad (J \Subset I).$$

Let $\ell > 0$. Then $\mathcal{P}_\ell f_j \to \mathcal{P}_\ell f$ uniformly on every compact set of centers $z = (x,t)$ with $t \in I$ and $t + \ell \in I$, and $\mathcal{M}_\ell f_j \to \mathcal{M}_\ell f$ uniformly on every compact set of centers with $t \in I$ and $t - \ell \in I$, the envelopes being those of `QuadEnvP` and `QuadEnvM`.

Proof. Write $\Omega = \{(y,\sigma) : \sigma \in I\}$, an open set on which f and every f_j are continuous.

Step 1: uniform localization. Let $\mathcal{K}$ be a compact set of centers $z = (x,t)$ with $t \in I$ and $t+\ell \in I$ for all $z \in \mathcal{K}$ (the case of the backward envelope is identical with $t-\ell$ in place of $t+\ell$, and we indicate the changes at the end). Put

$$R_0 = \max\{|x| : (x,t) \in \mathcal{K}\}, \qquad J = \{t : (x,t) \in \mathcal{K}\} \cup \{t+\ell : (x,t) \in \mathcal{K}\},$$

so $R_0 < \infty$ and J is a compact subset of I, being the union of two continuous images of $\mathcal{K}$.

Apply `UnifSubquadraticGrowth` with this J and $\varepsilon = 1/(8\ell)$: there is $M_1 > 0$ with

$$|f_j(y,\sigma)| \le \frac{|y|^2}{8\ell} \qquad (j \in \mathbb{N},\ |y| \ge M_1,\ \sigma \in J), \tag{869}$$

and apply `SubquadraticGrowth` with the same J and ε to get $M_2 > 0$ with the same bound for f. Put $M_3 = \max\{M_1, M_2, 4R_0, 1\}$.

For $|y| > M_3$, any x with $|x| \le R_0$ and any $\sigma \in J$ we have $|y-x| \ge |y| - R_0 \ge \frac{3}{4}|y|$, so, exactly as in the corresponding computation for a single function,

$$g(y,\sigma) - \frac{|y-x|^2}{2\ell} \le \frac{|y|^2}{8\ell} - \frac{9|y|^2}{32\ell} = -\frac{5|y|^2}{32\ell} \tag{870}$$

for $g = f$ and for every $g = f_j$.

The set $\mathcal{C}_3 = \{(y,\sigma) : |y| \le M_3,\ \sigma \in J\}$ is a compact subset of Ω. Since f is continuous on Ω, $N_f := \max_{\mathcal{C}_3} |f|$ is finite; and since $f_j \to f$ uniformly on $\mathcal{C}_3$, there is j_0 with $\sup_{\mathcal{C}_3} |f_j - f| \le 1$ for $j \ge j_0$, while for each of the finitely many $j < j_0$ the continuous f_j is bounded on $\mathcal{C}_3$. Hence

$$N := \sup_{j \in \mathbb{N}} \max_{\mathcal{C}_3} |f_j| \ \vee\ N_f < \infty. \tag{871}$$

Put $M = \max\{M_3,\ \sqrt{\frac{32}{5}N\ell} + 1\}$. For $|y| > M$, $|x| \le R_0$ and $\sigma \in J$ we have $\frac{5|y|^2}{32\ell} > N$, so (870) and (871) give, for $g = f$ and every $g = f_j$,

$$g(y,\sigma) - \frac{|y-x|^2}{2\ell} < -N \le g(x,\sigma),$$

the last inequality because $|x| \le R_0 \le M_3$. In particular no competitor with $|y| > M$ exceeds the competitor $y = x$.

Step 2: both envelopes are maxima over one fixed compact ball. Let $z = (x,t) \in \mathcal{K}$. By Step 1 applied with $\sigma = t+\ell \in J$, the set of competitor values in `QuadEnvP` for g at z and length ℓ is nonempty and bounded above, and its supremum equals the maximum over $\{|y| \le M\}$, which exists because $y \mapsto g(y,t+\ell) - |y-x|^2/(2\ell)$ is continuous on that compact ball. Hence

$$\mathcal{P}_\ell g(z) = \max_{|y| \le M} \Big(g(y,t+\ell) - \frac{|y-x|^2}{2\ell} \Big) \qquad (g = f \text{ or } g = f_j,\ z \in \mathcal{K}). \tag{872}$$

Step 3: conclusion. For bounded real functions G, H on a nonempty set Q one has $|\sup_Q G - \sup_Q H| \le \sup_Q |G - H|$. Applying this to (872) with $Q = \{|y| \le M\}$, the quadratic terms cancelling,

$$\big|\mathcal{P}_\ell f_j(z) - \mathcal{P}_\ell f(z)\big| \le \max_{|y| \le M} \big| f_j(y,t+\ell) - f(y,t+\ell) \big| \le \sup_{\mathcal{C}} |f_j - f|,$$

where $\mathcal{C} = \{(y,\sigma) : |y| \le M,\ \sigma \in J\}$ is a compact subset of Ω, and the bound is independent of $z \in \mathcal{K}$. By hypothesis $f_j \to f$ uniformly on $\mathcal{C}$, so the right side tends to 0. Hence $\mathcal{P}_\ell f_j \to \mathcal{P}_\ell f$ uniformly on $\mathcal{K}$.

For the backward envelope, repeat Steps 1–3 with $t-\ell$ in place of $t+\ell$ in the definition of J, with $-f$ and $-f_j$ in place of f and f_j in (870) — these satisfy the same hypotheses, the growth conditions of `SubquadraticGrowth` and `UnifSubquadraticGrowth` being invariant under a change of sign — and with infima and minima in place of suprema and maxima, using $\inf A = -\sup(-A)$ and `QuadEnvM`. This gives $\mathcal{M}_\ell f_j \to \mathcal{M}_\ell f$ uniformly on every compact set of centers $z = (x,t)$ with $t \in I$ and $t-\ell \in I$. □

2.193. `SlopeRigidity`.

Lemma 2.193 (Rigidity from constant envelope slopes)**.** *Let F be a finite-dimensional real inner-product space, let $1 \le C_0 < \infty$ and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`. Suppose that for some $(b,t_0) \in F \times \mathbb{R}$ and some $B \in \mathbb{R}$*

$$\mathcal{S}_\ell^+ v(b,t_0) = \mathcal{S}_\ell^- v(b,t_0) = B \qquad \text{for every } \ell > 0, \tag{873}$$

with the slopes of `EnvSlopeP` and `EnvSlopeM`. Then v is affine in the sense of `IsSpaceTimeAffine`.

Proof. Write $z_\star = (b,t_0)$ and, as in `CenteredGrowth`,

$$L(K) = 1 + \log K, \qquad \mathcal{B}_K = \big\{(x,t) \in F \times \mathbb{R} : |x| \le K,\ |t| \le K\big\} \qquad (K \ge 2),$$

so that $L(K) \ge 1 + \log 2 > 1$ and $KL(K)^2 \ge 2$ for $K \ge 2$. Throughout, $\mathcal{P}_\ell$ and $\mathcal{M}_\ell$ are the quadratic envelopes of `QuadEnvP` and `QuadEnvM` and $\mathcal{S}_\ell^\pm$ are the envelope slopes of `EnvSlopeP` and `EnvSlopeM`. The steps are ordered so that the subquadratic growth of V and of v is in hand before any envelope is compared with a competitor and before any constant is moved through a supremum: by the convention of the preamble such a comparison is available only together with the boundedness hypothesis, and the node that supplies it is `EnvelopeCompetitorBounds`.

Step 1: translation to the origin, and the reduction to V. Apply `FrameInvariance` with $z_0 = z_\star$, $R = 1$, $p_0 = 0$ and $a_0 = 0$. By `FrameChange` the resulting function is

$$V(y,s) = \frac{v\big(b + 1\cdot(y + 0\cdot s),\, t_0 + 1\cdot s\big) - v(b,t_0)}{1} - 0\cdot y - \Big(0 + \frac{|0|^2}{2}\Big)s = v(b+y,\, t_0+s) - v(b,t_0),$$

and `FrameInvariance` gives $V \in \mathcal{A}_{m,C_0}$. Note $V(0,0) = 0$, and record the two forms of the relation between v and V, each obtained from the other by the substitution $x = b+y$, $t = t_0+s$:

$$V(y,s) = v(b+y,\, t_0+s) - v(z_\star), \qquad v(x,t) = V(x-b,\, t-t_0) + v(z_\star) \qquad \big((y,s),(x,t) \in F\times\mathbb{R}\big). \tag{874}$$

If V is affine then so is v: from the second form of (874) and $V(y,s) = \langle q,y\rangle + \beta s + \gamma$ we get $v(x,t) = \langle q,x\rangle + \beta t + \big(\gamma - \langle q,b\rangle - \beta t_0 + v(z_\star)\big)$, which is the form required by `IsSpaceTimeAffine`. So it suffices to prove that V is affine, and that is what Steps 2–11 do. The hypothesis (873) is not used before Step 5.

Steps 2–4: growth, and the hypotheses of the envelope lemmas. By Step 1 the function V lies in $\mathcal{A}_{m,C_0}$, so `FitClassGrowth` applies to it and supplies a constant $C_V > 0$ with $\big|V(\zeta) - V(0)\big| \le C_V KL(K)^2$ for $\zeta \in \mathcal{B}_K$ and $K \ge 2$; since $V(0,0) = 0$ that reads

$$|V(\zeta)| \le C_V\, K\, L(K)^2 \qquad (\zeta \in \mathcal{B}_K,\ K \ge 2). \tag{875}$$

The same node also gives that V has subquadratic spatial growth on compact time subintervals of $I = \mathbb{R}$ in the sense of `SubquadraticGrowth`. Applying `FitClassGrowth` a second time, now to v itself — which lies in the class by hypothesis — gives that v too has subquadratic spatial growth on compact time subintervals of $\mathbb{R}$. No centering of either function is needed: that node is stated for an arbitrary member of the class, measuring growth from the value at the origin.

By `IsFitClass` both V and v are continuous on all of space-time and are viscosity solutions of the limiting equation there. Hence the hypotheses of `QuadEnvAttained` and of `EnvHamiltonianSolution` hold for V with $I = \mathbb{R}$, for which every time and every shifted time is admissible, and the hypotheses of `EnvelopeCompetitorBounds` hold for V and for v with $I = \mathbb{R}$ as well: that lemma asks only that the function be continuous on $\Omega_I = \{(y,\tau) : \tau \in I\}$, which for $I = \mathbb{R}$ is all of space-time, and have subquadratic spatial growth on compact time subintervals of I in the sense of `SubquadraticGrowth`. Since $I = \mathbb{R}$, its admissibility conditions $t \in I$ and $t \pm \ell \in I$ are vacuous, so parts (1) and (2) are available at every center and every $\ell > 0$. What they give is precisely what the preamble's supremum convention withholds from `QuadEnvP` and `QuadEnvM` on their own: part (1) says that the set of forward competitor values at a center is bounded above by the real number $\mathcal{P}_\ell V$, which is moreover its least upper bound, and part (2) says that the set of backward competitor values is bounded below by $\mathcal{M}_\ell V$, which is its greatest lower bound. Every comparison of an envelope with a chosen competitor below is an application of one of these two halves, and never of `QuadEnvP` or `QuadEnvM` alone, which are used only for the defining formulas. *Step 5: the envelope slopes of V at the origin equal B.* Fix

$\ell > 0$. By Steps 3 and 4, part (1) and part (2) of `EnvelopeCompetitorBounds` with $I = \mathbb{R}$ are available for V at the center $(0,0)$ and for v at the center $z_\star$, at this ℓ.

The forward envelopes. Let $y \in F$ and put $y' = b + y$, so that $|y| = |y' - b|$ and, by the first form of (874), $V(y,\ell) = v(y', t_0 + \ell) - v(z_\star)$. The domination half of part (1) for v at $z_\star$ gives $v(y', t_0 + \ell) - |y' - b|^2/(2\ell) \le \mathcal{P}_\ell v(z_\star)$, hence

$$V(y,\ell) - \frac{|y|^2}{2\ell} = \Big(v(y', t_0 + \ell) - \frac{|y' - b|^2}{2\ell}\Big) - v(z_\star) \le \mathcal{P}_\ell v(z_\star) - v(z_\star).$$

As $y \in F$ was arbitrary and $y \mapsto b + y$ is a bijection of F, the real number $\mathcal{P}_\ell v(z_\star) - v(z_\star)$ bounds every competitor for V at $(0,0)$ at length ℓ; the extremality half of part (1) for V therefore gives $\mathcal{P}_\ell V(0,0) \le \mathcal{P}_\ell v(z_\star) - v(z_\star)$. Conversely, let $y' \in F$ and put $y = y' - b$; the domination half of part (1) for V at $(0,0)$ gives $V(y,\ell) - |y|^2/(2\ell) \le \mathcal{P}_\ell V(0,0)$, that is

$$v(y', t_0 + \ell) - \frac{|y' - b|^2}{2\ell} \le \mathcal{P}_\ell V(0,0) + v(z_\star),$$

and as y' was arbitrary the extremality half of part (1) for v gives $\mathcal{P}_\ell v(z_\star) \le \mathcal{P}_\ell V(0,0) + v(z_\star)$. The two inequalities together are

$$\mathcal{P}_\ell V(0,0) = \mathcal{P}_\ell v(z_\star) - v(z_\star) \qquad (\ell > 0). \tag{876}$$

The backward envelopes. Let $y \in F$, $y' = b + y$, so $V(y,-\ell) = v(y', t_0 - \ell) - v(z_\star)$ by (874) and $|y| = |y' - b|$. The domination half of part (2) for v at $z_\star$ gives $\mathcal{M}_\ell v(z_\star) \le v(y', t_0 - \ell) + |y' - b|^2/(2\ell)$, that is $\mathcal{M}_\ell v(z_\star) - v(z_\star) \le V(y,-\ell) + |y|^2/(2\ell)$; as y was arbitrary, the extremality half of part (2) for V gives $\mathcal{M}_\ell v(z_\star) - v(z_\star) \le \mathcal{M}_\ell V(0,0)$. Conversely, for $y' \in F$ and $y = y' - b$ the domination half of part (2) for V gives $\mathcal{M}_\ell V(0,0) \le V(y,-\ell) + |y|^2/(2\ell) = v(y', t_0 - \ell) - v(z_\star) + |y' - b|^2/(2\ell)$, so $\mathcal{M}_\ell V(0,0) + v(z_\star) \le v(y', t_0 - \ell) + |y' - b|^2/(2\ell)$ for every y', and the extremality half of part (2) for v gives $\mathcal{M}_\ell V(0,0) + v(z_\star) \le \mathcal{M}_\ell v(z_\star)$. Hence

$$\mathcal{M}_\ell V(0,0) = \mathcal{M}_\ell v(z_\star) - v(z_\star) \qquad (\ell > 0). \tag{877}$$

Now use the hypothesis. By `EnvSlopeP`, $V(0,0) = 0$ and (876),

$$\mathcal{S}_\ell^+ V(0,0) = \frac{\mathcal{P}_\ell V(0,0) - V(0,0)}{\ell} = \frac{\mathcal{P}_\ell v(z_\star) - v(z_\star)}{\ell} = \mathcal{S}_\ell^+ v(z_\star) = B,$$

the last equality being (873). By `EnvSlopeM`, $V(0,0) = 0$ and (877),

$$\mathcal{S}_\ell^- V(0,0) = \frac{V(0,0) - \mathcal{M}_\ell V(0,0)}{\ell} = \frac{v(z_\star) - \mathcal{M}_\ell v(z_\star)}{\ell} = \mathcal{S}_\ell^- v(z_\star) = B.$$

So

$$\mathcal{S}_\ell^+ V(0,0) = \mathcal{S}_\ell^- V(0,0) = B \qquad (\ell > 0). \tag{878}$$

Step 6: the future paraboloid. Let $t > 0$. By (878), `EnvSlopeP` and $V(0,0) = 0$, $\mathcal{P}_t V(0,0) = Bt$. By the domination half of part (1) of `EnvelopeCompetitorBounds` for V with $I = \mathbb{R}$, at the center $(0,0)$ and the length $t > 0$, whose hypotheses were checked in Step 3, every competitor is at most $\mathcal{P}_t V(0,0) = Bt$: for every $y \in F$, $V(y,t) - |y|^2/(2t) \le Bt$, so

$$V(y,t) \le Bt + \frac{|y|^2}{2t} \qquad (t > 0). \tag{879}$$

Step 7: a bound for the far optimizers. Fix (x,t). For $\ell \ge 1$, part (1) of `QuadEnvAttained` gives a maximizer y_ℓ with

$$\mathcal{P}_\ell V(x,t) = V(y_\ell, t + \ell) - \frac{|y_\ell - x|^2}{2\ell}. \tag{880}$$

The domination half of part (1) of `EnvelopeCompetitorBounds` for V with $I = \mathbb{R}$, at the center (x,t) and the length $\ell > 0$, applied to the competitor $y = x$, gives $V(x, t + \ell) = V(x, t + \ell) - |x - x|^2/(2\ell) \le \mathcal{P}_\ell V(x,t)$, so by (880)

$$V(x, t + \ell) \le V(y_\ell, t + \ell) - \frac{|y_\ell - x|^2}{2\ell}.$$

That is exactly the hypothesis of FarOptimizerBound for the function V, the constant C_V of (875), the center (x,t), the length ℓ and the position y_ℓ; the box-bound hypothesis of that node is (875) itself, with the same $C_V>0$. It therefore supplies a constant $C_{x,t}<\infty$ with

$$|y_\ell| \le C_{x,t}\,\ell\,(1+\log \ell)^2 \qquad (\ell \ge 1). \tag{881}$$

Step 8: $\mathcal{S}_\ell^+ V(x,t)\le B$ for every $\ell>0$. Fix (x,t) and let $\ell \ge \max\{1,\,2|t|+1\}$, so $t+\ell \ge \ell/2>0$. By (879) applied at $(y_\ell, t+\ell)$ and (880),

$$\mathcal{P}_\ell V(x,t) \le B(t+\ell) + \frac{|y_\ell|^2}{2(t+\ell)} - \frac{|y_\ell - x|^2}{2\ell}.$$

Expanding $|y_\ell - x|^2 = |y_\ell|^2 - 2\langle x, y_\ell\rangle + |x|^2$,

$$\frac{|y_\ell|^2}{2(t+\ell)} - \frac{|y_\ell - x|^2}{2\ell} = -\frac{t\,|y_\ell|^2}{2\ell(t+\ell)} + \frac{\langle x, y_\ell\rangle}{\ell} - \frac{|x|^2}{2\ell},$$

since $\frac{1}{2(t+\ell)} - \frac{1}{2\ell} = -\frac{t}{2\ell(t+\ell)}$. By (881) and $t+\ell \ge \ell/2$,

$$\frac{|t|\,|y_\ell|^2}{2\ell(t+\ell)} \le \frac{|t|\,C_{x,t}^2\ell^2(1+\log\ell)^4}{2\ell\cdot(\ell/2)} = |t|\,C_{x,t}^2(1+\log\ell)^4, \qquad \frac{|\langle x, y_\ell\rangle|}{\ell} \le |x| C_{x,t}(1+\log\ell)^2,$$

and $|x|^2/(2\ell) \le |x|^2/2$. Hence

$$\mathcal{P}_\ell V(x,t) \le B(t+\ell) + O_{x,t}\big((1+\log\ell)^4\big),$$

and by EnvSlopeP

$$\mathcal{S}_\ell^+ V(x,t) = \frac{\mathcal{P}_\ell V(x,t) - V(x,t)}{\ell} \le B\frac{t+\ell}{\ell} + \frac{O_{x,t}\big((1+\log\ell)^4\big)}{\ell} - \frac{V(x,t)}{\ell} \longrightarrow B$$

as $\ell\to\infty$, because $(1+\log\ell)^4/\ell \to 0$. Therefore $\limsup_{\ell\to\infty} \mathcal{S}_\ell^+ V(x,t) \le B$.

By part (1) of EnvHamiltonianSolution, applied to V with $I=\mathbb{R}$ whose admissibility hypothesis is vacuous, $\ell \mapsto \mathcal{S}_\ell^+ V(x,t)$ is nondecreasing. Hence for every fixed $\ell>0$ and every $\ell' \ge \ell$, $\mathcal{S}_\ell^+ V(x,t) \le \mathcal{S}_{\ell'}^+ V(x,t)$; letting $\ell'\to\infty$ and using the previous paragraph,

$$\mathcal{S}_\ell^+ V(x,t) \le B \qquad (x\in F,\ t\in\mathbb{R},\ \ell>0). \tag{882}$$

Step 9: the directed comparison. Let $x,y\in F$, $t\in\mathbb{R}$ and $\ell>0$. By the domination half of part (1) of EnvelopeCompetitorBounds for V with $I=\mathbb{R}$, at the center (x,t) and the length ℓ, the competitor y gives $V(y,t+\ell) - |y-x|^2/(2\ell) \le \mathcal{P}_\ell V(x,t)$, and by EnvSlopeP and (882), $\mathcal{P}_\ell V(x,t) = V(x,t) + \ell\,\mathcal{S}_\ell^+ V(x,t) \le V(x,t) + B\ell$. Hence

$$V(y,t+\ell) - V(x,t) \le B\ell + \frac{|y-x|^2}{2\ell} \qquad (x,y\in F,\ t\in\mathbb{R},\ \ell>0). \tag{883}$$

Step 10: equality points and a single velocity. Let $\ell>0$. By (878) and $V(0,0)=0$ we have $\mathcal{P}_\ell V(0,0) = B\ell$, and part (1) of QuadEnvAttained gives a maximizer; writing it as ℓq_+ with $q_+\in F$,

$$V(\ell q_+, \ell) = B\ell + \frac{\ell|q_+|^2}{2}. \tag{884}$$

Let $r>0$. By (878) and EnvSlopeM, $\mathcal{M}_r V(0,0) = V(0,0) - Br = -Br$, and part (1) of QuadEnvAttained gives a minimizer; writing it as $-rq_-$,

$$V(-rq_-, -r) = -Br - \frac{r|q_-|^2}{2}. \tag{885}$$

Apply (883) from the point $(-rq_-,-r)$ to $(\ell q_+,\ell)$, i.e. with $x=-rq_-$, $t=-r$, $y=\ell q_+$ and the length $\ell+r>0$, for which $t+(\ell+r)=\ell$:

$$V(\ell q_+,\ell) - V(-rq_-,-r) \le B(\ell+r) + \frac{|\ell q_+ + rq_-|^2}{2(\ell+r)}.$$

By (884) and (885) the left side equals $B(\ell+r)+\frac{1}{2}\big(\ell|q_+|^2+r|q_-|^2\big)$, so after cancelling $B(\ell+r)$ and multiplying by $2(\ell+r)>0$,

$$(\ell+r)\big(\ell|q_+|^2+r|q_-|^2\big)-|\ell q_++rq_-|^2\le 0.$$

Expanding, the terms $\ell^2|q_+|^2$ and $r^2|q_-|^2$ cancel and the left side becomes $\ell r\big(|q_+|^2+|q_-|^2-2\langle q_+,q_-\rangle\big)=\ell r|q_+-q_-|^2$. Since $\ell,r>0$ this forces $q_+=q_-$.

Fix one backward equality point for $r=1$ and call the resulting vector q. The previous paragraph, applied with this $r=1$ and any $\ell>0$ and any choice of forward maximizer, shows that every forward velocity equals q; and applied with $\ell=1$, the forward velocity being q, it shows that every backward velocity equals q. Consequently (884) and (885) give, together with $V(0,0)=0$,

$$V(qs,s)=\Big(B+\frac{|q|^2}{2}\Big)s\qquad(s\in\mathbb{R}); \tag{886}$$

for $s>0$ take $\ell=s$ in (884), and for $s<0$ take $r=-s$ in (885).

Step 11: V is affine. Put $A(x,t)=\langle q,x\rangle+\big(B-\frac{1}{2}|q|^2\big)t$ and fix (x,t).

Let $s<t$ and apply (883) from (qs,s) to (x,t) with the length $\tau=t-s>0$:

$$V(x,t)\le V(qs,s)+B\tau+\frac{|x-qs|^2}{2\tau}=Bs+\frac{s|q|^2}{2}+B\tau+\frac{|x-qs|^2}{2\tau},$$

using (886). Now $x-qs=(x-qt)+\tau q$, so $|x-qs|^2=|x-qt|^2+2\tau\langle q,x-qt\rangle+\tau^2|q|^2$ and

$$\frac{|x-qs|^2}{2\tau}=\frac{|x-qt|^2}{2\tau}+\langle q,x\rangle-t|q|^2+\frac{\tau|q|^2}{2}.$$

Substituting and using $s+\tau=t$,

$$V(x,t)\le Bt+\frac{|x-qt|^2}{2\tau}+\langle q,x\rangle-t|q|^2+\frac{(s+\tau)|q|^2}{2}=A(x,t)+\frac{|x-qt|^2}{2(t-s)}.$$

Letting $s\to-\infty$, so $\tau=t-s\to+\infty$, gives $V(x,t)\le A(x,t)$.

Let $s>t$ and apply (883) from (x,t) to (qs,s) with the length $\tau'=s-t>0$:

$$V(qs,s)-V(x,t)\le B\tau'+\frac{|qs-x|^2}{2\tau'},$$

so by (886)

$$V(x,t)\ge Bs+\frac{s|q|^2}{2}-B\tau'-\frac{|x-qs|^2}{2\tau'}=Bt+\frac{s|q|^2}{2}-\frac{|x-qs|^2}{2\tau'}.$$

Now $x-qs=(x-qt)-\tau' q$, so $|x-qs|^2=|x-qt|^2-2\tau'\langle q,x-qt\rangle+\tau'^2|q|^2$ and

$$\frac{|x-qs|^2}{2\tau'}=\frac{|x-qt|^2}{2\tau'}-\langle q,x\rangle+t|q|^2+\frac{\tau'|q|^2}{2}.$$

Substituting and using $s-\tau'=t$,

$$V(x,t)\ge Bt-\frac{|x-qt|^2}{2\tau'}+\langle q,x\rangle-t|q|^2+\frac{(s-\tau')|q|^2}{2}=A(x,t)-\frac{|x-qt|^2}{2(s-t)}.$$

Letting $s\to+\infty$ gives $V(x,t)\ge A(x,t)$.

Hence $V=A$, which is affine in the sense of `IsSpaceTimeAffine`, and by the reduction of Step 1 so is v. □

2.194. `PowerDecay`.

Theorem 2.194 (Uniform power decay)**.** *Fix an integer $d\ge 1$, put $n=d+1$ and identify $\mathbb{R}^n$ with the space-time $\ell^2(\mathbb{R}^d\times\mathbb{R})$. Let b_d be the family flatness modulus of `FlatnessModulus`. Then there are $B\in(0,\infty)$, $\gamma\in(0,1]$ and $s_0\in(0,1/8]$ such that*

$$b_d(s)\le Bs^{\gamma}\qquad(0<s\le s_0). \tag{887}$$

Proof. Write $E = \ell^2(\mathbb{R}^d \times \mathbb{R})$ for the space-time, $e_b = (0,1) \in E$ for its added unit direction, and $\langle \cdot, \cdot \rangle$, $|\cdot|$ for the inner product and norm of E. Note $|e_b|^2 = |0|^2 + 1^2 = 1$, so e_b is a unit vector.

Step 1: the negation is the hypothesis of the selection lemma. Suppose, for contradiction, that no such triple (B, γ, s_0) exists. Then for every $B > 0$, every γ with $0 < \gamma \le 1$ and every s_0 with $0 < s_0 \le 1/8$, the implication in (887) fails for some s; that is, there is s with $0 < s \le s_0$ and $b_d(s) > Bs^\gamma$. This is exactly the hypothesis of `BadScales`, so that lemma provides real sequences $\gamma_i, s_i, r_i, \lambda_i$, functions U_i on E and centers $z_i \in E$ such that $(\gamma_i, s_i, r_i, \lambda_i, U_i, z_i)_{i \ge 0}$ is selected bad-scale data in dimension d in the sense of `IsBadScaleData`. By clauses 1 and 3 of `IsBadScaleData` we have, for every i,

$$\lambda_i > 0, \qquad r_i > 0. \tag{888}$$

Step 2: the slopes P_i and the index i_1. Apply clause 5 of `IsBadScaleData` with the fixed value $M = 2$, which is admissible since $2 \ge 2$. It provides an index i_0 such that for every $i \ge i_0$, every center w with $|w - z_i| \le 2r_i$ and every radius $0 < \rho \le 2r_i$, the parameter $32\lambda_i$ is good for U_i at (w, ρ). Put $i_1 = i_0$ and let $i \ge i_1$. The choice $w = z_i$ is admissible because $|z_i - z_i| = 0 \le 2r_i$ by (888), and the choice $\rho = r_i$ is admissible because $0 < r_i \le 2r_i$, again by (888). Hence $32\lambda_i$ is good for U_i at (z_i, r_i): by `IsCylinderFit` there is a vector P_i realizing the cylinder fit of parameter $32\lambda_i$ for U_i at (z_i, r_i). Fix one such P_i for each $i \ge i_1$, and put $P_i = e_b$ for $i < i_1$, a value never used below. By `IsCylinderFit` the realizing vector satisfies $P_i \ne 0$, so

$$c_i = |P_i| > 0 \qquad (i \ge i_1). \tag{889}$$

Step 3: the anisotropic frame Φ_i, constructed explicitly. Let $i \ge i_1$ and put

$$w_i = \frac{P_i}{c_i} = c_i^{-1} P_i \in E,$$

which satisfies $|w_i| = c_i^{-1}|P_i| = 1$ by (889). We construct a linear isometry Φ_i of E onto itself with $\Phi_i(e_b) = w_i$.

If $w_i = e_b$, take $\Phi_i = \mathrm{id}_E$, which is a linear isometry onto E and satisfies $\Phi_i(e_b) = e_b = w_i$.

Suppose $w_i \ne e_b$. Then $e_b - w_i \ne 0$; put

$$\nu_i = |e_b - w_i| > 0, \qquad \varepsilon_i = \frac{e_b - w_i}{\nu_i}, \qquad \Phi_i(y) = y - 2\langle \varepsilon_i, y \rangle \varepsilon_i \quad (y \in E).$$

Note $|\varepsilon_i| = 1$, hence $\langle \varepsilon_i, \varepsilon_i \rangle = 1$. Four verifications, each a direct computation in the inner product of E.

(a) Linearity. The map $y \mapsto \langle \varepsilon_i, y \rangle$ is linear and $\Phi_i(y) = y - 2\langle \varepsilon_i, y \rangle \varepsilon_i$ is a difference of the identity and a linear map composed with a fixed vector, so Φ_i is linear.

(b) Isometry. For every $y \in E$, expanding and using $\langle \varepsilon_i, \varepsilon_i \rangle = 1$,

$$\big|\Phi_i(y)\big|^2 = \langle y, y \rangle - 4\langle \varepsilon_i, y \rangle \langle y, \varepsilon_i \rangle + 4\langle \varepsilon_i, y \rangle^2 \langle \varepsilon_i, \varepsilon_i \rangle = |y|^2 - 4\langle \varepsilon_i, y \rangle^2 + 4\langle \varepsilon_i, y \rangle^2 = |y|^2,$$

so $|\Phi_i(y)| = |y|$, both sides being nonnegative.

(c) Involution, hence bijectivity. For every y, $\langle \varepsilon_i, \Phi_i(y) \rangle = \langle \varepsilon_i, y \rangle - 2\langle \varepsilon_i, y \rangle \langle \varepsilon_i, \varepsilon_i \rangle = -\langle \varepsilon_i, y \rangle$, hence

$$\Phi_i\big(\Phi_i(y)\big) = \Phi_i(y) - 2\langle \varepsilon_i, \Phi_i(y) \rangle \varepsilon_i = y - 2\langle \varepsilon_i, y \rangle \varepsilon_i + 2\langle \varepsilon_i, y \rangle \varepsilon_i = y.$$

So Φ_i is its own inverse and is in particular a bijection of E onto itself. With (a) and (b) it is a linear isometry of E onto itself.

(d) It carries e_b to w_i. Since $|e_b| = |w_i| = 1$,

$$2\langle e_b, e_b - w_i \rangle = 2\langle e_b, e_b \rangle - 2\langle e_b, w_i \rangle = 2 - 2\langle e_b, w_i \rangle,$$

while

$$\nu_i^2 = |e_b - w_i|^2 = \langle e_b, e_b \rangle - 2\langle e_b, w_i \rangle + \langle w_i, w_i \rangle = 2 - 2\langle e_b, w_i \rangle,$$

the cross terms combining by the symmetry of the real inner product. Hence $2\langle e_b, e_b - w_i \rangle = \nu_i^2$ and therefore

$$2\langle \varepsilon_i, e_b \rangle = \frac{2\langle e_b - w_i, e_b \rangle}{\nu_i} = \frac{\nu_i^2}{\nu_i} = \nu_i,$$

so

$$\Phi_i(e_b) = e_b - 2\langle \varepsilon_i, e_b\rangle \varepsilon_i = e_b - \nu_i \cdot \frac{e_b - w_i}{\nu_i} = e_b - (e_b - w_i) = w_i.$$

In both cases, then, Φ_i is a linear isometry of E onto itself with

$$\Phi_i(e_b) = \frac{P_i}{c_i} \qquad (i \geq i_1). \tag{890}$$

For $i < i_1$ put $\Phi_i = \mathrm{id}_E$. No orthonormal-basis extension and no appeal to any unstated standard fact is used: (a)–(d) are the complete verification.

Step 4: the normalized functions v_i, obtained by solving. Let $i \geq i_1$. By (888) and (889) the three numbers r_i, c_i and λ_i are positive, so we may define, for every $(x,t) \in \mathbb{R}^d \times \mathbb{R}$,

$$v_i(x,t) = \frac{1}{\lambda_i}\left[\frac{U_i\big(z_i + r_i\Phi_i(\sqrt{\lambda_i}\,x, t)\big) - U_i(z_i)}{r_i c_i} - t\right], \tag{891}$$

and put $v_i = 0$ for $i < i_1$. Multiplying (891) by $r_i c_i \lambda_i > 0$ gives

$$r_i c_i \lambda_i v_i(x,t) = U_i\big(z_i + r_i\Phi_i(\sqrt{\lambda_i}\,x, t)\big) - U_i(z_i) - r_i c_i t,$$

and adding $U_i(z_i) + r_i c_i t$ to both sides yields, for every (x,t) and every $i \geq i_1$,

$$U_i\big(z_i + r_i\Phi_i(\sqrt{\lambda_i}\,x, t)\big) = U_i(z_i) + r_i c_i\big(t + \lambda_i v_i(x,t)\big), \tag{892}$$

which is the anisotropic normalization required by `NonaffineLimit`.

Step 5: the nonaffine entire solution. All hypotheses of `NonaffineLimit` are now in place: $d \geq 1$; $(\gamma_i, s_i, r_i, \lambda_i, U_i, z_i)_{i\geq 0}$ is bad-scale data for d in the sense of `IsBadScaleData`, by Step 1; the index i_1 and the vectors P_i realize, for every $i \geq i_1$, a cylinder fit of parameter $32\lambda_i$ for U_i at (z_i, r_i) in the sense of `IsCylinderFit`, by Step 2; Φ_i is a linear isometry of the space-time onto itself with $\Phi_i(e_b) = P_i/c_i$ for $i \geq i_1$, by (890); and the functions v_i satisfy the normalization for every (x,t) and every $i \geq i_1$, by (892). Applying `NonaffineLimit` gives a constant C_0 with $0 \leq C_0 < \infty$, a strictly increasing sequence of indices $j \mapsto \psi(j)$ with $\psi(j) \geq i_1$, and a continuous function v on the space-time with $v_{\psi(j)} \to v$ uniformly on every compact set, such that $v \in \mathcal{A}_{d,C_0}$ in the sense of `IsFitClass` and v is not affine on space-time in the sense of `IsSpaceTimeAffine`.

Step 6: the contradiction. The function v satisfies the hypotheses of `Liouville` with the constant C_0, since $0 \leq C_0 < \infty$ and $v \in \mathcal{A}_{d,C_0}$; here the ambient finite-dimensional real inner-product space is $\mathbb{R}^d$, of dimension d. Hence v is affine on space-time in the sense of `IsSpaceTimeAffine`. This contradicts the conclusion of Step 5 that v is not affine.

Therefore the assumption of Step 1 is false: there are $B \in (0,\infty)$, $\gamma \in (0,1]$ and $s_0 \in (0, 1/8]$ for which (887) holds. Since d was the only datum fixed before the argument began, these constants depend only on d. □

2.195. `TaylorToHolderGradient`.

Lemma 2.195 (From a uniform Taylor estimate to a Hölder gradient)**.** *Let E be a finite-dimensional real inner-product space, let $\alpha \in (0,1]$, let $C > 0$ and let $\rho_0 > 0$. Then there is a real number C' with $0 < C' < \infty$, depending only on C, α and ρ_0 and on nothing else, with the following property.*

Let $S \subseteq E$, let $U : E \to \mathbb{R}$ and let $g : E \to E$ be arbitrary, and assume

$$|U(z+h) - U(z) - g(z)\cdot h| \leq C|h|^{1+\alpha} \qquad (z \in S,\ |h| \leq \rho_0),$$

and $|g(z)| \leq C$ for $z \in S$. Then U is differentiable at every $z \in S$ with gradient $g(z)$, and

$$|g(z) - g(z')| \leq C'|z - z'|^{\alpha} \qquad (z, z' \in S).$$

Proof. Put

$$C' = \max\{3\cdot 2^{1+\alpha}C,\ 2C(2/\rho_0)^{\alpha}\},$$

which is a positive real number determined by C, α and ρ_0 alone, since $C > 0$, $\alpha > 0$ and $\rho_0 > 0$. Now fix S, U and g satisfying the two hypotheses, and let us verify the two conclusions for this C'.

Differentiability. Let $z \in S$. For $h \in E$ with $0 < |h| \le \rho_0$, the hypothesis gives

$$\frac{|U(z+h) - U(z) - g(z)\cdot h|}{|h|} \le C|h|^{\alpha}.$$

Since $\alpha > 0$, the right side tends to 0 as $h \to 0$. Hence U is differentiable at z with derivative $h \mapsto g(z)\cdot h$, that is, with gradient $g(z)$.

The Hölder estimate. Let $z, z' \in S$ and set $\ell = |z - z'|$.

If $\ell = 0$ then $z = z'$ and the inequality is trivial.

Suppose $0 < \ell \le \rho_0/2$. Fix any unit vector $e \in E$. We estimate $U(z' + \ell e) - U(z')$ in two ways.

First, apply the Taylor estimate at z' with the increment $h = \ell e$, whose norm is $\ell \le \rho_0$:

$$\big|U(z' + \ell e) - U(z') - \ell\, g(z')\cdot e\big| \le C\ell^{1+\alpha}. \tag{893}$$

Second, apply the Taylor estimate at z with the two increments $h_1 = z' - z$ and $h_2 = z' + \ell e - z$. Their norms satisfy $|h_1| = \ell \le \rho_0$ and $|h_2| \le |z' - z| + \ell = 2\ell \le \rho_0$, so both are admissible, and

$$\big|U(z') - U(z) - g(z)\cdot h_1\big| \le C\ell^{1+\alpha}, \qquad \big|U(z' + \ell e) - U(z) - g(z)\cdot h_2\big| \le C(2\ell)^{1+\alpha}.$$

Subtracting, and using $h_2 - h_1 = \ell e$,

$$\big|U(z' + \ell e) - U(z') - \ell\, g(z)\cdot e\big| \le C\ell^{1+\alpha} + C(2\ell)^{1+\alpha} \le 2C\, 2^{1+\alpha}\ell^{1+\alpha}, \tag{894}$$

since $1 \le 2^{1+\alpha}$.

Subtracting (893) from (894) and using the triangle inequality,

$$\ell\,\big|(g(z) - g(z'))\cdot e\big| \le C\ell^{1+\alpha} + 2C\, 2^{1+\alpha}\ell^{1+\alpha} \le 3C\, 2^{1+\alpha}\ell^{1+\alpha}.$$

Dividing by $\ell > 0$ gives $\big|(g(z) - g(z'))\cdot e\big| \le 3C\, 2^{1+\alpha}\ell^{\alpha}$. The constants are independent of e, so taking the supremum over unit vectors e, which computes the norm of $g(z) - g(z')$, yields

$$|g(z) - g(z')| \le 3C\, 2^{1+\alpha}\ell^{\alpha} \le C'\ell^{\alpha}.$$

Finally suppose $\ell > \rho_0/2$. Then, by the bound $|g| \le C$ on S,

$$|g(z) - g(z')| \le 2C = 2C\left(\frac{2}{\rho_0}\right)^{\alpha}\left(\frac{\rho_0}{2}\right)^{\alpha} \le 2C\left(\frac{2}{\rho_0}\right)^{\alpha}\ell^{\alpha} \le C'\ell^{\alpha},$$

using $(\rho_0/2)^{\alpha} \le \ell^{\alpha}$, which holds because $\alpha > 0$ and $\rho_0/2 < \ell$. This completes the proof. □

3. Main results

3.1. `Liouville` (paper, Theorem 3.1).

Theorem 3.1 (Liouville Theorem)**.** *Let F be a finite-dimensional real inner-product space, write m for its dimension, let $0 \le C_0 < \infty$ and let $v \in \mathcal{A}_{m,C_0}$ in the sense of `IsFitClass`. Then v is affine on space-time in the sense of `IsSpaceTimeAffine`.*

Proof. Suppose, for contradiction, that v is not affine. Write $\delta_j = 2^{-j}$ and $z_* = (0, 1)$. Throughout, $\mathcal{P}_\ell$ and $\mathcal{M}_\ell$ are the quadratic envelopes of `QuadEnvP` and `QuadEnvM`, $\mathcal{S}_\ell^{\pm}$ the envelope slopes of `EnvSlopeP` and `EnvSlopeM`, $\mathcal{C}_s$ the barycentric envelope of `BaryEnv`, and H_f the envelope Hamiltonian of `EnvHamiltonian`.

Step 0: we may assume the constant is at least one. Put $C_1 = \max\{C_0, 1\}$, so $1 \le C_1 < \infty$ and $C_0 \le C_1$. If $R > 0$ and (p, a) is a fit for v at a center z of radius R with constant C_0 in the sense of `IsFit`, then for all admissible increments $|\ell| \le R$, $|h - \ell p| \le R$,

$$\Big|v\big(z + (h, \ell)\big) - v(z) - \langle p, h\rangle - \Big(a - \frac{|p|^2}{2}\Big)\ell\Big| \le C_0 R \le C_1 R,$$

the last inequality because $C_0 \le C_1$ and $R > 0$; so (p, a) is also a fit of radius R with constant C_1. By `IsFitClass` the function v is continuous on all of space-time and a viscosity solution of the limiting equation there, and it has a fit at every center and every positive radius with constant C_0, hence with constant C_1. Therefore $v \in \mathcal{A}_{m,C_1}$, and v is still not affine. So $\mathcal{A}_{m,C_1}$ contains a member that is not affine, with $1 \le C_1 < \infty$. Everything below uses only this, and derives a contradiction; that contradiction refutes the assumption and proves the theorem.

Step 1: the extremal contact pair. By `EnvelopeContact`, applied with the constant C_1 and the nonaffine member v, there are $\kappa \in (0,\infty)$, a real number H and a function $u \in \mathcal{A}_{m,C_1}$ centered at z_* with constant C_1 in the sense of `IsCenteredAt`, satisfying the horizontal Jensen bound with constant κ at every time in the sense of `HasJensenBound` with $I = \mathbb{R}$, whose contact envelope $W = \mathcal{C}_t u - \kappa^2 t/2$ of `ContactEnv` has the six listed properties. We record the three we use:

(E3) $u(x,t) - \kappa^2 t/2 \le W(x,t) \le u(x,t)$ for $t > 0$, and $W(z_*) = u(z_*) = 0$;
(E5) if $W(x,t) = u(x,t)$ and $t > 0$ then $\big(\mathcal{C}_t u(x,t) - u(x,t)\big)/t = \kappa^2/2$;
(E6) $H_u(z_*) = H$ with $\mathcal{S}^+_\ell W(z_*) \to H$ as $\ell \downarrow 0$, the Hamiltonian being that of `EnvHamiltonian` and the slope that of `EnvSlopeP`.

The remaining three, continuity and subquadratic growth of W on $\{t > 0\}$ in the sense of `SubquadraticGrowth`, the subsolution property in the sense of `IsLimitSubOn`, and the concavity of $x \mapsto W(x,t) - |x|^2/(2t)$, are carried along unchanged.

Since $u \in \mathcal{A}_{m,C_1}$ is centered at z_* with constant C_1, it is continuous on all of space-time and a viscosity solution of the limiting equation there by `IsFitClass`, and it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ by `UniformSubquadratic`, applied with the centering point z_* and the constant C_1. Hence `QuadEnvAttained` and `EnvHamiltonianSolution` are available for u with $I = \mathbb{R}$, every time and every shifted time being admissible there.

Step 2: the terminal point, the levels, and the contacts with fits. The thirteen hypotheses of `ContactPropagation` are exactly the data of Step 1: $1 \le C_1$, $0 < \kappa$, $u \in \mathcal{A}_{m,C_1}$ centered at z_* with constant C_1, the Jensen bound with constant κ, and the six properties of W. Applying it gives a constant $K \ge 0$, a point $x_{\mathrm{end}} \in F$ and an infinite set $\mathcal{J} \subseteq \{1,2,3,\dots\}$ with

$$H_u(x_{\mathrm{end}}, 0) > -\infty, \tag{895}$$

and such that for every $j \in \mathcal{J}$ there are $x_j, p_j \in F$ and $a_j \in \mathbb{R}$ with (p_j, a_j) a fit for u at (x_j, δ_j) of radius δ_j with constant C_1 in the sense of `IsFit`, and

$$W(x_j,\delta_j) = u(x_j,\delta_j), \qquad \big|x_{\mathrm{end}} - x_j + \delta_j p_j\big| \le K\delta_j. \tag{896}$$

Let $j_1 < j_2 < \cdots$ enumerate $\mathcal{J}$, which is possible because $\mathcal{J}$ is an infinite set of positive integers, and abbreviate

$$r_k = \delta_{j_k} = 2^{-j_k}, \qquad \xi_k = x_{j_k}, \qquad q_k = p_{j_k}, \qquad \alpha_k = a_{j_k}.$$

Then $r_k > 0$ and, since $j_k \ge k$ for every k,

$$0 < r_k \le 2^{-k} \longrightarrow 0 \qquad (k \to \infty). \tag{897}$$

Step 3: the fitted frames. Fix k. The pair (q_k, α_k) is a fit for u at (ξ_k, r_k) of radius r_k with constant C_1, and $r_k > 0$, so `FrameEnvelopeCovariance` applies with $r = r_k$, $x_0 = \xi_k$ and $(p_0, a_0) = (q_k, \alpha_k)$; its hypotheses $1 \le C_1 < \infty$, $0 \le \kappa < \infty$, $u \in \mathcal{A}_{m,C_1}$ and the Jensen bound with constant κ all hold by Step 1. Let $\bar v_k$ be the resulting contact frame,

$$\begin{aligned} s &= \frac{t}{r_k}, \qquad y = \frac{x - \xi_k - q_k(t - r_k)}{r_k}, \qquad x = \xi_k + r_k\big(y + q_k(s-1)\big), \qquad t = r_k s, \\ \bar v_k(y,s) &= \frac{u(x,t) - u(\xi_k, r_k)}{r_k} - q_k \cdot y - \Big(\alpha_k + \frac{|q_k|^2}{2}\Big)(s-1). \end{aligned} \tag{898}$$

By `FrameEnvelopeCovariance` we have $\bar v_k \in \mathcal{A}_{m,C_1}$, it is centered at $(0,1)$ with constant C_1 in the sense of `IsCenteredAt`, it satisfies the horizontal Jensen bound with the same constant κ, and

$$\mathcal{S}^{\pm}_\ell \bar v_k(y,s) = \mathcal{S}^{\pm}_{r_k \ell} u(x,t) - \alpha_k \qquad (\ell > 0) \tag{899}$$

at corresponding points, the slopes being those of `EnvSlopeP` and `EnvSlopeM`. In particular $\bar v_k(0,1) = 0$ for every k.

Put

$$b_k = \frac{x_{\mathrm{end}} - \xi_k + q_k r_k}{r_k} \in F.$$

The point $(b_k, 0)$ corresponds to $(x_{\text{end}}, 0)$ under (898): taking $y = b_k$ and $s = 0$ gives $t = r_k \cdot 0 = 0$ and

$$x = \xi_k + r_k\big(b_k + q_k(0-1)\big) = \xi_k + \big(x_{\text{end}} - \xi_k + q_k r_k\big) - r_k q_k = x_{\text{end}}.$$

By the second part of (896), $|b_k| \le K$ for every k.

Step 4: one limit function and one limit point. The sequence (b_k) lies in the closed ball of radius K in the finite-dimensional space F, which is a closed bounded set and hence compact; a compact metric space is sequentially compact, so some subsequence of (b_k) converges; replacing the enumeration (j_k) by the corresponding subsequence, we may and do assume

$$b_k \longrightarrow b \in F. \tag{900}$$

This replacement preserves everything established so far, since (897)–(899) hold for every k separately.

Every $\bar v_k$ lies in $\mathcal{A}_{m,C_1}$ and is centered at $(0,1)$ with constant C_1, so `ClassCompactness` with $z_c = (0,1)$ provides a strictly increasing $k \mapsto \iota(k)$ and a function v_∞ with $\bar v_{\iota(k)} \to v_\infty$ uniformly on every compact set, $v_\infty \in \mathcal{A}_{m,C_1}$, and v_∞ centered at $(0,1)$ with constant C_1. Along this further subsequence $b_{\iota(k)} \to b$ still, a subsequence of a convergent sequence having the same limit, and $r_{\iota(k)} \to 0$ by (897). To keep the notation light we rename, writing k for $\iota(k)$; thus from now on

$$\bar v_k \to v_\infty \text{ locally uniformly}, \qquad b_k \to b, \qquad r_k \to 0, \qquad v_\infty \in \mathcal{A}_{m,C_1} \text{ centered at } (0,1). \tag{901}$$

In particular $v_\infty(0,1) = 0$, because a function centered at $(0,1)$ vanishes there by `IsCenteredAt`; v_∞ is continuous by `IsFitClass`; and by `UniformSubquadratic` with centering point $(0,1)$ and constant C_1 it has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ in the sense of `SubquadraticGrowth`.

Step 5: the two growth hypotheses that the stability lemmas need. We check once and for all the hypotheses of `QuadEnvStability` and `BaryEnvStability` for the sequence $(\bar v_k)$ and the limit v_∞ with $I = \mathbb{R}$, so Ω is all of space-time. Each $\bar v_k$ and v_∞ is continuous on Ω by `IsFitClass`; the convergence $\bar v_k \to v_\infty$ is uniform on every compact subset of Ω by (901); and v_∞ has subquadratic spatial growth on compact time intervals of $\mathbb{R}$ by Step 4. For the remaining hypothesis, the common growth control of `UnifSubquadraticGrowth` on $I = \mathbb{R}$: let $J \subseteq \mathbb{R}$ be compact and let $\varepsilon > 0$. Every $\bar v_k$ belongs to $\mathcal{A}_{m,C_1}$ and is centered at $(0,1)$ with constant C_1, so `UniformSubquadratic` with $z_c = (0,1)$ supplies an $M > 0$, depending only on J, ε, C_1 and m, with $|w(x,t)| \le \varepsilon |x|^2$ for every member w of $\mathcal{A}_{m,C_1}$ centered at $(0,1)$ and all $|x| \ge M$, $t \in J$. Applying this to each $\bar v_k$ gives $|\bar v_k(x,t)| \le \varepsilon |x|^2$ for every k and all $|x| \ge M$, $t \in J$, with M independent of k. That is exactly the condition of `UnifSubquadraticGrowth` for the sequence on $I = \mathbb{R}$.

Consequently, for every fixed $\ell > 0$, `QuadEnvStability` gives

$$\mathcal{P}_\ell \bar v_k \to \mathcal{P}_\ell v_\infty \quad \text{and} \quad \mathcal{M}_\ell \bar v_k \to \mathcal{M}_\ell v_\infty \tag{902}$$

uniformly on every compact set of centers, all centers being admissible because $I = \mathbb{R}$; and for $s = 1$, `BaryEnvStability` gives

$$\mathcal{C}_1 \bar v_k \to \mathcal{C}_1 v_\infty \tag{903}$$

uniformly on every compact set of centers.

Step 6: evaluation along the moving points. Let g_k, g be real functions on space-time with g continuous and $g_k \to g$ uniformly on every compact set. Then $g_k(b_k, 0) \to g(b, 0)$. Indeed, let $\mathcal{C} = \{(x,0) : |x| \le K+1\}$, a compact set containing every $(b_k, 0)$ and $(b, 0)$ by Step 3 and (900); then

$$\big|g_k(b_k,0) - g(b,0)\big| \le \sup_{\mathcal{C}} |g_k - g| + \big|g(b_k,0) - g(b,0)\big|,$$

and both terms tend to 0, the first by uniform convergence on $\mathcal{C}$ and the second by continuity of g and (900).

We apply this three times. First with $g_k = \bar v_k$ and $g = v_\infty$, using (901) and the continuity of v_∞: $\bar v_k(b_k, 0) \to v_\infty(b, 0)$. Second and third with $g_k = \mathcal{P}_\ell \bar v_k$, $g = \mathcal{P}_\ell v_\infty$ and with $g_k = \mathcal{M}_\ell \bar v_k$, $g = \mathcal{M}_\ell v_\infty$ at a fixed $\ell > 0$, using (902) and the continuity of $\mathcal{P}_\ell v_\infty$ and $\mathcal{M}_\ell v_\infty$, which is part (2)

of `QuadEnvAttained` applied to v_∞ with $I = \mathbb{R}$ — its hypotheses of continuity and subquadratic growth hold by Step 4. Hence $\mathcal{P}_\ell \bar{v}_k(b_k, 0) \to \mathcal{P}_\ell v_\infty(b, 0)$ and $\mathcal{M}_\ell \bar{v}_k(b_k, 0) \to \mathcal{M}_\ell v_\infty(b, 0)$. By `EnvSlopeP` and `EnvSlopeM`, which express the slopes as $\mathcal{S}^+_\ell f = (\mathcal{P}_\ell f - f)/\ell$ and $\mathcal{S}^-_\ell f = (f - \mathcal{M}_\ell f)/\ell$ with the fixed positive factor ℓ,

$$\mathcal{S}^\pm_\ell \bar{v}_k(b_k, 0) \longrightarrow \mathcal{S}^\pm_\ell v_\infty(b, 0) \qquad (\ell > 0 \text{ fixed}). \tag{904}$$

Step 7: the shifts cancel and the limiting slopes are constant. By (895) and part (3) of `EnvHamiltonianSolution` applied to u with $I = \mathbb{R}$, which says that the Hamiltonian never takes the value $+\infty$, the value

$$H_\star = H_u(x_{\mathrm{end}}, 0)$$

is a real number. By part (2) of `EnvHamiltonianSolution` for u with $I = \mathbb{R}$, whose admissibility hypothesis $t + \sigma \in \mathbb{R}$ for all $\sigma \in \mathbb{R}$ is vacuous, the two monotone small-length limits exist and agree with the Hamiltonian:

$$\lim_{\lambda \downarrow 0} \mathcal{S}^+_\lambda u(x_{\mathrm{end}}, 0) = \lim_{\lambda \downarrow 0} \mathcal{S}^-_\lambda u(x_{\mathrm{end}}, 0) = H_\star \in \mathbb{R}.$$

Both are limits of real numbers to a finite value: each $\mathcal{S}^\pm_\lambda u(x_{\mathrm{end}}, 0)$ is a real number because, by part (1) of `QuadEnvAttained` for u with $I = \mathbb{R}$, the suprema and infima defining $\mathcal{P}_\lambda u$ and $\mathcal{M}_\lambda u$ are attained, and the two families are monotone in λ by part (1) of `EnvHamiltonianSolution`.

Fix $\ell > 0$. Evaluating (899) at the corresponding pair $(b_k, 0) \leftrightarrow (x_{\mathrm{end}}, 0)$ of Step 3, once at the length ℓ with either sign and once at the length 1 with the plus sign, and subtracting, the shift α_k cancels:

$$\mathcal{S}^\pm_\ell \bar{v}_k(b_k, 0) - \mathcal{S}^+_1 \bar{v}_k(b_k, 0) = \mathcal{S}^\pm_{r_k \ell} u(x_{\mathrm{end}}, 0) - \mathcal{S}^+_{r_k} u(x_{\mathrm{end}}, 0). \tag{905}$$

As $k \to \infty$ we have $r_k \to 0$ by (901), hence $r_k \ell \to 0$ and $r_k \to 0$ with $r_k \ell > 0$ and $r_k > 0$; so both terms on the right converge to $H_\star$ and the right-hand side tends to 0. By (904), applied at the length ℓ with either sign and at the length 1 with the plus sign, the left-hand side tends to $\mathcal{S}^\pm_\ell v_\infty(b, 0) - \mathcal{S}^+_1 v_\infty(b, 0)$. A sequence has at most one limit, so

$$\mathcal{S}^\pm_\ell v_\infty(b, 0) = \mathcal{S}^+_1 v_\infty(b, 0) \qquad (\ell > 0).$$

Note that this argument uses each fixed length ℓ along the one locally convergent subsequence fixed in Step 4, and that no convergence of the individual shifts α_k is required, only their cancellation in (905).

Put $B = \mathcal{S}^+_1 v_\infty(b, 0)$, a real number, since $\mathcal{P}_1 v_\infty(b, 0)$ is a real number by part (1) of `QuadEnvAttained` for v_∞ with $I = \mathbb{R}$, which asserts that the defining supremum is attained. Then

$$\mathcal{S}^+_\ell v_\infty(b, 0) = \mathcal{S}^-_\ell v_\infty(b, 0) = B \qquad (\ell > 0). \tag{906}$$

Step 8: the limit is affine. `SlopeRigidity`, applied with the constant $C_1 \geq 1$, the function $v_\infty \in \mathcal{A}_{m,C_1}$, the point $(b, t_0) = (b, 0)$ and the number B, whose hypothesis (906) we have just verified, shows that v_∞ is affine in the sense of `IsSpaceTimeAffine`: there are $\varrho \in F$ and $\beta, c \in \mathbb{R}$ with

$$v_\infty(x, t) = \langle \varrho, x \rangle + \beta t + c \qquad \text{for all } (x, t). \tag{907}$$

Step 9: the contacts keep a positive envelope gap in the limit. Fix k. By the first part of (896) at the level j_k we have $W(\xi_k, r_k) = u(\xi_k, r_k)$, and $r_k > 0$, so (E5) of Step 1 gives

$$\frac{\mathcal{C}_{r_k} u(\xi_k, r_k) - u(\xi_k, r_k)}{r_k} = \frac{\kappa^2}{2}. \tag{908}$$

The hypotheses of `FrameGapCovariance` are the same data as those of `FrameEnvelopeCovariance` in Step 3, namely $1 \leq C_1 < \infty$, $0 \leq \kappa < \infty$, $u \in \mathcal{A}_{m,C_1}$ with the Jensen bound at constant κ, $r = r_k > 0$, $x_0 = \xi_k$ and the fit (q_k, α_k) for u at (ξ_k, r_k) of radius r_k; and the frame $\bar{v}_k$ there is the same function (898). Its first identity, at the parameter $\sigma = 1$ and at the

corresponding pair $(y,s)=(0,1)\leftrightarrow(x,t)=(\xi_k,r_k)$ — the correspondence holding because $x=\xi_k+r_k(0+q_k(1-1))=\xi_k$ and $t=r_k\cdot 1=r_k$ — reads

$$\mathcal{C}_1\bar v_k(0,1)-\bar v_k(0,1)=\frac{\mathcal{C}_{r_k}u(\xi_k,r_k)-u(\xi_k,r_k)}{r_k}=\frac{\kappa^2}{2},$$

using (908). Since $\bar v_k(0,1)=0$ by Step 3,

$$\mathcal{C}_1\bar v_k(0,1)=\frac{\kappa^2}{2}\qquad\text{for every } k. \tag{909}$$

By (903) the functions $\mathcal{C}_1\bar v_k$ converge to $\mathcal{C}_1 v_\infty$ uniformly on every compact set of centers, in particular at the single center $(0,1)$, so letting $k\to\infty$ in (909) and using $v_\infty(0,1)=0$ from Step 4,

$$\mathcal{C}_1 v_\infty(0,1)-v_\infty(0,1)=\frac{\kappa^2}{2}>0, \tag{910}$$

the strict positivity because $\kappa>0$.

Step 10: an affine function has no barycentric gap, and the contradiction. Fix a time t and a point $x\in F$, and let (θ,y) be any finite horizontal probability configuration with mean x in the sense of `IsBaryConfig`, so $\theta_i>0$, $\sum_i\theta_i=1$ and $\sum_i\theta_i y_i=x$. By (907),

$$\sum_i\theta_i v_\infty(y_i,t)=\Big\langle\varrho,\sum_i\theta_i y_i\Big\rangle+(\beta t+c)\sum_i\theta_i=\langle\varrho,x\rangle+\beta t+c=v_\infty(x,t),$$

the spatial part being linear and the time part constant along the slice. Hence the competitor value appearing in `BaryEnv` at the parameter $s=1$ is

$$\sum_i\theta_i v_\infty(y_i,t)-\frac{\sum_i\theta_i|y_i-x|^2}{2}=v_\infty(x,t)-\frac{\sum_i\theta_i|y_i-x|^2}{2}\le v_\infty(x,t),$$

because $\sum_i\theta_i|y_i-x|^2$ is the square of the radius of `BaryRadius` and is therefore nonnegative. So the set of competitor values is bounded above by $v_\infty(x,t)$. It also contains $v_\infty(x,t)$: the single-point configuration $n=1$, $\theta_1=1$, $y_1=x$ is admissible by `IsBaryConfig` and its radius is 0. A nonempty set of reals that is bounded above by one of its own elements has that element as supremum, so by `BaryEnv`

$$\mathcal{C}_1 v_\infty(x,t)=v_\infty(x,t)\qquad\text{for all } (x,t).$$

Taking $(x,t)=(0,1)$ and comparing with (910) gives $0=\kappa^2/2>0$, which is absurd.

This contradiction refutes the assumption of Step 0 that v is not affine. Hence v is affine in the sense of `IsSpaceTimeAffine`. □

3.2. `MainEstimate` (paper, Theorem 1.1).

Theorem 3.2 (Uniform interior estimate)**.** *Fix an integer $d\ge 3$. Then there are real numbers α and C with*

$$0<\alpha\le\tfrac13,\qquad 0<C<\infty,$$

depending only on d, with the following property. Let $u:\mathbb{R}^d\to\mathbb{R}$ be infinity-harmonic on the unit ball $B_1\subseteq\mathbb{R}^d$ in the sense of `IsInfHarmonicOn`, and assume that $u(B_1)$ is bounded above and bounded below. Then there is a map $g:\mathbb{R}^d\to\mathbb{R}^d$ such that

(i) *u is differentiable at every $x\in B_{1/2}$ with gradient $g(x)$;*
(ii) *the set $\{|g(x)|:x\in B_{1/2}\}$ is bounded above;*
(iii) *the set $Q_\alpha(g;B_{1/2})$ of Hölder quotients of `HolderQuotSet` is bounded above;*
(iv) *with the Hölder seminorm of `HolderSeminormOn` and the oscillation of `OscOn`,*

$$\sup_{x\in B_{1/2}}|g(x)|+[g]_{C^{0,\alpha}(B_{1/2})}\le C\,\operatorname{osc}_{B_1}u. \tag{911}$$

Conclusions (i) and (iii) together are exactly the membership $u\in C^{1,\alpha}(B_{1/2})$, and (ii) and (iii) are the boundedness hypotheses that the two suprema in (911) require.

Proof. Write $n = d+1$ and identify $\mathbb{R}^n$ with the space-time $E = \ell^2(\mathbb{R}^d \times \mathbb{R})$, whose added unit direction is $e_b = (0,1)$. Let

$$\iota : \mathbb{R}^d \to E, \qquad \iota(y) = (y, 0),$$

which is linear and isometric, since $|\iota(y)|^2 = |y|^2 + 0^2 = |y|^2$, and satisfies

$$\langle P, \iota(h)\rangle = \langle P_x, h\rangle \qquad \big(P = (P_x, P_t) \in E,\ h \in \mathbb{R}^d\big), \tag{912}$$

because the inner product of E is $\langle (a,s),(b,t)\rangle = \langle a, b\rangle + st$. Write $\mathcal{B} = \overline{B}_{1/2} \subseteq E$ for the closed half ball of the space-time, $G = B_{3/4} \subseteq E$ and $D(z) = \operatorname{dist}(z, \partial G)$, and let $\mathcal{U}$ be the normalized family of `NormalizedInfFamily` in dimension d. Since $d \geq 3$ we have $d \geq 1$, which is the hypothesis the cited lemmas in dimension d carry.

Step 1: the constants, all depending only on d. By `PowerDecay` there are $B \in (0,\infty)$, $\gamma \in (0,1]$ and $s_0 \in (0, 1/8]$ with

$$b_d(s) \leq B s^{\gamma} \qquad (0 < s \leq s_0), \tag{913}$$

b_d being the family flatness modulus of `FlatnessModulus`. By `NormalizedFamilyLipschitz` there is $L \geq 0$ such that every $U \in \mathcal{U}$ satisfies

$$|U(a) - U(b)| \leq L|a-b| \qquad \big(a, b \in \overline{B}_{7/8} \subseteq E\big). \tag{914}$$

Put

$$d_0 = \tfrac{1}{4}, \qquad A_0 = B\,4^{\gamma} + 1, \qquad r_0 = \min\Big\{\tfrac{1}{4}\min\{s_0, \tfrac{1}{8}\},\ (2A_0)^{-1/\gamma},\ \tfrac{3}{8}\Big\},$$

and

$$\alpha = \frac{\gamma}{2+\gamma}, \qquad \lambda(r) = A_0 r^{\gamma}, \qquad \rho(r) = \sqrt{A_0}\, r^{1+\gamma/2} = \sqrt{\lambda(r)}\, r, \qquad \rho_0 = \rho(r_0).$$

All of these depend only on d. We record their elementary properties. Since $B > 0$ and $4^{\gamma} > 0$ we have $A_0 > B\,4^{\gamma} = B d_0^{-\gamma}$ and $A_0 \geq 1$; $r_0 > 0$ because each of the three entries is positive; and $r_0 \leq (2A_0)^{-1/\gamma}$ gives, the map $r \mapsto r^{\gamma}$ being increasing on $(0,\infty)$,

$$0 < \lambda(r) \leq \lambda(r_0) = A_0 r_0^{\gamma} \leq A_0 \cdot \frac{1}{2A_0} = \frac{1}{2} < 1 \qquad (0 < r \leq r_0). \tag{915}$$

Consequently $\rho(r) = \sqrt{\lambda(r)}\, r < r$ for $0 < r \leq r_0$, so $0 < \rho_0 < r_0 \leq 3/8$. Finally $\gamma \in (0,1]$ makes $\alpha = \gamma/(2+\gamma)$ positive and, the map $t \mapsto t/(2+t)$ being increasing on $(0,\infty)$, $\alpha \leq 1/(2+1) = 1/3$; in particular $0 < \alpha \leq 1/3 \leq 1$.

Step 2: $D(z) \geq d_0$ on $\mathcal{B}$. Let $z \in \mathcal{B}$, so $|z| \leq 1/2$. In the normed space E the closure of $B_{3/4}$ is $\overline{B}_{3/4}$ and its interior is $B_{3/4}$, so $\partial G = \{w \in E : |w| = 3/4\}$. This set is nonempty: E has dimension $n = d+1 \geq 4 \geq 1$, so it contains a unit vector e, and $\frac{3}{4}e \in \partial G$. For every $w \in \partial G$ the triangle inequality gives $|z - w| \geq |w| - |z| \geq 3/4 - 1/2 = 1/4$. Taking the infimum over the nonempty set ∂G,

$$D(z) \geq d_0 = \tfrac{1}{4} > 0 \qquad (z \in \mathcal{B}). \tag{916}$$

Also $|z| \leq 1/2 < 3/4$, so $z \in G$.

Step 3: $\lambda(r)$ is good at every center of $\mathcal{B}$ and every radius up to r_0. Fix $U \in \mathcal{U}$, $z \in \mathcal{B}$ and $0 < r \leq r_0$, and put $s = r/D(z)$, which is legitimate by (916). Then $s > 0$, and the first entry in the definition of r_0 reads $r_0 \leq \frac{1}{4}\min\{s_0, \frac{1}{8}\} = d_0 \min\{s_0, \frac{1}{8}\}$, so, using $D(z) \geq d_0 > 0$ and $r \leq r_0$,

$$s = \frac{r}{D(z)} \leq \frac{r}{d_0} \leq \frac{r_0}{d_0} \leq \min\{s_0, \tfrac{1}{8}\} \leq s_0.$$

The triple (U, z, r) is admissible for the supremum defining $b_d(s)$ in `FlatnessModulus`: $U \in \mathcal{U}$, $z \in G$ by Step 2, $r > 0$, and $r \leq s\,D(z) = r$. Moreover that supremum is over a set of values $\beta_U(z,r)$, each of which lies in $[0,1]$ by `BetaFlatness`, so the set is bounded above by 1 and the supremum dominates each of its members. Hence

$$\beta_U(z,r) \leq b_d(s) \leq B s^{\gamma} \leq B\Big(\frac{r}{d_0}\Big)^{\gamma} = B d_0^{-\gamma} r^{\gamma} < A_0 r^{\gamma} = \lambda(r),$$

using (913) at $s \le s_0$, then $s \le r/d_0$ with $t \mapsto t^\gamma$ increasing, and finally $A_0 > B d_0^{-\gamma}$ together with $r^\gamma > 0$. By (915) we also have $\lambda(r) < 1$, so $\lambda(r) \in (\beta_U(z,r), 1)$ and `BetaFlatness` makes $\lambda(r)$ good for U at (z,r): by `IsCylinderFit` there is $P \neq 0$ with

$$|U(Z) - U(z) - P \cdot (Z - z)| \le \lambda(r)\, r \qquad \big(Z \in Z_{\lambda(r)}(z,r;P)\big). \tag{917}$$

Step 4: the fit holds on the inscribed ball. Keep the data of Step 3 and write $\lambda = \lambda(r)$, $N_P = P/|P|$, which is a unit vector because $P \neq 0$. If $h \in E$ has $|h| \le \rho(r) = \sqrt{\lambda}\, r$, then by Cauchy–Schwarz the axial component satisfies $|h \cdot N_P| \le |h| \le \sqrt{\lambda}\, r \le r$, using $\sqrt{\lambda} \le 1$ from (915), and the transverse component satisfies

$$\big|h - (h \cdot N_P) N_P\big|^2 = |h|^2 - (h \cdot N_P)^2 \le |h|^2 \le \big(\sqrt{\lambda}\, r\big)^2,$$

so $|h - (h \cdot N_P) N_P| \le \sqrt{\lambda}\, r$. By `FitCylinder` this says $z + h \in Z_\lambda(z,r;P)$. Hence (917) gives

$$\big|U(z+h) - U(z) - P \cdot h\big| \le \lambda(r)\, r = A_0 r^{1+\gamma} \qquad \big(|h| \le \rho(r)\big). \tag{918}$$

Note that $z + h$ is unconstrained apart from $|h| \le \rho(r)$; in particular it need not lie in $\mathcal{B}$.

Step 5: the dyadic scales and the limiting slope. Fix $U \in \mathcal{U}$ and $z \in \mathcal{B}$. Put

$$r_j = 2^{-j} r_0, \qquad \lambda_j = \lambda(r_j), \qquad \rho_j = \rho(r_j) = \sqrt{A_0}\, r_j^{1+\gamma/2} \qquad (j \ge 0),$$

so $0 < r_j \le r_0$, and by Step 3 choose, for each $j \ge 0$, a vector $P_j = P_j(z)$ realizing the cylinder fit of parameter λ_j for U at (z, r_j); by Step 4 it satisfies

$$\big|U(z+h) - U(z) - P_j \cdot h\big| \le A_0 r_j^{1+\gamma} \qquad (|h| \le \rho_j). \tag{919}$$

The sequences r_j, λ_j, ρ_j are strictly decreasing and tend to 0, and

$$\rho_{j+1} = \sqrt{A_0}\Big(\frac{r_j}{2}\Big)^{1+\gamma/2} = 2^{-(1+\gamma/2)} \sqrt{A_0}\, r_j^{1+\gamma/2}. \tag{920}$$

Compare two consecutive fits on the smaller ball. Let $|h| \le \rho_{j+1}$; then also $|h| \le \rho_j$, so (919) at j and at $j+1$ and the triangle inequality give

$$\big|(P_j - P_{j+1}) \cdot h\big| \le A_0 r_j^{1+\gamma} + A_0 r_{j+1}^{1+\gamma} \le 2 A_0 r_j^{1+\gamma},$$

using $r_{j+1} \le r_j$. If $P_j \neq P_{j+1}$, take $h = \rho_{j+1}(P_j - P_{j+1})/|P_j - P_{j+1}|$, which has $|h| = \rho_{j+1}$ and makes the left side $\rho_{j+1}|P_j - P_{j+1}|$; dividing by $\rho_{j+1} > 0$ and using (920),

$$|P_j - P_{j+1}| \le \frac{2A_0 r_j^{1+\gamma}}{\rho_{j+1}} = \frac{2A_0 r_j^{1+\gamma}\, 2^{1+\gamma/2}}{\sqrt{A_0}\, r_j^{1+\gamma/2}} = C_1 r_j^{\gamma/2}, \qquad C_1 = 2^{2+\gamma/2}\sqrt{A_0}, \tag{921}$$

and if $P_j = P_{j+1}$ the bound is trivial. Put $q = 2^{-\gamma/2} \in (0,1)$, so $r_j^{\gamma/2} = q^j r_0^{\gamma/2}$ and, for $k \ge j$, $r_k^{\gamma/2} = q^{k-j} r_j^{\gamma/2}$. Hence

$$\sum_{k \ge j} |P_k - P_{k+1}| \le C_1 r_j^{\gamma/2} \sum_{i \ge 0} q^i = C_2 r_j^{\gamma/2}, \qquad C_2 = \frac{C_1}{1-q} < \infty,$$

the geometric series converging because $0 < q < 1$. In particular the tails of $\sum_k (P_k - P_{k+1})$ tend to 0, since $r_j^{\gamma/2} = q^j r_0^{\gamma/2} \to 0$, so for $i \ge j$ we have $|P_i - P_j| \le \sum_{k \ge j} |P_k - P_{k+1}| \le C_2 r_j^{\gamma/2}$ and $(P_j)_{j \ge 0}$ is a Cauchy sequence in E. It converges: being Cauchy it is bounded, hence contained in some closed ball of E, which is compact because E is finite-dimensional, so some subsequence converges, and a Cauchy sequence with a convergent subsequence converges to the same limit. Let $P_\infty = P_\infty(z)$ be that limit. Summing the telescoping series from j onwards,

$$|P_j - P_\infty| \le \sum_{k \ge j} |P_k - P_{k+1}| \le C_2 r_j^{\gamma/2} \qquad (j \ge 0). \tag{922}$$

The constants C_1 and C_2 depend only on A_0 and γ, hence only on d; they do not depend on U or z.

Step 6: the uniform Taylor estimate with exponent $1 + \alpha$. Keep U and z and let $h \in E$ with $|h| \le \rho_0$. If $h = 0$ the estimate below is $0 \le 0$, so assume $h \neq 0$. The sequence (ρ_j) is strictly

decreasing to 0 and $\rho_0 \ge |h| > 0$, so $\{j \ge 0 : \rho_j \ge |h|\}$ is a nonempty finite set; let j be its largest element, so that

$$\rho_{j+1} < |h| \le \rho_j. \tag{923}$$

By (919), (922) and Cauchy–Schwarz,

$$\big|U(z+h) - U(z) - P_\infty \cdot h\big| \le \big|U(z+h) - U(z) - P_j \cdot h\big| + \big|(P_j - P_\infty)\cdot h\big| \le A_0 r_j^{1+\gamma} + C_2 r_j^{\gamma/2}|h|.$$

From (923) and (920), $2^{-(1+\gamma/2)}\sqrt{A_0}\, r_j^{1+\gamma/2} < |h|$, that is

$$r_j^{1+\gamma/2} < \frac{2^{1+\gamma/2}}{\sqrt{A_0}}\,|h|. \tag{924}$$

Multiplying (924) by $A_0 r_j^{\gamma/2} > 0$ and using $r_j^{1+\gamma} = r_j^{\gamma/2}\, r_j^{1+\gamma/2}$,

$$A_0 r_j^{1+\gamma} < 2^{1+\gamma/2}\sqrt{A_0}\, r_j^{\gamma/2}|h|,$$

so with $C_3 = 2^{1+\gamma/2}\sqrt{A_0} + C_2$,

$$\big|U(z+h) - U(z) - P_\infty \cdot h\big| \le C_3 r_j^{\gamma/2}|h|.$$

Finally, $\alpha = \gamma/(2+\gamma) = (\gamma/2)/(1+\gamma/2)$, so $r_j^{\gamma/2} = \big(r_j^{1+\gamma/2}\big)^\alpha$; raising (924) to the power $\alpha > 0$, which preserves the inequality between positive numbers,

$$r_j^{\gamma/2} \le \Big(\frac{2^{1+\gamma/2}}{\sqrt{A_0}}\Big)^\alpha |h|^\alpha.$$

Therefore, with $C_4 = C_3\big(2^{1+\gamma/2}A_0^{-1/2}\big)^\alpha$,

$$\big|U(z+h) - U(z) - P_\infty(z)\cdot h\big| \le C_4|h|^{1+\alpha} \qquad \big(z \in \mathcal{B},\ |h| \le \rho_0\big), \tag{925}$$

and C_4 depends only on d.

Step 7: the limiting slope is bounded by $L+1$. Keep U and z, and fix $j \ge 0$. For $|h| \le r_j$ we have $|z+h| \le |z| + |h| \le \frac12 + r_0 \le \frac12 + \frac38 = \frac78$ and $|z| \le \frac12 \le \frac78$, so (914) applies to the pair $a = z+h$, $b = z$ and gives $|U(z+h) - U(z)| \le L|h|$. With $r = r_j > 0$, $\lambda = \lambda_j$ satisfying $0 < \lambda_j \le 1$ by (915), and the fit P_j of Step 5, `CylinderFitSlopeBounds` gives

$$|P_j| \le L + \sqrt{\lambda_j} \le L + 1,$$

again by (915). Letting $j \to \infty$ and using the continuity of the norm,

$$|P_\infty(z)| \le L + 1 \qquad (z \in \mathcal{B}). \tag{926}$$

Step 8: one Hölder constant for the whole family. Put $C_{\mathrm{tay}} = \max\{C_4, L+1\}$, so $C_{\mathrm{tay}} \ge L+1 \ge 1 > 0$, and recall $0 < \alpha \le 1$ and $\rho_0 > 0$ from Step 1. Apply `TaylorToHolderGradient` with the ambient space E, the exponent α, the constant C_{tay} and the radius ρ_0. It produces a real number C' with $0 < C' < \infty$, depending only on C_{tay}, α and ρ_0 — hence only on d — such that, for every $S \subseteq E$, every $\widetilde{U} : E \to \mathbb{R}$ and every $\widetilde{g} : E \to E$ satisfying the Taylor estimate with constant C_{tay} and radius ρ_0 on S and the slope bound $|\widetilde{g}| \le C_{\mathrm{tay}}$ on S, the function $\widetilde{U}$ is differentiable at every point of S with gradient $\widetilde{g}$, and $|\widetilde{g}(z) - \widetilde{g}(z')| \le C'|z - z'|^\alpha$ for all $z, z' \in S$.

This is the point at which the order of quantifiers matters: C' is produced before the data $S, \widetilde{U}, \widetilde{g}$, so one and the same C' serves every member of $\mathcal{U}$.

Now let $U \in \mathcal{U}$ and take $S = \mathcal{B}$, $\widetilde{U} = U$ and $\widetilde{g} = P_\infty$ from Steps 5–7. The Taylor hypothesis holds by (925), since $C_4 \le C_{\mathrm{tay}}$, and the slope hypothesis holds by (926), since $L+1 \le C_{\mathrm{tay}}$. Hence

$$U \text{ is differentiable at every } z \in \mathcal{B} \text{ with gradient } P_\infty(z), \qquad |P_\infty(z) - P_\infty(z')| \le C'|z - z'|^\alpha \quad (z, z' \in \mathcal{B}). \tag{927}$$

Step 9: the time-zero slice. Let $w : \mathbb{R}^d \to \mathbb{R}$ be continuous and infinity-harmonic on $B_1 \subseteq \mathbb{R}^d$ with $w(0) = 0$, with $w(B_1)$ bounded above and below and $\operatorname{osc}_{B_1} w \le 1$, and let

$$U(y, b) = w(y) + b \qquad \big((y, b) \in \mathbb{R}^d \times \mathbb{R}\big).$$

By `NormalizedInfFamily` we then have $U \in \mathcal{U}$: its three clauses are exactly the listed properties of w and the displayed formula for U. Apply Step 8 to this U and set

$$\mathfrak{q}(y) = \big(P_\infty(\iota(y))\big)_x \in \mathbb{R}^d \qquad (y \in \mathbb{R}^d),$$

the spatial component of the limiting slope at the time-zero point. For $y \in B_{1/2} \subseteq \mathbb{R}^d$ we have $|\iota(y)| = |y| < 1/2$, so $\iota(y) \in \mathcal{B}$ and (927) applies at $\iota(y)$.

Fix $y \in B_{1/2}$ and write $P = P_\infty(\iota(y))$. For $h \in \mathbb{R}^d$,

$$U\big(\iota(y) + \iota(h)\big) = U(y + h, 0) = w(y + h), \qquad U(\iota(y)) = w(y), \qquad \langle P, \iota(h)\rangle = \langle \mathfrak{q}(y), h\rangle$$

by (912). Differentiability of U at $\iota(y)$ with gradient P means $U(\iota(y) + k) - U(\iota(y)) - \langle P, k\rangle = o(|k|)$ as $k \to 0$ in E; restricting to $k = \iota(h)$ and using $|\iota(h)| = |h|$ gives

$$w(y + h) - w(y) - \langle \mathfrak{q}(y), h\rangle = o(|h|) \qquad (h \to 0 \text{ in } \mathbb{R}^d),$$

that is, w is differentiable at y with gradient $\mathfrak{q}(y)$. Moreover, for $(a, s) \in E$ one has $|a|^2 \le |a|^2 + s^2 = |(a, s)|^2$, so $|\mathfrak{q}(y)| \le |P_\infty(\iota(y))| \le L + 1$ by (926); and for $y, y' \in B_{1/2}$, since ι is linear and isometric,

$$|\mathfrak{q}(y) - \mathfrak{q}(y')| \le \big|P_\infty(\iota(y)) - P_\infty(\iota(y'))\big| \le C'\big|\iota(y) - \iota(y')\big|^\alpha = C'|y - y'|^\alpha,$$

using (927). Summarizing,

$$\begin{gathered} w \text{ is differentiable at every } y \in B_{1/2} \text{ with gradient } \mathfrak{q}(y), \\ |\mathfrak{q}(y)| \le L + 1, \qquad |\mathfrak{q}(y) - \mathfrak{q}(y')| \le C'|y - y'|^\alpha \qquad (y, y' \in B_{1/2}). \end{gathered} \tag{928}$$

Step 10: the theorem. Put $C = L + 1 + C'$, which is finite and, since $L \ge 0$ and $C' > 0$, satisfies $C \ge 1 > 0$; it depends only on d. We also fix α from Step 1, so $0 < \alpha \le 1/3$. Let u be as in the statement and put $M = \operatorname{osc}_{B_1} u$. The set $u(B_1)$ is nonempty, because $0 \in B_1$, and bounded above and below by hypothesis, so by `OscOn` the numbers $\sup u(B_1)$ and $\inf u(B_1)$ are real, $M = \sup u(B_1) - \inf u(B_1)$, and $M \ge 0$ because $\inf u(B_1) \le \sup u(B_1)$.

The case $M = 0$. Then $\sup u(B_1) = \inf u(B_1) =: c_0$, and for every $x \in B_1$ we have $c_0 = \inf u(B_1) \le u(x) \le \sup u(B_1) = c_0$, so $u \equiv c_0$ on B_1. Take $g \equiv 0$. For $x \in B_{1/2}$ the set B_1 is a neighbourhood of x on which u is constant, so $u(x + h) - u(x) - \langle 0, h\rangle = 0$ for all small h and u is differentiable at x with gradient 0; this is (i). The set $\{|g(x)| : x \in B_{1/2}\} = \{0\}$ is bounded above with supremum 0, which is (ii). Since $d \ge 3 \ge 1$, the ball $B_{1/2} \subseteq \mathbb{R}^d$ contains two distinct points, and by `HolderQuotSet` every element of $Q_\alpha(g; B_{1/2})$ has the form $|g(x) - g(y)|/|x - y|^\alpha = 0/|x - y|^\alpha = 0$ with $|x - y| > 0$; so $Q_\alpha(g; B_{1/2}) = \{0\}$, which is bounded above, giving (iii), and by `HolderSeminormOn` $[g]_{C^{0,\alpha}(B_{1/2})} = \sup\{0\} = 0$. Hence the left side of (911) is $0 + 0 = 0 = C \cdot 0 = C\,M$, which is (iv).

The case $M > 0$. Put $c = u(0)$ and $w = (u - c)/M$. By `InfHarmonicNormalize` with $\Omega = B_1$, the constant c and the factor $M > 0$, the function w is infinity-harmonic on B_1; in particular it is continuous there by `IsInfHarmonicOn`. Also $w(0) = (c - c)/M = 0$. The map $T(t) = (t - c)/M$ is an increasing affine bijection of $\mathbb{R}$, and $w(B_1) = T\big(u(B_1)\big)$; an increasing bijection of $\mathbb{R}$ carries suprema to suprema and infima to infima of nonempty bounded sets, so $w(B_1)$ is bounded above and below with

$$\sup w(B_1) = \frac{\sup u(B_1) - c}{M}, \qquad \inf w(B_1) = \frac{\inf u(B_1) - c}{M},$$

whence, by `OscOn`, $\operatorname{osc}_{B_1} w = \big(\sup u(B_1) - \inf u(B_1)\big)/M = M/M = 1 \le 1$.

So w satisfies all the hypotheses of Step 9. Let $\mathfrak{q}$ be as produced there and put

$$g(x) = M\,\mathfrak{q}(x) \qquad (x \in \mathbb{R}^d).$$

Since $u = Mw + c$ on all of $\mathbb{R}^d$, for $x \in B_{1/2}$ and $h \in \mathbb{R}^d$

$$u(x + h) - u(x) - \langle g(x), h\rangle = M\big(w(x + h) - w(x) - \langle \mathfrak{q}(x), h\rangle\big) = M \cdot o(|h|) = o(|h|),$$

so u is differentiable at every $x \in B_{1/2}$ with gradient $g(x)$, which is (i). By (928) and $M > 0$,

$$|g(x)| \le M(L + 1), \qquad |g(x) - g(x')| \le MC'|x - x'|^\alpha \qquad (x, x' \in B_{1/2}). \tag{929}$$

The first bound shows that $\{|g(x)| : x \in B_{1/2}\}$ is bounded above, which is (ii), and that its supremum is at most $M(L+1)$, the set being nonempty because $0 \in B_{1/2}$. The second bound shows, via `HolderQuotSet`, that every element $|g(x)-g(x')|/|x-x'|^\alpha$ of $Q_\alpha(g; B_{1/2})$ is at most MC', so that set is bounded above — which is (iii) — and nonempty, $B_{1/2}$ having two distinct points; by `HolderSeminormOn` its supremum satisfies $[g]_{C^{0,\alpha}(B_{1/2})} \le MC'$. Adding the two estimates,

$$\sup_{x\in B_{1/2}} |g(x)| + [g]_{C^{0,\alpha}(B_{1/2})} \le M(L+1) + MC' = (L+1+C')M = C \operatorname{osc}_{B_1} u,$$

which is (iv). In both cases α and C were fixed in Step 1 and Step 10 before u was introduced, so they depend only on d. This completes the proof. □

3.3. `InteriorC1` (paper, Corollary 1.2).

Corollary 3.3 (Interior C^1 regularity)**.** *Fix an integer $d \ge 3$, let $\Omega \subseteq \mathbb{R}^d$ be a nonempty connected set and let $u : \mathbb{R}^d \to \mathbb{R}$ be infinity-harmonic on Ω in the sense of `IsInfHarmonicOn`, which in particular makes Ω open and u continuous on Ω. Then u is continuously differentiable on Ω: there is a map $g : \Omega \to \mathbb{R}^d$, continuous on Ω, such that u is differentiable at every $x \in \Omega$ with gradient $g(x)$. Equivalently, since Ω is open, $u \in C^1_{\mathrm{loc}}(\Omega) = C^1(\Omega)$.*

Proof. By `IsInfHarmonicOn` the set Ω is open and u is continuous on Ω. The gradient of u at a point, when it exists, is unique, being the Riesz representative of the Fréchet derivative there; so it suffices to produce, for every $x_0 \in \Omega$, an open ball $V \ni x_0$ with $V \subseteq \Omega$ such that u is differentiable at every point of V with a gradient depending continuously on that point. Indeed, the resulting gradient map is then well defined on all of Ω — any two of the local maps agree on the overlap, both being the gradient of u — it is defined at every point of Ω, and continuity is a local property, so it is continuous on Ω. Note that the hypotheses that Ω is nonempty and connected are not used below; they are carried because the statement being formalized carries them.

Step 1: a good ball and the rescaled function. Fix $x_0 \in \Omega$. Since Ω is open there is $R > 0$ with $B_R(x_0) \subseteq \Omega$. Put $r = R/4 > 0$, so

$$\overline{B}_{2r}(x_0) \subseteq B_R(x_0) \subseteq \Omega,$$

because $|x - x_0| \le 2r = R/2 < R$. Let $A = 2r > 0$, let $T(y) = x_0 + Ay$ and put

$$v = u \circ T, \qquad v(y) = u(x_0 + 2ry).$$

By `InfHarmonicRescale`, applied with this Ω, this u, the point x_0 and the factor $A > 0$, the function v is infinity-harmonic on $T^{-1}(\Omega)$.

Step 2: v is infinity-harmonic on the unit ball. First, $B_1 \subseteq T^{-1}(\Omega)$: if $|y| < 1$ then $|T(y) - x_0| = 2r|y| < 2r$, so $T(y) \in \overline{B}_{2r}(x_0) \subseteq \Omega$.

Next, infinity-harmonicity passes to open subsets. Let $W \subseteq \Omega'$ with W open, Ω' open and f an infinity subsolution on Ω' in the sense of `IsInfSubsolutionOn`. Let $z \in W$ and let ϕ be C^2 near z with $f - \phi$ having a local maximum on W at z, i.e. there is a neighbourhood N of z with $(f-\phi)(x) \le (f-\phi)(z)$ for all $x \in N \cap W$. Then $N \cap W$ is itself a neighbourhood of z, W being open and containing z, and for every $x \in (N \cap W) \cap \Omega'$ we have $x \in N \cap W$ and hence $(f-\phi)(x) \le (f-\phi)(z)$; so $f - \phi$ has a local maximum on Ω' at z, and `IsInfSubsolutionOn` gives $\Delta_\infty \phi(z) \ge 0$. As W is open and f is continuous on $\Omega' \supseteq W$, hence on W, this shows that f is an infinity subsolution on W. The same argument with inequalities reversed, using `IsInfSupersolutionOn`, shows that an infinity supersolution on Ω' is one on W. By `IsInfHarmonicOn` — which is the conjunction of the two — infinity-harmonicity on Ω' implies infinity-harmonicity on every open $W \subseteq \Omega'$.

Applying this with $\Omega' = T^{-1}(\Omega)$, which is open because T is a homeomorphism of $\mathbb{R}^d$ and Ω is open, and $W = B_1$, we conclude that v is infinity-harmonic on B_1 in the sense of `IsInfHarmonicOn`.

Step 3: v is bounded on the unit ball. By `IsInfHarmonicOn` the function u is continuous on Ω, so it is continuous on the compact set $\overline{B}_{2r}(x_0) \subseteq \Omega$, which is closed and bounded in $\mathbb{R}^d$ and

hence compact. By the Weierstrass item of the preamble u attains a maximum and a minimum on it; call them m^+ and m^-. Every point of $v(B_1)$ has the form $u(T(y))$ with $T(y) \in \overline{B}_{2r}(x_0)$, as in Step 2, so $m^- \le u(T(y)) \le m^+$. Thus $v(B_1)$ is bounded above and bounded below.

Step 4: the estimate for v and the Hölder continuity of its gradient. Let α and C be the constants `MainEstimate` provides for this d, so $0 < \alpha \le 1/3$ and $0 < C < \infty$. By Steps 2 and 3 the function v satisfies its hypotheses, so there is a map $g_v : \mathbb{R}^d \to \mathbb{R}^d$ such that v is differentiable at every $y \in B_{1/2}$ with gradient $g_v(y)$, the set $Q_\alpha(g_v; B_{1/2})$ of Hölder quotients of `HolderQuotSet` is bounded above, and the two remaining conclusions hold as well; we use only these two.

Put $K = [g_v]_{C^{0,\alpha}(B_{1/2})}$, the Hölder seminorm of `HolderSeminormOn`, which is the supremum of $Q_\alpha(g_v; B_{1/2})$. That set is nonempty, because $d \ge 3 \ge 1$ makes $B_{1/2} \subseteq \mathbb{R}^d$ contain two distinct points, and bounded above, so K is a real number and dominates every member of the set; also $K \ge 0$, every member being a quotient of nonnegative numbers. By `HolderQuotSet`, for all $y, y' \in B_{1/2}$ with $y \ne y'$,

$$\frac{|g_v(y) - g_v(y')|}{|y - y'|^\alpha} \le K, \qquad \text{hence} \qquad |g_v(y) - g_v(y')| \le K|y - y'|^\alpha,$$

and the last inequality also holds trivially when $y = y'$. Since $\alpha > 0$, the right-hand side tends to 0 as $y' \to y$, so g_v is continuous at every $y \in B_{1/2}$ relative to $B_{1/2}$; as $B_{1/2}$ is open, g_v is continuous on $B_{1/2}$.

Step 5: unscaling. Put $V = B_r(x_0)$, an open ball with $x_0 \in V$ and $V \subseteq \overline{B}_{2r}(x_0) \subseteq \Omega$, and define

$$S(x) = \frac{x - x_0}{2r}, \qquad g(x) = \frac{1}{2r}\, g_v\big(S(x)\big) \qquad (x \in V).$$

For $x \in V$ we have $|S(x)| = |x - x_0|/(2r) < r/(2r) = 1/2$, so $S(x) \in B_{1/2}$ and g is defined on V. Note $T(S(x)) = x_0 + 2r \cdot (x - x_0)/(2r) = x$, so $v(S(x)) = u(x)$, and more generally $v\big(S(x) + k\big) = u(x + 2rk)$ for every $k \in \mathbb{R}^d$, since $T(S(x) + k) = x + 2rk$.

Fix $x \in V$ and put $y = S(x) \in B_{1/2}$. For $h \in \mathbb{R}^d$ write $k = h/(2r)$, so $h = 2rk$ and $|k| = |h|/(2r)$. Then

$$u(x + h) - u(x) - \Big\langle \frac{1}{2r} g_v(y), h \Big\rangle = v(y + k) - v(y) - \big\langle g_v(y), k \big\rangle,$$

because $u(x + h) = v(y + k)$, $u(x) = v(y)$ and $\langle (2r)^{-1} g_v(y), 2rk \rangle = \langle g_v(y), k \rangle$. By Step 4 the right side is $o(|k|)$ as $k \to 0$, and $|k| = |h|/(2r)$ with $2r$ a fixed positive constant, so the left side is $o(|h|)$ as $h \to 0$. Hence u is differentiable at x with gradient $g(x) = (2r)^{-1} g_v(S(x))$.

Finally g is continuous on V: it is the composition of the continuous affine map S, which carries V into $B_{1/2}$, with the map g_v, continuous on $B_{1/2}$ by Step 4, followed by multiplication by the constant $(2r)^{-1}$.

So $V = B_r(x_0)$ is an open ball containing x_0 and contained in Ω on which u is differentiable with continuous gradient. Since $x_0 \in \Omega$ was arbitrary, the reduction at the start of the proof applies and u is continuously differentiable on Ω. $\square$